WORKSHOP MANUAL

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660s AGILE ***770s AGILE***
880s AGILE ***990s AGILE***

660v AGILE ***775v AGILE***

introduction

This publication is intended for the trained technician who must operate on our tractors.

It contains all general information relating to our tractor range, and in particular it highlights the inspection, overhauling and adjustment procedures as well as the main instructions for dismantling and reassembling operations.

The workshop manual is a natural summary for the mechanic who has attended the vocational training and specialization courses, which are held every year at our Service School, to permit him to perform a precise and qualified work on tractor.

Its contents are therefore an exhaustive reference book for the experienced mechanic who desires to refresh his memory on the sequence of the operations to be done. It is then good practice for every authorized dealer mechanic to have at his disposal this publication, so that it may be consulted quickly when necessary.

We wish to thank in advance for the cooperation all thos people, who will let us have their suggestions in order to make this publication more complete.

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9 - APPENDIX

Engine electronic unit I
Power lift tester version XLVII

TRACTOR CONFIGURATIONS:

- *660s 770s 880s 990s AGILE*
- *660v 775v*

660s 770s 880s 990s AGILE 660v 775v”	***2WD*** *WITH PLATFORMA* ***2WD*** *WITH CAB* ***4WD*** *WITH PLATFORMA* ***4WD*** *WITH CAB*

CAB
- ventilation + heating
- with ventilation + heating + air conditioning

GEARBOX Fully synchronised:

For tractors with driving position with footsteps only
- 16 Forward + 8 Reverse: 4 speeds x 2 ranges + mini-reduction + shuttle
- 24 Forward + 12 Reverse: 4 speeds x 3 ranges + mini-reduction + shuttle

For tractors with driving position with plataform/cab only
- 20 Forward 10 Reverse: 5 speeds x 2 ranges (L-V) + mini-reduction-shuttle
- 30 Forward 15 Reverse: 5 speeds x 3 ranges (SR-L-V) + mini-reduction-shuttle
- 45 Forward 45 Reverse: 5 speeds x 3 ranges (L-N-V)+ Reverser + version *MULTISPEED*

CONTROLS
- electro-hydraulic MULTISPEED control with 2 pushbuttons located on gear lever
- rear P.T.O. clutch with electro-hydraulic control and mechanical speed selector
- groundspeed P.T.O. with mechanical engagement/disengagement control
- front P.T.O. with electrohydraulic control
- 4WD with mechanical control or optional electro-hydraulic control
- differential locks with electro-hydraulic control for tractors equipped with platform or cab
- differential locks with mechanical control for tractors with footplates
- with electronic engine throttle (for tractors equipped with electronic governor)

MECHANICALLY OPERATED REAR POWER-LIFT
- with supplementary rams
- without supplementary rams

ELECTRONIC REAR LIFT
- with auxiliary rams
- without auxiliary rams
- with auxiliary control pushbuttons mounted on fender

HYDRAULIC SYSTEM

The hydraulic system is equipped with two pumps:

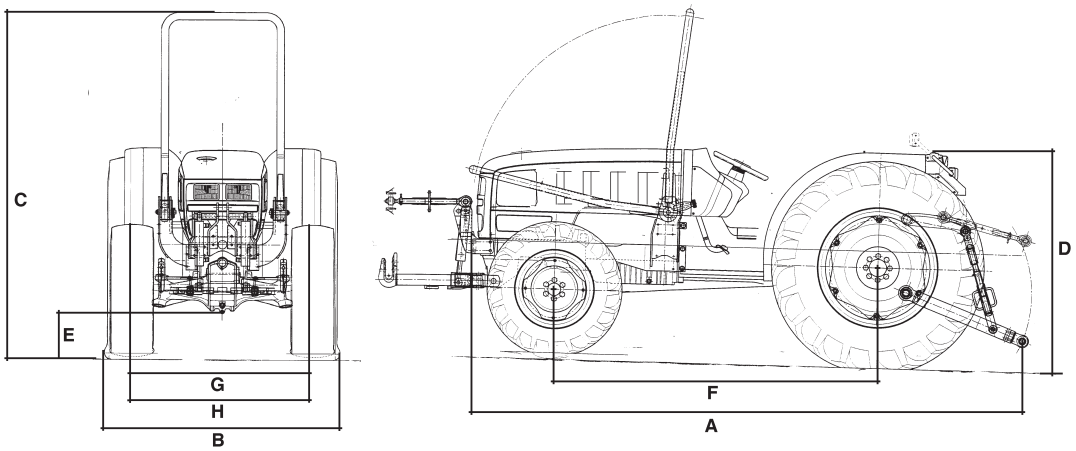
Standard:
- 11 cc pump with 27 l/min capacity. Supplies power steering, electro-hydraulic control unit and gearbox lubrication circuits.
- 14 cc pump with 34 l/min capacity. Supplies auxiliary service control valves and hydraulic lift.

Optional:
- 11 cc pump with 27 l/min capacity. Supplies power steering, MULTISPEED control unit, electro-hydraulic control unit and gearbox lubrication circuits.
- 19 cc pump with 47 l/min capacity. Supplies the hydraulic trailer breaking control valve, auxiliary service control valves and hydraulic lift.

MAIN EQUIPMENT
- front P.T.O.
- Economy P.T.O.
- front lift
- Electronic lift
- Electronic engine governor
- 3-point linkage with hydraulically adjustable right-hand lift rod and stabilisers
- 4-way or 6-way control valves with “Flow Divider”

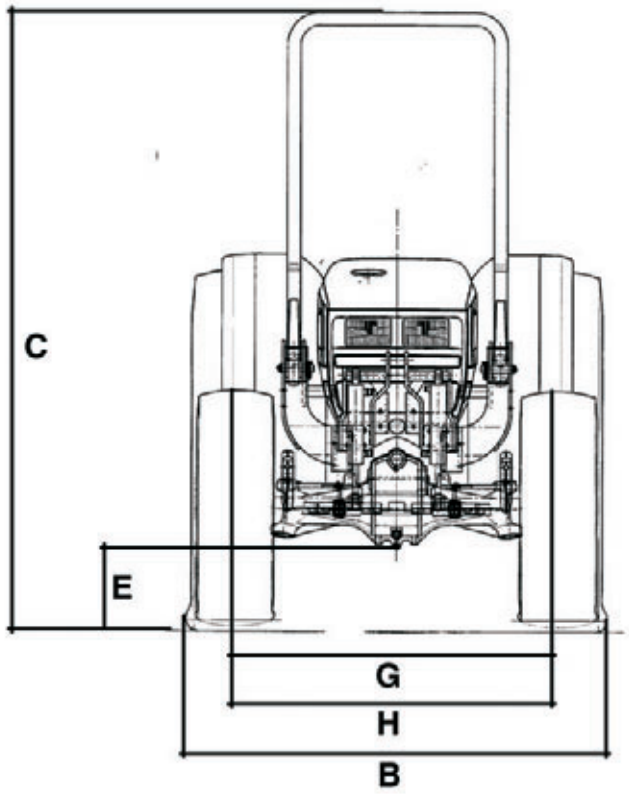
DIMENSIONS AND WEIGHTS

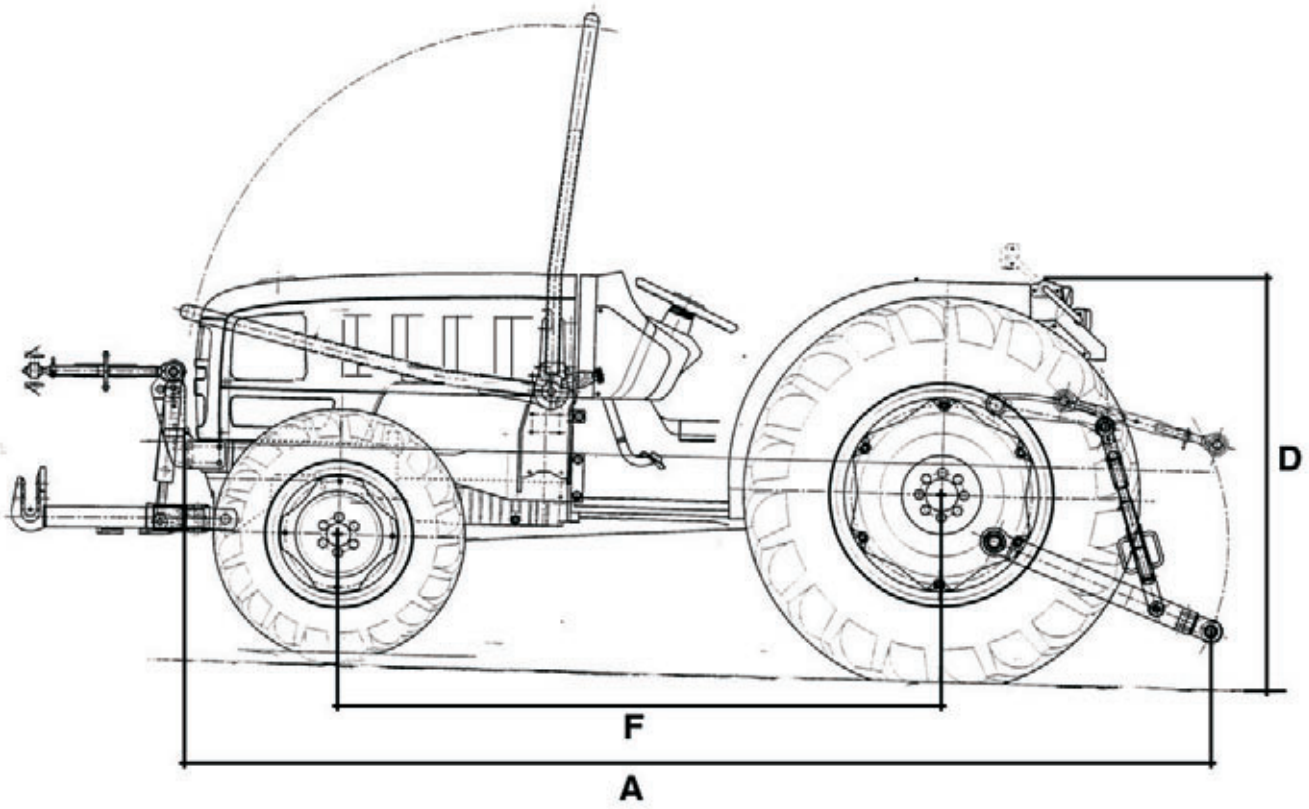
		660v AGILE		775v AGILE	
		2 WD	***4 WD***	***2 WD***	***4 WD***
Tyres:					
- front		7.50-16	8.25/16	7.50/16	280/70R18
- rear		13.6R28	13.6R28	420/70R28	420/70R28
Max. length - without front linkage	**(A)** mm	3000	3000	3150	3150
Min./max. width	(B) mm	1063/1067	1063/1067	1063/1667	1063/1667
Max. height - to hood	**(C)** mm	1185	1185	1185	1185
- to steering wheel	**(D)** mm	1236	1236	1236	1236
Ground clearance	(E) mm	225	225	225	225
Wheelbase	(F) mm	1926	1926	2056	2056
Front track width	(G)				
standard	mm	853	863	853	863
minimum/maximum	mm	803/1079	793/1089	803/1079	793/1089
Rear track width	(H)				
standard	mm	902	902	902	902
minimum/maximum	mm	822/1426	822/1426	822/1426	822/1426
Front tyres		6.50-16	210/80R16	6.50-16	210/80R16
Rear tyres		9.5R28	9.5R28	9.5R28	9.5R28
Minimum turning radius (without brakes)	mm	3060	3060	3230	3230
Kerb weight (with driver platform)	kg	1950	2100	1950	2100
Weight of ballast - front cast iron plates	kg	140 (70x2)	140 (70x2)	140 (70x2)	140 (70x2)
- one-piece block (4WD only)	kg	-	200	-	200



DIMENSIONS AND WEIGHTS

		660s	*770s*	*880s*	*990s*
		2 WD	*4 WD*	*2 WD*	*4 WD*
Max. length					
- without front linkage					
Without ballast	(**A**) mm	3431	3431	3686	3686
With ballast	(**A**) mm	3671	3671	3801	3801
- with front and rear linkage					
Without ballast	(**A**) mm	4036	4036	4201	4201
With ballast	(**A**) mm	4136	4136	4301	4301
Width min.-max	(**B**) mm	1207/1507	1207/1507	1495/1691	1495/1691
Max. height.					
- at safety frame	(**C**) mm	2210	2210	2210	2210
- at cab (standard)	(**C**) mm	2220	2220	2170	2170
Ground clearance	(**E**) mm	230	230	230	230
Wheel base	(**F**) mm	1926	1926	2056	2056
Front track	(**G**)				
- standard	mm	1050	1050	1050	1050
- minimum/maximum	mm	954/1476	954/1476	1076/1476	1076/1476
Rear track	(**H**)				
- standard	mm	974	974	1076	1076
- minimum/maximum	mm	974/1476	974/1476	1076/1476	1076/1476
Front tyres		7.50-16	320/70R20	7.50-16	320/70R20
Rear tyres		420/70R28	420/70R28	420/70R28	420/70R28
Operating weight (with front lift)					
- *with driver platform*					
Without ballast	kg	2150	2180	2290	2370
With ballast	kg	2255	2285	2395	2475
- *with cab*					
Without ballast	kg	2300	2380	2455	2600
With ballast	kg	2405	2485	2560	2705
Operating weight (with front lift)					
- *with driver platform*					
Without ballast	kg	-	2350	-	2540
With ballast	kg	-	2600	-	2790
- *with cab*					
Without ballast	kg	-	2500	-	2730
With ballast	kg	-	2750	-	2970
Min. turning radius (without brakes)	mm	2905	3115	2905	3115
Weight of front ballast					
- front cast iron plates	kg	140 (70x2)	140 (70x2)	140 (70x2)	140 (70x2)
- monolithic block (4WD only)	kg	-	200	-	200


C
E
G
H
B


D
F
A

PRESCRIBED LUBRICANTS AND FUELS

(amounts in litres)

<table>
<tr><th>Part to be supplied</th><th>Amt</th><th>Oil type</th><th>Product</th></tr>
<tr><td>Engine</td><td>6,7
60-70-HP

11
75 - 80 -87HP</td><td>Multigrade engine oil
ACEA E3-96
API CF
SDFG OM-1991A
MIL-L-2104 E leve
SAE 15W40 (15 ÷ +40 °C)
SAE 10W30 -25 ÷ +30 °C)</td><td>AKROS SUPER
Special Formula
LAMBORGHINI</td></tr>
<tr><td>Gearbox and
Rear axle
Power-lift
Auxiliary Systems
Hydrostatic steering</td><td>41</td><td rowspan="3">SDFG OT - 1891
API GL 4

SAE 10W30</td><td rowspan="3">AKROS MULTI</td></tr>
<tr><td>Front P.T.O.</td><td>2,5</td></tr>
<tr><td>Front - wheel drive
Central axle
Side reductions</td><td>6
1,5x2</td></tr>
<tr><td>Brakes control and clutch</td><td>max. level</td><td>ATF DEXRON II</td><td>AKROS MATIC</td></tr>
<tr><td>Lubrication points</td><td></td><td>NLGI 2
LITIO/Ca</td><td>AKROS GREASE T2</td></tr>
<tr><td>Radiator antifreeze (version with water-cooled engine</td><td>11</td><td></td><td>AKROS FREEZE</td></tr>
<tr><td>Fuel tank</td><td colspan="3">For tractors 660s - 660v - 770s - 775v - 880s - 990s AGILE
69 litres

For tractors 660s - 660v - 770s - 775v - 880s - 990s AGILE
For tractors with frontal P.T.O. and front lift
58 litres</td></tr>
</table>

WARNING: The engine oil sump has been replenished after the first 50 work hours.

It is advisable to always use the some type of oil when replenishing.

CONVERSION TABLE FROM

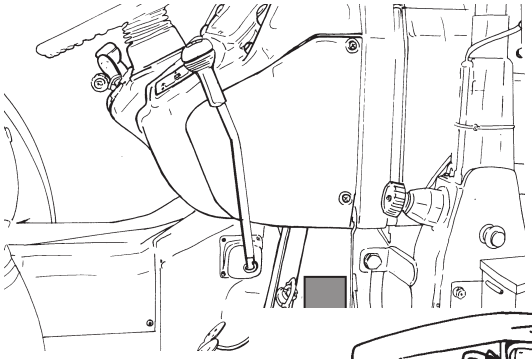
FROM	TO	multiply by:
inch	cm	2.540
cm	inch	0.394
foot	m	0.305
m	foot	3.281
yard	m	0.914
m	yard	1.094
Eng. miles	km	1.609
km	Eng. miles	0.622
Sq.in.	cm2	6.452
cm2	Sq.ft.	0.155
Sq.ft.	m2	0.093
m2	Sq.ft.	10.77
Sq.yard	m2	0.835
m2	Sq.yard	1.197
Cu.in.	cm3	16.39
cm3	Cu.in.	0.061
Cu.ft.	Liter	28.36
Liter	Cu.ft.	0.035
Cu.yard	m3	0.763
m3	Cu.yard	1.311
Imp.gall.	Liter	4.547
Liter	Imp.gall.	0.220
US gall.	Liter	3.785
Liter	US gall.	0.264
pint	Liter	0.568
Liter	pint	1.762
quart	Liter	1.137
Liter	quart	0.880
oz.	kg	0.028
kg	oz.	35.25
lb.	kg	0.454
kg	lb.	2.203
lb.ft.	kgm	0.139
kgm	lb.ft.	7.233
lb/in.	kg/m	17.87
kg/m	lb/in.	0.056
lb./sq.in.	kg/cm2	0.070
kg/cm2	lb/sq.in.	14.22
lb./Imp.gall.	kg/l	0.100
kg/l	lb./Imp.gall.	10.00
lb./US gall.	kg/l	0.120
kg/l	lb./US gall.	8.333
lb./cu.ft.	kg/m3	16.21
kg/m3	lb./cu.ft.	0.062
cu.ft./lb.	m3/kg	0.062
m3/kg	cu.ft./lb.	16.21
Nm	kgm	0.102
kgm	Nm	9.81
kW	PS	1.36
PS	kW	0.736
bar	kg/cm2	1.014
kg/cm2	bar	0.981
dm3	l	1
l	dm3	1

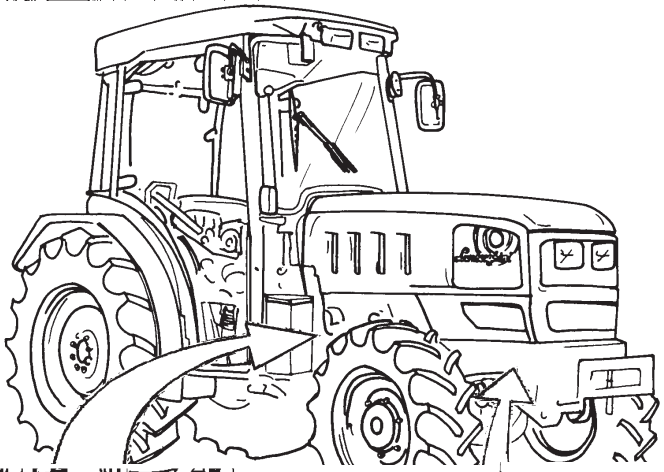
HOW TO ORDER SPARE PARTS

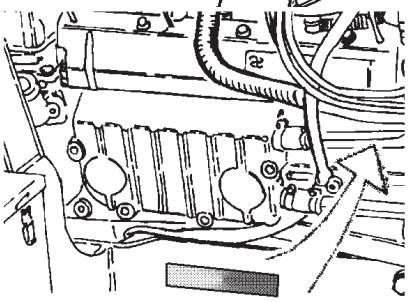
To ensure perfect tractor efficiency thus avoiding serious drawbacks, and to optimize your investment and the operational expenses, the use of “ORIGINAL SPARE PARTS” is recommended.
Spare parts orders must specify the following:

— **Tractor serial number and engine serial number (if the engine is concerned).**
— **Spare part name and reference code.**

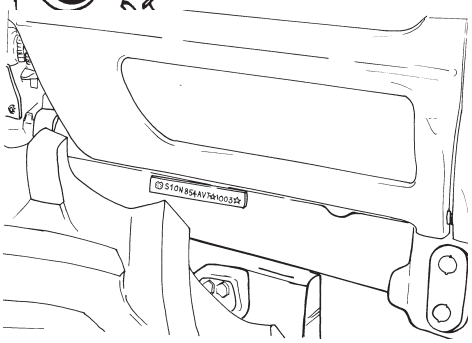
TRACTOR IDENTIFICATION DATA PLATE







ENGINE TYPE AND SERIAL NUMBER



TRACTOR FRAME TYPE AND SERIAL NUMBER

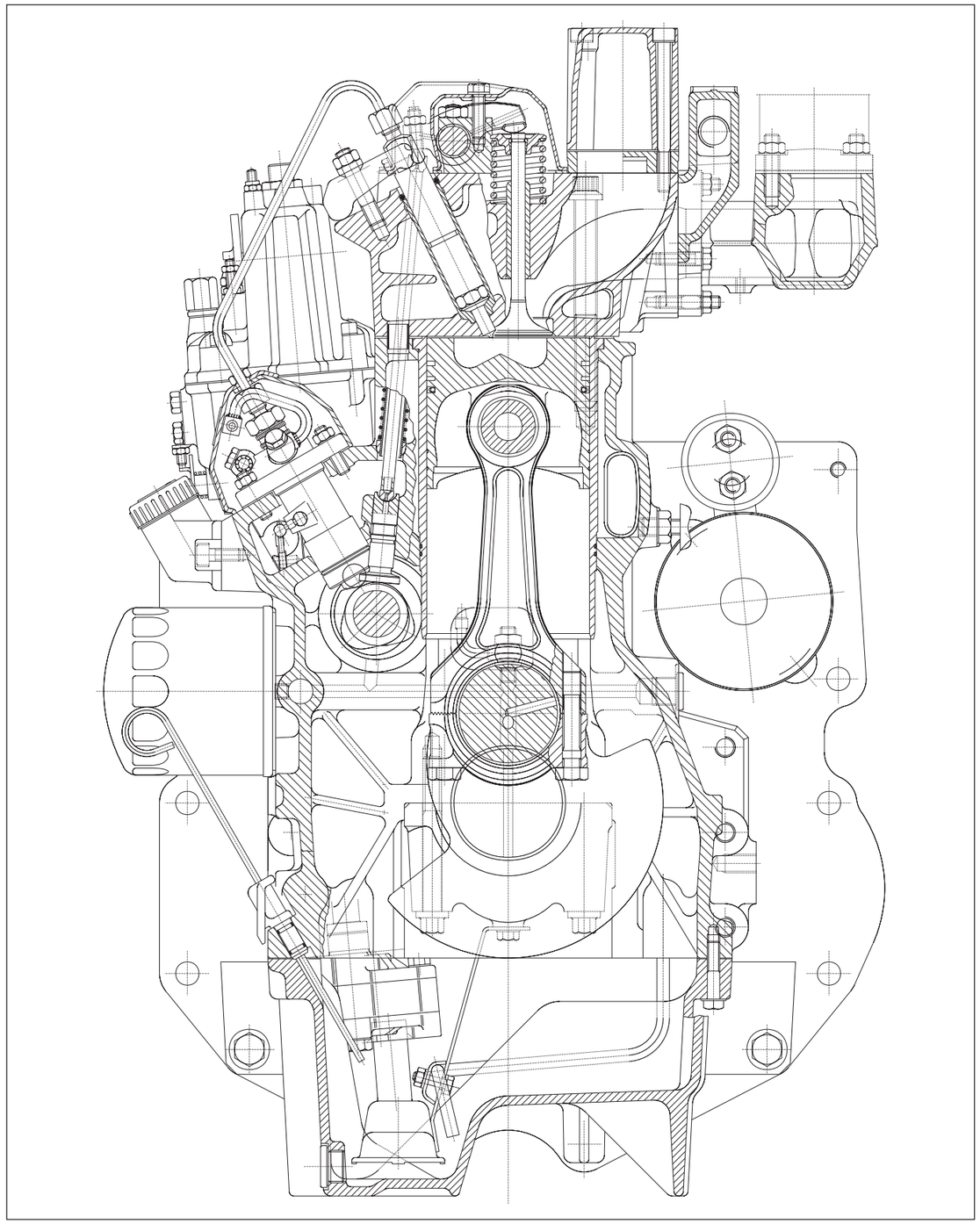
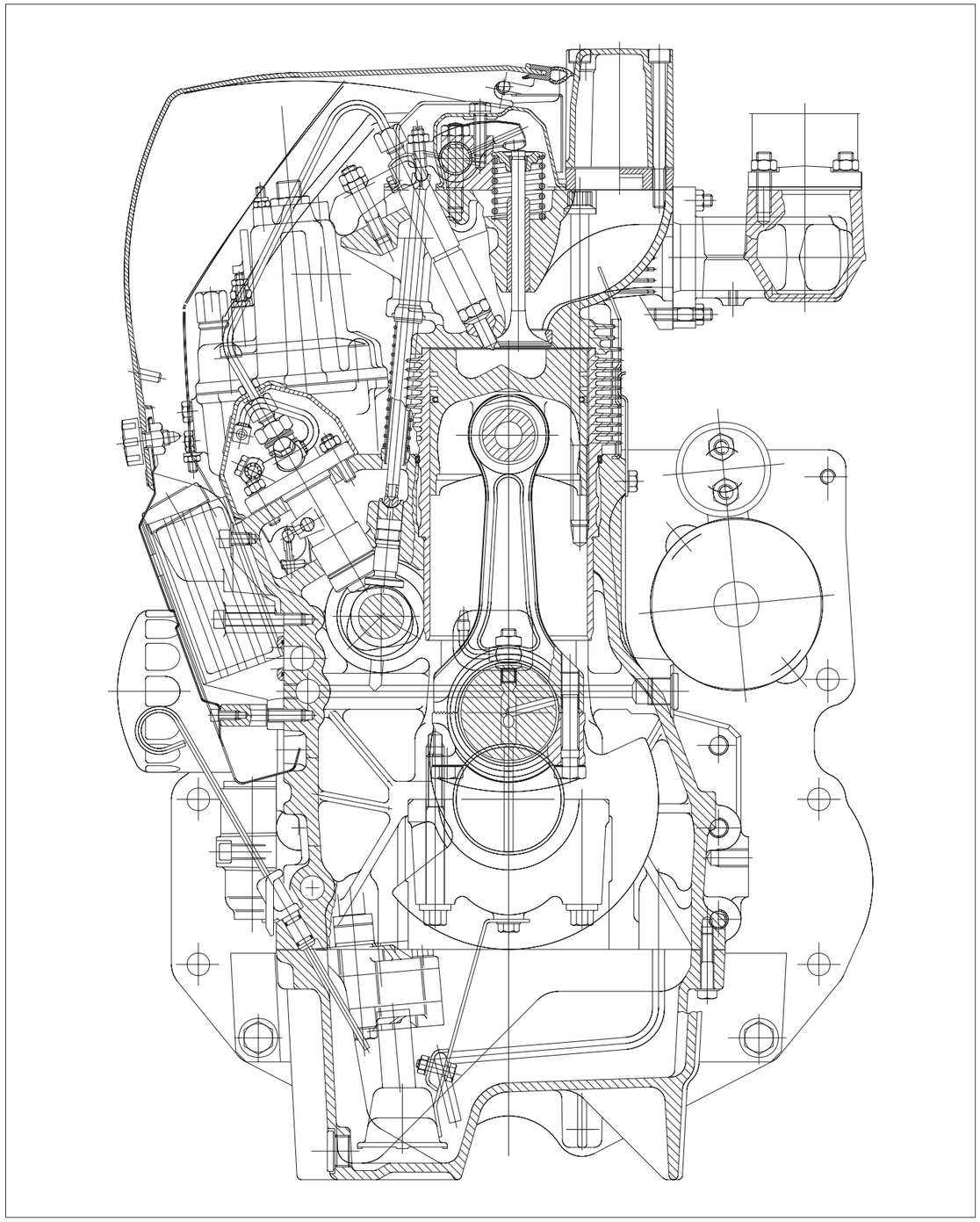


Fig. 1 - ***660s AGILE*** - Cross section of 3-cylinder engine



660v AGILE - Cross section of 3-cylinder engine

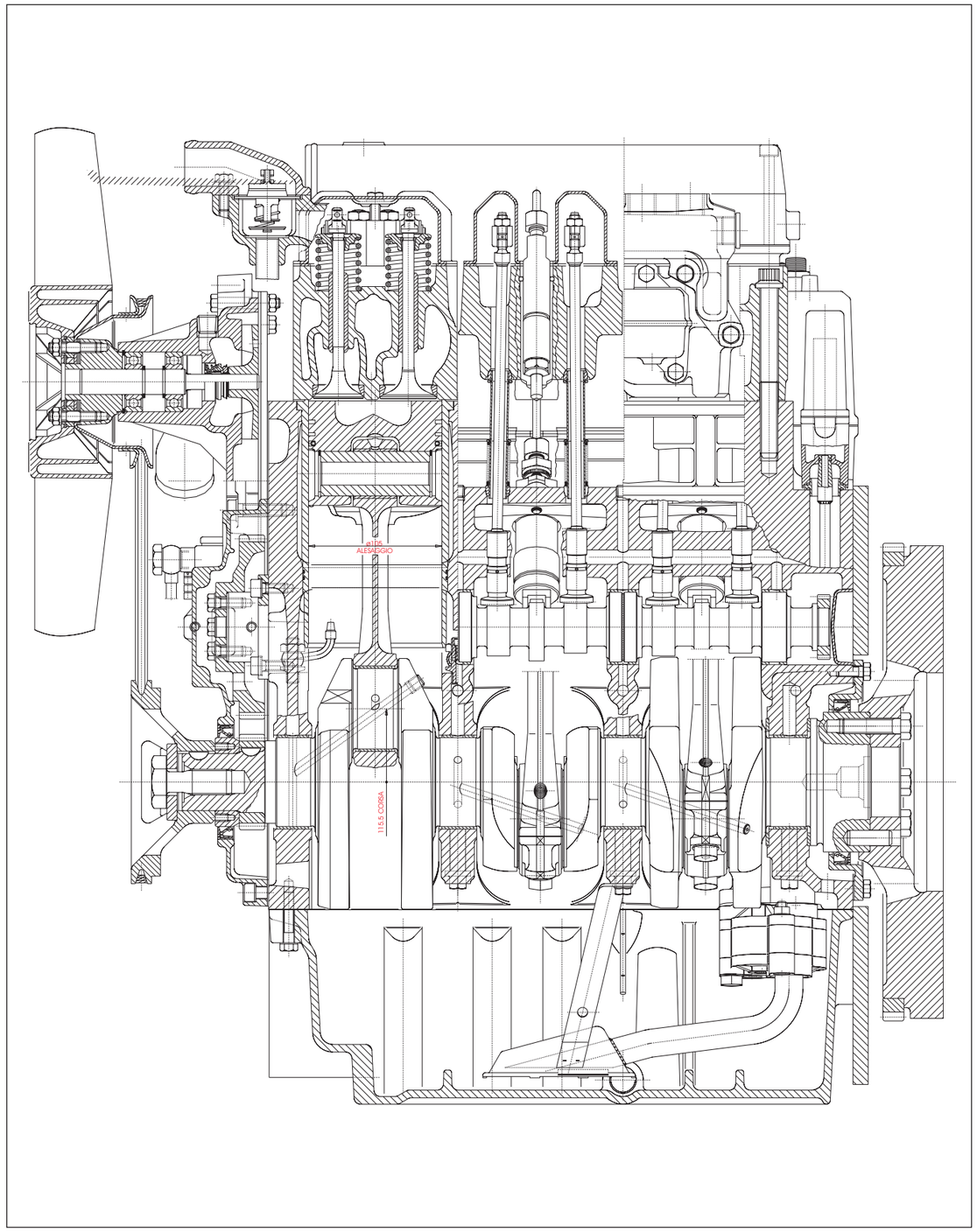
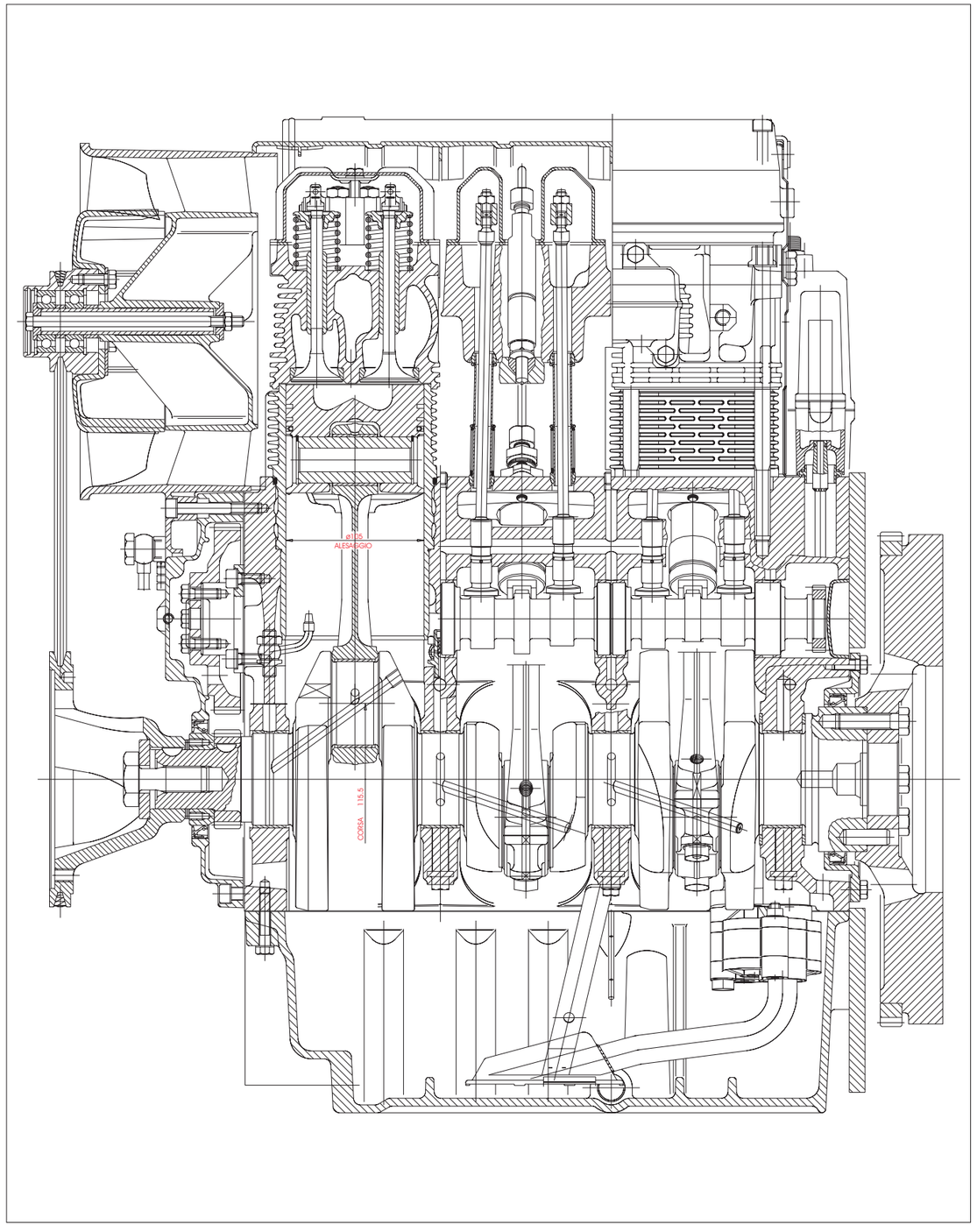


Fig. 2 - ***660s AGILE*** - Longitudinal section of 3-cylinder engine



660v AGILE - Longitudinal section of 3-cylinder engine

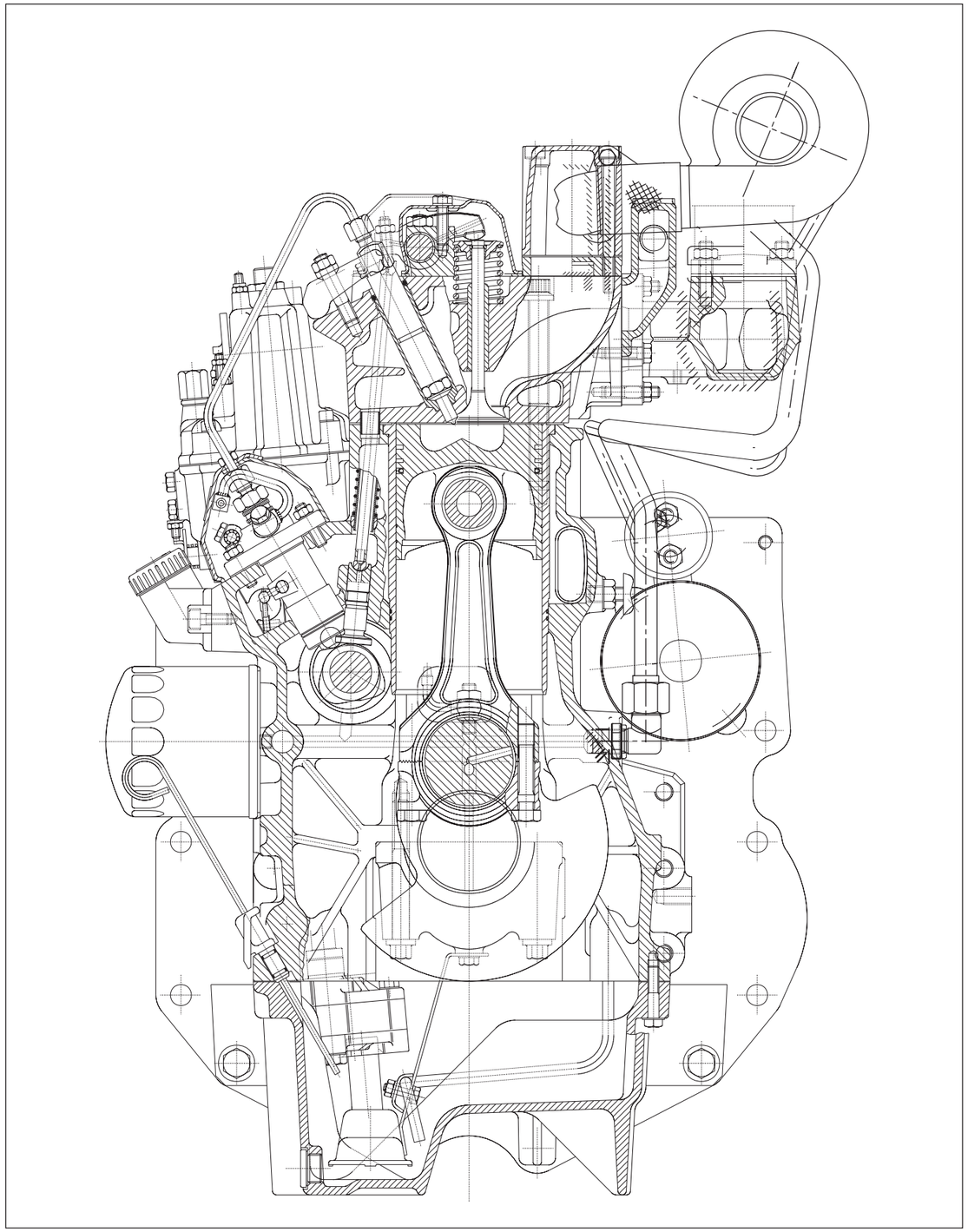


Fig. 3 - ***770s AGILE*** - Cross section of 3-cylinder TURBO engine.

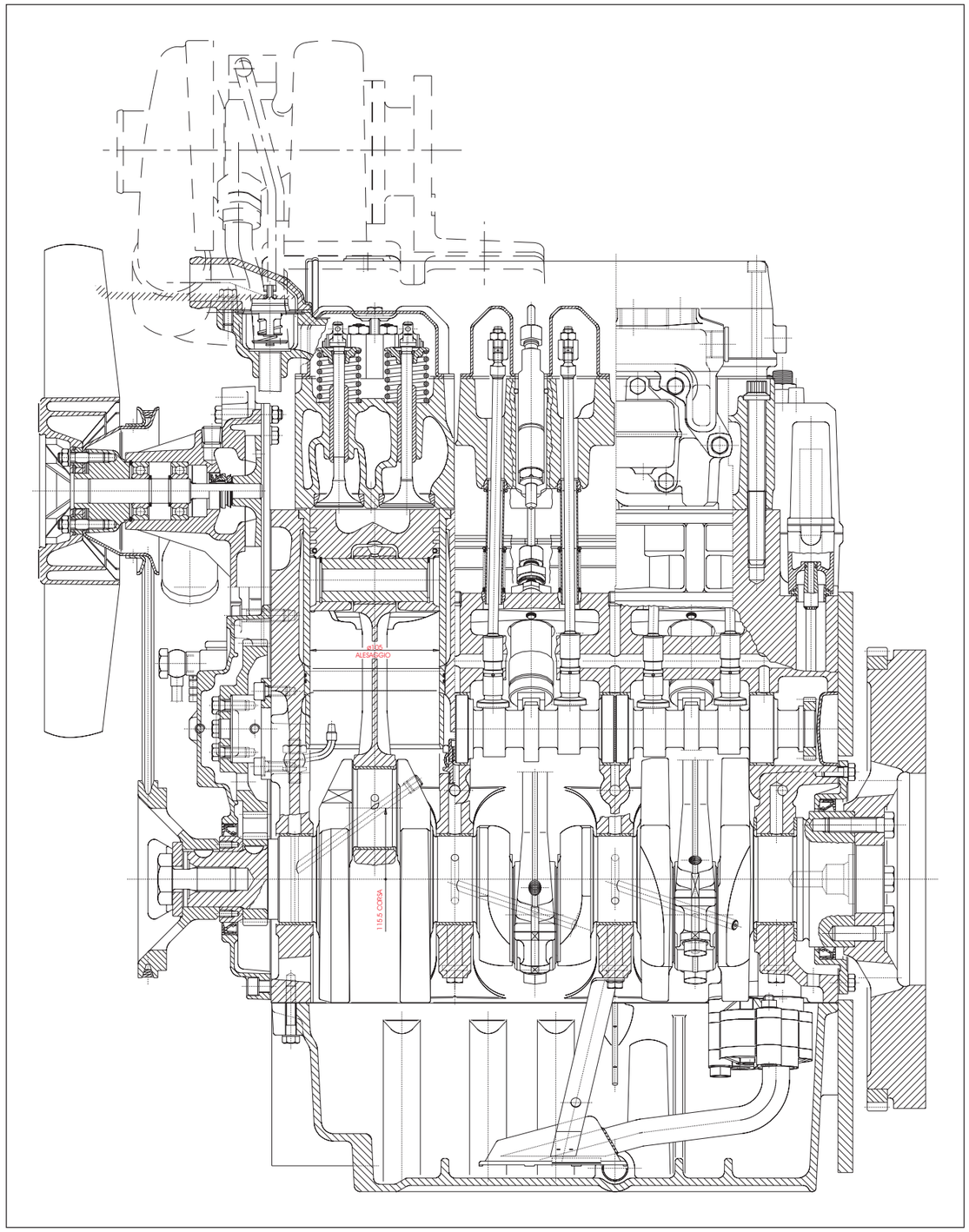


Fig. 4 - ***770s AGILE*** - Longitudinal section of 3-cylinder TURBO engine

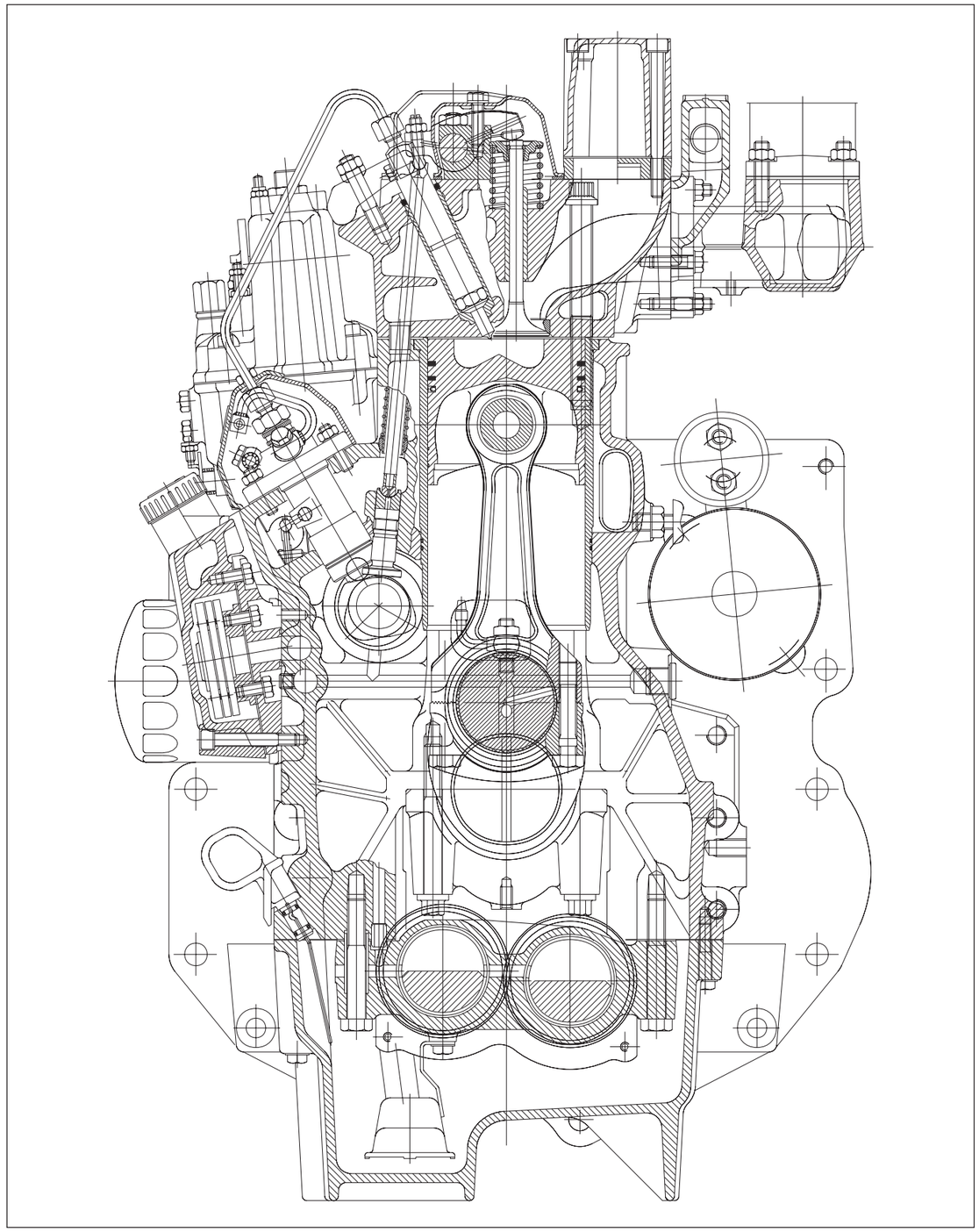
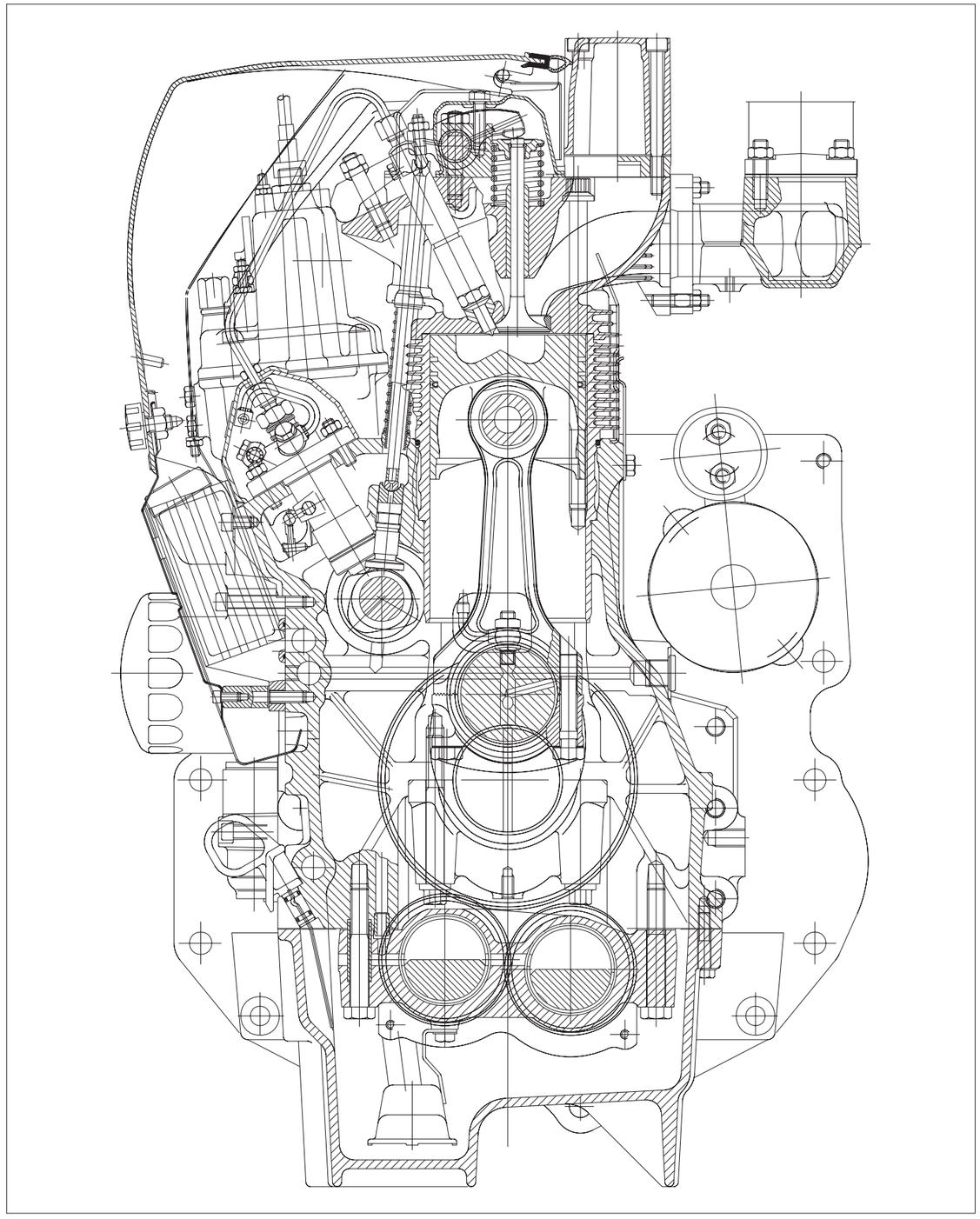


Fig. 5 - ***880s AGILE*** - Cross section of 4-cylinder engine



775v AGILE - Cross section of 4-cylinder engine

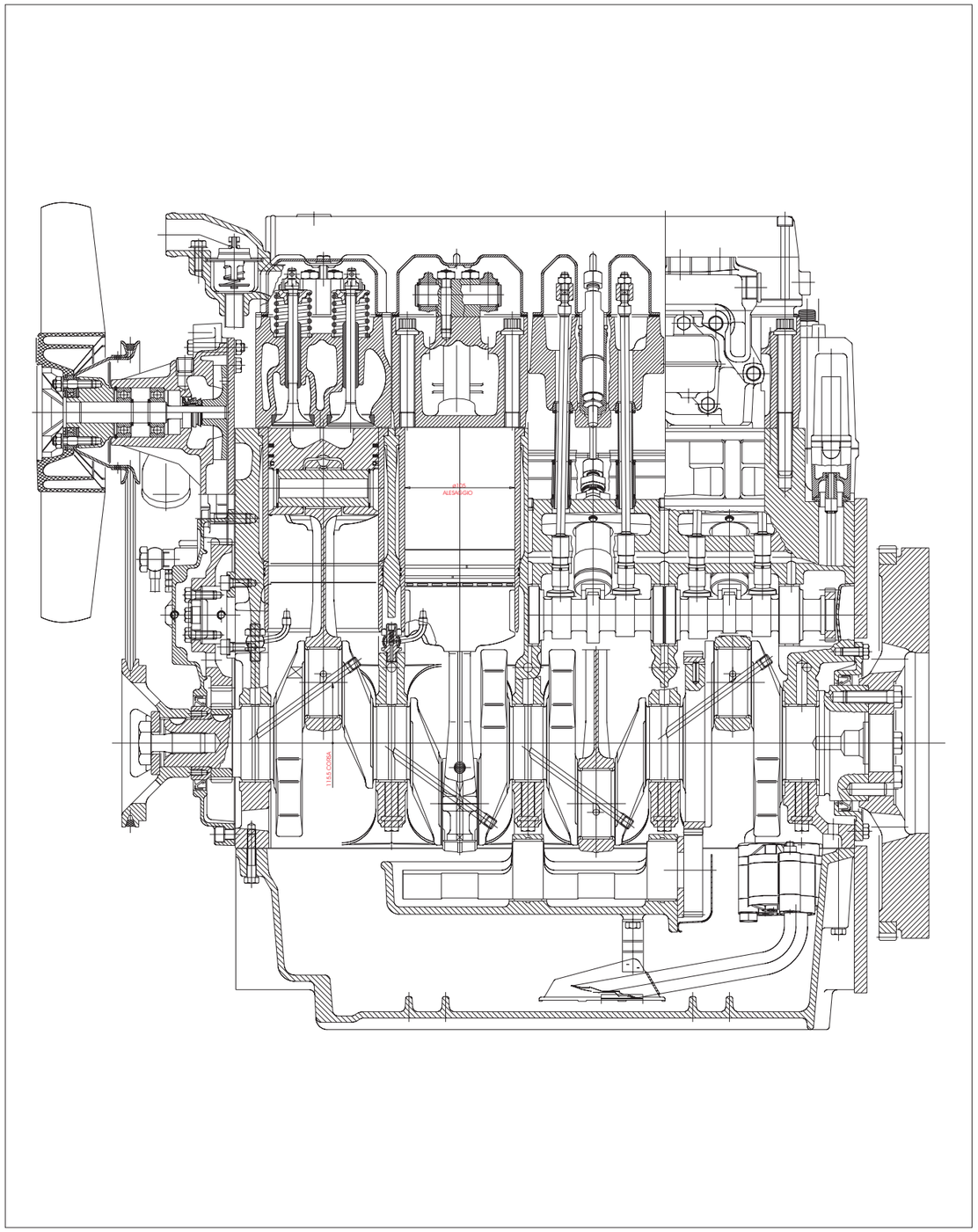
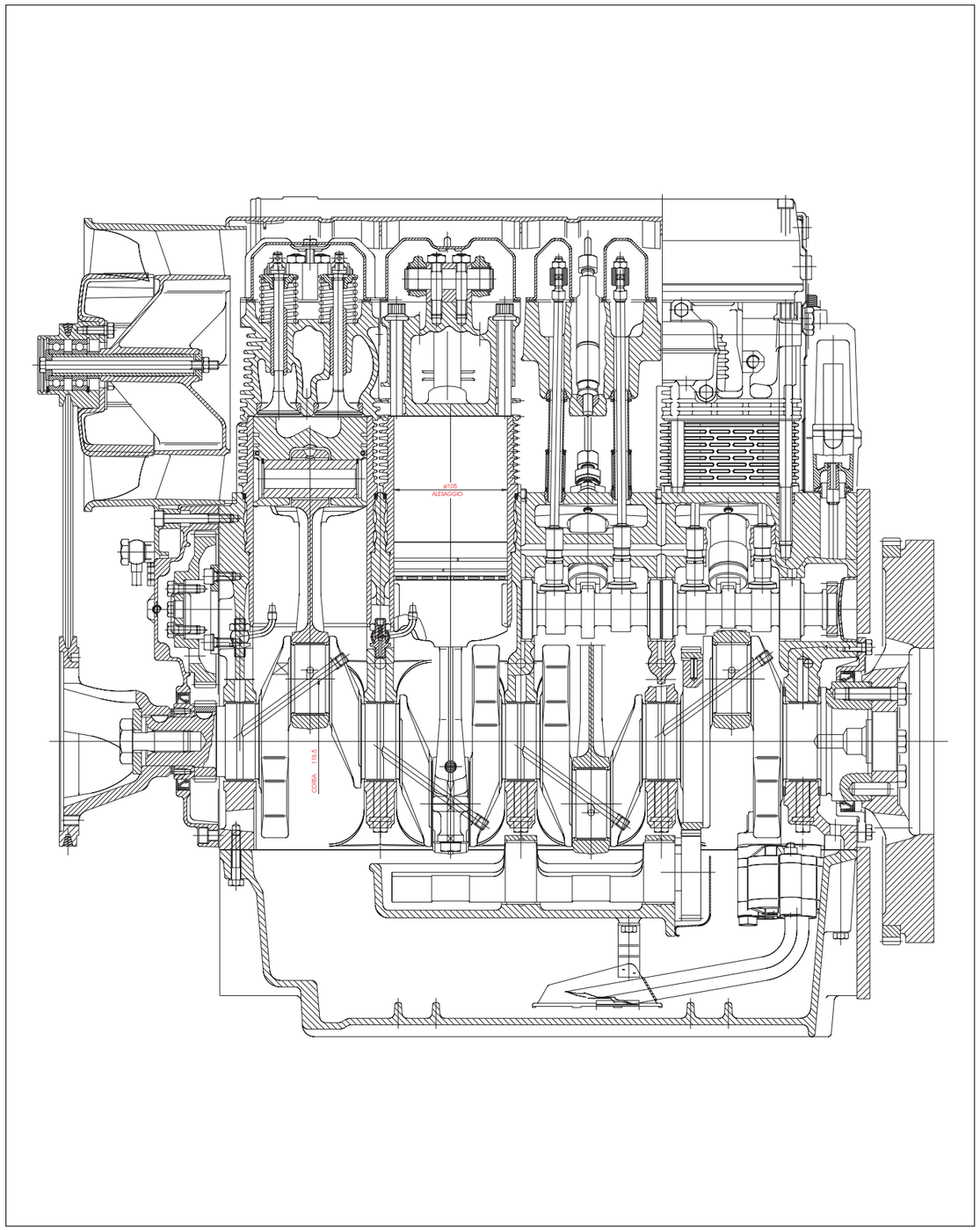


Fig. 6 - **880s *AGILE*** - Longitudinal section of 4-cylinder engine



775v AGILE - Longitudinal section of 4-cylinder engine

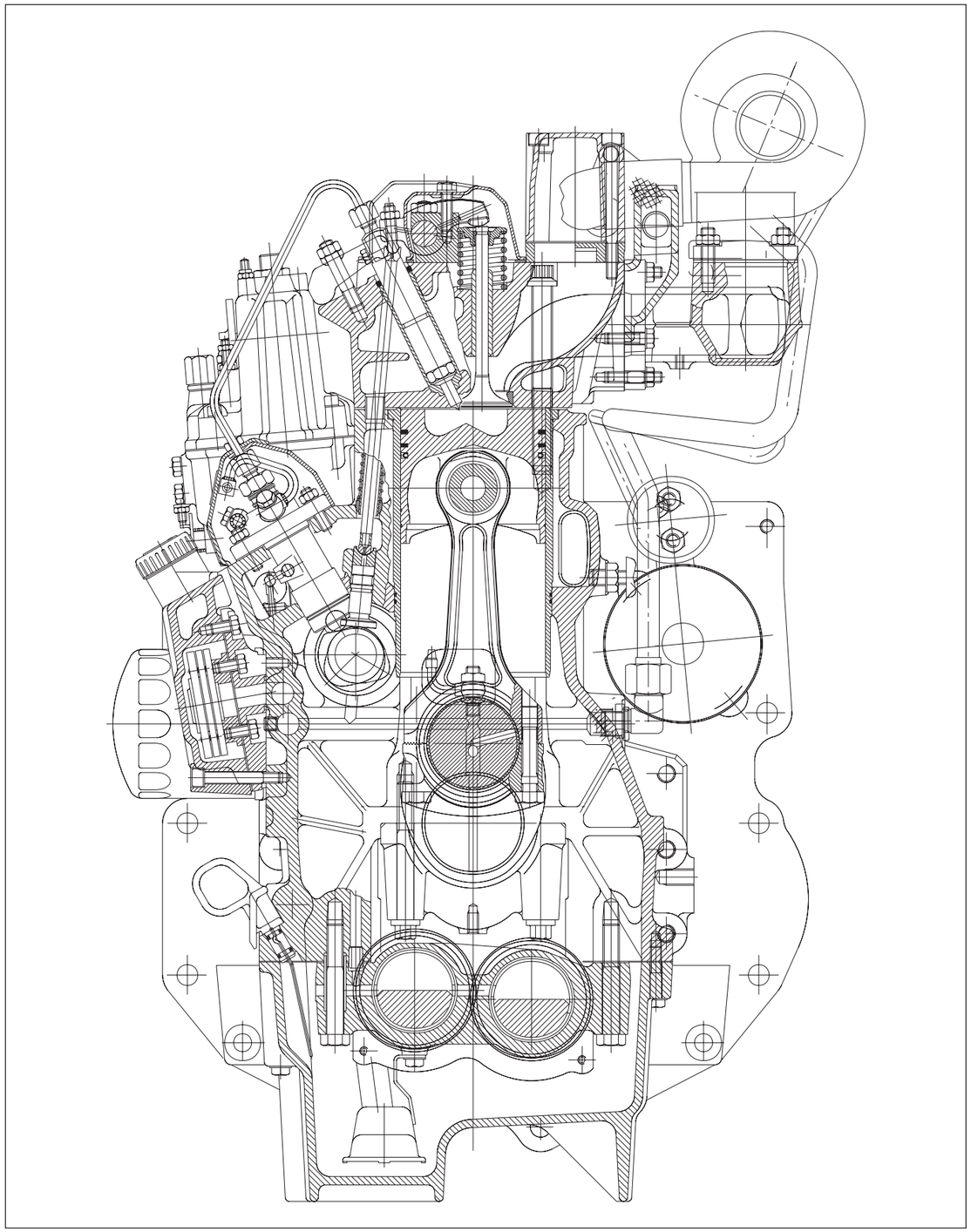


Fig. 7 - ***990s AGILE*** - Cross section of 4-cylinder TURBO engine

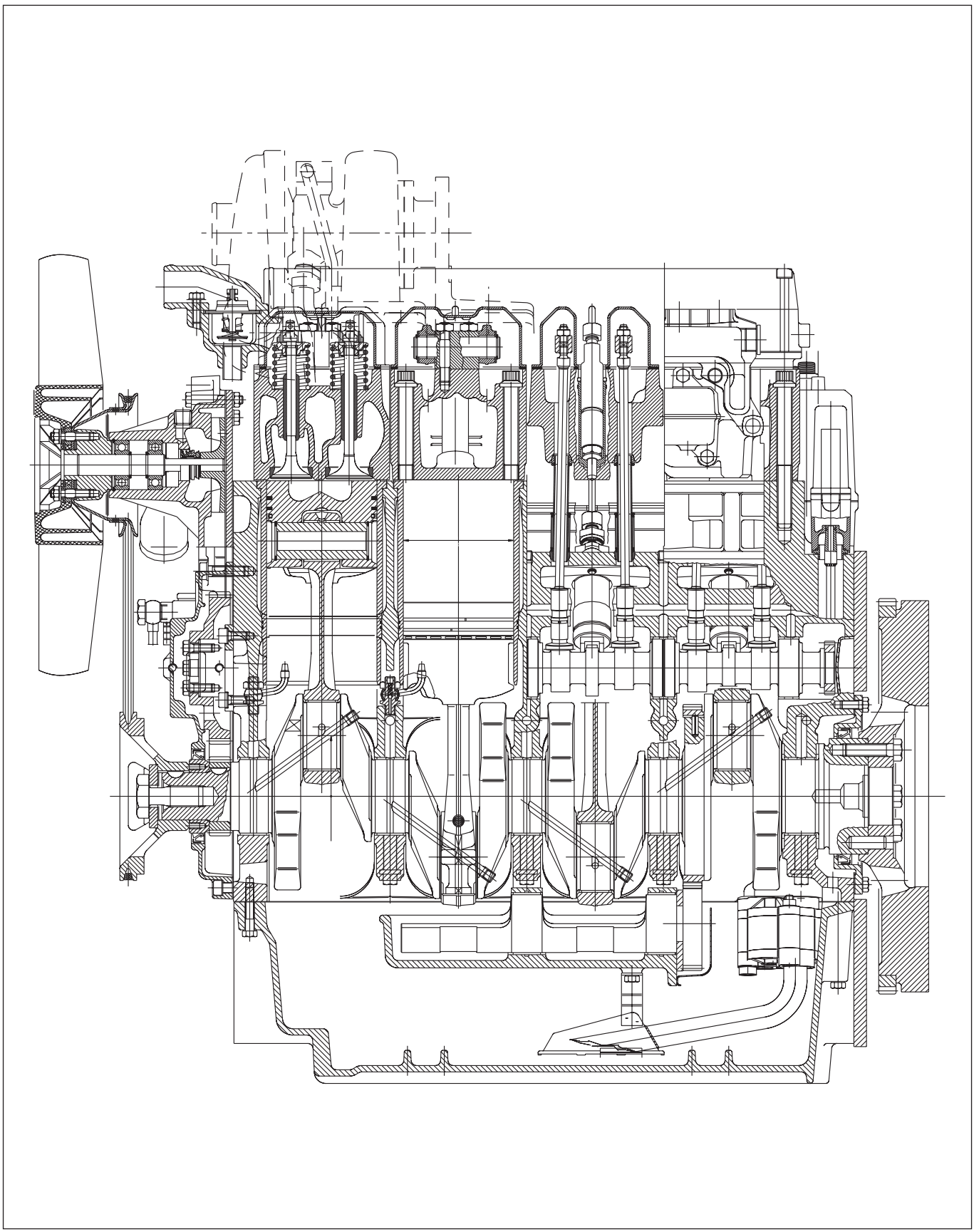
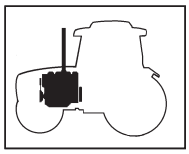


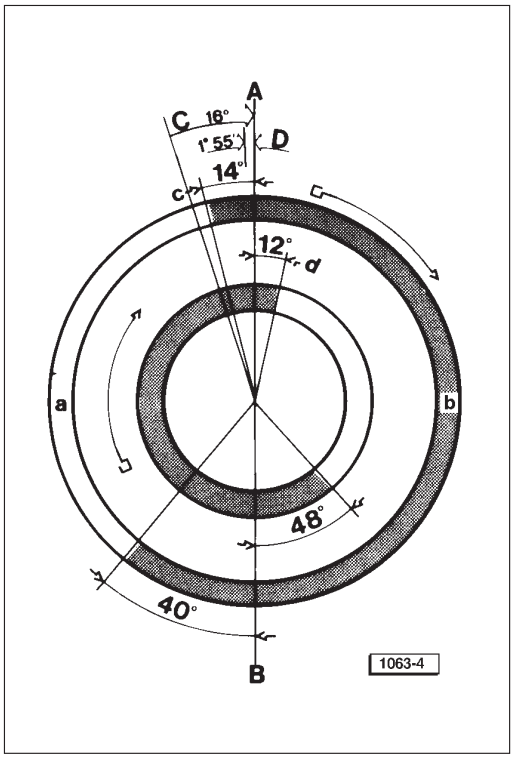
Fig. 8 - ***990s AGILE*** - Longitudinal section of 4-cylinder TURBO engine



1 Engine

General information

		660s AGILE	770s AGILE	880s AGILE	990s AGILE	660v AGILE	775v AGILE
engine							
type		1000.3-W1	1000.3-WT1	1000.4-W3	1000.4-WT1	1000.3-A5	1000.4-A3
ciyle		diesel	diesel	diesel	diesel	diesel	diesel
strokes		4	4	4	4	4	4
turbocharging		-	turbo	-	turbo	-	-
injection		DIRECT	DIRECT	DIRECT	DIRECT	DIRECT	DIRECT
cylinder No		3	3	4	4	3	4
cylinder arrangement		IN LINE	IN LINE	IN LINE	IN LINE	IN LINE	IN LINE
bore and stroke	mm	105 x 115,5	105 x 115,5	105 x 115,5	105x115,5	105x115,5	105x115,5
stroke / bore ratio		1,1	1,1	1,1	1,1	1,1	1,1
displacement	cm	3000,44	3000,44	4000,44	4000,44	3000,44	4000,44
compression ratio		17:1	16:1	17:1	16:1	17:1	17:1
max. output	cv CUNA	60	70	80	87	60	75
	Kw	44	51,5	58,8	64	44	55,5
peack horsepower speed	r.p.m.	2350	2350	2350	2350	2350	2350
max. torque	nm	207	250	296	325	202	28
	kgm	21	25,5	30	33	20,6	274,2
max. torque speed	r.p.m.	1400	1500	1450	1450	1400	1450
specific horsepower	cv/l	20	23,3	20	21,7	20	18,7
cooling		by water	by water	by water	by water	by air	by air
low idling speed	r.p.m.	650-700	650-700	650-700	650-700	650-700	650-700
peak speed	r.p.m.	2520/2550	2520/2550	2520/2550	2520/2550	2520/2550	2520/2550
minimum lubricating oil pressure							
low idling speed (hot oil)	bar	≥ 0,5	≥ 0,5	≥ 0,5	≥ 0,5	≥ 0,5	≥ 0,5
high idling speed (hot oil)	bar	≥ 3,5	≥ 3,5	≥ 3,5	≥ 3,5	≥ 3,5	≥ 3,5
oil filter with replaceable cartrige	n.	1	1	1	1	1	1
filtering capacity	µ	15	15	15	15	15	15
fuel filter		replaceable cartridge					
valve arrangement		vertical in-line					



Timing diagram

A - Top dead centre (TDC)
B - Bottom dead centre (BDC)
C - Injection advance
D - Valve overlap

a - Intake
b - Exhaust
c - Opening
d - Closing

Timing		by overhead valves and camshafited into engine block
valve arrangement		vertical in-line
intake valve		
-opening before TDC		14°
-closing after BDC		40°
exhaust valves		
-opening before BDC		48°
-closing after TDC		12°
clearance between valves and rockers (cold engine)	mm	0,30 with water-cooled engine
	mm	0,20 with air-cooled engine
injection advance (geometric) before TDC		16°
piston stroke as to injection advance	mm	2,92

Lubrication system - specifications

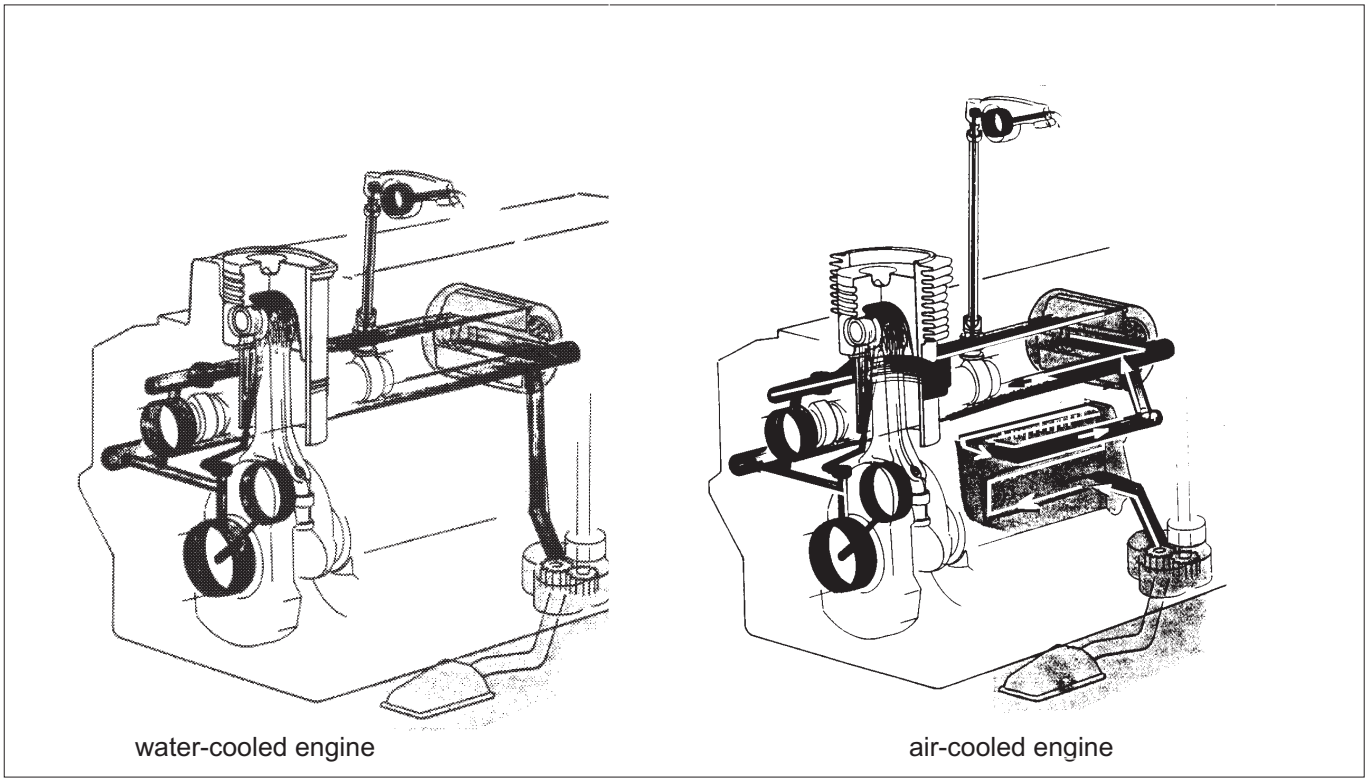
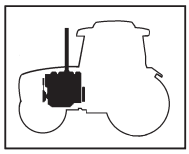


Fig. 9 - Lubrication diagram.

engine type		1000.3-A5 1000.3-W1 1000.3-WT1	1000.4-A3 1000.4-W3 1000.4-WT1
lubrication		forced-type, gear pump-dri-ven by camshaft	
oil pump		010.5481.4	010.5441.4/10
pump delivery rate at 2500 engine r.p.m. (dm/60 secretary)	l/min	38 ÷ 42	48 ÷ 51
pressure relief valve calibration	bar	3,5 ÷ 4,5	3,5 ÷ 4,5
piston cooling nozzle calibration	bar	1,5	1,5
minimum lubricating oil pressure (hot oil)			
at low idling speed	bar	≥0,5	≥0,5
at high idling speed	bar	≥0,3	≥0,3
oil sump capacity	(dm3) l	6.7	11



Fuel system - specifications

fuel supply		by diaphragm pump
fuel pump ref. Code		2.4519.300.0
pump delivery when pressure inside the circuit changes from 0 to 0.4 bar and engine is at peak speed	(dm/h) l/h	100
injection		by single-cylinder, immersed-type pump and camshaft-controlled plungers
- manufacturer		BOSCH
- type		PFR 1K 90 A 543 PFR 1K 90 A 555 (for USA)
- number of pumps used	3 cylinder engine 4 cylinder engine	3 4
- injection order	3 cylinder engine 4 cylinder engine	1-3-2 1-3-4-2
- plunger diameter	mm	9
- injection system residual calibrating pressure	bar	70
injectors		
- manufacturer		BOSCH
- injector type		DLLA 150 S 925
- injector holder type		KBEL 100S 31
- rated calibration pressure	bar	180
- injection pipe inside diameter	mm	1,5
cylinder pressure at sea level at 150 r.p.m. (with hot oil) measured by equipment no. 5.9030.500.6/10	bar	25 ÷ 30
dry air filter		
- code	8"	2.4249.600.0
fuel filter		with replaceable paper cartridge
- code		2.4319.060.0/10
- capacity	μ	1,5 ÷ 2

Turbocharging - specifications

Only for 770s 990s

The engine is supercharged through a turbocharger, whose characteristics are here you of succession bring again.

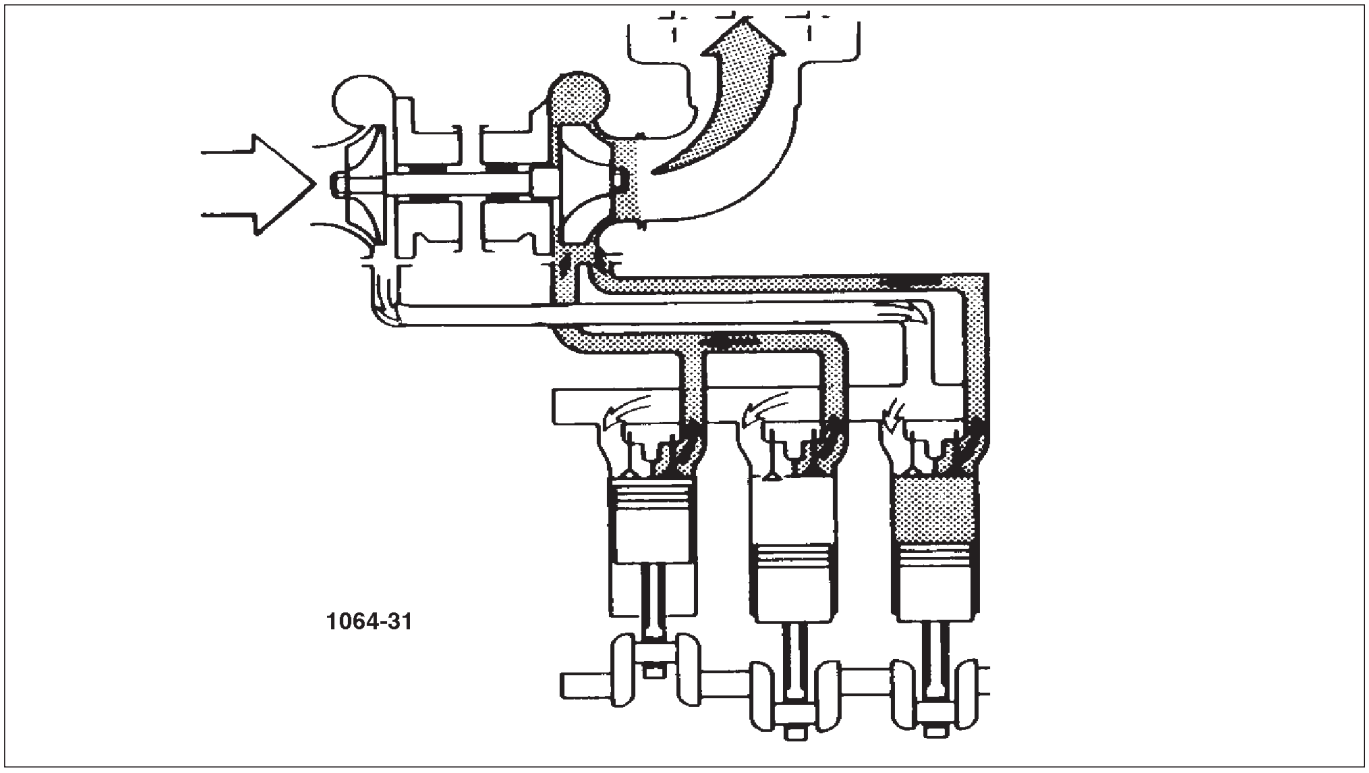


Fig. 10 - Turbocharger.

		770s AGILE	*990s AGILE*
type/turbine manufacturer		GARRETT T25-45/0.68 A/R 62 SCHWITZER S 1A/4808DB/0,81 MM1	GARRETT T25-50/0.68 A/R 76 SCHWITZER S 1B/5309W6/0,81 GG1
code		007.0720.4	007.1443.4
compressor and turbine		with radial wheel	
impeller shaft		on floating bearings with oil seals	
lubrication system		linked to engine oil lubrication system	
operating revolving speed	r.p.m.	55.000 ÷ 85.000	
mean pressure in the duct at compressor exit:			
- at 1400 engine r.p.m	bar	0.25	
- at 2500 engine r.p.m.	bar	0,60	
air flow rate under operating conditions:			
- at 1400 engine r.p.m	m³/sec	0,06	
- at 2500 engine r.p.m.	m³/sec	0,10	
air temperature inside intake manifold:			
- at 1400 engine r.p.m	°C	60°C	
- at 2500 engine r.p.m.	°C	90°C	

Engine cooling system

By water for 660s - 770s - 880s - 990s

The water cooling system is composed of a conventional radiator, a water pump and a thermostat.
An external manifold routes the water flow separately to each single cylinder-head-liner unit.Subsequently the water streams through a special inlet opening in the engine block and enters a heat exchanger thus providing for enine lubricating oil cooling.

The thermostat provides uniform temperature control inside the circuit. When the operating temperature is exceeded, it ensures proper temperature regulation by permitting water to flow into the radiator gradually and in small volumes.

A fan, constantly driven by the engine, projects the air stream being necessary for cooling the water which flow across the radiator.

The cooling system should be filled with AKROS FREEZE

	3 cylinders	3 cylinders turbo	4 cylinders	4 cylinders turbo
Fan		6 polypropylene blades		
fan diameter	500	500	500	500
water pump	centrifugal blade	centrifugal blade	centrifugal blade	centrifugal blade
ratio of transmission between motor and pomp for basic version	1,179	1,179	1,179	1,179
delivery at 2,500 r.p.m.	147	147	147	147
delivery head	0,34	0,34	0,34	0,34
thermostat				
quantità	1		1	
type			WAHLER 4015	
start opening temperature			85÷ 2	
maximum opening temperature		95÷2		
valve stroke			5,5	
water cooler		**copper-type with 3 rows**		
manufacturer	CURLI	CURLI	CURLI	CURLI
type	with pipe blunles	with pipe blunles	with pipe blunles	with pipe blunles
rated pressure	0,7	0,7	0,7	0,7

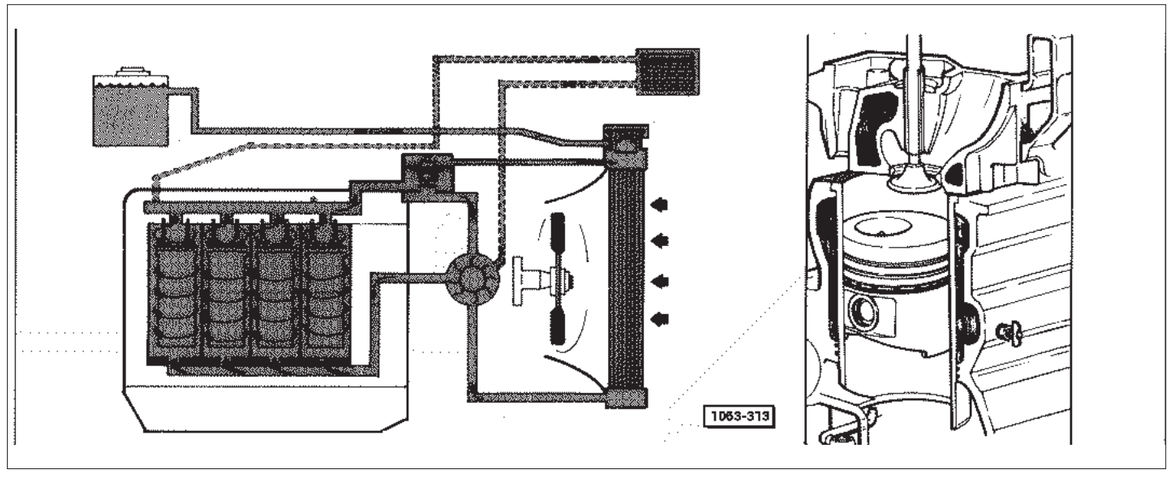


Fig. 11 - Cooling system , (in the picture is shown the version for 4 cylinders).

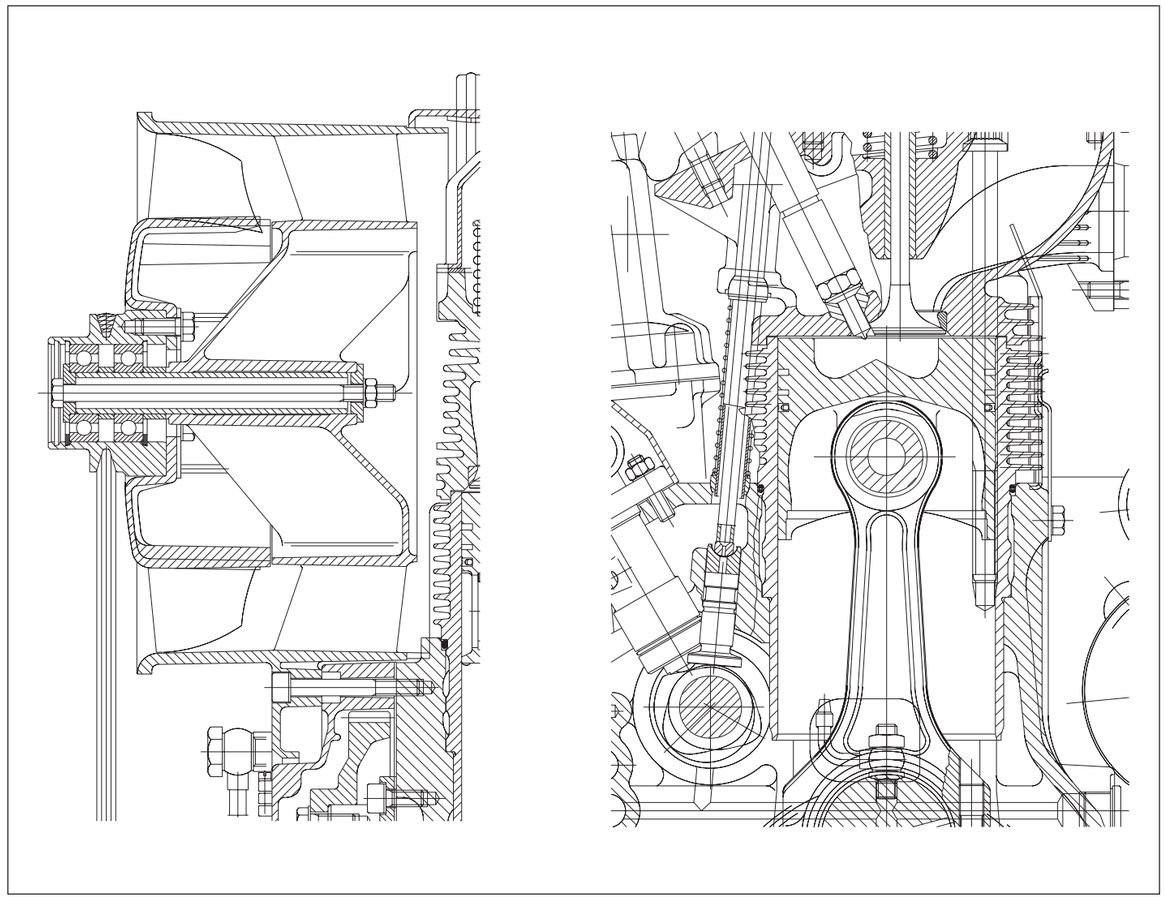
Engine cooling system

By air for 660v - 775v

The cooling system consists of a ventilation unit composed by a fan, a stator and an air conveyor.

This system ensures high thermodynamic efficiency as well as prompt engine power availability after a cold start.

	1000.3-A5	**1000.4-A3**
fan	with blades	with blades
fan diameter	250	250
n. blades	13	13



Cooling system.

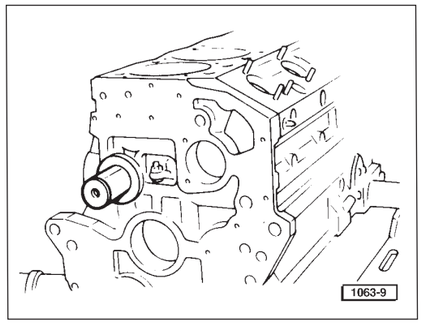
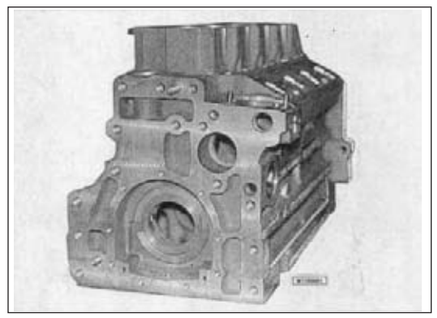


Fig. 1 - Engine cylinder block *660v 775v*.


Engine cylinder block *660s 770s 880s 990s*.

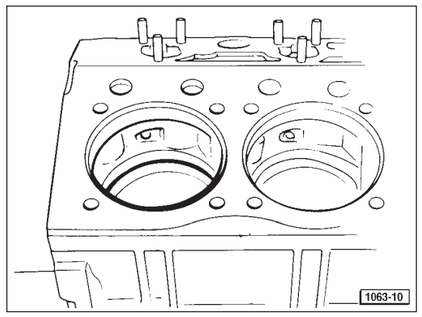
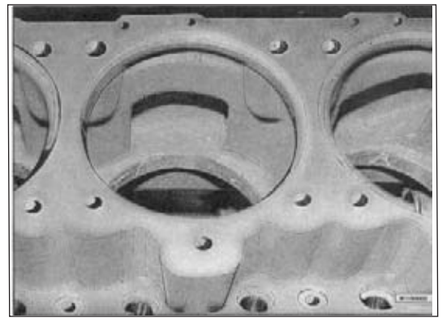


Fig. 2 - Oil chamber for cylinder cooling. *660v 775v*.


Oil chamber for cylinder cooling. *660s 770s 880s 990s*

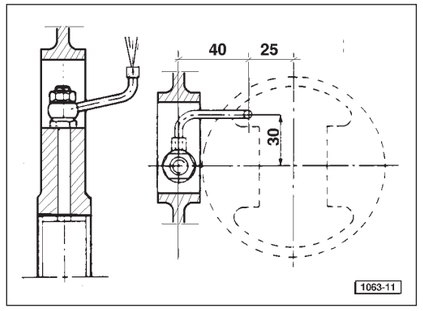
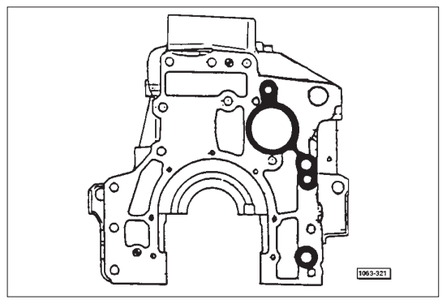


Fig. 3 - Piston cooling nozzles.



Area on engine block in which sealant is to be applied for water-cooled engines

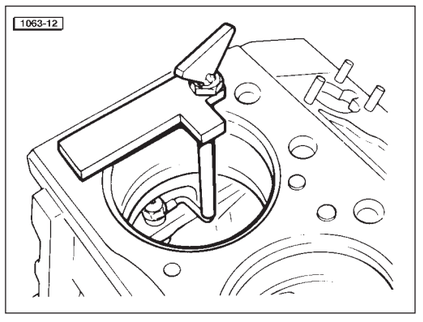


Fig. 4 - Piston cooling nozzle assembly with 5.9030.731.4 tool.

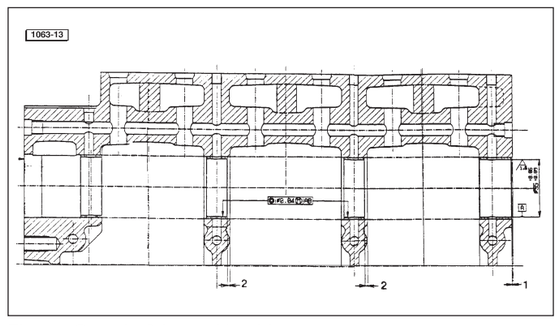


Fig. 5 - Camshaft journal bushings.

Engine cylinder block

The one-piece engine cylinder block is a particularly sturdy casting. All oil passages necessary to piston lubrication and cooling are directly machined in it.

Whenever the engine is stripped, ensure that all passages are not obstructed. If necessary perform a thorough cleaning by compressed air after the block has been soaked in a water and soda or Diesel oil bath for some time, and all passage caps have been removed.

It is most important to make sure that neither the piston cooling nozzles nor the cylinder bottom cooling throats are obstructed as this could prevent the engine from running smoothly.

If the piston cooling nozzles have been removed, pay particular attention on reassembly so as to avoid any interference with piston stroke, since this would certainly result in a serious engine damage. For this reason we recommend using the special tool no. 5.9030.731.4, enabling the nozzles to be correctly positioned.

NOTE: When installing the fitting of the piston cooling nozzle, apply a small quantity of Loctite 242 to the thread and tighten to a torque of 3 kgm. Fit the nozzle to the fitting, tightening the retaining nut to a torque of 2.5 kgm.

The tappet seat surfaces should be completely smooth.

Tappets must be fitted before installing the camshaft and with the cylinder block overturned. After fitting tappets, ensure these can move freely.

Installing bushings into the camshaft journals

Ascertain the camshaft journal bushings are correctly positioned, i.e. thoroughly aligned with the oil passages machined in the cylinder block.

Bushings should be positioned so that they may recess about 2 mm from the holes in the block, except the last bushing on flywheel side which should be flush with the block wall.

Adjusting backlash between the gear teeth of the auxiliary engine drive.

Maximum horsepower drawn should never exceed 30 HP.

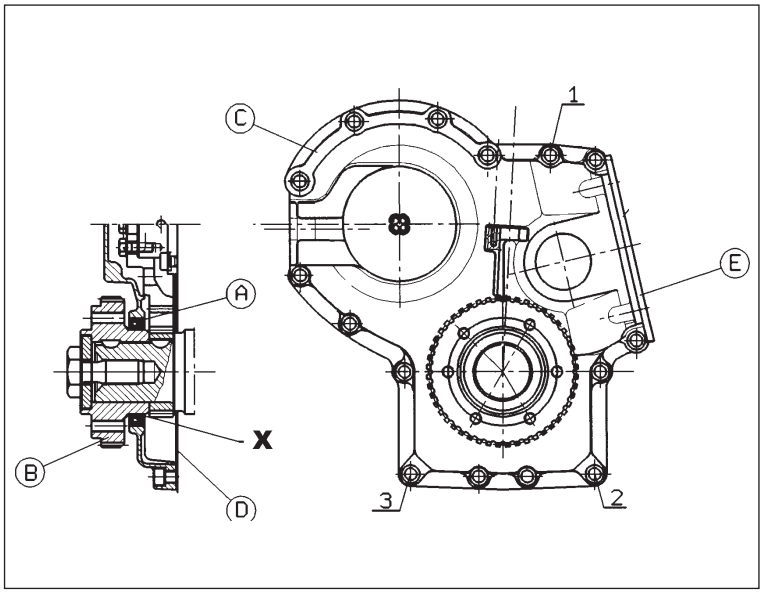
Between engine plane and application base fit a number of shims to allow a 0.1 to 0.2 mm backlash may take place between the gear teeth.
The pack of shims should be put together by alternating metallic and isogene shims, the first and the last shims being of isogene (the first metal shim on engine side shall be 0.5 mm).
After dismantling the timing case gears, provision shall be made for a new timing keeping to the instructions given on page 77 prior to reassembly.

Make sure the idler gear shoulder rings are not excessively worn; also check the pin surface and the gear inside face for scoring.

Support for hydraulic pumps or air compressor located between engine block and timing cover

The timing cover is made of lightweight aluminium alloy.
To fit the timing cover, it is essential to use tool 5.9030.634.0 to ensure a perfect alignment between the oil seal and the crankshaft and between the mating surfaces of the auxiliary power take-off.

Fit the oil seal **A** in the timing cover with the locating tabs oriented towards the inside of the cover.
Fit the gasket **D** and timing cover **C** to the block. Screw in the cover bolts but do not tighten at this stage.
Fit the hub **B**.
Align the mating face of the timing cover with that of the engine block using tool **E** 5.9030.634.0, which is to be secured with two bolts to the engine block.
Tighten the timing cover bolts in the sequence 1-2-3 and then the all remaining bolts to a torque of 24.5 Nm (2.5 kgm).
Remove the tool **E** and the strip of excess gasket **D**.



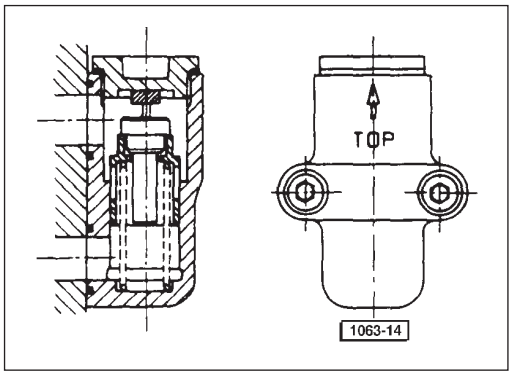


Fig. 6 - Engine oil thermostat.-I begin closing to 83°C -elegant closing to 100°C.

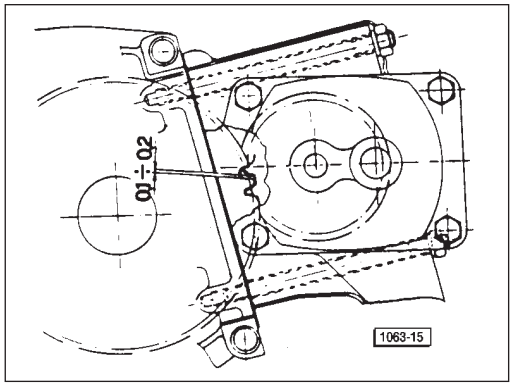


Fig. 7 - Drive gear toothing backlash.

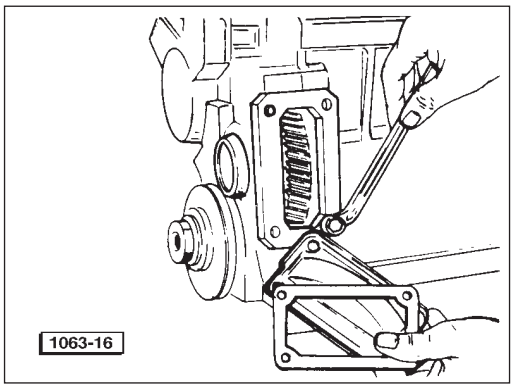


Fig. 8 - Aligning the timing case with the cylinder block

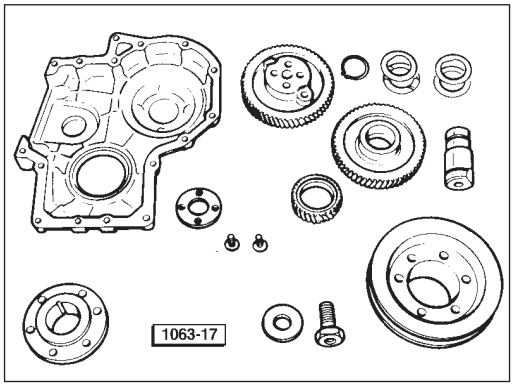


Fig. 9 - Timing case stripped components.

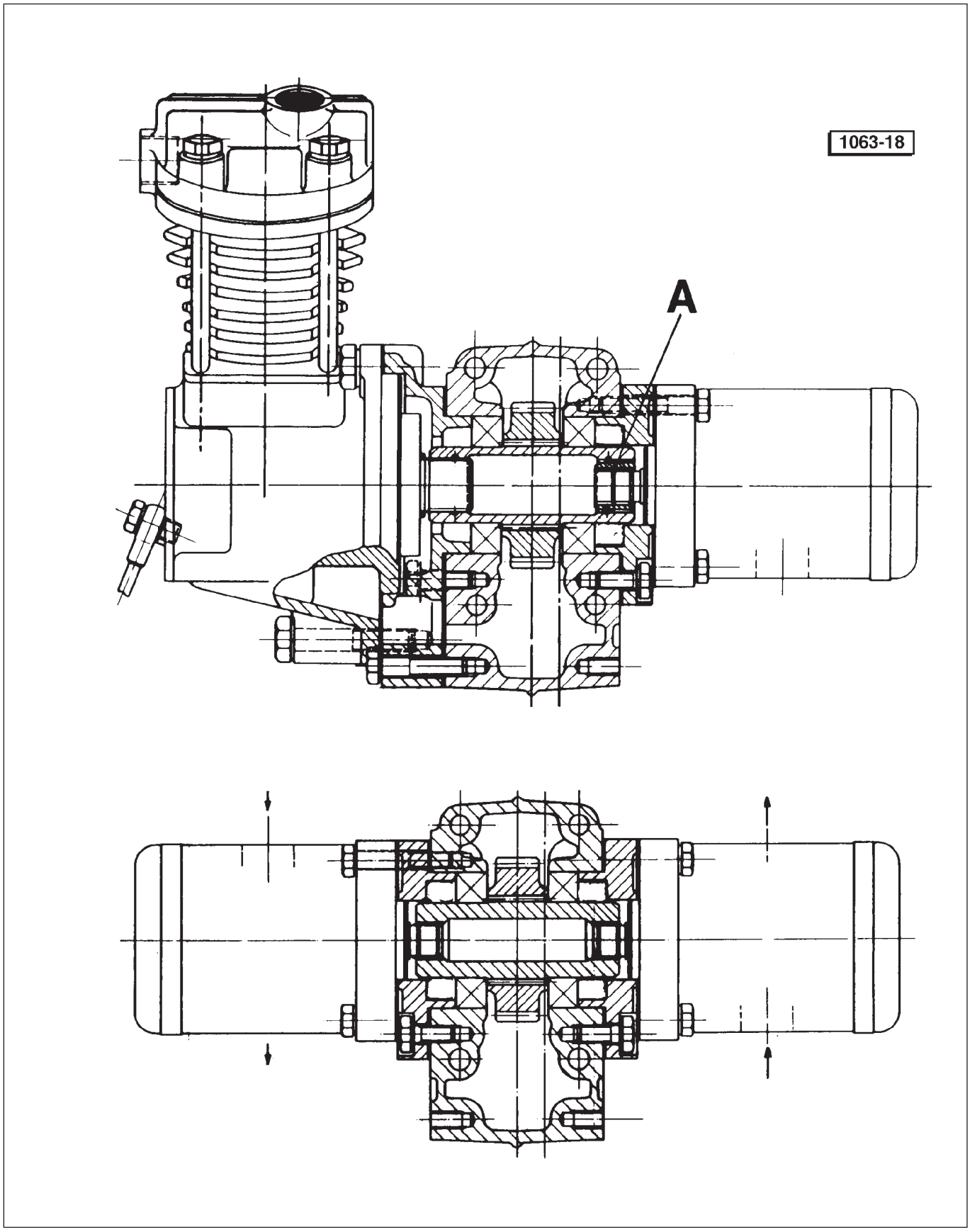


Fig. 10 - Auxiliary drive for either hydraulic pumps or air compressors.

Timing idler gea

Use a magnetic base dial gauge to check for correct tooth backlash between idler, engine gear and timing gear.

This backlash should not exceed 0.10 mm; otherwise the idler gear should be replaced by another one having the same thickness as the tooth located on the different pitch diameter, so that a correct tooth backlash may be established.

Identification among gears is provided with marks of different colours (either RED or YELLOW or GREEN) as illustrated in figure 11.

To each colour corresponds a well-defined tooth thickness value.

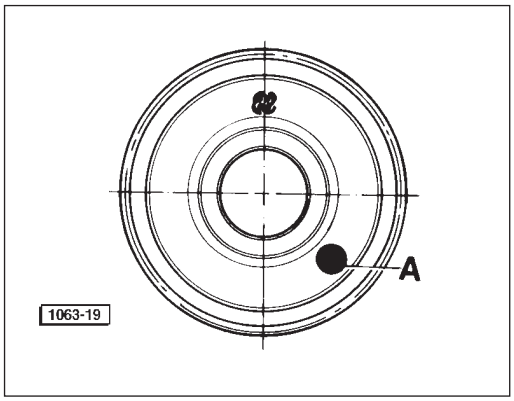


Fig. 11 - Timing idler gear
A - Identification mark

tooth thickness on the pitch diameter

ref. code 007.1177.0	RED colour	=3,829 $^{-0,088}_{-0,044}$
ref. code 007.1178.0	YELLOW colour	=3,829 $^{-0,053}_{-0,009}$
ref. code 007.1179.0	GREEN colour	=3,829 $^{+0,026}_{-0,018}$

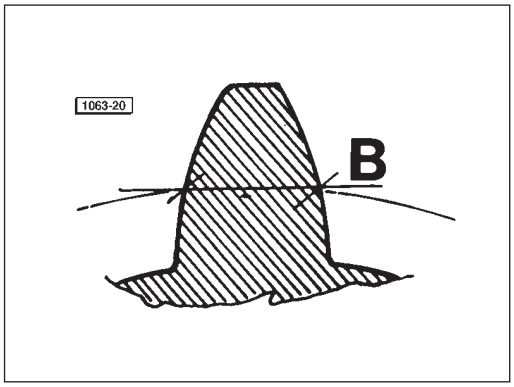


Fig. 12 - Gear toothing
B - Gear tooth thickness.

A - Idler gear
B - Cranshaft driving gear
C - Timing driving gear

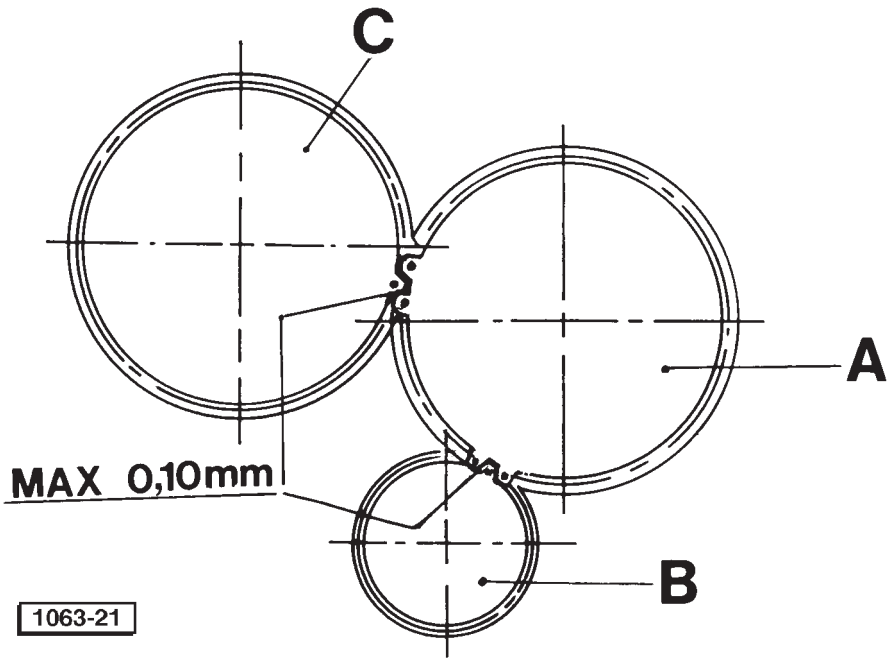
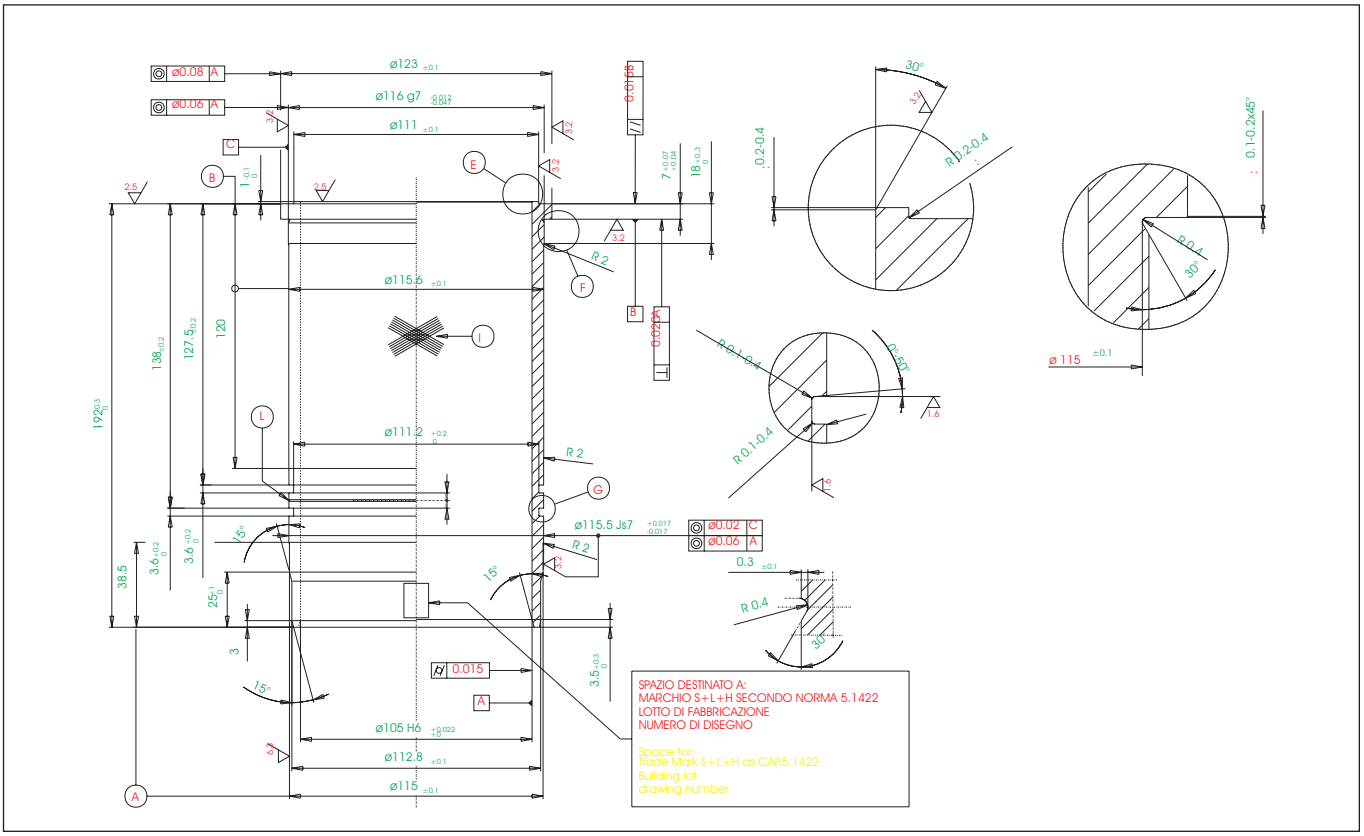
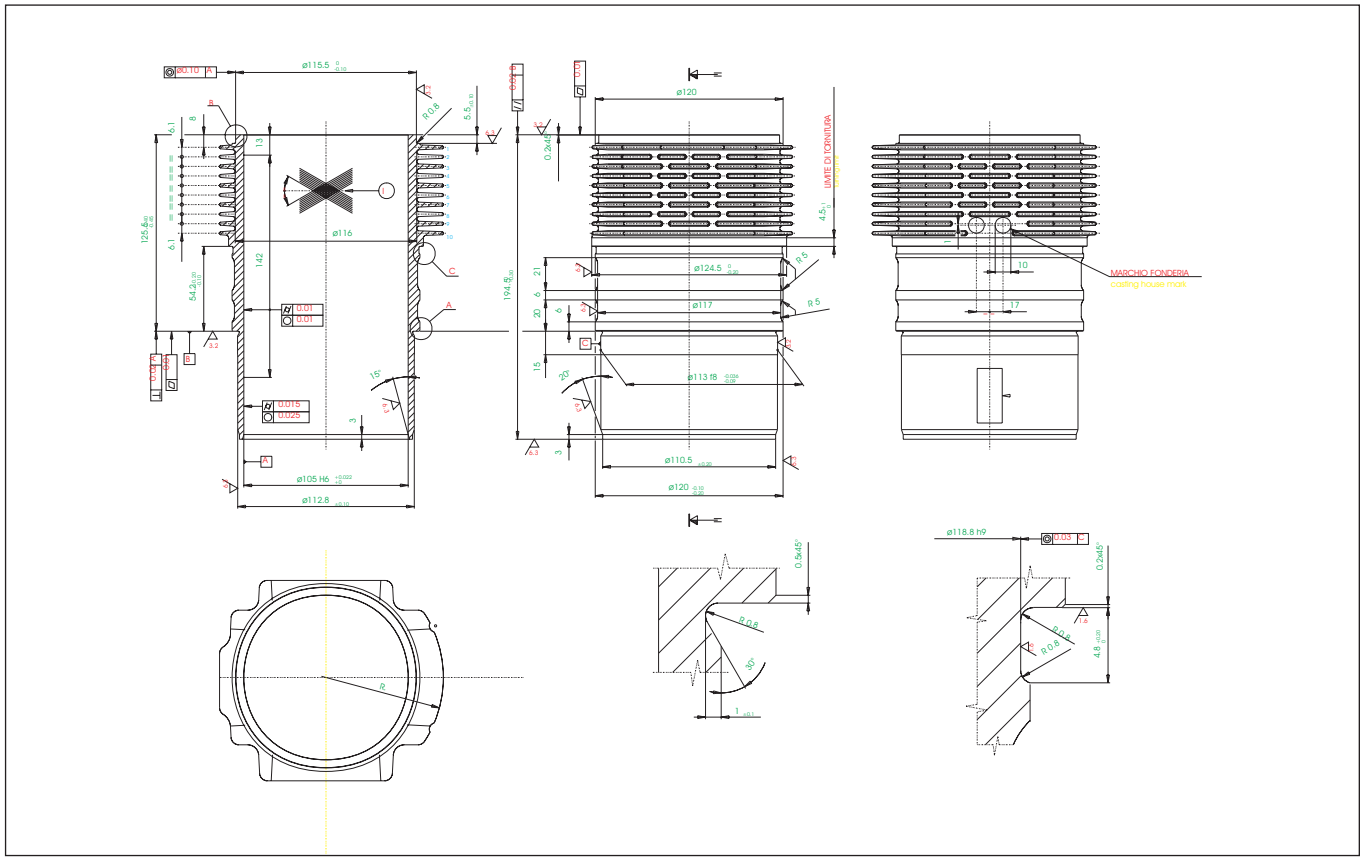


Fig. 13 - Idler gear



Cylinder for engine 1000-W.



Cylinder for engine 1000.A

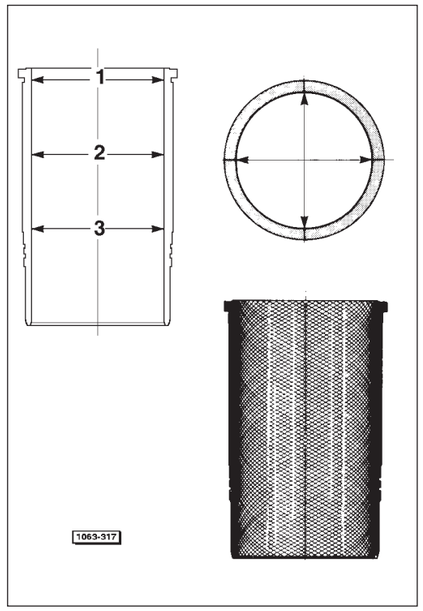


Fig. 1 - Cylinder wear check diagram.

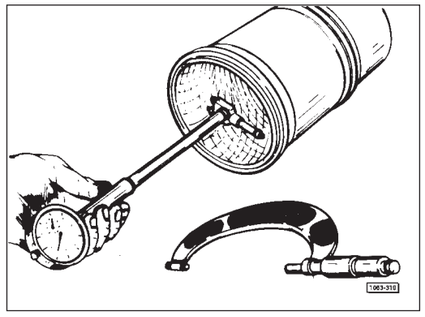


Fig. 3 - Measuring cylinder diameter.

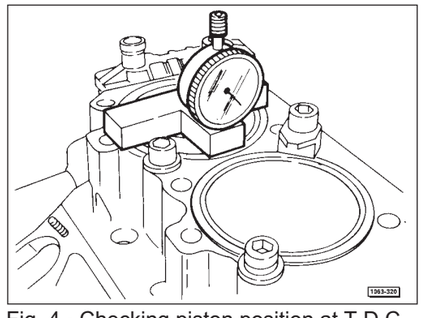


Fig. 4 - Checking piston position at T.D.C.

Cylinder

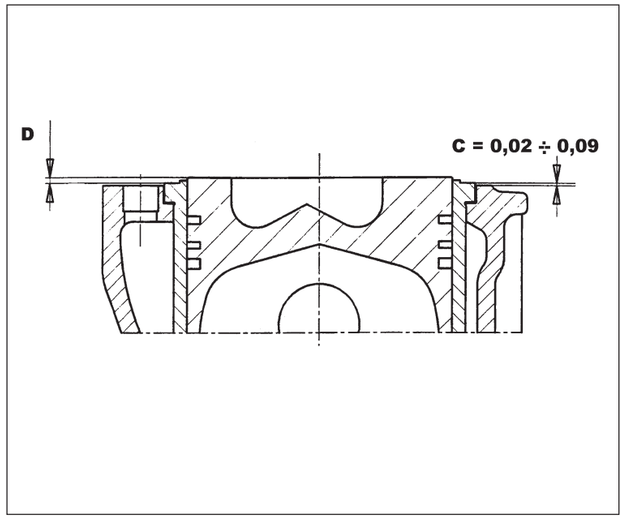


Fig. 2 - Cylinder-to-piston position.
C = Cylinder protrusion from the crankcase
D = Piston protrusion from the bearing surface of the cylinder head gaskets on the cylinder.

Overall cleaning
After removal, clean the cylinders with water and soda or Diesel oil; also perform a torough cleaning of the cylinder block parts forming the oil chamber for cylinder cooling.

Checking cylinders
Examine the cylinder surfaces for scoring, ovalization, taper and excessive wear.

An acceptable cylinder wear gives the cylinder a light tapered shape; but if barrel-shape wear is noticed, the cylinder should be replaced.

Measurements are to be taken on each single cylinder at three different levels, and also on to planes perpendicular to each other.

If the diameter measured is above the maximum wear limit, the cylinders should be replaced.

Whenever the piston rings are replaced because of excessive wear, also inspect the cylinder internal surface; if this is completely smooth, cylinder reboring will be necessary to permit the new piston rings to be properly bedded. Use the special tool to perform a swift to-and-fro movement along the cylinder internal wall in order to score the inner cylinder surface as shown in figure 1.

Using the following equipment:
5.9030.627.0 centesimal dial gauge
5.9030.433.0 dial gauge base
5.9030.631.4/10 cylinder pressing tool

Perform the following checks:
- cylinder protrusion from the crankcase: 0.02 to 0.08 mm, also refer to dimension C in figure 7.
- piston protrusion from the bearing surface of the cylinder head gaskets on the cylinder: 0,14 to 0,78 mm, see dimension D in figure 7 and select the proper gaskets ad to reference D in the table here below.
WARNING: Gasket thickness selection should be made according to dimension **D** measured (piston protrusion from the bearing surface of the cylinder head gaskets).

Fitting the cylinder liner in the engine block
Using a suitable applicator, apply a bead of Loctite AUX 935 all around the circumference in zone **A** as indicated in figure 6.
Align the cylinder liner with the axis of the cylinder bore, and while holding it in a vertical position, insert it in the engine block, taking care not to damage the sealant bead or contaminate the seal rings with Loctite.

Use a press tool to facilitate insertion of the seal rings in the bore. (Do not use a mallet for this operation).
Fully drive home the cylinder liners and clamp them in position with specific tool 5.9030.481.4 for about 4 hours.
Before fitting the gasket, carefully clean the contact surfaces **B** and **C** for the O-ring seals located between the cylinder heads and the block around the coolant passages and for the head gaskets.
Warning: both O-rings must be renewed each time the cylinder liners are removed from the block.

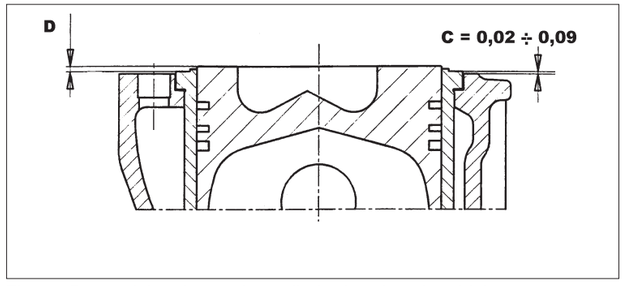


Fig. 7 - Determining the engine cylinder head gasket thickness.

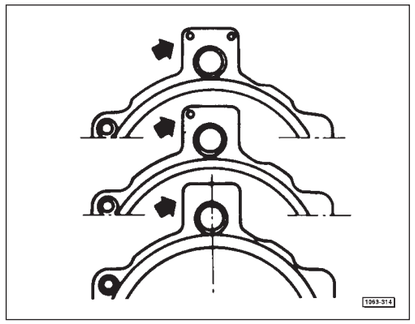


Fig. 5 - Cylinder head gasket.

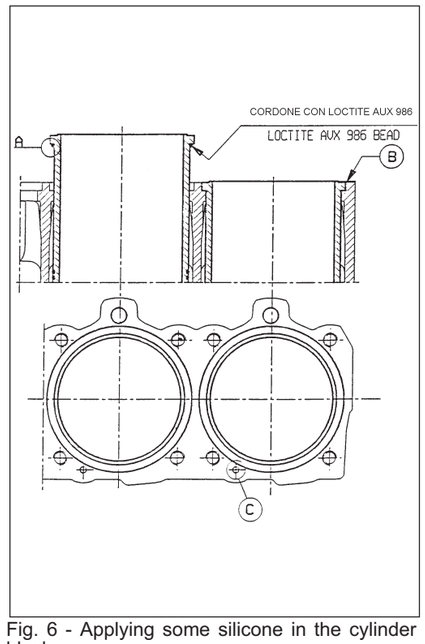


Fig. 6 - Applying some silicone in the cylinder block.

A - inside diameter measured half-way along the cylinder	mm	Ø machining	Ø max wear
B - roundness error	mm	$105{,}000\ ^{-0}_{+0,022}$	105,100
C - piston recess from the head gasket bearing face on cylinder	mm	0,020	0,080
D - determination of head gasket		0,14÷0,78	
- gasket ref. code 078.1451.0	mm	thickness **D** reading	gasket thickness
- gasket ref. code 078.1452.0	mm	-0,14÷0,40	1,2 without locating notches
- gasket ref. code 078.1253.0	mm	-0,41÷0,60	1,4 with 2 locating notch
		-0,61÷0,78	1,6 with 1 locating notch

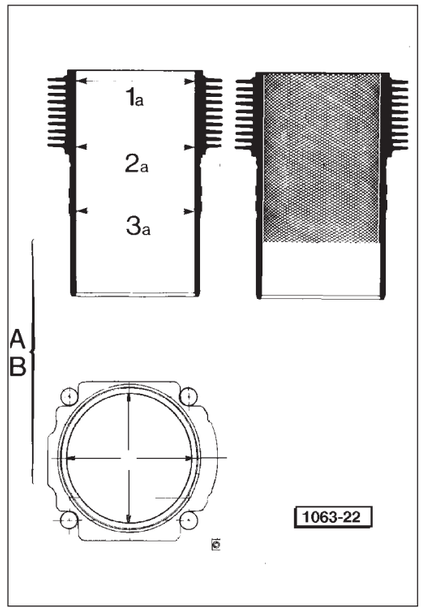


Fig. 1/a - Cylinder wear check diagram.

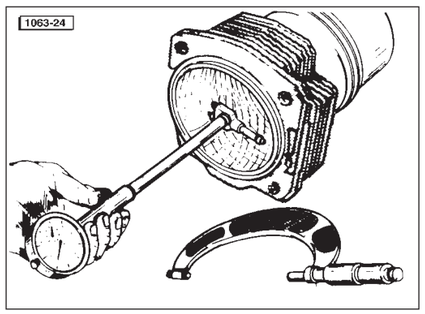


Fig. 3/a - Measuring cylinder diameter.

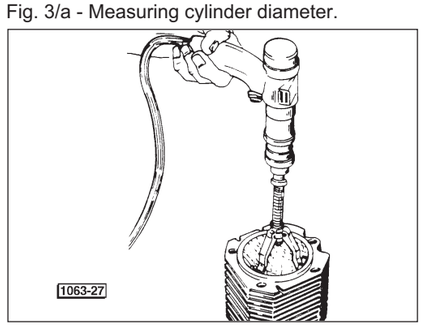


Fig. 4/a - Cylinder inside roughening with No. 5.9030.349.0 equipment.

Cylinders

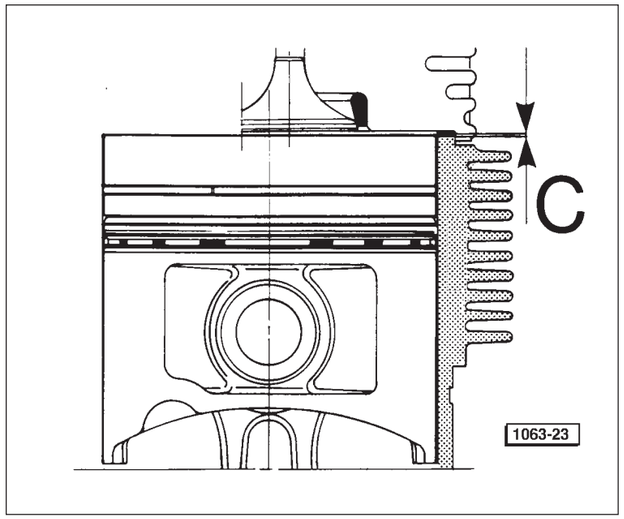


Fig. 2/a - Cylinder-to-piston position.

Overall cleaning

After removal, clean the cylinders with water and soda or Diesel oil; also perform a torough cleaning of the cylinder block parts forming the oil chamber for cylinder cooling.

Checking cylinders

Examine the cylinder surfaces for scoring, ovalization, taper and excessive wear.

An acceptable cylinder wear gives the cylinder a light tapered shape; but if barrel-shape wear is noticed, the cylinder should be replaced.

Measurements are to be taken on each single cylinder at three different levels, and also on to planes perpendicular to each other.

If the diameter measured is above the maximum wear limit, the cylinders should be replaced.

Whenever the piston rings are replaced because of excessive wear, also inspect the cylinder internal surface; if this is completely smooth, cylinder reboring will be necessary to permit the new piston rings to be properly bedded. Use the special tool to perform a swift to-and-fro movement along the cylinder internal wall in order to score the inner cylinder surface as shown in figure 1/a.

Using the following equipment:

5.9030.627.0 centesimal dial gauge;
5.9030.433.0 dial gauge base;
5.9030.631.4/10 cylinder pressing tool,

Ensure the piston crown at T.D.C. recesses from the head gasket bearing surface (see Fig. 6).

Move the dial gauge base until the gauge feeler lies on the cylinder surface and take reading from the dial.

Reset the dial gauge.

Place the base so that the gauge feeler lies on the piston head at T.D.C. and take reading.

Choose the gasket to be fitted according to reading taken (also refer to item **D** in the table here below).

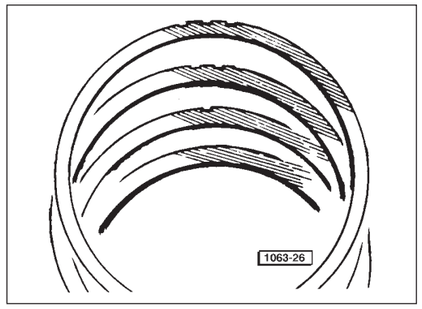


Fig. 5/a - Cylinder head gasket.

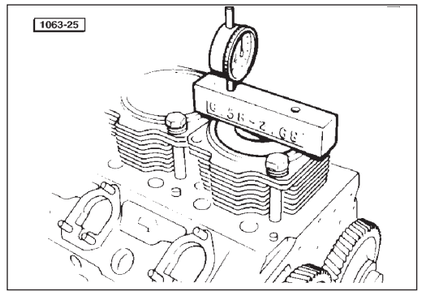


Fig. 6/a - Checking piston position at T.D.C.

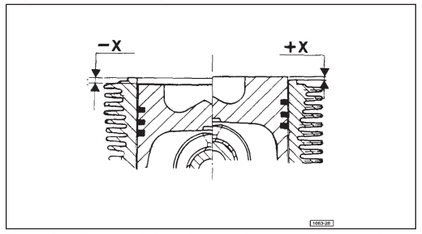


Fig. 7/a - Determining the engine cylinder head gasket thickness.

		Ø machining	Ø max wear
A - inside diameter measured half-way along the cylinder	mm	105,000 $^{-0}_{+0,022}$	105,100
B - roundness error	mm	0,020	0,080
C - piston recess from the head gasket bearing face on cylinder	mm	0,30÷0,40	
D - determination of head gasket		thickness **X** reading	gasket thickness
- gasket ref. code 085.1450.0	mm	-0,562÷0,30	0,5 without locating notches
- gasket ref. code 085.1451.0	mm	-0,290÷0,11	0,7 with 1 locating notch
- gasket ref. code 085.1253.0	mm	-0,100÷0,02	0,8 with 3 locating notch
- gasket ref. code 085.1482.0	mm	-0,030÷0,178	1 with 2 locating notch

Main bearings

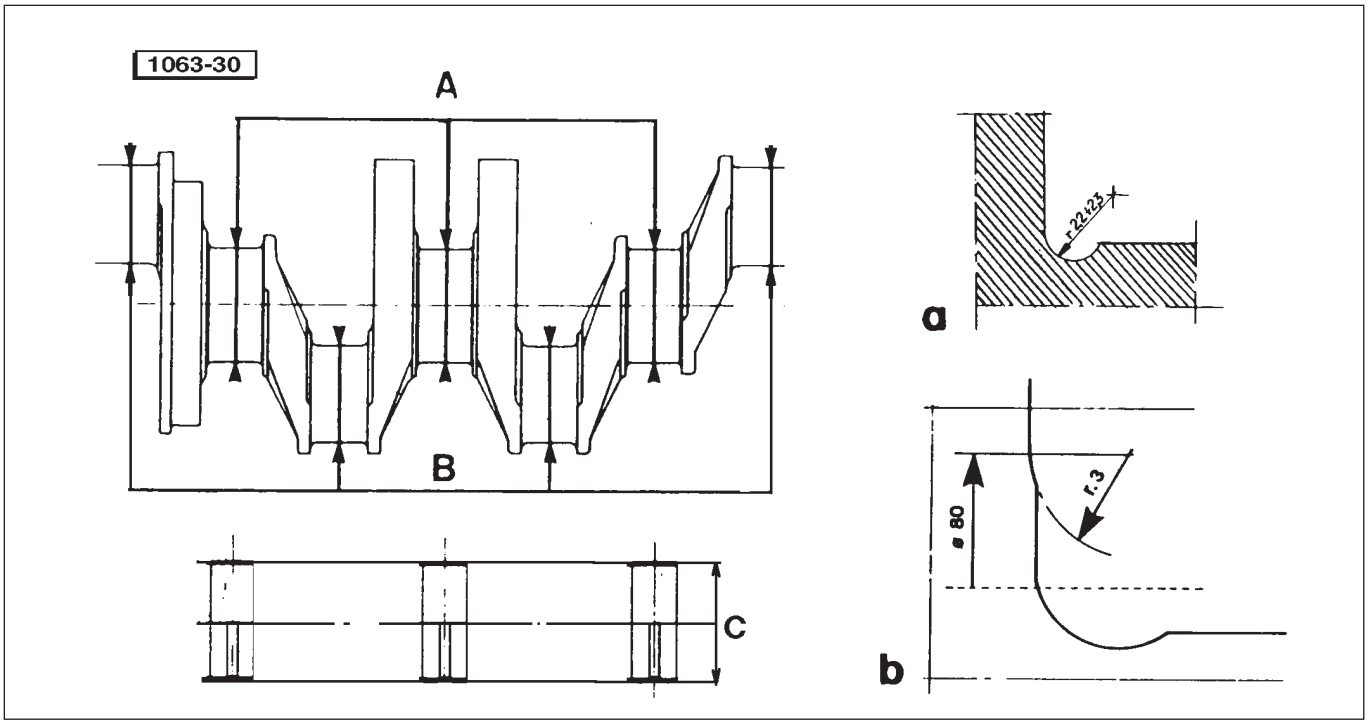


Fig. 9 - Crankshaft and main bearing dimensions. (The scheme is relative to the motor to 4 cylinders).
a crankshaft main journal and big end journal machining.
b crankshaft tail journal machining.
c main bearing inside diameter

			Ø machining	Ø max. wear
A	**main journal diameter**	**mm**	**70,000 $^{-0,020}_{-0,040}$**	**69,900**
	1st grinding	mm	69,750 $^{-0,020}_{-0,040}$	69,650
	2nd grinding	mm	69,500 $^{-0,020}_{-0,040}$	69,400
	max. journal taper allowed	mm	0,020	0,050
	max. main journal ovality allowed	mm	0,005	0,050
	max. journal boring diameter	mm	74 $^{0}_{-0,030}$	
B	**B big end journal diameter**	**mm**	**63,500 $^{0}_{-0,020}$**	**63,400**
	1st grinding	mm	63,250 $^{0}_{-0,020}$	63,150
	2nd grinding	mm	63,000 $^{0}_{-0,020}$	62,900
	big end journal cilindricity	mm	0,020	0,050
	big end journal out-off-roudness	mm	0,005	0,050
C	**mean bearing inside diameter**	mm	70,000 $^{+0,018}_{+0,068}$	70,130
	1st grinding	mm	69,750 $^{+0,018}_{+0,068}$	69,880
	2nd grinding	mm	69,500 $^{+0,018}_{+0,068}$	69,650

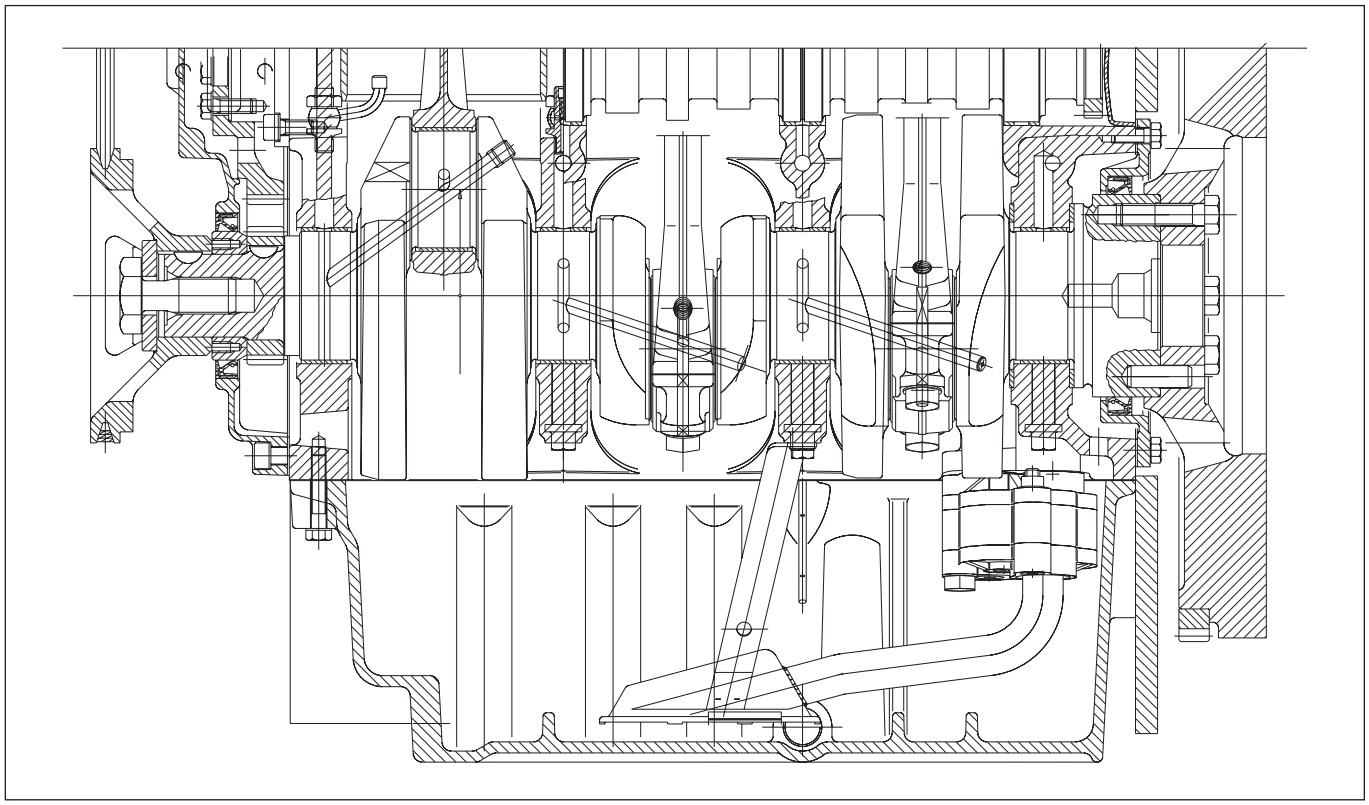


Fig.10 - Crankshaft assembly - 3-cylinder engine

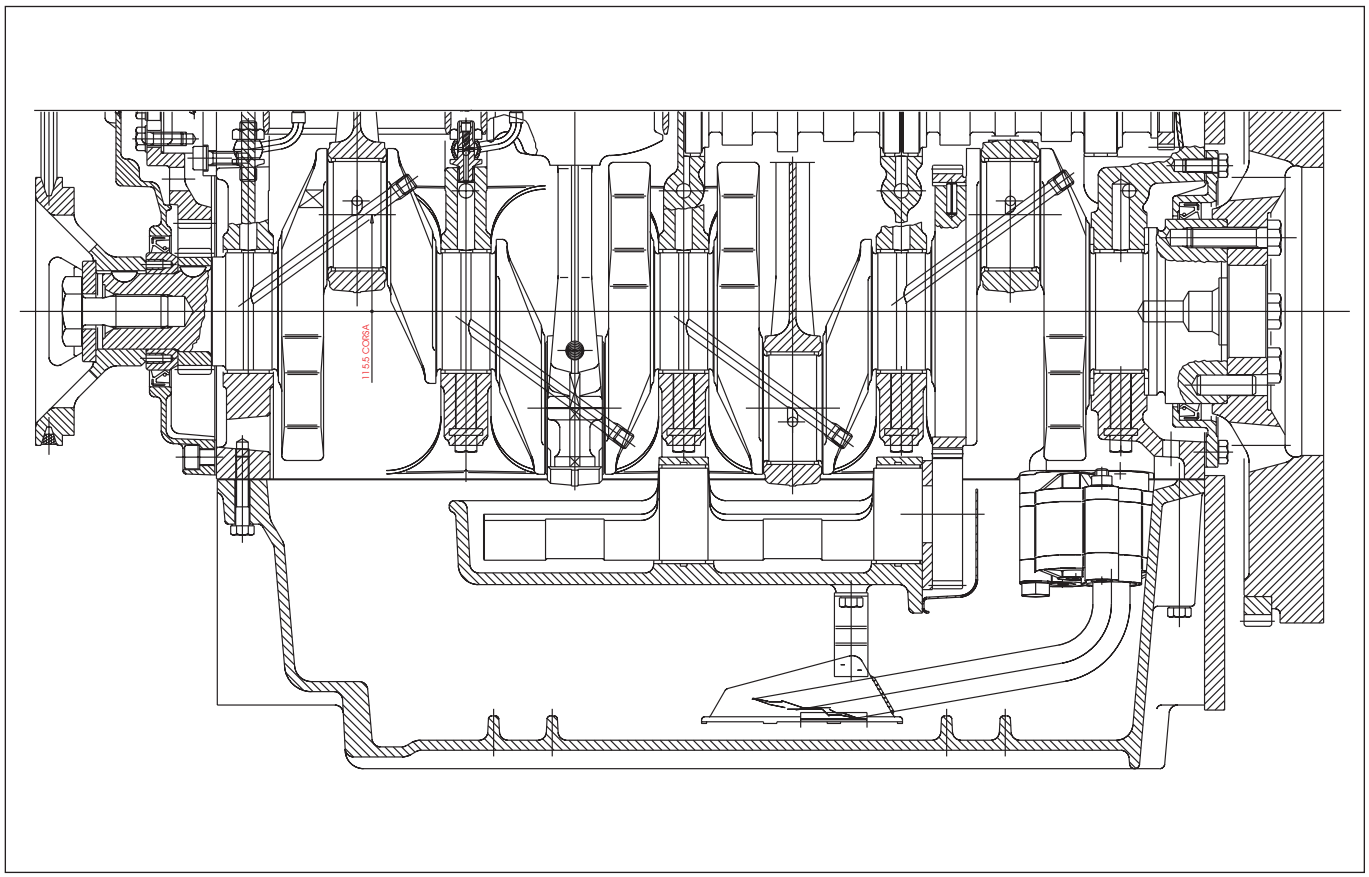


Fig. 11 - Crankshaft assembly - 4-cylinder engine

Crankshaft

The crankshaft is in nodular cast iron.
All operations must be carried out in compliance with the following standards:
1 - Metalloscope inspection of 100% of pieces; cracks are not permissible.
2 - When grinding the crankshaft, it should preferably rotate in the opposite direction to its normal rotation in the engine, but rotation in the same direction is permissible. During the polishing operation, however, the crankshaft must rotate only in the same direction as it normally rotates in the engine.
The surfaces must be polished and totally free of machining marks.
NB: the permitted values for crankshaft grinding are given on page 33.
3 - The emerging Ø6 mm oilways must be chamfered with a radius of 1.5 mm, and these chamfers must be free from machining marks.
4 - Round off all sharp edges.
5 - The crankshaft (complete with ring gear, in the case of 4-cylinder engines) and Ø12 mm dowel, must be balanced both statically and dynamically. The maximum permissible unbalance on the end supports is 150 g cm.
6 - Before assembly, degrease and thoroughly clean the crankshaft, including the oilways.

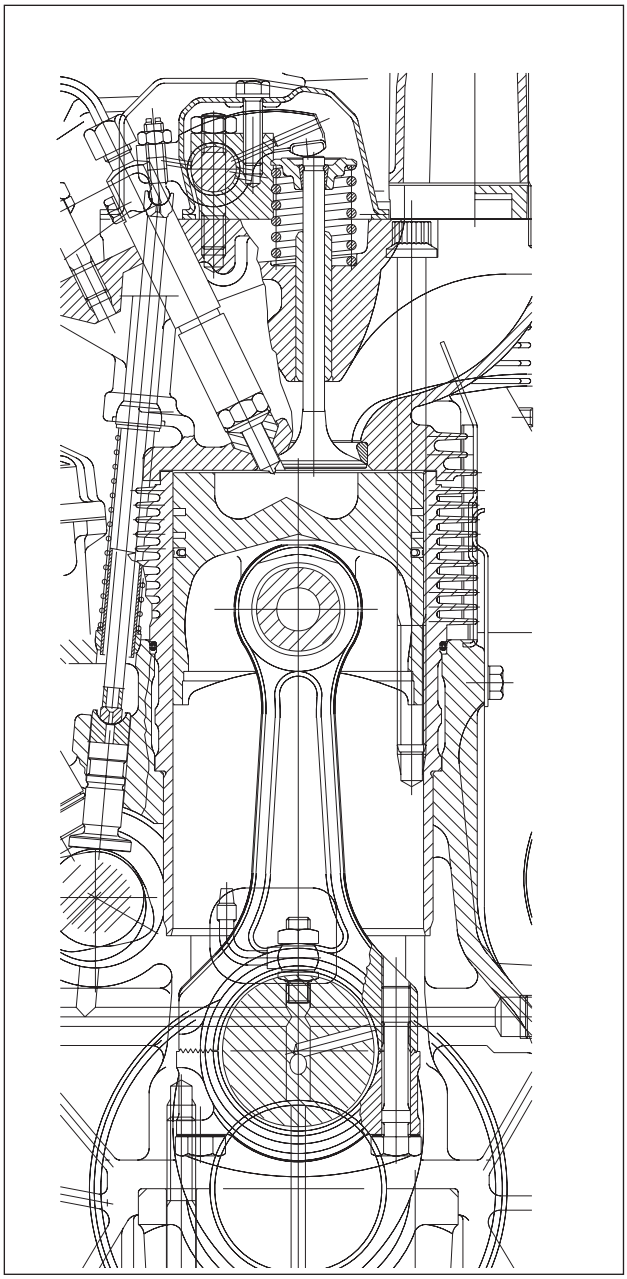


Fig. 11 - 4-cylinder engine crankshaft cross-section 1000-A.

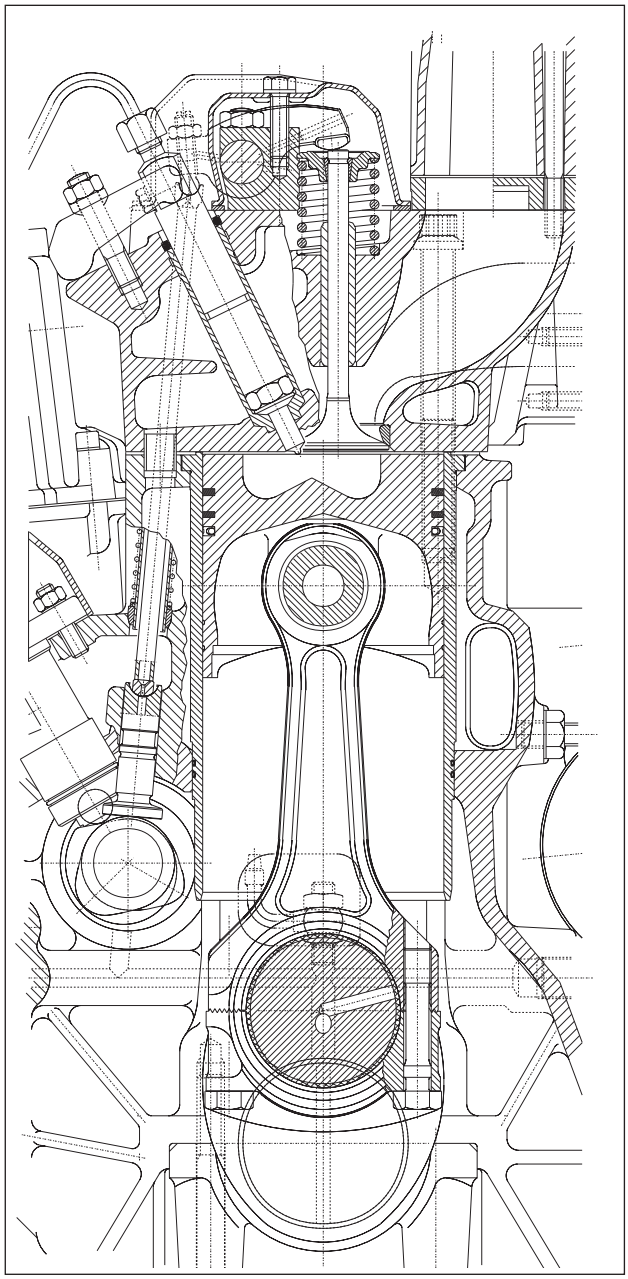


Fig. 12 - 4-cylinder engine crankshaft cross-section 1000-W.

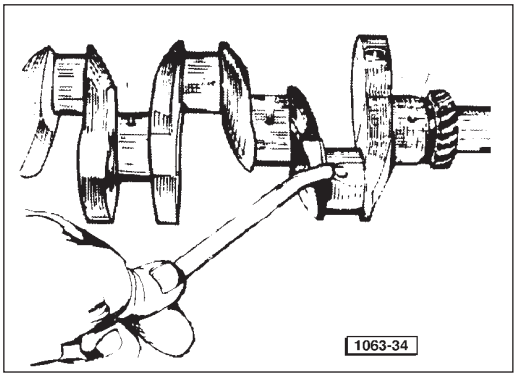


Fig. 13 - Cleaning crankshaft lubricating holes.

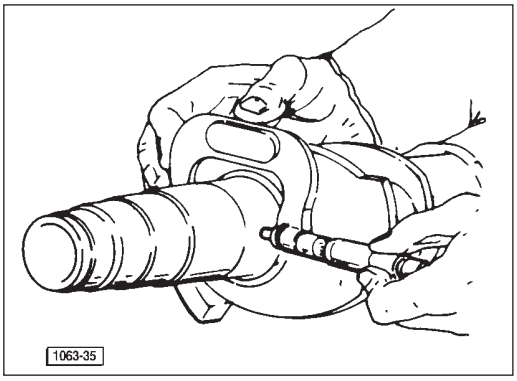


Fig. 14 - Checking crankshaft dimensions.

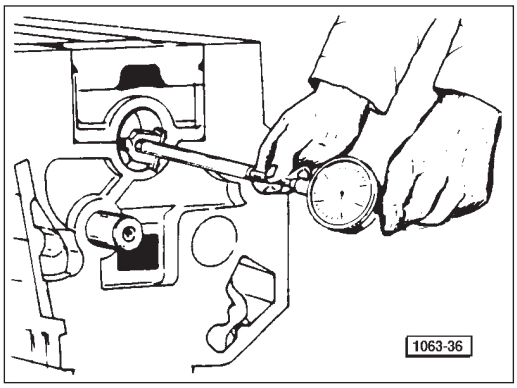


Fig. 15 - Checking main bearing inside diameter.

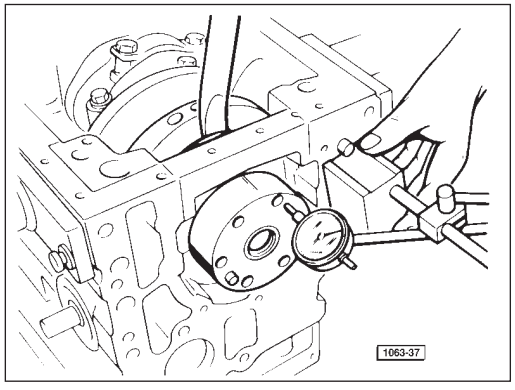


Fig. 16 - Checking crankshaft end play.

Crankshaft end play is adjusted by two-part spacer rings (see also specifications table) fitted onto the rear main bearing sides (flywheel side); these are provided with a projection preventing faulty assembly, the rings can be mounted after crankshaft installation too.

Each main bearing consists of two shells each being provided with a small tongue permitting it to be held in position inside the bearing cap seat. The upper bearing shell is provided with a cavity for oil flow. Main bearing caps shall be positioned according to the numbering printed on their bodies and fixed to the block with two special securing screws, which can only be replaced with other original screws.

Checking crankshaft

After the crankshaft has been thoroughly cleaned examine it carefully.

Meticulously ascertain the main and big end journals do not show signs of seizure, otherwise rigrinding will be necessary. If the journals are cracked the crankshaft shall be replaced.

Using a micrometer gauge make sure the main journal diameters are not below the specifications given in the related table, otherwise a regrinding should be performed.

Checking crankshaft journal out-of-roundness and tape

With a micrometer gauge measure crankshaft journal taper and out-of-roundness amounts; if reading exceed the maximum allowances specified a regrinding shall be performed. (Refer to specification table).

Checking main bearings

Clean main bearings carefully and then examine the internal surfaces for indentation, scuffing, scratching or evident antifriction lining wear. If any replace the main bearings.

Check main bearing inside diameters with an internal comparator. If the diameters are found in excess of the maximum wear limits, the main bearings shall be replaced.

Main bearings are supplied as spare parts either with normal size or undersize inside diameters. The inside diameter undersize range is 0.25 to 0.50 mm.

Be very careful when installing the main bearings in order they can be correctly positioned. The bearing shells provided with a lubricating hole must be placed in the upper side (i.e. into the block forging and not into the main bearing caps).

Mounting main bearing caps

The numbers stamped on the main bearing caps must be on the same side as those stamped on the block.
Carefully check main bearing cap fixing screws for stretching, if so replace them with original screws only.

The screws are to be tightened progessively to 1 kgm (9.8 Nm) torque and subsequently to 3 kgm (30 Nm) torque then using no. 5.9030.640.0 tool furtherly tighten each single screw to an angle of 55°±1'.

Cleaning crankshaft and cylinder block internal passages

Cleaning of the lubricating oil passage should be made after all crankshaft and cylinder block overhauling operations have been accomplished.
This cleaning can be carried out by using Diesel fuel or compressed air blasted into the passages

Checking crankshaft end play (Fig. 16).

This check is performed with a magnetic base comparator operating as follows:

— place the magnetic base together with the comparator onto the crankcase and then make the comparator feeler contact one of the crankshaft ends.
— use a lever to move the crankshaft axially towards the comparator side until the comparator hand stops moving.
— reset the comparator and push the crankshaft with the lever towards the opposite side and then take the end play reading from the comparator dial.

This reading should be 0.10 to 0.27 mm; if higher than specified, some spacing half-rings with 0.10 to 0.20 mm oversize with respect to the nominal thickness shall be fitted. These half-rings are regularly supplied as spare parts.

Triangular-section gaskets should protrude as shown in figure 17.
(Dopo il posizionamento nel caso di sostituzione delle guarnizioni, tagliare la parte eccedente)
Sigillare la zona lato coppa olio Engine e la zona carter anteriore con silicone tipo DIRKO TRASPARENT della ditta Erling.

In its rear portion the crankshaft is provided with a 3x30° chamfer. In addition the surface contacting the sealing ring has undergone a hardening process.
The oil seal must be fitted using tool 5.9030.628.0.

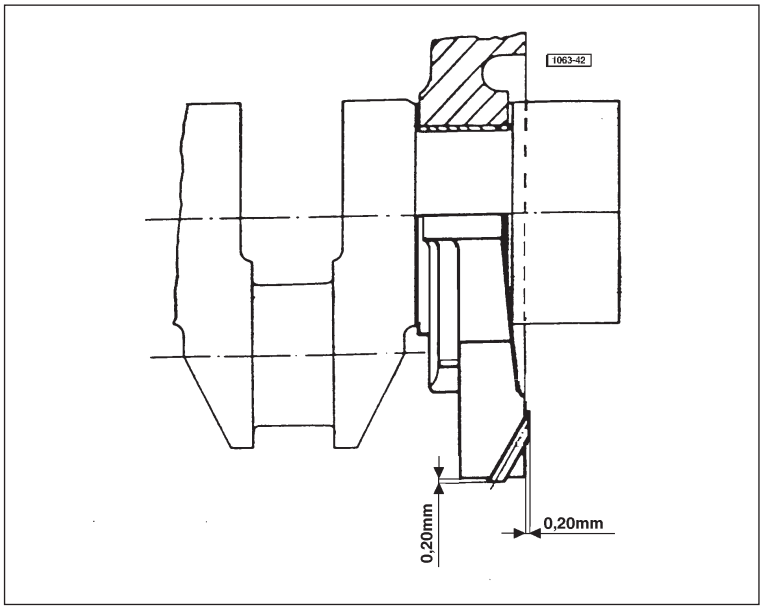


Fig. 17 - Main bearing cap gasket installation.

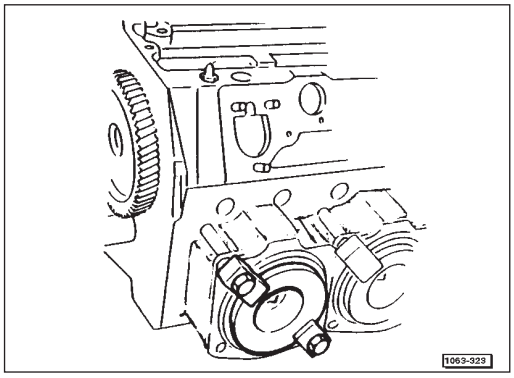


Fig. 18 - Special fixture for piston positioning during crankshaft installation.

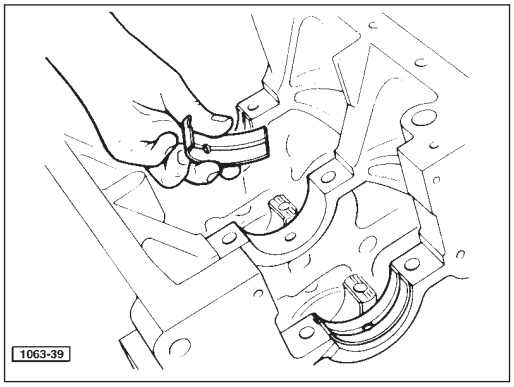


Fig. 19 - Mounting one holed main bearing shell into the cylinder block forging.

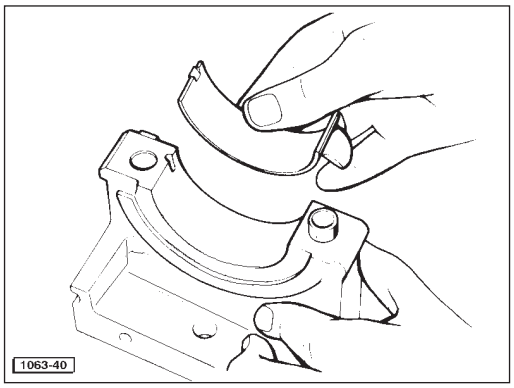


Fig. 20 - Fitting the one main bearing shell into the main bearing cap.

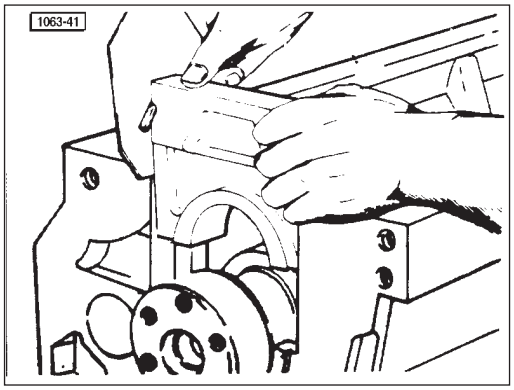


Fig. 21 -Installing the crankshaft main bearing caps.

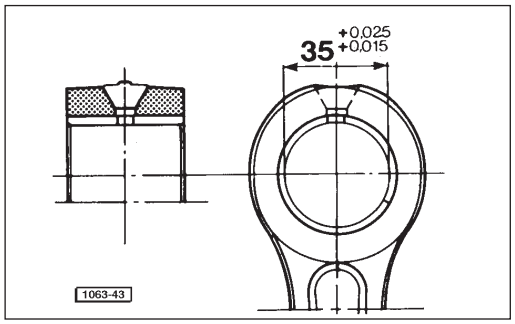


Fig. 22 - Fitting bushing into the con-rod small end.

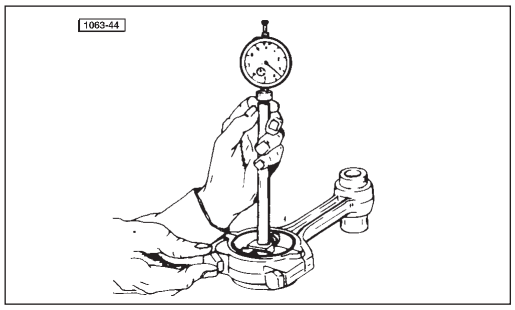


Fig. 23 - Checking con-rod for wear.

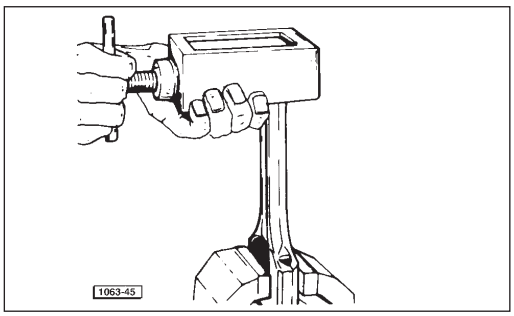


Fig. 24 - Pulling out/driving in con-rod bushing.

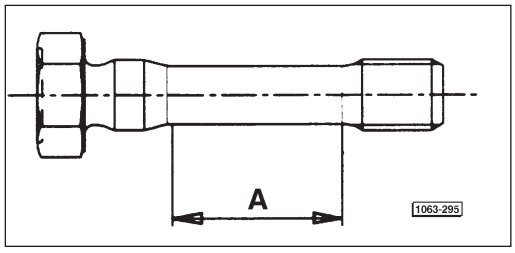


Fig. 25 - Checking the diameter of the connecting rod clamp bolts.

Connecting rods - connecting rod bearings and bushings (Fig. 26)

Checking connecting rod axis parallelism
(using no. 5.9030.651.4 special tool)
Place the connecting rod on the special tool and mount the comparator by loading the feeler 1 mm against the gudgeon pin.
Move the connecting rod from right to left and reset the comparator to zero when the comparator hand indicates the maximum reading. Remove the connecting rod from the tool and replace it after a 180° rotation along its longitudinal axis.
Repeat this operation and note the difference in reading compared with the previous taken. If the difference is above the specified tolerances, the rod shall be straightened by hand press.

Checking connecting rod bushing and bearings

Check connecting rod bushings and bearings for scoring or scuffing. Wear amount should not exceed specifications, otherwise they must be replaced. After fitting the bushing, check that the diameter is as specified in figure 22.

Checking connecting rod weight

The difference in weight between one engine connecting rod and another must never exceed 20 gr.

Installing connecting rods

The screws are to be tightened progessively to 1 kgm (9.8 Nm) torque and subsequently to 3 kgm (29.5 Nm) torque then using no. 5.9030.640.0 tool furtherly tighten each single screw to an angle of 90°±1'
If these bolts are to be reused, check that the difference between the maximum and minimum diameter measured at the part denoted **A** (Fig 25) is less than 0,1 mm. If not, the bolt must be replaced.

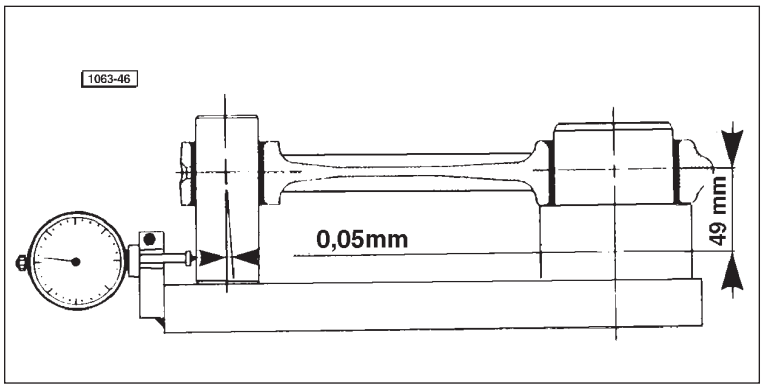


Fig. 26 - Checking con-rod axis parallelism.

		Ø machining	Ø max wear
connecting rod bearing inside diameter	mm	63,500 $^{+0,035}_{+0,075}$	63,620
1st undersize	mm	63,250 $^{+0,035}_{+0,075}$	63,370
2st undersize	mm	63,000 $^{+0,035}_{+0,075}$	63,120
connecting rod bushing inside diameter	mm	35 $^{+0,015}_{+0,025}$	35,060
maximum misalignment of the connecting rod smoll end axis (measured at 49 mm from con-rod centre line)			0,05 mm

pistons - gudgeon pins - piston rings

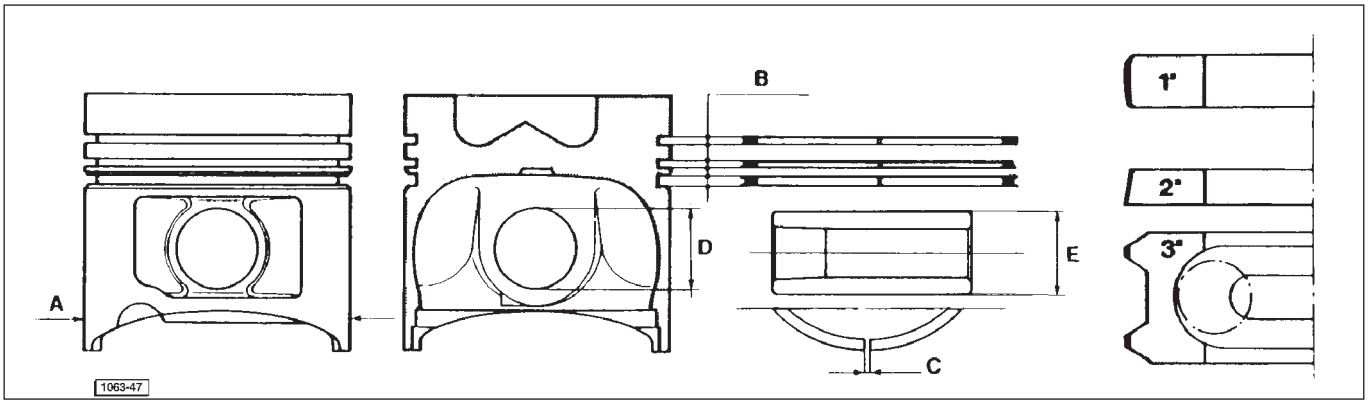


Fig. 27 - Piston, piston ring and gudgeon pin dimension.

Pistons

			Ø machining	Ø max. wear
A	**piston diameter at 90° of the gudgeon pin axis and at 10 mm from the bottom base**	mm	104,9 ± 0,007	104,680
B	**clearance betwen piston ring and piston**			
	1st piston ring	mm	0,110 ÷ 0,145	0,250
	2st piston ring	mm	0,050 ÷ 0,085	0,150
	3st piston ring	mm	0,030 ÷ 0,650	0,150
C	**piston ring and gap**			
	1st piston ring	mm	0,400 ÷ 0,650	1,500
	2st piston ring	mm	0,400 ÷ 0,650	1,500
	3st piston ring	mm	0,300 ÷ 0,600	1,500
D	**inside diameter of gudgeon bore in piston**	mm	naturally aspirated 35 $^{+0,006}_{+0,012}$	35,020
		mm	turbo 35 $^{0}_{-0,006}$	35,020
E	**gudgeon pin diameter**	mm	35	34,970

Pistons are specially forged hypereutectic aluminium-silicone-alloy. The air-fuel mixing chamber machined in piston crown has toroidal shape. Both the part number and reference A (meaning air) are printed in piston top, i.e. the piston identification code.

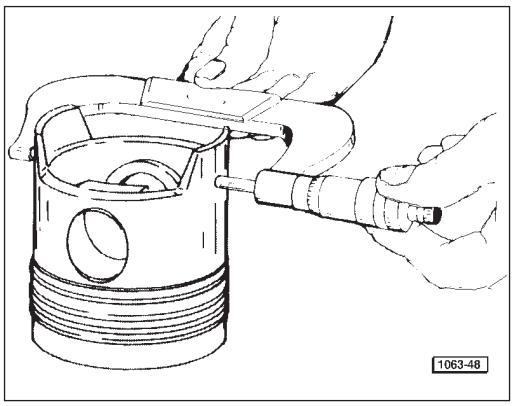


Fig. 28 - Measuring piston diameter.

Piston rings

Pay particular attention to the identification marks on the piston rings for correct assembly.
Inserting the first or the second piston ring upside down will result in either engine uneven operation or serious damage, even though the piston groove is the correct one. The figure above illustrates how assembly should be performed correctly.
Figure 30 on page 41 shows the correct assembly.
Warning: for turbocharged engines, when replacing the piston rings check that the second piston ring is also chromium faced (do not fit the non-chromed ring used for naturally aspirated engines).

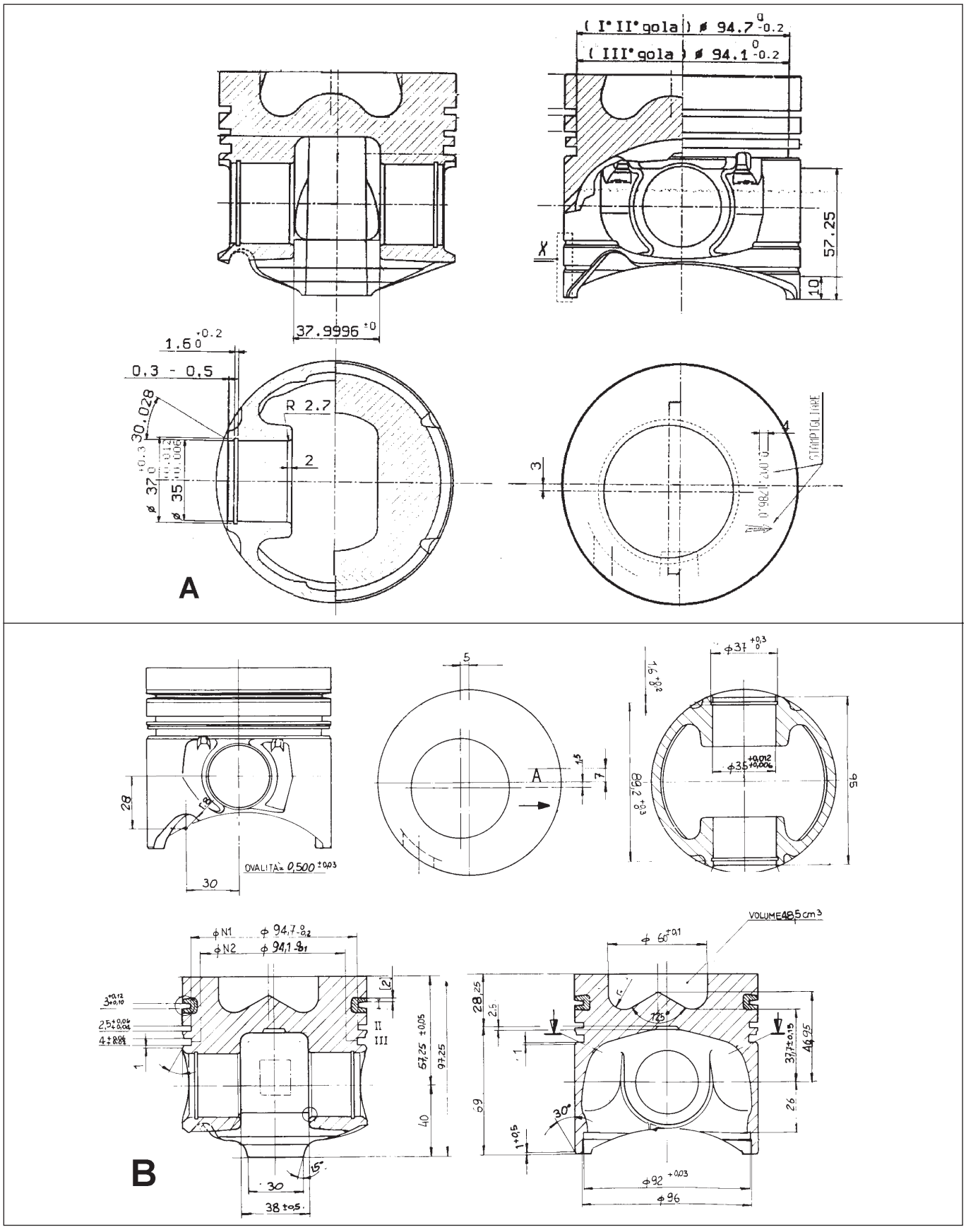


Fig. 29 - Pistons. **A** - naturally aspirated engines **B** - turbo engines.

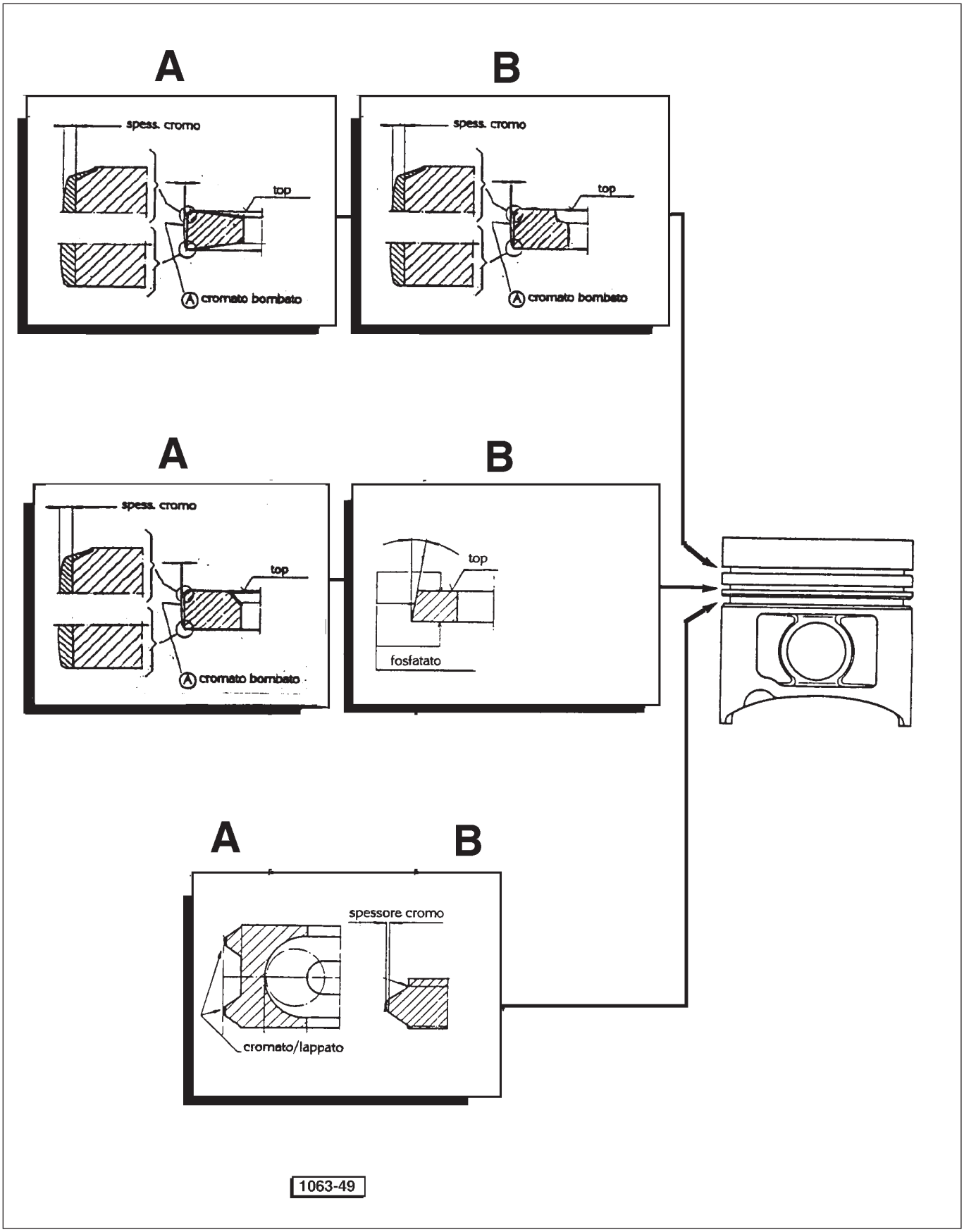


Fig. 30 - Instructions for fitting piston rings to pistons. **A** - naturally aspirated engines **B** - turbo engines

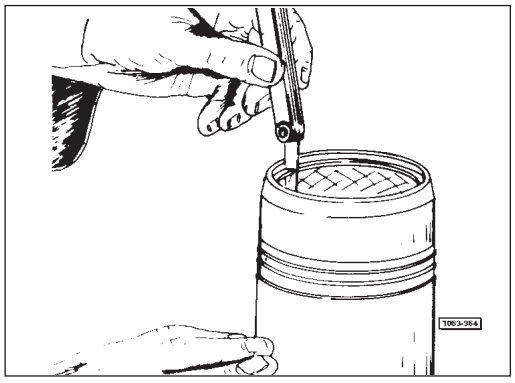


Fig. 31 - Checking clearance between piston rings and piston ring grooves in piston.

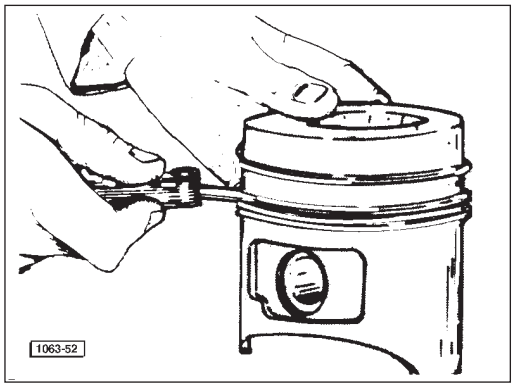


Fig. 32 - Checking clearance between the piston rings and grooves

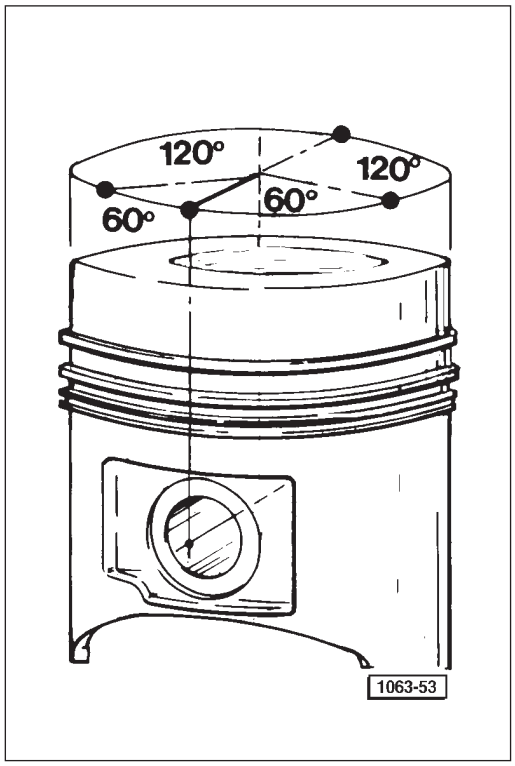


Fig. 33 - Correct assembly of piston rings.

Cleaning pistons

Descale piston crown and piston ring grooves using an emery cloth.

Remove any buildups from gudgeon snap ring seats.

After all parts have been thoroughly cleaned and before further checking is performed, examine them carefully for cracking or damage which may require replacement.

Light scoring o seizing signs can be removed with very fine emery cloth.

Checking pistons

Measure piston diameter with a micrometer gauge as shown in figure 28.

If the diameter measured is below the specifications given in table on page 39, the piston needs to be replaced.

Checking gudgeon pin and gudgeon pin seat in piston

If either gudgeon or related seat diameter measurement differs from specifications given in table on page 39, the replacement of the worn-out part is recommended.

Check also gudgeon pin snap rings for proper elasticity or damage. If they are found not conforming to the operating conditions replace them.

Checking piston ring end gap

Both the compression ring and the oil control ring ends installed in the cylinder should not have a gap exceeding the specifications shown in tableon page 39.

This check is carried out by inserting the piston rings into the cylinder so as the ring end gap can be measured with no. 5.9030.270.0 special thickness gauge, as shown in figure 31.

NOTE - Excessive clearance between piston rings and piston ring grooves in piston may cause high lubricant consumption as well as engine power loss.

Piston rings must be replaced each time piston is removed or cylinders are replaced.

Checking clearance between piston rings and piston ring grooves in piston

This check is to be made with new piston rings, as shown in figure 32, by inserting the piston ring into the groove and then the blade of no. 5.9030.270.0 thickness gauge. Refer to specifications table on page 39, for maximum clearance allowance. If the clearance measured exceeds the maximum allowance, piston shall be replaced.

Installing piston rings

During cylinder assembly perform a proper orientation of the first piston ring so that its cut be moved by 60° from the gudgeon pin axis.
The other piston rings should be inserted with their cuts moved by 120° from each other.

Installing pistons into cylinders

Use no. 5.9030.654.0 ring clamping band as shown in figure 34 to install piston into the cylinder.
Afterwards insert piston and cylinder assemblies into the cylinder block.

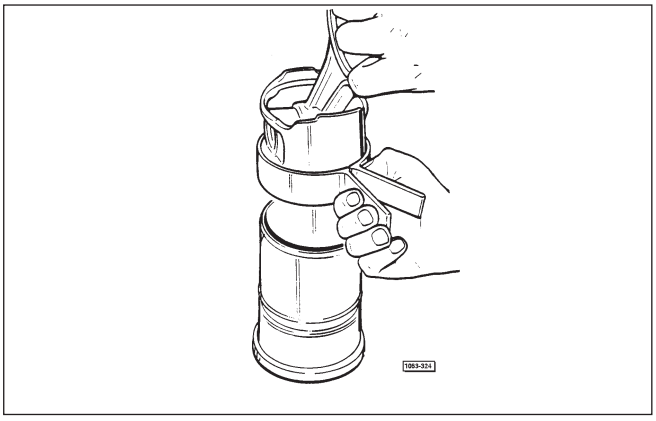
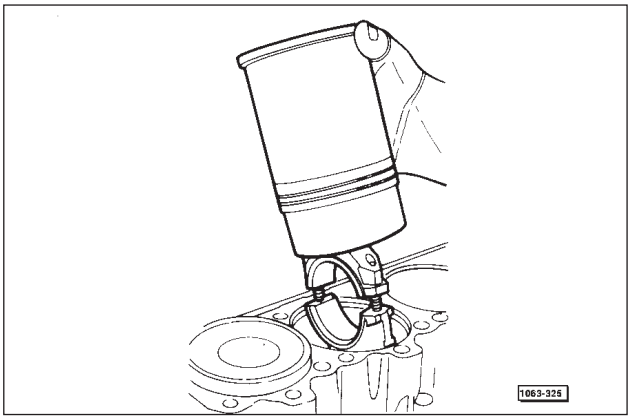


Fig. 34 - Installing piston into cylinder 1000-W.



Installing piston and cylinder assembly 1000-W.

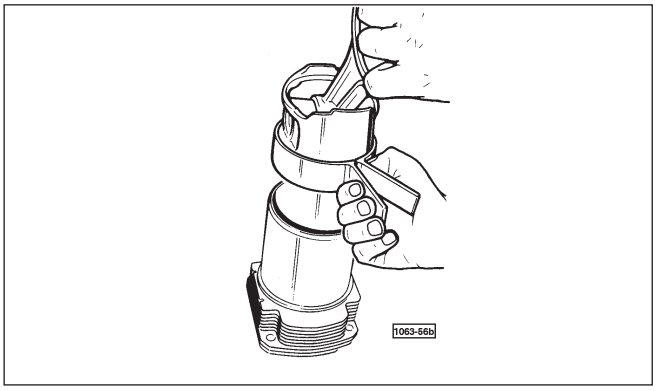
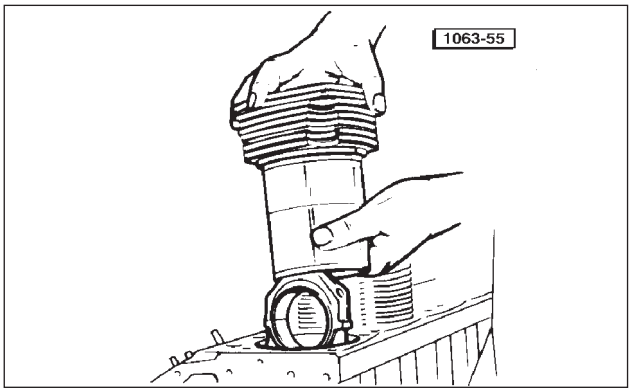


Fig. 35 - Installing piston into cylinder 1000-A



Installing piston and cylinder assembly 1000-A

NOTE - Pistons can be taken without removing the crankshaft, just withdrawing along with the connecting rods directly from the cylinders after removing both cylinder heads and connecting rod caps.

Before installing the cylinder liners, fit the O-ring seals code 2.1519.096.0/20.

The connecting rod-piston assembly must be assembled with the piston crown recess oriented towards the tappets (Fig. 36).

The big-end caps must be fitted so that the punched number appears on the same side as the number on the connecting rod.

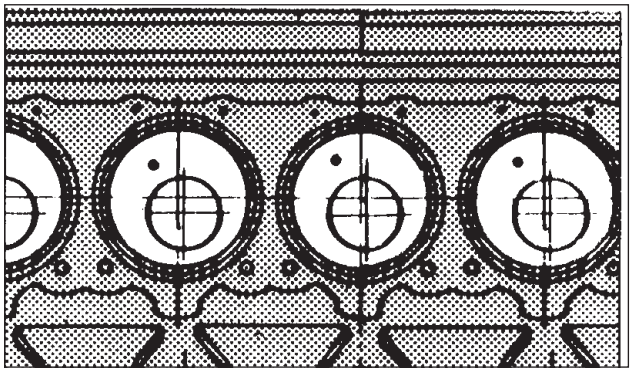


Fig. 36 - Correct orientation of piston and con-rod assembly

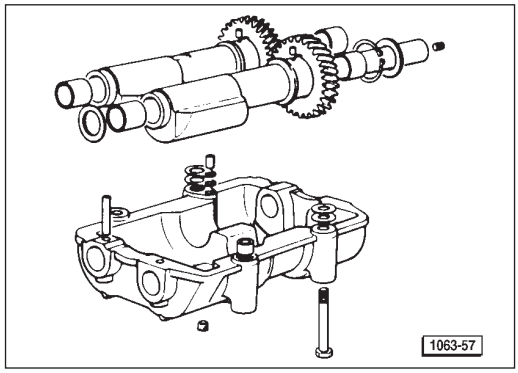


Fig. 37 - Counterweight assembly components.

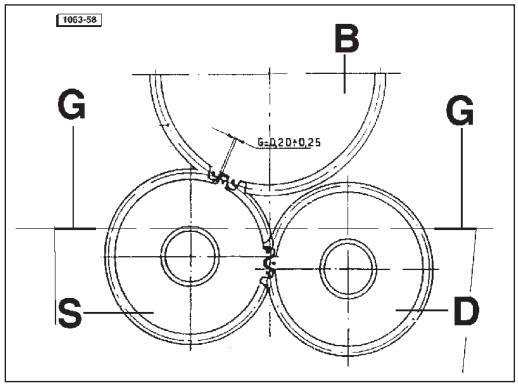


Fig. 38 - Counterweight assembly.

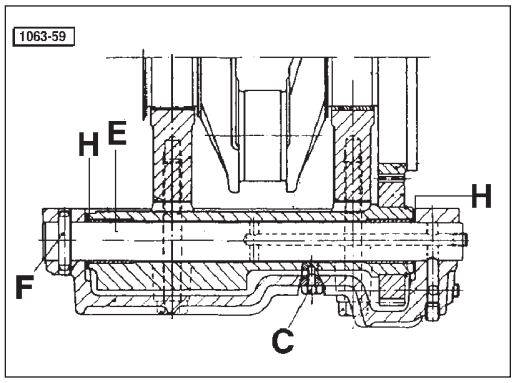


Fig. 39 - Counterweight assembly cutaway-view.

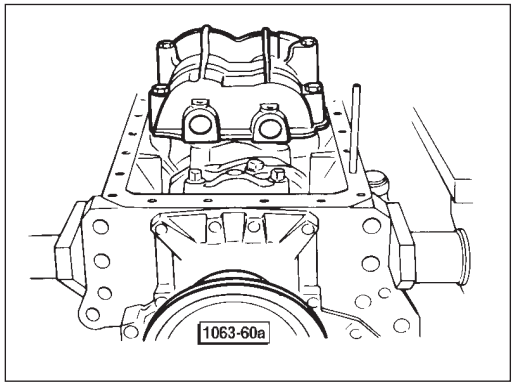


Fig. 40 - Engine-installed counterweights.

Counterweights for 4-cylinder engines

Specifications

counterweight end play	mm	0,1 ÷ 0,43
backlash allowance between counterweight and crankshaft crown wheel teeth	mm	0,20 ÷ 0,25
bushing inside diameter	mm	26 $^{+0,050}_{+0,075}$
max. wear	mm	26,150

Checking counterweights

Check counterweight bushing inside surface conditions. Ensure the inside diameter is not above specifications; otherwise replace bushings.
With bushing installed perform boring according to specifications given in the table above.

Installing and adjusting counterweights

Position the weight marked with **S** in the related support seat, then place weight **D** too, so that the engraved teeth coincide as shown in figure.

Fit gudgeons **E** and insert spacer washers **H**.
Be sure the counterweight end play is 0.1 to 0.43 mm. Complete installation by fixing gudgeons through special pins **F** and the weights with securing screw **C**.

Apply the support and weight assembly underneath the cylinder block so as the engraved counterweight **S** tooth engages between the two engraved teeth of crankshaft crown wheel **B**. Then fit a number of shims **G** until a 0.20 to 0.25 mm backlash between crankshaft crown wheel and weight **S** teeth is obtained.

Remove screw **C** and finally tighten the weight support securing screws to cylinder block also applying some Loctite 242.

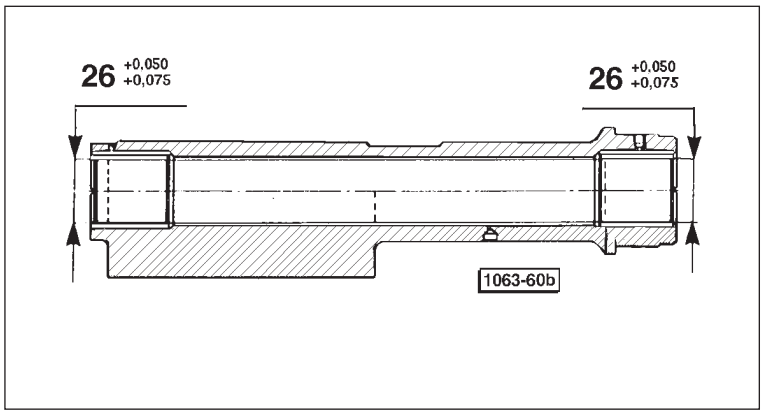


Fig. 41 - Installed counterwight bushings.

Engine flywheel

Before dismounting the flywheel it is advisable to remove two diametrically opposed screws replacing them with two stud bolts.

To replace the flywheel ring gear operate as follows:

— Immerse the flywheel into a water-filled container. Keep the flywheel lifted a few centimeters from the container bottom using some small iron blocks fitted under the ring gear teeth. The ring gear should be turned upwards and project a few millimeters from the water surface.

— Using a flame heat the ring gear all around the toothed sector until the flywheel is dropped to the container bottom due to heat expansion.

— Heat the new ring gear up to a temperature of about 200° C and position into flywheel housing. Ensure the tooth chamfers are turned upwards as shown in figure 43.

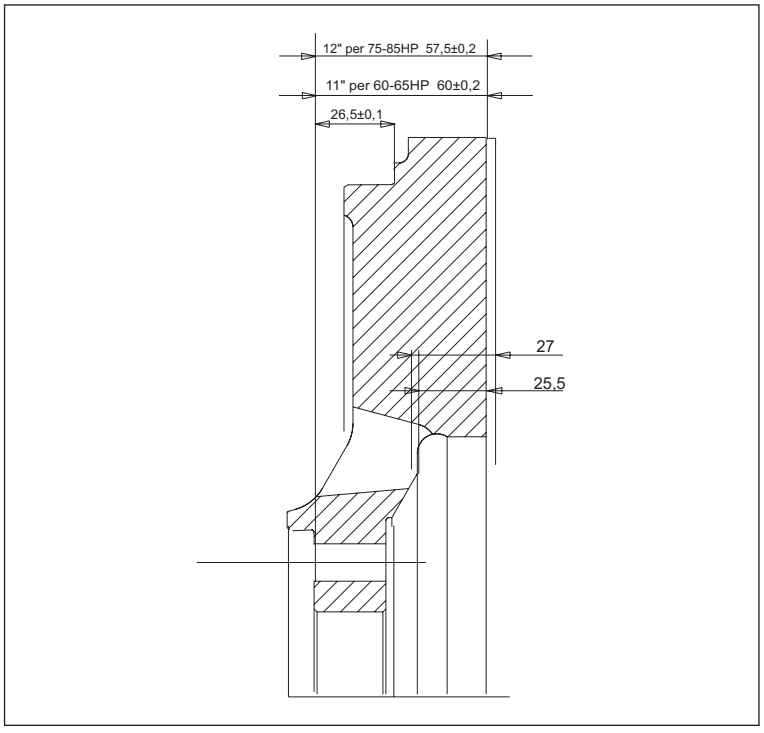


Fig. 42 - Machining depths for clutch matching surface of the flywheel.

Checks

Make certain that the sliding surface of the flywheel is not scored; material can be removed, if necessary, to a maximum depth of 0,5 mm, (surface roughness 1,6 mm).

The flywheel contact surface should thoroughly be in the same plane as the clutch plate. This can be verified by fitting a magnetic base comparator as illustrated in figure 44 and rotating the flywheel by hand.

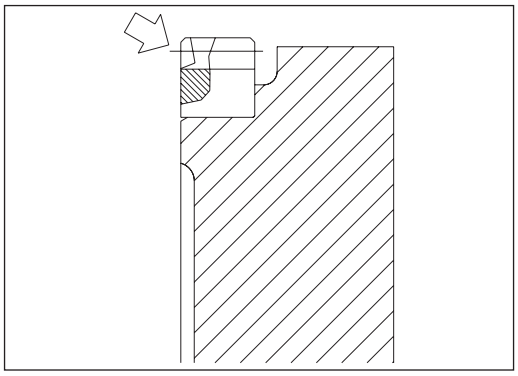


Fig. 43 - Ring gear position on engine flywheel.

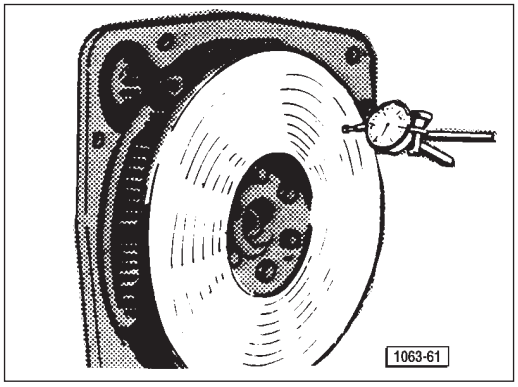


Fig. 44 - Checking flywheel face coplanarity
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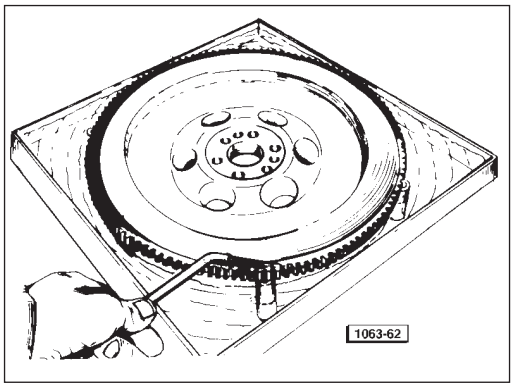


Fig. 45 - Removing flywheel ring gear.

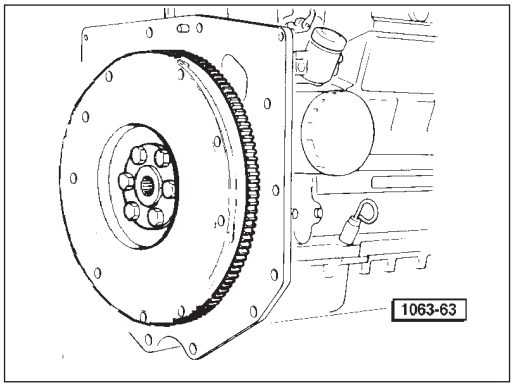


Fig. 46 - Engine flywheel.

Crankshaft removal procedure

— Remove engine from tractor and install on no. 5.9030.002.0 traveling stand for transfer on no. 5.9030.491.4 swivel bench.
— Remove the oil pan and the oil pump suction rose.
— After detaching the pulley belt take the pulley from the crankshaft along with the pulley hub.
— Dismount engine front case, rear flywheel, engine-to-gearbox connecting flange and sealing ring holding flange. (In 4-cylinder engines also the counterweight assembly is to be removed).
— Remove big end caps and related bronze bushings.
— Remove main bearing caps and related bronze bushings.
— Remove the crankshaft and take the upper bronze bushings.

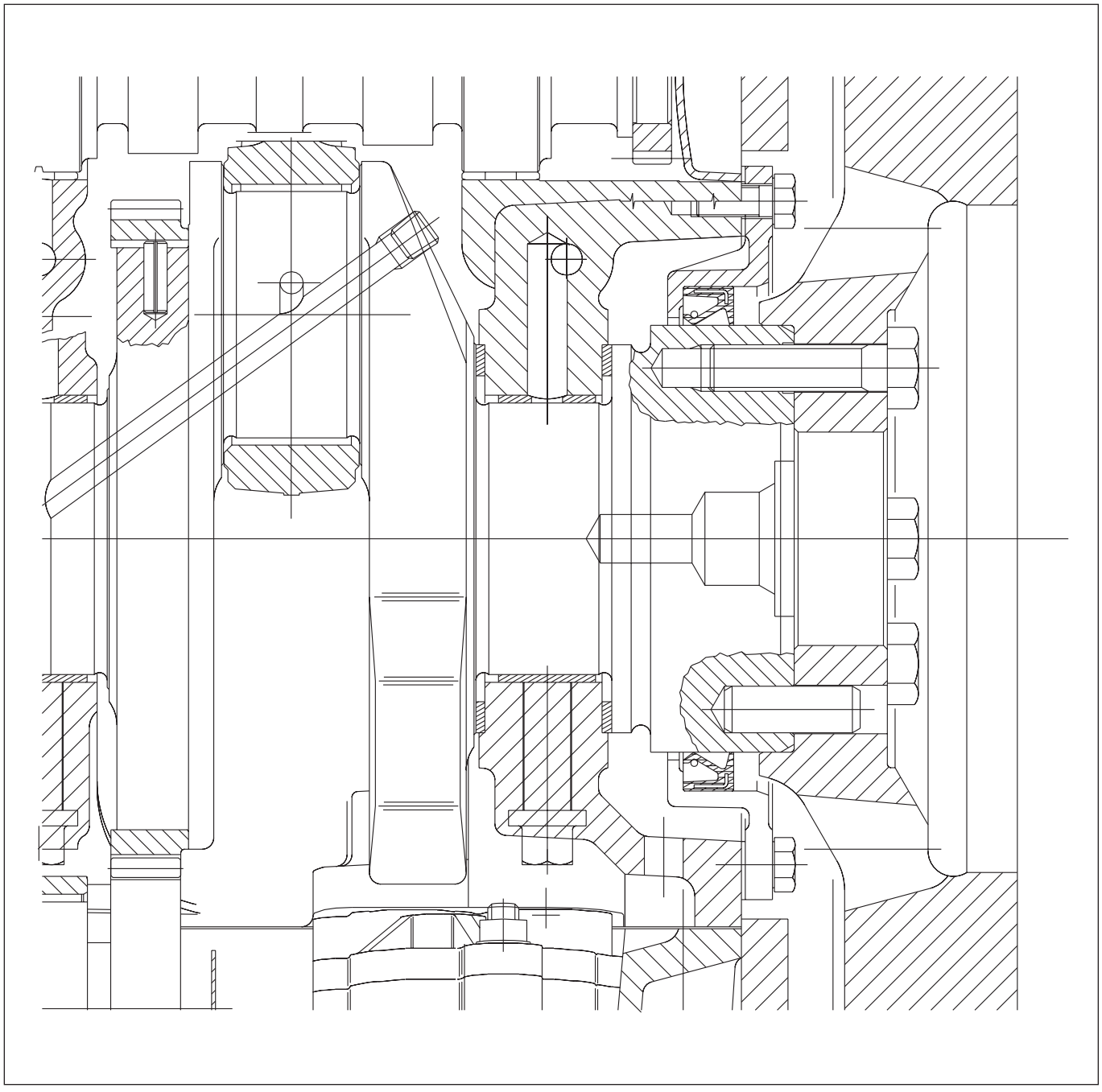
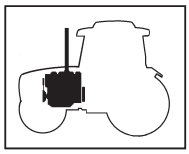


Fig. 47 - Rear flange cross-section.



		Ø machining	Ø max. wear
Camshaft cam lift			
intake cam	mm	6,86 $^{+0}_{-0,087}$	6,30
exhaust cam	mm	6,71 $^{+0,05}_{-0,05}$	6,20
injection cam	mm	8 $^{+0,05}_{-0,05}$	7,50
Camshaft journal diameter	mm	55 $^{0}_{-0,030}$	54,950
Camshaft bushing diameter	mm	55 $^{+0,060}_{-0,090}$	55,180
Idler gear bushing diameter	mm	40 $^{+0,025}_{+0,064}$	40,125
Idler gear pin diameter	mm	40 $^{0}_{-0,025}$	39,950

The installation of single-cylinder injection pumps besides ensuring an even load distribution on the camshaft has permitted two bearings per camshaft journal to be mounted in the engine block, thus minimizing camshaft flexure.

A bimetallic bushing is inserted in each bearing to ensure very smooth camshaft operation. This can be replaced when wear exceeds the limits specified above.

All these features guarantee precise cam movement thus providing smooth timing system operation and regular fuel supply as well.

The camshaft is held in position by a forked plate fitted on engine front side.

The engine oil pump and engine governor driving gear is mounted at the rear camshaft end.

Prior to installing camshaft, the engine block should be turned upside down and the tappets inserted in the related seats in the block after having been properly lubricated with oil.

Make sure the tappets slide in their seats freely.

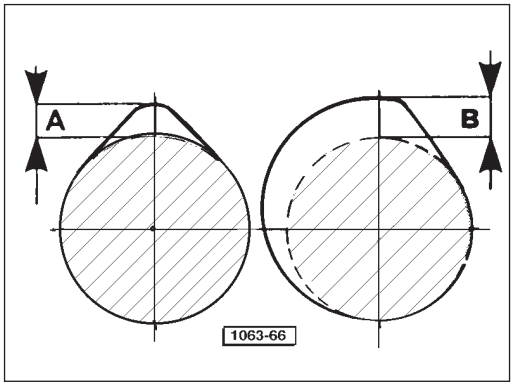


Fig. 1 - Timing system cams:
A - Valve cam; **B** - Injection cam..

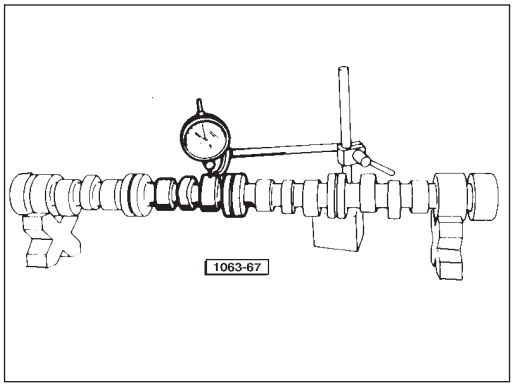


Fig. 2 - Checking cam wear.

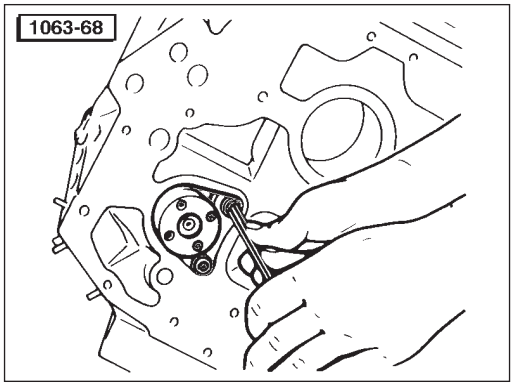


Fig. 3 - Camshaft fixing forked plate.

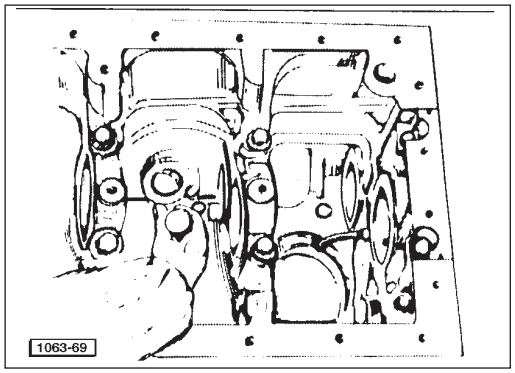


Fig. 4 - Installing tappets into the engine block.

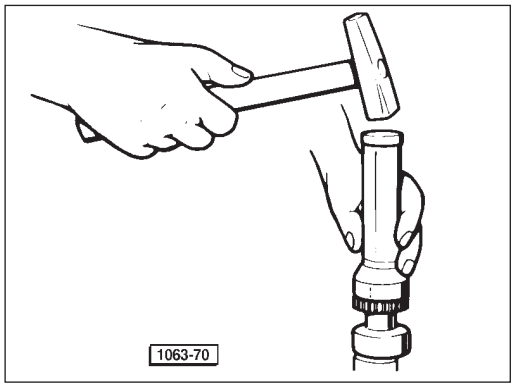


Fig. 5 - Installing the 1000.4 engine governor driving gear.

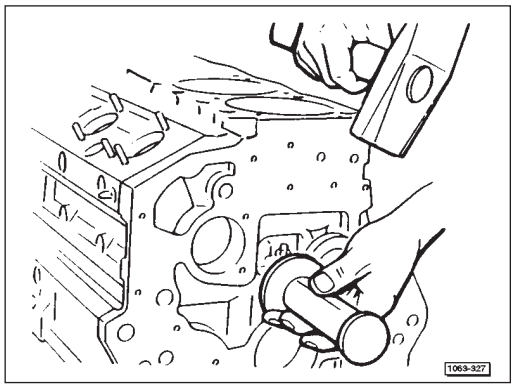


Fig. 6 - Inserting the camshaft bushings.

Checking camshaft

Examine both camshaft journal and cam surfaces: they must be thoroughly smooth and in perfect operating conditions.

If on the contrary cams show any sizing or scoring, it is convenient that both the camshaft and the bushings installed in the block be replaced. Only if a very small evidence of deterioration is noticed, this might be eliminated using a very fine grain abrasive stone.

To check for wear, place the camshaft on the special cross-shaped rests and apply a comparator feeler on each single cam then rotate the camshaft.

Minimum cam lift should never be below specifications, otherwise the camshaft should be replaced.

Checking camshaft bushings

Carefully examine bushing inside surface; should any seizing or evident wear of the antifriction lining be found replace the bushings.

If normal conditions are noticed, check the bushing inside diameters by means of an internal comparator. If the bushing inside diameters measured are in excess of the maximum wear limits specified, these shall be replaced.

Each time the bushings are replaced they must be bored.

Checking timing gear

Be sure the gear teeth are not spalled or excessively worn. Replace the gear, as necessary.

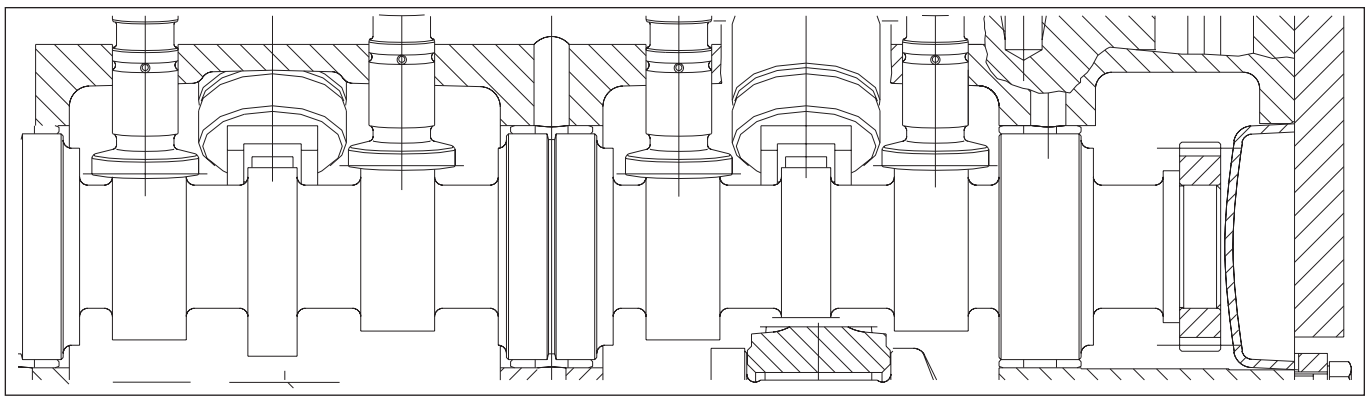


Fig. 7 - Camshaft.

Cylinder heads - valves - valve rockers

			Ø machining		Ø max. wear x
			naturally aspirated	turbo	
A	valve stem diameter	intake mm	9 $^{-0,013}_{-0,028}$	8,98 $^{-0,04}_{0}$	8,940
		exhaust mm	9 $^{-0,013}_{-0,028}$	8,96 $^{-0,02}$	8,940
B	inside diameter the valve guides installed in the heads mm		9		9,100
C	valve seat angle with respect to valve				
		intake	29°30' $^{0}_{-15}$		
		exhaust	44°30' $^{0}_{-15}$	44°22' $^{-7'30''}$	
D	valve seat width on the valve seat insert				
		intake mm	0,8 $^{+0,5}_{-0,1}$		1,8
		exhaust mm	1 $^{+0,5}_{-0,1}$		2
E	valve recess from cylinder head level				
		intake mm	1,55		1,7
		exhaust mm	2,20		2,35
F	valve seat taper	intake	60° $^{0}_{-15}$		
		exhaust	45° $^{0}_{-15}$		
G	valve diameter	intake	44,2	44,1	
		exhaust	39,5	39,4	

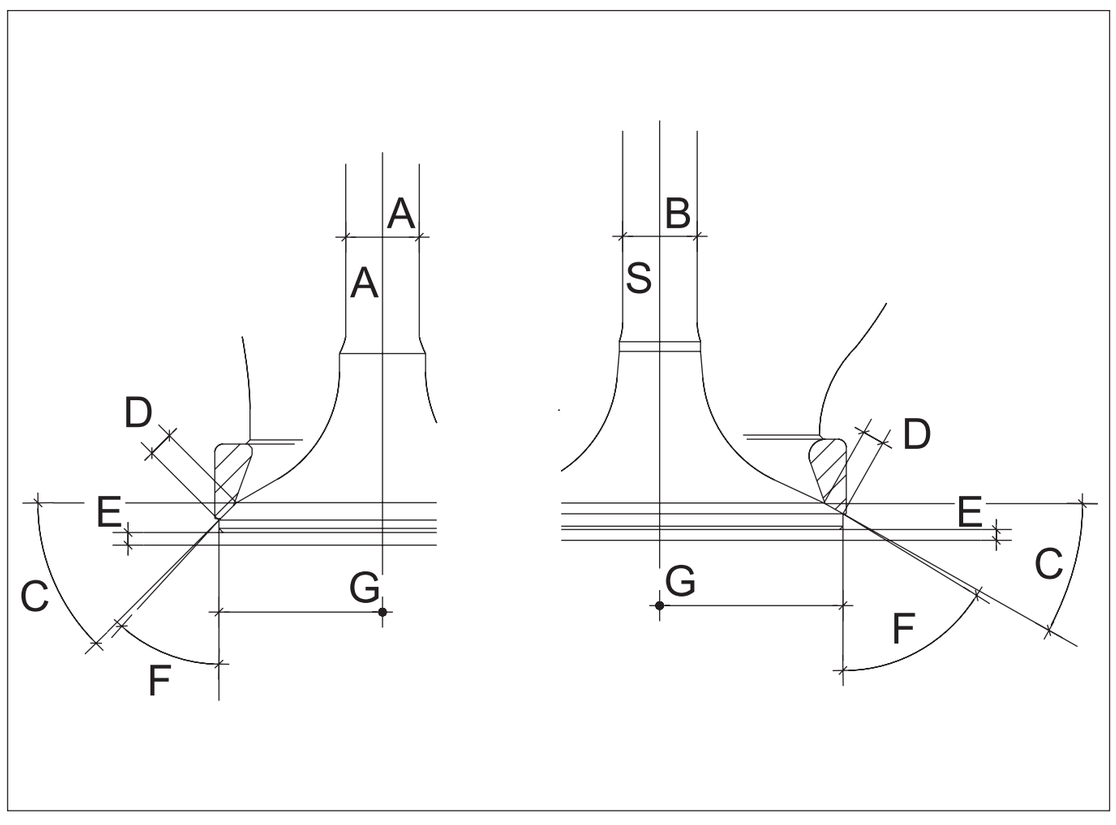


Fig. 1 - Main dimensions of valves and valve seat inserts.

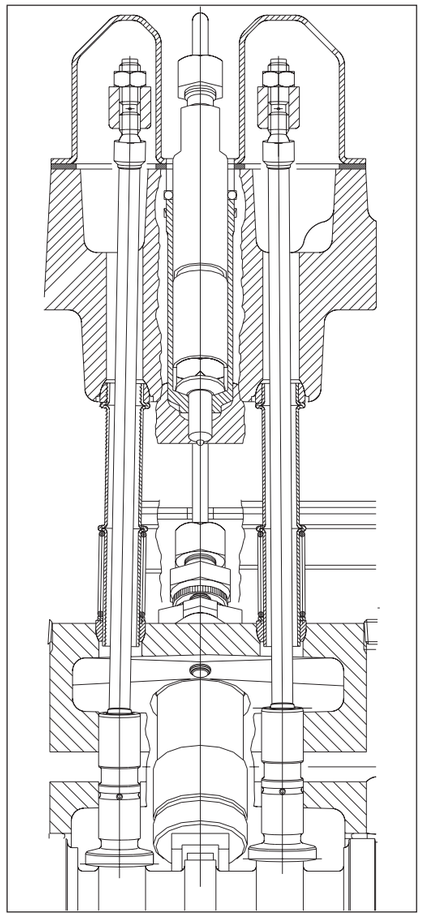


Fig. 2 - Tappet assembly. 1000-W.

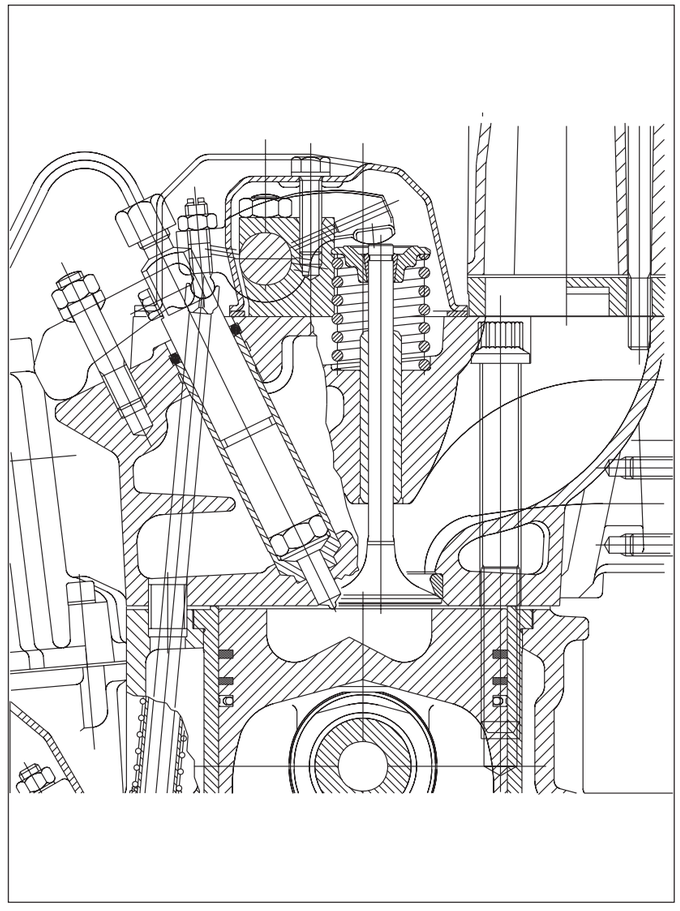


Fig. 3 - Cylinder head cross-section. 1000-W

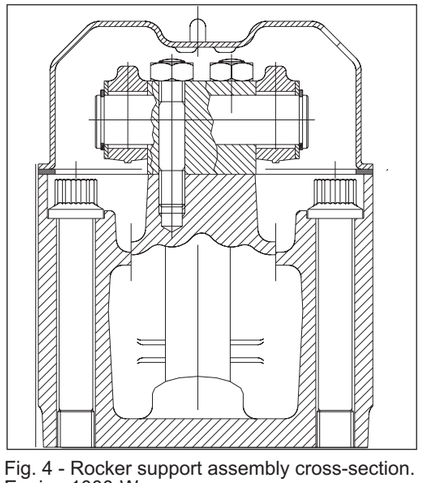


Fig. 4 - Rocker support assembly cross-section. Engine 1000-W

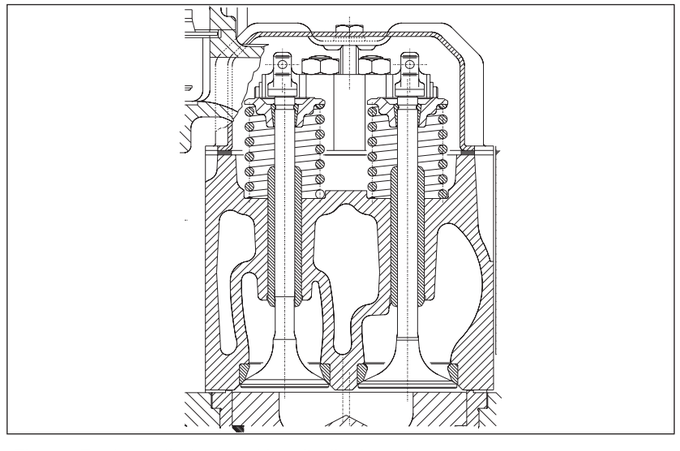


Fig. 5 - Engine cylinder head longitudinal section. Engine 1000-W

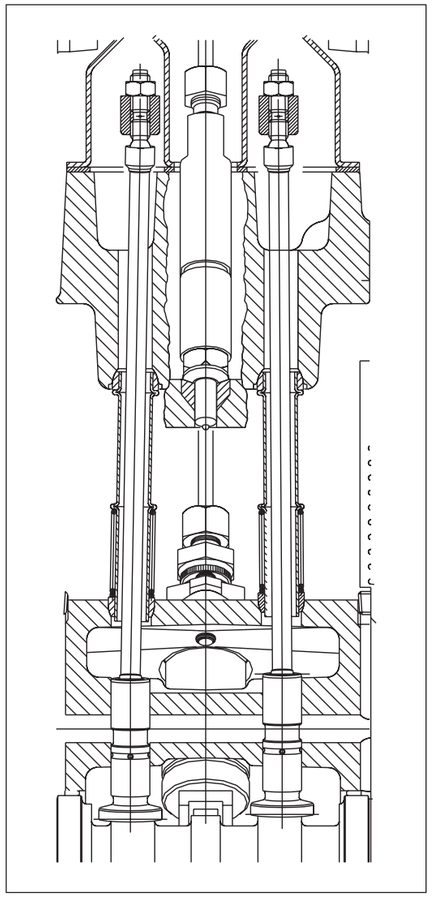


Fig. 2/a - Tappet assembly. 1000-A.

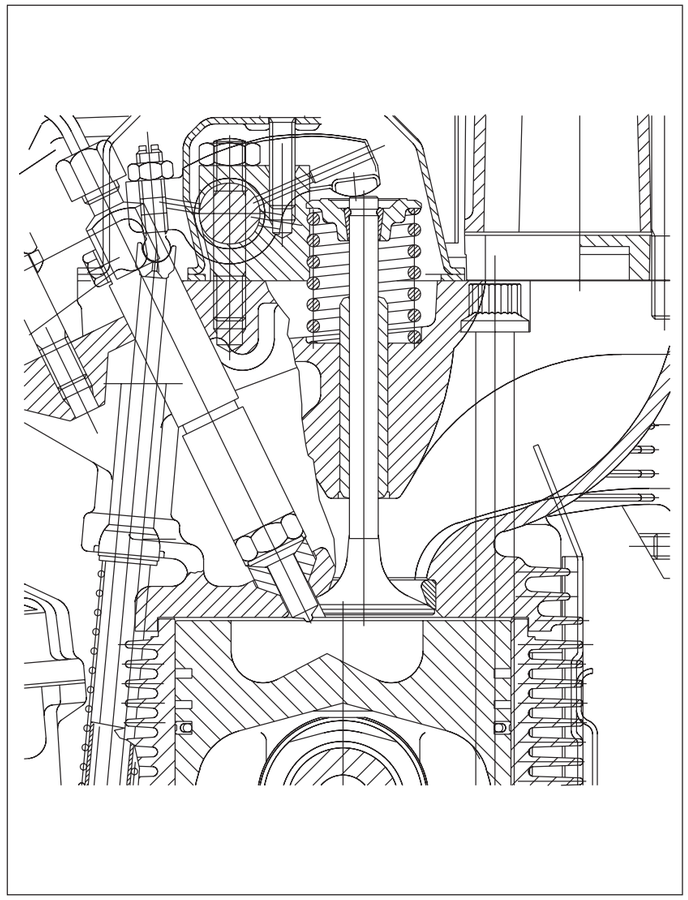


Fig. 3/a - Cylinder head cross-section. 1000-A

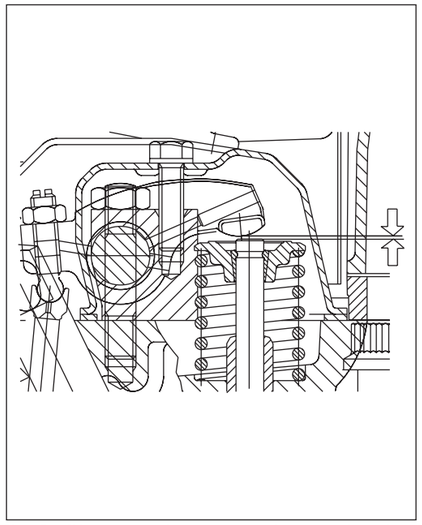


Fig. 6 - Clearance between valves and rockers.

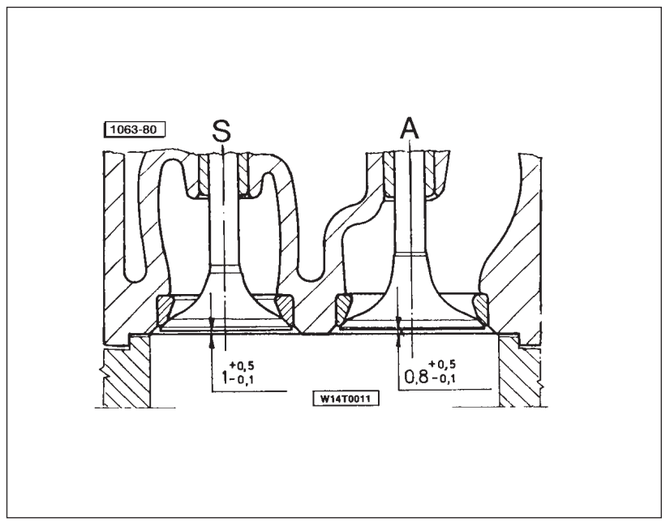


Fig. 7 - Valve recess from the cylinder head gasket bearing face.

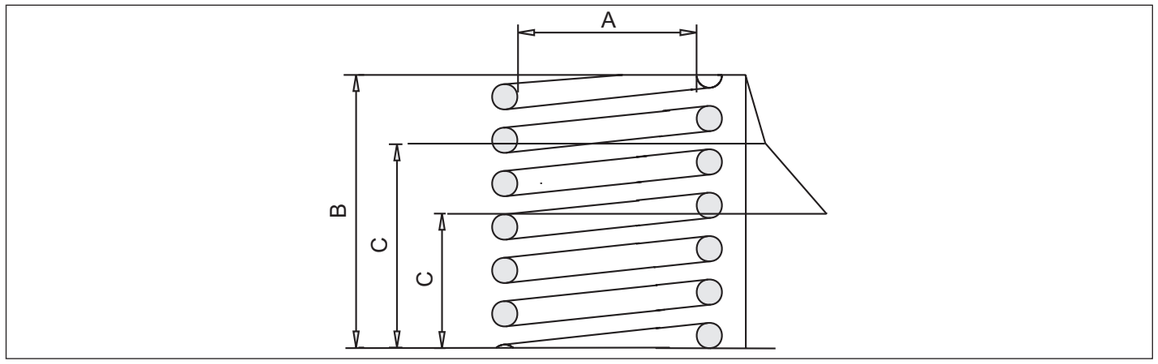


Fig. 8 - Main valve spring dimensions.

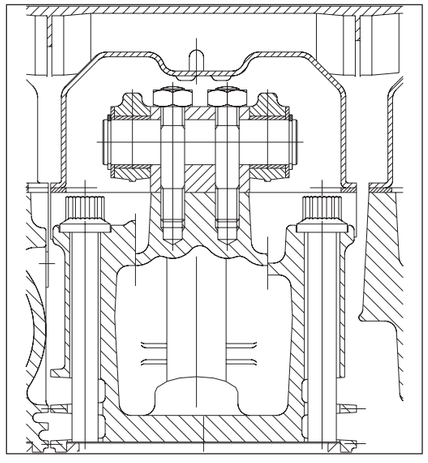


Fig. 4/a - Rocker support assembly cross-section
Engine 1000-A

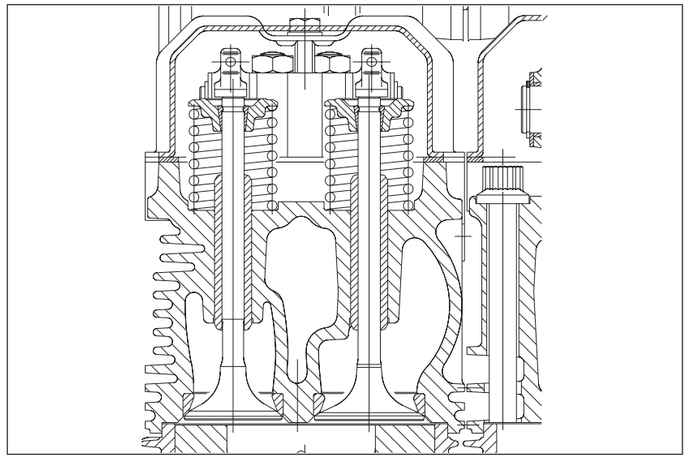


Fig. 5/a - Engine cylinder head longitudinal section.
Engine 1000-A

			Ø machining	Ø max. wear
Valve springs				
a inside diameter mm				27,7
b free spring length mm				53,6
c loaded spring length				
	Kg 15,85 ± 5% mm			42
	Kg 19,60 ± 5% mm			39,6
rocker arm bushing inside diameter (see fig. 10 on page 51)		mm	19 $^{+0,015}_{-0,011}$	19,060
cker arm pin diameter (see fig. 10 on page 51)		mm	19 $^{0}_{-0,011}$	18,980
clearance adjustment between valve and rocker arm (cold)		mm	0,30 for eng. 1000-W	0,20 for eng. 1000-A

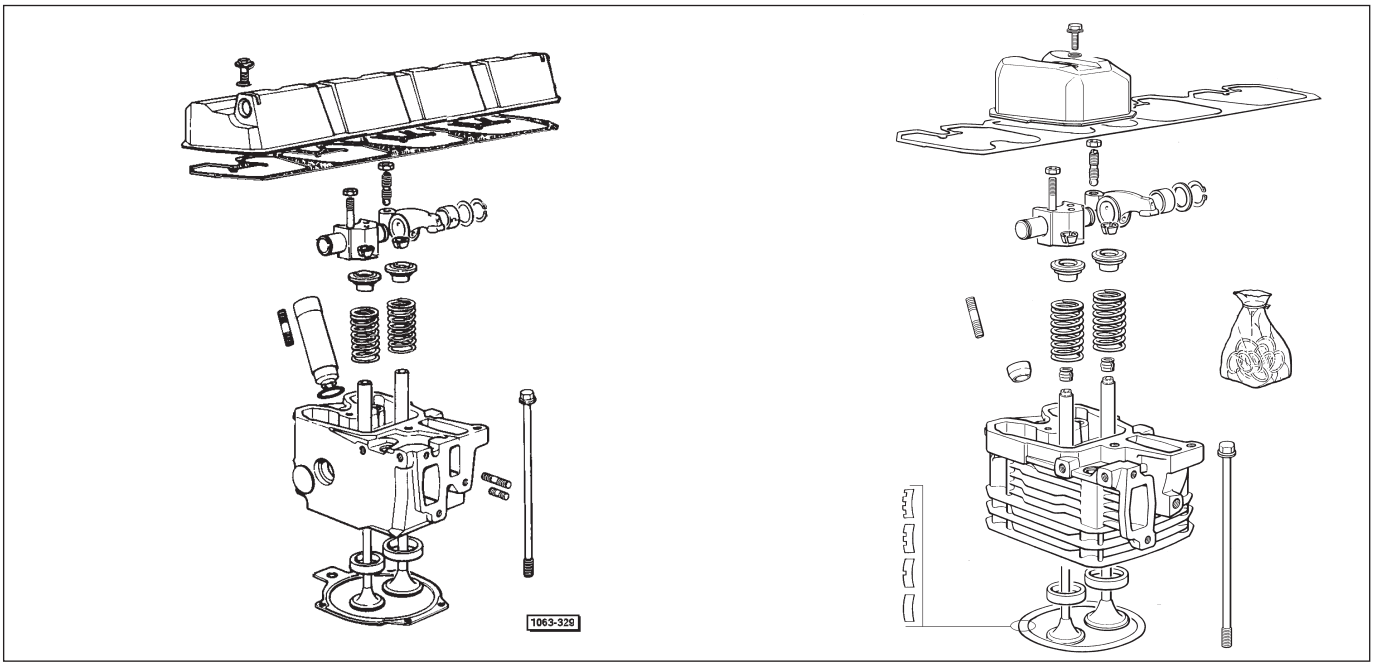


Fig. 9 - Cylinder heads, cylinders, valves and rocker arms.(it has presented in figure the gasket for 4 cylinders. 1000-W (Sx) 1000-A (Dx)

Cleaning cylinder heads

Clean the cylinder heads thoroughly by removing any coking inside both exhaust and intake ducts. Also clean cylinder head cooling fins carefully.

Checking valve seats

First descale and clean valve seats and then inspect for either pitting or corrosion in the valve contact area, otherwise a new grinding (if light wear is found) or a replacement shall be provided.
To install new valve seat inserts operate as follows:
chill the valve seat inserts in liquid nitrogen for easier installation in the cylinder head by running fit.
Valve seat inserts are supplied as finished parts and do not require any further machining. On installation avoid knocking on the valve seats to prevent distortion.
NOTE - If no liquid nitrogen is available for valve seat insert chilling before installation, it is also possible to heat up the cylinder head to a temperature of 200° to 300°C.

Checking valves

Descale the valves with the special brush.
Make sure the valve seat is thoroughly intact, otherwise replace.
Check the valve stem for deformation, be sure the stem outside diameter is not below specifications.

Testing valve tightness

After grinding the valve seats, test valve tightness as follows: plug the intake and exhaust ports through the related valves and then pour in some petrol and check for any leaks (if new valves are installed a slight dripping is allowed).

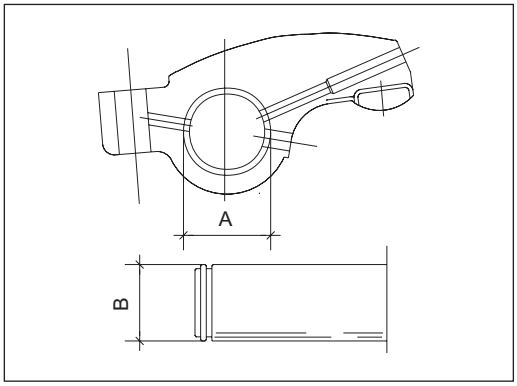


Fig. 10 - Bushing and rocker arm positions and wear check dimensions.
A - Rocker arm bushing inside diameter.
B - Rocker arm pin outside diameter.

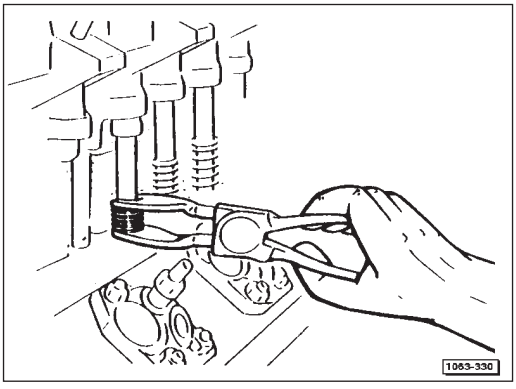


Fig. 11 - No. 5.9030.635.0 tool for tappet rod cap removal.

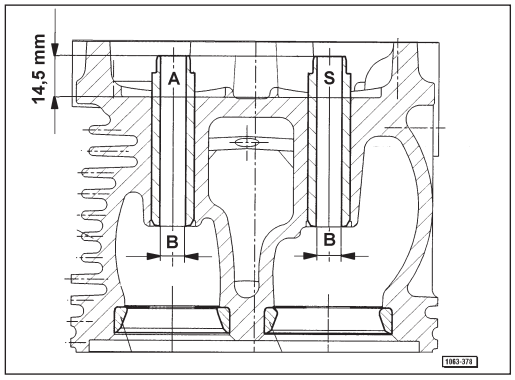


Fig. 12 - Valve seat inserts and valve guides. 1000-A.

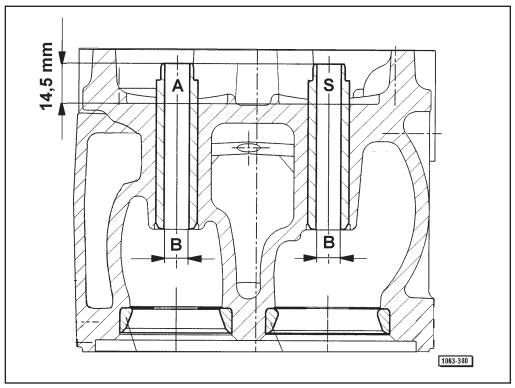


Fig. 14 - Valve seat inserts and valve guides. 1000-W.

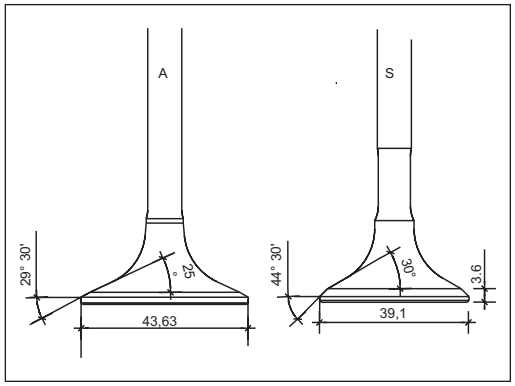


Fig. 15 - **A** - Intake valve
S - Exhaust valve.

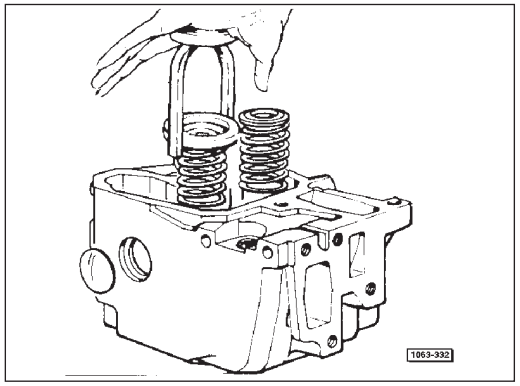


Fig. 16 - Using no. 5.9030.012.0 tool to remove valves.

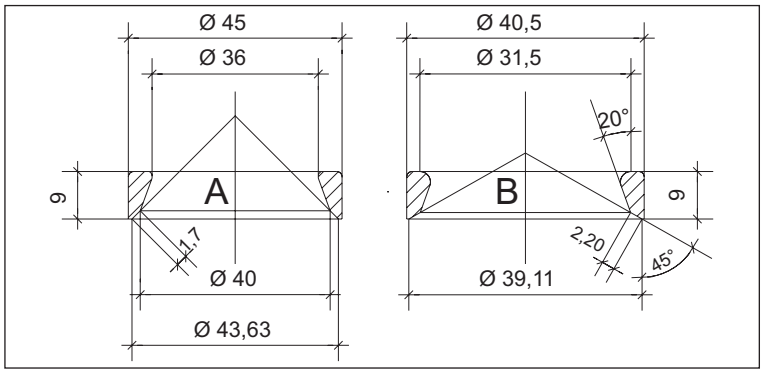


Fig. 13 - Valve seat inserts.

Checking valve guides

Inspect valve guide boring inside surface. It should be thoroughly smooth and show no evidence either of seizing or scoring.
Use no. 5.9030.020.0 gauge (fig. 17) for measuring the valve guide inside diameter. Excessively worn guides are to be replaced.
When re-installing the valve guides they should protrude 14.5 ± 0.2 mm from the seat in the cylinder head.
Using tool specific for installation will enable this dimension to be automatically obtained.

Checking push rods

Examine rocker arm push rods for deformation and the rocker arm contact ball seat for signs of seizing or roughness, if any, replace.
Make sure also the other push rod end in contact with the tappet does not show excessive wear or nicks.

Checking valve springs

Make sure the valve springs have not lost their elasticity.
Also inspect valve springs for rust or damaged lacquering.

Checking rocker arms

Ensure the rocker arm working area is thoroughly smooth and no nicks are shown.
Use a micrometer gauge to measure the rocker arm pin diameter; reading should not be lower than specifications, otherwise replace.
Check the rocker arm bushing for excessive wear, replace as necessary.
Check both rocker arms and valves for proper lubrication by running the engine at idling speed, ensure the oil flow rate is regular; should this not be the case the rocker arm bushing should be inspected for proper installation or the ducts checked for clogging.

Adjusting valve clearance

This adjustment is to be performed with a cold engine and piston at T.D.C. at the end of the compression stroke (both rocker arms shall be in uppermost position and detached from the valve stems).

Rotate the crankshaft until the above mentioned conditions are obtained, then use a thickness gauge to take clearance measurement.

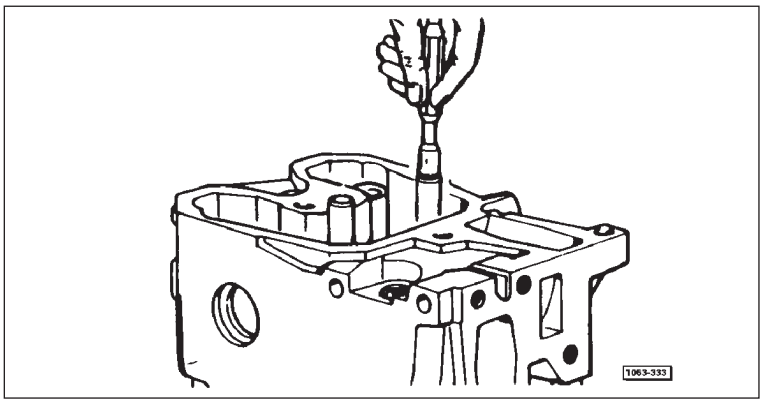


Fig. 17 - Checking valve guide wear.

Installing cylinder heads

Cylinder head securing screws should be tightened evenly following the cross sequence shown in figure to a 2 kgm (19 Nm) torque, then repeat tightening in the shown sequence to a 5 kgm (49 Nm) torque, finally use n. 5.9030.640.0 equipment to furtherly tighten the cap screws until each of them is locked by an angle of 90°÷ 3'.
Before performing screw tightening, the use of plate **A** is recommended to align the cylinder heads correctly, as shown in figure 21.

Checking engine compression

Run the engine until the operating temperature is attained.
Remove the injection nozzle from the cylinder head and fit the special engine compression tester.
Ensure the injection pump is in STOP position.
Start the engine and let it turn until the peak pressure in the cylinders is obtained and then check the engine r.p.m.
Be sure the same checking procedure is applied for each single cylinder so as to provide even measurements.
Cylinder pressure should be 25 to 30 bar at sea level at 150 engine r.p.m. and with hot oil.
Any difference in readings should be no more than 10%. Altitude affects engine compression: 4% pressure loss every 300 m above sea level is usually recorded. To avoid incorrect readings, the battery should be thoroughly efficient. Lack of compression may be due either to piston ring or valve and valve seat or cylinder wear.

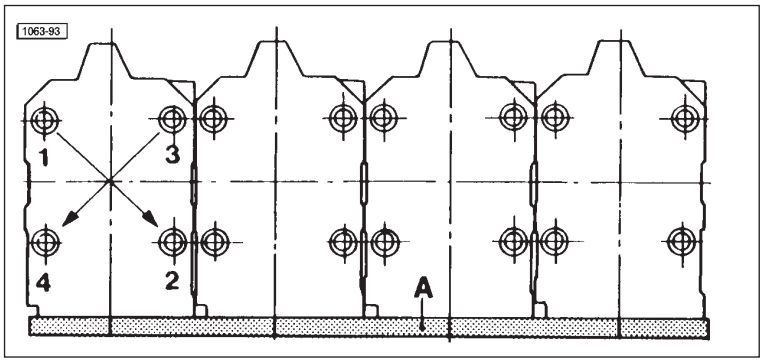


Fig. 21 - Using alignment plate A to install the cylinder heads and cap screw tightening order.

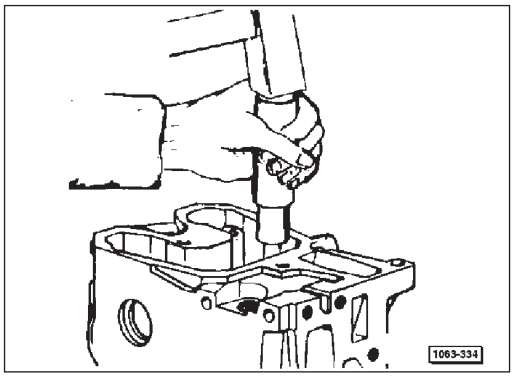


Fig. 18 - Inserting valve guides with the special tool.

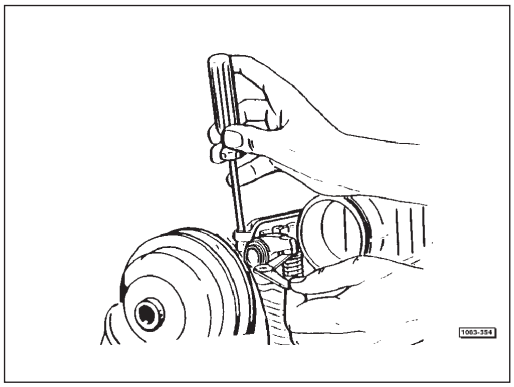


Fig. 19 - Adjusting valve clearance.

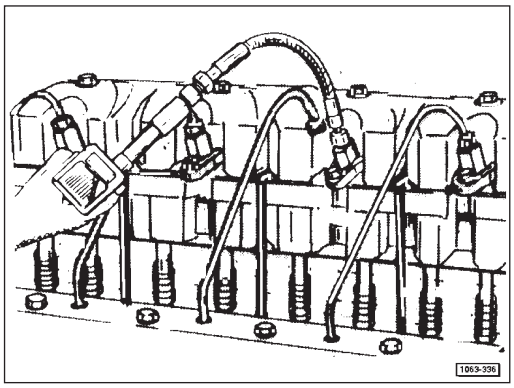


Fig. 20 - Checking cylinder compression.

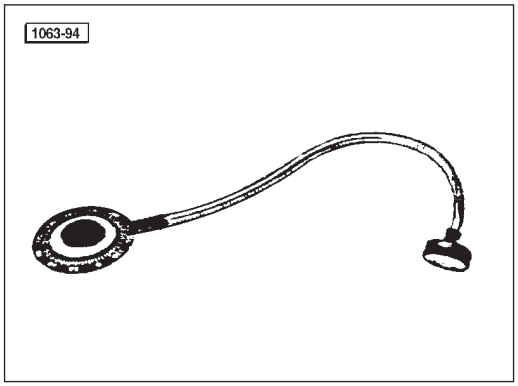


Fig. 22 - No. 5.9030.640.0 equipment for tightening the cylinder head securing screws angularly.

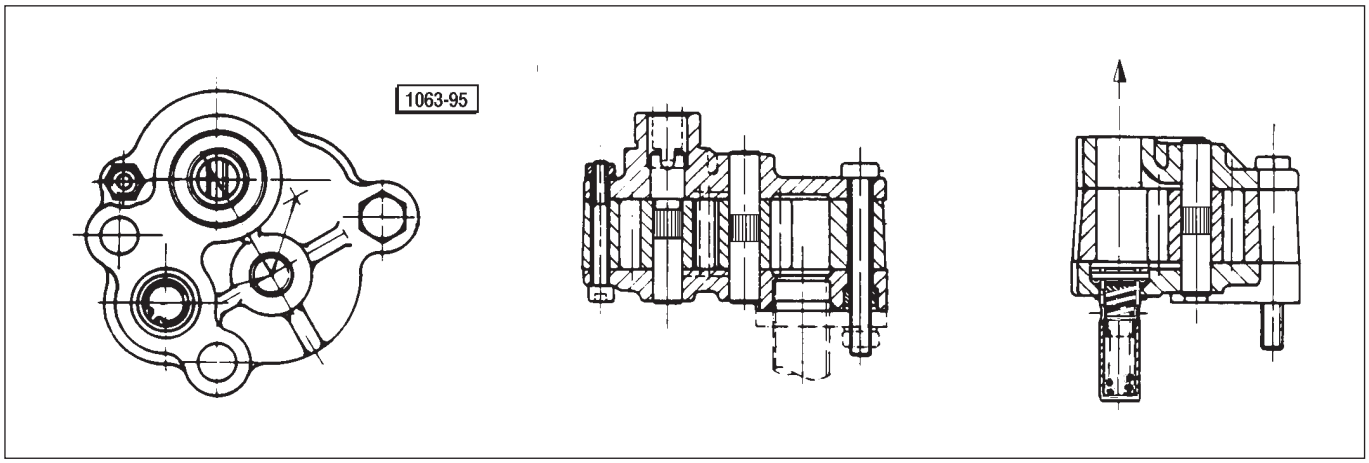


Fig. 1 - Oil pump assembly cross-section.

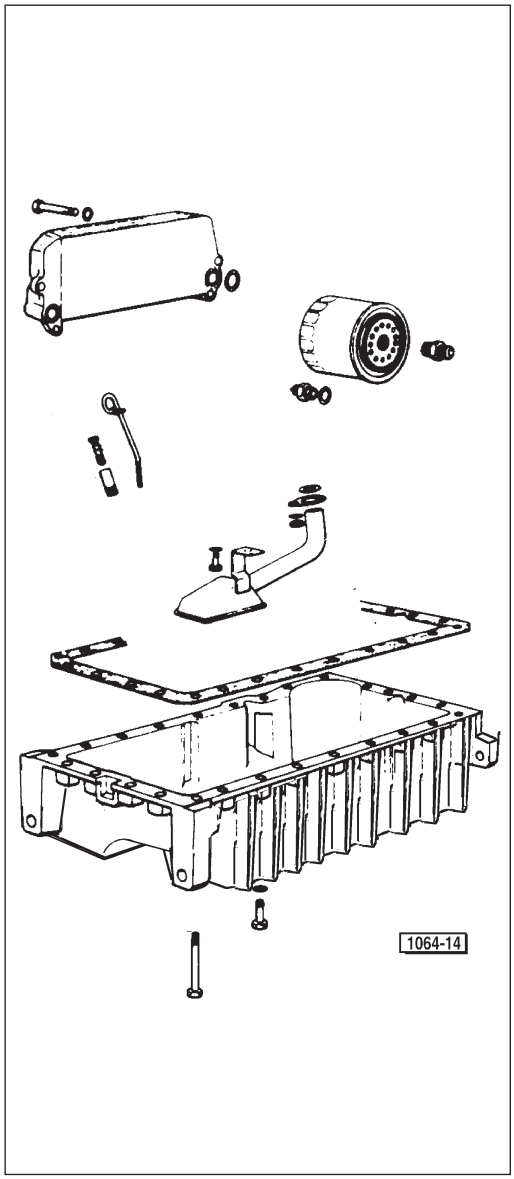


Fig. 2 - Oil radiator, oil filter and oil pan.

Oil pump

Each time the oil pump is removed or installed, ensure the gears can rotate freely and there is no evidence of tool spalling or wear.

Otherwise the whole pump assembly must be replaced.

To perform a complete oli pump efficiency test insert an oil flow rate meter along with an oil pressure gauge between cylinder block and oil filter. Ensure readings correspond to specifications on page 22.

Checking pressure relief valve

Using the special fitting connect the valve to no. 5.9030.520.4 equipment and make sure the valve calibrating pressure is 3.5 to 4.5 bar.
If this is not the case the valve should be replaced.

Warning: whenever the oil pan is removed, ensure the prefilter wire mesh located under the oil rose pipe is thoroughly clean.
Engine oil pressure can also be checked on the special dashboard-mounted indicator.
Engine oil level should never be below the minimum level notch on dipstick.

Fitting shims between engine oil pan and front support

Between engine oil pan **C** and front support **A** it is required that shims be fitted so that a ±0.1 mm alignment with the engine block plane may be obtained.

Mount the oil pan onto engine block paying particular attention the rear planes are thoroughly aligned (engine flywheel side).

With no. 5.9030.272.0 centesimal gauge and no. 5.9030.433.0 gauge base measure the difference in dimension between oil pan **C** front plane and engine block plane **B**. This difference in reading should be 0.35 to 0.95 mm (make sure the oil pan plane always recesses from engine block plane).

Install the front support placing a number of shims which permits both planes to be aligned within a ± 0.1 mm allowance.

WARNING: Reading and shimming should be performed both on the front right-hand and the front left-hand side of the oil pan as shown in figure.

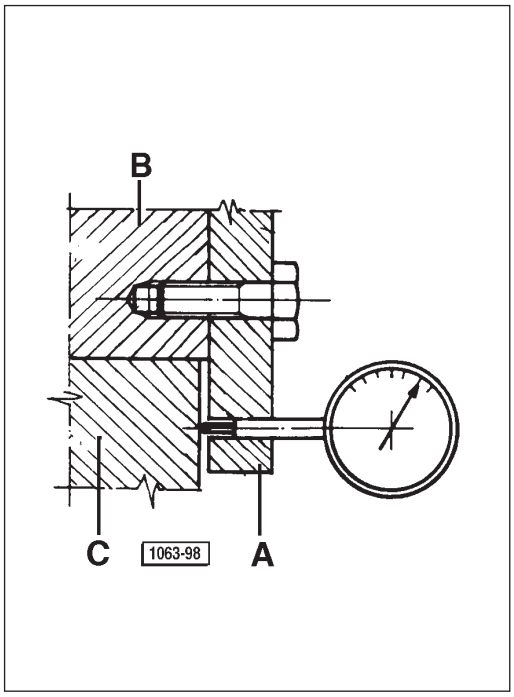


Fig. 3 - Measuring the difference in dimension between oil pan and engine block planes.

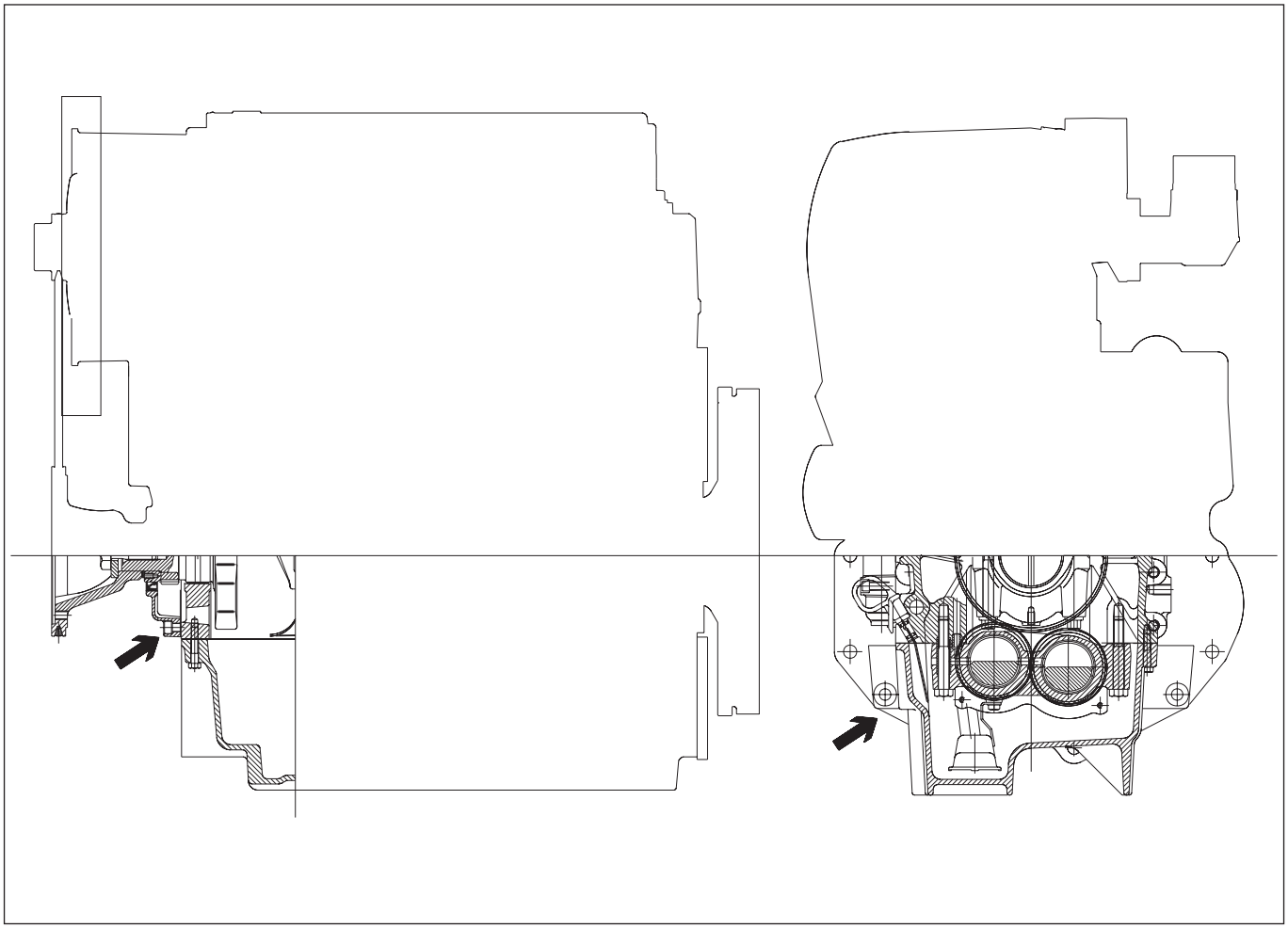


Fig. 4 - Positions at which to measure the difference in height between the front face of the sump pan and the engine block.

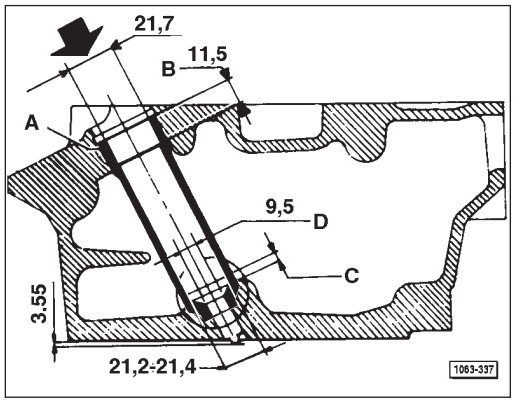


Fig. 1 - Injection nozzle assembly.

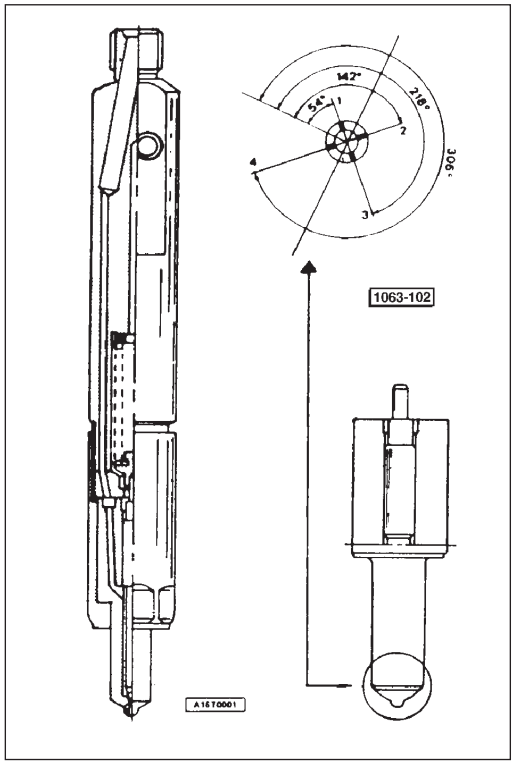


Fig. 2 - Setion of injection

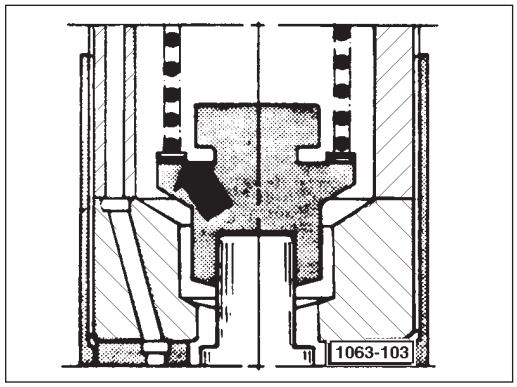


Fig. 3 - Spring load adjustment.

Fuel injection nozzles

Checking fuel injection nozzles

The checks on injection nozzle operating conditions include injection nozzle spraying efficiency and calibration.

However, it is recommended that the utmost cleanliness rules be strictly observed and a thorough test on the hydraulic seal of the different system fittings be performed. As a matter of fact even a small leakage would invalid the tests carried out.

Installing injector bushing into the cylinder head of the water-cooled engine

Apply some silicone type DIRKO from ELRING onto surface A (figure 1) (between application and installation should elapse no more than 15 minutes).
Install the bushing into the head using a press so as it may thoroughly adhere to its seat.
First roll surface B and thereafter surface C using tool no. 5.9030.639.0, also machining the tapred surface so that the diameter may be corrected to 21.2 to 21.4 mm paying attention the bushing be pressed according to the direction shown by the arrow.
Then roll surface B again so that a diameter of 21.7 mm be obtained.
Machine surface D in such a way that a diameter of 9.5 mm may be **obtained**.

Injection nozzle calibration

Check pressure reading at which the nozzle needle begins to be lifted from its seat thus determining the injection starting.

This pressure should be within the recommended specifications. For this reason, each time that calibrating pressure readings differ from specifications the injection nozzle should be adjusted as required.

To perform the above checks, the use of a pump connected to the injection nozzle to be tested by a piping is unavoidable.
(Prior to performing the test the fuel recovery plug should be removed).

The following procedure shall apply:
Operate the pump lever (fig. 4) a few times to force any air out of system.

Slowly operate the pump lever and notice the pressure reading right in the moment the injection begins. This pressure should be 180 bar. If readings differing from that specified are indicated, it will be necessary to adjust the pressure spring by simply adding or removing the special spring preloading shims.

To gain access to spring adjustment, remove first the lower cap of the injection nozzle holder and then add or remove shims bewteen spring and flat cap until the desired nozzle calibrating pressure is obtained.

Checking injection nozzle seat tightness

Checking the injection nozzle seat tightness is very important as this permits to verify whether there is any fuel dripping from the injection nozzle before the calibrating pressure is reached.

This test is carried out by operating the hand pump until a pressure reading about 10% lower than the calibration pressure is attained. Take care this pressure be maintained by oppotunely pressing on the hand pump lever, so as to compensate the pressure drop which otherwise would be caused by the fuel dripping between pin and nozzle holder body.

Under these conditions the injection nozzle should never drip; otherwise consult an injection pump specialist.

WARNING: on refitting, position the injector as shown in figure 56 of page 103.

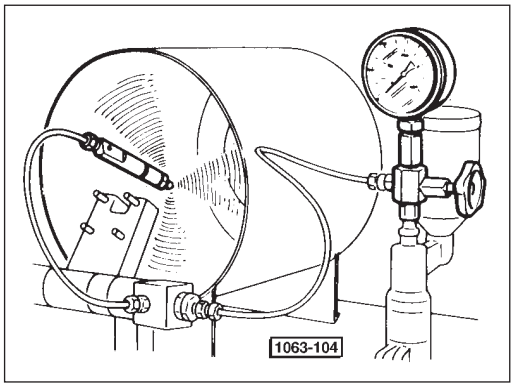


Fig. 4 - Checking injection nozzle spraying efficiency.

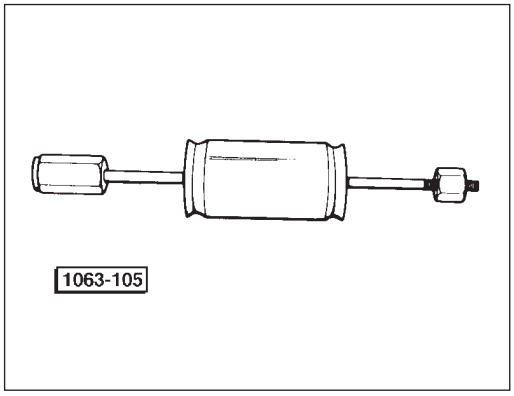


Fig. 5 - No. 5.9030.618.4 tool used to remove injection nozzle from seat.

Checking injection nozzle spraying efficiency

Connect the injection nozzle to a pump and operate the pump lever repeatedly so that the operating conditions may be reproduced.

Fuel must be sprayed from the nozzle holes in form of finely atomized jets, which should all be equal in size. There should be as many jets as there are holes in the injection nozzle.

Otherwise, clean the nozzles by removing first the residual carbon deposits from the nozzle holes with a Ø 0.26 mm steel wire and then clean them thoroughly with a Ø 0.28 mm wire, this having the same diameter as the nozzle spraying holes.

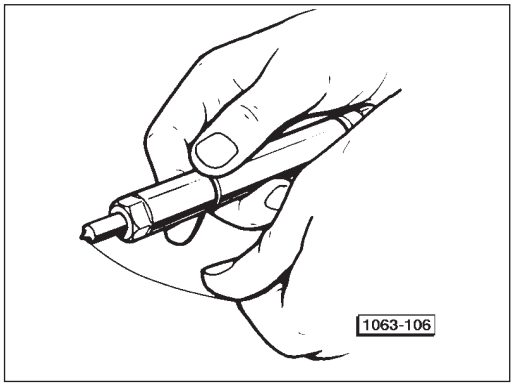


Fig. 6 - Cleaning injection nozzle with a piano wire.

Injection nozzle projection from the cylinder head face plane

Make sure the injection nozzle projects 3.40 ÷ 3.88 mm from the cylinder head face plane.

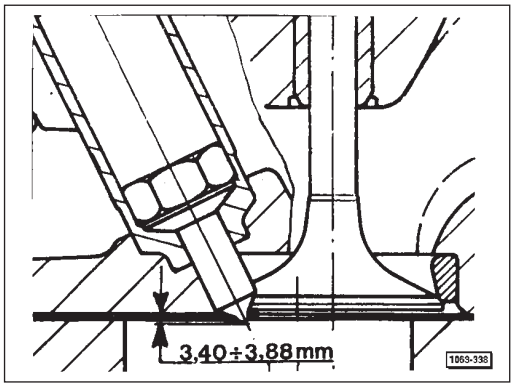


Fig. 7 - Injection nozzle projection from the cylinder head face.

MECHANICAL-TYPE ENGINE GOVERNOR

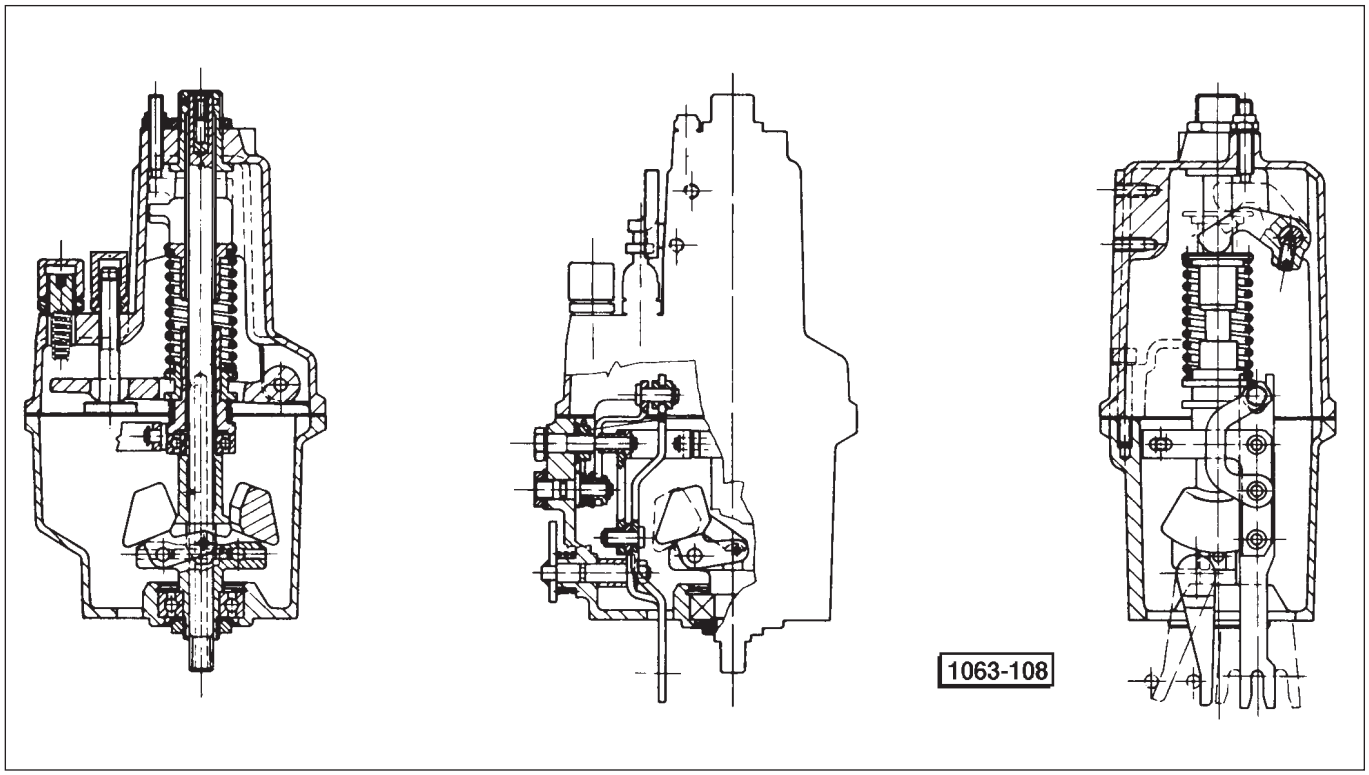


Fig. 8 - Engine governor assembly longitudinal sections.

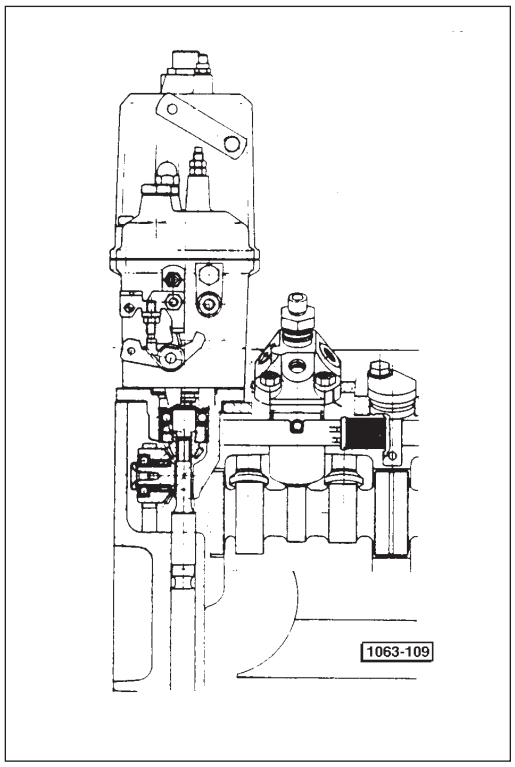


Fig. 9 - Engine governor transmission assembly.

Mechanical-type engine governor

The engine injection pump governor assembly is a very advanced design both for specific operating features and easy servicing.
The engine governor has been developed as a complete unit on which either repairs or adjustments can be performed after removal from engine.
Besides facilitating operations, also it offers the advantage of being swiftly replaced or temporarily interchanged with a unit of same type and specifications, thus preventing the machine from lying idle in the event of mechanical interventions on it.
The following equipment is also provided with the governor assembly:

Automatic extra fuel supply
Whenever the engine is turned off, extra fuel is automatically supplied to the combustion chamber by this unit; this ensures a prompt engine starting response. (To be sure this automatic supply function is not prevented in very cold weather, we recommend releasing the engine fuel shut-off control - STOP - immediately after the engine has been stopped).

Engine fuel shut-off
This control permits the engine to be stopped effortlessly, as it directly operates the pump control rod and no connection to the governor assembly is provided.

Anti-hunting
This device limits the engine speed variation when this turns at fast idling speed.

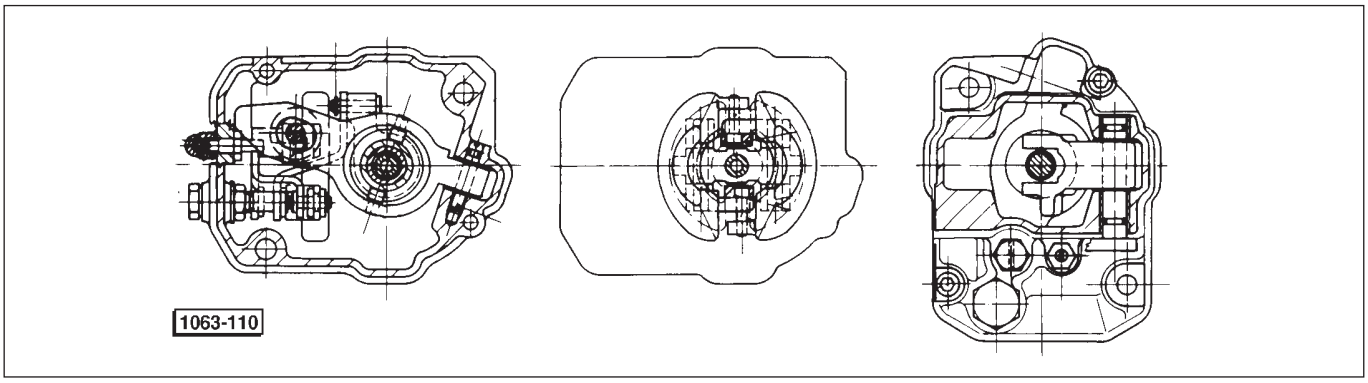


Fig. 10 - Engine governor assembly cross-section.

Mounting governor weights (Fig. 11)

Apply Loctite 601 to the crankshaft surface **A** in contact with support **B**, then fit the support to the crankshaft, positioning the two pieces to allow fitting of dowel **F**.

Fit the weights **C** and **E** to the support by means of plates D inserted in the groove of the sliding sleeve.

Open weight **C** against the stop thereby holding the weight whose plate rests on the coupling and releasing the other one, then move the coupling against the locked plate. Be sure the dimension reading is 61 ± 0,1 mm as shown in Fig. 11, otherwise add or remove shims (ref. code 2.1589.160.0 and 2.1589.161.0) to/from the position indicated by the arrow.

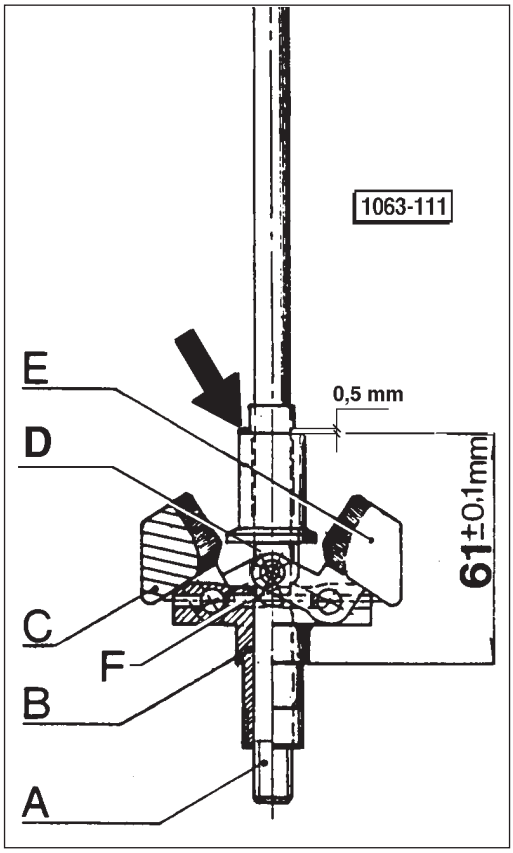
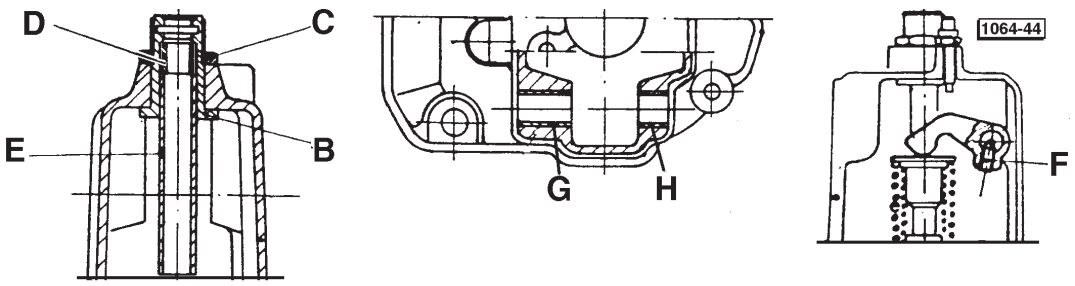


Fig. 11 - Installing weights on the governor shaft.



Before assembly, apply Loctite 601 to the outer surfaces of bush **D**, tube **E** and fitting **B**.
Before assembly, apply Loctite 601 to the outer surfaces of bush **G** and bush **H**.
Before tightening nut **C**, coat the thread with Loctite 270.
Before assembly, apply Loctite 270 to thread of the control lever retaining bolt **F**.

Fig. 12 - LOCTITE application points.

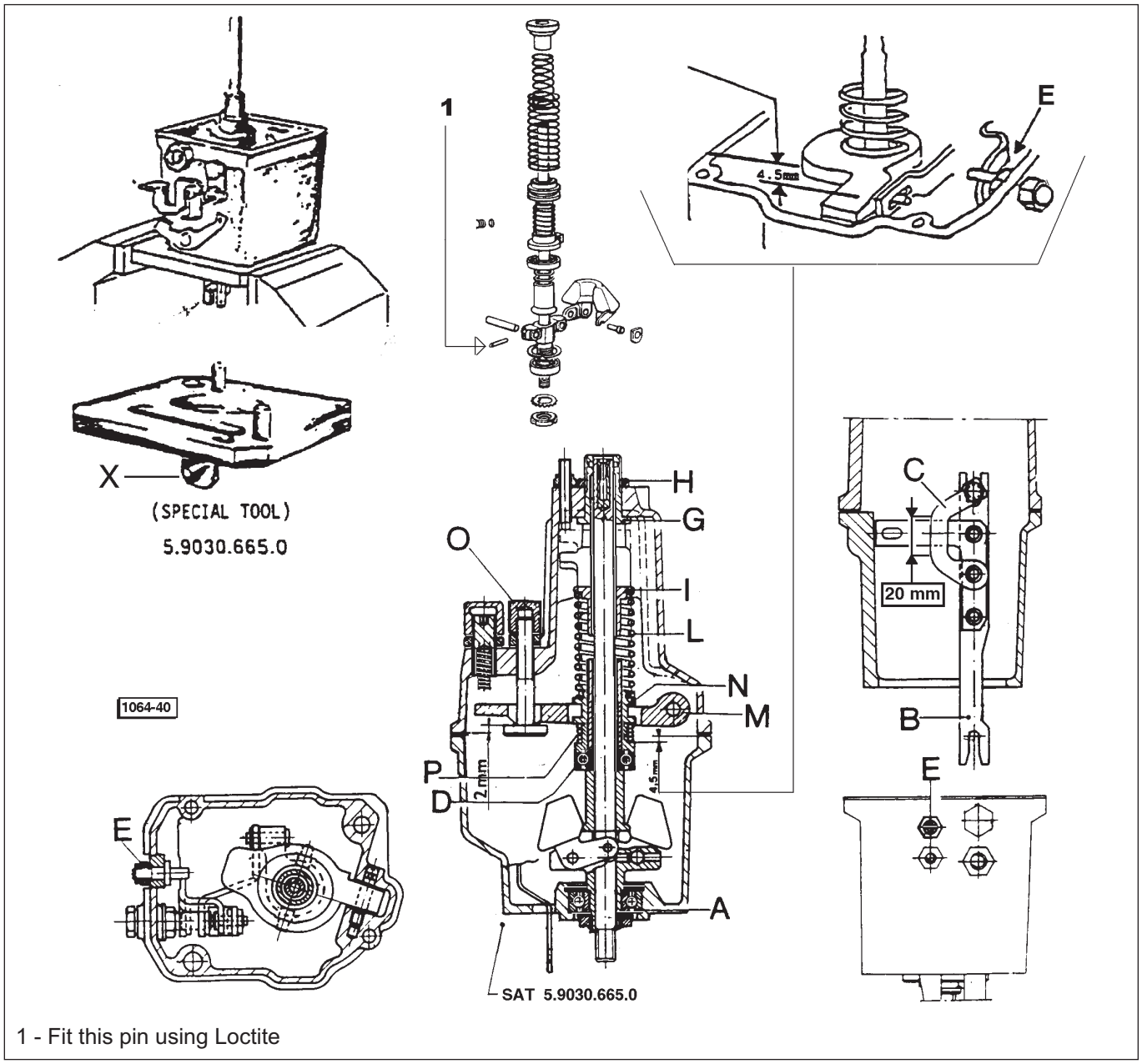


Fig. 13 - Engine governor assembly.

Assembly of the governor unit (fig. 13)

Lower casing
Fit the internal and external STOP control levers and fit the plate and stop screws. Fit bearing **A** in the casing and fit levers **B** and **C** with complete unit.

WARNING: before fitting lever **C**, check that 20 mm section indicated in the figure is completely free of burrs or indents.
Insert the shaft complete with the governor weights, securing it to the bearing with the nut.
Fit bearing **D** and the sliding sleeve and fit eccentric screw **E** to the casing so that the chasing on the head is positioned at the top.ent of the actuator
Position the governor on the special tool, taking care to centre the fork with the fixed pin **X** (fig. 13).
Turn the cam screw **E** to adjust the distance between the surface of the lower half-casing (without gasket) and the sliding sleeve to obtain a distance of 4.5+0.1. Then lock cam screw **E** in position.

Upper casing (Fig. 13)
Fit the minimum and maximum accelerator control lever travel stop screws into the upper casing.
Mount bushing **G** with the relevant O-ring using nut **H**; install the accelerator internal and external control levers.
Position spring-holder cap **I** and springs **L**; secure lever **M** and inserted coupling **N** into casing. Install travel stop rod **O**.

Positioning governor lever M
Operate travel stop rod **O** with an Allen wrench until the distance between the lower casing edge and the travel stop rod head corner is about 2 mm, as shown in figure 13.
Connect the upper to the lower casing, first positioning the automatic extra fuel supply unit spring **P** on coupling **N**.

Calibrating engine governor

Engine governor calibration is very easy and requires no special equipment. The procedure recommended permits all lever plays produced during engine running to be removed, since this adjustment can be performed with engine governor assembled and engine running.This setting is to be effected when external operations which may involve engine governor, injection pump or injection pump control rack replacement are performed.
Important - Whenever the governor is removed from engine, the cable connected to the battery positive pole should be disconnected so that a sudden engine start may be prevented.
Observe the following procedure:
— start the engine and then bring to a 2000± 100 r.p.m. by means of the hand throttle (avoid subjecting the engine to dragging loads and ensure it is not hunting).
— Using an Allen wrench (fig. 16) loosen the fuel maximum delivery adjusting screw (turn it anticlockwise) until the engine is about to stop (or keeps on running through the automatic operation of the extra fuel supply unit stabilizing at 300 to 500 r.p.m.).

Warning: locating this condition requires most accuracy and meticulous operating procedure and namely:
— locate approximately the point where the engine r.p.m. starts drooping;
— screw in the adjusting screw of about 3 turns to permit the engine to be stabilized at 2000 r.p.m.;
— loosen first the adjusting screw two full turns and then continue loosening by about 30° movements at a time and wait 2 - 3 seconds after each screw movement until the engine stop point is located.
With the engine stopped, place the contact point of a dial gauge on the head of screw **A** (fig. 16), zero the gauge, then screw in the screw the number of turns indicated in the table, to obtain the gauge reading indicated below:

ENGINE TYPE	R.P.M.	HP	NO OF TIGHTENING TURNS OF THE ADJUSTING SCREW
1000.3 W1	2350	60	7
1000.3-WT1	2350	70	7,25
1000.4- W3	2350	80	6,75
1000.4-WT1	2350	87	7,1
1000.4- A3	2350	60	6 ,2
1000.4-A3	2350	75	7,5

NOTE: This adjustment can be performed at an engine oil temperture not less than 80°C.
Tighten the lock nut.

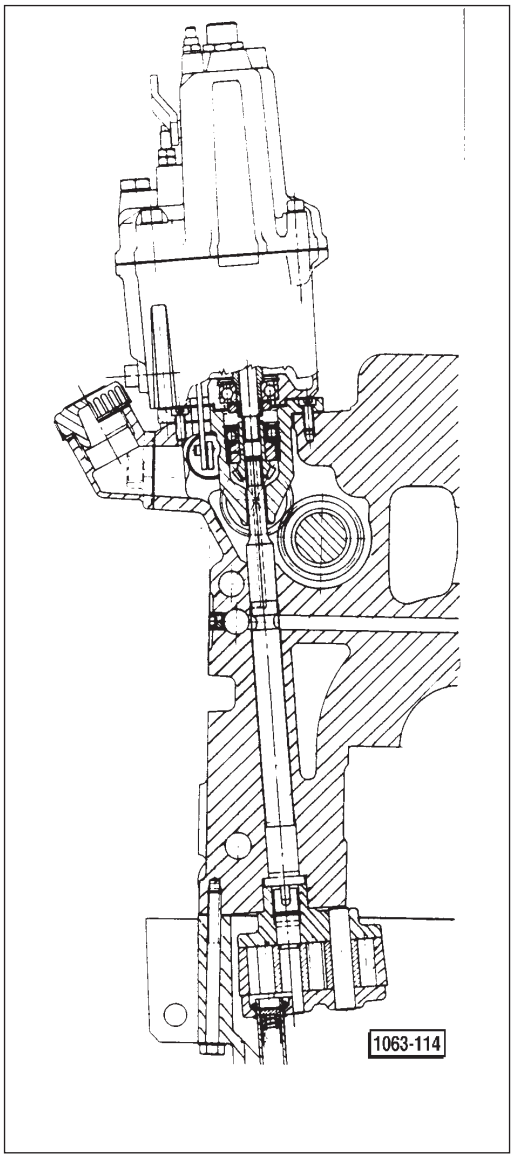


Fig. 14 - Engine-mounted governor unit. (For engine 4-cylinders with governor mechanical type).

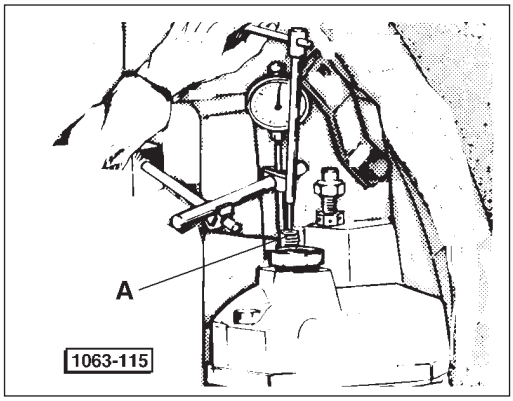


Fig. 15 - Tractor-installed engine governor unit.

FITTING AND ADJUSTMENT OF THE EXTERNAL CONTROLS OF THE MECHANICAL GOVERNOR

Accelerator pedal (fig. 16).
Attach the throttle cable **A** to bracket **C** and then to lever **D**, then fit spring **P**. Adjust the position of screw **G** to obtain a pedal travel of 75 mm as shown in figure 16 (corresponding to an angle of approximately 32°) and secure in position with locknut **Q**.

Hand throttle
Attach the throttle cable **B** to bracket **E** by tightening screw N and positioning the locknuts **O** and then attach the cable to lever **V**.

Connection of the accelerator pedal and hand throttle cables to the governor.
Attach the throttle cable to bracket **H** of the governor by way of screw **L**, and secure with locknuts **I**;
Connect the throttle cable to bracket **M** of the governor.
Adjust the freeplay of the throttle cable by turning the screw **L** (after having undone the locknuts **I**) so that the cable is correctly tensioned, then re-tighten the locknuts.
Attach the throttle cable **S** of the hand throttle to bracket **T** and then attach it to lever **U**.
Then apply a force **f = 5** to lever **F** and screw in self-locking nut **R** until the force applied is no longer capable of moving the lever. Move the hand throttle lever to the "MAX" position and manually push lever **F** to the end of its travel, then screw in screw **N** until the throttle cable is fully tensioned, then tighten locknuts **O**.

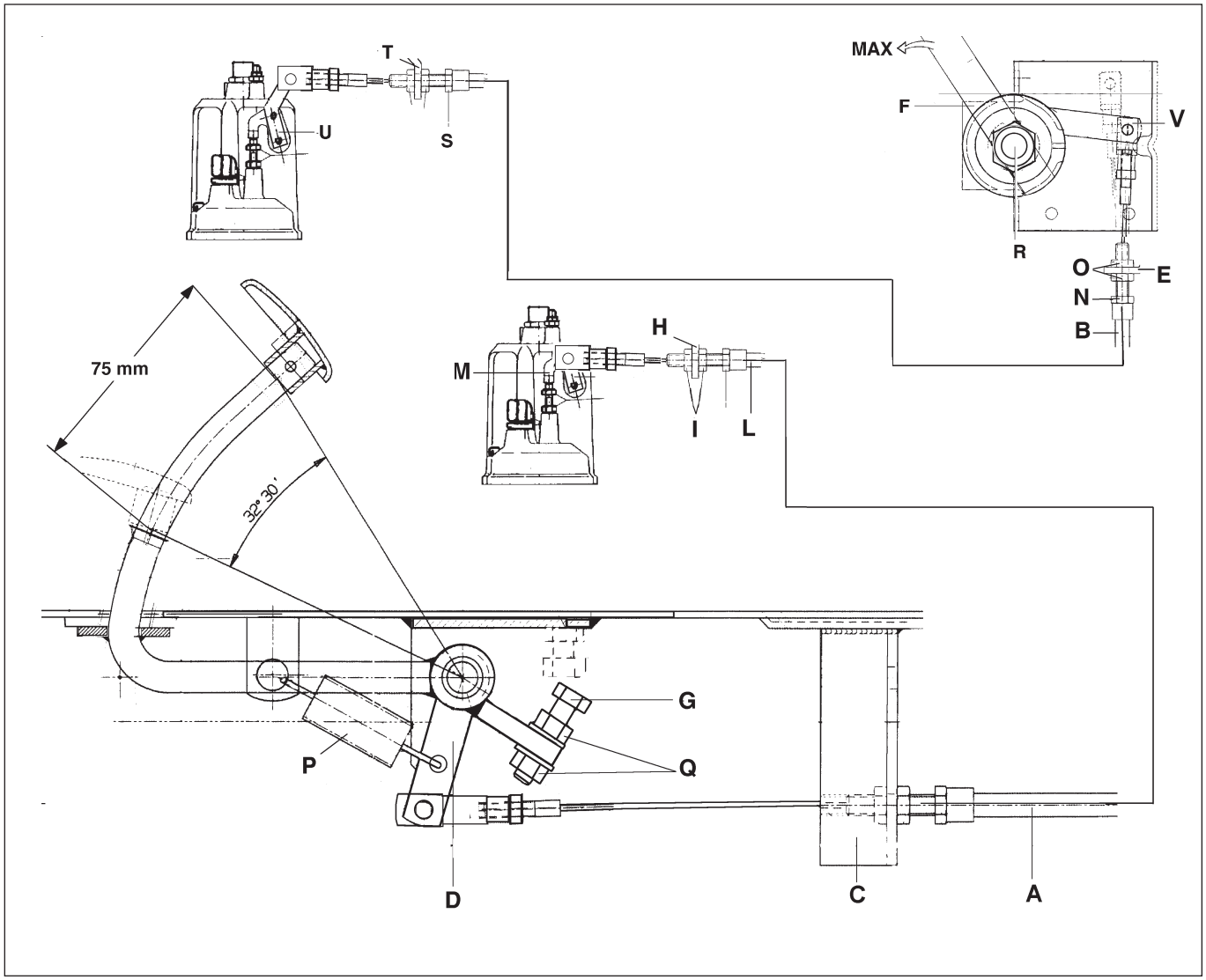


Fig. 16 - Adjustment of the external controls of the mechanical governor.

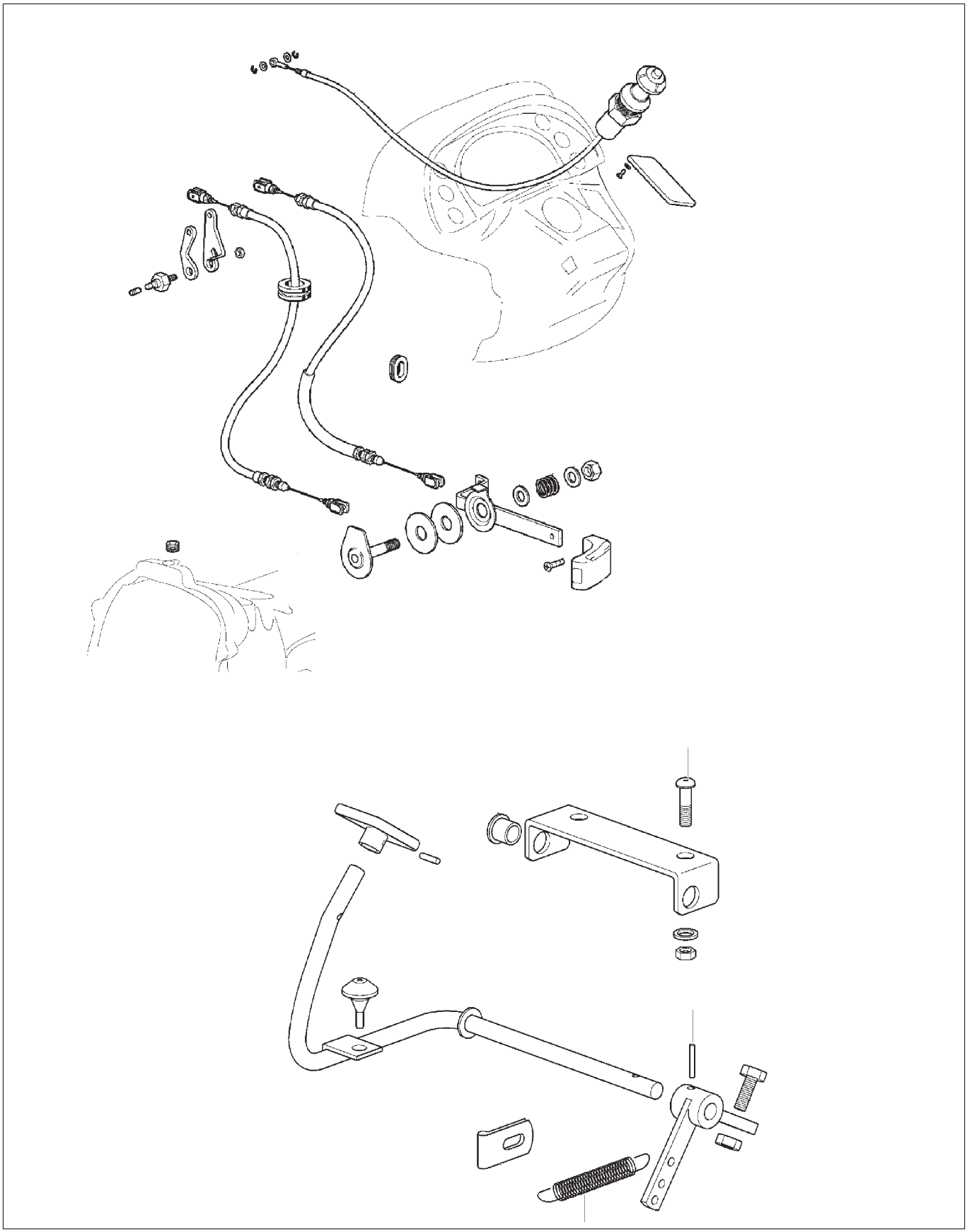


Fig. 17 - Component for control accelerator pedal and manual.

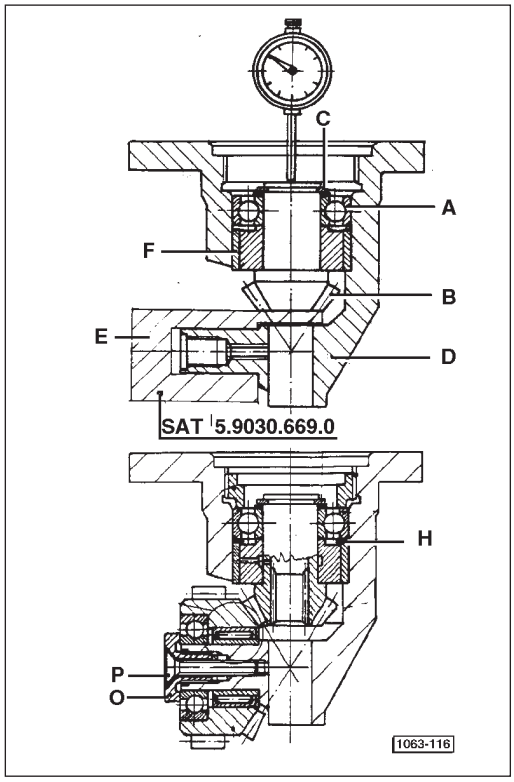


Fig. 18 - Installing pinion and crown whee.

Engine governor control assembly

Installing pinion (Fig. 18)

First insert bearing **A** in pinion **B** placing a pack of shims **C** (ref. codes 2.1589.146.0, 2.1589.147.0, 2.1589.153.0) between both until assembly clearance has been fully removed.
Position the preinstalled assembly in support **D** so that pinion head rests on tool **E** as shown in figure 18.

Fit a comparator resting its feeler on pinion end and then set it to zero.
Remove the tool previously fitted and push the pinion until bearing **A** stops against ring **F**, then take comparator reading.
This reading corresponds to the thickness shims **H** to be placed in the position indicated in figure 19.

Tighten the ring nut fully and hit in three different points to prevent it from becoming loose.

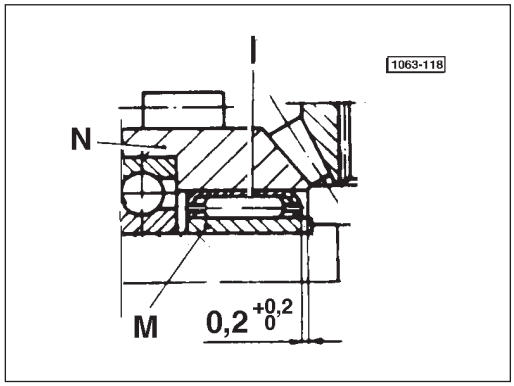


Fig. 19 - Install cage with rollers.

Installing crown gear (Fig. 18 and Fig. 19)

Install cage **I** with rollers and place as shown in figure.

Insert ring **M** and crown gear **N** in the support then tighten ring nut **O** to allow for a tooth radial backlash of 0.03 ÷ 0.08 mm; then tighten screw **P** ensuring the ring nut remains in position.

Warning: screw **P** is to be mounted after applying a little amount of Loctite 270.

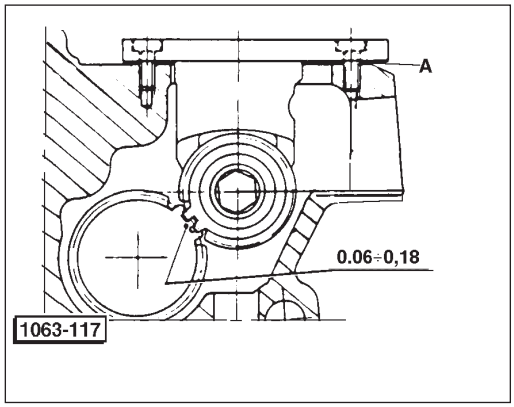


Fig. 20 - Installing the control assembly into the cylinder block.

Installing the control assembly into the cylinder block (Fig. 20)

Fit some shims **A** (ref. code 065.2560.0 - figure 22), so as a 0.06 ÷ 0.018 mm backlash between the governor drive gear teeth is obtained.

In any case one shim at least needs to be fitted.

Adjusting anti-hunting device (fig. 21)

This adjustment is to be made only in the event of engine hunting at engine maximum idling speed.

Loosen bush **A** until being sure that spring **B** is not protruding inside the governor casing.
Start the engine and run to peak speed.
Screw in the bush so as to bring the spring close to the hunting system until hunting dampening is felt.

Warning: the engine idling speed variation obtained through the anti-hunting device should never exceed 20 r.p.m.

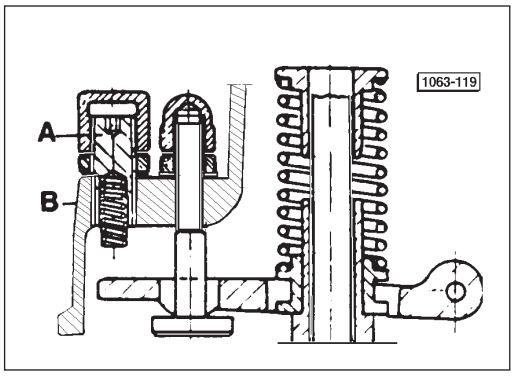
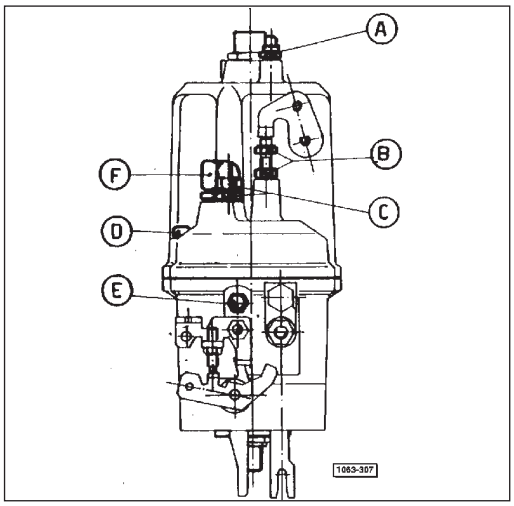


Fig. 21 - Engine anti-hunting system.

Sealing engine governor

The governor is fitted with sealing points, indicated in figure 23, with the letters **A** - **B** - **C** - **D** - **E** - **F**.

Before starting any operations during the warranty period consult the local After-Sales Service.



Installing the governor (fig. 22).

Fit the governor by engaging the control fork **A** in the pawl **B** of the pump control shaft and securing it in position with the relative screws.
Then manually check through the injection pump control rod access window that the control rod moves freely and that the STOP control operates correctly.

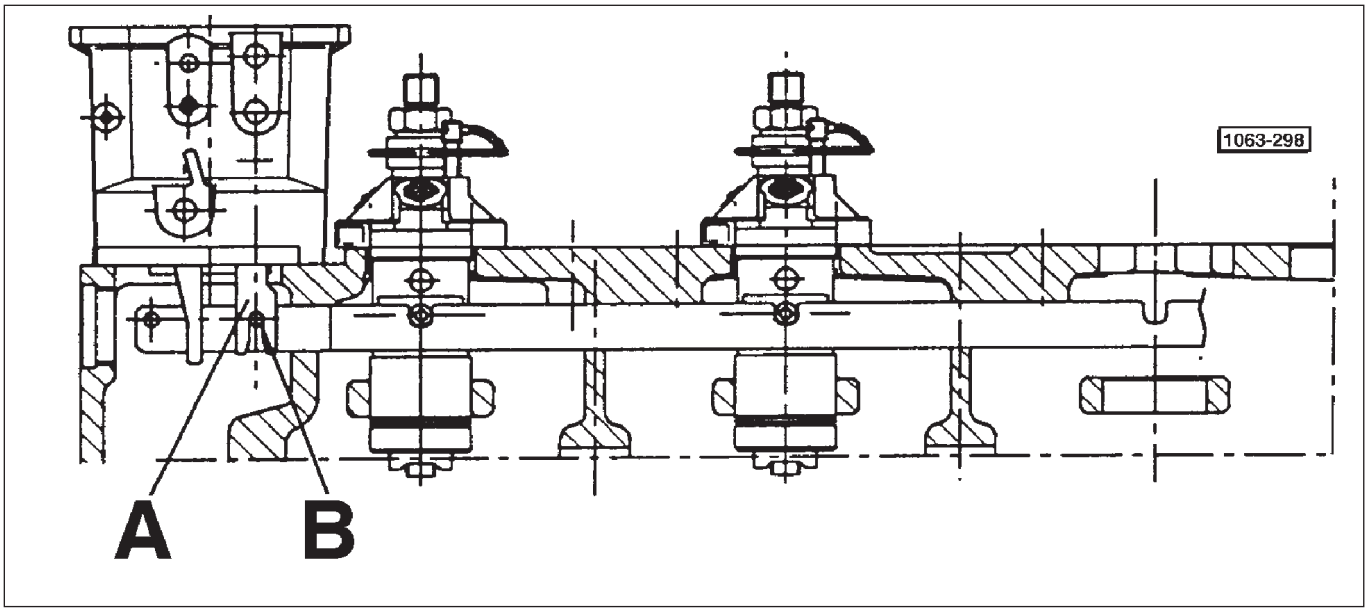


Fig. 22 - Installing the mechanical governor.

RELECTRONIC ENGINE GOVERNOR

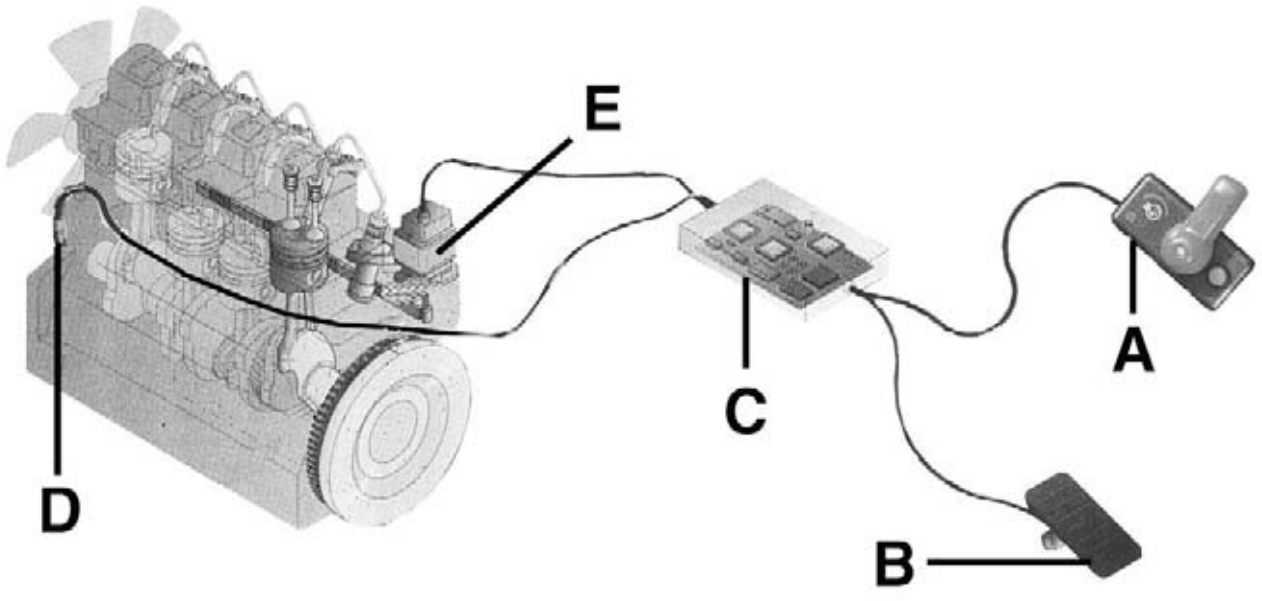
General information

he electronic engine speed governor consisting of a REG2MK microprocessor permits an accurate and stable engine r.p.m. control to be obtained. The system includes an ELECTRIC ACTUATOR, a MAGNETIC PICK-UP acting as a speed recorder, a PUSH-BUTTON PANEL with LED indication (hand throttle) and a POTENTIOMETER located on the accelerator pedal.

Specifications

power supply	8 to 15 Vdc
card electrical input	35 mA + actuator current
maximum actuator current output	6 A
PICK-UP input	0,5 ÷ 4,5 V / 0,5 ÷ 1 mA
operating temperature range	-40 to +120 °C
control	isochronous or with statism
speed range	600 to 2700 r.p.m.
overspeed	set at 3200 r.p.m.
reliability	70 °C

The engine governor receives the speed signal in form of electric signal from a pick-up facing the engine flywheel ring gear and controls the actuator installed on engine by an electronic control unit.
The control unit is fully digital type and contains the mode of operation defined for each engine model in form of permanent storage programs (EPROM).
Communications towards the control unit may be entered by an EXTERNAL PROGRAMMER equipped with specific functions enabling certain parameters to be diagnosed and set.



A - Electronic throttle hand control
B - Accelerator pedal with potentiometer
C - Injection control unit
D - Engine speed sensor
E - Electronic actuator

Fig. 23 - Configuration of the electronic engine control system

OPERATION (1st version):
Engine r.p.m. governor consists of the following parts:
— electronic control card on microprocessor;
— magnetic pick-up on engine flywheel;
— potentiometer on accelerator pedal;
— hand throttle push-button control panel and self-diagnosis LEDs;
— magnetic actuator controlling the engine injection pump rack rod.

The engine r.p.m. is selected by the operator through the accelerator pedal: the potentiometer connected to pedal supplies the card 1 ÷ 4 V voltage corresponding to a r.p.m. varying from minimum to maximum r.p.m. available.
The magnetic pick-up located against the flywheel ring gear teeth generates a voltage whose frequency is proportional to the engine r.p.m.
The electronic control unit compares the engine r.p.m. from the pick-up with the potentiometer reference thus setting up a current in the actuator so that the desired engine r.p.m. may be established.
The adjustment constants are preset in governor storage and for this reason cannot be altered.
The push-button panel permits further adjusting functions to be selected.

Three push buttons are provided in the push-button panel:
— UP
— HOLD/RESET
— DOWN

In addition, close to HOLD/RESET push button it is located a special LED providing signalling and self-diagnosing functions.

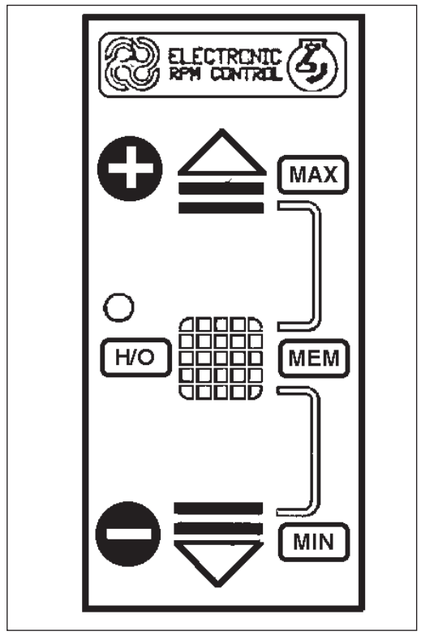


Fig. 24 - Electronic accelerator manual control.

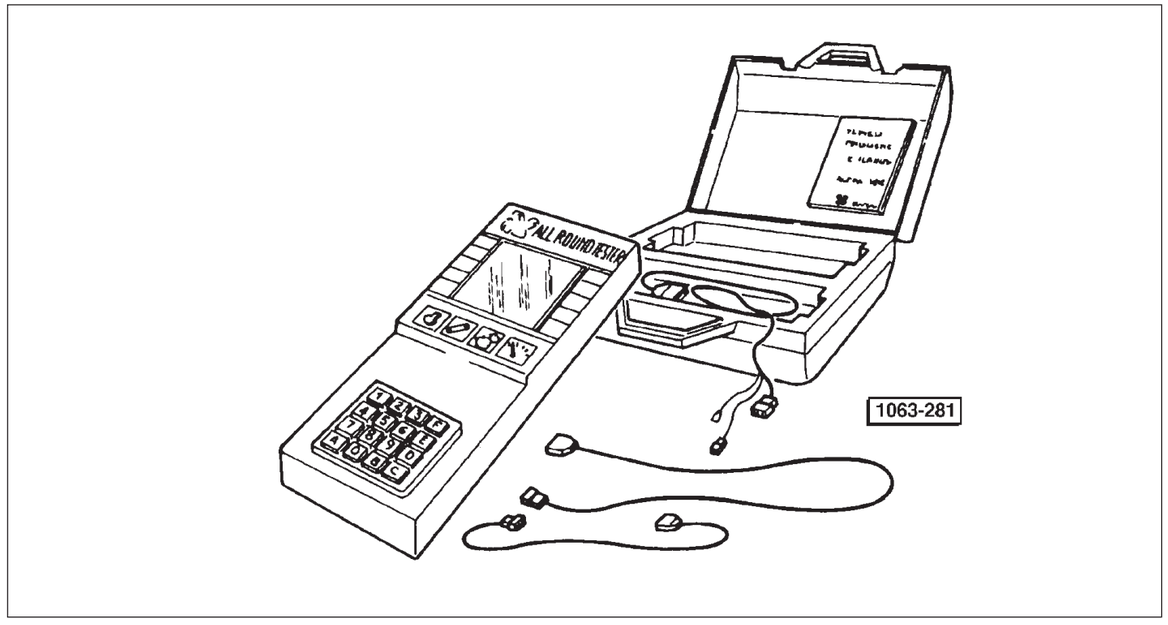
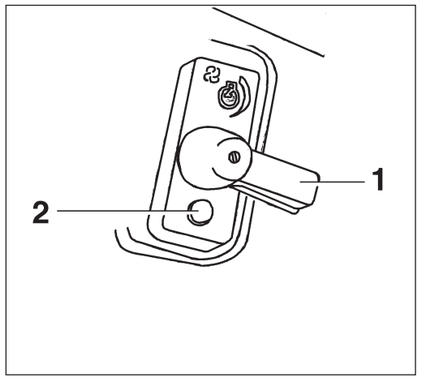


Fig. 25 - Tester (SAT 5.9030.730.6) for checking the governor.

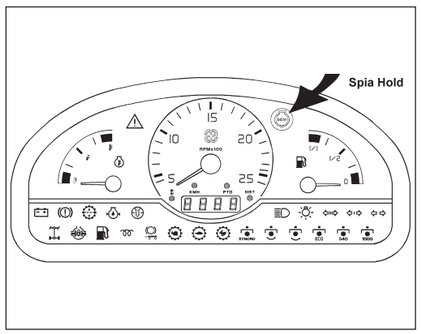
OPERATION (2nd version):
The engine rpm control comprises the following parts:
- electronic control unit with microprocessor;
- magnetic pickup;
- potentiometer on accelerator pedal;
- hand throttle with pushbutton controls and diagnostic LEDs;
- magnetic actuator that operates the injection pump control rod.



The operator controls the engine speed by depressing the accelerator pedal; the potentiometer connected to the pedal sends a signal to the ECU in the range of 1 to 4 volts, which correspond respectively to the minimum and maximum engine speeds.

The magnetic pickup, located adjacent to the teeth of the flywheel ring gear, generates a signal at a frequency proportional to the engine rpm.

The ECU compares the information received from the engine pickup and the accelerator position potentiometer and accordingly supplies a current to the actuator in order to bring the engine to the required speed. The regulation constants are preset in the ECU memory and cannot be altered.



Controlling the engine speed
There are three controls by which the operator may adjust the engine speed:
— Accelerator pedal
— Hand throttle lever
— Memory buttons

— Accelerator pedal
This control is permanently activated; the position of the pedal corresponds to the engine speed.
If a memorised speed setting is active or if the hand throttle is in any position other than the minimum position, the accelerator pedal will have no effect until it is depressed beyond the position corresponding to the speed determined by the other control devices; once this position is exceeded, control priority is returned to the accelerator pedal.
Hand throttle
Moving the hand throttle lever controls engine speed in the same way as depressing the accelerator pedal.
Memory button
Pressing this button simultaneously commands:
— "Isochronous" operation (constant engine speed maintained regardless of varying load conditions).
— Recall of the previously set engine speed from memory

On engine start-up, the "Hold" light is off, and engine speed is determined by the position of the accelerator pedal or hand throttle lever.

If the button is pressed within 3 seconds of start-up:
— the previously set engine speed is recalled;
— the "Hold" light is illuminated;
— "isochronous" operation is activated.

If the memory button is held pressed for more than three seconds (regardless of whether the "Hold" light is on or off), the following condition is obtained:
— the current engine speed is stored in memory;
— the "Hold" light comes on (if off);
— the "Hold" light flashes for 1 second;
— isochronous operation is activated;
— the engine speed (now stored in memory) is maintained.

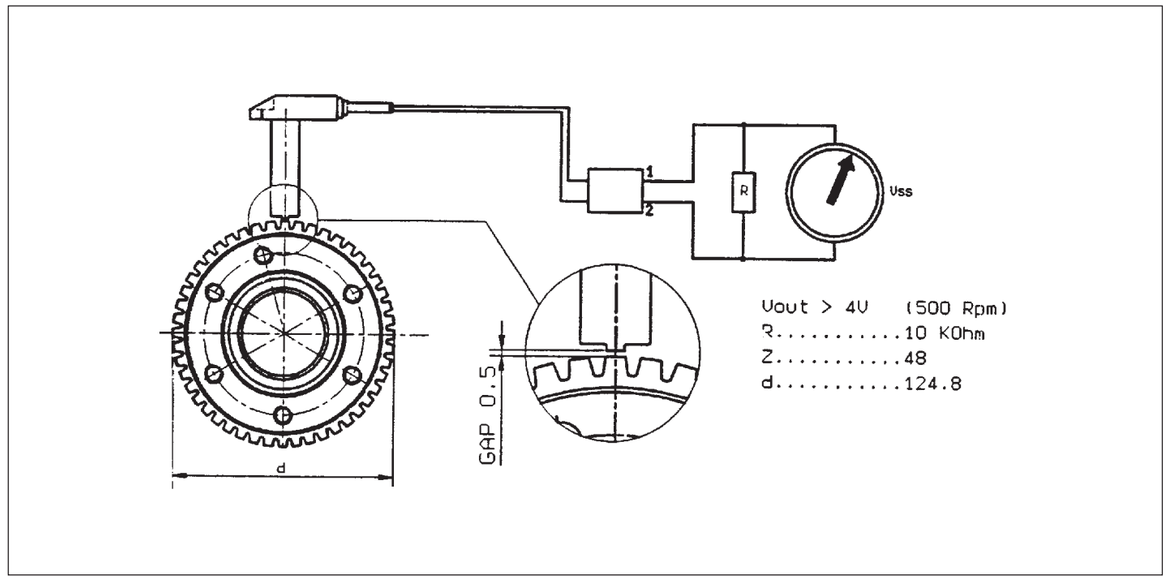
If the "Hold" light is on and the memory button is pressed within 3 seconds, the following condition is obtained:
— control of the engine speed is returned to the accelerator pedal/hand throttle;
— the "Hold" light is extinguished;
— governor droop operation.

Installing and checking the pick-up

Pick-up engine 1st version

The pick-up should be installed in the following manner:
Adjustment of a new pick-up
Screw the pick-up in the 1.5 pitch threaded hole M16 placed on the bell until contacting the ring gear then tighten the locknut.
Adjustment of an operating pick-up
If the pick-up has already been operated, after screwing against the ring gear loosen by 1/2 turn and then lock so as 0.65 ÷ 0.75 mm gap may be obtained.

Pick-up engine 2nd version



The pick-up should be installed in the following manner:

Position the pickup in its seating and secure with retaining screw.
The distance between the pickup and the pulse wheel must be 0.5 mm; check using feeler gauge 5.9030.270.0.

isochronous operation: depressing push button HOLD/RESET makes the LED light up and the governor keep the r.p.m. rigorously constant.
operation with statism: if the HOLD/RESET push button is pressed again the LED is switched off and the governor operates with a r.p.m. droop of about 7% from idling to full-load speed. This engine running mode is useful when driving tractor on public roads.
fixed-speed hand control: this helps reproduce the hand throttle control.

functions:
— isochronous operating mode;
— speed statism operating mode;
— 600 ÷ 2700 RPM speed adjustment;
— hand throttle digital control;
— dling speed under variable statism of 600 to 1500 r.p.m.;
— speed under variable isochronous operating mode from previous idling speed to 2500 r.p.m.;
— limitation of the peak horsepower varying with the speed;
— self-diagnosis of protection intervention and faults

protections:
— pick-up signal loss protection;
— actuator terminal short-circuit protection;
— overspeed protection;
— battery polarity reversal protection;
— power supply loss protection;
— microprocessor fault protection.

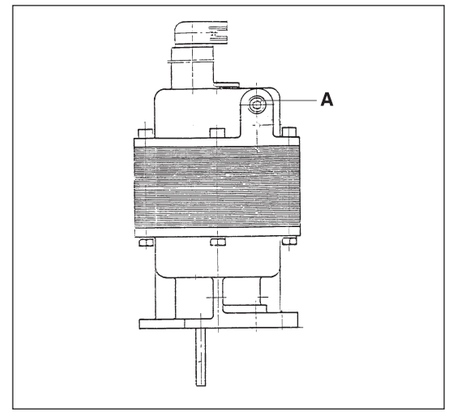


Fig. 26 - Protection against electronic governor actual access.

Installing components

The installation of the various components should be carried out with battery disconnected.

— the positive of the key subsequently supplying power to the card is directed towards battery +.
— the card power supply in-series fuse is 7.5 A.

Note: the metal box housing the card is connected to battery negative through a card-internal connection.

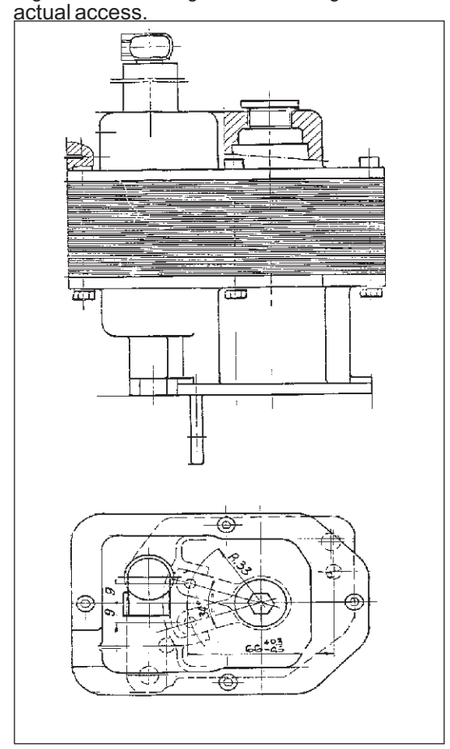


Fig. 27 - Views of the electronic governor actuator.

Installing and checking actuator

When installing the actuator take note of the following:
— **the actuator lever should move the pump rod freely and without friction;**
— a free rod must return in stop position;
— the actuator rod must be matched to the pump rod fork with a slight play.

WARNING:
The actuator is provided with cover **A** (Fig. 29) to protect the access to the injection pump control rod stroke adjusting device.

Calibration on a specific test bench:
power supply 2A ± 0,001

rod stroke 11,5 mm

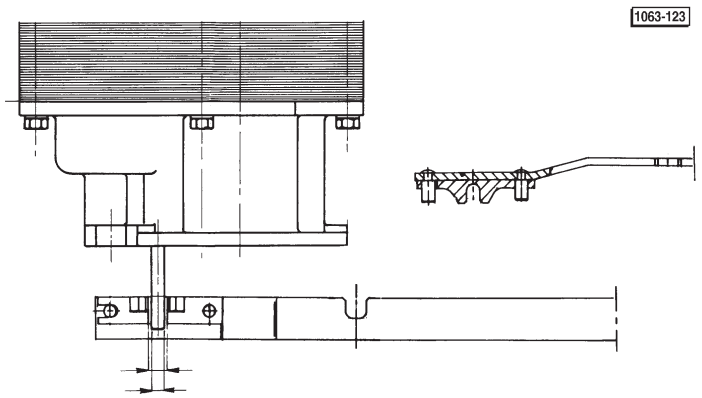


Fig. 28 - Positioning the injection pumps actuator and control rod clevis.

Calibrating the governor actuator

Calibrating the actuator is a most delicate operation.
This should be performed in case of external interventions including actuator replacement.
Operate as follows:
— Turn on the engine and run to 2000± 100 r.p.m. using the hand throttle. The engine should never be subjected to dragging loads, make sure the red LED on hand throttle control lever is illuminated.
— Remove plug **A** (Fig. 29) to gain access to the actuator spring for spring load setting and loosen screw **B** (Fig. 29) until first the engine r.p.m. droops and then the engine stops.
WARNING: locating this condition requires most accuracy and meticulous operating procedure and namely:
— locate approximately the point where the engine r.p.m. starts drooping;
— with engine stationary lock the screw **B** (Fig. 29) of a number of turns as shown in table below.

TRACTOR TYPE	ENGINE R.P.M.	HP	ADJUSTING SCREW B TIGTHENING TURN NO.
660S	2100	60	3,25
770S	2100	70	3,75
880S	2100	80	2,90
990S	2100	87	3,25
660V	2100	60	3,15
775V	2100	75	2,90

WARNING: when refitting plug **A**, take care not to alter the position of adjuster screw **B**.

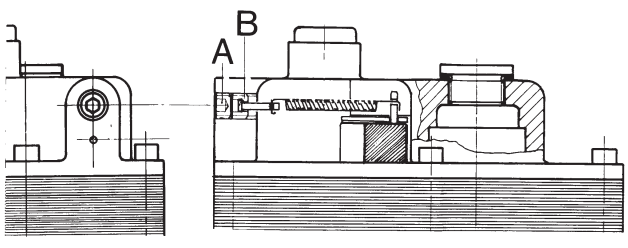


Fig. 29 - Actuator calibration

FITTING AND CHECKING THE ACCELERATOR PEDAL (Fig. 31) 1 version.

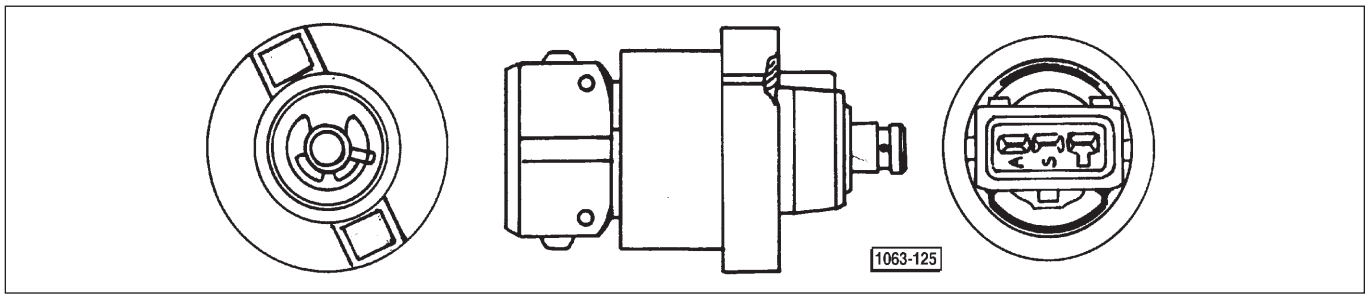


Fig. 30 - accelerator pedal potentiometer (1 version).

Fit lever **A**, and fit potentiometer **B** on support **C** securing it in position with plate **D** and pin **S**.
Connect the electrical circuit as indicated in the figure and hook up to a 5V battery.

Engine idling speed adjustment
Turn screw **V2** to obtain a voltage **VU** =1000 ± 30 mV, then tighten nut **E2**.

Maximum speed adjustment
Move lever **A** up against screw **V1** and turn this screw to obtain a voltage of **VU** = 4000 ± 30; then tighten nut **E1**

Pedal travel adjustment
Attach the sheath of the cable **F** to bracket **G** with screw **H** and position locknuts **I**. Then connect the other end of the sheath to bracket **T**. Then apply spring **L**. Adjust screw **M** to obtain a pedal travel of 75 mm as shown in figure 33 (corresponding to an approximate angle of 33°) and tighten locknut **N**. Feed cable **P** through sheath **F** and connect it to lever **R** via the clevis **Q**. Fully depress the accelerator pedal **O** (maximum throttle position). Then turn adjuster screw **H** to move lever **A** up against the stop screw and tighten locknuts I. Loosen nut **E1**, unscrew screw **V1** by two turns and then re-tighten nut **E1**

Putting into operation the electronic governor

Switch on dashboard with the ignition key.
If the LED on the accelerator control card blinks, decode the fault (refer to SELF-DIAGNOSIS section). Depressing push button HOLD the LED should illuminate, depressing HOLD button a second time the LED should go out.

Switch on dashboard with the ignition key.
If the LED on the accelerator control card blinks, decode the fault (refer to SELF-DIAGNOSIS section). Depressing push button HOLD the LED should illuminate, depressing HOLD button a second time the LED should go out.

If the LED does not light up operate as follows:
Make sure the card is properly supplied (12 V, no. 20 fuse in fusebox).
Check for card operation measuring actuator voltage. This voltage should be 0.1 ÷ 0.6V direct voltage.
Otherwise, either the card is not operating or the actuator is disconnected or defective.
After these checks are concluded and being sure all is in good order, start the engine, remove the upper actuator cap and insert a 5 mm Allen wrench on the injection pump control rod drive pin so that the injection pump control rod may be moved to stop position, when irregular engine running is noticed after starting the engine.
Keep on checking ensuring the pick-up voltage be at least a 10V alternate voltage and the direct current measured on the actuator with engine running unloaded at low idling speed be 1.5 ÷ 1.7A.

If the engine does not start operate as follows:
Ensure pick-up voltage during starting attempt be at least 3V, otherwise check for pick-up correct installation.
Make sure the battery voltage during starting stage never drops below 5.5V.
Ascertain the actuator current during starting stage is at least 2.5A and the actuator moves the injection pump rod.

Check hand control functions through the push-button panel and the hand throttle LED.
Ensure the pedal potentiometer voltage is 1 ÷ 4V.

Note: The hand throttle control push-button panel LED, besides signalling the HOLD condition also helps in diagnosing malfunctions

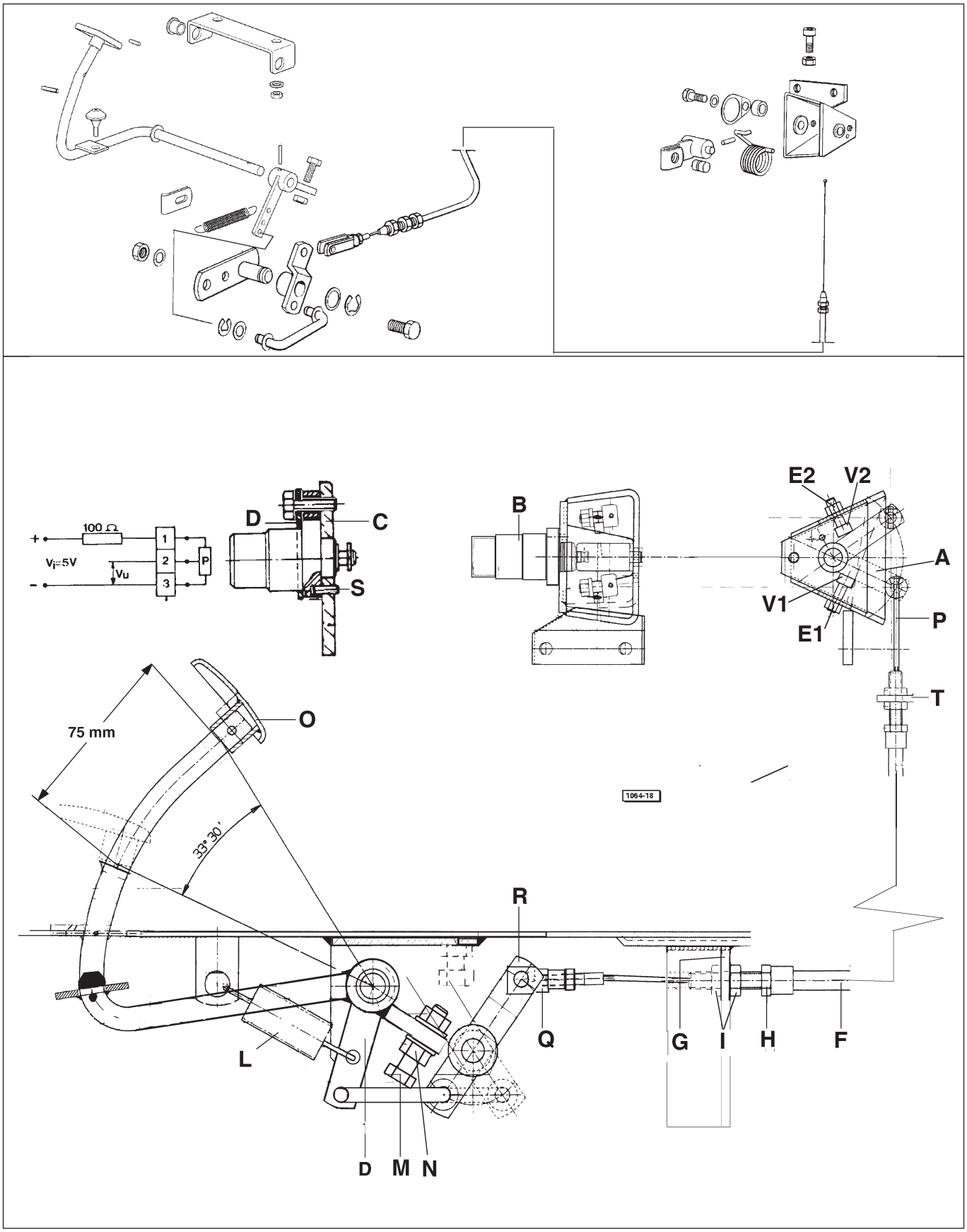


Fig. 31 - Accelerator pedal adjustment diagram (1 version)

FITTING AND CHECKING THE ACCELERATOR PEDAL (Fig. 32) 2 version.

Fit lever **A**, and fit potentiometer **B** on support **C**, securing it in position with the screws **D**.

Connect the electrical circuit as indicated in the figure and connect a 5V battery.

Idle speed adjustment
Turn screw **V2** to obtain a voltage **U1 10 = 0.5V**, then tighten locknut **E2**.

Maximum speed adjustment
Bring lever **A** up against screw **V1** and turn the screw to obtain a voltage **U 90 = 4.5 V**, then tighten locknut **E1**.

Pedal travel adjustment
Attach the outer sheath of the cable **F** to bracket **G** with bolt H and fit the locknuts I. Then attach the opposite end of the outer sheath to bracket **T**. Fit spring **L**. Adjust the position of screw **M** to obtain the pedal travel indicated in Fig. 32 (corresponding to an angle of approx. 33*) and tighten locknut **N**. Feed the cable **P** through sheath **F** and attach it to lever **R** via clevis **Q**. Fully depress the accelerator pedal **O** (maximum throttle position), then turn adjuster screw **H** to bring lever **A** up against the stop screw and tighten locknuts I. Back off the locknut **E1**, unscrew the screw **V1** by two full turns, and then re-tighten the locknut **E1**.

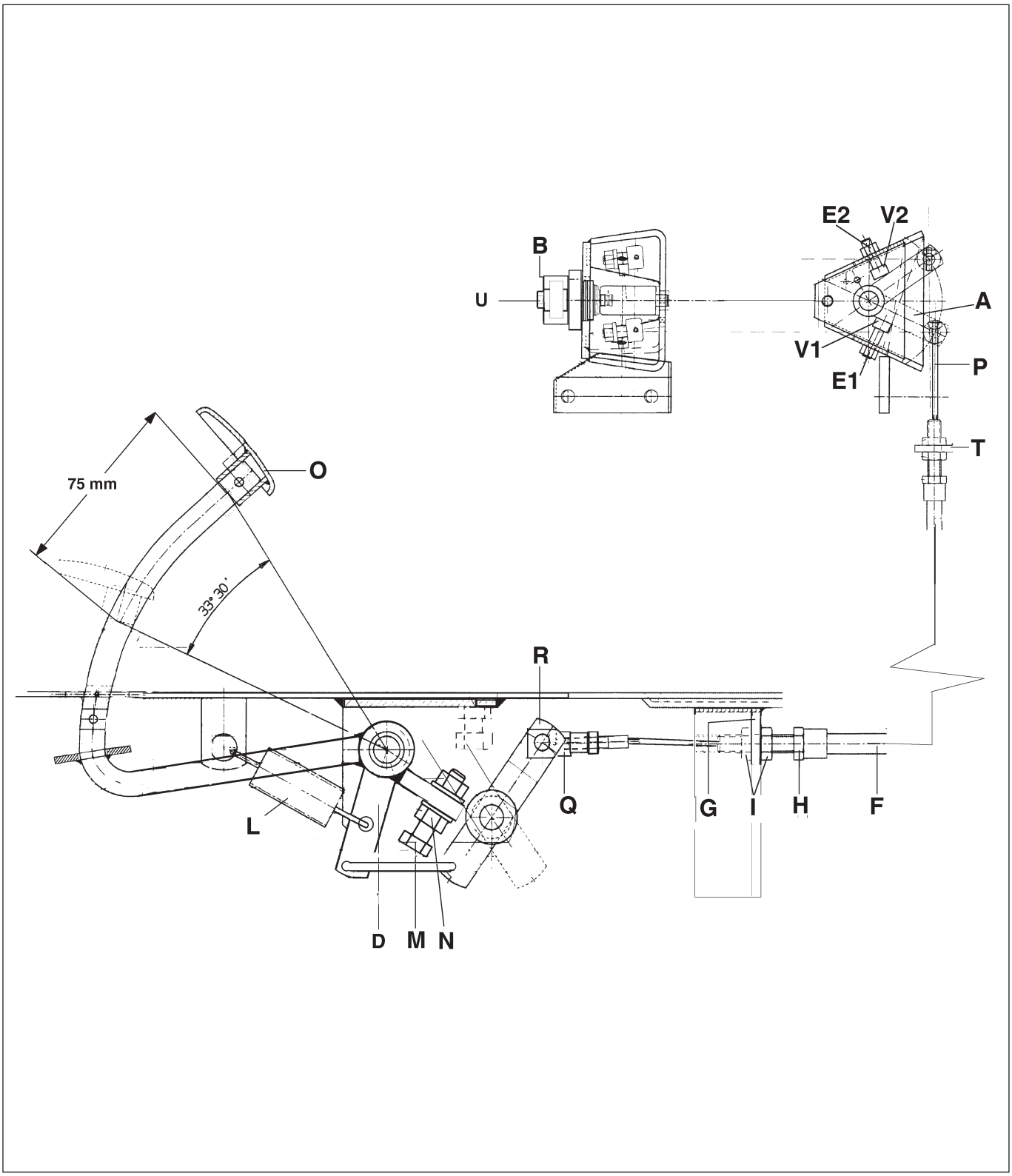


Fig. 32 - Accelerator pedal adjustment diagram (2 version).

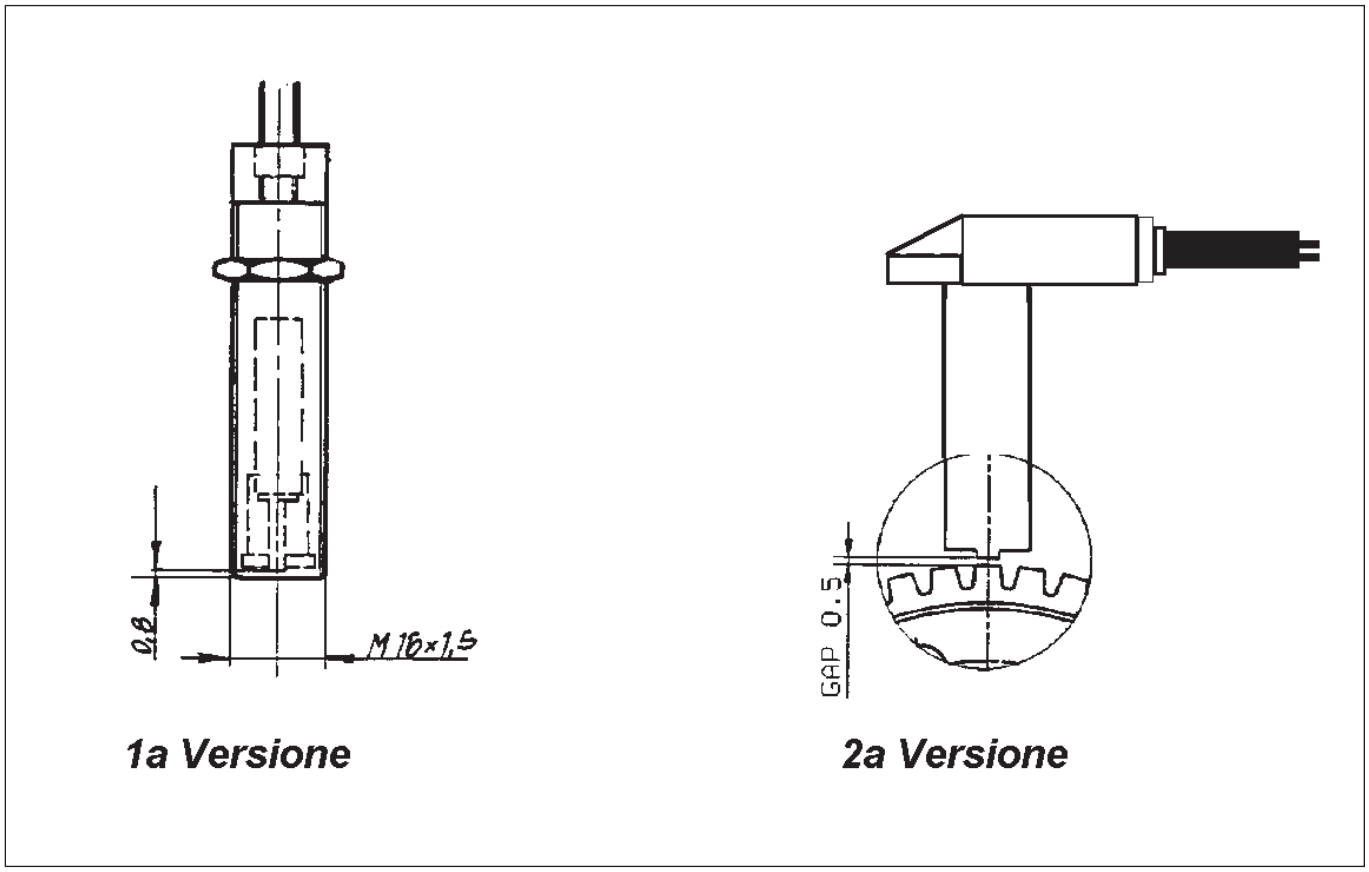


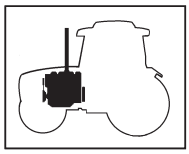
Fig. 33 - Pick-up.

DIAGNOSING MALFUNCTIONS (1 version)

Operating as follows the self-diagnosing system is in a position to diagnose any failures and warn the operator through the led located on the hand throttle control push-button panel.

FAILURE	NO. OF LED BLINKS	LIKELY CAUSE
pick-up	4	pick-up failure, pick-up disconnected
actuator	2	actuator short-circuited, actuator disconnected
potentiometer pedal accelerator	3	potentiometer failure, potentiometer disconnected
overspeed	1	attuatore locked, injection pump locked
storage	5	parameter storage not programmed or defective

WARNING:
Should any failure be found take the number of LED blinks from the control push-button;
This LED blink number will permit the repairing technician to readily perform replacement through the correct spare part needed.



Setting the maximum speed to 30 km/h

When so required by local regulations, the maximum speed can be limited to **30 km/h** by means of the engine control unit and a vehicle speed sensor mounted on the gearbox.

Maximum speed is set as follows:
Connect the **"ALL ROUND TESTER"** and calibrate the engine speed following the normal procedure.
Access menu "1" Engine (enter the password "A5555" if required).
- Press 6 "Wheels" and 1 "Configuration" and enable only the parameter for 30 km/h and 40 km/h tractors, if fitted with 13.6R36 tyres.

Consult the table below and enter the appropriate wheel constant value.

60 - 70 - 75 - 80 - 90 CV	
Pneus	**Parameter**
320/70 R 24	1512
12.4 R 24	1442
360/70 R 24	1442
13.6 R24	1390
380/70 R 28	1390
12.4 R 28	1320
420/70 R 24	1320
360/70 R 28	1320
13.6 R 28	1276
380/70 R 28	1276
420/70 R 28	1216
380/70R20	1448
11.2 R 24	1448

Press **C** to confirm the selection, and then press **E** twice to exit the programme.

Checking the calibration.
With the tractor stationary, check that the engine speed reaches the maximum rpm;
Then, with the tractor moving, engage top gear and accelerate to maximum speed; in this condition the engine speed should be between **1700 and 1900 rpm.**

NB: the control unit's diagnostic procedure includes the alarm **"speed sensor faulty or not connected"**. This alarm is signalled with six flashes of the LED on the hand throttle.

Fuel injection pumps

The single-cylinder fuel injection pumps are equipped with a constant-pressure backflow valve (short G.D.V. valve).

The plunger load is evenly distributed over the whole camshaft, since an equal-distance distribution is provided.

The G.D.V. valve keeps pressure inside the injection pipes to a high level during the intermediate pump delivery phases.

This ensures engine smooth running even at low operating speed, higher engine performance as well as more favourable torque curves, besides suppressing any nozzle dribbles resulting in unburnt gas, poor injection nozzle efficiency and engine detonation.

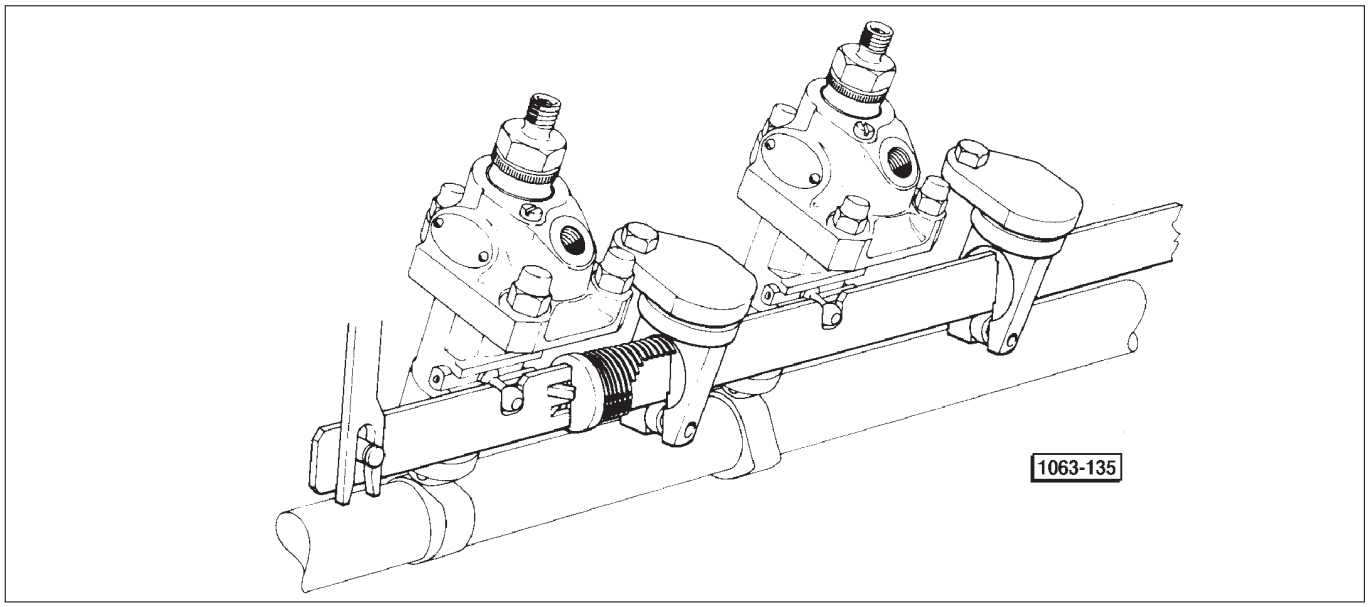


Fig. 1 - Injection pump control assembly.

Injection pump control system

The injection pumps are simultaneously controlled by a blank sheet steel bar which besides ensuring the required stiffness has particular lightweight properties to prevent excessive friction and inertia under operating conditions.
This bar controls the injection pumps operating through specially arranged slots and is connected to the engine governor by a ratchet.
The injection pump control bar is guided by a couple of block-mounted supports provided with slide rings.

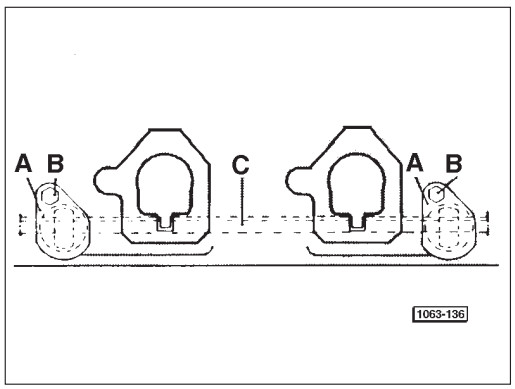


Fig. 2 - Mounting positions of the injection pump control bar guide supports.

Installing injection pump control bar guide supports (Fig. 2)

Install supports **A** with related O-rings, rollers and lockpins using screws **B** into the engine block. Insert the injection pump control bar **C** into the supports and tighten screws **B** and ensure that the bar is free to slide inside the support. (The two supports equipped with sliding rollers should be placed at both bar ends).
Warning - When the injection pumps are removed operate the engine shutoff control (STOP) so as to bring the rack rod dog back to the injection pump centre line. This permits it to be passed through the special aperture in the engine block.
Check is made visually through the sight windows in the engine block.

Engine timing

Valve timing

This should be carried out before injection pump calibration.

Raise the first cylinder piston to upper position; remove the rocker arm support, remove intake valve spring and drop valve onto piston head then insert a snap ring into the valve stem.

Fasten no. 5.9030.616.4 equipment on piston head as shown in figure 4 (the comparator indicating 25 mm stroke reading should be placed on the engine valve).

After raising piston to T.D.C. position set to zero the comparator fixed to the valve stem. When the concerned cam rest race has been located also reset the comparator applied to the intake valve tappet head.

Turn the crankshaft clockwise until a 17.27 mm piston stroke has been completed. Then rotate the camshaft until a 3.93 ± 0.1 mm intake cam lift is obtained.

Fit the timing gear and bring the timing gear marked tooth into contact with both idler gear marked teeth. Press on the gear anticlockwise when tightening the securing screws until any backlash between teeth is taken up.

Take the equipment and proceed with the injection pump setting operations.

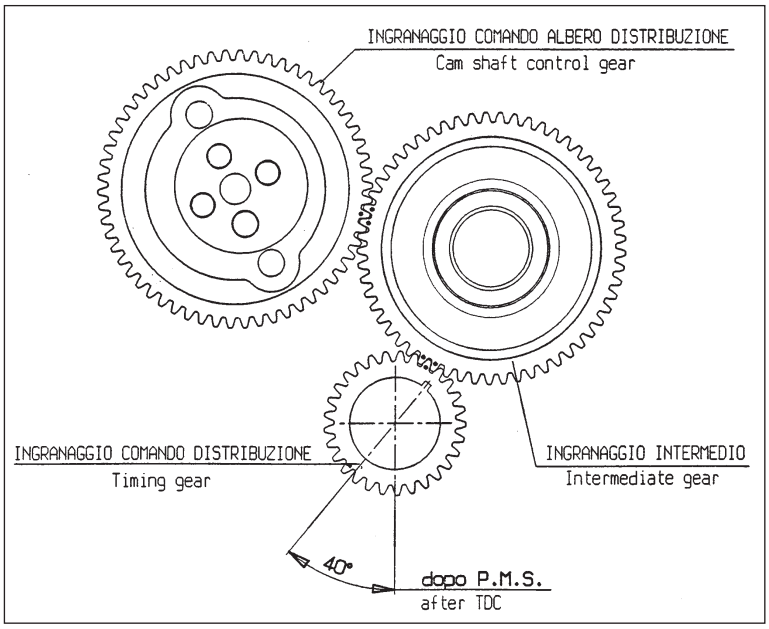


Fig. 6 - Timing gear reference marks for engine timing

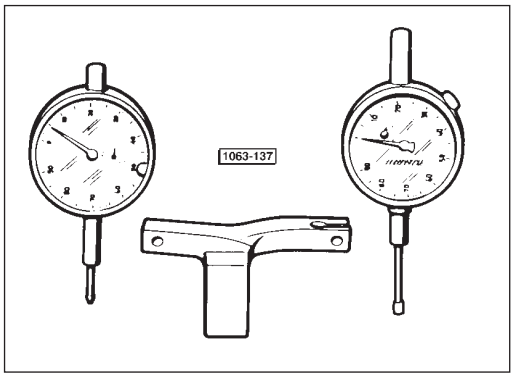


Fig. 3 - No. 5.9030.616.4 engine timing equipment.

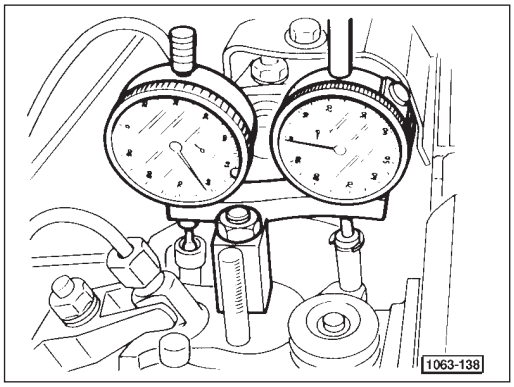


Fig. 4 - Equipment mounting position for 1st cylinder timing.

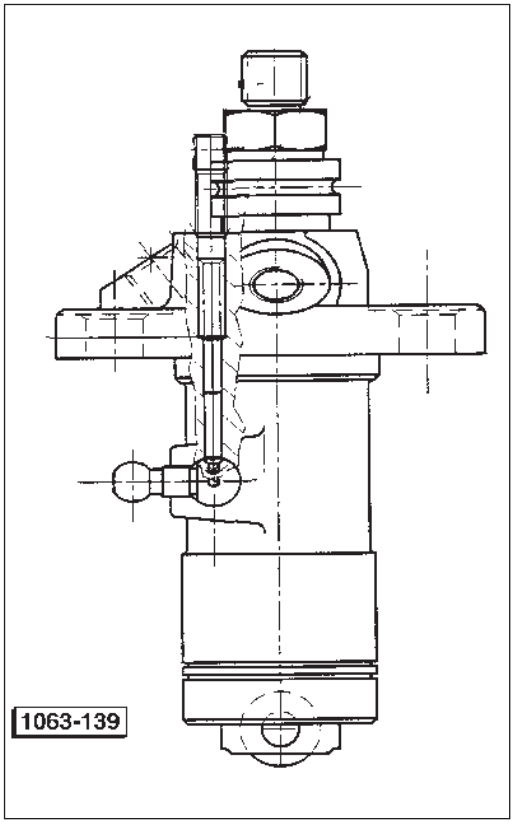


Fig. 5 - Injection pump

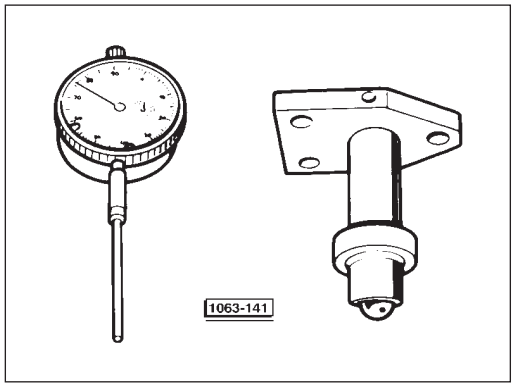


Fig. 7 - No. 5.9030.617.4 /10 injection pump timing equipment.

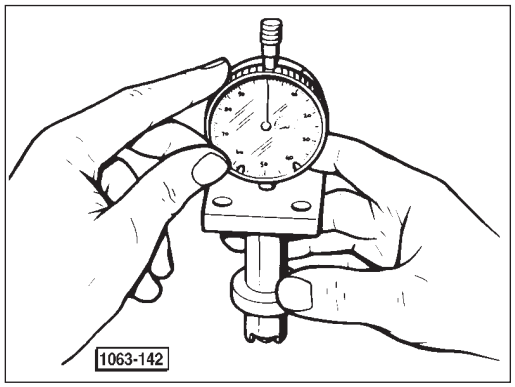


Fig. 8 - Resetting no. 5.9030.617.4 /10 comparator-equipped tool.

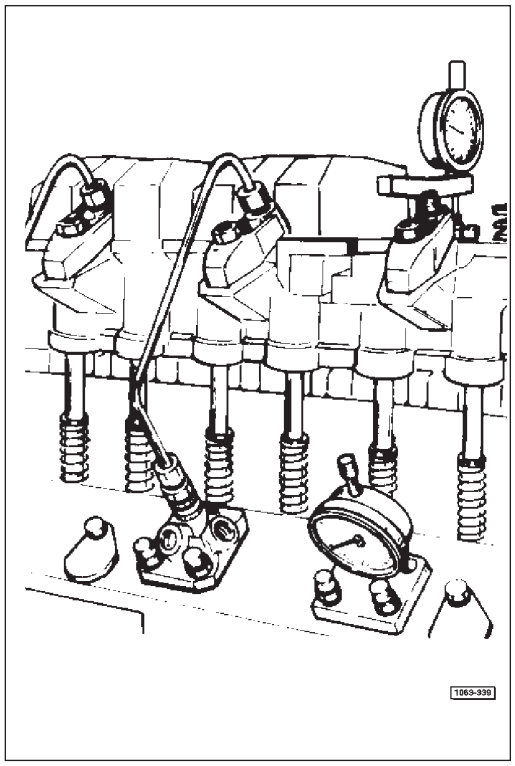


Fig. 9 - Mounting position of the no. 5.9030.617.4/10 equipment fitted with comparator.

Injection pump calibration (BOSCH injection pump equipped with a lock pin for the control rod)

After valve timing has been successfully concluded undertake this calibration by using no. 5.9030.617.4/10 equipment in addition to the equipment already positioned on the first cylinder and excepting the comparator on the valve tappet rod.

To each shim pack add a pack of shims equal to the value stamped on the pumps. (example: +2 must be added a thickness of 0,2mm).

Injection pump calibration is to determine a 80.4 mm dimension between the pumps-to-block fixing surface and the corresponding cam races, considering the position they find when the injection is being started; (Fig.10).

Rotate the crankshaft to move the 1st cylinder piston into T.D.C. position (compression stroke end, valves closed) and set to zero the comparator attached to the valve stem; the same procedure as described in valve timing section shall apply.

Rotate the shaft in a counter-clockwise direction until a reading of 2.92 mm is shown on the dial gauge; this value corresponds to a timing advance of 16.
Correct positioning takes place when through anticlockwise rotation either the 2.92 mm set point is exceeded, soon after coming back to such a dimension through clockwise crankshaft rotation so that any clearances may be taken up.

Fit the special no. 5.9030.617.4 comparator. To reset the comparator rest the comparator measuring roll onto a flat surface.

Insert the tool into the first injection pump seat and take measurement reading from the comparator dial to align the bearing face with the engine block. This reading corresponds to shim pack **X**.

To each pack of shims **X** add a number of shims **Y** being equal to the dimension stamped on injection pump flange, so as to obtain a total shim packing (**X** + **Y**) to be put between injection pump and engine block, this permitting a 80.4 mm standard dimension to be determined.

Repeat this operation for each injection pump, fitting tool 5.9030.616.4 to the corresponding cylinder.

Warning - Install the injection pump after applying some Silastik 738 silicone between shims, engine block and injection pump.

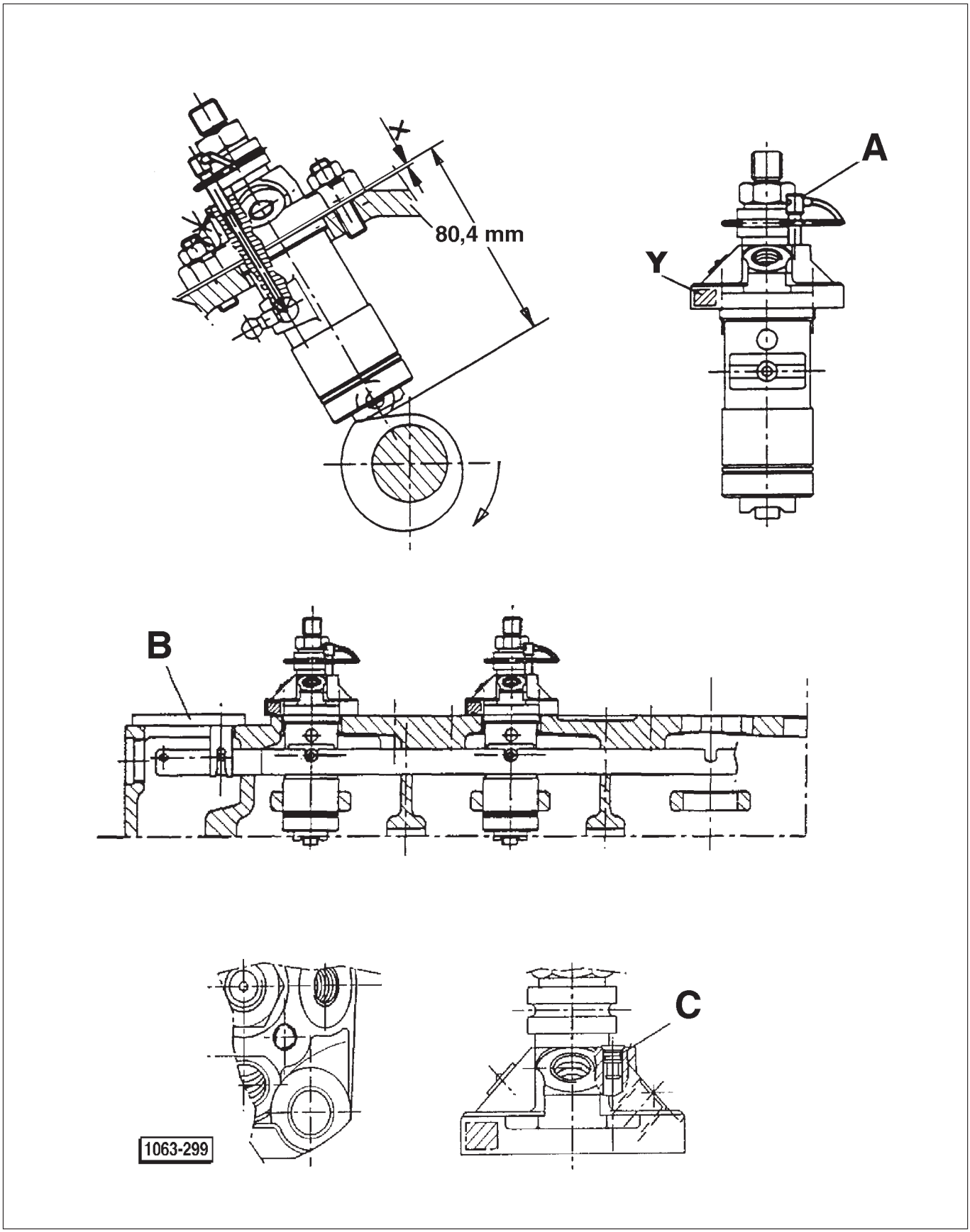


Fig. 10 - Installing the injection pumps.

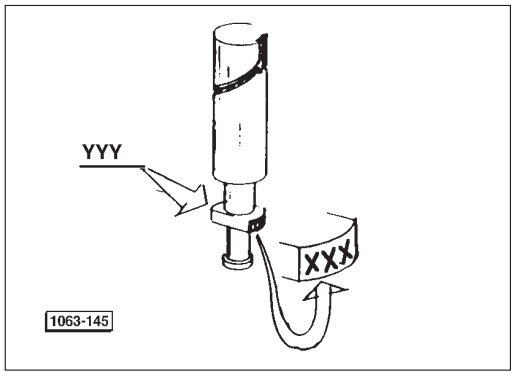


Fig. 11 - Injection pump plunger identification marks.

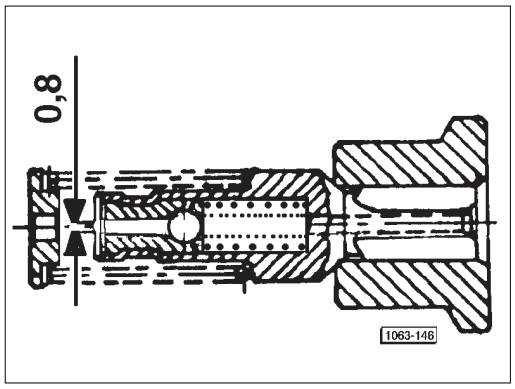


Fig. 12 - Injection pump G.D.V. valve.

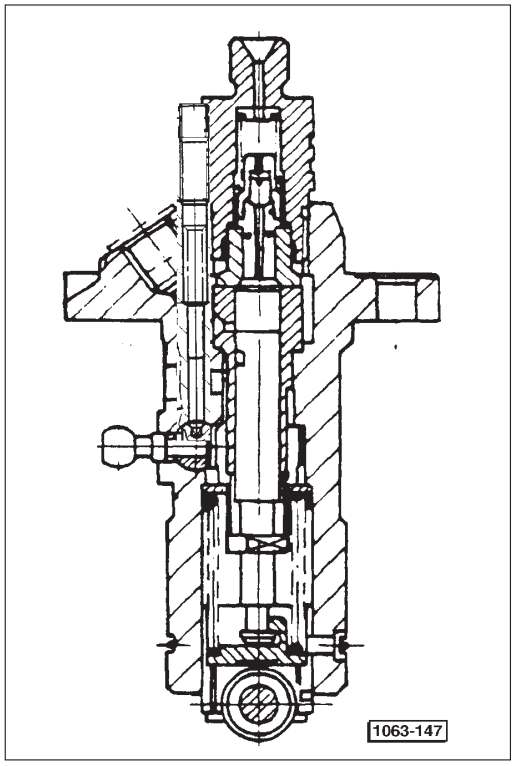


Fig. 13 - Injection pump section

Positioning the pumps (fig. 10).

Before positioning the pumps, lock the control rod using tool 5.9030.728.0, which is to be positioned with the governor removed and as shown in figure 10.
Then position each injection pump (interposing a shim pack of the predetermined thickness), so that the dog engages the fork in the control rod.

Rotate the pump clockwise until you feel the dog fully engage the fork, then tighten the retaining screw.

NB: The pump must be positioned with the injection cams in b.d.c. position.
Remove tool (**B** Fig. 10), release the pump by removing the lock pin **A** (Fig. 10 - tool SAT 5.9030.728.0); fit the governor and the rear plug (flywheel end).
Check that the rod moves freely and that the STOP control operates correctly.

Servicing

All the pumps are interchangeable as all the plungers are of the same type.
N.B: new plungers can be identified from the markings in the positions shown in figure 11.

Removal and refitting of an injection pump without replacing components.
This operation does not require the use of any special tools. Refit the pump using a shim pack of the original thickness.
Hold the pump control rack in position using lock pin **A** (fig. 10) to. To do this, operate the STOP control lever so that the pin engages the relative notch on the pump control rod rack.

Install the pump (also equipped with a rack lock pin) in the cylinder block, making sure that the dog engages in the fork of the control rod; then rotate the pump in a clockwise direction whilst simultaneously pushing the control rod through the engine inspection flange towards the front of the engine.

Removal and refitting of a pump after replacing one or more components
Before installation the pump must be re-calibrated on a test bench.

Remove any previous timing marks and stamp a new notch and dimension (**Y**) on the pump.

Refit the pump using a shim pack (see fig. 17) according to the new dimension stamped on the pump (always make a note of the old value stamped on the pump before erasure, in order to facilitate calculation of the new shim pack thickness) and follow the instructions given in the previous point.

Replacing a pump
The thickness of the shim pack (**X** +**Y**) must be corrected by first removing shims to the thickness indicated on the previous pump and then adding shims to the value indicated on the new pump.

Replacement of camshaft or cylinder block
In the event of camshaft or cylinder block replacement, the valve and injection pump timing procedures must be repeated, but the injection pumps will not require re-calibration.
Install the new pump in the cylinder block with the control rack locked in position, making sure that the dog engages the control rod and following the instructions on page 80 (pump positioning).
Replacing the injection pumps control rod
Proceed to refit the new rod following the directions on page 80 (pumps), and complete all operations relative to the positioning of the injection pumps.

Calibration of injection pumps 2.4619.140.0 (type BOSCH PFR 1 K 90 A 543) e **2.4619.030.0** (type BOSCH PFR 1 K 90 A 555)

Injection pump calibration is of crucial importance to the correct engine running. Injection pump calibration procedure should be carried out by specialized workshops provided with the following BOSCH equipment:

— Injection 2.4719.200.0/10
— Injection pipe Ø6x1.5x267
— Test oil ISO 4113 a temp.40°C
— Fuel supply fuel 0,4 ± 0,05 bar
— Camshaft N. 1 686 101 021 (with 7 mm cam lift)
— Outlet with metering orifice Ø 1mm

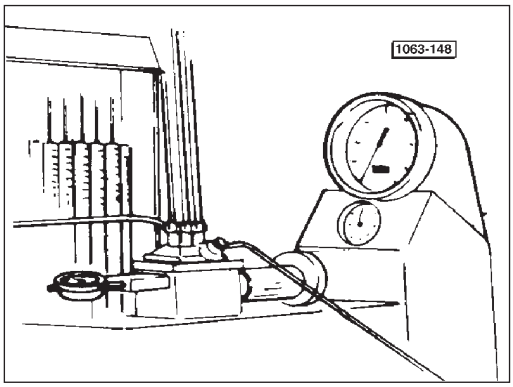


Fig. 14 - Injection pump calibration test bench.

Calibration specifications

radjustments		60 - 85 HP	
test conditions	r.p.m.	from X* in mm	delivery mm/lift
1°test	300	-3,5	12,5 ÷ 13,5
2°test	750	3	99 ÷ 105
3°test	1300	3	84 ÷90
4°test	100	9	76 ÷95

***X** Reference position for the pawl of the injection pump determined using the BOSCH special tool.

Characteristics of the injection pump

Nominal prelift for delivery start at X	3 X+3= 2,4 ÷ 2,5
Spill valve bar	70
Depth of delay notch mm	0,5
Start of delay notch intervention mm	X+7

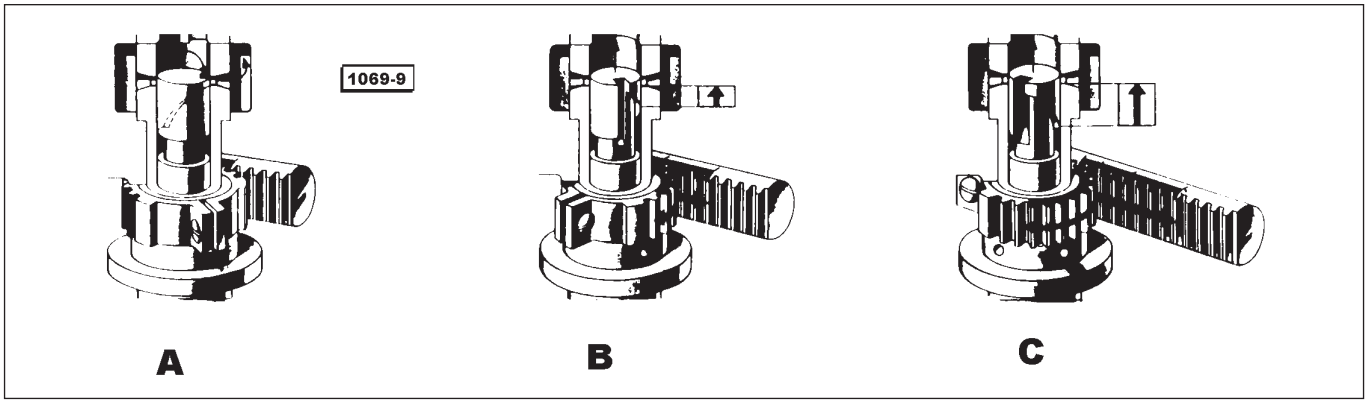


Fig. 15 - Injection pump plungers
A - no delivery; the fuel flows through the longitudinal slot.

B - partial delivery; correct adjustment opens flow at mid-stroke position.

C - full delivery; the plunger stroke corresponds to maximum delivery.

Synchronisation of injection pumps using the method of balancing the exhaust gas temperatures

1 - Bring the engine up to operating temperature.

2 - Remove the screws in the lower part of the exhaust manifold.

3 - Run the engine at idling speed (700 rpm).

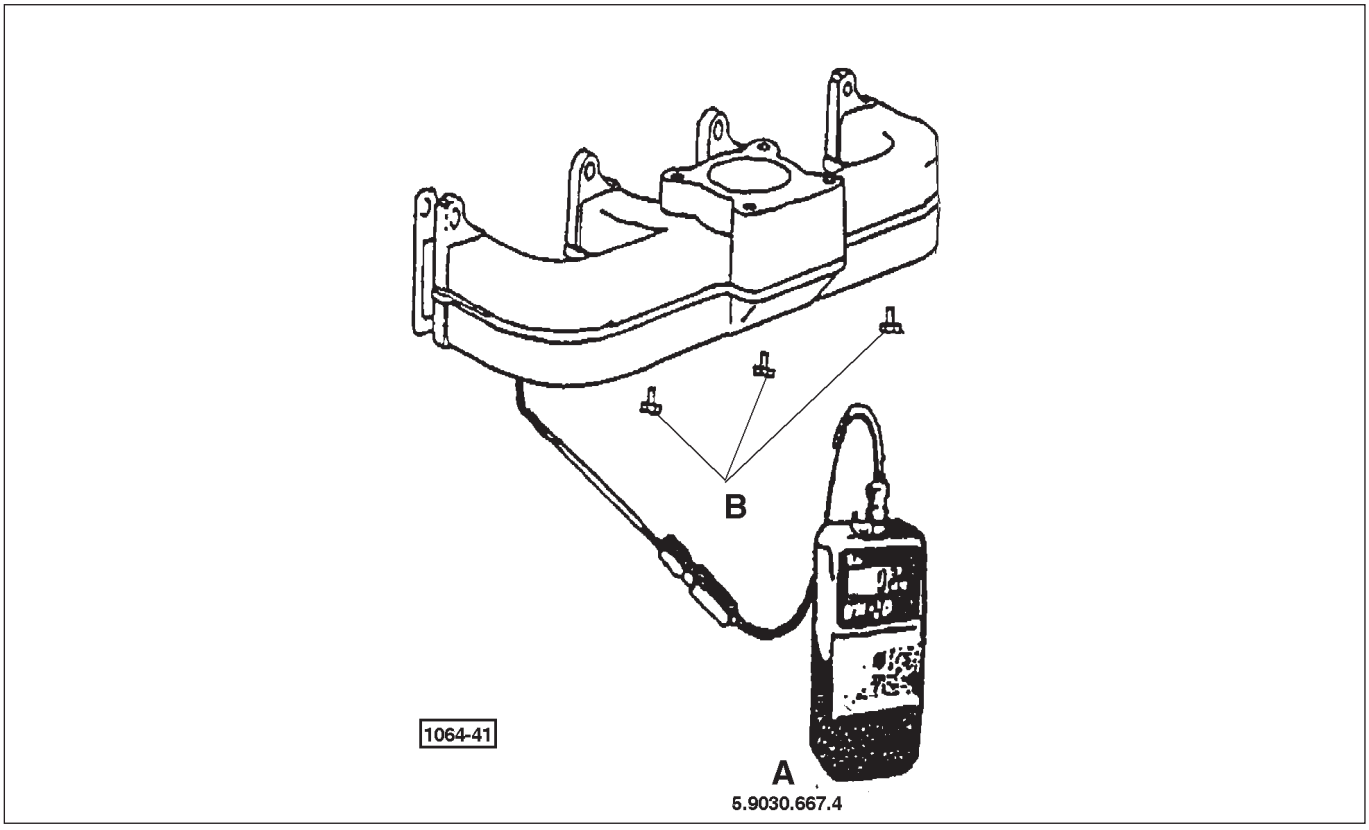
4 - Insert the probe of the digital thermometer 5.9030.667.4 in the hole of the exhaust manifold corresponding to the first cylinder and allow the temperature reading to stabilise. Repeat this operation for the other cylinders, noting the temperature readings obtained for each.

5 - The difference in temperature between one cylinder and another must not exceed 20°C. If the difference exceeds this value, it will be necessary to adjust the injection pumps.

6 - To adjust the injection pumps, proceed as follows:
Slightly loosen the 3 retaining nuts of the injection pump so that you can rotate the pump by tapping its sides with a punch while the engine is running (rotate the pump in a clockwise direction to increase the exhaust gas temperature and counter-clockwise to decrease it).

NB: after adjusting one of the pumps, you will need to re-check the exhaust gas temperature of the cylinders served by the other pumps.

7 - Tighten the injection pump retaining nuts to a torque of 1kgm (10Nm) and replace the screws in their holes in the exhaust manifold.



Synchronisation of injection pumps using the method of balancing the exhaust gas temperatures

A - Digital thermometer -30 ÷ 250 °C
B - Screws 10x6x15 exhaust manifold.

Fuel prefilter

Strip the fuel prefilter by releasing holding spring **A**.
Clean cup **B** and cartridge **C** with a compressed air jet and ensure each single component is fully sound, if not replace.
Should any water be found in the glass cup bottom, loosen securing dowel **D** and wait until fuel comes out without air bubbles.

Fuel filter

To replace the filter cartridge, remove the cartridge retaining screw in the upper body, remove the old cartridge and replace with a new cartridge of the original type.
Check the seal between the new cartridge and the cover. Position the cartridge and replace the retaining screw.

Draining water from fuel filter

To drain off any water from the filter, loosen the screw in the bottom of the filter housing and drain off all fuel mixed with water. When clean fuel appears, re-tighten the screw.

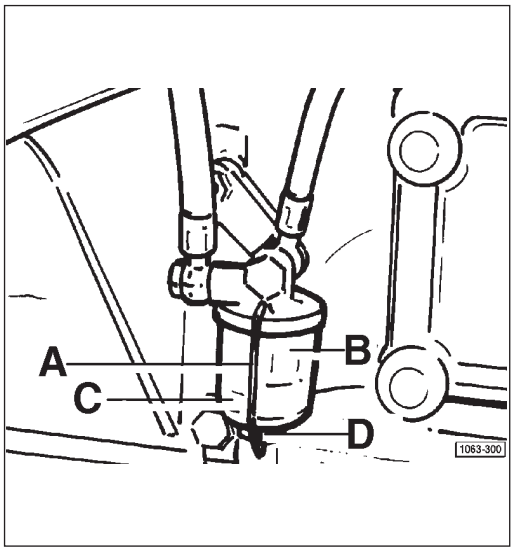


Fig. 18 - Fuel prefilters.
A - holding spring
B - cup
C - cartridge
D - dowel

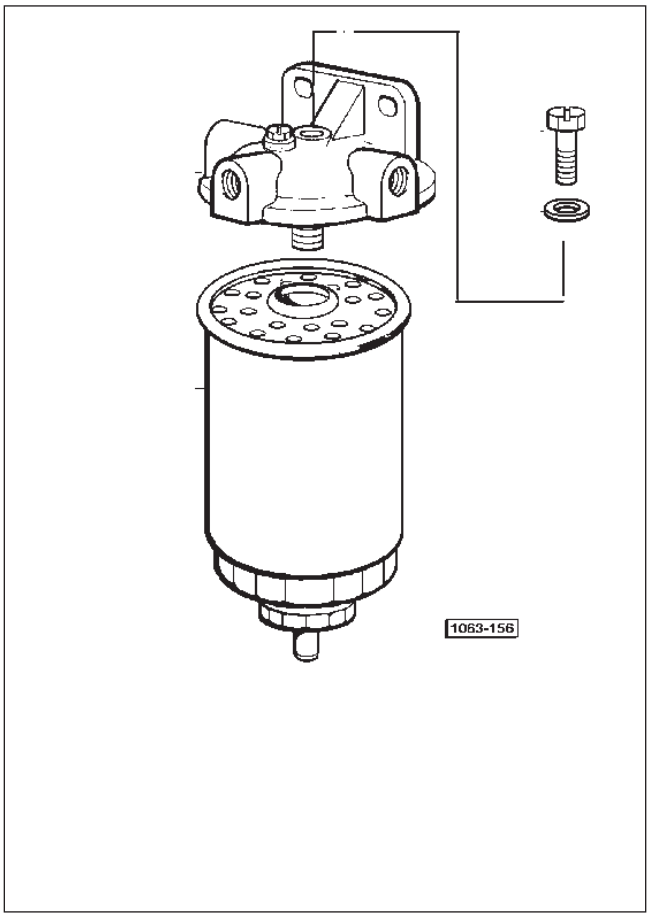


Fig. 19 - Fuel filters.

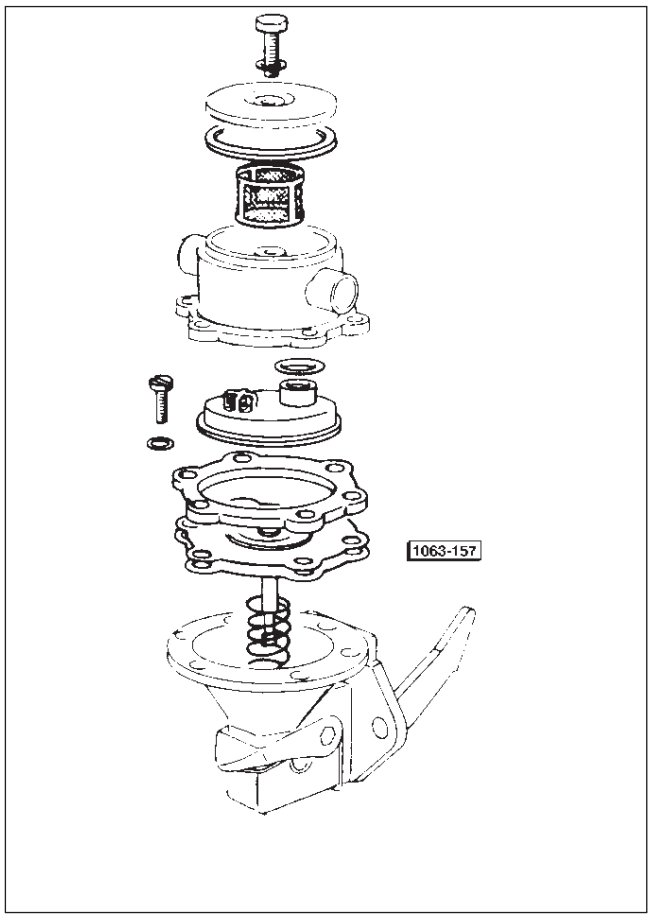


Fig. 20 - Fuel supply pump.

Checking the fuel supply pump efficiency (A.C. pump)

This check can be performed by measuring the pump delivery (see on page 22).

If necessary, carry out inner filter cleaning or replcae any items being part of the assembly under overhaul. Full pump assembly replacement is recommended only if pump casing cracks are noticed.

Bleeding the air from the fuel supply system

The system air-bleeding operation should be made whenever pipe lines are disconnected or there is air in the fuel supply system.

Operate as follows:
Be sure the fuel tank is filled, then loosen the bleeding valve located in the fuel filter upper side, press on the fuel pump hand control lever until the fuel flows out without air bubbles. Tighten the retaining screw.

NOTE: While operating the pump control lever, ascertain that a certain resistance is felt. Otherwise set starter going so as to locate the pump priming position (camshaft control cam at B.D.C.).

Loosen injector pipes only if necessary, set starter going until fuel free of air bubbles flows out of the loosened pipe fittings. Thereafter tighten fittings to the specified torque.

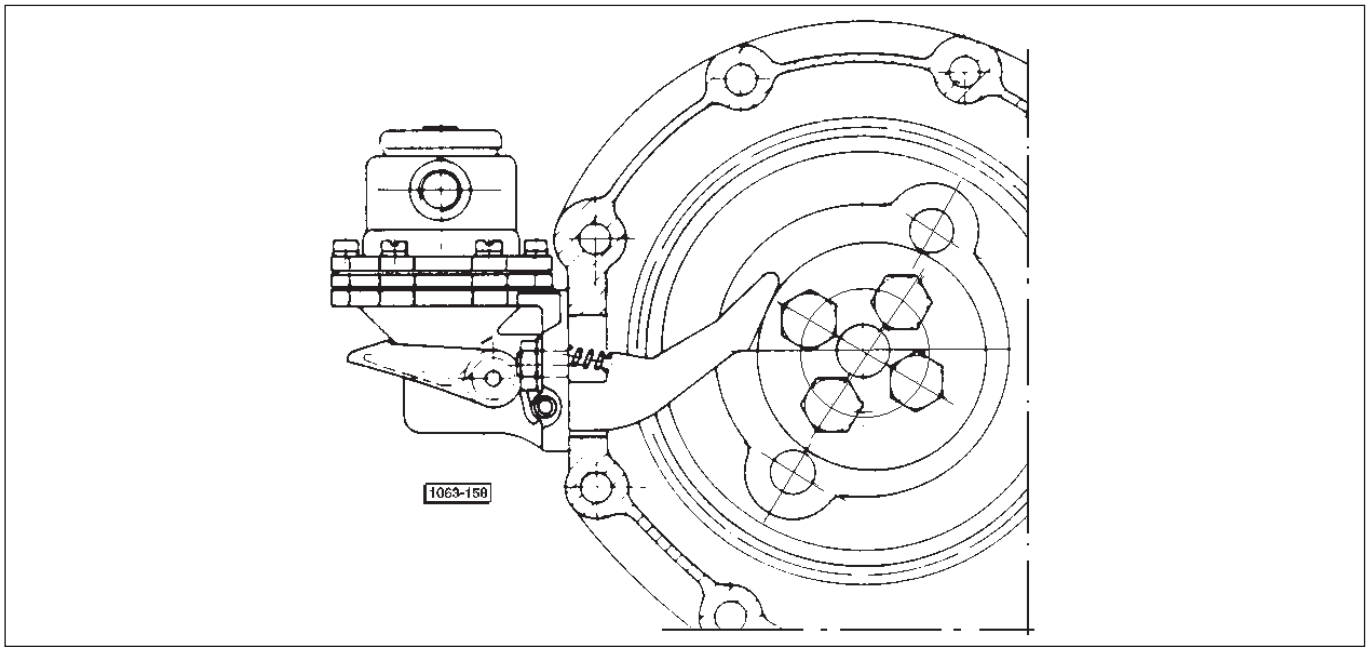


Fig. 21 - Fuel supply pump control.

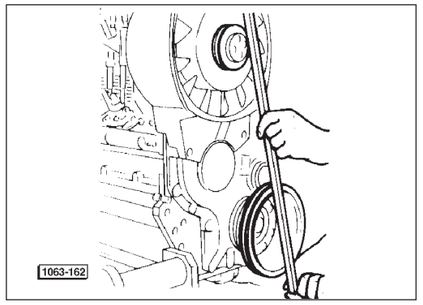


Fig. 1 - Fan assembly..

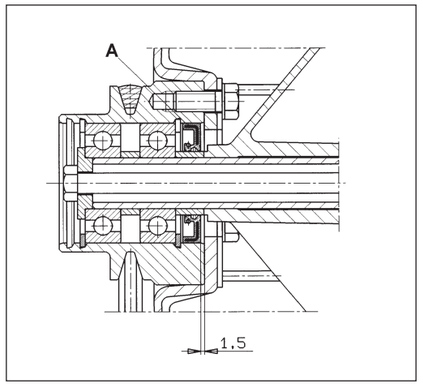


Fig. 3 - Fan bearings protection seal.

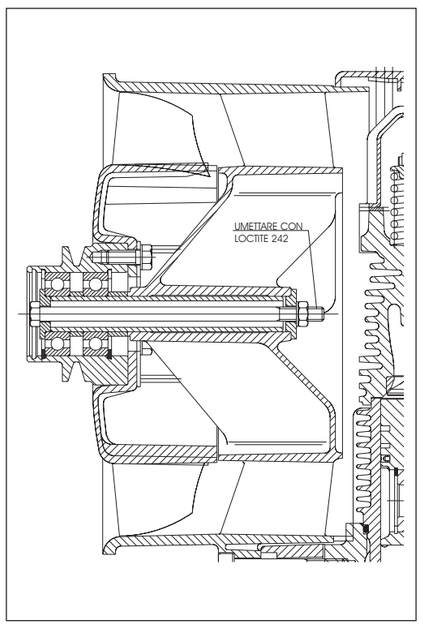


Fig. 4 - Fan assembly cutaway view.

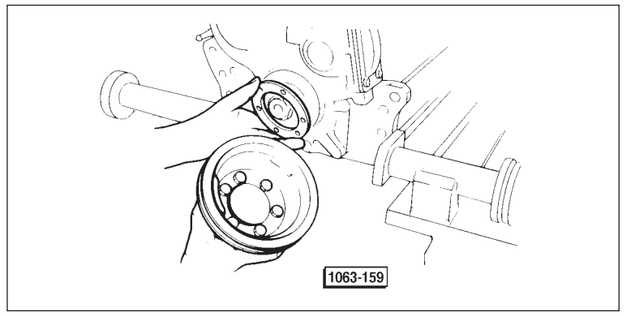


Fig.2 - Shimming engine pulley.

Fan assembly for engine 1000-A

No specific servicing is required for fan assembly. Occasionally check bearings for proper sliding, replace if necessary.

Fitting the fan bearings protection seal (Fig.3)

Position the seal **A** as shown in the figure; check that the seal is recessed 1.5 mm relative to the face of the hub.

Aligning fan pulley to engine pulley (Fig. 1 and Fig. 2).

Both pulleys should be thoroughly aligned. Fit a straight rod on the bigger pulley face and ensure it is aligned to that of the other pulley.
If not, remove or add shims to the crankshaft pulley in the position indicated by the arrow **B** (see figure 5).
Any extra shims are to be positioned under the heads of the pulley retaining bolts.

Aligning alternator pulley to engine pulley (Fig. 5)

Fit the alternator and measure the amount of misalignment using a straightedge across the face of the crankshaft pulley.
Fit shims of this value between the pulley and the support in position **A** and **C**.

Aligning compressor air conditioning pulley to engine pulley

Fit the compressor air conditioning on the he suport and measure the amount of misalignment using a straightedge across the face of the crankshaft pulley.
Fit shims of this value between the pulley and the support until the obtain the aligning.

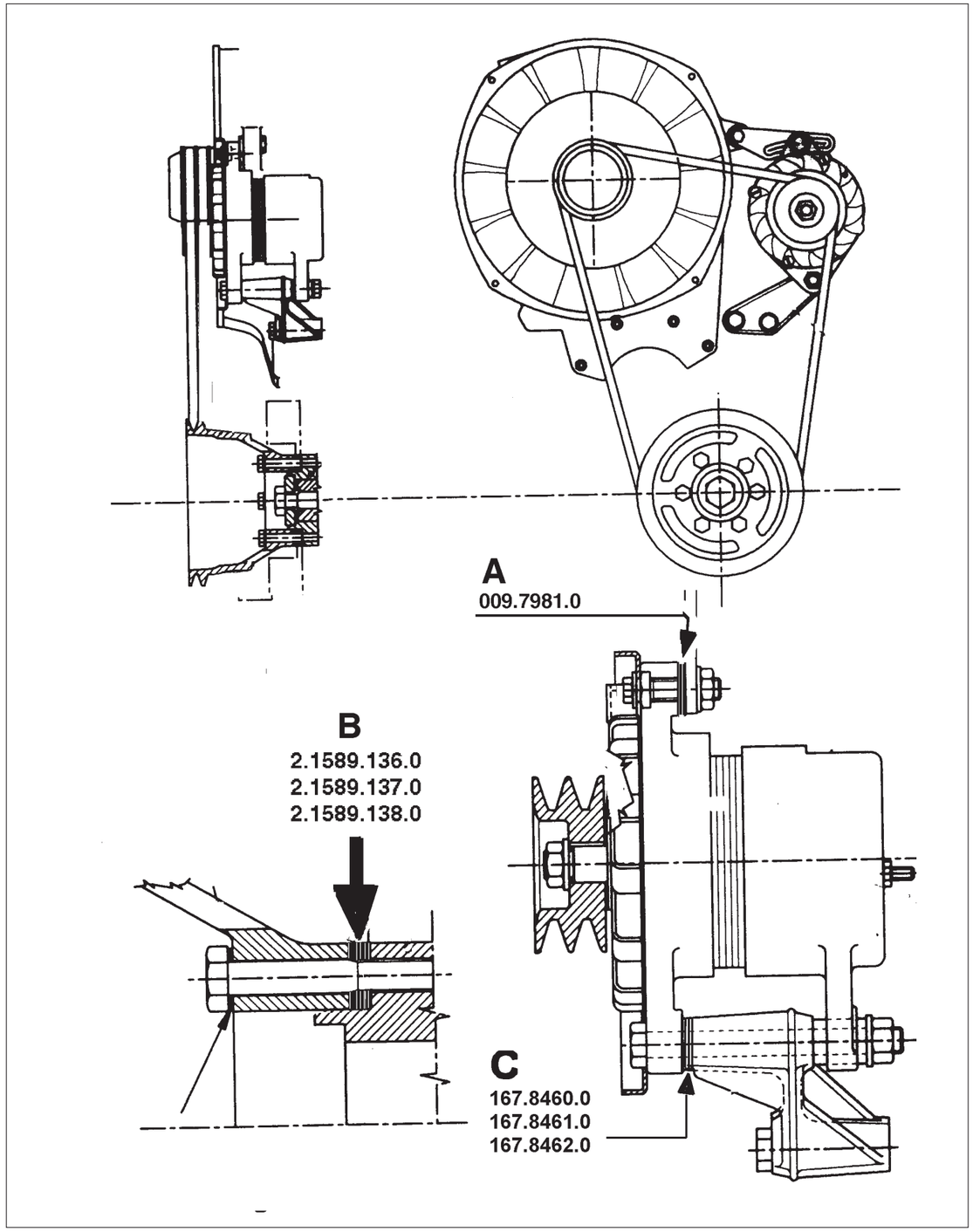
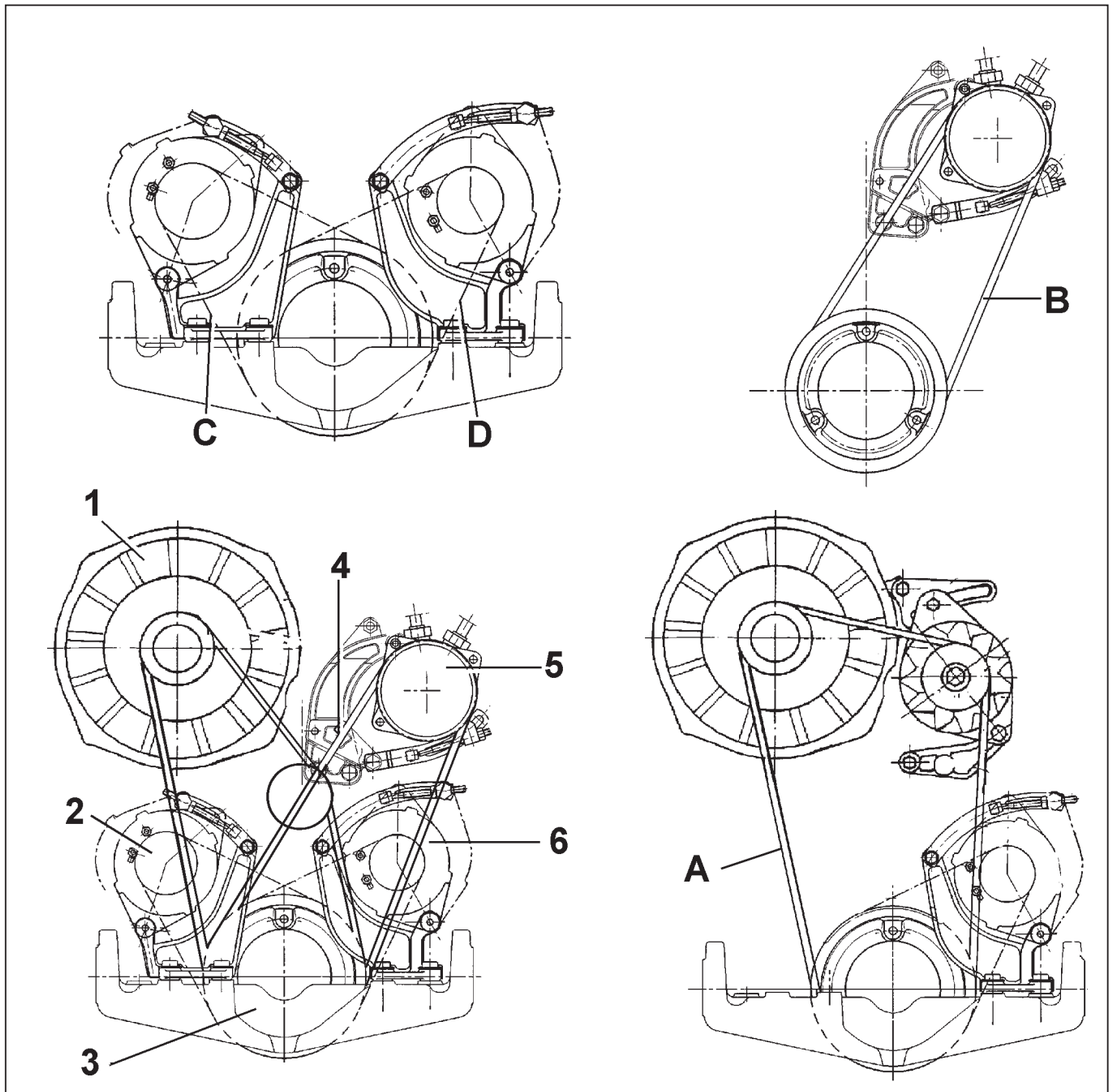


Fig. 5 - Installing the alternator pulley tightener and aligning engine, fan, alternator pulleys.

ADJUSTING OF DRIVE BELT TENSION FOR ENGINE 1000-A

Adjusting the belt tension turning screws the regolation, operate as follows:
Turn the adjusting screws until to obtain the belt settling **A** - **B**, exerting finger pressure to the middle of the longest section; the belt should deflect approximately 20 mm and 15 mm for belts with reference **C - D.**

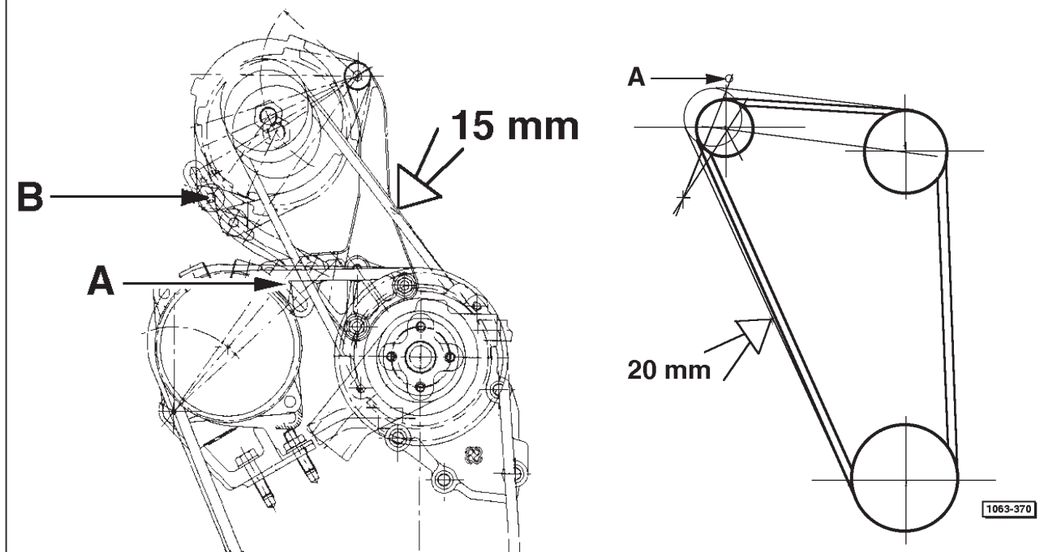


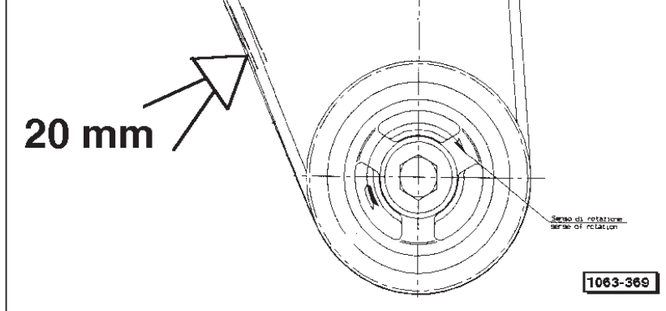
Front view of engine
(tractors equipped with heating and air conditioning)
1 - Engine fan unit
2 - Engine alternator
3 - Crankshaft pulley
4 - Belt tensioners
5 - Air conditioning system compressor
6 - Heating system alternator

ADJUSTING OF DRIVE BELT TENSION FOR ENGINE 1000-W

To adjust the belt tension operate as follows:
Tractor without air-conditioning system: loose the screw A (see on the picture below) and move the alternator in order to obtain a belt deflection of 20 mm approx, by exerting a finger pressure to the point of the longest section as indicated by the arrow.
NOTE - check with the hand on the middling point of the length tug of the belt.
Tractor with air-conditioning system: loose the screw A (see on the picture below) and move the compressor in order to obtain a belt deflection of 20 mm approx, by exerting a finger pressure to the point of the longest section,
For the second belt, loose the screw B and move the alternator in order to obtain a belt deflection of 15 mm as indicated by the arrow. approx, by exerting a finger pressure to the point of the longest section, see the picture.

VERSION WITHOUT AIR-CONDITIONING SYSTEM





VERSION WITH AIR-CONDITIONING SYSTEM

Water radiator (only for engine 1000-W)

Removal:

detach the water hoses;
detach the fixing brackes of the upper tank to radiator;
separate the air conveying wall;
remove radiator from the upper fixing bracket;
loosen both scews securing the radiator to the engine front support through the Silent-blocks.
remove the oil radiator front connected to water radiator.
Make sure the radiator is perfectly efficient by checking for obstructions both internally and externally.

COOLANT AKROS FREEZE

It consists of an antifreeze solution which, besides preventing freezing at temperatures as low as -39°C, also has anti-oxidizing, anti-corrosive, anti-foaming and anti-scaling properties.

Every 1200 hours (or every 2 years):

Replace the antifreeze solution.
Proceed as follows:
Empty the cooling circuit;
— Fill the circuit with AKROS FREEZE, let the engine run for a few minutes and then replenish.
— The quantity of anti-freeze solution contained inside the cooling circuit is 11 litres

ATTENTION - If the tractor cab is equipped with a heating system, the quantity of anti-freeze solution must be increased by 2 litres.

It's mandatory to use the antifreeze AKROS FREEZE; this antifreeze is supplied by our Part Departement.

Fan

Fan is dynamically and statically balanced to prevent vibrations which may result from rotation (about 45% higher than engine r.p.m.). It is installed on the pump and secured with four fixing screws.

Driving belts

Belt should be replaced as soon as it is suspected not to function properly. No engine part needs to be dismounted for that purpose:
loosen the the bolt securing the alternator to the belt tightener and then remove the belt fit the new belt and tighten it.

Thermostat

It is placed on the coupling hose at the end of the water manifold.
Removal is performed by simply detaching the rubber couplings.
After taking the cover, access to the thermostat is gained.
The thermostat has fixed setting, so no further adjustment is required.

Whenever it is disassembled, all gaskets should be replaced and any limestone deposits removed.

Aligning fan pulley to engine pulley. Both pulleys should be thoroughly aligned.
Fit a staight rod on the bigger pulley face and ensure it is flush with that of the other pulley.
If this is not the case, remove or add shims to the fan pulley as indicated in figure 3/a.

Aligning alternator pulley to engine pulley

Install the alternator and mesure misalignment applying a straight bar to the engine pulley face; fit shims between alternator and support (corresponding to the value measured) in the positions indicated in figure 3/a.

Engine water pump

The water pump is centrifugal type with an impeller.
The pump impeller is driven by a shaft supported by bearings and housed within the pump casing. A special sealing ring prevents water from seeping in.

Water pump assembly

Flange **A** and impeller **B** should be coupled after assembling all the items composing the unit. Be sure that between impeller **B** and pump casing **C** a 0,16 0,16÷1,15 mm clearance be provided.
Between flange **A** and impeller shaft apply some Loctite 601.
Fit the sealing ring **D** into pump casing **A** so that a 8,89 mm dimension may be restored, as shown in figure 4/a.

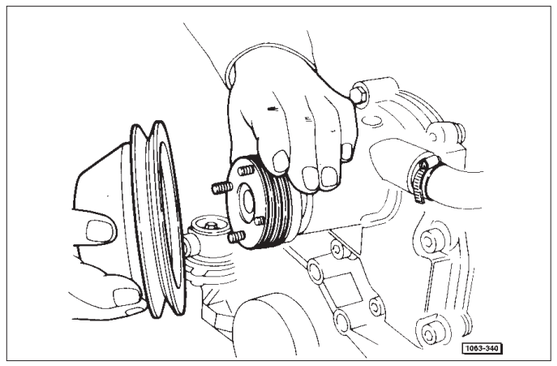


Fig - 1/a - Shimming up the fan pulley.

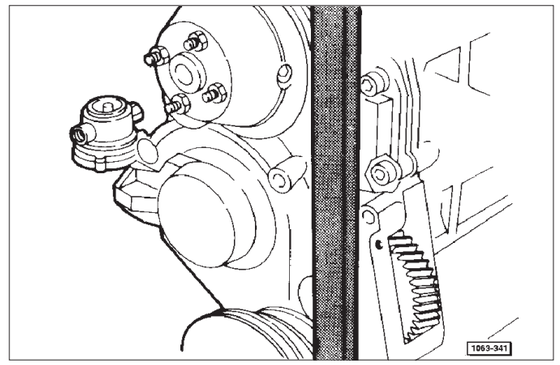


Fig - 2/a - Shimming up the fan pulley.

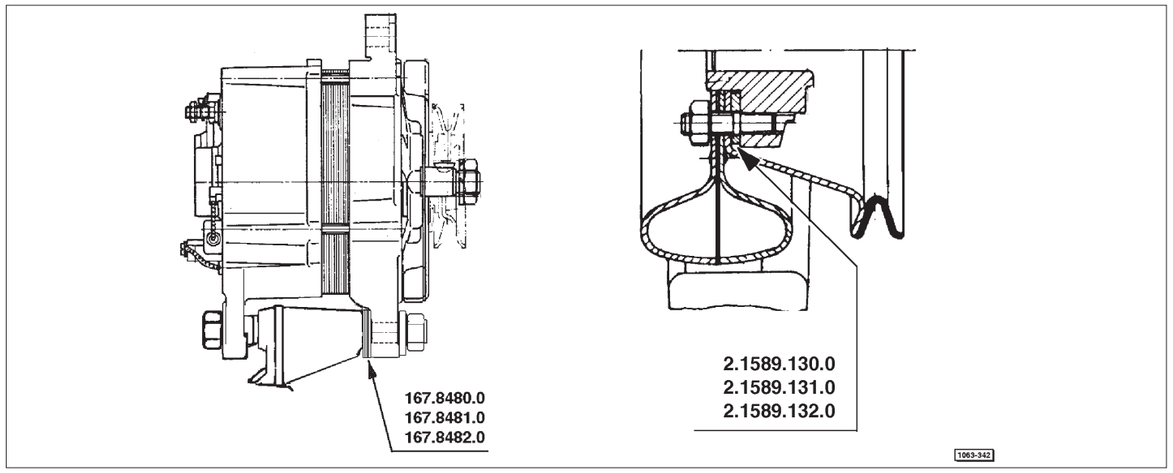


Fig. 3/a - Shimming up the fan pulley.

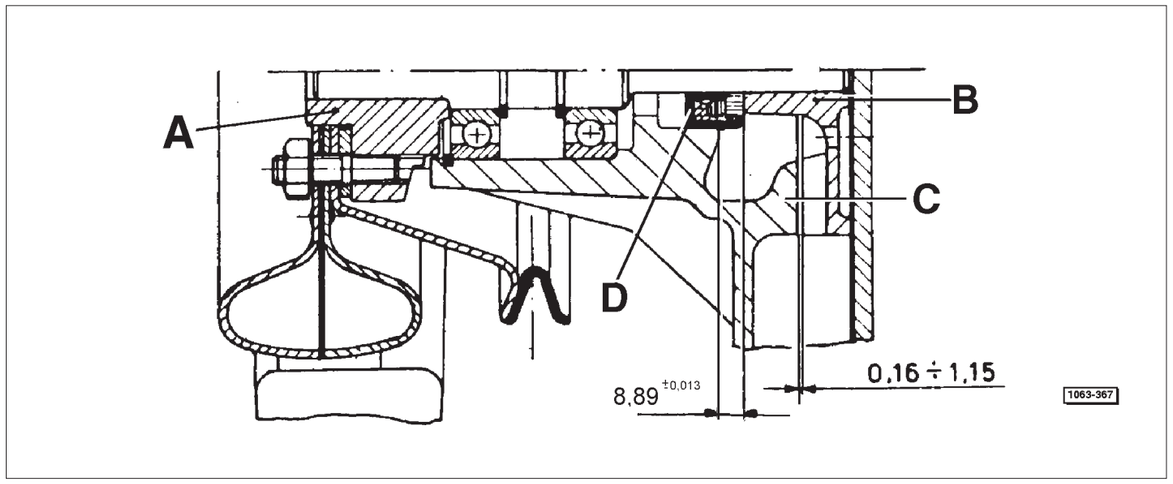


Fig. 4/a - Eater pump unit assembly.

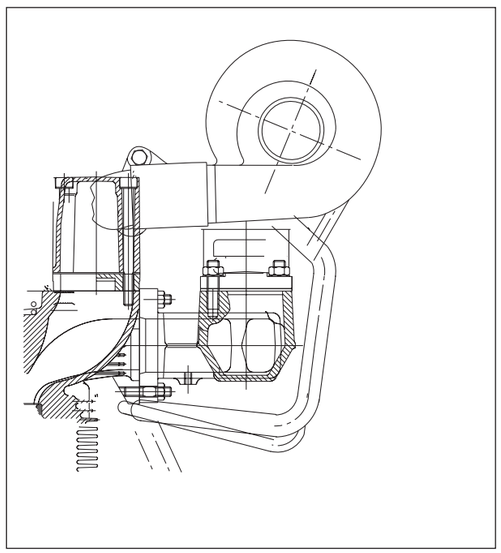


Fig.1 - Turbocharger.

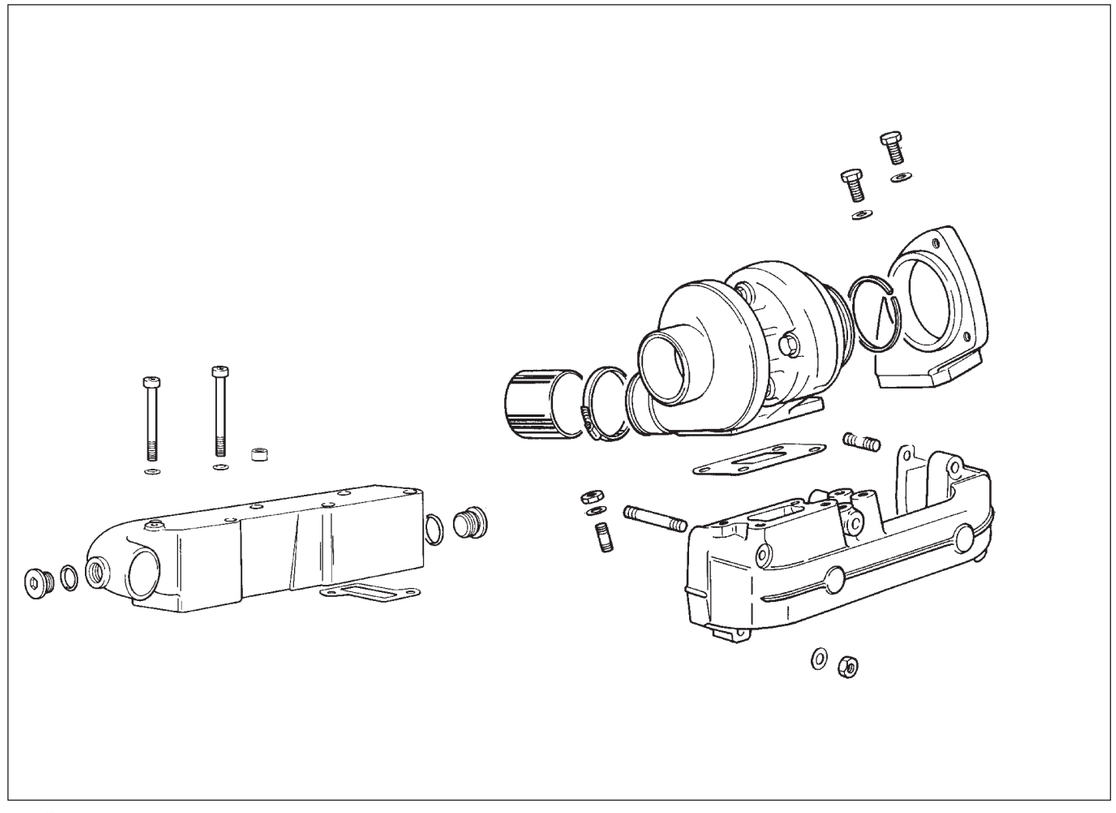


Fig. 2 - Turbocharger and exhaust manifolds

Supercharging 770s 990s

The turbocharger assembly covers the function of forcing more air into the cylinders. This also allows the diesel fuel delivery capacity to be proportionally increased.

Mixture increase is necessary to obtain greater engine power.

The turbocharger assembly is an extremely simple and rational design and consists of a turbine and a compressor.

The turbine is operated by the exhaust gases from the engine (in this way, a part of the gases' kinetic energy which otherwise would be wasted is recovered). The turbine drives the compressor, which compresses the air sucked in the cylinder intake duct through the filter.

Tests

A precise test on turbocharger efficiency can only be performed on a test bed.

Tests performed with turbocharger mounted on engine can only give approximate results.

Before starting checking operations make sure the engine fuel supply system is operating correctly.

Checking pressure in the duct at turbocharger exit (Fig. 3).

Remove the intake pipe plug and fit no. 5.9030.515.0 pressure gauge.

Start the engine and ensure that both peak horsepower speed and maximum torque speed pressures correspond to the specifications indicated in related table to page 24.

Out of specifications readings may be due to poor air filter efficiency which should therefore be checked.

Also check couplings for distorsion or tearing; replace as necessary.

If trouble persists, remove turbocharger side covers and check impeller blades for dirt build-up.

All carbon deposits should be removed using a plastic brush.

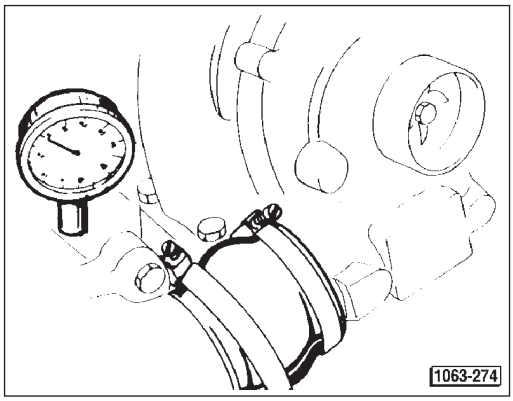


Fig. 3 - Checking pressure inside the intake pipe.

Checking exhaust pipes

Be sure the exhaust pipes enable the exhaust gases to be properly expelled; otherwise remove any carbon deposits.

If after performing the above checks the turbocharger is not operating smoothly a thorough overhaul shall be carried out.

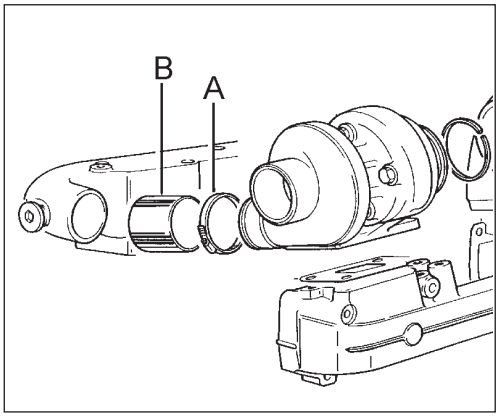


Fig. 4 - Manifolds-to-turbocharger connecting couplings.

Removal (Fig. 4 and Fig. 5)

Disconnect the lubricating pipes from cylinder block.

Loosen the pipe-to-turbocharger securing screws and remove together with gaskets.

Ease-off clamps **A** and remove coupling **B** connecting turbocharger to intake manifold, (Fig, 4).

Loosen the two coupling fixing clamps placed between air filter and turbocharger and remove coupling.

Warning: conveniently plug the turbocharger ducts to prevent any foreign matters from entering.

Unloose the 4 screws securing the turbocharger **A** manifolds to exhaust manifold **B** as well as the nuts fastening the muffler support **C** both to turbocharger and exhaust pipe, then remove the turbocharger, (Fig, 5).

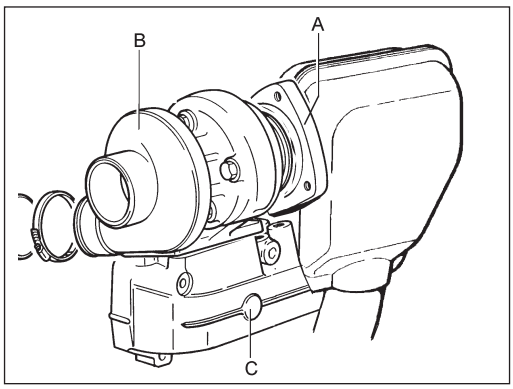


Fig. 5 - Turbocharger separation.

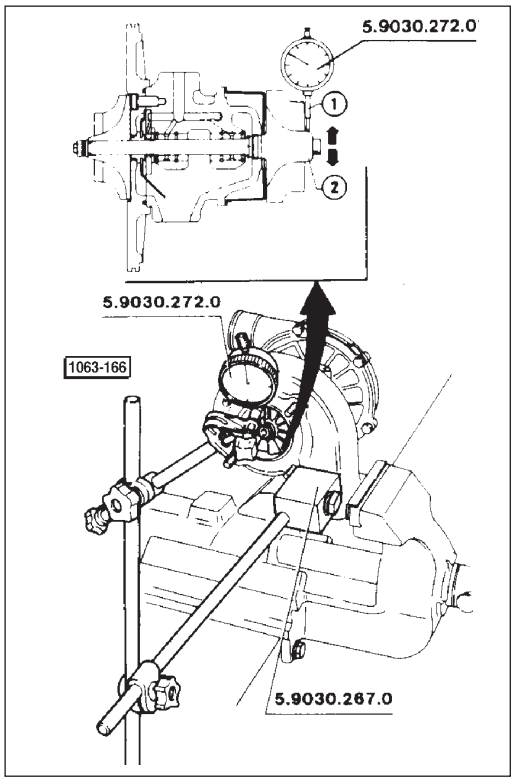


Fig. 6 - Checking rotor shaft radial play.

Checking rotor shaft radial play - turbine side (Fig. 6)

Place a dial gauge through the related magnetic base in such a way that feeler rests on the turbocharger rotor shaft as illustrated in figure 6.

Set dial gauge to zero and press the hub first downward and then upward taking the readings shown by the dial hand.

The difference in readings shall result in the rotor shaft radial play and this shall not exceed 0.46 mm.

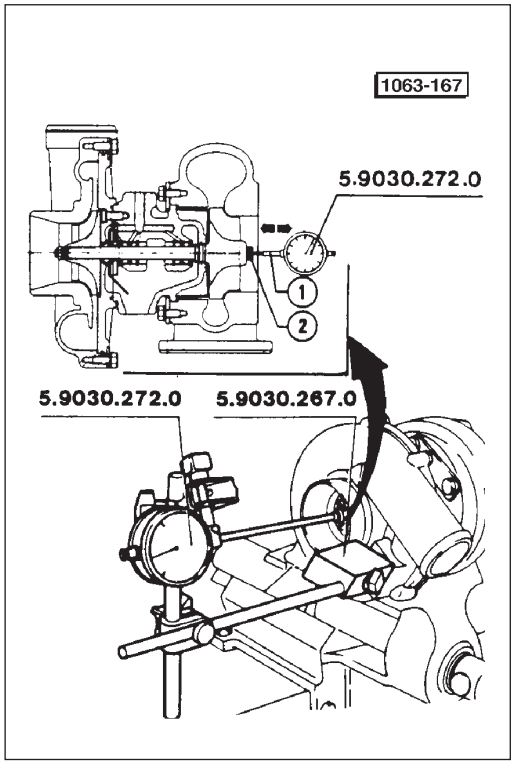


Fig. 7 - Checking rotor shaft end play.

Checking rotor shaft end play (Fig. 7)

Fit the dial gauge through the related magnetic base in such a way that feeler rests on the turbocharger rotor shaft end as illustrated in figure 7.

Set dial gauge to zero and move the shaft first to the right and then to the left taking readings shown by the dial hand.
The difference in both readings shall be the rotor shaft end play and this shall not exceed 0.16 mm.

If the amount of axial or radial play exceeds the prescribed value, remove the turbocharger and take it to a specialised service centre for a general overhaul.

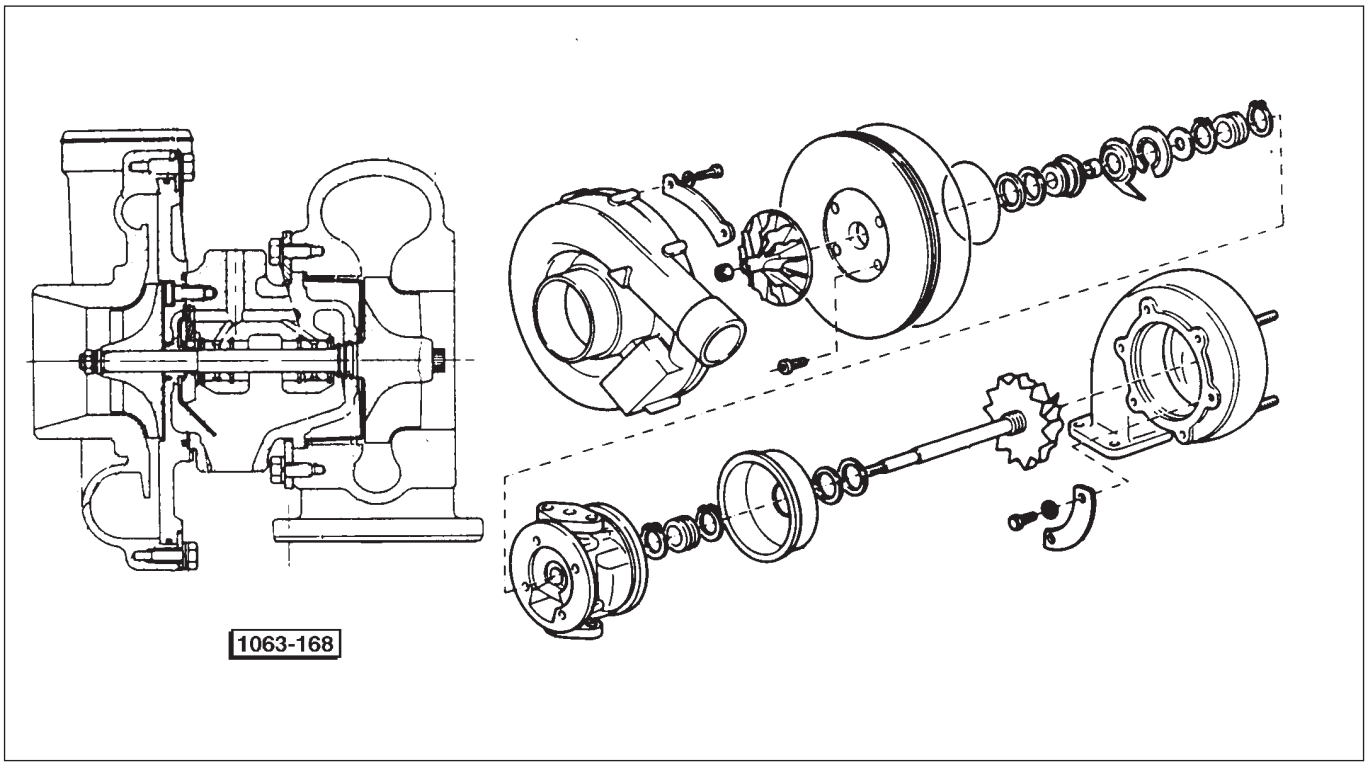


Fig. 8 - Turbocharger assembly components.

Installing turbocharger on engine (Fig. 9)
Secure the turbocharger assembly to its mounting. First fit coupling **A** to turbocharger assembly and next muffler support **B** to the exhaust manifold fixing with the securing screws.
Tighten first support **B** securing nuts and then the coupling securing nuts.
Connect the lubricating pipes.
Warning: Never race a cold engine.
Run the engine just above low idling speed for a minute or two so as to allow the oil to warm up progressively (the turbocharger si lubricated by the engine oil) thus ensuring the rotor to be thoroughly lubricated.
After some hours' heavy work, let it run idling for a few minutes before stopping. This is to prevent the rotor from continuing to rotate through inertia without receiving a sufficient lubrication.
Since the turbocharger under normal working rate rotates at very high speeds (55.000 ÷ 110.000 r.p.m.), even a few seconds' oil lack may lead to irreparable damage.

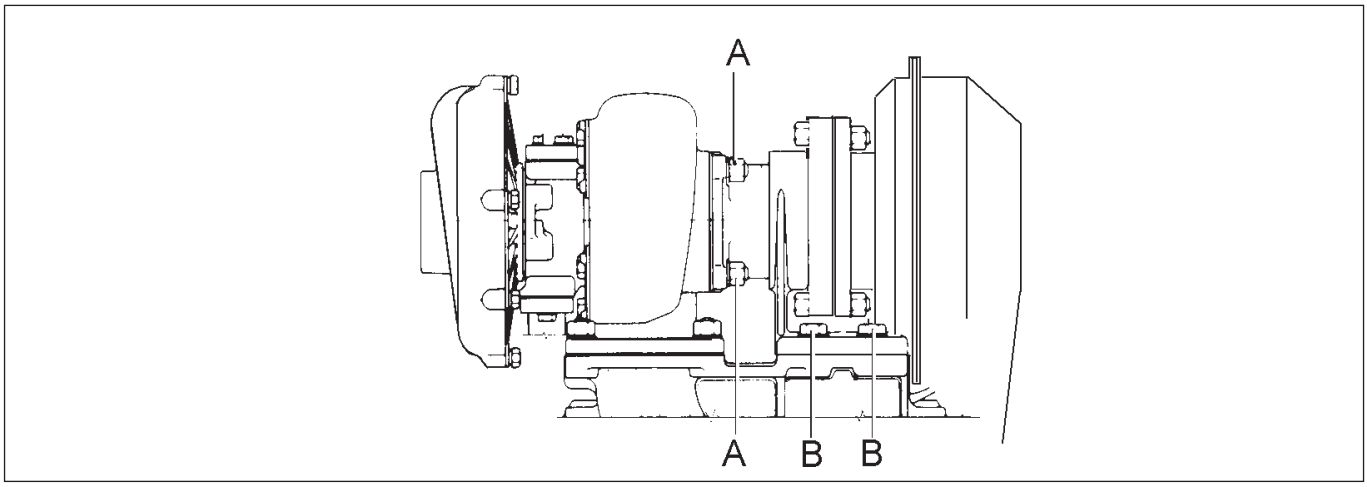


Fig. 9 - Turbocharger assembly on engine.

Engine air filter

If a blocked air filter indicator switch is to be replaced, make sure the new switch to be installed has same setting specifications checking the reference stamped on it.
Aspirated engine blocked air filter indicator switch setting is 0.94 ÷ 0.93 bar.

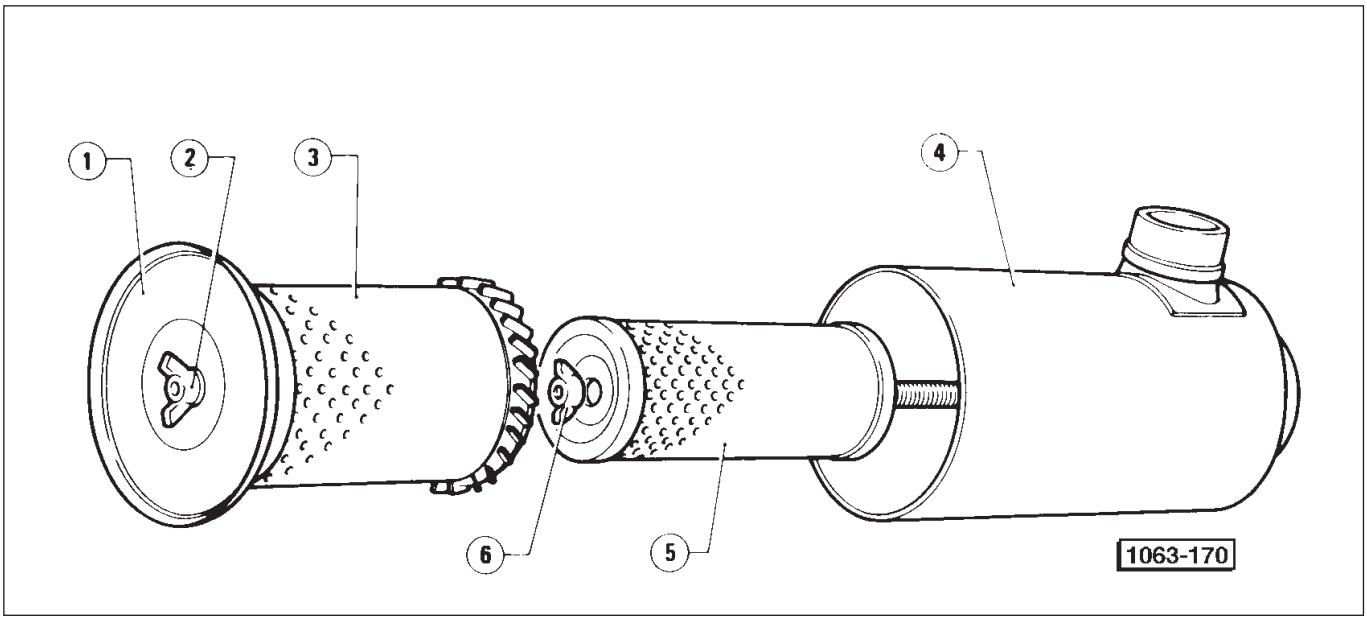


Fig. 10 - Engine air filter.

Tightening torques

Before tightening all screws should be degreased, cleaned and lubricated with engine oil.

	Kgm	Nm
engine flywheel securing screws	11	108
rocker arm support securing screws	4	39
injection nozzle bracket fixing nuts	4	39
inejction pipe fixing nuts	2,5 ÷ 3	24 ÷ 29
securing bolts for pressed steel rocker cover	0,5	5
counterweight housing securing screws (4-cylinder engines)	8	78
front pulley securing screws	34 ÷ 36	315 ÷350
oil pump securing screws	4	39
oil pan securing screws	3	30
rear oil seal cover retaining bolts	2,5	24,5
front casing retaining bolts	2,5	24,5
main bearing cap bolts	see page 36	
big-end bearing cap bolts	see page 38	
tightening the cylinder head bolts	see page 53	
water outlet manifold securing bolts	3	29
thermostat cap fixing screws	3	29
thermostat cap fixing screws	3	29
flange of water pump securing screw	3	29
water-pump-to block securing screws	5	49
engine flange retaining bolts	12	120

Engine air filter

Warning
It is important that the warning light, which shall be lit when the air filter is clogged, is always working efficiently. This check can be carried out by starting the engine and briefly blocking the air intake pipe by means of a plug (never touch it with your hands).

The filter cartridge should be thoroughly clean, if not, it shall be cleaned or washed.

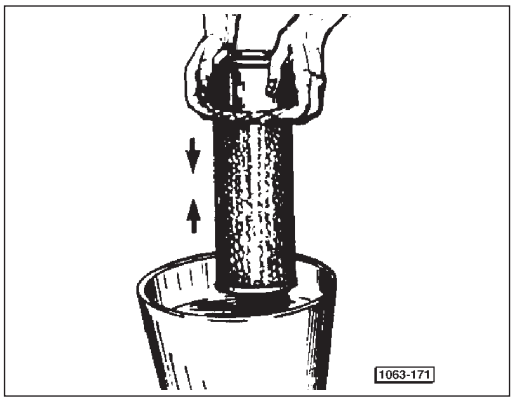


Fig. 11 - Washing the filter cartridge.

Cleaning with compressed air

This method is used when the filter cartridge is clogged with dust.

Direct a compressed air jet at a maximum pressure of 7 bar from the inside outwards of the filter cartridge, afterwards blow the air over the whole surface by moving the air jet along the foldings in the paper until all dust has been removed.

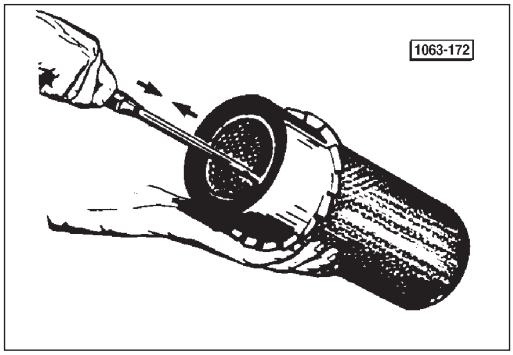


Fig. 12 - Cleaning the filter cartridge with compressed air.

Washing with water

This method is particularly recommended when the filter cartridge is obstructed by oily substances and consists of washing the filter cartridge in clean water.

Rinse the cartridge with a jet of water directed from the inside outwards at a maximum pressure of 3 bar.

Let the cartridge dry 24 hours at room temperature in a dust-free environment.

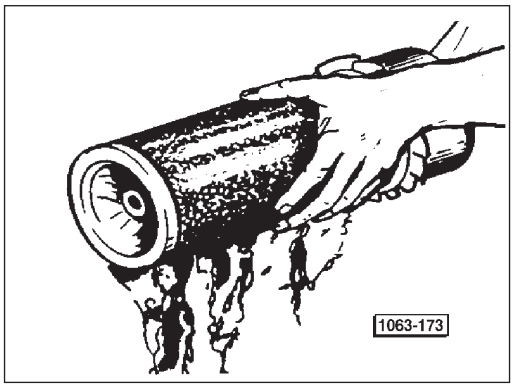


Fig. 13 - Using a water jet to remove any residual left on the filter cartridge.

Reassembling the cartridge

Clean the inside of the filter body with a dry cloth and make sure that the cartridge has not become deformed in any way.
Also check that the paper of the filtering cartridge is not torn by introducing a light source into the cartridge interior and visually inspecting its outside surface to see if any light shows through. If no light source is available, carefully examine the external surface of the cartridge folded paper.
Should any holes or tears be found, regardless of their size, the filter cartridge should be replaced at once.

The inner safety cartridge should never be cleaned but only replaced every six times the main cartridge is cleaned.

Fig. 14 - Using a light source to check the filter cartridge folded paper conditions.

Instructions for engine assembly

— Be sure all plugs have been removed, then carefully clean the cylinder block and the lubricating passages with cleaning solvent and afterwards perform drying using compressed air.
— All sealing rings and gaskets should be replaced on assembly.
— The use of proper tools is of prime importance to the correct performance of all mechanical operations.
— Prior to reassembly use some engine oil to lubricate the faces of all components subject to relative motion.
— The use of special grease is needed for some components so as they can be correctly held in position and assembly operations facilitated.
— Pay particular attention to the direction of assembly of some items by directly referring to the exploded views illustrated.

— **WARNING**: Before refitting the engine you must lubricate the following parts:
— cylinders (internal surface);
— pistons;
— crankshaft (main journals and crankpins);
— connecting rods (small end bearing for gudgeon pin);
— valves (stems);
— rocker arms (bushes);
— oil pump and governor drive shaft;
— cam shaft (journals and cams);
— oil pump control rod and bevel gear drive.

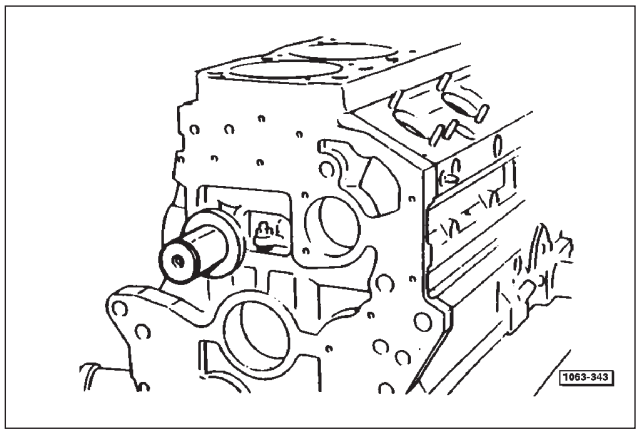


Fig. 1 - Clean the block, refit all plugs and attach it to the engine stand 5 9030.491.4. Apply PIANERMETIC DENSO 510 sealant around the plugs on the rear of the engine. Fit the piston cooling nozzles as indicated on page 22.

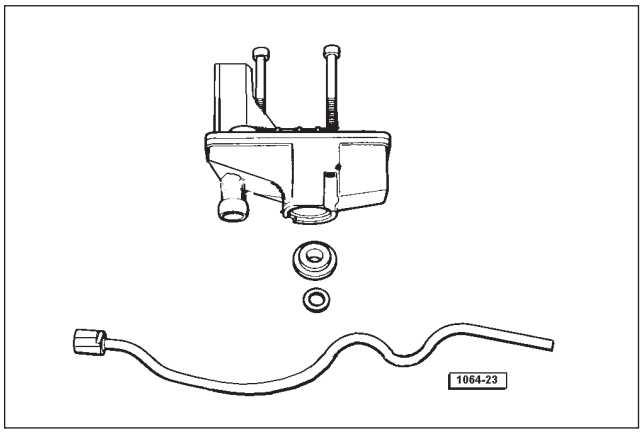


Fig. 2 - Components of the engine oil vapour breather unit;

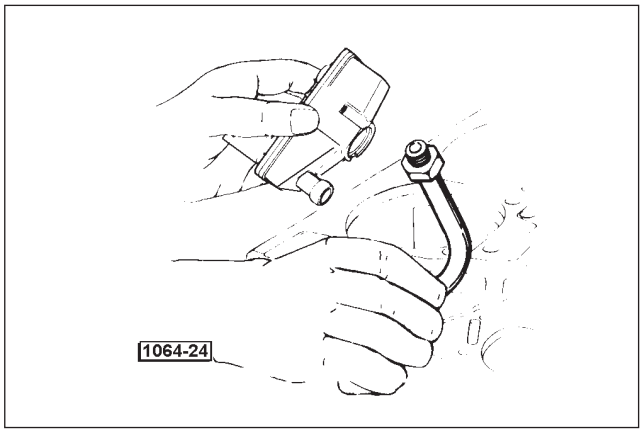


Fig. 3 - Fit the breather unit pipe providing oil return to engine into the cylinder block;

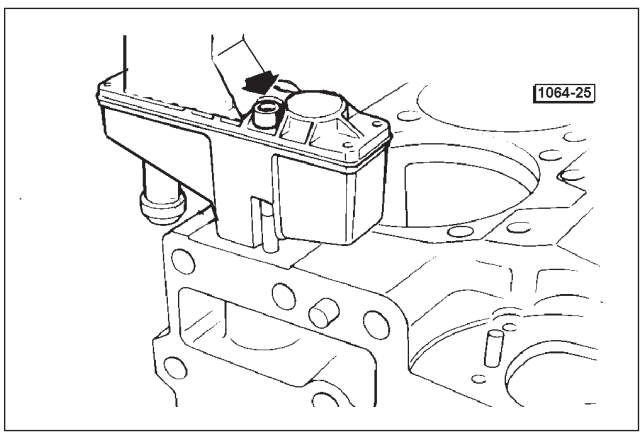


Fig. 4 - Install the oil vapour decantation unit in the engine block using the two securing screws;

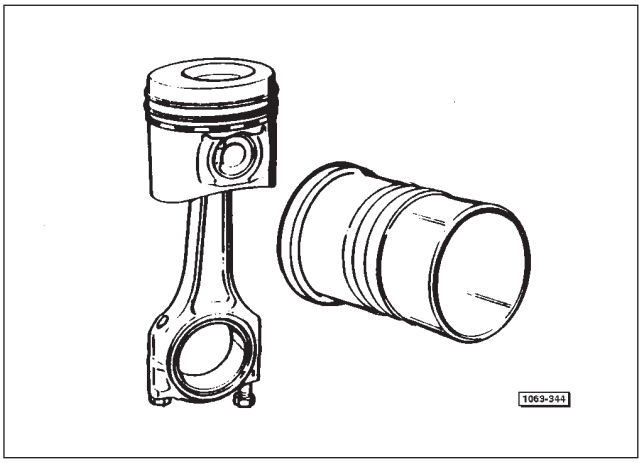


Fig. 5 - Install the piston rings on piston and then connect piston to connecting rod;

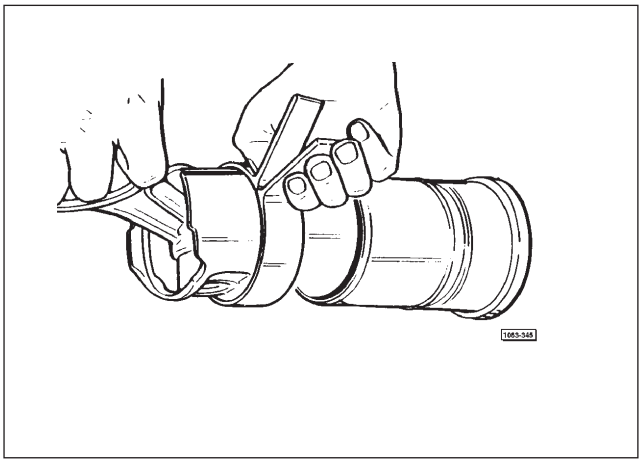


Fig. 6 - Fitting the piston-connecting rod assembly into the cylinder;

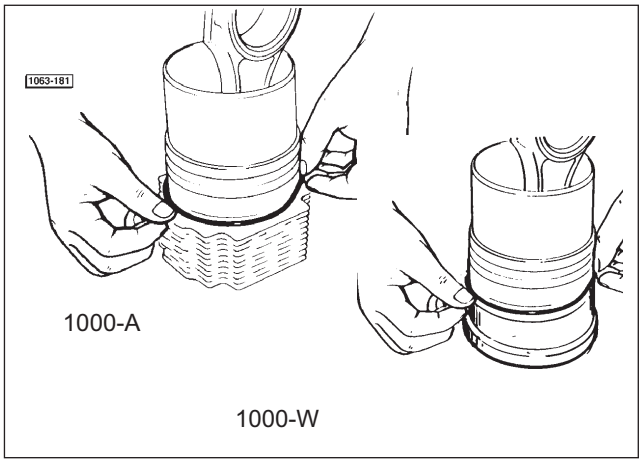


Fig. 7 - Fit the O-rings on the cylinder;

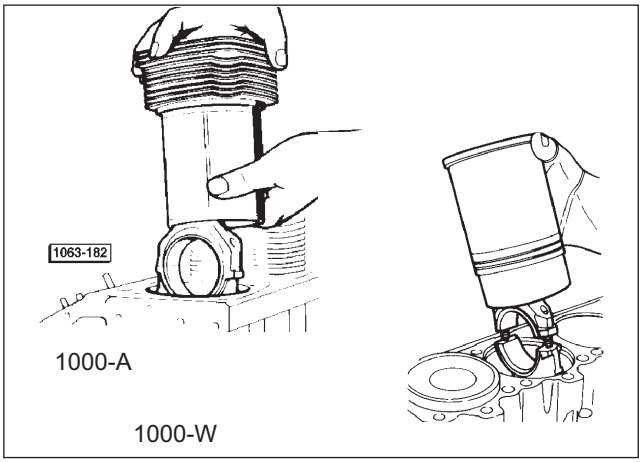


Fig. 8 - Installing the cylinder-piston-connecting rod assembly in the engine block;

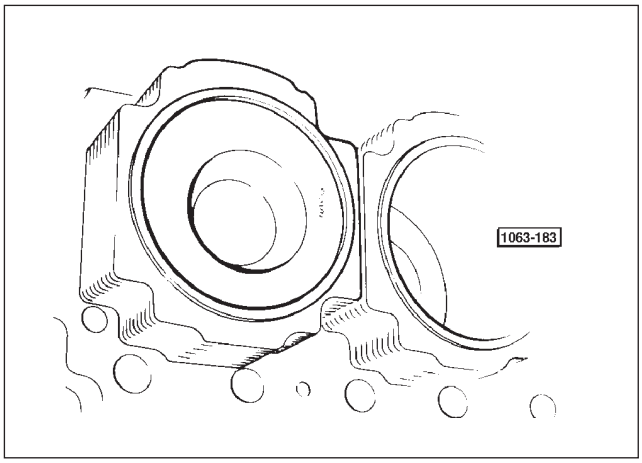


Fig. 9 - Orient the piston combustion chamber towards the injection pump side, secure the piston-connecting rod assembly by means of no. 5.9030.631.4/10 equipment, then turn the engine upside down;

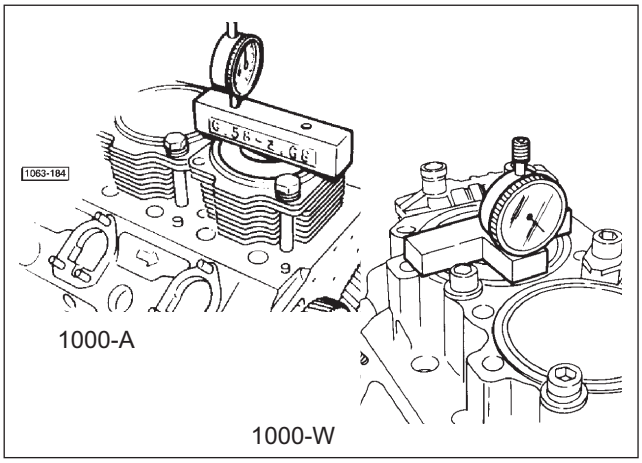


Fig. 10 - Checking for correct cylinder and assembled piston positioning at T.D.C. Measure the piston protrusion from cylinder and select the suited gasket type to be fitted between cylinder and cylinder head as shown on page 31-32.

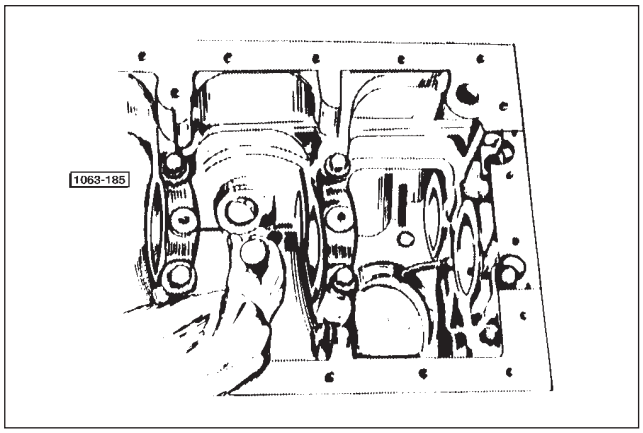


Fig. 11 - Fit the tappet plates into their seats in the block after lubricating with engine oil;

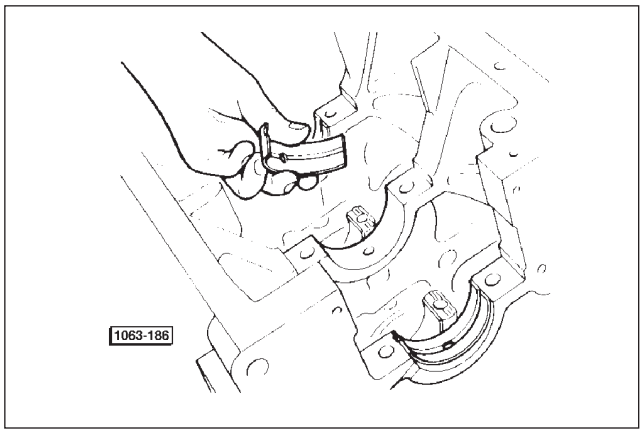


Fig. 12 - Place the upper crankshaft main bearing shells in their seats in the engine block. These main bearing shells are distinguished from the lower shells as they are provided with a lubricating hole;

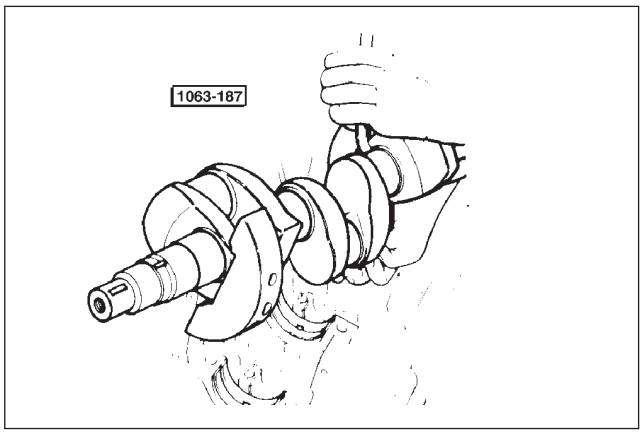


Fig. 13 - Install the crankshaft in the engine block. We recommend using a hoist for accomplishing this operation;

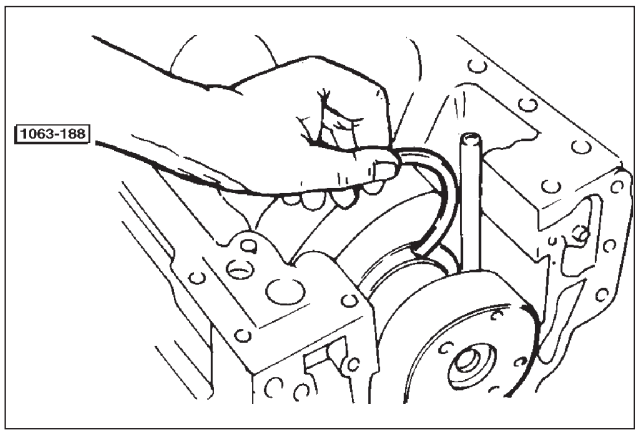


Fig. 14 - Fit the two crankshaft half shoulder rings into the rear block support;

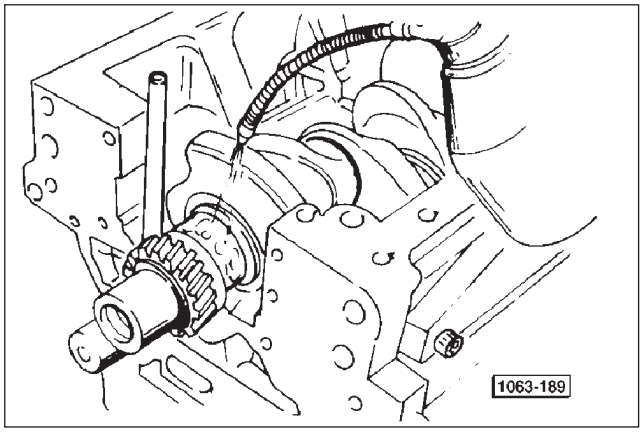


Fig. 15 - Oil the crankshaft main journals and big end journals;

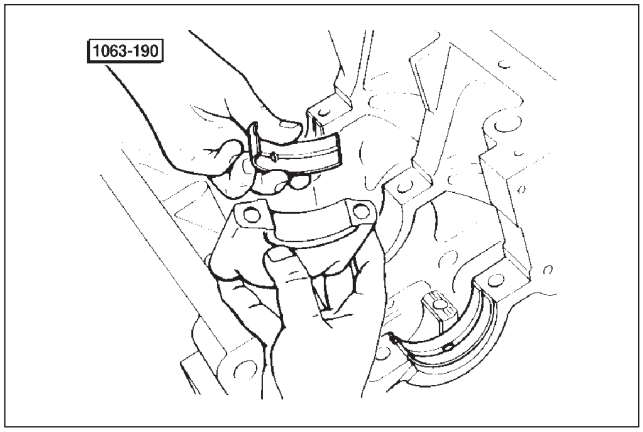


Fig. 16 - Fit the main bearing shells into the crankshaft main bearings. (All main bearing shells are provided with a ratchet. This ratchet should be completely engaged in its seat);

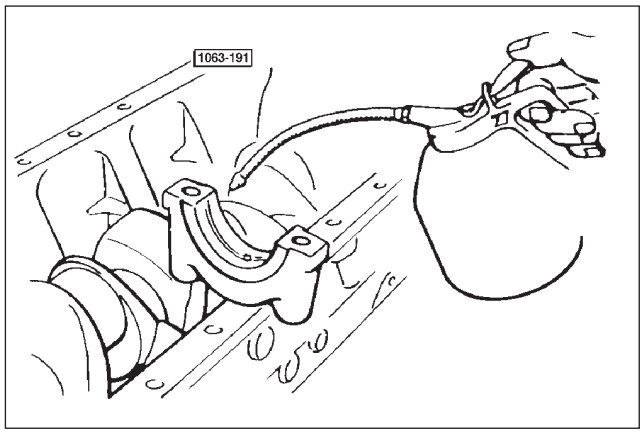


Fig. 17 - Oil the crankshaft main bearing shells;

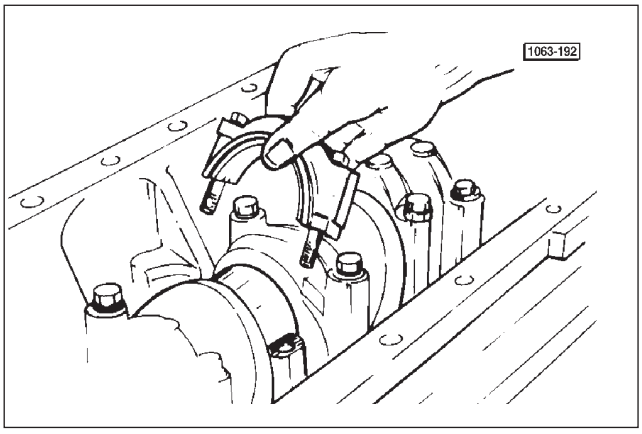


Fig. 18 - Fit the big end cap bearing shells in the related seats and then install the big end caps;

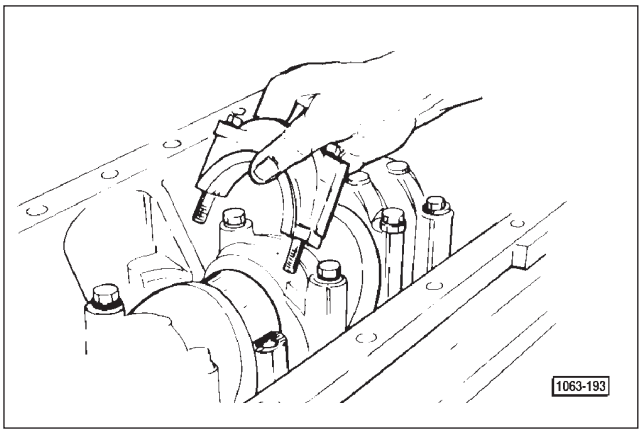


Fig. 19 - Install the intermediate crankshaft main bearing taking care the numbers stamped on the bearing correspond to those stamped on the crankcase. (Numbers should all be oriented in the same direction);

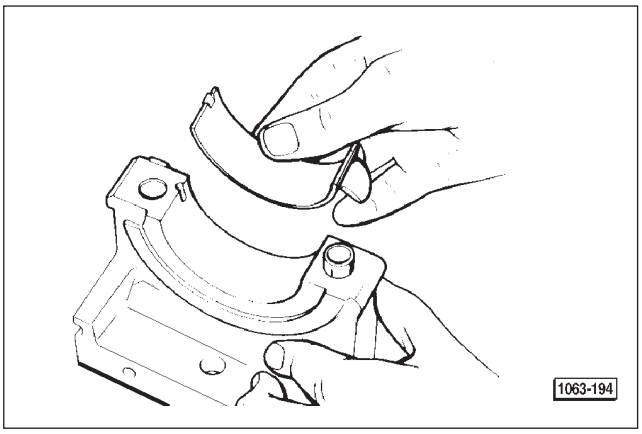


Fig. 20 - Install the main bearing shells in the front and rear crankshaft bearings. Shoulder half rings shall be secured in the rear bearing after applying some grease;

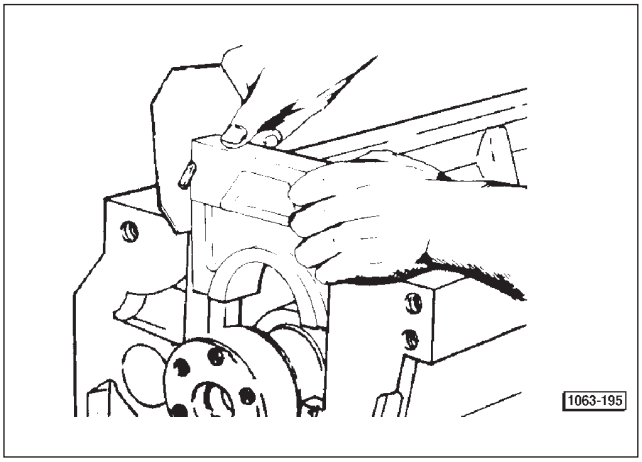


Fig. 21 - Install the gaskets in the front and rear main bearings; then fit bearings easing them into the engine block using sheet metal slides and tapping gently;

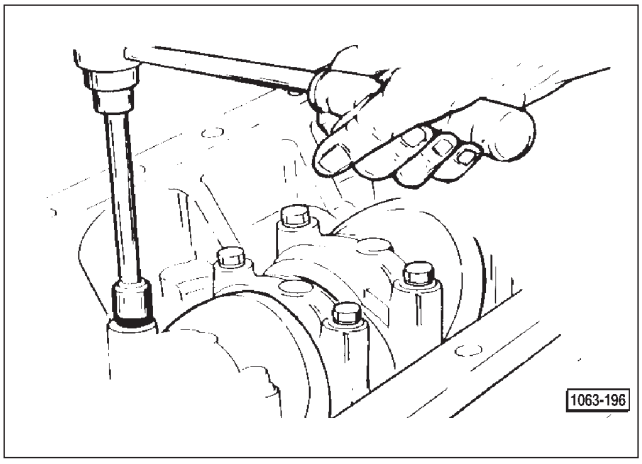


Fig. 22 - The main and big end bearings securing screw threads should be covered with LOCTITE 242. Thereafter tighten the securing screws using tool no. 5.9030.640.0; following the instructions on pages 36 - 38.

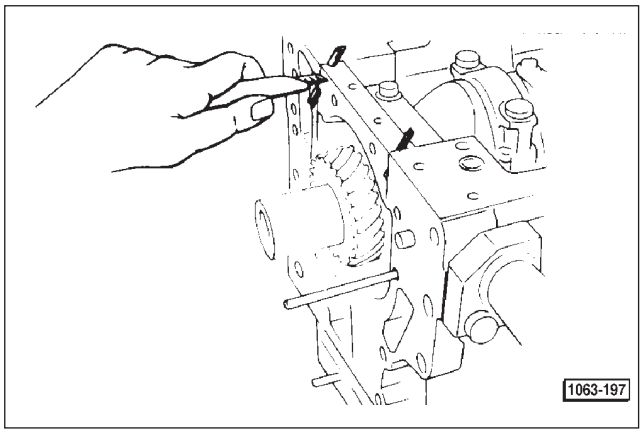


Fig. 23 - Protruding gaskets should be cut flush with the engine block at the front and rear side;

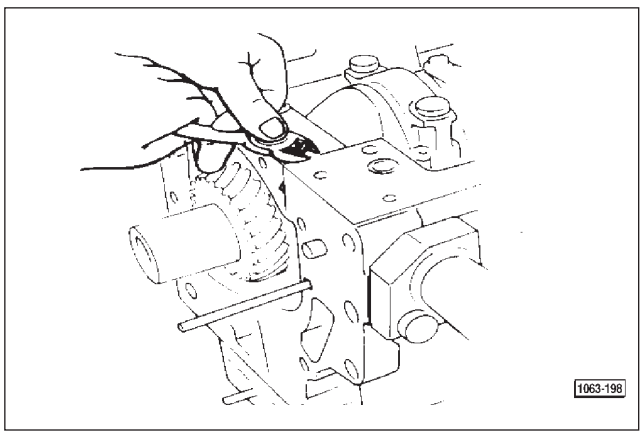


Fig. 24 - Using a suitable tool cut the rubber gaskets protruding from block oil sump face according to specifications given on page 37. Smear some sealant on the rubber gaskets in the block;

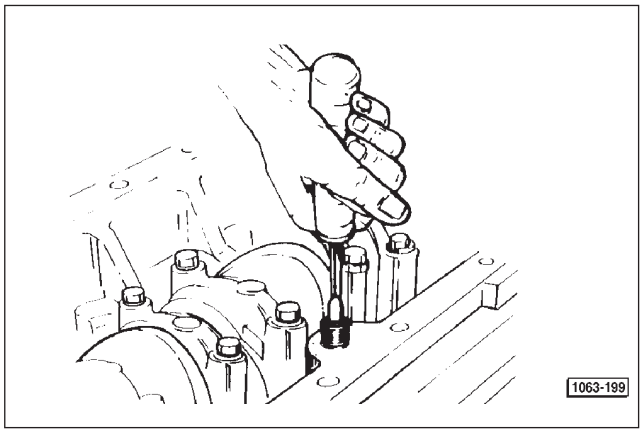


Fig. 25 - Use a screwdriver to fit the screw plug in the position shown in figure;

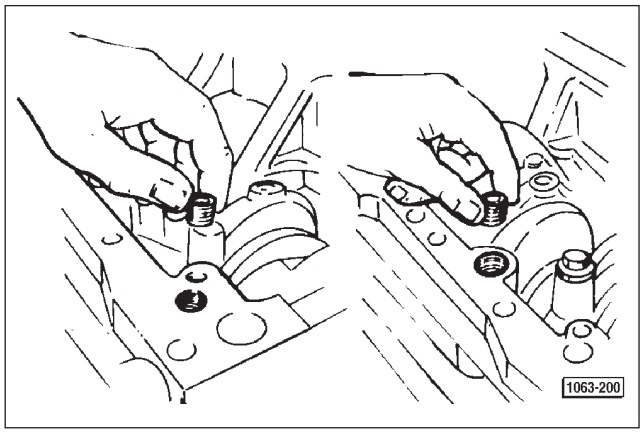


Fig. 26 - Install the socket head screw plugs in the positions shown in figure;

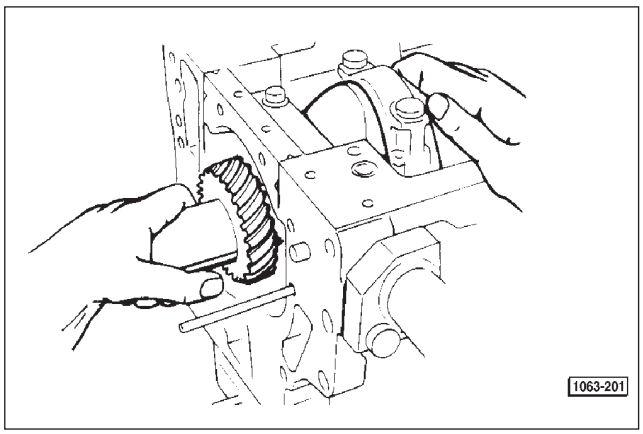


Fig. 27 - Turn the crankshaft and make sure it can rotate freely and no interference between pistons and spray nozzles occur, then measure the crankshaft end play;

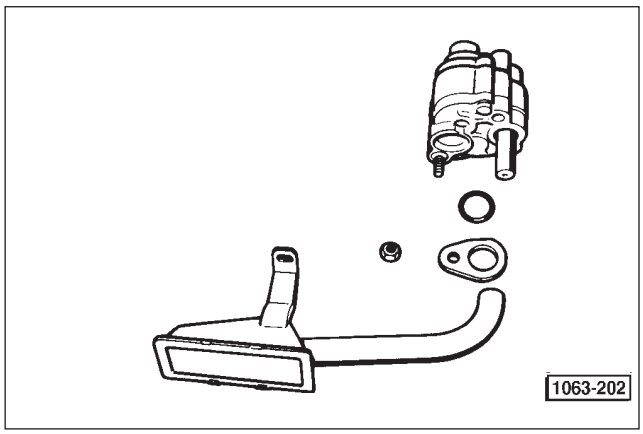


Fig. 28 - Fit the oil pump and after securing to engine block check gears for free sliding. Install the oil pipe rose;

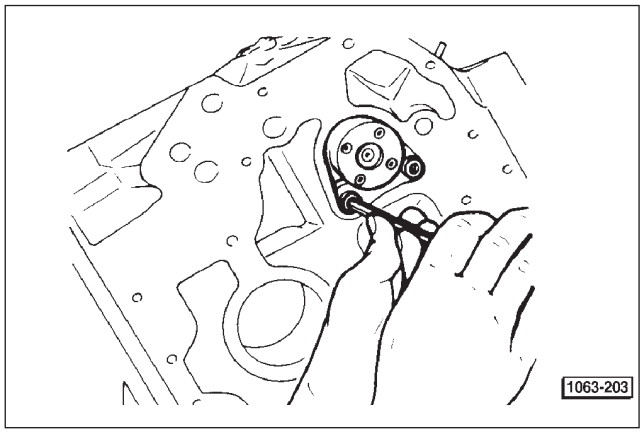


Fig. 29 -Install the camshaft in crankcase. Mount the forked flange permitting the camshaft to be positioned. Make sure the camshaft is free to rotate with a slight amount of end float. **Warning** - If the timing gear has been removed or replaced a new valve timing should be performed;

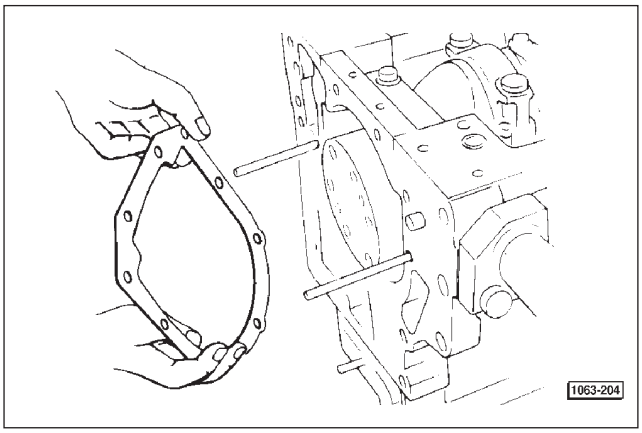


Fig. 30 - Fit the crankshaft rear flange gasket;

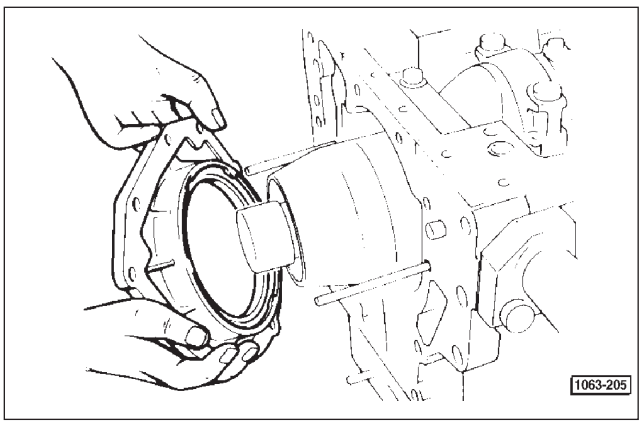


Fig. 31 - Using a guard ring install the flange. Warning - The ring inserted into flange should never be greased. Cover the securing screws with a sealing compound. Interpose the copper washers and tighten to the specified torque as indicated on page 98;

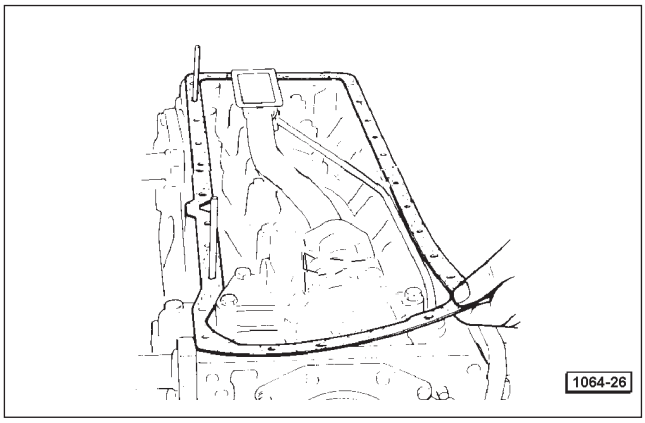


Fig. 32 - Place the engine oil pan gaskets;

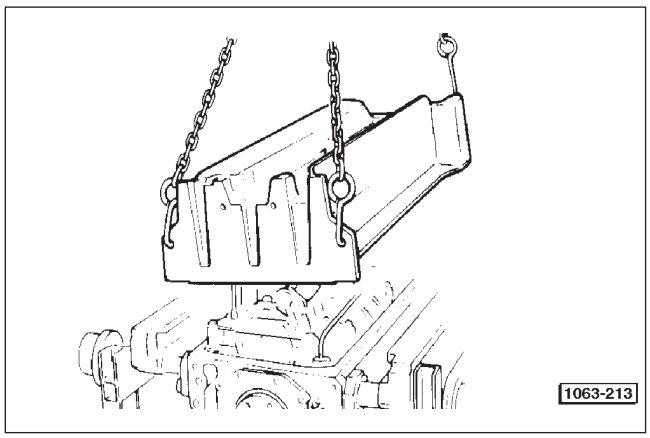


Fig. 33 - Install the engine oil pan;

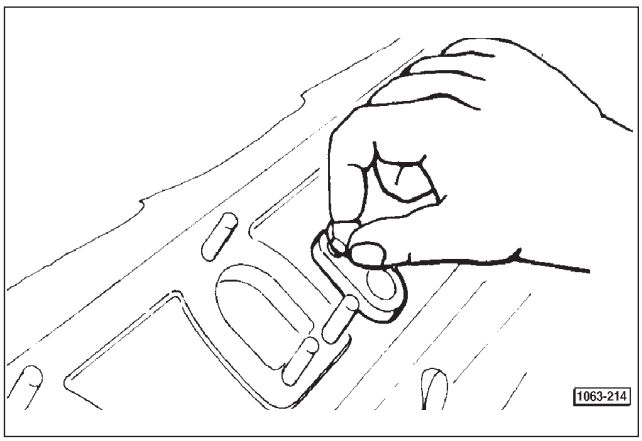


Fig. 34 - Place the injection pump control rod guide supports in the block and lock with the related securing screws;

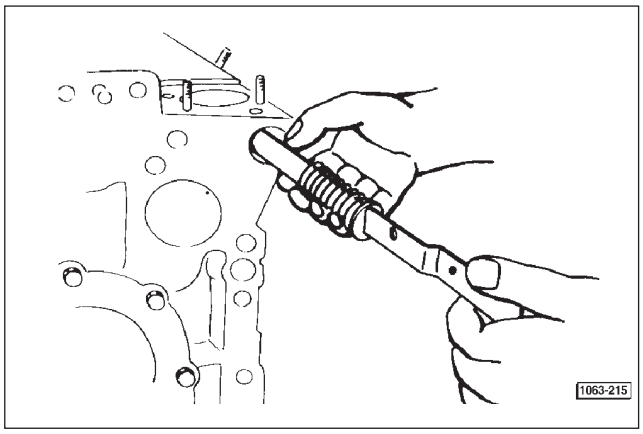


Fig. 35 - Insert the injection pump control rod in the its seat in the cylinder block along with pressure spring and stop cap. Make sure it is free to move inside the guide supports;

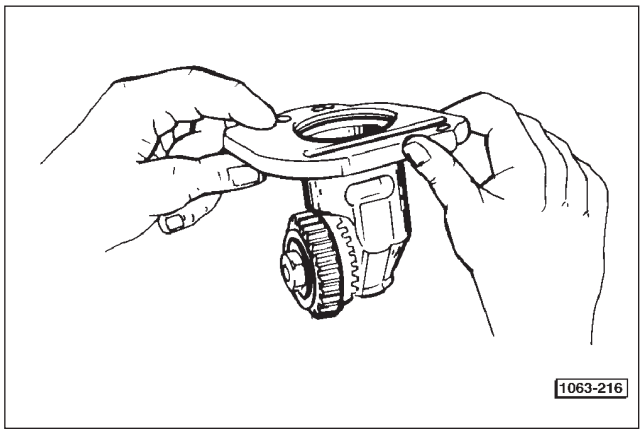


Fig. 36 - ONLY FOR TRAKTORS WITH MECHANICAL GOVERNOR ONLY - Assemble the square transmission controlling both engine governor and oil pump. As to assembly and adjustment procedures see instructions on page 60;

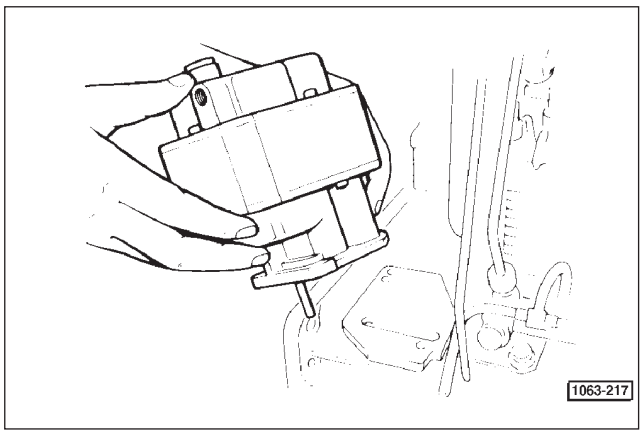


Fig. 37 - FOR TRAKTORS WITH ELECTRONIC GOVERNOR ONLY - After inserting a sealing ring on the stem fit the engine actuator support in the block;

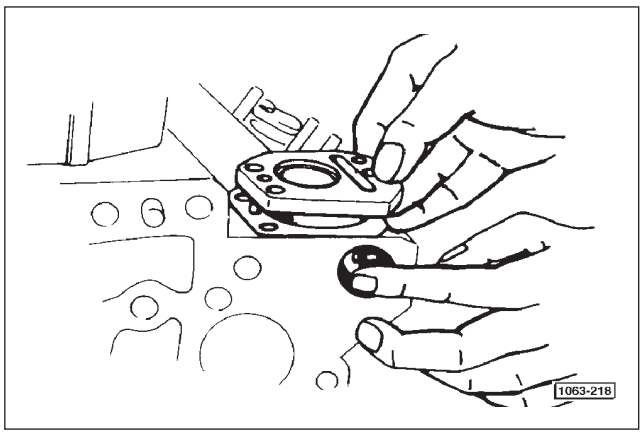


Fig. 38 - FOR TRAKTORS WITH MECHANICAL GOVERNOR ONLY - Install the square transmission in its seat in the block placing the special backlash adjusting shims between the gear teeth (see instructions on page 64);

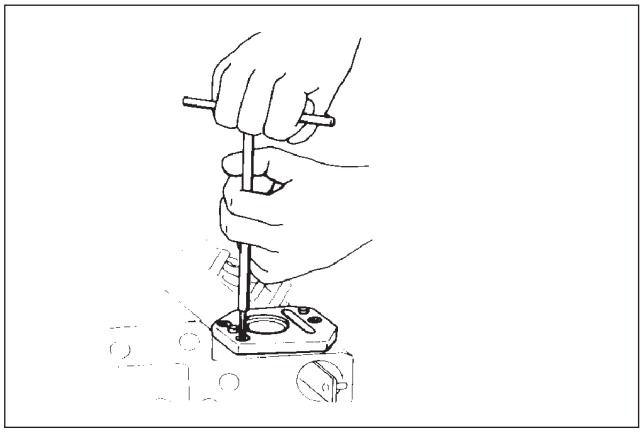


Fig. 39 - Fix the square transmission to the block tightening the related securing screws;

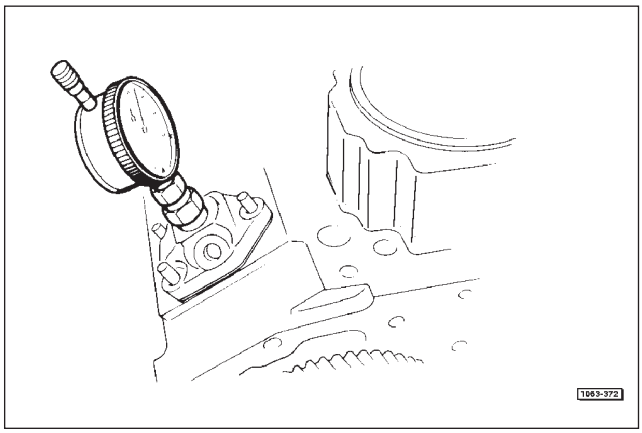


Fig. 40 - Only if necessary, determine the shim pack amounts to be fit between injections pumps and engine block (see instructions on page 78);

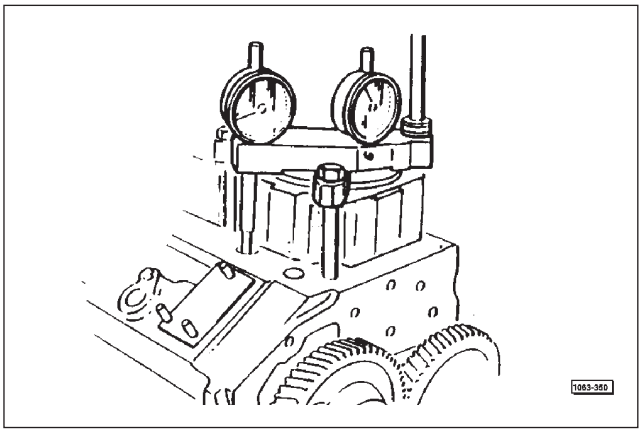


Fig. 41 - If the timing gear has been removed from the shaft carry out valve timing for timing the engine as indicated on page 77;

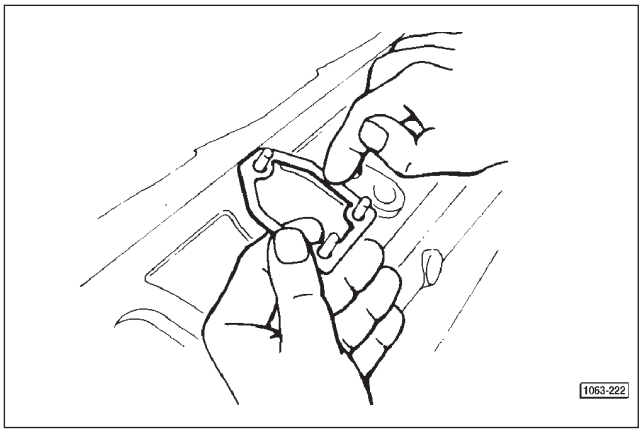


Fig. 42 - After smearing the shim faces with some sealing adhesive place the shim packs between pumps and block. Subsequently perform injection pump installation;

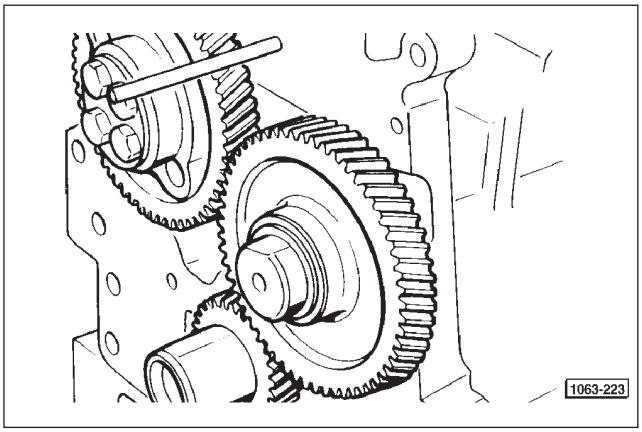


Fig. 43 - After the timing gear has been secured to shaft, fit the idler gear taking care the chiselings on teeth be in the same line. If one of these three gears must be replaced, valve timing procedure shall be repeated;

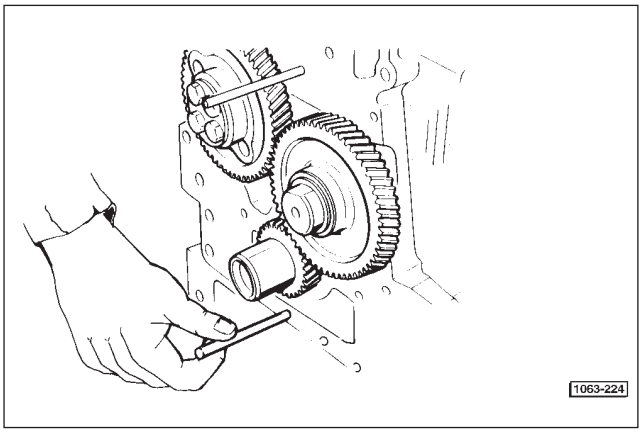


Fig. 44 - Fit some locating pins or studs to centre the timing case correctly during assembly;

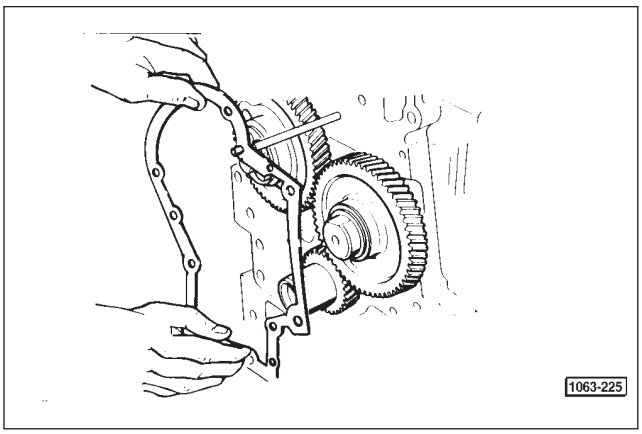


Fig. 45 - Place the isogene gasket between engine block and timing case;

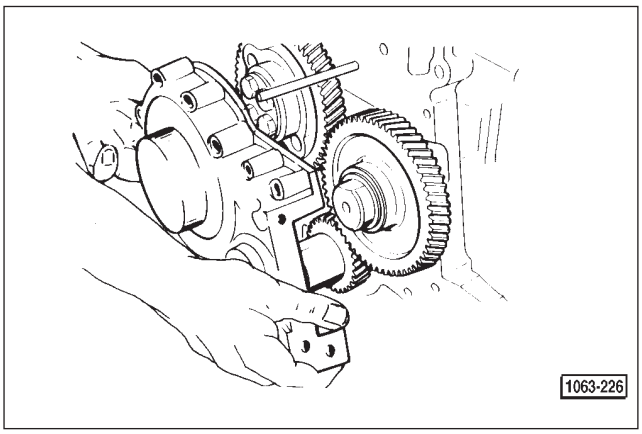


Fig. 46 - The timing case should be installed paying attention not to damage the sealing ring lip fitted inside;

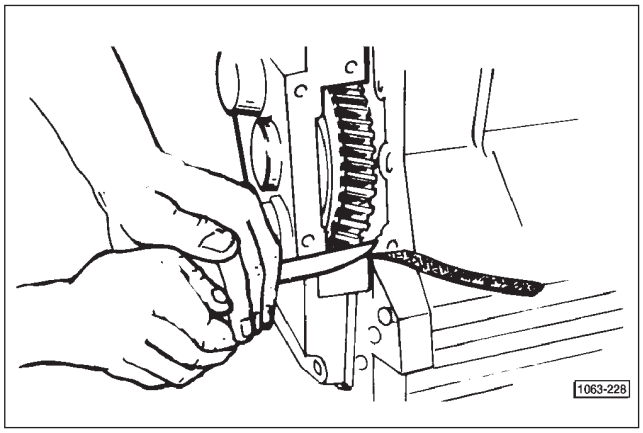


Fig. 47 - Remove excess gasket from the face used to fix external applications;

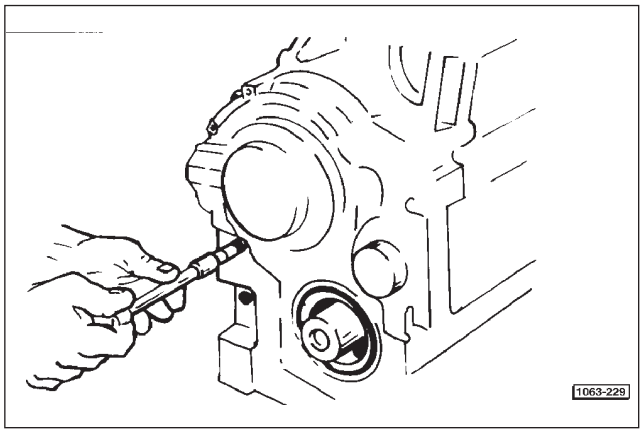


Fig. 48 - Insert all timing case securing screws but do not tighten fully;

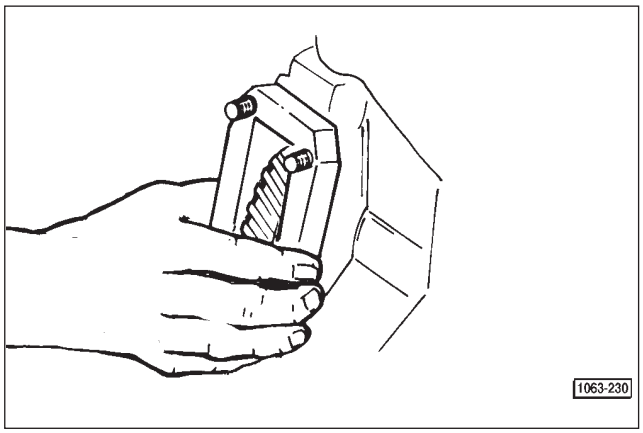


Fig. 49 - Fit no. 5.9030.634.0 tool as shown in figure to obtain thorough concentricity between sealing ring and crankshaft as well as a full coplanarity of the external application fixing faces, then fully tighten the timing case securing screws. Remove the tool;

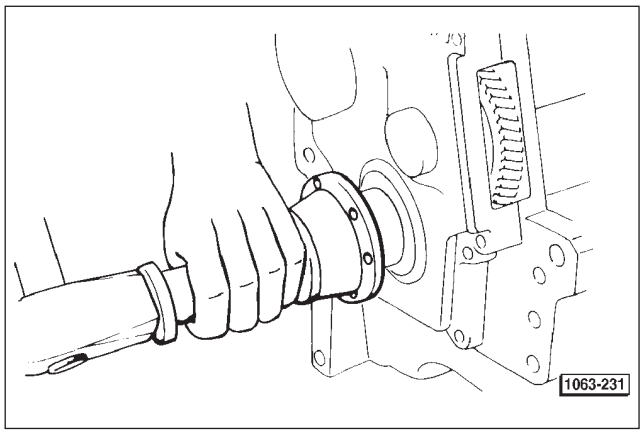


Fig. 50 - Using a suitable plug fit the pulley hub to the crankshaft;

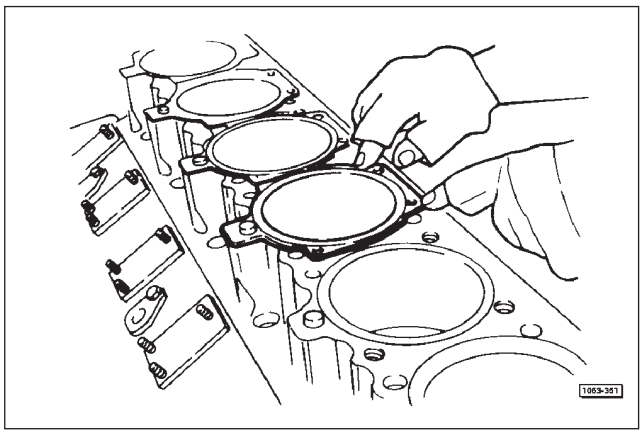


Fig. 51 - Fit gaskets between cylinder heads and cylinders (see instructions on page 31);

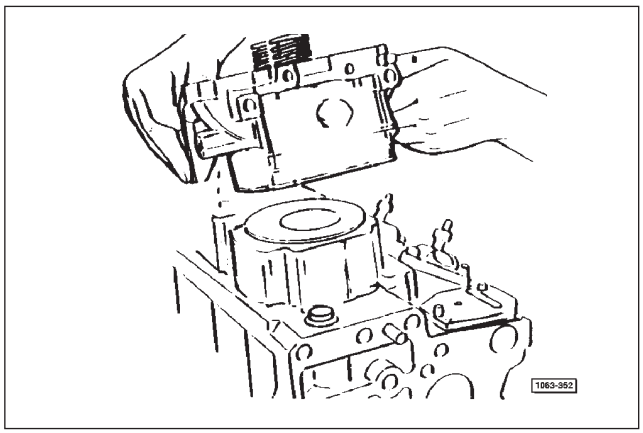


Fig. 52 - After preassembling the heads install on engine;

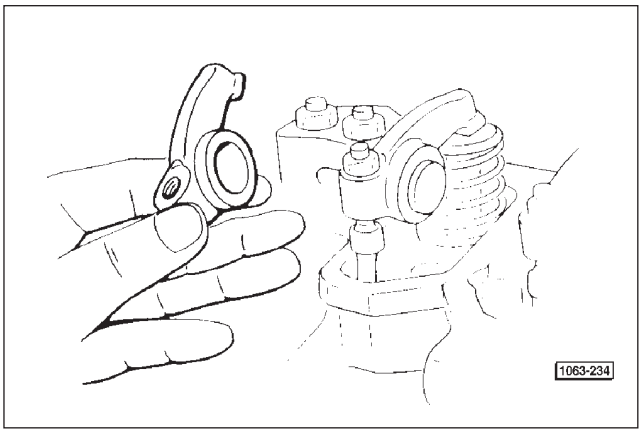


Fig. 53 - Insert the tappet rods, fit the rocker arms and the rocker arm supports;

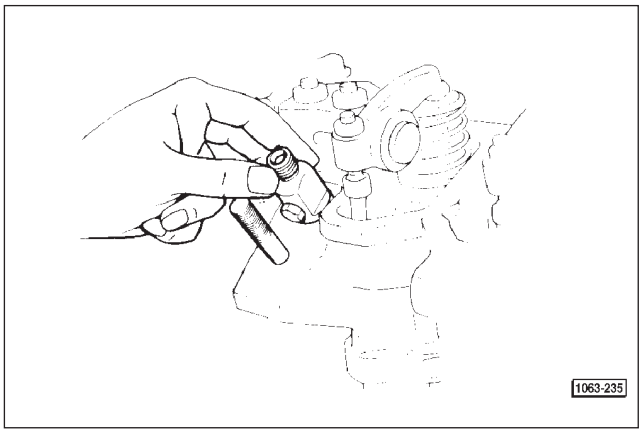


Fig. 54 - Install the injection nozzles in their seats;

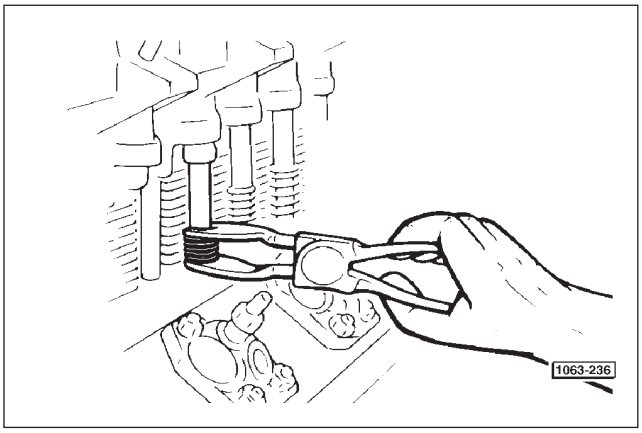


Fig. 55 - Using a special tool fit the rod cover pipes and the positioning springs;

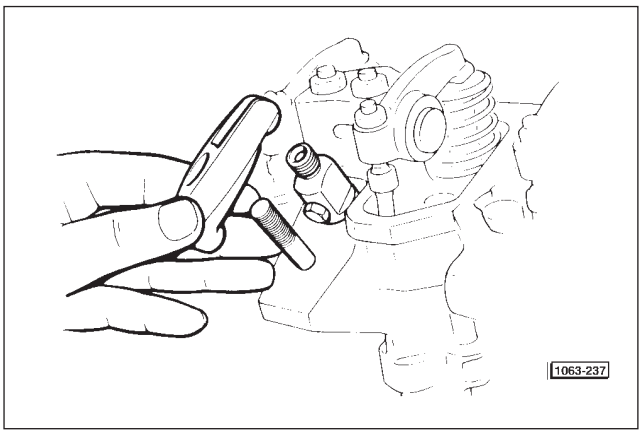


Fig. 56 - Using a plug thoroughly insert the injection nozzles in their seats orienting the fuel rejection hole towards the fixing bracket. Install the injection nozzle fixing brackets tightening the securing screws to the torque specified in the table on page 92;

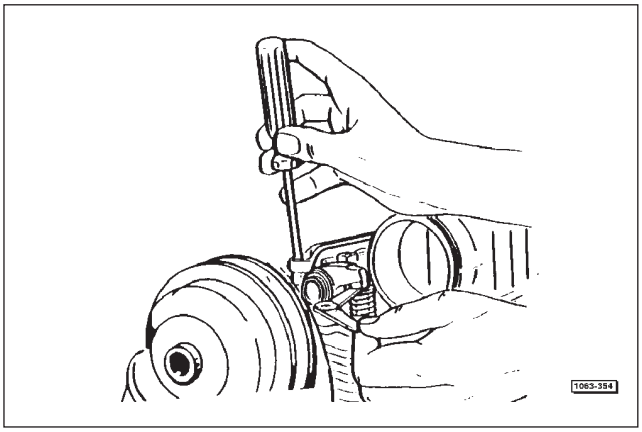


Fig. 57 - Adjust valve and rocker arm clearance;

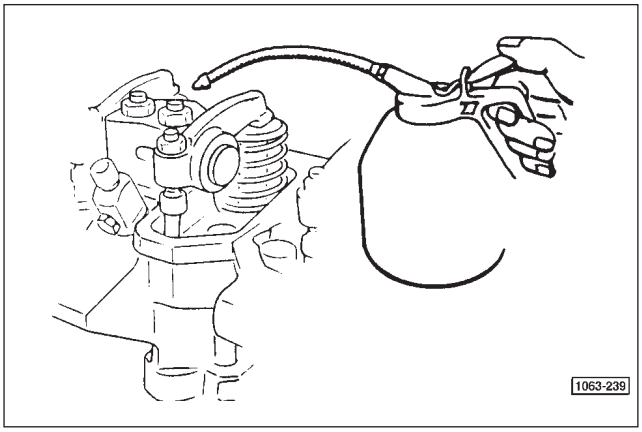


Fig. 58 - Oil valve caps and valve springs;

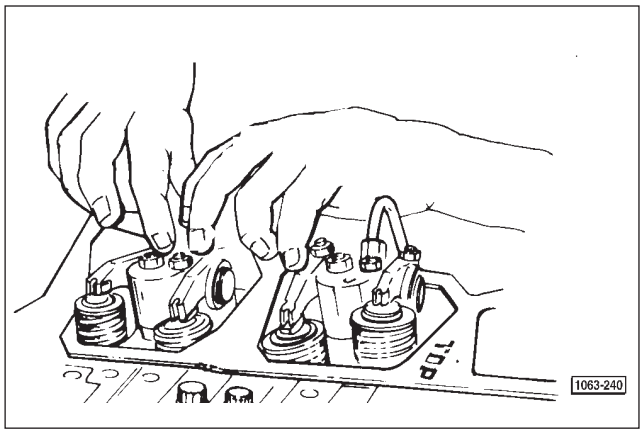


Fig. 59 - Fit gaskets between heads and caps. Orientation shall be provided according to “TOP” indication on them;

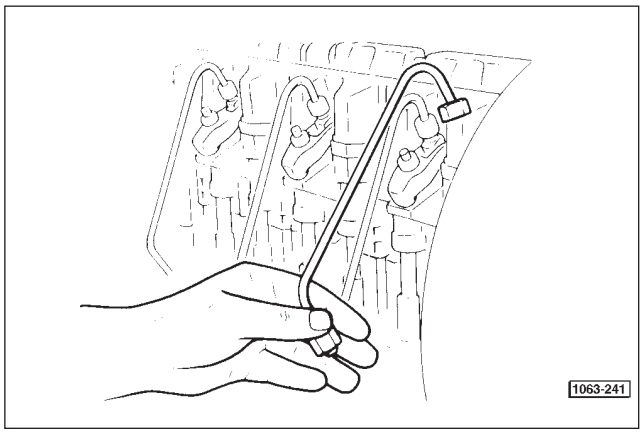


Fig. 60 - Install pump-to-injector pipes as well as the fuel recovery pipes from injection pumps and injectors;

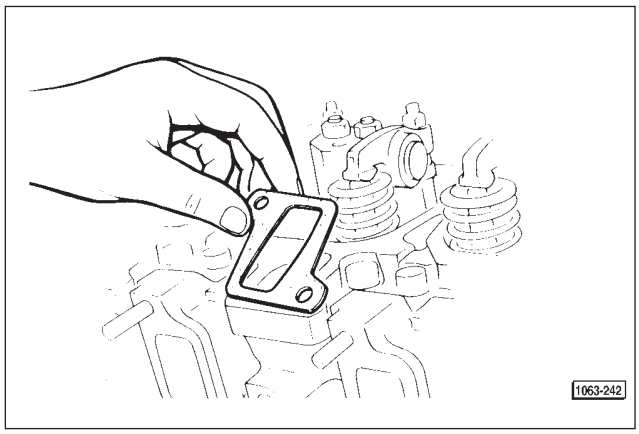


Fig. 61 - Install gaskets between heads and intake manifold;

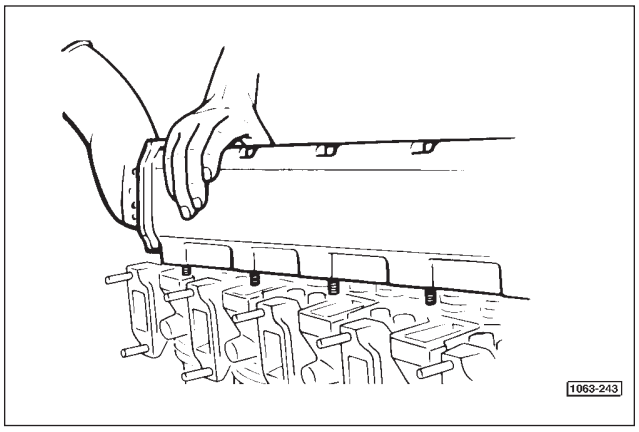


Fig. 62 - Install the intake manifold;

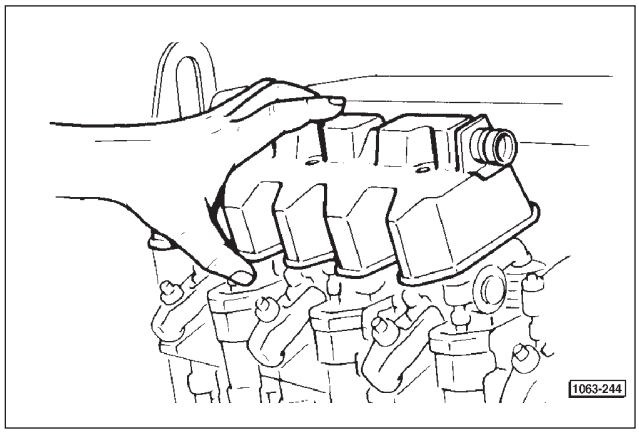


Fig. 63 - Install the head caps;

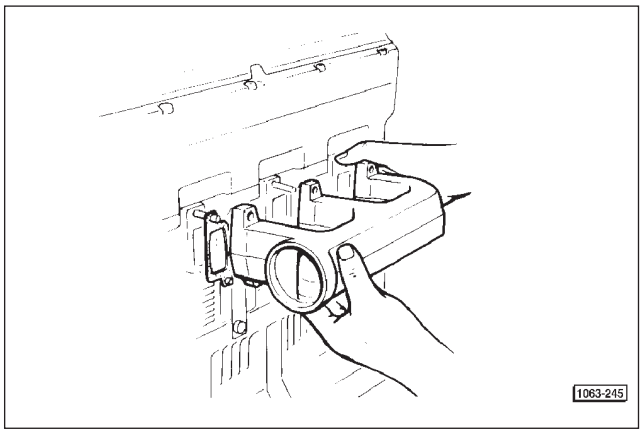


Fig. 64 - Fit the exhaust manifold gaskets. Install the exhaust manifold;

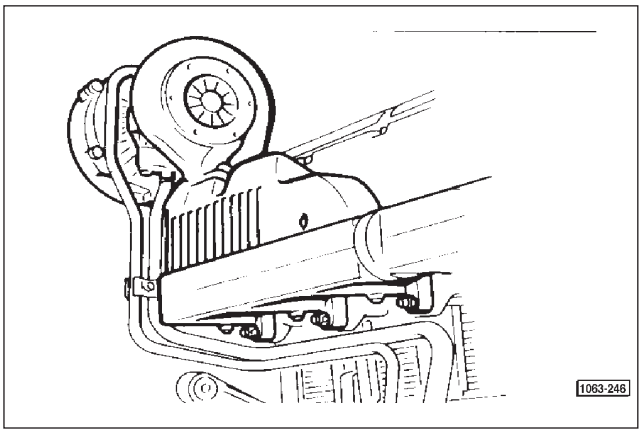


Fig. 65 - On turbocharged engines: install the turbocharger assembly following the instructions given on page 91;

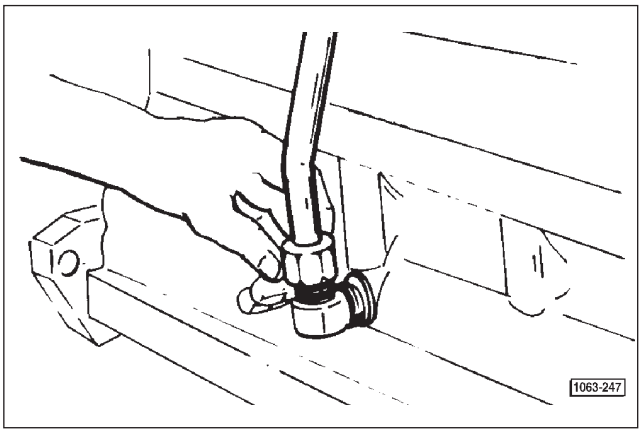


Fig. 66 - On turbocharged engines: connect the turbocharger oil delivery and return pipes;

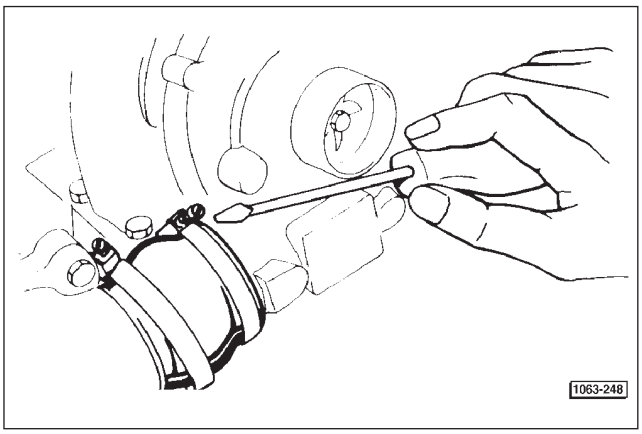


Fig. 67 - On turbocharged engines: place the rubber hose connecting the turbocharger to the intake manifold and secure with two clamps;

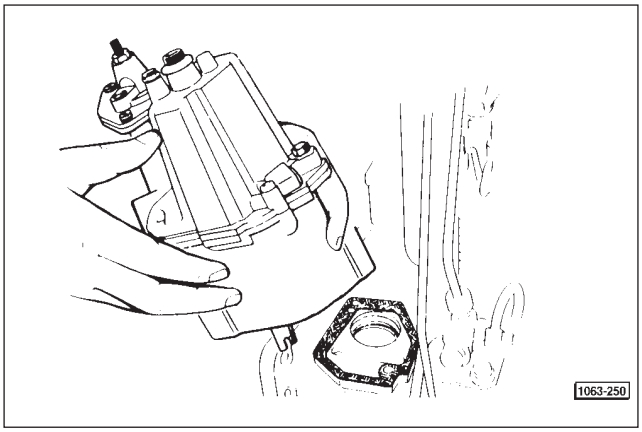


Fig. 68 - FOR TRAKTORS WITH MECHANICAL GOVERNOR ONLY - Install the engine governor taking care the control frok engages in the pump control rod ratchet correctly; make sure it is properly fit in its seat and then tighten with both securing screws;

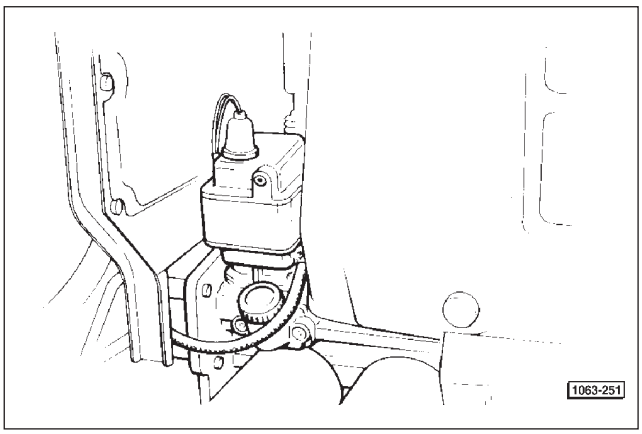


Fig. 69 - For tractors equipped with electronic governors only: fit the actuator assembly;

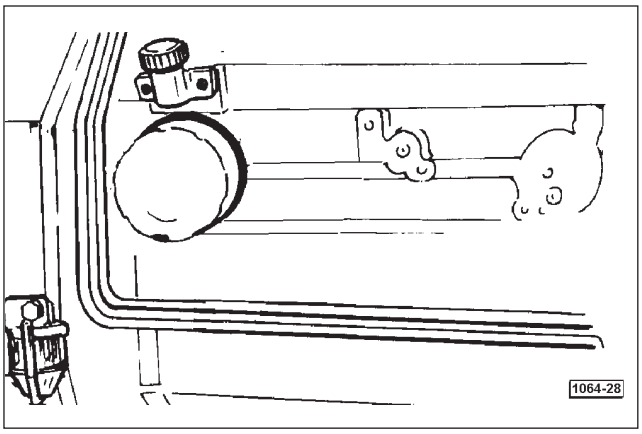


Fig. 70 - Fitbthe oil fuel filter and engine oil filters; then fit the fuel filters and the fuel prefilter. Connect the fuel lines

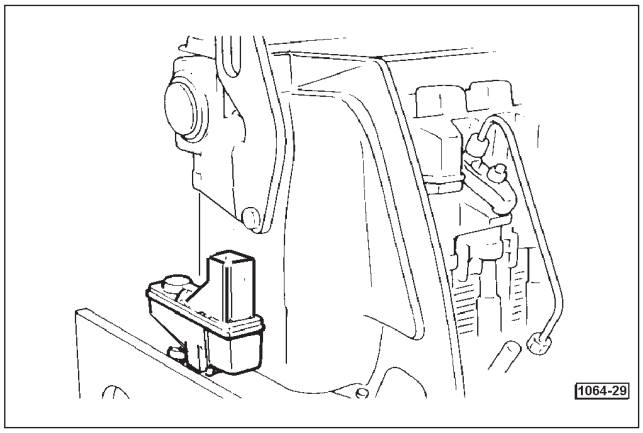


Fig. 71 - Fit the head vapour breather pipe and the rear deflector;

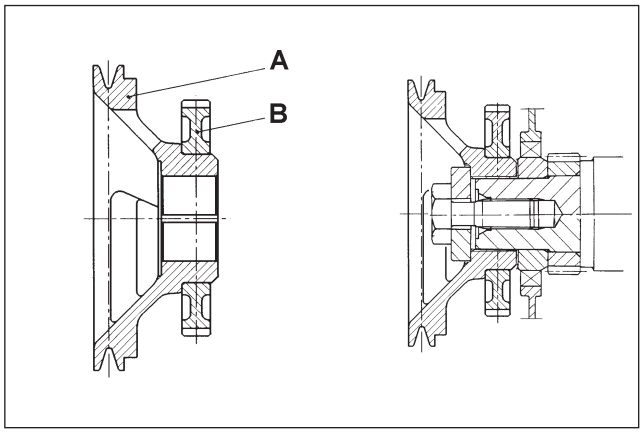


Fig. 72 - On engines equipped with electronic governor, fit pulse wheel B to the crankshaft pulley A. before fitting the pulse wheel, apply Loctite 648 to the contact surfaces of the two parts.
Fit the retaining bolt with the relative thrust washer, lightly lubricating the thread with engine oil. Tighten the bolt first to a torque of 60 to 70 Nm (6 to 7 kgm) and then to a final torque of 340 Nm (34 kgm).

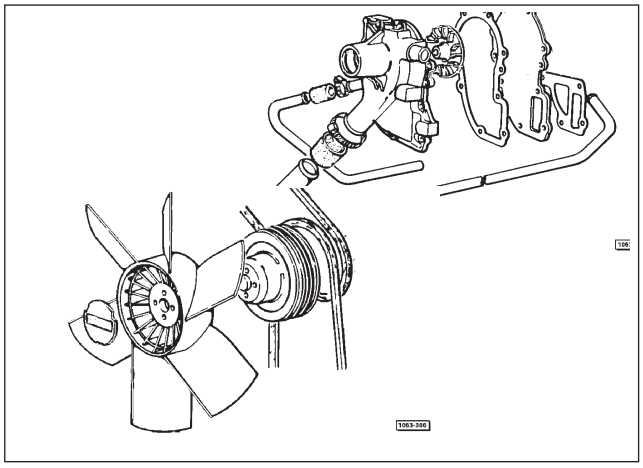


Fig. 73 - Install the fan assembly and pump the cooling system. For engine1000-W

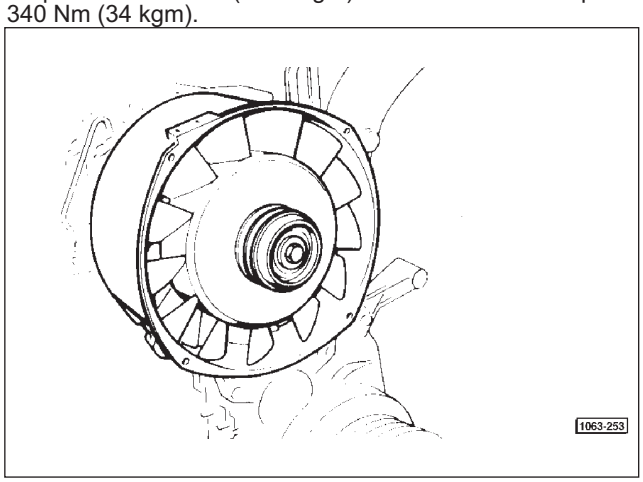


Fig. 74 - Fan assembly; For engine 1000-A

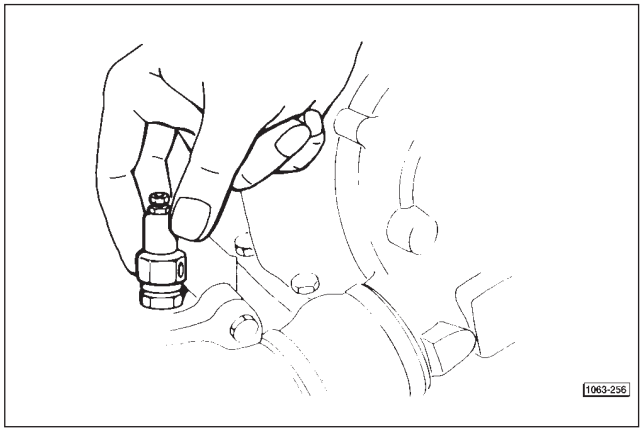


Fig. 75 - Install the thermostarter;

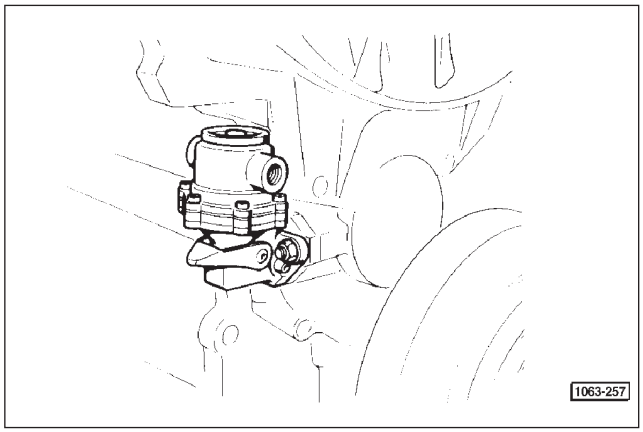


Fig. 76 - Install the fuel pump and the fuel filters and when required the fuel prefilter.
Fit the fuel system lines;

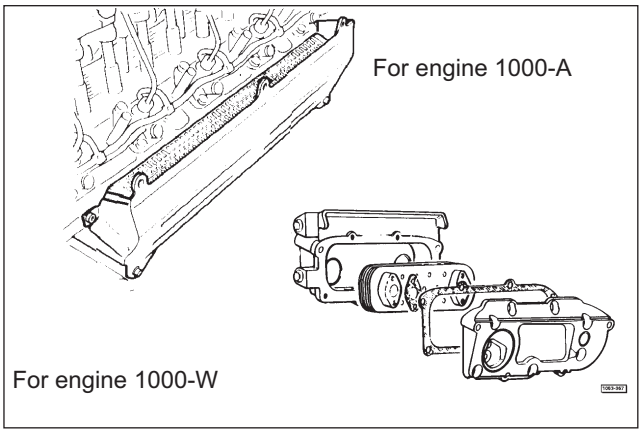


Fig. 77 - Install the engine oil radiator;

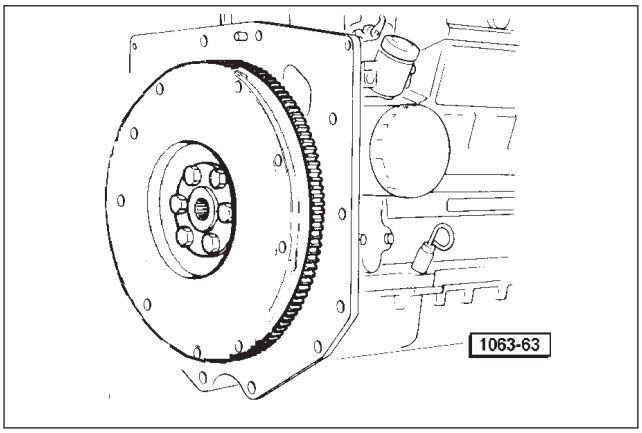


Fig. 78 - Fit the injection pump cover; where required, fit the auxiliary power take-off, interposing shims to obtain the specified gear backlash (see instruction of p. 29). Fit the engine flywheel; on engines with electronic governor, fit the speed sensor pickup in its location of the governor housing and secure with the specific screw.

Reassembling the engine on tractor

To perform engine reassembly on tractor two operators are required.
— Install the clutch assembly onto the engine flywheel.
— Secure two eye-bolts to engine and hook it to a hoist, then lift from swivel bench.
— Position the engine so as it may be centred on the studs inserted in gearbox front part.

Warning: when carrying out these operations take great care to prevent any shock between engine and gearbox. After truing engine onto gearbox studs, use a suitable spanner to rotate the crankshaft until the gearbox shafts are centred in both the clutch plate and the engine flywheel flange broachings.
— Tighten all securing screws and nuts to the specified torques.

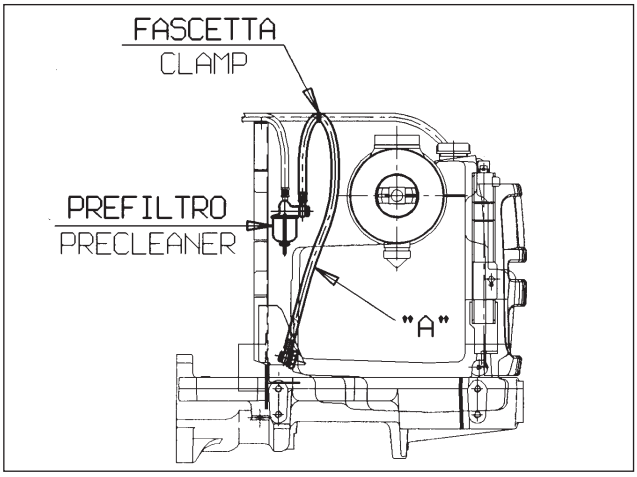
— Place a suitable stand under the engine (insert a squared wooden piece between both parts and always keep the hoist tension wires taut).

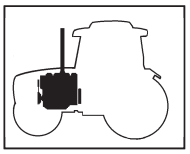
— Bring the front tractor section close to engine and adjust both assembly positions so as to make connection easier.

Between engine oil pan and front support insert a suitable pack of shims so as both parts may contact fully. Centre the front support onto the engine studs (should it be hard to do so the use of a lever is recommended).
Complete reassembly following in the reversed order the operating procedures described in the preceding pages. Fill the engine oil pan with oil keeping to the amount, type and specifications as described on page 9.

Carry out the following operations:

— bleed the air from the hydrostatic steering hydraulic control;
— adjust accelerator control;
— bleed the air from both the fuel supply and fuel injection systems; (pipe **A** in the figure on the right must be secured with a nylon retaining strap as shown in the figure; this is to prevent fuel spillage when cleaning the prefilter)
— start the engine, let it warm up suitably and check engine r.p.m.
— top up brake fluid and bleed the air from the braking system
— on tractors equipped with an air-conditioning system, provide filling as required

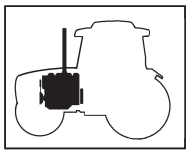




1 Engine

Diagnosing malfunctions

mengine won't run when started	check battery charge	charged battery	battery terminals oxidized	clean battery contacts and tighten terminals
		discharged battery	recharge battery	
	starter failure	consult a motor vehicle electrician for having the defective parts replaced		
engine difficult to start or won't restart after a stop	check fuel level	top up fuel level in tank		
		inspect fuel filter for cleanliness	clean filter, replace if necessary	
	inefficient injection system	a metal noise is heard when engine idles	out-of-adjustment pumps, excessive advance	recalibrate the pumps
		air in the fuel system	perform air bleeding	
			defective fuel pump or pump control	replace defective parts
		out-of-calibration injection nozzles	calibrate	
	Check thermostarter	clean or replace		
	Insufficient compression	check clearance between valves and rockers	restore correct backlash	grind or replace valves, cylinder head gaskets and piston rings
			check valves for efficient operating conditions	replace pistons and cylinders
clear exhaust smoke	check injection pump setting	perform a new setting	check piston rings and cylinders for wear	replace worn-out parts
		make sure there is no excessive clearance between valves and valve guides	replace worn-out parts	
	check injectors for proper cleanliness	clean scrupulously		



dark exhaust smoke	check air filter cartridge for blocking	clean or replace		
		check injectors for proper efficiency	restore calibration according to specifications	
	out-of-calibration engine governor	adjust governor and provide a new calibration		
poor engine response after a sudden acceleration	check air filter cartridge for clogging	check for proper fuel filter cleanliness	clean or replace filters	bleed the system
	make sure no air is within the fuel system	bleed the air in system		
	insufficient compression in the cylinders	check clearance between valves and rockers	restore correct clearance	grind or replace valves, cylinder head gaskets and piston rings
			check valve efficiency	replace pistons and cylinders
engine won't keep constant r.p.m.	make sure neither air nor water are in the injection system	empty the tank and fill with decanted diesel fuel	bleed air from system	
		check injectors for correct calibration		
engine overheating	check belt tension	operate the related belt tightener		
	check cooling system efficiency	clean carefully		
		retime as necessary		
	inspect pumps for proper timing	check injectors for correct calibration	restore specified calibration	
	measure pressure in the duct at compressor exit	clean the turbocharger		

Gearshift clutch

General specifications

The clutch, a single stage type, comprises a friction disc, a pressure plate and a diaphragm spring.
The hydrostatic control is self-adjusting: a master cylinder operated by the pedal directs oil to the slave cylinder mounted to the left hand side of the intermediate housing, which in turn operates the clutch release lever.

Technical specifications		***660s-660v-770s***	***775v-880s-990s***	***660s-770s-880s-990s (USA) 6 blades***
Type of clutch		single disc dry organic facing	single disc dry organic facing	single disc dry cerametallik facing
Type of operation		hydrostatic with automatic take-up of free travel		
Disc p/n		009.6913.3	009.6924.3/20	009.9770.3
Disc diameter	mm	279,4	310	310
Minimum permissible thickness of disc	mm	6	6	9,7
Thickness of friction disc	mm	$8{,}5^{+0,3}_{-0,3}$	$8{,}5^{+0,3}_{-0,3}$	$10^{+0.2}_{-0.2}$
Type of facing material on friction disc		TEXTAR T385	TEXTAR T385	UA 330 DX
Type of master cylinder		Benditalia 3/4"		
Type of oil		AKROS MATIC		

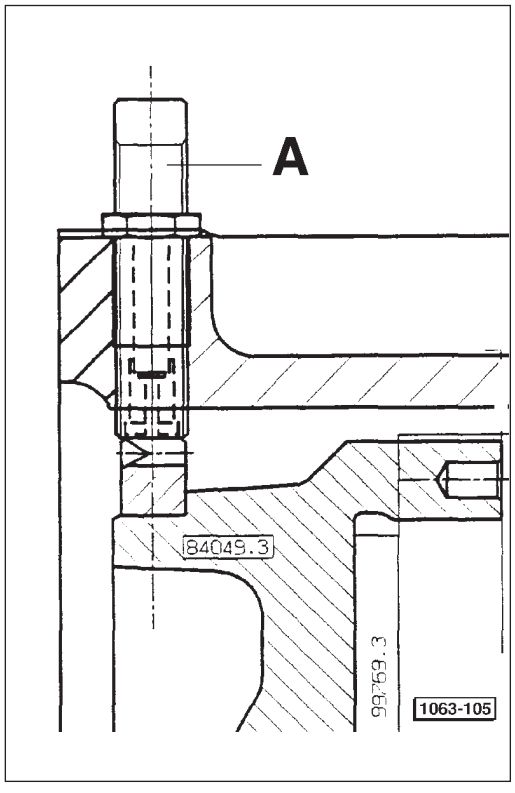


Fig. 1 - Engine rpm pickup. (1st version)

Spring specifications to Belleville washer for the clutch engagement

Load on the pressure plate	Nm	11000

CAUTION: In the event the transmission being split to gain access to the clutch assembly, the pickup (A, fig. 1) must be removed to avoid its being damaged by the teeth of the engine flywheel.

IMPORTANT: *If there are signs of scoring on the surface of the flywheel, the surface must be machined smooth following the directions of page 45.*

IIMPORTANT: In the event of the friction disc being removed, care must be taken during refitment to position the components correctly, as indicated, since the disc is not symmetrical.

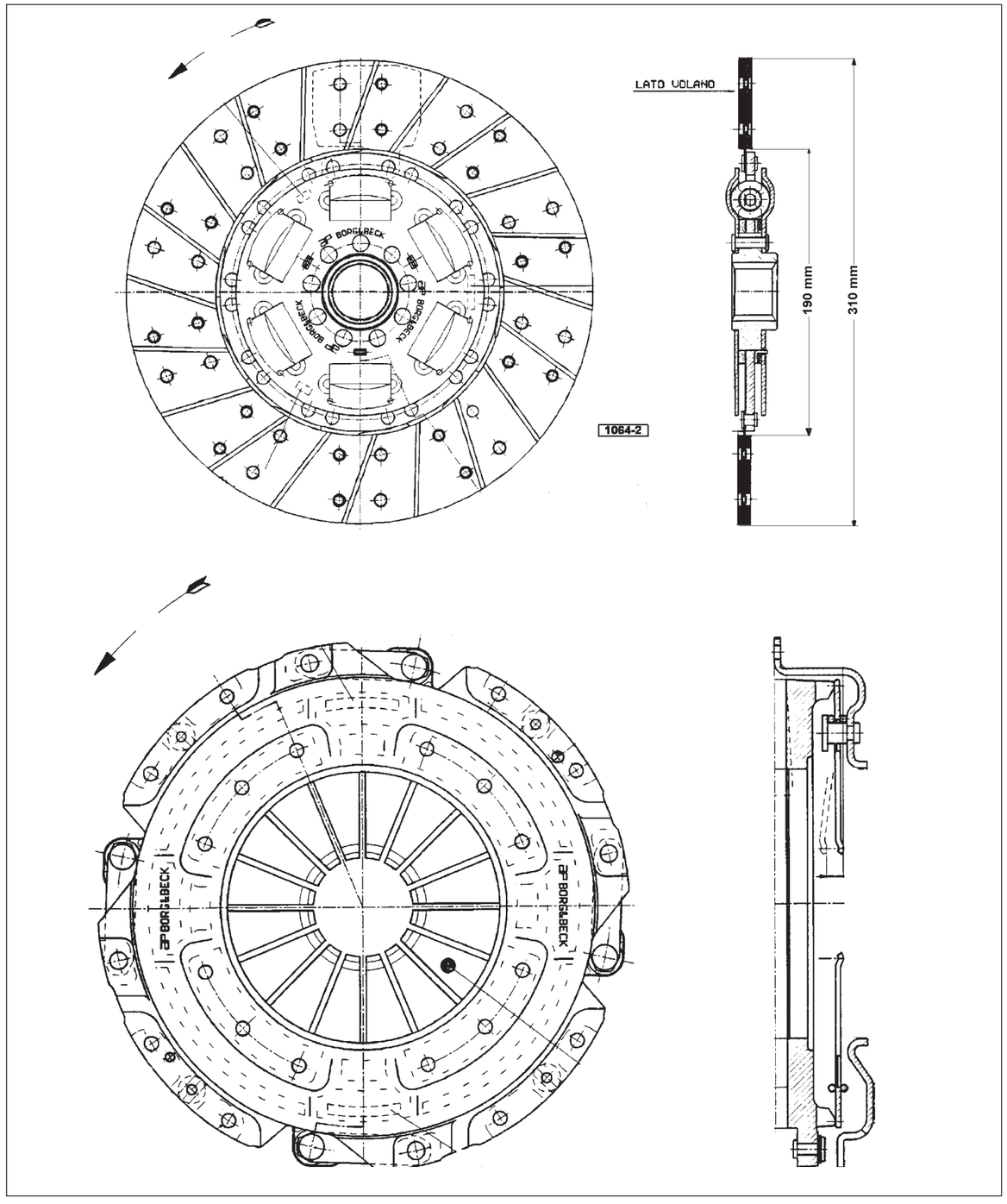


Fig. 2 - Clutch unit for ***660s - 660v - 770s AGILE***

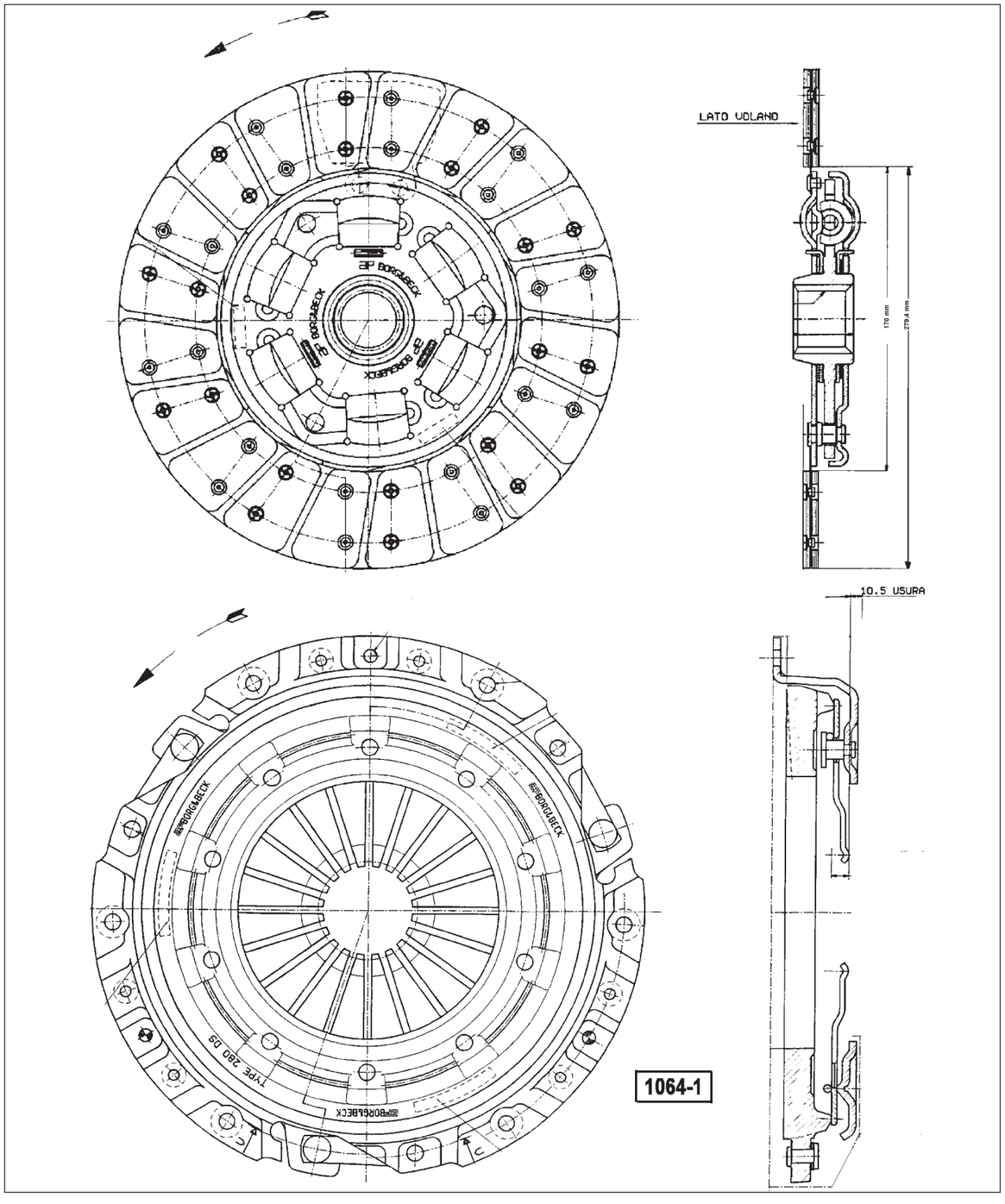
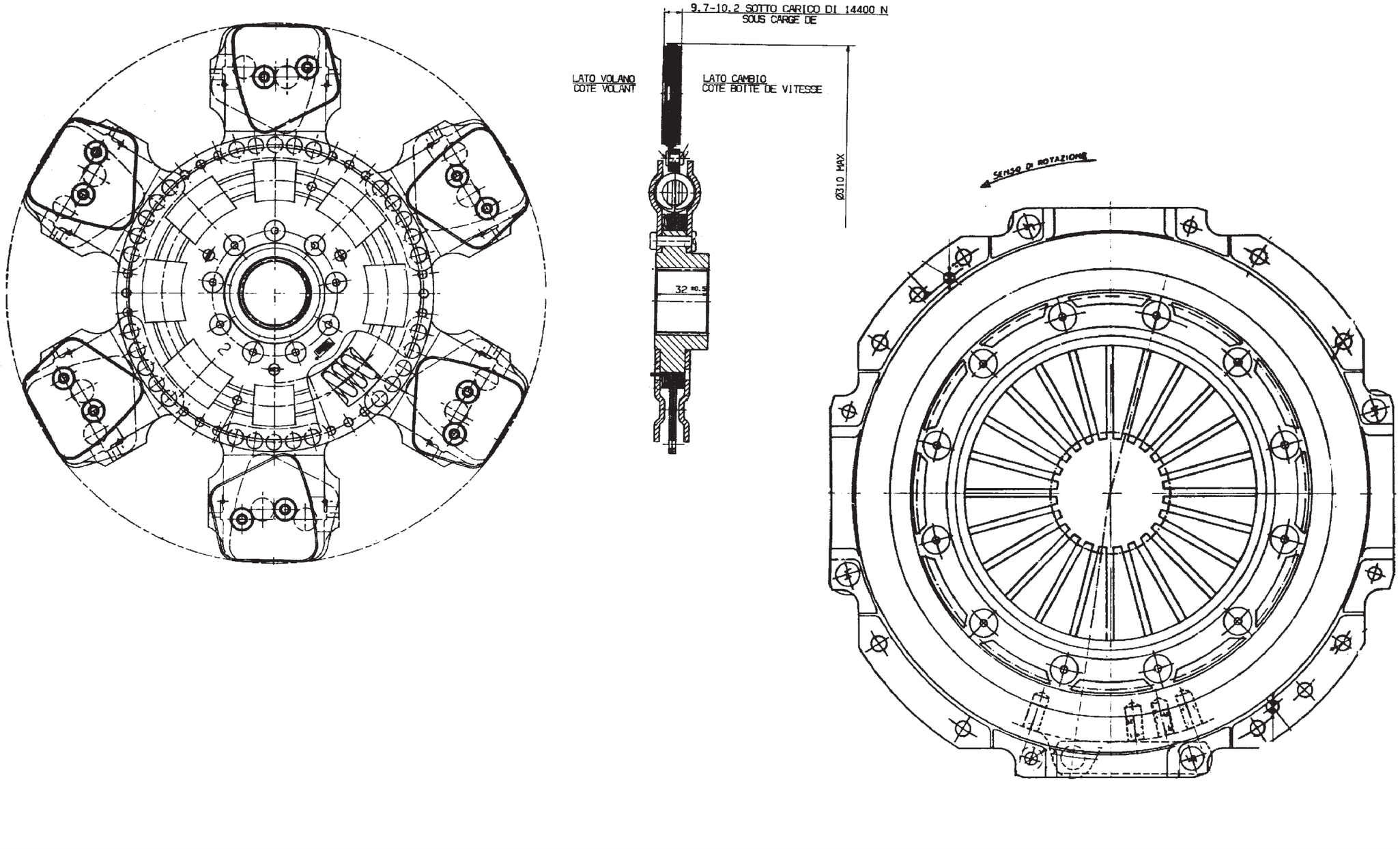


Fig. 3 - Clutch unit for ***775v - 880s - 990s AGILE***



Clutch unit for ***666s - 770s - 880s - 990s AGILE (USA)*** 6 blades

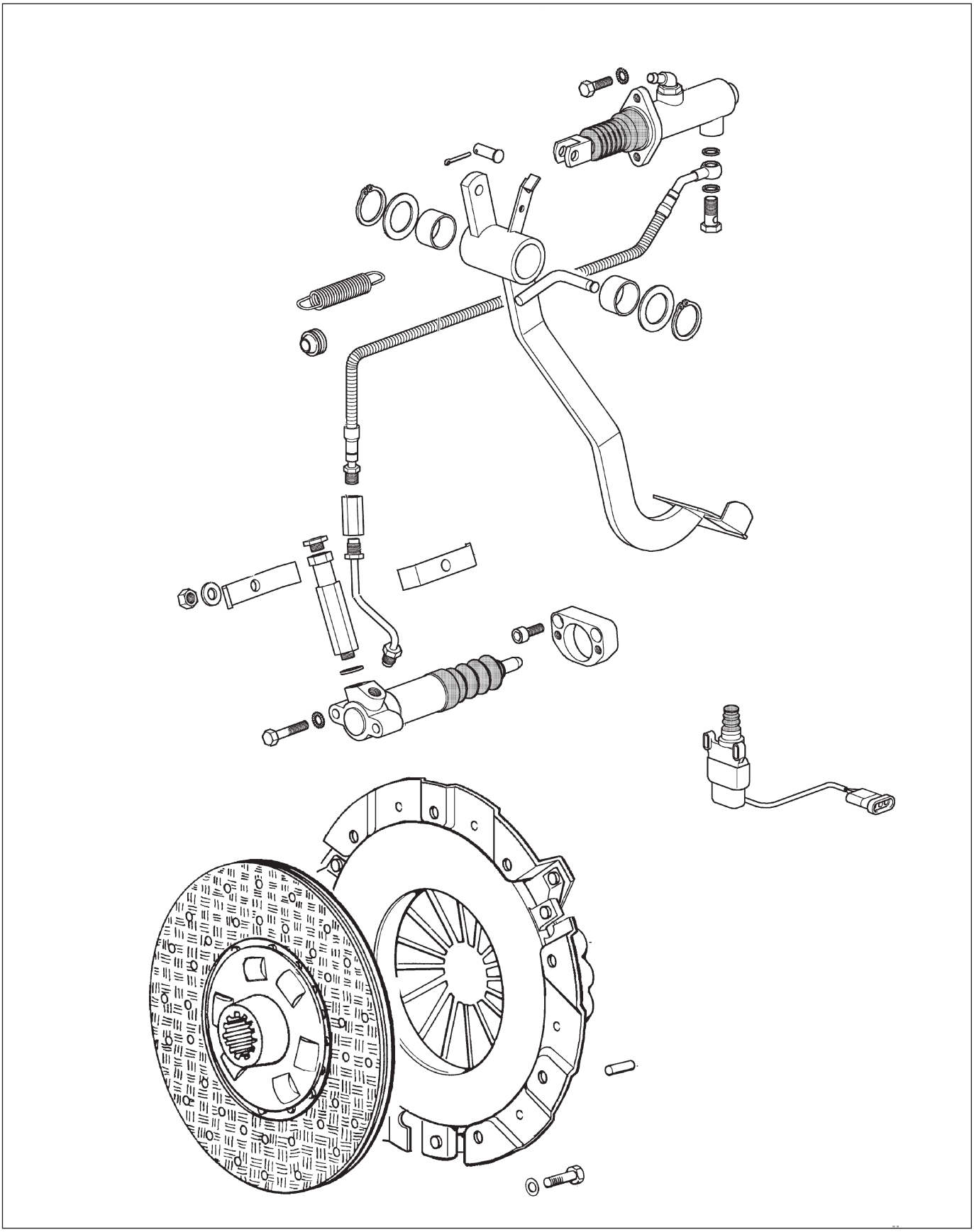


Fig. 4 - Components of clutch assembly

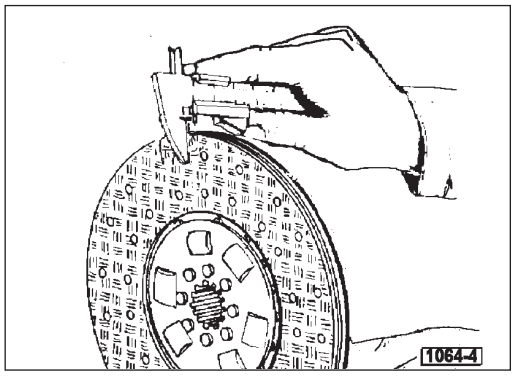


Fig. 5 - Checking clutch disk thickness.

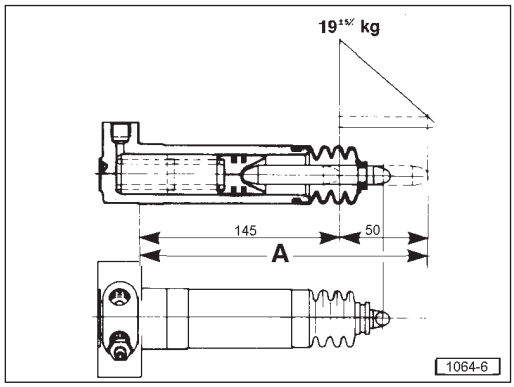


Fig. 6 - Clutch assembly hydraulic operating cylinder.

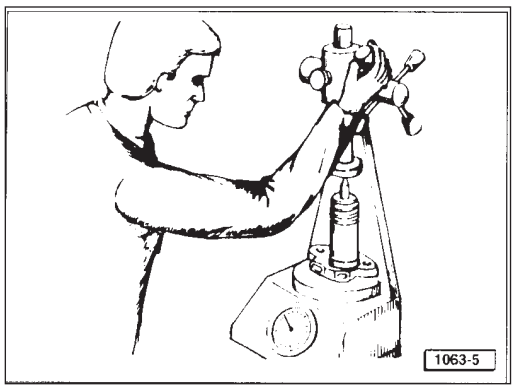


Fig. 7 - Checking operating cylinder inside spring efficiency.

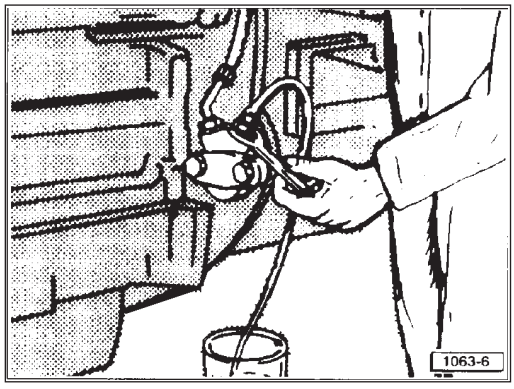


Fig. 8 - Bleeding the air from the clutch hydraulic circuit.

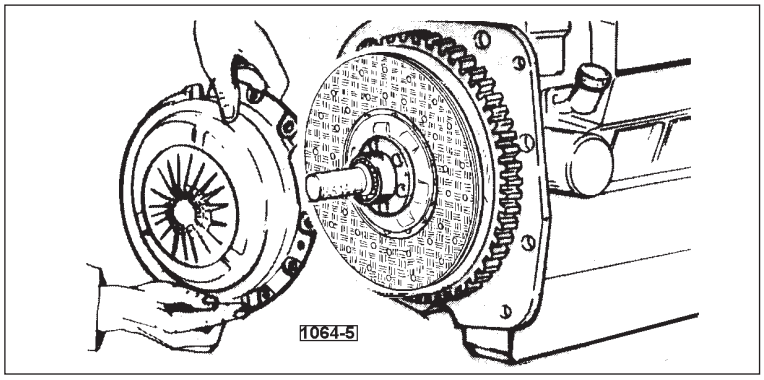


Fig. 9 - Installing clutch assembly through n. 5.9030.256.4/10 equipment.

Cecking clutch

Check the disk lining for signs of chipping and the disk friction face for scoring which may prevent the clutch from operating properly.

Check that the sliding surface of the engine flywheel is not scored; if signs of scoring are evident, machine the surface as indicated on page 45.
Check the disk thrust plate for scoring or bluish areas caused by tempering and also ensure the diaphragm-type spring has not lost its efficiency; if so the whole clutch assembly shall be replaced.

Be sure the clutch disk is free to move in its housing and the friction lining securing rivets are duly riveted.
Should any wear be found on the thrust bearing or the diaphragm-type spring, an exhaustive check over the spring operating conditions installed in the disk thrust lever operating cylinder (see Fig. 6) shall be performed; replace if necessary.
On reassembly ascertain dimension **A** (see Fig. 6) between the operating cylinder fixing face and the push rod contacting the engagement lever is 195 mm.

NOTE: To facilitate correct clutch disk assembly the use of no. 5.9030.256.4/10 centering tool is recommended.
Warning: With engine running, never ride the clutch pedal with your foot to prevent the clutch disk from being damaged because of overheating.
Important: The thrust bearing is prelubricated, and must never on any account be cleaned with fuel oil or other solvents as these will render the prelubrication treatment ineffective.

Adjusting clutch control pedal

Make sure the distance between pump fixing surface and fork hole centre is **95** ± **1** (Fig. 15) otherwise loosen nut **A** and operate adjusting stay rod **B** (Fig. 15).

Bleeding air from the hydraulic circuit

Operate the clutch pedal several times, then keeping the clutch pedal in fully depressed position, slightly unscrew and soon after tighten the air bleeding screw valve again (this being located on thrust lever operating cylinder). This operation should be repeated as many times as the oil flows out of the bleeding screw valve without air bubbles.

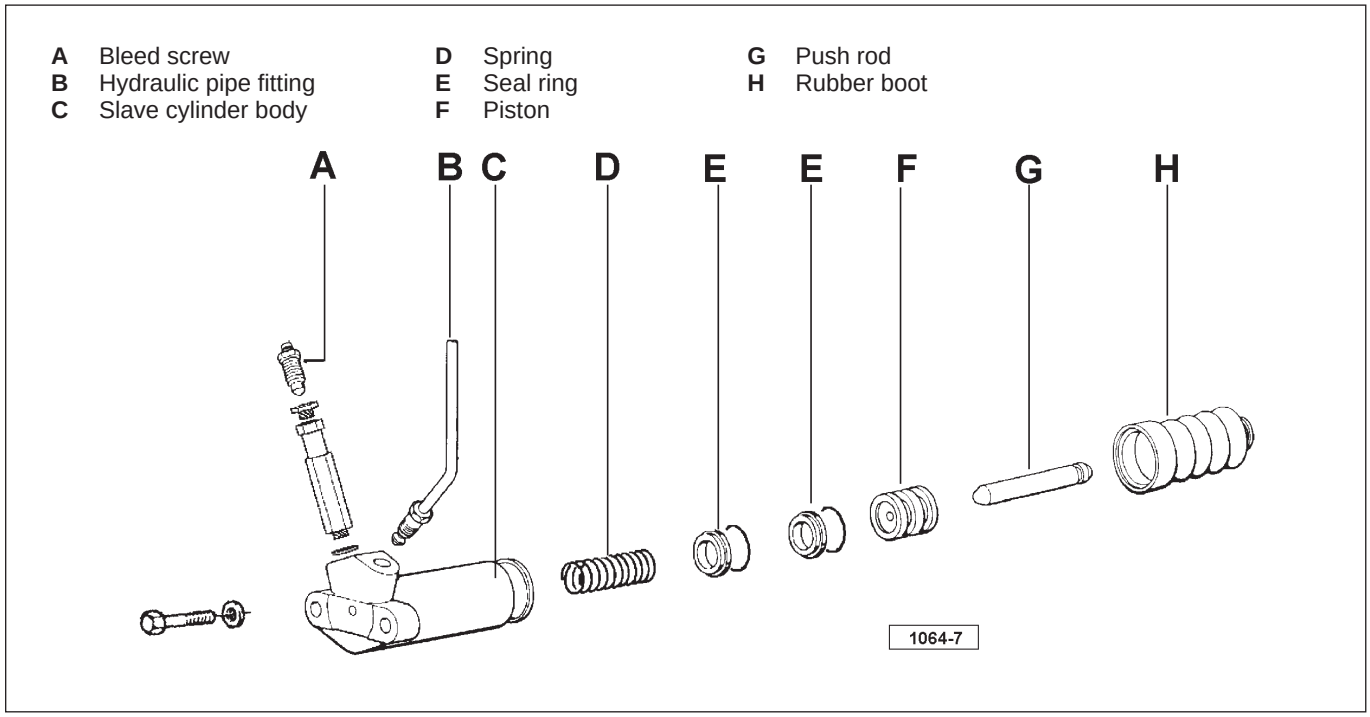


Fig. 10 - Clutch operating cylinder

Stripping the slave cylinder (Fig. 10)

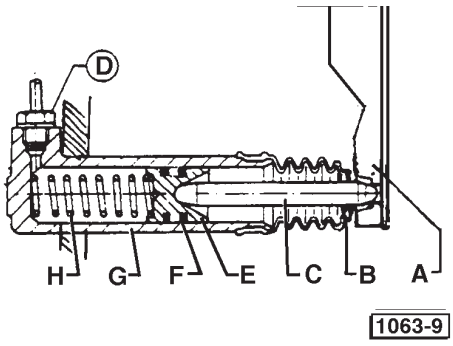
Remove the boot **H**. Withdraw the piston **F** carefully from the cylinder **C**, blasting with compressed air at low pressure to assist removal.
Remove the spring **D** from the cylinder and loosen the bleed screw **A.**
Remove the seals **E** fom the piston **F.**
CAUTION: When cleaning the components of the cylinder, use only specifically formulated brake and clutch fluids (see page 110). Do not use petrol, paraffin or other mineral oils as these will damage parts in rubber.

Inspections

— Check both internal and external piston surfaces for scratching. Replace if required.
— Make sure the seal ring grooves are duly clean; blow the grooves with compressed air if necessary.
— Inspect seal ring, dust boot conditions as well as spring efficiency, worn-out parts should be replaced.
— Ensure the air bleeding hole is free from impurities.

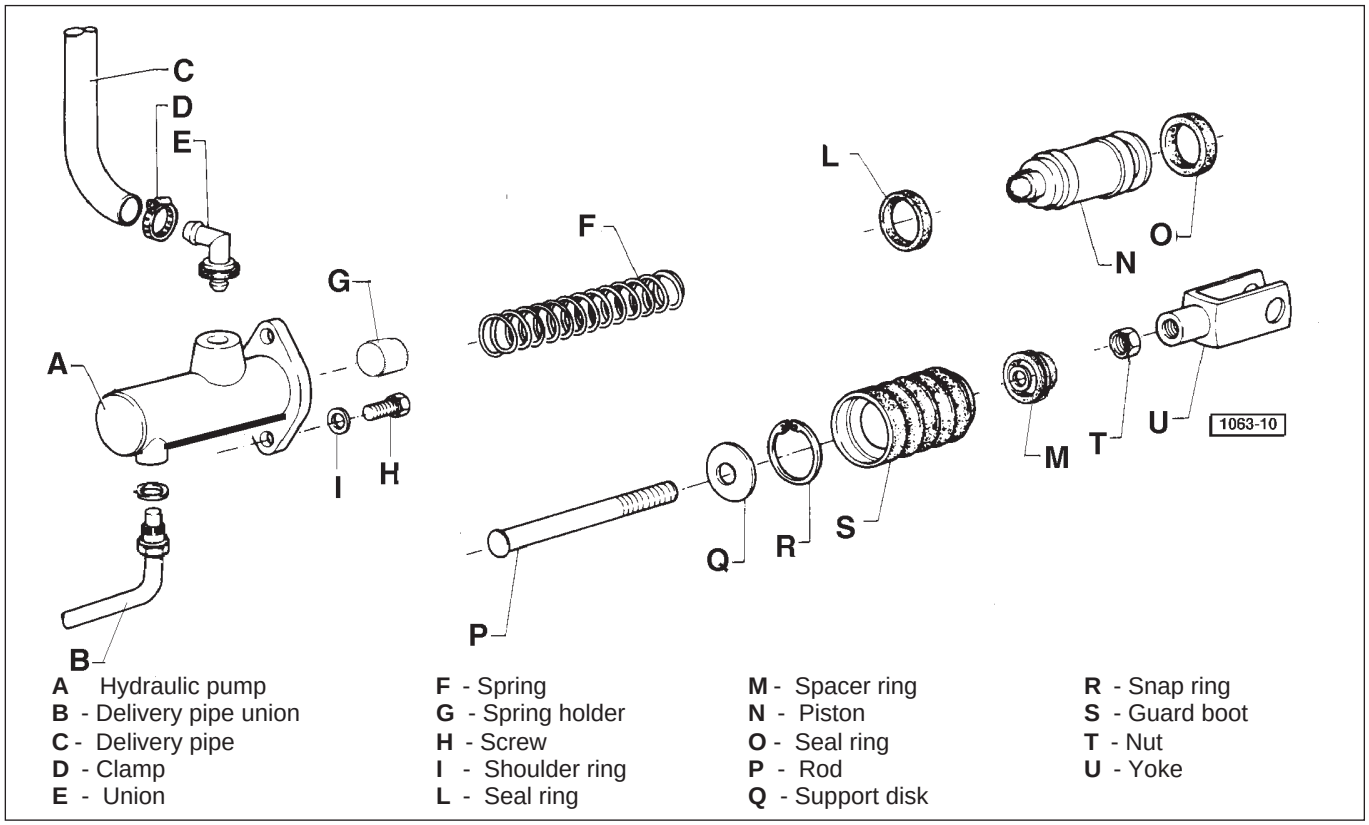
Notes on refitment

— When reconnecting the transmission housing, check that the clutch fork remains correctly positioned and free to rock on its fulcrum pivot. This can be ensured by removing the side plug from the flange of the intermediate housing and viewing the fork through the hole. In the event of the fork pivot being unseated, reposition correctly with the aid of a screwdriver inserted through the hole vacated by the plug.
— Before refitting the slave cylinder, fill with the recommended oil so as to facilitate the subsequent bleeding procedure.
— Once all components are correctly and securely in place, bleed the hydraulic circuit.



A - Control lever
B - Dust boot
C - Pushrod
D - Union
E - Piston
F - Seal ring
G - Cylinder casing
H - Spring

Fig. 11 - Clutch operating cylinder cutaway view.



A Hydraulic pump
B - Delivery pipe union
C - Delivery pipe
D - Clamp
E - Union
F - Spring
G - Spring holder
H - Screw
I - Shoulder ring
L - Seal ring
M - Spacer ring
N - Piston
O - Seal ring
P - Rod
Q - Support disk
R - Snap ring
S - Guard boot
T - Nut
U - Yoke

Fig. 12 - Clutch hydraulic pump parts.

Stripping the master cylinder

Referring to figure 14, remove the protective boot **E**, dislodge the circlip **B** and withdraw the rod **D** together with the disc **C**.
Remove the piston together with the spacer, the seal, the backup washer and the spring beneath.

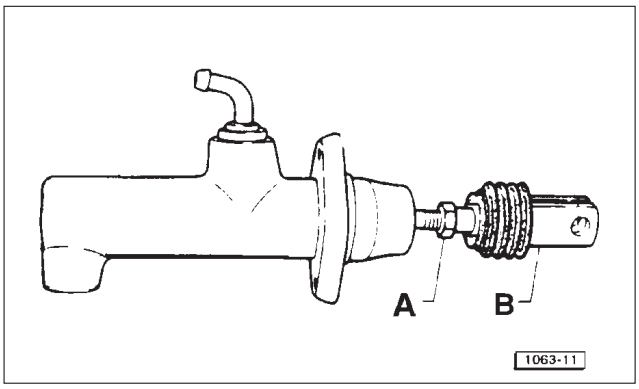


Fig. 13 - Clutch pump control positioning.
A - Locknut
B - Yoke

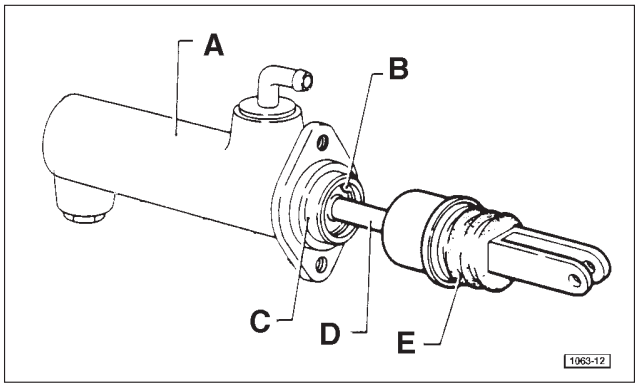


Fig. 14 - Pump control seal ring.
A - Pump
B - Snap ring
C - Support disk
D - Rod
E - Guard boot

Inspections and checks

WARNING: To clean and wash the hydraulic pump components use only the oil type recommended for brakes and clutch (see page 110). Never use petrol, kerosene or other mineral oils to prevent damaging the rubber parts.
Inspect both internal and external piston sliding surfaces for scratching. Replace if required.
Make sure the seal ring grooves are duly clean; blow the grooves with compressed air if necessary.
Inspect seal ring, dust boot conditions as well as spring efficiency, worn-out parts should be replaced.
Inspect all pump internal compartments, apertures and passages and make sure these are properly clean and free from foreign matters.
Check that the spring is neither lazy nor deformed; replace if necessary.

Reassembly

Reassemble the cylinder, repeating the disassembly steps in reverse order and observing the following directions:
— Lubricate surfaces engaged in relative sliding contact, using the recommended oil (see page 110).
— Verify correct operation of the cylinder, making certain that the piston is able to complete its full stroke unimpeded.
In the event that the fork linking the master cylinder with the pedal has been removed, check that with the push rod fully extended, the distance between the reference surface of the cylinder and the centre of the hole in the fork is as indicated in figure 15.
If not (referring to fig 13), remove the boot, loosen the lock nut **A** and screw or unscrew the fork **B** to obtain the prescribed clearance, then retighten the lock nut and reposition the boot.

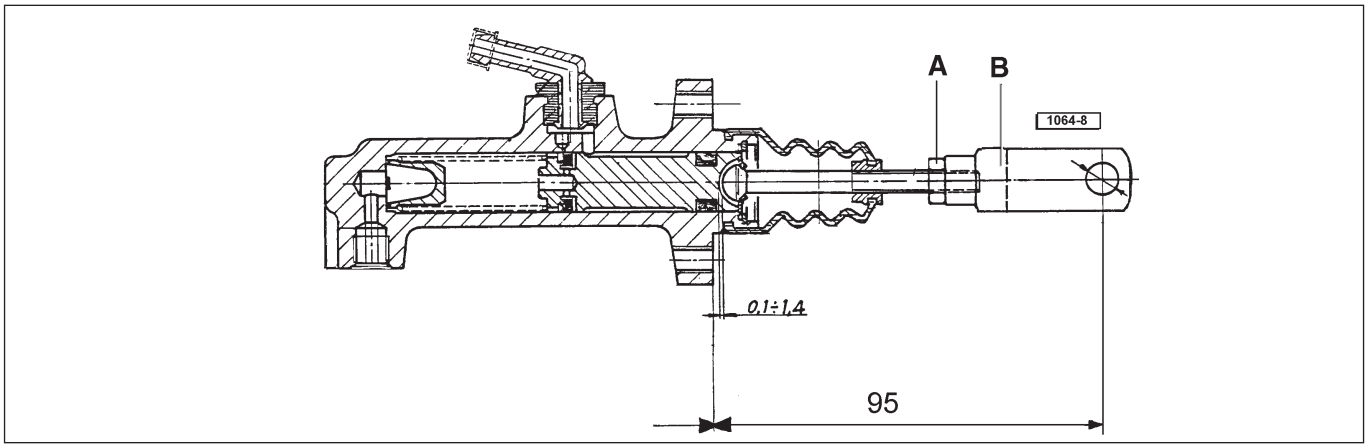
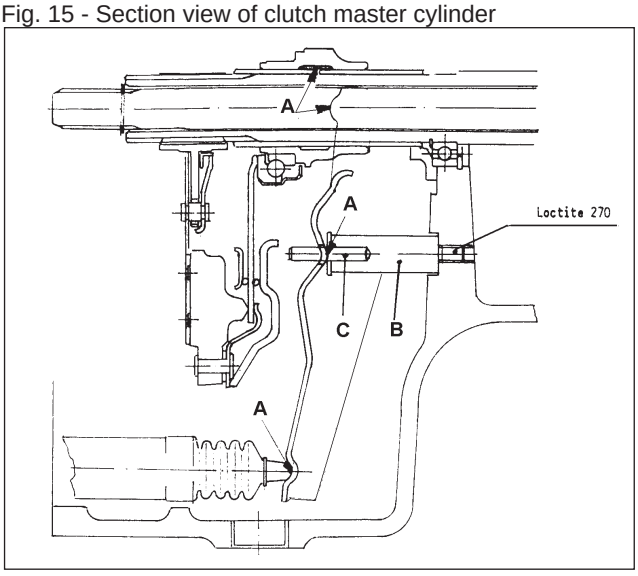
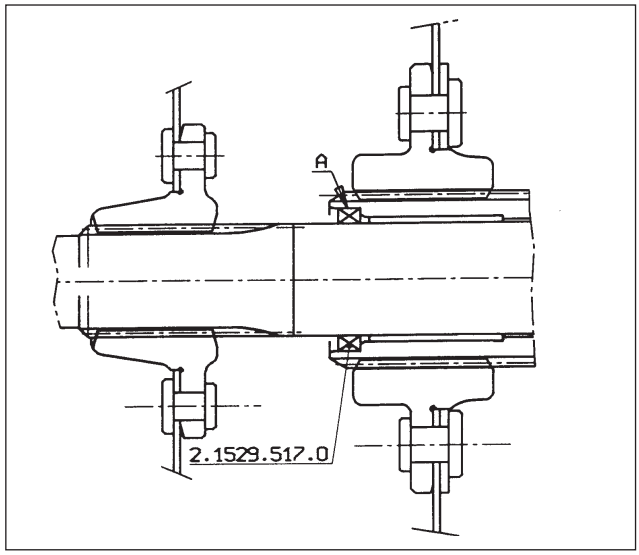


Fig. 15 - Section view of clutch master cylinder



Clutch greasing points.
Apply Molikote Gn-plus at all the points indicated in the figure with letter **A**.
Fit part **B** with Loctite 270.
Fit pin **C** with Loctite 601.



Fitting the P.T.O. shaft oil seal
Before fitting the oil seal 2.1529.517.0 apply Loctite 222 to the outer face **A** indicated in the figure.

	lubricant in clutch housing	renew the front gearbox oil seal and the rear engine oil seal	clean oil seal contact surfaces with petrol	replace disk
clutch slips	clutch worn	check condition of clutch disk	check condition of the spring disk	fit new clutch assembly
	thrust bearing sticking	clean surfaces and apply grease	replace thrust bearing	clean or replace the disk
clutch jerks	clutch disk surfaces dirty	clean the friction surfaces		
	clutch disk warped	clutch disk surfaces dirty	replace clutch disk	
		clutch disk worn	replace clutch disk	
		plate seals loose	replace clutch disk	
clutch fails to disengage	difficulty in engaging gears when engine running	clutch disk warped	replace disk	
		hydraulic pump inefficient	check the stroke of the clutch control piston and replace any worn parts	
		clutch disk stuck to flywheel	clean contact surfaces with a wire brush and petrol	
clutch noisy when disengaged	worn parts in clutch engagement mechanism	replace parts		

Multispeed unit

The **Multispeed** unit is composed of an **epicyclic speed reducer** and **three oil-immersed multiple disc clutches** (**LOW - MED - HIGH**).
The unit is located between the clutch-coupled shaft and the gearbox.
The speed reducer is engaged and disengaged selectively by way of the three clutches, which are operated electrohydraulically from a button mounted to the knob of the shift lever.

Selecting the control, the **MED** clutch releases as the force of the belleville discs is overcome by hydraulic pressure; at the same time, the **HIGH** clutch engages, locking the planet carrier of the epicyclic train to the relative housing so that drive is transmitted to the range input shaft with no speed reduction whatever.

Selecting the control, the **MED** clutch releases as the force of the belleville discs is overcome by hydraulic pressure; at the same time, the **LOW** clutch engages, and the shaft carrying the sun wheel is locked to the structure of the transmission housing. As a result, the planet carrier is made to rotate as one with the epicyclic housing and drive is transmitted to the layshaft at a speed reduction of 0.687 (the ratio between the annulus and the sun wheel gear teeth).

Selecting the control, the Multispeed unit is isolated from hydraulic pressure altogether and the belleville discs are able to engage the **MED** clutch, with the result that the shaft in mesh with the larger planet wheel is locked to the transmission housing. This gives a reduction of 0.825, by reason of the compound ratio between the planet wheels and the flange of the speed reducer in mesh with the range input shaft.

Technical specifications

Clutch				
maker		SAME DEUTZ - FAHR GROUP		
number of friction discs		7	4	2
diameter of friction discs		129,5	129,5	129,5
overall thickness of assembled friction discs complete the intermediate discs	mm	25,2 ÷ 25,4	17,50 ÷ 17,60	11,70 ÷ 12,14
number of intermediate discs		3+1	2+1	2+1
disc lubrication pressure	bar	5	5	5
maximum pressure	Bar	16	16	16
piston thrust	Kg	1986	1986	1986
	Nm	19463	19463	19463
Epicyclic speed reducer				
- LOW		1 + (30/66) = 1,4545		
- MEDIUM		1 + (21x18) : (66x27) = 1,2121		
- HIGH		1		

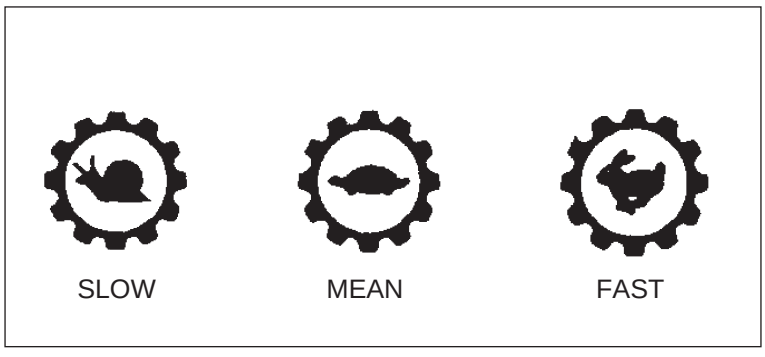


Fig. 1 - Range selection indicators mounted to the instrument panel.

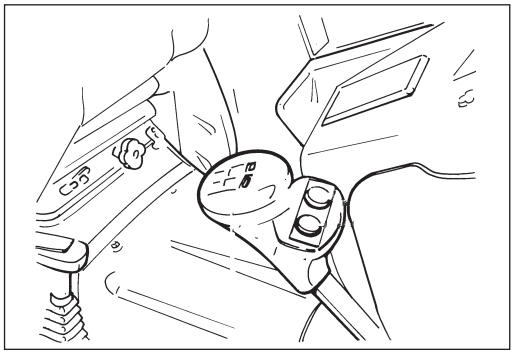


Fig. 2 - Plate indicating selection of electrohydraulic control.

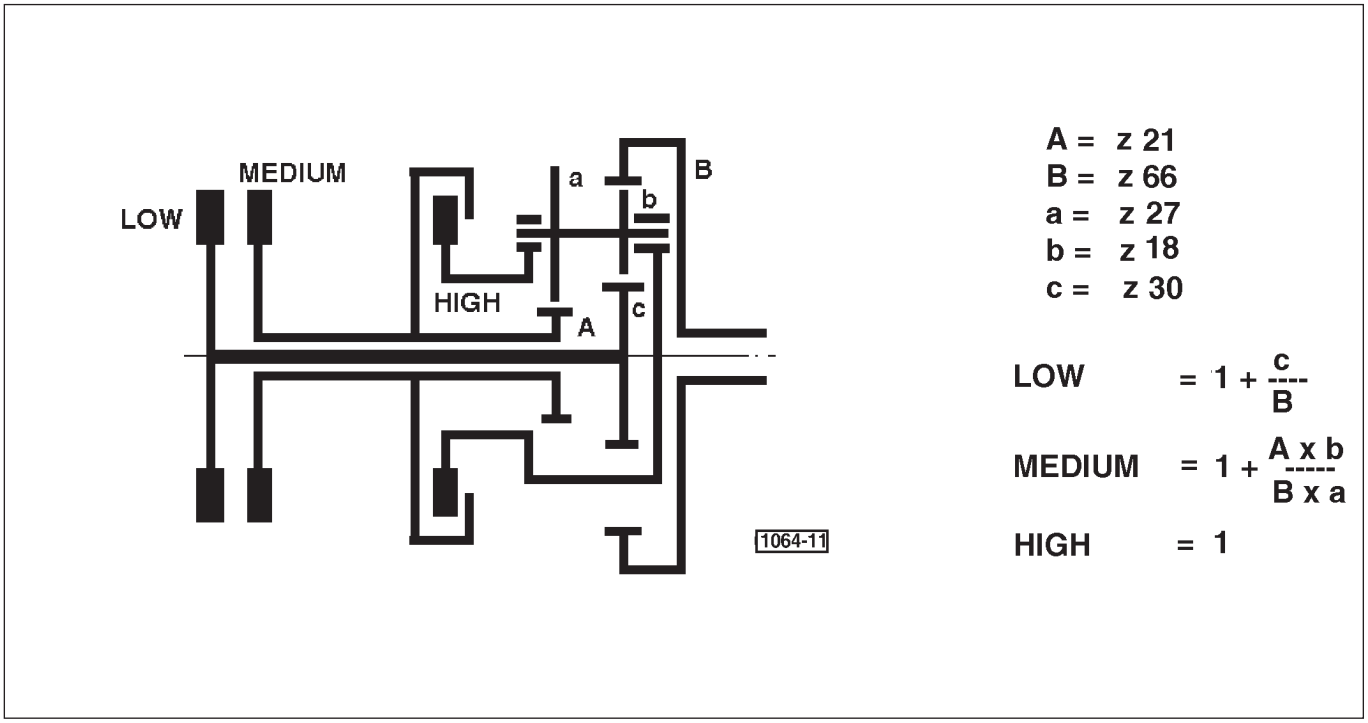


Fig. 3 - Diagram illustrating operation of the MULTISPEED system.

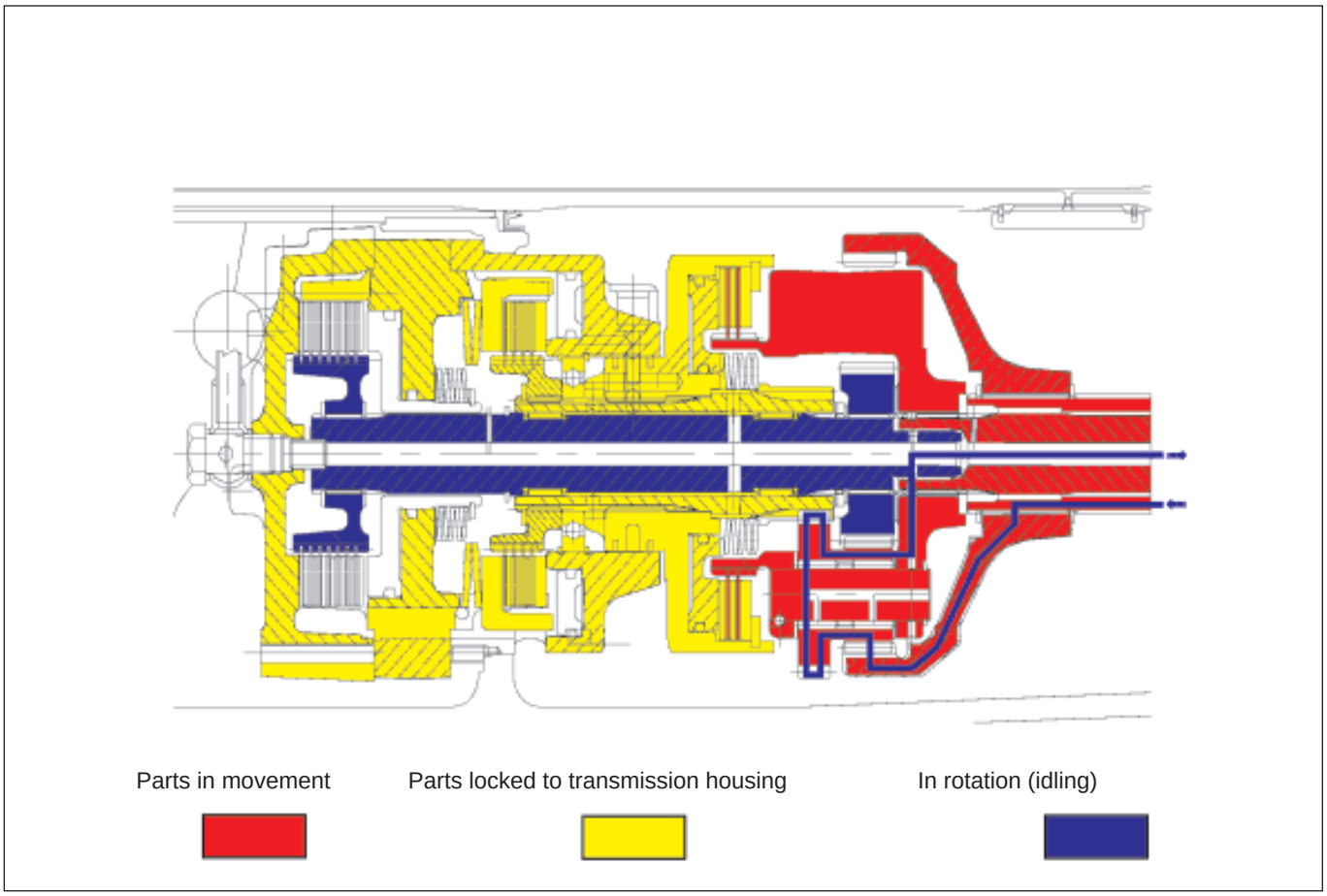


Fig. 4 - Diagram showing engagement of MULTISPEED with MED range selected.

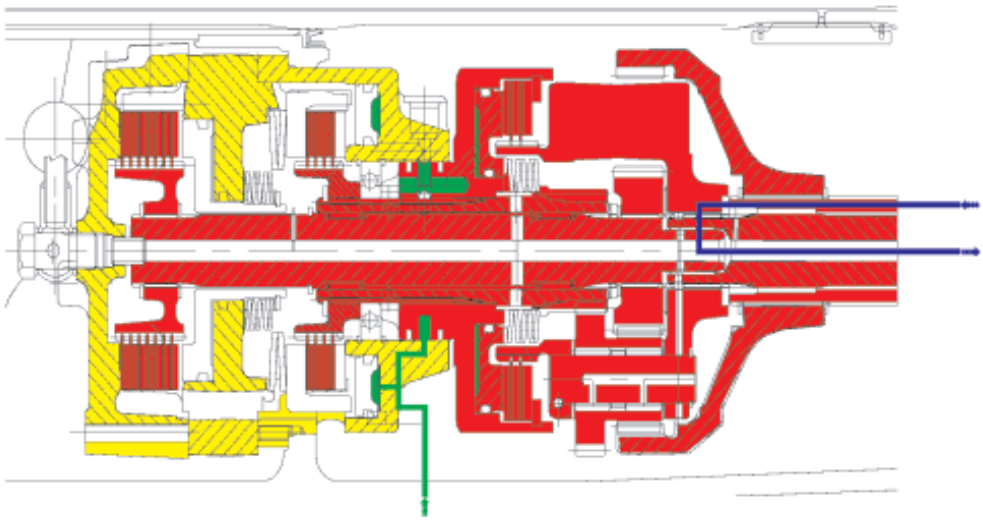
Parts locked to transmission housing

Parts in movement

Oil circuit for operation of clutches

In rotation (idling)

Fig. 5 - Diagram showing engagement of MULTISPEED with LOW range selected.



Parts locked to transmission housing

Parts locked to epicyclic speed reducer

Oil circuit for operation of clutches

In rotation (idling)

Fig. 6 - Diagram showing engagement of MULTISPEED with HIGH range selected.

MULTISPEED UNIT DETACH FROM THE GEAR BOX

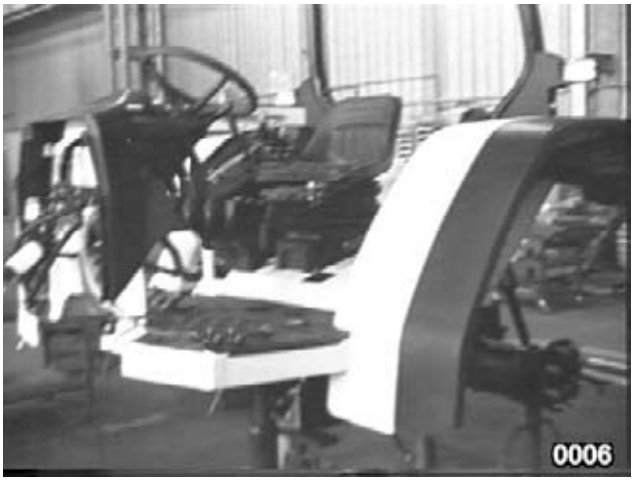


Fig. 7 - Detach the driving platform (or the cab, if fitted) from the tractor.

Fig. 8 - Detach the front part of the tractor complete with engine from the gearbox.

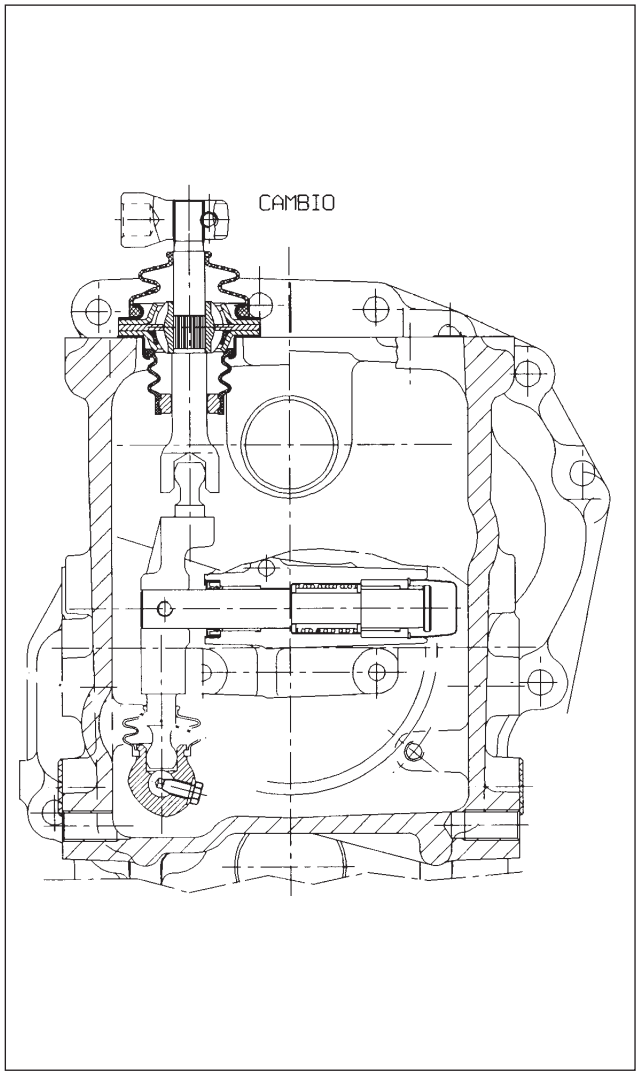


Fig. 9 - Remove the internal clutch control from the front gearbox and then remove the entire gearshift control.

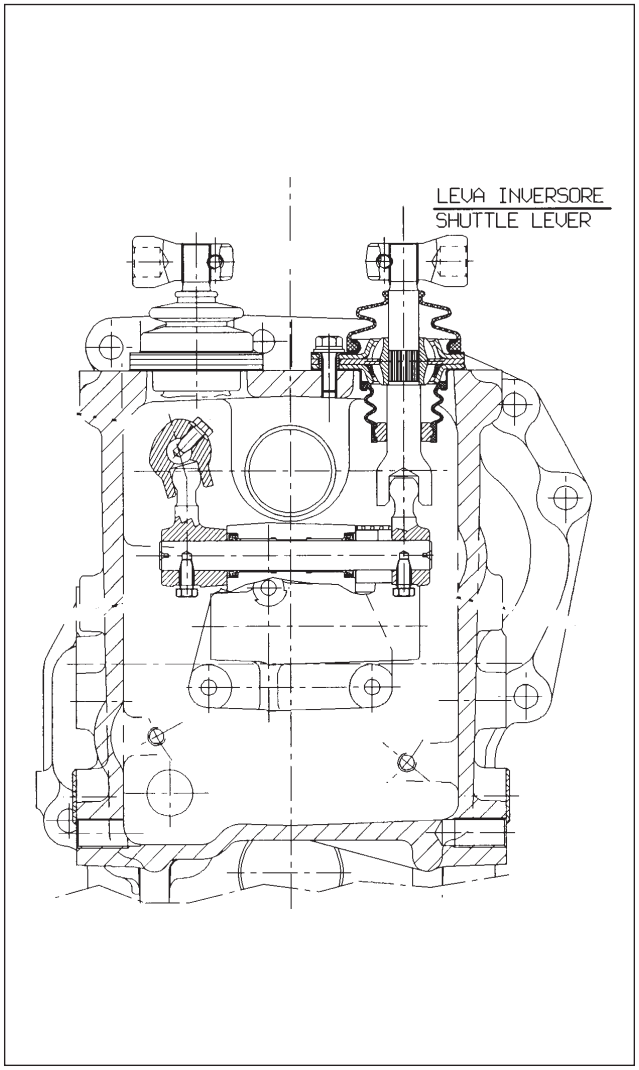


Fig. 10 - Remove the entire shuttle control.

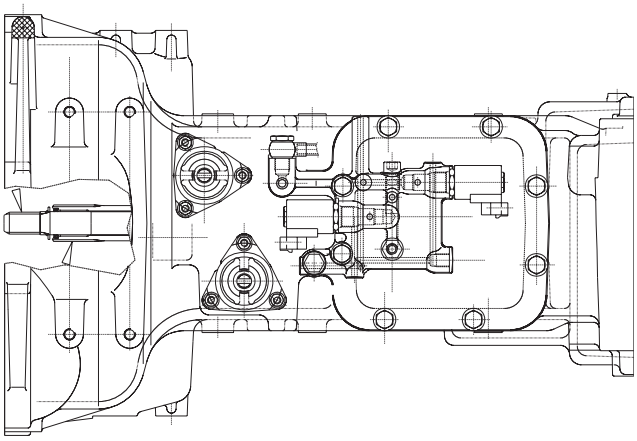


Fig. 11 - Remove the upper cover from the front gearbox. Remove the hydraulic lubrication pipe of the Multispeed unit by first disconnecting the part on the outside of the gearbox.

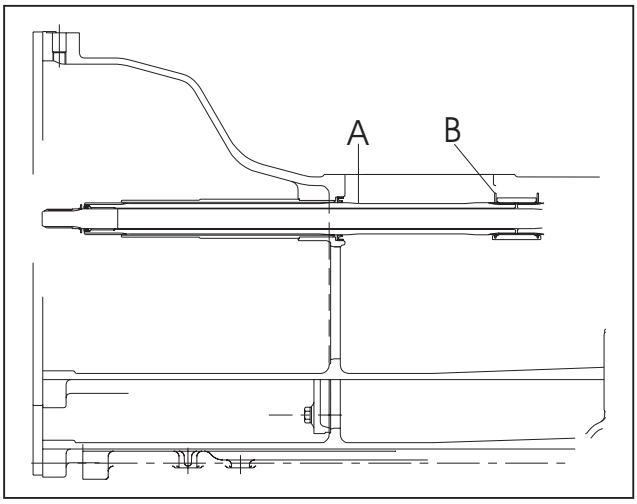


Fig. 12 - Detach the front gearbox from the rear gearbox.

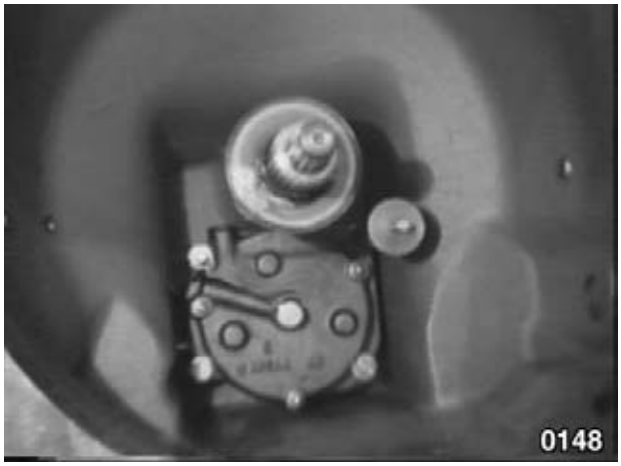


Fig. 13 - Multispeed unit installed in front gearbox.

Fig. 14 - Using a drift as shown in the figure, drive the Multispeed unit out of the front of the gearbox.

Fig. 15 - Remove the Multispeed unit from the front end of the gearbox.

Fig. 16 - After having removed the Multispeed unit, proceed with its disassembly, if necessary. Then remove the epicyclic reduction unit.

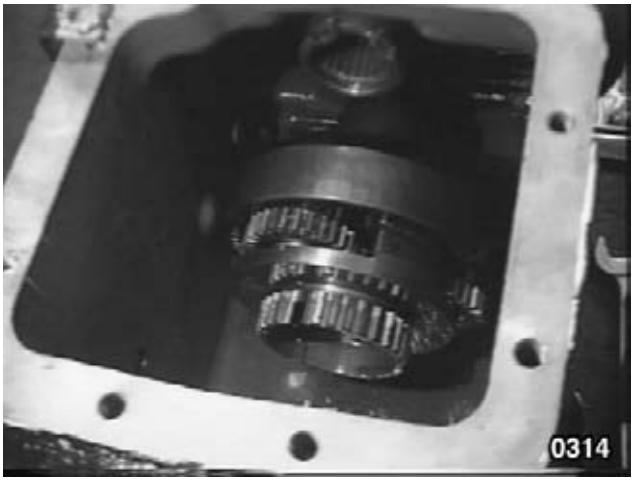


Fig. 17 - Remove the planet carrier flange from the epicyclic reduction unit.

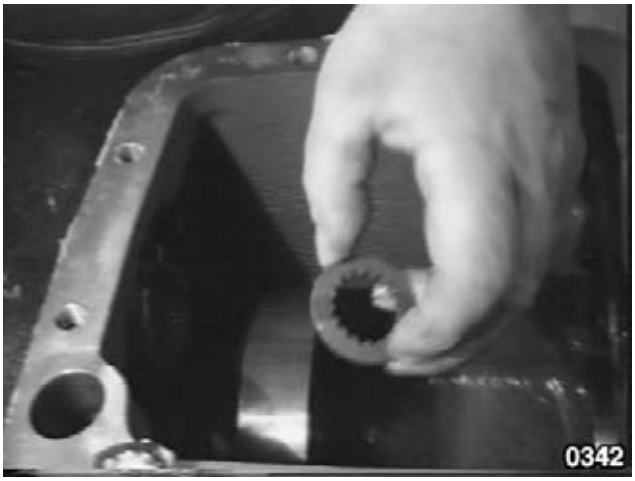


Fig. 18 - Remove the circlip and the thrust washer. On re-assembly, the thrust washer must be fitted with the lubrication channels facing inwards towards the planet carrier flange.

Fig. 19 - The play adjustment shim for the Multispeed shafts is located between the housing and the epicyclic reduction unit.

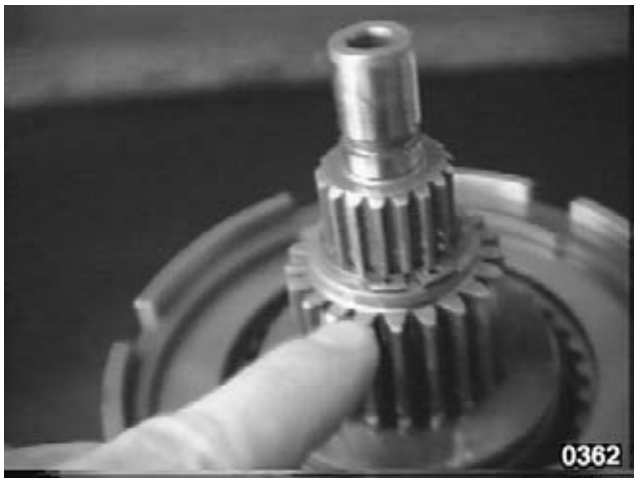


Fig. 20 - To dismantle the Multispeed unit, remove the circlip indicated in the figure.

Fig. 21 - Remove the circlip indicated in the figure using a pair of bent nose pliers.

Fig. 22 - Remove the thrust washer.

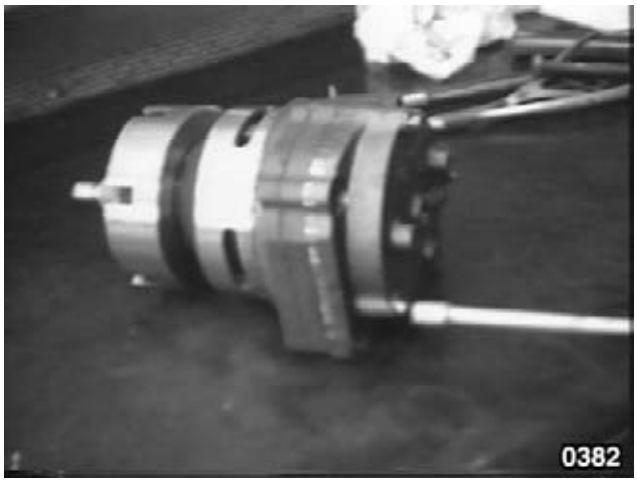


Fig. 23 - Unscrew the screws securing the LOW clutch housing to the Multispeed unit.

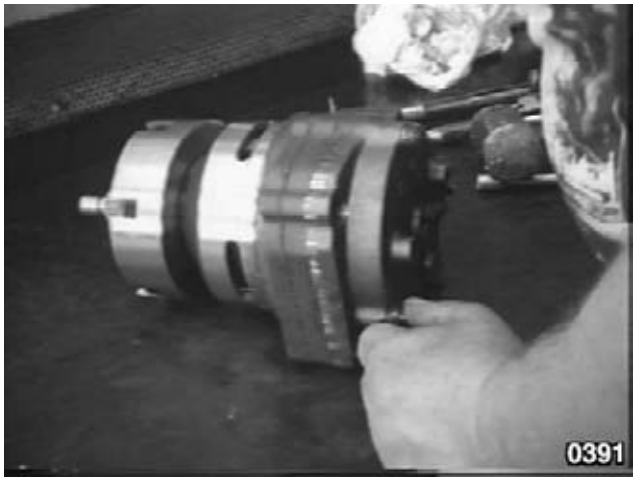


Fig. 24 - Remove the retaining screws.

Fig. 25 - It is advisable to mark the relative positions of the parts prior to disassembly.

Fig. 26 - Using a screwdriver, separate the housing from the unit.

Fig. 27 - Remove the LOW clutch disc pack (7+7).

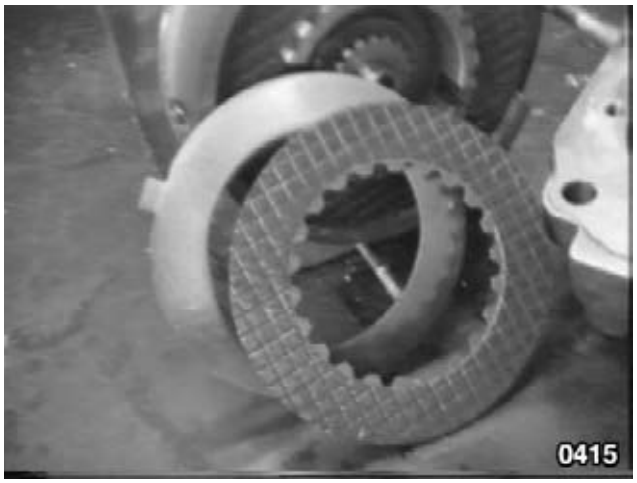


Fig. 28 - Inspect the clutch discs for wear.

Fig. 29 - Remove the circlip securing the hub of the LOW clutch.

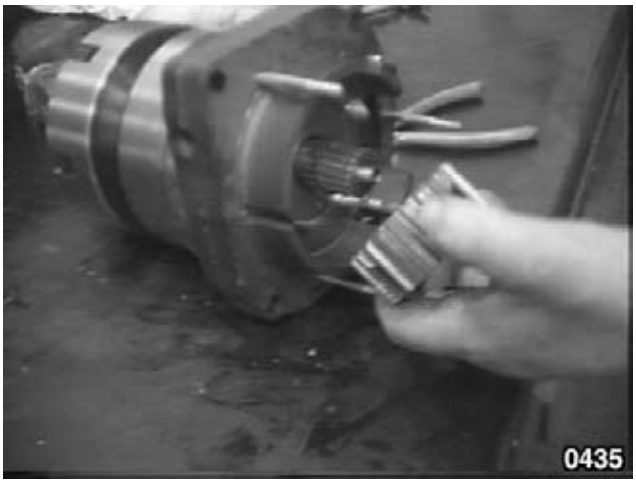


Fig. 30 - Remove the hub of the LOW clutch.

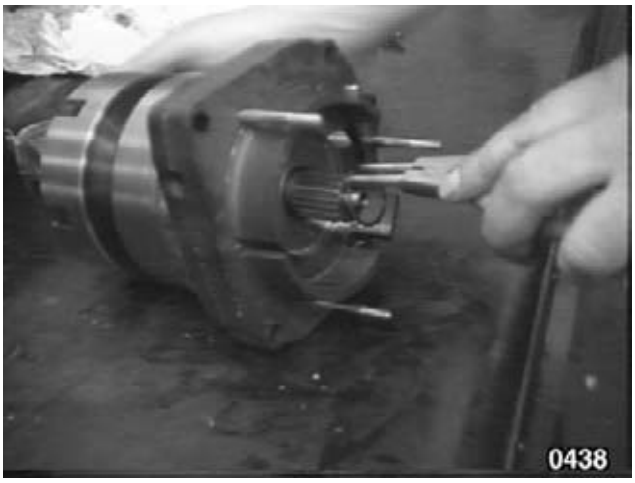


Fig. 31 - Remove the circlip located behind the hub of the LOW clutch.

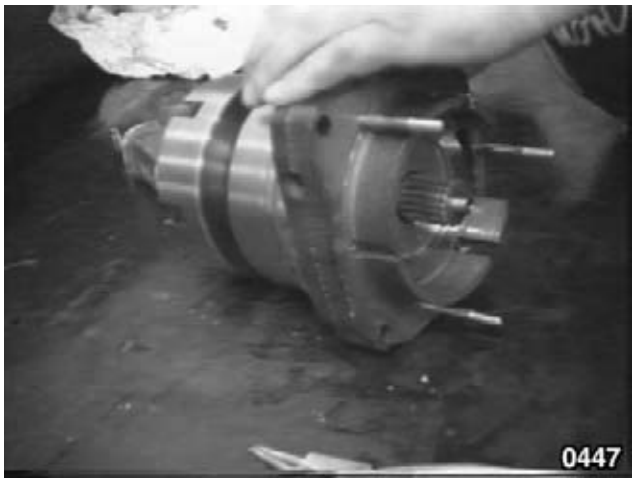


Fig. 32 - Proceed with the disassembly of the housing of the MEDIUM-LOW clutches.

Fig. 33 - Using a drift, separate the MEDIUM-LOW housing from the rest of the unit.

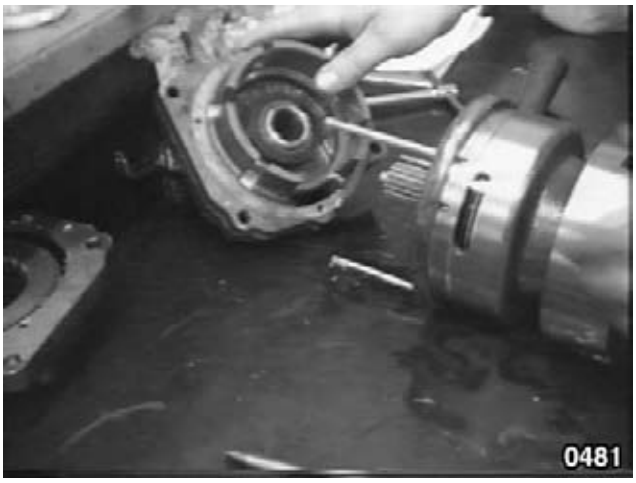


Fig. 34 - Detach the MEDIUM-LOW housing from the unit.

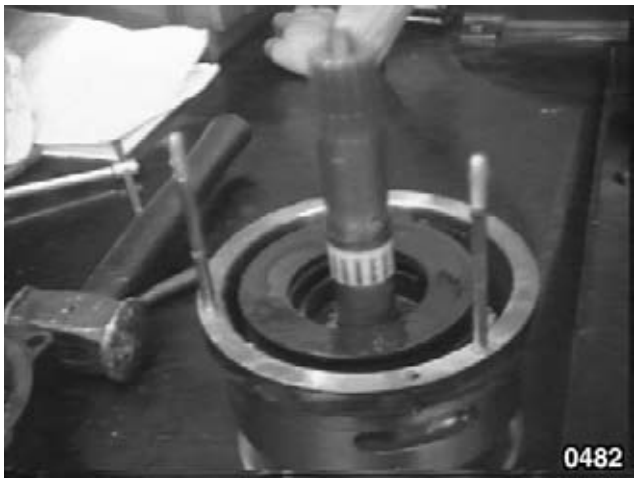


Fig. 35 - Remove the Belleville springs from the MEDIUM clutch.

Fig. 36 - Remove the discs from the MEDIUM clutch.

Fig. 37- MEDIUM clutch discs (4 friction discs+3 counter-discs+2 thrust plates).

Fig. 38 - The control piston of the LOW clutch may be removed from the “MEDIUM-LOW” bell-housing using a press of the type illustrated below.

Fig. 39 - This piston can be removed by removing the circlip of the piston return spring located on the opposite end of the housing.

Fig. 40 - Remove the LOW clutch shaft.

Fig. 41 - Check that the oil ways in the shaft are not obstructed.

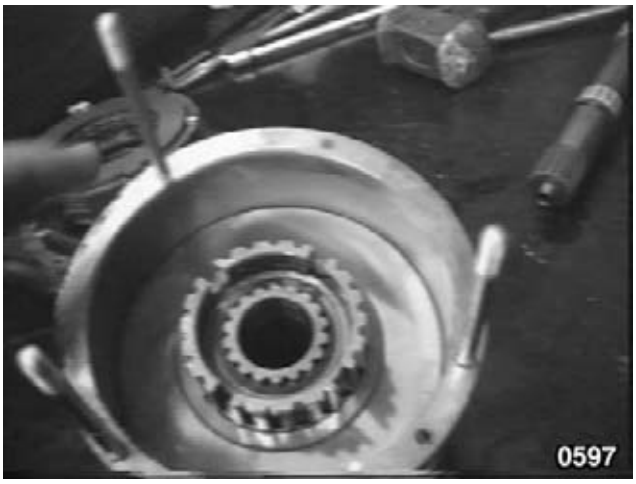


Fig. 42 - The LOW clutch control piston can be removed from the housing using a press of the type illustrated below.

Fig. 43 - This piston can be removed by removing the circlip of the piston return spring located on the opposite end of the housing.

Fig. 44 - Remove the circlip securing the HIGH clutch discs.

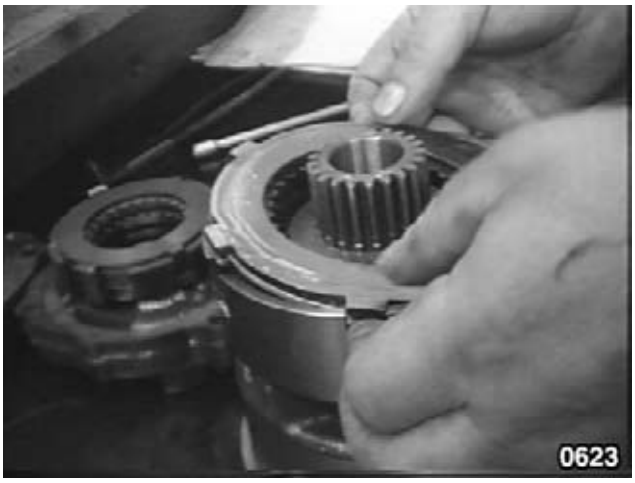


Fig. 45 - Remove the thrust washer of the HIGH clutch.

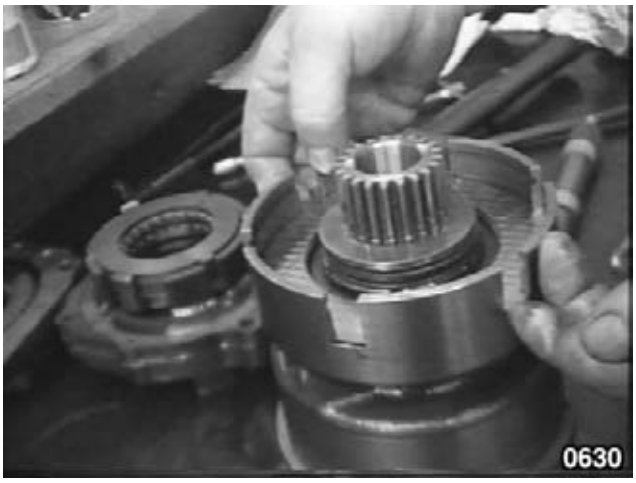


Fig. 46 - Remove the HIGH clutch disc pack (2 friction discs + 2 counterdiscs + 1 thrust plate).

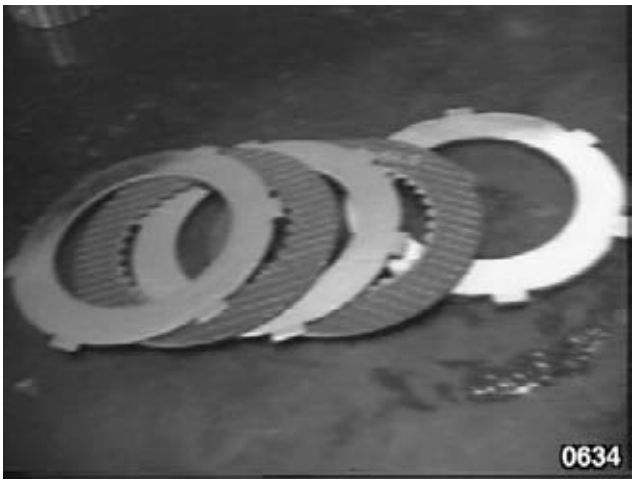


Fig. 47 - HIGH clutch discs.

Fig. 48 - HIGH clutch control piston.

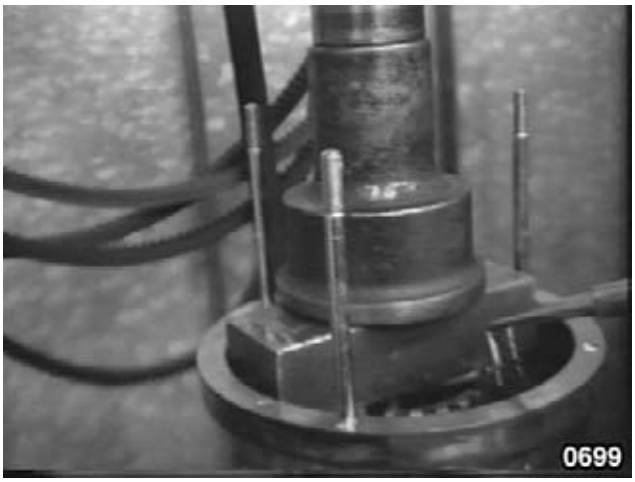


Fig. 49 - Separate the HIGH clutch housing from the MEDIUM clutch housing using a press to compress the springs and remove the circlip.

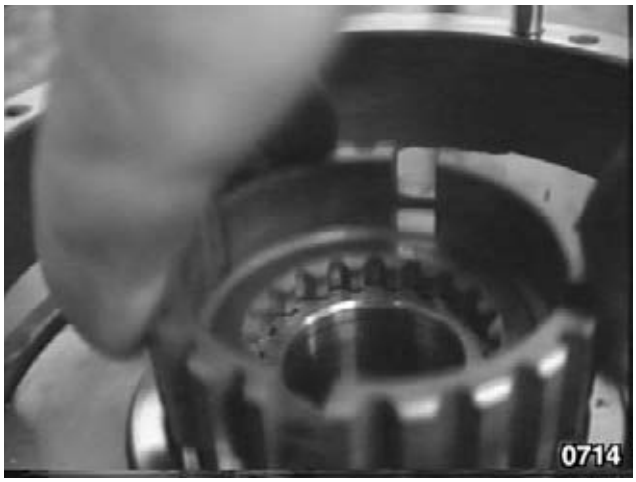


Fig. 50 - Remove the MEDIUM clutch hub and then the relative housing.

Fig. 51 - “ VESPEL” seals of the HIGH clutch housing.

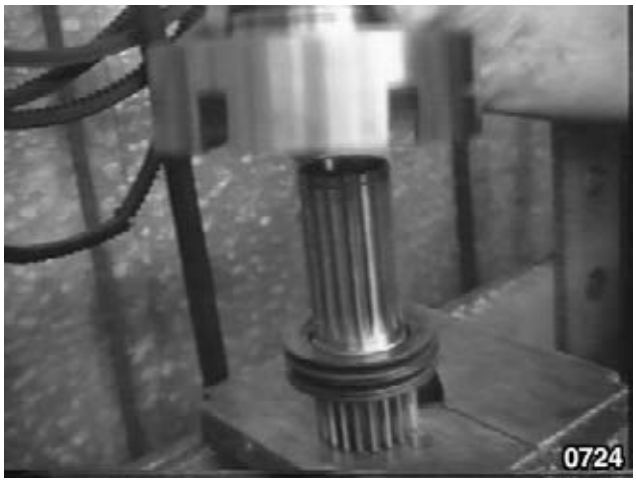


Fig. 52 - Remove the HIGH clutch housing.

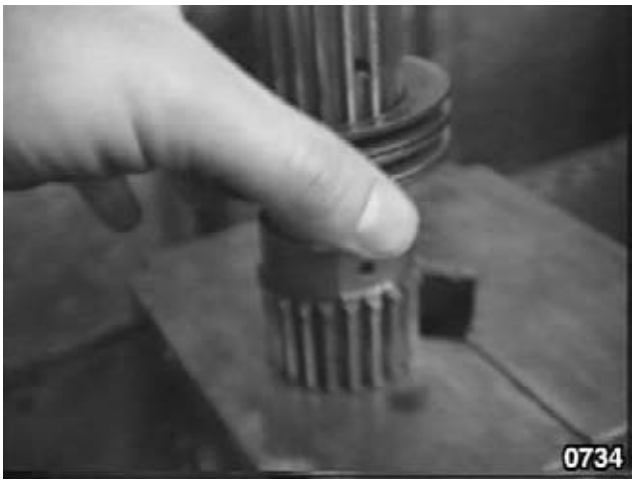


Fig. 53 - Remove the Belleville return springs of the HIGH clutch control piston from the MEDIUM shaft.

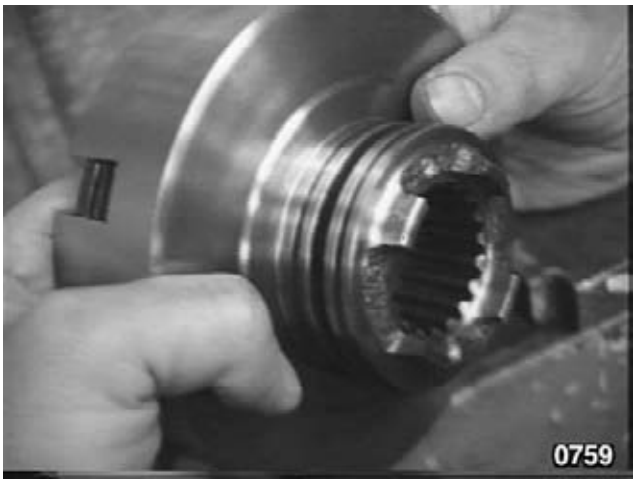


Fig. 54 - Examine the condition of the “VESPEL” seals on the HIGH clutch housing.

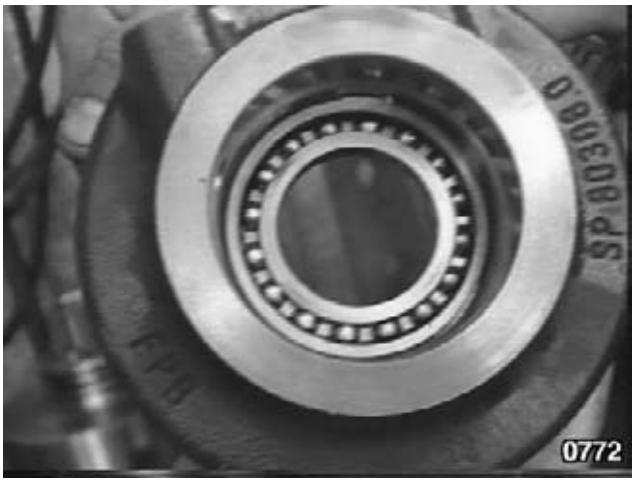


Fig. 55 - If necessary, remove the circlip and remove the bearing from the MEDIUM clutch housing.

Fig. 56 - Remove the piston from the MEDIUM clutch housing.

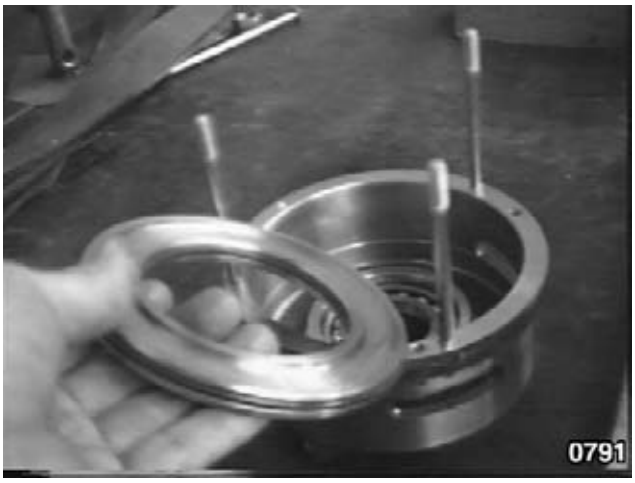


Fig. 57 - Examine the condition of the seal ring of the piston from the MEDIUM clutch housing.

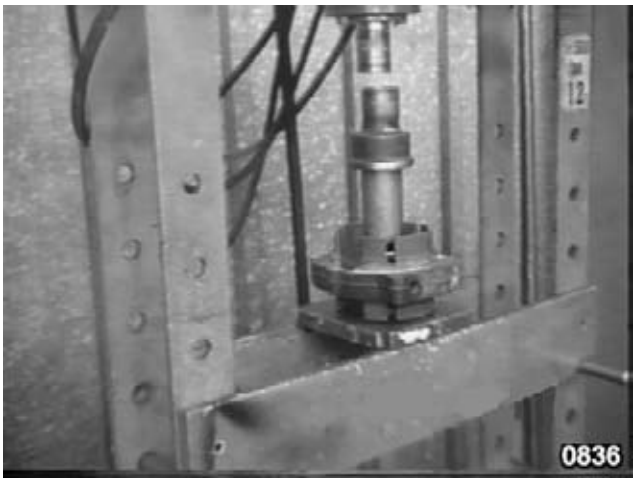


Fig. 58 - Proceed to dismantle the LOW clutch control piston with the aid of a press.

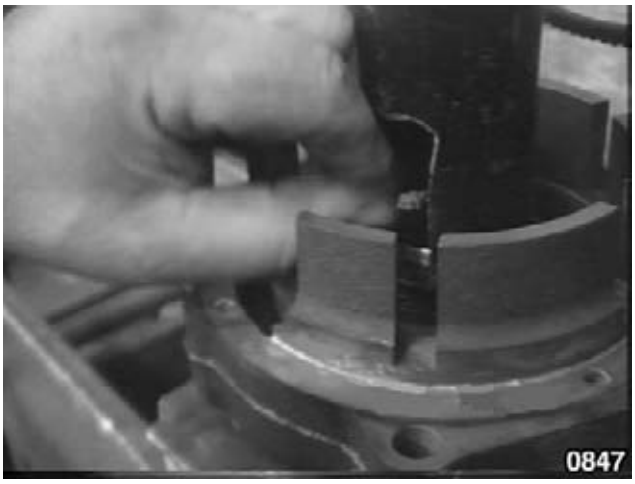


Fig. 59 - Remove the circlip securing the springs.

Fig. 60 - Remove the assembly from the press and remove the Belleville springs.

Fig. 61 - Disassembly sequence for the 5 Belleville washers.

Fig. 62 - Remove the piston.

Fig. 63 - Groove for O-ring in the LOW clutch housing.

Fig. 64 - Examine the condition of seal ring of the piston from the LOW clutch housing.

Clutch and transmission

Multispeed

A - Cover
B - LOW clutch
C - LOW clutch control piston
D - Clutch housing
MEDIUM - LOW
E - LOW clutch belleville springs
F - MEDIUM clutch belleville springs
G - Actuator disc
H - MEDIUM clutch
I - Seeger
L - Clutch hub
M - Thrust plate
N - Bearing
O - Thrust plate
P - MEDIUM clutch control piston
Q - HIGH clutch housing
R - HIGH clutch control piston
S - HIGH clutch
T - LOW sun wheel
U - Planet carrier
V - Thrust washer
Z - Ring gear
X - Planet gears
J - LOW shaft
Y - MEDIUM shaft
W - Secondary shaft
K - Reduction unit shaft

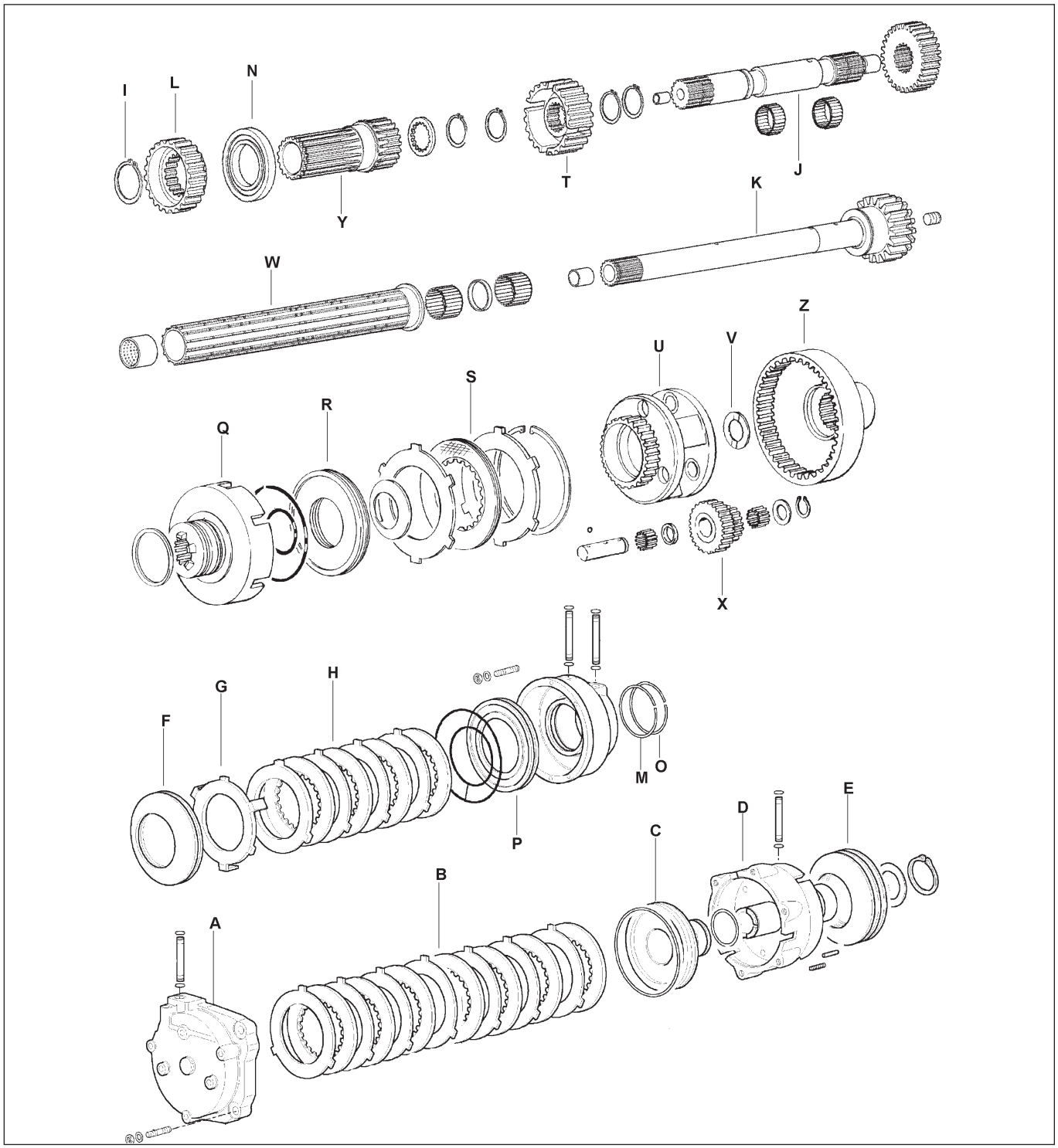


Fig. 65 - Multispeed unit

Assembly of MULTISPEED unit

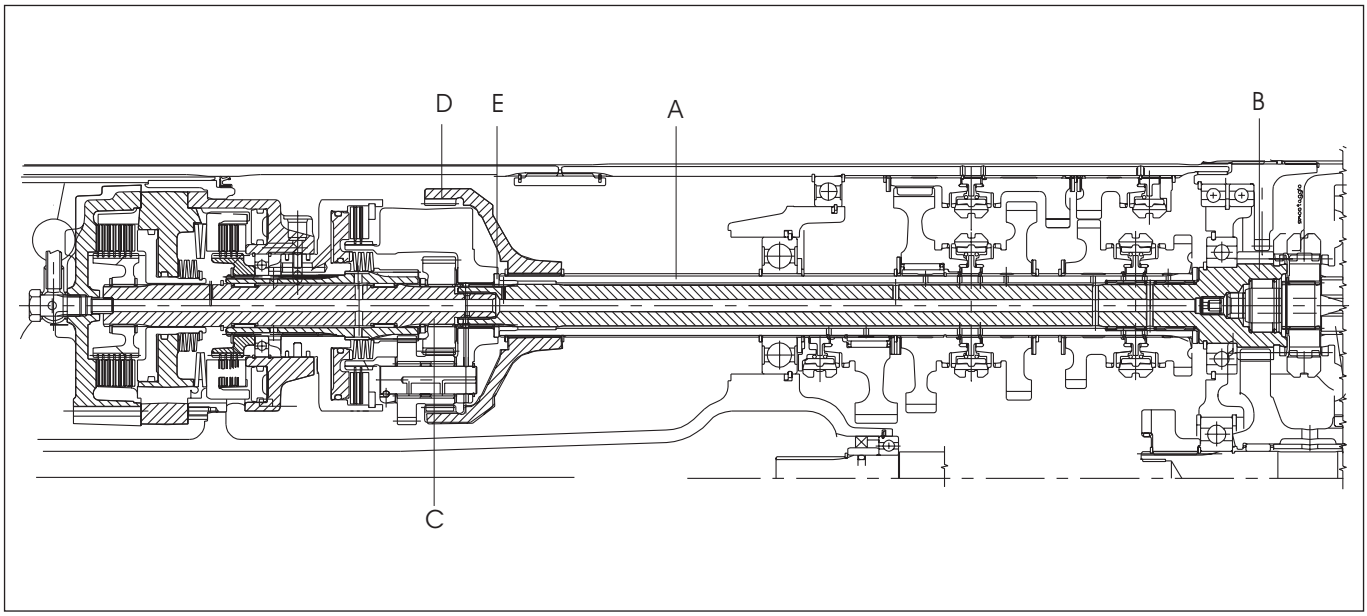


Fig. 66 - Components of Multispeed unit

The Multispeed unit requires that the basic gearbox, with respect top the standard version, is equipped with a **secondary shaft A** and a r**ange input shaft B**, of specific type (**A** = code 010.0307.3 and **B** = code 009.9925.3/30).

Warning: no special tools are required to re-assemble this unit, just an M8x1.25x30 screw to hold the central shaft **C** in position during **Multispeed** play adjustment.

Assemble the unit as follows:
— Fit the input ring gear **D** code 008.0266.0/30 to the secondary shaft, securing it with circlip **E**;
— Separately assemble LOW gear **F** 008.0311.0 with the planet carrier flange **G** code 008.0263.0. When fitting gear **F**, the planets must be positioned so that their punch marks are aligned with the position reference marks on the carrier flange, (Fig. 67).

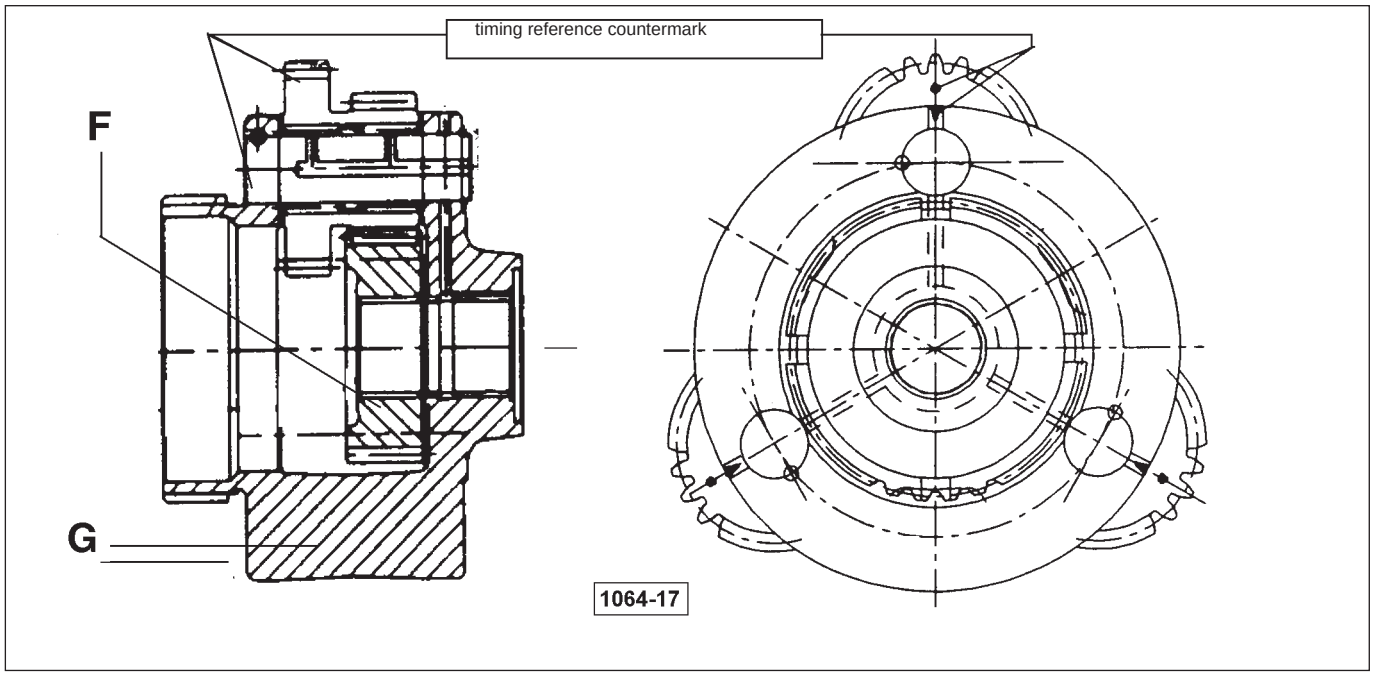
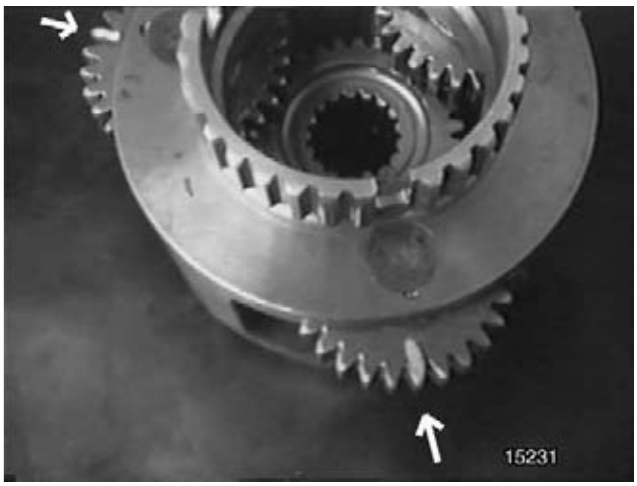
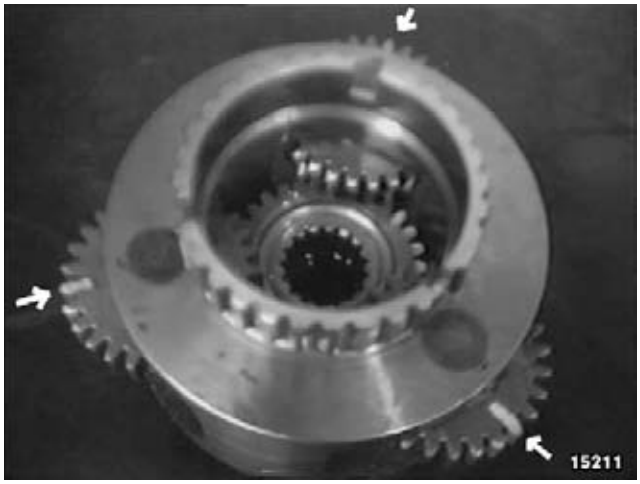


Fig. 67 - Timing the epicyclic speed reducer of the Multispeed unit.





Timing marks on gear teeth of the epicyclic reduction of the Multispeed unit.

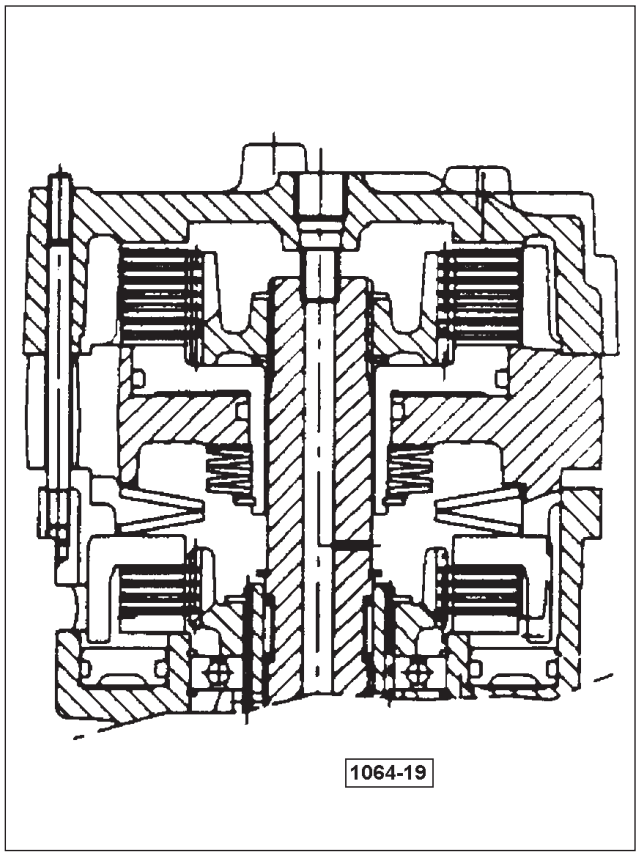


Fig. 68 - Pre-assembly of parts on MEDIUM shaft.

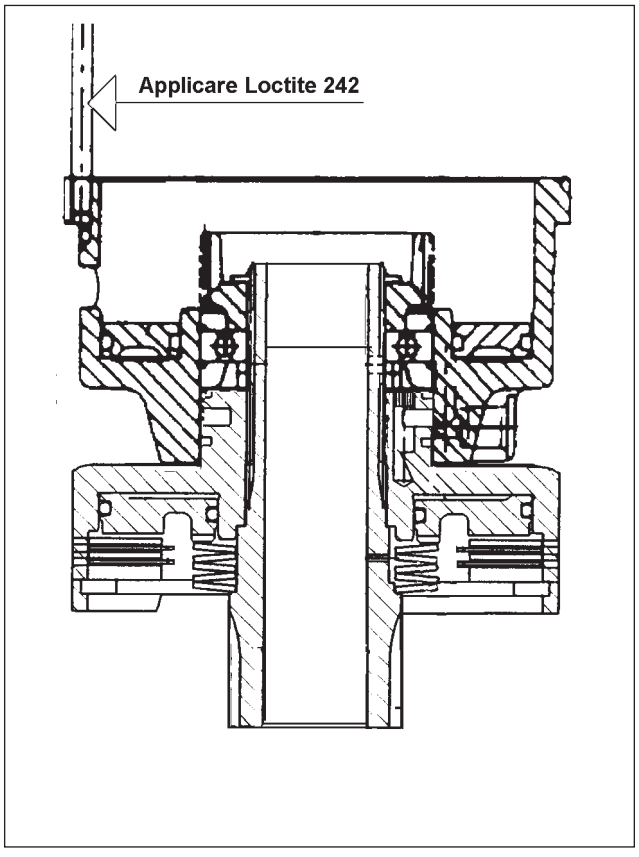


Fig. 69 - Pre-assembly of MEDIUM-LOW clutches.

Re-assembly of the Multispeed unit.

Fig. 70 - Assemble the HIGH clutch. Fit the bearing with the relative snap rings in the MEDIUM clutch housing; fit the hub securing it with the relative circlip and fit the three service circlips to the housing. Fit the O-ring seals to the piston , then fit the piston.

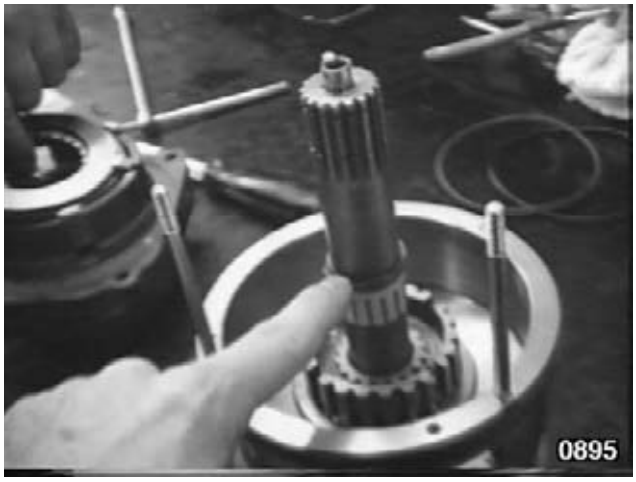


Fig. 71 - Fit the LOW shaft, the relative roller bearings, and position the snap ring.

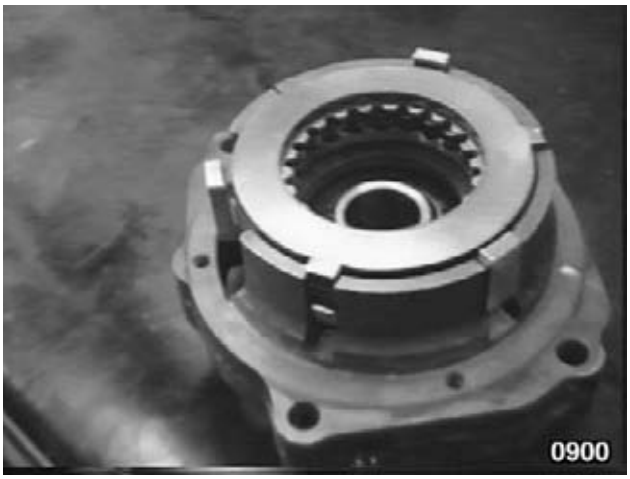


Fig. 72 - Fit the MEDIUM clutch discs.

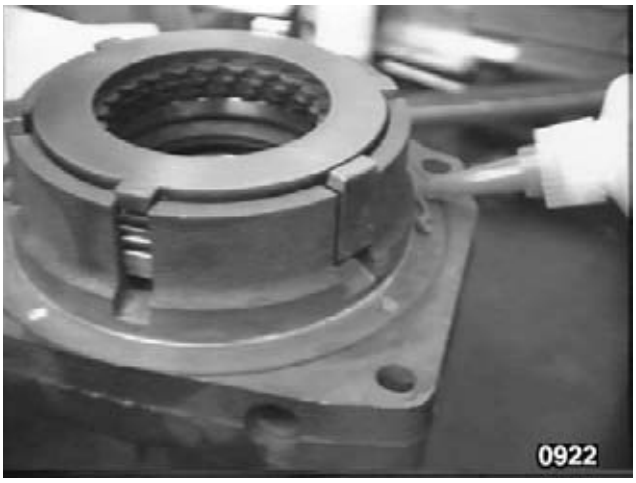


Fig. 73 - Apply Loctite 510 sealing compound to the housing contact surfaces as shown in the figure.
WARNING: when applying sealing compound take care not to obstruct the oil ways.

Fig. 74 - Fit the circlip and check that the shaft has certain amount of play.

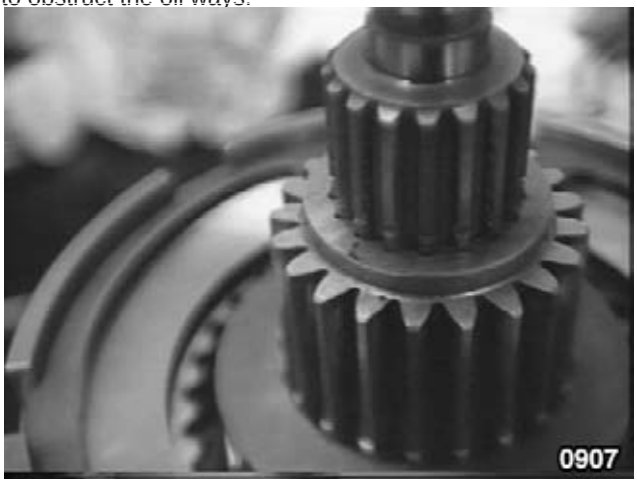


Fig. 75 - Fit the MEDIUM clutch thrust washer with outlets oriented towards the circlip.

Fig. 76 - Join the two housings together; the studs will locate in the 3 holes to ensure the correct relative positions of the two housings. Rotate the shaft during this operation so as to centre the clutch discs.

Fig. 77 - Note that a small space will be left between the two housings; this is to allow preloading of the Belleville springs when the screws are tightened.

Fig. 78 - Fit the LOW clutch, first positioning the circlip on the shaft as indicated in the figure.

Fig. 79 - Then fit the clutch hub and position the second circlip.

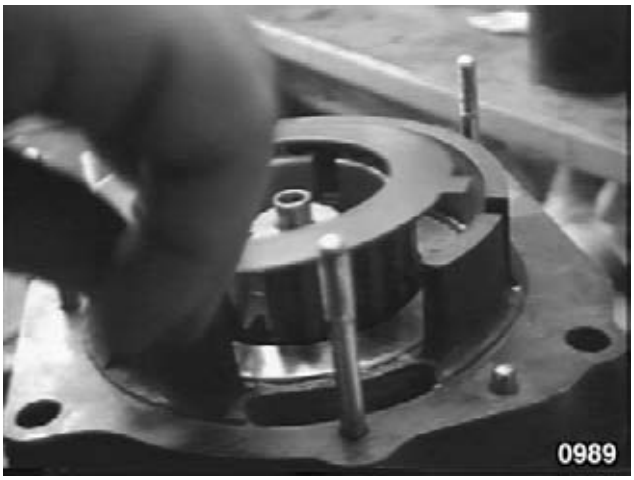


Fig. 80 - Then install the discs of the LOW clutch.

Fig. 81 - The disc pack is completed with a final friction disc, the tabs of which do not engage the splines on the hub, and therefore, when refitting the cover, it is advisable not to rotate the shaft so as to allow the disc tabs to mesh correctly with the hub.

ADJUSTMENT OF END FLOAT IN THE MULTISPEED UNIT (fig. 82)

Screw one M8x1,25x30 bolt ***finger tight*** into the cover **W** (fig 82), so as to hold the centre shaft of the H-M-L unit in position during the adjustment.

Measure dimensions **X** and **Y** with a gauge then calculate **(X - Y - 1 mm)** , which gives the thickness of the thrust washer **K** to fit.

Example: values measured **X = 191,88 Y = 188,24**

(191,88 - 188,24 - 1) = 2,64 ~ 2,6 mm (2.1599.727.0/10 washer required is p/n 2.1599.729.0/10)

IMPORTANT: the thrust washer must be selected from those available to give either the exact thickness, or the next thickness down from the value calculated.
washer p/n:

2.1599.725.0/10	thickness 2.2 mm
2.1599.726.0/10	thickness 2.4 mm
2.1599.727.0/10	**thickness 2.6 mm**
2.1599.728.0/10	thickness 2.8 mm
2.1599.729.0/10	thickness 3 mm
2.1599.730.0/10	thickness 3.2 mm
2.1599.731.0/10	thickness 3.4 mm
2.1599.732.0/10	thickness 3.6 mm
2.1599.733.0/10	thickness 3.8 mm
2.1599.734.0/10	thickness 4 mm

— Fit a shim of the predetermined thickness on the LOW shaft with the relative circlip
— Fit the epicyclic reduction unit in the gearbox and then fit the MULTISPEED clutches unit, applying Silastic sealing compound to the contact surfaces.

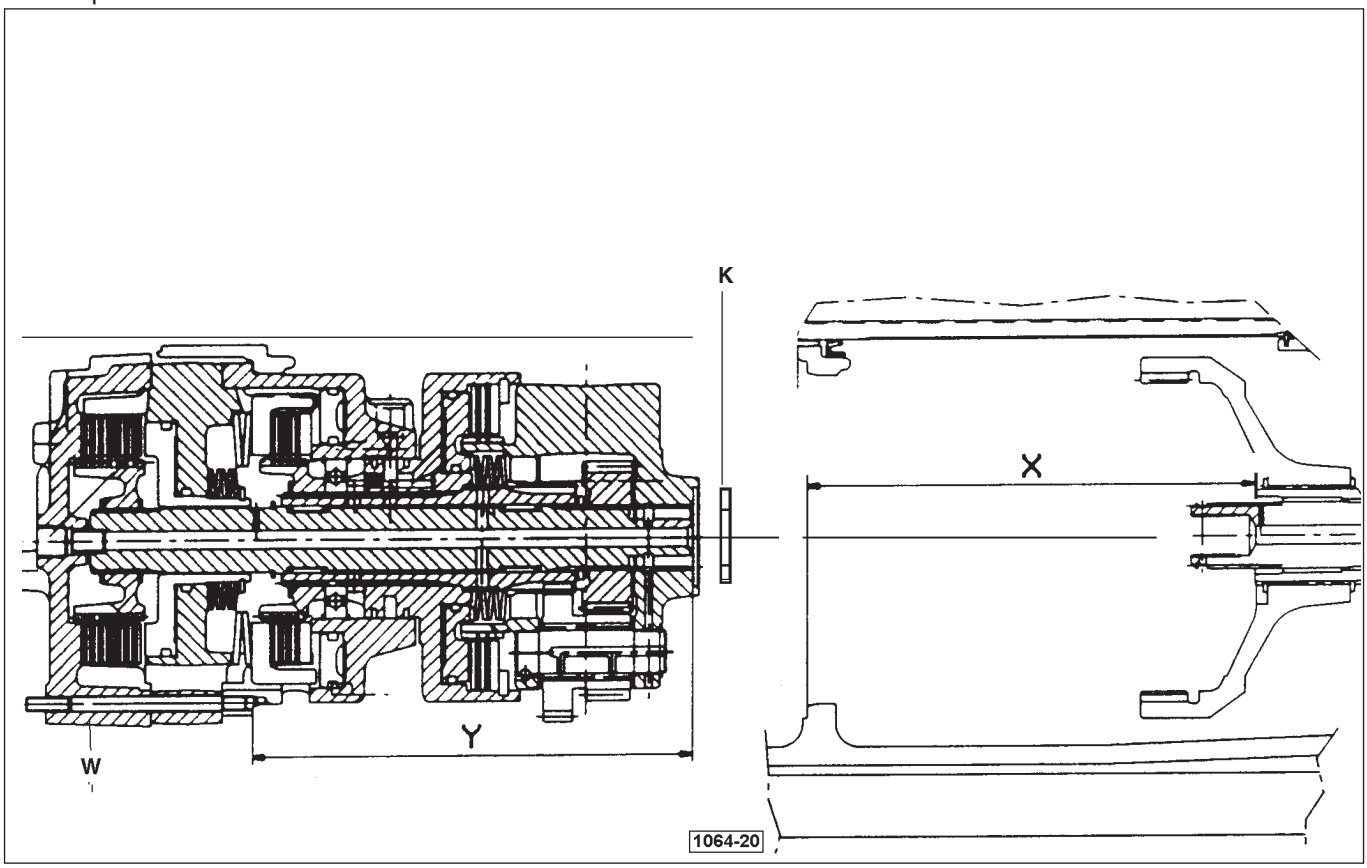
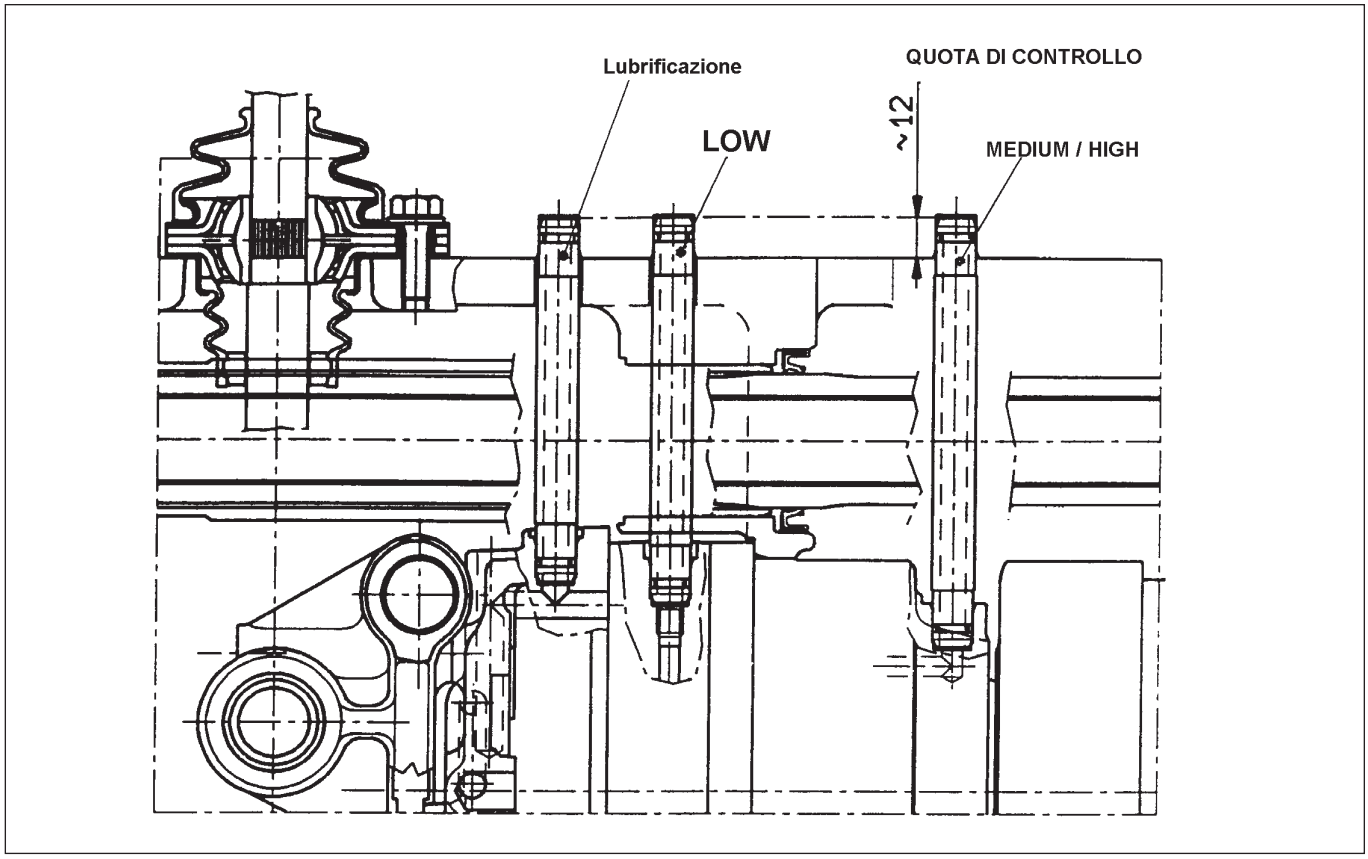
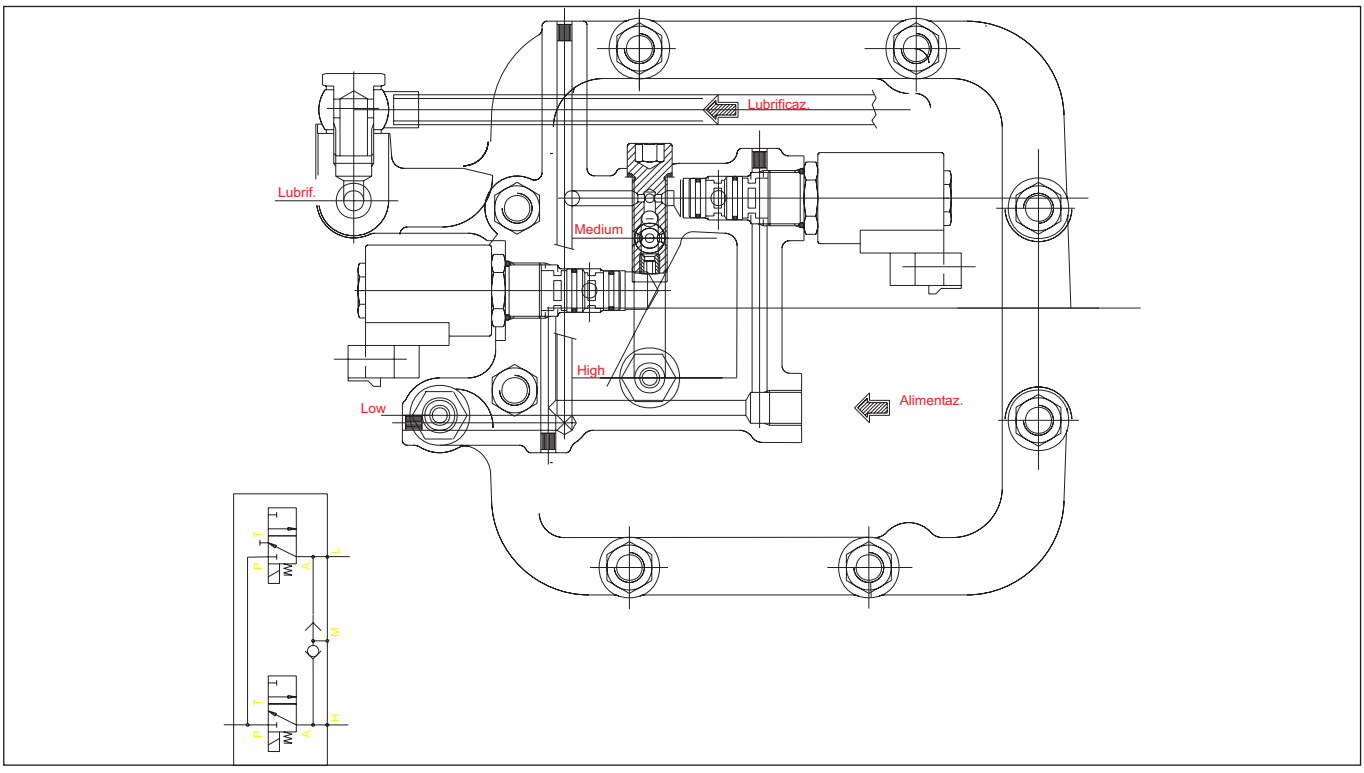


Fig. 82 - Location of the preassembled MULTISPEED unit in its housing.

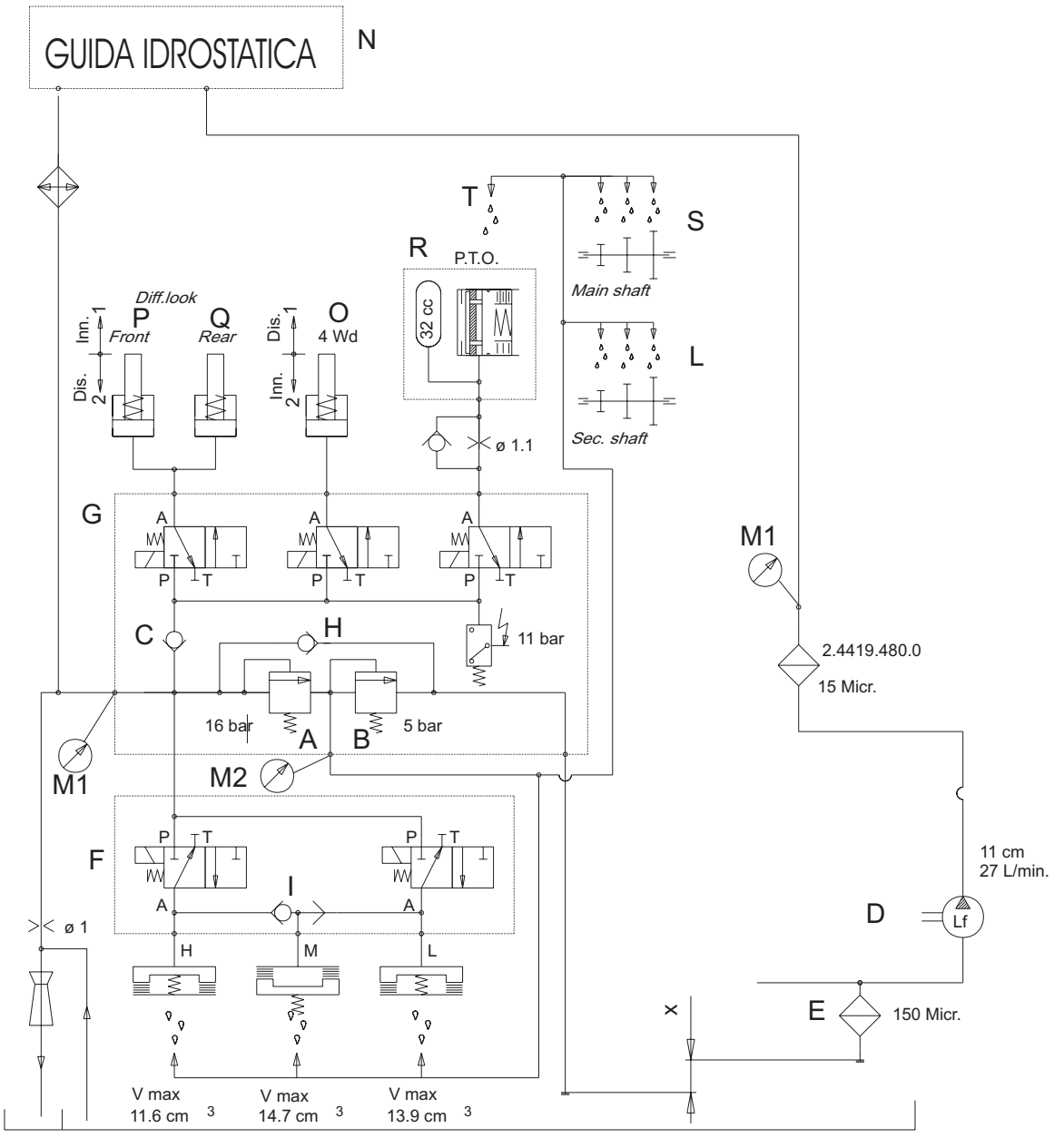
Fitting the oil manifolds of the Multispeed unit



Fit the manifolds in their respective seats so that they protrude 12 mm relative to the upper face of the gearbox, as indicated in the figure.



Hydraulic control circuit (version without hydraulic shuttle).



A pressure regulator valve (operating pressure 16 bar)
B gearbox lubrication oil pressure setting valve (setting 5 bar)
C anticavitation valve for safety steering (30 l/min.)
D oil pump (capacity 10 to 27 l/min.)
E suction filter
F solenoid control valves
G electro-hydraulic control unit
H anticavitation valve for safety steering
I bistable valve
L secondary shaft
M1 pressure gauge with scale 0 ÷ 40 bar
M2 pressure gauge with scale 0 ÷ 10 bar
N hydrostatic steering
O 4-wheel drive
1 - disengaged
2 - engaged
P front differential lock
1 - disengaged
2 - engaged
Q rear differential lock
1 - disengaged
2 - engaged
R P.T.O.
S primary shaft
T lubrication

Fig. 83 - Hydraulic diagram illustrating operation.

Clutch and transmission

Multispeed

PROCEDURE FOR CHECKING THE H-M-L CONTROL UNIT INLET PRESSURE

1 - Connect pressure gauge 5.9030.514.0 (0 ÷ 40 bar) at position M1.
2 - Connect pressure gauge 5.9030.515.0 (0 ÷ 10 bar) at position M2.
Start the engine and throttle up to the revolutions indicated in the table; now verify that the inlet pressures are as prescribed.
NB: Pressure tests should be conducted using oil at a temperature of 80 °C.

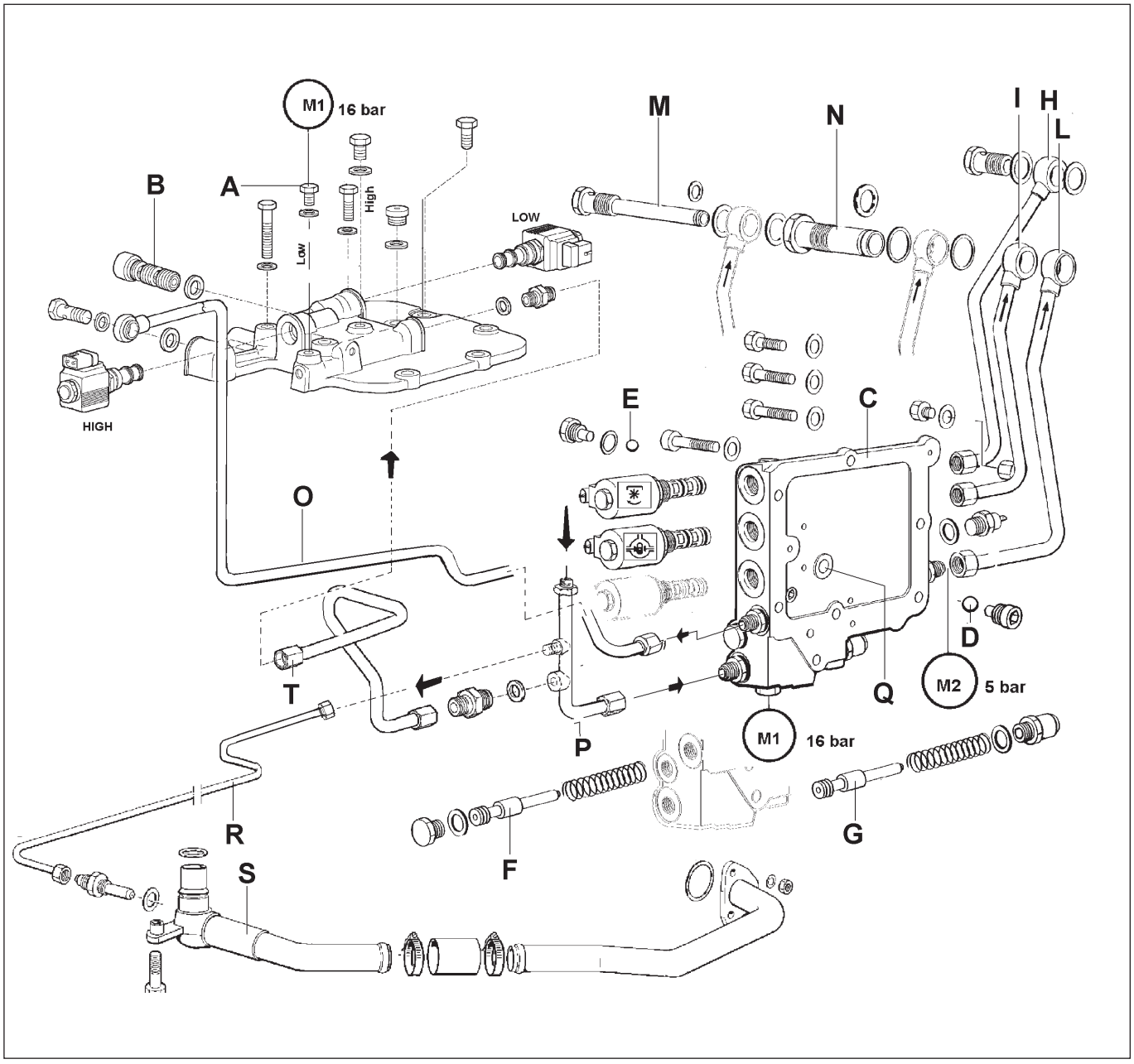
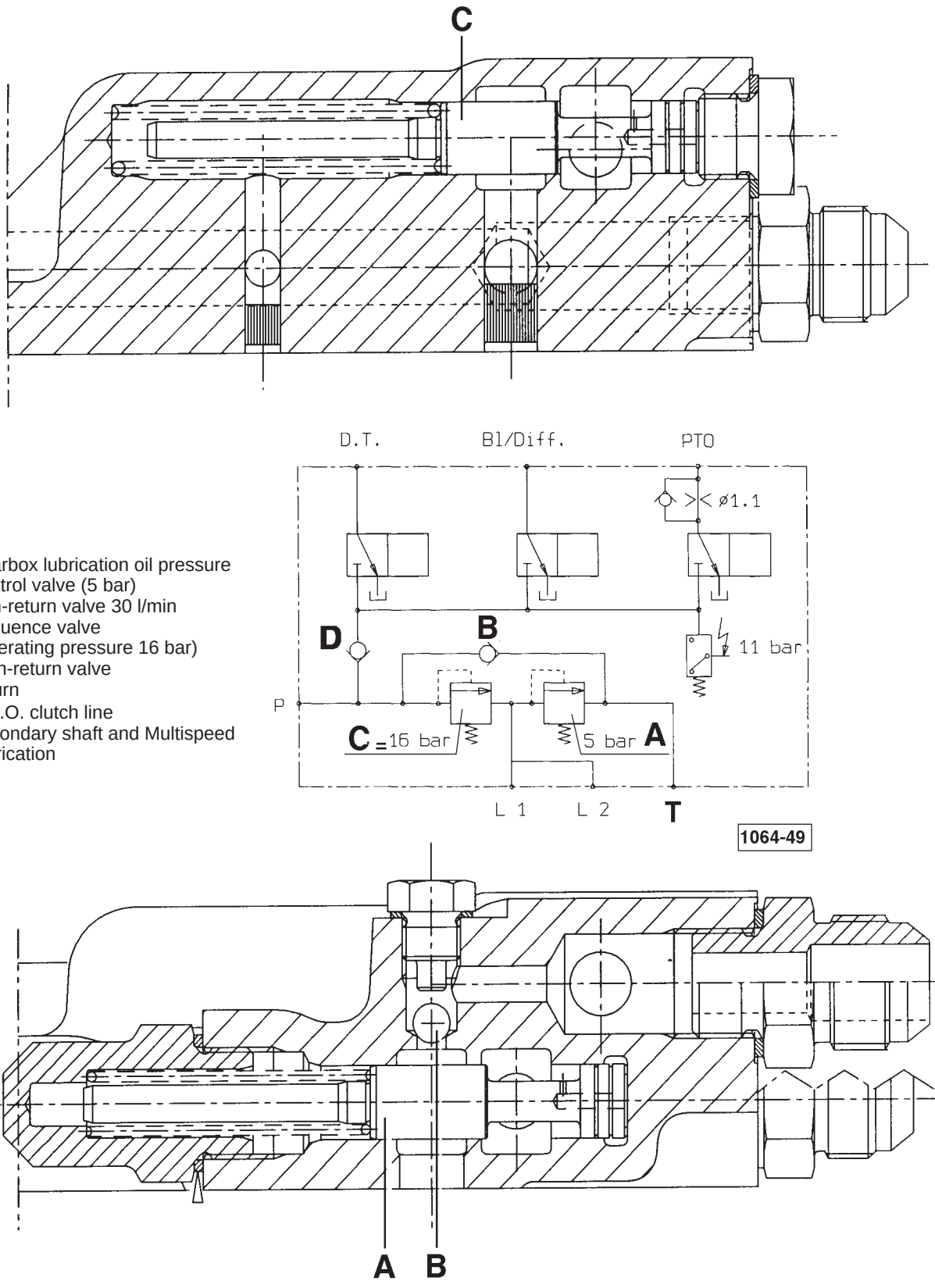


Fig. 84 - Electrohydraulic control unit for MULTISPEED system.

A Test fitting for testing pressure of LOW clutch hydraulic circuit
B Bistable valve
C Valve body
D Non-return valve
E Non-return valve
F Oil pressure regulator valve
G Lubrication pressure control valve
H Delivery line to P.T.O.
I Delivery line to differential lock
L P.T.O. lubrication pipe
M P.T.O. disk lubrification manifold
N Manifold for P.T.O. clutch oil pipe
O Secondary shaft and Agroshift lubrification pipe
P Pipe from radiator oil to the gearbox
Q Delivery line to front-wheel drive engagement piston
R Hydraulic unloader
S Gearboxes oil levelling pipe
T Multispeed oil delivery pipe
M1 Pressure gauge - scale 0 to 40 bar
M2 Pressure gauge scale 0 to 10 bar



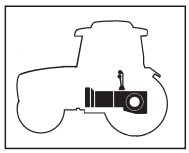
A gearbox lubrication oil pressure control valve (5 bar)
B non-return valve 30 l/min
C sequence valve (operating pressure 16 bar)
D non-return valve
T return
L1 P.T.O. clutch line
L2 secondary shaft and Multispeed lubrication

Fig. 85 - MULTISPEED pressure control valves assembly

Clutch and transmission

Diagnosing malfunctions

clutch slips	check the clutch assembly hydraulic system	replace the defective parts		
		examine piston and manifold seal rings	replace defective parts	
	check piston for free stroke	remove any roughness preventing piston from sliding freely		
	check the clutch plates for wear	replace the clutch disks		
clutch won't disengage	inspect the control mechanism	replace defective parts		
	inspect the disks	burnt disks	replace the disks	
		jammed disks		remove any roughness preventing disks from sliding freely
	piston jammed in seat	relpace the piston		



General specifications

The basic transmission utilizes 5 speed synchromesh gearbox with 2 or 3 speed ranges.

A shuttle, fitted as standard, makes the tractor easy to handle and manoeuvre.

Options include Creeper, and Underdrive (in place of or in addition), giving a total of:

For tractors equipped with driving position with footplates only
16 FWD + 8 REV: 4 gears x 2 ranges + mini-reduction + reverser shuttle
24 FWD + 12 REV: 4 gears x 3 ranges + mini-reduction + reverser shuttle

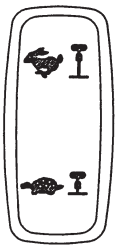
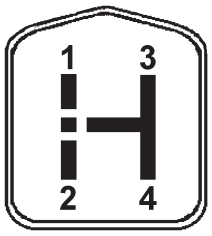
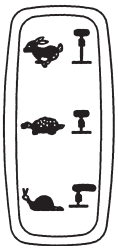
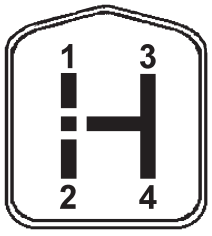
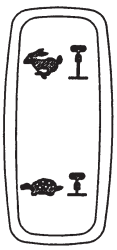
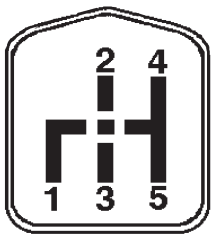
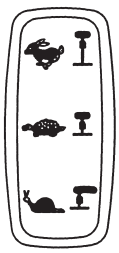
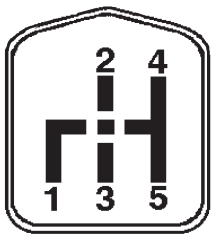
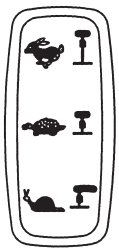
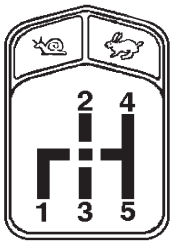
For tractors equipped with platform or cab
20 FWD + 10 REV: 5 gears x 2 ranges (L-V) + mini-reduction + reverser shuttle
30 FWD + 15 REV: 5 gears x 3 ranges (SR-L-V) + mini-reduction + reverser shuttle.
45 FWD + 45 REV: 5 gears x 3 ranges (SR-L-V) + reverser shuttle + MULTISPEED

Each version is also available with 30 or 40 km/h top speed.

Technical specifications

number of speeds		16 AV + 8 RM	20 AV + 10 RM
gearbox ratios			
	1st gear	19/59 = 1/3,1053	19/59 = 1/3,1053
	2nd gear	24/47 = 1/1,9583	23/50 = 1/2,1739
	3rd gear	34/42 = 1/1,2353	29/44 = 1/1,5172
	4th gear	41/32 = 1/0,7805	35/38 = 1/1,0857
	5th gear	-	42/30 = 1/0,7143
shuttle:	forward	29/37 = 1/0,7838	
	reverse	23/19 = 1/0,8261	
range ratios:	low	(19/50)x(22/46)= 1/5,5024	
	high	1	
rear axle ratios			
bevel gear pair	30 km/h	8/36 = 1/4,5000	-
	40 km/h	12/45 = 1/3,7500	12/45 = 1/3,7500
- with tyres 380/79R20 and 240/70R16	30 km/h	-	-
- with tyres 380/79R20 and 240/70R16	40 km/h	9/32 = 1/3,5556	9/32 = 1/3,5556
epicyclic final drive		(12/12+69) = 1/6,7500	
overall ratio	30 km/h	1/30,3750	-
	40 kmIh	1/25,3125	1/25,3125
backlash between crownwheel and pinion teeth	mm	0,18 - 0,24	
Mini-reduction		32/34 = 1/0,9412	
Super-reduction		(19/50)x(22/46)x(19/50)x(20/48) = 1/34,7520	
axial play of the secondary shaft	mm	0,09	
axial play of the gears of the change, of the reverser and mini-reduction	mm	0,15 ÷ 0,30	
reference distance for adjustment of bevel gear pair	mm	2,5 ± 0,1	

Speed change configurations

Speed range Control	Gearshifting Control	Reversing Control	NO. OF SPEED (forward + reverse)
			***16 + 8***
			***24 + 12***
			***20 + 10***
			***30 + 15***
			***45 + 45***

DURING TRANSPORT OPERATIONS ON PUBLIC ROADS BOTH BRAKE PEDALS MUST BE OBLIGATORILY COUPLED BY MEANS OF THE SPECIAL LOCKING LATCH. IN ANY CASE, ALWAYS OBSERVE LAWS AND REGULATIONS IN FORCE.

* see technical characteristics of gearbox on p. 143
** see technical characteristics of 4wd front axle on p. 217

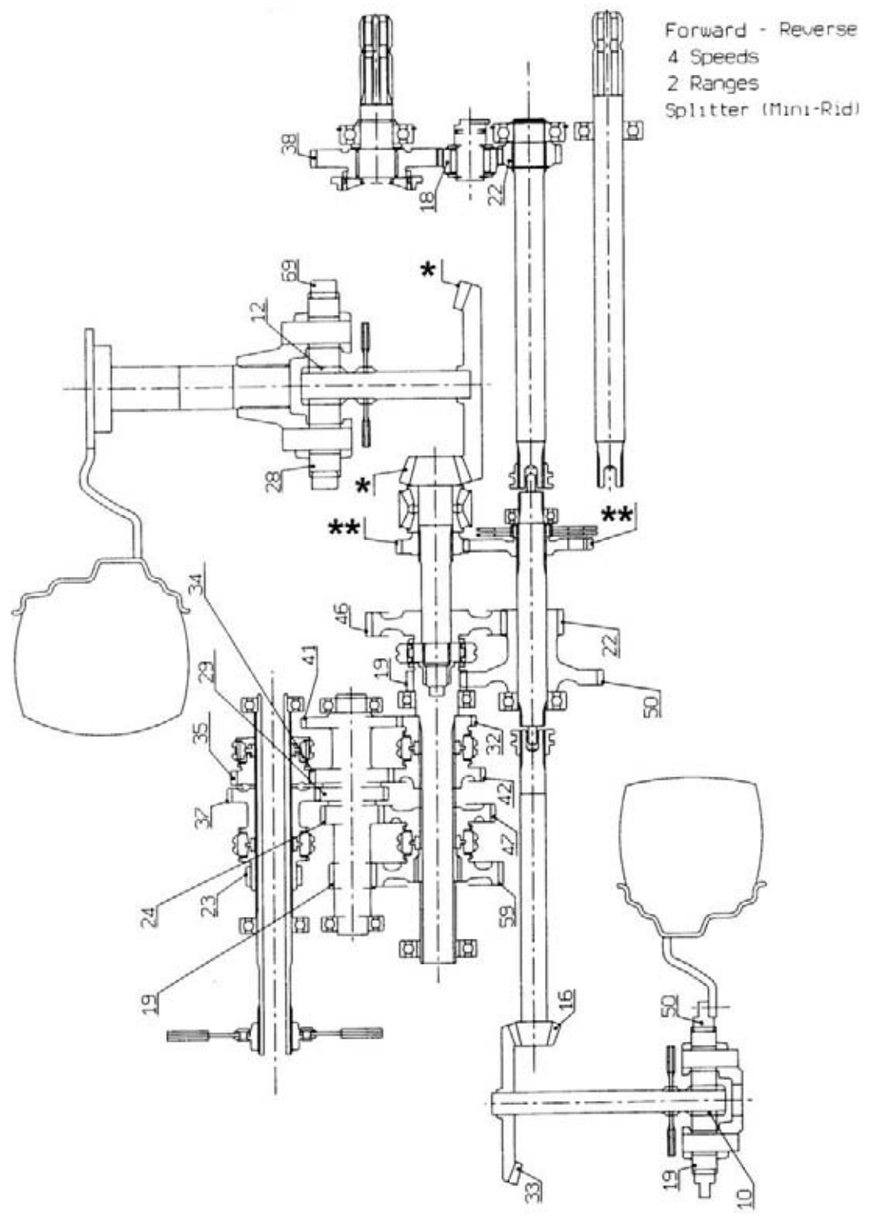
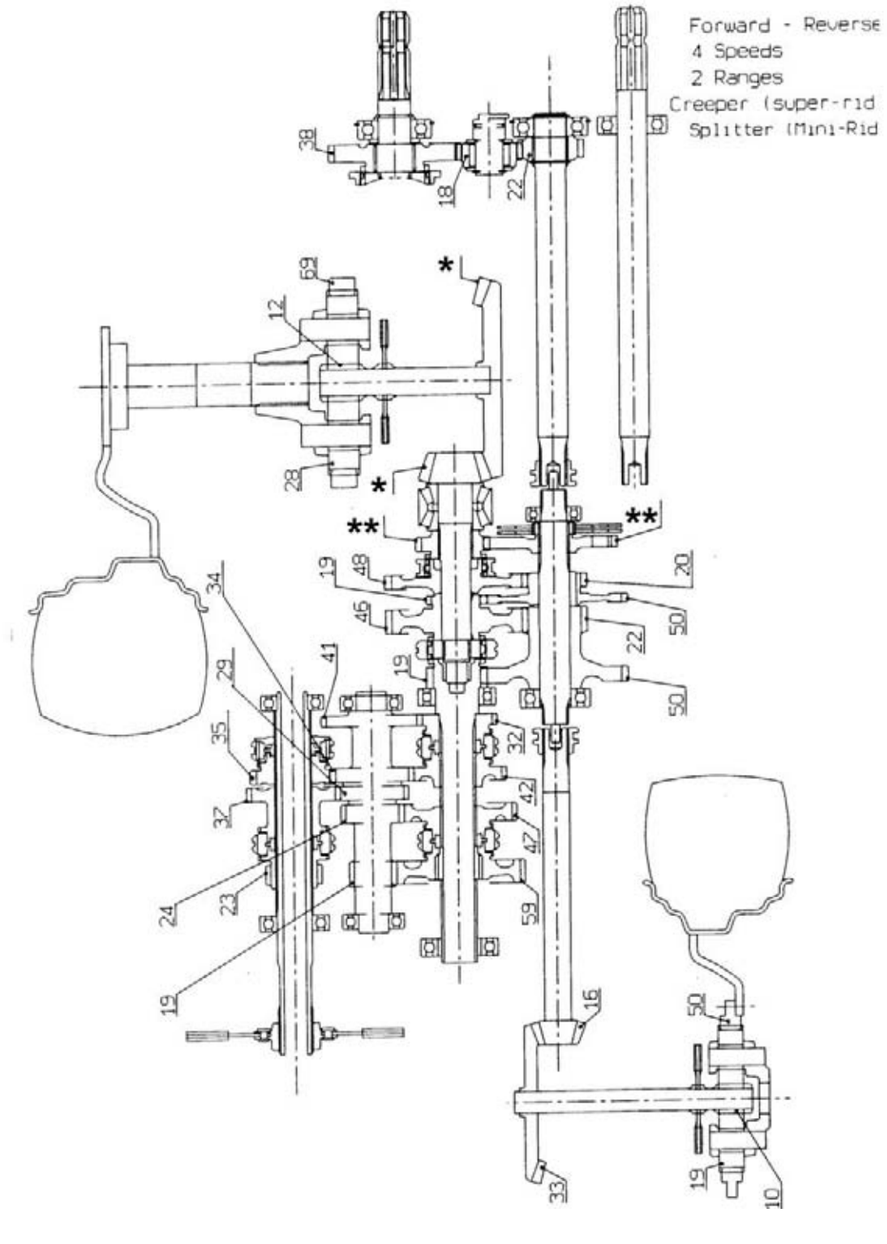


Fig. 1 - Ratios of 4-speed x 2 range transmission with mini-reduction (16 FWD + 8 REV).

* see technical characteristics of gearbox on p. 143
** see technical characteristics of 4wd front axle on p. 209



Ratios of 4-speed x 2 range transmission with mini-reduction and super-reduction (12 FWD + 12 REV).

* see technical characteristics of gearbox on p. 139
** see technical characteristics of 4wd front axle on p. 217

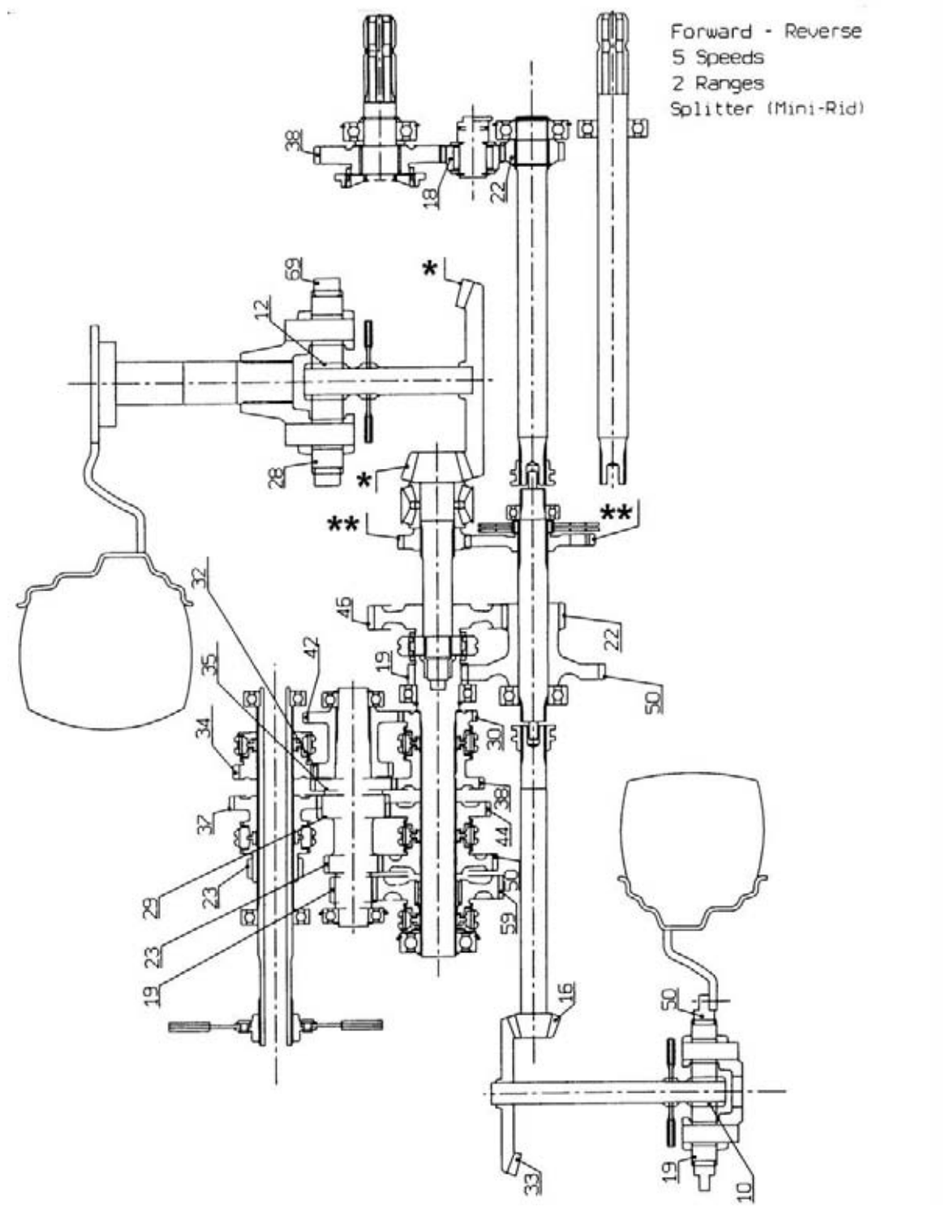
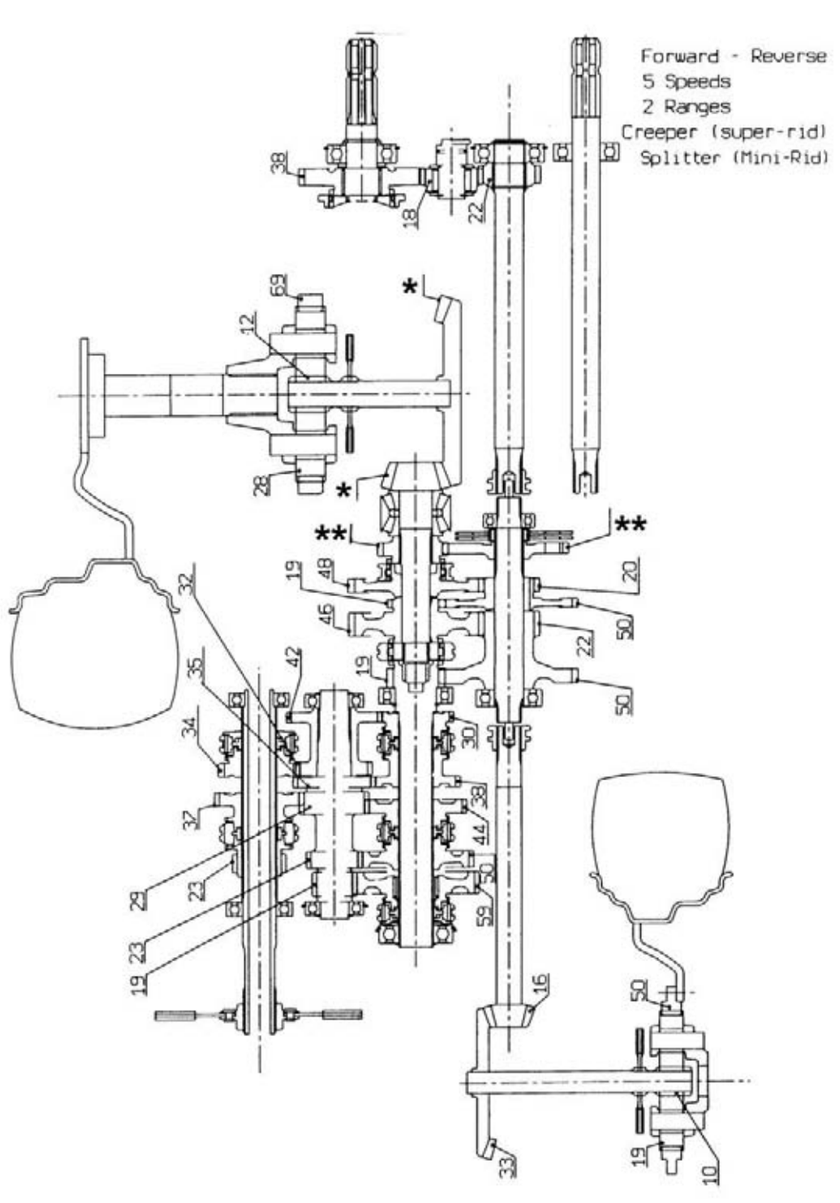


Fig. 2 - Ratios of 5-speed x 2 range transmission with mini-reduction (20 FWD + 10 REV).

* see technical characteristics of gearbox on p. 143
** see technical characteristics of 4wd front axle on p. 209



Ratios of 5-speed x 2 range transmission with mini-reduction and super-reduction (30 FWD + 15 REV).

* see technical characteristics of gearbox on p. 143
** see technical characteristics of 4wd front axle on p. 217

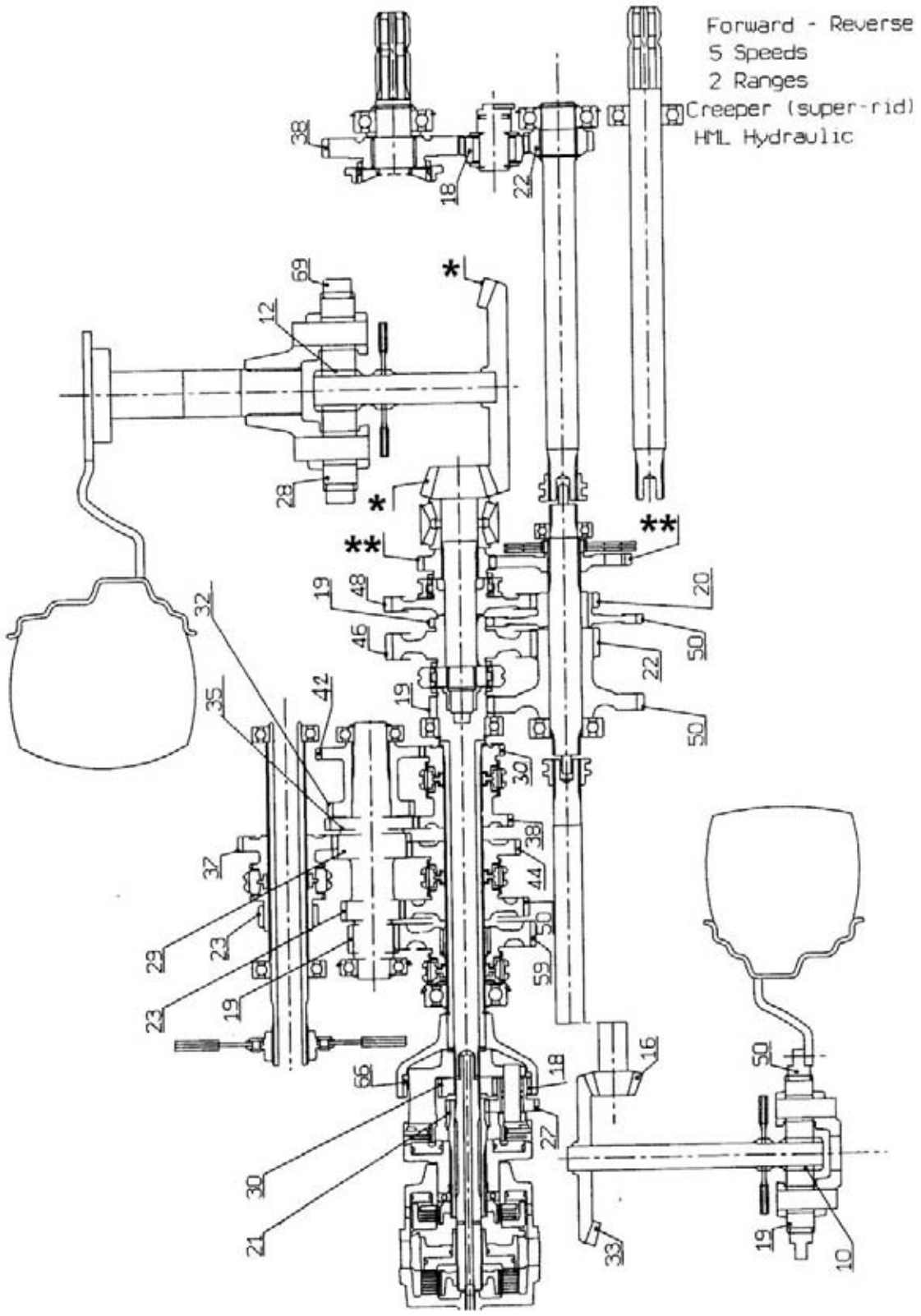
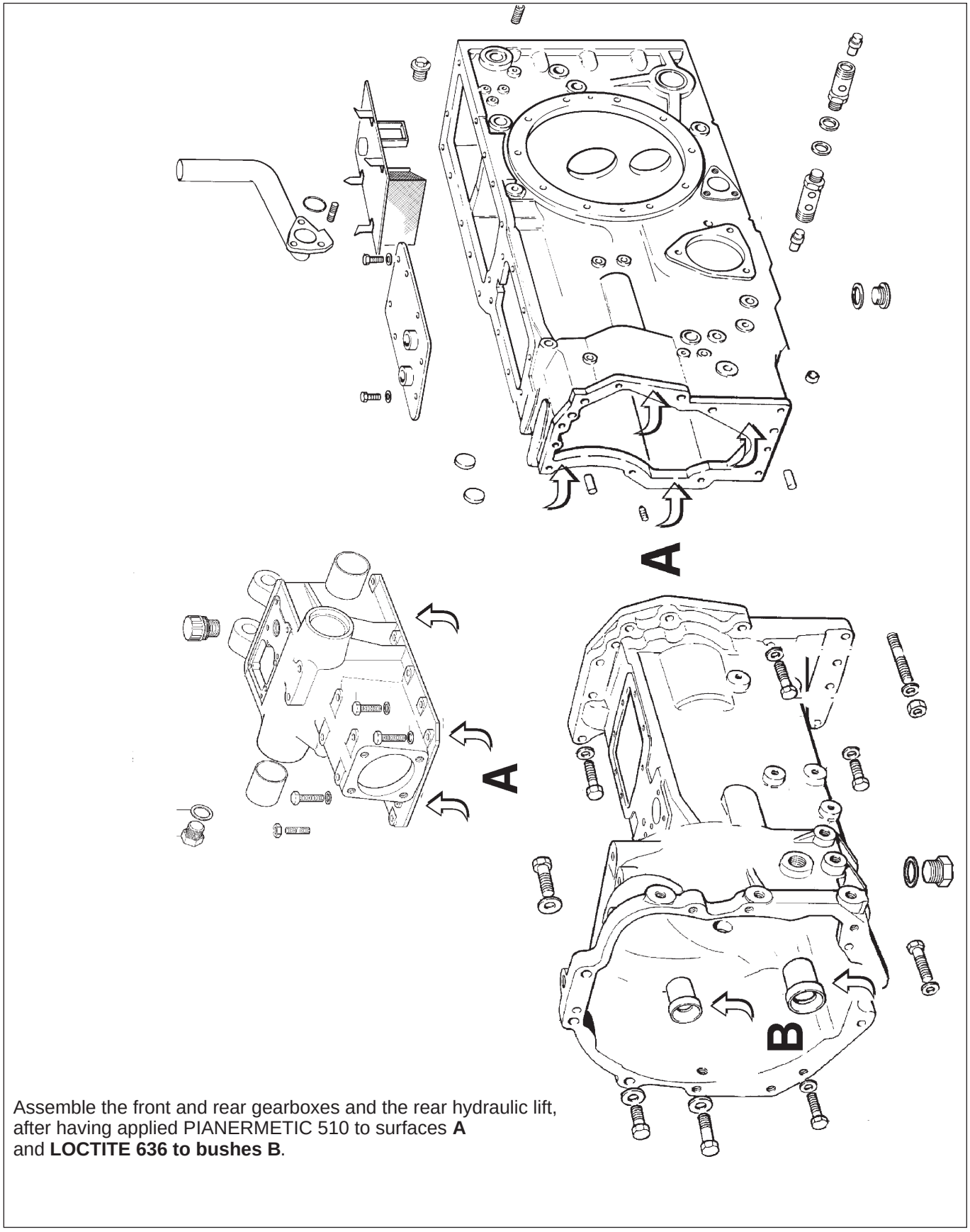


Fig. 3 - Ratios of transmission with 3 ranges (Super-reduction) + MULTISPEED (45 FWD + 45 REV).



Assemble the front and rear gearboxes and the rear hydraulic lift, after having applied PIANERMETIC 510 to surfaces **A** and **LOCTITE 636 to bushes B**.

Fig. 4 - Front and rear gearbox.

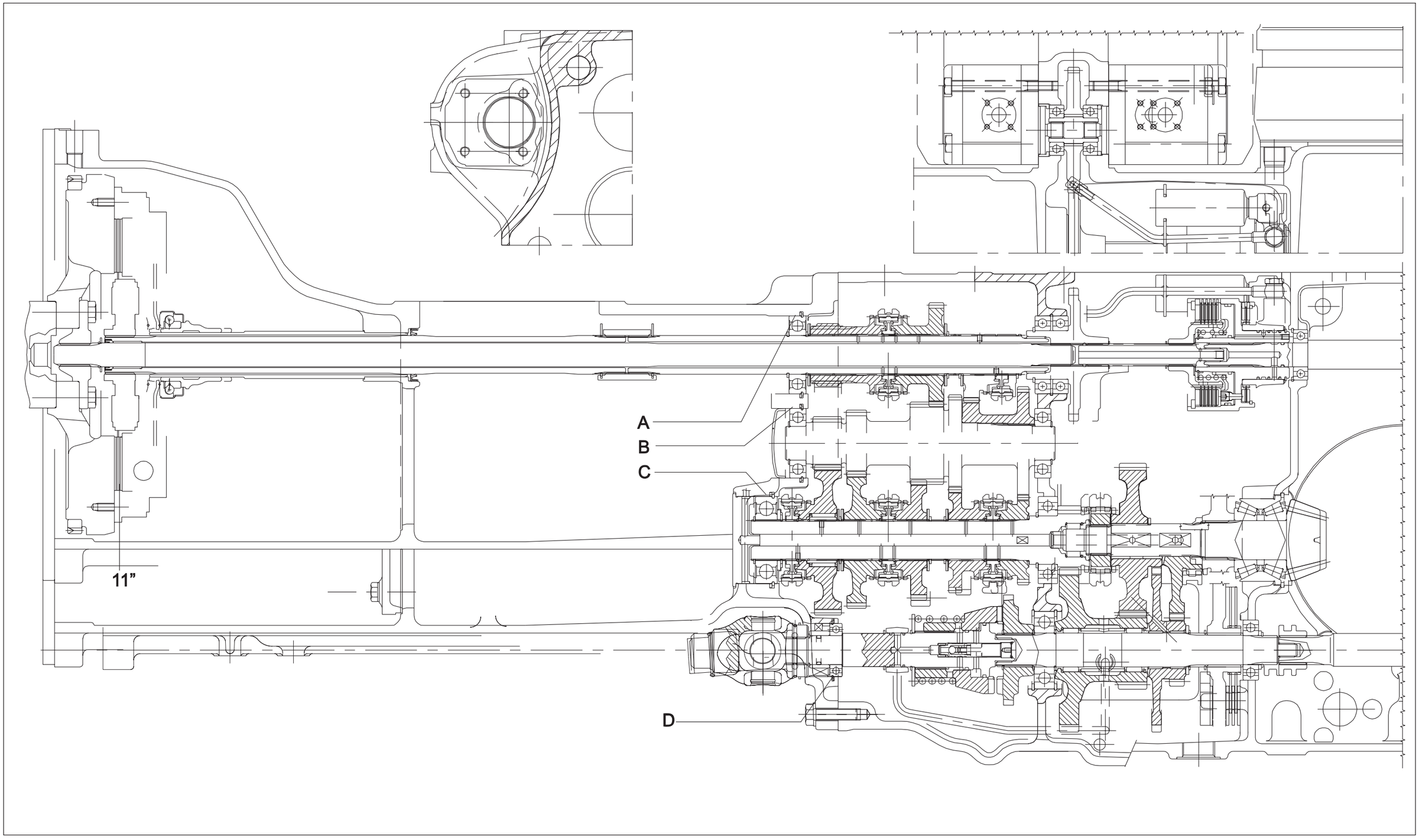


Fig. 5 - Longitudinal section through transmission.

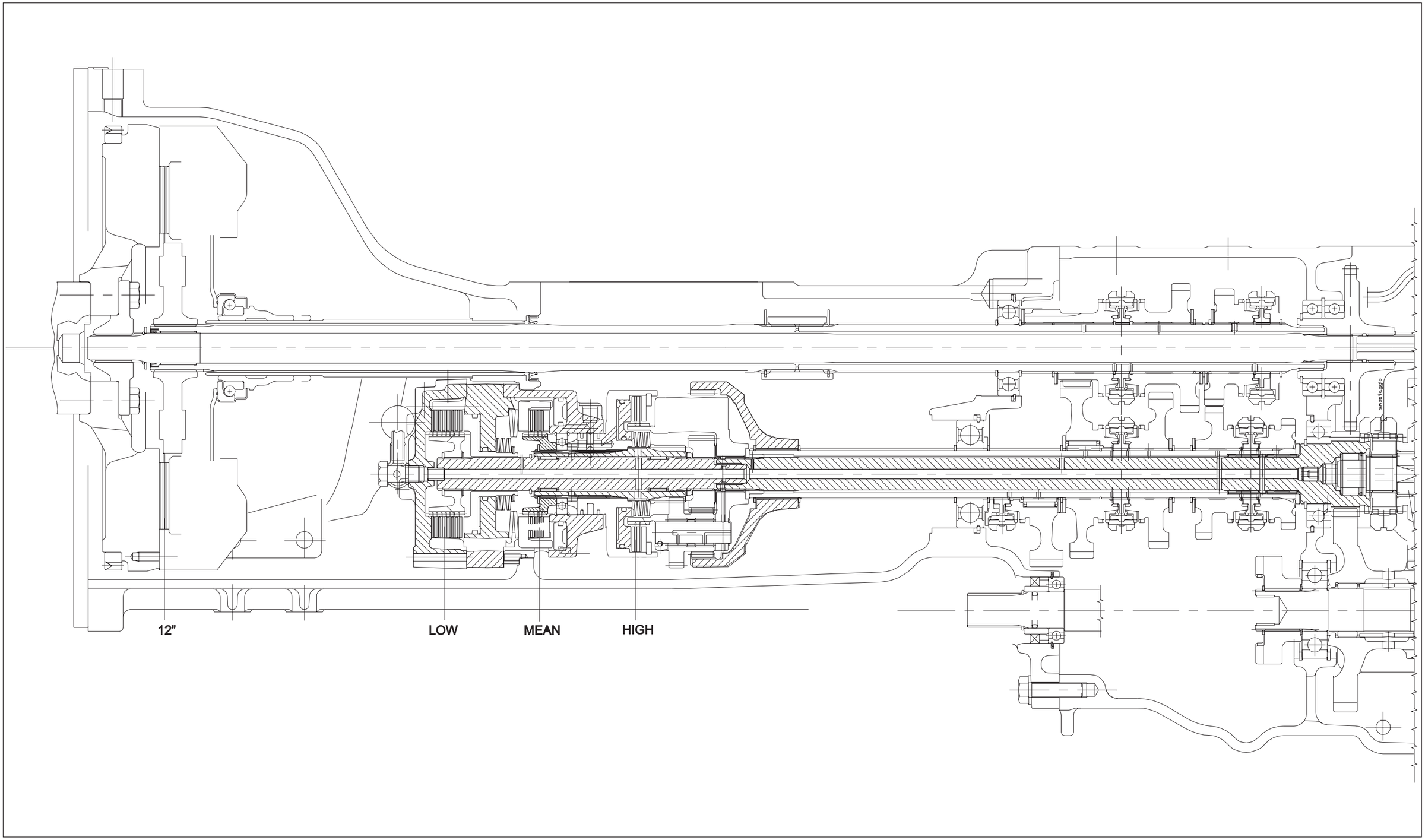


Fig. 6 - Longitudinal section through transmission with Multispeed unit.

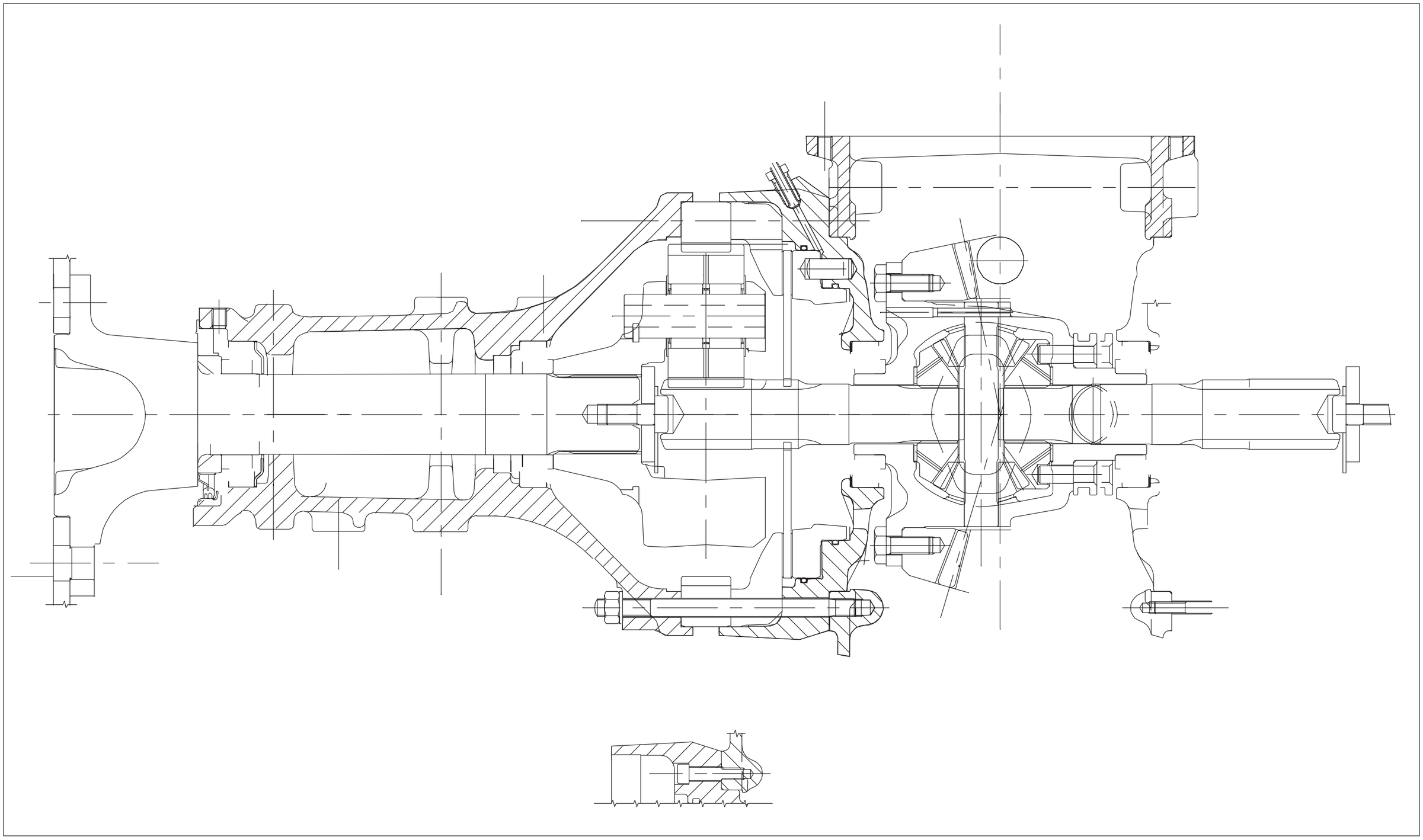


Fig. 7 - Rear axle longitudinal section.

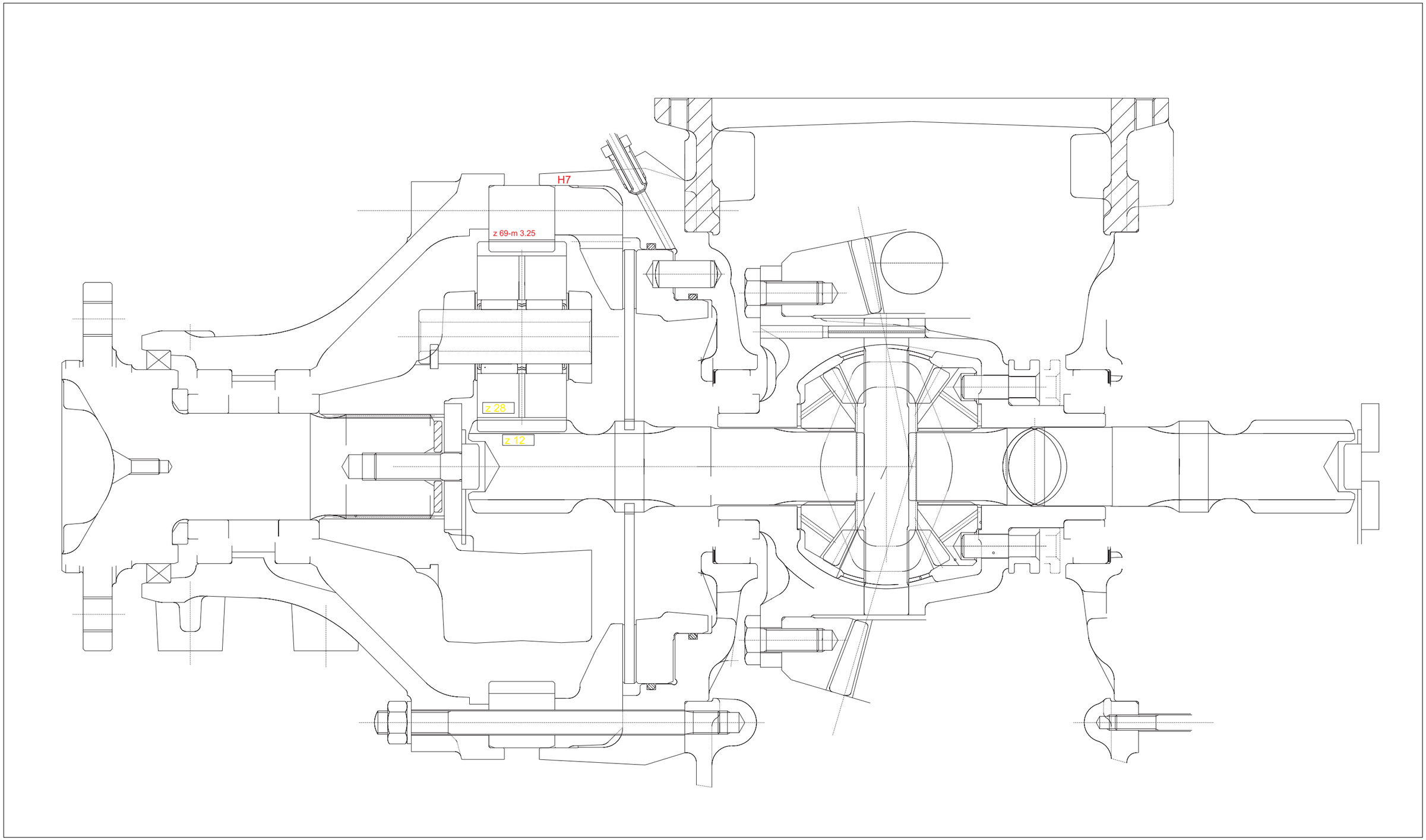


Fig. 8 - Rear axle longitudinal section (vigneto version).

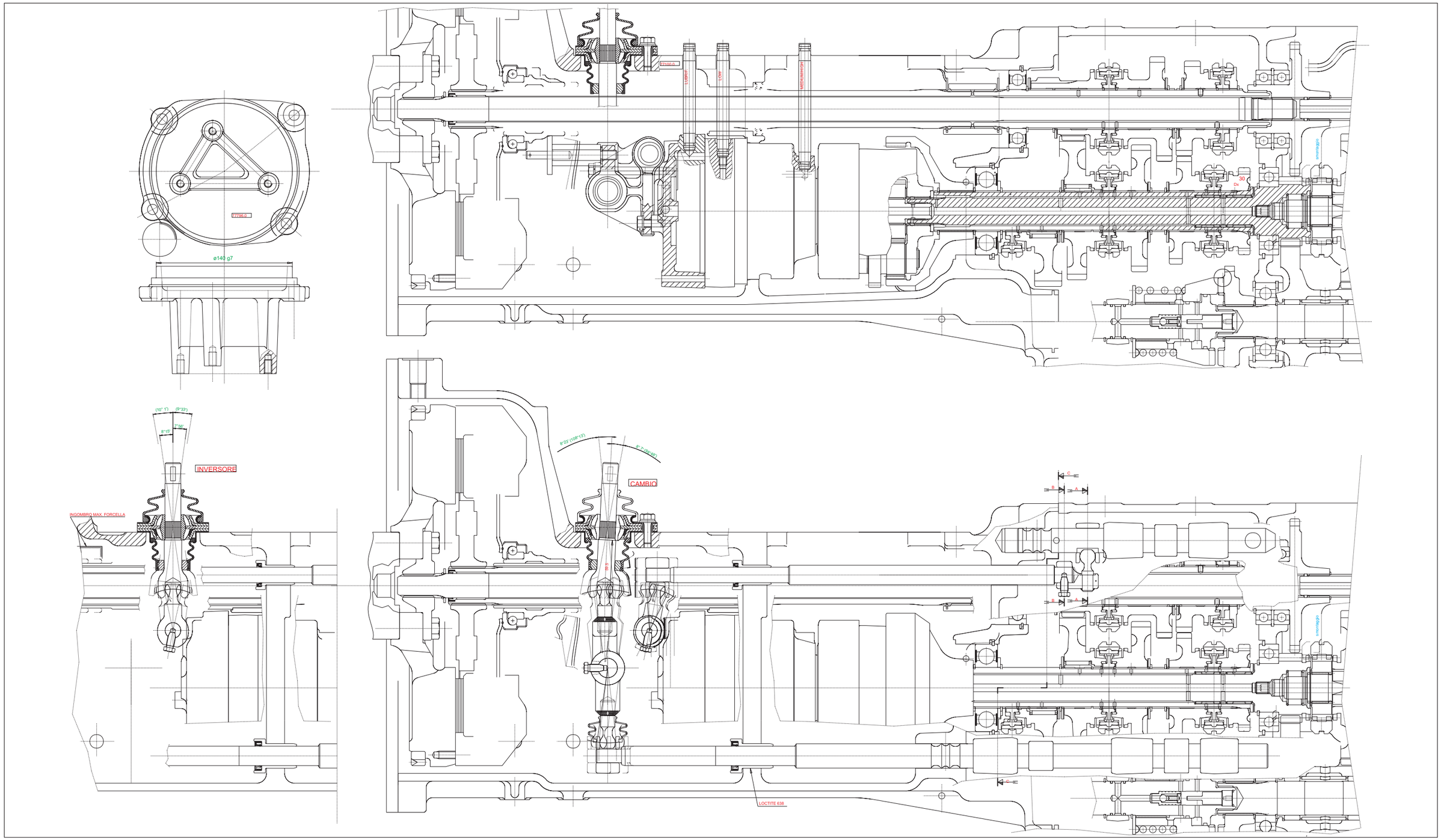


Fig. 9 - Gear lever.

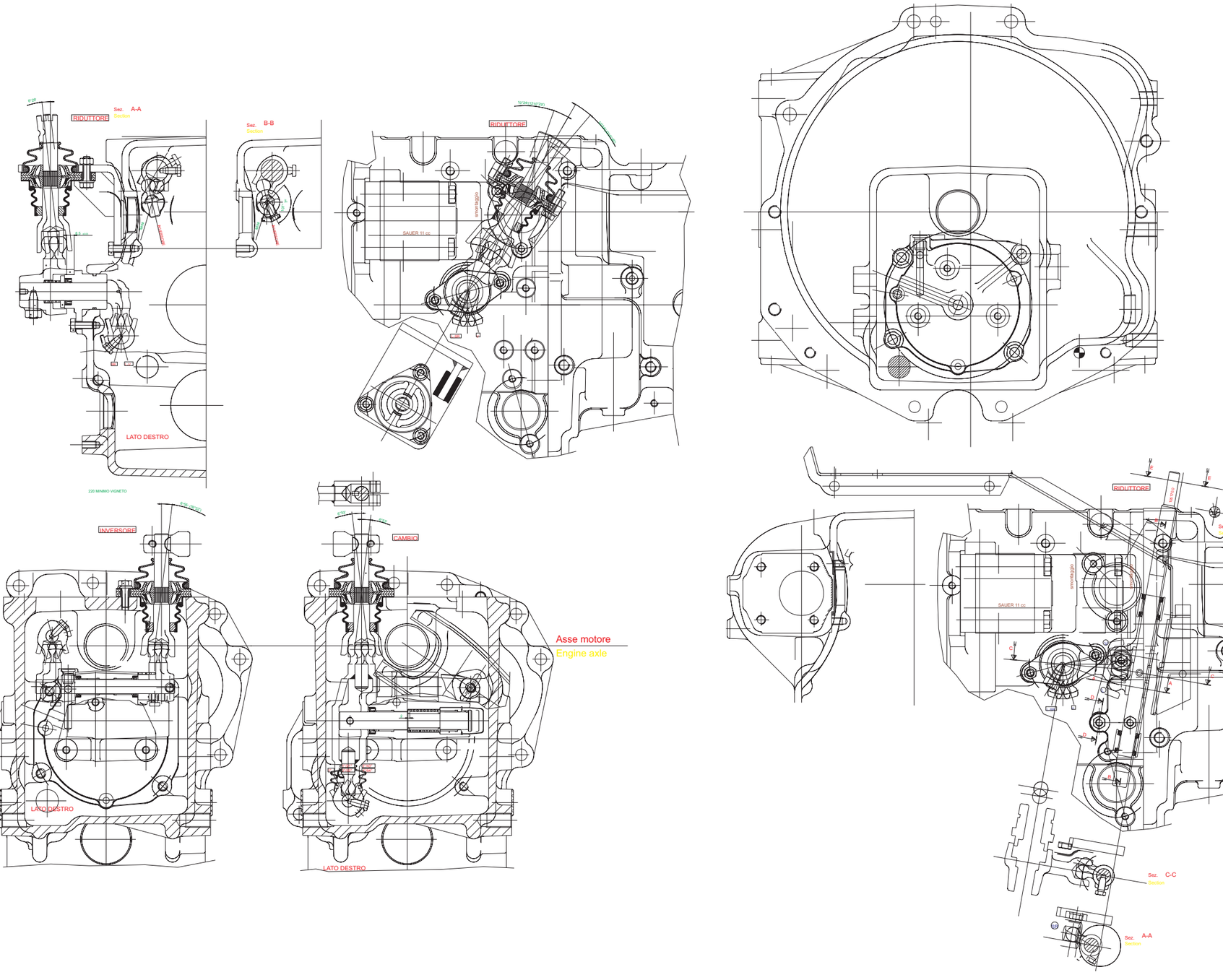


Fig. 10 - Views of the gearbox.

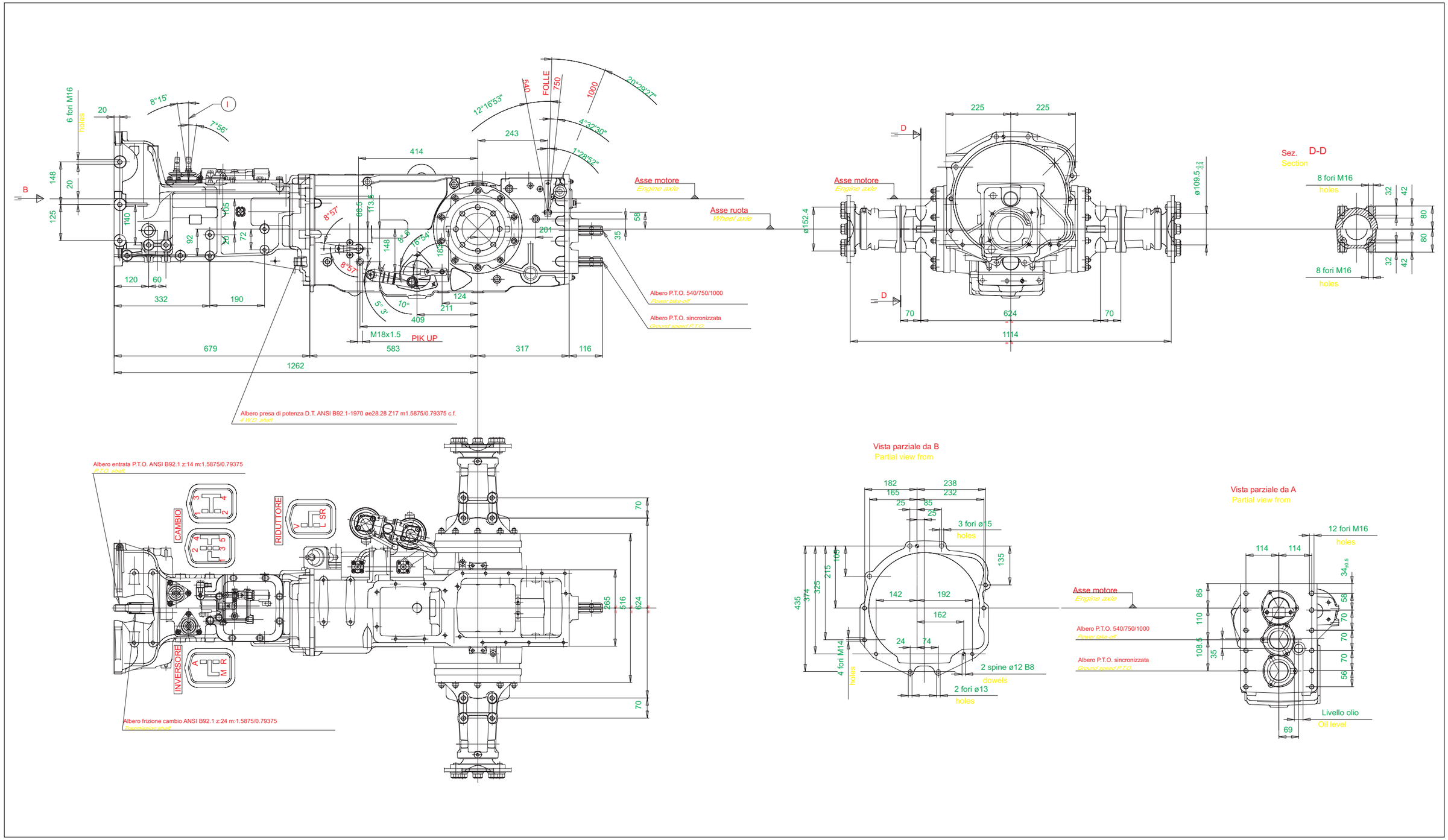


Fig. 11 - Views of the gearbox.

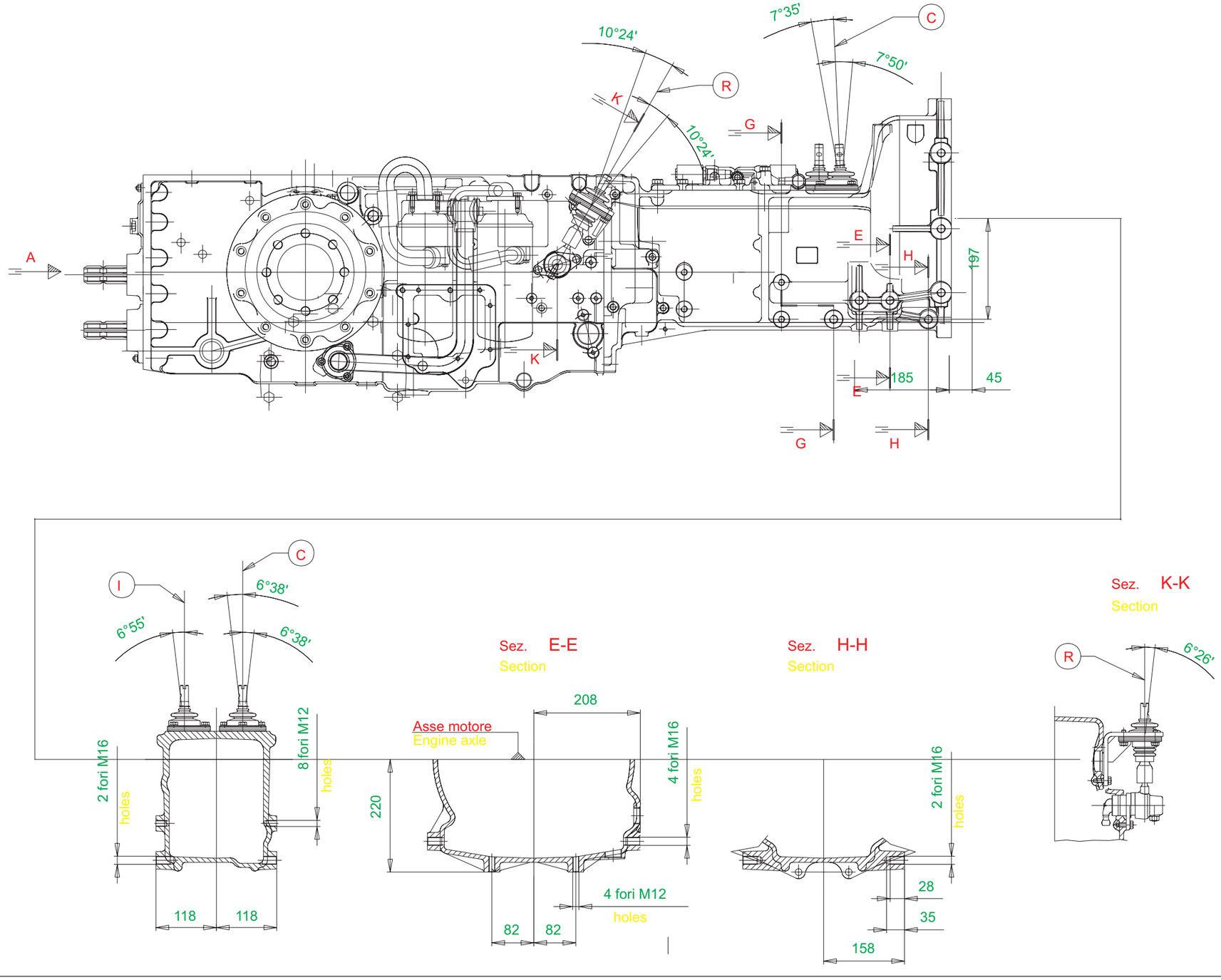


Fig. 12 - Views of the gearbox.

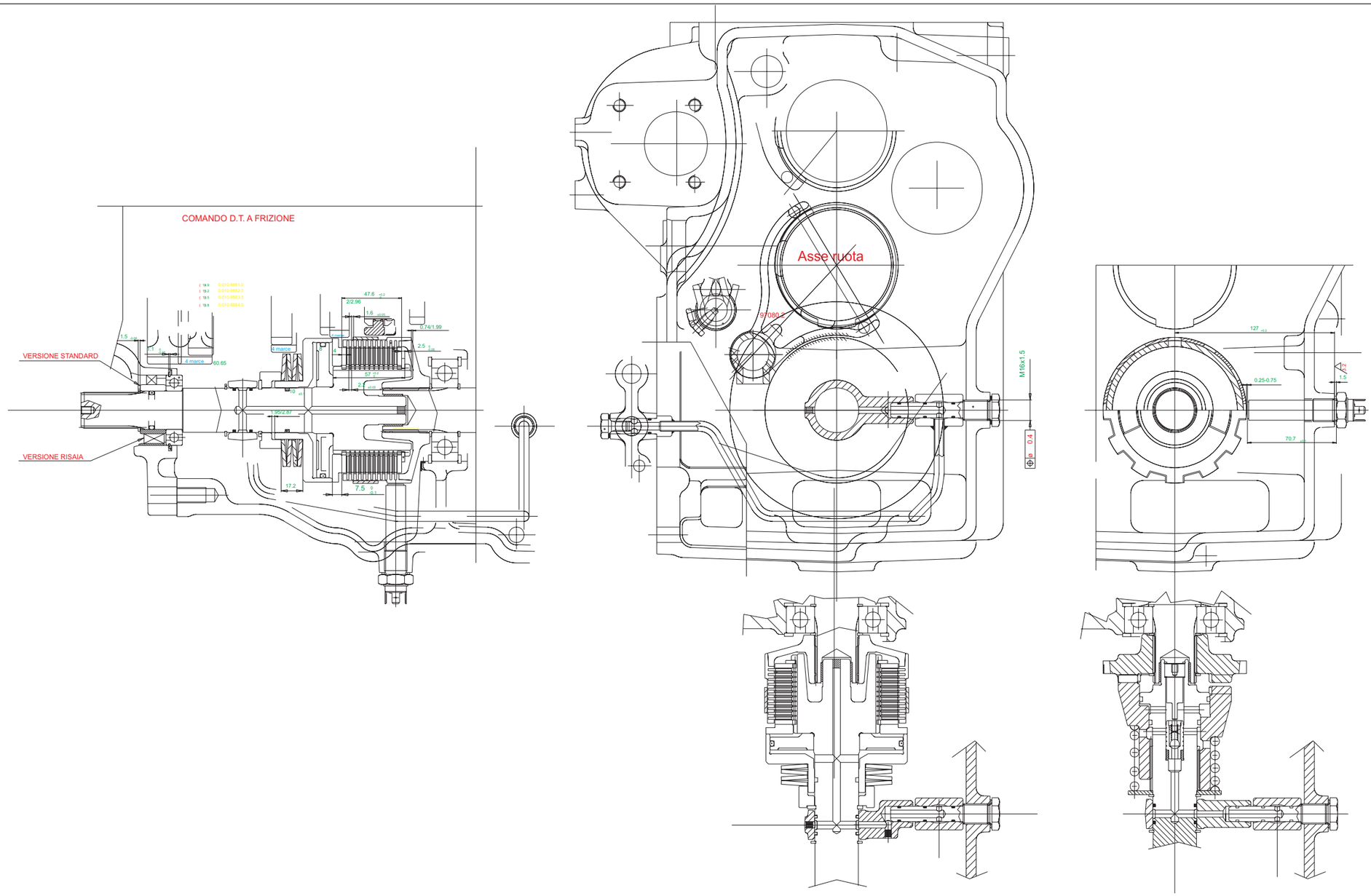


Fig. 13 - Electro - hydraulic control for front-wheel drive engagement/disengagement.

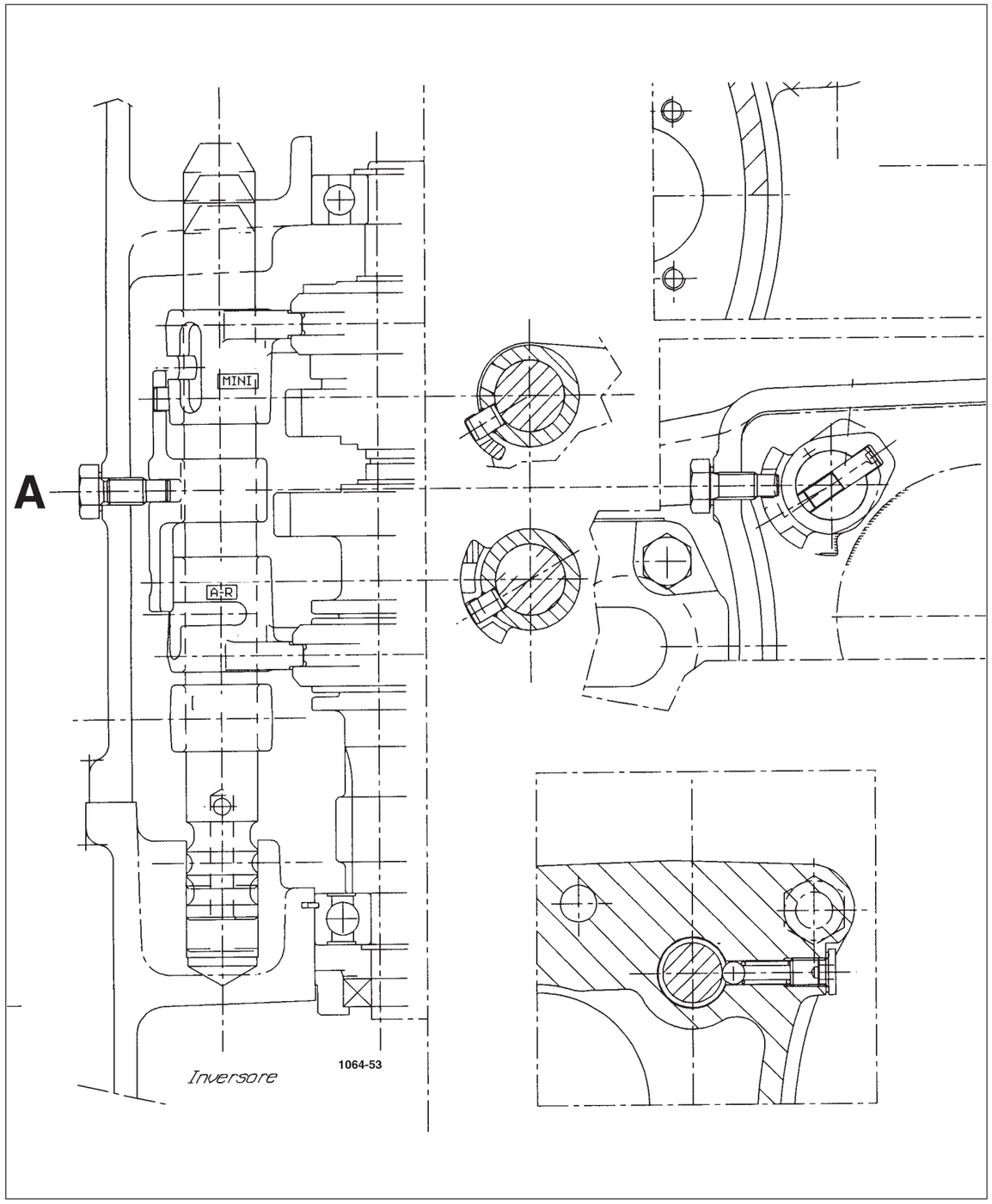


Fig. 14 - Gear selector rod. **A** - bolts securing hubs to rod.

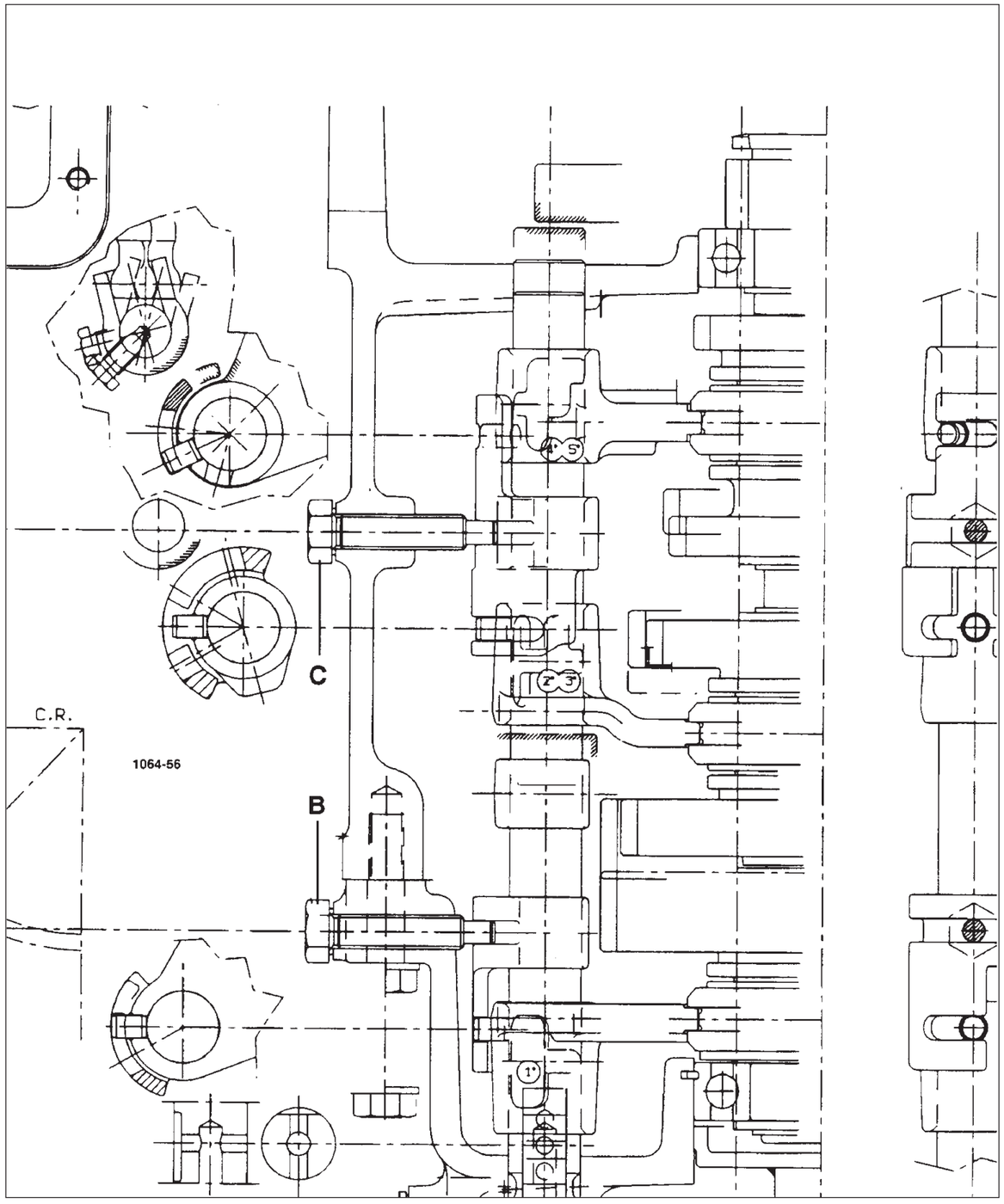


Fig. 15 - Gear selector rod. **B - C** - bolts securing hubs to rod.

Range selector rods and forks.

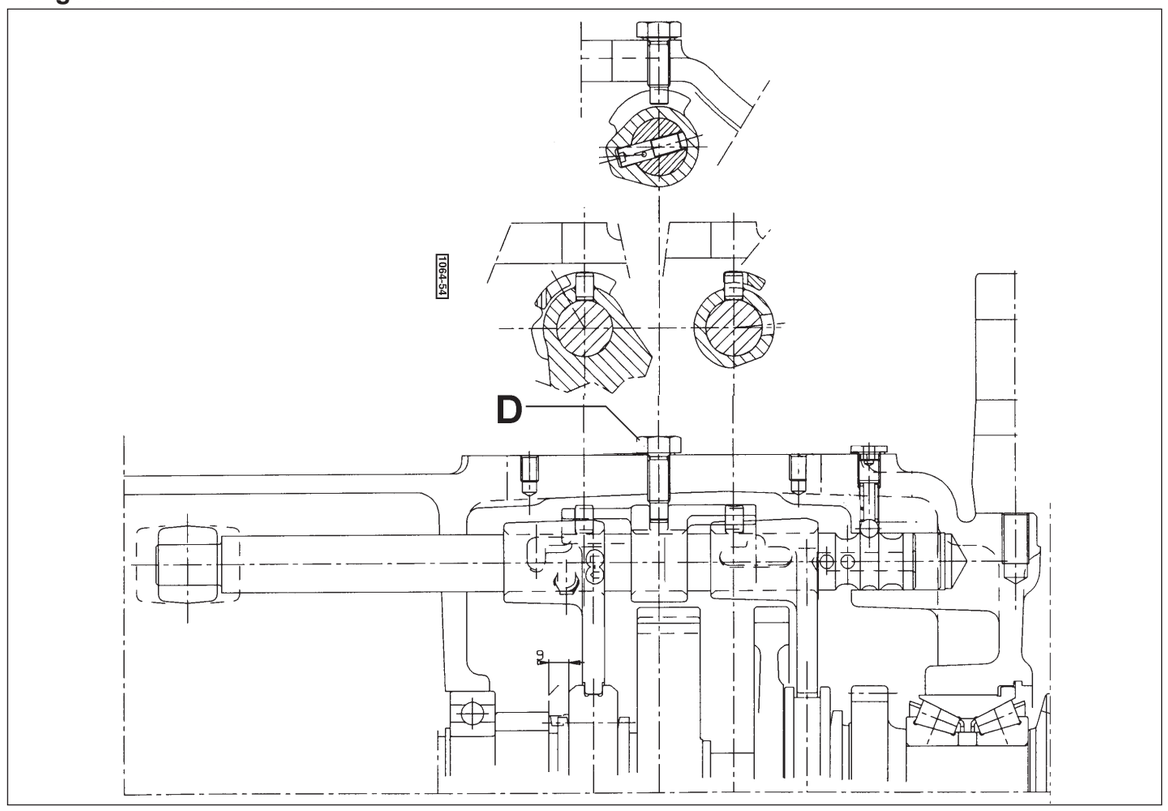
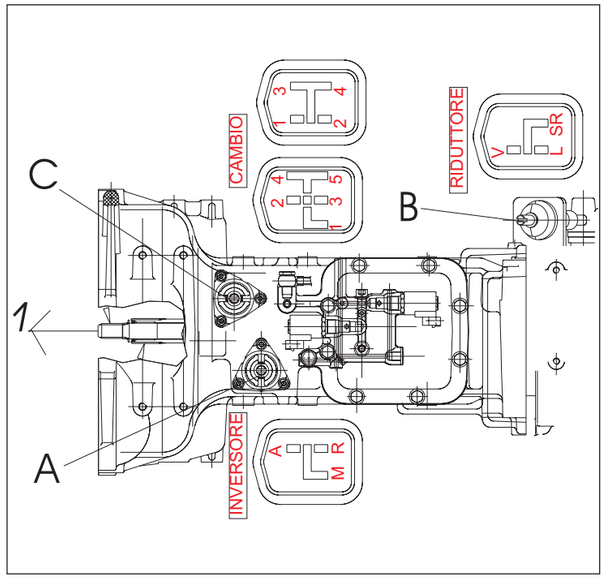
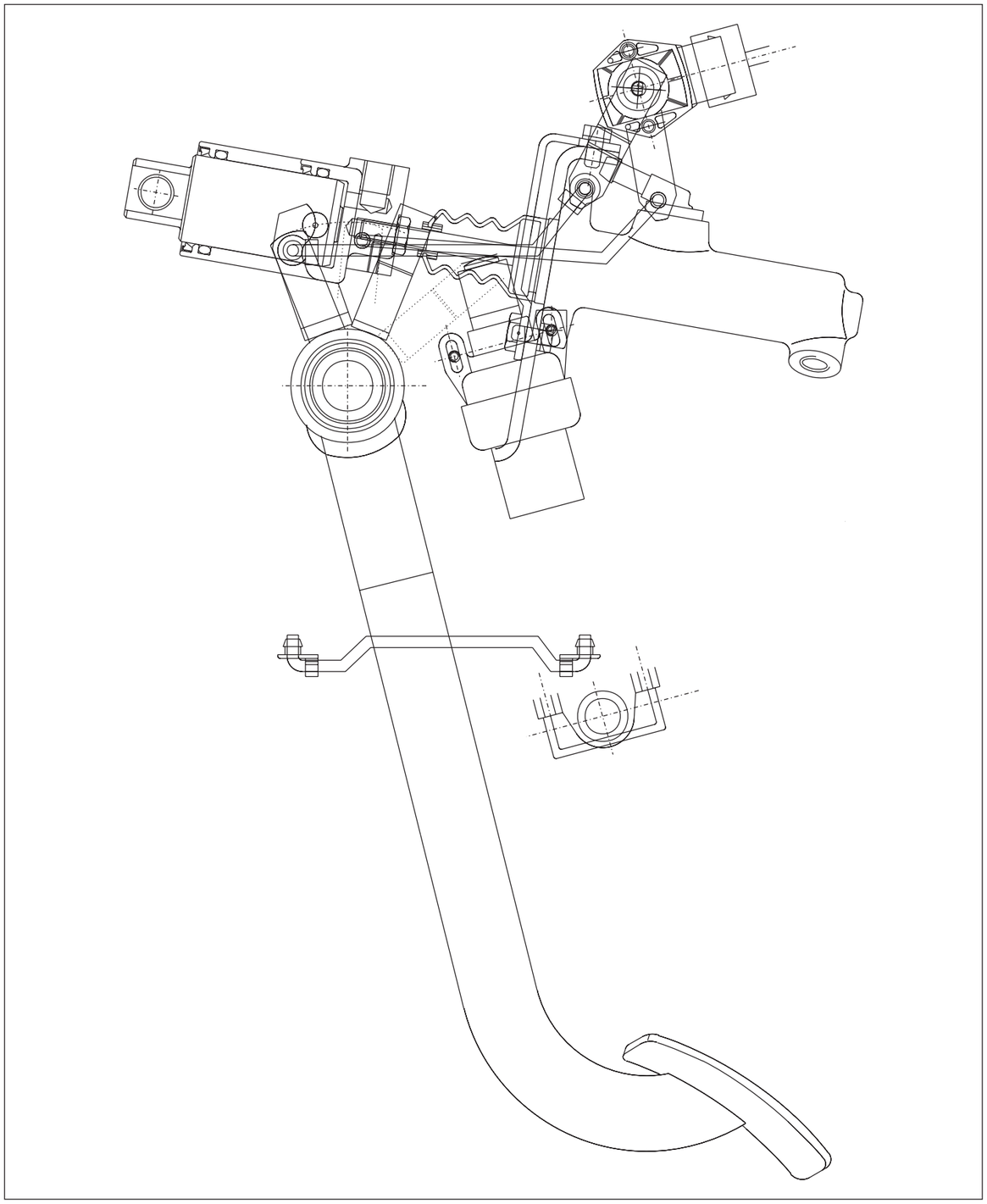


Fig. 16 - Range selector rod. **D** - bolt securing hub to rod.

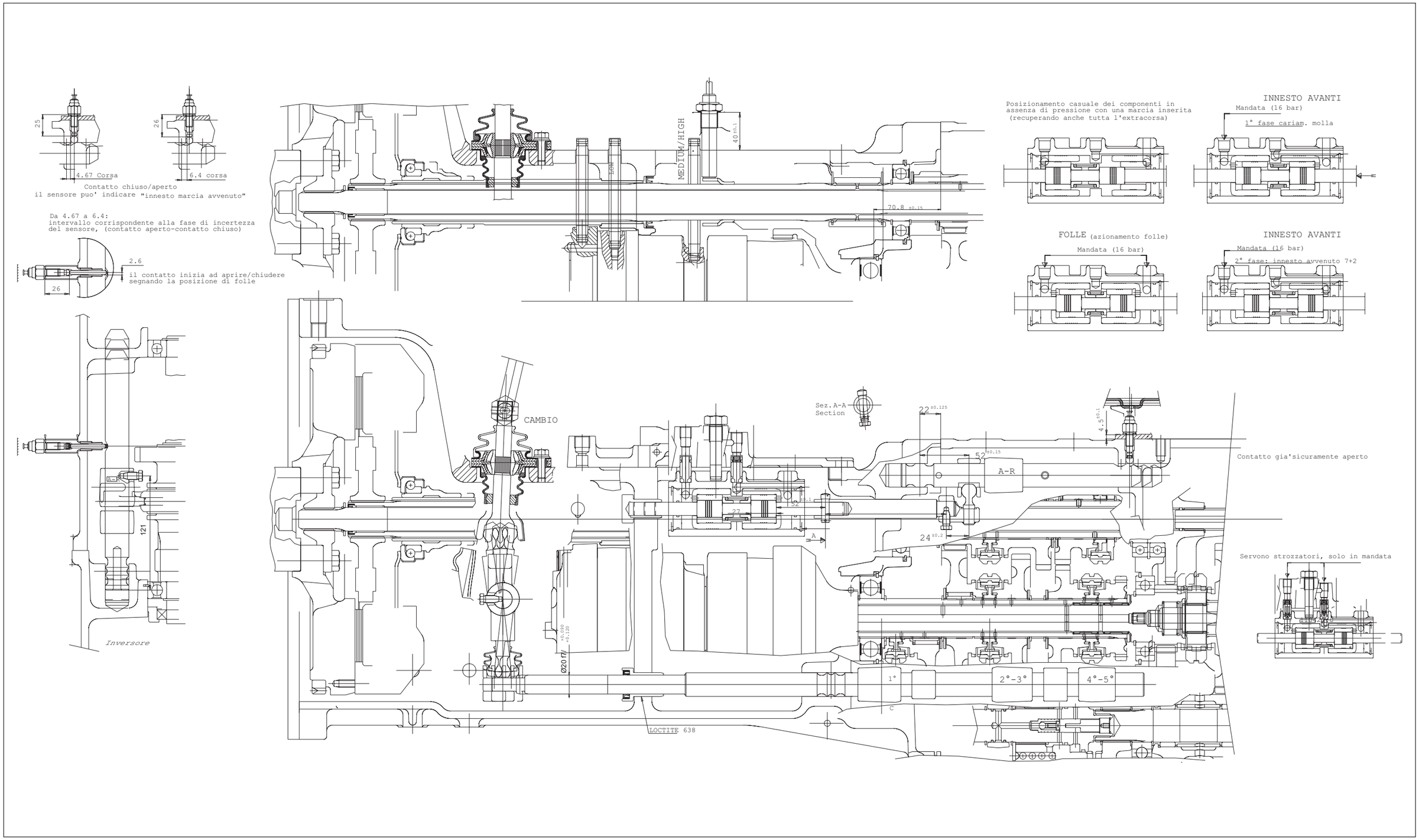


Correct positioning of the gearbox, shuttle and range control levers.
Fit the gearbox, shuttle and range control levers so that the grooves are oriented as shown in the figure.

1	direction of travel
A	shuttle lever
B	range lever
C	gear lever



Hydraulic actuator and position sensor for clutch control with hydraulic shuttle.



Hydraulic shuttle control.

REMOVAL AND REFITTING

Removal of the rear gearbox without removing the platform (tractors equipped with platform or cab only).

— Jack up the tractor, place wedges under the front axle support and the rear axle and remove the rear wheels;

— If the tractor is equipped with an auxiliary fuel tank, disconnect the fuel pipes and the fuel level sensor lead and then remove the tank;

— Disconnect the rear brake pipes from the separate brakes valve and also from the rear gearbox, loosening the support bracket attached to the front gearbox (position a suitable container to collect the brake fluid as all the fluid in the reservoir will drain out during this operation).

— Disconnect the hydraulic unloader pipe and the oil transfer pipe from under the two gearboxes.

— Disconnect the intermediate section of the front differential control pipe on the right-hand side of the tractor.

— Recover the cover of the platform tunnel, then loosen off the two uppermost bolts securing the two gearboxes.

— On four-wheel drive models, to remove the front-wheel drive transmission shaft, release the sleeve coupling the shaft to the gearbox by removing the circlip securing the lock pin and then removing the pin. Move the shaft towards the front of the tractor to disengage it from the output shaft of the gearbox.

— Using a pair of long-nose pliers, remove the circlip securing the front-wheel drive output shaft, thereby allowing the shaft to be withdrawn from the front gearbox when the gearbox is removed.

— Disconnect the oil delivery and return pipes form the oil cooler located underneath platform and to the left of the rear gearbox.

— Disconnect the oil line between the MULTISPEED control unit and the electro-hydraulic services control valves.

— Disconnect the oil delivery pipe from the solenoid valve block of the electro-hydraulic controls to the front gearbox (Multispeed).

— Loosen the oil drain plug located under the rear gearbox (a container of at least 33 litres will be required to collect the oil).

— Disconnect all the electrical leads from the starter motor and the lead connected to the engine oil pressure switch.

— Unplug the bulkhead connector for the wires to the group of indicator lights on the dashboard.

— Disconnect the front lift delivery line from the valve located above the rear lift.

— On tractors equipped with an auxiliary hydraulic control valve mounted centrally on the left-hand side of the tractor, disconnect the intermediate sections of the three pipes connected to the valve. Also disconnect the intermediate sections of the pipes of the hydraulic lines located centrally on the right-hand side of the tractor and connected to the rear hydraulic control valve.

— Position tool 5.9030.002.0 (mobile stands on tracks to support the front and rear gearboxes, then remove all the bolts connecting the two gearboxes together.

— Put the range gear lever in neutral and loosen the two bolts securing the hub complete with the rod which enters the gearbox and remove this hub.

— Loosen the hex socket screw (or the two hex socket screws on tractors equipped with mini-reduction) securing the gear selector rod (see fig. 33 p. 166).

— While holding the rear gearbox in position, move the front wheels to separate the two gearboxes (time required 3.5 to 4 hours).

WARNING: under no circumstances undo union **B** Fig. 19 (hex head), located on the left-hand side of the gearbox.

This union holds the oil manifold **D** Fig. 19 for engagement of the front-wheel drive transmission shaft. If this union is removed, the manifold would rotate downwards, thereby making it necessary to dismantle the gear hub in order to reposition it and refit the union.

The delivery pipe **C** Fig. 19 of the front-wheel drive engagement actuator is connected at the opposite end, inside the gearbox, to the group of solenoid control valves for the hydraulic services **E** Fig. 19; if this group is to be removed, undo union **A** Fig. 19, to prevent deformation of the internal pipe.

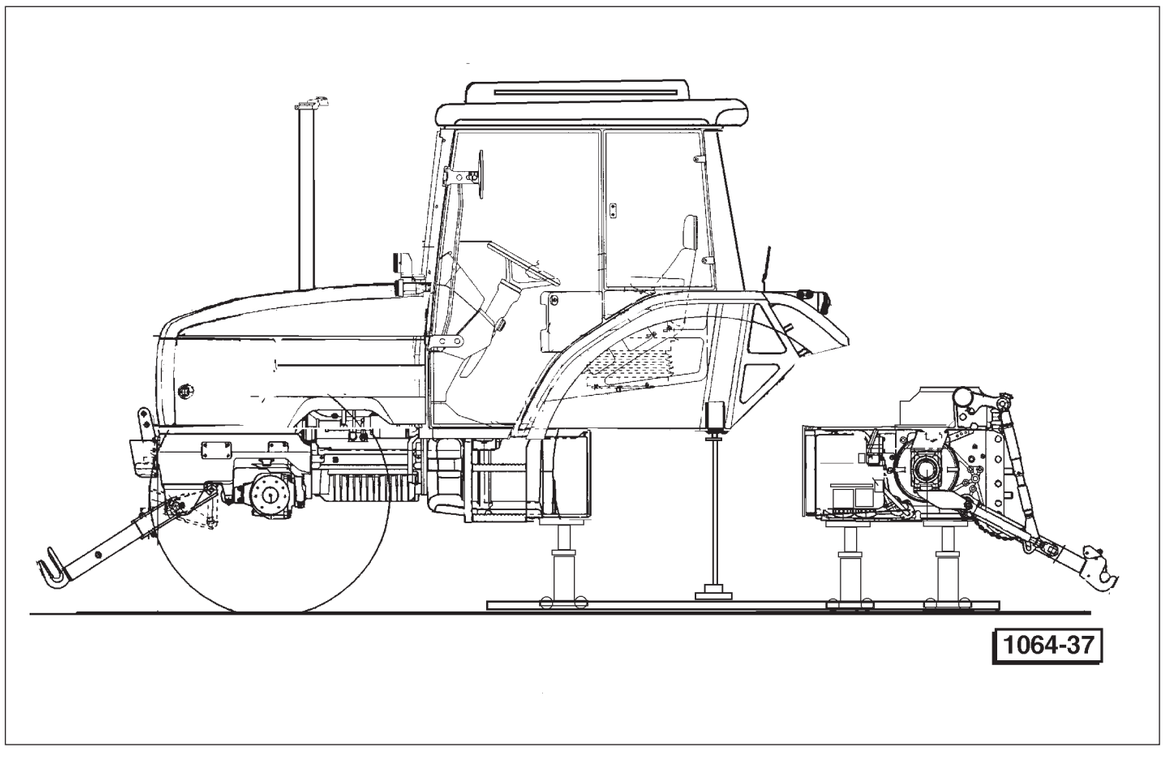


Fig. 17 - Removal of the rear gearbox from the tractor

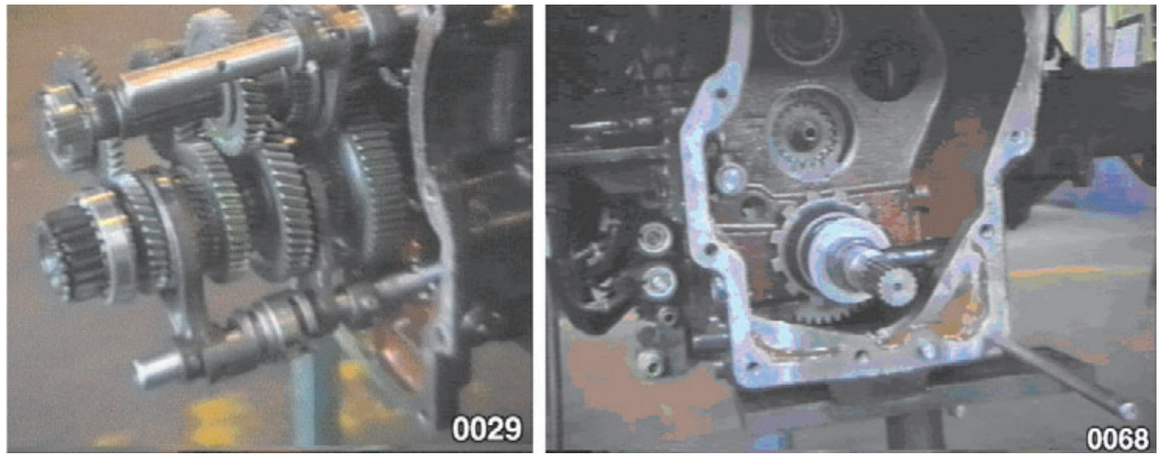


Fig. 18 - Separate the rear gearbox from the front gearbox

Separating the front gearbox from the engine.

(this operation is necessary in order to access the MULTISPEED unit)

— Remove the engine pick-up from the top of the gearbox (in the area of the spacer housing).

— Place wedges under the front axle support.

— On four-wheel drive models, to remove the front-wheel drive transmission shaft, release the sleeve coupling the shaft to the gearbox by removing the circlip securing the lock pin and then removing the pin. Move the shaft towards the front of the tractor to disengage it from the output shaft of the gearbox.

— Disconnect the oil delivery and return pipes form the oil cooler.

— Disconnect the two front brake pipes and the differential lock control pipe.

— Loosen the oil drain plug located under the rear gearbox (a container of at least 33 litres will be required to collect the oil).

— Disconnect all the electrical leads from the starter motor and the lead connected to the engine oil pressure switch.

— Unplug the bulkhead connector for the wires to the group of indicator lights on the dashboard.

— Disconnect the front lift delivery line.

— Support the front gearbox on a stand.

— Remove the 10 screws securing the front gearbox to the engine flange, then detach the gearbox.

Dismantling the gearbox

— Remove the 8 screws securing the top cover and then remove the cover itself.

— Remove the gearbox clutch control located in the spacer housing as follows:

— Remove the thrust bearing.

— Remove the lock pin and withdraw the control lever.

— Undo the union and remove the oil delivery pipe to the MULTISPEED unit.

— Unscrew the 2 screws and the 2 nuts securing the MULTISPEED unit;
NOTE: one of the screws has an extension with a small plate to support the clutch control lever; these parts are to be removed together with the screw.

Removal of the gearbox input and P.T.O. shafts.

— Remove the circlip attaching the gearbox input shaft to the inversor shaft coupling sleeve.
— Withdraw the 2 shafts from the front of the spacer housing (the shafts should be withdrawn together to prevent damage to the seal ring positioned between them).
— **NOTE:** To replace the seal on the gearbox input shaft, it will be necessary to remove the cab or platform to gain access to the upper cover of the gearbox, from which the seal housing may be accessed.

Separating the MULTISPEED unit from the gearbox

— Using a copper drift, tap the housing of the MULTISPEED unit in order to extract from the front part of the spacer housing; (if the purpose of this operation is solely to inspect or replace the hydraulic clutches, it will not be necessary to remove the epicyclic reduction unit from the gearbox or to remove the cab/platform).

— If instead the epicyclic reduction unit is to be removed, simply withdraw it from the front of the gearbox, then remove the calibrated thrust washer and the circlip, and then remove the housing complete with the ring gear.

— The MULTISPEED clutch units are to dismantled on the bench.

— **NOTE:** for instructions on dismantling the MULTISPEED unit, see page 122.

Removal of the gear train positioned in the front gearbox

WARNING: On tractors equipped with Multispeed, first follow the instructions to remove the Multispeed unit and the gearbox input and P.T.O. shafts.
For the following operations the driving platform or cab must be removed in order to gain access to the upper gearbox cover.

Before proceeding with the removal of the gear train, it is necessary to remove the selector rods and forks. These operations are only possible after having separated the rear gearbox from the front; before separating the two gearboxes, remove **D** fig. 5 on page149.
Detach the front gearbox from the rear gearbox. To do this, it will be necessary to remove the 5 speed relay hub together with the intermediate support, located on the right-hand side of the rear gearbox. Remove screw **A** (fig. 14) screws **B** (fig. 15) and screw **D** (fig. 16).

NB: Remove the selector rod detent balls and springs by unscrewing the relative plugs located on the right-hand side of the gearbox. Then proceed with the separation of the two gearboxes.
The 3 gearbox shafts (inversor, primary and secondary) and the gear selector and inversor control rods and forks will remain attached to the front gearbox. To remove these rods, it will first be necessary to remove the 3 shafts by removing the 3 circlips **A** - **B** - **C** fig. 5 on page 149 and then moving all 3 shafts simultaneously backwards to disengage them from their housings; the first shaft to be detached will be the primary shaft, followed by the secondary, and then the input shaft located at the uppermost part of the gearbox is to be withdrawn. Each selector rod will come away together with its shaft.

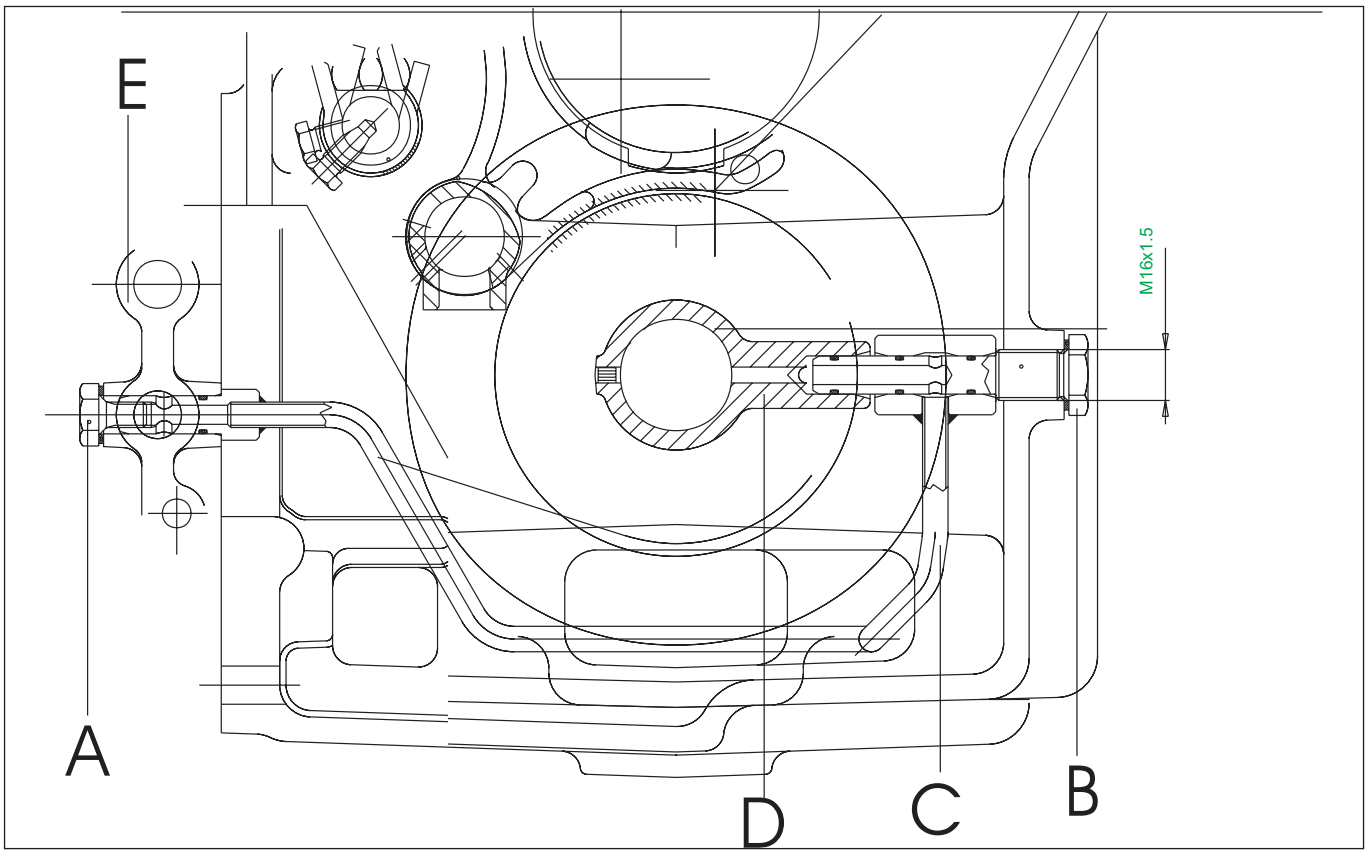


Fig. 19 - Oil pipe from the solenoid control valve to the hydraulic actuator responsible for engaging/disengaging front-wheel drive.
A Pipe union on solenoid valve end **B** Pipe union on actuator ende **C** Connection pipe.
D Manifold **E** Solenoid valve assembly

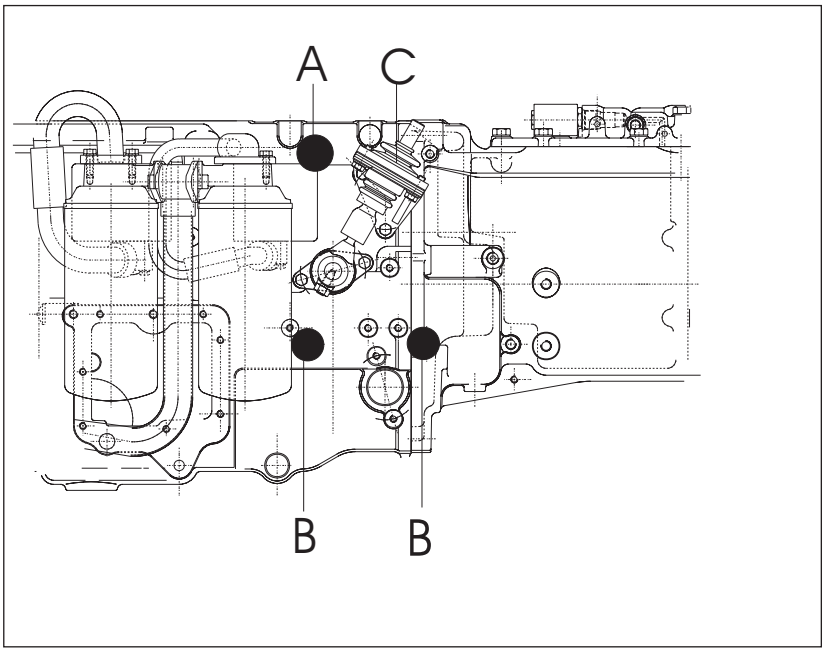


Fig. 20 - Retaining screws of the gear rods to be removed prior to detaching the front gearbox from the rear gearbox.

A retaining screw of the inversor control rod hub.

B retaining screws of the gear control rod hubs.

C Range gear control rod.

Disassembly of the inversor control rods and forks

Disengage the fork from the sleeve and rotate it so that the hole in the fork is aligned with the retaining pin; then, using a drift as shown in figure 23, tap the pin out of the hole.
On tractor equipped with a minireduction unit, repeat the above operations to remove the second selector fork.

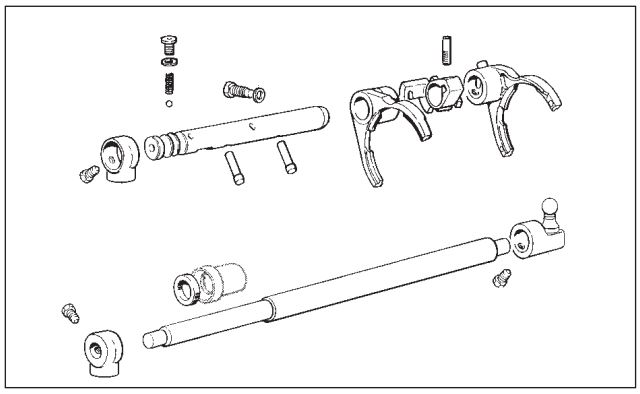


Fig. 21 - Components of inversor rod and fork assemblye for gearbox with minireduction.

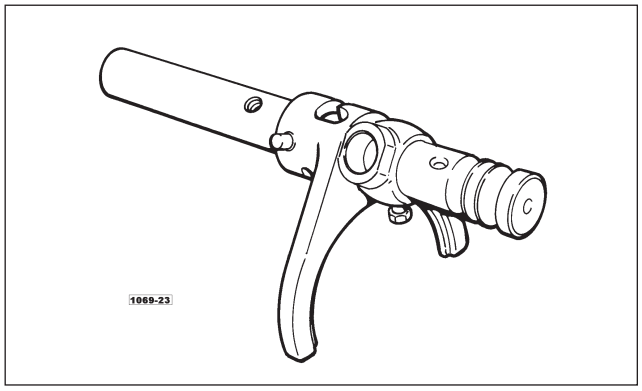


Fig. 22 - Rod and fork for gearbox without minireduction.

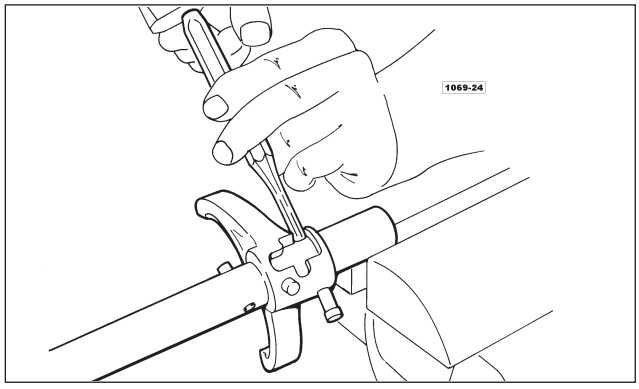


Fig. 23 - Removal of the fork retaining pin.

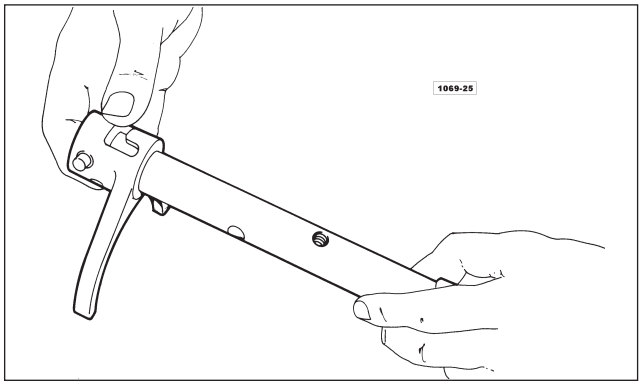


Fig. 24 - Removal of fork from control rod.

Dismantling of the gearchange rod and fork assembly.

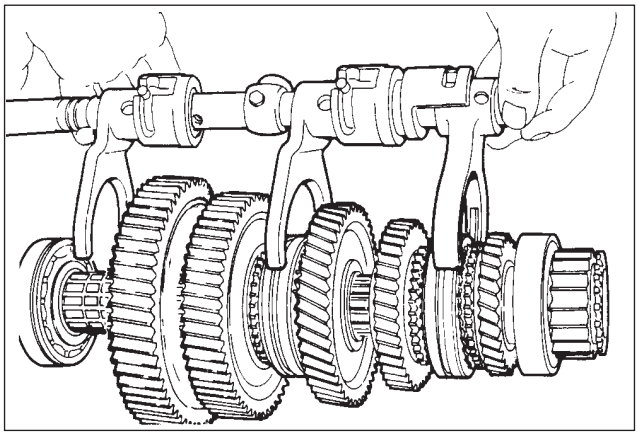


Fig. 25 - Detach the rod with forks from the secondary shaft by moving it laterally

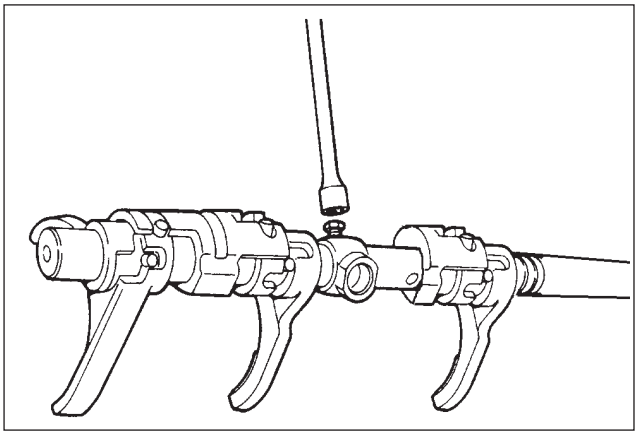


Fig. 26 - Remove the screw securing the rod hub and remove the hub from the rod.

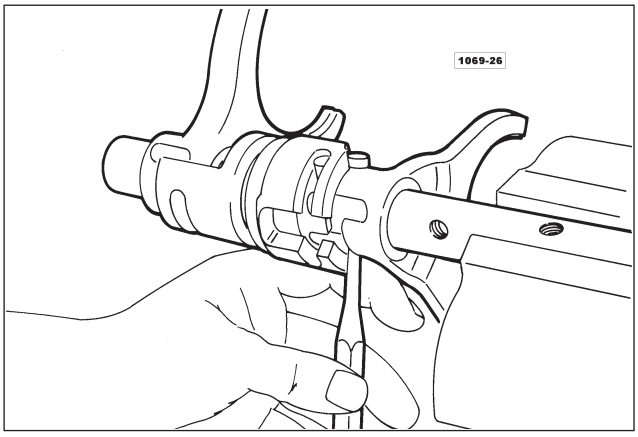


Fig. 27 - Disengage the 1st gear selector fork from the sleeve and rotate it to align the hole on the fork with the retaining pin. Then, using a suitable drift, drive the pin out of the hole. Detach the 2nd and 3rd gear selector forks and sleeves in the same way.

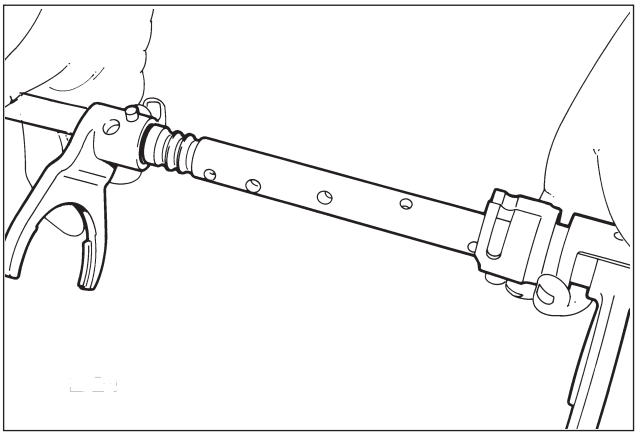


Fig. 28 - Withdraw the gearchange control rod.

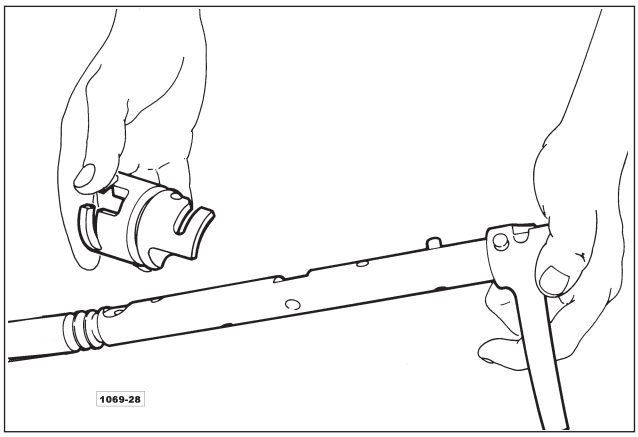


Fig. 29 - Remove the sleeve located on the rod between selector fork.

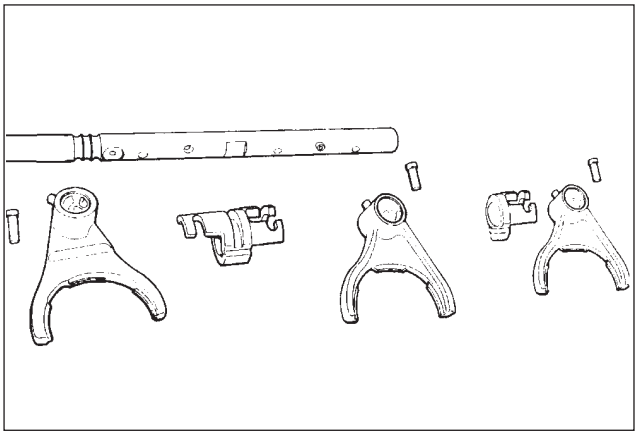


Fig. 30 - Rotate the 4th and 5th gear the two forks. so that the hole on the fork is aligned with the retaining pin, then using a drift and a hammer as shown in figure 23, drive the pin out of the hole (the figure shows the components of the selector rods and forks assembly).

Dismantling of the gearchange selector rods and forks assembly

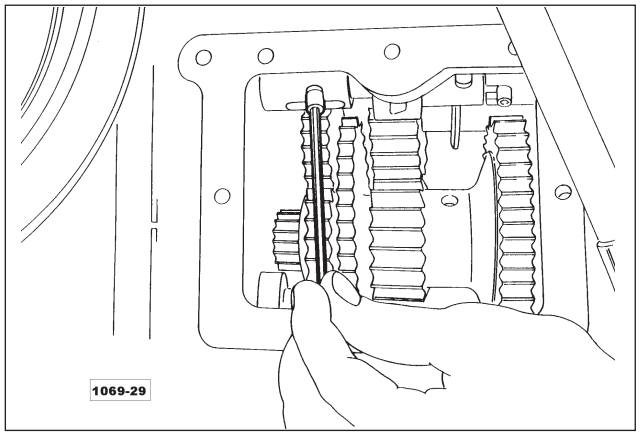


Fig. 31 - Unscrew the 6 retaining screws and remove the cover giving access to the gearchange selector rods and forks assembly. Then using a spanner, unscrew the guide bolt of the super-reduction control fork.

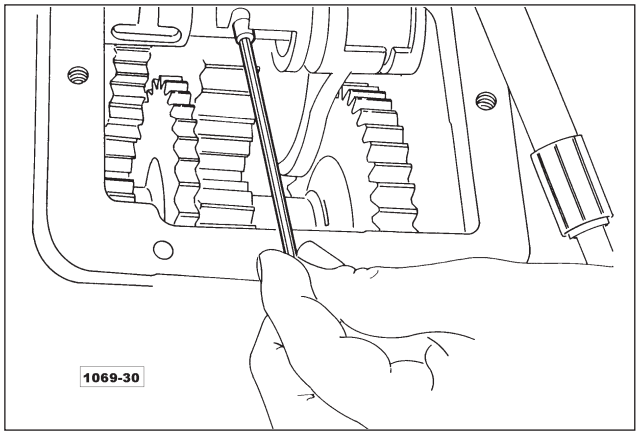


Fig. 32 - Unscrew the guide screw of the central sleeve located between the 2 forks. (NB. Loctite has been applied to these screws; we recommend that they are heated using an oxyacetylene torch to facilitate removal).

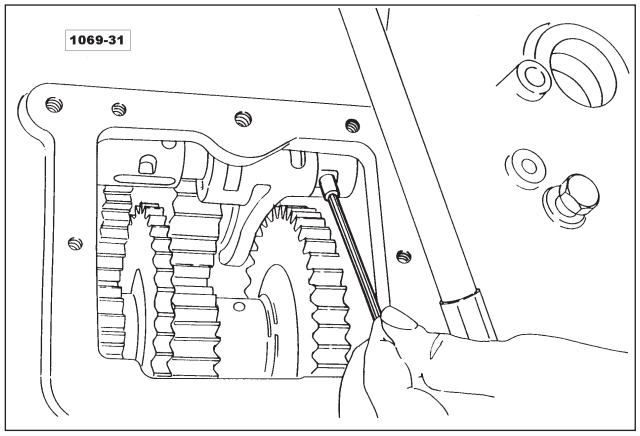


Fig. 33 - Unscrew the guide screw of the L-V selector fork.

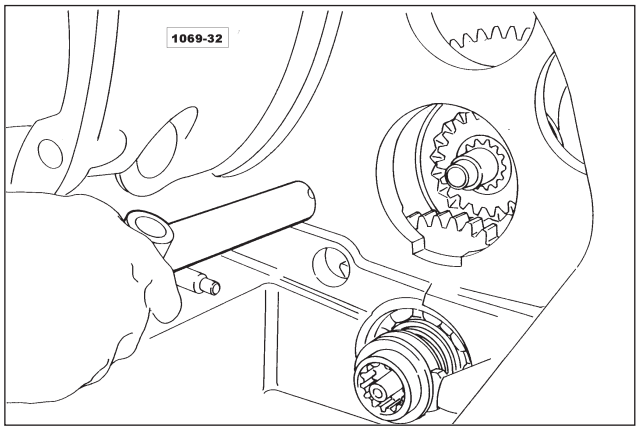


Fig. 34 - Access the front compartment of the gearbox extract the rod by pulling the hub.

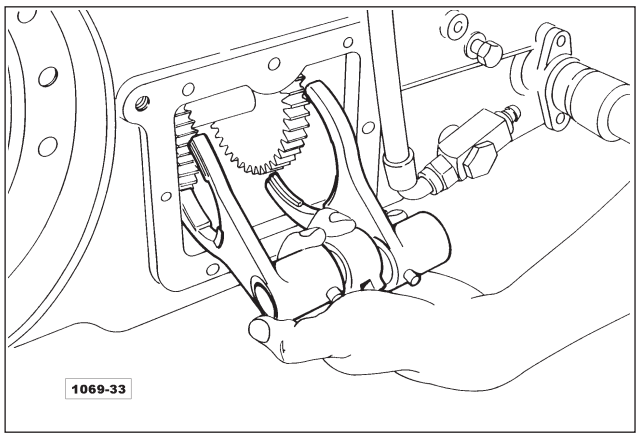


Fig. 35 - Remove all the parts mounted on the rod, recovering them from the lateral access hole.

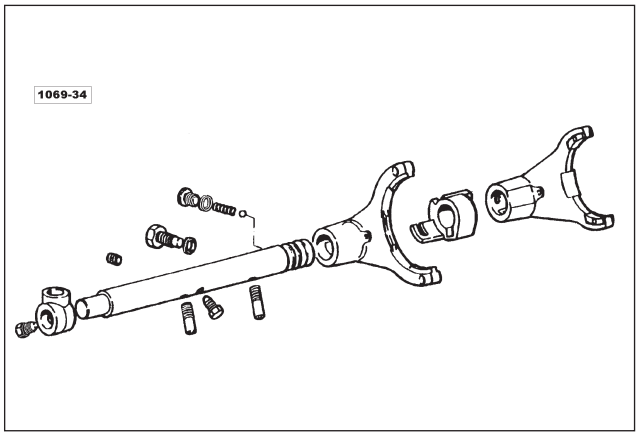


Fig. 36 - Parts comprising the selector rod and forks them from the lateral access hole assembly for the super-reduction unit and range gears.

Removal of the shaft with the actuator for engagement/disengagement of the front-wheel drive
Separate the rear gearbox from the front gearbox, (it is first necessary to remove circlip (**A** fig 37). Unscrew union **B** (fig 19 page 163) located on the left-hand side of the rear gearbox and remove the front shaft complete with the front-wheel drive actuator.

IMPORTANT: Only unscrew union **B** fig. (19 page 163) located on the left-hand side of the rear gearbox when the two gearboxes are separated, as refitting is only possible in this condition.

Disassemble the unit, removing the circlip securing the oil manifold and the circlip securing the thrust plate of the helical spring; then disassemble the unit into its component parts, inspecting for wear the two oil seals (on the shaft and in contact with the control piston) and the areas in contact with the oil manifold.

WARNING: the valve located inside the shaft must be positioned as shown on page 173.

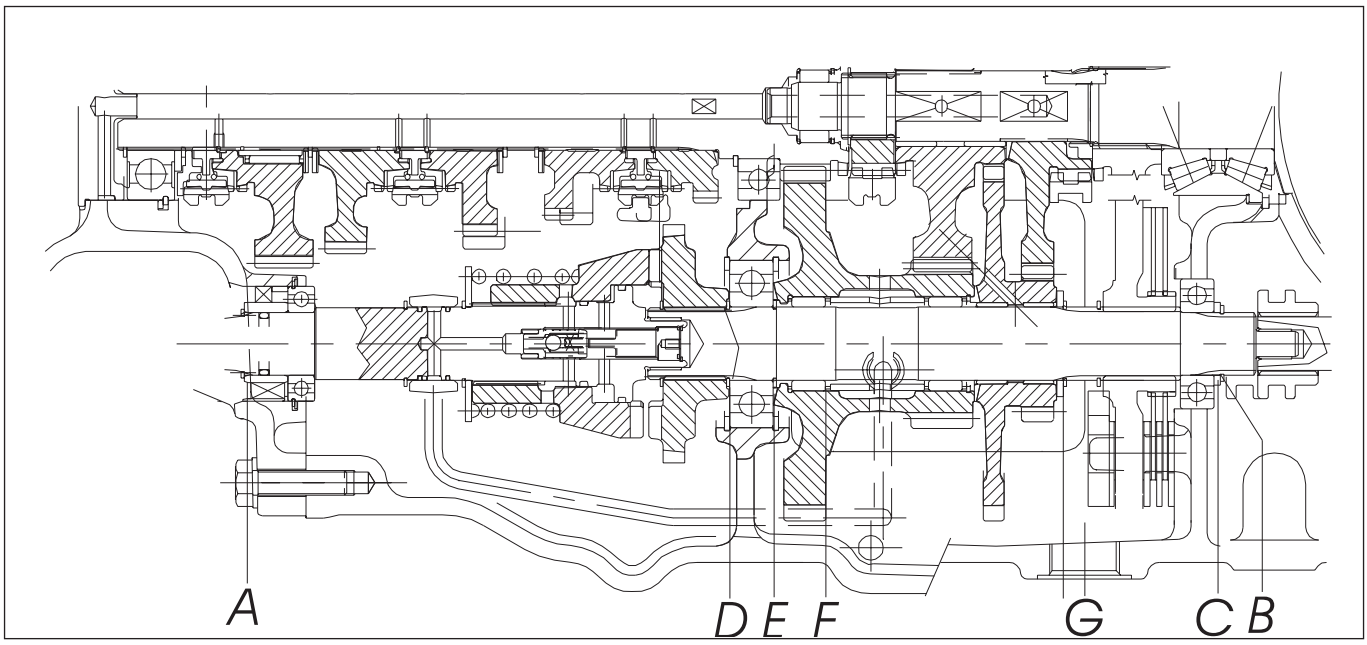


Fig. 37 - Front-wheel drive output.

Removal of the range gear shaft
Separate the rear gearbox from the front gearbox, (it is first necessary to remove circlip **A** fig. 37) remove the right rear wheel support and the cover on which the electro-hydraulic control valves are mounted, detach the oil pipe for 4WD disengagement by undoing union **B** (fig. 19 on page 163) located on the solenoid valve mounting on the right-hand side of the gearbox. Remove the cover supporting the internal parking brake control.
Then remove the synchronised P.T.O. shaft by removing the retaining flange mounted on the rear of the gearbox and remove the synchronised P.T.O. engagement sleeve mounted on the 4WD shaft, which can be accessed through the space left void by removal of the wheel support.
Remove circlip **B** fig. 37 and thrust plate **C** fig. 37, which can be accessed through the space created by removal of the wheel support.
Working through the aperture created by removal of the cover on which the electro-hydraulic control valves are mounted, remove circlips **D** - **E** - **F** - **G** fig. 37 and pull shaft towards the front of the gearbox.

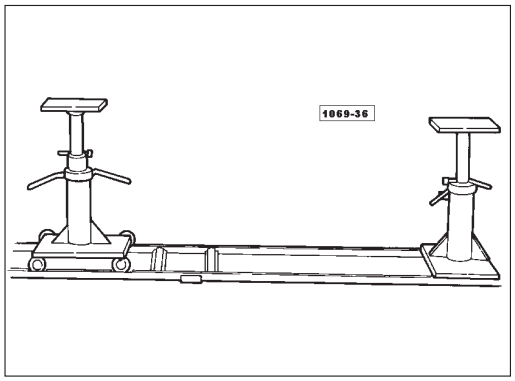


Fig. 38 - Equipment 5.9030.002.0 for the dismantling the gearbox.

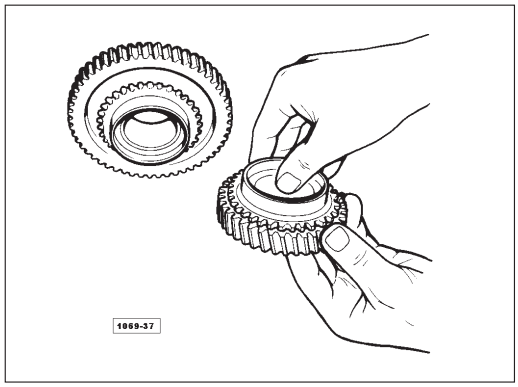


Fig. 39 - Visual control of the tooth of the gears.

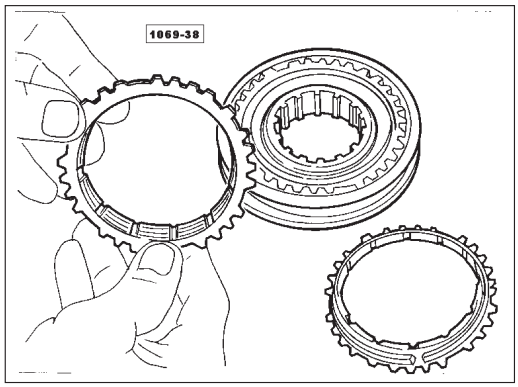


Fig. 40 - Unit synchronizer.

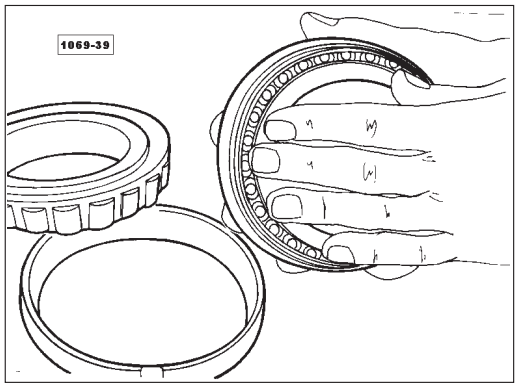


Fig. 41 - Checking slide of the bearing.

Examining parts removed

Gearbox case

Gearbox case should show no cracks, bearing seats must not be worn or damaged.

If seriuos damage or excessive wear is occured replace the parts concerned.

Whenever the gearbox is disassembled clean all sealing surfaces removing old adhesive and applying some new on all surfaces evenly when reassembling.

Shafts

Examine shafts for excessive wear and shaft splines for pitting, these should allow the gears to slide freely.

Gears

Examine gear teeth for wear or damage and make sure these work on the whole contact surface.

Check also the gear-mounted bushings for seizing: if so replace them.

Synchronizers

Check the ring inside tapered portion for excessive wear or damage and be sure the gear part being frictioned by the synchronizer does not show signs of scoring which may prevent the gear from meshing correctly.

To check the synchronizer ring for wear, measure the distance from the friction cone; reading should be $1{,}25^{+0,3}_{-0,15}$ mm with a new synchronizer or drop to zero when the ring is maximally worn.

Bearings

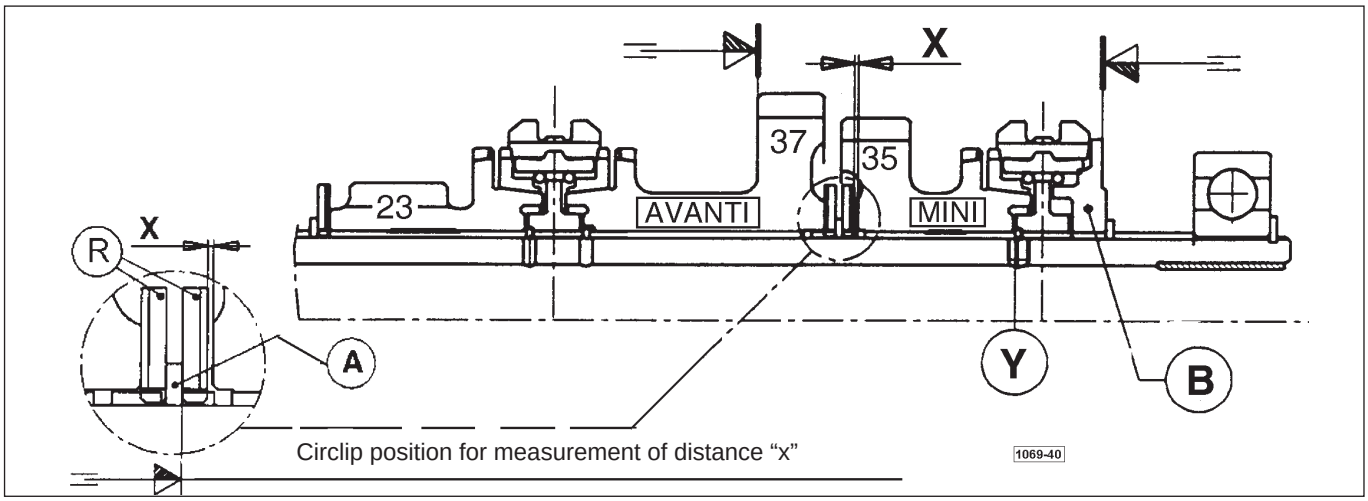
Bearings should be in perfect conditions without showing excessive radial or end play.
Holding the bearings pressed by hand and making them simultaneously turn in both directions of rotation a free sliding as well as no roughness at all should be felt.

Examine the taper roller bearings for proper working conditions, these should be neither worn nor overheated, replace as soon as poor working efficiency is suspected, as an abnormal bearing operation may result in either gear tooth seizure or noisy gearbox.

Warning: oil leakages from the clutch-disk-to-gearbox shaft or the engine-flywheel-to-P.T.O.-clutch shaft may be stopped when the assembly procedure is performed with the utmost care and operating in such a way that the splined parts are prevented from damaging the O-ring and also providing a thorough cleaning of the parts prior to being installed.

ADJUSTING PLAY OF THE GEARBOX SHAFTS BY MEANS OF THE THRUST PLATES ON THE MINI/INVERSOR SHAFT AND THE SECONDARY SHAFT

- **MINI/INVERSOR SHAFTS**

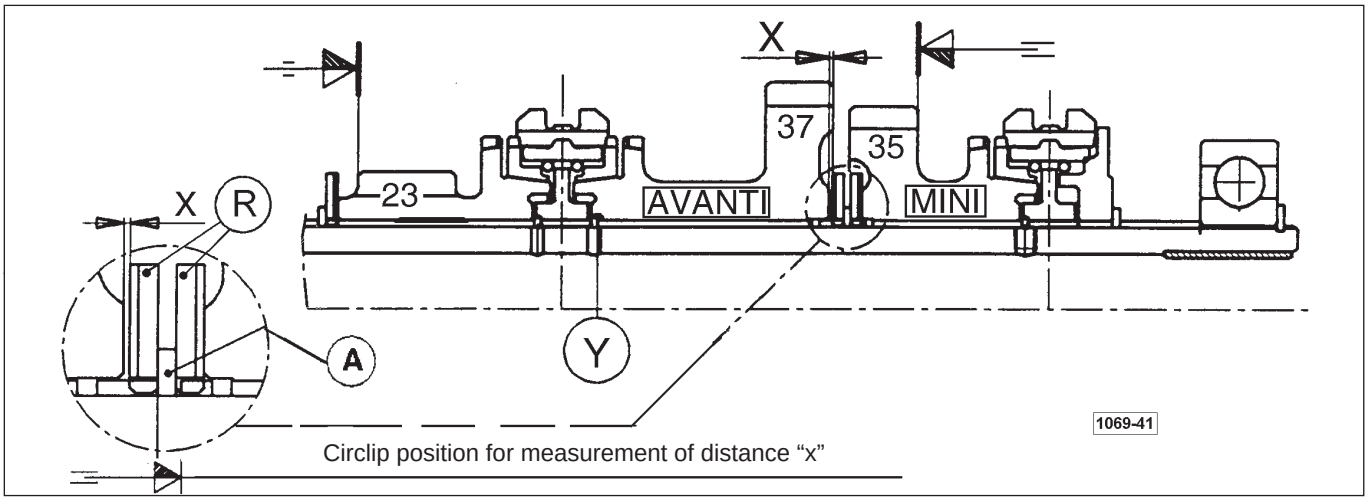


Circlip position for measurement of distance “x”

Adjusting the play of the minireduction gear mounted on the inversor shaft.
Obtain an play **X** of **0.15 ÷ 0.30 mm** of minireduction gear (z=35) by selecting the thrust plate **R** of the appropriate thickness from among those available:

cod.	**008.0251.0**	**010.0629.0**	**010.0630.0**	**010.0631.0**	**010.0632.0**

Play **X** is to be measured by pushing the spacer **B** in the direction indicated by the arrow up to the point where the synchroniser is pushed up against point **Y**, while simultaneously pushing the forward speed gear (z=37) in the direction indicated by the arrow up against circlip **A**.



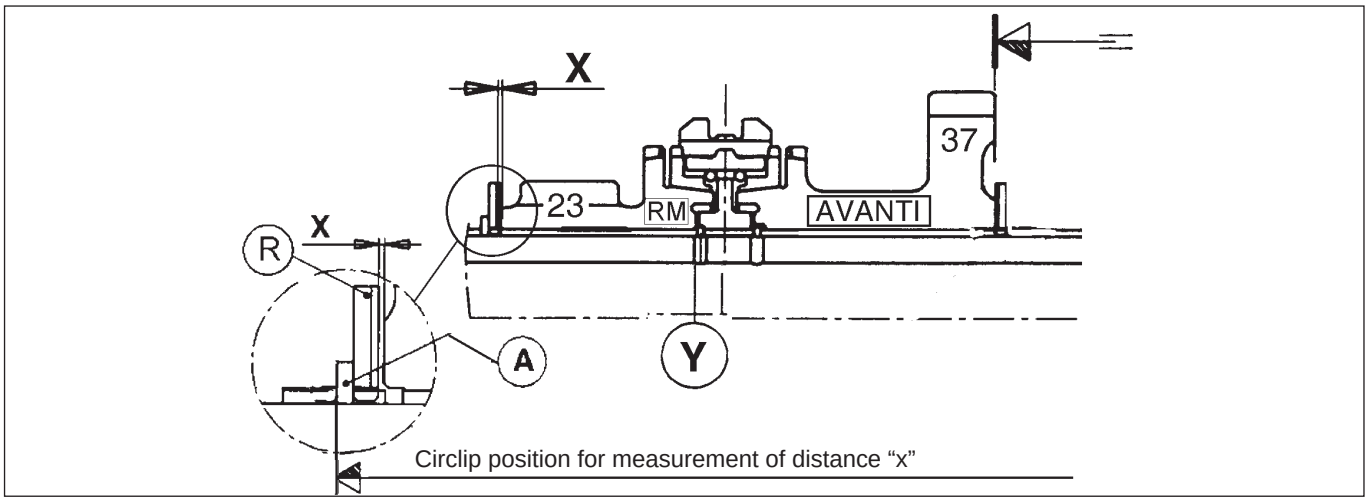
Circlip position for measurement of distance “x”

Adjusting the play of the forward speed gear on the inversor shaft (version with minireduction).
Obtain an play **X** of **0.15 ÷ 0.30 mm** of the forward speed gear (z=35) by selecting the thrust plate **R** of the appropriate thickness from among those available:

cod.	**008.0251.0**	**010.0629.0**	**010.0630.0**	**010.0631.0**	**010.0632.0**

Play **X** is to be measured by pushing the gear (z=23) in the direction indicated by the arrow so that the synchroniser is pushed up against point **Y**, while simultaneously pushing the minireduction gear (z=35) in the direction indicated by the arrow up against circlip **A.**

- **REVERSE SHAFT**

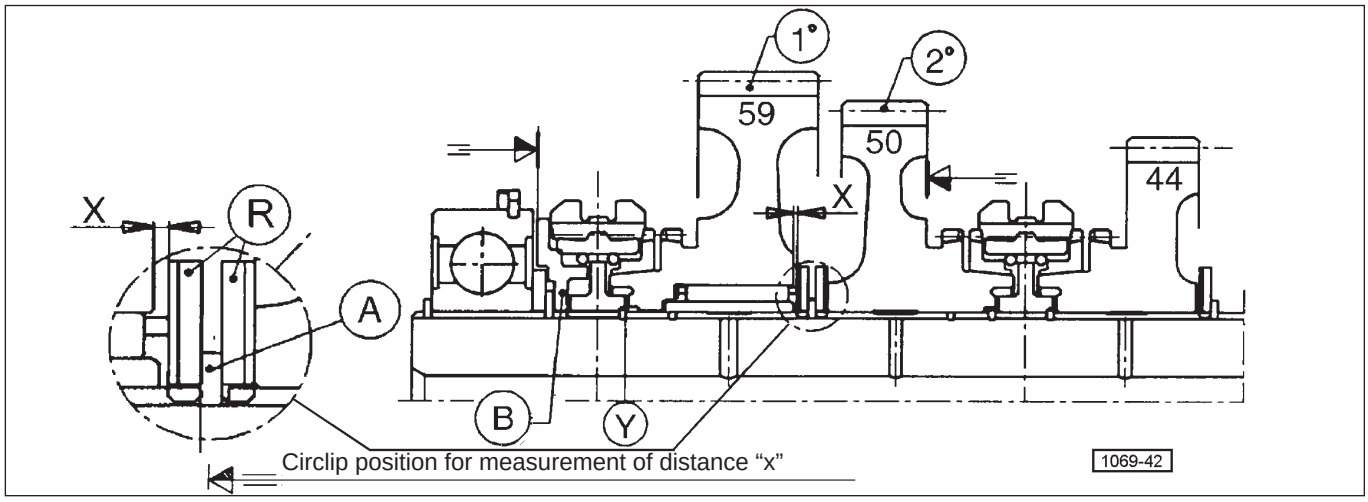


Adjusting the play of the reverse gearwheel mounted on the shuttle shaft.
Obtain play **X** of **0.15** to **0.30 mm** of reverse gearwheel (z = 23) by selecting and fitting a thrust washer **R** of suitable thickness from those available:

code **008.0251.0** **010.0629.0** **010.0630.0** **010.0631.0** **010.0632.0**

Play **X** is measured by pushing the forward gearwheel (z=37) in the direction indicated by the arrow, thereby pushing the synchronizer up against point **Y** and pushing the circlip A as indicated in the detail.

- **SECONDARY SHAFT 1st SPEED**

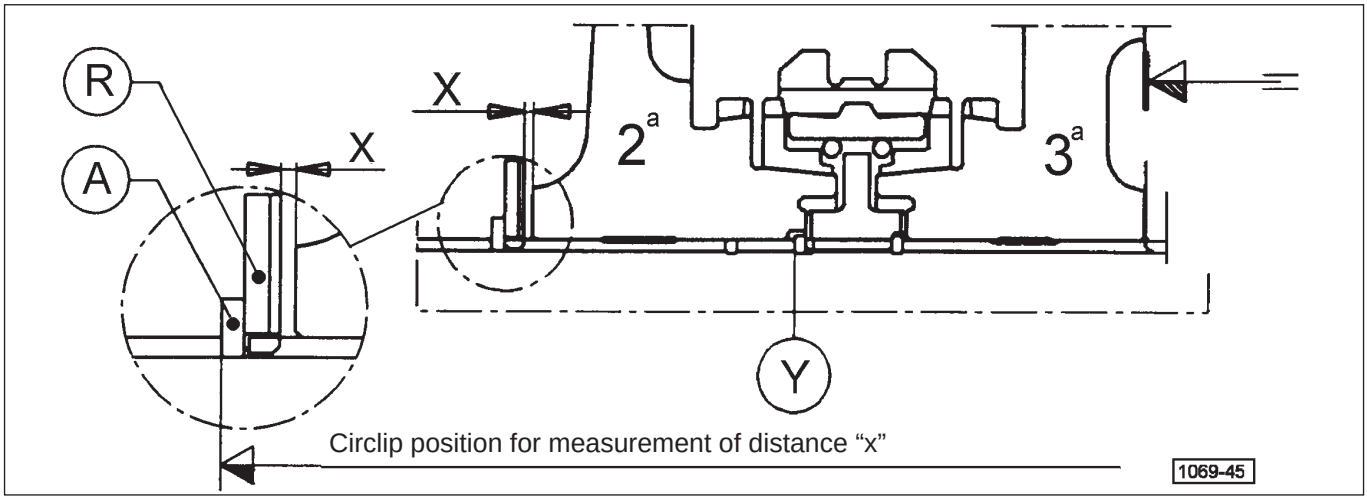


Adjusting the play of the **1st speed gear (z=59)**.
Obtain an play **X** of **0.15 ÷ 0.30 mm** of the 1st speed gear (z=59) by selecting the thrust plate **R** of the appropriate thickness from among those available:

cod. **008.0251.0** **010.0629.0** **010.0630.0** **010.0631.0** **010.0632.0**

Play **X** is to be measured with a feeler gauge while pushing spacer **B** in the direction indicated by the arrow so that the synchroniser is pushed up against point **Y**, while simultaneously pushing the 2nd speed gear (z=50) in the direction indicated by the arrow up against circlip **A**.

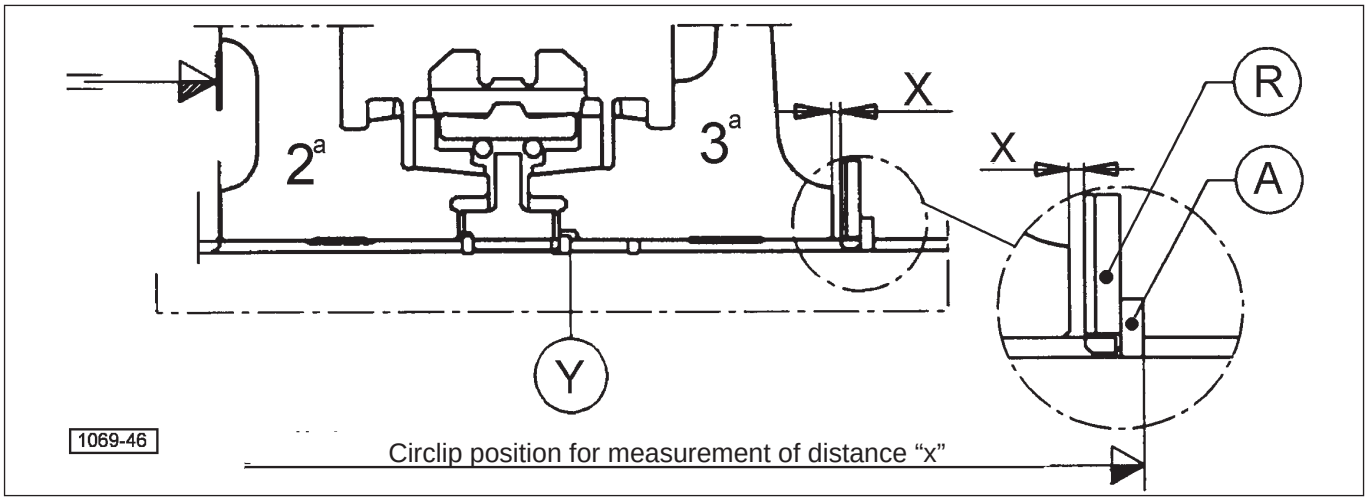
- **SECONDARY SHAFT 2nd/3th SPEED**



Adjusting play of **2nd speed gear** cluster
Obtain play **X** of **0.15 ÷ 0.30 mm** of the 2nd speed gear cluster by selecting a thrust washer **R** of the appropriate thickness from among those available:

cod.	**008.0251.0**	**010.0629.0**	**010.0630.0**	**010.0631.0**	**010.0632.0**

Play **X** is to be measured using a feeler gauge while pushing the **3rd** speed gear in the direction of the arrow, thereby pushing the synchronizer up against point **Y** and pushing the circlip **A** as indicated in the detail.

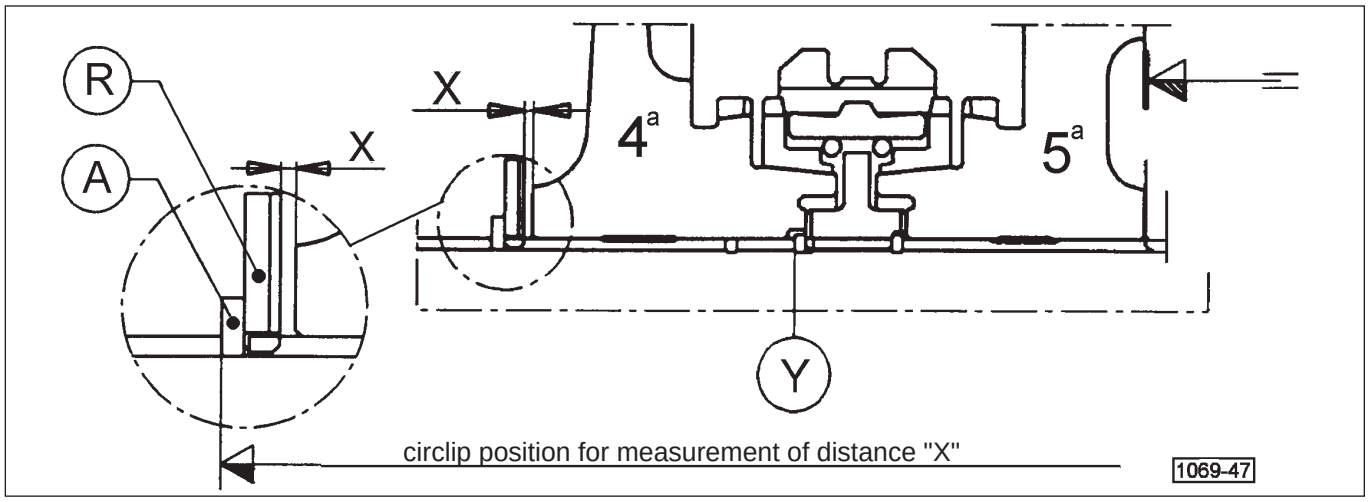


Adjusting play of **3rd speed gear**
Obtain a play **X** of **0.15 ÷ 0.30 mm** of the 3rd speed gear by selecting a thrust washer **R** of the appropriate thickness from among those available:

cod.	**008.0251.0**	**010.0629.0**	**010.0630.0**	**010.0631.0**	**010.0632.0**

Play **X** is to be measured using a feeler gauge while pushing the **3rd** speed gear in the direction of the arrow, thereby pushing the synchronizer up against point **Y** and pushing the circlip **A** as indicated in the detail.

- **SECONDARY SHAFT 4th/5th SPEED**

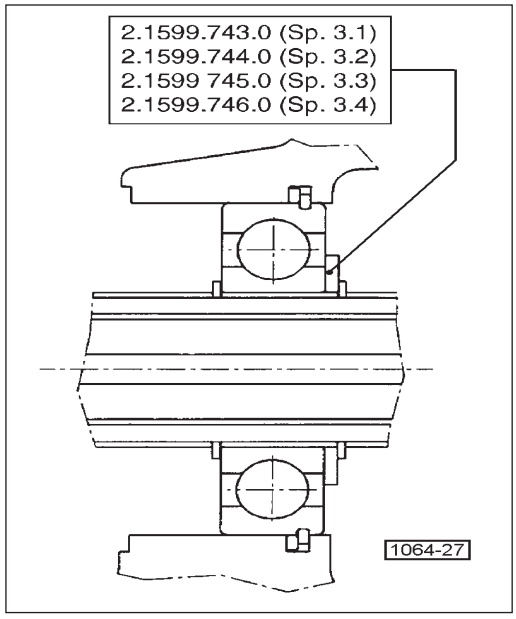


Adjusting the play of the **4th speed gear** cluster.
Obtain a play **X** of **0.15** to **0.30 mm** of the **4th speed** gear cluster by selecting and fitting a thrust washer **R** of the appropriate thickness from those available:

code 008.0251.0 010.0629.0 010.0630.0 010.0631.0 010.0632.0

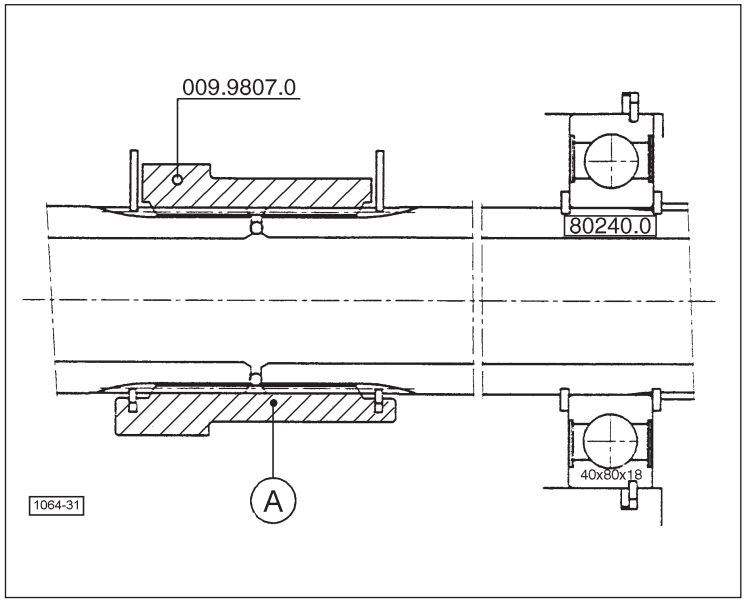
Play **X** is to be measured using a feeler gauge while pushing the **3rd** speed gear in the direction of the arrow, thereby pushing the synchronizer up against point **Y** and pushing the circlip **A** as indicated in the detail.

SECONDARY SHAFT

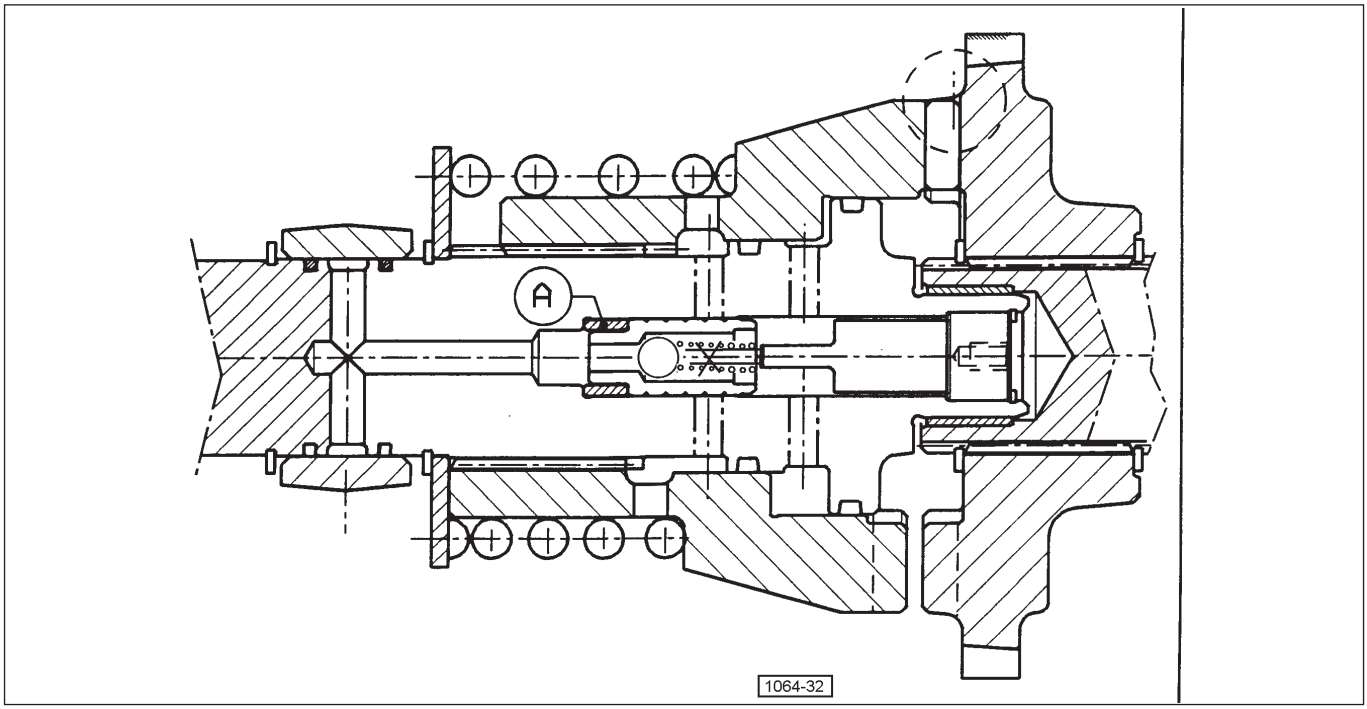


Shimming the front bearing of the secondary shaft
Fit shims of appropriate thickness to reduce end-float between the shaft and the bearing to the minimum possible amount. By fitting the correct shims you can obtain a **maximum play** of **0.09 mm.**

INPUT SHAFT AND INVERSOR SHAFT



Fitting the bush joining the input shaft to the inversor shaft
Fit bush **A** oriented as shown in the figure to facilitate subsequent removal.



Fitting the spacer determining the stroke of the valve for hydraulic disengagement of front-wheel drive.
To allow correct disengagement of the front-wheel drive, bush **A** must be installed as shown in the figure above.

WARNINGS RELATED TO ASSEMBLY OF THE GEARS OF THE P.T.O. UNIT, THE RANGE REDUCTION UNIT AND THE SYNCHRONISED P.T.O. SHAFT

Assembly of the P.T.O.

Install:
— The synchronised P.T.O. output shaft (to be installed after the range reduction unit);
— The P.T.O. output shaft and relative gears;
— The rear P.T.O. shaft.

Installation of the range reduction unit, the gear for the front-wheel drive shaft and the parking brake discs

— Position on the bottom of the gearbox the L-V range reduction unit (complete with bushes and roller cage) and the super-reduction gear (complete with roller cage).
— Slide the shaft, complete with circlip, through the front end of the gearbox, and fit the previously positioned gears onto the shaft.
— Fit the thrust washer and 2 circlips.
— Position the drive gear of the front-wheel drive complete with parking brake discs.
— Fit the ball bearing for the rear end of the shaft, inserting it in its housing in the gearbox.
— Insert the thrust washer and position the circlip.
— Position the bearing and the bearing circlip in the housing;
— Fit the thrust washer and the circlip;
— Install the synchronised P.T.O. shaft in the rear end of the gearbox, and fit the control sleeve to the shaft.
— Fit the bevel pinion and the gears of the range reduction unit with the relative synchroniser;
— Fit the differential housing complete with the crown wheel and the internal diff. lock control.

Warnings:
The spacer installed on the rear output shaft is fitted with an O-ring on the outer surface and an internal oil seal.
To remove this spacer requires the use of a puller to be attached to the spacer using the two holes provided.
At each dismantling of the pins of the brake and clutch control levers, check the bushes for wear.

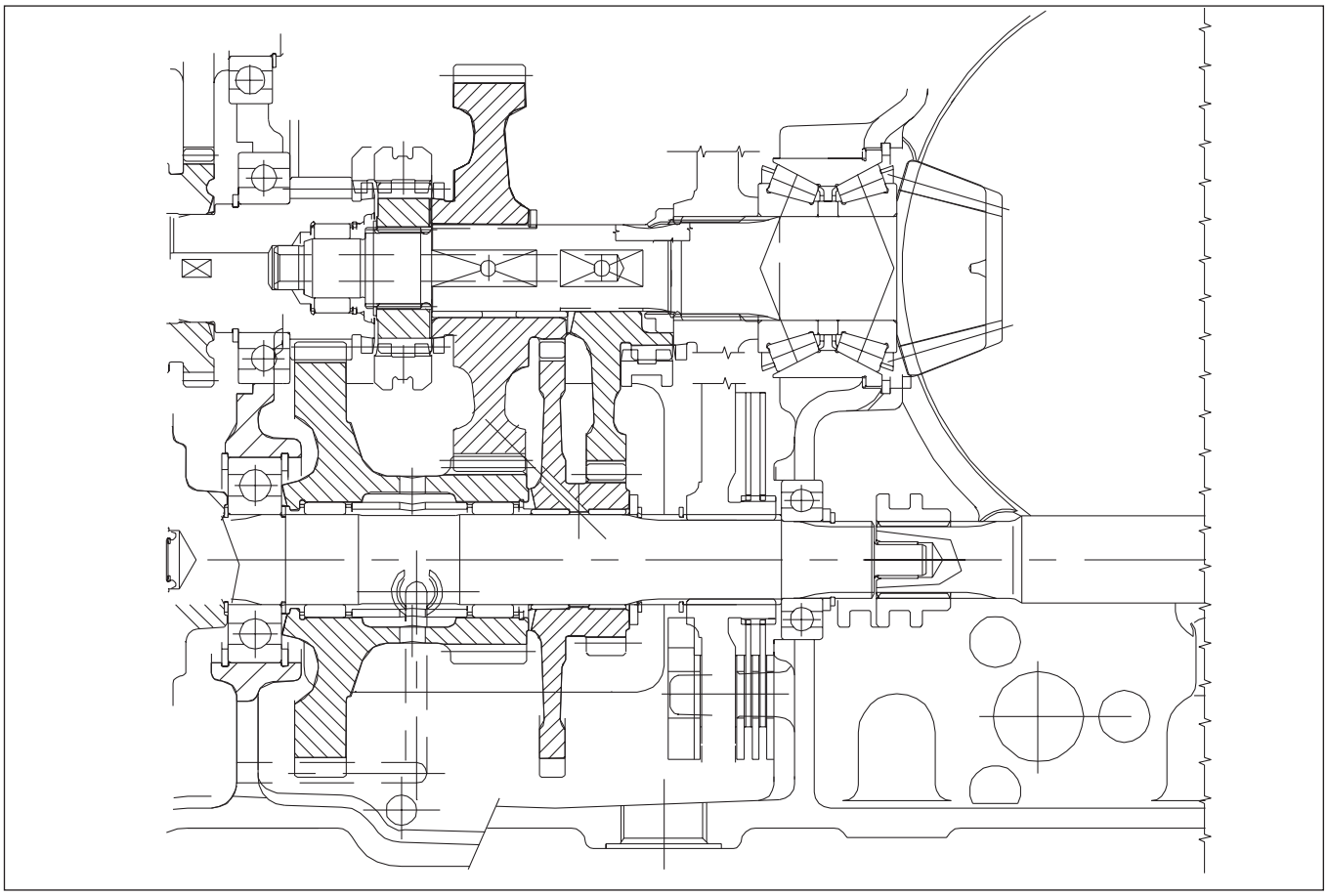


Fig. 42 - Range gearbox

Fig. 43 - Removal of the groundspeed P.T.O. sleeve.

Fig. 44 - Removal of the circlip securing the shaft.

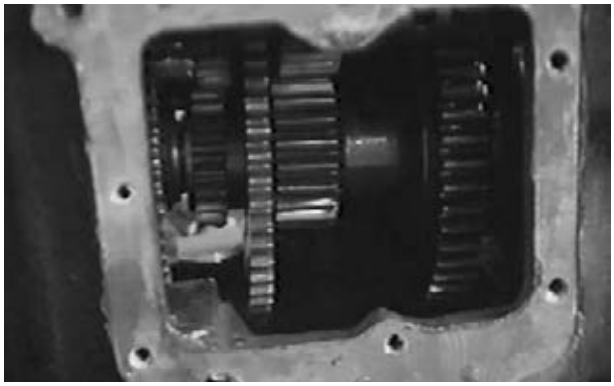


FIg. 45 - Gears on range gearbox shaft.

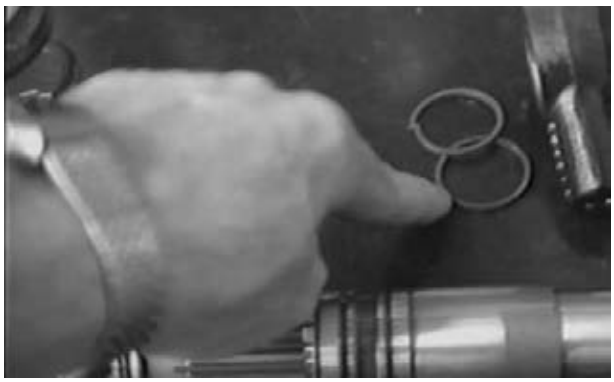


Fig. 46 - Checking the oil seals on the 4WD shaft.

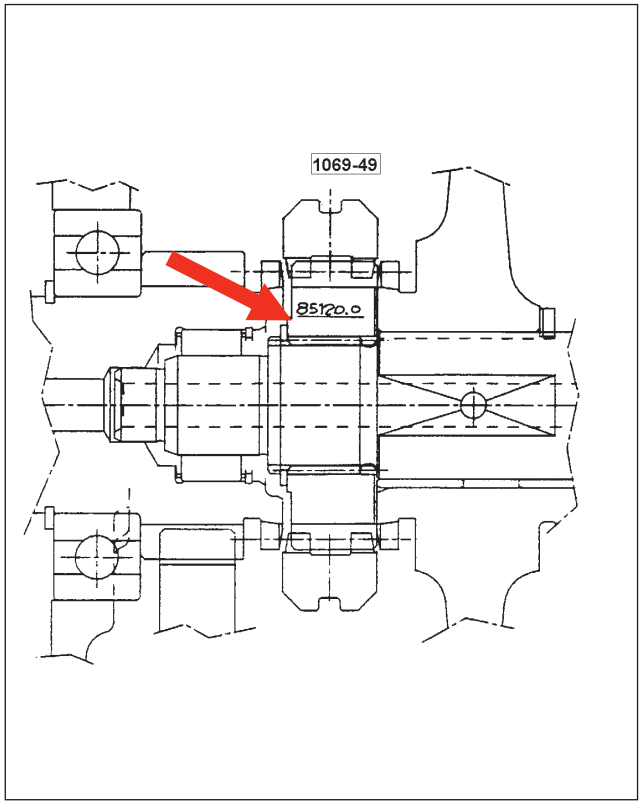


Fig. 47 - Range gearbox synchronizer.

Range gearbox synchronizer.
Warning: the synchronizer hub must installed so that the indicated by the arrow is positioned as shown in the figure.

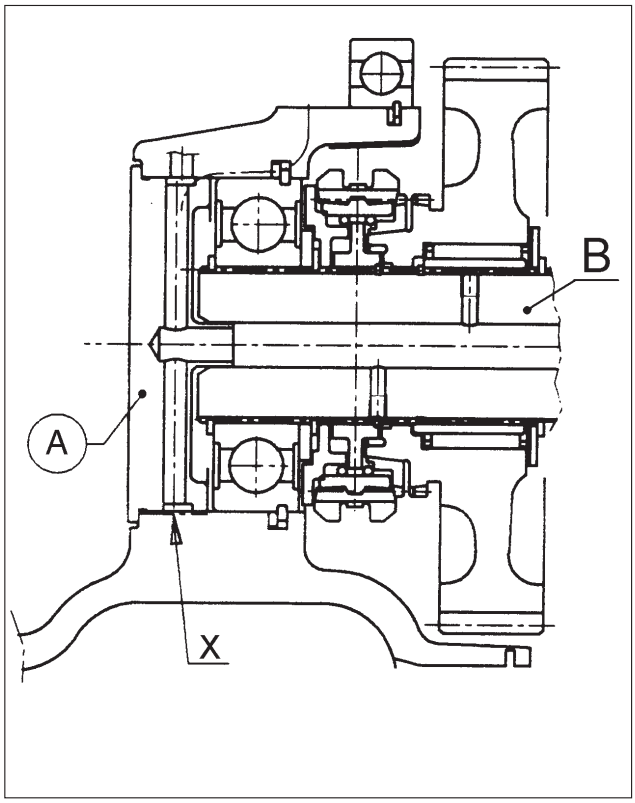


Fig. 48 - Secondary shaft cover plate

Fitting the secondary shaft cover plate. Fit the cover **A** applying LOCTITE 601 around the hole of the front gearbox casing indicated **X** in the figure.
B - secondary shaft

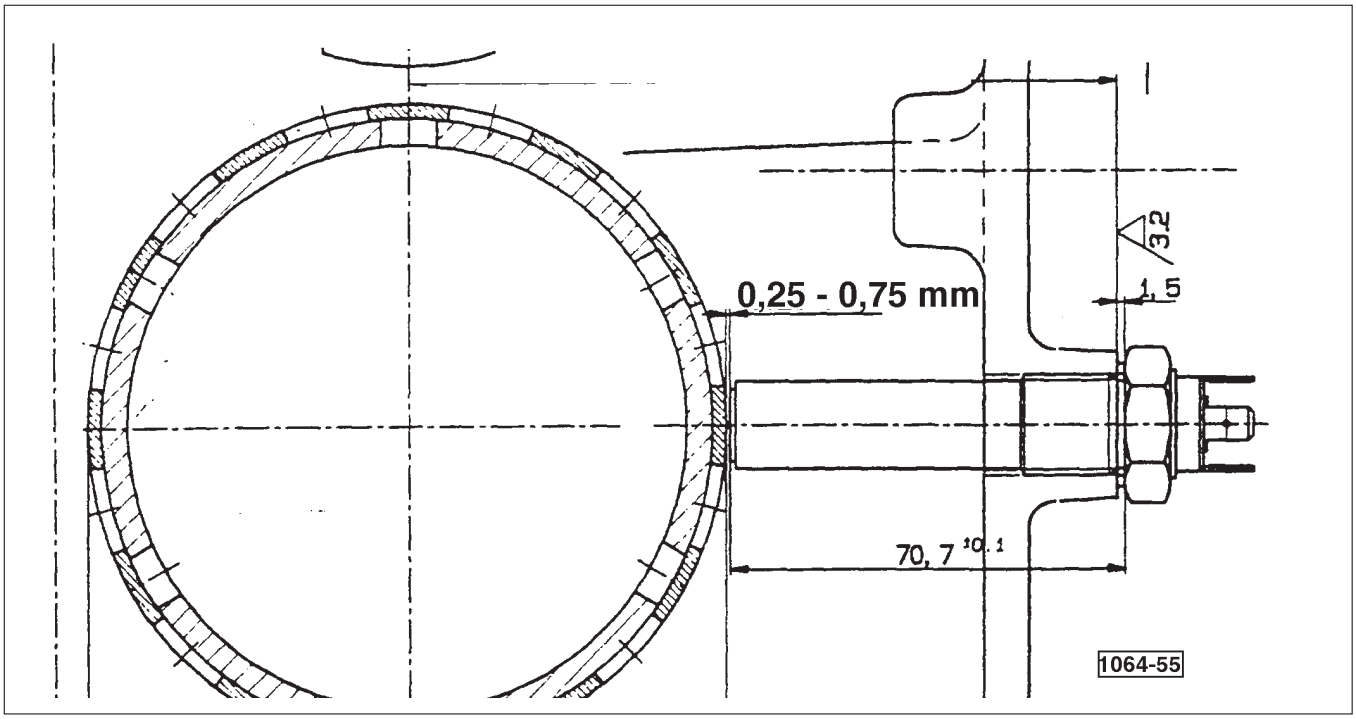
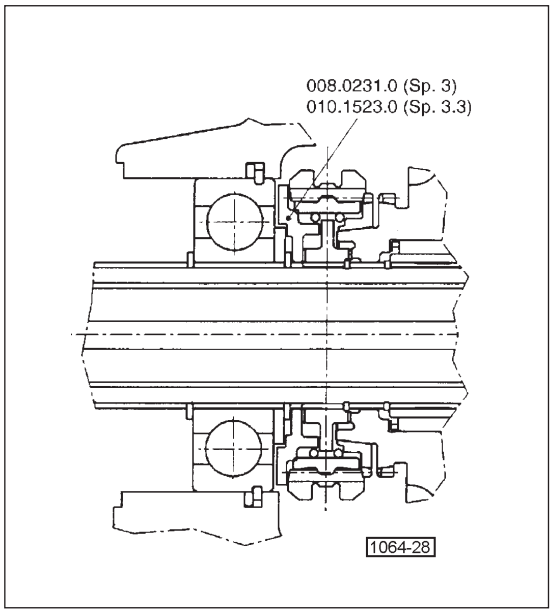


Fig. 49 - Fitting the transmission speed sensor pickup.



Fitting the baulk ring of the synchronizer for the 1st **speed gear mounted on the secondary shaft.**
Should it be necessary to restore the correct play **X of 0.15 to 0.30 mm** of the 1st speed gear (z=59), it is possible to select a synchroniser baulk ring of suitable thickness from the two available, as shown in the figure.

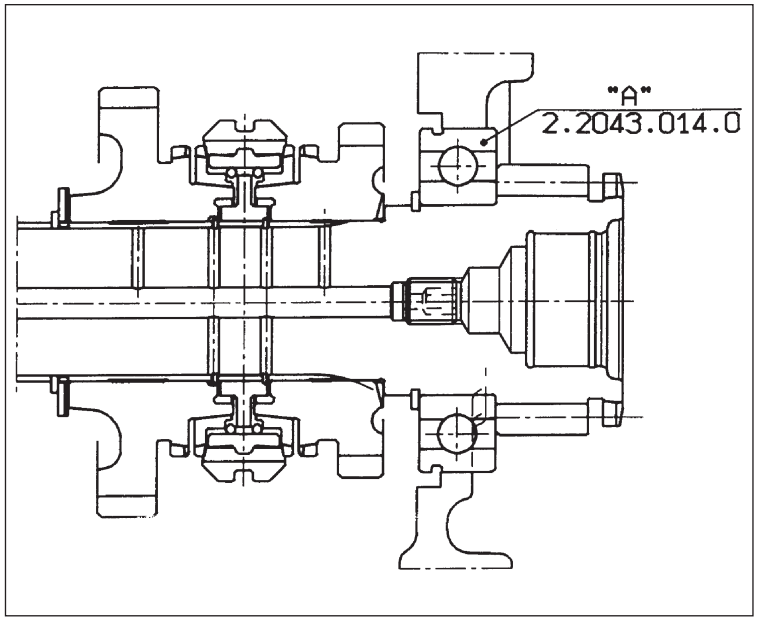


Fig. 50 - Fit the 50x90x20 ball bearing, indicated A in the figure, without the retaining ring.

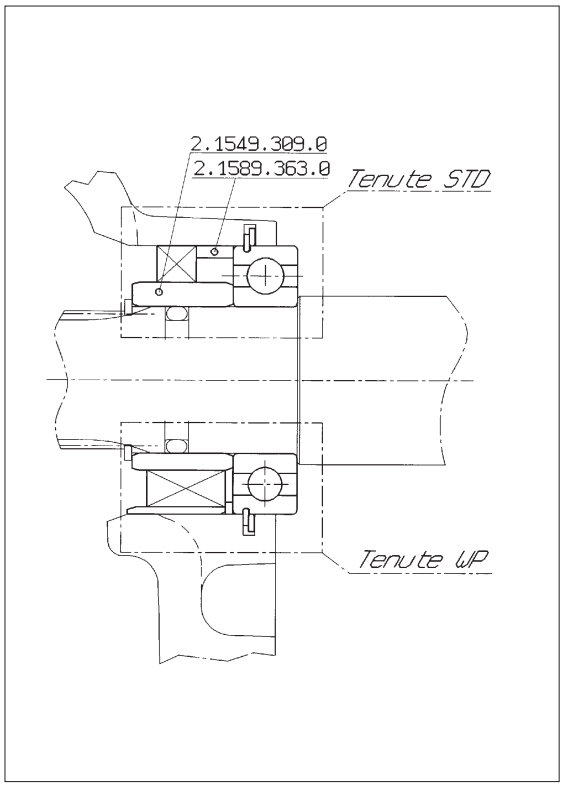


Fig. 51 - Bearing for 4WD shaft.

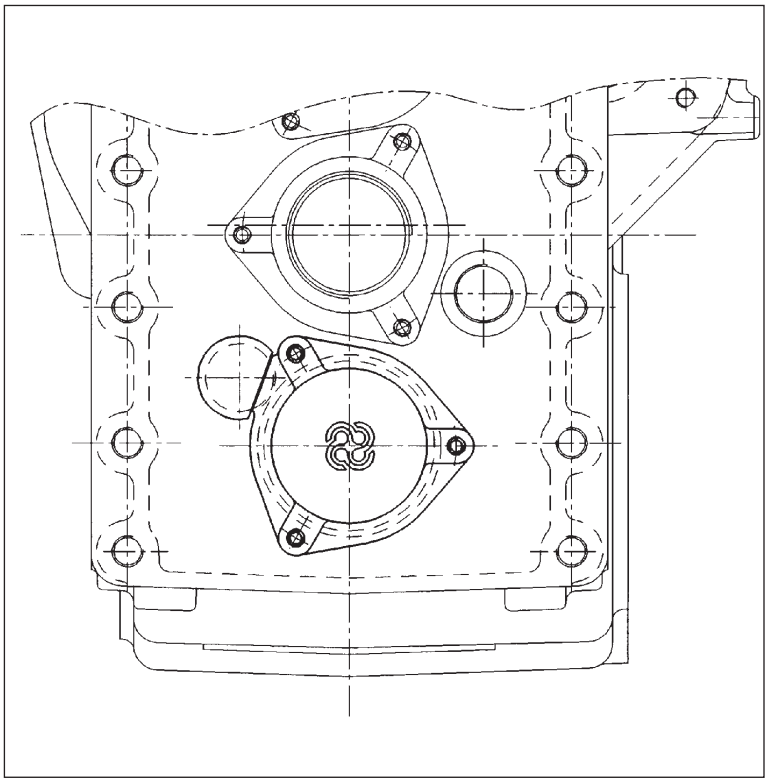


Fig. 52 - Position the groundspeed P.T.O. cover plate as indicated in the figure.

Points where sealant is to be used

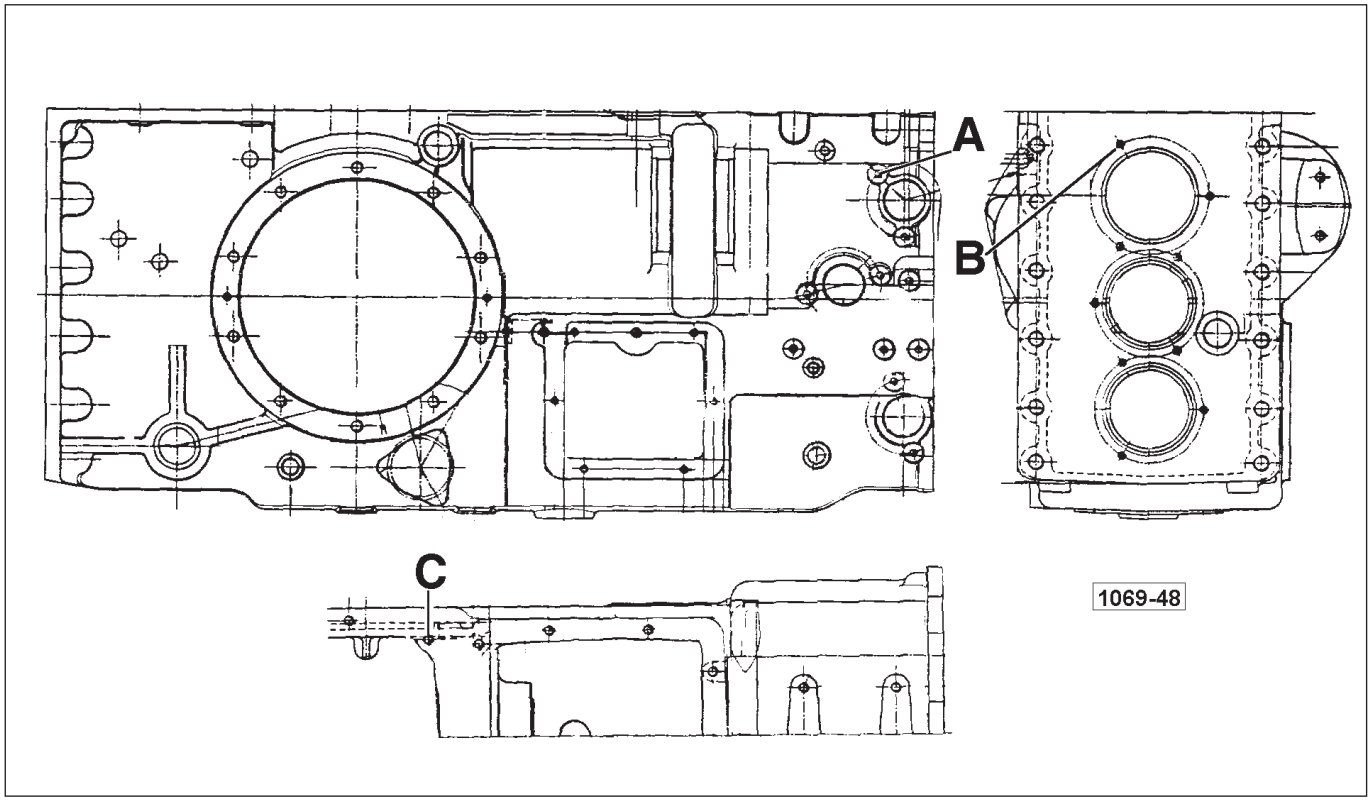


Fig. 53 - Rear gearbox.
Apply LOCTITE n° 601 to the base of studs **A** (n° 9) - **B** (n° 1) - **C** (n° 1), as shown in the figure, prior to assembly.

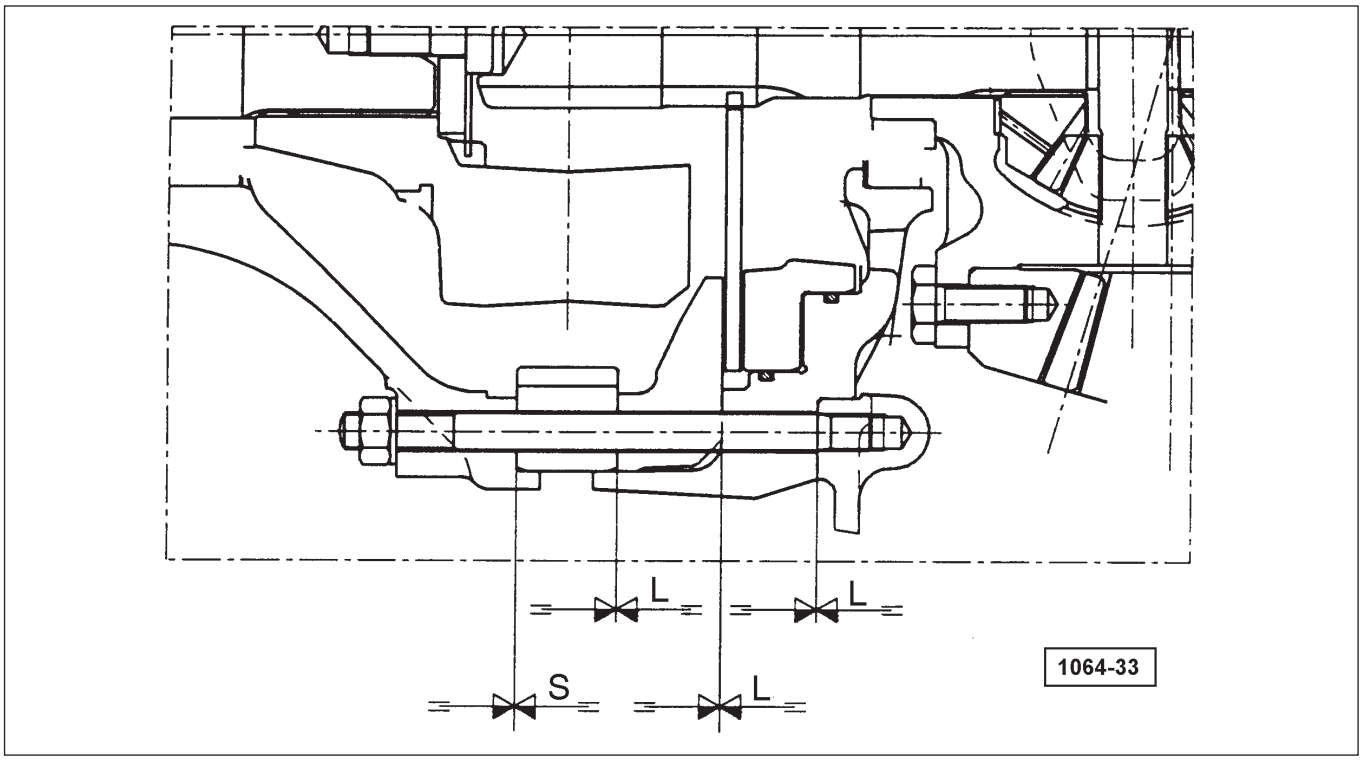


Fig. 54 - Fitting the wheel support.
Apply PIANERMETIC 510 sealant to the surfaces indicated with the letter **L** in the figure; apply SILASTIK sealant to the surfaces indicated with the letter **S**.

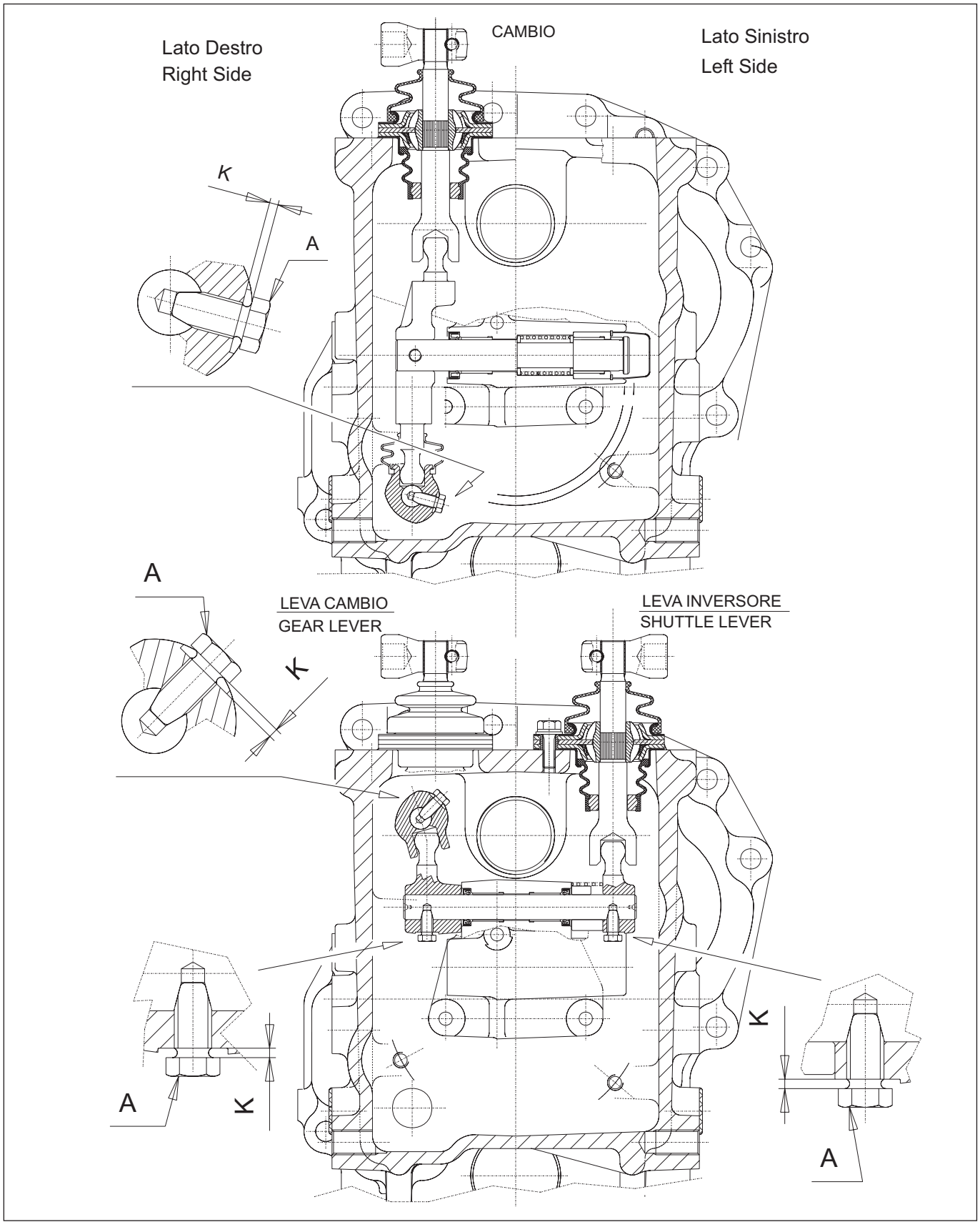


Fig. 55 - Tightening the bolts of the gear selector linkage.
After fully tightening the bolts **A** indicated in the figure, check that distance **K** is equal to **3 mm**.

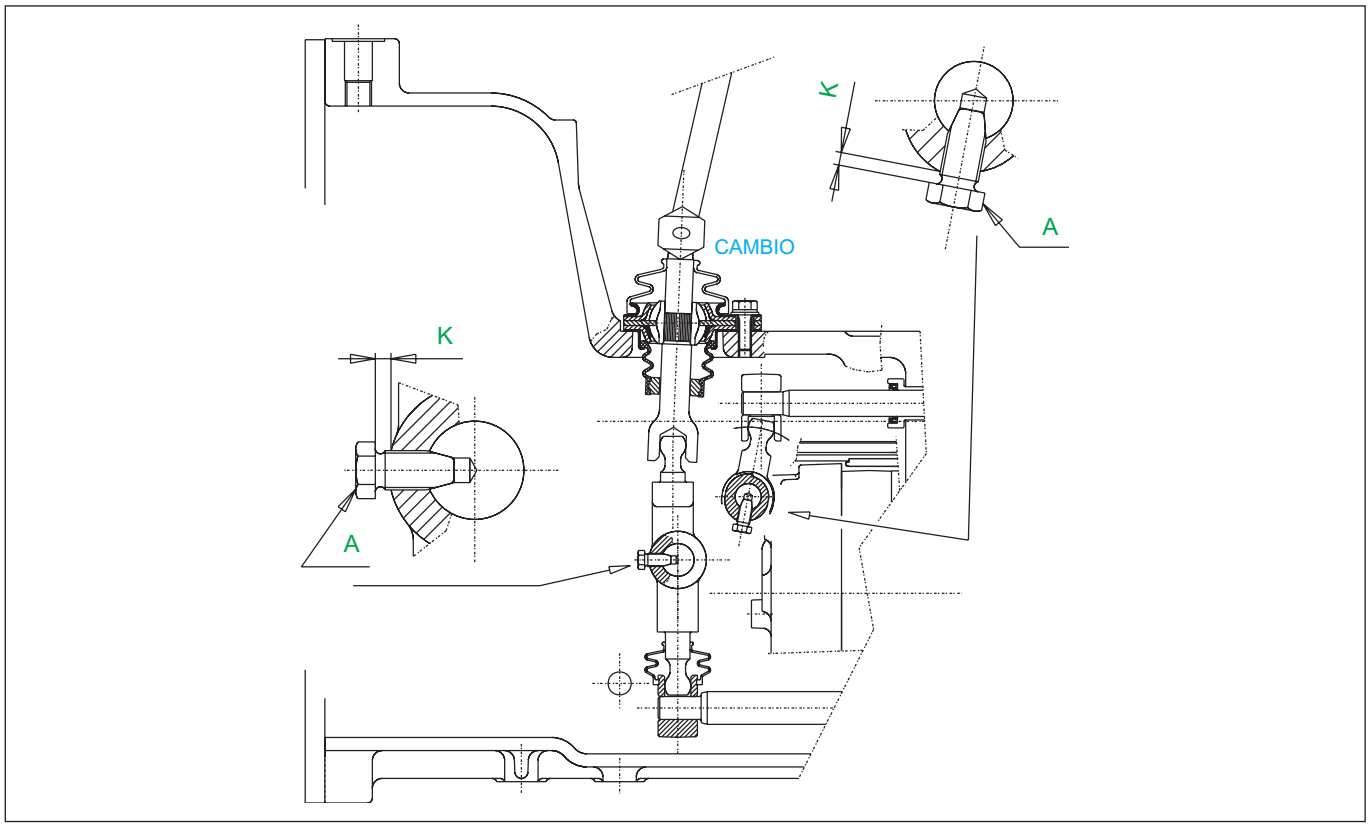


Fig. 56 - Tightening the bolts of the gear selector linkage.
After fully tightening the bolts **A** indicated in the figure, check that distance **K** is equal to **3 mm**.

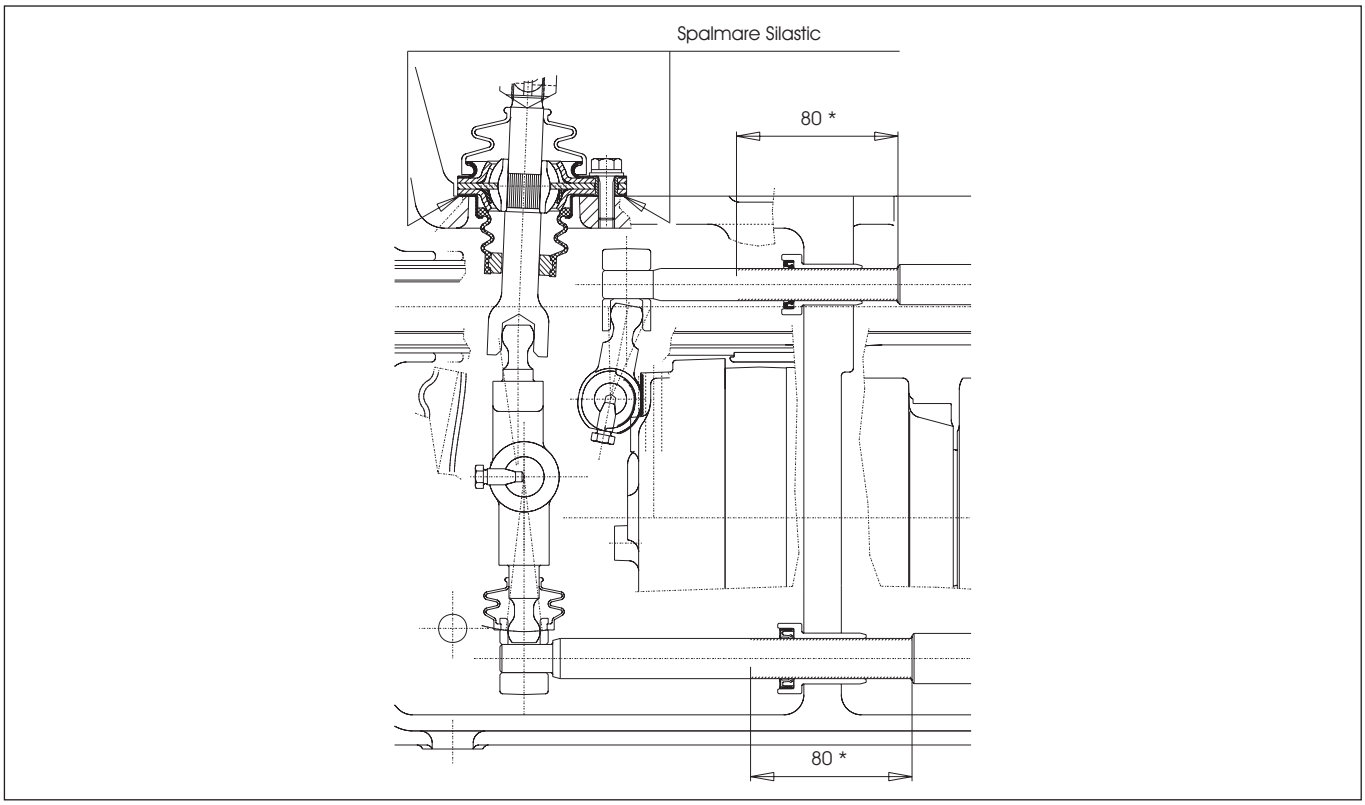


Fig. 57 - Silastik and grease application. Apply a coating of Silastik between the face of the front gearbox and the retaining face of the rubber gasket. Apply grease to the sections of the selector rods indicated * in the figure.

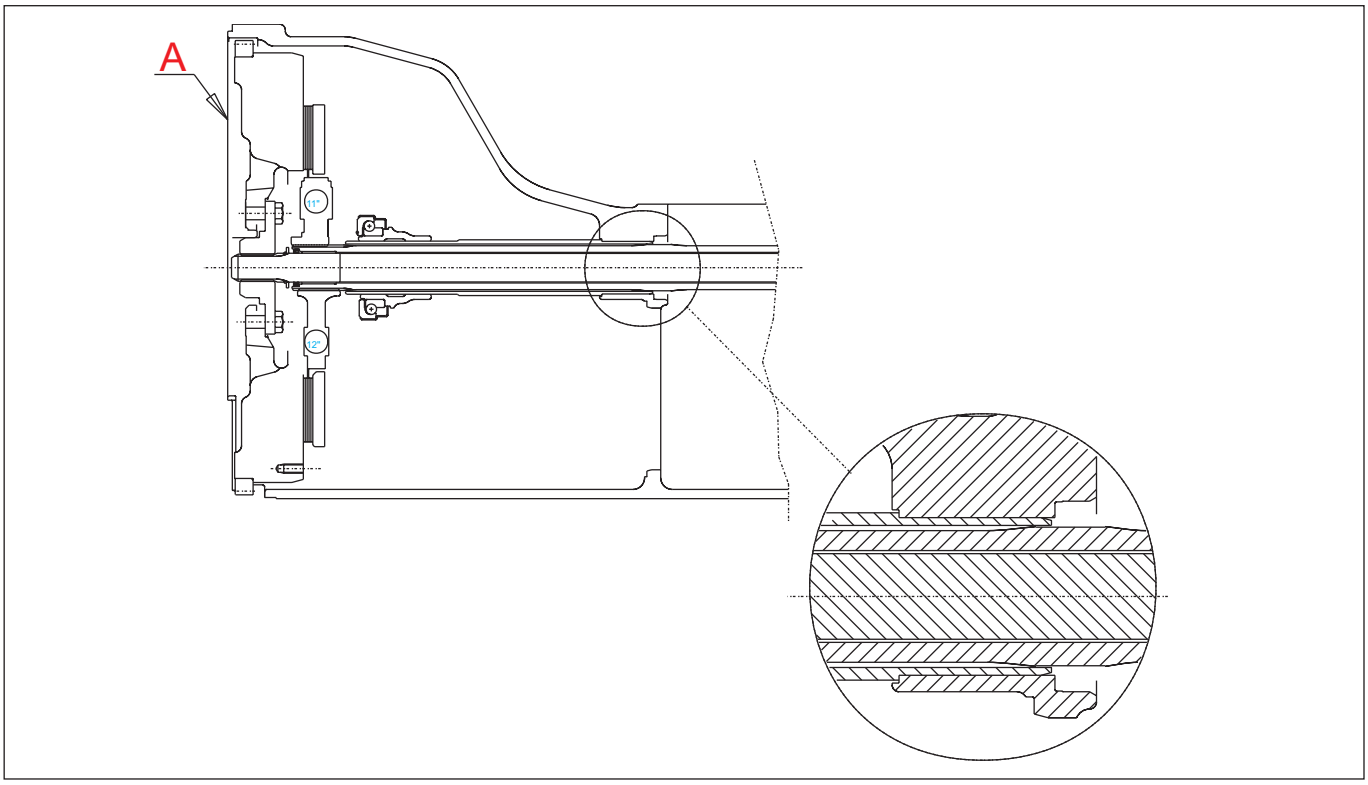


Fig. 58 - Correct installation of the support tube for the clutch control sleeve.
Position the gearbox vertically with side **A** facing upwards. Apply Loctite 601 to the seat of the tube and fit the tube.
Wait about 1 hour before returning the gearbox to the horizontal position.

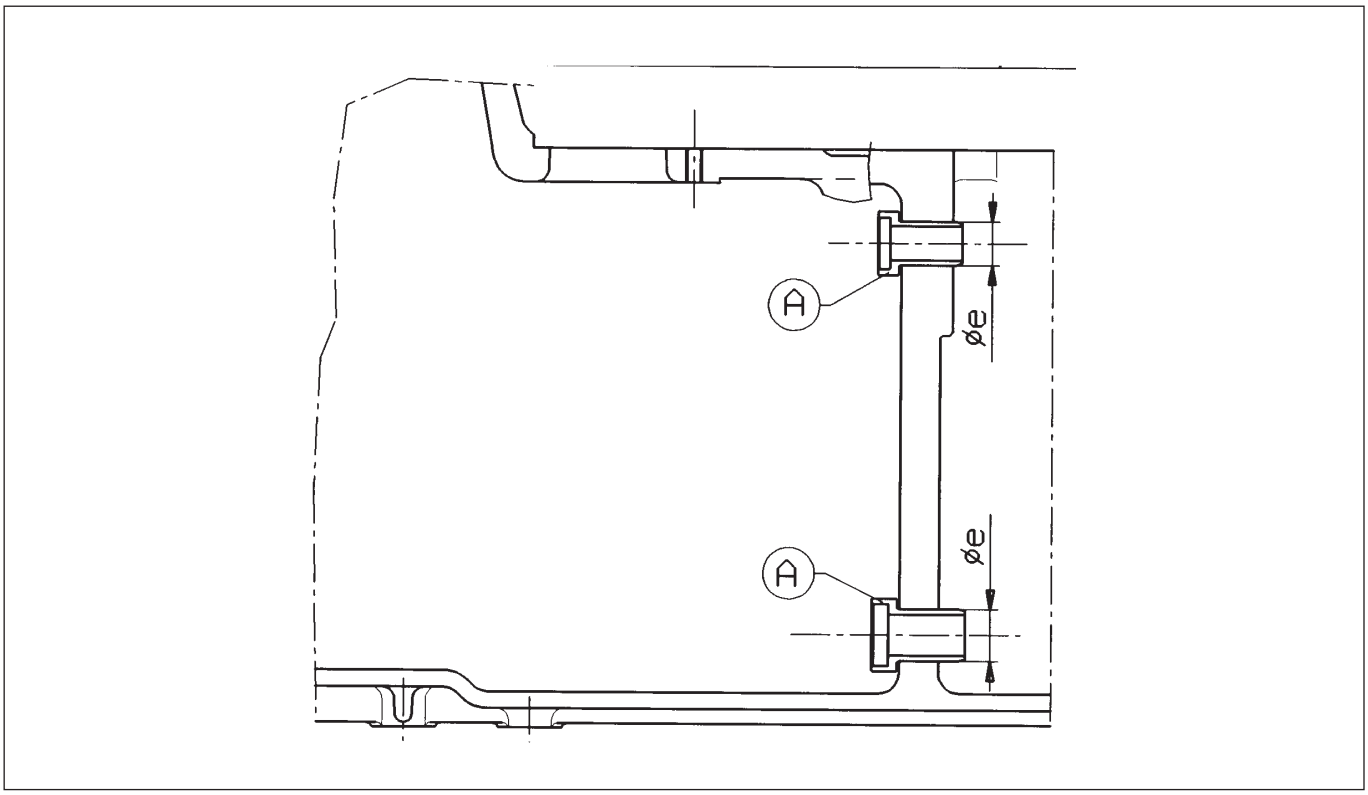


Fig. 59 - Fitting the selector rod support bushes.
Apply Loctite 638 to the contact surfaces of the bushes A, and fit them pressing them up against the wall of the casing.

IMPORTANT INSTRUCTIONS

Fig. 60 - Positioning of the 4WD shaft complete with engagement/disengagement system. After positioning the shaft, connect the oil manifold to solenoid control valve (located on the mounting on the right-hand side of the gearbox) using the specially shaped pipe.

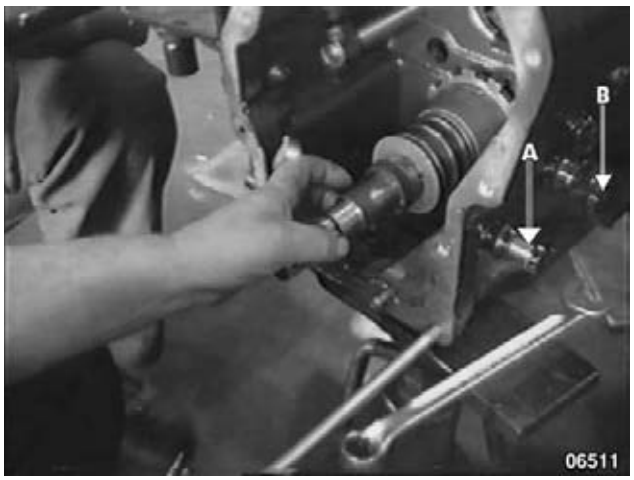


Fig. 61 - After positioning the pipe, connect the oil manifold by inserting union **A** from the left-hand side of the gearbox. The PICK-UP speed sensor must be installed in hole **B** indicated by the arrow (see instructions on page 175).

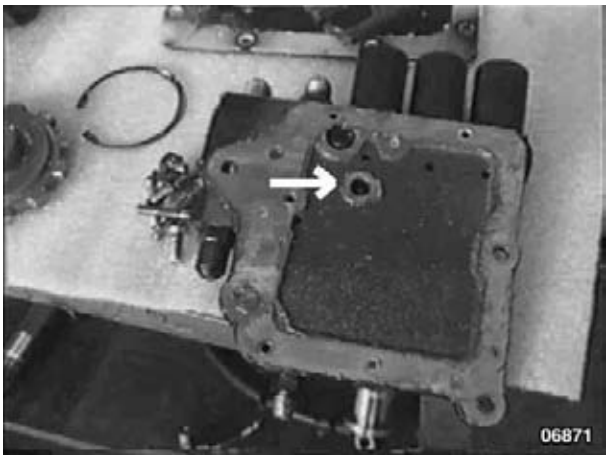


Fig. 62 - The pipe connecting the hydraulic control of the front-wheel drive must be connected to the solenoid valve block at the external fitting indicated by the small arrow.

Fig. 63 - The MULTISPEED unit is to be installed in the gearbox and the shaft end-float adjusted with the shaft engaged with the epicyclic reduction unit.

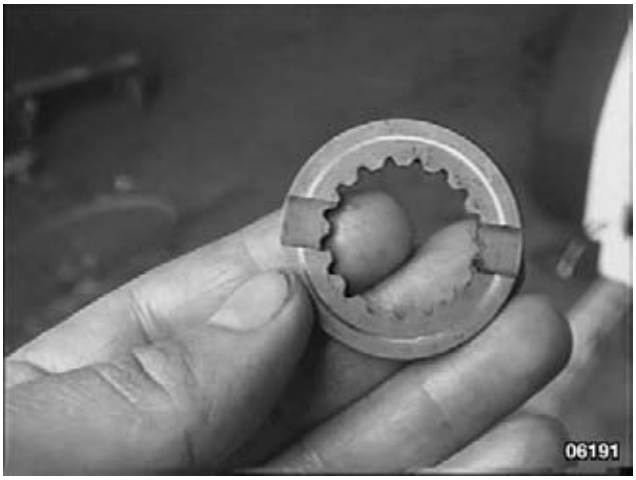


Fig. 64 - End-float is to be adjusted by fitting a shim of appropriate thickness following the instructions given on page 137.

Fig. 65 - Checking shim thickness using a micrometer.

Fig. 66 - Fitting the shim to adjust end-float of the MULTISPEED shaft.

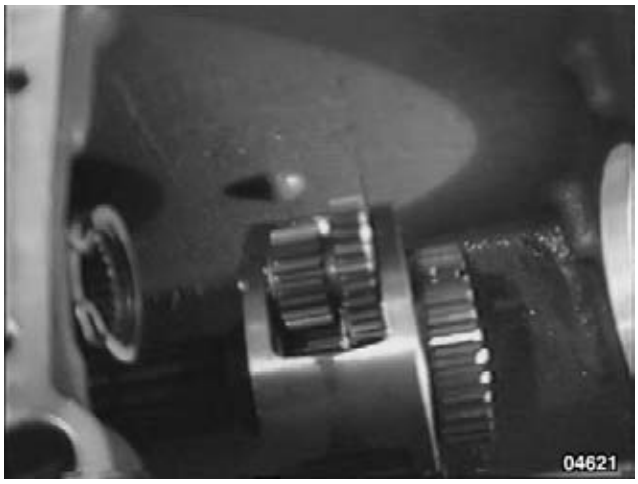


Fig. 67 - Positioning the epicyclic reduction unit of the MULTISPEED in the front gearbox.

Fig. 68 - Fitting the epicyclic reduction unit inside the housing previously attached to the gearbox output shaft.

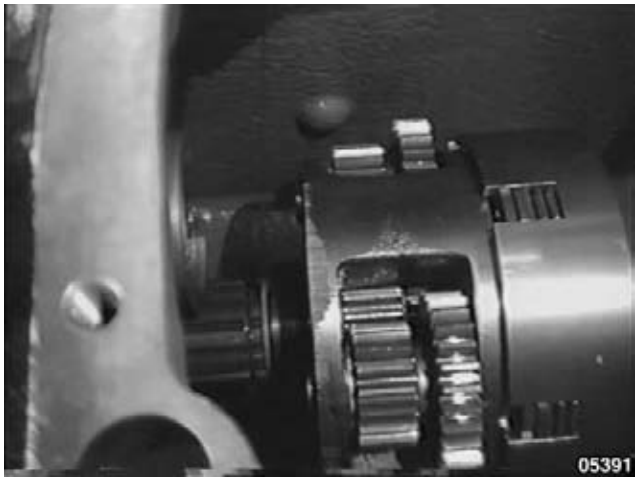


Fig. 69 - MULTISPEED unit installed in the gearbox.

Fig. 70 - Oil seal on shaft between clutch and gearbox. This seal can be renewed by accessing the inside of the front gearbox.

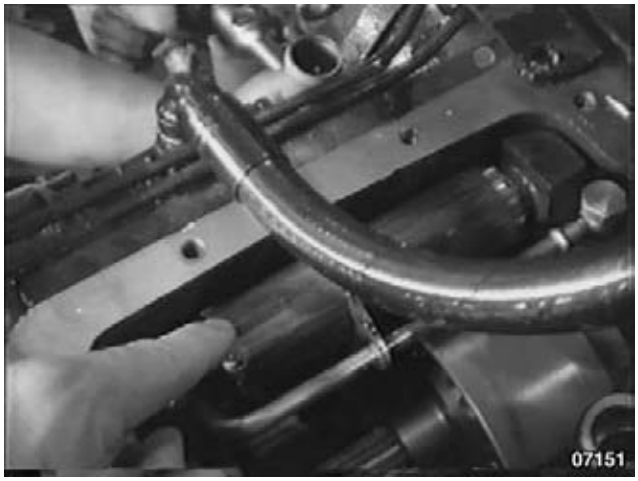


Fig. 71 -Positioning the P.T.O. clutch unit inside the gearbox.

Bevel drive adjustment (see figs. 1 and 3).

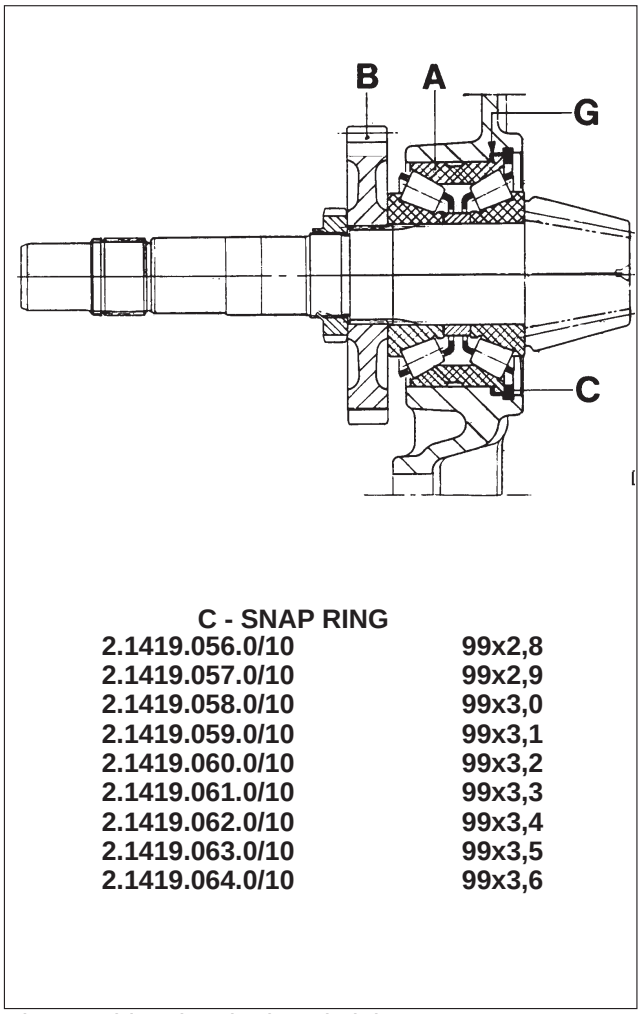
Install the pinion in the gearbox together with the bearings pack **A** (without interposing shims); position gear **B** and tighten ring nut to a torque of: 28 ÷ 30 Kgm (275÷294 Nm).

Fit the differential housing without the crown wheel; shim up the tapered bearings **D** and **E** so that they rotate freely in the housings yet a slight preloading can be felt; then add a 0.1 mm. shim to the pack in order to preload the bearings.

Adjust the distance **F** between the differential housing and the head of the pinion, interposing shims in the position **G** indicated in figure 1; the exact measurement for this adjustment is obtained by adding or subtracting the value stamped on one of the pinion teeth to/from the base value of 2.5 mm.

Secure the bearings pack by inserting a circlip (select one from those supplied in thicknesses of 2.8 ÷ 3.6 mm. (see figure 1) which fits perfectly in the seating preventing any bearing play.

Install the crown wheel in the differential housing and, with a dial gauge, measure the crown wheel/pinion backlash; the correct value range is 0.18÷ 0.24 mm (see figure 4 on page 184). If the measured backlash does not come within this range, remove shims from pack **D** and add them to pack **E**, to move the crown wheel nearer to the pinion, or vice versa to move it further away.
NB.: On completion of the bevel drive adjustment operations, loosen off the ring nut, apply Loctite 270, then tighten to a torque of 28 ÷ 30 Kgm (275 ÷ 294 Nm) and then stake the pinion retaining nut.
Tighten the crown wheel retaining screws to a torque of 7÷8 kgm (68÷78Nm).



C - SNAP RING

2.1419.056.0/10	**99x2,8**
2.1419.057.0/10	**99x2,9**
2.1419.058.0/10	**99x3,0**
2.1419.059.0/10	**99x3,1**
2.1419.060.0/10	**99x3,2**
2.1419.061.0/10	**99x3,3**
2.1419.062.0/10	**99x3,4**
2.1419.063.0/10	**99x3,5**
2.1419.064.0/10	**99x3,6**

Fig. 1 - Shimming the bevel pinion.

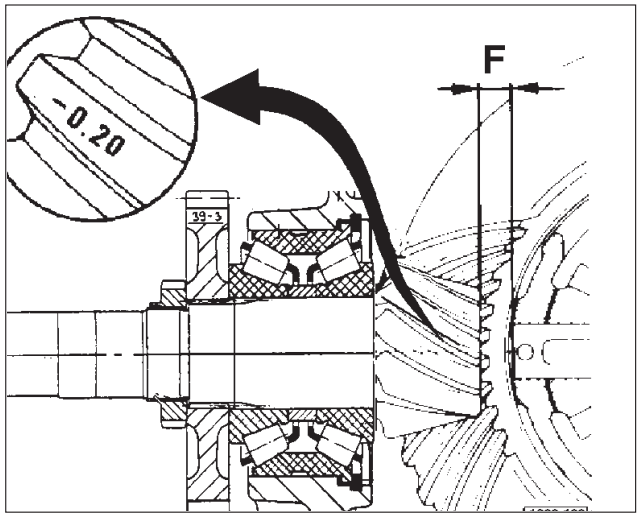


Fig. 2 - Value stamped on a tooth of the pinion.

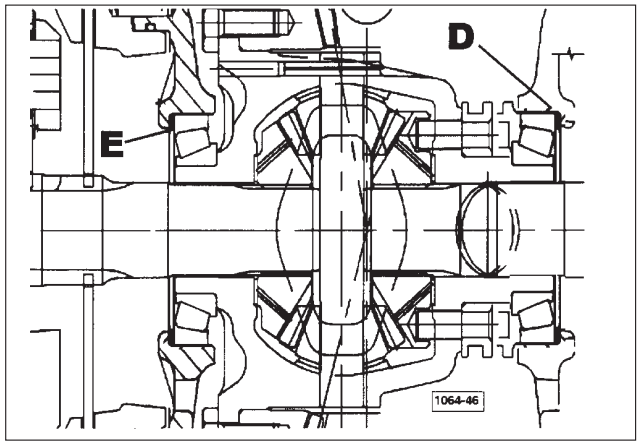


Fig. 3 - Shimming the crown wheel.

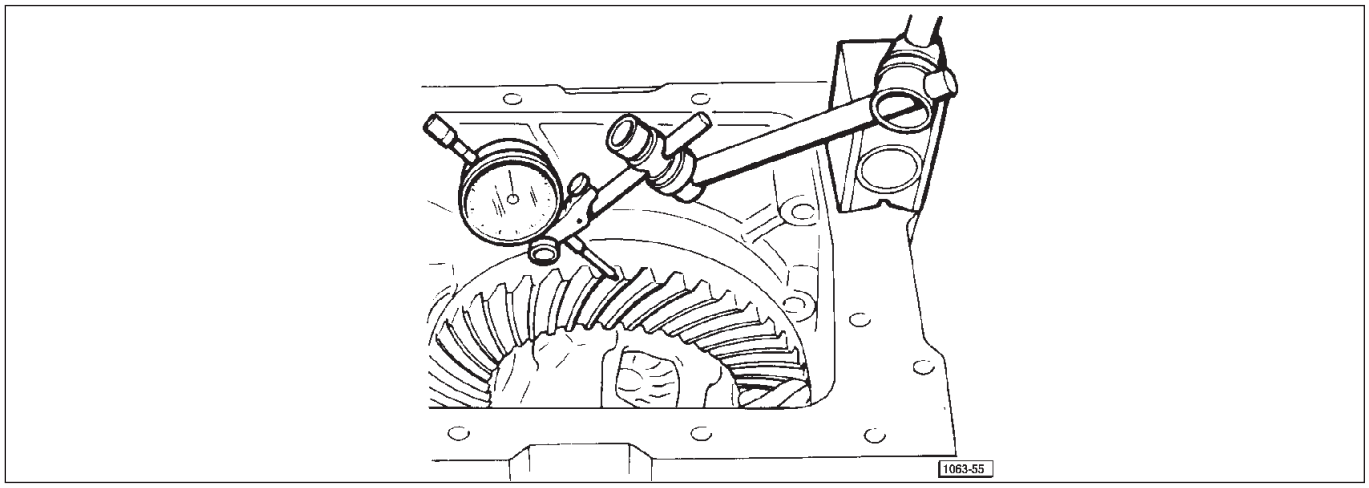


Fig. 4 - Checking the backlash between the crown wheel and pinion.

Servicing operations

Installation of the bevel drive without renewing any parts.
New adjustment is not required; make sure that the same shims packs are refitted in the same places, and check that the distance between the head of the pinion and the differential housing is as prescribed.

Renewal of crown wheel bearings
Shim the bearings to obtain a preloading of 0.1 mm; then adjust backlash between the crown wheel and pinion.

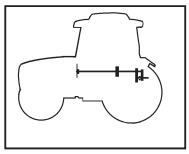
Renewal of pinion bearings
The bevel drive will require adjustment.

Installing the rear differential lock
The rear differential lock system does not require any adjustment.

Tightening torques

Degrease and clean all bolts/screws before tightening.

	Kgm	Nm
crown wheel retaining bolts	7 to 8	68 to 78
differential flange retaining bolts	6	58
bolts securing differential cage to half-shafts	6	58
rear carrier retaining nuts (on assembly apply PIANERMETIC 510 sealant to threads)	9,2 to 10,3	90 to 100
P.T.O. shaft retaining nut	10 to 12	98 to 115
gearbox-engine flange bolts	9 to 12	88 to 107
bevel gear retaining nut	28 to 30	274 to 294
nuts securing gearbox to intermediate transmission casing	10 to 12	98 to 107
hex socket screws for reduction unit ring gear (on assembly apply LOCTITE 242 to screws)	7 ÷ 8	68 ÷ 78



Rear power take-off

Rear P.T.O. gears

The rear P.T.O. shaft, directly coupled to the engine, is equipped with a selector allowing selection of P.T.O. speeds of 540/1000 rpm or, on request, 540-750-1000 rpm.
The selector is of the sliding sleeve type. The selector is mechanically controlled and can only be operated when the moving parts are stationary. The P.T.O. output shaft rotates in a counter-clockwise direction, as viewed from the rear of the tractor.

Fitting the rear P.T.O. output shaft

Before installing the studs, remove any oil/grease from the threads, then apply “LOCTITE 242".

Technical characteristics

P.T.O. speeds

540 rpm	**60** with "ECONOMY" P.T.O.	14/53	=	1/3,7857
540 rpm	**60 - 70 - 75 - 80 - 90 HP** without "ECONOMY" P.T.O.	13/50	=	1/3,8462
750 rpm	**60**	16/50	=	1/3,1250
750 rpm	**70 - 75 - 80 -90 HP**	45/46	=	1/3,0660
1000 rpm	**60 - 70 - 75 - 80- 90 HP**	20/46	=	1/2,3000

power take-off	P.T.O. r.p.m. / engine r.p.m. at max horsepower speed	engine r.p.m. / P.T.O. r.p.m	P.T.O. r.p.m. / engine r.p.m. at fas idling speed	P.T.O. r.p.m. / engine r.p.m. at fas idling speed
for 60 HP with ECO P.T.O. 540 r.p.m. (2045 engine r.p.m.)	620/2350	3,7857	(657/2490) ÷ (665/2520)	(171/650) ÷ (184/700)
for 60 HP without **ECO** P.T.O. **70-75-80-90**HP 540 r.p.m. (2077 engine r.p.m.)	611/2350	3,8462	(647/2490) ÷ (655/2520)	(169/650) ÷ (181/700)
for 60 - 70 - 75 - 80 - 90 HP 1000 r.p.m. (2033 engine r.p.m.)	1022/2350	2,3000	(1082/2490) ÷ (1095/2520)	(282/650) ÷ (304/700)
for 70-75-80-90 HP 750 r.p.m. (2343 engine r.p.m.)	752/2350	3,1250	(797/2490) ÷ (806/2560)	(208/650) ÷ (224/700)
for 65 - 75 - 85 HP 750 r.p.m. (2300 engine r.p.m.)	766/2350	3,0660	(812/2490) ÷ (822/2520)	(212/650) ÷ (228/700)

Synchronised P.T.O. P.T.O. revolutions for each wheel revolution 14,7400

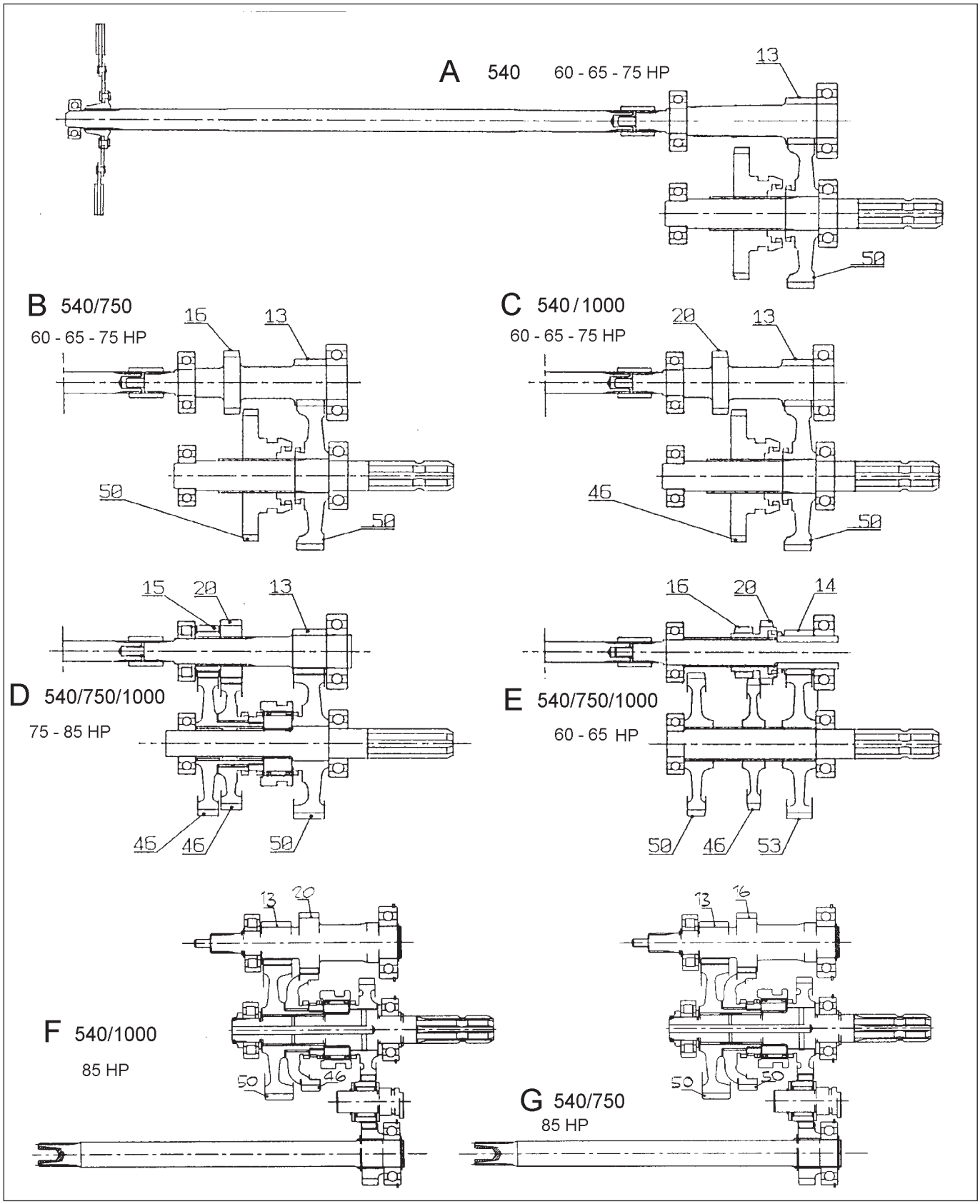
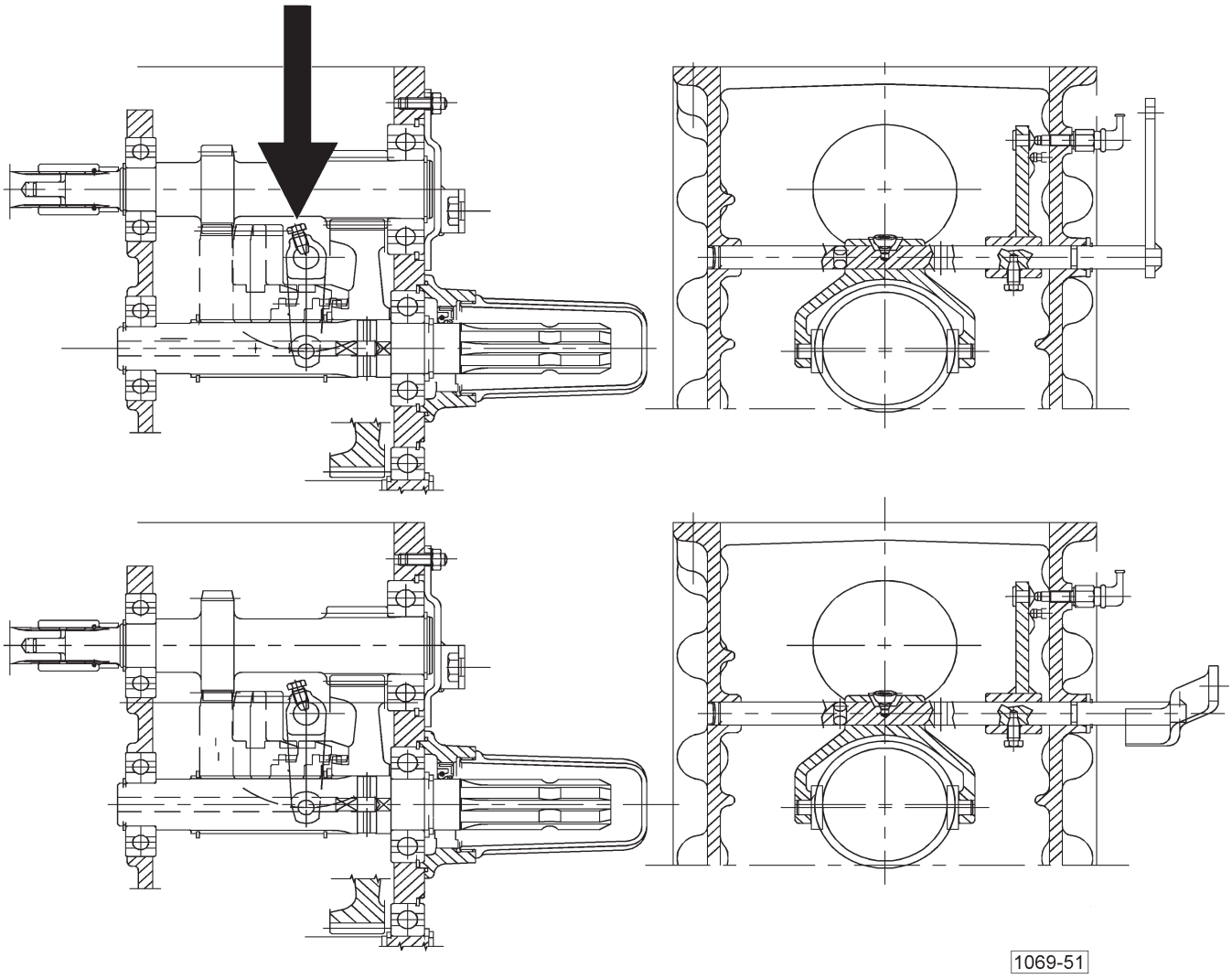


Fig. 1 - Rear P.T.O. speed configurations.

2-speed version 540 – 750 rpm

WARNING: on assembly, position the lever indicated with the arrow so that head of the lock screw is facing the front of the tractor.



2-speed version 540 – 1000 rpm

Fig. 2 - P.T.O. controls and output shafts (2-speed version) for **60 - 75 HP** models

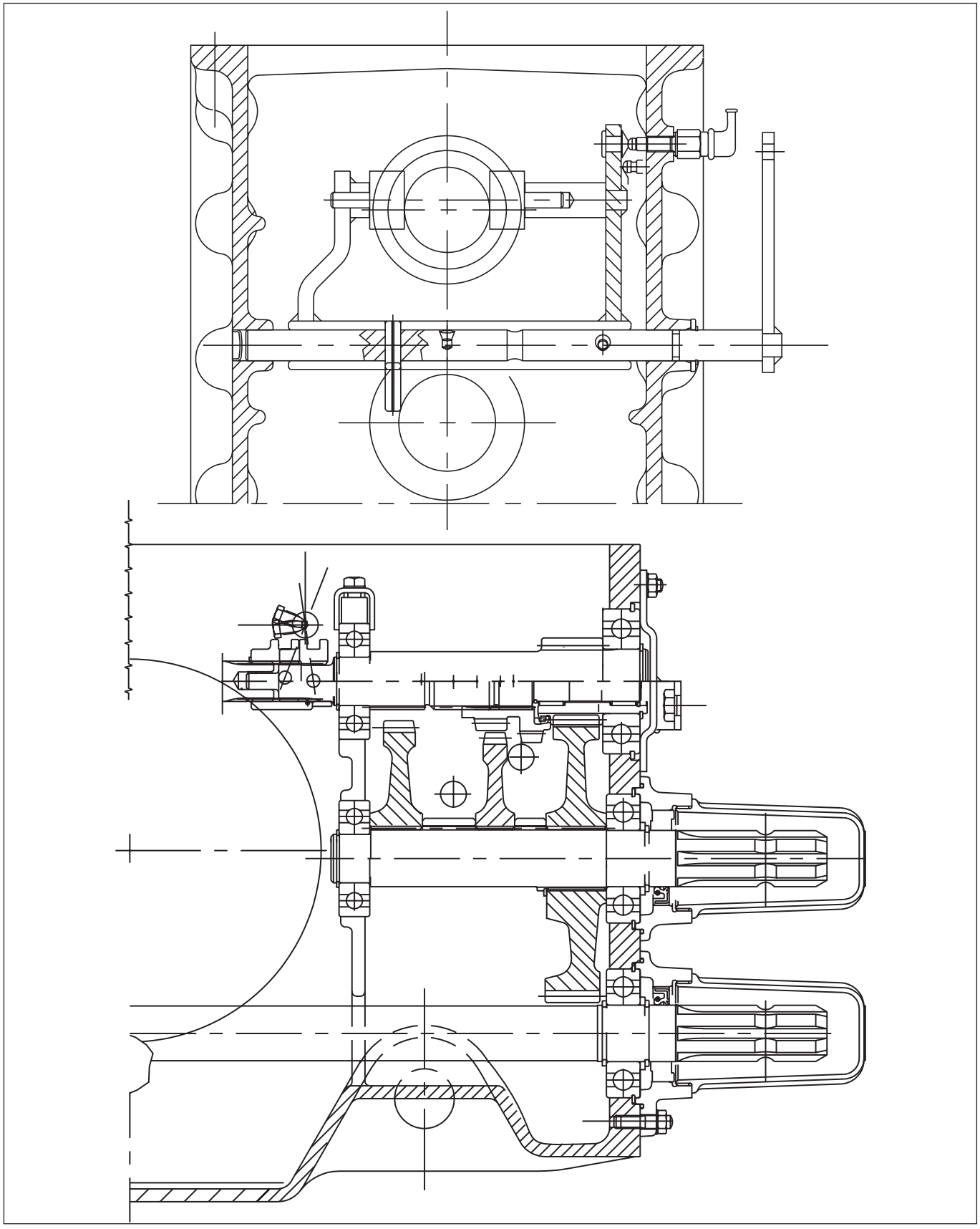


Fig. 3 - P.T.O. controls and output shafts for **60 - 75 HP** models (3-speed version)

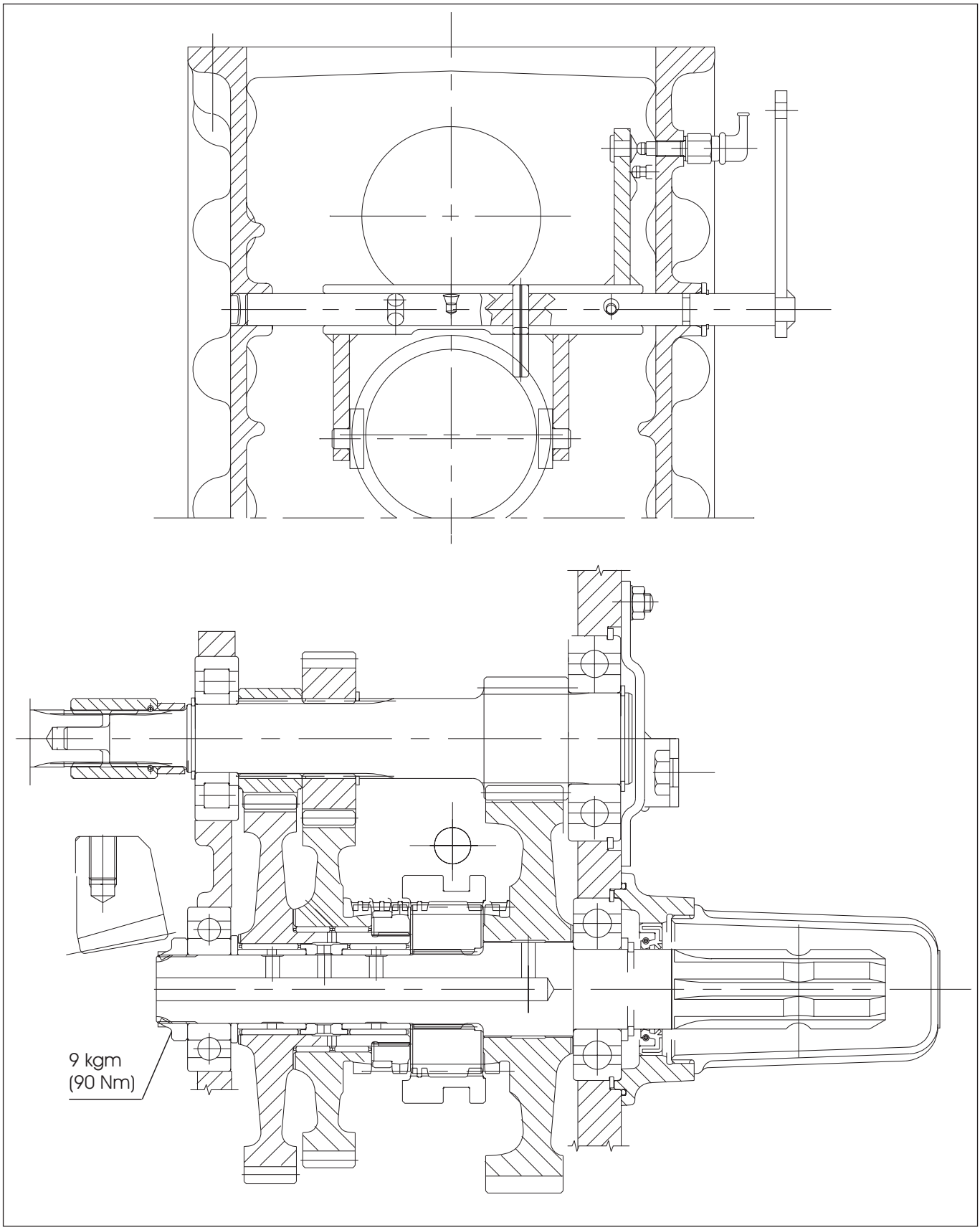


Fig. 4 - P.T.O. controls and output shafts for **70 - 80 - 90 HP** models (3-speed version).

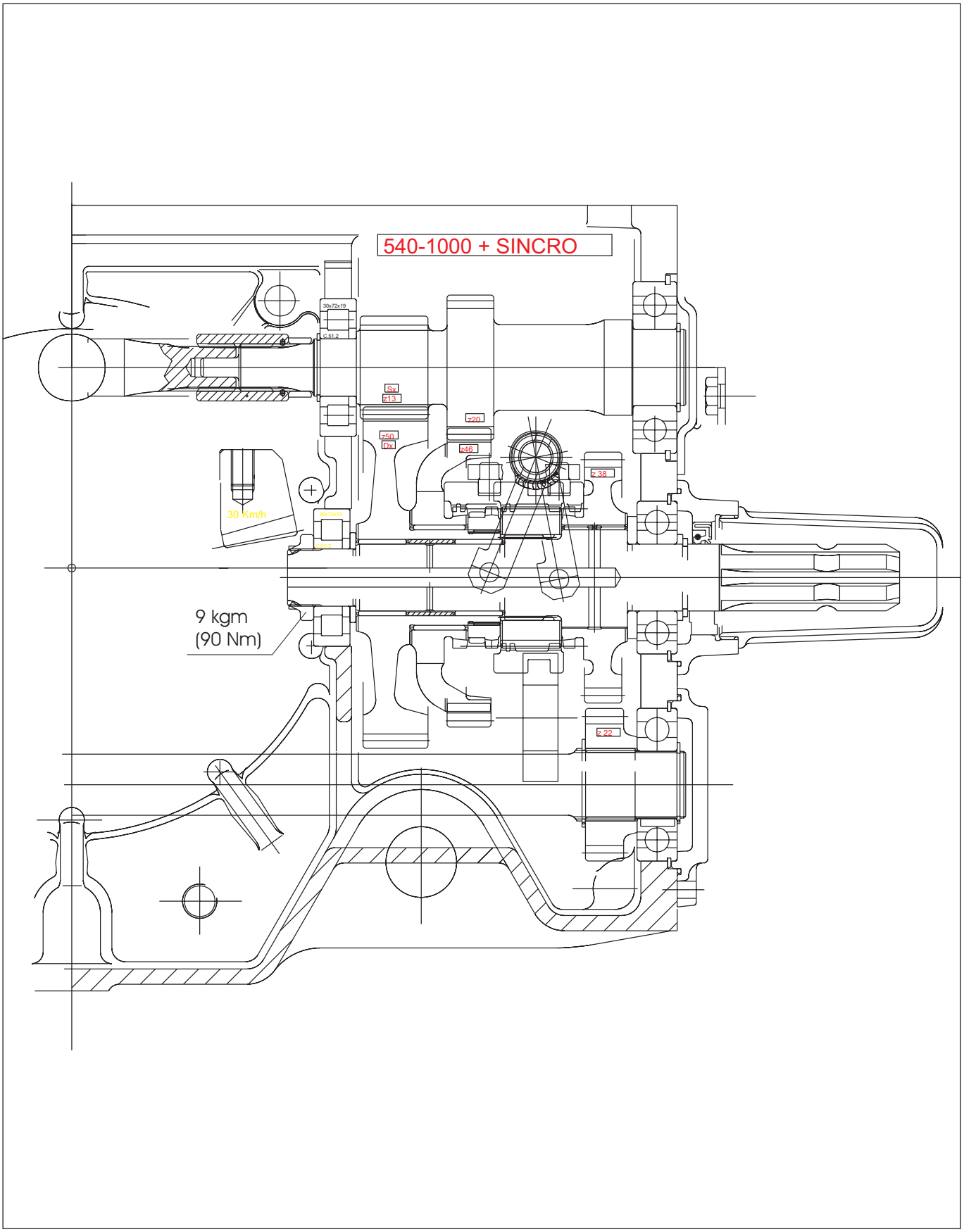


Fig. 5 - P.T.O. controls and output shafts for **70 - 80 - 90 HP** models (2-speed 540/1000 + groundspeed version).

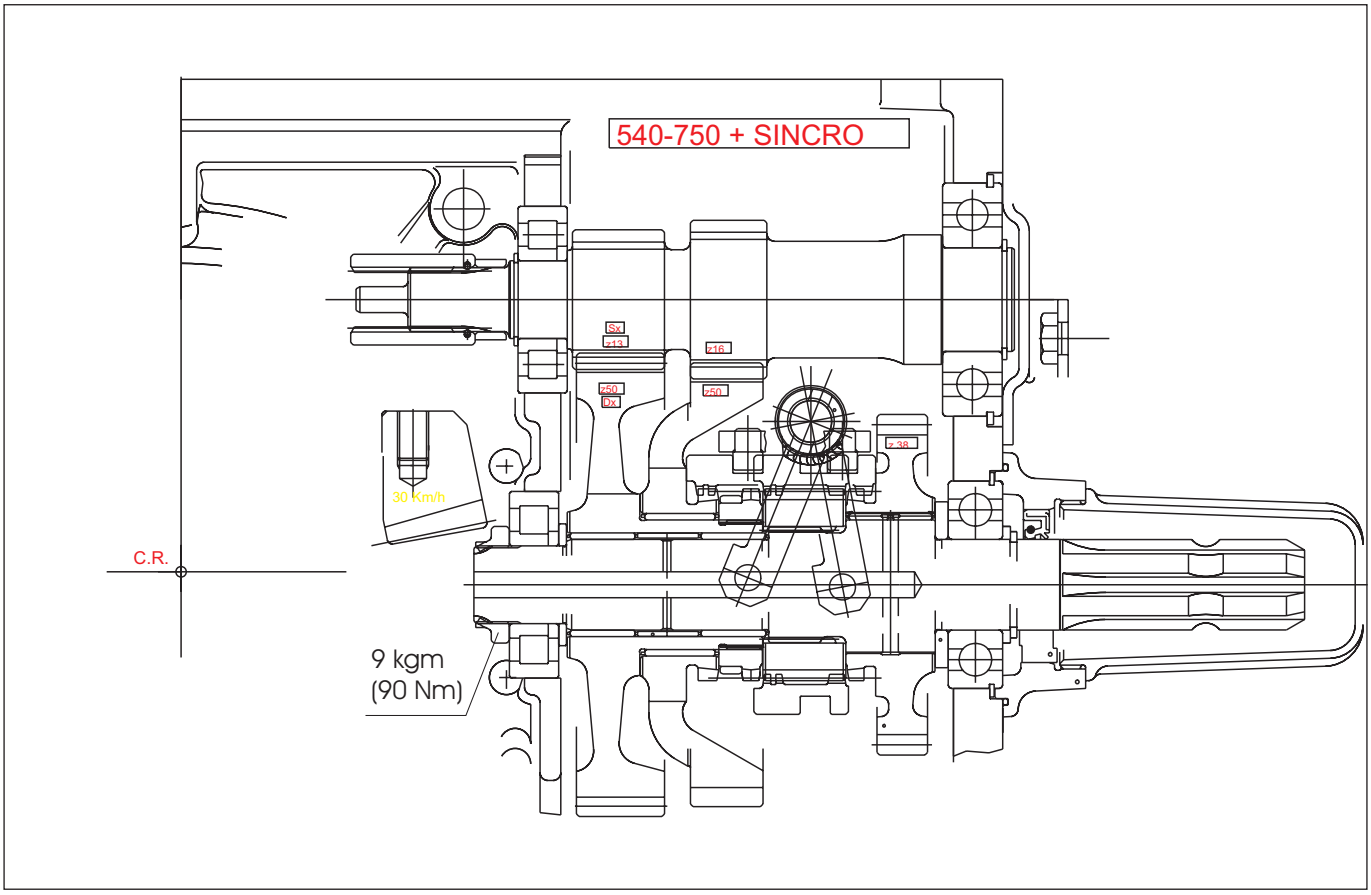


Fig. 6 - P.T.O. controls and output shafts for **70 - 80 - 90 HP** models (3-speed version).

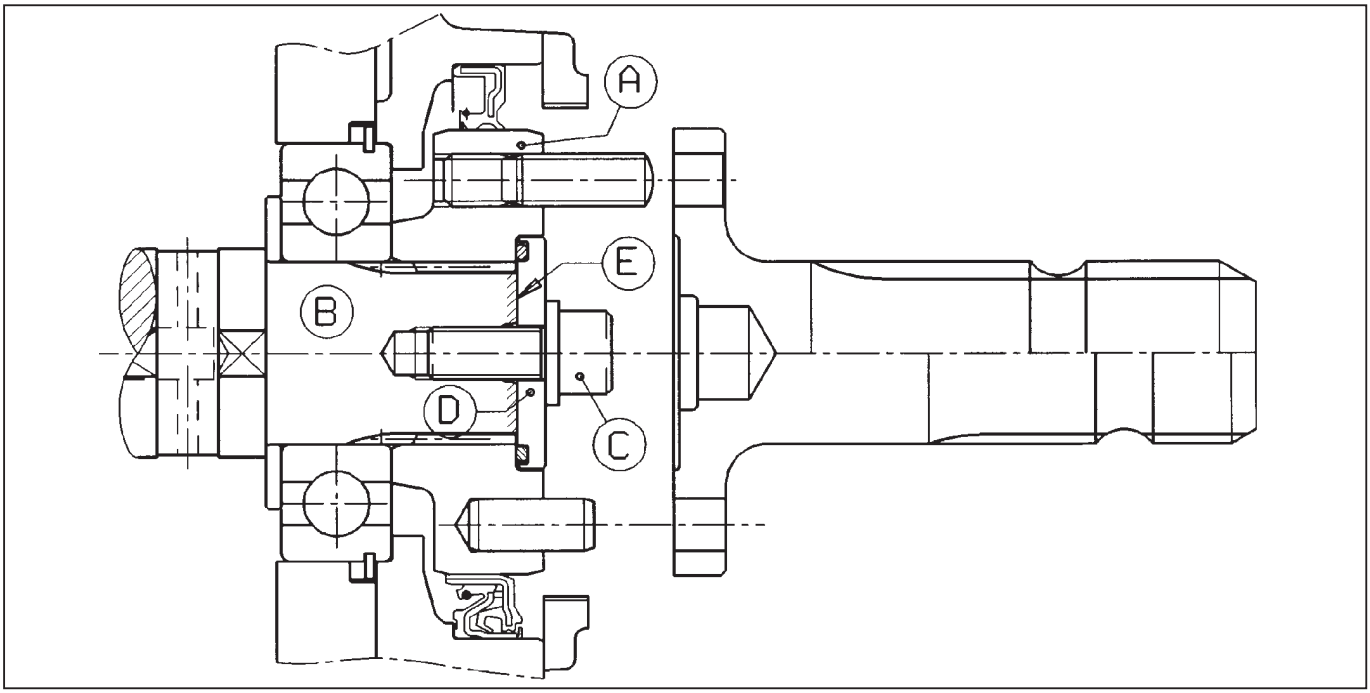


Fig. 7 - Fitting the P.T.O. shaft with interchangeable output shaft (on certain models only).
In the event of interference between shaft **B** and flange **A**, heat the flange to allow coupling. Before fitting plate **D**, apply Pianermetic 510 to the surfaces **E** of shaft **B**. Tighten bolt **C** to a torque of 68 Nm (7 kgm).

P.T.O. clutch

General characteristics

The P.T.O. clutch is a wet-type multidisc clutch and is operated by hydraulic pressure.

The oil required for the P.T.O. hydraulic control is drawn from the gearbox and filtered before it arrives at the gear pump located on the right-hand side of the gearbox.

The clutch control is electro-hydraulic. The electro-hydraulic system controls the circuit by means of a solenoid valve, which supplies oil under pressure to the clutch control piston. In both cases, the driving discs and driven discs are forced together, thereby transmitting drive to the output shaft.

The P.T.O. clutch brake is operated mechanically by the piston during the return stroke determined by the helical clutch release spring. Three pins connected to the piston press against the brake counter disc.

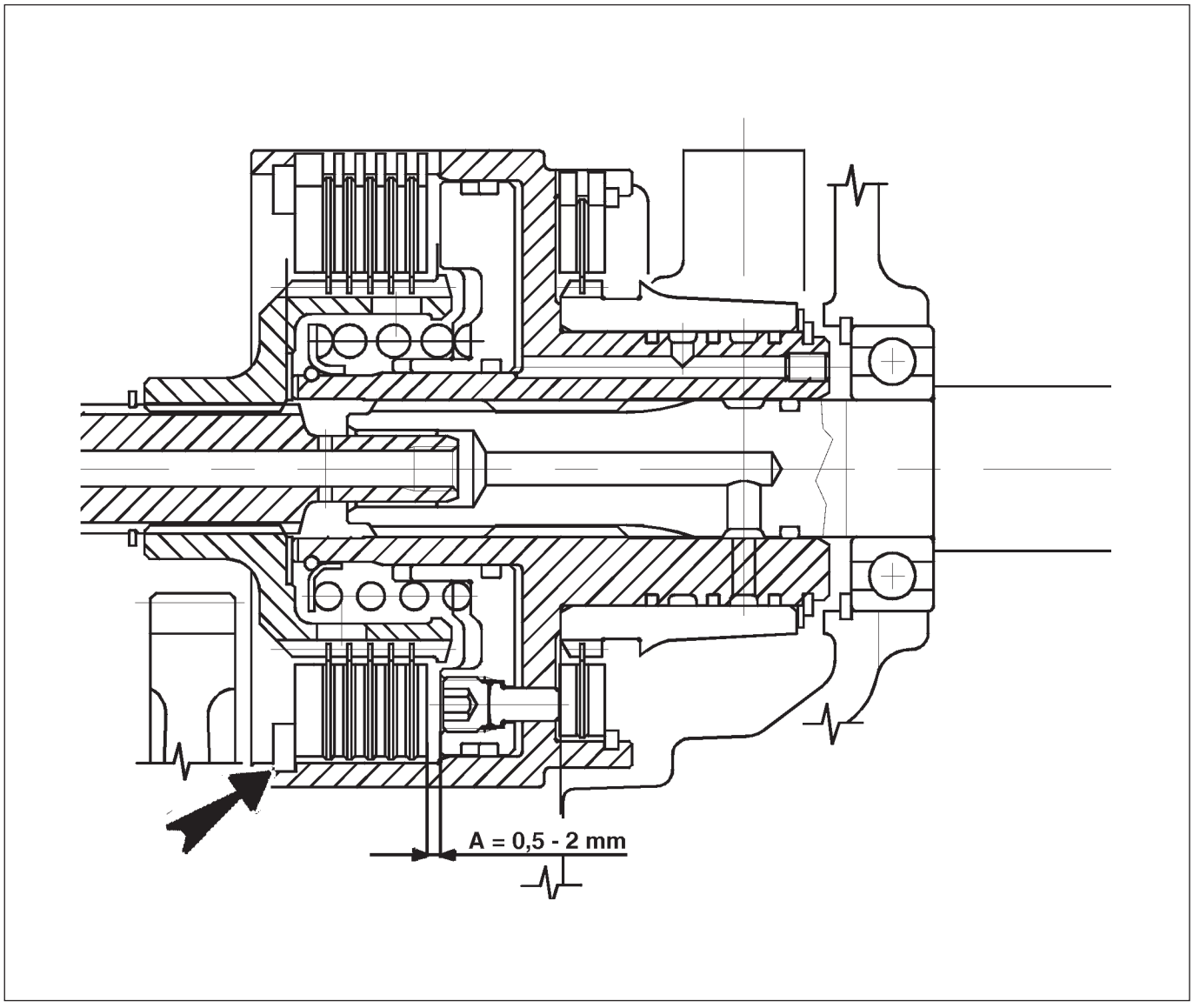


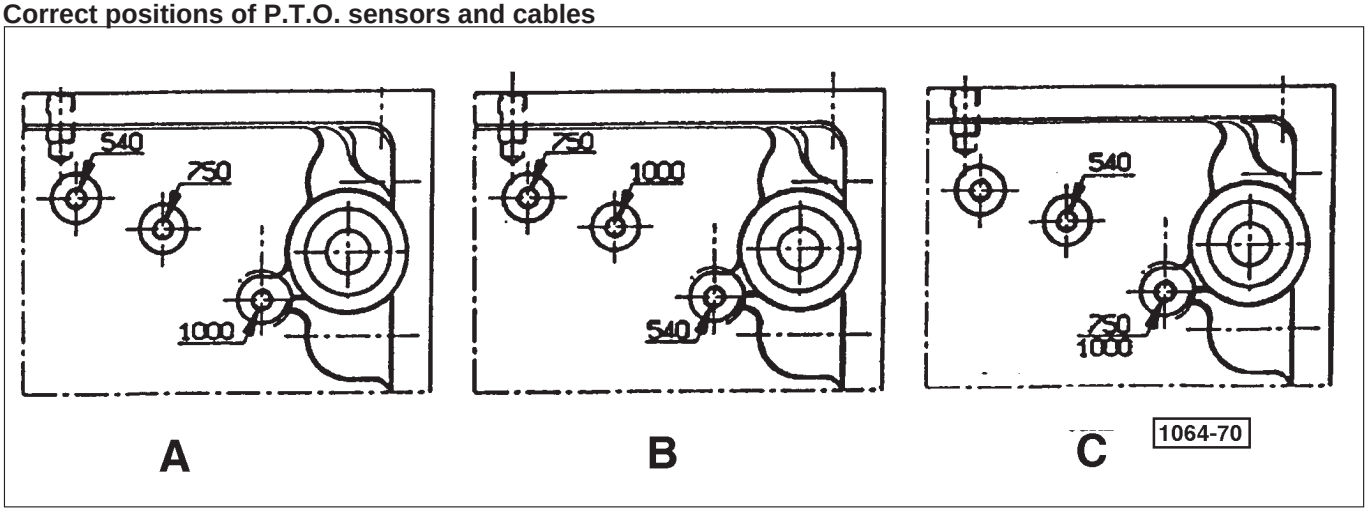
Fig. 8 - Cross section view of the P.T.O. clutch.

Technical specifications

P.T.O. Clutch

Clutch	oil-immersed multiple disc with axial piston	
number of friction discs		5
diameter of discs		105
thickness of assembled friction discs		
minimum	mm	27.32
maximum	mm	29.90
number of intermediate discs		5
disc lubrication pressure	bar	5
P.T.O. clutch relief pressure setting	bar	11
P.T.O. pressure control valve	bar	16 $^{+2}_{0}$
restrictor valve	mm	1.1

Correct positions of P.T.O. sensors and cables



A - 3-speed P.T.O. for 75-85 hp models **B** - 3-speed P.T.O. for 60-65 hp models **C** - 2-speed P.T.O.

Springs technical data:

Piston return spring			
diameter of wire		mm	6
external diameter		mm	52,5
spring relaxed		mm	50
spring compressed	kg 82 (808 Nm)	mm	28
	kg 98 (978 Nm)	mm	23,4

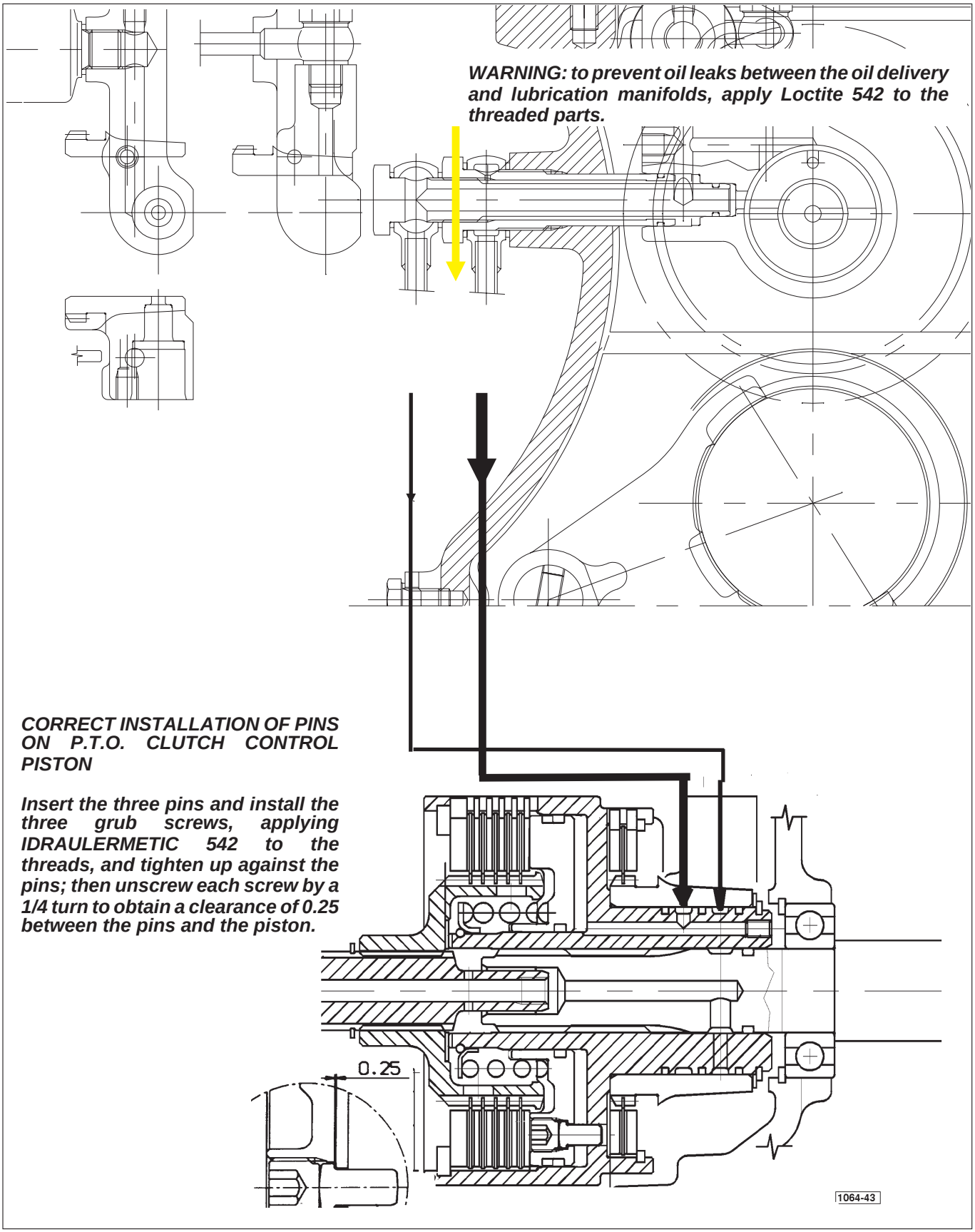


Fig. 9 - Rear P.T.O. control.

Clutch inspection

Inspect the contact surfaces of the clutch friction discs. If worn or torn, the discs must be replaced.
Inspect the control piston and replace if scored.
Inspect the clutch hub. If depressions are worn into the tops of the splines on which the friction discs slide, the hub must be replaced.

Check the efficiency of the piston return spring. The spring characteristics should match those specified in the relative table; if not, replace the spring.

Examine the seals of the clutch control piston and the clutch brake piston and the brake disc. Replace if worn.

At each assembly of the P.T.O. clutch, carefully clean the oil ways in the clutch housing; use a compressed air jet if necessary.

Checking clutch disc clearance
With new discs installed, check that clearance **A** (Fig. 8) is 0.5÷2mm .
If the clearance exceeds the maximum value, fit one shim (code 2.1599.499.0) in the position indicated by the arrow.

Warning: Apply IDRAULERMETIC 542 to the threads of the three screws **A** before fitting; fully tighten the screws and then back off each screw a 1/4 turn to obtain a clearance of 0.25 mm between the head of the pin and the piston.

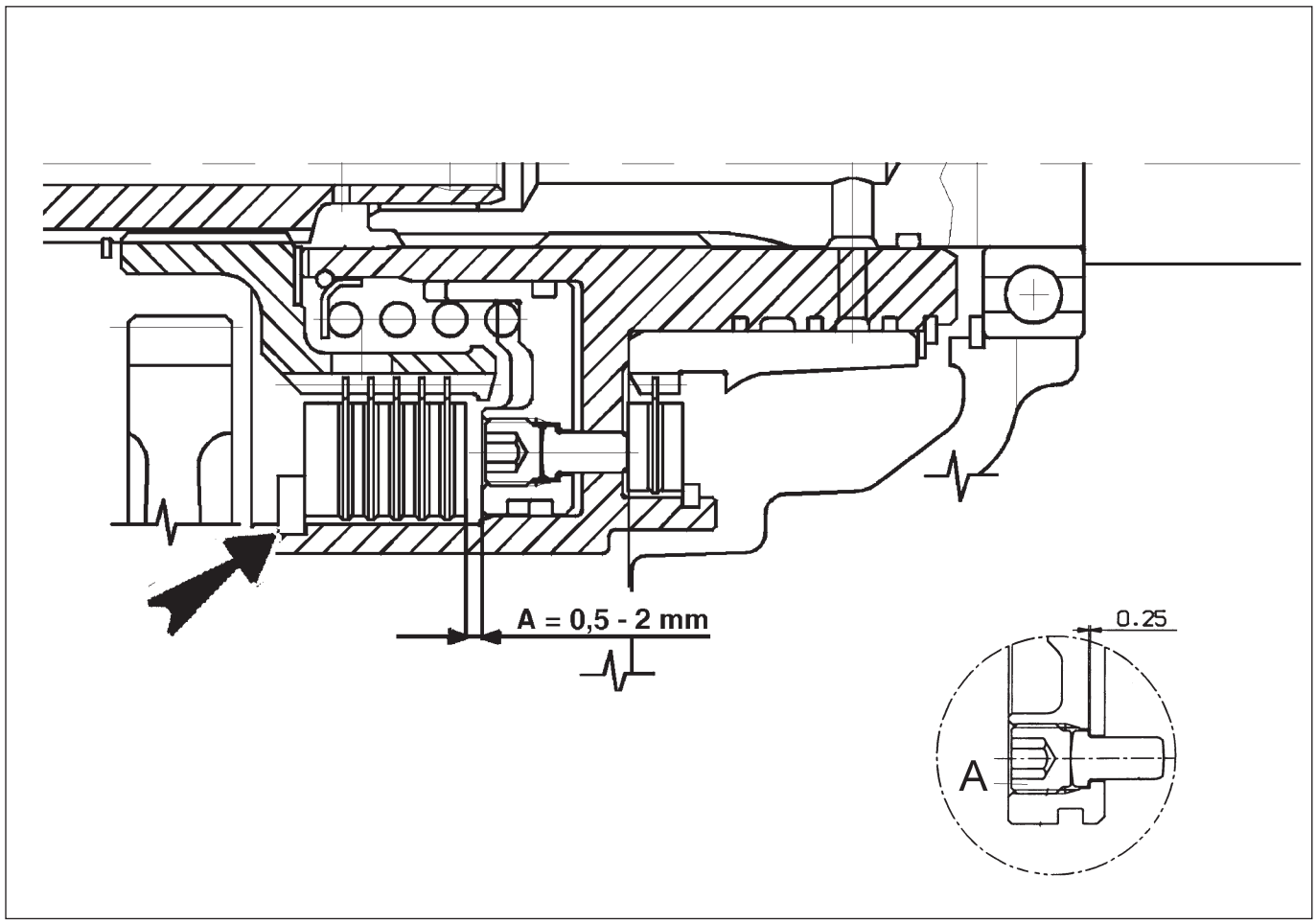


Fig. 10 - Checking clutch disc clearance.
A - If the clearance exceeds the maximum prescribed value, fit one shim code 2.1599.499.0

Checking clutch hydraulic pressures

Connect the mini fitting code 009.2923.4 (with M4 12x1.5 threaded fitting) to the electro-hydraulic control unit as shown in figure 11.
Connect tool 5.9030.517.4 to the fitting and attach the pressure gauge 5.9030.514.0.
Operate the control valve and check that the pressure gauge shows a reading of approximately 16 bar.
For instructions on measuring pressure and relative values see the chapter “electro-hydraulic controls”.

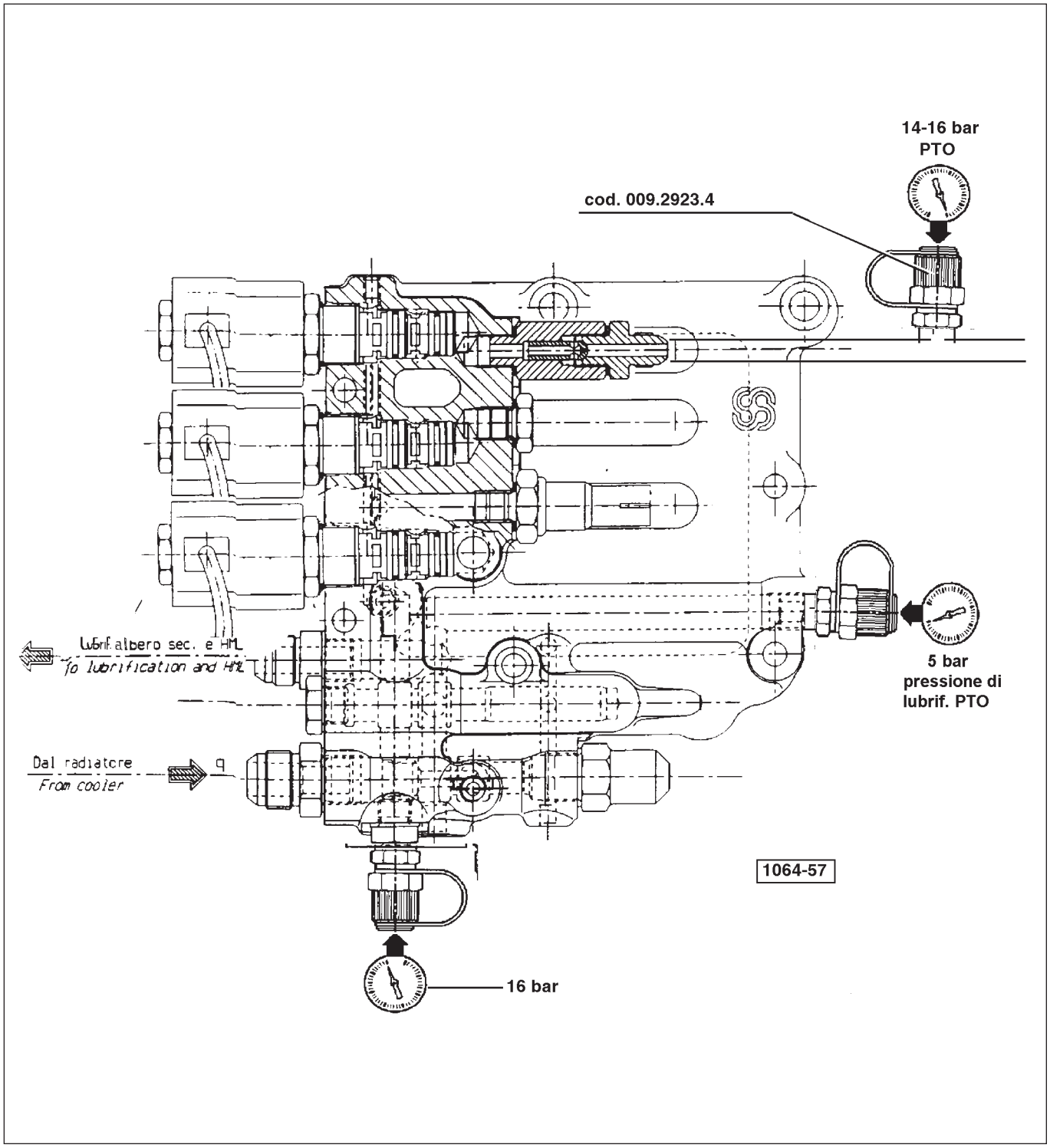


Fig. 11 - Checking the operating pressure in the P.T.O. clutch control circuit.

Checking the end-play of the front shaft of the P.T.O. clutch

On the front shaft of the P.T.O., install circlip **A** and thrust washer **B** code 2.1599.761.0; position the shaft in the clutch unit and push it towards the front of the tractor so that it rest up against the oil seal shown in figure at **C**, then check that clearance between the thrust washer **B** and the clutch hub is not less than 1 mm.
If the clearance is less than 1 mm, remove thrust washer **B**.

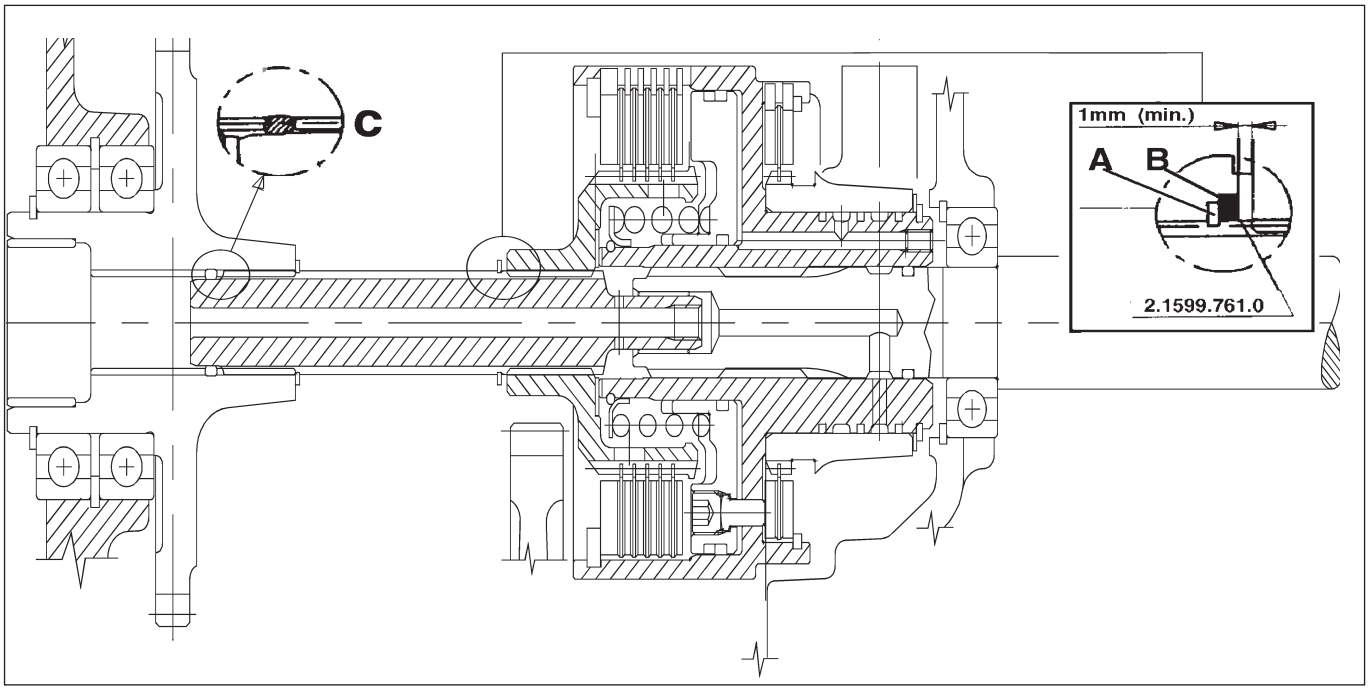
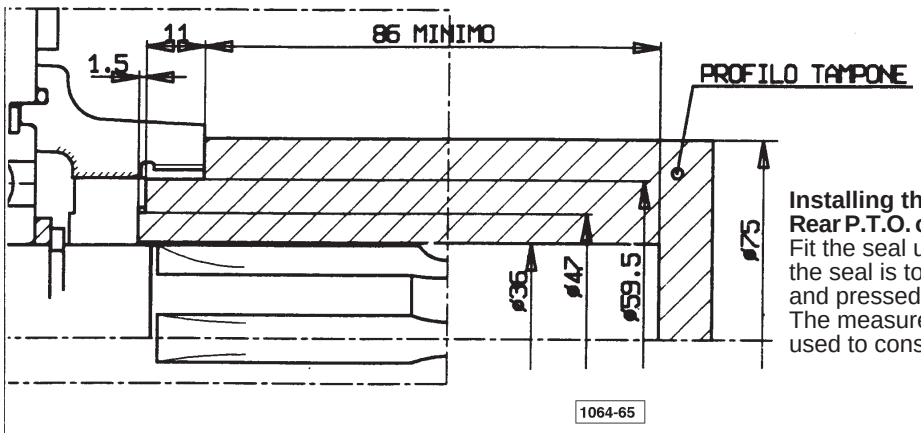


Fig. 12 - Checking the end-play of the front shaft of the P.T.O. clutch.



Installing the waterproof seals
Rear P.T.O. output shaft and synchronised P.T.O.
Fit the seal using a suitable installation tool;
the seal is to be oriented as shown in the figure
and pressed in to the full depth of the tool.
The measurements indicated here are to be
used to construct the oil seal installation tool.

Fig. 13 - Installation of the waterproof seals.

RENEWAL OF THE REAR P.T.O. CLUTCH

The P.T.O. clutch unit can be removed without having to remove the driving platform.

proceed as follows:

— Remove the top link;
— Detach the lift rods of the 3-point linkage from the lift arms;
— Detach the lift control rods from the control levers;
— Disconnect the oil delivery pipe to the auxiliary control valves from the rear pump;
— On models with 4-way control valves, detach the oil pipe from the lift control valve to the auxiliary hydraulic control valves. On models with 6-way control valves, detach also the pipe from hydraulic control valves to the upper gearbox cover plate (for 4-way control valves);
— Disconnect the control levers of the hydraulic control valves from the relay levers connected to the control valves, by removing the relative cotter pins and retaining pins;
— Remove the 3 screws securing the hydraulic control valve from the support bracket;
— Remove the 9 screws and the two nuts securing the lift unit to the rear gearbox;
— Attach the lift unit to a hoist and lift it clear of the tractor (when removing the lift unit, take care not to damage the threads on the two studs);
— Unscrew the screws securing the P.T.O. access cover (located above the gearbox) and remove this cover;
— Using a suitable lever, push towards the front of the tractor the sleeve coupling the P.T.O. clutch output shaft to the P.T.O. rear shaft;
— Loosen the two screws and remove the rear shaft bearing stop plate;
NOTE: when removing this plate, take care that the spacers positioned underneath it do not fall into the gearbox.
— Unscrew the 3 retaining screws and remove the top flange located at the rear of the gearbox. Then withdraw the bearing together with the upper 540 rpm gear.
NOTE: when withdrawing the gear, take care that the roller bearing located in the gear does not fall into the gearbox.
— Disconnect the pipe supplying oil for lubrication of the inversor shaft and to the oil-bath for the P.T.O. clutch discs, located on the right-hand side of the gearbox, by undoing the relative union; then remove the union of the oil supply pipe to the P.T.O. clutch;
— With the aid of a lever, push the P.T.O. shaft a few centimetres to the rear;
This will only be possible if the shaft is positioned in such a way that there is no interference between the teeth of the gear and those on the shaft;
— Slide the sleeve onto the rear shaft after having first pushed the bearing located in the wall of the gearbox back a sufficient distance;
— Push the clutch unit together with the front shaft and the output shaft to the rear, so as to be able to withdraw the front shaft to the front P.T.O. unit;
— Extract the complete clutch unit, disengaging it from the rear shaft;
— Inspect the clutch discs and renew them if necessary, removing the large circlip and the spacer.

WARNING: on re-assembly, after having positioned the clutch unit complete with the front shaft, refit the two oil unions in order to ensure that the unit is positioned correctly, then insert the rear shafts and complete the assembly.

MAIN OPERATIONS FOR REMOVAL OF THE REAR P.T.O. UNIT.

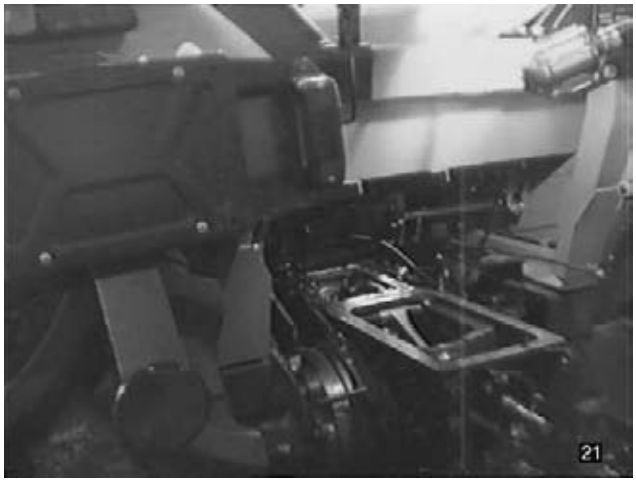


Fig. 14 - The rear P.T.O. clutch may be removed without having to remove the cab from the tractor.
Disconnect the controls of the auxiliary control valves.

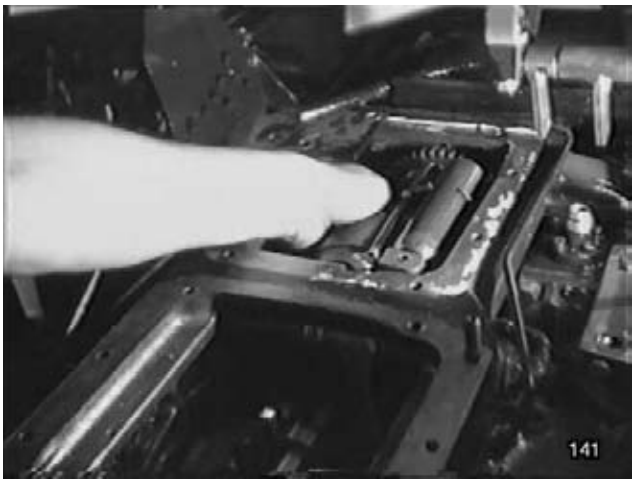


Fig. 15 - Remove the lift and the front cover of the rear gearbox to gain access to the P.T.O. clutch unit.

Fig. 16 - Release the rear shaft of the P.T.O. towards the rear.

Fig. 17 - Remove the circlip on the front clutch shaft.

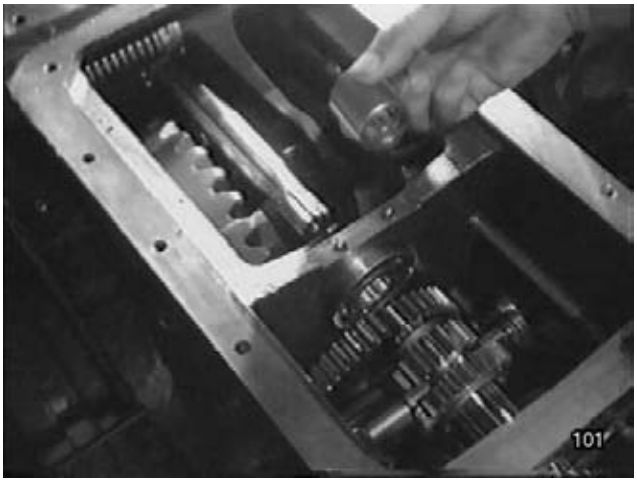


Fig. 18 - If necessary, push the rear shaft support bearing forward and withdraw the sleeve coupling the 2 rear shafts of the P.T.O. together.

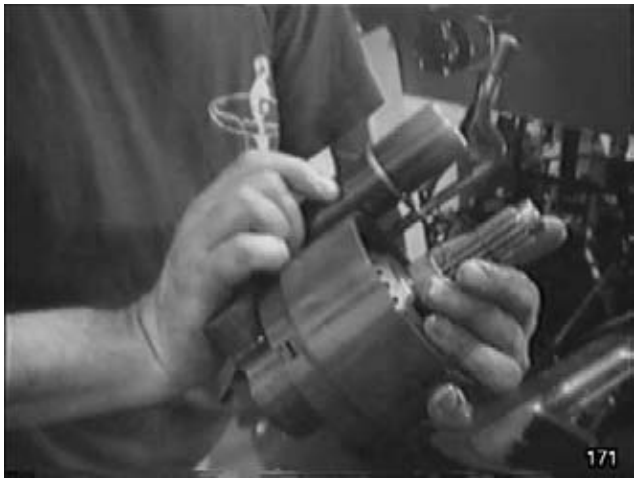


Fig. 19 - Disconnect the oil supply pipes and remove the complete P.T.O. clutch unit from the gearbox.

OPERATIONS FOR REMOVAL OF THE REAR P.T.O. UNIT

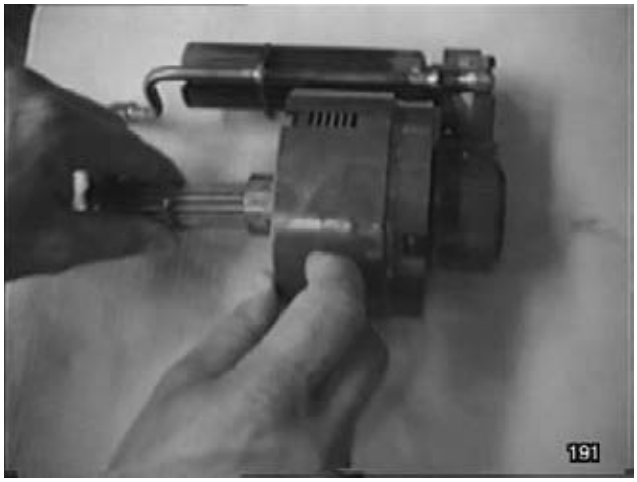


Fig. 20 - Withdraw the front shaft from the clutch unit.

Fig. 21 - Check that the clutch hub slides freely in its housing.

Fig. 22 - Check the condition of the lube pipe and nozzle for the hydraulic pumps drive gears.

Fig. 23 - Remove the circlip securing the clutch discs.

Fig. 24 - Remove the clutch discs.

Fig. 25 - The complete disc pack can be extracted along with the clutch hub.

Fig. 26 - Inspect the piston; note the three pins of the clutch brake, which is engaged by the action of the clutch release spring in the absence of hydraulic pressure.

Fig. 27 - Remove the thrust washer from between the spring and the clutch hub.

Fig. 28 - Inspect the clutch discs for wear.

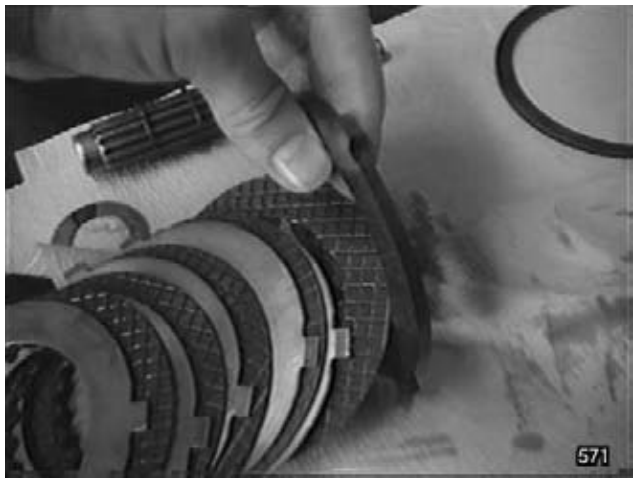


Fig. 29 - Examine the condition of the counter disc.

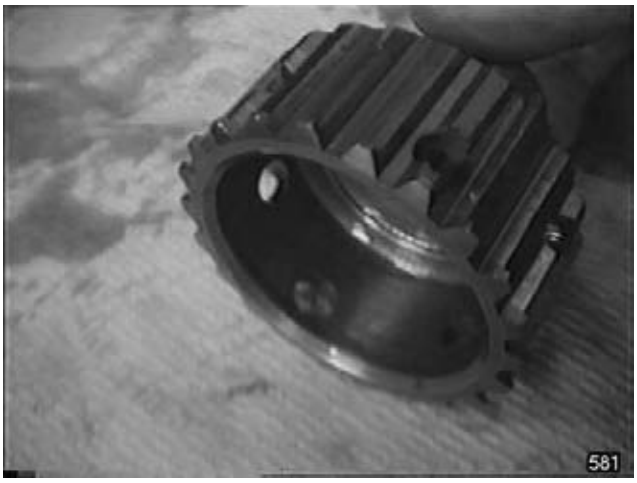


Fig. 30 - Examine the teeth of the clutch hub.

Fig. 31 - Remove the circlip securing the hydraulic manifold of the clutch unit.

Fig. 32 - Remove the shoulder plate.

Fig. 33 - Remove the lube pipe for lubrication of the hydraulic pump gears.

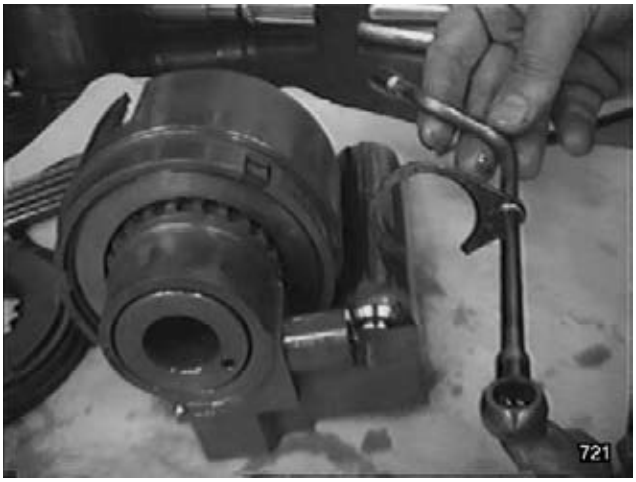


Fig. 34 - Completely disconnect the lube pipe.

Fig. 35 - Remove the clutch oil supply manifold.

Fig. 36 - Check the manifold oil ways and check the manifold for wear.

Fig. 37 - Remove the snap rings and check for signs of wear.

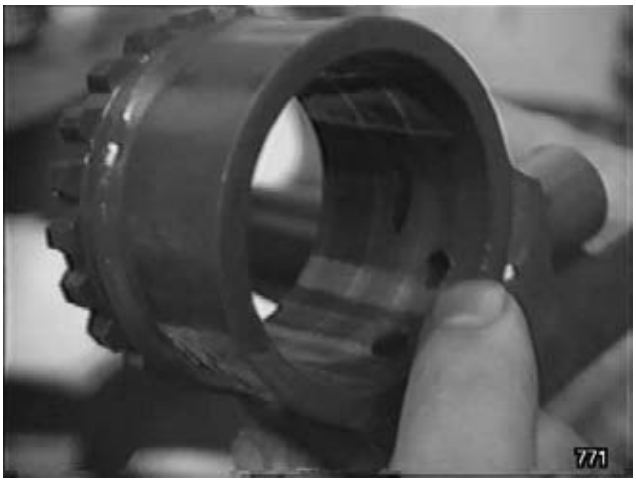


Fig. 38 - The two holes at the front of the manifold as shown in the figure serve for lubrication of the clutch discs, while the large hole to the rear of these supplies the oil to operate the clutch control piston.

Fig. 39 - Examine the seals on the oil supply fittings.

Fig. 40 - Dismantle the two oil supply fittings and examine them.

Fig. 41 - Remove the brake disc of the clutch unit.

Fig. 42 - Remove the circlip securing the brake disc thrust plate.

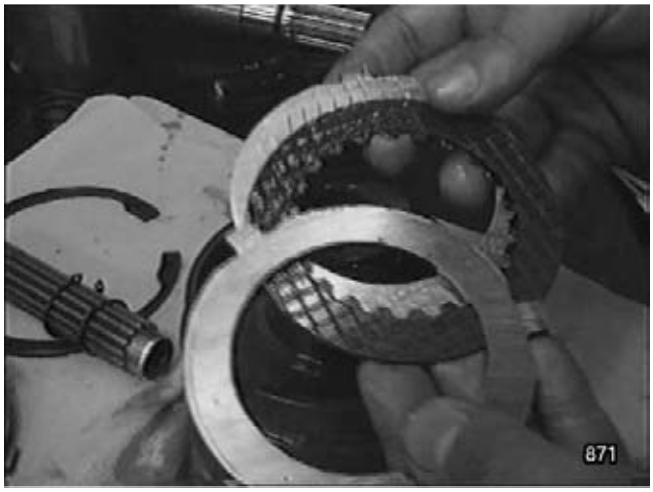


Fig. 43 - Extract the two thrust plates together with the brake disc.

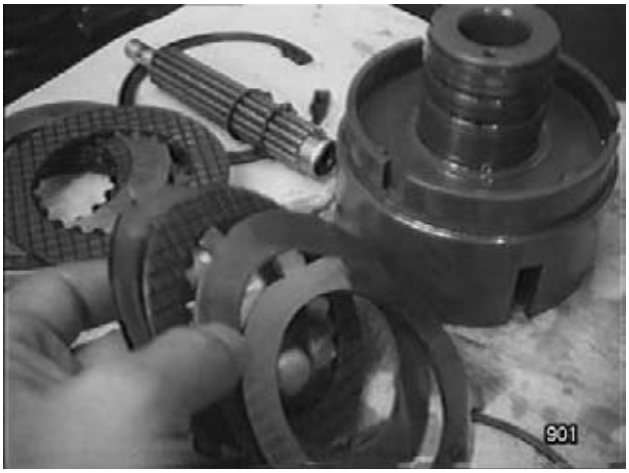


Fig. 44 - Examine the clutch brake disc pack.

Fig. 45 - Check the three pressure pins for wear.

Diagnosing malfunctions

the clutch slips	check the hydraulic system of the solenoid valve block	replace the springs of incorrectly calibrated valves		
		check the seals on the piston and manifold	if necessary, replace the defective parts	
	check that the piston slides freely	eliminate any roughness preventing free movement		
	check the clutch plates for wear	replace the plates		
the clutch fails to disengage	incomplete control valve stroke, solenoid valve faulty	check the control linkage	replace the defective parts	replace the plates
		check the plates	plates burned	eliminate any roughness preventing free movement
			plates jammed	
	piston stuck	replace the piston		

Rear axle

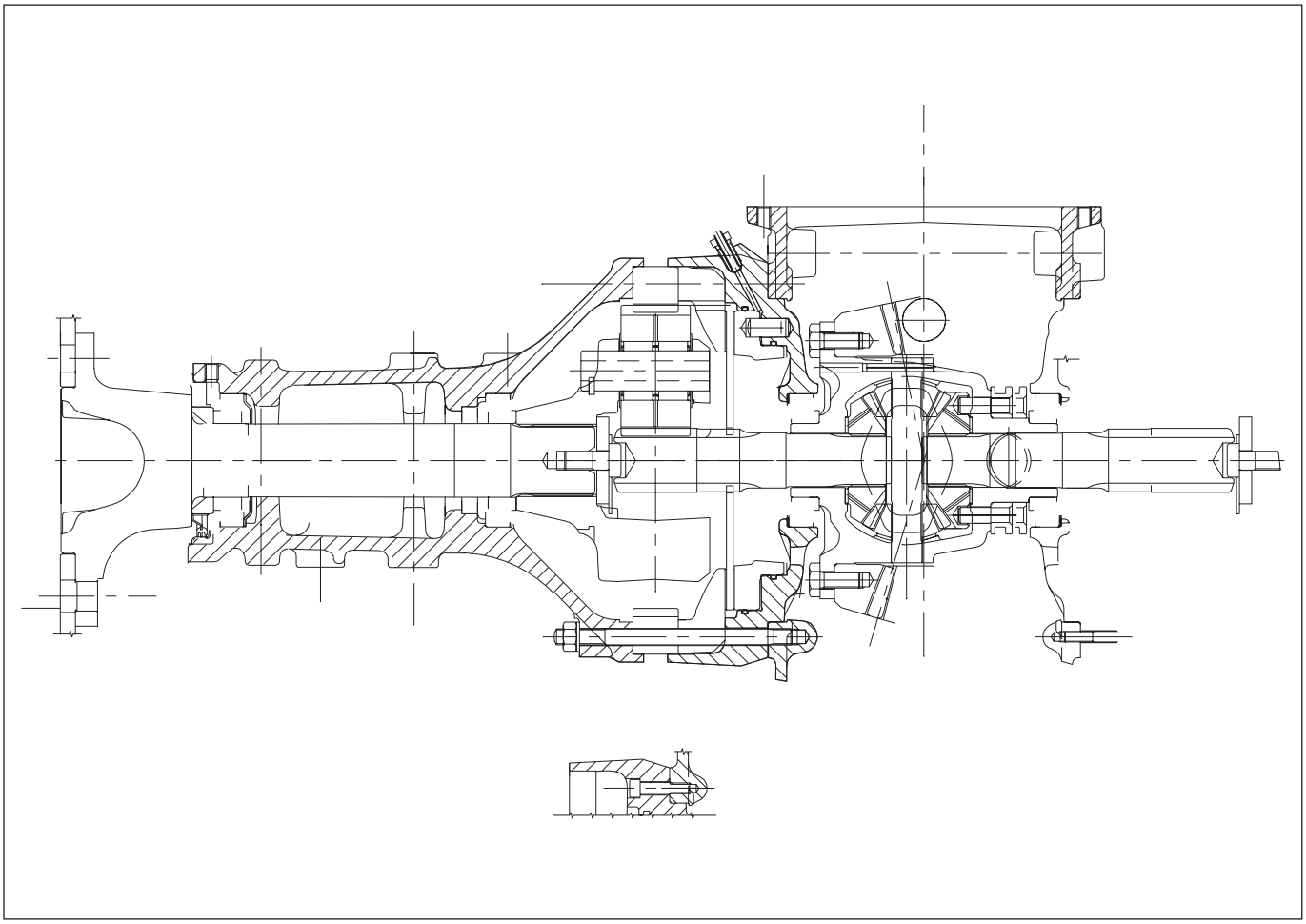


Fig. 1 - Rear axle.

Installing the rear half-shafts (fig. 2)

— The half-shafts are fixed to the planet gear carrier flange of the epicyclic reduction unit by means of a plate inserted in the flange and attached to the internal face of the half-shaft by a screw.

— The correct position of the half-shaft in the trumpet housing is determined by the shim pack **A** indicated in figure 2.

— To install the half-shafts, proceed as follows:

— Secure the half-shaft in a vertical position to prevent its rotation;

— Press the rings of bearings **B** and **C** into the respective housings in the trumpet housing; (before installing the ring of bearing **C**, position the cover underneath to retain the grease);

— Then fit spacer **I**, oil seal **D** and bearing **E** to the half-shaft. Slide the trumpet housing onto the half-shaft, fit oil seal **H** and bearing **F**, then position the planet carriers;

— Secure the assembly by way of the plate and screw **G**, interposing a shim pack **A** between the plate and the half-shaft to obtain a rolling drag torque of 8 to 12 Nm;

— Apply LOCTITE 270 to screw **G** and tighten to a torque of 18 kgm (177 Nm).

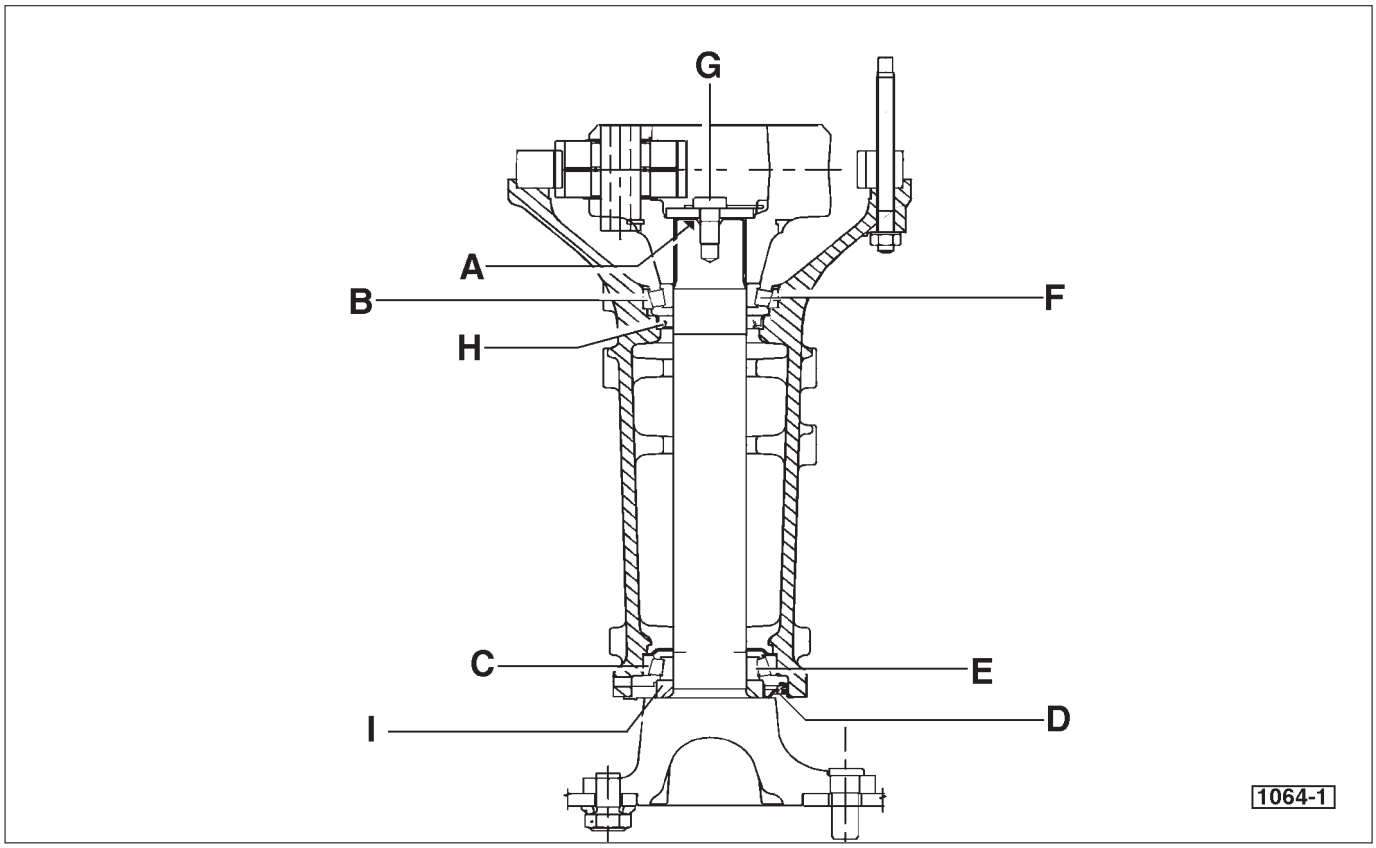


Fig. 2 - Fitting of rear stub axles.

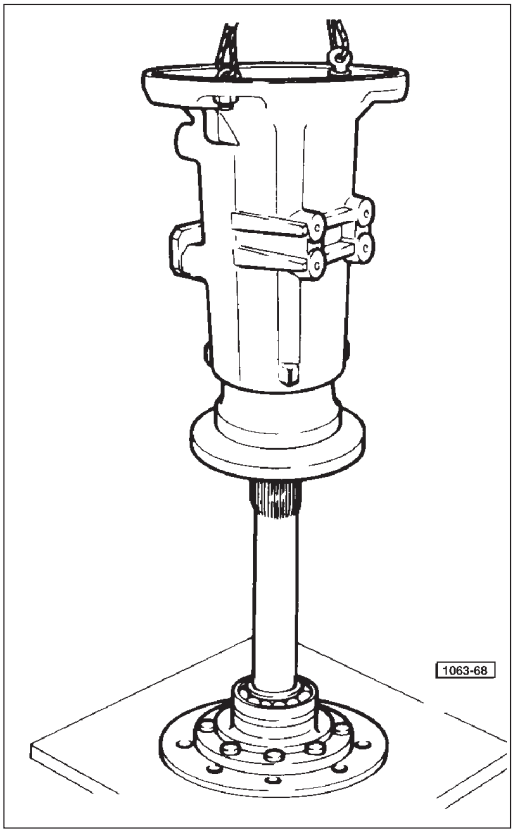


Fig. 3 - Fitting of rear stub axles.

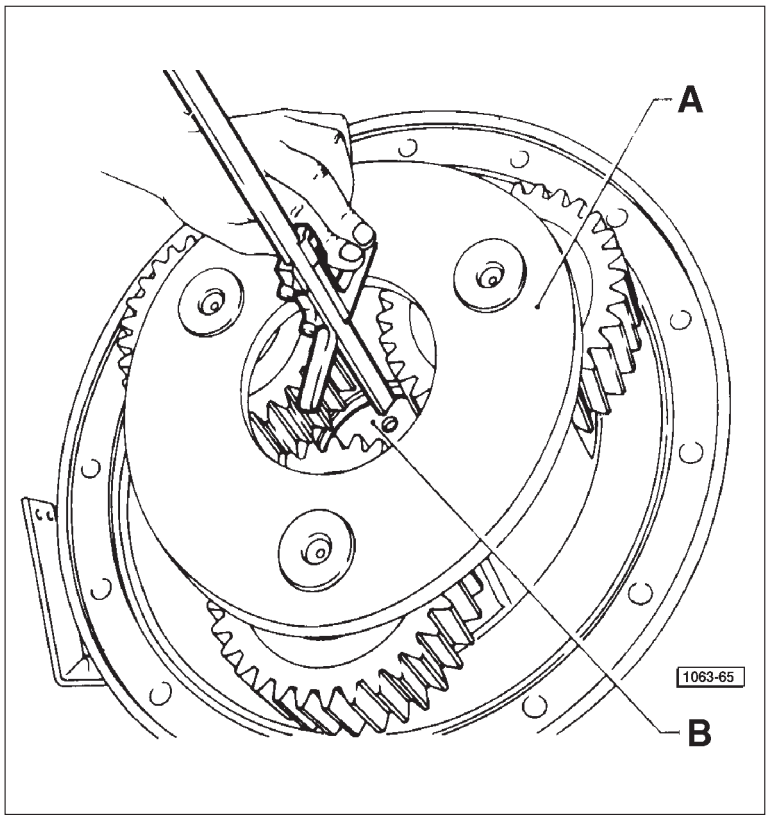


Fig. 4 - Halfshaft securing plate.
A - Planet carrier housing
B - Spacer

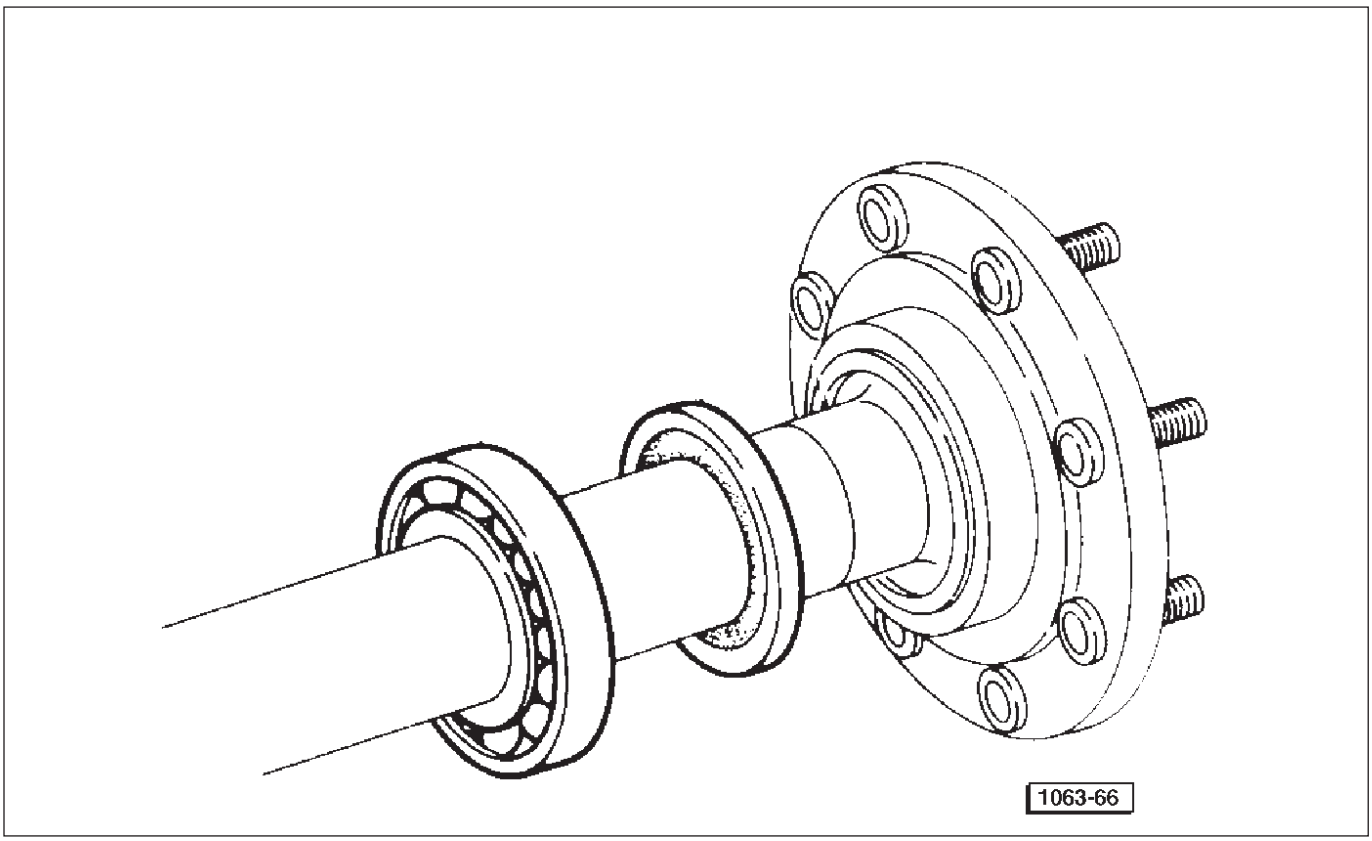


Fig. 5 - Fitting of bearing to halfshaft.

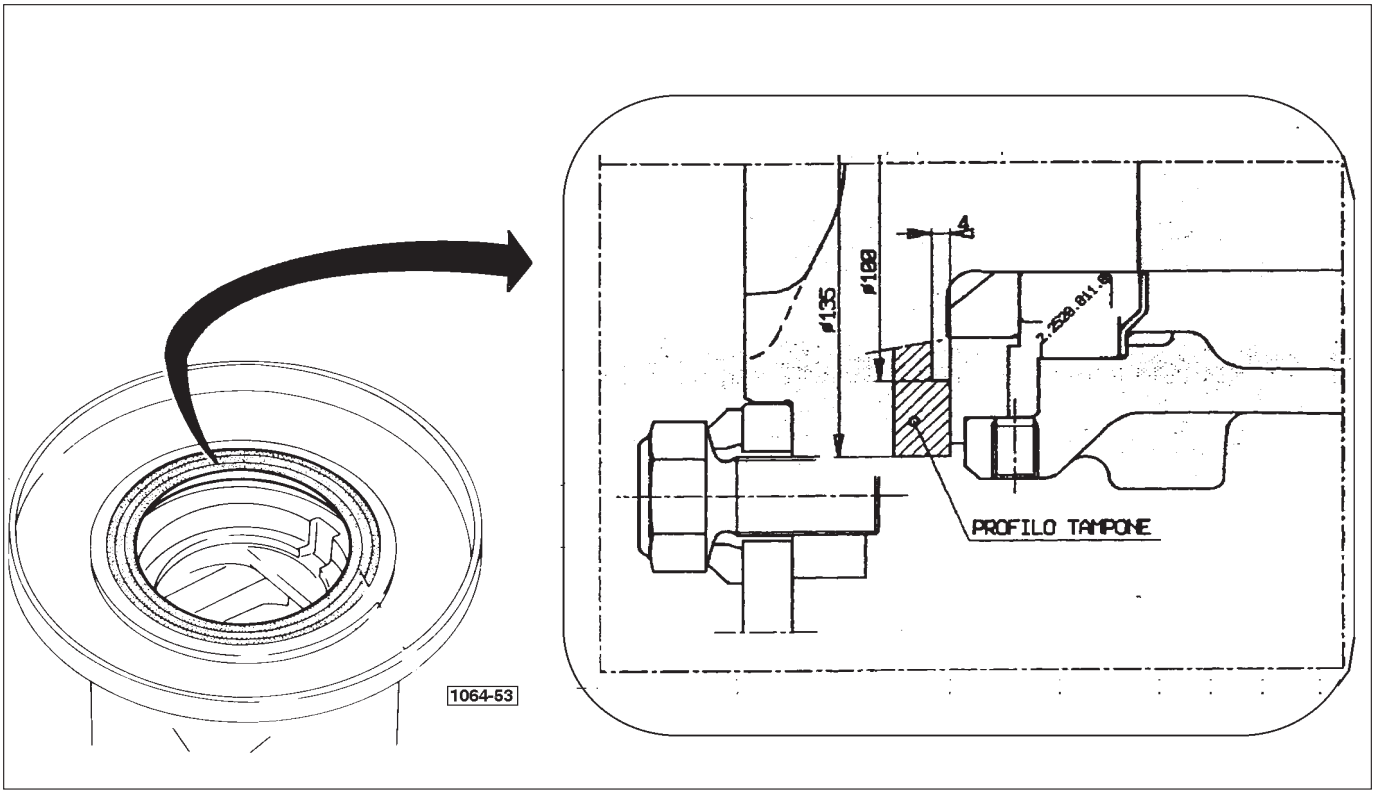


Fig. 6 - Fitting of O-ring to stub axles.

Removal and disassembly of the epicyclic reduction unit

Unscrew the three screws securing the planet carrier **B** (Fig. 7) to the half-shaft.

Then remove the spacer and the flange, and recover the underlying shim pack.

Remove the epicyclic reduction unit and disassemble it into its component parts as follows:

1 - Using a hammer and drift, remove the pin securing journal **A** of planet gear **B** (Fig. 8).

2 - Withdraw the journal and remove the planet gear together with the roller cage and the two thrust washers.

3 - Remove bearing **A** from the flange of epicyclic reduction unit **B**, using a suitable puller (Fig. 9).

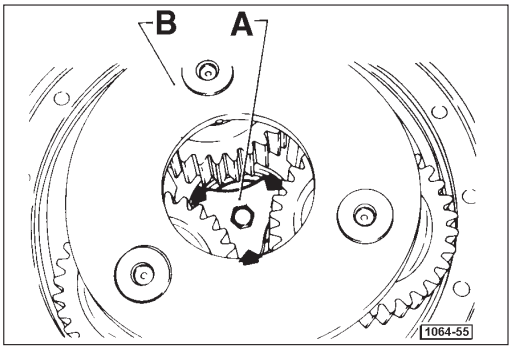


Fig. 7 - Removing the spacer.
A - Flange
B - Planet carrier

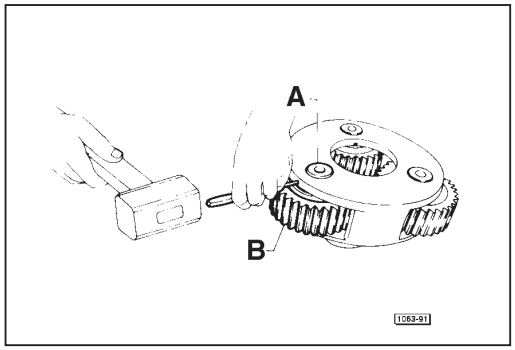


Fig. 8 - Epicyclic reduction unit.
A - Journal
B - Planet

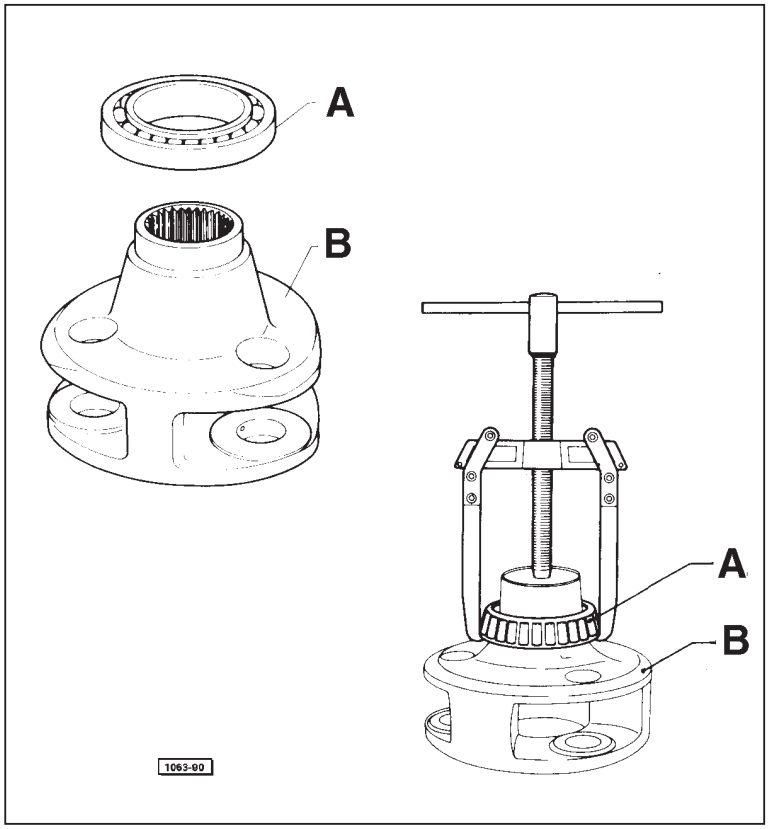


Fig. 9 - Removing the bearing from the epicyclic reduction unit flange.

FITTING LATERAL STUB AXLES OF THE WHEEL

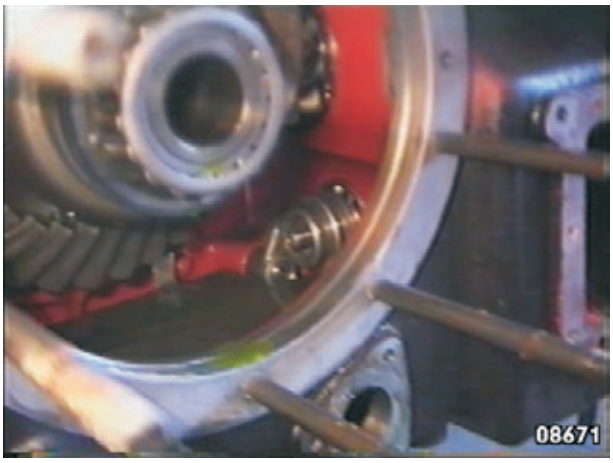


Fig. 10 - Before installing the trumpet housing, position sleeve A of the synchronised P.T.O..

Fig. 11 - Position the inner half-shaft in the differential housing.

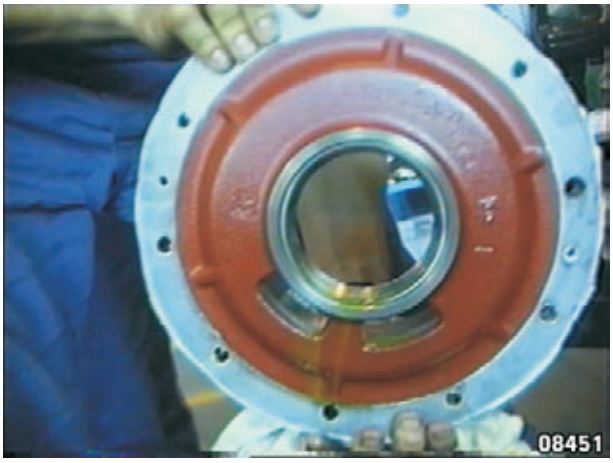


Fig. 12 - Fit the differential housing support flange and then install the brake piston and brake disc.

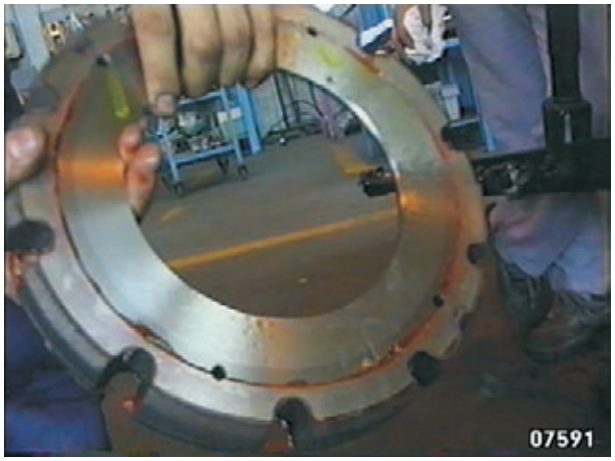


Fig. 13 - Install the thrust plate of the brake disc. The side shown in the figure is to be positioned in contact with the brake disc.

Fig. 14 - View showing the side of the brake disc thrust plate to be positioned in contact with the ring gear of the epicyclic reduction unit.

Fig. 15 - Fit the trumpet housing with the aid of a hoist.

2WD extendible axle

General information

The wide sideways front axle swinging obtained through excellent anchoring, not only makes it possible to operate the tractor nimbly on steep slopes but also permits the axle to be easily removed from the tractor when any repair is to be performed. End play check is the only operation required on axle reassembly.

The telescopic-type axle is extremely rugged and enables the tread to be adapted to any work requirements without removing the hydrostatic steering cylinders.

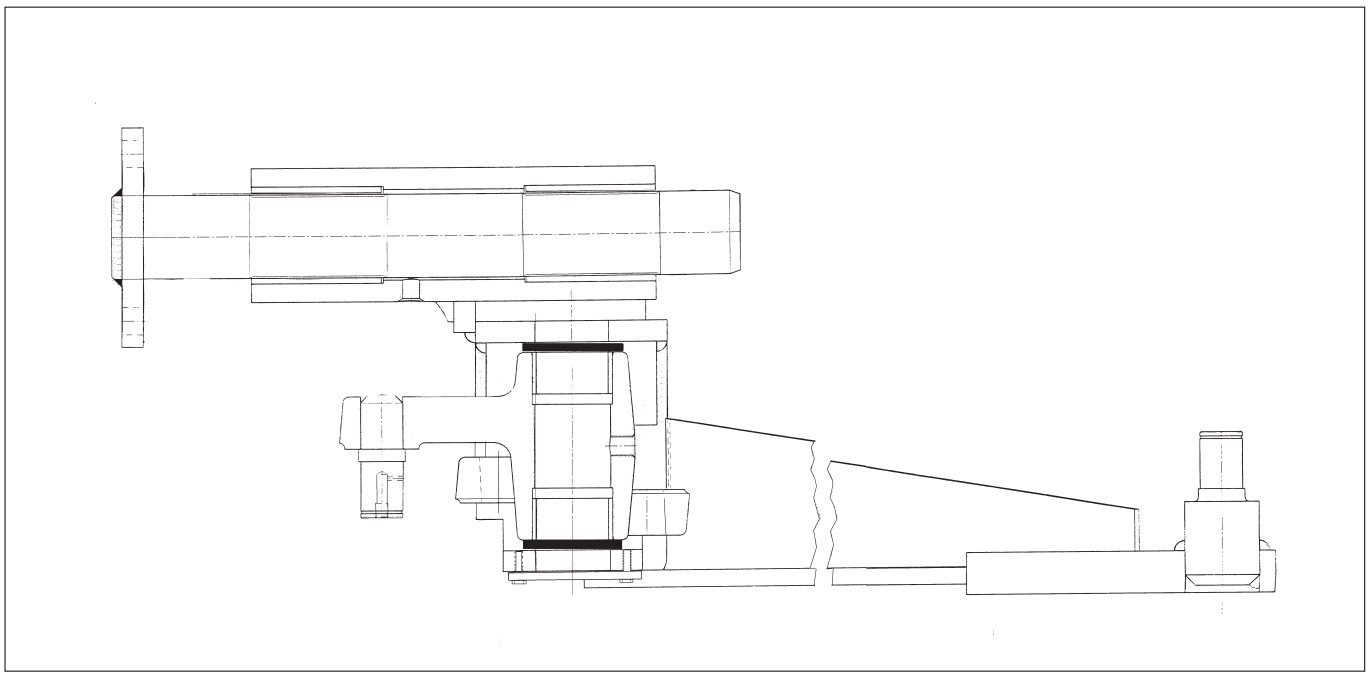


Fig. 1 - Extendible front axle pivot system.

Tecnical specifications

front axle		***660s - 770s - 880s - 990s AGILE***	***660v - 775v AGILE***
tread			
- minimum	mm	993	803
- base	mm	1050	853
- maximum	mm	1271	1079
toe-in	mm	2 - 6	2 - 6
wheel caster angle		3°	3°
axle swinging angle		12°	12°
tyre inflating pressure	bar	2,4	2,4
lubricating grease type		NLGI 2 LITIO/Ca	
steering angle		70°	70°

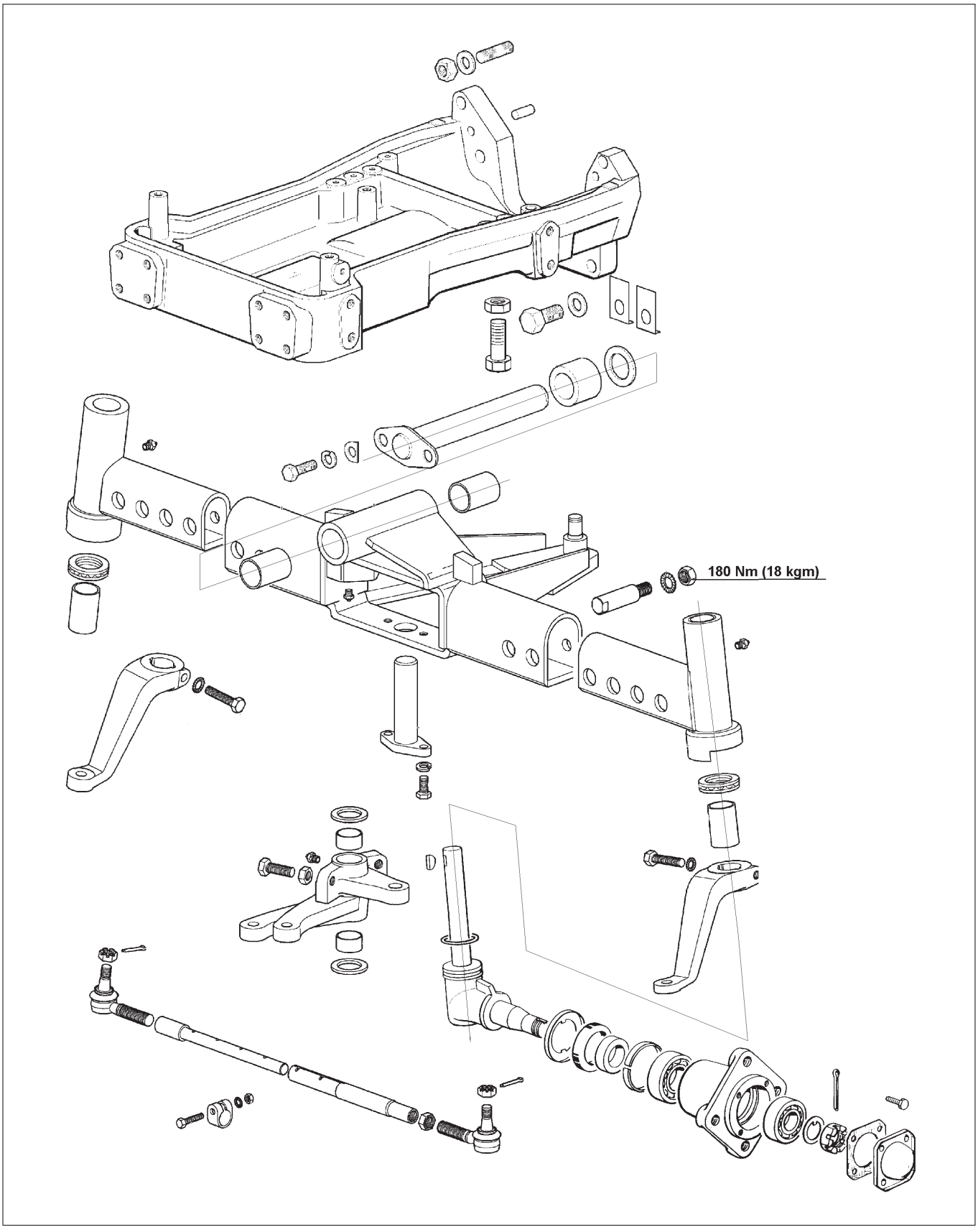


Fig. 2 - Components of the front axle assembly.

Removing the axle from the front support

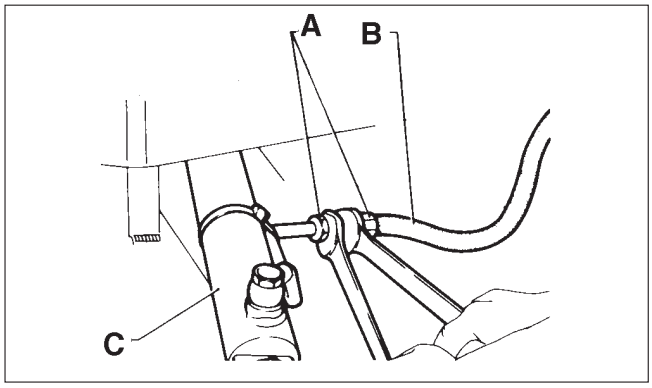


Fig. 3 - Place some suitable vessels beneath the steering control cylinders, then loosen unions **A** of control cylinders **C** from hoses **B** side.

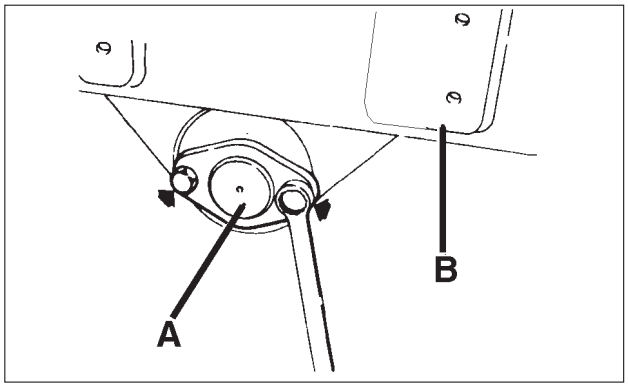


Fig. 4 - Loosen both securing screws of pin **A** to axle support **B**.

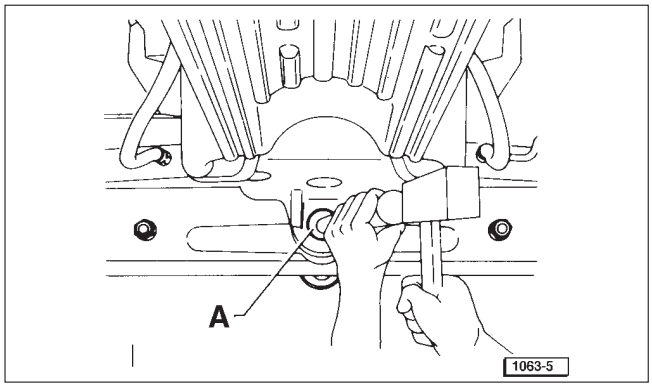


Fig. 5 - Using a hammer and light-alloy punch tap on rear end of pin **A**.

WARNING: Keep the two shim packs separated so that the correct position can be restored on reassembly.

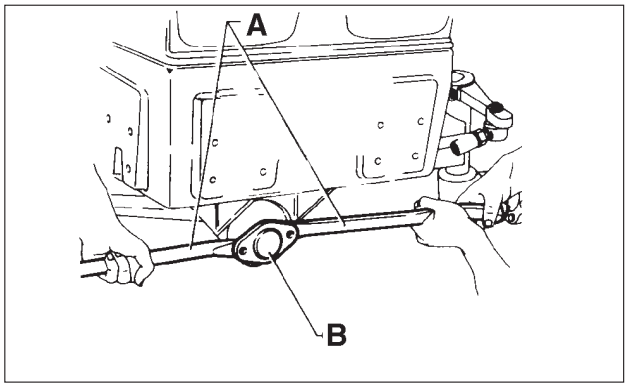


Fig. 6 - Fig. 6 - By way of both levers **A** withdraw pin **B** forward and recover the shims interposed between flange and front support.
Ensure the securing pin surface shows no flattening, craking or chipping, replace the pin if necessary.

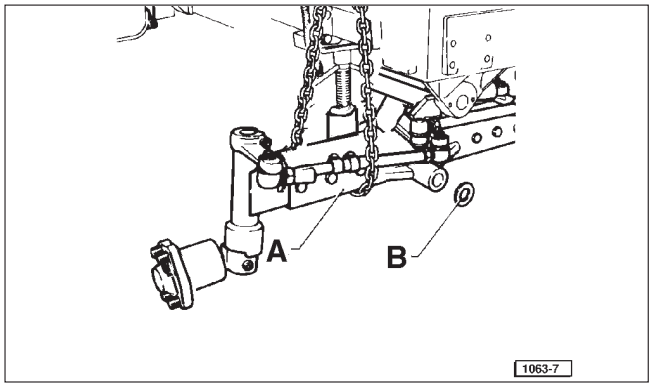


Fig. 7 - Remove the axle from the front support by tapping a few times the axle centre portion with the hammer.
Lower axle **A** very slowly and make sure that it does not touch other parts in the process, then remove shoulder rings **B**.

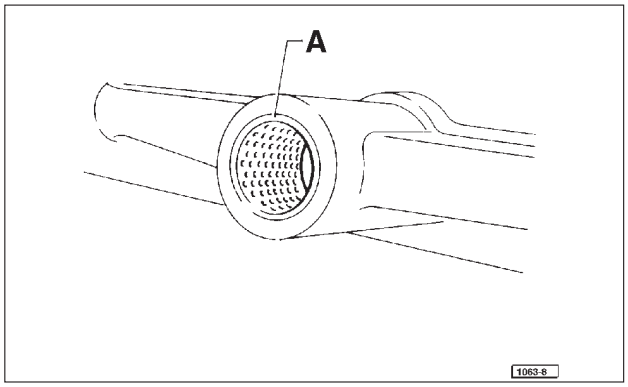


Fig. 8 - Ensure that bushes **A** of the centre axle are not worn, checking for well-evident indentations, remove the bushes by hammer and punch if necessary.

Carry out installation according to the following procedure:

— Use very fine abrasive paper to remove any signs of oxidation from both pin and front support;

— Bushes and pin should be properly lubricated with the specified grease type;

— Coat the front support shoulder rings with grease to make positioning easier;

— Making use of a hammer and light alloy punch tap on spacer **F** so as it may be moved toward the front side;

— Insert pin **E** into the spacer and fit front shoulder ring **B;**

— Lift the axle and insert the pin into bush **C;**

— Place rear shoulder ring **D** and push the pin almost fully inward;

— Fit shims **G** adjusting as instructed on page 216 and then fasten front axle pin securing screws **H**;

— Top up the oil in the hydrostatic steering system and bleed the air from the system (see chapter on hydrostatic steering).

— Check steering rod ball-and-socket joints for damage or wear and make sure the are able to rotate freely and without excessive play inside their seats.

— Ensure the rubber ball joint guard is intact, otherwise replace the ball joints.

— Examine the operating rods for damage or warping. Replace if required.

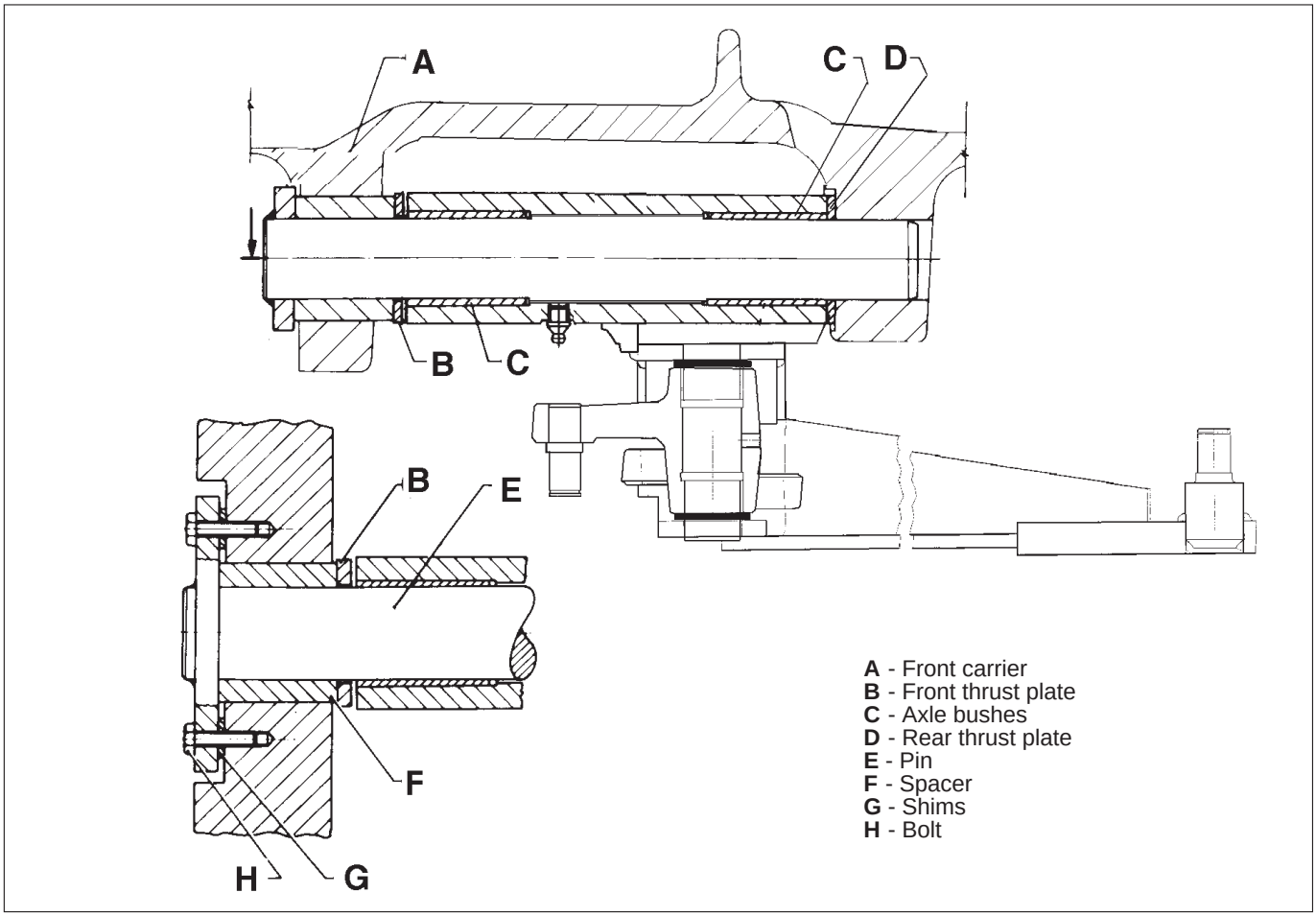


Fig. 9 - Front axle longitudinal section.

Drive axles - axles

2 W.D. extendible axle

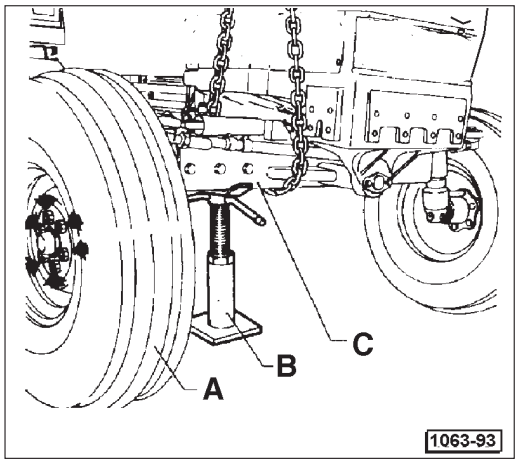


Fig. 10 - Removing the wheel hub.
A - Tractor wheel
B - Safety stand
C - Front axle

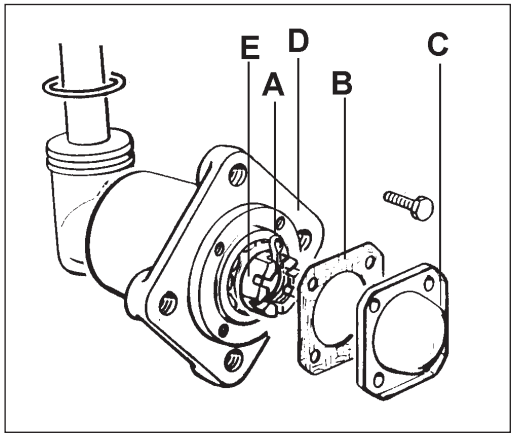


Fig. 11 - Removing the guard cover.
A - Check pin
B - Gasket
C - Cover
D - Hub
E - Nut

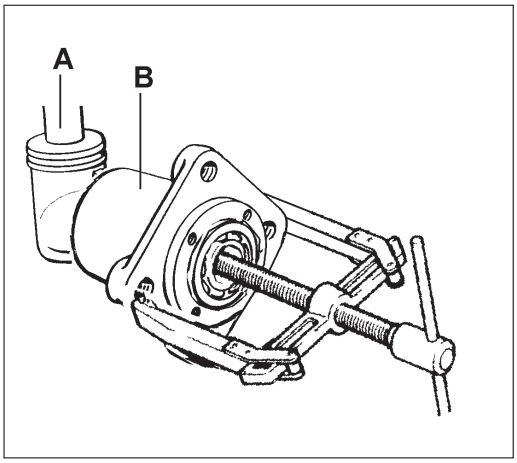


Fig. 12 - Using a puller to remove the wheel hub.
A - Wheel hub shaft pin
B - Whole wheel hub

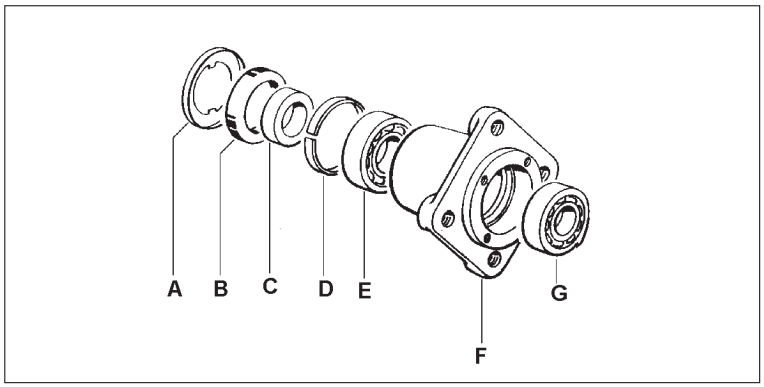


Fig. 13 - Wheel hub components.
A - Cover
B - Oil seal
C - Spacer
D - Spacer
E - Ball bearing
F - Hub
G - Ball bearing

Wheel hub

— Using a universal puller withdraw whole wheel hub **B** from pin, (Fig. 12).
— Make use of a suitable puller to remove front ball bearing **G** (Fig. 13)**.**
— Remove retaining ring **A** by prising, then using a proper puller remove rear ball bearing **E**, (Fig.13).
— Before assembly coat bearings **B** and **D** and new retaining ring **A** with the recommended grease, (Fig. 14).
Carefully insert retaining ring **A** using a hammer and light-alloy punch.
— Install wheel hub **C** onto pin **G** tapping slightly with the hammer for proper bedding
— Tighten castellated nut **E** to the prescribed torque.
Make sure the nut cut is duly aligned with the check pin hole in the pin, otherwise tighten the nut further to make them coincide and re-install the check pin.
— Ensure wheel hub **C** end play is 0.12 to 0.48 mm with respect to wheel shaft pin **G**.

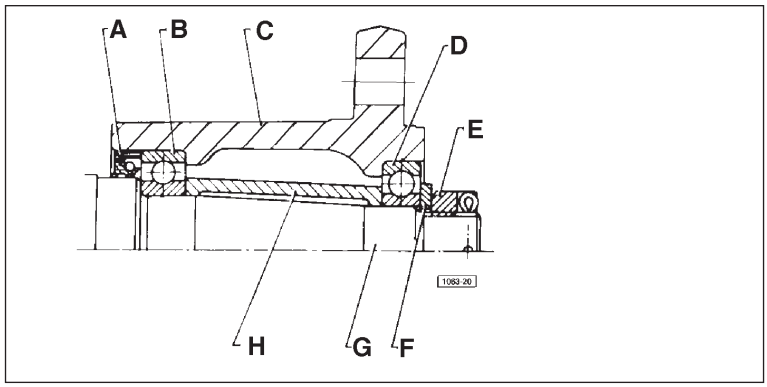


Fig. 14 - Section view of the wheel hub.
A - Oil seal
B - Innner ball bearing
C - Whel hub
D - Outer ball bearing
E - Castellated nut
F - Thrust plate
G - Whel shaft pin
H - Space

Inspections and checks

Clean bearings and the other parts carefully, then dry using compressed air.
Carefully examine the bearings turning them slowly, if the bearing is sound no vibrations, noise nor slight jam-ups shall occur.

Carefully examine bearing race and rolling member conditions ensuring no scoring, marking, grinding signs due to foreign matter abrasion are noticed, otherwise replace the bearing.

WARNING: Should a bearing be no longer serviceable, both the internal and the external bearings must be replaced.

Make sure the wheel shaft has suffered no damage and the threading is whole, replace if required.

Be sure no signs of oxidation are visible on the wheel shaft, if so remove with very fine abrasive paper.

Check the shaft for either warp or damage, if so replace the shaft.
Carefully inspect the axial bearing making sure no grinding, scoring or marking are evident either on the rolling members or on the bearing races, otherwise replace the worn parts.

Check side axle upper and lower bushes **A** for wear, ascertaining they still show evidence of the purpose-made indentations, otherwise replace.

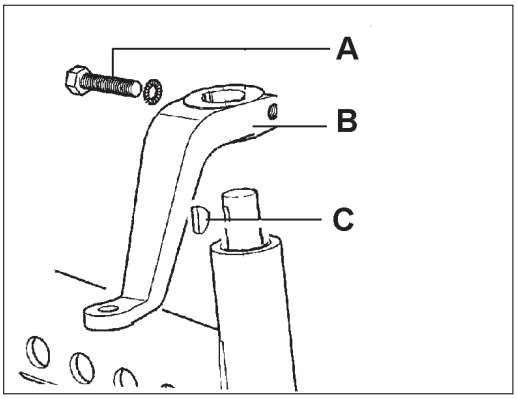


Fig. 16 - Removing the steering lever.
A - External steering lever
B - Gasket
C - Wheel shaft

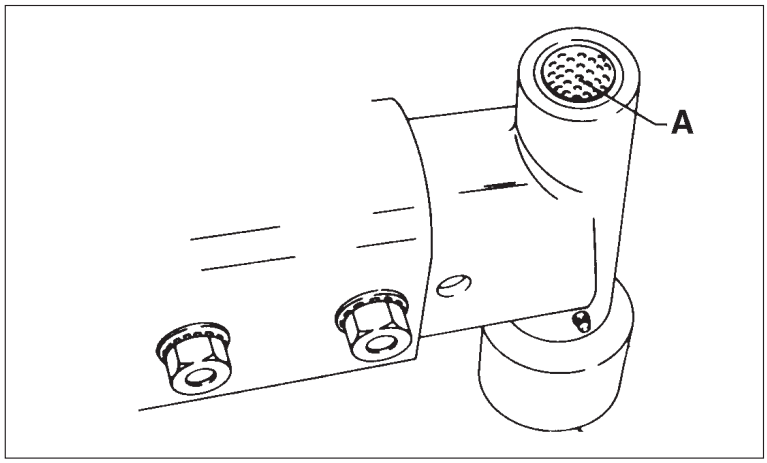


Fig. 15 - Wheel pivot bushes.
A - Clearance hole

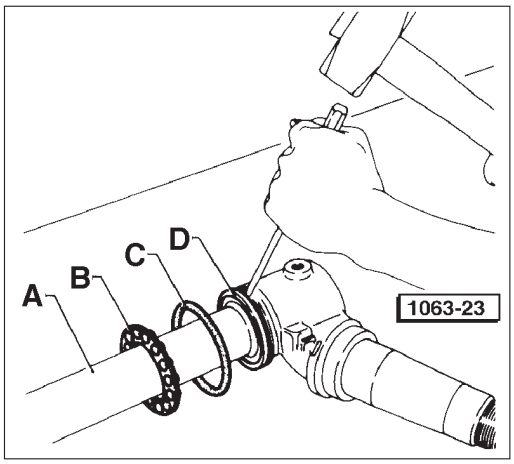


Fig. 17 - Removing the wheel pivot axial bearing.
A - Wheel shaft
B - Axial bearing
C - O-Ring
D - Axial bearing race

Drive axles - axles

2 W.D. extendible axle

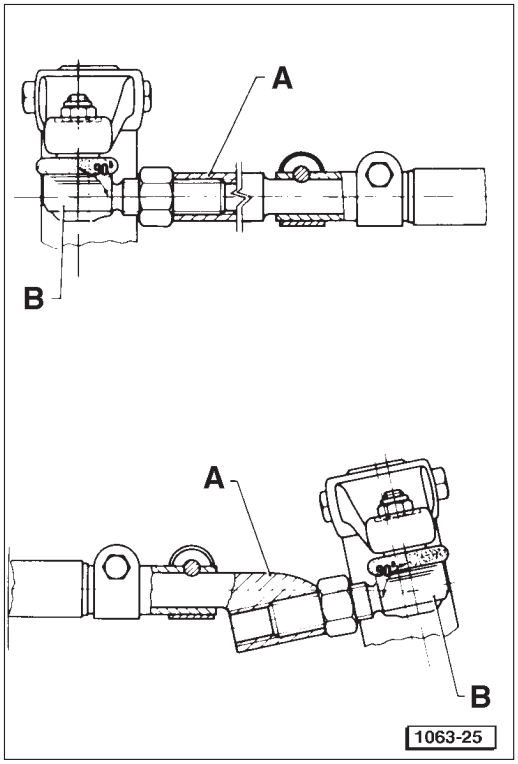


Fig. 18 - Track rods.
A - Sleeve
B - Steering knuckle

Correct track rod assembly (Fig. 18)
Rotate sleeve **A** around its axis so that knuckle **B** is in the position shown in figure, i.e. make sure the knuckle working angle is 90°, then insert and tighten both the sleeve securing bolts.

End float adjustment (Fig. 19).
Push the axle towards the rear and, using a feeler gauge, check that the maximum end float **"G"** does not exceed 0.4 mm.

If the end float exceeds this value, proceed as follows.

— Unscrew bolts **E** securing pin **B** to the front carrier **A**;

— Slightly move the pin and remove some of shims **D** to restore correct end float.

— Re-tighten bolts **E** and check end float again.

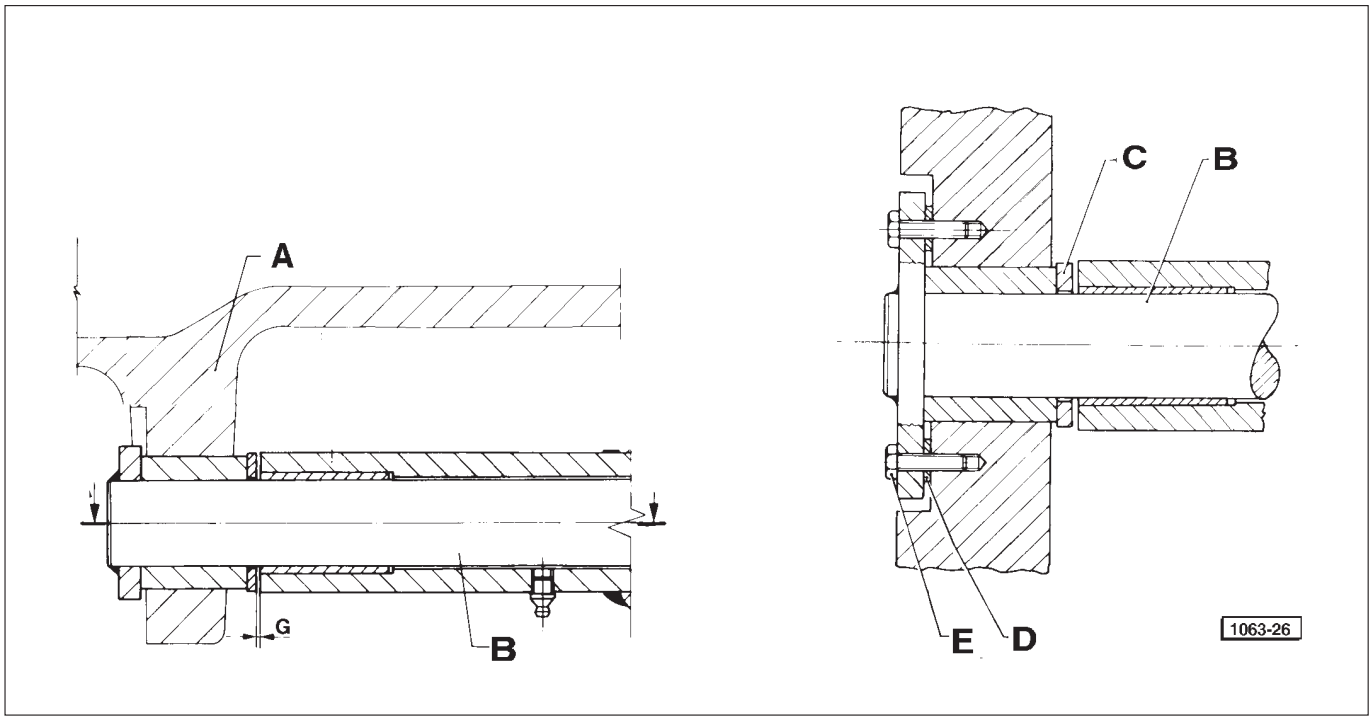


Fig. 19 - Adjusting the endfloat of the front axle .
A - Front carrier
B - Pin
C - Thrust plate
D - Shims
E - Bolt

FRONT-WHEEL DRIVE

Specifications

front drive axle ratios

reduction gear-front-wheel drive	30 km/h	21/47 = 1/2.2381
	40 km/h	28/47 = 1/1.6786
bevel gears		15/33 = 1/2.2000
final epicyclic reducer		(13/13+50)=1/4.8462
total reduction	30 km/h	1/23.8618
	40 km/h	1/17.8966
mechanical ratio (number of front wheel turns for each rear wheel turn)		
	30 km/h	1.3790
	40 km/h	1.3830
bevel gear teeth backlash	mm	0.15 ÷ 0.20
front drive axle end play	mm	0.1 ÷ 0.4
front drive axle swinging		10°
steering angle		50°
toe-in		see specification table here below
wheel caster angle		7°
camber		1°
Check value for the adjusting the bevel gears	mm	1

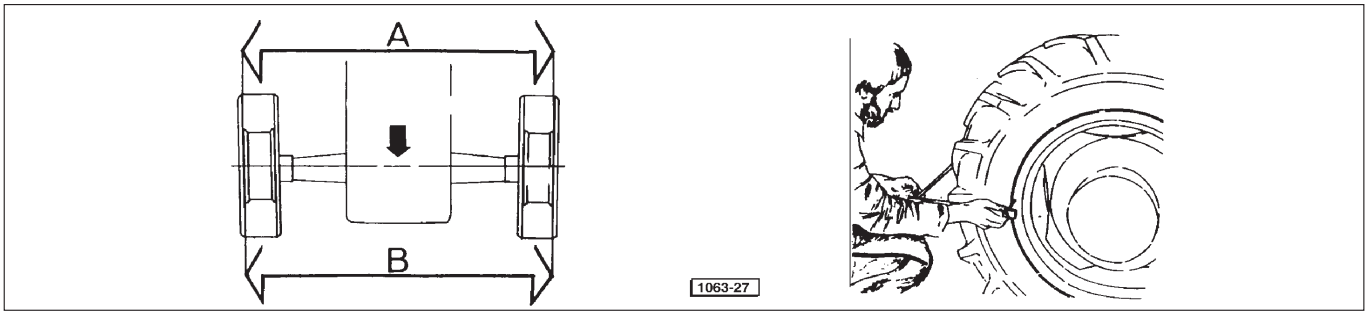


Fig. 1 - Measuring front wheel toe-in and front wheel geometry.

Adjusting toe-in
Adjust the front wheel tie rod so as the difference A-B shown in figure 1 is:

wheel rim keying diameter		**A-B** mm	
		2RM	**4RM**
up to 20"	(508 mm)	2 to 6	0±2
from 20'' to 30"	(509 ÷ 762 mm)	2 to 6	0±3
over 30"		2 to 6	0±3.5

Adjusting steering angles
Make sure the front drive wheel steering angle is as specified in the table above.
Otherwise adjust by way of the special setscrews or spacers.

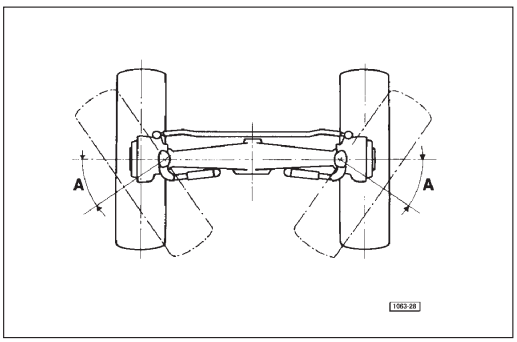


Fig. 2 -Front-wheel drive steering angles.

Removal of the axle from the front carrier

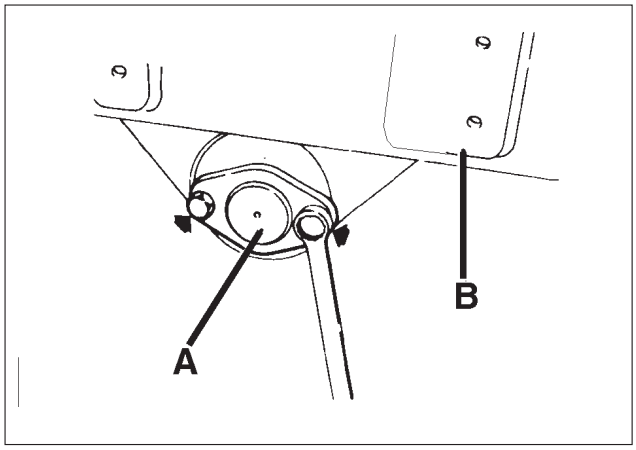


Fig. 3 - Dismantling the axle pivot pin.

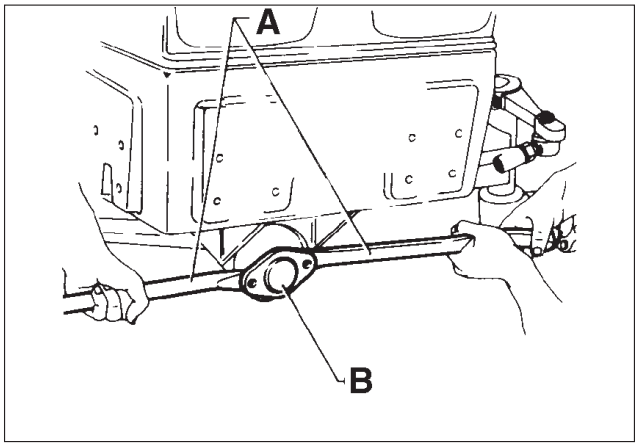


Fig. 4 - Removal of the axle pivot pin.

— Place suitable containers under the steering cylinders, then unscrew the fittings **A** of the cylinders **C** on the side of the flexible hoses **B**.
— Unscrew the two bolts securing the pivot pin **A** to carrier **B**.
— Using a hammer and a drift made of soft material.
WARNING: keep the two shim packs separated so that the correct position can be restored on reassembly.
— Using two levers **A** withdraw the pivot pin **B** towards the front of the tractor and recover the shims from between the flange and the front carrier.
Inspect the pivot pin for wear, cracks, or nicks and renew if necessary.
— Detach the axle from the front carrier by tapping the central part of the axle with a hammer.
Slowly lower the axle **A**, making sure that it does not contact other components; remove the thrust plates **B**.
— Inspect the bushes **A** for wear and that the indentations are still present; if necessary, remove the bushes using a hammer and drift.

NB: On models equipped with front P.T.O., the pin securing the front axle to the front axle is held in position by bolt A (fig. 5). To remove the pin, first remove the bolt and then withdraw the pin from the rear of the carrier (towards the engine).

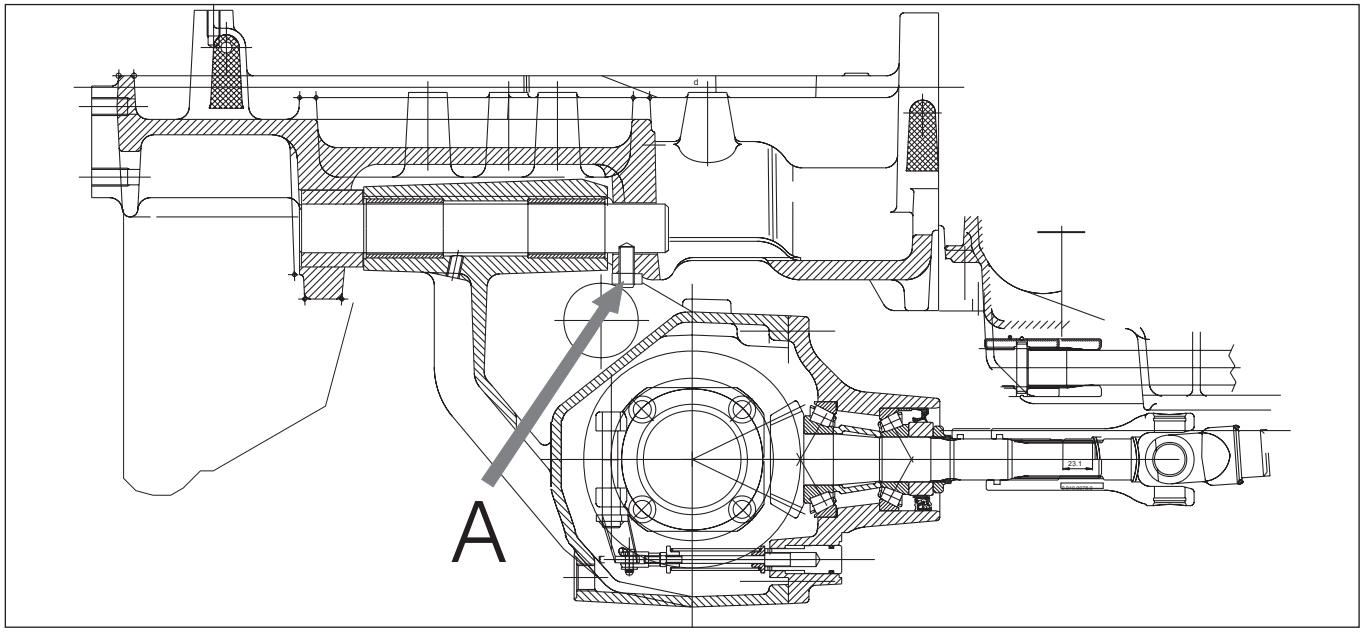


Fig. 5 - Front axle pivot pin for models equipped with front P.T.O..
Warning: apply Loctite 242 to the bolt before refitting.

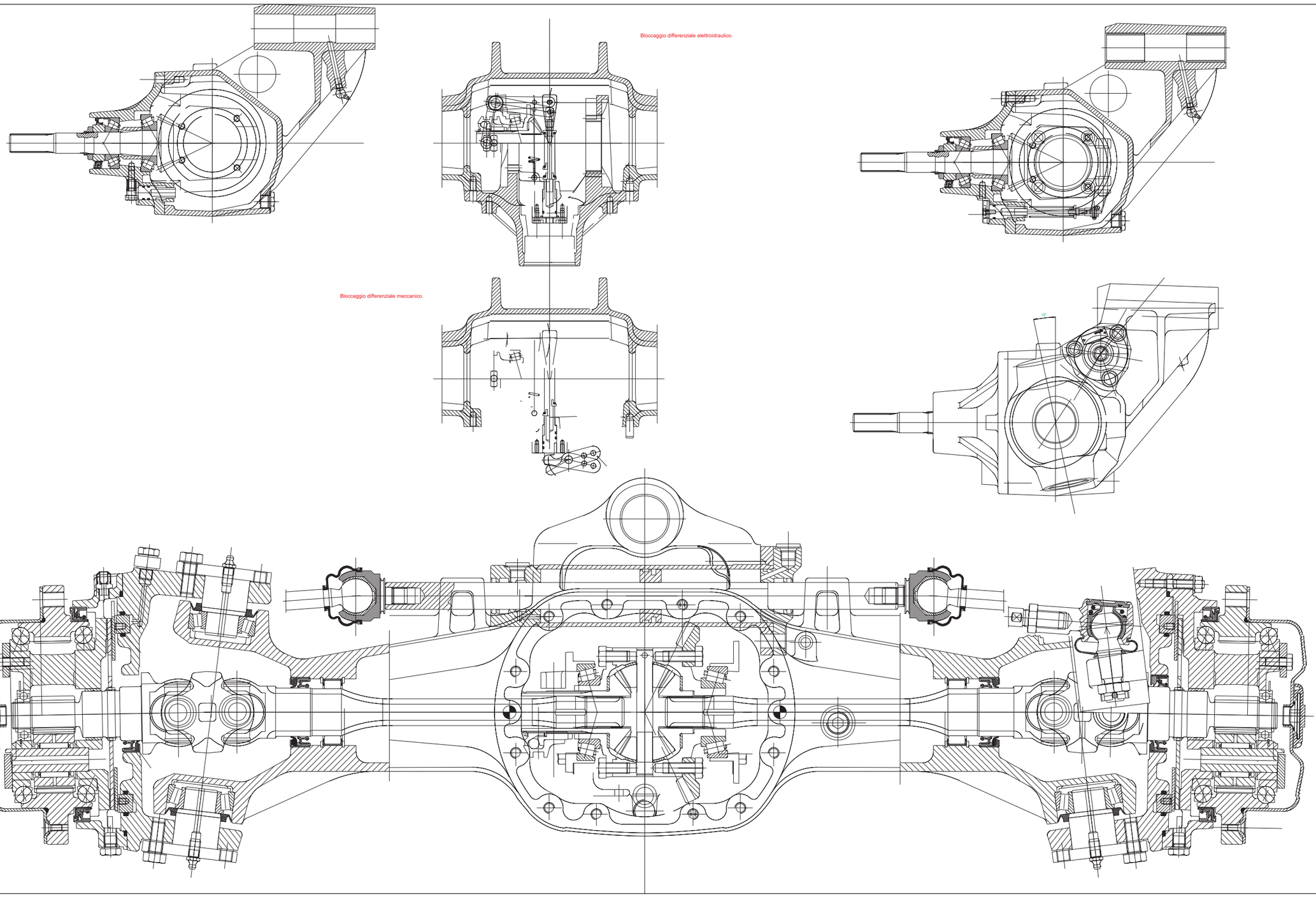


Fig. 6 - Front axle - longitudinal section.

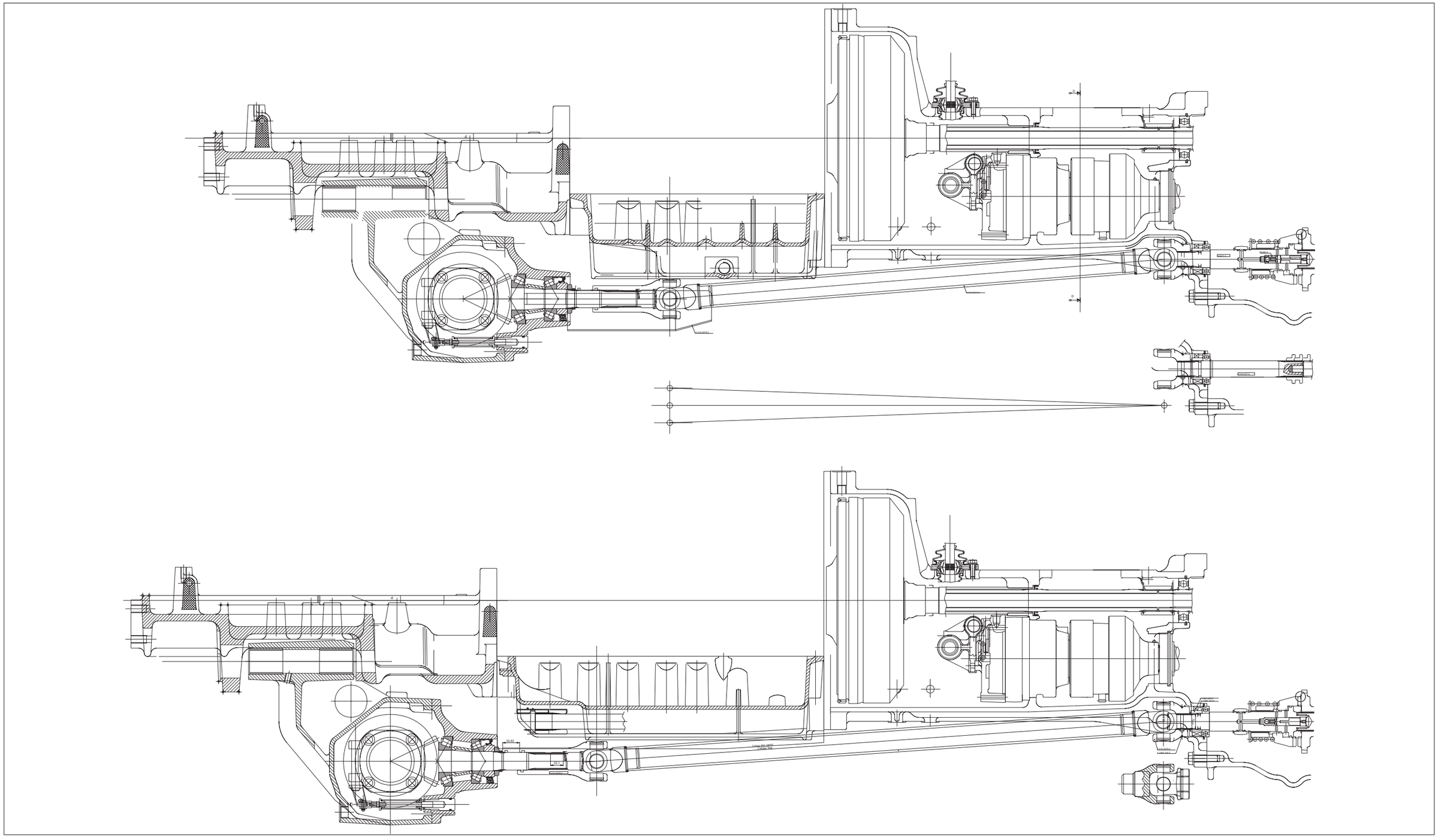


Fig. 7 - Views of front axle.

Epicyclic reduction unit

dismantling the hub

Remove the 6 hex socket bolts **A** securing the hub cover **B** and remove the cover, and collect the oil in a suitable container.

WARNING: always use a mallet of drift made of soft material on metal parts.

Unscrew the 6 bolts **C** securing the reduction unit flange **D** and remove the flange.

Using a small lever as shown in the figure, remove the hub from the brake housing flange **I**.

Remove the 3 planet pins **H** and then withdraw the 3 planet pinions along with their needle roller bearings.

Re-assembly
Fit the planet pinions **E** complete with roller bearings **F** in the planet carrier, with the teeth of the pins facing the centre of the hub.
Mesh the teeth of the planet pinions with the sun wheel, taking care not to damage them while inserting the planet carrier in the outer casing.

Press fit the hub in the brake housing flange **I**; position flange **D** and secure with the 6 bolts **C**.

WARNING: if the hub seal ring is to be renewed, install the seal in its seating using a suitable tool as shown in figure 8.
N.B. the sun wheel of the epicyclic reduction unit must be inserted dry into the seal: do not apply grease.

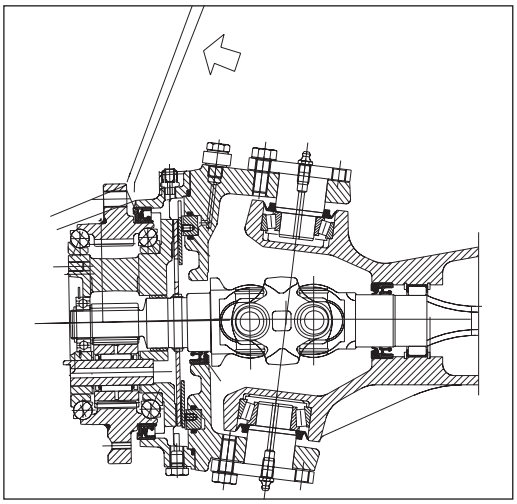


Fig. 8 - Dismantling the hub

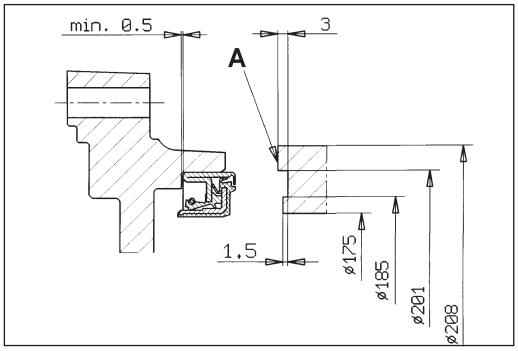


Fig. 9 - Installing the hub seal ring.
A - Reference shoulder on the flange.

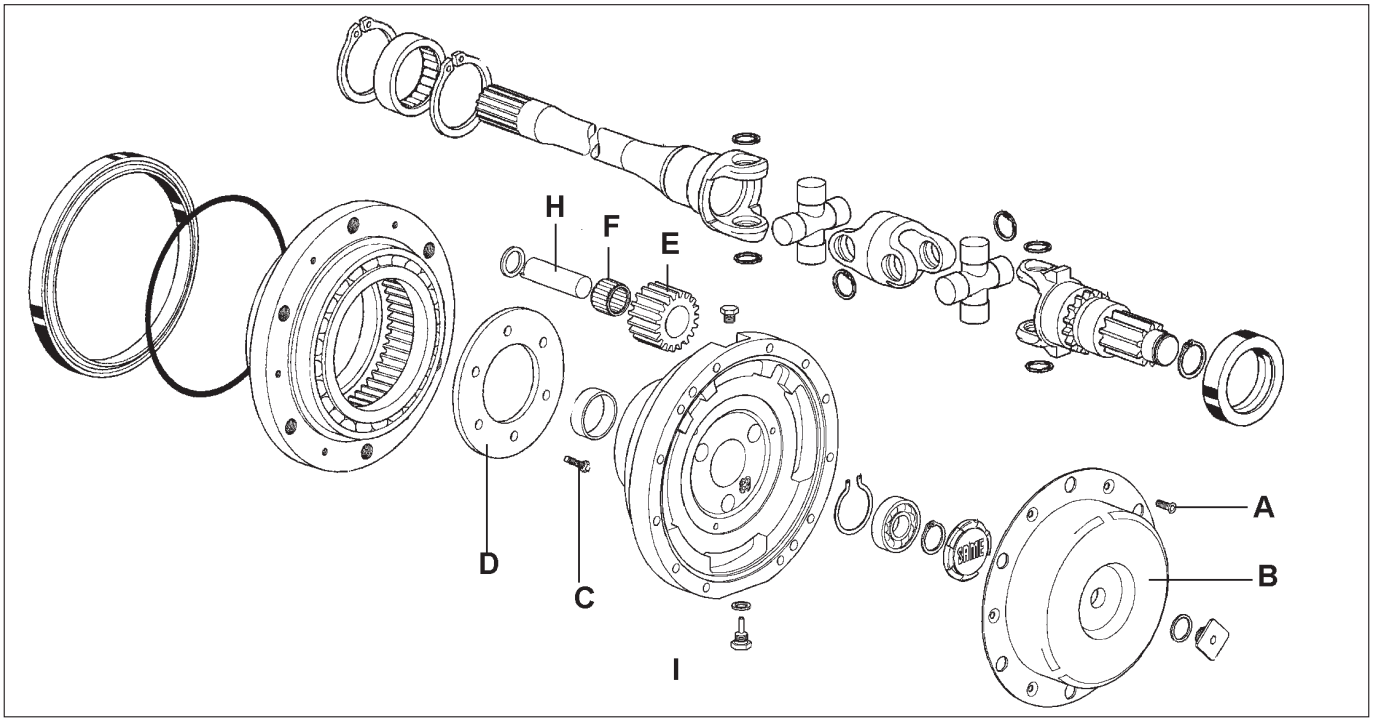


Fig. 10 - Components of the front halfshaft and hub.

Drive axles - axles

4 W.D. front axle

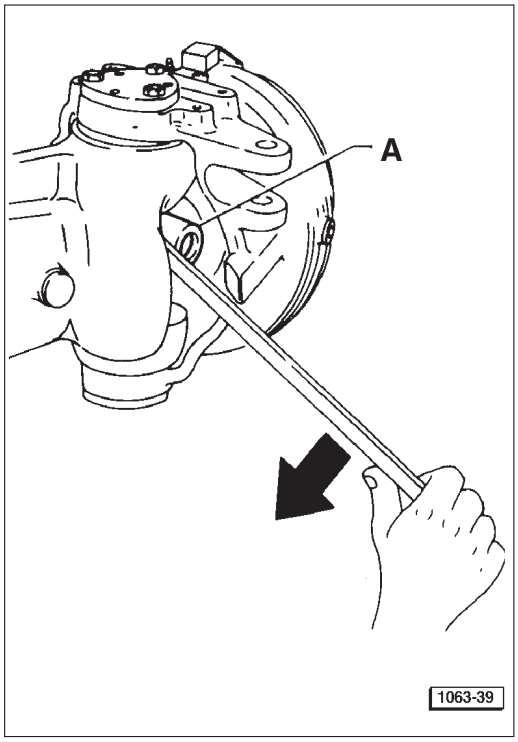


Fig. 11 - Fixing the halfshaft for assembly of the epicyclic reduction unit
A - Spider

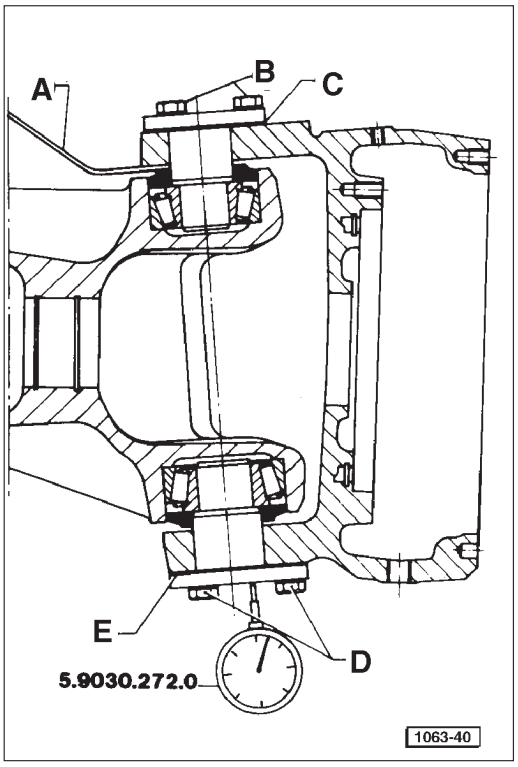


Fig. 12 - Checking wheel hub clearance.
A - Lever
B - Screws
C - Pack of shims
D - Screws
E - 0.5 mm shim.

The following procedure should be observed on re-assembly

Brakes
Replace the brake control piston O-rings and insert piston in its seat with the oil grooves turned outwards.

Final epicyclic reducer
Install the final epicyclic reducer holding the axle shaft in position as shown in figure 12 by prising outwards on cross journal **A**; in order to prevent the axle shaft from moving inwards.

Axle shafts
When inserting the axle shaft pay attention not to damage the roller bearing or the seal ring.
The axle shaft end should be correctly introduced into the differential gear planetaries.
Make sure the axle shaft is free to rotate without any hindrance.

Forks
After installing the bearings and the dust rings mount the fork by placing one 0.5 mm shim under the lower pin, then fit the pin, use a hammer if necessary, finally tighten the securing screws. Fit a pack of shims thicker than the one taken on removal under the upper pin. Install the pin tightening the securing screws.

Adjusting the preloading of the hub carrier bearings
Loosen the screws and install a shim pack of thickness 0.5 mm under the lower pivot pin and a shim pack of greater thickness than that previously installed under the upper pivot pin.
In the pack of shims to be used for adjustment it is always advisable to group more shims in one: as an example, it is better one 0,2 mm shim be used instead of two 0.1 mm shims.

Fit no. 5.9030.267.0 magnetic base with no. 5.9030.272.0 centesimal dial gauge onto the drive axle and then place the gauge feeler perpendicularly to the lower pin, close to centre, and set to zero.

Using lever **A** as shown in figure, move the fork fully upwards and read the clearance on the gauge dial.
Loosen the two screws **B** and remove shims from pack **C** so that any clearance may be taken up without preloading the bearings.

WARNING: The clearance amount should be reduced gradually by repeating reading with the dial gauge each time so that the bearings are not preloaded.
After all clearance has been taken up, remove a 0.10 to 0.15 mm shim pack, so that a correct bearing preloading can be obtained.
Tighten screws **B** and **D** to the specified torque.

After correctly performing the adjustment, ensure the shaft taper roller bearing slide in their seats freely, even though a slight preloading is felt.

Grease the hub bearings.

Side hubs

Disassemble the twin universal joints if required, keeping to the following procedure:

— Remove both snap rings **A** from one cross journal **B**. A non-ferrous metal hammer and punch may help in carrying out this operation, (see Fig. 13).

— Fix the joint fork in a vice provided with protective jaws.

— Tap with a hammer on wheel shaft fork **A** (see Fig. 15), so that the bearing may be removed upwards from its seat in fork **A**. Separate cross joint **D** from the fork.

— Following the same procedure separate the cross joint from the universal joint fork as well as from the axle shaft fork.

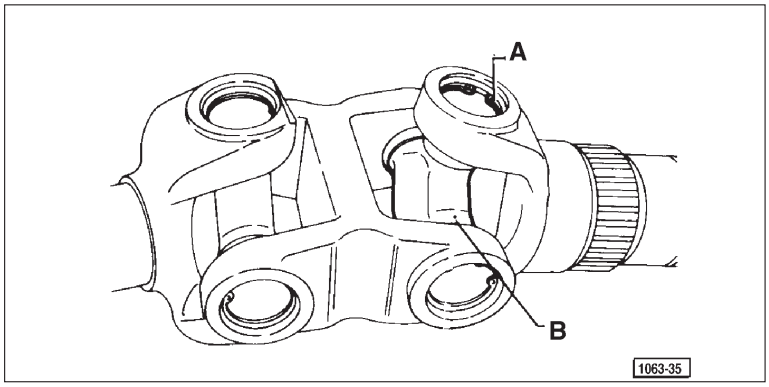


Fig. 13 - Twin universal joint.
A - Snap ring
B - Twin universal joint cross journal

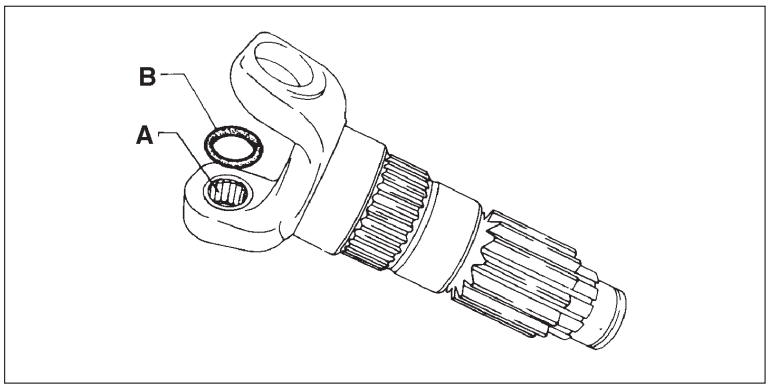


Fig. 14 - Axle shaft twin-type universal joint bearing.
A - Rollers
B - Dust rings

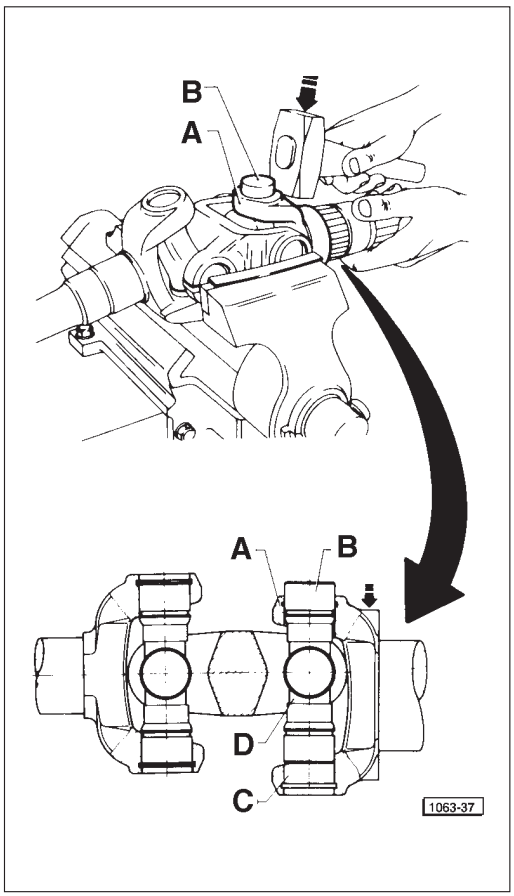


Fig. 15 - Disassembling the twin-type universal joint from the axle shaft.
A - Wheel shaft forks
B - Roller bearing
C - Roller bearing
D - Cross joint

Fitting the front axle studs

Fit the 2 studs code 2.0432.161.7 in holes 1 and 2 in the figure.

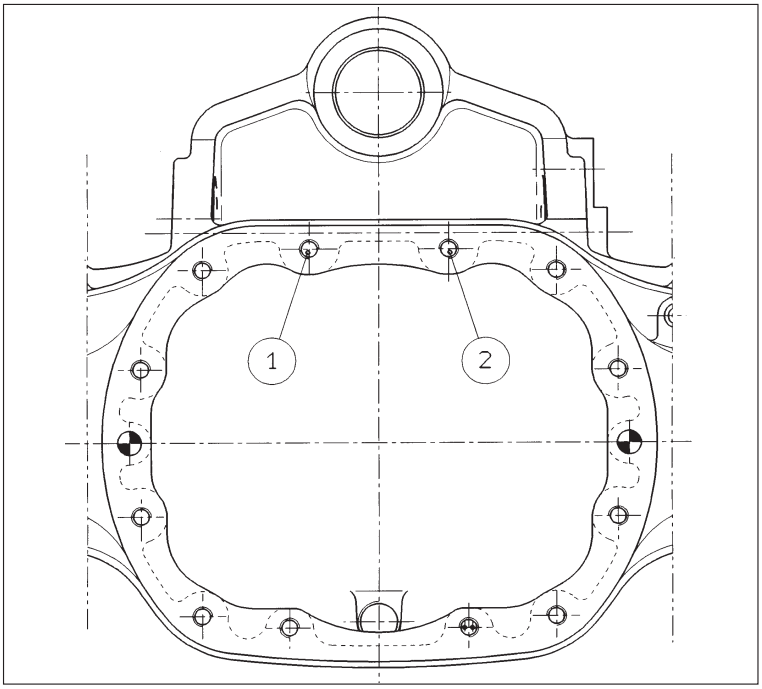


Fig. 16 - Fitting the front axle studs.

Axle shafts

Check for excessive wear: splines should not be nicked and permit a free gear movement.
Spider forks should not be warped and bearings should slide freely.

Gears

Make sure all gear toothing is neither worn nor damaged, teeth should work on the whole contact face.

Bearings

Examine bearings for proper working conditions, ensure they show neither excessive radial nor end play.
Pressing the bearings by hand and simultaneously turning them in both directions, they should slide freely.

TIGHTENING TORQUES

Before tightening, all screws should be degreased and cleaned

	kgm	Nm
Differential/bevel gear securing screws	7,5	73
Bevel pinion fixing nut	21 ÷ 23	206 ÷ 226
Bearing holding flange securing screws	9,7	95
Hydraulic cylinder fixing pin	5,5	54
Ball-and-socket joint castellated nuts	12	119
Transmission shaft flange securing screws *	5	49
Differential half case securing screws	11.5 ÷ 12	113 ÷ 118

* Coat the nut with a small amount of Loctite 242.

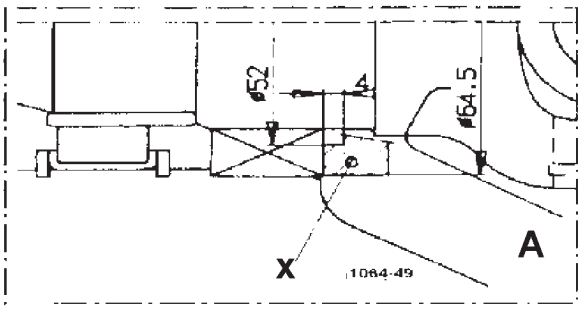
Check periodically that the wheel bolts are correctly tightened.

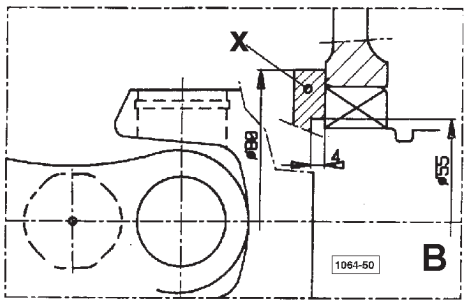
Front wheel bolts	2RM	(M20x1.5)	50 kgm (490 Nm);
Front wheel bolts	4RM	(M18x1.5)	36.8 kgm (360 Nm)
Rear wheel bolts		(M20x1.5)	50 kgm (490 Nm).

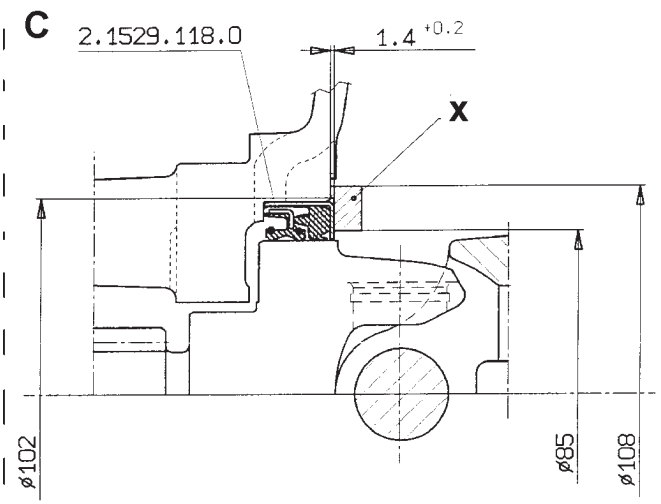
Fitting the waterproof seals

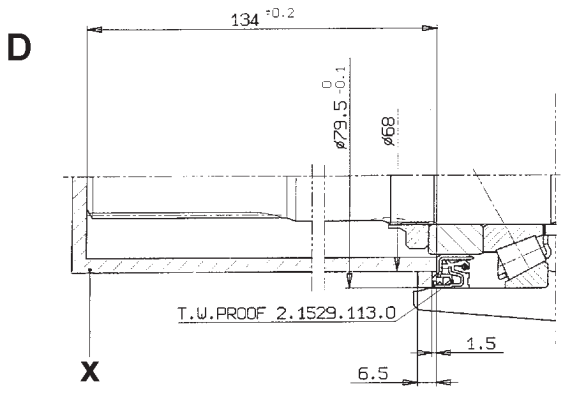
A - Inner half-shaft
B - Outer half-shaft
C - Hub
D - Bevel pinion (standard version)
E - Bevel pinion (version with waterproof seal)
X - Installation tool

Fit the seals using specially-made installation tools, positioning the seals as shown in the figure and pressing them in until the tool is fully seated.
The measurements required to make the installation tools are given in the figure.









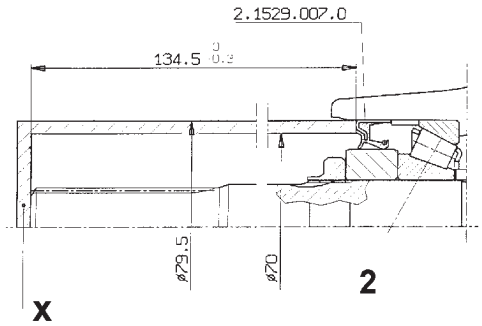


Fig. 17 - Waterproof seals.

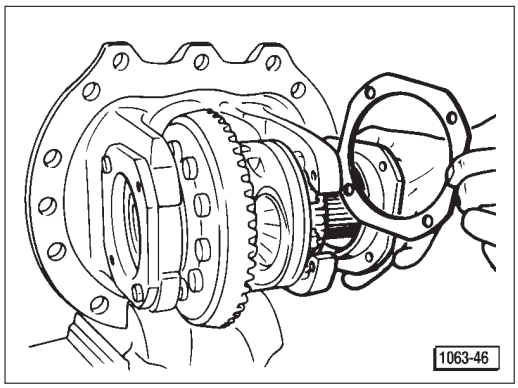


Fig. 18 - Shimming the bevel pinion bearings.

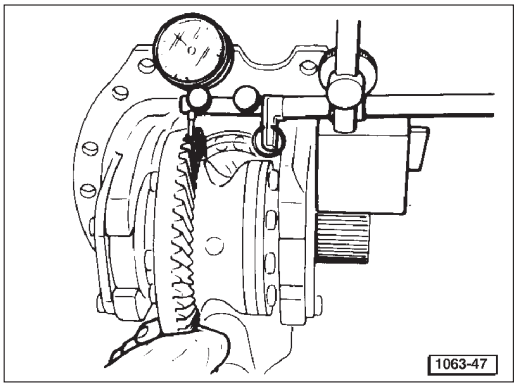


Fig. 19 - Measuring the backlash between bevel pinion and ring gear teeth for adjusting bevel gears.

Adjusting bevel gears

1 - Assemble the differential housing, inserting a number of shims between the bearing flanges and the carrier such as will give a preload of 0.04 mm at the bearings.

2 -Remove the differential housing again and fit the pinion, inserting a first pack of shims **A** to a thickness of 0.50 mm and a further pack **B** of thickness such that when the lock nut is torqued to 206 to 226 Nm (21 to 23 kgm), the bearings are able to turn freely in their seats while discernibly subject to a slight preload (no more than 0.04 mm).

3 - Adjust the distance **E** between the differential housing and the end of the pinion by adding or removing shims at pack **A**.

The exact measurement for this adjustment is obtained by adding or subtracting the value stamped on one tooth of the pinion to or from a value of 1 mm.
Warning: having adjusted the distance between the differential housing and the end of the pinion, the preload adjustment must be repeated to re-establish the thickness of the pack of shims **B**.

4 - Refit the differential housing and measure the backlash between the crownwheel and pinion teeth with a comparator: the gap must be between 0.15 and 0.20 mm; if not, the crownwheel can be brought closer to the pinion by removing shims from **C** and adding to **D**, or distanced by removing from **D** and adding to **C**.

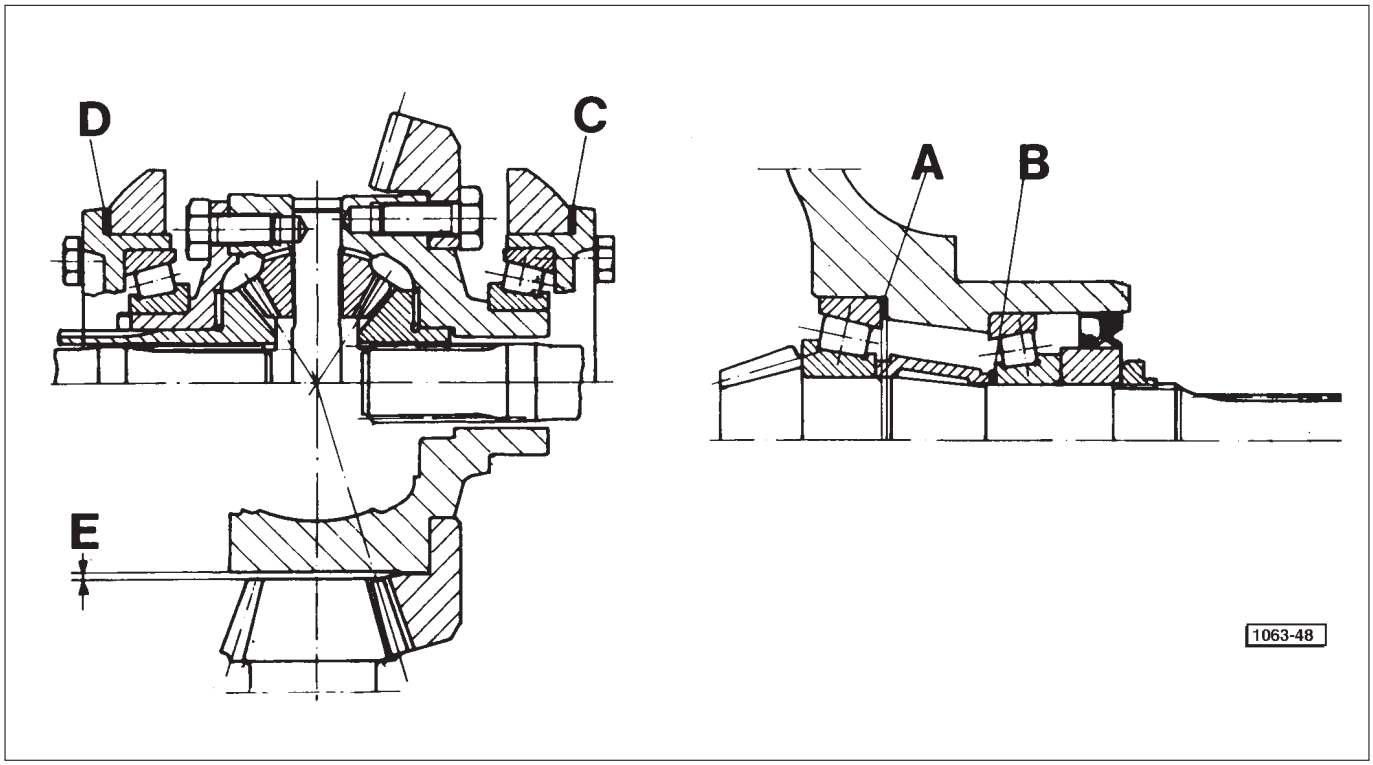


Fig. 20 - Adjusting the bevel gears.

Adjustment of the internal control of the mechanical differential lock (Fig. 23).

(For adjustment of the external control see chapter “controls”).

1 - Assemble the parts, installing shims **A** on the side of circlip **B** to obtain play of spacer **C** equal to 0 ± 0.05 mm.

2 - Move a number of the shims in pack **A** to position **G** in order to move sleeve **D** into the engaged position.

WARNING - shims of packs equal to or less than 0.20 mm are to be moved. Sleeve **D** is correctly engaged when it is able to overcome balls **F** without forcing to occupy the position **E** in Fig. 23.

Installing the differential assembly into the drive axle

Swivel the assembly so that the differential lock sleeve is at the same side as the internal control lever.

The sleeve should be moved outwards to enable the lever pad to be housed inside the groove.
After positioning the differential into the drive axle, check the differential lock for proper engagement by operating the control lever and rotating the bevel pinion by hand.

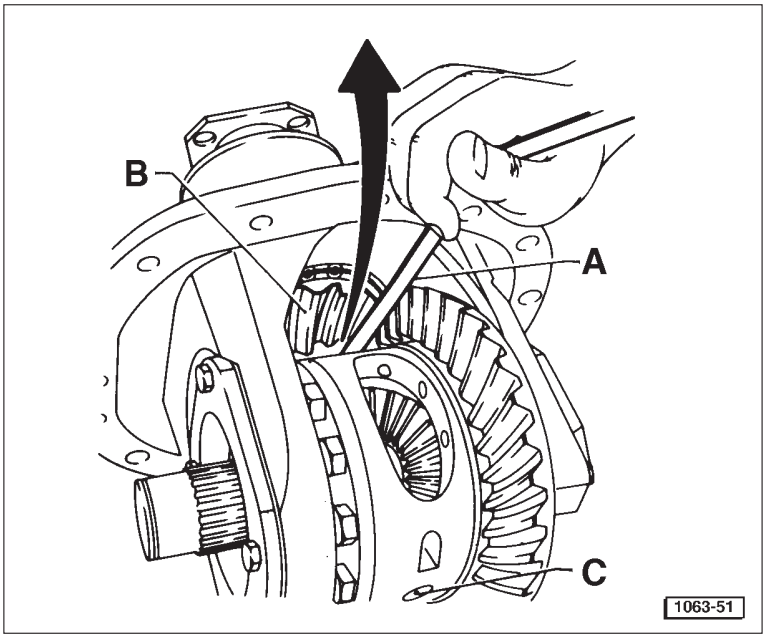


Fig. 21 - Checking the difference between the bevel pinion head and the differential housing ground surface.
A - Thickness gauge
B - Bevel pinion head
C - Differential planetary carrying pin

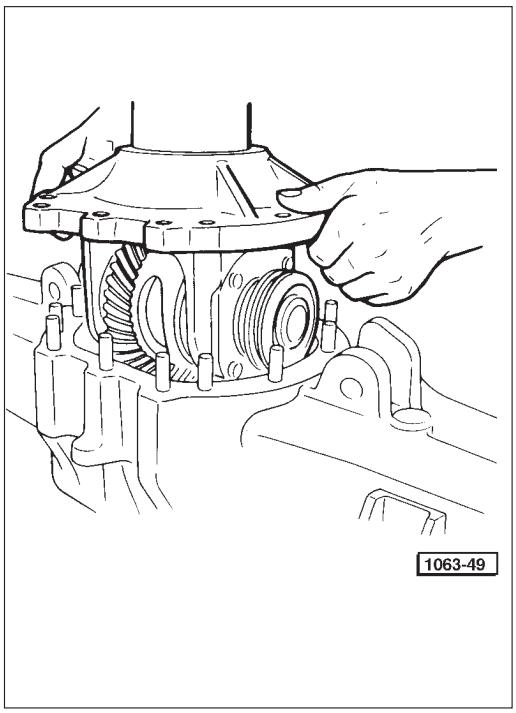


Fig. 22 - Swivelling the differential assembly in the drive axle.

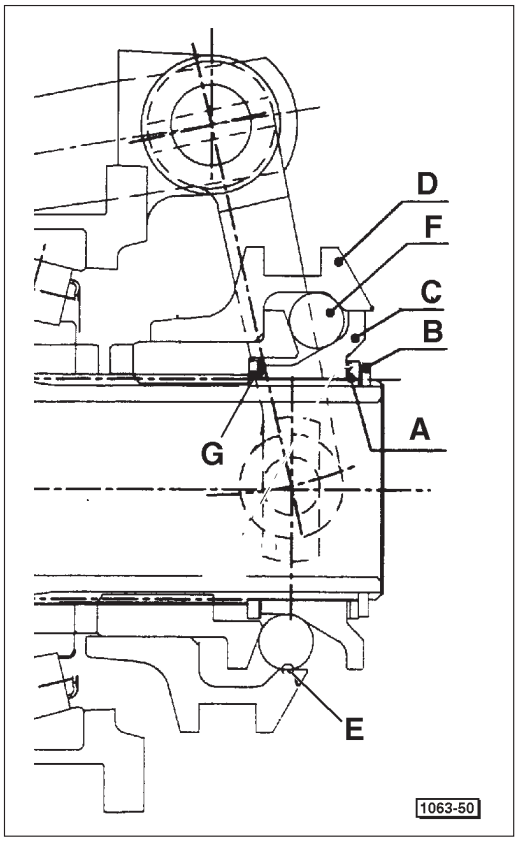


Fig. 23 - Differential lock adjusting references.

Fitting the steering angle limiting bolts

On models not equipped with front fenders, the maximum internal steering angle must be within the range 44° to 60°. Adjust the steering angle by way of bolts **A**.

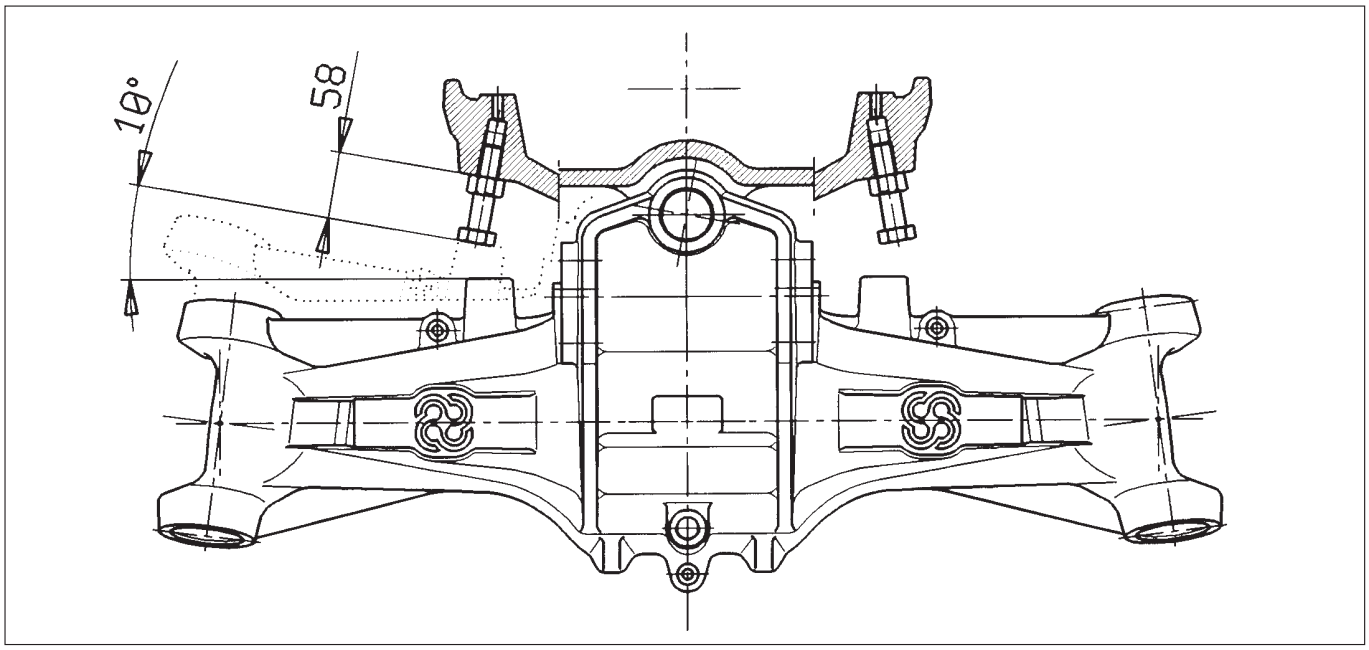


Fig. 24 - Fitting the steering angle limiting bolts

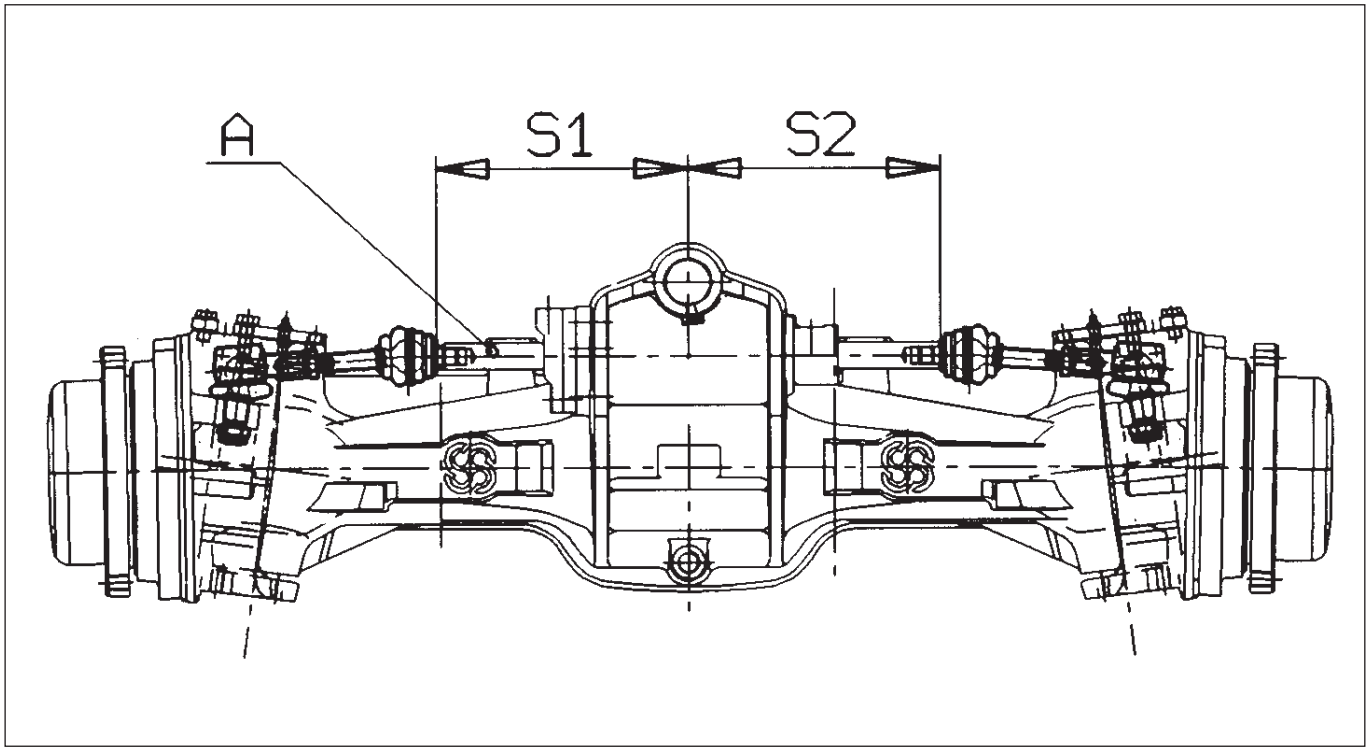


Fig. 25 - Adjusting the length of the track rods

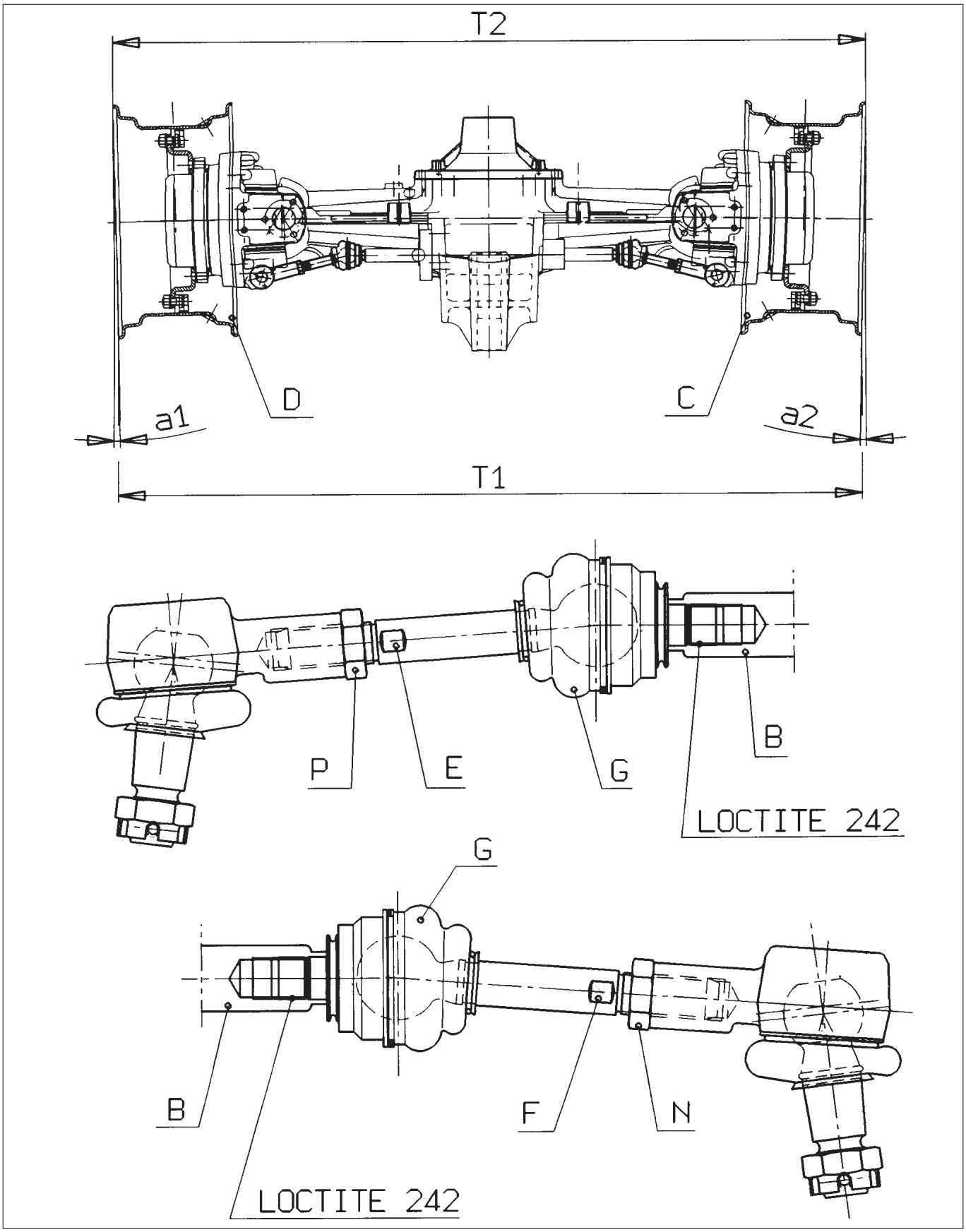


Fig. 26 - Adjusting the steering angle on models equipped with front fenders.

Drive axles - axles

4 W.D. front axle

Diagnosing malfunctions

Wear on the universal joints	front axle overloaded	only use front loaders recommended by the tractor manufacturer	fit rear ballast
Oil leaks	check oil seals for wear	replace worn seals	
	bleeding the obstructed oil	Clean	
Tyre wear	even wear	use 4WD only on farmland	excessive use of the tractor for road transport fit suitable tyres
	uneven wear	Check wheels toe-in	Adjust
Wheel oscillation	check tyre wear	replace if necessary	
	check steering joints for wear	replace if necessary	
	check hub bearings	adjust or replace if necessary	

Brakes

General information

The service brakes are located between gearbox and final epicyclic reducers and assure a precise and safe braking action. In addition, the four-wheel drive tractor models are also equipped with integral braking system providing braking on all tractor wheels.

The braking system is sintered-lining, oil-immersed disk type. The right-hand brake control is completely independent from the left-hand brake control; this allows the tractor turning radius to be conveniently reduced (this operation is only possible when working on farm land and never during transport operations on public roads). The braking system is provided with the “SEPARATE BRAKES” valve which enables the front wheel braking to be excluded.

Each brake pedal controls a hydraulic pump which delivers oil under pressure to a disk thrust plate, which causes the brake disk to lock.
Operating both brake pedals simulteneously by coupling them with the special joining latch, opens an oil duct connecting both hydraulic circuits to balance the baking pressure on the wheels.
Brake maintenance is limited to an easy adjustment and a system air bleeding if necessary.

The parking brake has mechanical control and acts on transmission downstream of gearbox. Pulling the control lever upwards makes the brake disks pack together thus locking the shaft being constantly meshed with the wheels.
Maintenance is very easy and consists of replacing the friction pads when worn or adjusting the control lever travel.

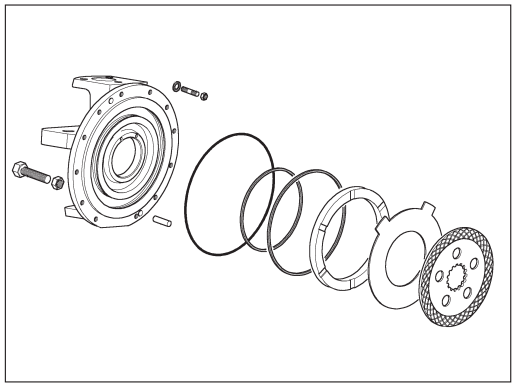


Fig. 1 - Front service brakes.

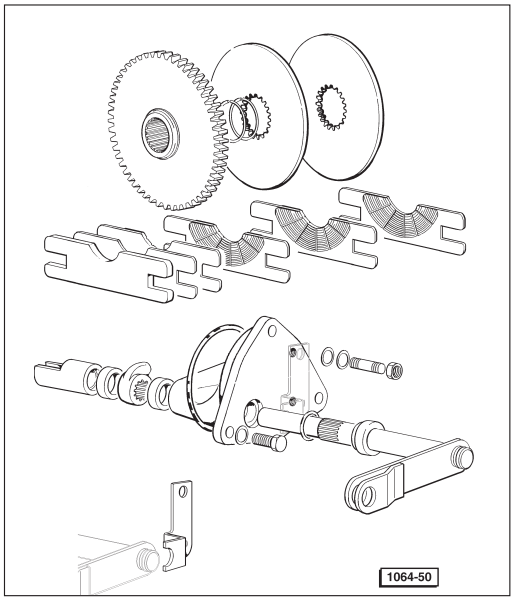


Fig. 2 - Parking brake parts.

Specifications

service brakes		**front**		**rear**
		2RM	**4RM**	
Manufacturer		SAME - DEUTZ FAHR GROUP		
brake type		oil-immersed disks		
number of disks each brake		2 (each side)	1 (each side)	1 (each side)
brake disk outside diameter	mm	223,4	223,4	280
original brake disk thickness	mm	4,80	4,80	7
brake disk minimum thickness allowable	mm	4,40	4,40	6
brake pedal free travel	mm	40		
braking piston max. Play	mm	1,15		
hydraulic pump type		benditalia Ø 1’’		
parking brake				
brake type		oil-immersed disks		
number of braking pads	double braking surface	n° 2		
	single braking surface	n° 2		
original braking pad thickness	double	mm	5	
	single	mm	3,5	
braking pad minimum thickness allowable	double	mm	4,3	
	single	mm	3,2	
brake disk number		3		
parking brake control lever travel	mm	100		

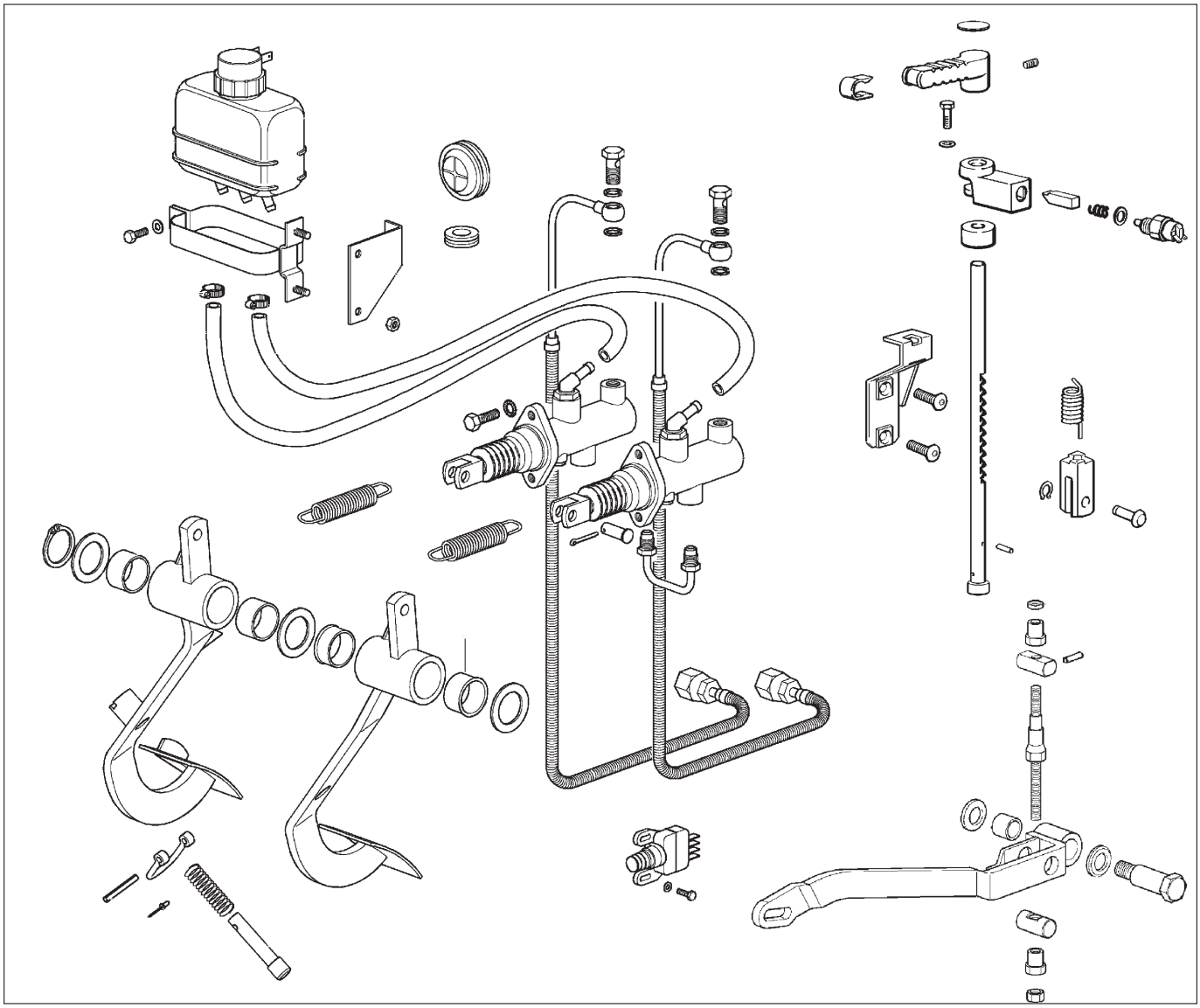


Fig. 3 - Brake control assembly parts.

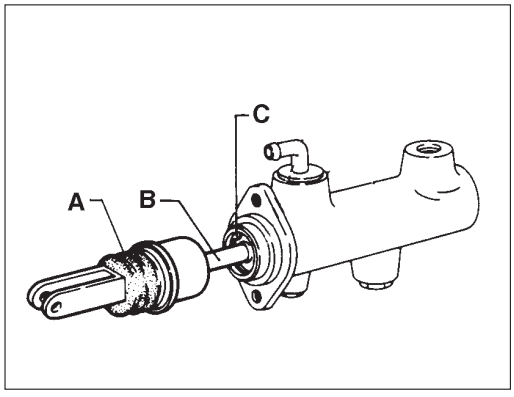


Fig. 4 - Brake pump control unit.

Hydraulic pump

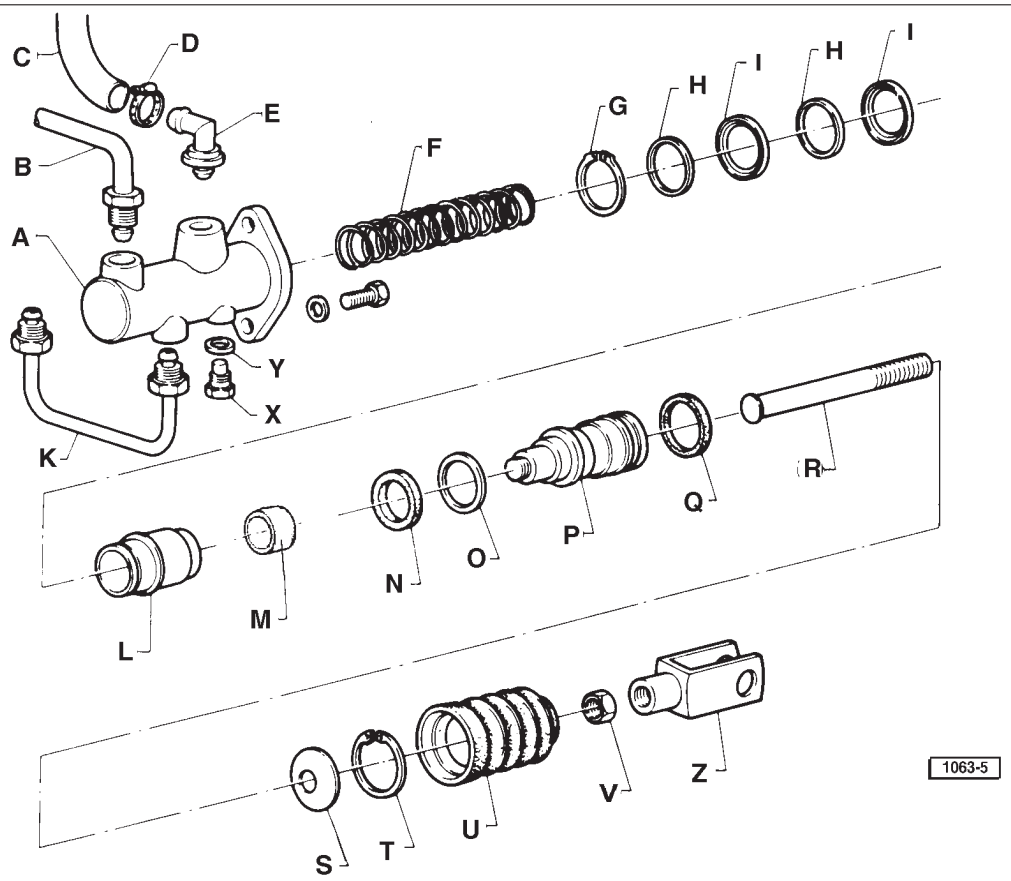
Disassebly and checking procedure

Referring to fig 4:

— Remove guard boot **A**, take snap ring **B** and withdraw rod **C** along with the support disk.

Referring to fig 6:

— Fix the pump in a vice provided with protective jaws and pushing pistons into the pump partly, as shown in figure 6, loosen the piston retaining screws and then remove the pistons together with the spring down below.



A - Pump
B - Delivery pipe union
C - Delivery pipe
D - Clamp
E - Pipe fitting
F- Spring
G - Snap ring
H - Spacer
I - Seal ring
L- Intermediate piston
M - Spring
N - Seal ring
O - Washer
P - Piston
Q - Seal ring
R - Rod
S - Support disk
T - Snap ring
U - Guard boot
V - Nut
Z - Fork
X - Stop screw
Y - Gasket
K - Pump union pipe

Fig. 5 - Brake pump assembly parts.

— Examine the cylinder interior and the pistons for either scoring or rust. Replace if necessary.

— Check cylinder and pistons for wear. If excessive plays are noticed replace either the whole piston or the whole cylinder assembly.

— Inspect sealing rings and dust guard boot, replace any worn parts.

— Inspect all pump internal compartments, apertures and passages and make sure all is properly clean and free from foreign matters.

— Ensure the springs are neither yielded nor warped. Replace if necessary.

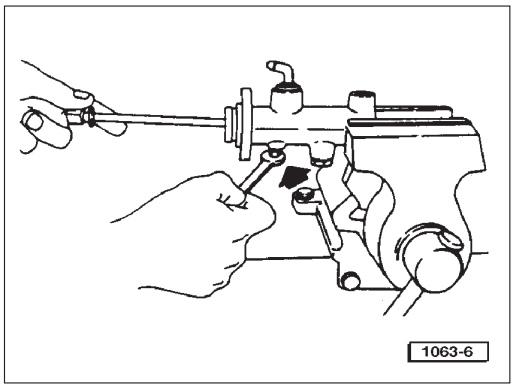


Fig. 6 - Piston retaining screw.

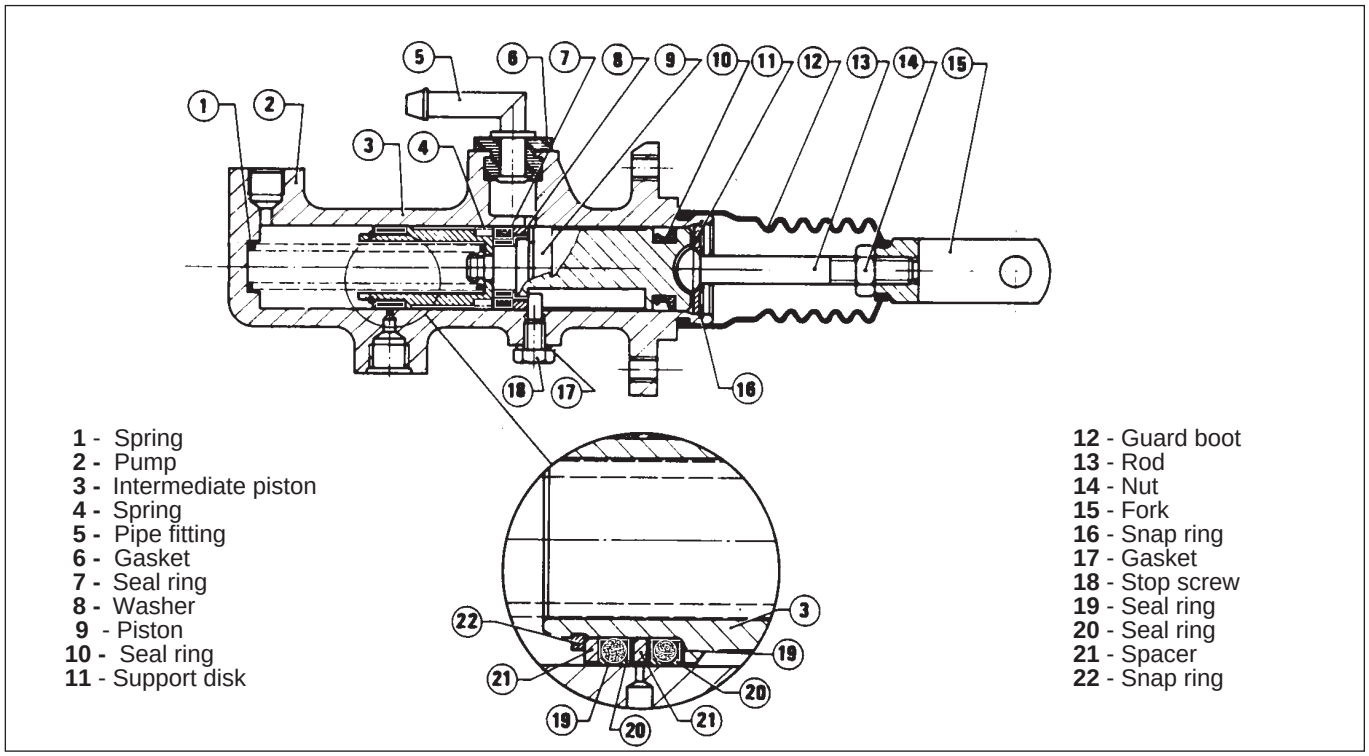


Fig. 7 - section of the brake pump.

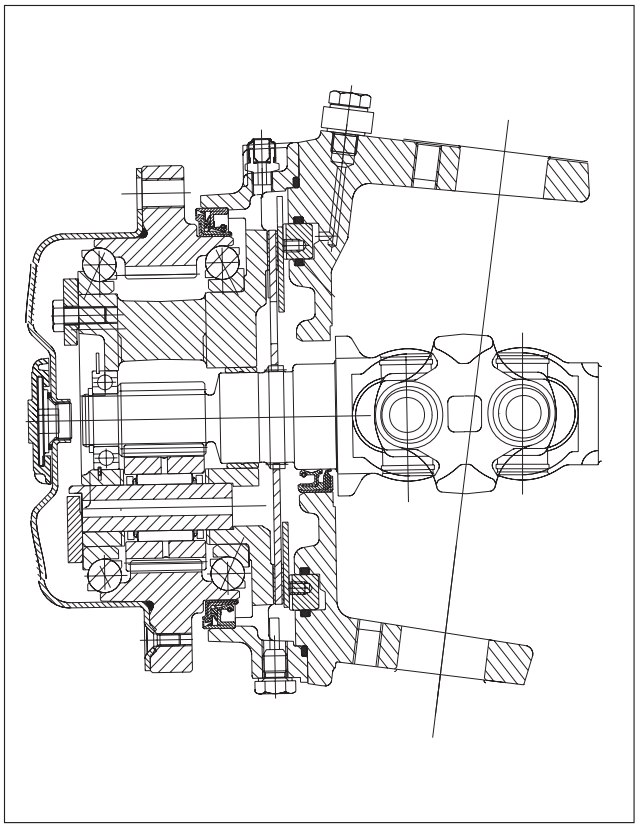


Fig. 8 - Installing the front brake disk pressure piston seal ring.

Assembly of brake master cylinder (see fig 5).

Screw the front piston **L** and the rear piston **P** together, then check that there is clearance between the two.
Insert the pistons into the cylinder, checking that the recess afforded by the outer piston **P** is aligned correctly with the seat of the limiter screw **X**.
Verify correct operation of the cylinder, making certain that the pistons complete their full travel freely.

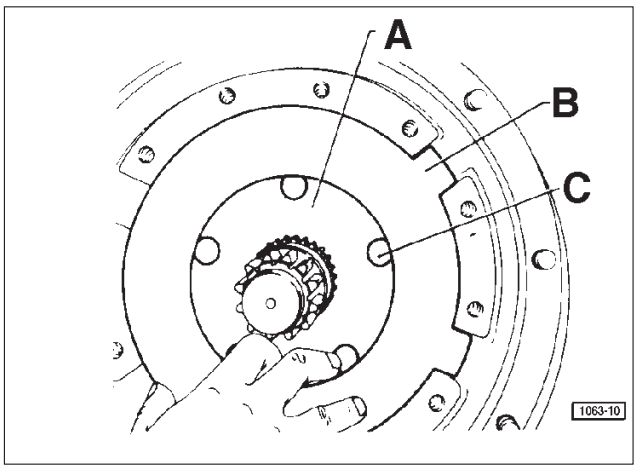


Fig. 9 - Brake assembly.
A - Brake disks
B - Intermediate disk
C - Brake control piston

Checking the front brake disks on 2WD and 4WD front axles and the rear brake disks.

Check the ground surfaces contacting the brake disks for excessive scoring.
If excess wear is found replace the worn-out parts.

Check disk conditions and tickness, compare with specifications table.

Examine the brake disk broaching for wear or damage.

If excessive oil consumption is noticed, check the piston rings as follows:

Connect no. 5.9030.520.4 hydraulic pump to the oil delivery circuit; if the hydraulic circuit is not fully oil-tight under a 1,5 bar pressure, the rings must be replaced. (To check the effectiveness of the seals more accurately, blow compressed air around the perimeter of the seal; if air bubbles appear, the seal is imperfect).

Check each brake.

For 4WD front brakes and rear brakes only
Mark piston and brake housing case with a reference near to a locating pin so as to make reassembly easier.

Coat the brake housing case seal ring contacting surfaces of piston **A** with recommended grease, see figure 11.

Carefully mount piston into the brake housing case.

Fit the brake housings and halfshaft trumpet housings, repeating the removal operations in reverse order and observing the following directions.

Coat the brake housing case inner surface as well as both epicyclic gear crown wheel surfaces with recommended sealant.

Tighten the securing nuts of the brake housings and trumpet housings to the prescribed torque (see values below).

- brake housing bolts: 3,3 ÷ 4,1 (32 ÷ 42)
- trumpet housing bolts: 8,6 ÷ 9,6 (84 ÷ 94)

Adjusting service brake pedals

Operate right-hand pump fork A (figure 13) to adjust brake pedal position, until the most suited position for the operator is attained, and in such a way that the pedal may complete its whole travel freely when braking.

Operate left-hand pump fork B (figure 13) to adjust the related brake pedal so that the coupling latch holes are in the same line.

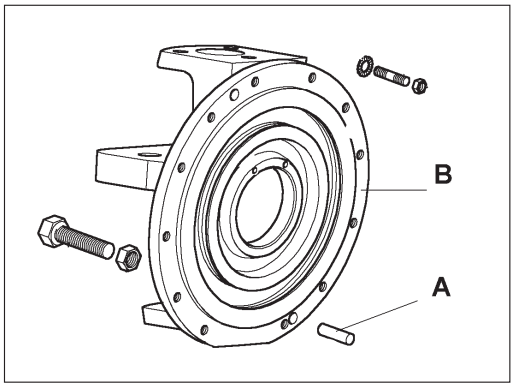


Fig. 10 - Brake housing.
A - Locating dowels.
B - Brake housing

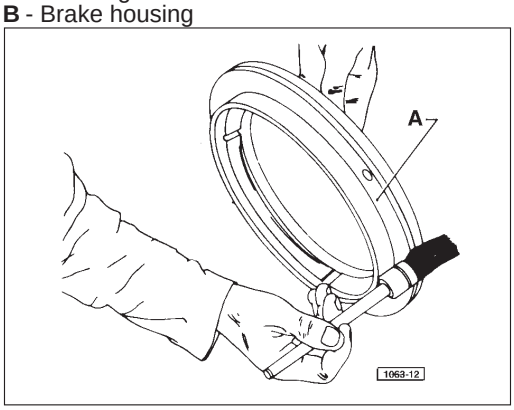


Fig. 11 - Greasing the seal contact surface of the brake piston.
A - Piston

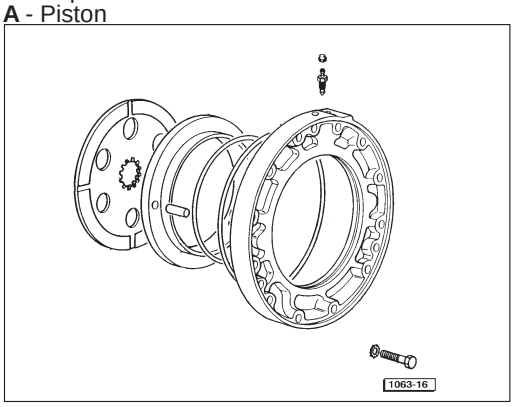


Fig. 12 - Rear brake assembly.

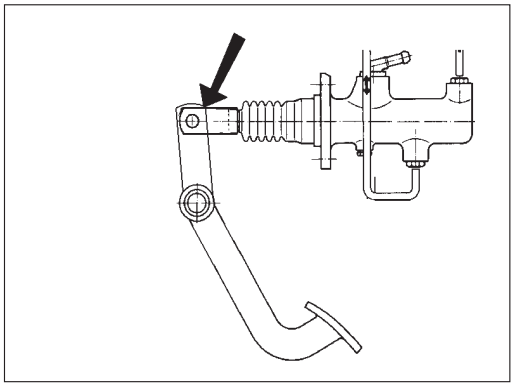


Fig. 13 - Adjusting the brake pedals.

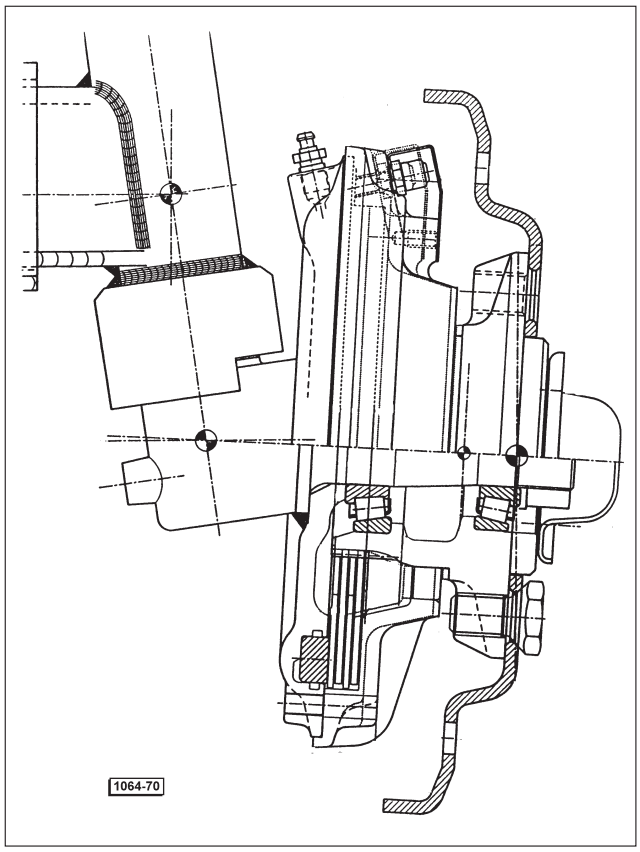


Fig. 14 - Rear brakes for 2WD front axle.

Correct installation of inspection cover for parking brake discs

Before fitting the cover, apply the sealant PIANERMETIC 510 to the surfaces in contact with the gearbox (marked **A** in the figure), then tighten the three screws marked **B** in the figure to a torque of 5.8 kgm (55 Nm).
Check that with the control lever positioned at angle of approx. 40° relative to the horizontal, the brake play is that indicated in the figure. If the play exceeds this value, replace disc **C** with the single braking surface with a discs with a double braking surface (to recover 1mm of play). Alternatively insert a shim (code 009.9648.0) between the disc **C** and the reaction plate (to recover 2mm of play).

Checking parking brake pads

Examine the brake pads for wear. If brake pad thickness is out of specifications these shall be replaced.
Brake pads can be dismounted after removing the lower brake pad lockpin, this permits access through the cover placed under the gearbox to be gained.
Before installing new brake pads, check lockpin seal ring conditions, replace if necessary.

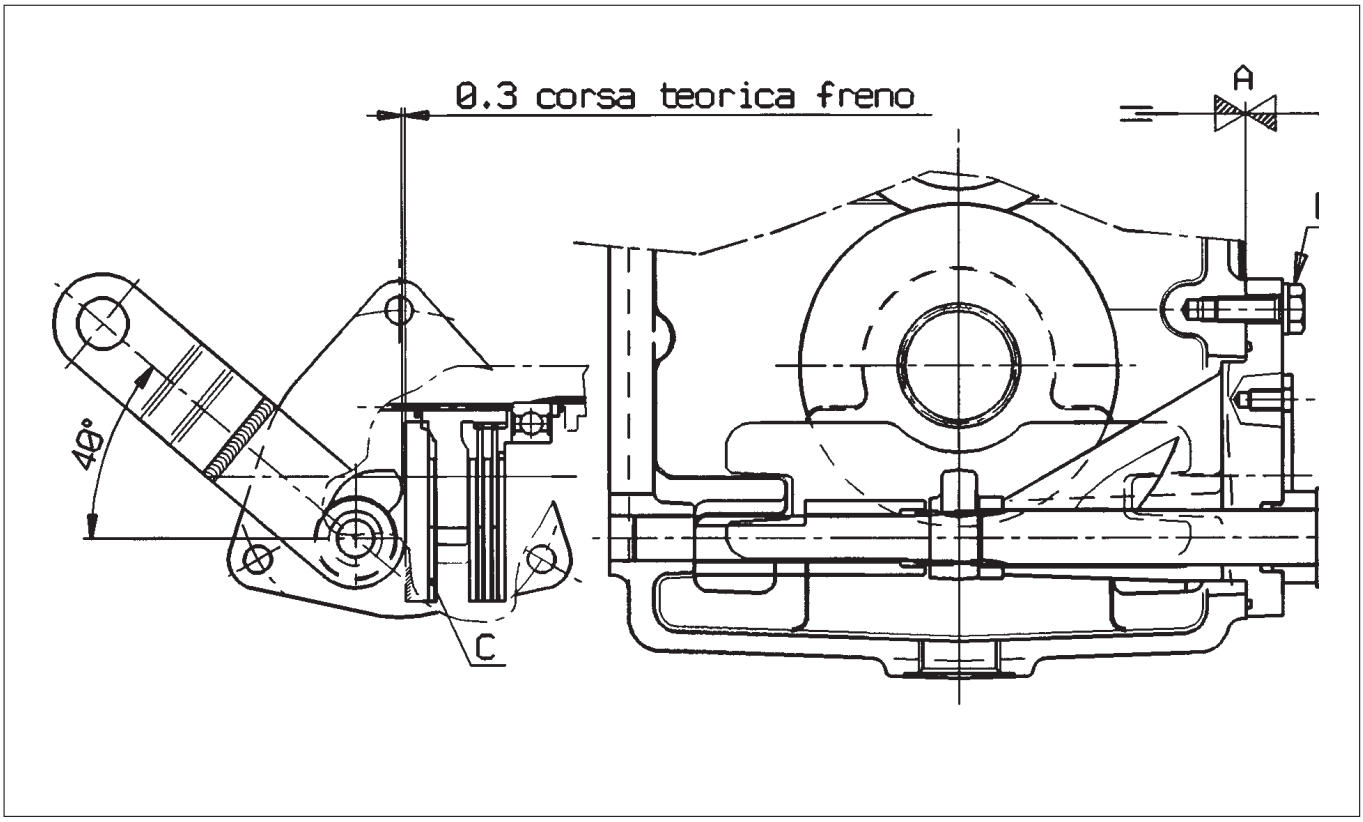


Fig. 15 - Parking brake inspection window

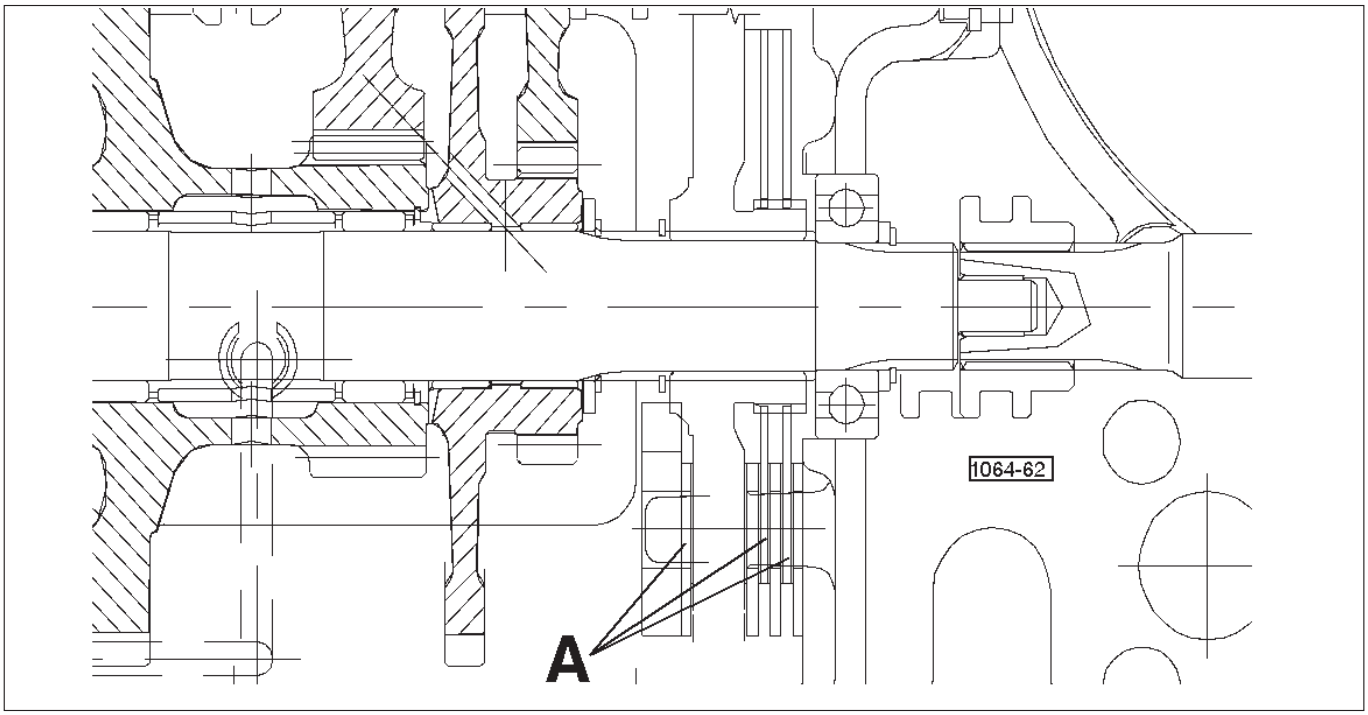


Fig. 16 - Parking brake assembly; (**A**- Brakes disk).

Bleeding air from the brake hydraulic system

Operate as follows:

— unhook pedals by removing the coupling latch: afterwards place the “SEPARATE BRAKES” valve control in OFF position;
— operate the right-hand brake pedal several times;
— by holding the pedal in fully depressed position, slightly unscrew and tighten soon after the relevant air bleeding valve on the rear right-hand brake.

This operation should be repeated until such time as the oil flows out of the bleeding valve without air bubbles.

Repeat the same procedure on the front right-hand brake by acting on the related air bleeding valve.

Follow the same procedure on the front and rear left-hand brakes making use of the related air bleeding valves.

We recommend that only oil with international specifications ATF DEXRON II I used in the braking system.

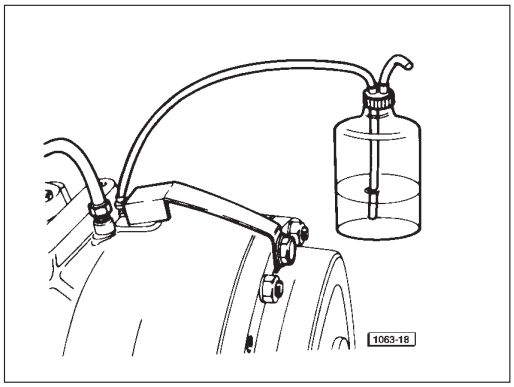


Fig. 17 - Bleeding the air from the rear service brake hydraulic system.

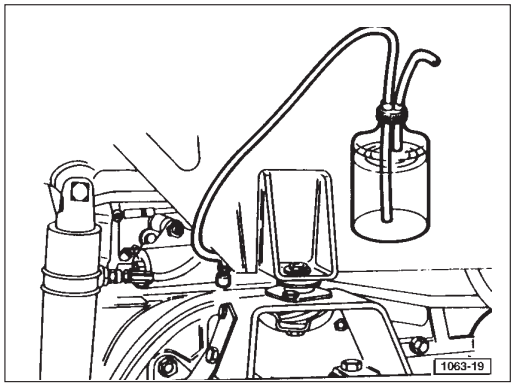


Fig. 18 - Bleeding the air from the front service brake hydraulic system.

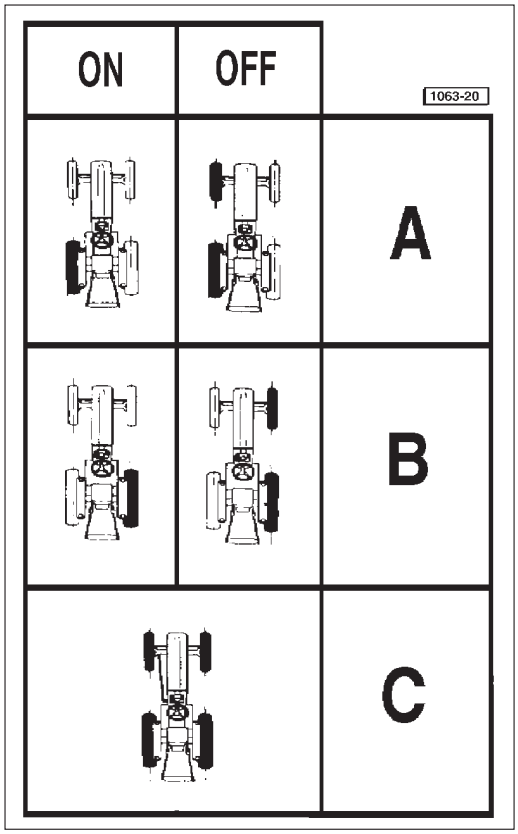


Fig. 19 - Different braking action diagram.

"Separate Brakes" valve

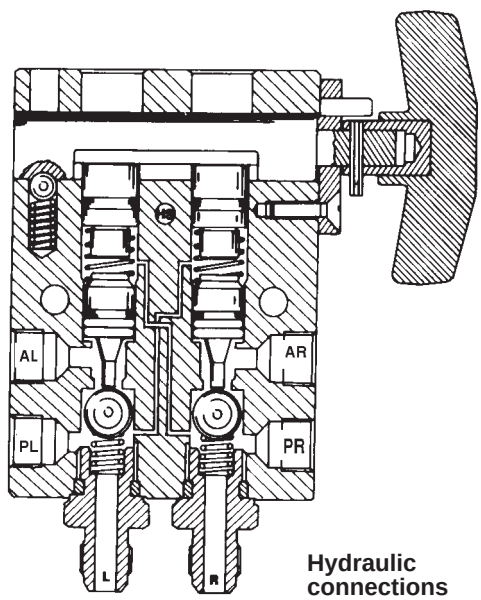
The "SEPARATE BRAKES" valve permits the front wheel side-slip to be removed when performing independent braking. (This prevents cultivation damage as the front wheel is under no braking action).

The valve operation can be stopped by operating the special control tap **A**.

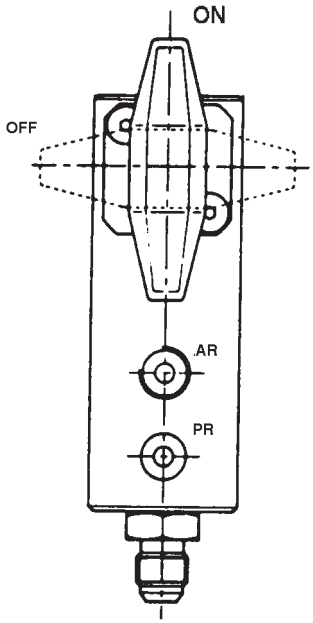
When both brake pedals are operated at the same time (coupled pedals), the braking of the four wheels will be always obtained.

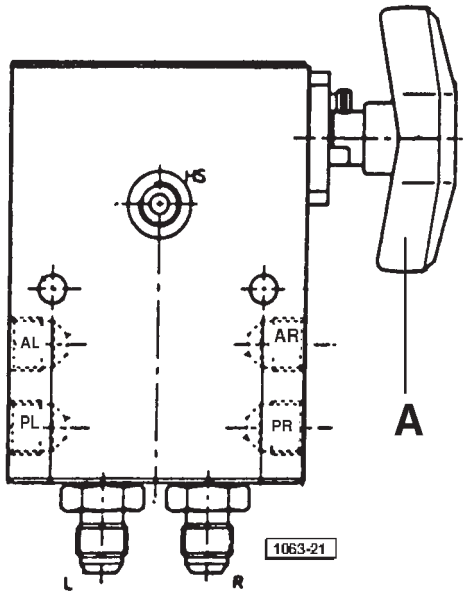
IMPORTANT: The "idrostop" sensor must be installed in the SEPARATE BRAKES valve housing, smearing Loctite 542 on the screw thread.

A = LH brake pedal
B = RH brake pedal
C = Both when coupled



Hydraulic connections





PR - Rear RH brake
L - LH brake pedal pump
AL - Front LH brake
R - RH brake pedal pump
AR - RH front brake
PL - Rear LH brake
NS - STOP pressure switch

Fig. 20 - "SEPARATE BRAKES" valve.

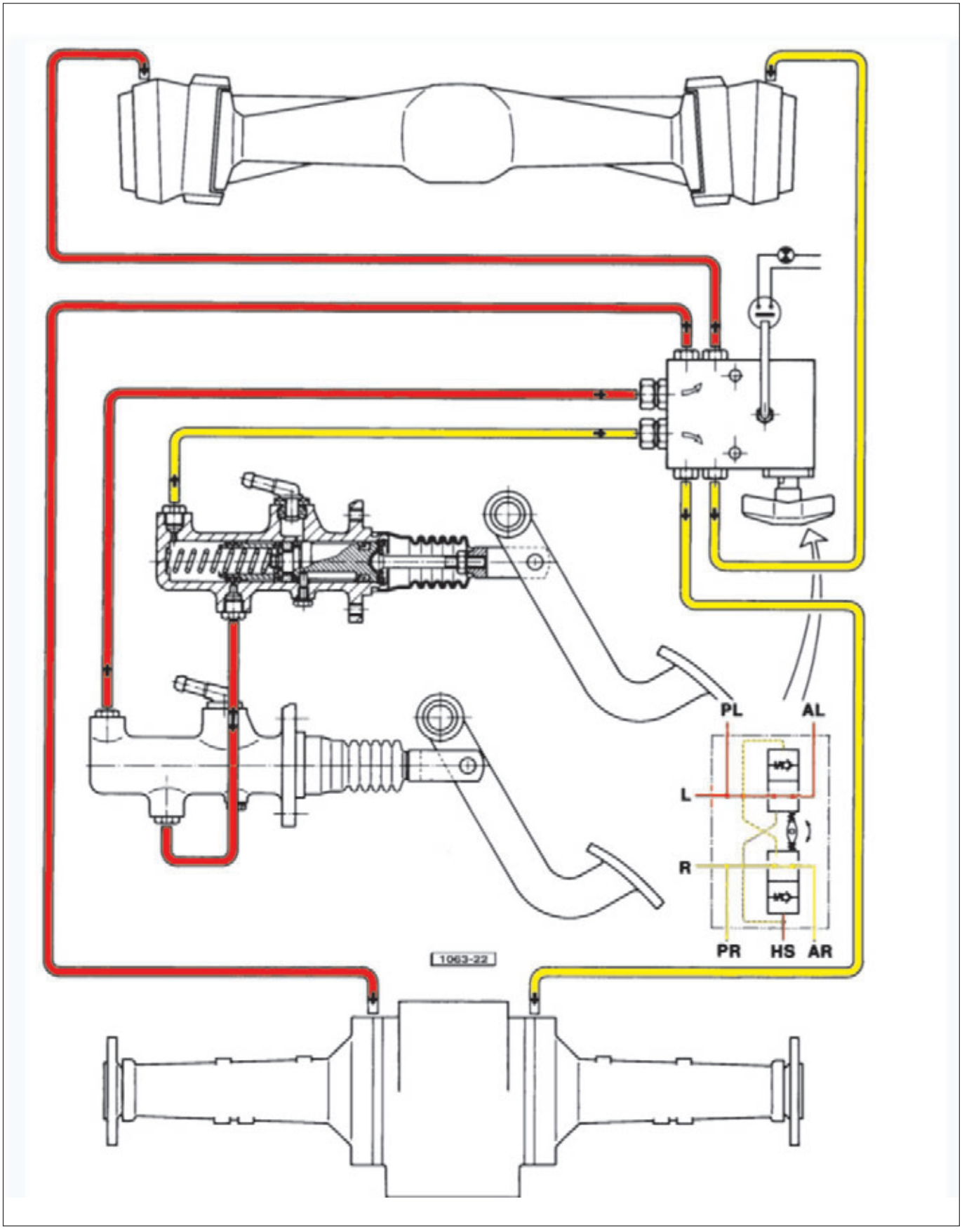


Fig. 21 - Hydraulic circuit diagram for SEPARATE BRAKES system

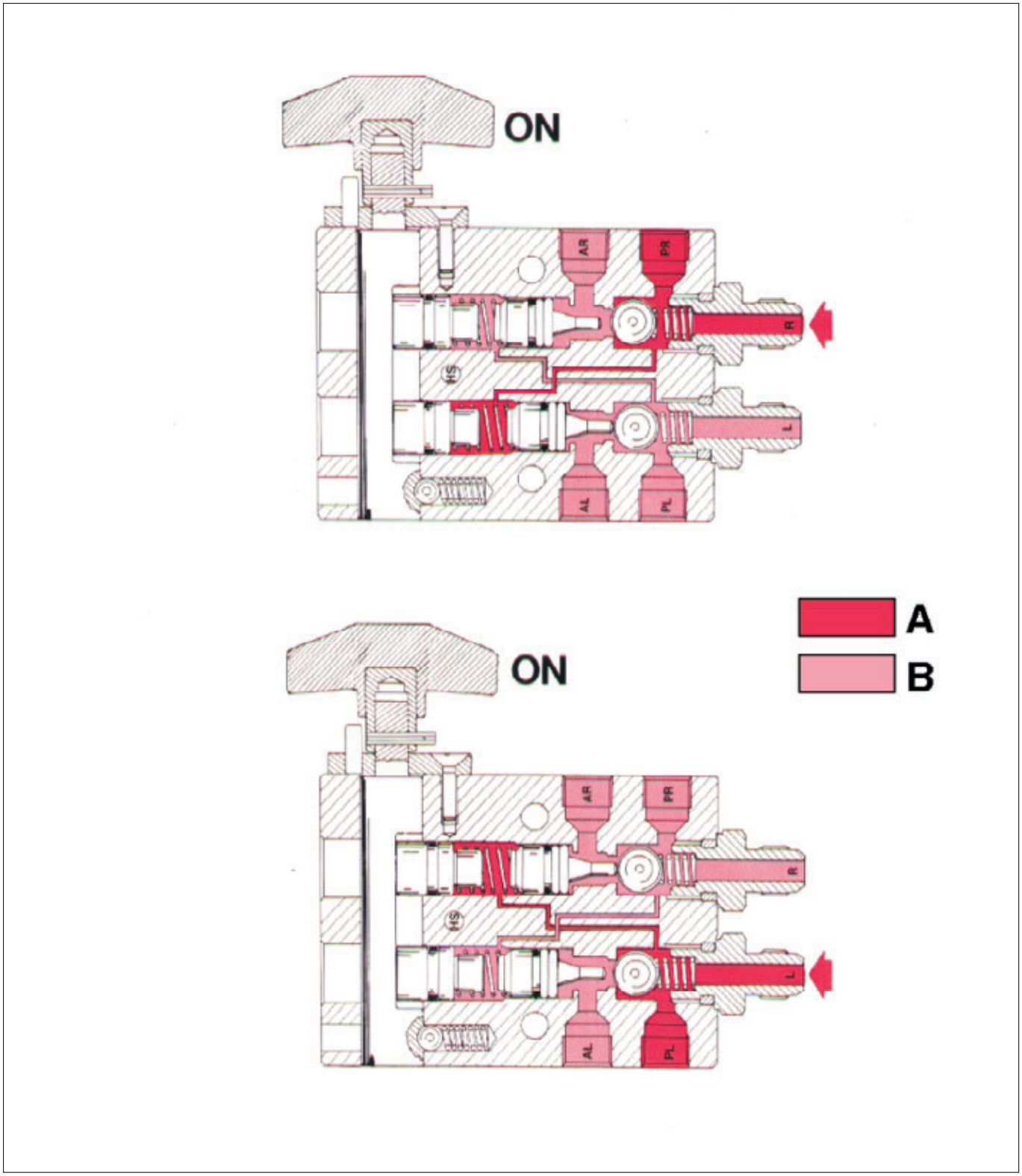


Fig. 22 - Brake hydraulic system schematic diagram with tap in position “ON”.
A - oil pressure
B - discharge oil

Hydraulic connections

PR - Rear RH brake
L - LH brake pedal pump
AL - Front LH brake
R - RH brake pedal pump
AR - RH front brake
PL - Rear LH brake
HS - STOP pressure switch

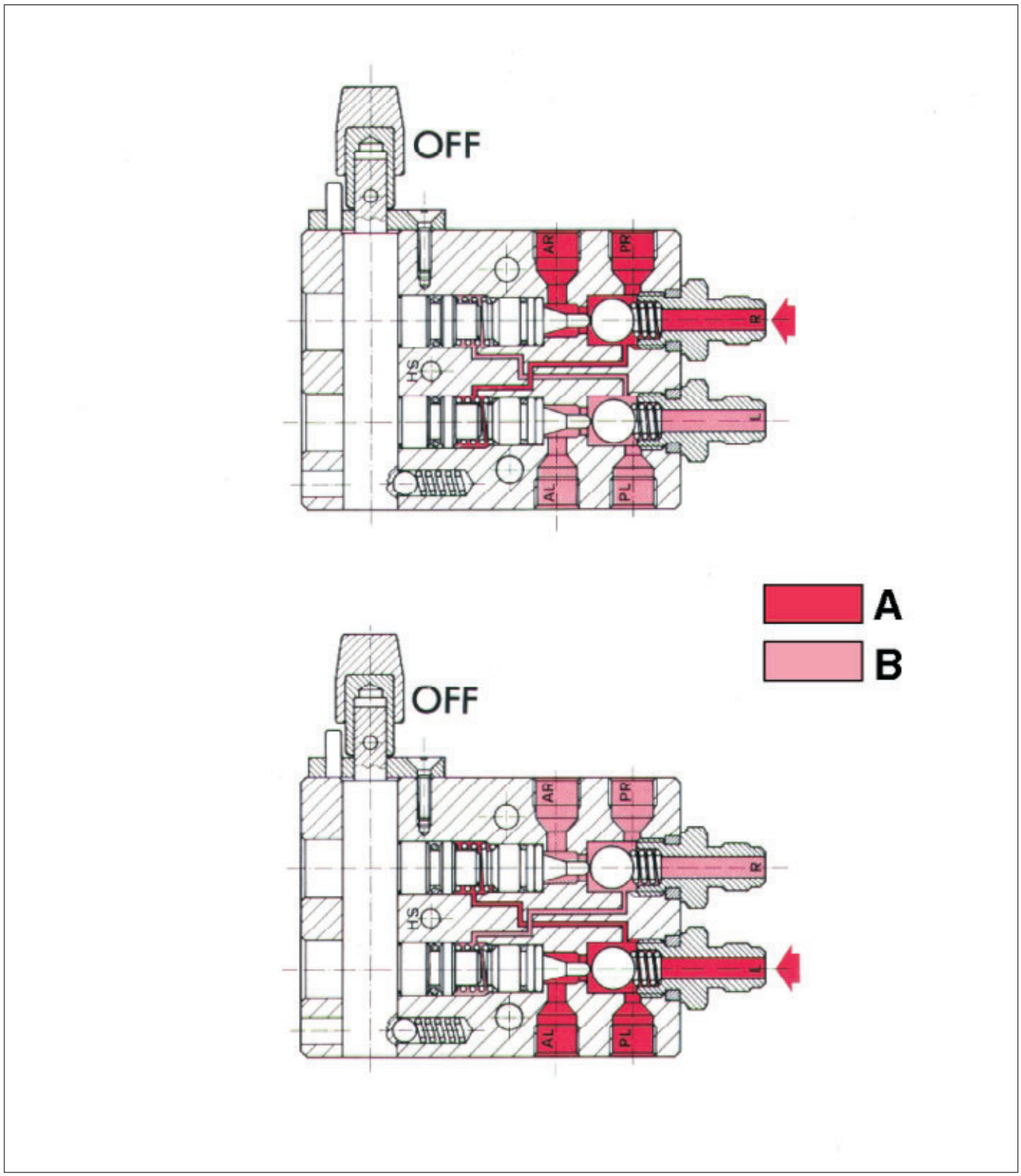


Fig. 23 - Brake hydraulic system schematic diagram with tap in position “OFF”
A - oil pressure
B - discharge oil

Hydraulic connections

PR - Rear RH brake
L - LH brake pedal pump
PL - Front LH brake
R - RH brake pedal pump
AR - RH front brake
AL - Rear LH brake
HS - STOP pressure switch

Diagnosing malfunctions

	make sure no air is inside system	bleed the air
poor braking	check for correct adjustment	adjust
	check brake disks for wear	replace
uneven braking	make sure the recommended oil type is used	bleed the air
		replace
	check for correct adjustment	adjust
noisy braking	make sure the recommended oil type is used	replace
	check brakes disk for wear	replace

Hydraulic lift with load sensing

The automatic draft control device, connected directly to the top link, guarantees rapid lift operation and is highly sensitive to even minimal variations in the load on the implement.

The hydraulic lift is equipped with an internal cylinder positioned horizontally.
The hydraulic lift is equipped with an open-centre control valve.

The oil in the suction line is drawn from the gearbox by a high-capacity pump (the same pump supplies the auxiliary control valves) ‘and filtered through a filter with an interchangeable.
The oil under passes through the auxiliary control valves before arriving at the lift.

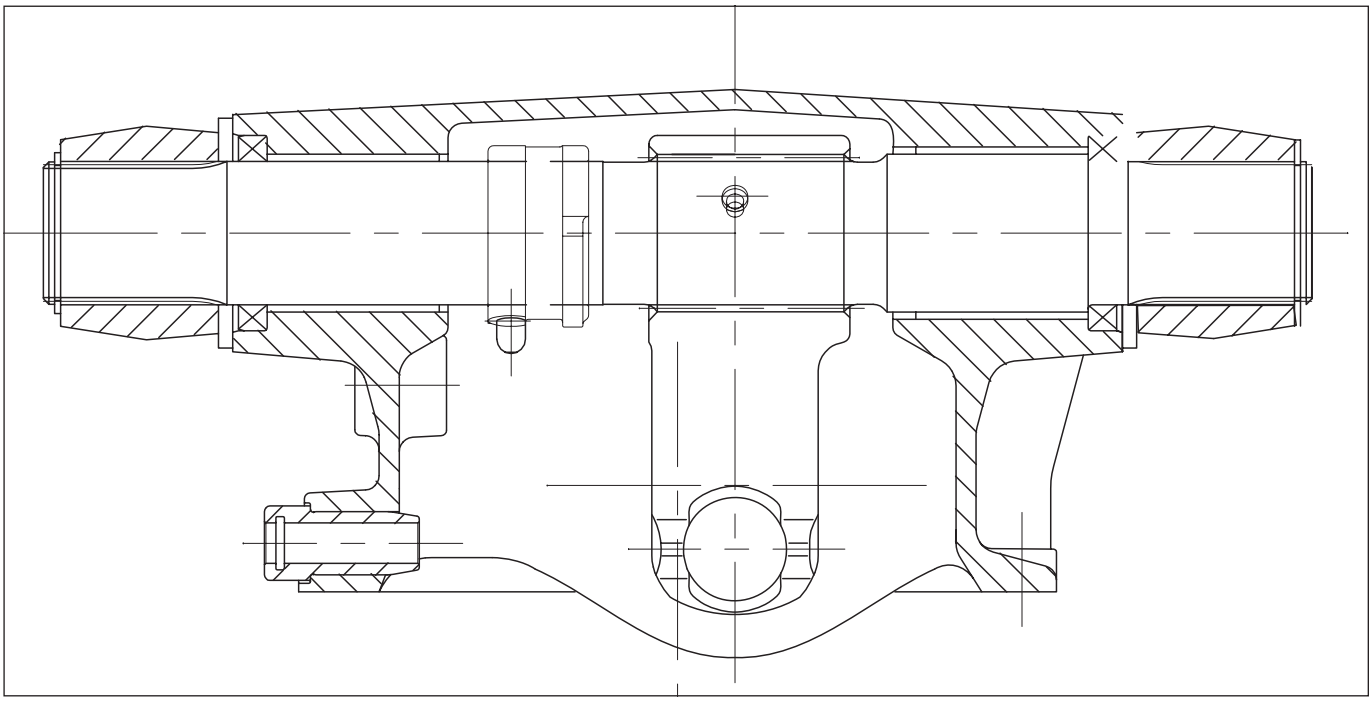


Fig. 1 - Lift lever shaft

Technical characteristics

manufacturer		SAME DEUTZ-FAHR GROUP
lift type		with open-centre control valve
safety valve setting	bar	$190\ ^{+10}_{0}$
minimum operating pressure	bar	8
minimum permissible piston diameter	mm	89,900
maximum permissible cylinder diameter	mm	90,050
lifting capacity with load concentrated on lower link ball ends at height of rear wheel centres	kg	1600
tightening torque for cylinder retaining bolts	kgm Nm	15 -16,5 150 - 165

Installing the lift and front cover plate of the gearbox

Install the two lift mounting studs (**A** in figure) and the two cover plate retaining screws (**B** in figure), applying Loctite 242 to the threads in both cases.

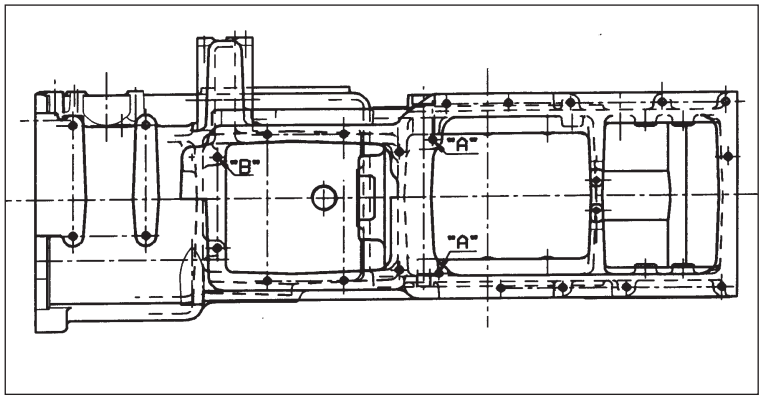


Fig. 2 - Fixing the front lift and the front cover to the gearbox

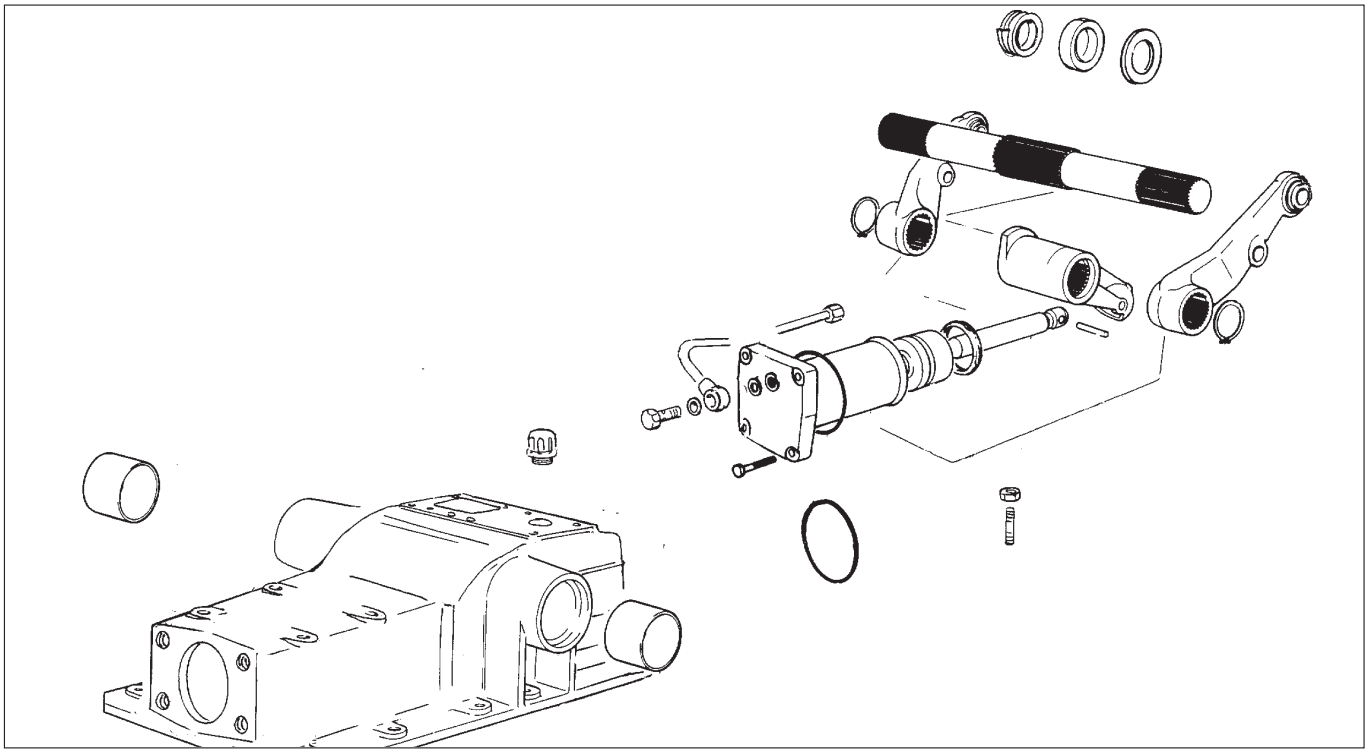


Fig. 3 - Lift unit components.

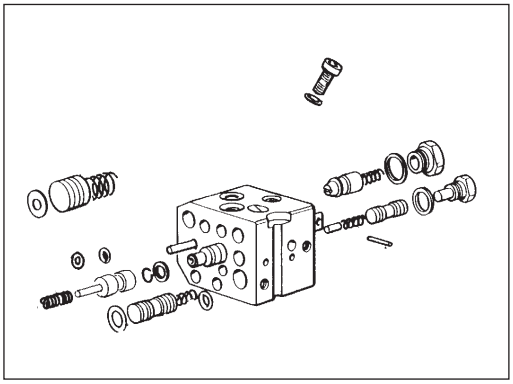


Fig. 4 - Hydraulic control valve.

Lift mechanism
Inspect the splines for grooves or excessive wear; replace the levers if necessary.
Inspect the splines of the lift shaft for excessive wear and check that the machined surfaces in contact with the supports are not damaged; carefully clean the lubrication slots for the shaft supports.
At each dismantling, remove the seals and gaskets. Check that there are no oil leaks from the lift cylinder.

Checking the safety valves
Check that the safety valve incorporated in the auxiliary control valves is set to 190 bar.
Remember that to ensure the correct operation of the lift, the operating pressure in the lift circuit must not exceed 150 bar.

Checking the protrusion of the non-return valve control pin B

Move the ball **A** so that rests on the valve body **D**, and move the valve body **D** so that it rests on the control valve housing, then insert pin **B** up against ball **A** (while making sure that the ball **A** does not move).

Check that the pin **B** protrudes 15.8 ÷ 0.05 mm as indicated in the figure.

Check that the moving parts slide freely

For the control vale to function properly, the following parts must slide freely:
G - control valve spool
K - pilot valve
L - enabling valve

Checks to be carried out after re-assembly of the control valve.

— Check that the control valve spool slides freely.
After assembly of the control vale housing with the connection housing, check that the spool **G** slides freely in its bore.
— Check coverage of lifting and lowering ports.
Push the spool **G** in the direction of arrow **E** until the point where lifting commences. From this point, move the spool back in the direction of arrow **F** until lowering commences.
Check that the spool travel between the lifting and lowering positions is 0.6 to 0.8 mm.

— Check for leaks
Check that no part of the control valve leaks (plugs, check valve, etc.) at a pressure of 150 bar.

Apply Loctite 542 to all the control valve plugs before fitting.
The spring calibration values must match those specified in the tables.
For bench testing of the control valve use oil of the recommended type and a pump with a capacity of 10 to 15 l/min.
Warning: *the presence of emulsified oil may prevent correct operation of the control valve.*

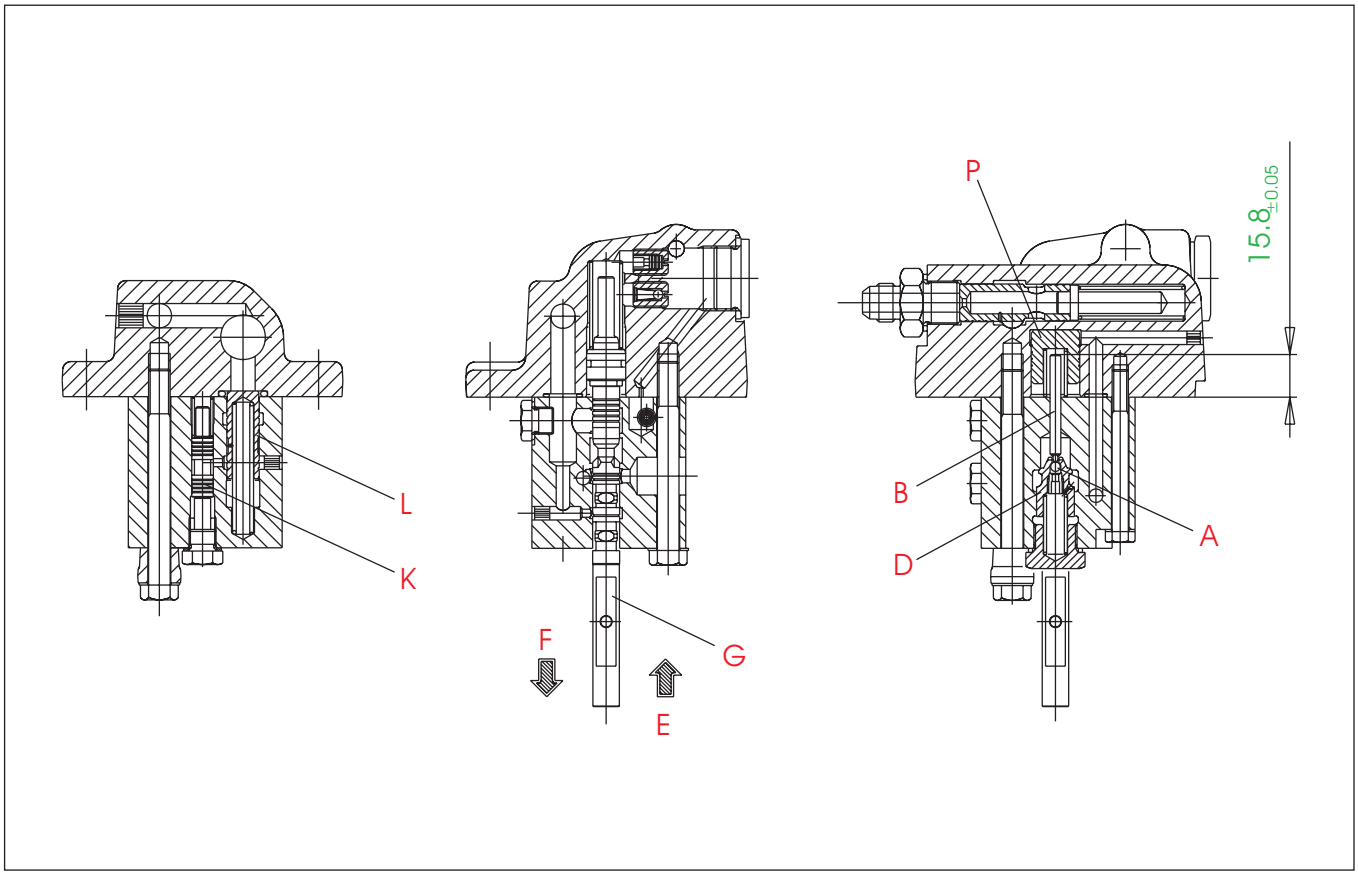


Fig. 5 - Hydraulic control valve.

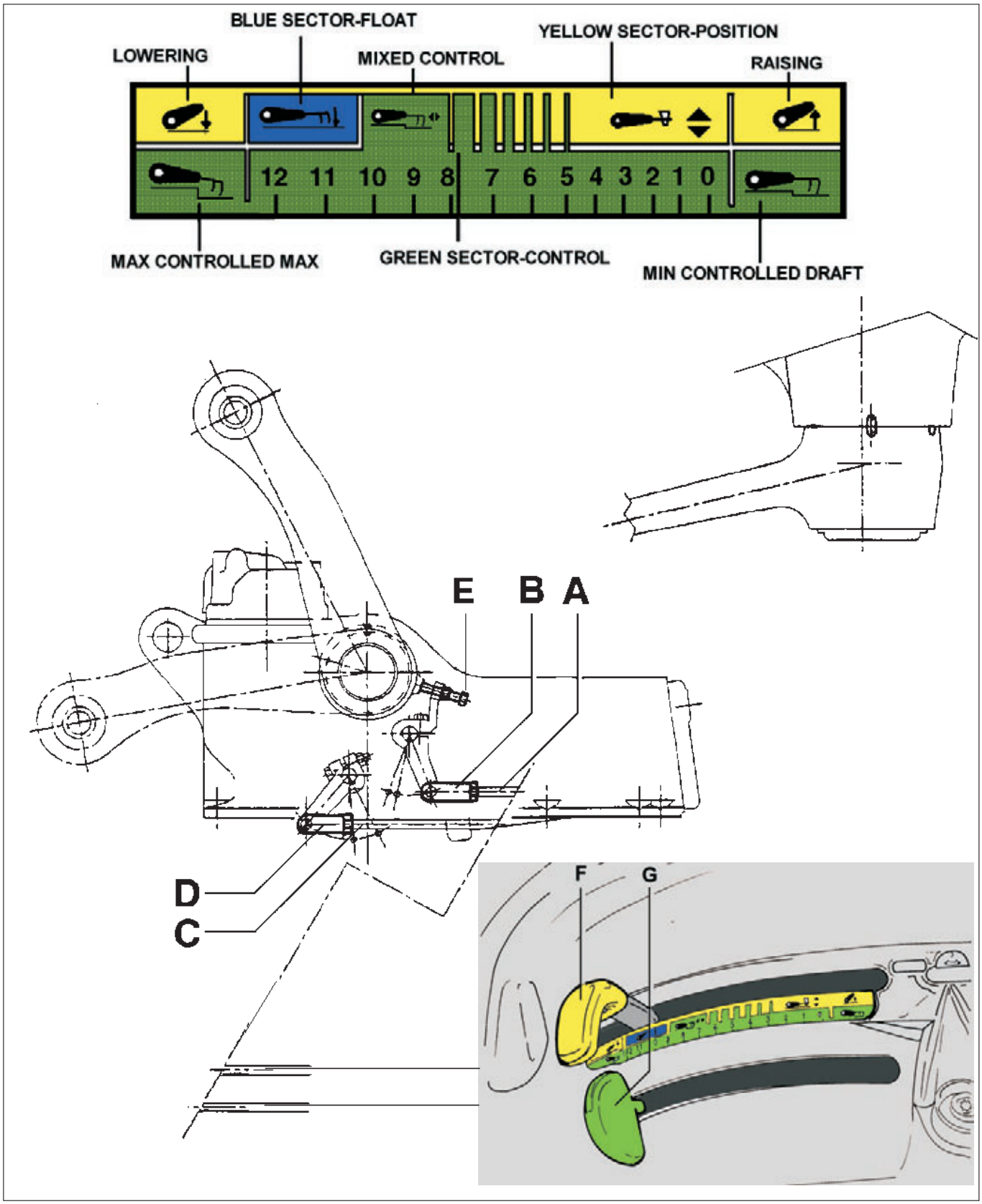


Fig. 6 - Lift control adjustment diagram.
F - Position control lever (yellow knob)
G - Draft control lever (green knob)

Adjusting the lift (Fig. 6)

Warning: before adjustment, attach a weight of approx. 200 kg to the 3-point hitch.

Adjust the forks of the external tie-rods of the lift control linkage so that the yellow lever and the green lever are free to move the entire length of the control slot.

Adjusting the lift travel - YELLOW LEVER -

Start the engine, and move the yellow lever to the maximum lift position and move the green lever to position 12 in its control slot.
Adjust screw **E** (fig. 6), so that it is up against its stop when the reference marks on the right-hand side of the lift housing and the lift arm coincide.
If necessary, also adjust fork **B** of tie-rod **A**.

Adjusting the GREEN LEVER

Move the yellow lever to the position **CONTR** of its control slot, and check that when the green lever is moved from position **12** towards **0**, the lifting action commences when the lever is in correspondence with position **4**.
If not, adjust fork **D** of tie-rod **C**.

Checking operation of the safety stop.

Move the green lever to the maximum lift position (position **0**). The lift arms will be raised, rotating through a further angle of approx. 4° (corresponding to a movement of the notch on the lift arm of 6 mm).

Checking balance.

With the yellow lever in the **CONTR** position, move the green lever from position **12** to **4** to find the point of balance (before the lifting action begins); check that when the counterweight applied to the lift arms is pulled upwards, the lift raises the arms, and that when the counterweight is pushed downwards, the lift lowers the arms.

Adjusting the lift control levers

Screw the self-locking screws **A - B** fully home, unscrew screws **A** one and a half turns and screws **B** a half turn.

Operate the levers to check the effort required.

If necessary, adjust the screws until the levers function correctly.

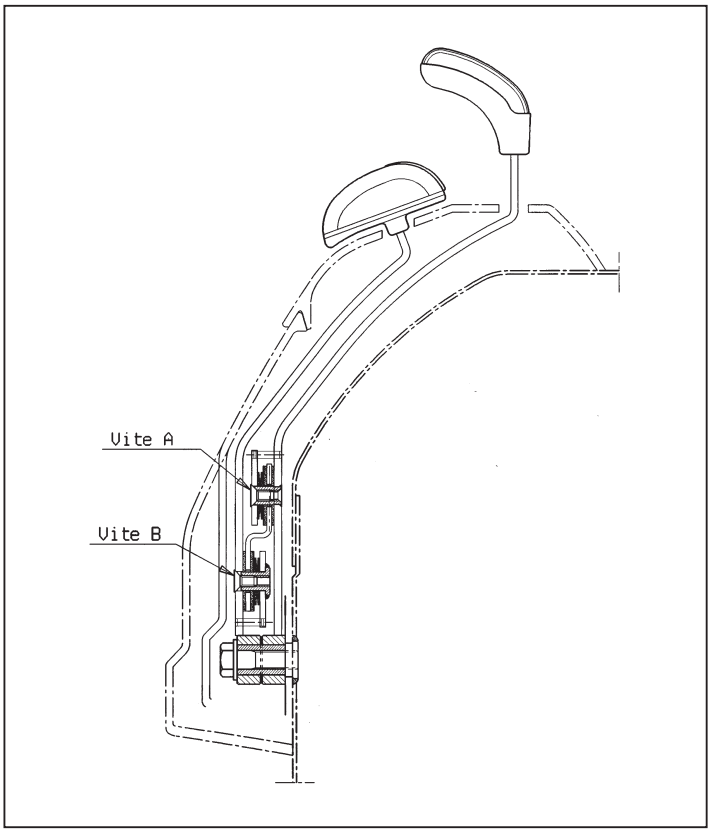
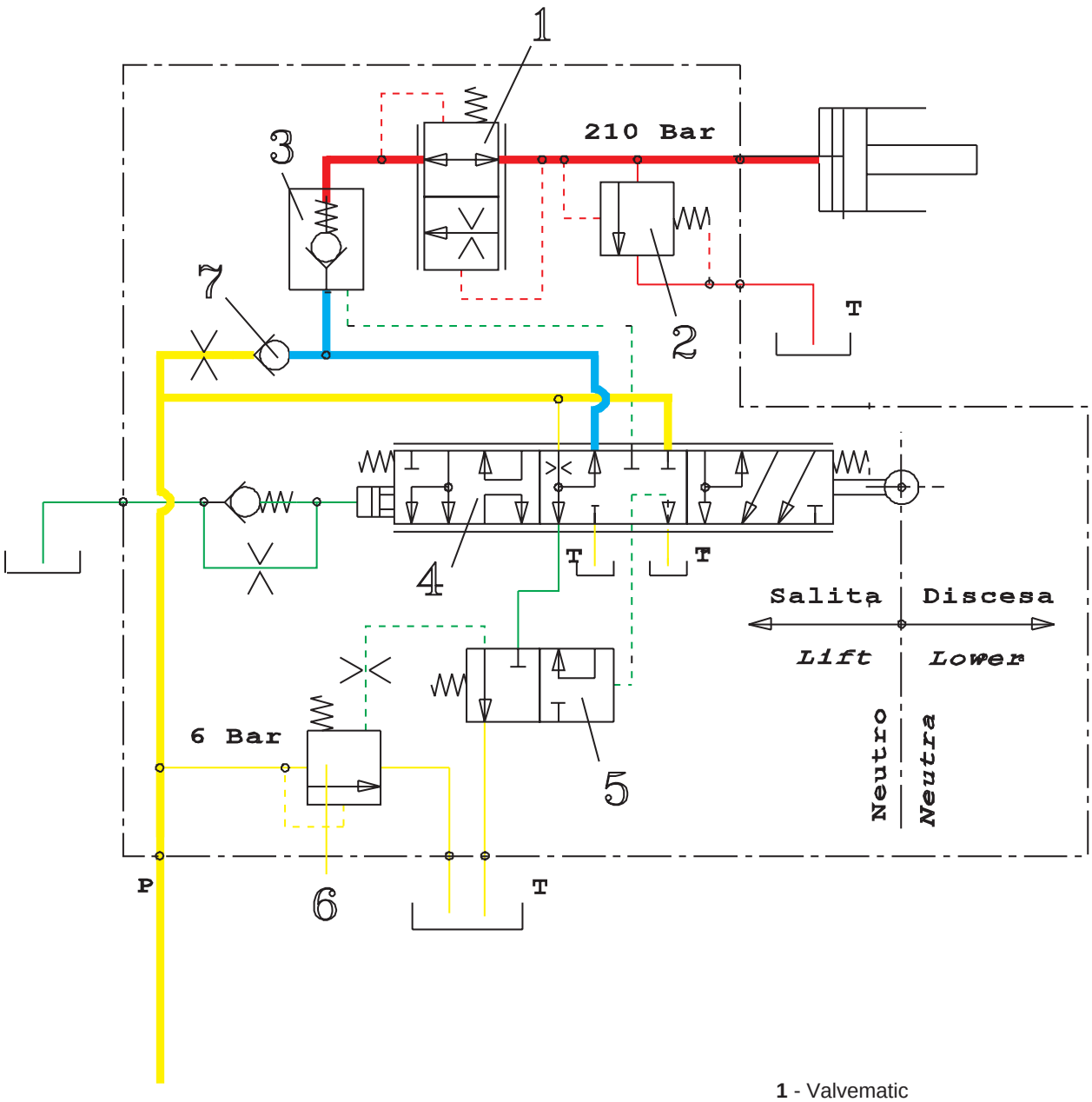


Fig. 7 - Adjusting the lift control levers

Yellow = Oil delivery to the hydraulic control valve
Blue = Oil flow from the control valve to the lowering valve
Green = Oil for piloting the valves
Red = Pressurised oil to the lift piston



1 - Valvematic
2 - Anti-shock valve
3 - Down valve
4 - Control valve
5 - Pilot valve
6 - Enabling valve
7 - Inlet valve
P - Pump
T - Return

Fig. 8 - Lift hydraulic circuit.

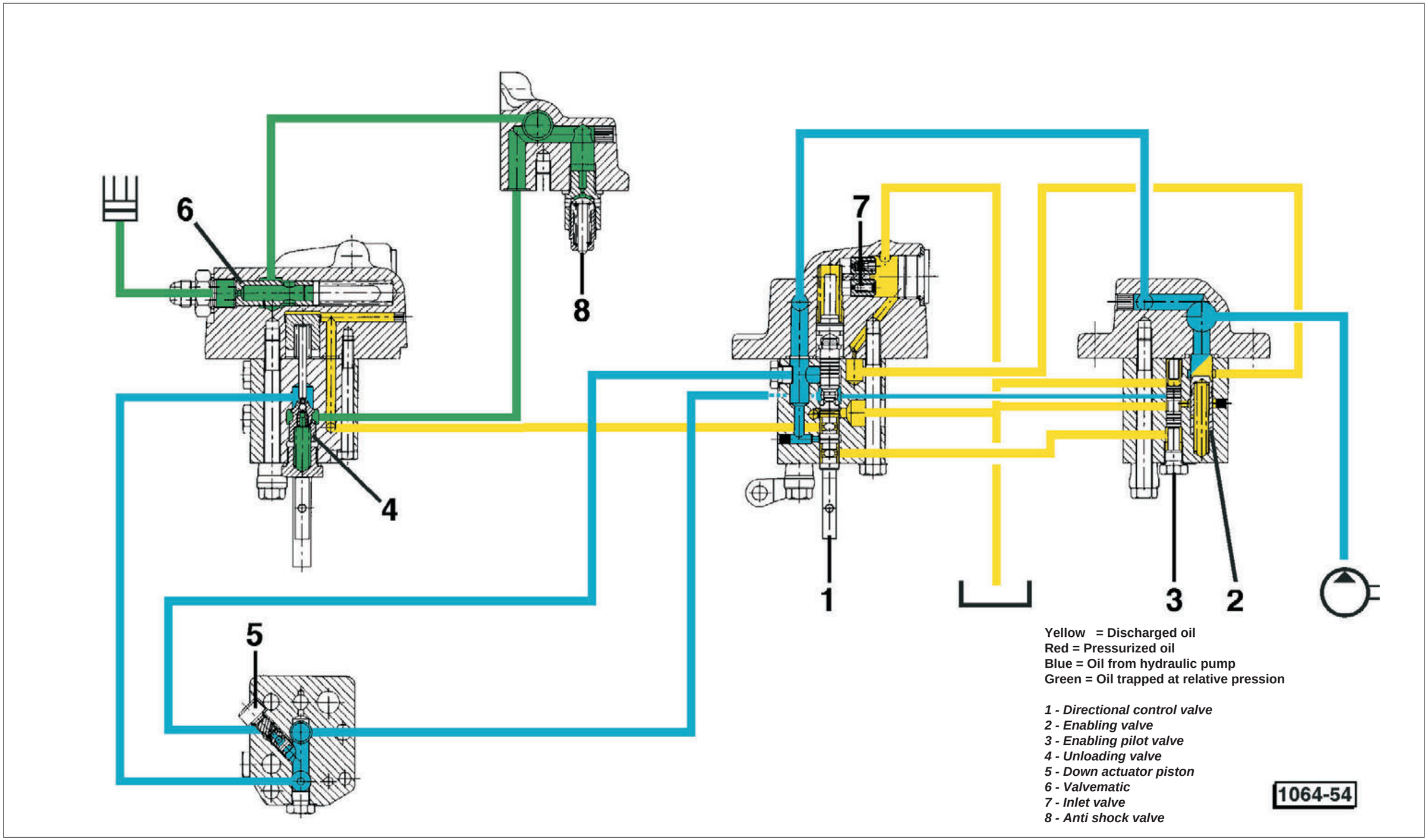


Fig. 9 - Hydraulic lift circuit - **NEUTRAL**

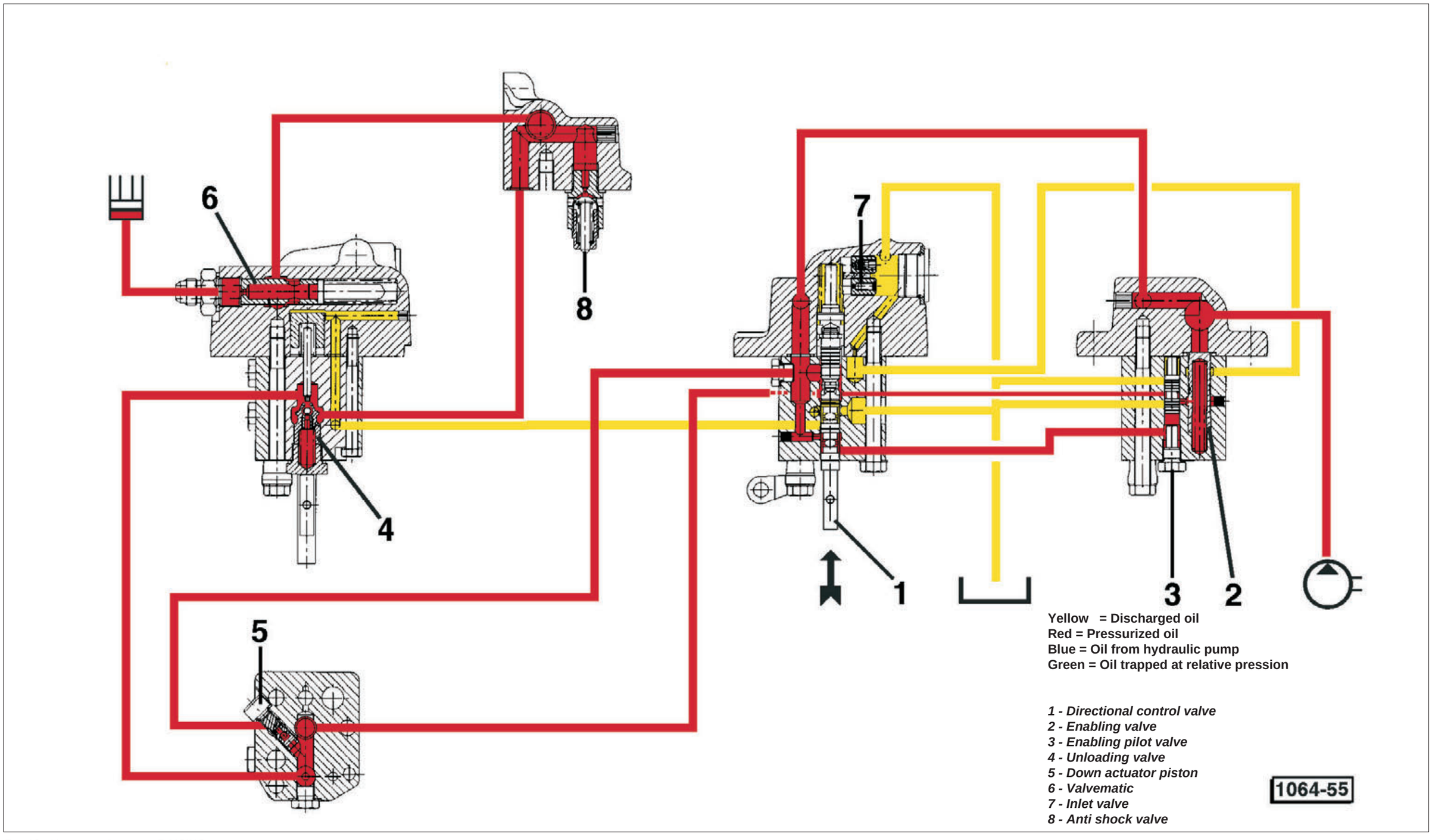


Fig. 10 - Hydraulic lift circuit - **RAISING**

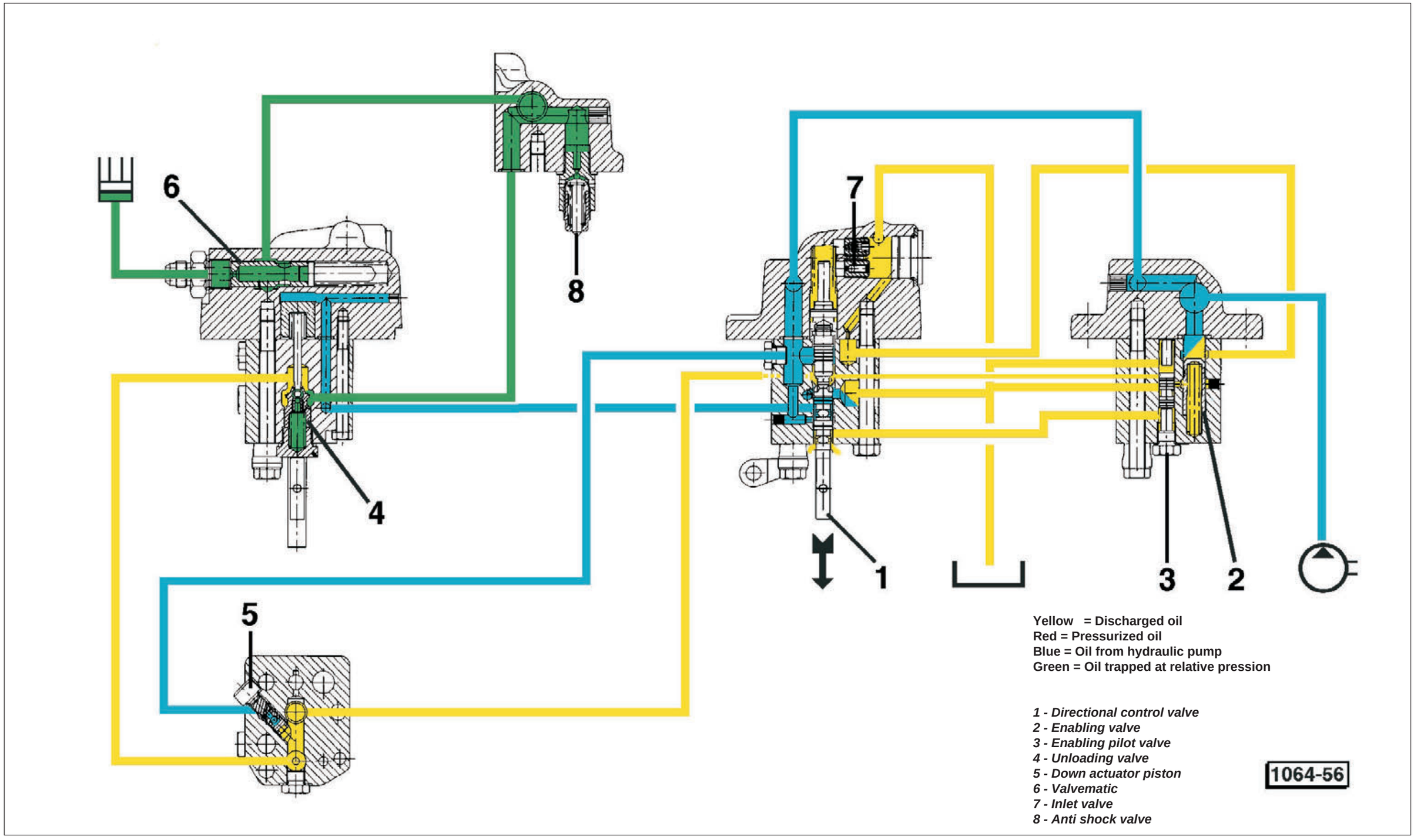


Fig. 11 - Hydraulic lift circuit - **LOWERING**

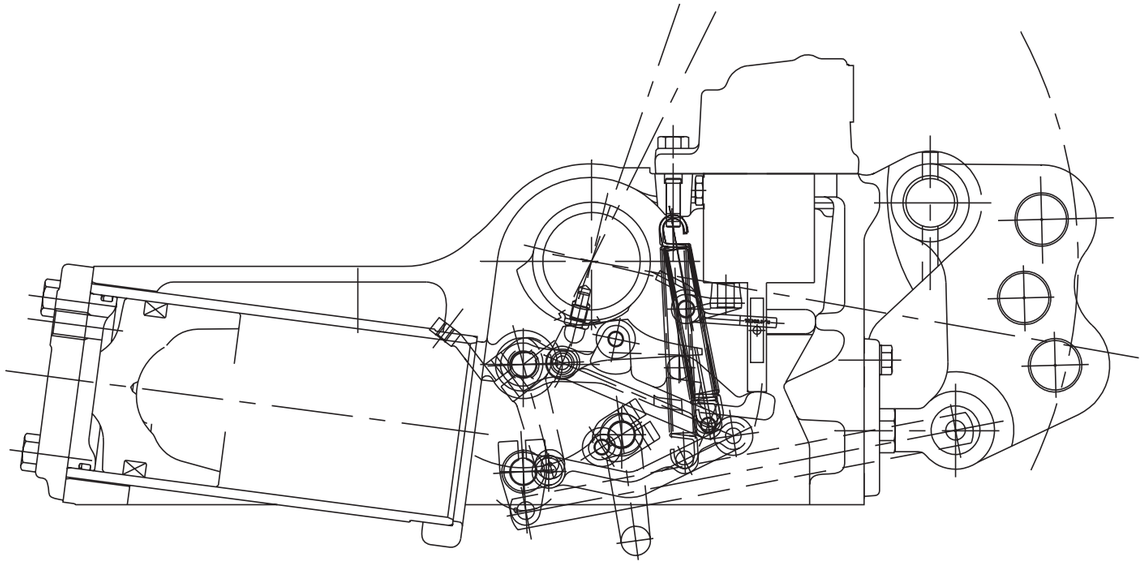


Fig. 12 - Control linkage for draft control.

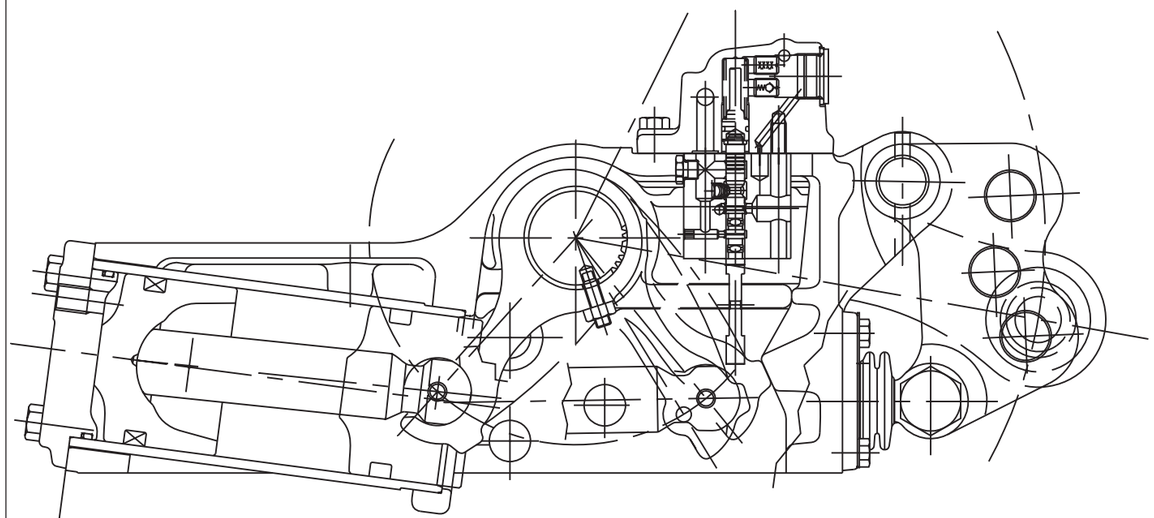


Fig. 13 - Lift hydraulic control valve.

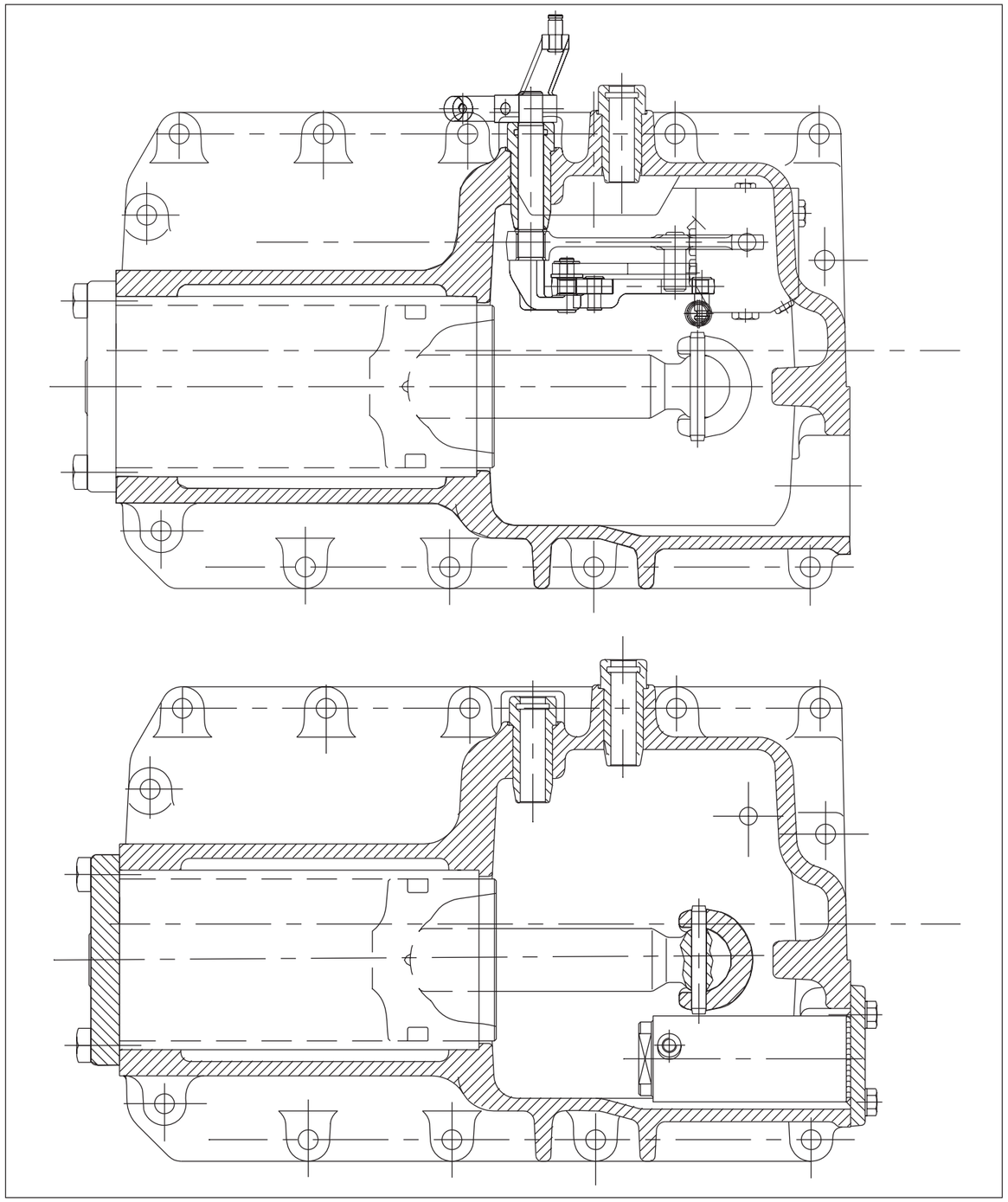


Fig. 14 - Section views of the lift

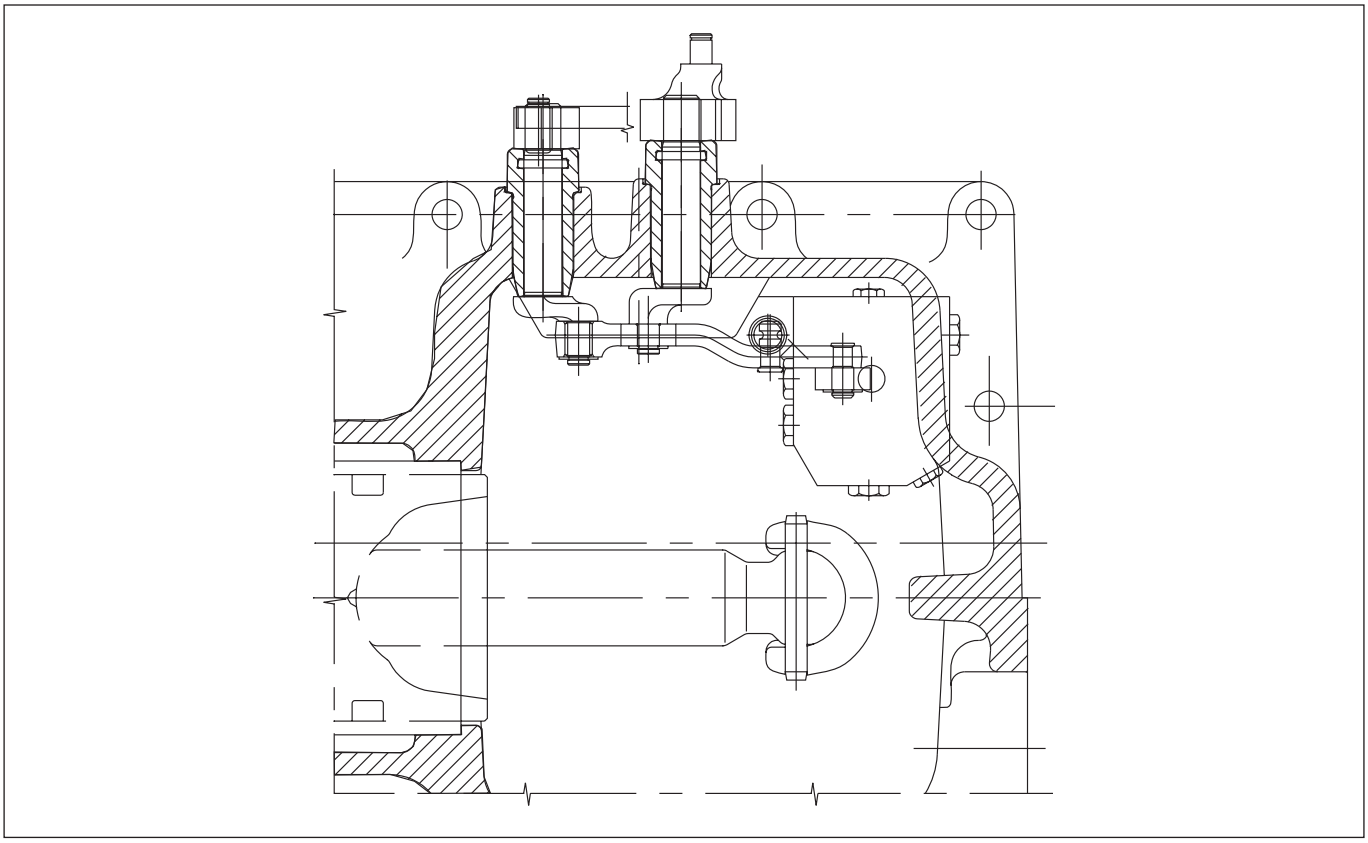


Fig. 15 - Pins of the draft control and position control levers

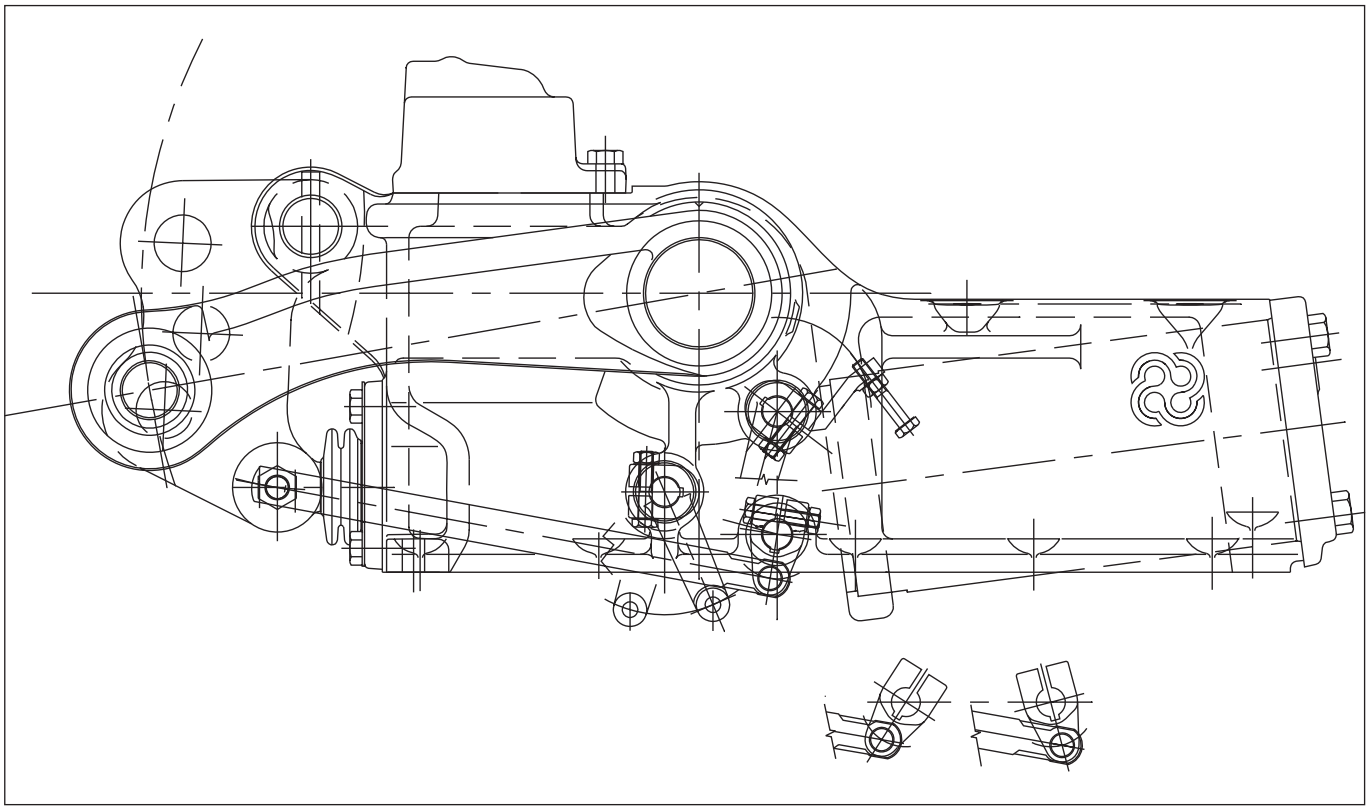


Fig. 16 - Control linkage for draft control and stop screw for the position control lever

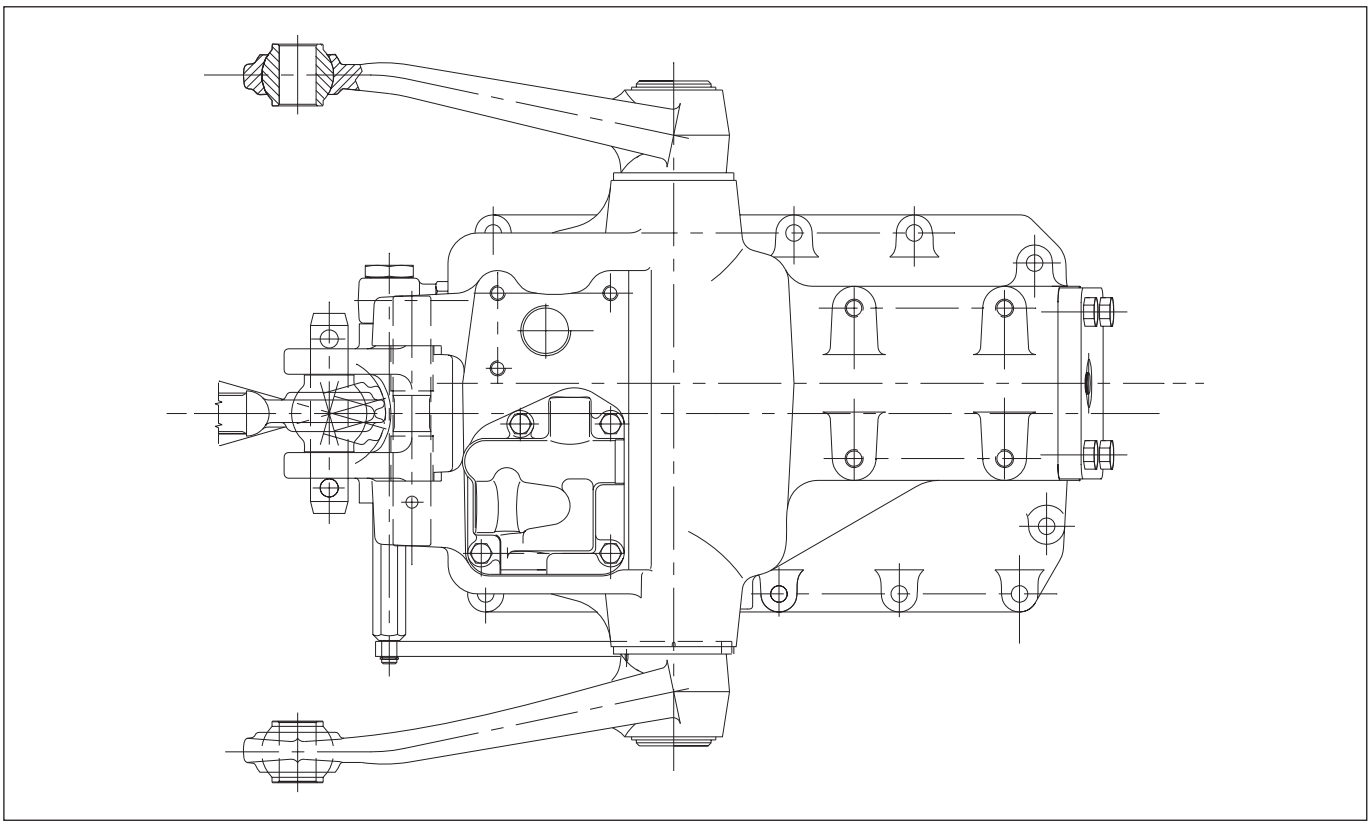


Fig. 17 - Top view of the lift.

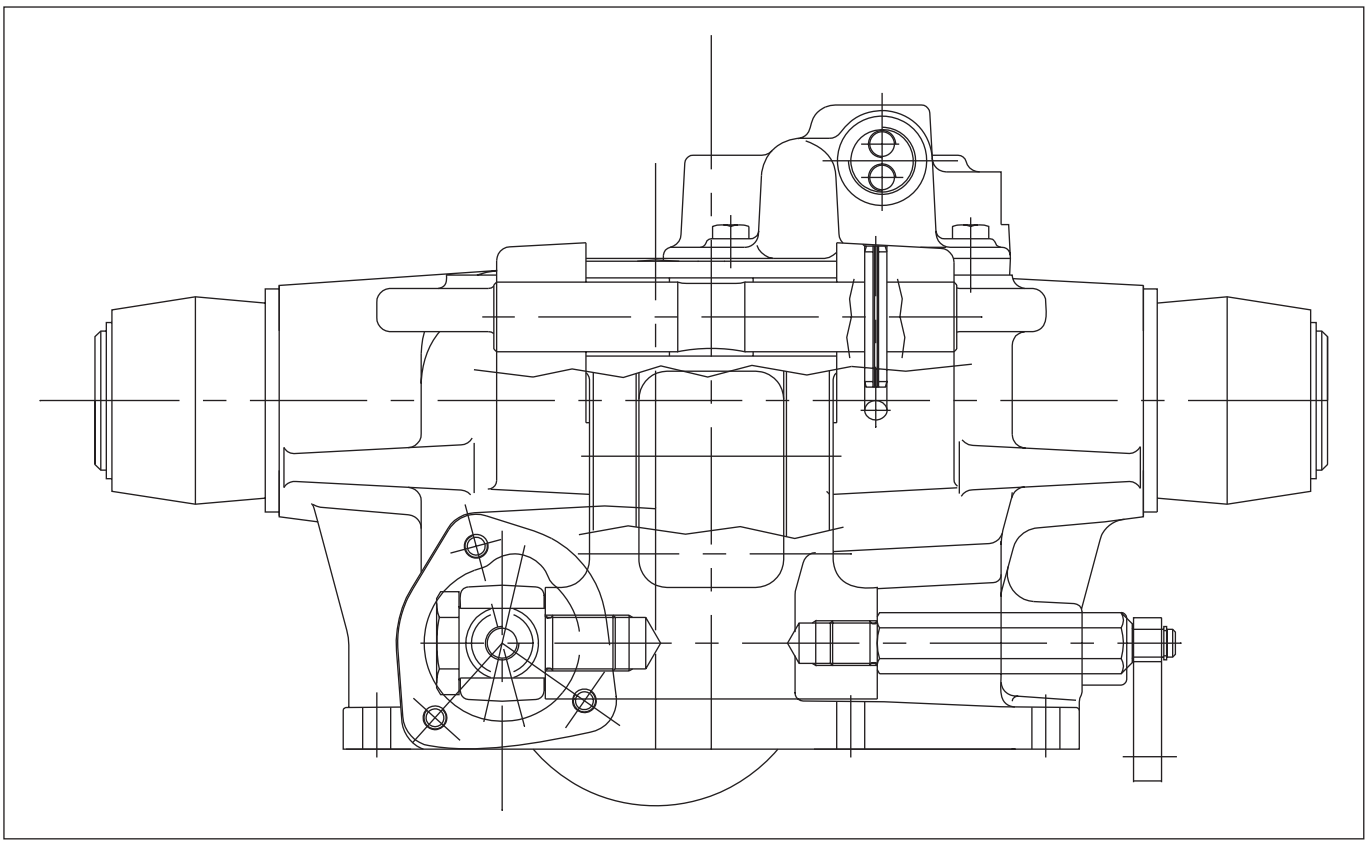


Fig. 18 - Rear view of the lift.

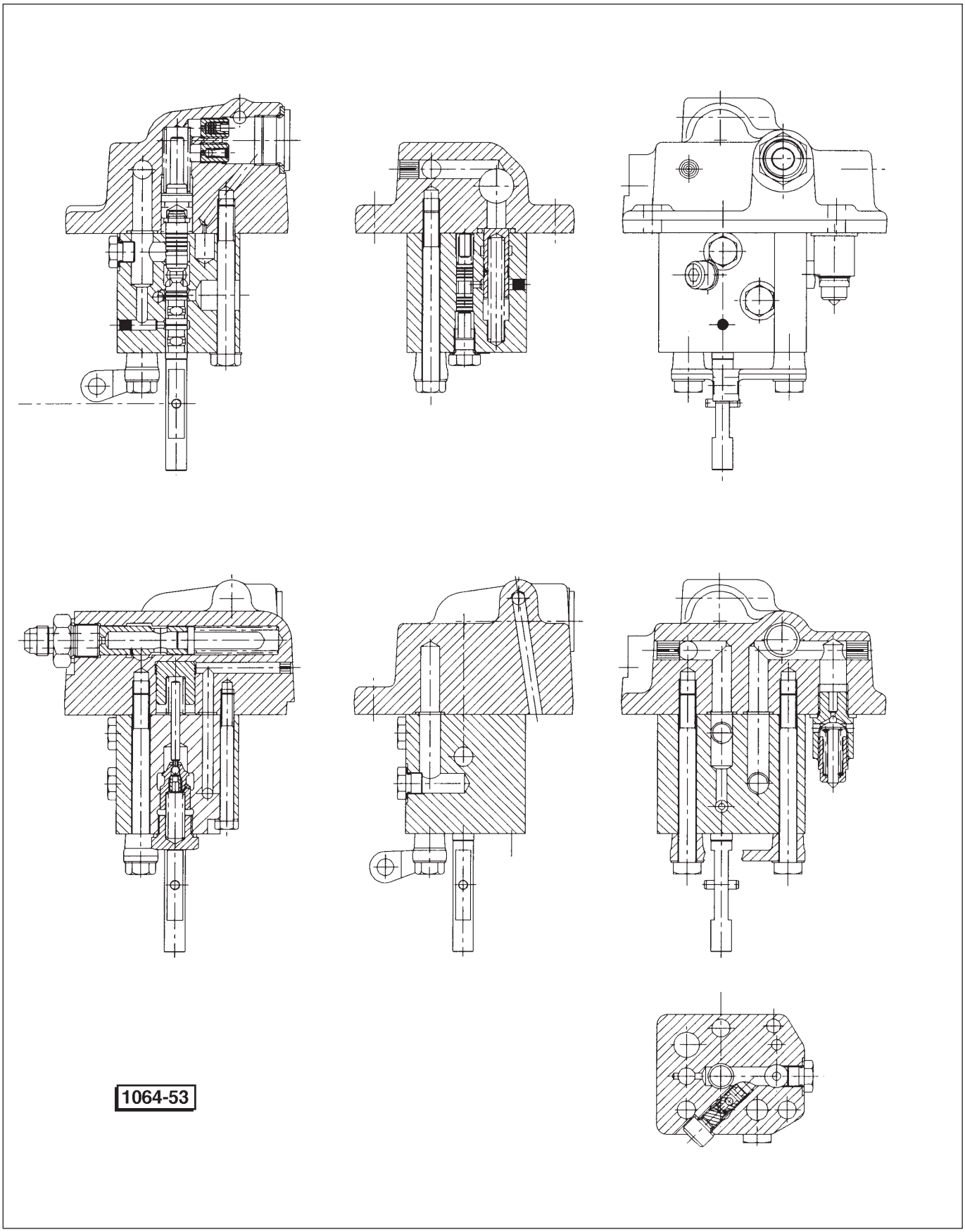


Fig. 19 - Section views of the lift hydraulic control valve

Sensing arm assembly

- Install on screw **A**:
- ring **B**,
- spacer **C**,
- spring **D**,
- ring **E**,
- screw in nut **F** to take up all play.

– Screw in nut **F** a further quarter turn to preload the spring.

– Slide this assembly into spring holder **G**.

– Fit eye **H** and tighten nut **F** against it to a torque of 100 ÷105 Nm (~10 Kgm), taking care that nut **F** does not turn on screw **A**.

– Screw plug **I** into spring holder **G** to prevent axial movement of the previously assembled pack.

– Apply a small quantity of LOCTITE 242 before fitting plug **I**.

– Apply a small quantity of LOCTITE 242 before fitting nut **F** and eye **H**.

– Grease spring **D** before fitting

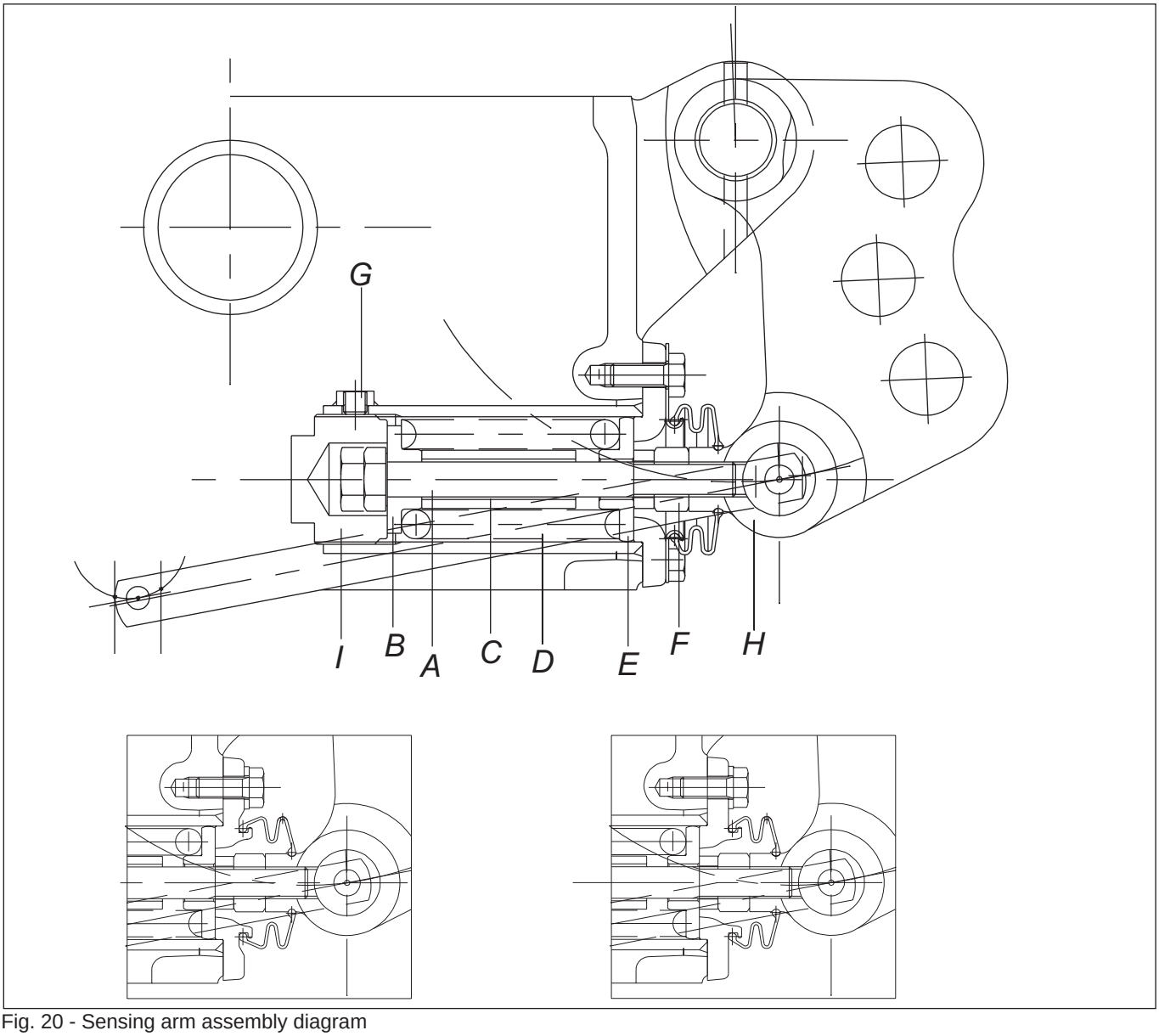
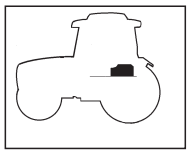


Fig. 20 - Sensing arm assembly diagram



Power-lift distributor valve spring setting specifications

oil flow control valve spring

no. of springs used		n.	1
wire diameter		mm	1,4
external diameter		mm	7,8
released spring		mm	54
loaded spring	(N 40,4) -kg 4,115	mm	47,5
	(N 66) - kg 6,727	mm	43,37

pilot valve spring

no. of springs used		n.	1
wire diameter		mm	0,9
external diameter		mm	7,8
released spring		mm	25,5
loaded spring	(N 21) - kg 2,133	mm	15,2
	(N 28,5) - kg 2,899	mm	11,5

hydraulic distributor spring

no. of springs used		n.	1
wire diameter		mm	1,2
external diameter		mm	10,4
released spring		mm	110
loaded spring	(N 43) - kg 4,4	mm	65
	(N 39) - kg 4	mm	60

nonreturn valve spring

no. of springs used		n.	2
wire diameter		mm	1,1
external diameter		mm	8,8
released spring		mm	25
loaded spring	(N 29) - kg 2,96	mm	17
	(N 43) - kg 4,43	mm	13

valvematicspring

no. of springs used		Nº	1
wire diameter		mm	1,2
external diameter		mm	9,5
released spring		mm	57,3
loaded spring	(N 41,3) - kg 4,2	mm	35
	(N 54,3) - kg 5,5	mm	25,8

ELECTRONIC LIFT

The electronic lift is controlled by an electronic control unit and a control panel.

The electronic control unit processes the signals from the input devices and generates the output signals for the electro-hydraulic control of the lift control valve.

The system provides Draft and Position control functions.

IMPORTANT
The "ALL ROUND TESTER" provides the interface for electronic control unit self-diagnostic and programming operations. All the display messages are given in the chapter "LIFT TESTER Control unit version 1.24A" of the Appendix (see table of contents).

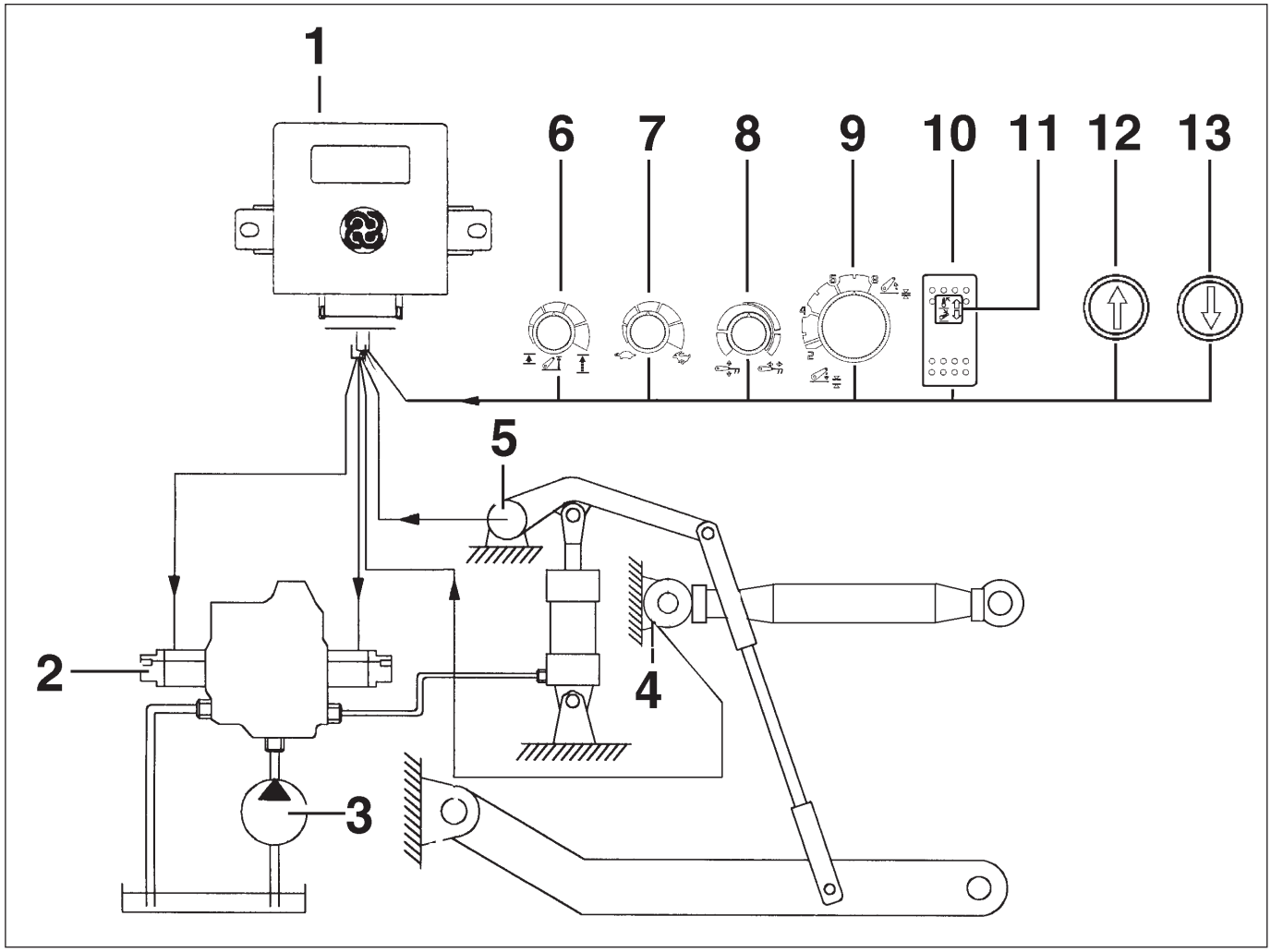


Fig. 1 - Configuration of the electronic lift system
1 - Control panel
2 - Electro-hydraulically operated control valve (2 proportional solenoid valves)
3 - Hydraulic pump
4 - Draft sensor
5 - Position sensor
6 - Maximum lift height control knob
7 - Lowering speed control knob
8 - Draft/wheelslip intermix control knob
9 - Depth control knob
10 - Up/Down control
11 - Lock and alarm warning light
12 - Manual UP control
13 - Manual DOWN control

Control panel

The control panel comprises a series of potentiometers, lights and switches which serve to set the system operating mode, raise or lower the lift and to provide information about system status.

The control panel is separate from the electronic control unit, to which it is connected by way of a multi-pin connector.

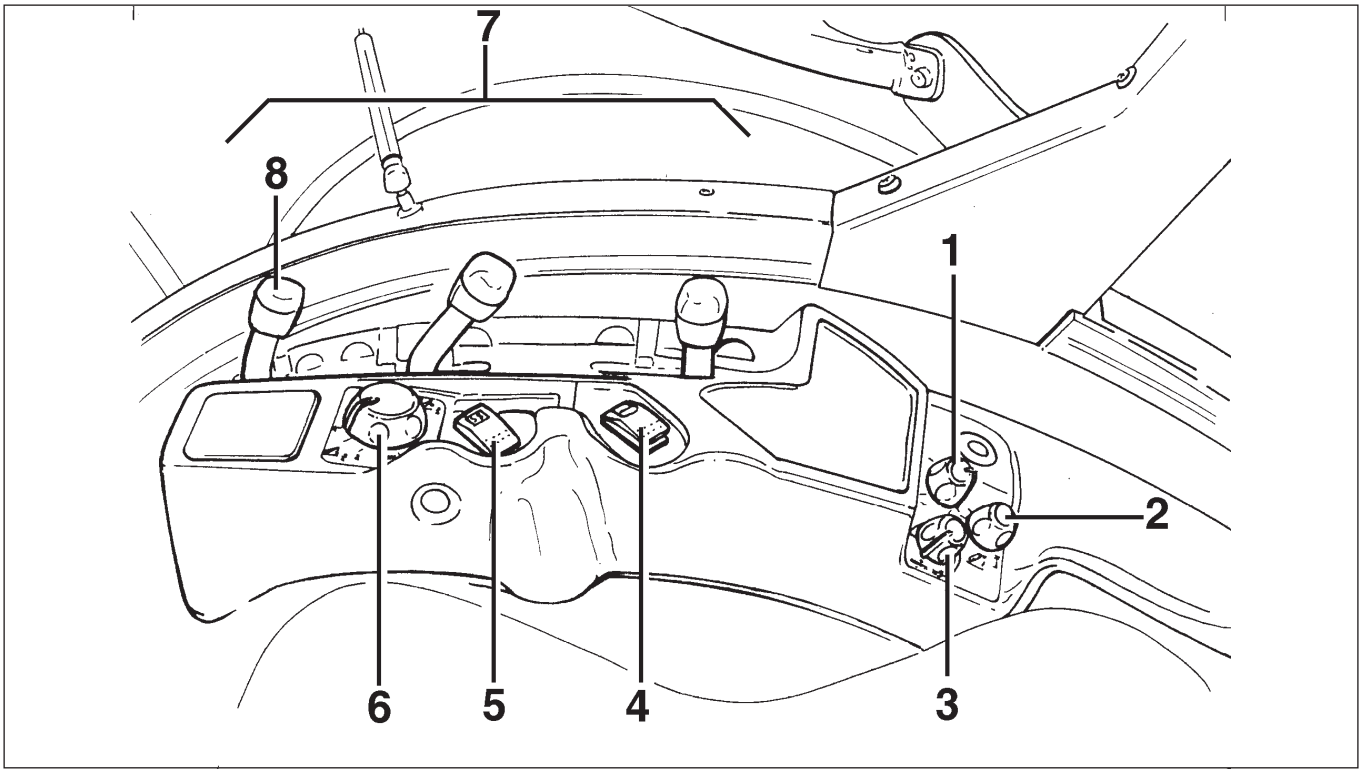


Fig. 2 - Control panel
1 - Lowering speed control knob
2 - Maximum lift height control knob
3 - Draft/wheelslip intermix control knob
4 - Rear P.T.O. engagement control
5 - Up/Down control switch with integral Lock and Alarm warning light
6 - Depth control knob
7 - Auxiliary services control valve levers
8 - Lever for hydraulic adjustment of right-hand lift rod and/or stabilisers of 3-point linkage

Control level or depth control knob (6)
Used to set the reference value for the control function.
Informs the electronic control unit of the required level of working depth or draft control for respectively the position and draft control functions. This knob is fitted with a mechanical lock to prevent excessive force being applied at the limits of potentiometer travel.

- Turning the knob to the left increases the working depth. The first sector on the left permanent Float control.

Mix position/draft control (3)
Used to select the control mode.

- When turned fully to the left, pure draft control is selected.
- When turned fully to the right, pure draft control is selected.
- In the intermediate sector, sets the minimum depth for draft control: the function is disengaged by turning the knob to the right, the depth is decreased by turning the knob to the left.

Lowering speed control knob (1)
Used to vary the rate at which the linkage is lowered.
Sets the maximum aperture of the proportional solenoid valve that controls the lowering speed.
Turning the knob to the left reduces the lowering speed.

Maximum lift height control knob (2)
Used to set the maximum height to which the linkage is raised when the driver operates the lift control.
Turning the knob to the left reduces the maximum lift height.

Up/Down control switch (5)
Rocker switch used to raise or lower the linkage, incorporating a safety lockout/release.

Up control
Pressing this button for at least 0.2 seconds raises the lift to the maximum lift height.

Control/Float mode
If the Down button is pressed for less than 0.2 seconds, Control mode is activated.
If the button is held pressed for longer, Float mode is activated. In the latter case, the implement will be lowered at the rate set by the lowering speed control knob and the lowering solenoid valve will be operated for as long as the button is held pressed.
The moment the button is released, the system will exit Float mode and automatically resume Control mode.

Lock/release
Activates/deactivates total lock out of the control panel. When the lock is activated, the warning light illuminates.
On starting, the system is always locked; to release the lock, hold the Up control button pressed for 3 seconds until the light goes out. The light will flash rapidly while the button is pressed.
To activate the lock, hold the Up control button pressed for 3 seconds after which the light will illuminate; again, the light will flash rapidly while the button is being pressed.
On activating the lock, the linkage is immediately raised to the height determined by the maximum lift height control knob. Any subsequent alteration to the position of the lift controls (including the maximum lift height control knob) will have no effect on the position of the linkage until the lock is released.

Lift status indicator light
This light provides information about the operating status of the lift.
If the light is permanently illuminated, this indicates that the system is locked.
If the light is off, this indicates that the system is operating normally.
If the light is flashing, this indicates an alarm condition.

WARNING: only use the Up control button located on the solenoid valve of the lift control valve in cases of extreme necessity and when the engine is running at idle speed. Release the button before the linkage reaches the maximum height, as the automatic stop determined by the both position sensor and the safety switch is excluded during this operation.

Remote pushbuttons for lift operation from ground

Two remote pushbuttons are located on the rear fender.

To enable these pushbuttons it is first necessary to release the lift control system.

Operation of the manual controls overrides any current lift functions and places the control unit in STOP condition, awaiting new control signals.

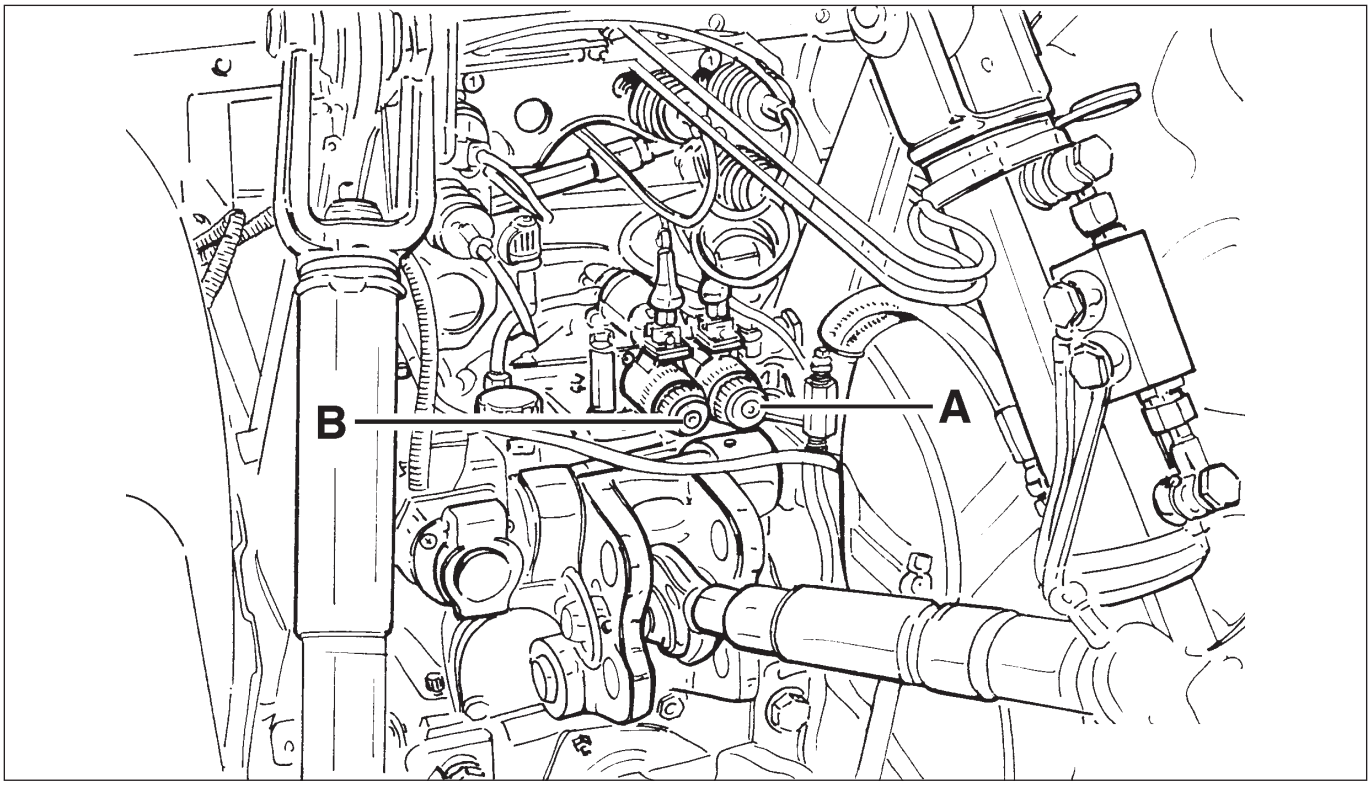


Fig. 3 - Lift control pushbuttons ***A - Up B - Down***

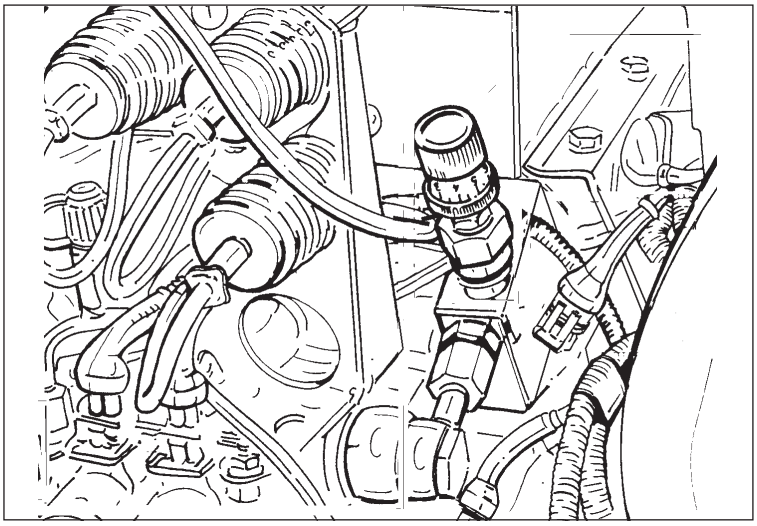


Fig. 4 - Flow control valve used to regulate the rate of oil flow to the lift cylinder.

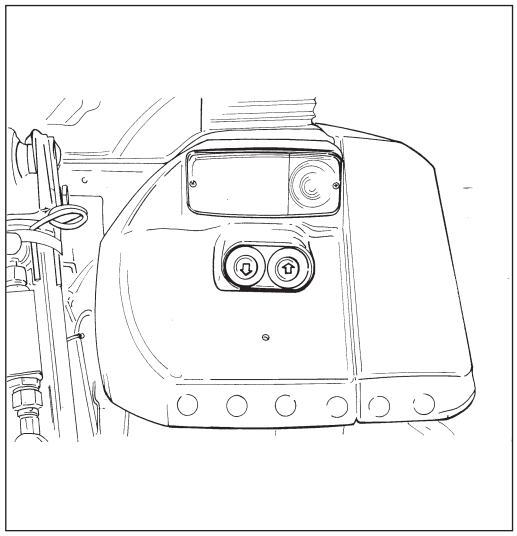


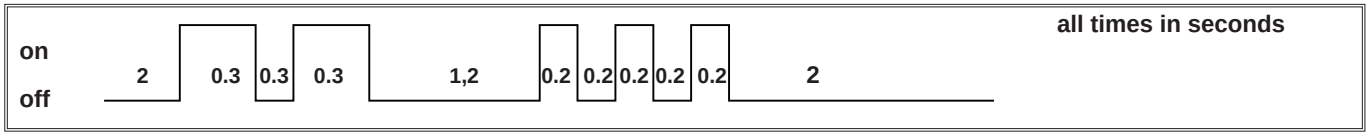
Fig. 5 - Ground-operated lift controls.

Lift operation
On starting the engine, the lift is in Lock condition; to release the lift, press the UP switch for at least 3 seconds.
The lift control system enters STOP status in which condition the lift may be raised to the maximum lift height by pressing the switch for at least 0.5 seconds but no more than 3 seconds.
To activate the lift control system, press the DOWN switch for less than 0.3 seconds; if the switch is held pressed for longer, FLOAT mode is activated and CONTROL mode will only be resumed when the switch is released.
The functions of the three potentiometers are as follows:
1 - MAXIMUM LIFT HEIGHT (used to set the maximum height to which the linkage can be raised)
2 - LOWERING SPEED (used to set the rate at which the implement is lowered).
3 - CONTROL (used to select the control type: when turned fully to the left POSITION control is selected, when turned fully right DRAFT control is selected).
When turned to the intermediate sector, mixed draft and position control is selected.

LIFT ALARMS

CODE	MESSAGE ON TESTER	*Alarm description*
11	UPSV DISCONN.	UP SOLENOID DISCONNECTED
12	UPSV S.C.	UP SOLENOID SHORT CIRCUIT
13	DWSV DISCONN.	DOWN SOLENOID DISCONNECTED
14	DWSV S.C.	DOWN SOLENOID SHORT-CIRCUIT
15	-	-
21	POS.SENS.S.C.	POSITION SENSOR SHORT CIRCUIT
22	POS.SENS. DIS.	POSITION SENSOR DISCONNECTED
23	GEN.FAIL.CPU	GENERAL FAILURE OF CPU.
24	-	
25	-	
31	-	
41	DRAFT SENS. DIS.	DRAFT SENSOR DISCONNECTED
42	DRAFT SENS. S.C.	DRAFT SENSOR SHORT CIRCUIT
45	EEPROM CHECK	EEPROM ERROR

Example display of code 23:

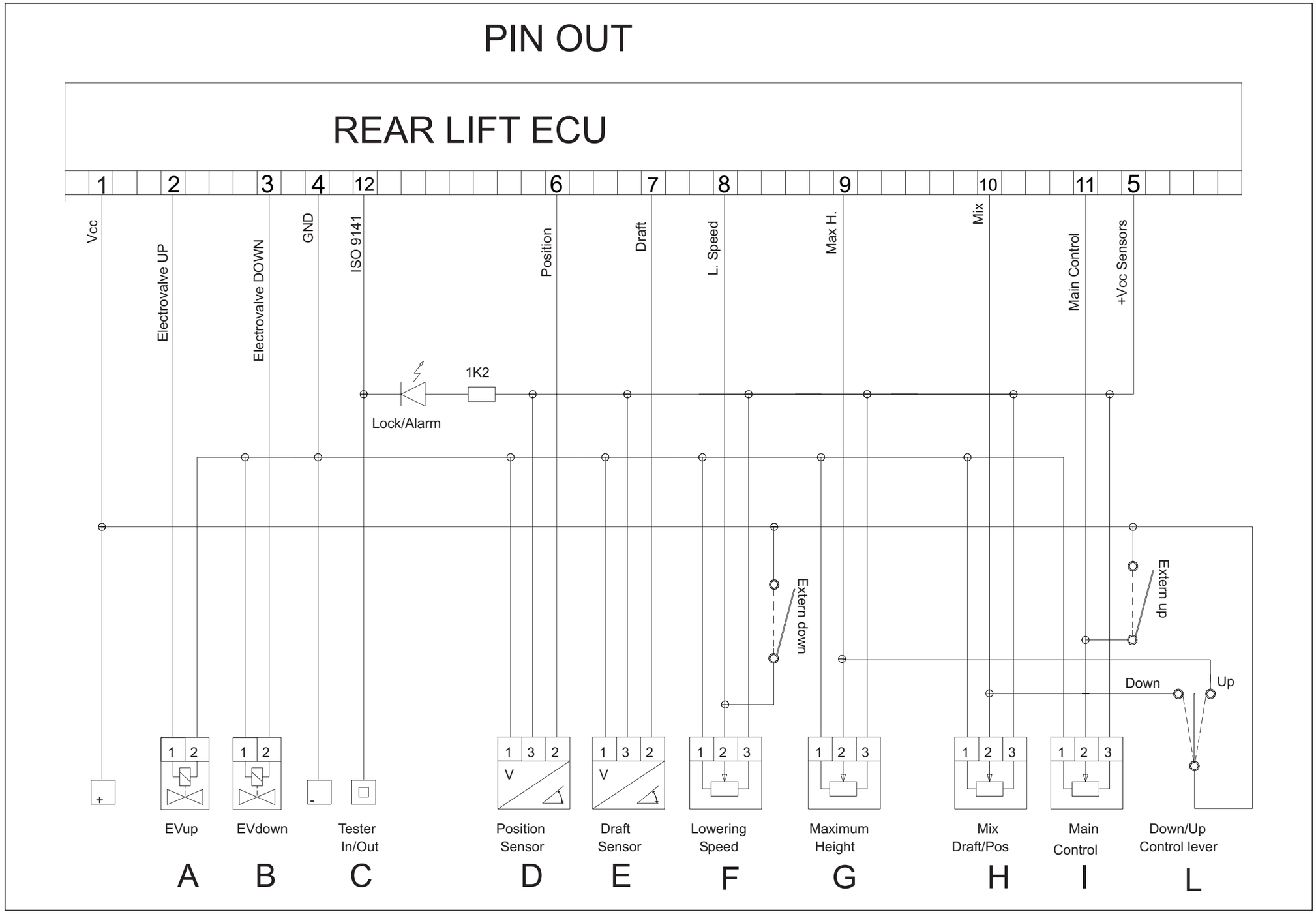


The criteria for encoding are as follows:
— light off for 2 seconds
— number of 0.3 second flashes corresponding to the tens digit
— light off continuously for 1.2 seconds
— number of 0.2 second flashes corresponding to the units digit
— repetition of the cycle until the alarm ceases

Electronic circuit diagram of the ECU

1 – Vdc
2 –UP solenoid valve
3 - DOWN solenoid valve
4 – GND
5 – Sensor Vdc
6 – Position
7 – Draft
8 – Lowering speed
9 – Maximum height
10 – Mix
11 – Main control
12 – ISO 9141

A - UP solenoid valve
B - DOWN solenoid valve
C - Tester IN/OUT
D - Position sensor
E - Draft sensor
F - Lowering speed
G - Maximum height
H - DRAFT/POSITION mix control
I - Main control
L - DOWN/UP control



Calibration of the AUTOMATIC
1 - Unlock the lift.
2 - Move the lift arms to the required height using the ground-operated UP/DOWN controls.
3 - Press the P.T.O. AUTO switch.
4 - Press the P.T.O. engagement switch for approx. 10 seconds.

The selected height is set as the maximum height limit for P.T.O. operation.

Emergency manual lift control
In the event of a malfunction of the electronic system, the lift can still be controlled by way of the mechanical pushbuttons located on the solenoid valves of the lift control valve (see figure below). However this operation should only be carried in an emergency and with the engine running at idle speed.
Release the pushbutton before the lift arms reach their maximum height, as the automatic stop determined by the position sensor is disabled during this manoeuvre.

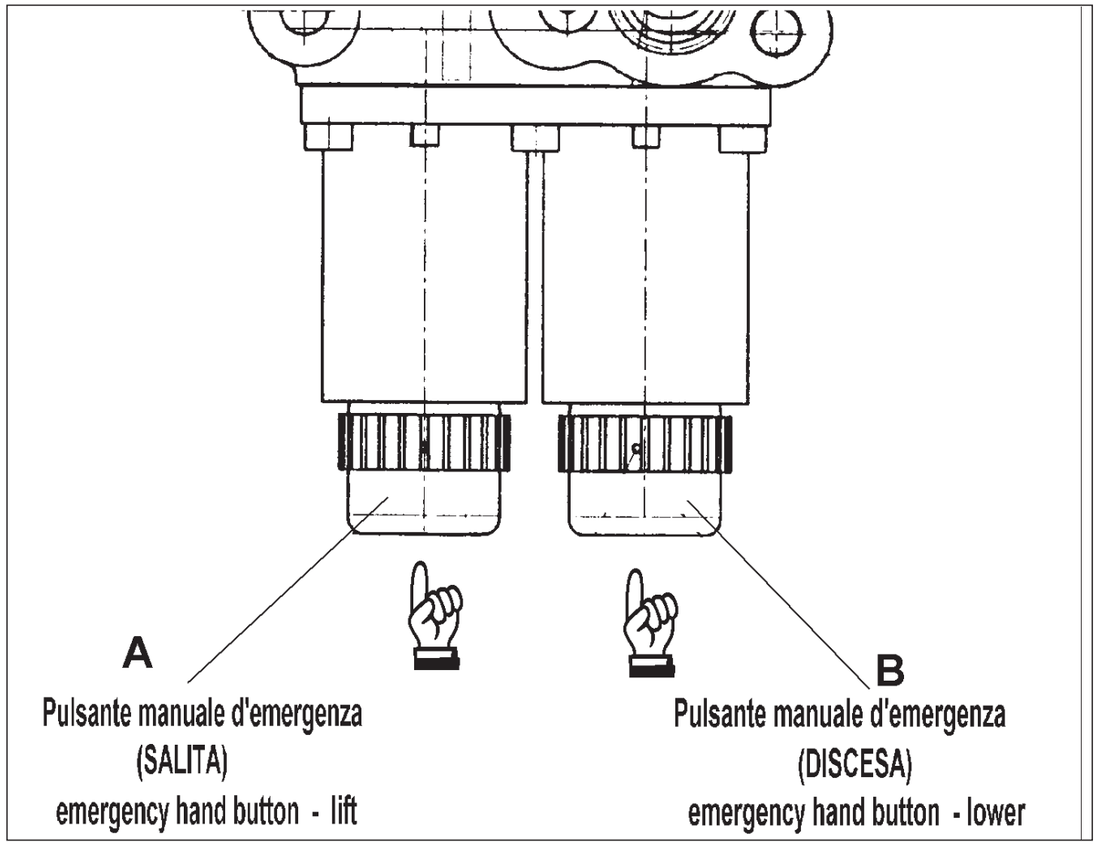


Fig. 6 - Emergency manual lift control pushbuttons.
HYDRAULIC CONTROL VALVE
(located on lift casing)

A - Up
B - Down

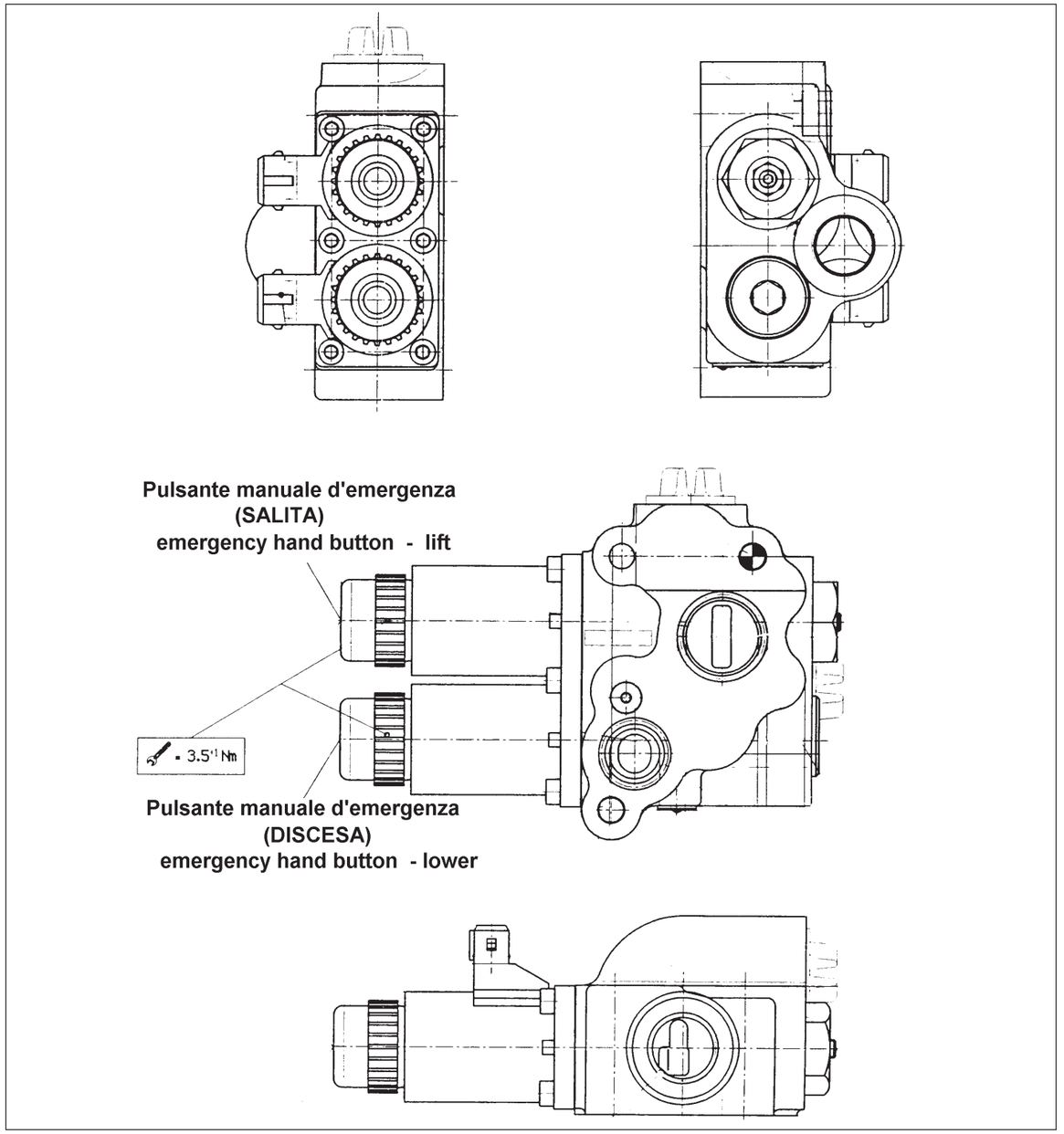


Fig. 7 - Diagram of the hydraulic control valve.
Hydraulic control valve of the electronic lift
This control valve is of the open centre type and has two solenoid valves for raising and lowering the lift arms.
The two solenoid valves are controlled directly by the ECU.
Do not connect the solenoid valves directly to an electrical power source, as they would be irreparably damaged.

LIST OF ELECTRONIC LIFT TESTS release 1.24 A

Connect the ALL-ROUND TESTER to the diagnostics socket and select option (2) [REAR LIFT CONTROL].
Press (0) twice and enter the password [BACO] in order to modify the parameters.
Press [E] twice to display the first screen page.
The first screen indicates the selected tractor type (10N or 10S).
If the selected tractor type does not correspond to the machine being tested, change the type by pressing the number alongside the required model and follow the on-screen instructions to store the new type in memory.
Release 1.23A does not display the tractor type selection screen and the parameters are those relative to the 10N type.

Enter the MENU screen and press (0) to enter the modification menu.
2 - Modification menu.

1 - Monitor
2 - Parameters
3 - Tests
4 - Max/min height calibr.
5 - Alarms
6 - Calibr. PWMmin

- MONITOR -
Displays data related to the control algorithm and the PWM controls (sensor and potentiometer status).

- Parameters **10N** - Programming parameters of the control solenoid valves (SV).

- Tests - allows you to test ground-operated pushbuttons and display sensor values.

- MAXIMUM/MINIMUM LIFT HEIGHT CALIBRATION
Allows you to calibrate the maximum lift height and the lifting speed.

DEFAULT VALUES **10N**:
1- max. Height 850
2 - Lifting speed 315

To calibrate the maximum height, put the lift control in STOP condition (see **LIFT OPERATION**) and select "CALIBRATION" from the main menu. Then select option 1 from the "CALIBRATION" menu, turn the MAXIMUM LIFT HEIGHT knob to the maximum position and, using the ground-operated **UP/DOWN pushbuttons**, position the lift arms at the required maximum height.
Press "E" and "C" to save and confirm the new maximum height value. Exit the "CALIBRATION" menu and continue the tests.

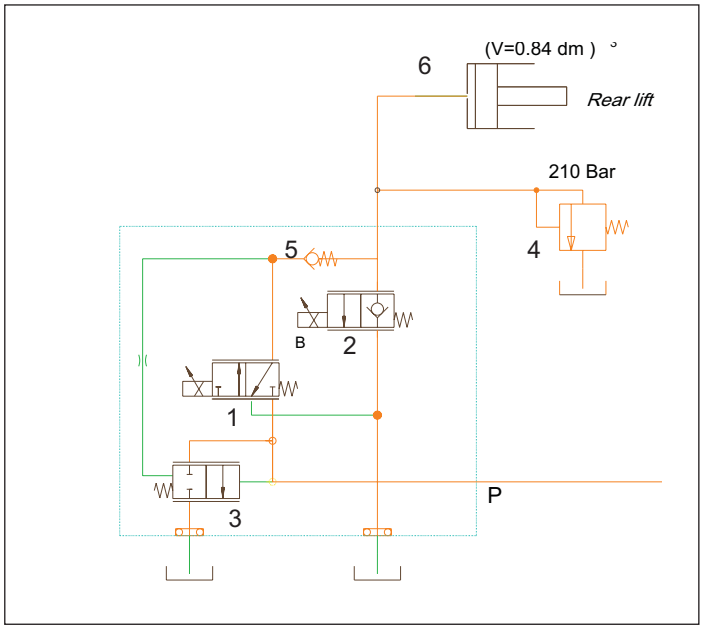


Fig. 8 - Functional diagram of the lift control valve.

1 - UP valve
2 - DOWN valve
3 - Enabling valve
4 - Antishock valve
5 - Check valve
6 - Lifting cylinder

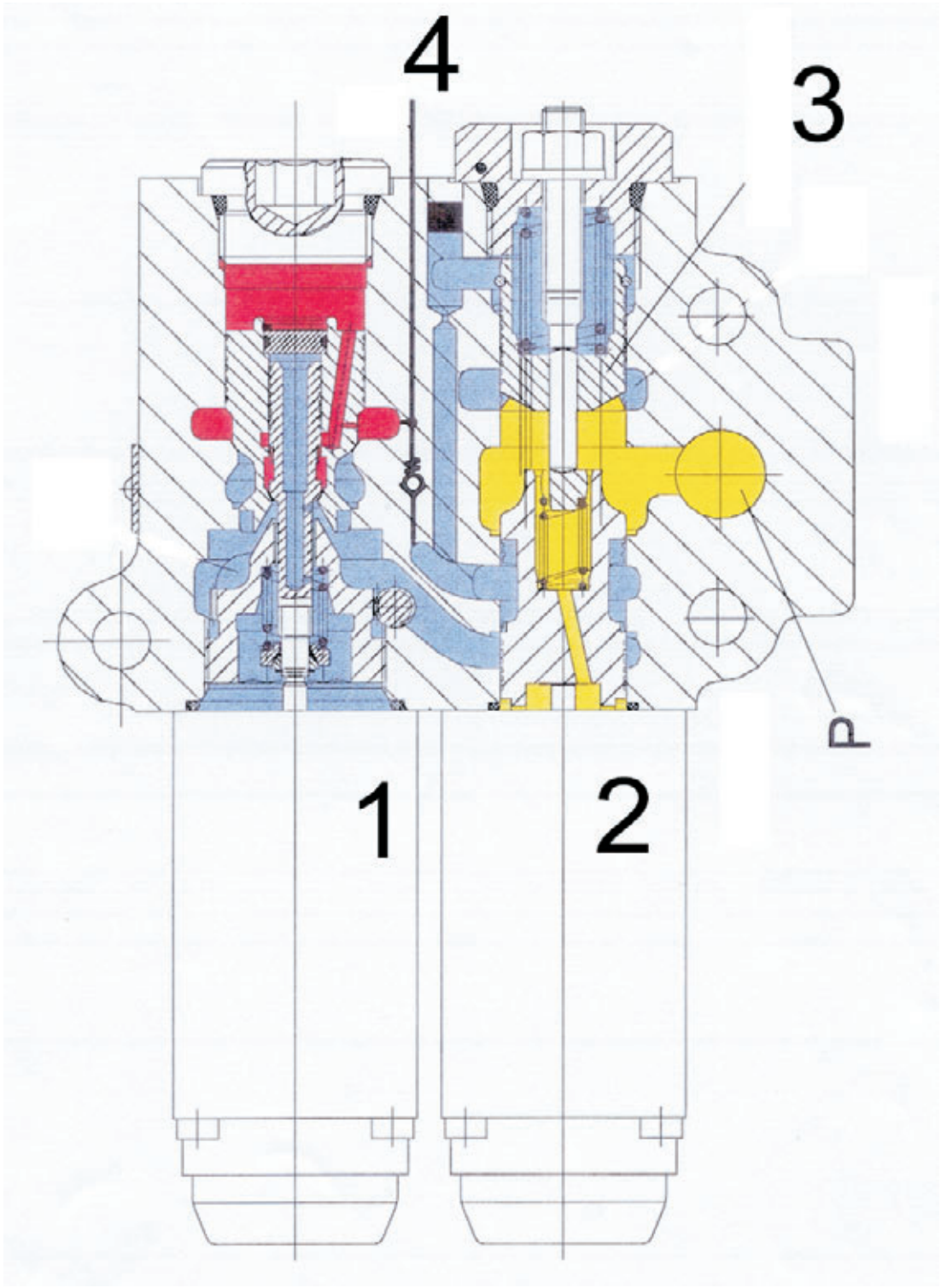


Fig. 9 - Control valve hydraulic circuit - **NEUTRAL**
1 - UP solenoid valve
2 - DOWN solenoid valve
3 - Enabling valve
4 - Check valve
P - Oil flow

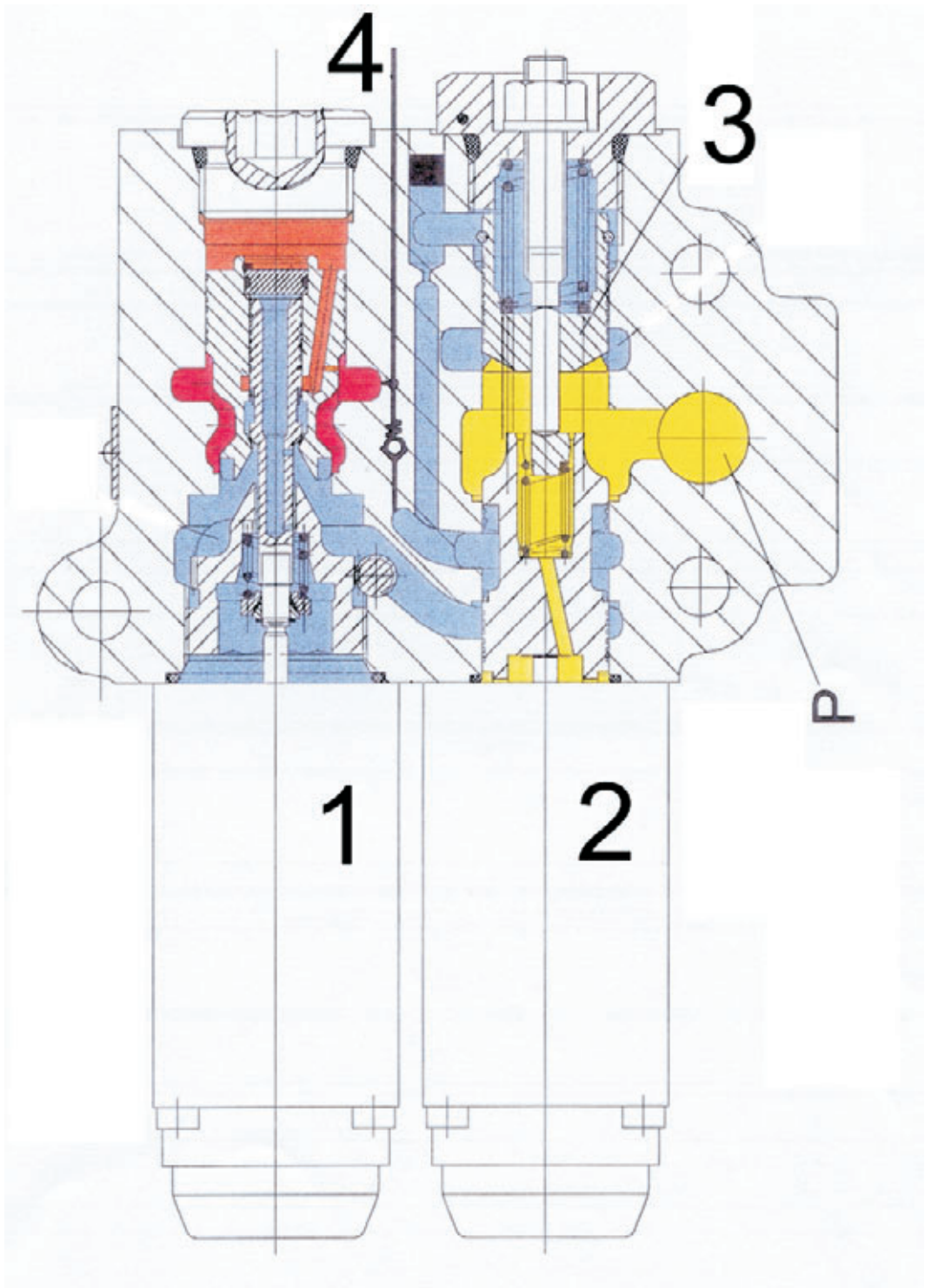


Fig. 10 - Control valve hydraulic circuit - **LIFTING**
1 - UP solenoid valve
2 - DOWN solenoid valve
3 - Enabling valve
4 - Check valve
P - Oil flow

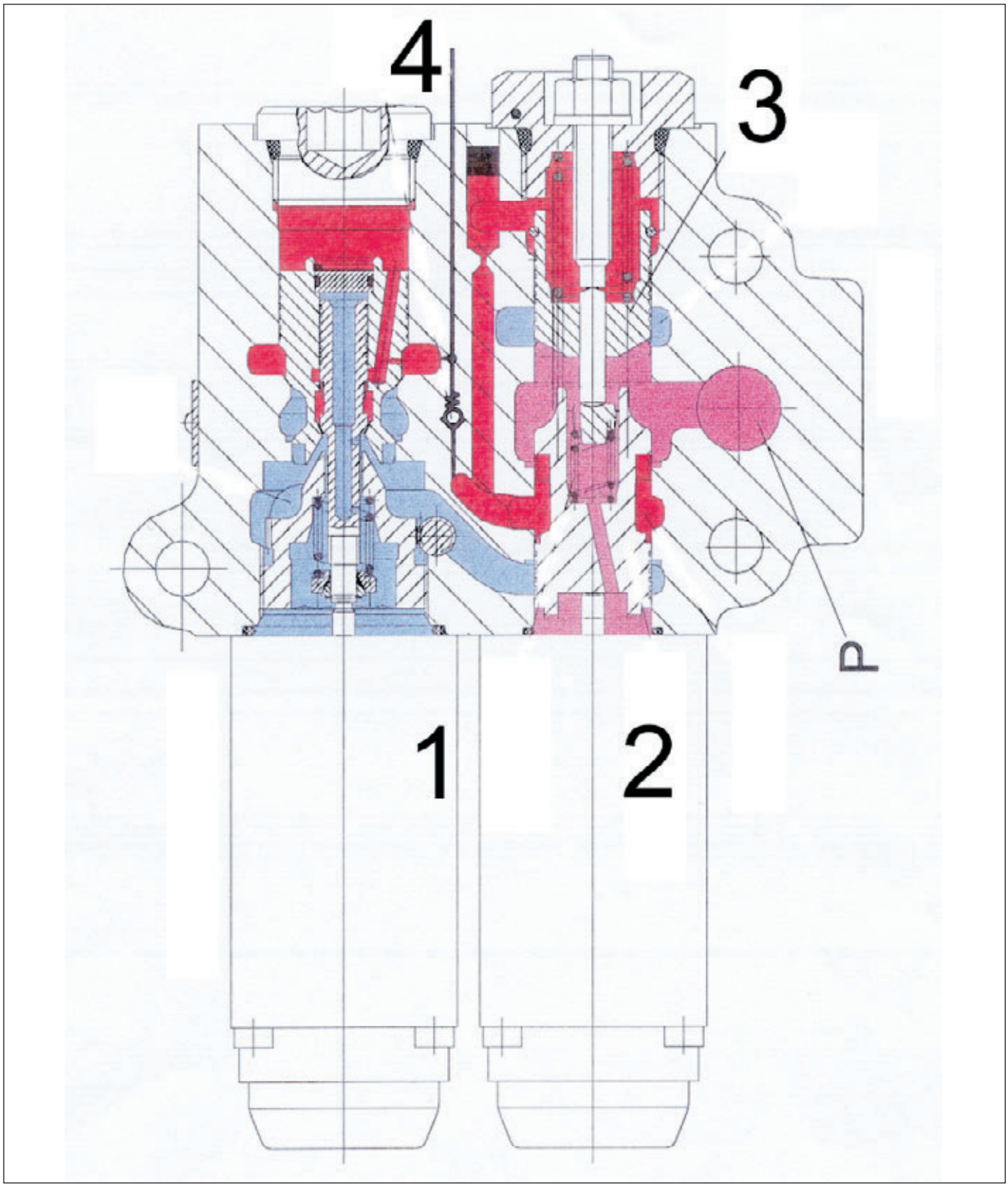


Fig. 11 - Control valve hydraulic circuit - **LOWERING**
1 - UP solenoid valve
2 - DOWN solenoid valve
3 - Enabling valve
4 - Check valve
P - Oil flow

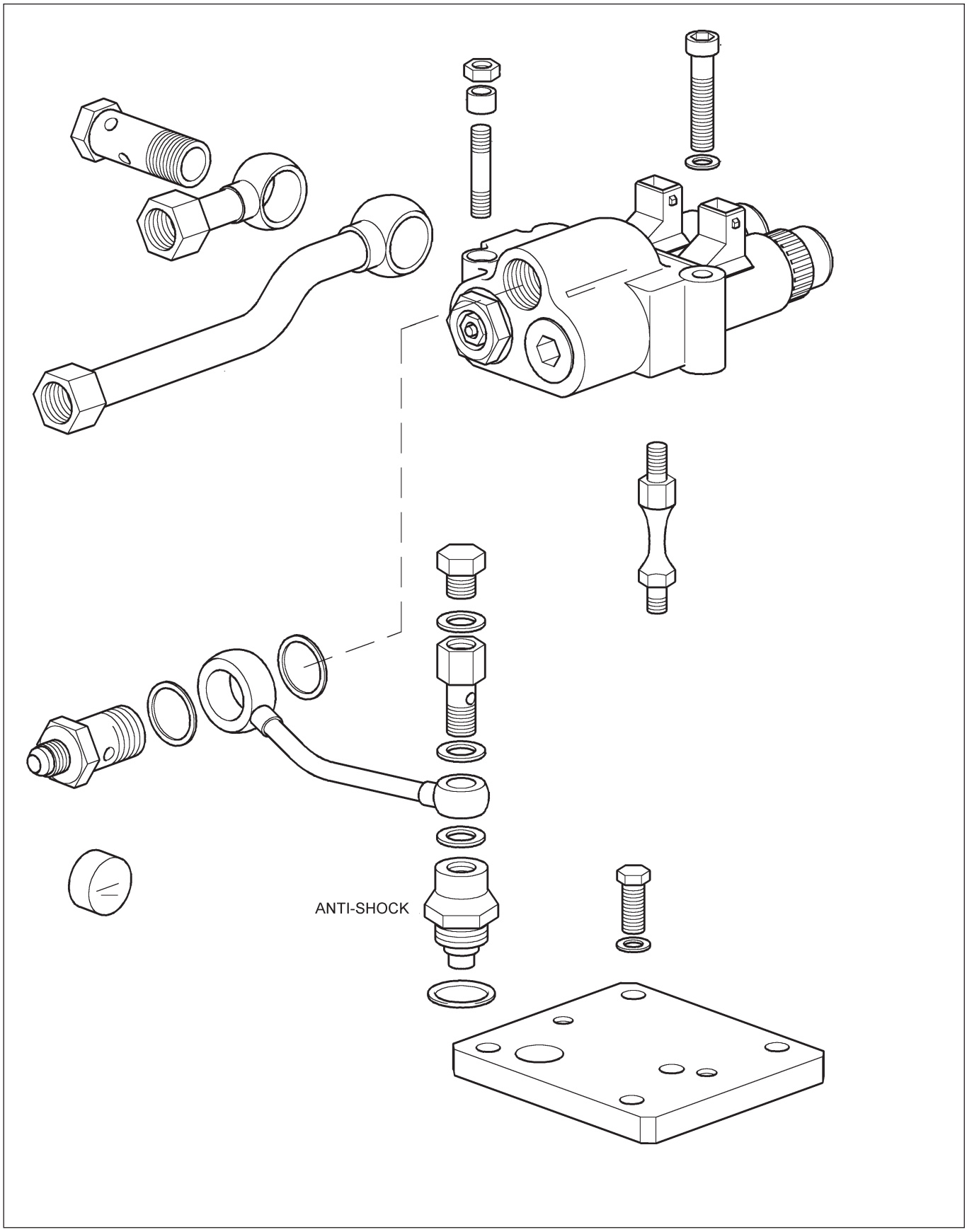


Fig. 12 - Hydraulic control valve for electronic lift.

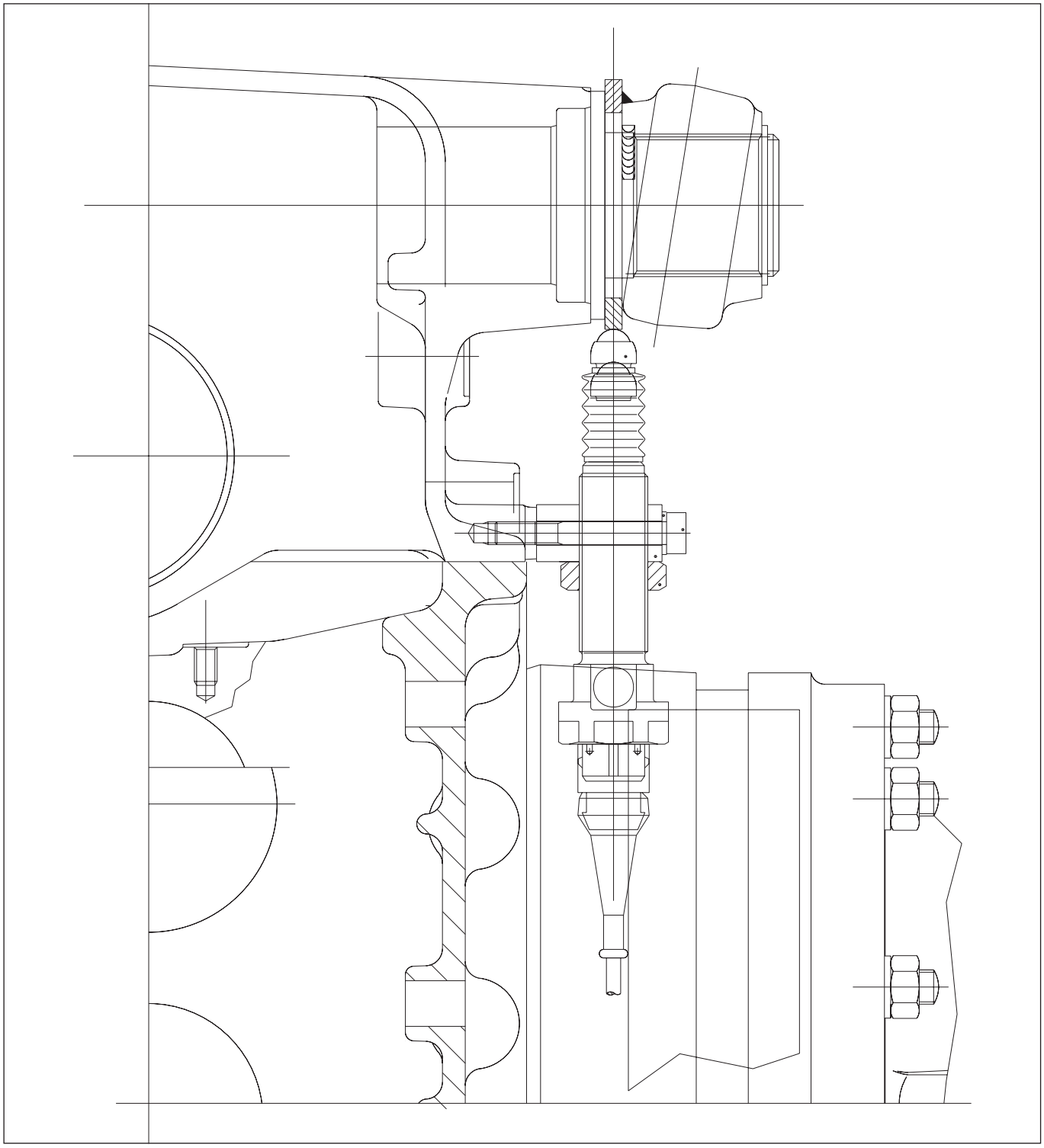


Fig. 13 - Position sensor calibration of the electronic lift.

POSITION SENSOR CALIBRATION.
If a new position sensor is fitted, it will have to be adjusted. Proceed as follows:
1 – Move the lift arms to the minimum height position, connect the "ALL AROUND TESTER" and access the maximum height calibration screen; screw the sensor in or out until a number between 4 and 10 is displayed.
2 – Tighten down the sensor
3 – Proceed with the calibration of the maximum height.

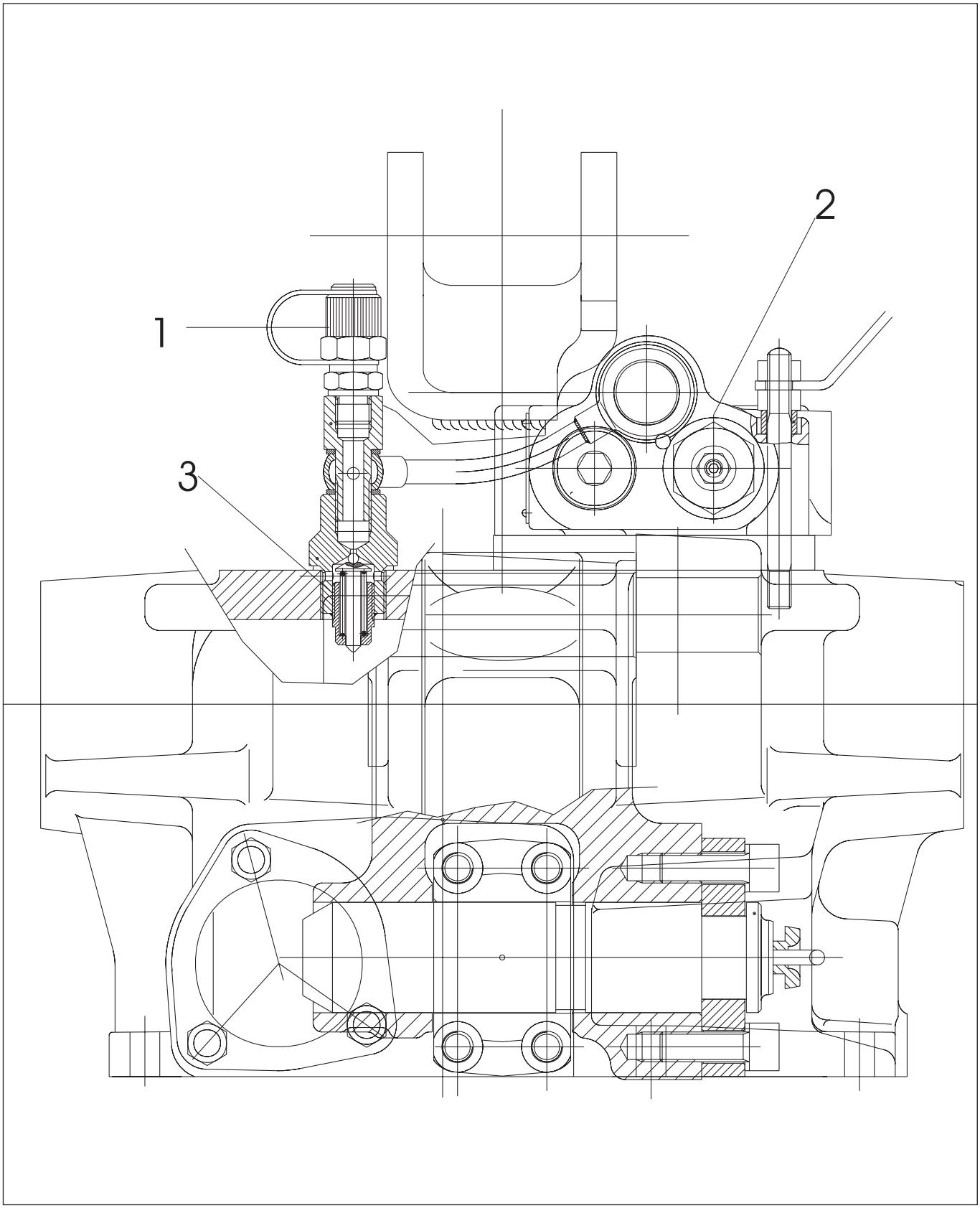


Fig. 14 - Pin for draft control of the electronic lift
1 - Pressure test fitting for lift cylinder pressure
2 - Anti-shock valve
3 - Lift control valve

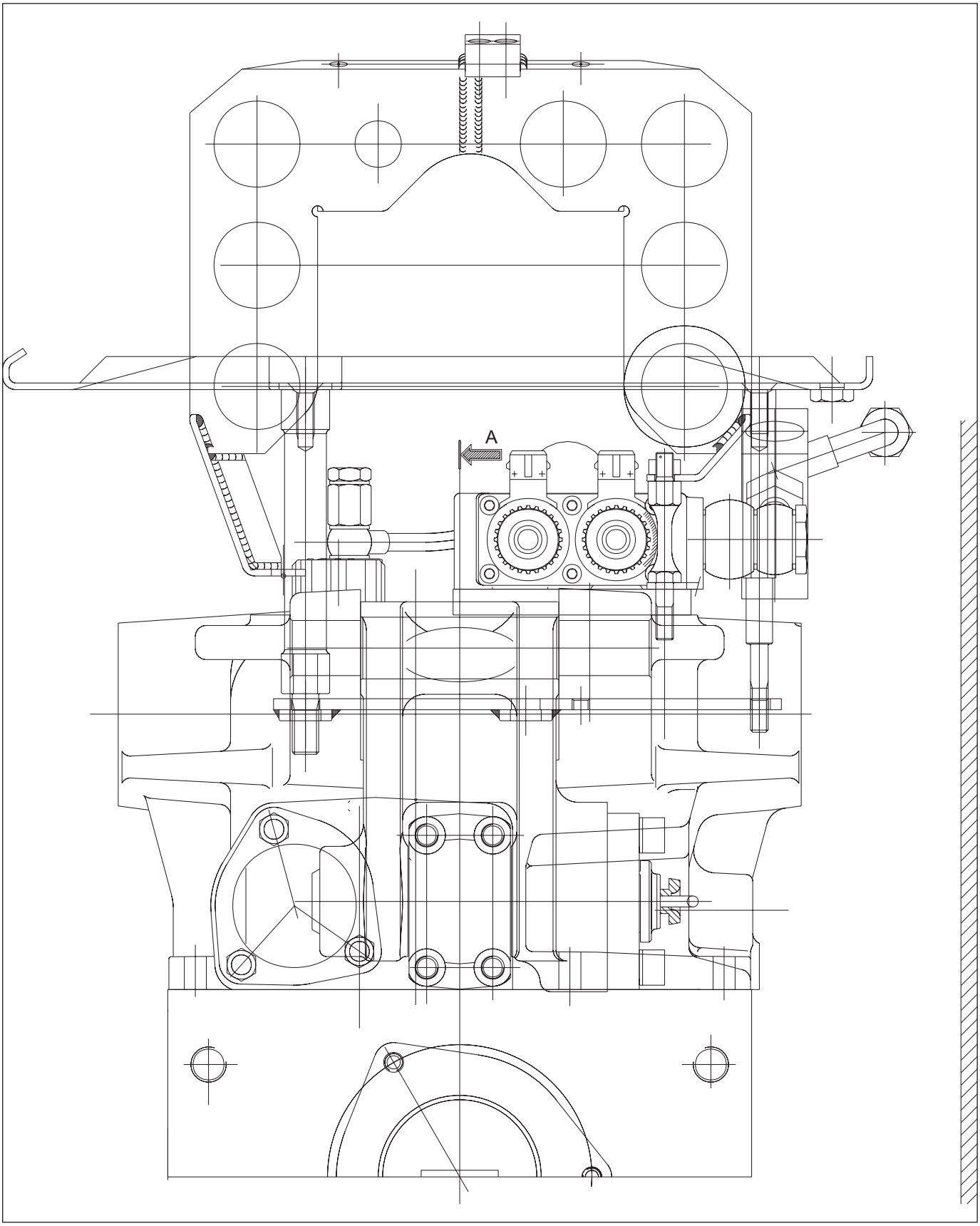


Fig. 15 - Rear view of the electronic lift.

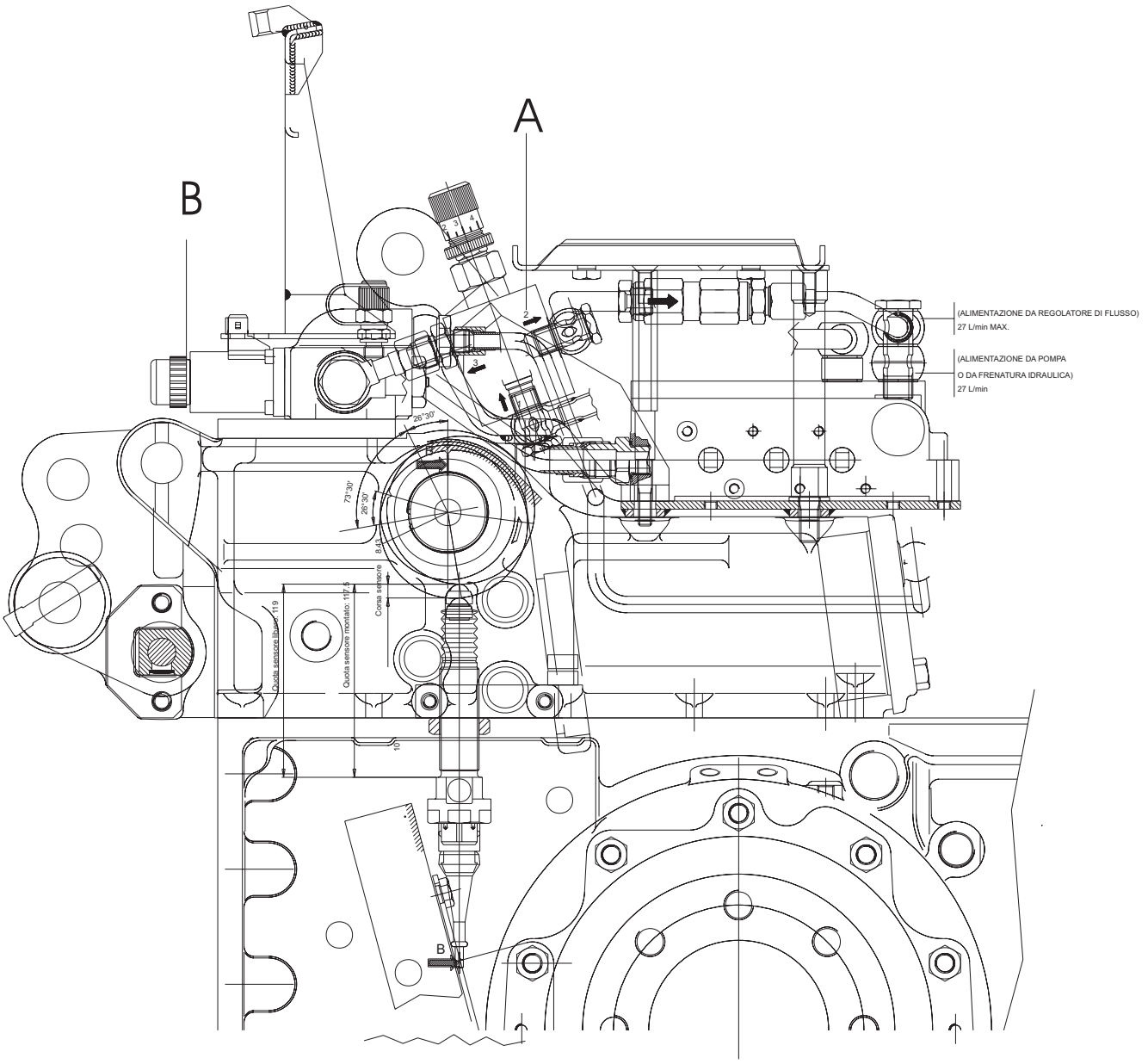


Fig. 16 - External hydraulic circuit of the lift.
1 - Flow divider valve
2 - Lift control valve

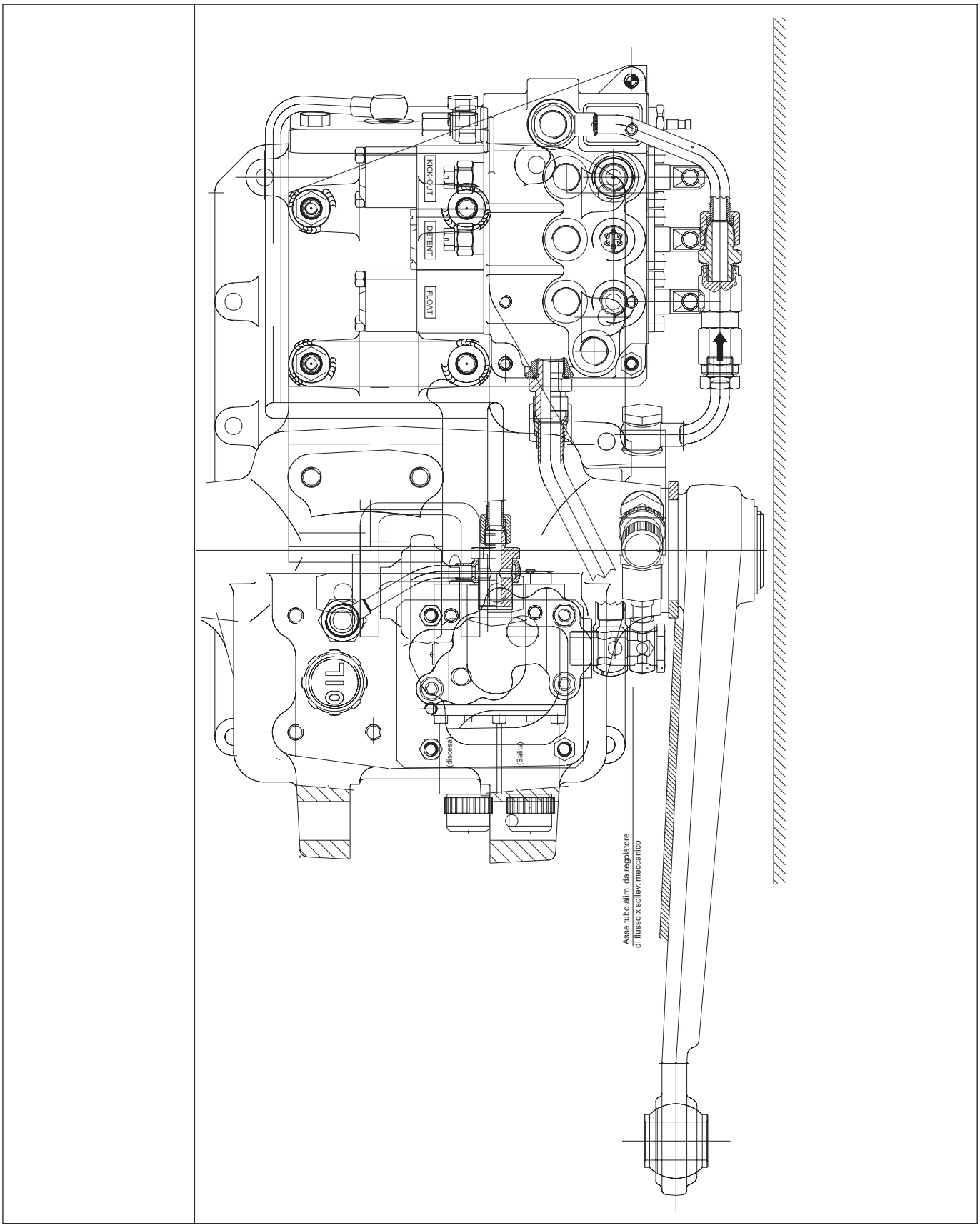
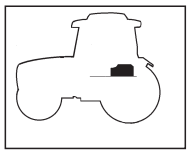


Fig. 17 - Top view of the electronic lift.



Precautions for electronic equipment

The tractor is equipped with a number of sensitive electrical devices and electronic components. In particular, electronic components do not tolerate polarity inversion, over-current, connection errors, etc..
The electronic components are provided with internal and external protections, but the following precautions are still necessary:

— Switch off the control panel before carrying out any work on electrical components.
— Do not use a screwdriver of light bulb to test for electrical current; only use the appropriate diagnostic equipment (tester).
— Do not apply excessive force when connecting and disconnecting electrical connectors; do not prise connectors apart using a screwdriver, and always respect the polarity of the connection.
— In the event of a fault in a circuit controlled by an electronic control unit, do not immediately replace the control unit, but first check that the sensors and actuators are functioning correctly.
— When replacing a defective electronic control unit, only unscrew the retaining screws; avoid interfering with other screws as these could be calibration screws.
— Inspect contacts on sensors and connectors for corrosion.
— Do not use arc welding in the vicinity of electrical circuits.
— Avoid short-circuits and polarity inversion.
— When working on components controlled by microprocessors, always disconnect the power supply by turning the ignition key to the STOP position before disconnecting the component.
— If the tractor is to be resprayed and passed through a paint baking oven, remove all electronic components.
— Do not insert the probes of the tester into multi-pole sockets on connectors or other devices (this deforms the contacts and thus leads to poor connections).
— Do not test for the presence of electrical current by causing a spark, i.e. by short-circuiting components across wires or to earth.
— Do not reverse the polarity or swap over wires in multi-pole connectors.

CHECKING THE ELECTRONIC SYSTEM

- **Elements to check:**

— Sensors
— Electronic control units
— Actuators

1) Checking the system power supply:
— current present at input to the connector plugged into the device;
— current present at output from connector
— earth connection
— voltage correct.
2) Check downstream components: sensors.
3) Check upstream components.

CHECKING MECHANICAL COMPONENTS

Check the splines on the external and internal lift arms for damage or excessive wear; replace the levers if necessary.

Check the bushes of the lift arm shaft for scoring or excessive wear.
Check the bushes and the pins connecting the lift arms to the hydraulic cylinder are not worn and are firmly inserted in the seats on the levers.

Check the diameters of the rod and bushes in the cylinders and compare with the maximum wear limits in the table.

Check that the splines on the lifting shaft are not excessively worn and that the machines surfaces in contact with the support bushes are not damaged.

NOTE: whenever the lifting cylinders are dismantled, new seals must be fitted.

Front hydraulic lift

On request the tractor can be equipped with the front hydraulic lifting system.
It is operated by means of two hydraulic cylinders actuated by the related control lever placed at the right-hand side of the operator driving position.
The lever controls a single-acting distributor valve linked to the rear tractor power-lift case.

Inspections
Ensure that the cylinder pin bushings are not worn and be sure these are firmly secured in their seats.
NOTE: Replace gaskets after each hydraulic lifting cylinder disassembly.

Specifications

power-lift type	with two hydraulic cylinders and	removable arms
hydraulic cylinder type		single-acting
oil supply		from the rear auxiliary hydraulic distributor
peak operating pressure	bar	180
cylinder stem diameter	mm	50
lifting stroke	mm	170
distance between both lower link ends	mm	865
lifting capacity	kg	1500

A - Front lift cylinders (2)

B - Hydraulic accumulator
Volume 0.700 litres
Nitrogen precharge 130 bar
maximum operating pressure 250 bar

C - Antishock valve

D - Lowering speed regulator valve

E - Lockout cock (prevents oil from discharging from the cylinders when transporting raised implements on the road).

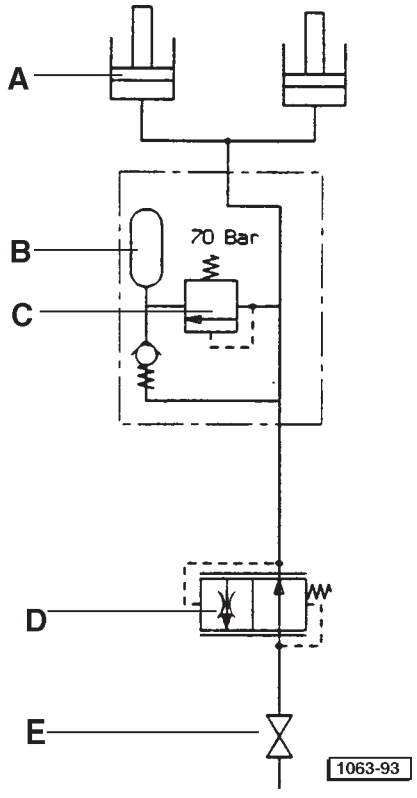


Fig. 1 - Lift operation.

Hydraulic accumulator and antishock valve for front lift

The front lift is equipped with a hydraulic accumulator with an antishock valve. This improves hydraulic system performance and reduces strain on the front axle.

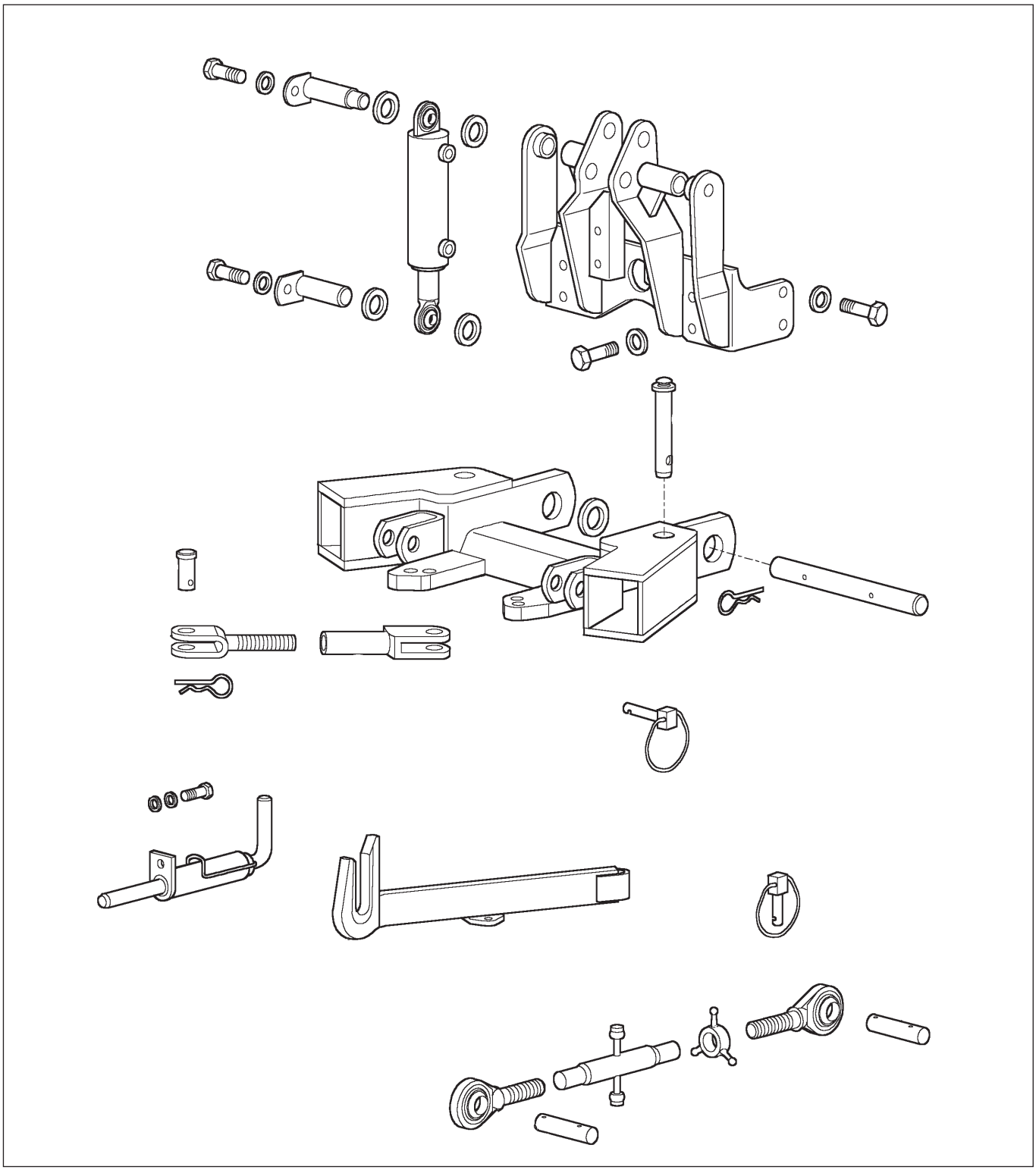


Fig. 2 - Front lift components.

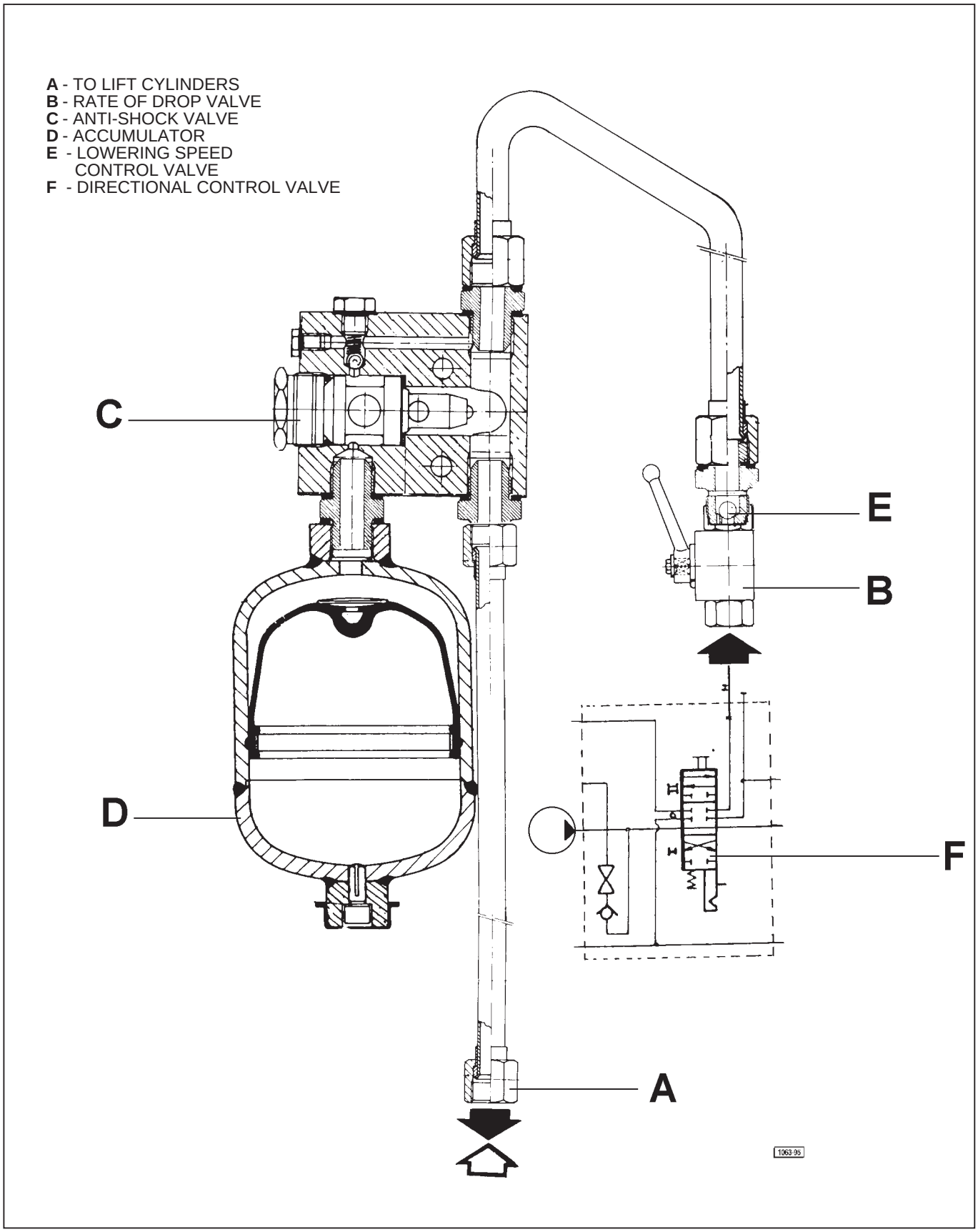
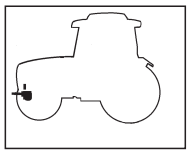


Fig. 3 - Oil pressure accumulator and anti-shock valve. (The directional valve to which is connecting, is convertable into single-acting. For the operations to see auxiliary systems chapter).



Front power take-off

General information

The power take-off clutch is a multidisk, oil-bath and electro-hydraulic control type.

The front power take-off is directly linked to the crankshaft by means of a flexing coupling; while the shaft-end is equipped by an universal joint.

The oil used for P.T.O. operation is contained in the P.T.O. Casing.

A specific gear pump installed on the P.T.O. shaft sucks the oil, which flows first through a filter for being subsequently routed under pressure to the clutch assembly through an electro-hydraulic distributor.

To avoid exceeding these values, we recommend that a suitable coupling is installed between the implement and the power take-off.

The P.T.O. shaft rear teminal has clockwise direction of rotation (with respect to the driving position).

The electro-hydraulic control operates the system by way of a solenoid valve. When the valve is open, pressurised oil acts on the pistons to force together the driving plates and driven plates of the clutch and thereby transmit drive to the output shaft.

The P.T.O. clutch brake is operated mechanically by the piston during the return stroke determined by the helical spring. Three pins attached to the piston push against the brake pressure plate.Control is electro-hydraulic type and is provided by the relevant switching knob.

specifications

power take-off clutch

Clutch		wet multi-plate with axial piston
number of plates		5
diameter of plates		105
thickness of clutch plate pack		
Minimum	mm	27.32
Maximum	mm	29.90
N° of thrust plates		5
plate lubrication pressure	bar	5
maximum engagement pressure setting	bar	$16\ ^{+2}_{0}$
P.T.O. pressure control valve	bar	$16\ ^{+2}_{0}$
Restrictor orifice	mm	1.1

clutch spring specifications

Piston return spring

wire diameter		mm	6
outside diameter		mm	52,5
free length		mm	50
length under compression	kg 82 (808 Nm)	mm	28
	kg 98 (978 Nm)	mm	23,4

Checking the clutch

– Inspect the clutch plate contact surfaces and replace the plates if worn or damaged.
– The thrust piston must be replaced if it shows signs of scoring.
– If the splined surface along which the plates slide is worn low in its upper part the hub should be replaced.
– Check the efficiency of the piston return spring; the spring characteristics should correspond to those specified in the relative table; replace the spring if necessary.
– The seal rings of the clutch piston and brake piston must be replaced if worn, as should the brake disc itself.
– At each disassembly, carefully clean the oil ways in the clutch housing; use a compressed air jet if necessary.

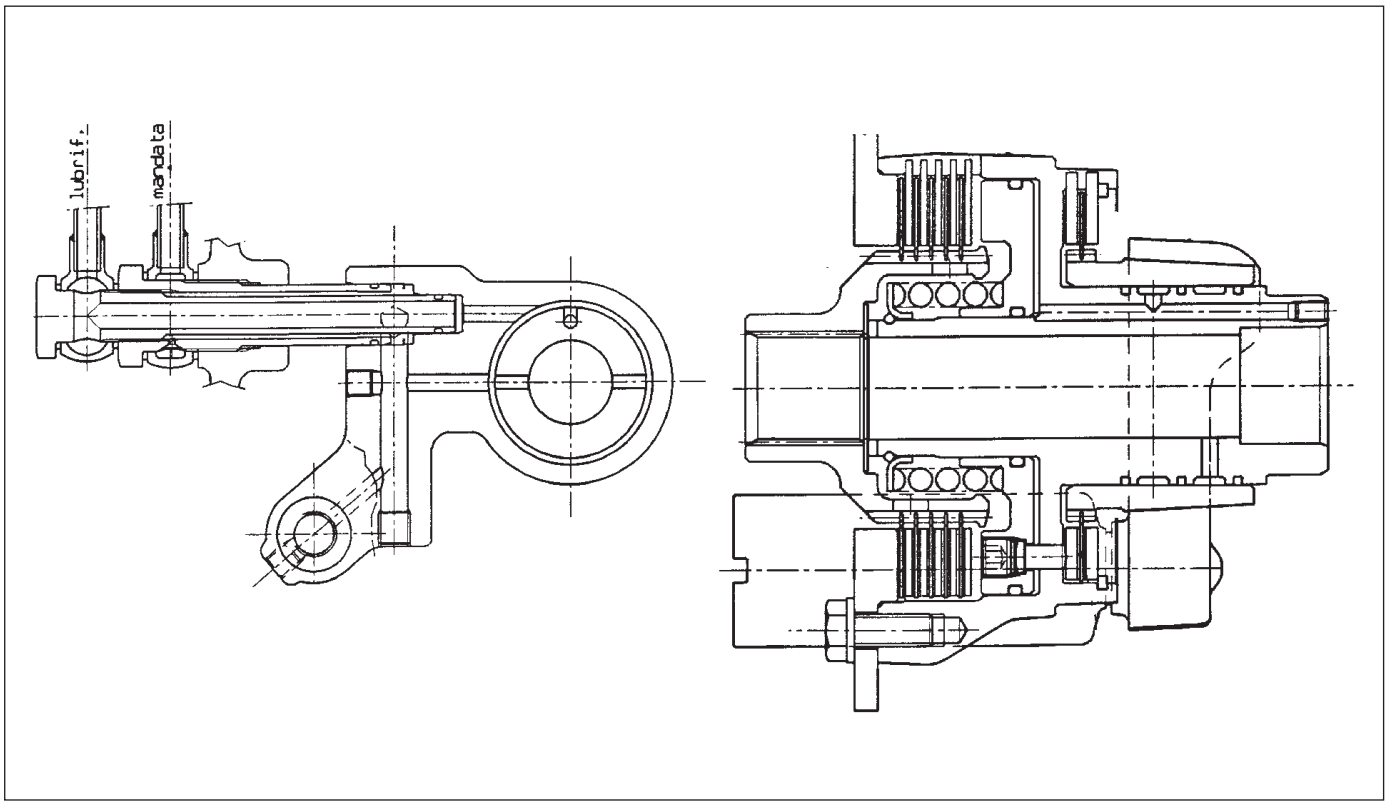


Fig. 1 - Checking clutch plate clearance.

Testing the pressure settings of the clutch control valve

Connect the test fitting code 009.2923.4 (M12x1.5 thread) to the valve block at position P as indicated figure 2. Then connect tool 5.9030.517.4 to the test fitting and attach pressure gauge 5.9030.514.0.
Connect a shut-off cock in position **A** (fig. 2).

Allow the oil to flow through the valve block, take the following pressure readings:

SOLENOID EV.4 NOT ENERGISED SHUT-OFF VALVE OPEN		SOLENOID EV.4 ENERGISED SHUT-OFF VALVE CLOSED	
Pressure control valve 2		**Pressure limiting valve 3**	
Pump capacity l/min	Pressure M bar	Pump capacity l/min	Pressure M bar
7	**5 max.**	**7**	**15 min. 18 max.**

1 - Filter (filtering capacity 25)
2 - Pressure control valve.
3 - Pressure limiting valve
4 - Solenoid valve ON / OFF

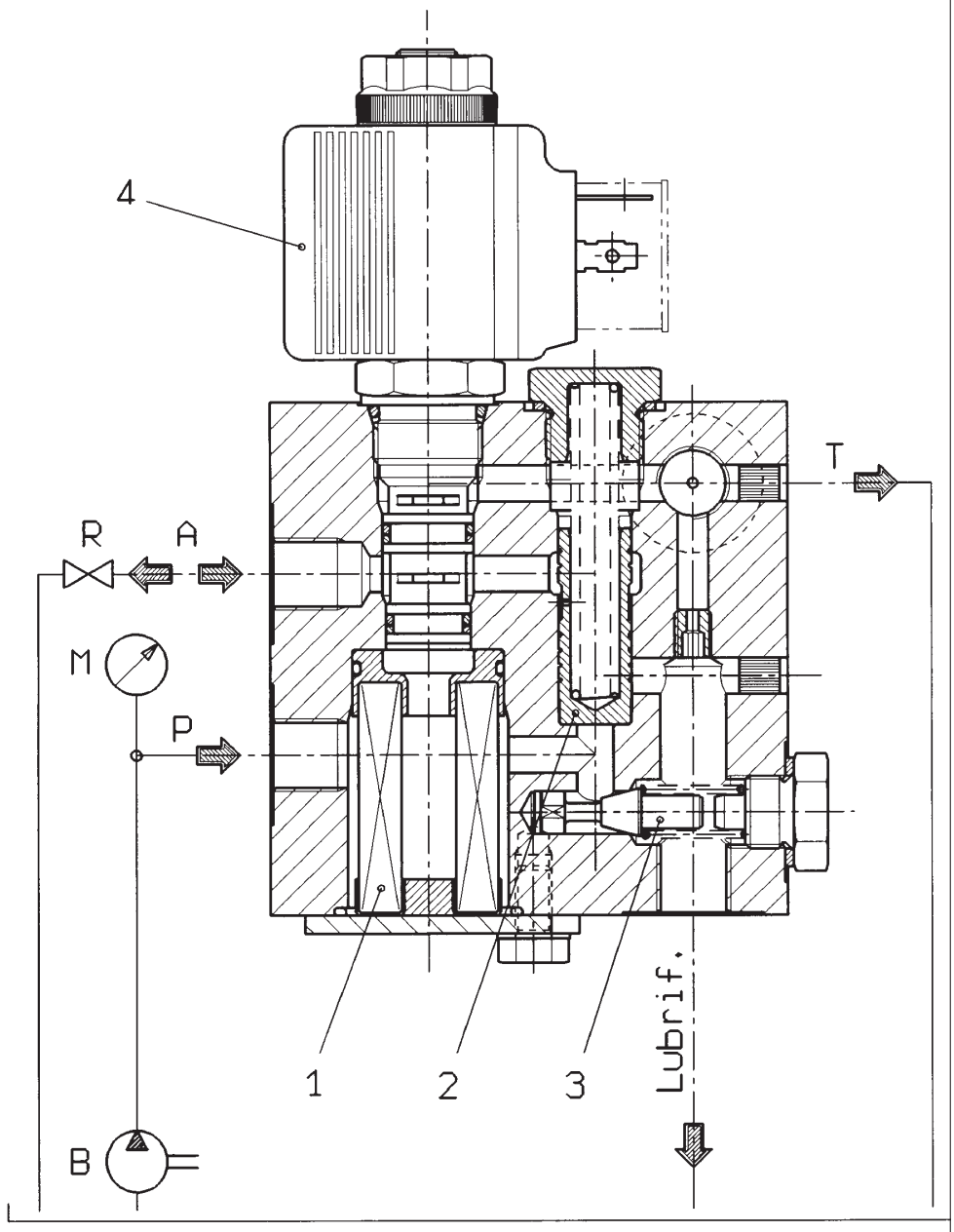


Fig. 2 - Testing the operating pressure in the P.T.O. clutch supply circuit.

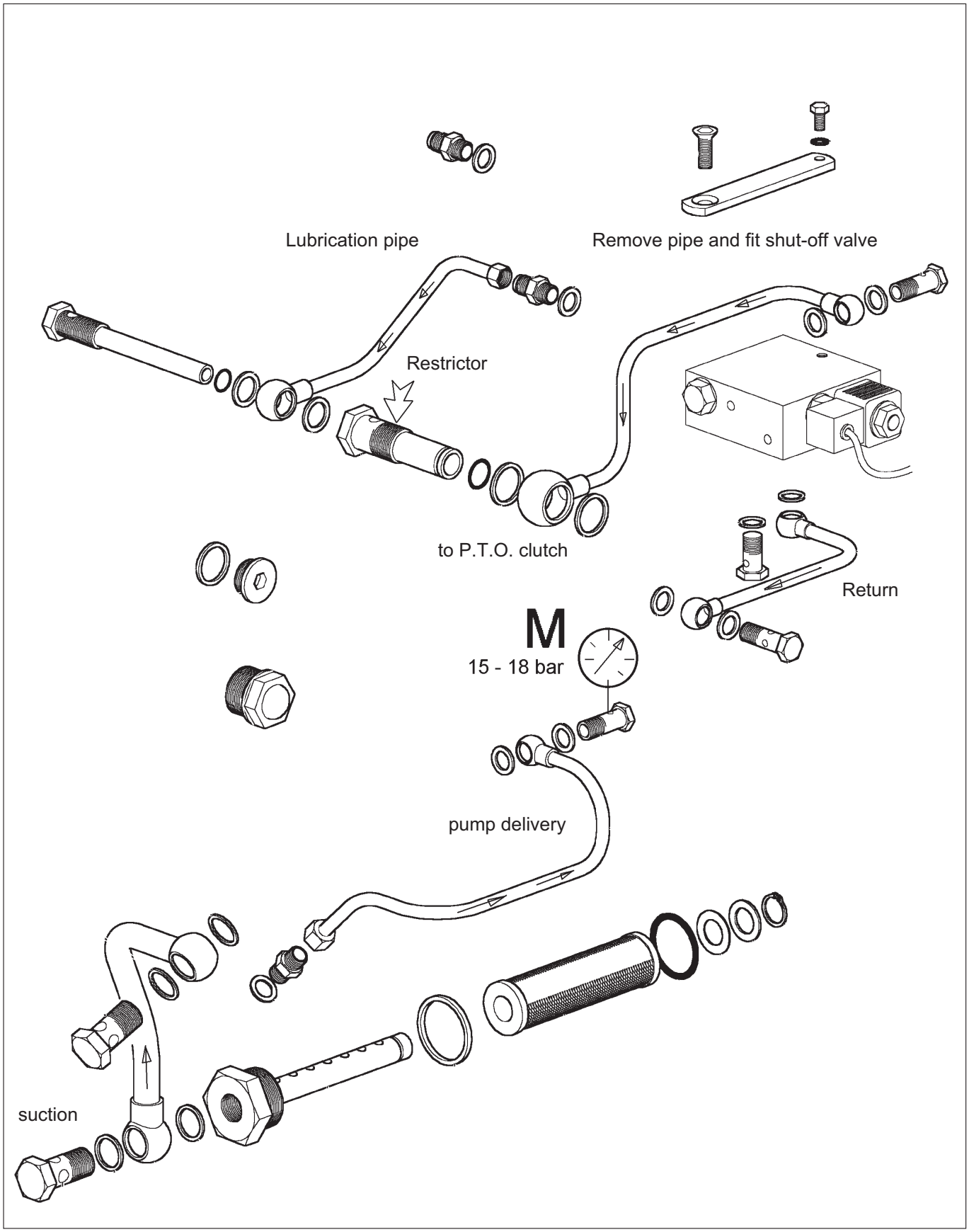


Fig. 3 - Hydraulic control and power system components.

Instructions for disengaging the drive to the front P.T.O.

When front P.T.O. is not to be used for long periods, it is recommended that the drive to the output shaft is disengaged.

Proceed as follows:
Remove cover **E**, then remove spacer **B** (fig. 4) and place in the bottom of cover **E** in position **D**, as indicated in the diagram on the right-hand side of figure **2** (drive shaft disconnected). Now pull the shaft **F** forward to disengage it from the drive coupling and position washer **A** in the groove **G**.
Replace cover **E**.

To re-connect the front P.T.O. drive, remove cover **E** and washer **A** and push the drive shaft in. Replace washer **A** as shown in the diagram on the left of the figure (drive shaft connected), then replace cover **E**.

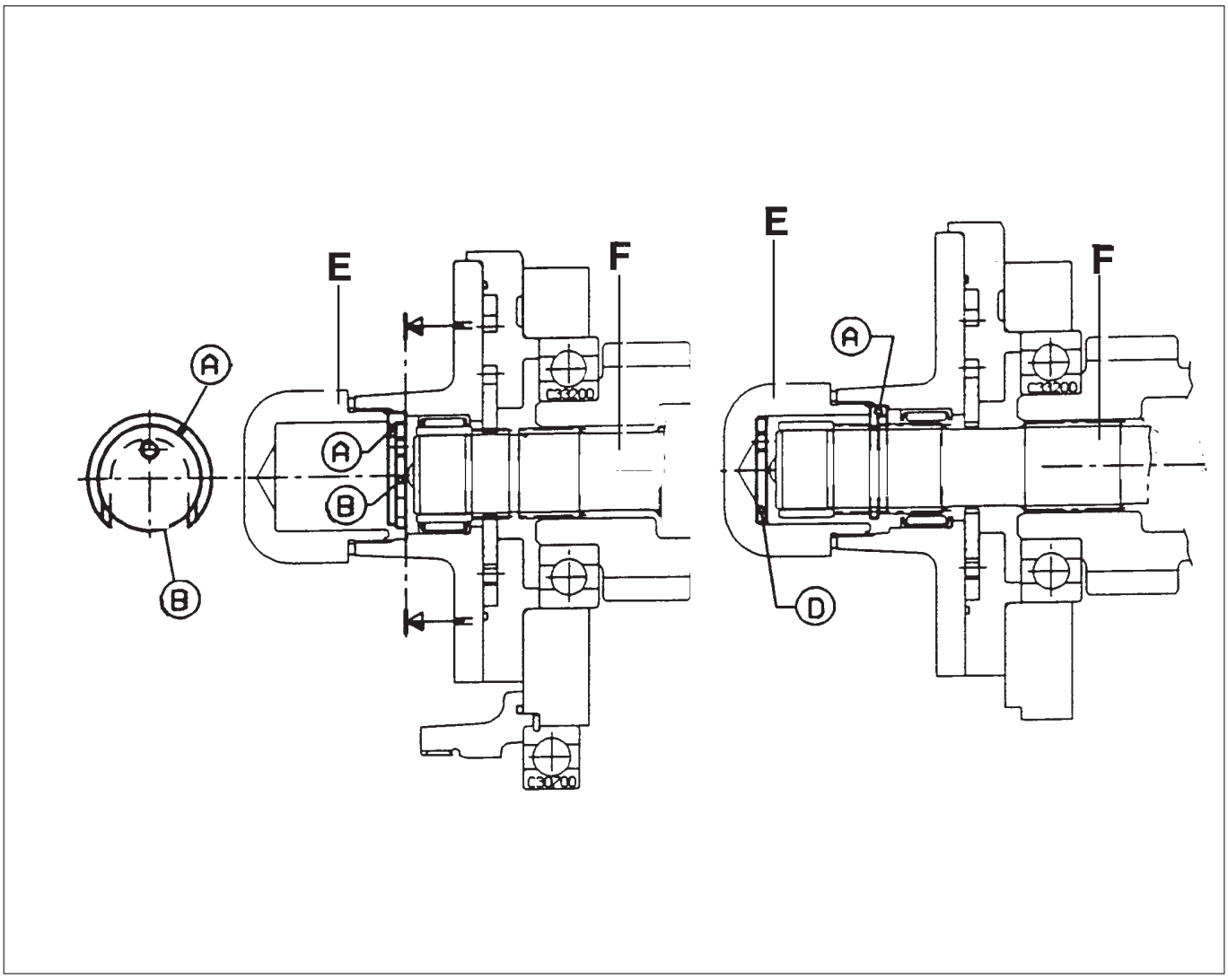


Fig. 4 - Drive shaft connected - Drive shaft disconnected

Fitting the "RING-FEEDER" rings - fig. 5

Fit the parts in the order shown in the figure.
Tighten down the pulley hub on the crankshaft applying a torque of 34 ± 1.5 kgm (333 ± 15 Nm).

Fitting the oil seal on the front P.T.O. shaft - fig. 6

To avoid damaging the seal during installation, first fit a protective sleeve **A** over the splines of the shaft as shown in figure 6.

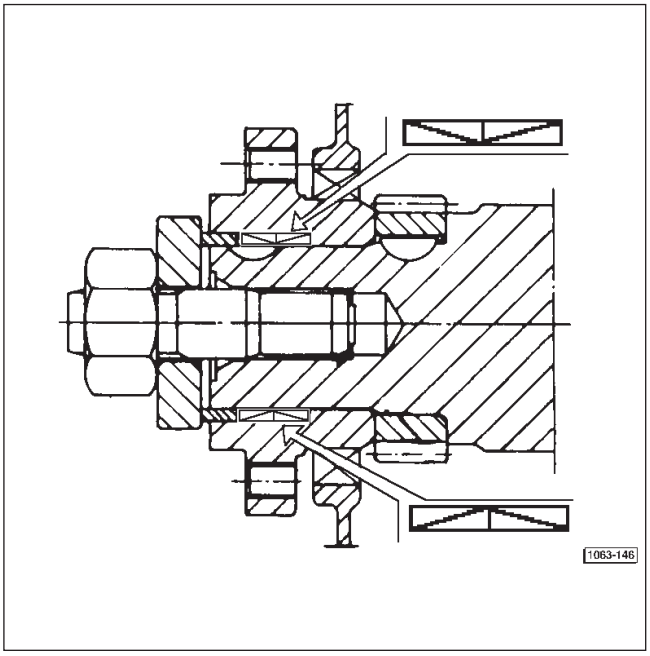


Fig. 5 - “RING-FEEDER” rings installed"

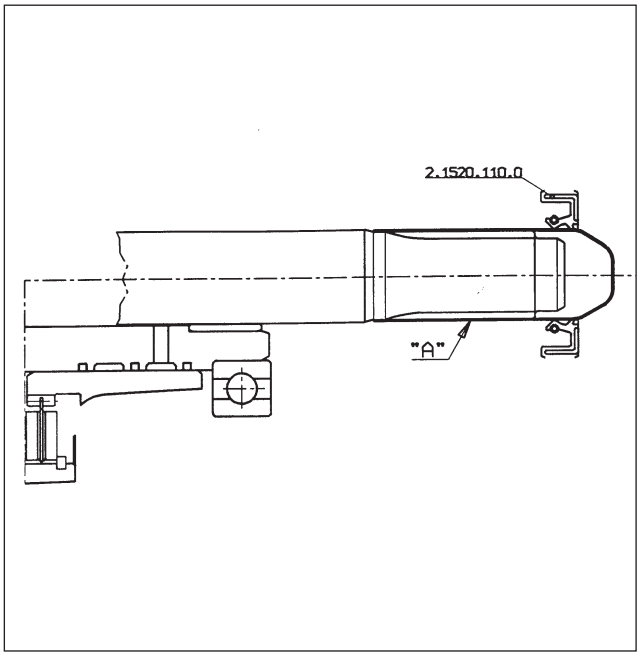


Fig. 6 - Front P.T.O. assembly parts.

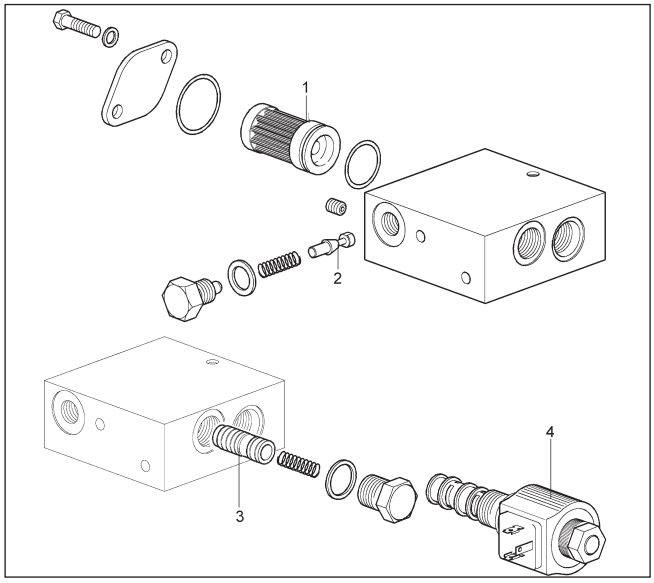


Fig. 7 Solenoid control valve block.
1 - Filter
2 - Pressure limiting valve
3 - Pressure control valve
4 - Solenoid control valve

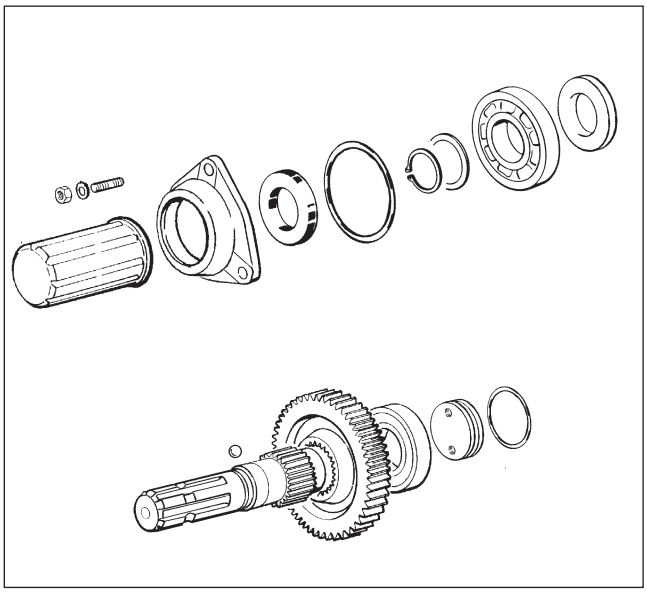


Fig. 8 - P.T.O. output shaft.

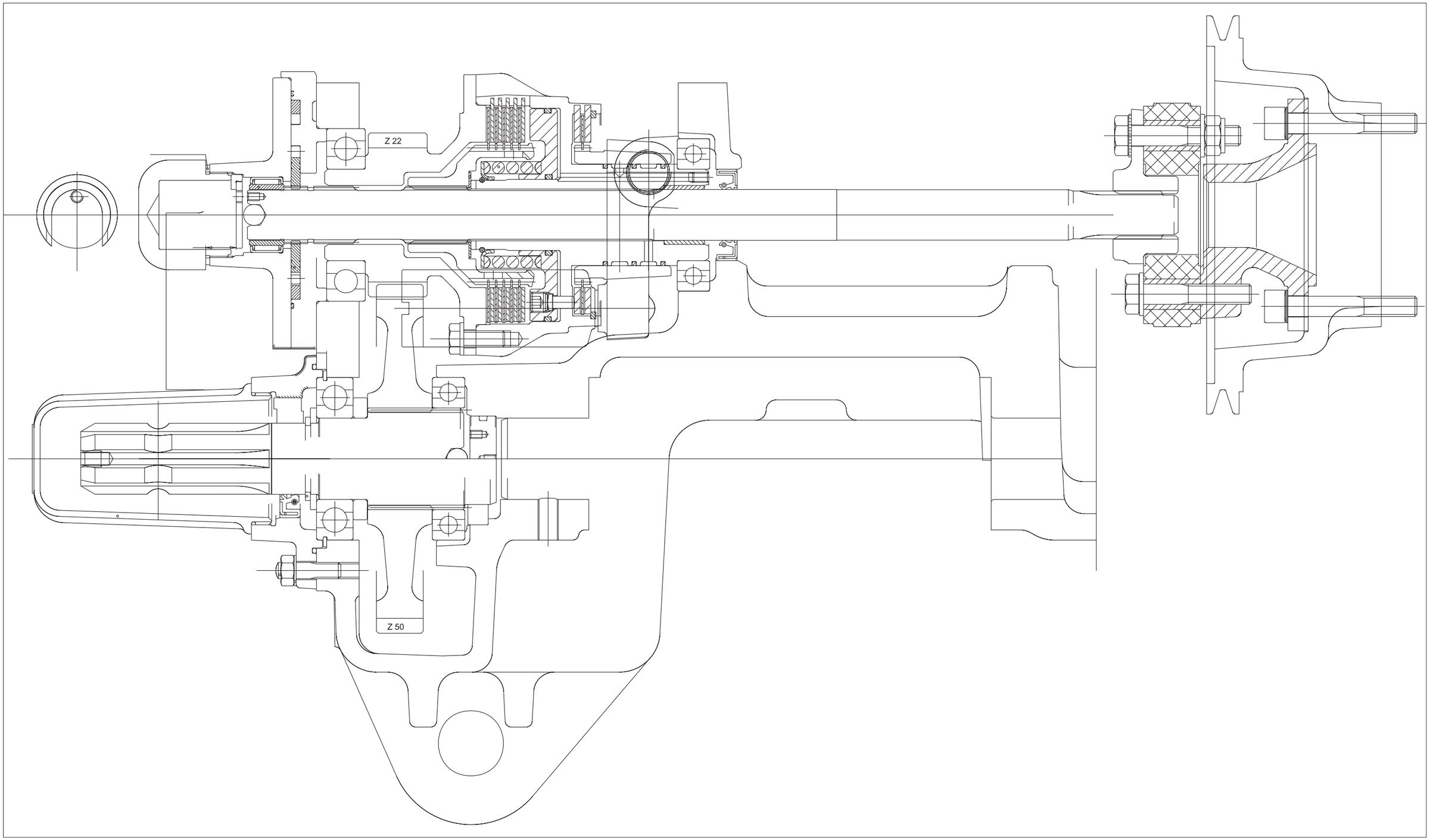


Fig. 9 - P.T.O. clutch hydraulic circuit.

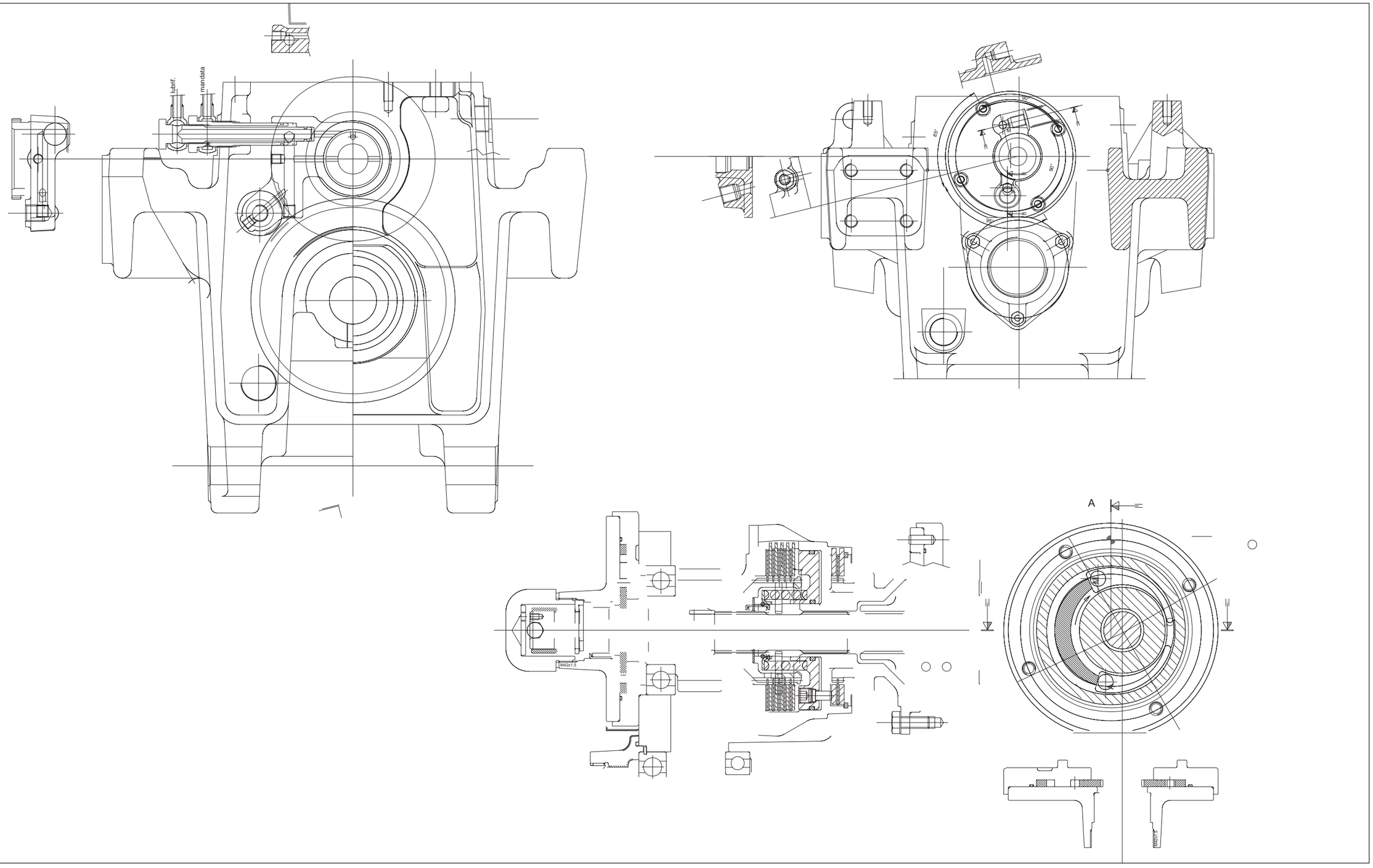


Fig.10 - P.T.O. clutch hydraulic circuit

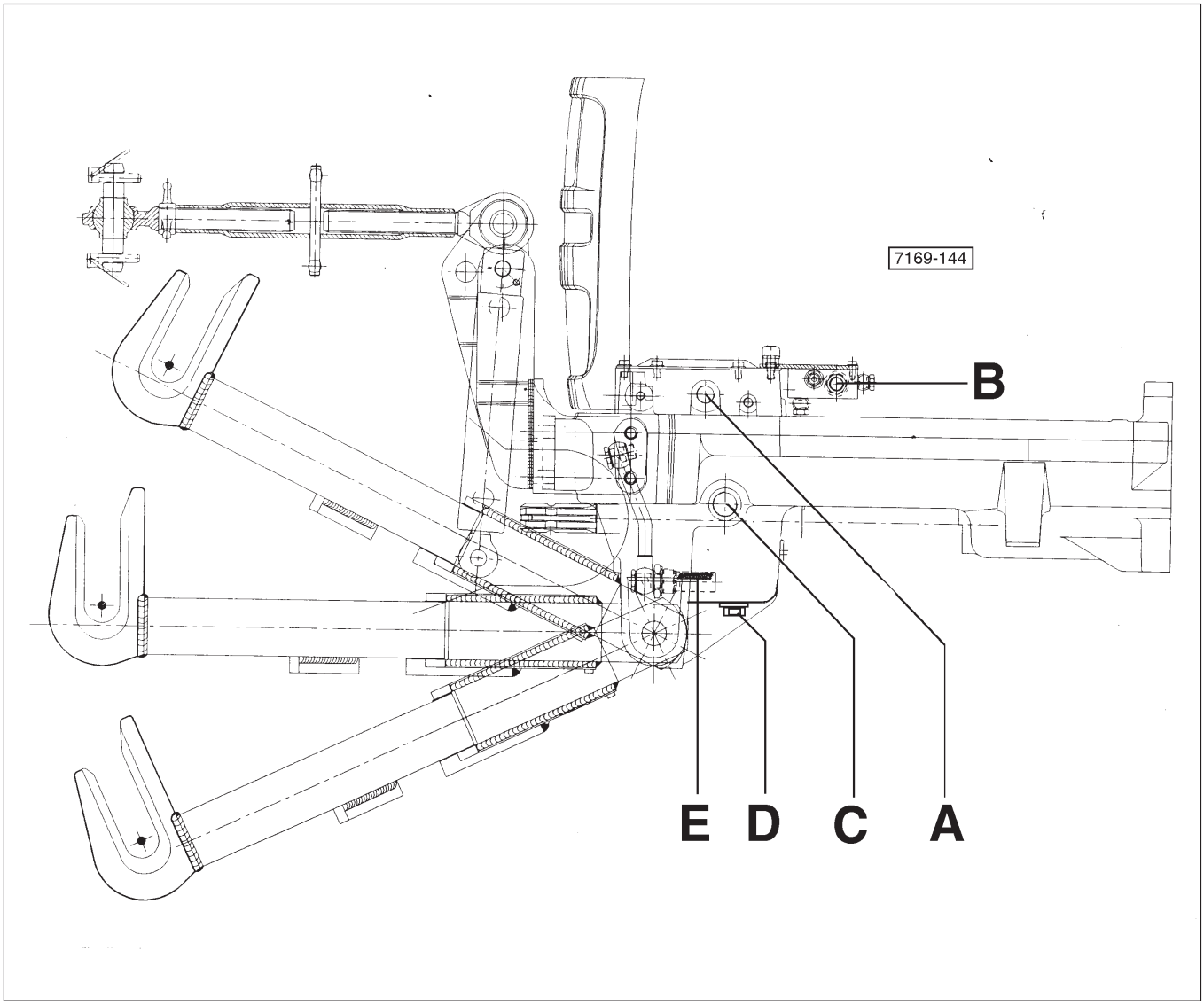


Fig. 11 - Servicing the front P.T.O. unit

A - Oil filler plug.

B - P.T.O. oil delivery line filter (to access for servicing, unscrew the two screws and remove the filter flange).

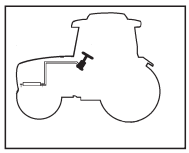
C - Oil level inspection window.
To access, remove the plastic plug located on the left0-hand side of the front carrier.

D - Oil drain plug.

E - Wire mesh filter on suction line (clean the filter if you notice any control malfunction; then, if necessary, also clean or renew the delivery line filter **B**).

Diagnosing malfuntions

	Verify activation of the solenoid valve	replace the solenoid valve if necessary	
	check piston for free stroke	replace the springs of inefficient valves	recheck pressure
		inspect O-rings on piston and manifold	replace the defective parts
clutch slipping			
	verify movement of piston	remove any roughness preventing smooth action	replace the solenoid valves if necessary
	inspect clutch discs for wear	replace discs	
	Check the solenoid operated control valve	replace the solenoid valve if necessary	
P.T.O. brake does not apply	Check engagement pressure	inspect O-rings and replace if necessary	
	piston jammed	replace out-of- setting valve springs	



Hydrostatic steering

Code 290.6310.4/10 Danfoss type for 4WD
Code 009.3141.4/10 for 2RM

General information

Turning the steering wheel operates a rotary distributor connected to an oil flow control shutter permitting the pressure oil to be directed to the hydraulic cylinder in the amount necessary to provide the desired steering action.

The steering system is an open-centre reactive type: the user can directly feel the steering response on the steering wheel and perform a prompt correction of the driving direction.
In addition, the remarkable wheel caster angle will help increase the wheel self-centering effect as the wheels will automatically get in a straight line when the steering wheel is released.

The hydraulic circuit is fully independent. The oil drawn from the gearbox flows first through a high-delivery filter before being routed under pressure to the distributor by a pump constantly in mesh driven by the engine and fitted on gearbox left-hand side.
Two powerful single-acting cylinders, connected to the front wheel hubs, ensure the thrust necessary to steering under any working conditions.

Oil returning from the power steering unit is cooled by a radiator in the engine compartment and then directed to the unit piloting the electrohydraulic controls and sole function to lubricate the gearbox.
Moreover this system offers the advantage that the tractor can be driven also in the event of either a pump failure or an engine stop. A valve within the distributor will allow the oil to be drawn from the discharge circuit through the oil control shutter, whenever a pressure drop occurs in the pump delivery circuit.

hydraulic control valve		
control valve		OSPC 80 OR, open-centre reaction-type valve
safety valve setting	bar	150 for 4WD
	bar	100 for 2WD
clearance between two bodies of control valve	mm	0,03
steering wheel rotation before wheels turn		
- with engine running		0° - 2°
- with engine off		0° - 6°

Steering cylinder - 2WD version		
working stoke of cylinder rod	mm	142
rod diameter	mm	22
piston diameter	mm	48
steering wheel turns to achieve full-lock	number	2,17

Steering cylinder - 4WD version		
working stoke of cylinder rod	mm	170 (85 mm for part)
rod diameter	mm	16,2
piston diameter	mm	58
steering wheel turns to achieve full-lock	number	2,17

tightening torques				
control valve cover bolts	kgm	3	Nm	2
cylinder retaining bolts	kgm	16	Nm	155

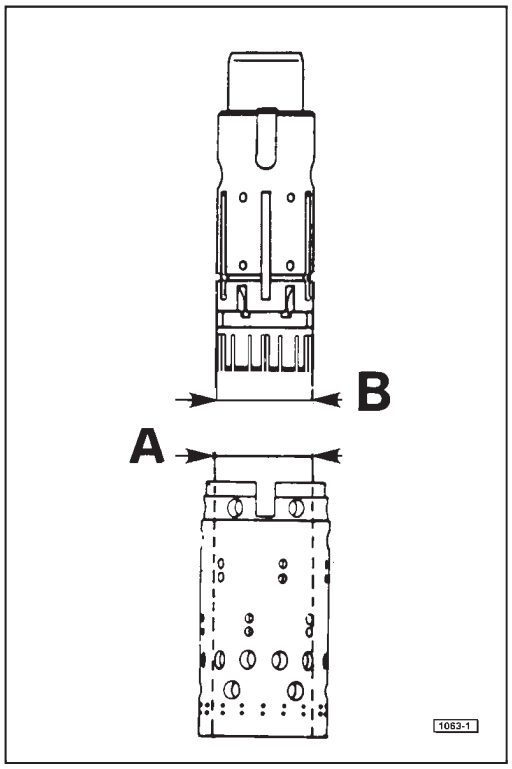


Fig. 1 - Fit diameter between internal and external elements of the control valve.

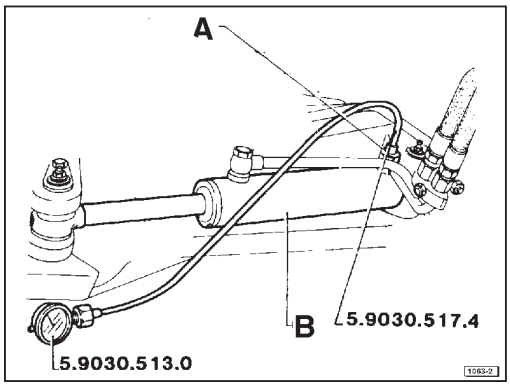


Fig. 2 - Testing pressure in the hydraulic steering circuit.
A - fitting
B - cylinder

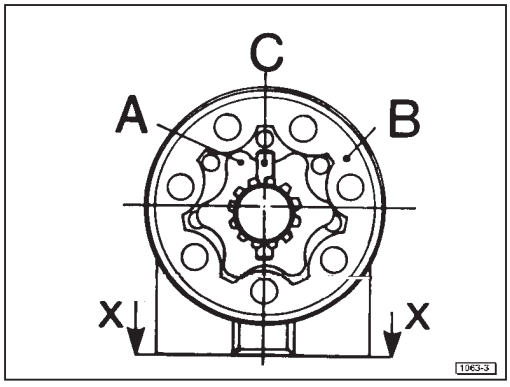


Fig. 3 - Correct assembly of orbital pump unit

Inspections and checks

Steering pump
Clean all component parts of the pump thoroughly, using petrol.

Check that the gears are not unduly worn, otherwise replace.

Directional control valve
Clean all parts of the valve thoroughly.
Check, utilizing a comparator for internal dimensions, that the clearance between spool and sleeve (**A - B** - Fig 1) is no greater that 0.03 mm: otherwise replace.
The two components are not available separately as spare parts.

Check the setting of the pressure relief valve
Couple pressure gauge 5.9030.513.0 by means of adaptor 5.9030.579.0 to the fitting of the left hand steering cylinder (as indicated in fig 2).

Start the engine and allow to idle for a few minutes; then, with the steering on full right hand lock, turn the wheel further until the hand of the gauge stabilizes at around the maximum pressure. If this is different to the maximum operating pressure (150 bar), adjust the setscrew (**A** - fig 10, page 300) of the valve until the requisite value is re-established in the circuit.

Repeat the same sequence of operations with the pressure gauge and adaptor fitted to the right hand steering cylinder, this time on the left hand lock.

Bleeding the hydraulic circuit
Start the engine and allow to idle at minimum rpm.

Loosen the two fittings at the cylinder and turn the steering wheel each way until oil emerges devoid of air bubbles.

Assembly of orbital pump unit
Locate the rotor **A** in the stator **B** as illustrated, making certain that the pin **C** is perpendicular to the plane **x - x** of the valve body as indicated in fig 3.

Teering wheel shaft

Verify the integrity of the steering column, and in particular, make certain that the surfaces of the bearings are not scored, also that the splines are neither damaged nor showing signs of excessive wear.
Do not grease or oil the bearing inside the sleeve.

Check that the steering column rotates freely, without sticking, but also without excessive play.
Before refitting the rubber boot **A** (Fig 4), smear the steering column **B** with the specified grease at the position indicated in fig 4.
Having tightened all components, check that the steering wheel continues to operate correctly even when at its two height adjustment limit positions (fully raised and fully lowered).

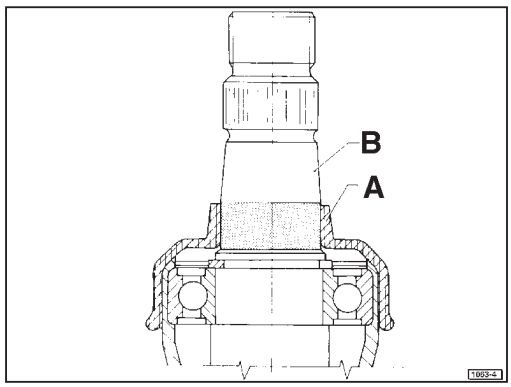


Fig. 4 - Section through steering wheel shaft
A - Rubber boot
B - Steering column

Steering cylinders

Hold the cylinder in a vice, unscrew ring nut **B** and withdraw the piston **L** by pulling the rod **N** (fig. 7).
If necessary, remove the circlip **F** (fig. 8) from its groove with the aid of a suitable tool to allow withdrawal of the ball end **G** (fig. 8).
If necessary, remove the seal (**F** - fig. 7) from the cyilinder with the aid of a screwdriver from the piston.

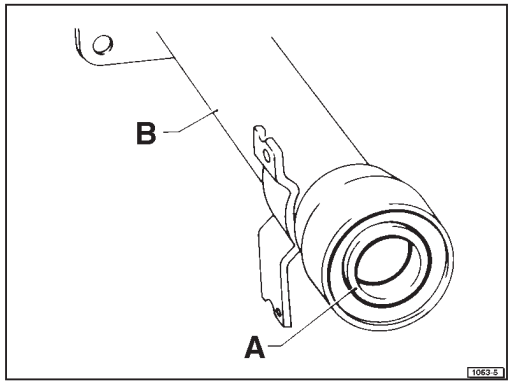


Fig. 5 - Steering column.
A - Bearing
B - Sleeve

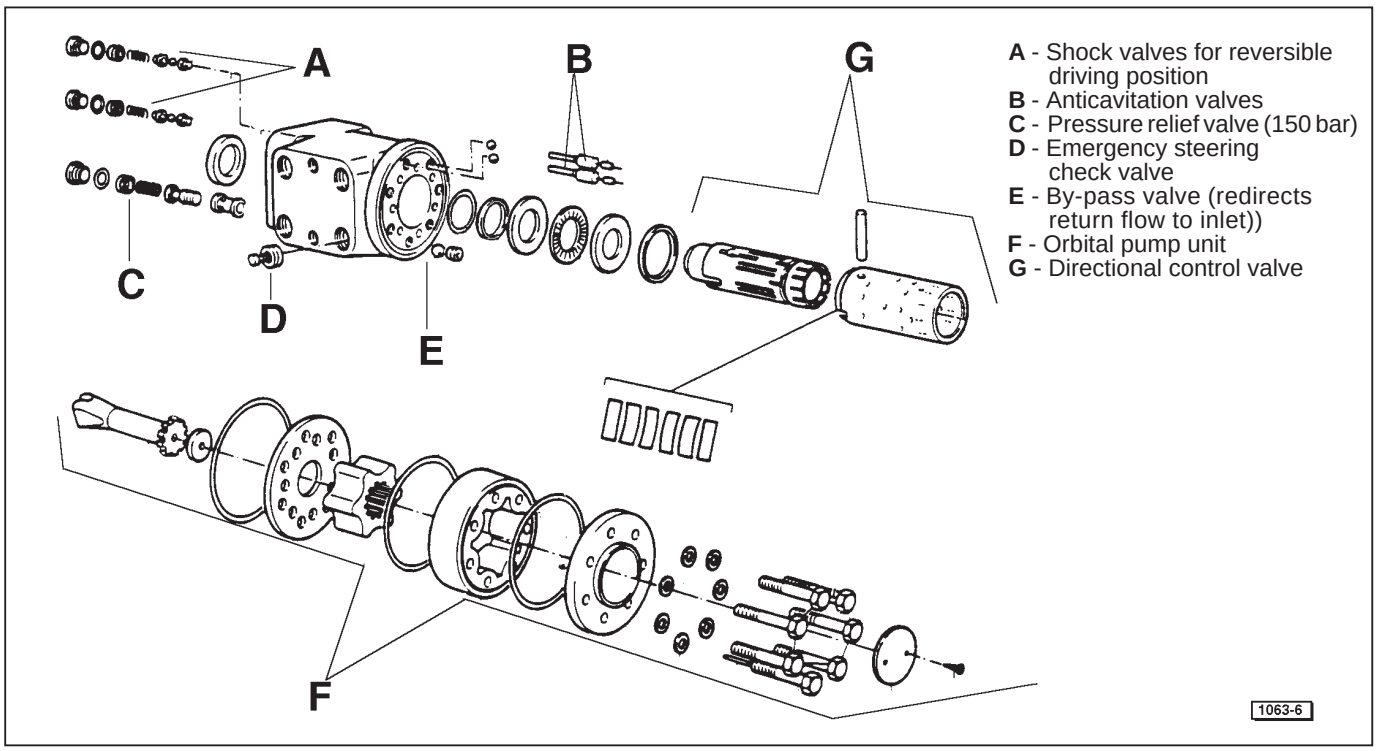


Fig. 6 - Components of directional control valve

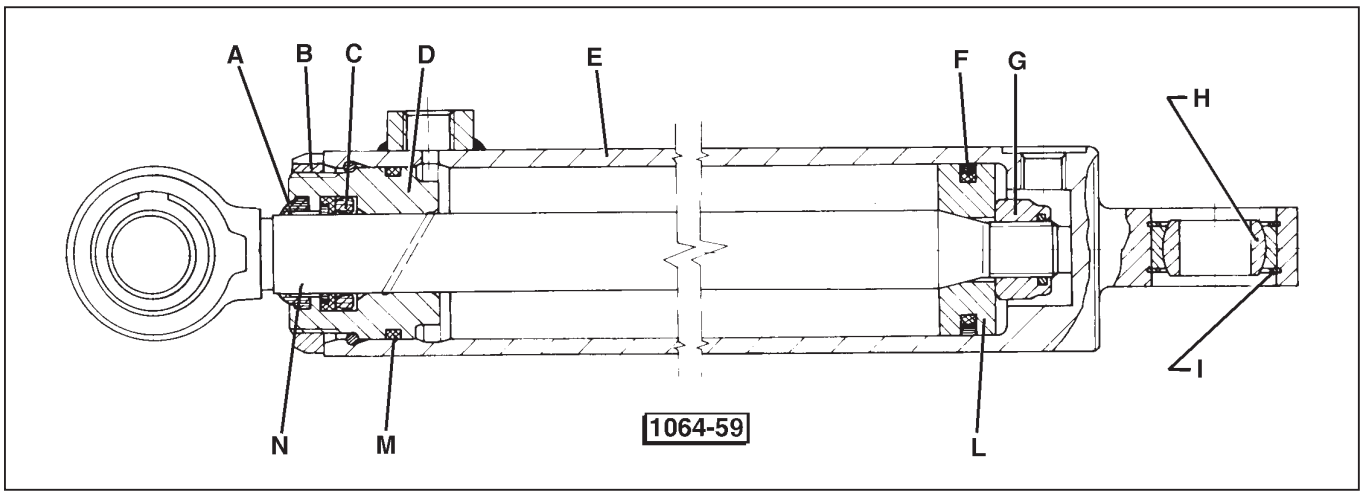


Fig. 7 - Double-acting cylinder - **2WD**.

A - Dust seal
B - Ring nut
C - Seal
D - End cap
E - Cylinder barrel
F - Seal
G -Bball joint
H - Ball end
I - Circlip
L - Piston
M - Seal
N - Rod

Servicing

Verify the integrity of the O-ring and wiper seals. Replace any components showing signs of wear or damage.
Inspect the sliding surfaces both of the rod and of the cylinder for signs of scoring, heavy wear or any kind of damage; if any of these are discernible, replace the affected component.
Check for correct operation of the ball end, ensuring that it betrays no signs of heavy wear, scoring or oxidation, the presence of which dictates replacement.
Lubricate the wiper seals and ball end with care.
Reassemble the cylinder, repeating the dismantling operations in reverse sequence.

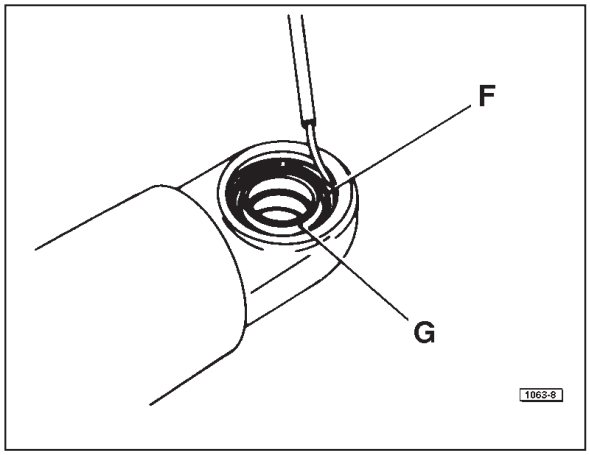


Fig. 8 - Ball joint
F - Circlip **G** - Ball joint

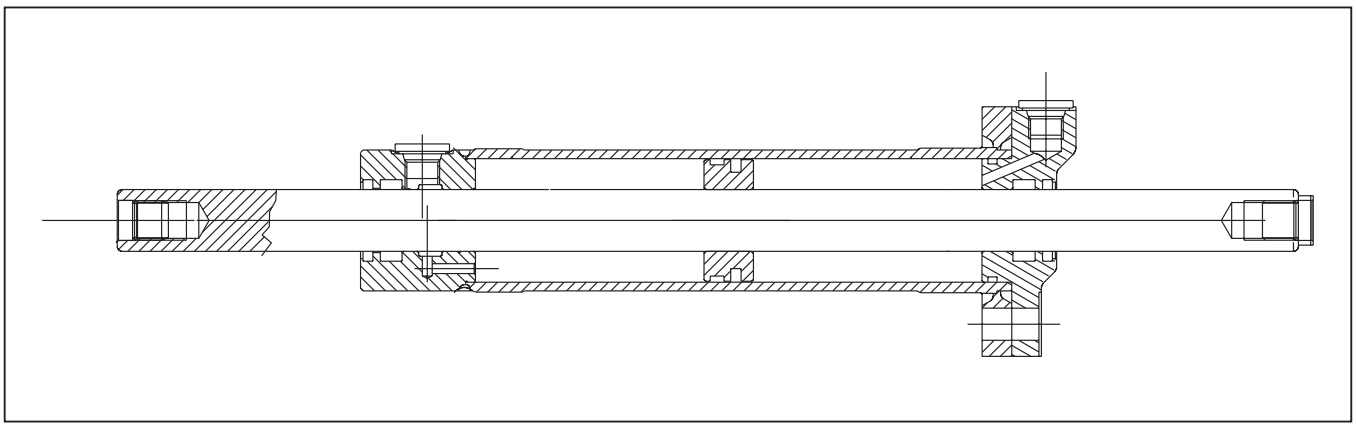
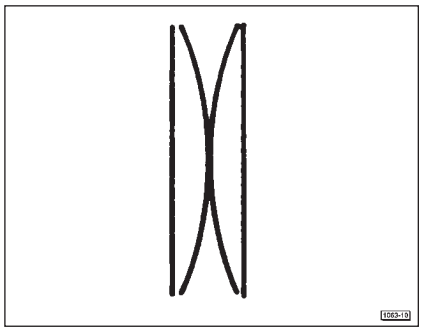
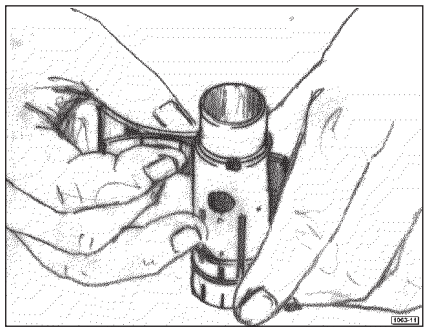


Fig. 9 - Section view of double-acting cylinder - **4WD**.

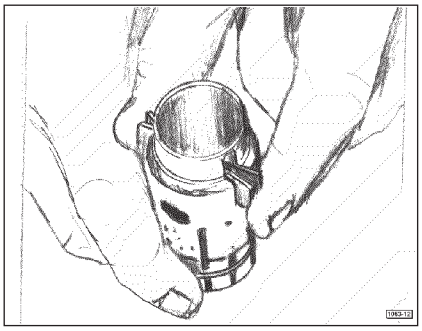
Instructions for the hydrostatic steering distributor assembly



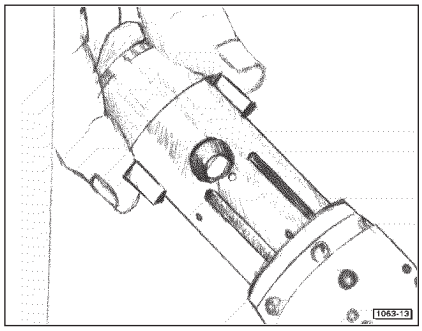
1 - Insert the spring set (blue) as shown in figure.



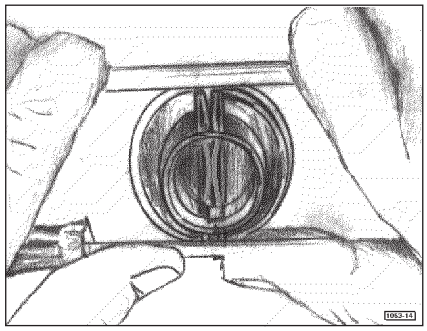
2 -Insert first the two external spring blades into the rotary distributor and subsequently the two internal ones.



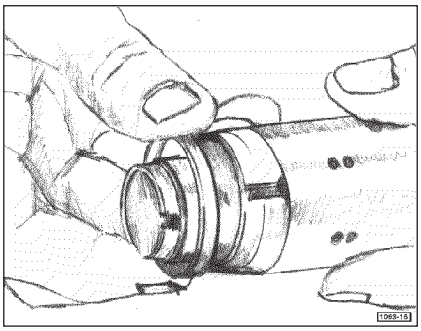
3 - Press the spring blades to align in centre position.



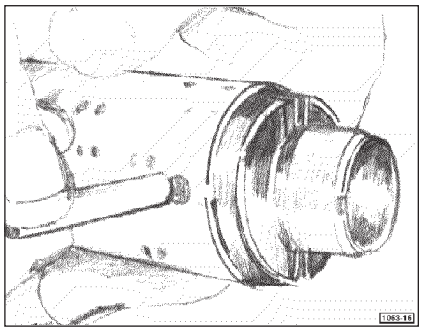
4 - Reassemble the rotary distributor.



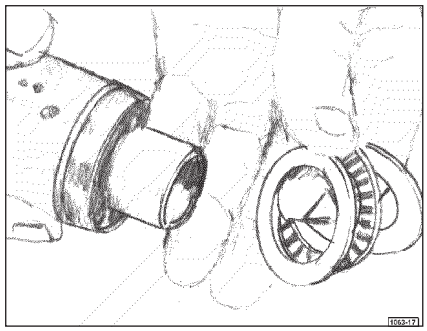
5 - Act on springs in such a way they are prevented from projecting outwards with respect to the rotary distributor outside surface.



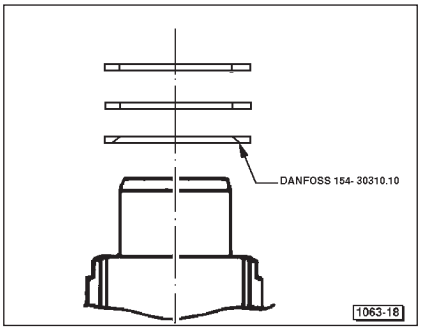
6 -Fit the spacer onte the distributor.



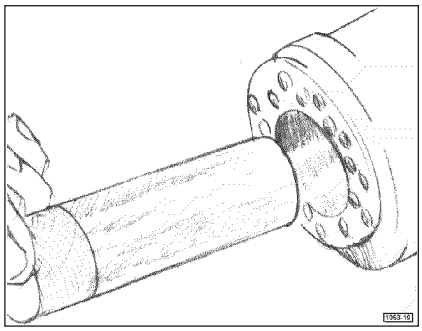
7 - Insert the drive pin into the cylinder



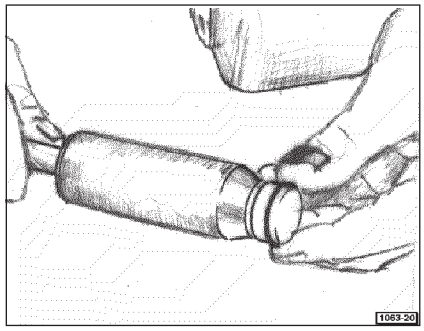
8 - Install the axial bearing following the assembly order shown in next figure.



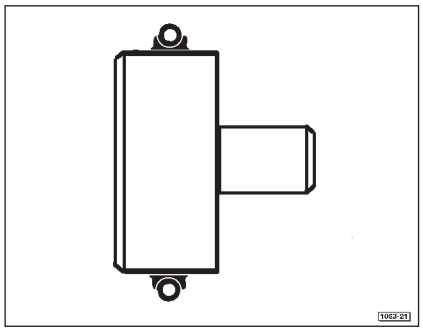
9 - Axial bearing installation.



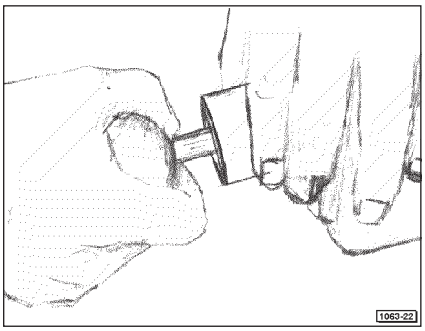
10 - Use no. 5.9030.480.0 tool to fit the seal ring.



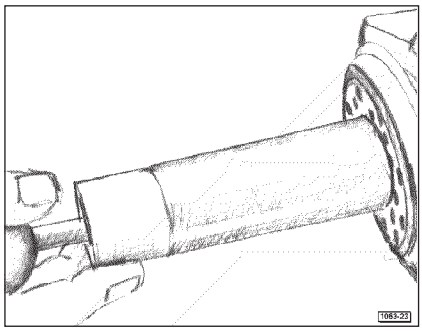
11 - Insert the O-ring after fitting the tool sleeve into the distributor case.



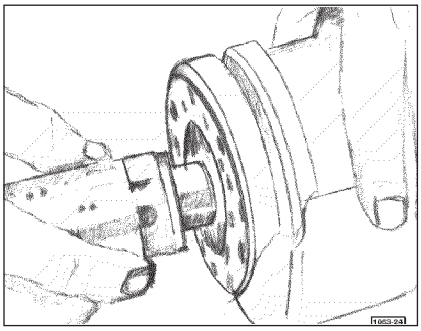
12 - O-ring as installed on the plug.



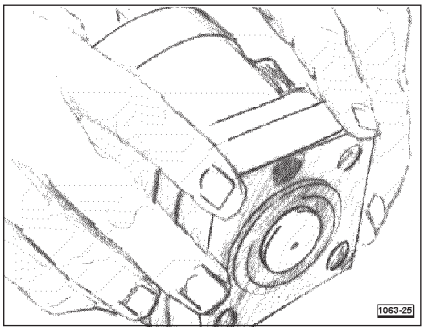
13 - Insert the O-ring into the sleeve and press until it is fully inserted.



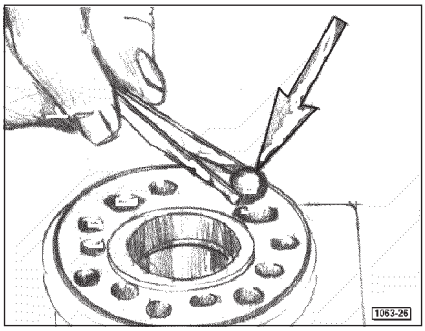
14 - Pull out the tool.



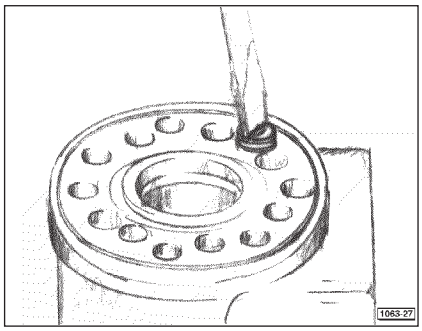
15 - Insert the distributor from the bottom trying to make installation easier alternately rotating in both directions.



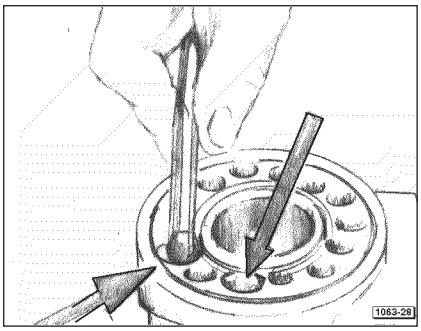
16 - Press the distributor until the plug is pushed out of the opposite side.



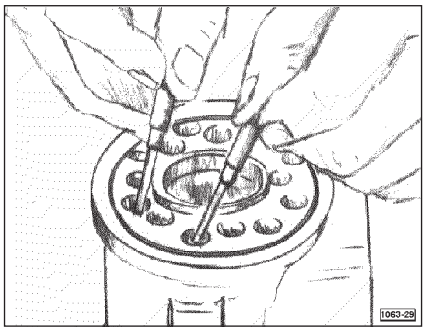
17 - Turn the distributor case upside down holding the rotary distributor in place and iserting the shunt valve as indicated by the arrow in figure.



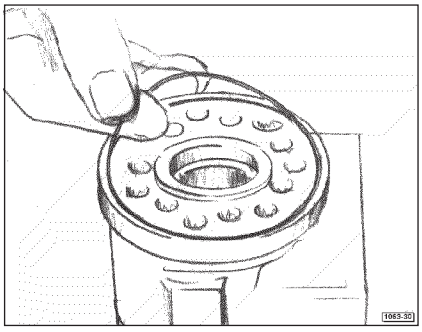
18 - Screw down the valve grub screw in its seat.



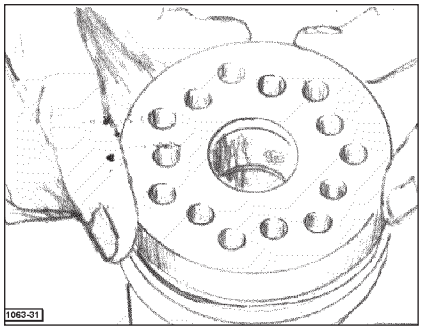
19 - Insert the balls as shown by the arrow.



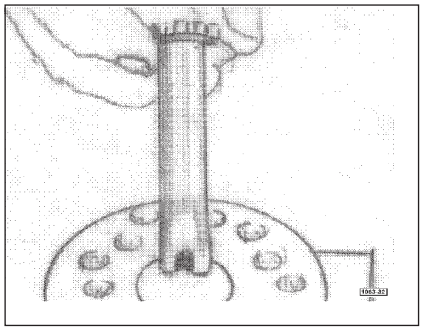
20 - Insert the two pins in the same positions as previously shown.



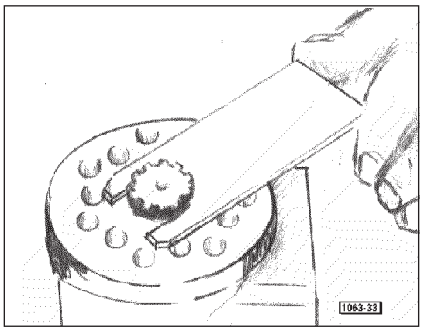
21 - Fit the seal ring in the related seat.



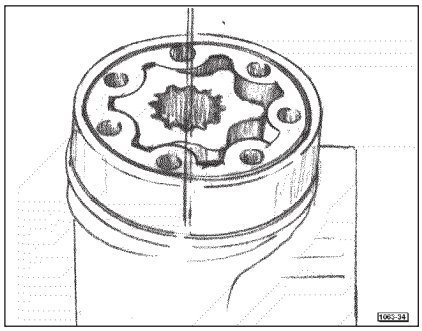
22 - Fit the spacer in such a way the holes in it are in the same line as the corresponding grooves.



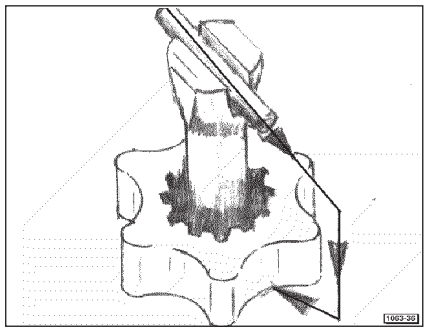
23 - Insert the drive shaft in the distributor so that the notching may engage into the pin in a position being parallel to the distributor face.



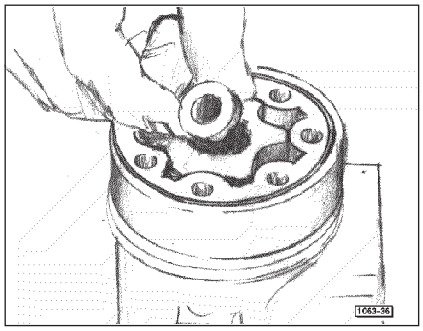
24 - Place the fork as shown in figure so as the drive shaft may be held in position.



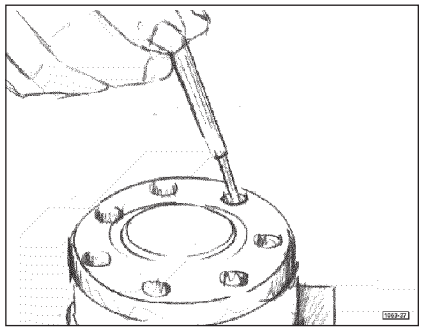
25 - After inserting the two O-rings properly lubricated into the oil flow control shutter fit the stator positioning as shown in figure.



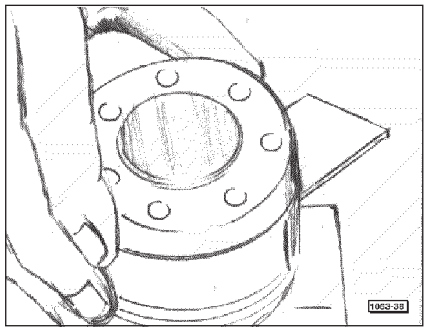
26 - The rotor should be placed to the stator and then swiveled with regard to the drive shaft as shown in figure.



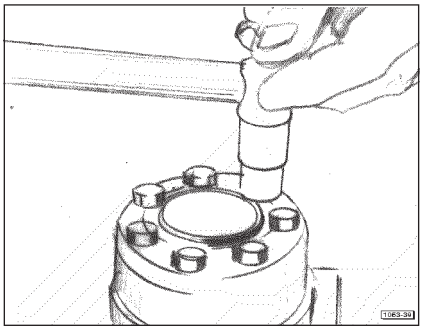
27 - Place the spacer ring, (only for those models which the spacer is fitted).



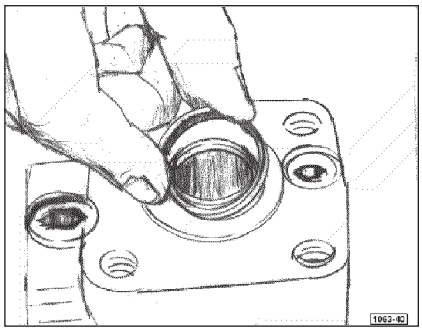
28 - Install the pin screw in the special hole (the ball valve hole).



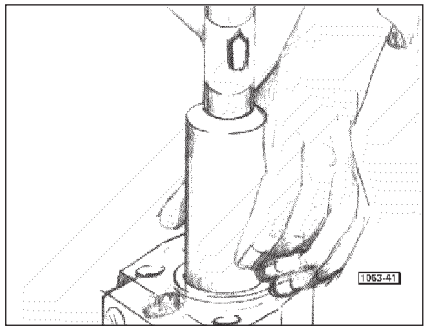
29 - Fit the cover.



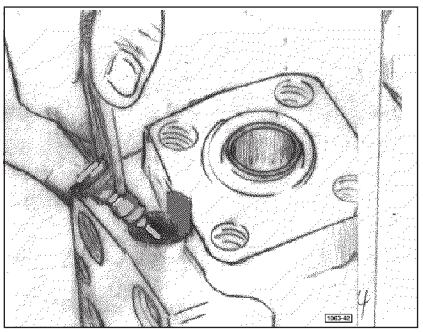
30 - Fix the cover with the securing screws by tightening to the recommended torque.



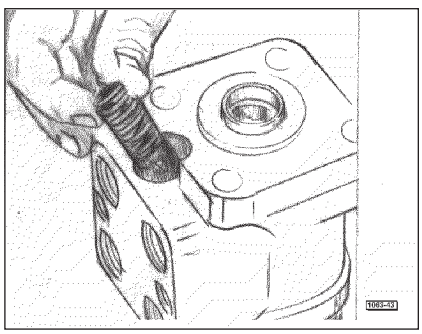
31 - Fit the seal ring.



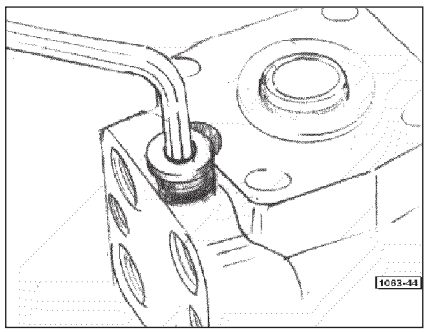
32 - Insert the seal ring into its seat.



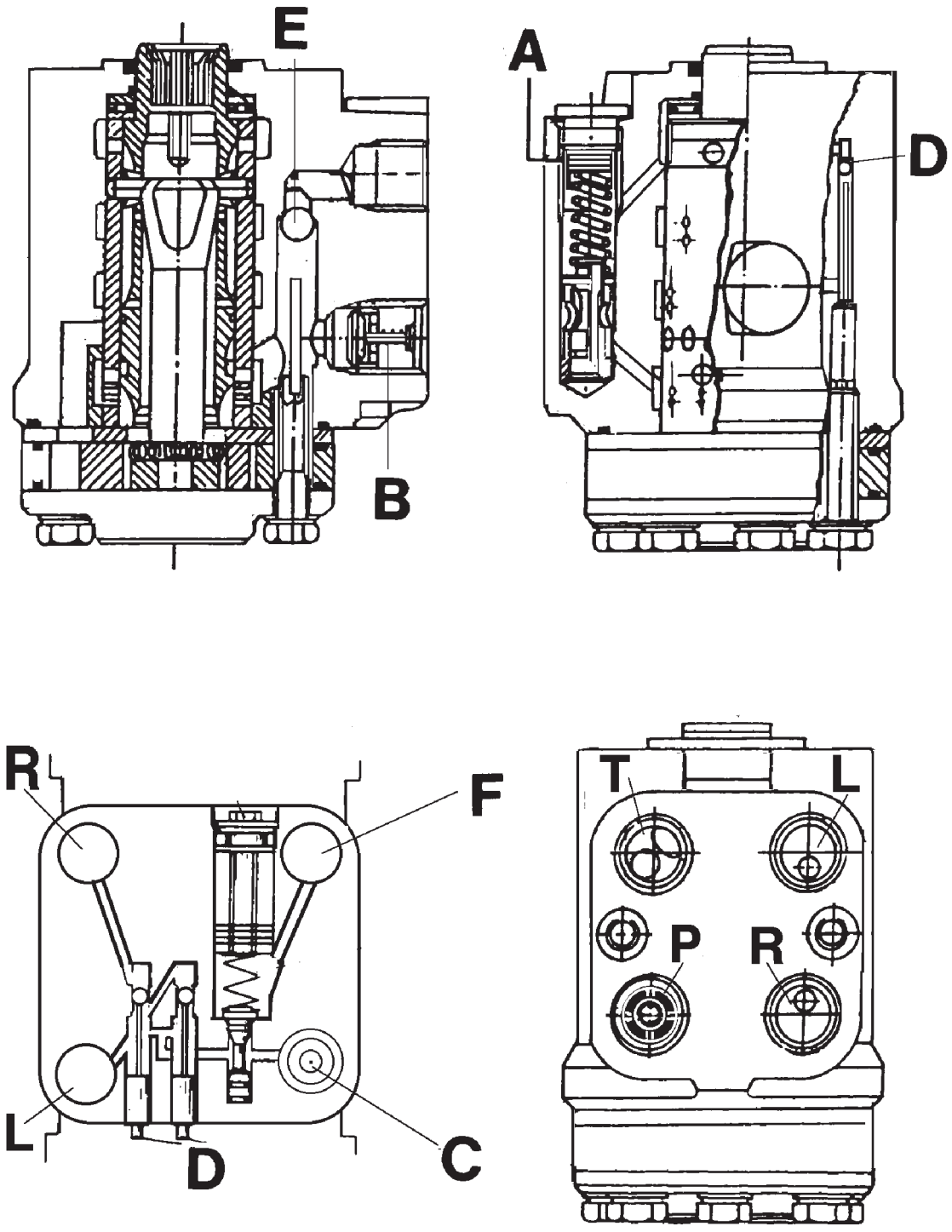
33 - Install the pressure relief valve.



34 - Insert the spring.



35 - Fit the pressure relief valve cap screw.



A - pressure relief valve
B - check valve
C - from pump
D - anticavitation valve
E - emergency steering check valve
F - return flow
L - to left hand cylinder
R - to right hand cylinder
P - pump
T - discharger

Fig. 10 - Distributor cross-section.

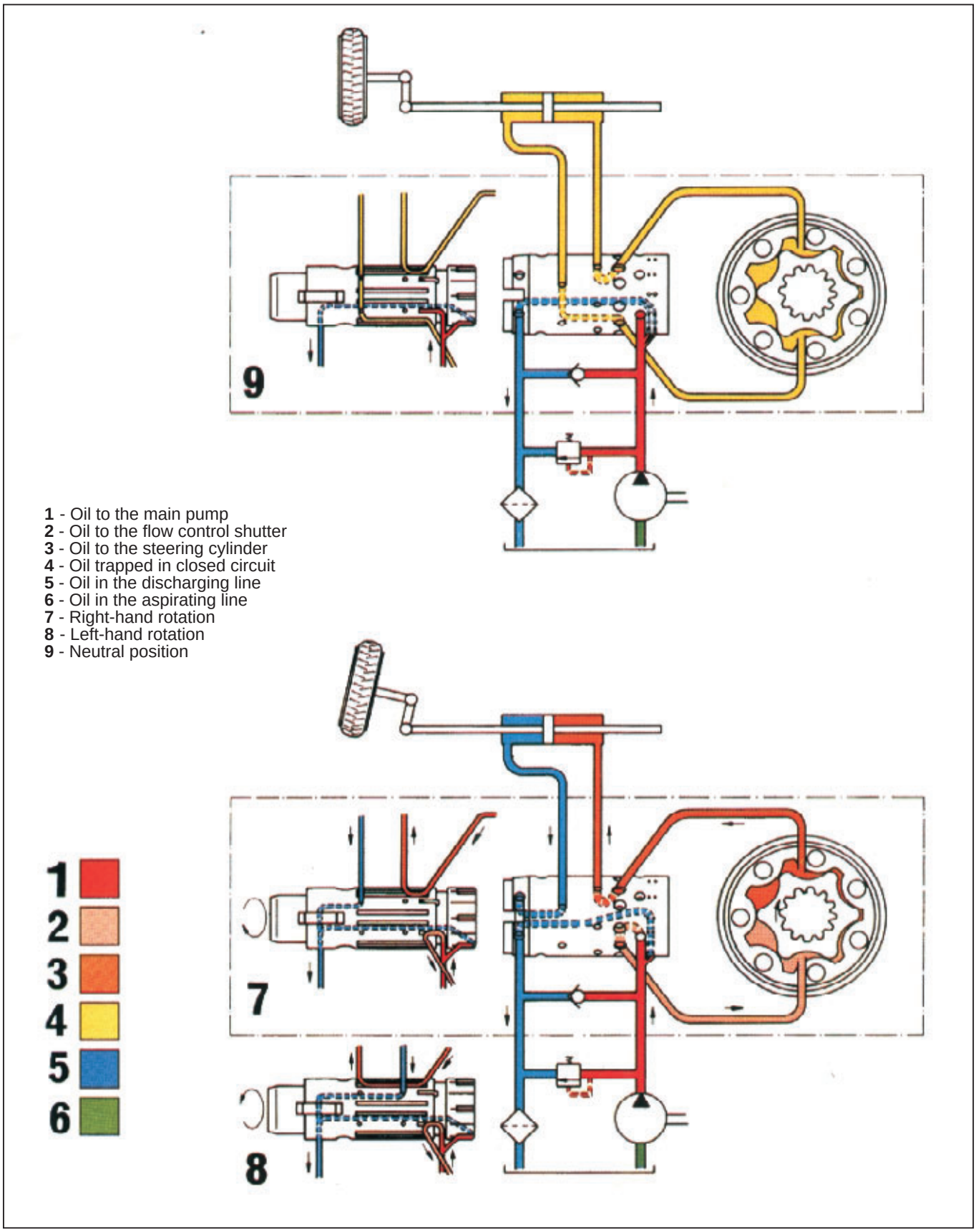
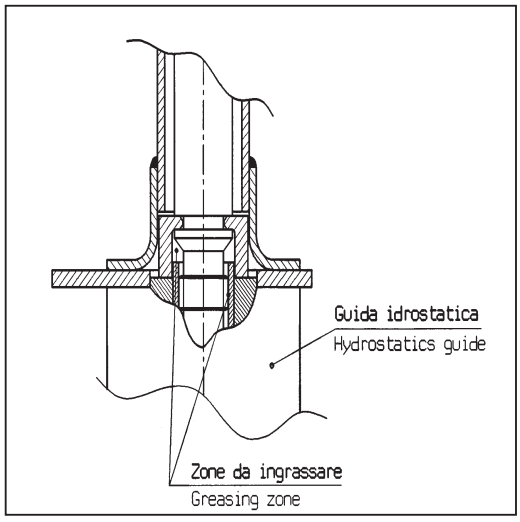


Fig. 11 - Reactive-type, open-centre hydraulic distributor schematic diagram.



Greasing the steering shaft

On assembly, grease the areas shown on the left with Molikote GN-PLUS

Fig. 12 - Greasing the steering shaft.

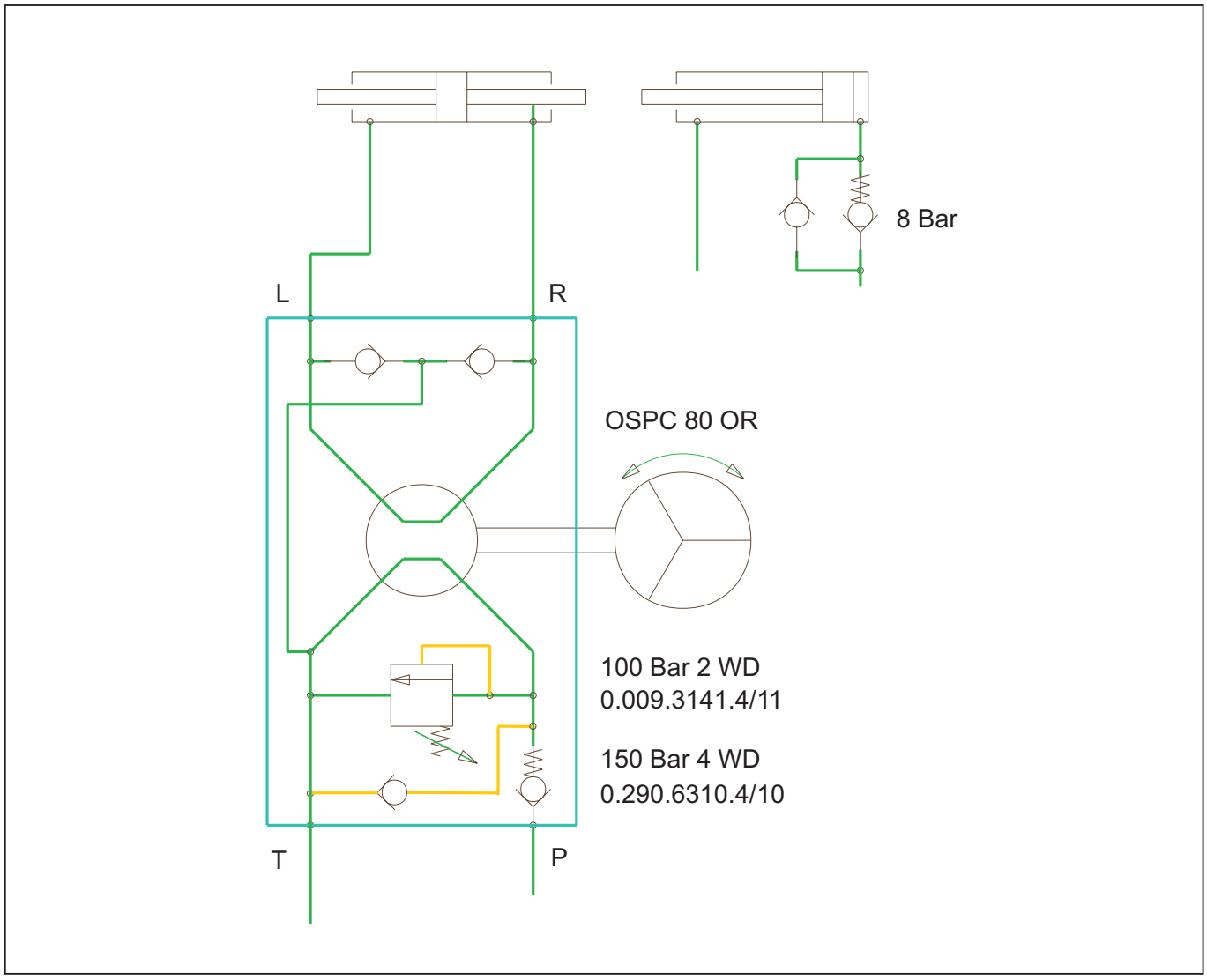
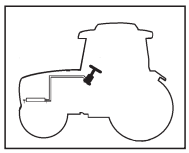


Fig. 13 - Hydrostatic steering operation diagram.



Diagnosing malfunctions

Unable to steer tractor	Check oil level in tank	Restore level	
		Check for air in system	Bleed the system
	Check safety valve setting	Check for shorn distributor securing pin	Replace pin
Hard steering		Reset or replace if necessary	
	Check safety valve setting	Check for air in the system	Bleed the system
			Check for leakage from the distributor and the piston seal rings in the hydraulic cylinder — Replace worn seal rings
Distributor and oil flow control shutter worn	Check oil filter for dirt	Change oil and replace filter	
		Check that the oil is of prescribed type	Change oil and replace filter
Blocked distributor	Check for dirty oil	Clean distributor and replace oil	
		Check that oil is not excessively hot	Stop tractor and let oil cool
Wheels oscillation		Bleed the system	
	Check for air in system	Check for oil leakage in the hydraulic cylinder	Replace seal ring
		Check valve setting	Adjust

Mechanical controls

Given the constructional simplicity of these controls, we present a series of figures illustrating the individual components of the controls.
Only some of these controls require adjustment; for instructions for these adjustments, refer to the chapter dealing with corresponding unit.

Fig. 1 Mini-reduction/shuttle control
Fig. 2 Gearshift controls
Fig. 3 Range gear control
Fig. 4 Internal gear controls
Fig. 5 Internal shuttle control
Fig. 6 Differential lock control
Fig. 7 Internal differential lock control
Fig. 8 Handbrake control
Fig. 9 Handbrake control (part under footplate)
Fig. 10 4WD - Front differential lock
Fig. 11 4WD - Drive shaft
Fig. 12 4WD - Power take-off
Fig. 13 4WD engagement control
Fig. 14 2-speed P.T.O.
Fig. 15 3-speed P.T.O. - 60 HP
Fig. 16 3-speed P.T.O. - 65 -75 - 80 - 90 HP
Fig. 17 2-speed + groundspeed P.T.O.
Fig. 18 2-speed + groundspeed P.T.O. (internal controls)
Fig. 19 Groundspeed P.T.O. control
Fig. 20 Groundspeed P.T.O. internal control
Fig. 21 Front P.T.O.
Fig. 22 Throttle controls
Fig. 23 Hand throttle control
Fig. 24 Electronic hand throttle control
Fig. 25 Accelerator pedal control
Fig. 26 Lift control levers
Fig. 27 Auxiliary service control valve - flow control valve
Fig. 28 Lift control levers under footplate
Fig. 29 Auxiliary service control valves - 6-way
Fig. 30 Auxiliary service control valves - 4-way

WARNING: It is important that those tie-rods fitted with adjustable forks are adjusted so that their respective control levers can be moved through their entire range from min. to max. without obstruction.

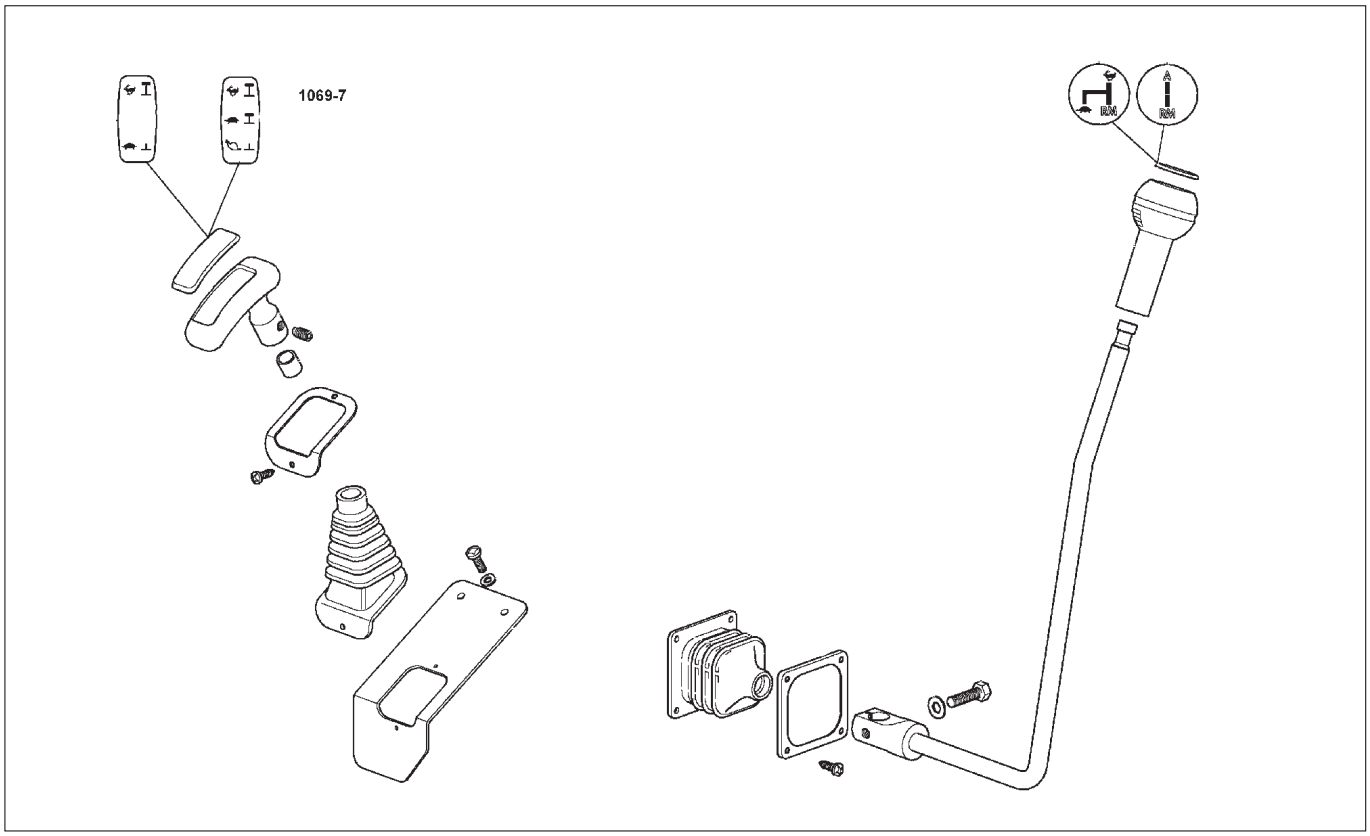


Fig. 1 - Mini-reduction/shuttle control

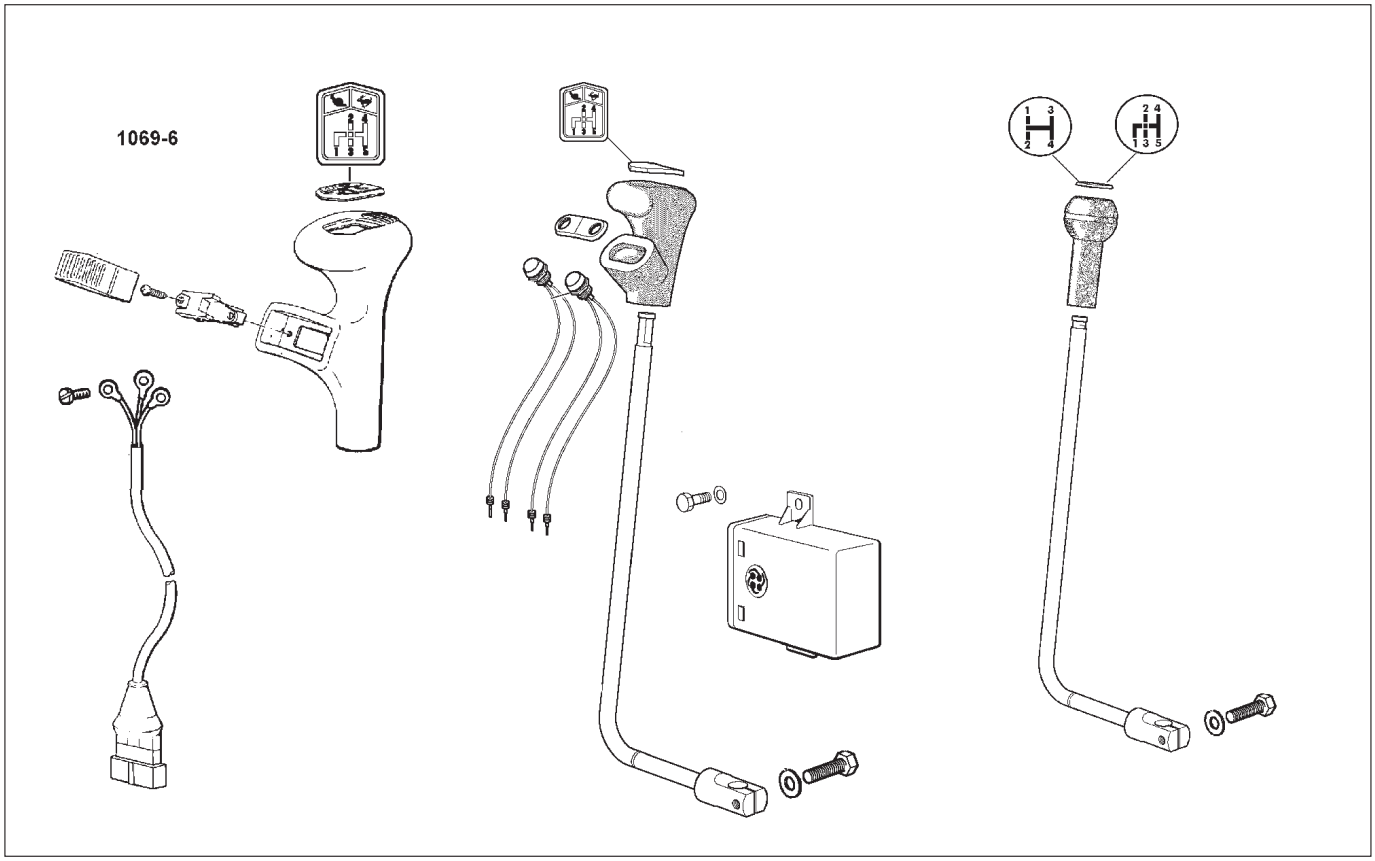


Fig. 2 - Gearshift controls

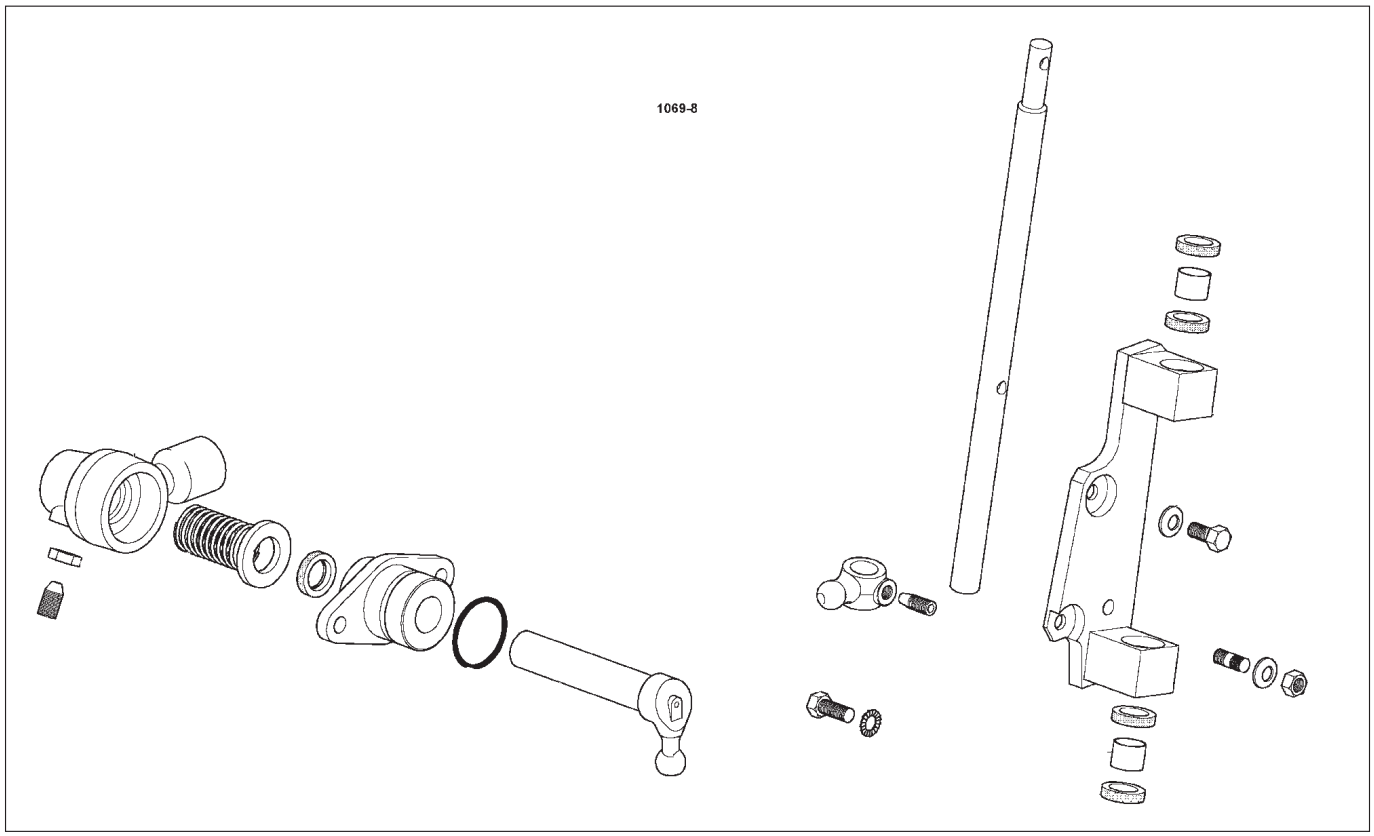


Fig. 3 - Range gear control

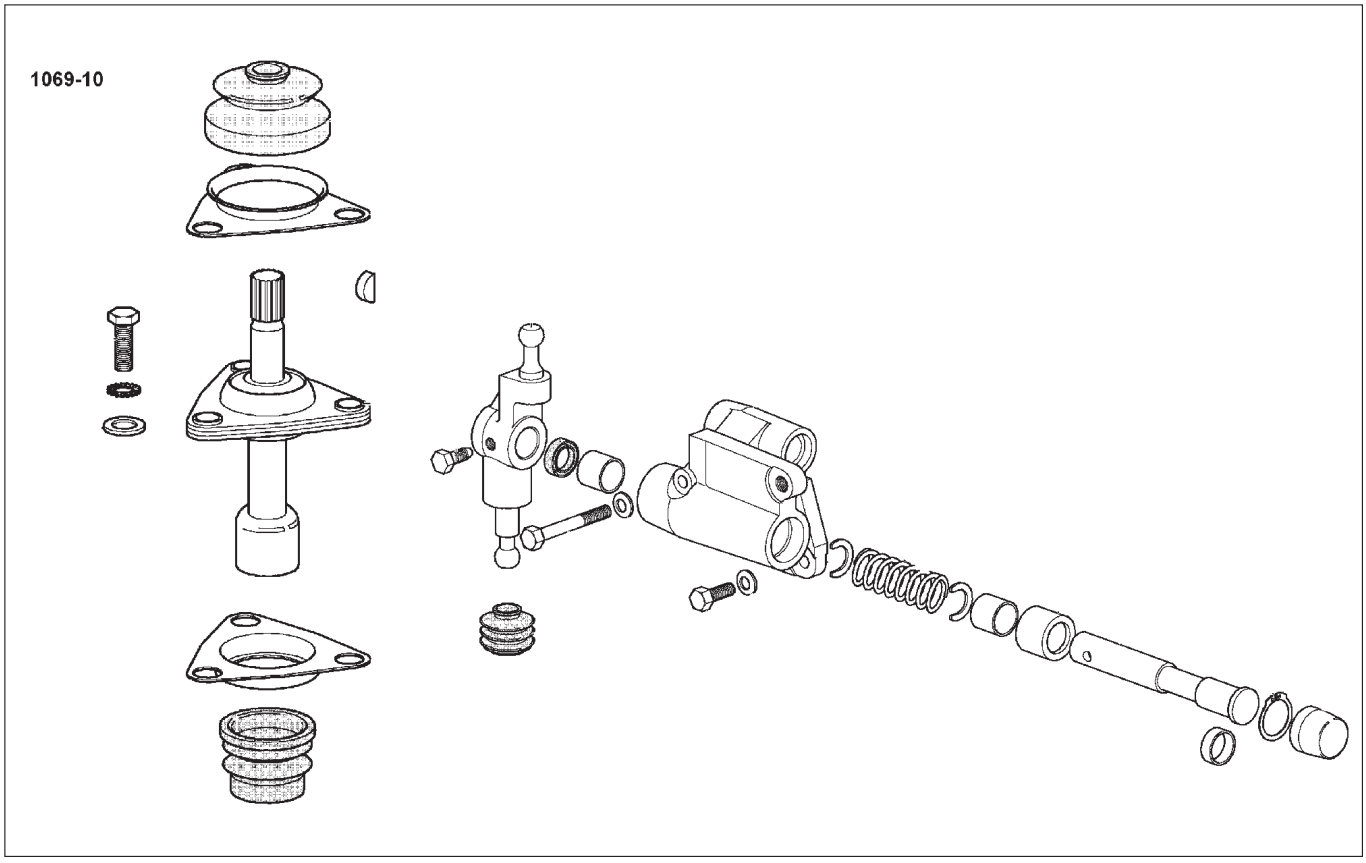


Fig. 4 - Internal gear controls.

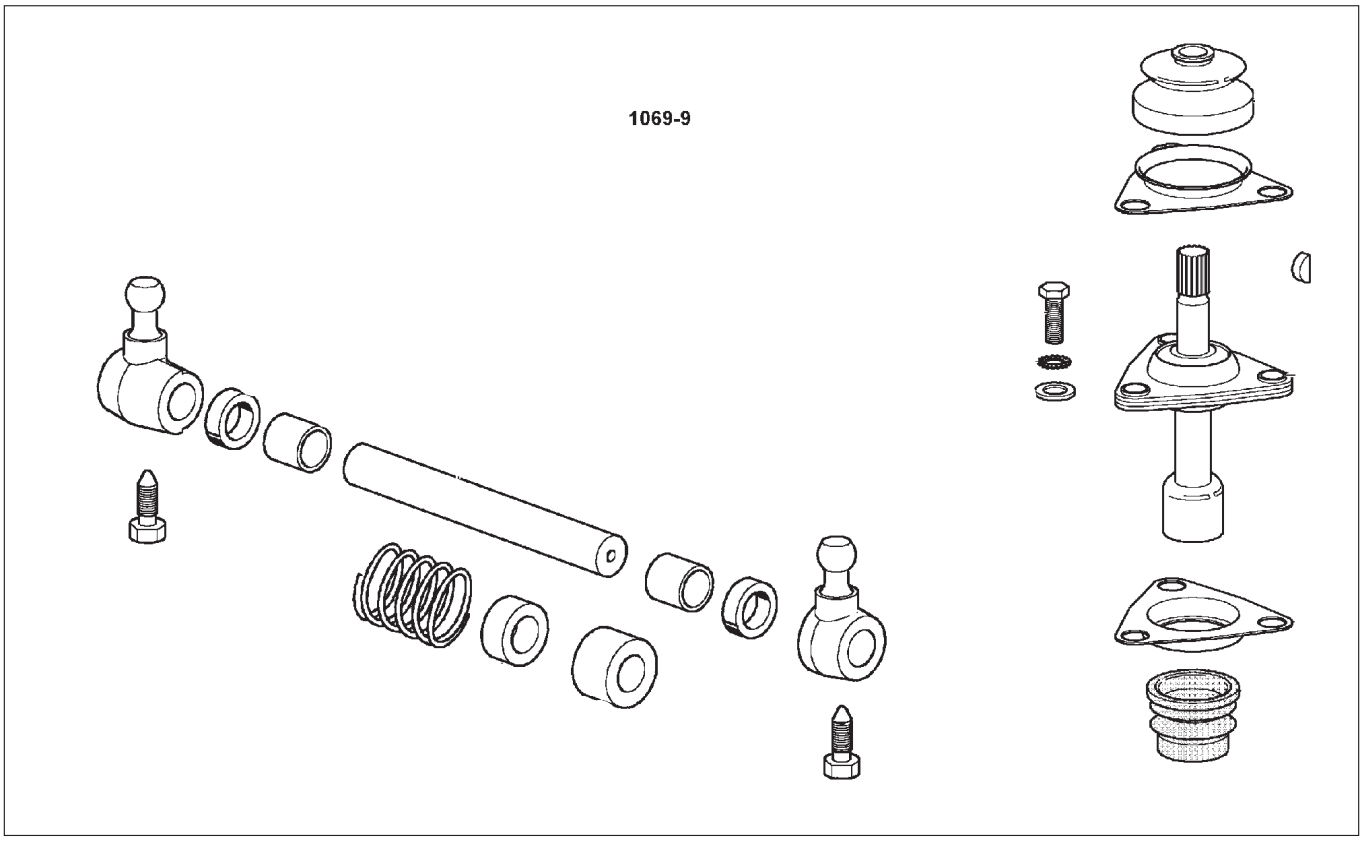


Fig. 5 - Internal shuttle control.

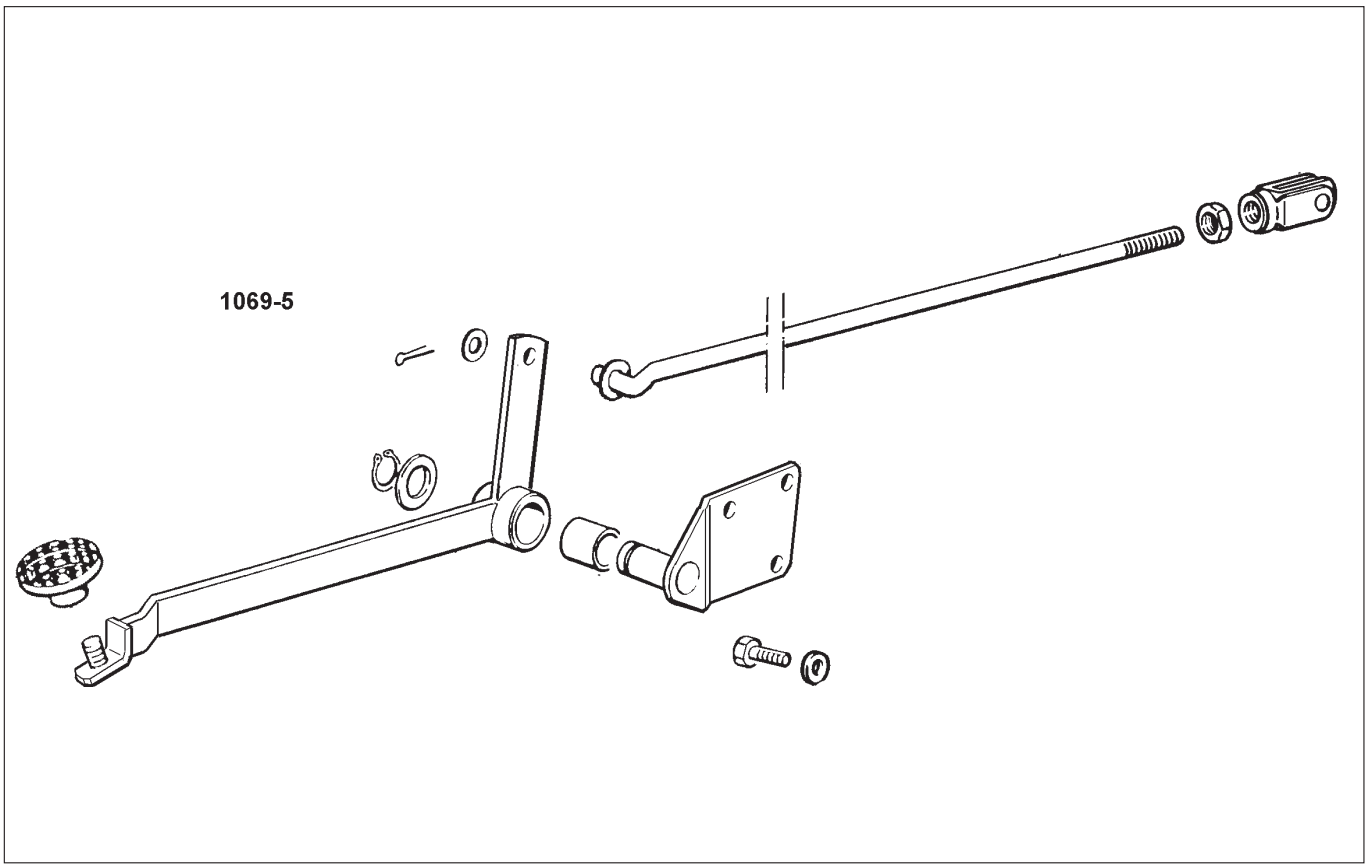


Fig. 6 - Differential lock control.

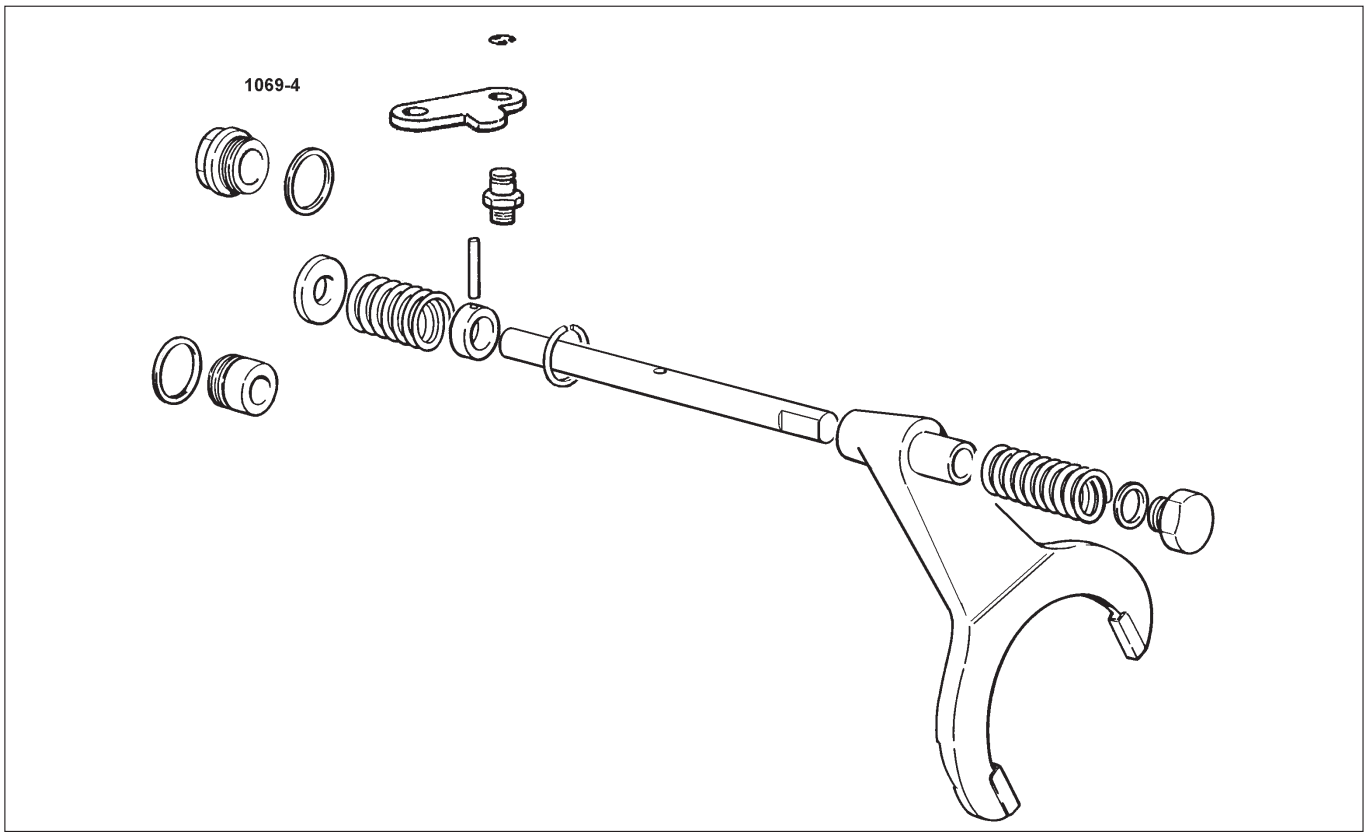


Fig. 7 - Internal differential lock control.

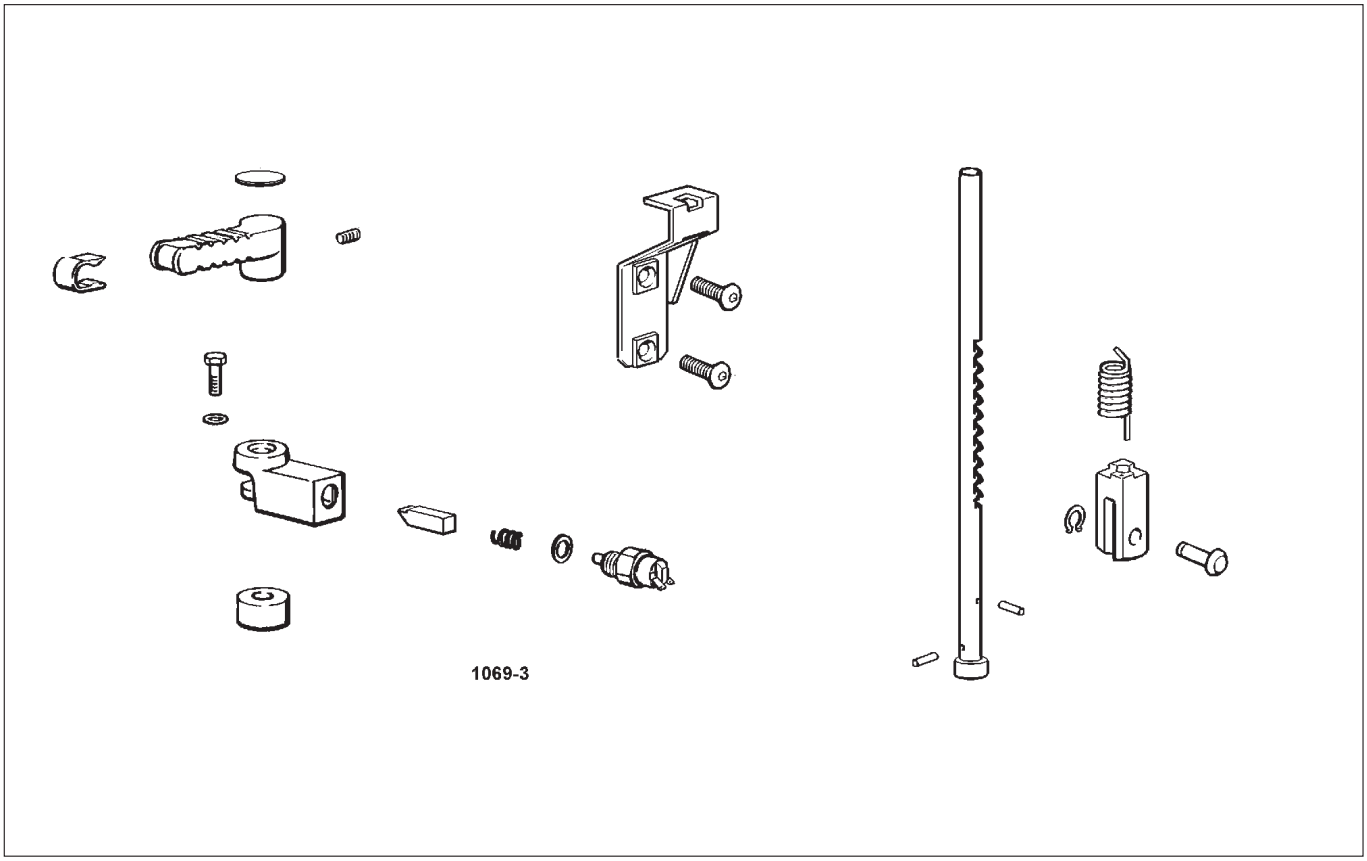


Fig. 8 - Handbrake control.

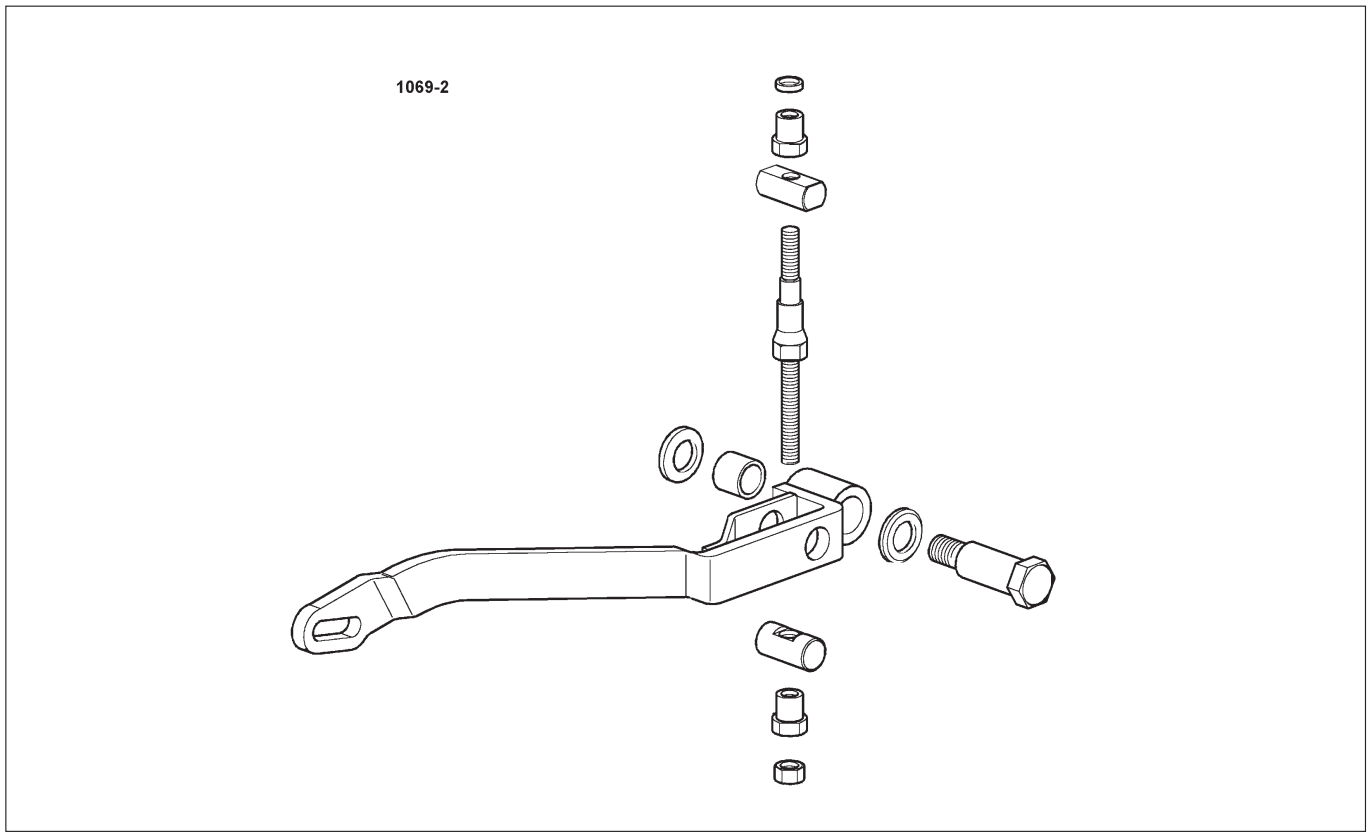


Fig. 9 - Handbrake control (part under footplate).

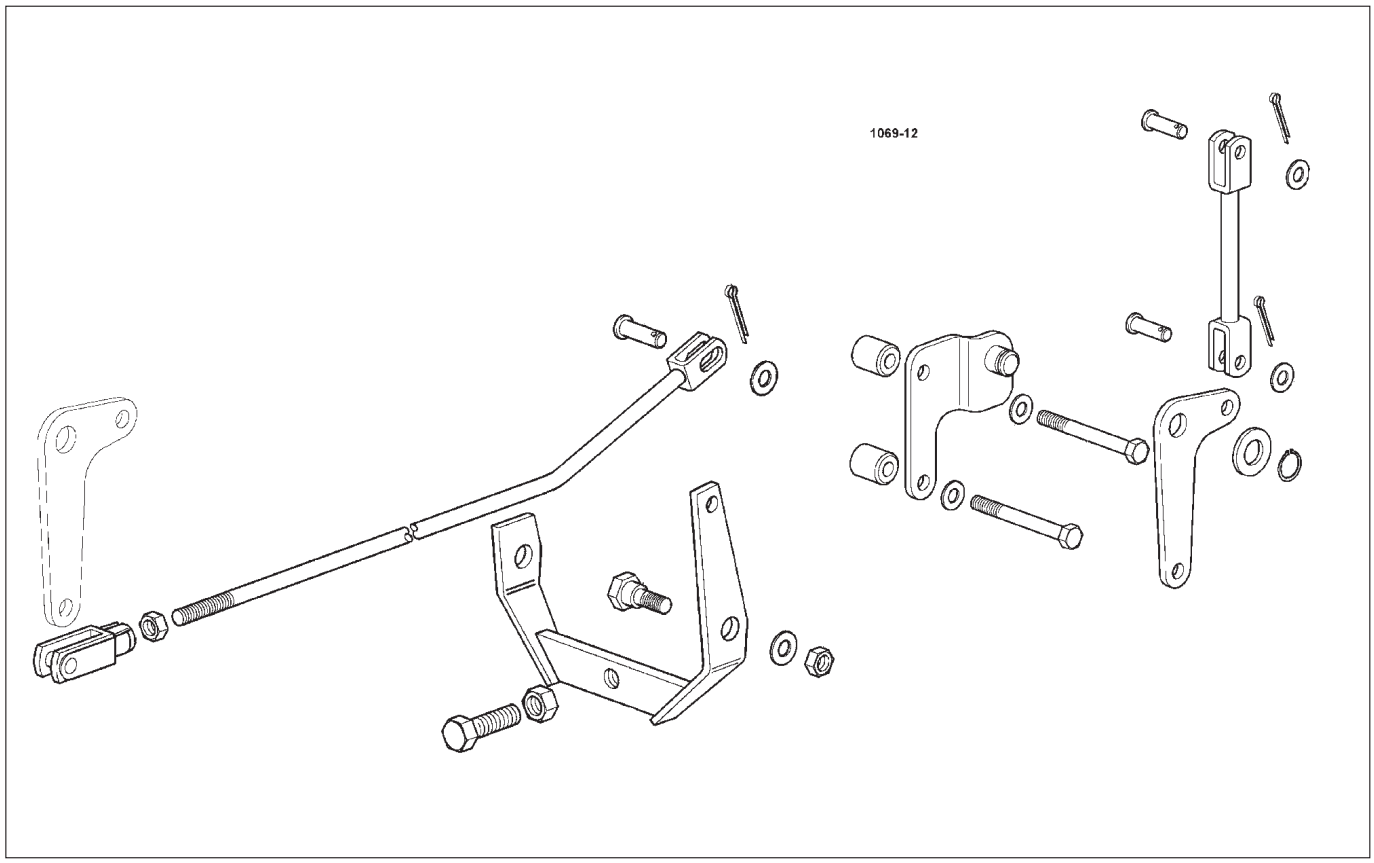


Fig. 10 - 4WD - Front differential lock.

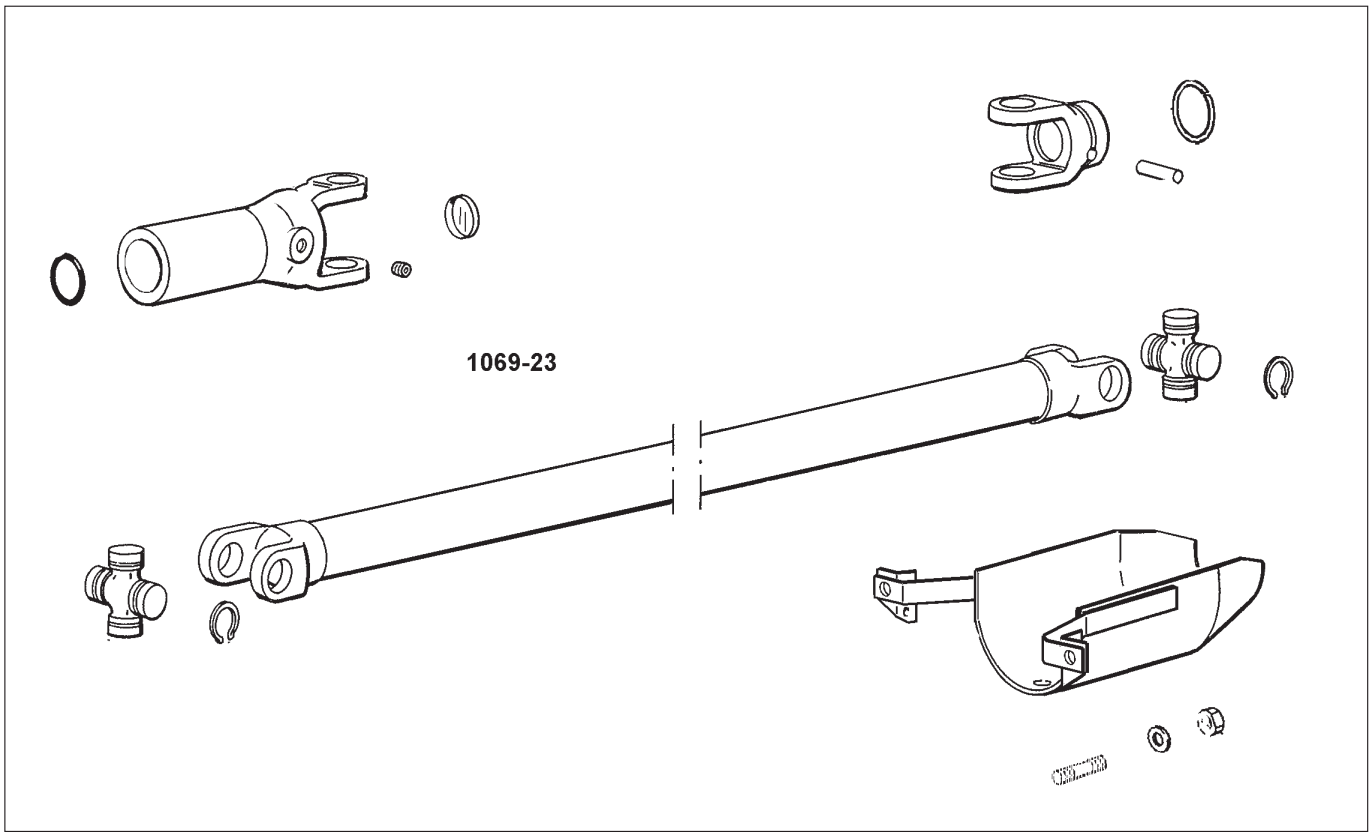


Fig. 11 - 4WD - Drive shaft

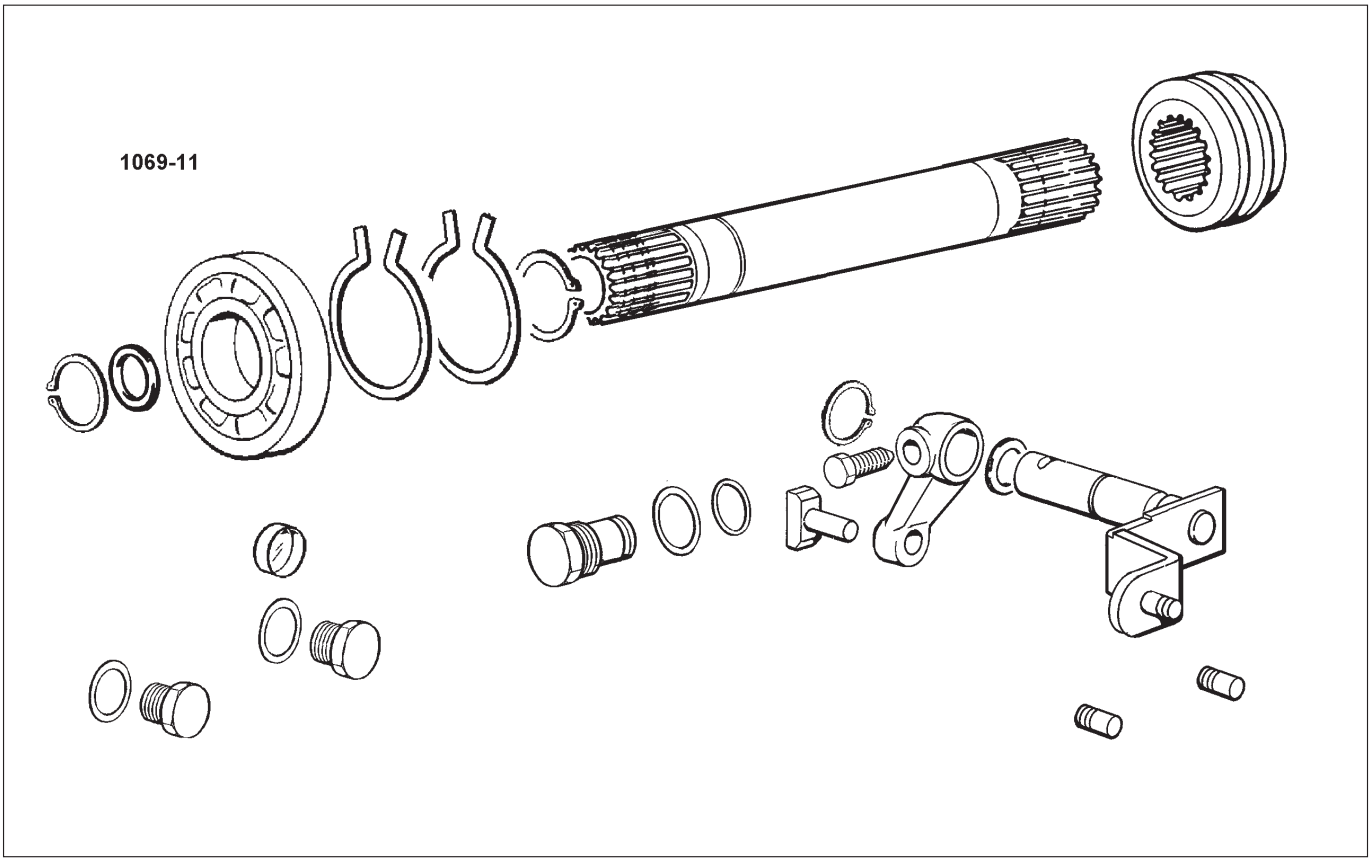


Fig. 12 - 4WD - Power take-off.

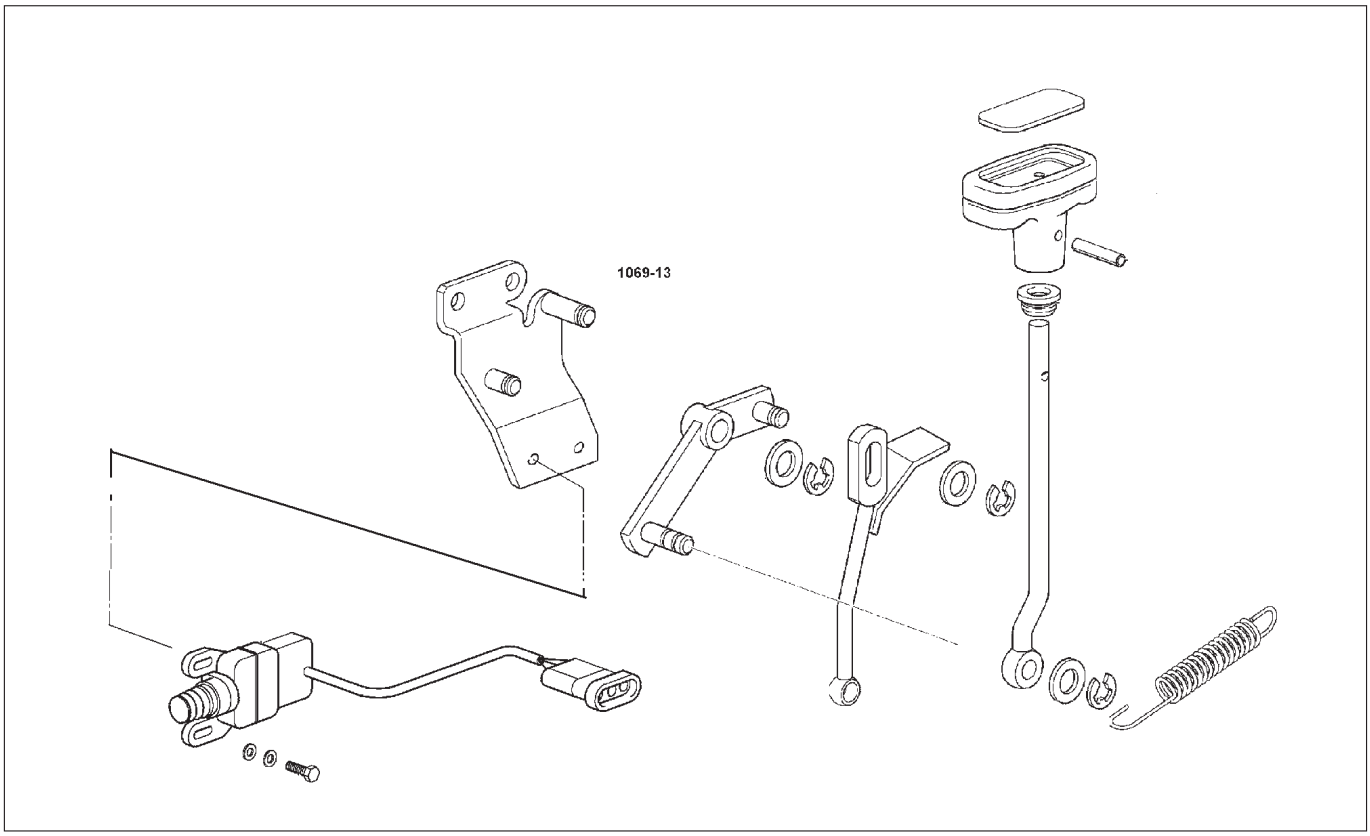


Fig. 13 - 4WD engagement control.

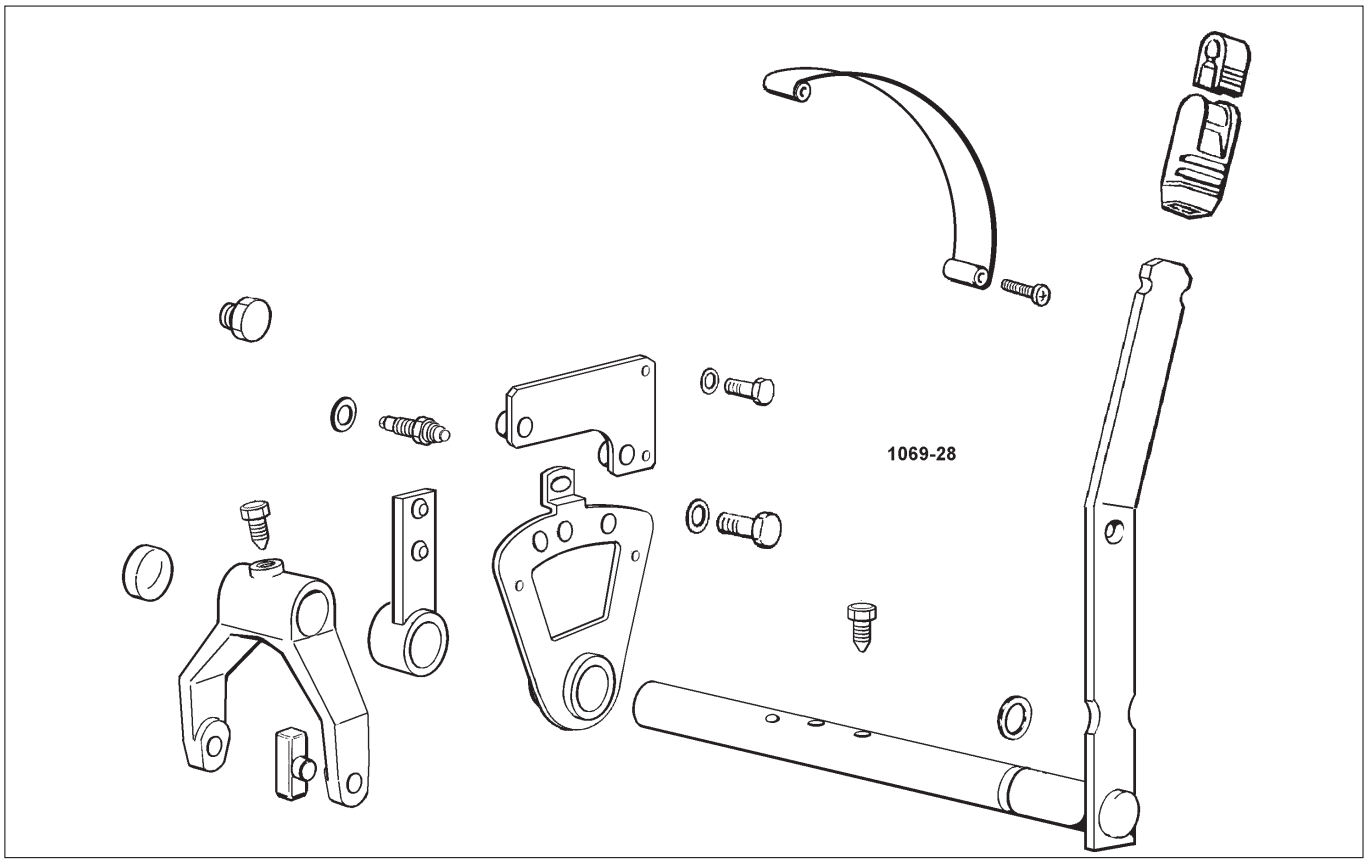


Fig. 14 - 2-speed P.T.O..

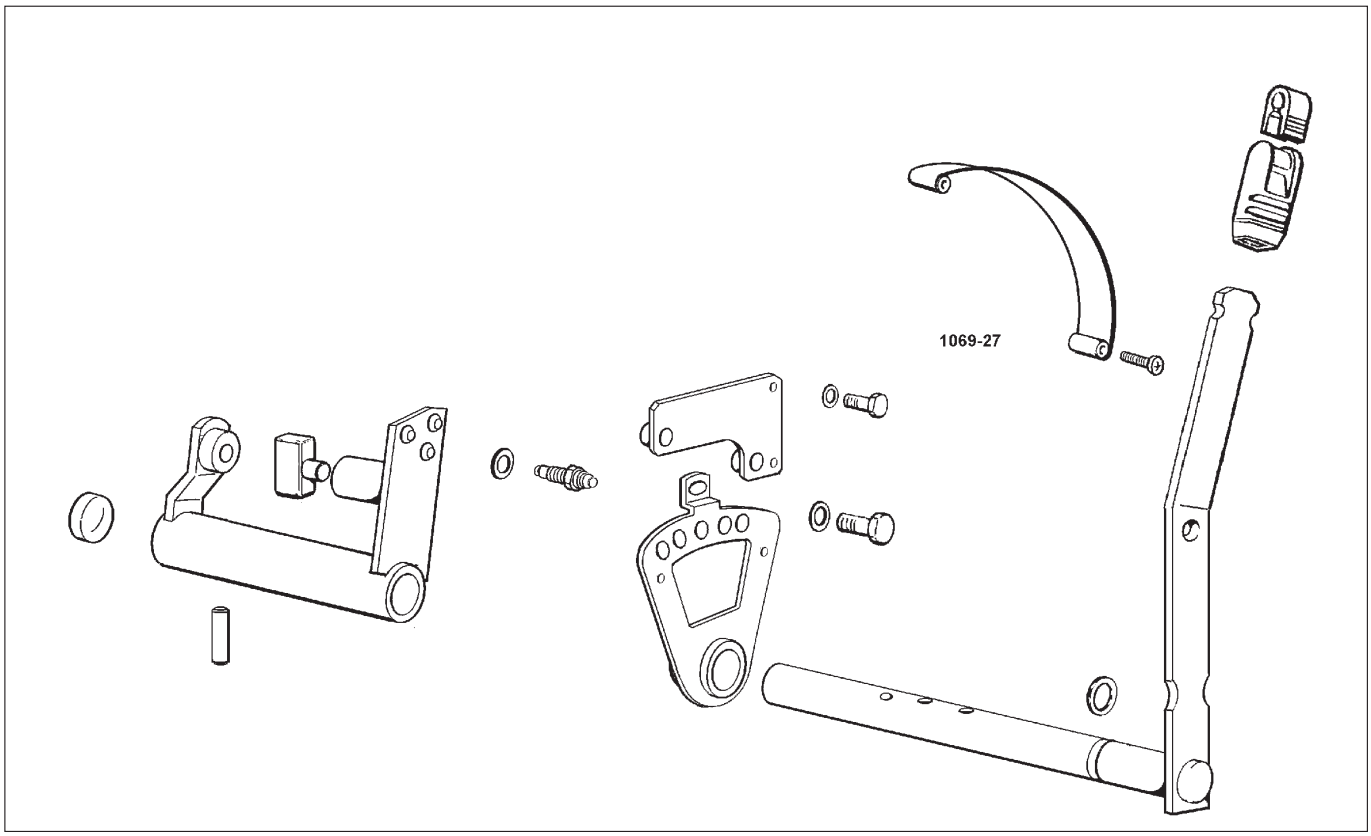


Fig. 15 - 3-speed P.T.O. - 60 HP.

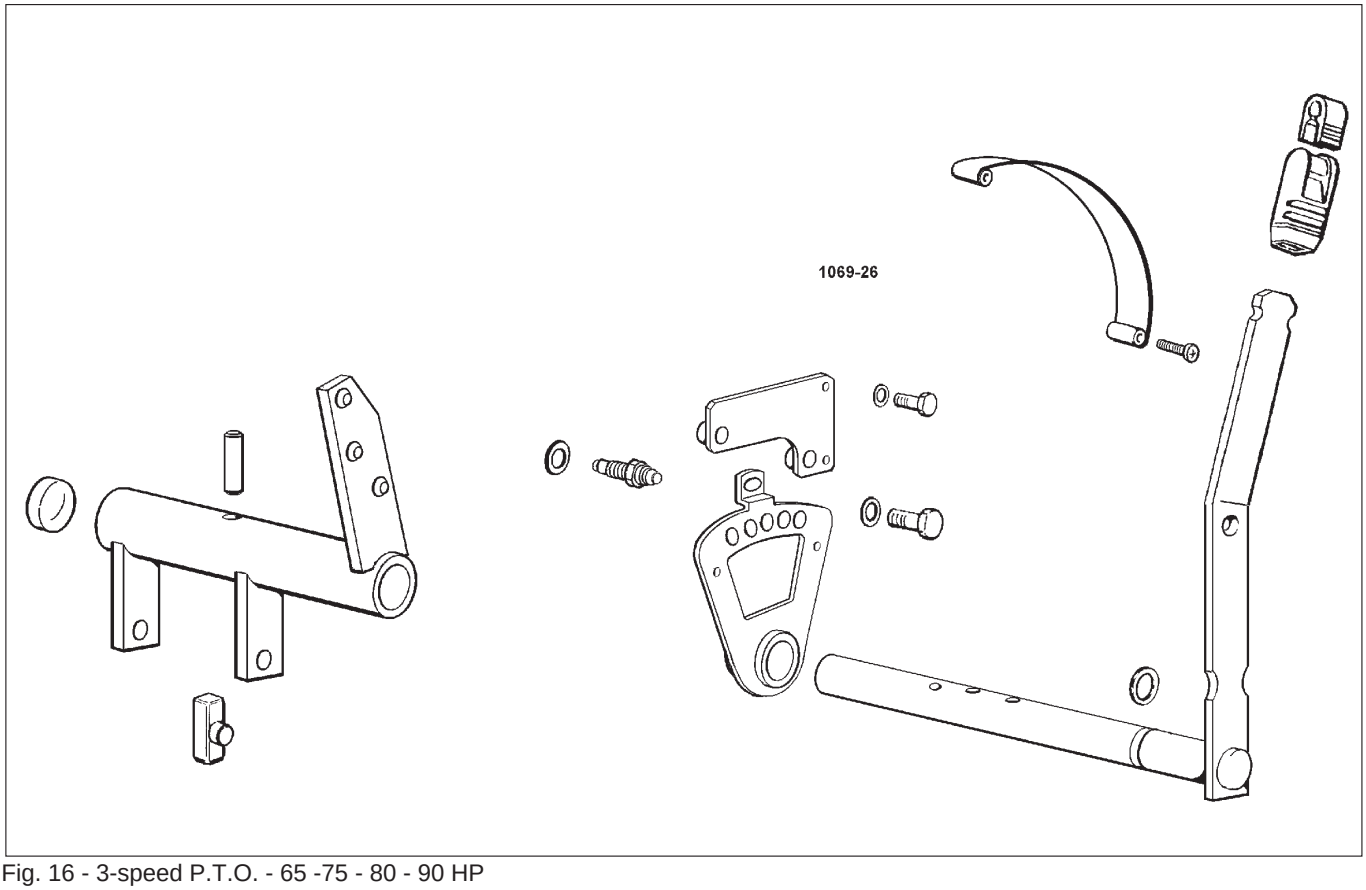


Fig. 16 - 3-speed P.T.O. - 65 -75 - 80 - 90 HP

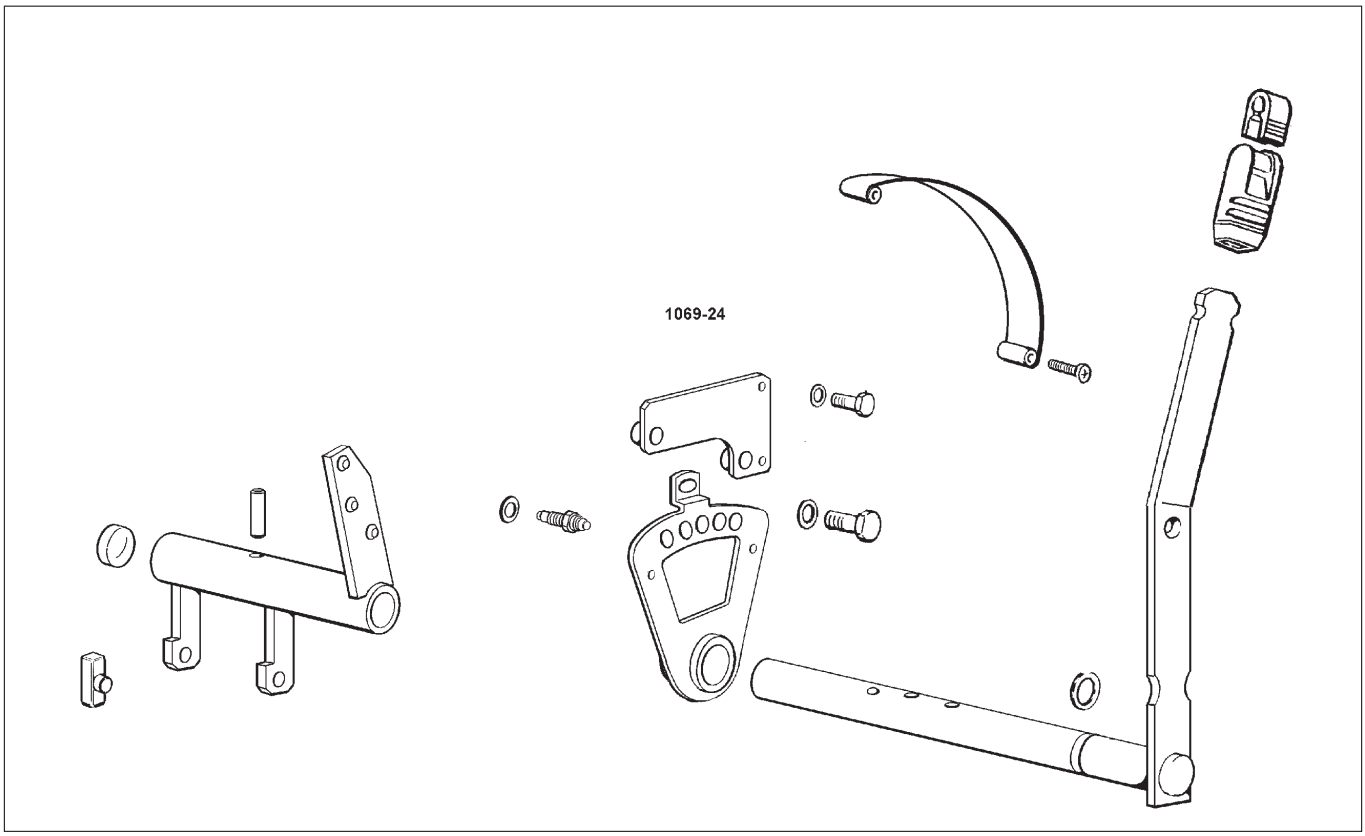


Fig. 17 - 2-speed + groundspeed P.T.O..

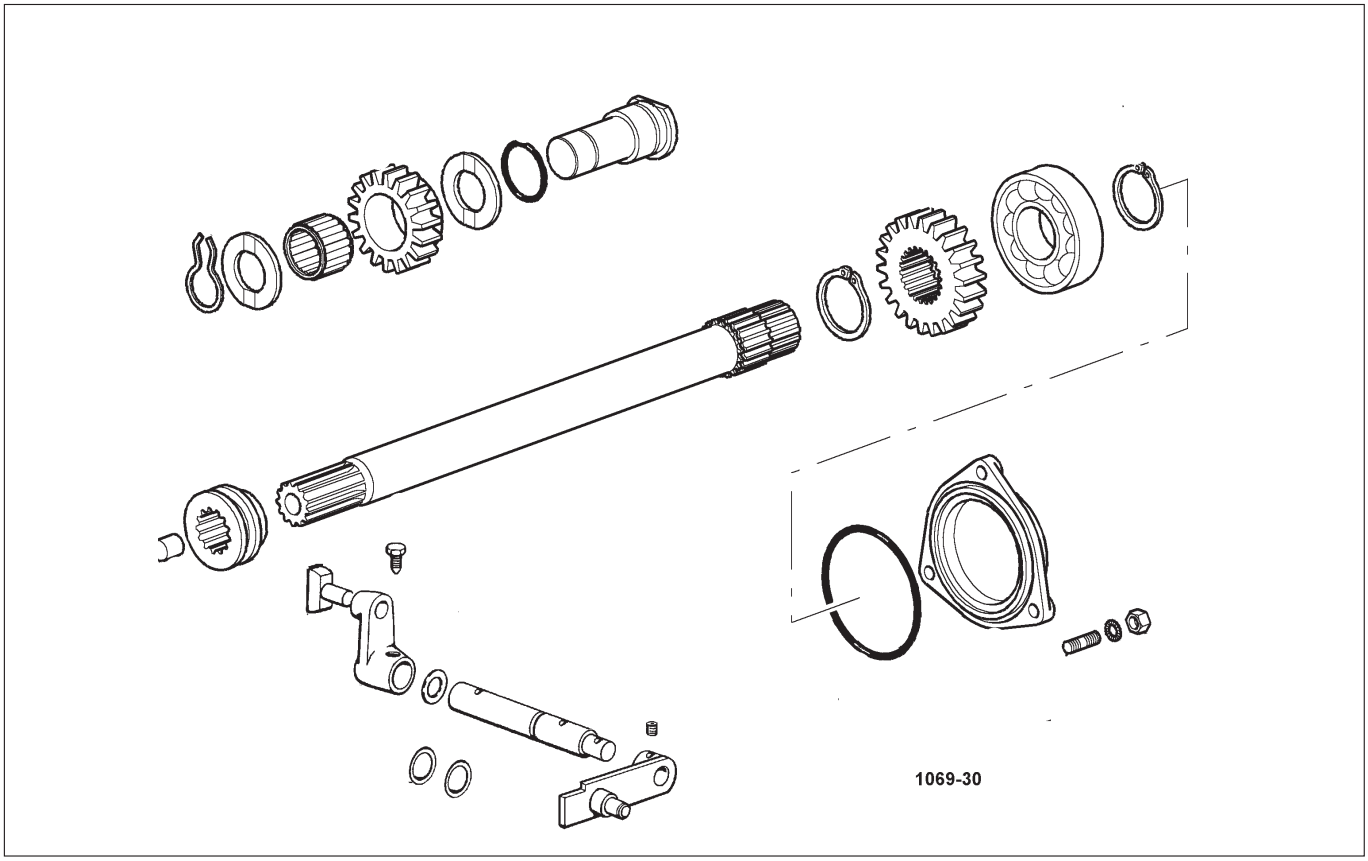


Fig. 18 - 2-speed + groundspeed P.T.O. (internal controls).

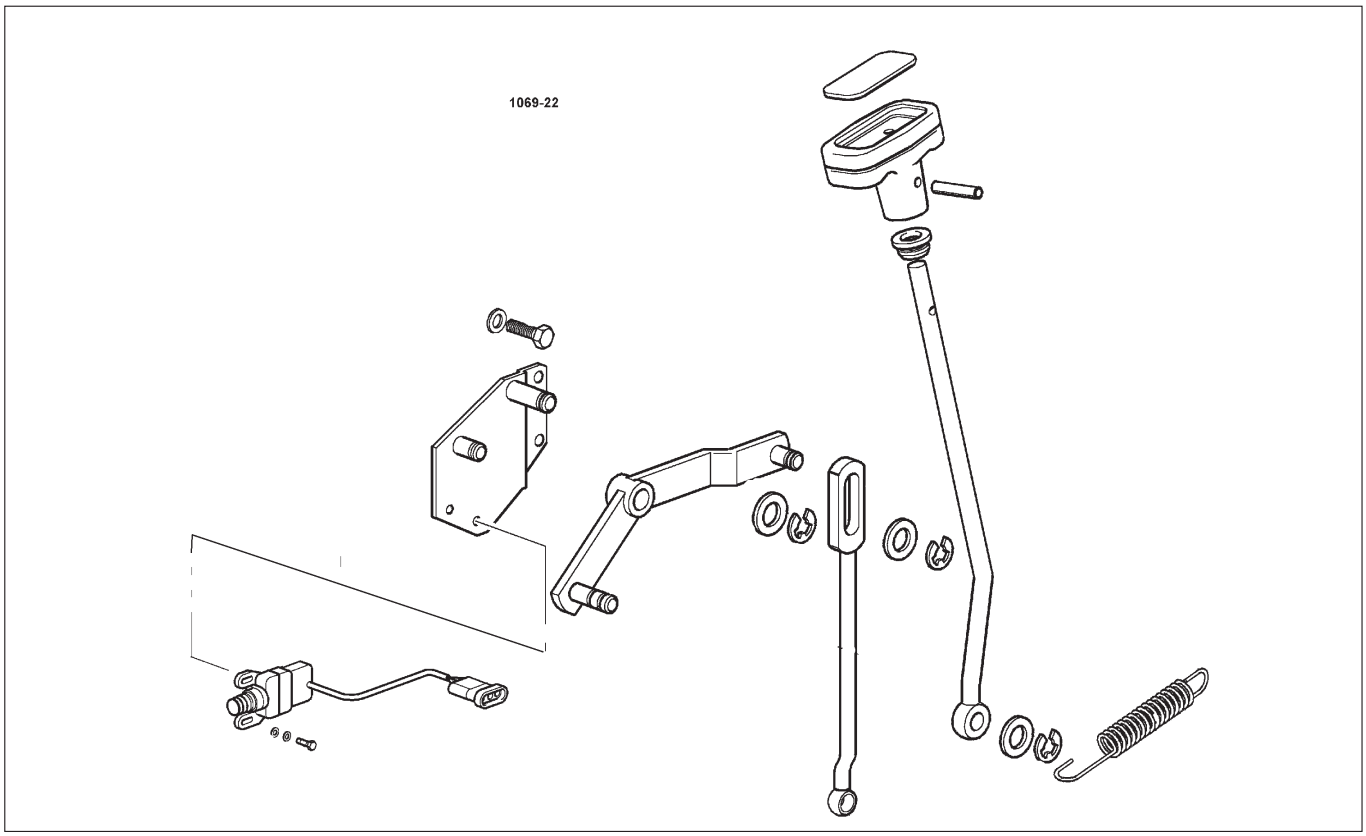


Fig. 19 - Groundspeed P.T.O. control.

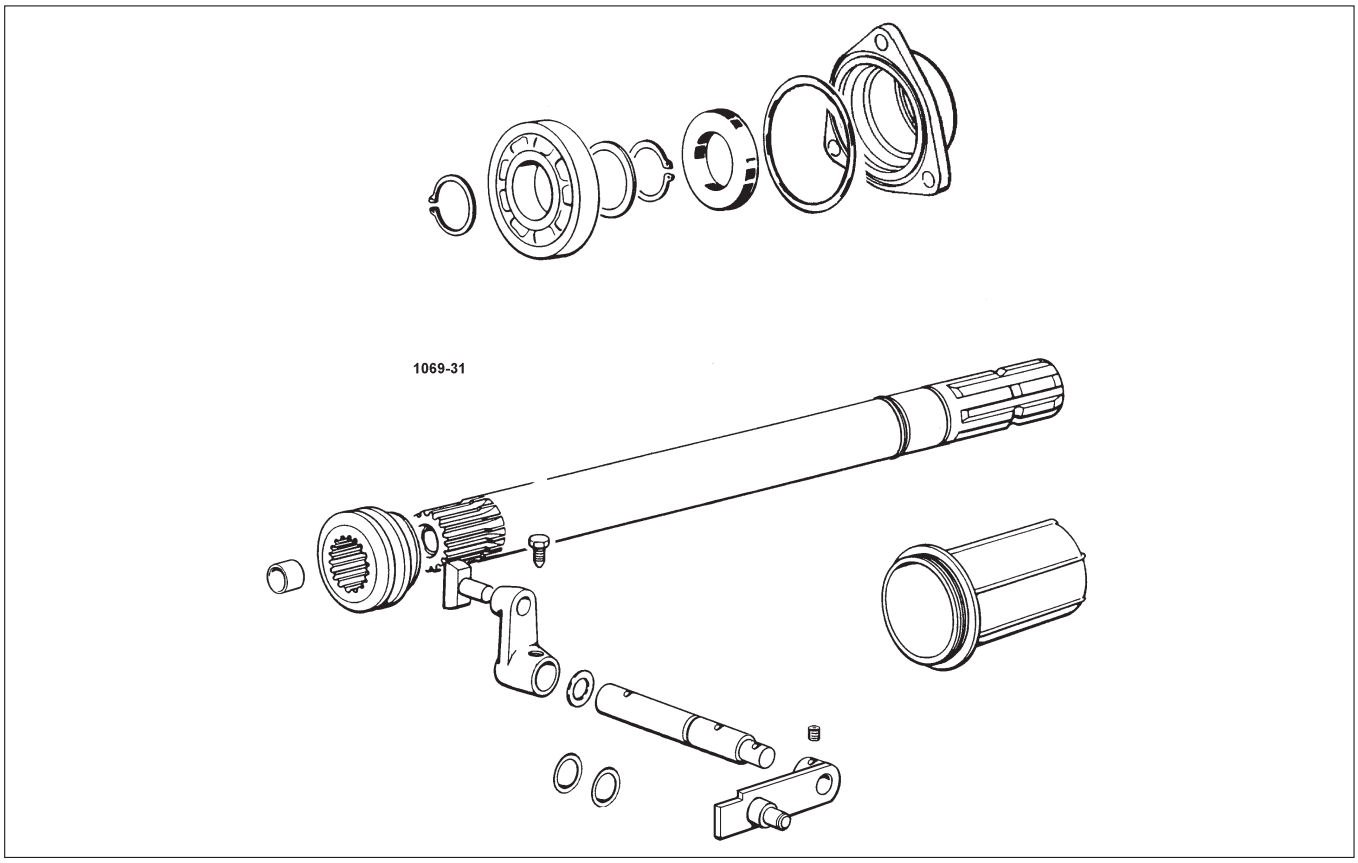


Fig. 20 - Groundspeed P.T.O. internal control.

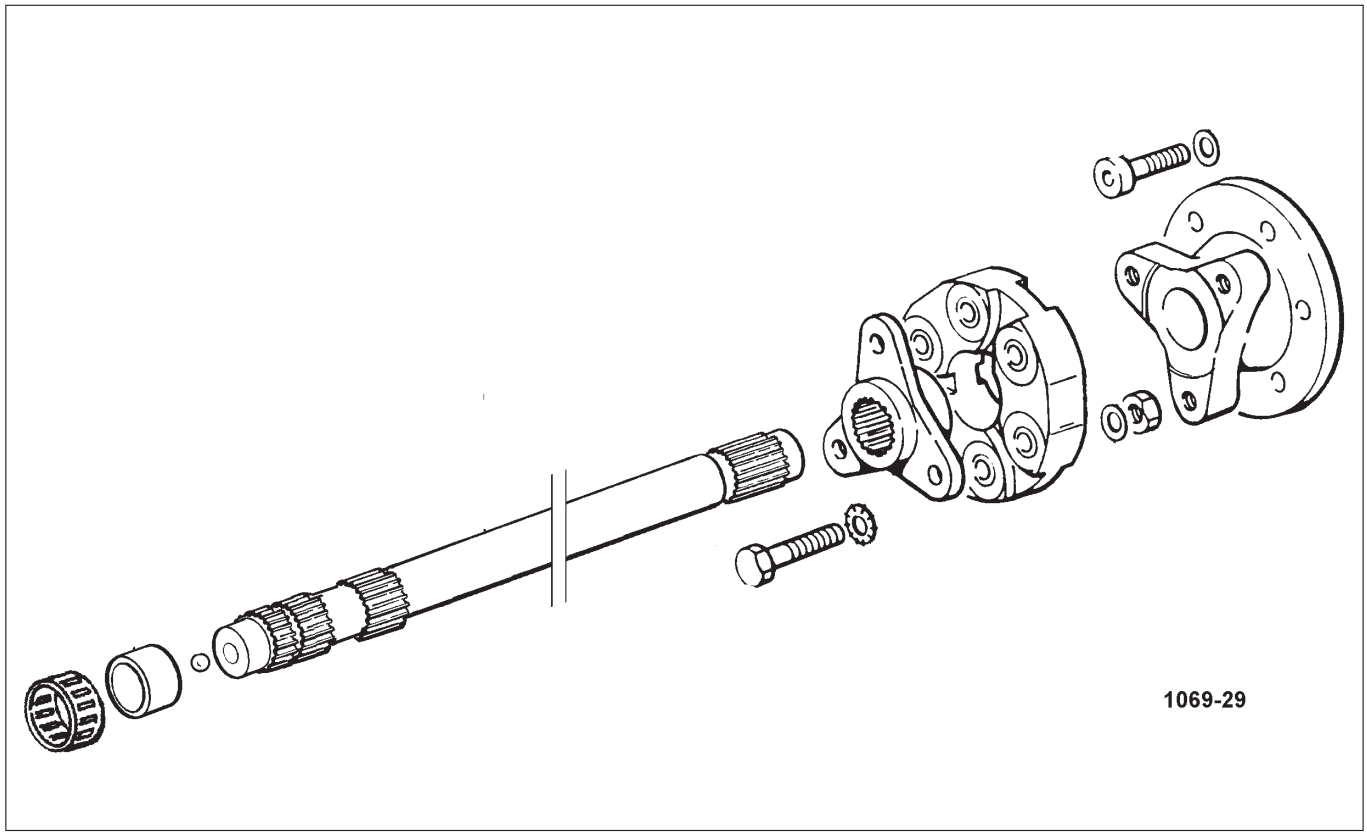


Fig. 21 - Front P.T.O..

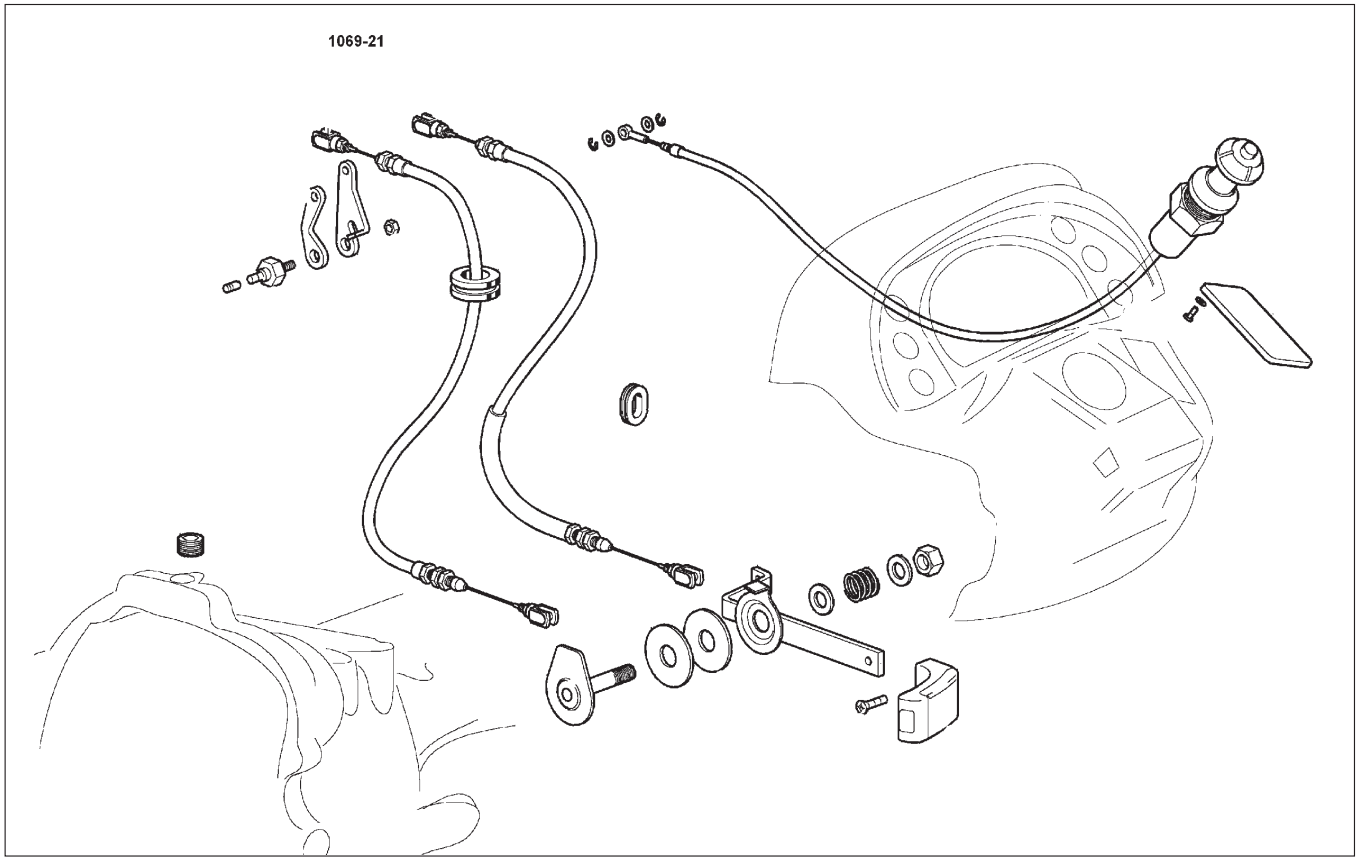


Fig. 22 - Throttle controls.

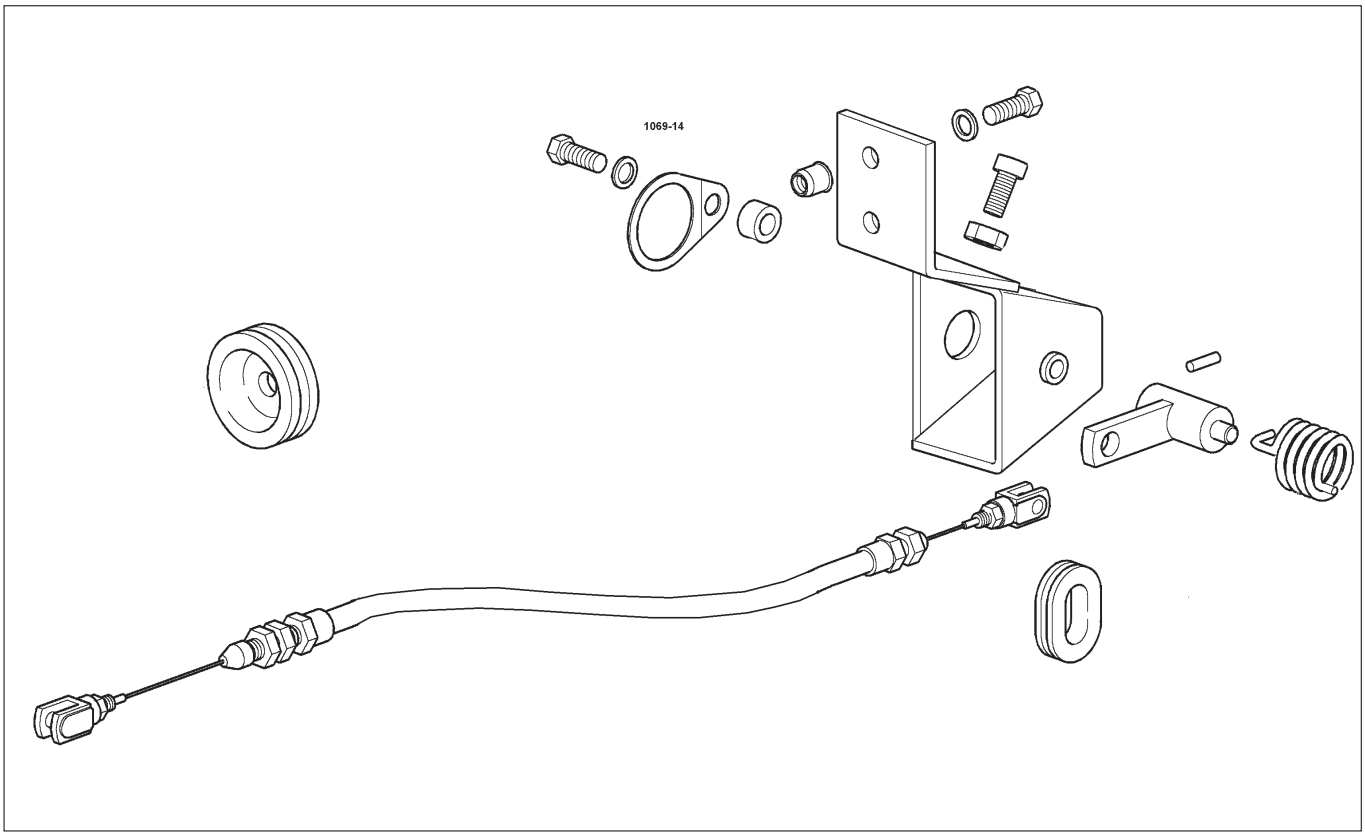


Fig. 23 - Hand throttle control.

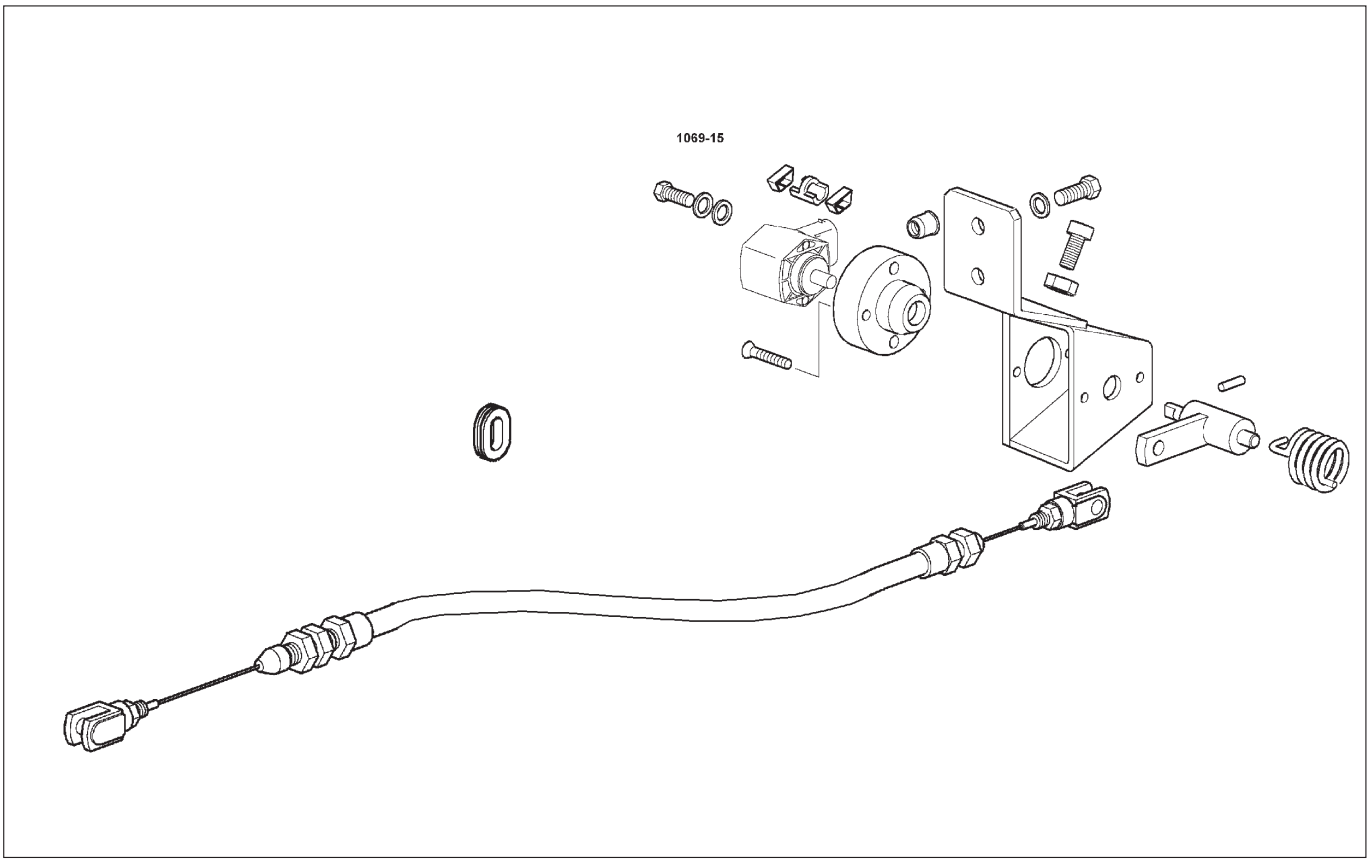


Fig. 24 - Electronic hand throttle control.

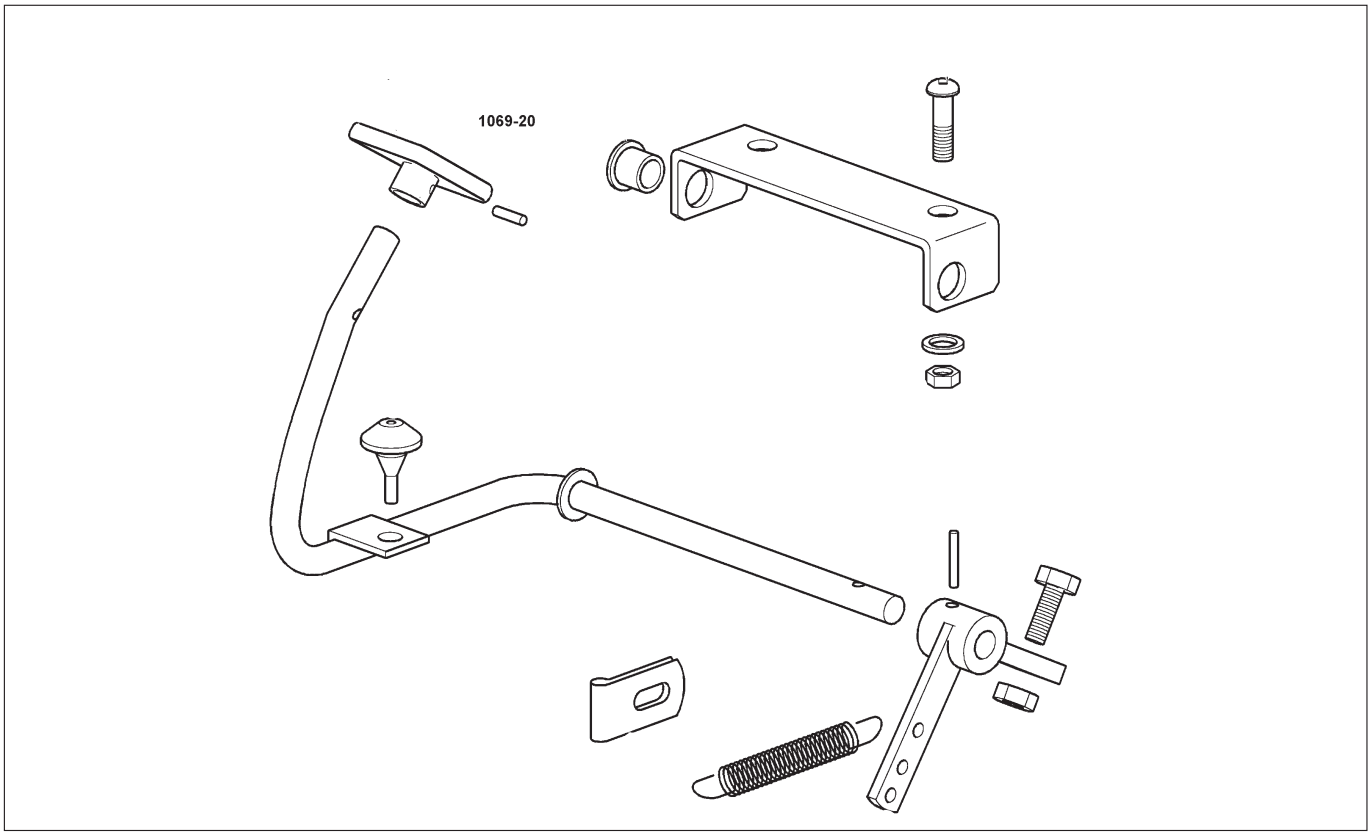


Fig. 25 - Accelerator pedal control.

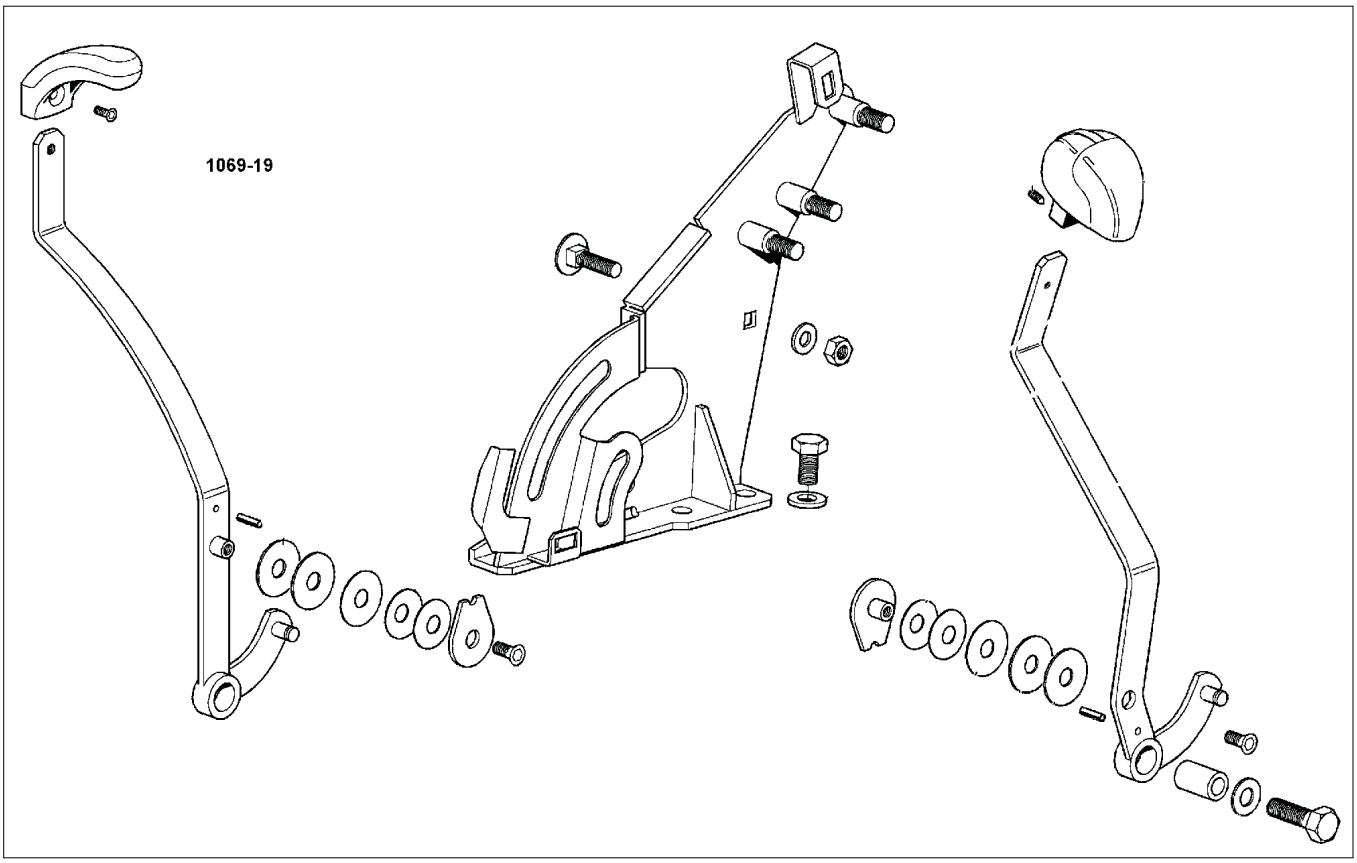


Fig. 26 - Lift control levers.

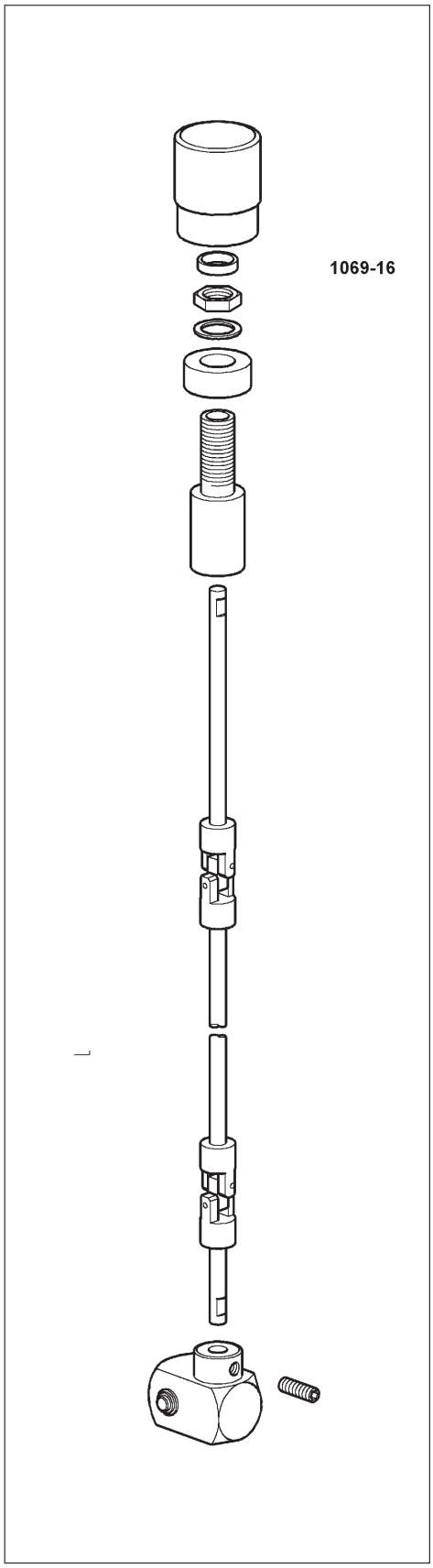


Fig. 27 - Auxiliary service control valve - flow control valve

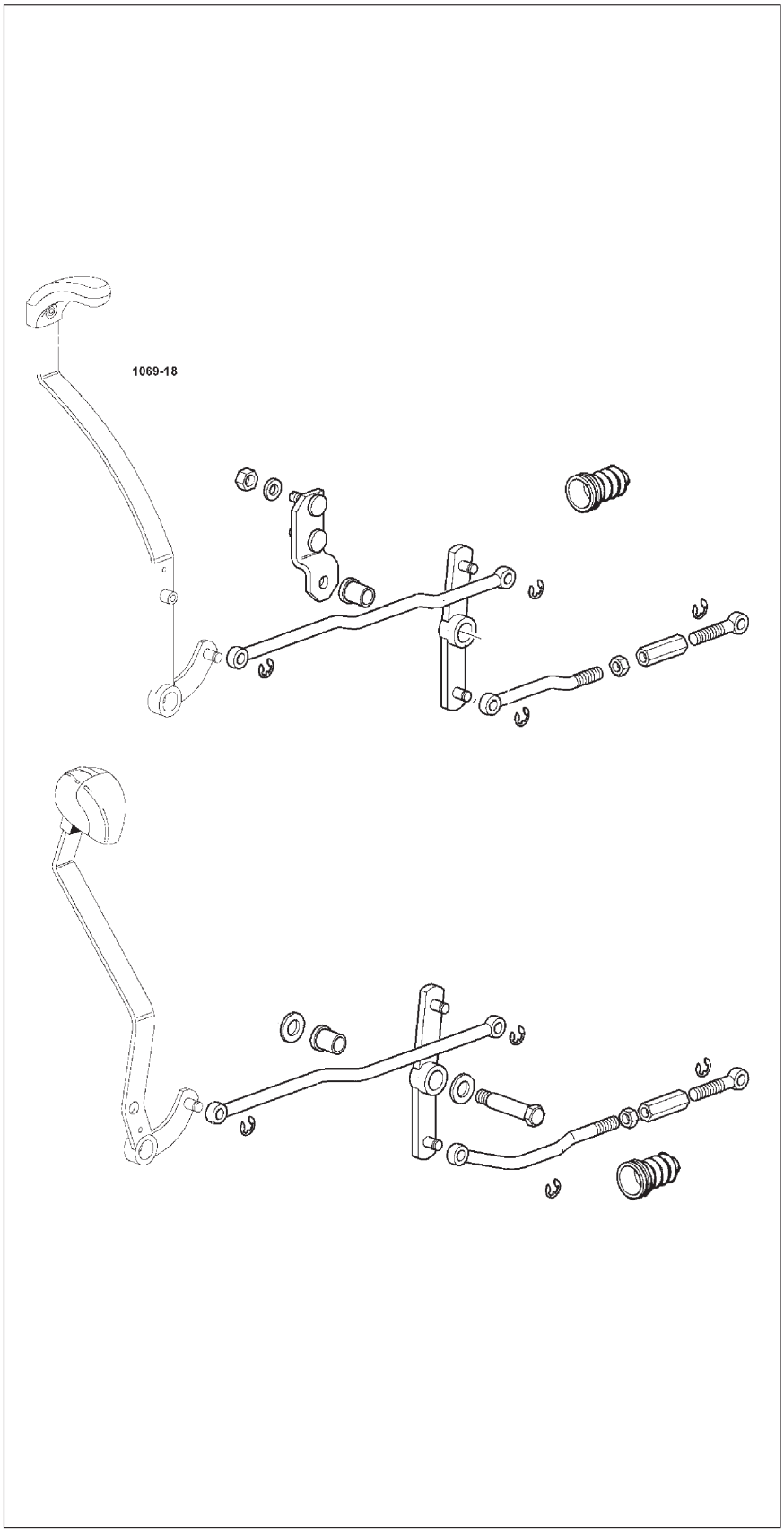


Fig. 28 - Lift control levers.

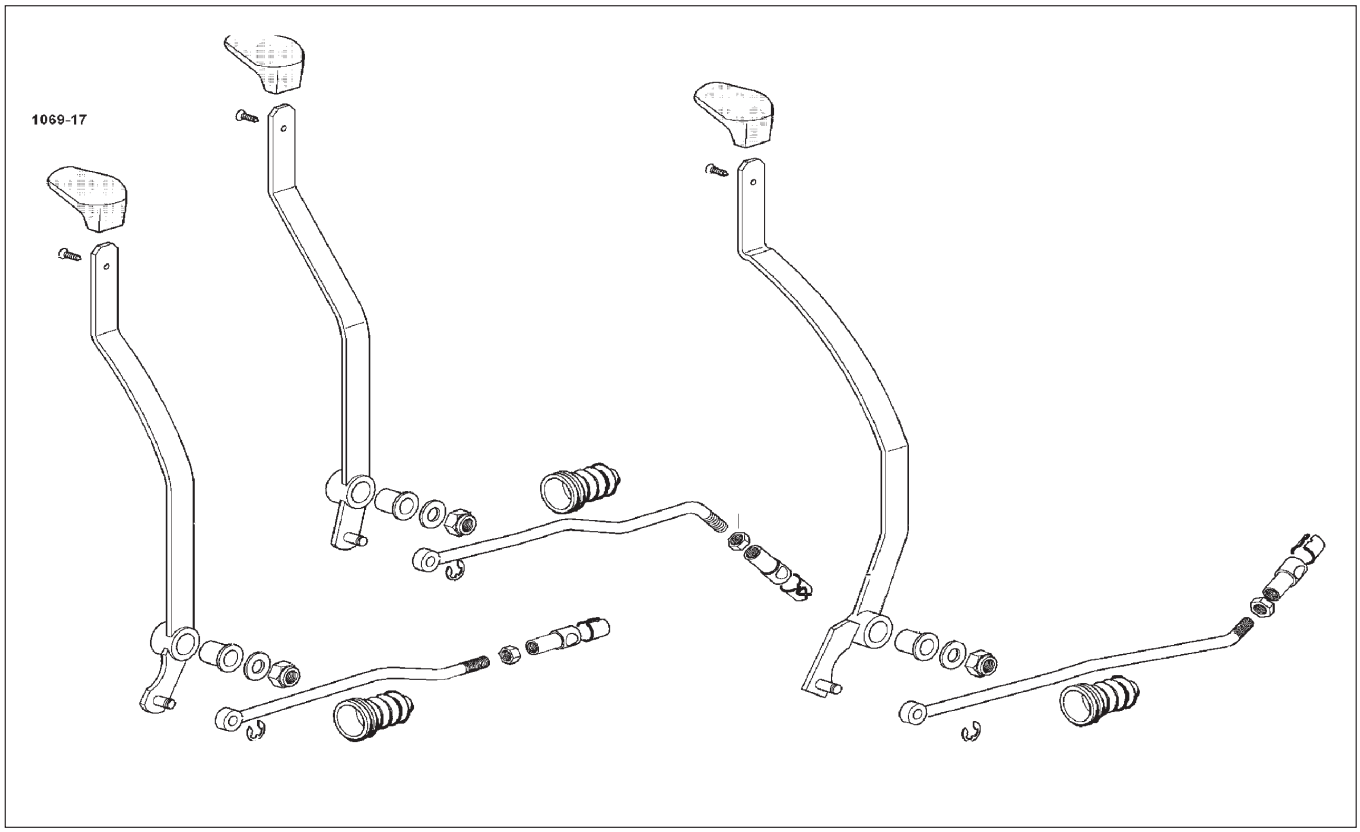


Fig. 29 - Auxiliary service control valves - 6-way.

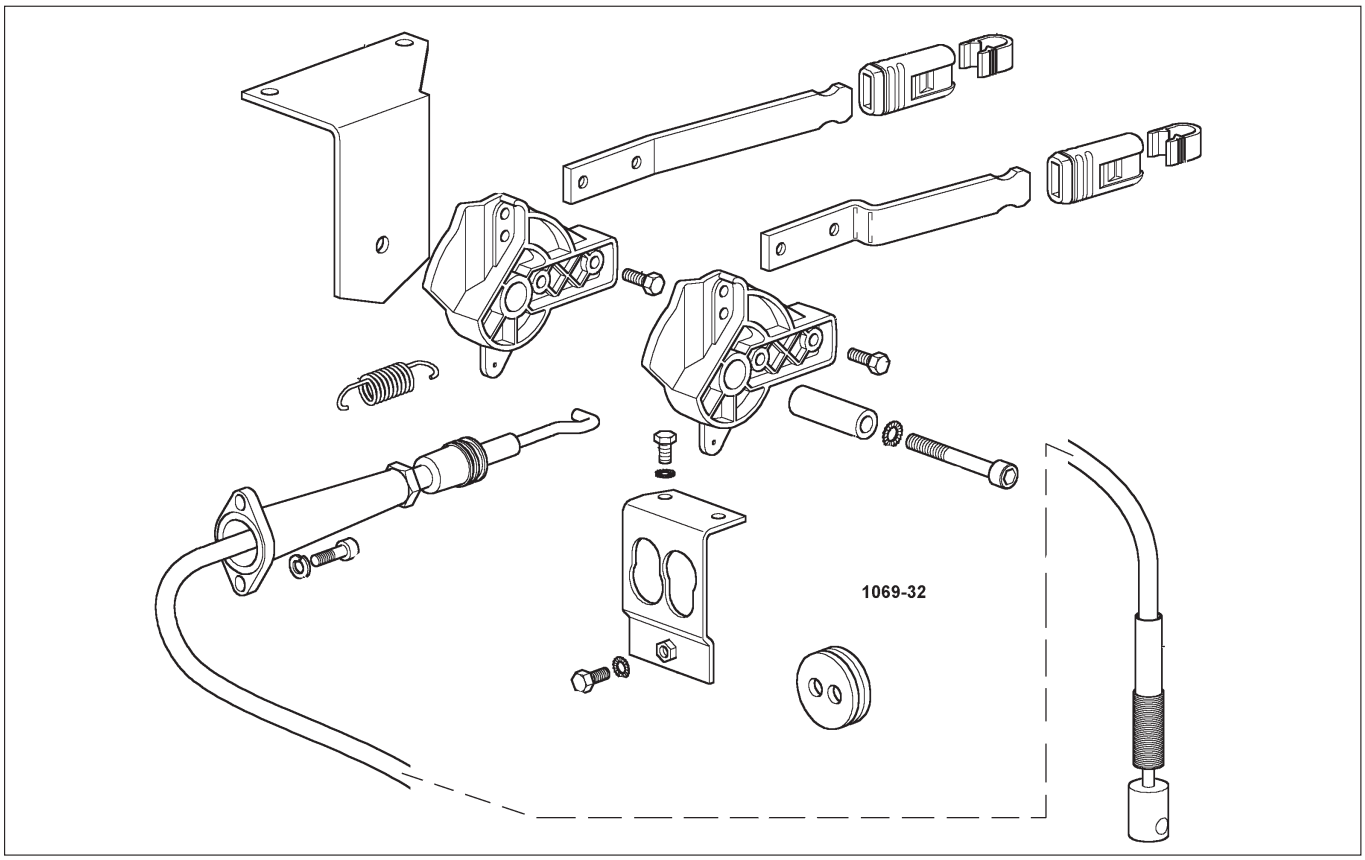


Fig. 30 - Auxiliary service control valves - 4-way.

Electro-hydraulic controls

Front P.T.O. clutch engagement control
The front P.T.O. can only be engaged when the engine is running. Pressing the pushbutton engages the front P.T.O.. The integral light illuminates to indicate that the P.T.O. is engaged.

Rear P.T.O. clutch engagement control
The rear P.T.O. can only be engaged when the engine is running. Pressing the pushbutton engages the rear P.T.O.. The integral light illuminates to indicate that the P.T.O. is engaged.

Differential lock engagement control
Pressing the pushbutton engages the front and rear differential locks. The integral light illuminates to indicate that the locks are engaged. Pressing the pushbutton again disengages the locks.

Front-wheel drive engagement control
Pushbutton with integral indicator light.
Press to engage front-wheel drive; press again to disengage.

N.B. Engagement of the above devices is also signalled by indicator lights on the console.

Rear P.T.O. engagement control
Pushbutton with integral indicator light.
Press to engage the rear P.T.O.; press again to disengage.
The P.T.O. may only be engaged when the engine is running.

Gearbox

The MULTISPEED transmission offers the possibility to select each speed in , or (using the gear and range lever) without having to depress the clutch pedal simply by pressing the electro-hydraulic control switch.
This is achieved by way of three multi-plate clutches.

Electro-hydraulic shuttle
As an alternative to the mechanical shuttle, all transmission types may be equipped with an electro-hydraulically controlled shuttle. This comprises an electronically controlled system, where two hydraulically controlled actuators act on the gearbox clutch and the direction selector rod. The shuttle control lever is located on the left below the steering column.

Front and rear lift
The front lift is operated by way of the control lever of the auxiliary service control valves located on the right of the driving seat, after first opening the shut-off cock in the oil delivery line to the front lift.
The mechanical rear lift is operated by controls located to the right of the driving seat.
For tractors equipped with electronic lifts, see p. 258 for a description of the controls.

Electro-hydraulic controls for adjustment of the right-hand lift rod and the lateral stabilisers.
The adjustable lift rod consists of a double-acting hydraulic cylinder, while the stabilisers are single-acting hydraulic cylinders.

In the version in which the oil is drawn from the auxiliary service control valves, the control system is equipped with a flow divider valve which, operated by a solenoid valve controlled from a switch, sends oil to the lift rod or the stabilisers. The linkage can then be adjusted using the first control valve lever.

In the version with full electro-hydraulic control, the system is equipped with a specific control unit, which allows direct adjustment of the linkage.

A - Main oil filter on supply circuit
B - Hydraulic pump for auxiliary service control valve and power lift
C - Hydraulic pump for hydrostatic steering and electro-hydraulic controls
E - Hydrostatic steering control valve
F - Steering cylinders
G - Oil cooler
H - Pressure control valves
I - Gearbox lubrication
L - Hydraulic power unit
M - Differential locks
N - 4WD engagement
O - Rear P.T.O. clutch
P - 6-way auxiliary service control valve
Q - Front lift hydraulic control valve
R - Lifting cylinder

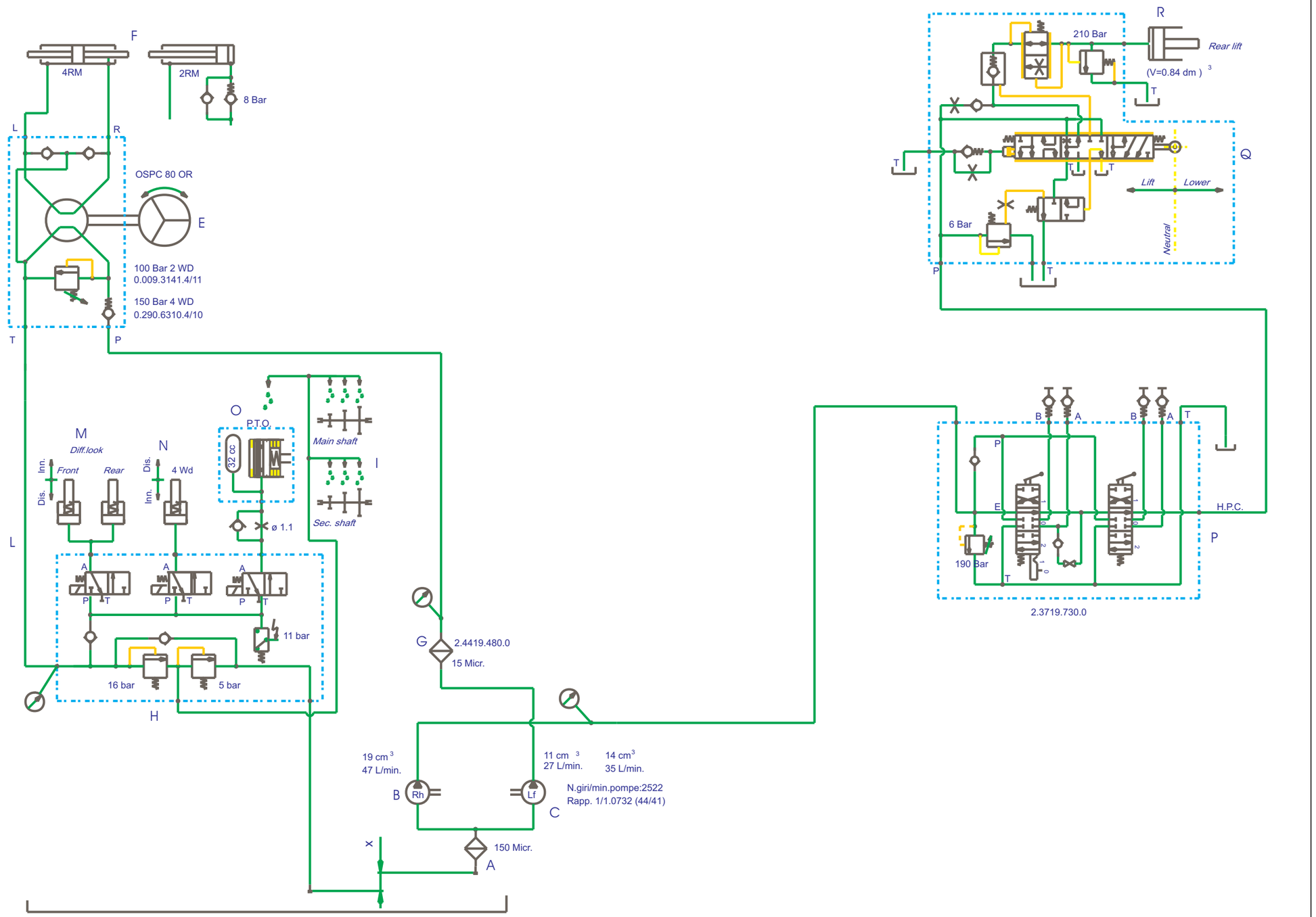


Fig. 1 - Hydraulic circuit diagram (with electronic lift - STANDARD version).

A - Main oil filter on supply circuit
B - Hydraulic pump for auxiliary service control valve and power lift
C - Hydraulic pump for hydrostatic steering and electro-hydraulic controls
D - HML
E - Hydrostatic steering control valve
F - Steering cylinders
G - Suction line filter
G1 - Delivery line filter
H - Pressure control valves
I - Gearbox lubrication
L - Hydraulic power unit
M - Differential locks
N - 4WD engagement
O - Rear P.T.O. clutch
P - 6-way auxiliary service control valve
Q - Front lift hydraulic control valve
R - Lifting cylinder
S - Shut-off valve for front lift
T - Control valve for hydraulic trailer braking
U - Front lift cylinders
V1 - Anti-shock valve for front lift
V2 - Lowering speed control valve for front lift
Z - Hydraulic cylinders for adjustable stabilisers
W - Hydraulic cylinder for adjustable lift rod
X - Solenoid valve for switching flow between stabilisers/lift rod
Y - Supplementary hydraulic control valve
J1 - Supplementary hydraulic line
J2 - Supplementary hydraulic line for free return
J 3 - Hydraulic unloader (oil level pump)

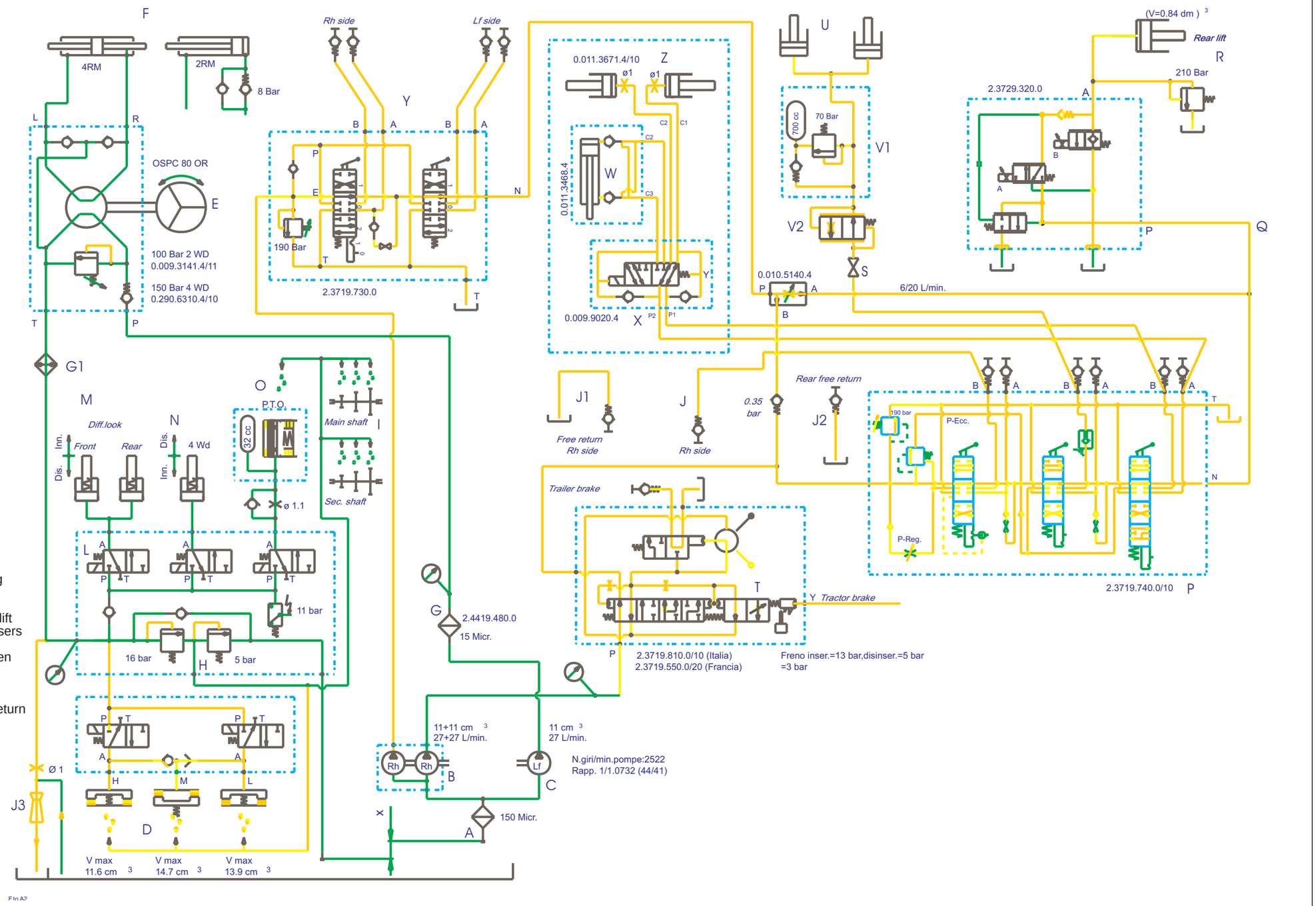


Fig. 2 - Hydraulic circuit diagram (version with full complement of optional equipment).

A - Main oil filter on supply circuit
B - Hydraulic pump for auxiliary service control valve and power lift
C - Hydraulic pump for hydrostatic steering and electro-hydraulic controls
D - MULTISPEED
E - Hydrostatic steering control valve
F - Steering cylinders
G - Suction line filter
G1 - Delivery line filter
H - Pressure control valves
I - Gearbox lubrication
L - Hydraulic power unit
M - Differential locks
N - 4WD engagement
O - Rear P.T.O. clutch
P - 6-way auxiliary service control valve
P1 -Supplementary 6-way auxiliary service control valve (with electro-hydraulic controls)
Q - Hydraulic unloader (oil level pump)
R - Lifting cylinder
S - Shut-off valve for front lift
T - Control valve for hydraulic trailer braking
U - Front lift cylinders
V1 - Anti-shock valve for front lift
V2 - Lowering speed control valve for front lift
Z - Hydraulic cylinders for adjustable stabilisers
W - Hydraulic cylinder for adjustable lift rod
X - Solenoid valve for switching flow between stabilisers/rh lift rod
Y - Supplementary hydraulic control valve with electro-hydraulic controls
J1 - Supplementary hydraulic line
J2 - Supplementary hydraulic line for free return
K - Shuttle control solenoid valve (FWD/REV)
K1 - Shuttle actuator cylinder
K2 - Proportional clutch valve
K3 - Clutch actuator cylinder

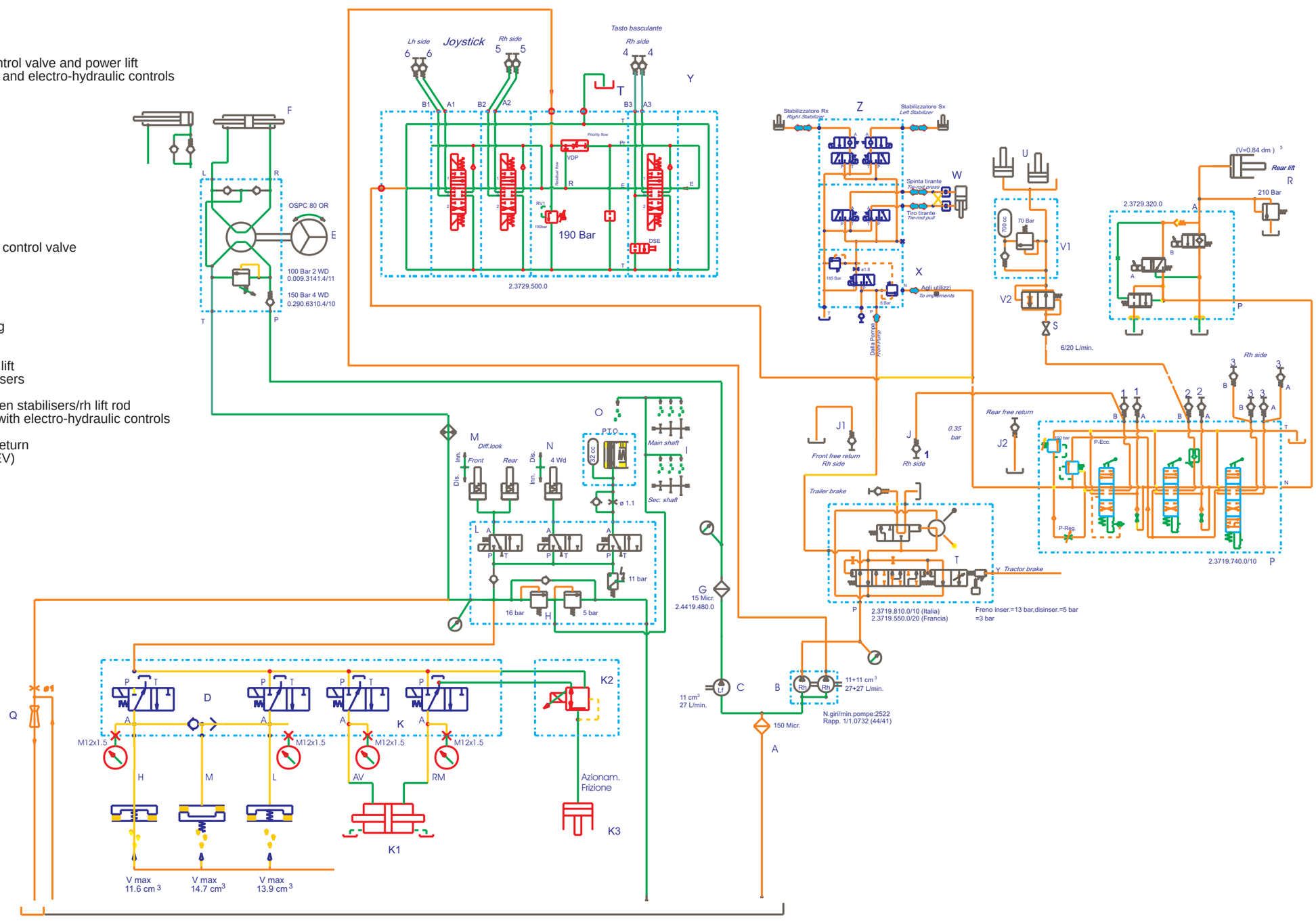


Fig. 3 - Hydraulic circuit diagram (version with MULTISPEED and hydraulic shuttle).

A - Check valve
B - Check valve
C - Gearbox lubrication pressure
D - Sequence valve (16 bar)
P - Pump
T - Oil return
L1 - Gearbox lubrication
L2 - P.T.O. lubrication
1 - Rear P.T.O.
2 - Differential lock
3 - 4WD

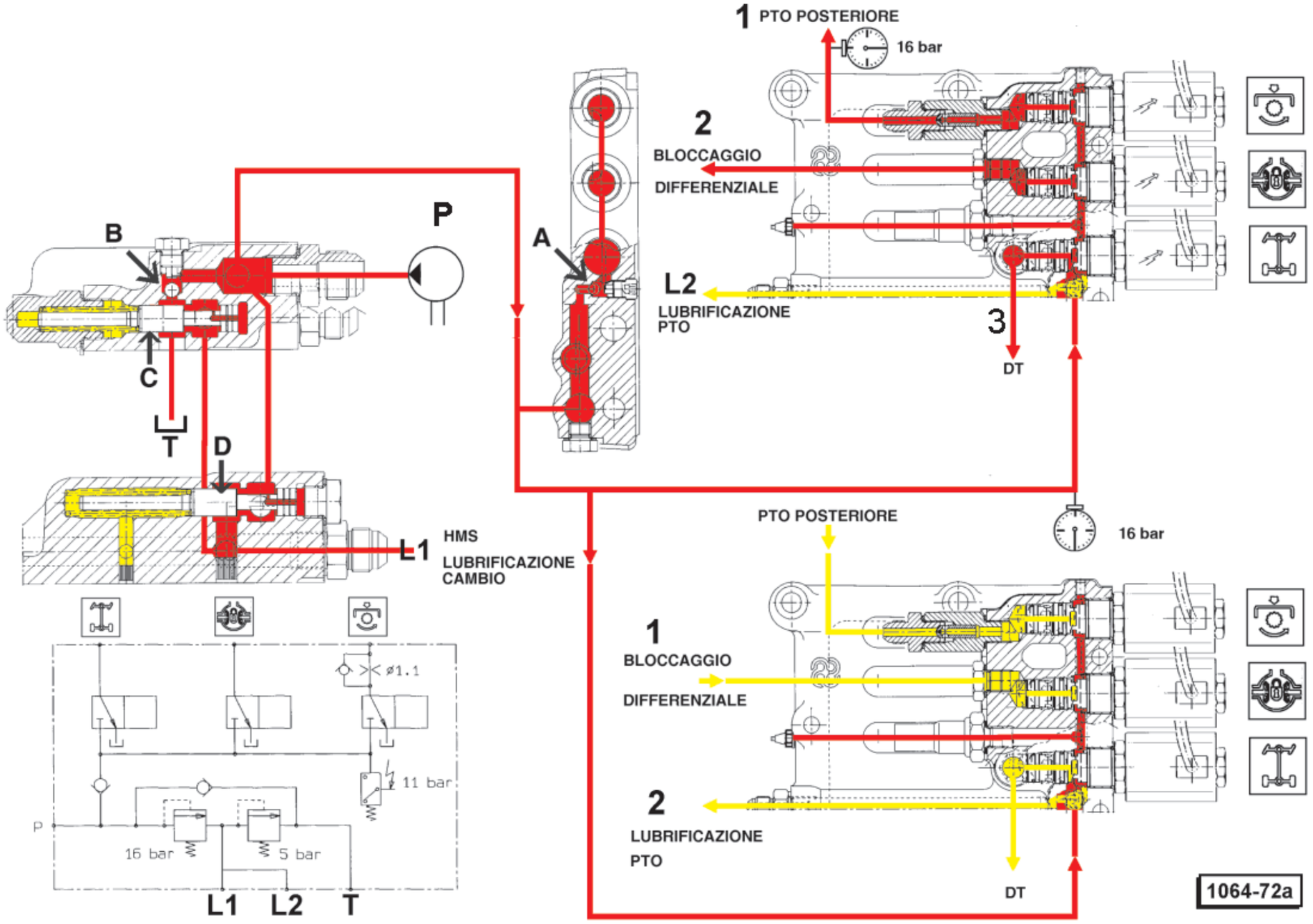


Fig. 4 - Hydraulic circuit on right-hand side of gearbox.

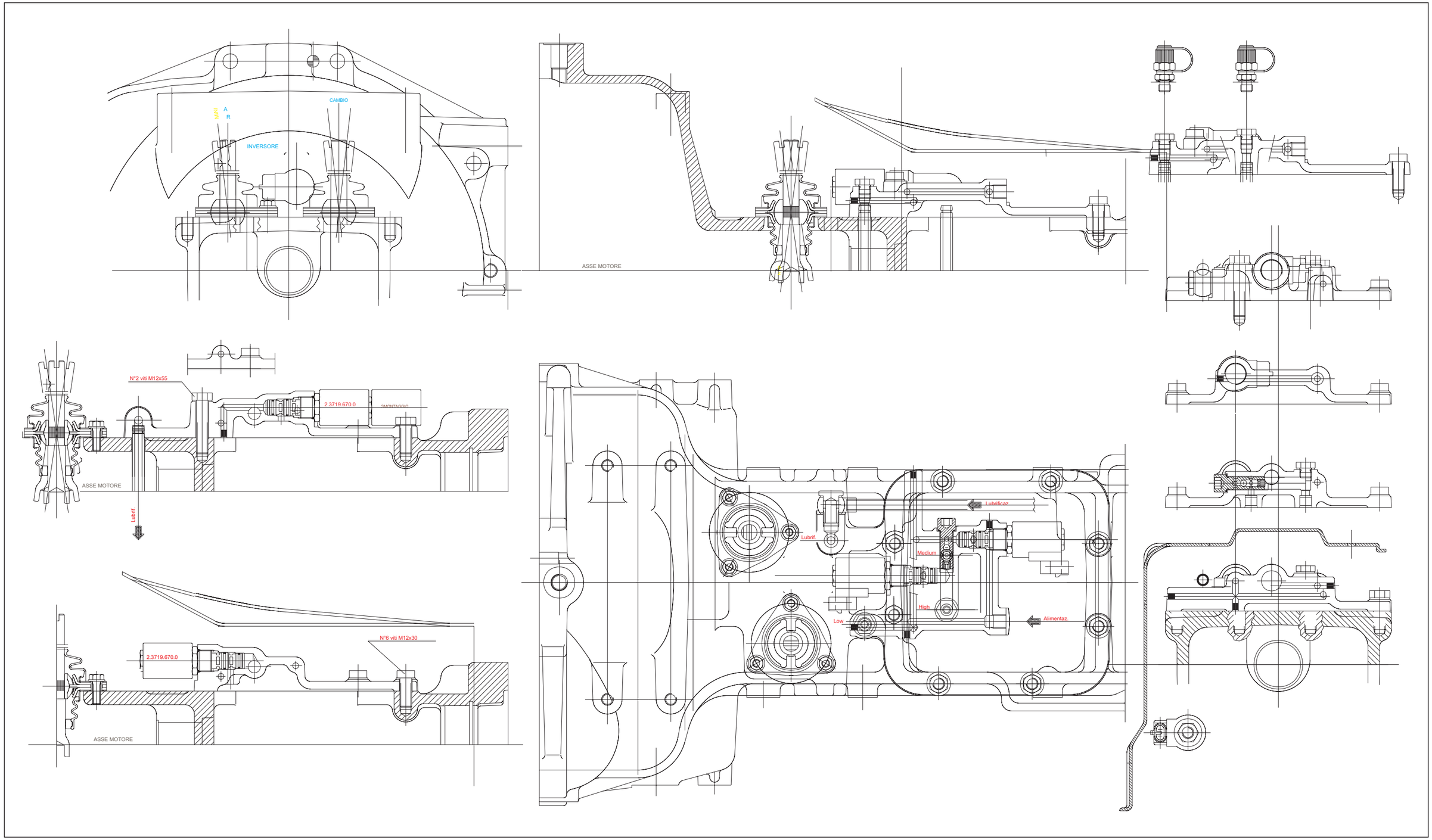


Fig. 5 - Hydraulic components on upper part of gearbox casing - Version with MULTISPEED.

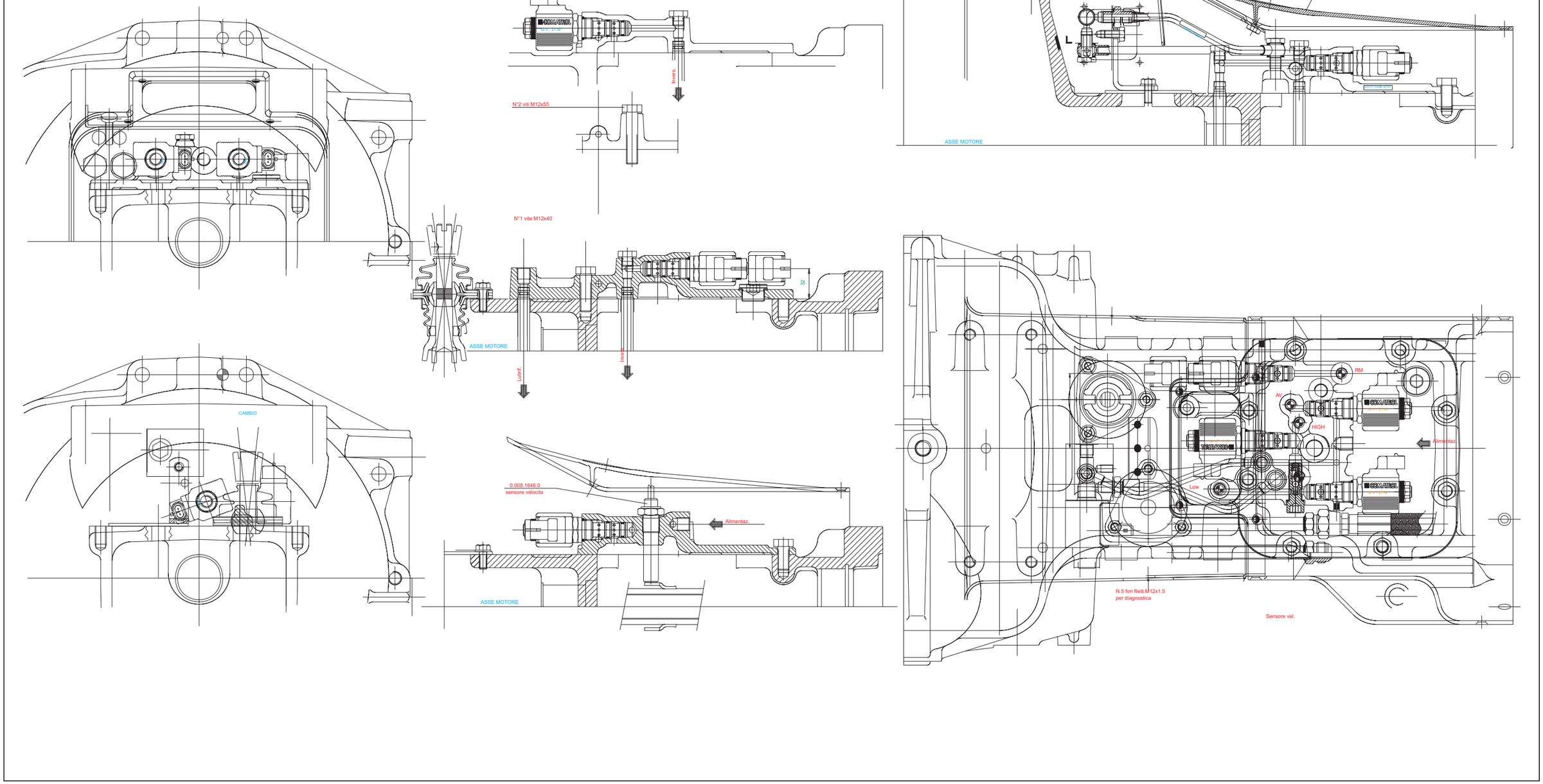
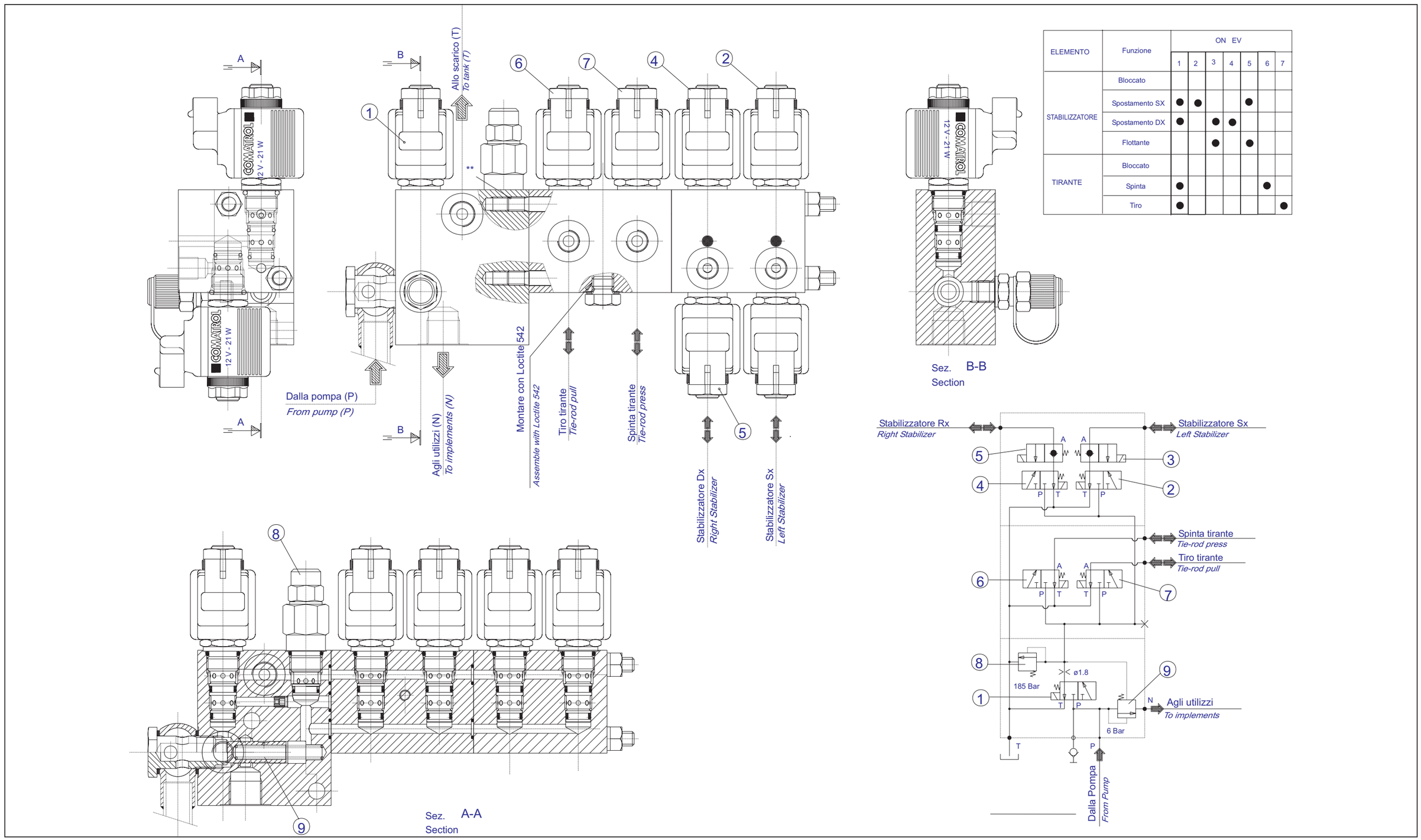


Fig. 6 - Hydraulic controls for shuttle and MULTISPEED.



ELEMENTO	Funzione	ON EV						
		1	2	3	4	5	6	7
STABILIZZATORE	Bloccato							
	Spostamento SX	●	●			●		
	Spostamento DX	●		●	●			
	Flottante			●		●		
TIRANTE	Bloccato							
	Spinta	●					●	
	Tiro	●						●

Fig. 7 - Electro-hydraulic controls for stabilisers and right-hand lift rod.

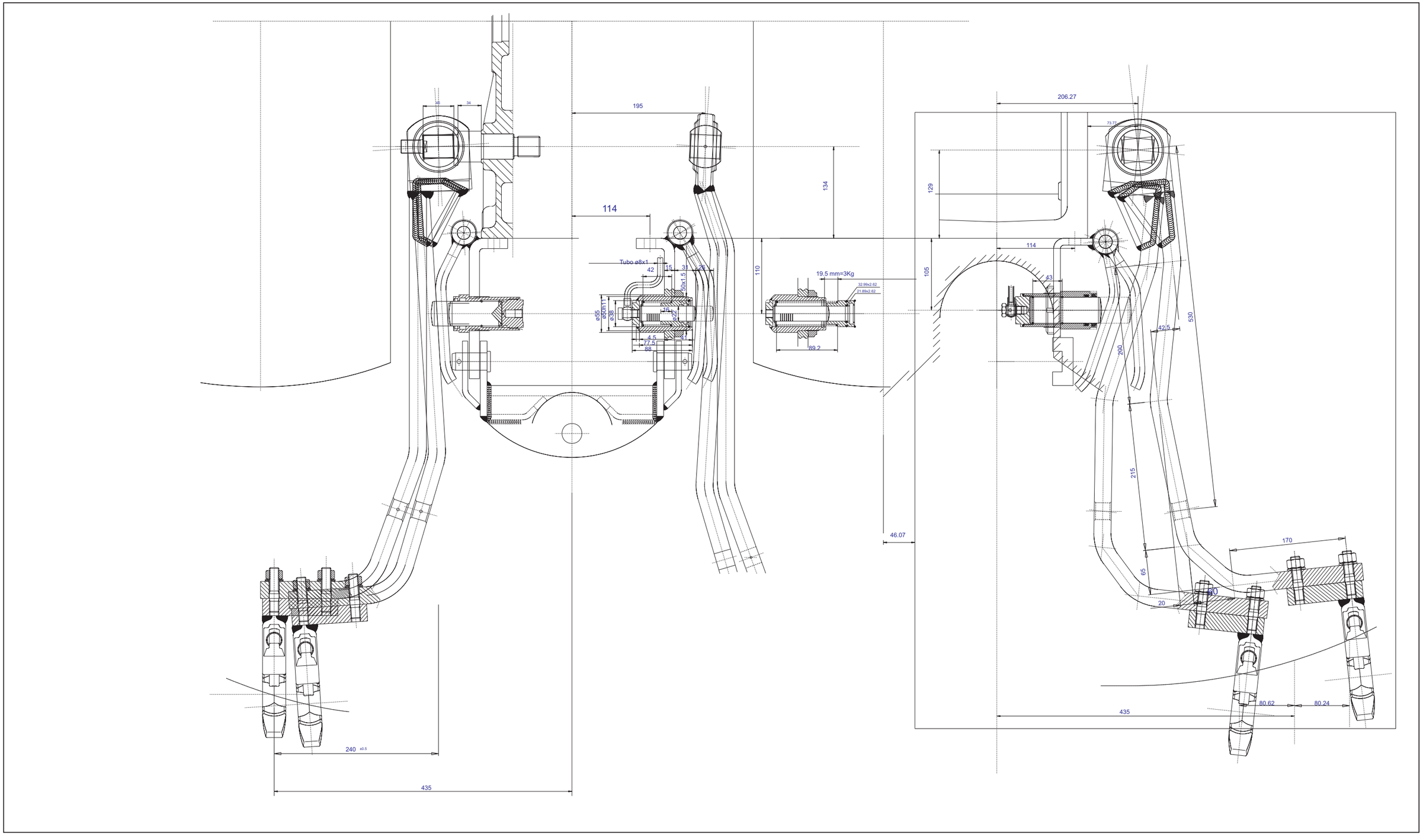


Fig. 8 - Hydraulic system for adjustment of stabilisers.

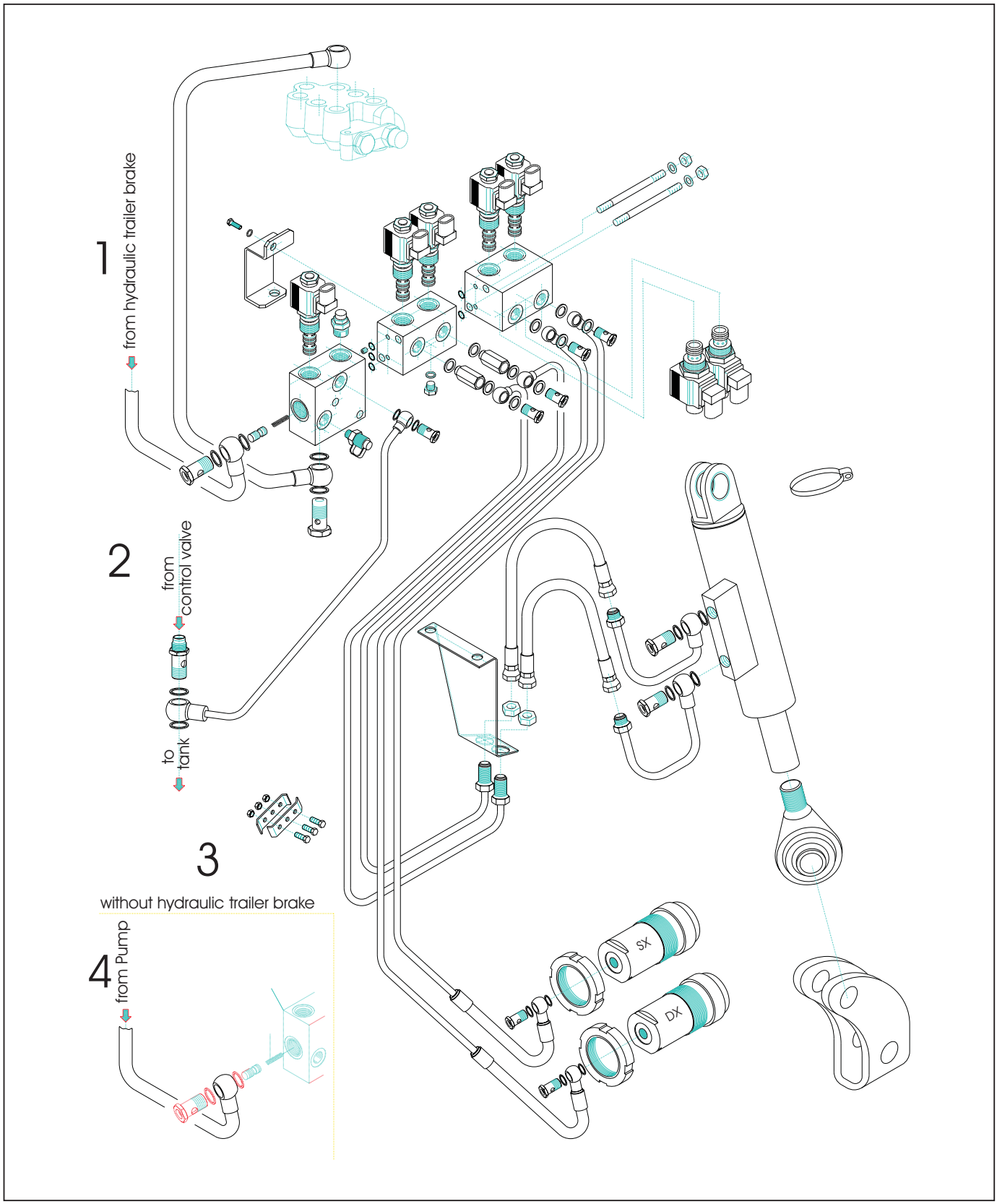


Fig. 9 - Hydraulic system for adjustment of stabilisers and right-hand lift rod.
1 - From trailer braking control valve
2 - Return oil (from pressure control valve)
3 - Version with hydraulic trailer braking
4 - From oil pump

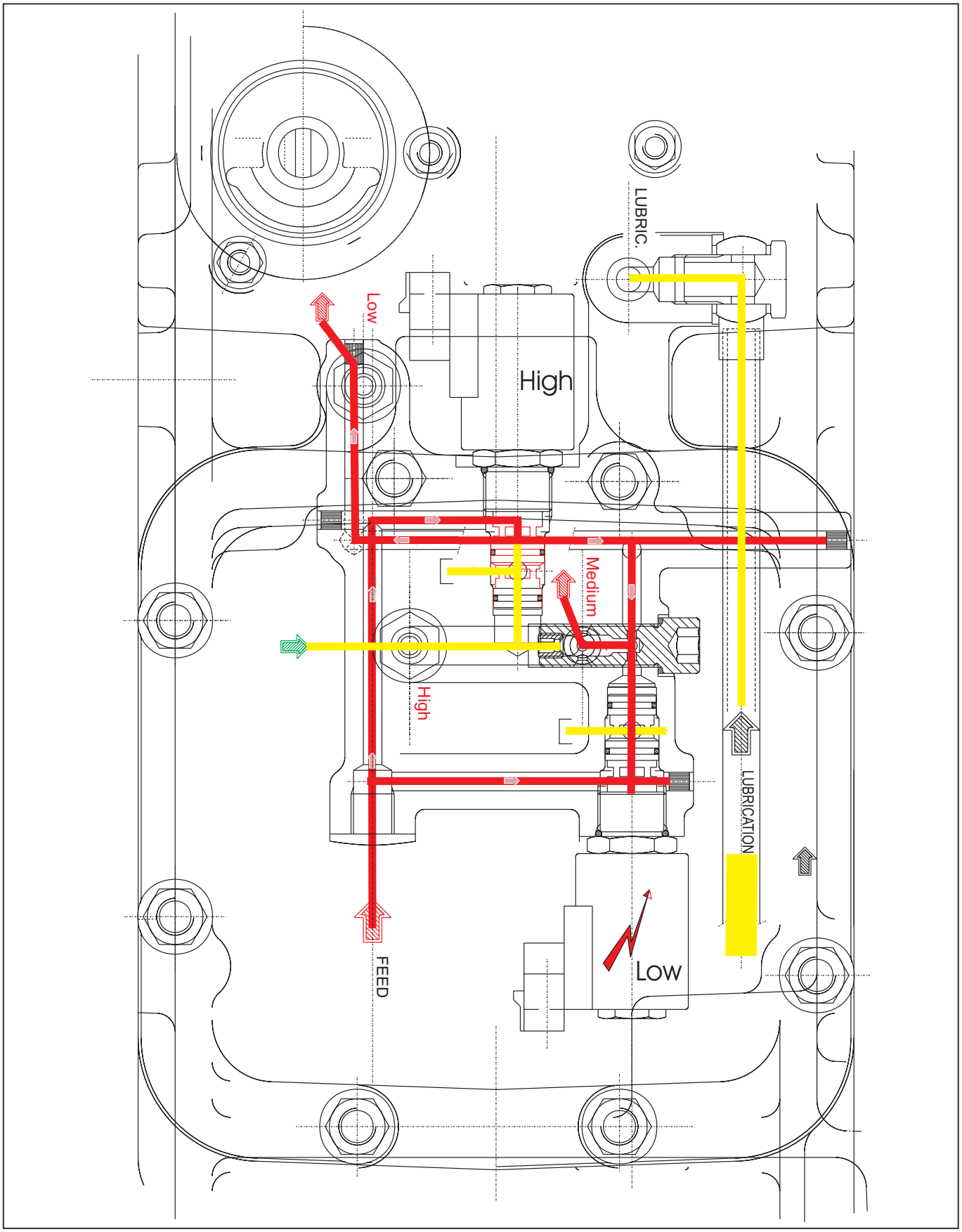


Fig. 10 - MULTISPEED solenoid valves - **LOW clutch engagement**

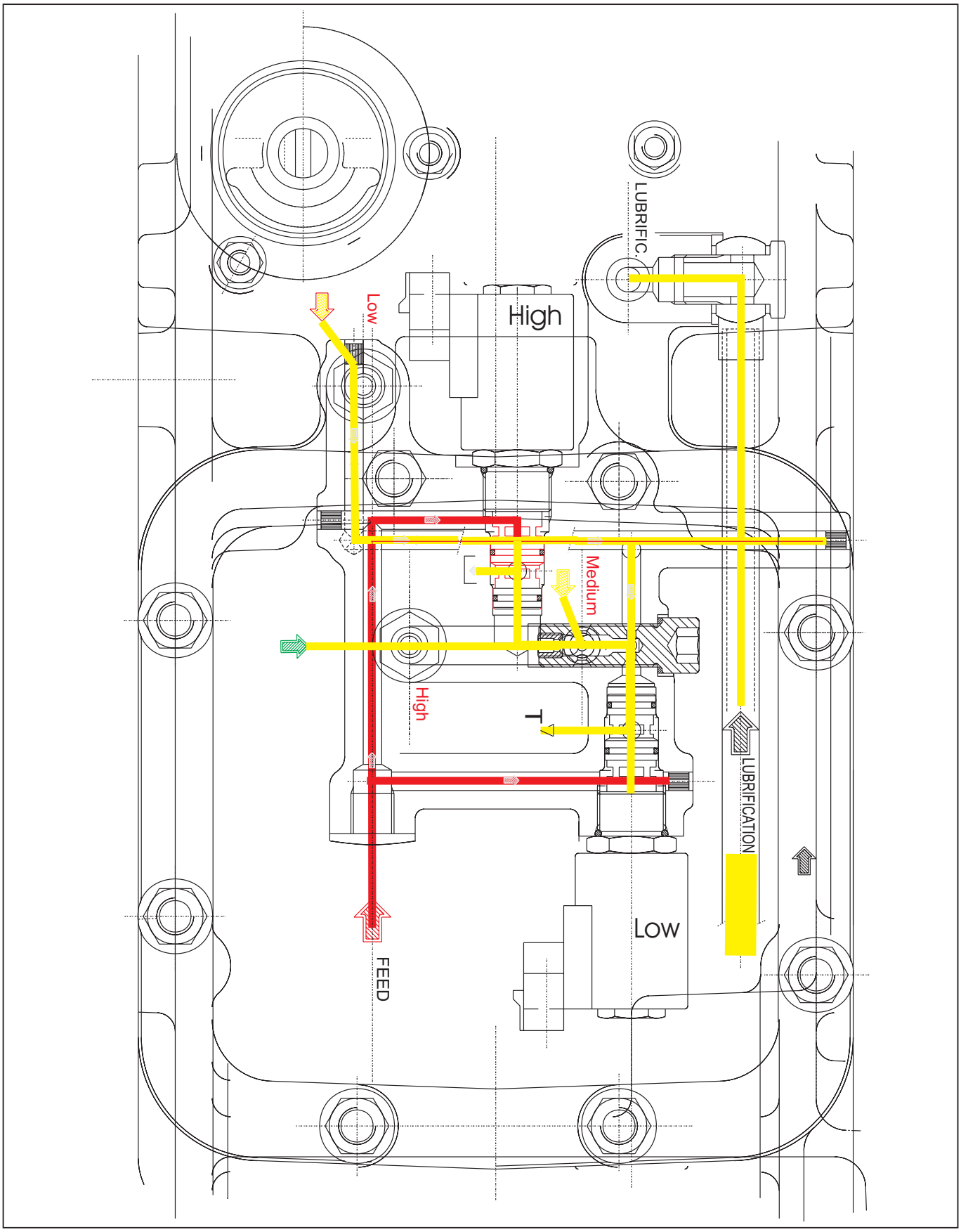


Fig. 11 - MULTISPEED solenoid valves - **MEDIUM clutch engagement**

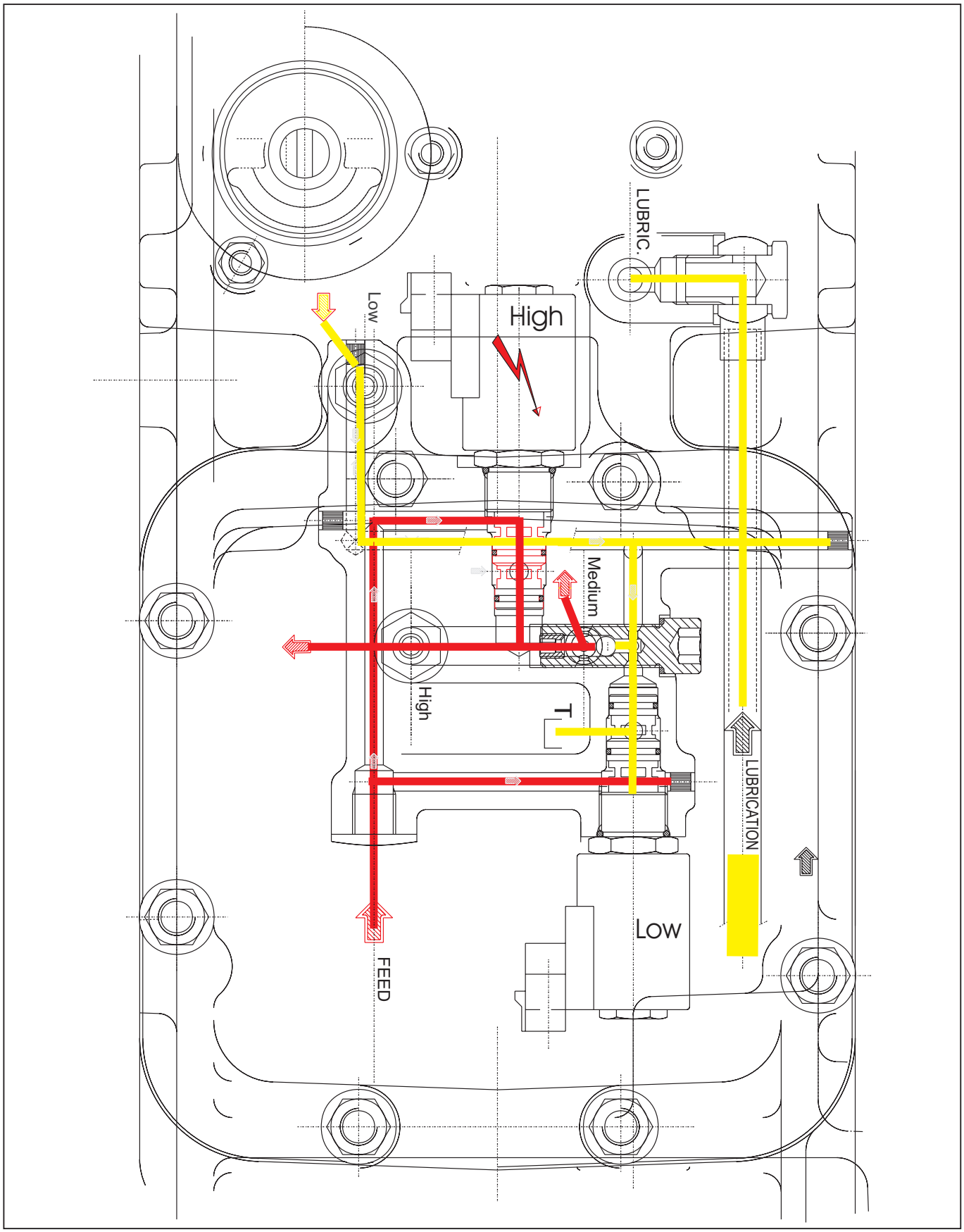
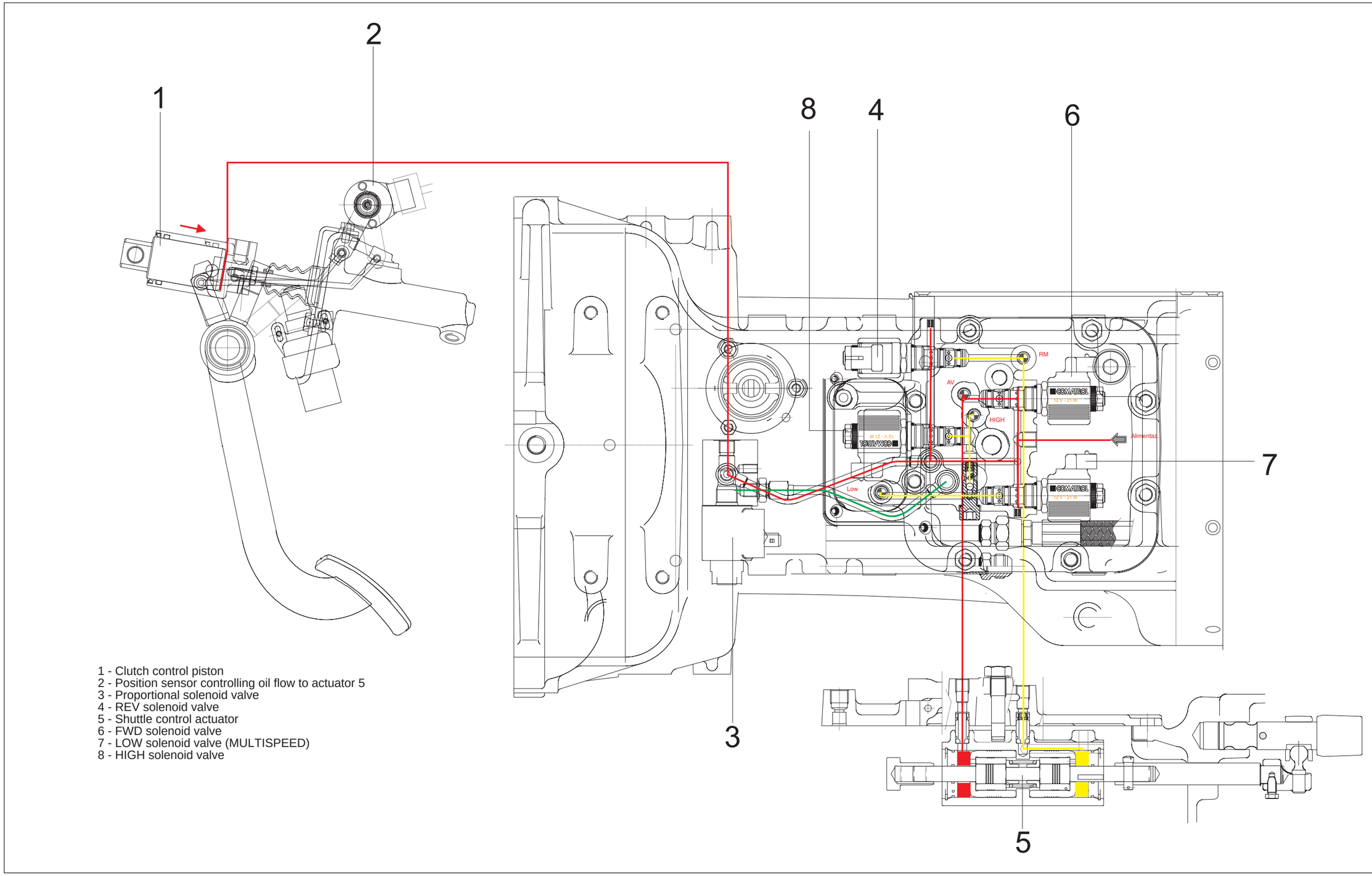


Fig. 12 - MULTISPEED solenoid valves - **HIGH clutch engagement**.



1 - Clutch control piston
2 - Position sensor controlling oil flow to actuator 5
3 - Proportional solenoid valve
4 - REV solenoid valve
5 - Shuttle control actuator
6 - FWD solenoid valve
7 - LOW solenoid valve (MULTISPEED)
8 - HIGH solenoid valve

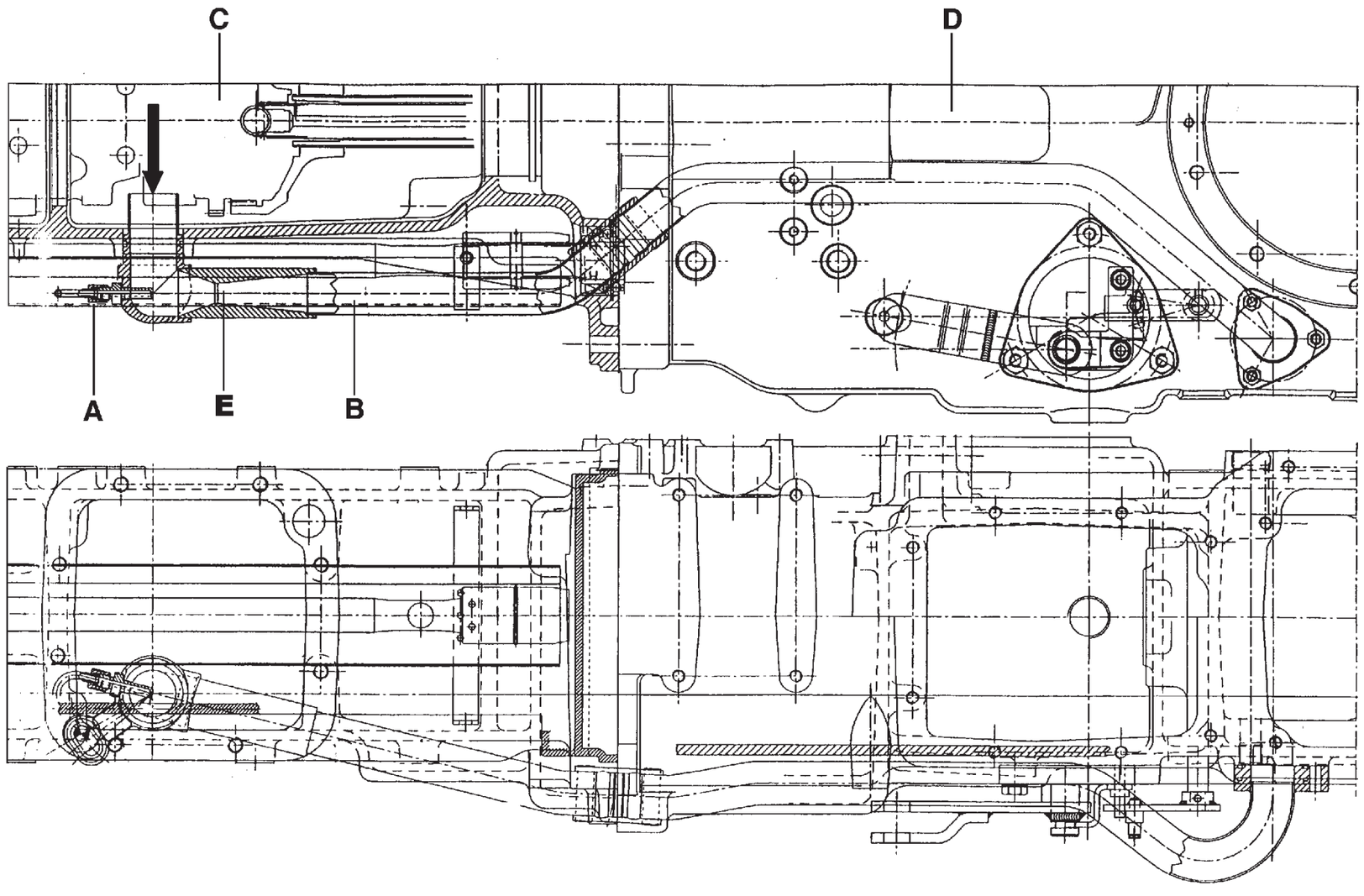
Fig. 13 - Control system for MULTISPEED with hydraulic shuttle.

The hydraulic injector **A** inserted in the transfer pipe **B** equalises the oil levels in the front and rear gearboxes.

The injector allows oil in the front gearbox to flow easily into the rear gearbox.

TECHNICAL CHARACTERISTICS

Flow used	1,8 l/min
pressure of jet	16 bar
flow transferred	25 l/min



A - Hydraulic injector
B - Transfer pipe
C - Front gearbox
D - Rear gearbox
E - Venturi pipe

Fig. 14 - Details of front and rear gearboxes.

Piston for hydraulic adjustment of stabilisers

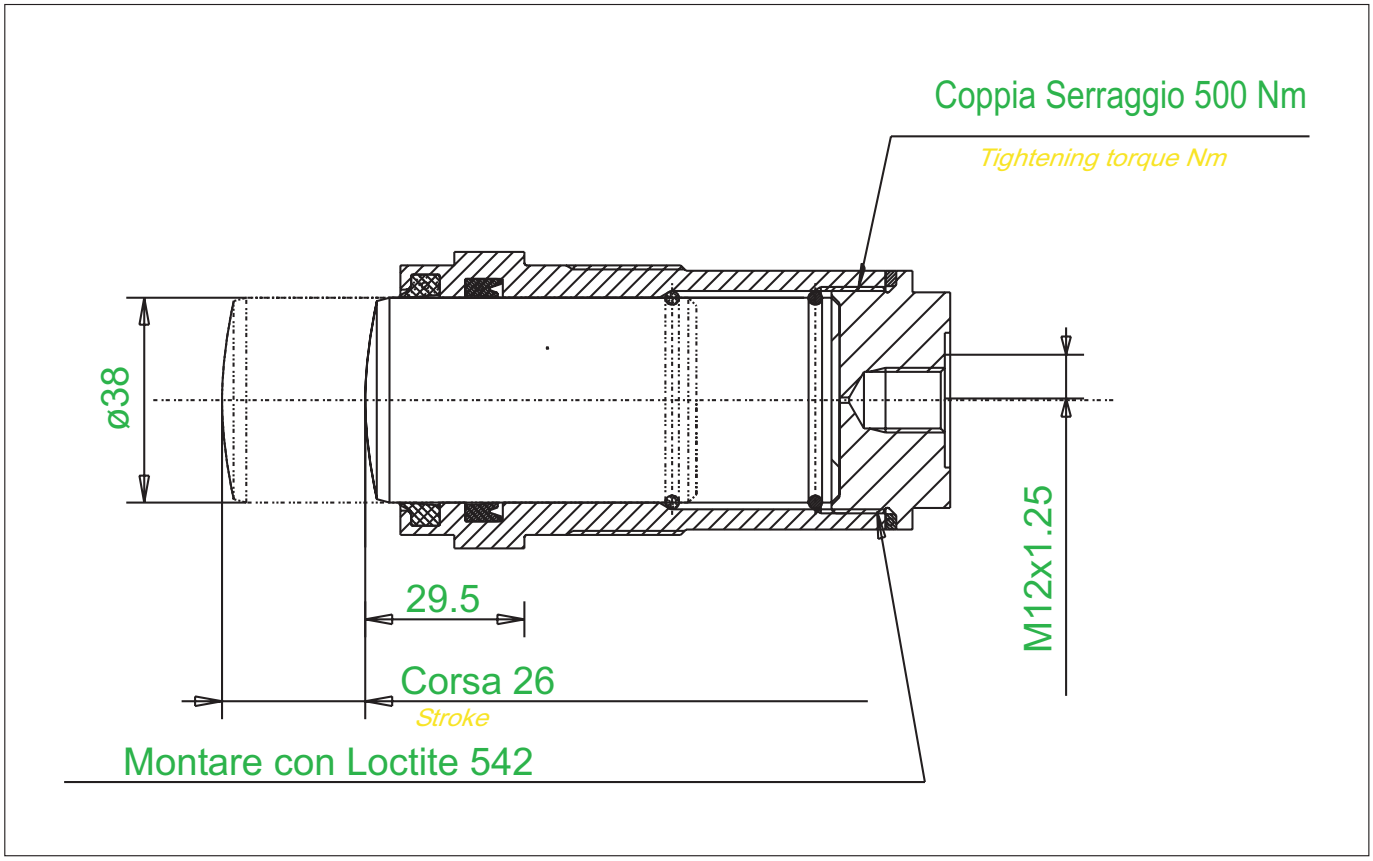


Fig. 15 - Section view of piston for hydraulic adjustment of stabilisers.

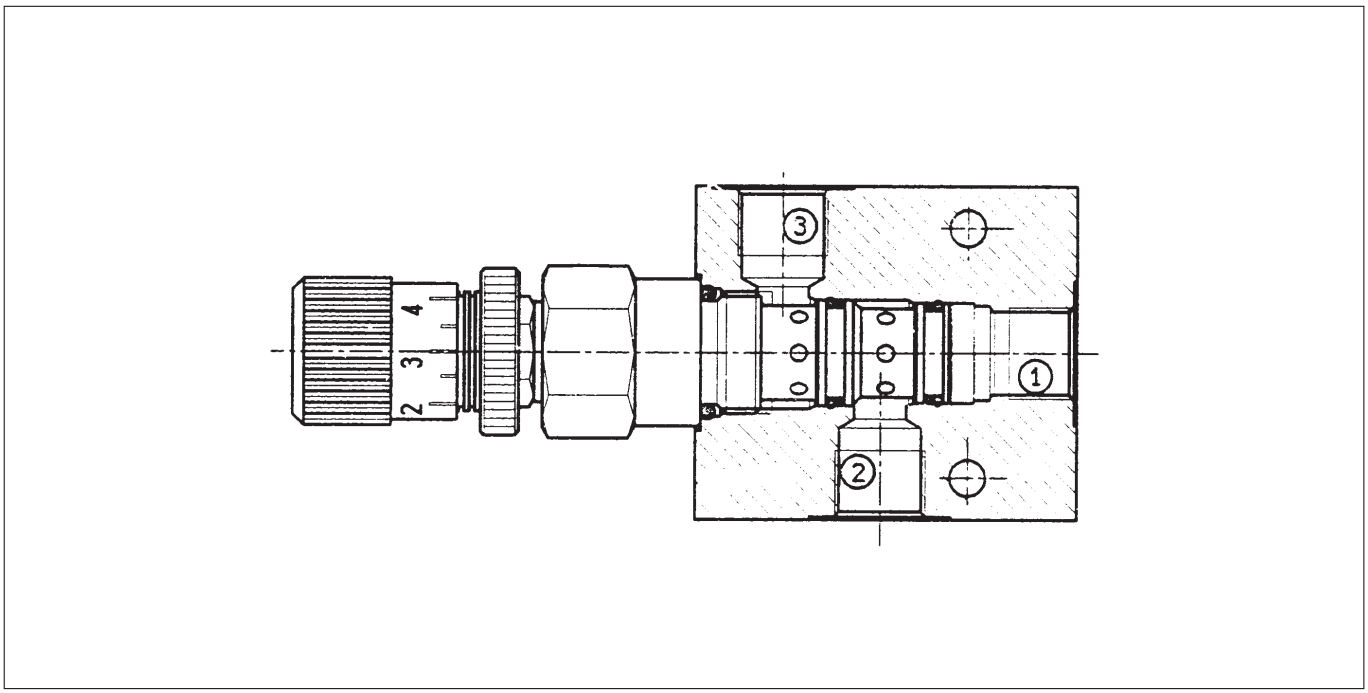


Fig. 16 - Flow divider valve with manual adjustment.

Adjustment of front and rear differential lock control

Turn fork **A** and lock nut **B** to preload spring **C**, so as to obtain a distance **X** allowing contact with piston **E** (with spring **D** attached) when the piston is positioned flush with surface **F** (on version with electrohydraulic controls) or protrudes 18.5 mm (on version with mechanical controls). Also check that the shoes of the fork do not rub against the sides of the engagement sleeve.
For adjustment of engagement sleeve, see chapter on front-wheel drive.

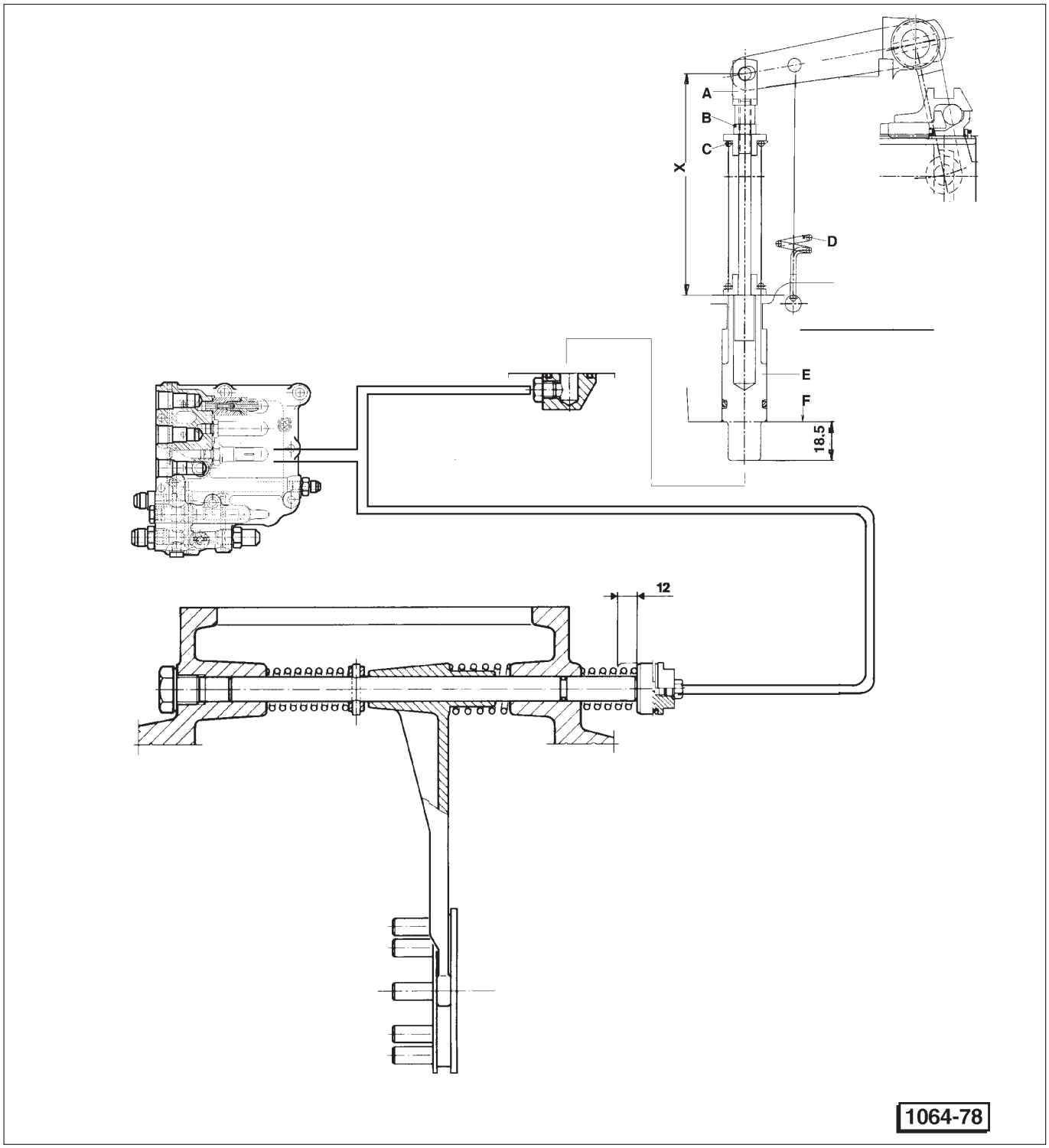


Fig. 17 - Front and rear differential lock control.

Platform

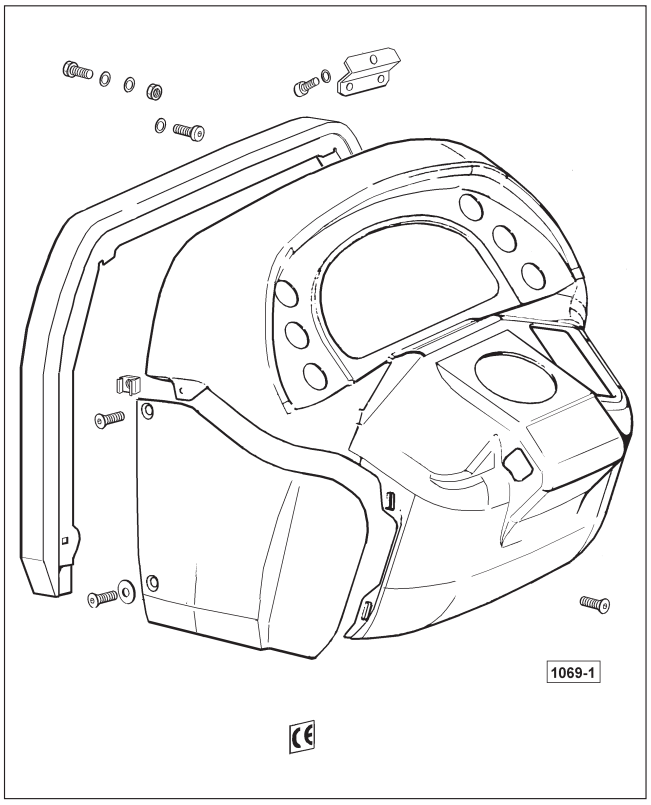


Fig. 1 - Instrument panel.

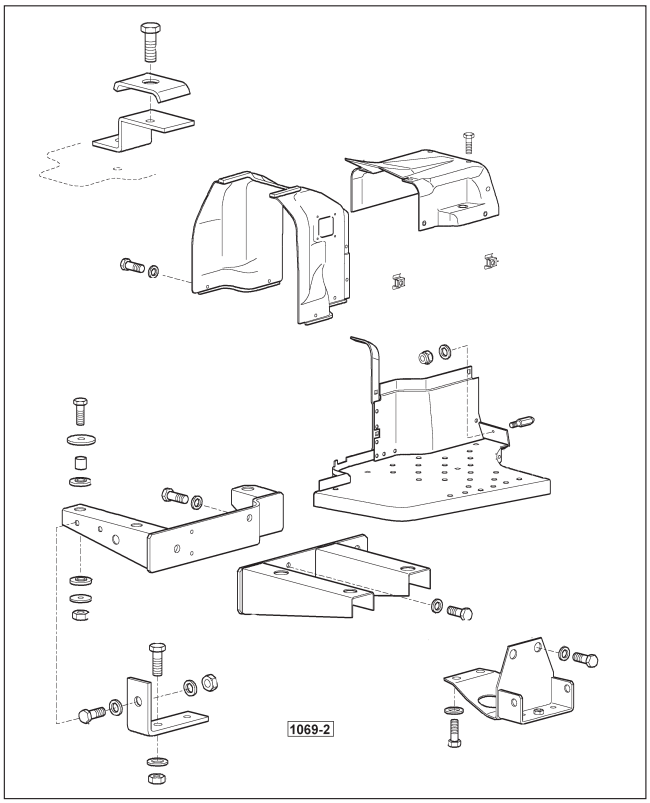


Fig. 2 - Footplates and covers.

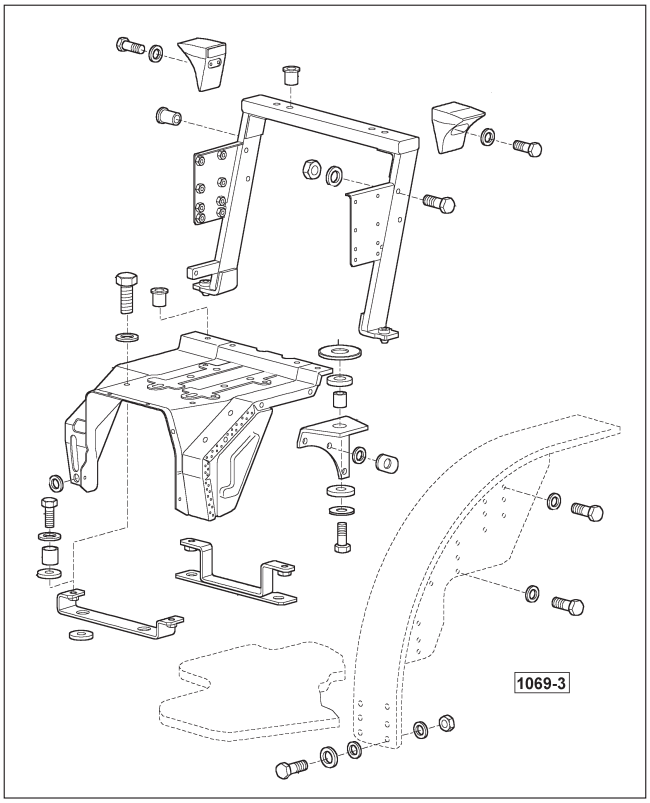


Fig. 3 - Footplates and covers.

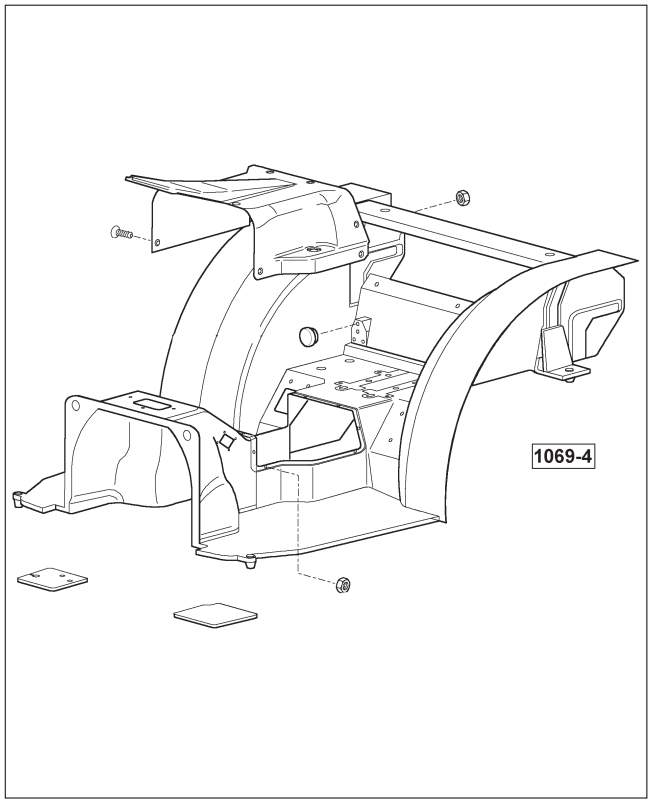


Fig. 4 - Platform.

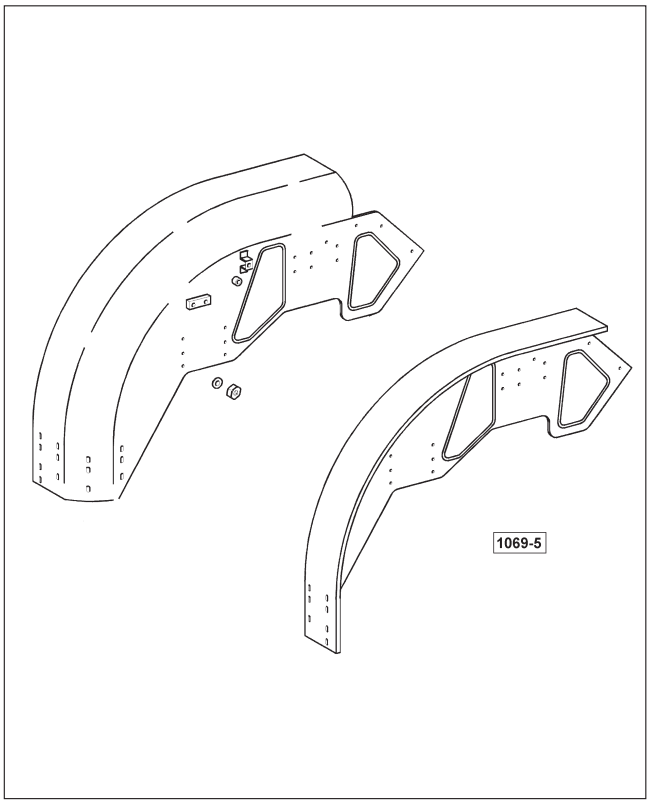


Fig. 5 - Rear mudguards.

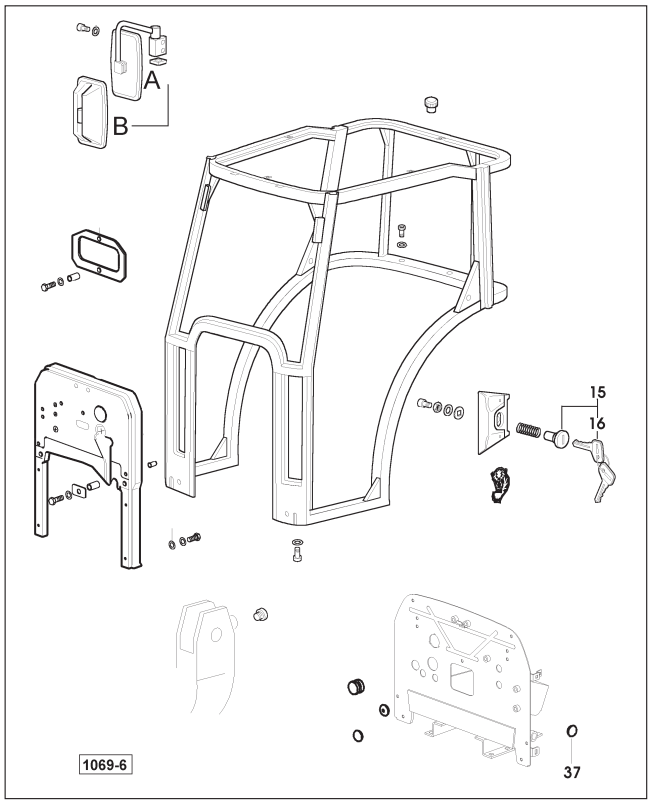


Fig. 6 - One-piece cab.

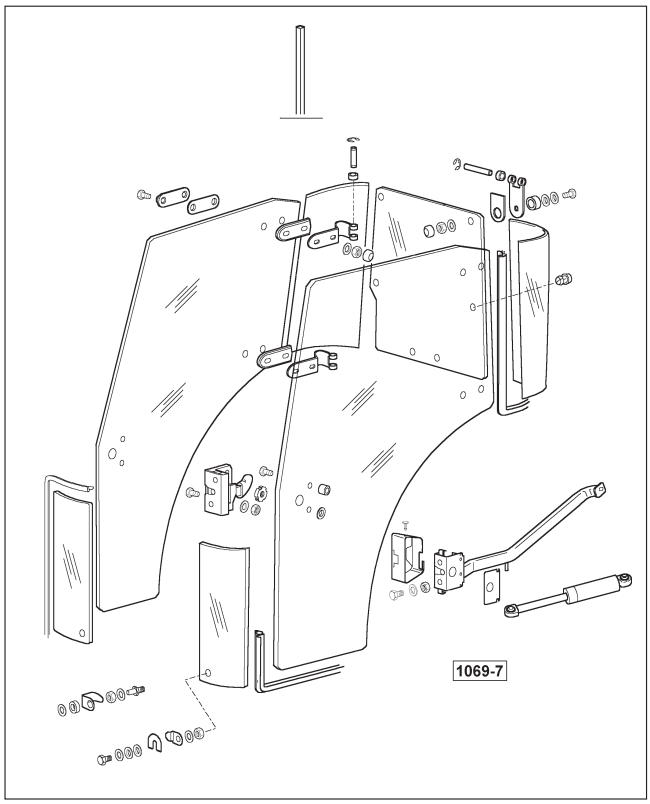


Fig. 7 - Glass and seals.

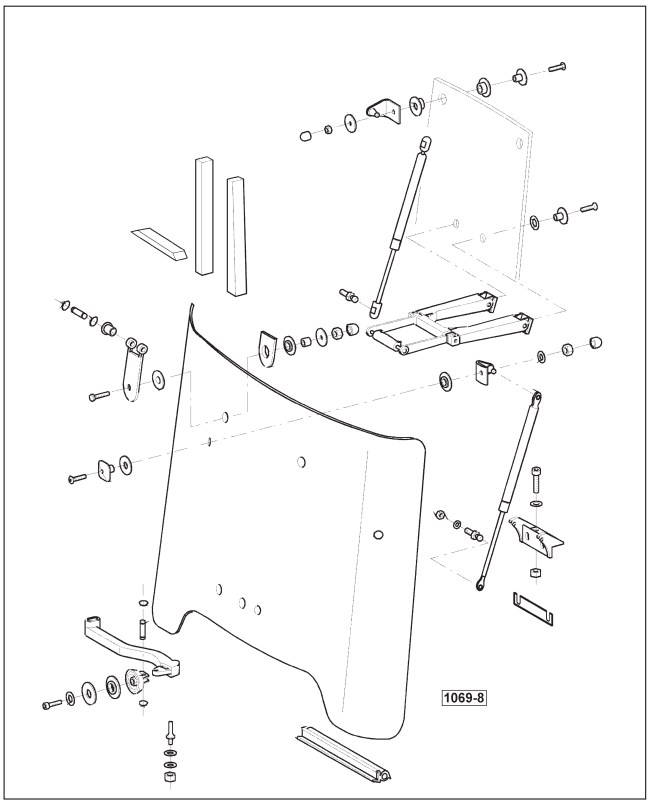


Fig. 8 - Glass and seals.

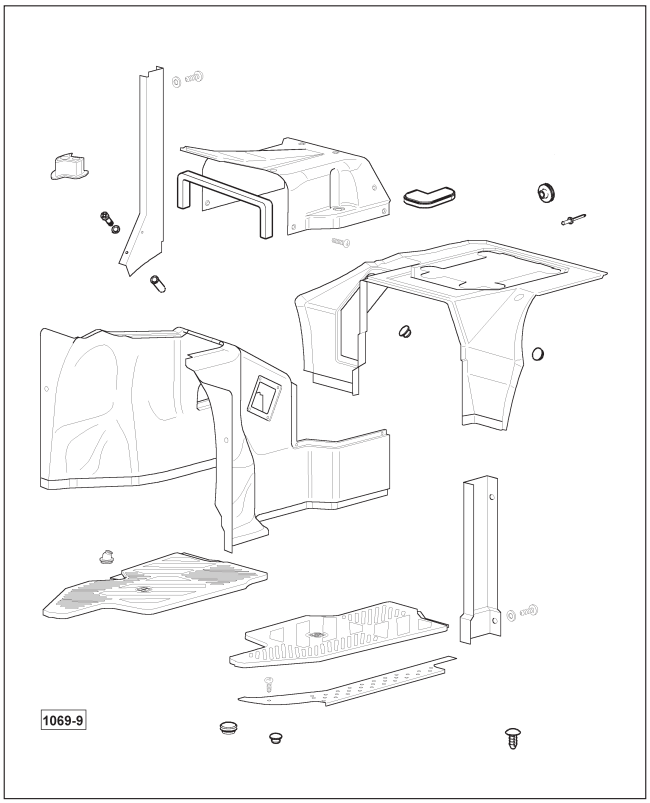


Fig. 9 - Interior trim panels.

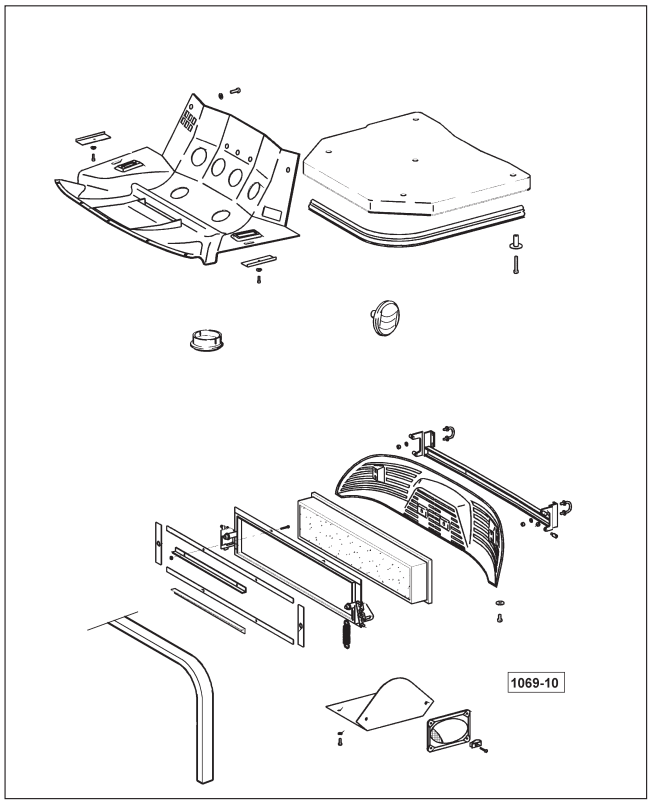


Fig. 10 - Interior trim panels.

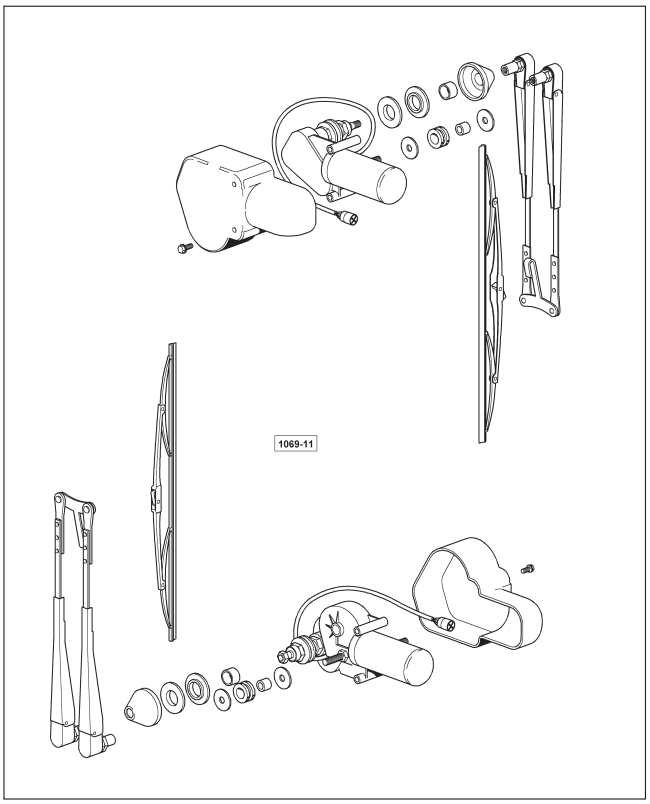


Fig. 11 - Windscreen wiper.

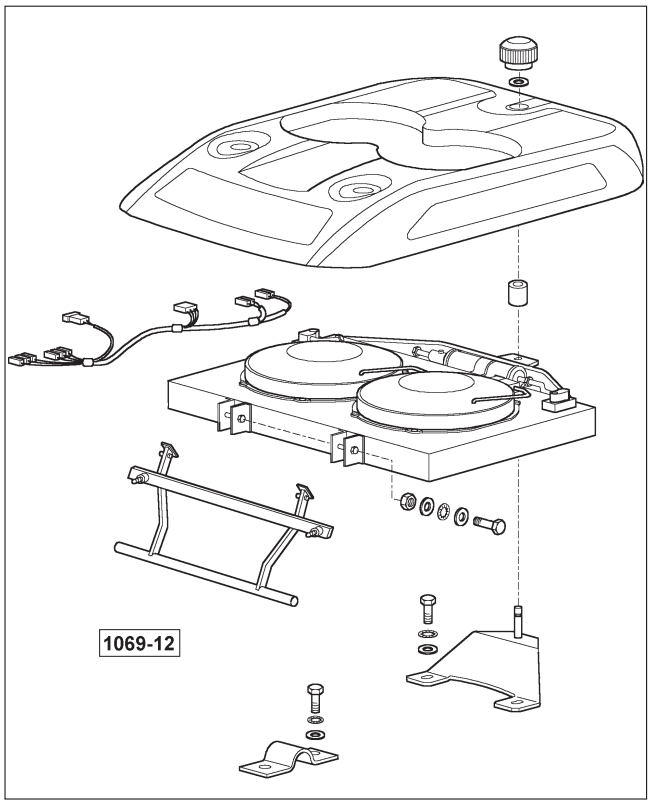


Fig. 12 - Ventilation and air conditioning.

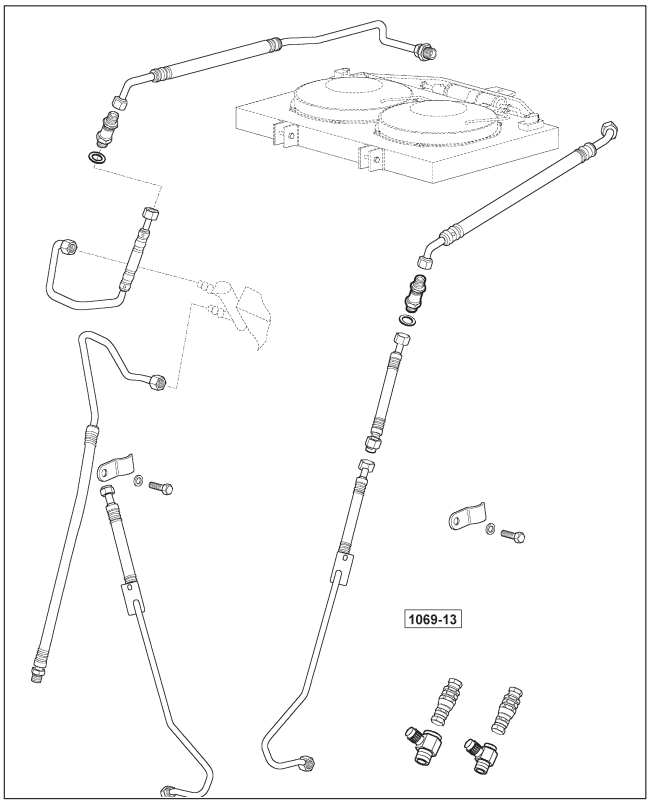


Fig. 13 - Ventilation and air conditioning.

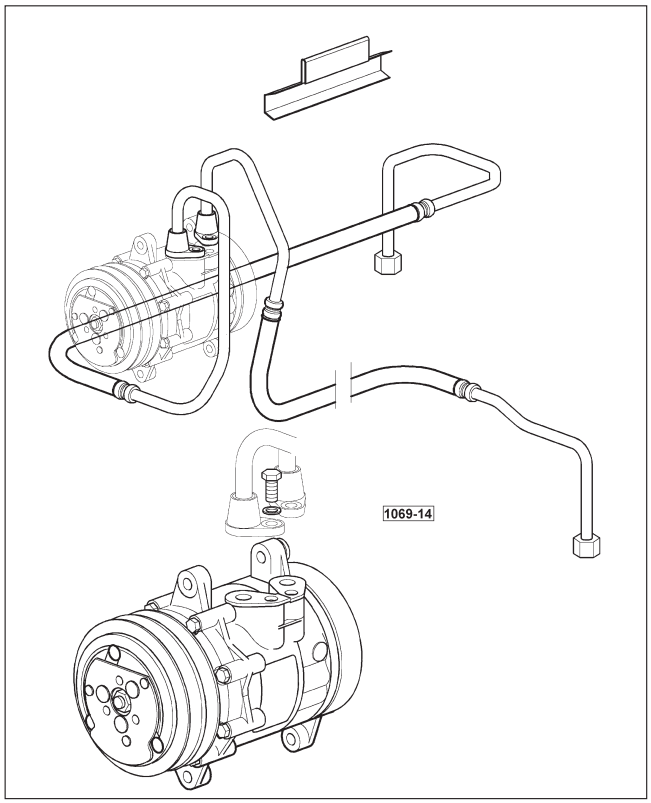


Fig 14 - Ventilation and air conditioning. (version with air-cooled engine

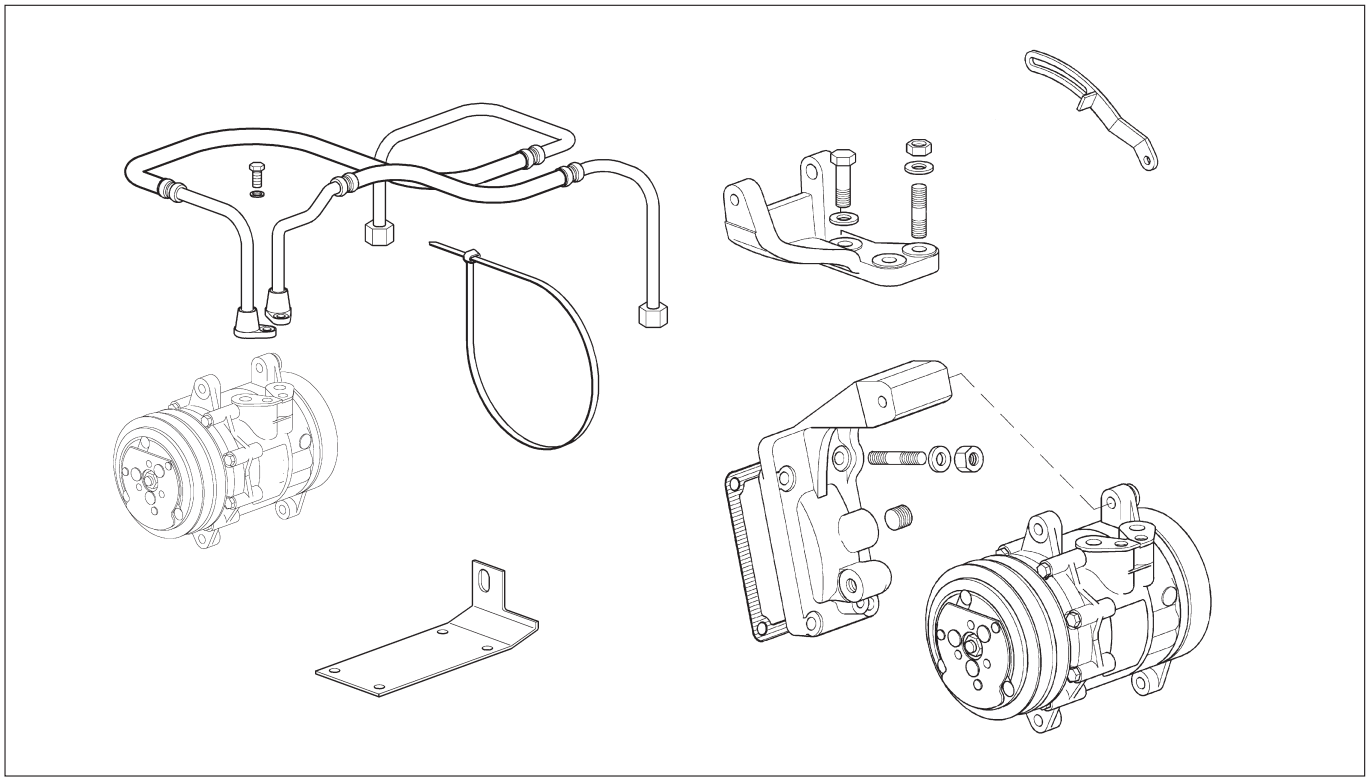


Fig. 15 - Ventilation - air conditioning - heating (version with water-cooled engine).

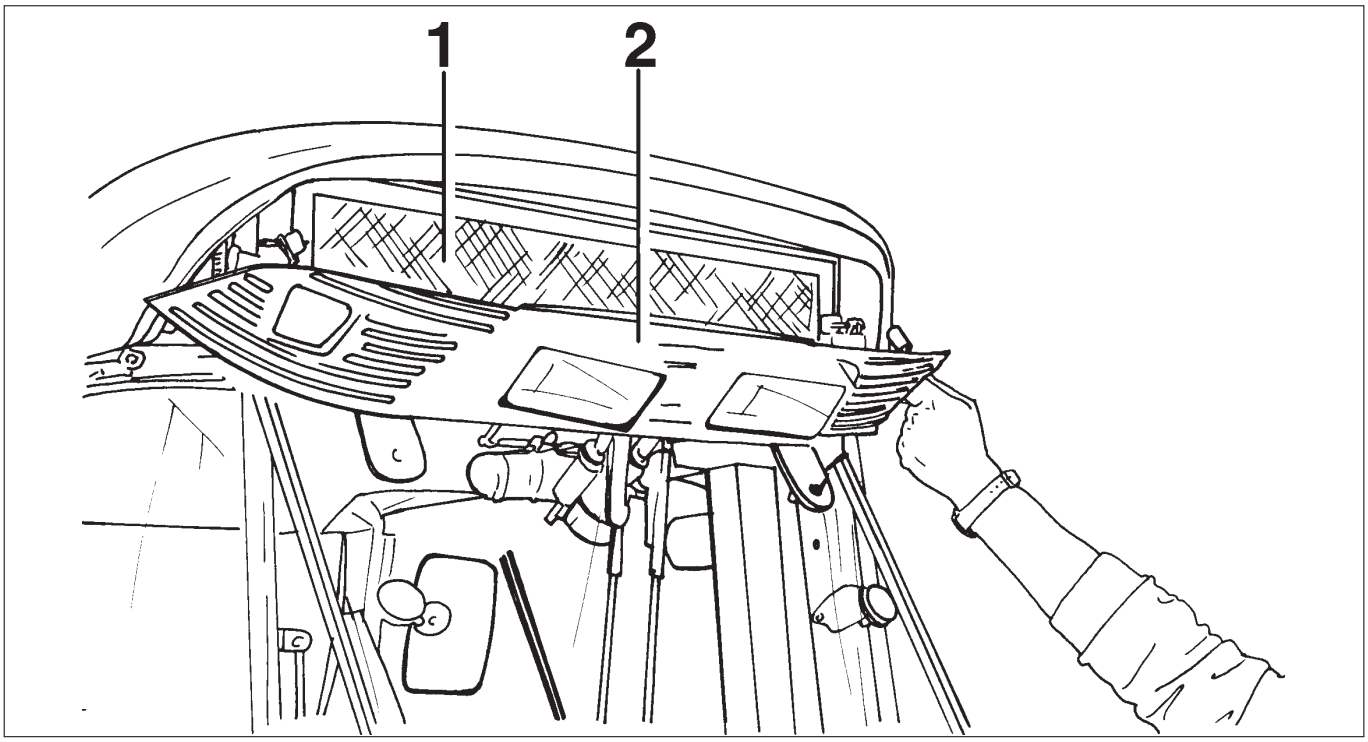


Fig. 16 - Cab air filter assembly.
1 - Air filter
2 - Grille

Cab air filter
Periodically (as dictated by operating conditions): clean the filter.
To gain access to the filter, lower the protective grille,release the two filter clamp knobs, and remove
The filter is cleaned as follows:
— blast with compressed air (6 bar max) in the direction opposite to that of normal filtration, until all accumulated dust has been dislodged;
— wash for about 15 minutes in a solution of detergent and water at 40 °C;
— rinse clean in running water;
— leave to dry naturally at ambient temperature.
Important: the filter must be replaced if damaged, or renewed routinely after cleaning 6 times.

Screen wash

Periodically (as dictated by operating conditions):
check the level of screenwash liquid in the plastic container located at the rear of the machine.
In the event of the jet being rendered defective by an obstruction, this will be remedied generally by freeing the nozzle outlet with a pin.
If necessary, reposition the nozzle so that the jet strikes the screen at the top of the sweep made by the wiper blade.
It is good policy during winter months to add antifreeze solution to the screenwash liquid, or methyl alcohol.
Warning: it is important that all glass in the cab is kept thoroughly clean.
The rear view mirrors must always be kept clean and properly positioned.

Screen wipers (front and rear)

The wiper blade has a bayonet fitting.
The blade is removed by lifting the catch of the central mounting.

Remove cab.

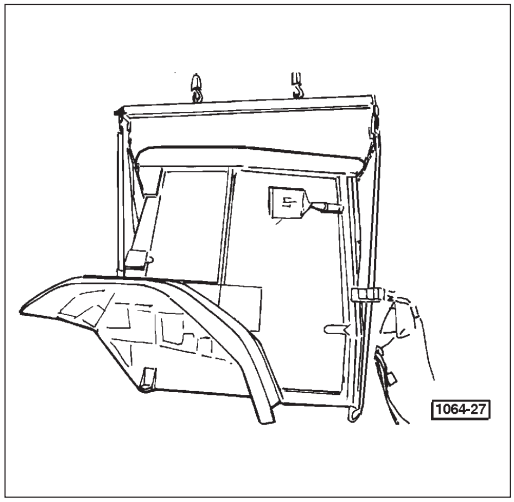


Fig. 17 - Position the cab on suitable supports and proceed with the operations on the tractor.

WARNING: during the cab hoisting operation, check carefully that all the parts have been disconnected and that the cab disengages from the tractor.

IMPORTANT: If the fuel tank is also to be removed, disconnect the flexible pipe from the metal leak-off pipe from the injection pump.

Breakage of the top hood release cable

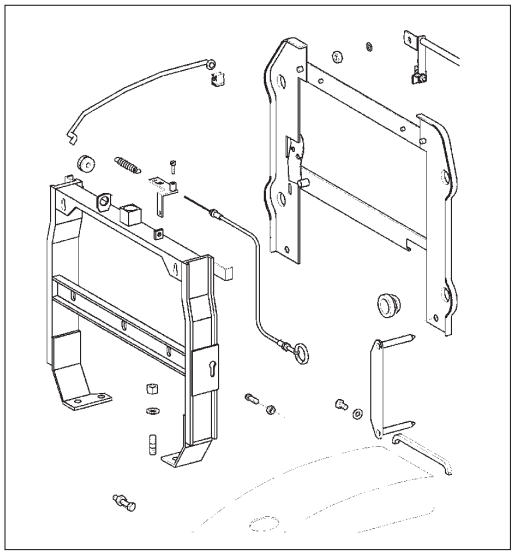


Fig. 18 - Component of the top hood release mechanism.

To open the hood, insert a screwdriver between the hood and the grille to operate the hood catch.

Push the screwdriver to the left (relative to the driving position) to release the hood.

Ventilation

The ventilation unit is housed in the cab ceiling.
To switch it on and adjust it, turn the electrical fan switch to the desired speed.
The cab becomes slightly pressurized when the ventilation system is in operation, so that fresh air can enter only by way of the filter installed in the rear section of the cab roof.
The fan switch can be operated only after the ignition key is inserted.

The air flow can be regulated and directed by suitably positioning the air diffusers.
Air can be taken in fresh from outside or recirculated from within the cab by way of the relative side inlets.
Recirculation inlets fully closed: air is taken in entirely from outside the cab through the rear grille and filtered through a paper element positioned behind the grille.
Either fully or partially open air recirculating grille: now the air is recirculated inside the cab.
N.B. - It is very important that the air diffusers never be completely closed so as to allow for a steady air flow.
To obtain a greater pressurization inside the cab, it is necessary to take the air from the outside, therefore the inside air recirculating grille should be fully closed.

Climate control system

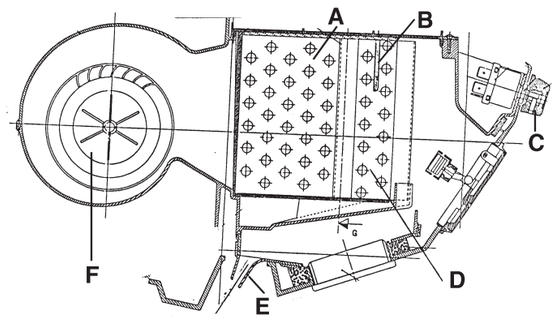
A - Condenser

B - Temperature sensor
(only for air-cooled engine)
C - Control switches

D - 2400 Watt electric heating element
(resistance across terminals 0.655 ohm)
(only for air-cooled engine)
D - Water radiator
(only for water-cooled engine)

E - Defroster

F - Ventilation unit



Heating System
The heater is switched on and adjusted by rotating the control knob at the roof console, then switching on the blower and setting the selector at the preferred speed.
To warm the cab up quickly, the knob should be rotated fully clockwise and the blower set to speed III.
The screen is demisted or defrosted by air directed through a slot vent.
For defrost or fast demist, all other vents should be closed off.

IMPORTANT: ventilation is provided by a single blower unit serving both the heating system and the air-conditioning system.
After reaching the desired temperature adjust the system to suit your needs.

N.B. - For ideal system operation, the engine must run at 1600 r.p.m.

WARNING: Before starting the engine, make sure the system is off (by turning off the ventilation fan) so as not to overload the battery.

After using the system at full power for a long period of time, never turn it off suddenly but let it first idle for about 20 seconds.

System configuration

(For air-cooled engine)
The heating system consist of two units:
1 - Electric heater 3 (3 fig. 1) and blower unit (4 fig. 1) installed behind roof console.
2 - Power supplyng set, consisting of an auxiliary alternator (1 fig.1) located front of the engine and driven by a belt directly linked to the engine pulley.

System configuration

(For water-cooled engine)
The heating system comprises a blower unit (3 fig. 1) and a heater (4 fig. 1) installed behind roof console.

The heating system utilized the same fluid as the engine cooling circuit.

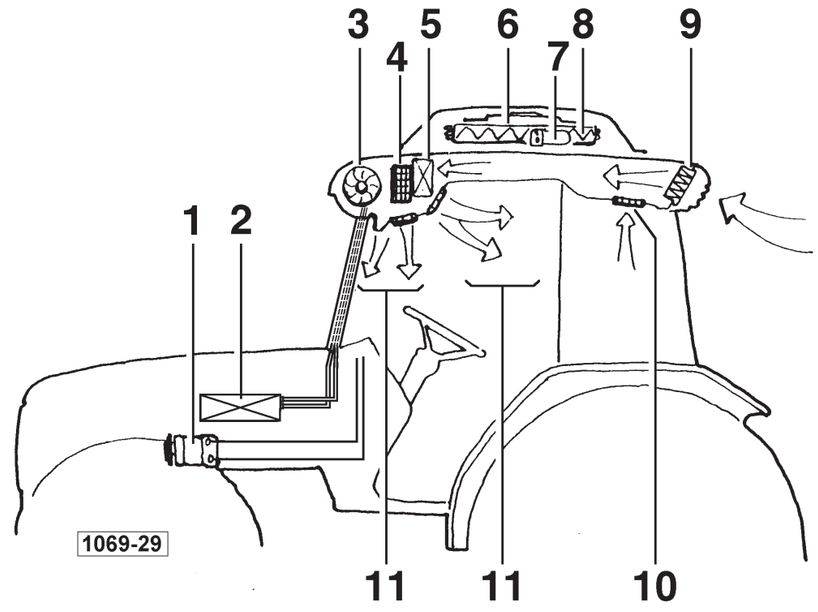


Fig. 1 - Configuration of ventilation and heating unit.

1 - Compressor (For air-cooled engine)
2 - Engine coolant manifold (For water-cooled engine)
- Alternator (For air-cooled engine)
3 - 3 speed fan
4 - Heater
5 - Evaporator
6 - Ventilator
7 - Dehydrating filter
8 - Condenser
9 - Air filter
10 - Recirculation inlets
11 - Pivotable air diffusers

If the air does not come out from the diffusers right away as soon as the system is started, turn off immediately and identify the fault.

N.B. Never turn on the heating system when working in dusty environments.

In the event of the system failing to operate, check the fuse in the roof terminal box.

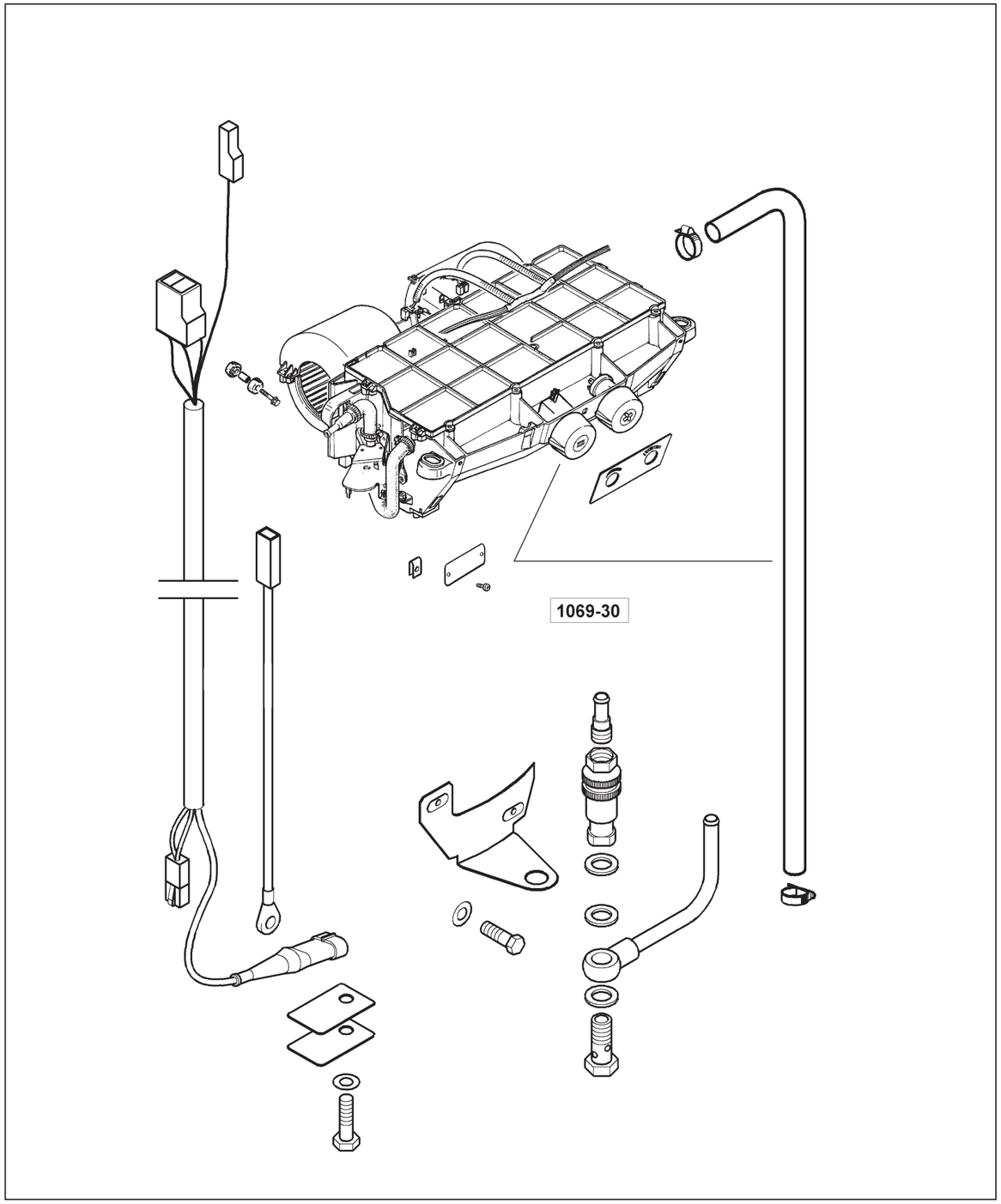


Fig. 2 - Ventilation - heating system. (for water-cooled engine)

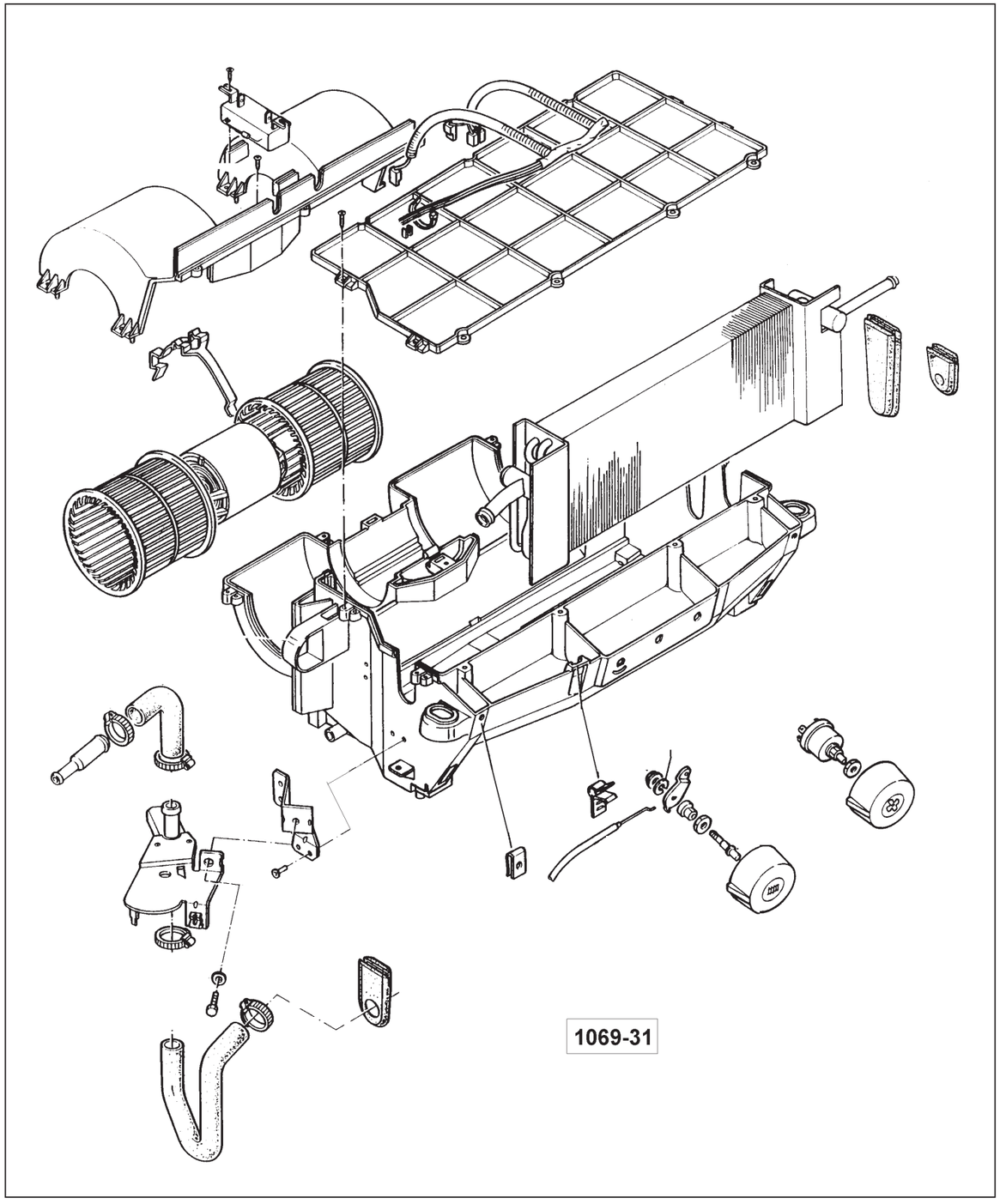


Fig. 3 - Components of the ventilation - heating system. (for water-cooled engine)

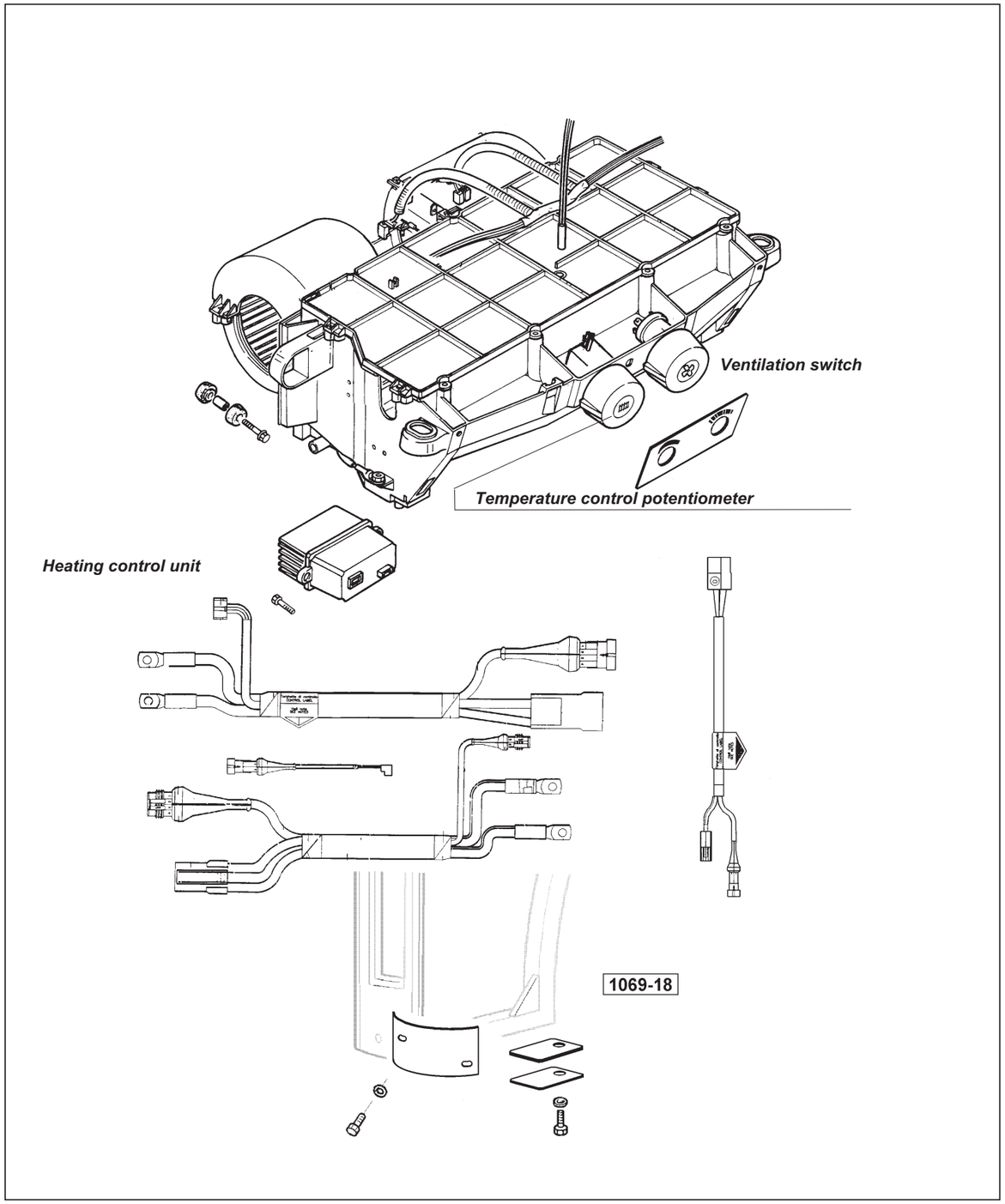


Fig. 2 - Ventilation - heating system. (for air-cooled engine)

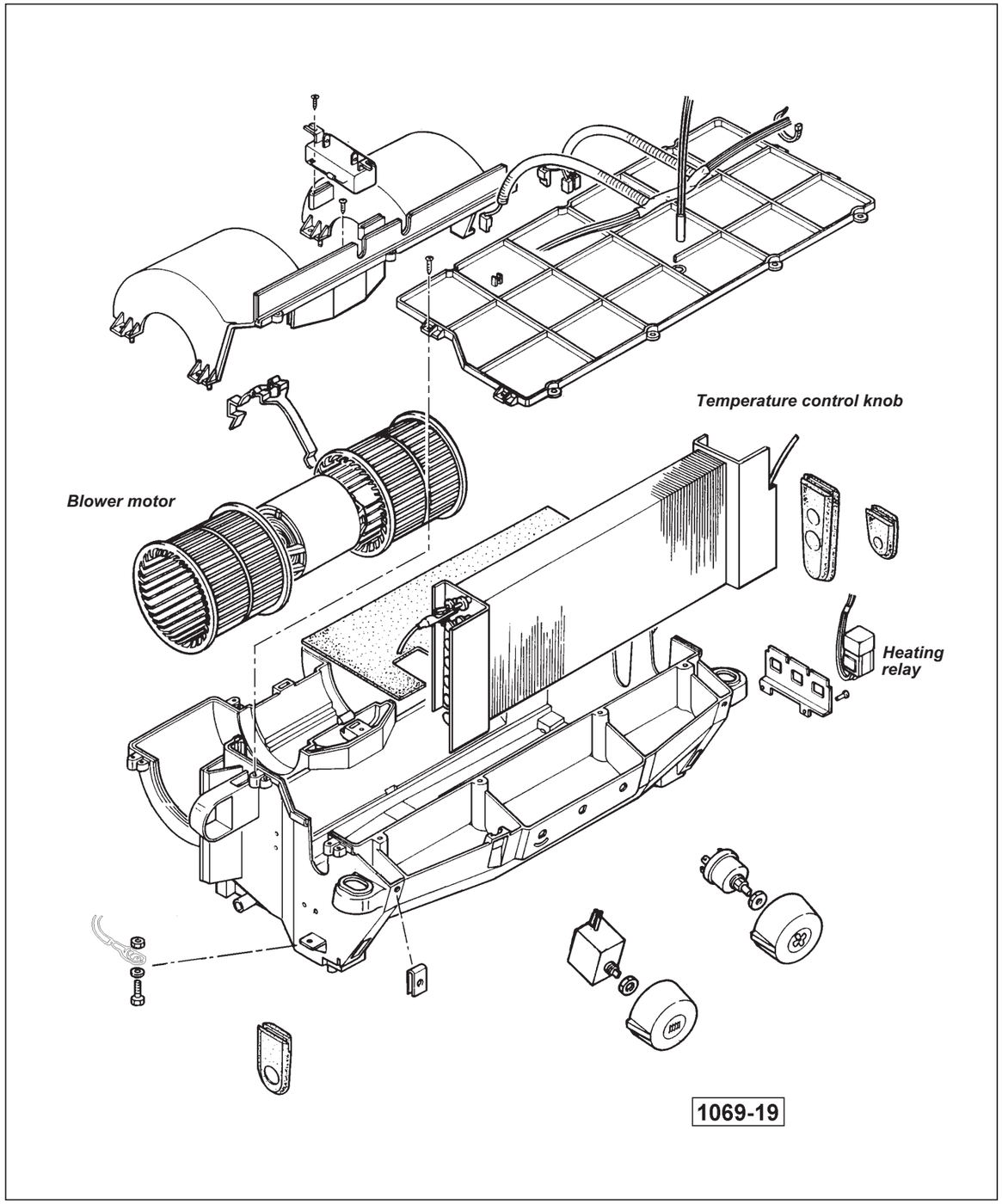
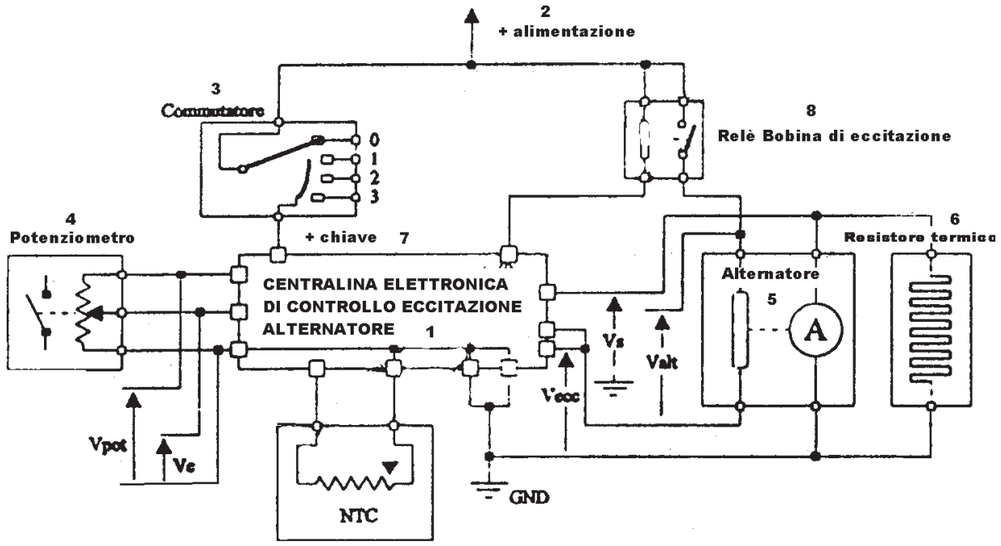


Fig. 3 - Components of the ventilation - heating system. (for air-cooled engine)

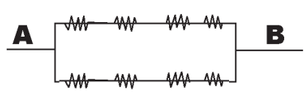
Description of the system



Key to block diagram

+ keyswitch: Supply voltage Vc: Control voltage Vs: Alternator output voltage
Vecc. Output voltage controlled by ECU Valt: Supply voltage Vpot: Potentiometer supply voltage
GND: Power and control signal earth 1 - Electronic control unit 2 - Power supply +
3 - Switch 4 - Potentiometer 5 - Alternator 6 - Heating element 7 - Keyswitch + 8 - Excitation relay

Electronic control unit
Functions: 1 - excitation control 2 - over-voltage protection 3 - over-temperature protection (TEMP-FET)
4 - thermal protection of heating element by NTC 5 - excitation relay control.



Heating element
Testing procedure: measure the resistance using a multimeter with a 5 1/2 digit display and 0.5 mW resolution.
Alternatively, apply 43V across the terminals A - B using a suitable power supplier and use a ammeter to measure the absorbed current, which should be 63.3 A

Over-temperature protection

The ECU is protected against over-heating of the TEMP-FET, which could occur in the event of a failure in the ventilation of the heat sink (e.g. blower failure).
This thermal protection is incorporated in the TEMP-FET and the trip thresholds for the protection are specified in the manufacturer's data sheet supplied by Siemens.
A temperature of below 95* is guaranteed at the ECU fixing points.

Over-voltage protection

Due to the variable factors related to the heating element and the alternator rpm, the voltage may rise to potentially dangerous levels.
To protect against this possibility, the ECU employs a closed loop control system for control of the alternator, with voltages above 43 Vc to the maximum excitation, thereby relating the excitation voltage to the current produced.
In order to prevent susceptibility of the control, the protection intervention delay time in the event of a variation in the Vs signal is 70 ± 20 msec.

Excitation relay control
Control of the excitation relay is achieved using as a reference the disabling due to intervention of the protective devices i.e. over-temperature and NTC circuit open. The voltage drop between the excitation relay output and earth (with relay on) is less than or equal to 1.5 V.

Excitation control
Excitation control is achieved by a closed loop by predicting the voltage supplied by the alternator and thereby controlling the excitation level in a linear way.
The voltage supplied by the alternator is therefore a function of the excitation level and the heating element connected to it.
The characteristics relative to Vc (START) and Vc (SAT) relative to variations in the supply voltage (keyswitch +) with the voltage drops are indicated in the following tables:

at normal temperature	keyswitch + = 10.0 V	keyswitch + = 13,5 V	keyswitch + = 16.0 V
Vc (start) ± 8%	0,666	0,657	0,651
Vc (sat) ± 8%	3,65	4,69	5,43

VOLTAGE DROPS

Between Vecc outputs and GND with: I=4.7 A; Vc=MAX; keyswitch + = 13.5 V	less than 1,4 V
Between relay outputs and GND with: I=200 mA; keyswitch + = 13.5 V	less than 1,5 V
Between common GND terminals with: I = 1A	less than 10 V

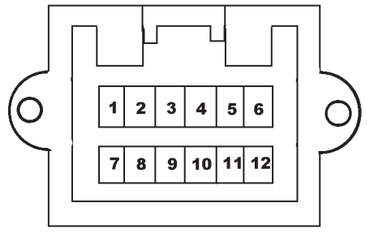
Connector pin-out
Pin-out of the AMP Multilock 040 12-way 90* connector:

Terminal n°	Function	Max. current absorption with eyswitch+ = 16V (mA)	Current absorption to be guaranteed wit hkeyswitch+ = 10V (mA)	Operating voltage range (V)
1	Excitation voltage 1 (Vecc)	3300	-	0 to keyswitch+
2	Excitation voltage 1 (Vecc)	3300	-	0 to keyswitch+
3	keyswitch+	100	-	0 to keyswitch+
4	Electronic GND 1	13350	-	0
5	Electronic GND 2	23350	-	0
6	Control voltage (Vc)	0,1	-	0÷V po
7	Over-voltage (Vc)	2	-	0 to 50
8	NTC sensor	1	-	0 to 7
9	NTC sensor GND	1	-	0
10	Potentiometer GND	0,7	-	0
11	Excitation relay control	280	-	0 to keyswitch+
12	Potentiometer power (Vpot)	0,8		-3 to 6

Keyswitch+ 10 to 16 V

Connector (AMP 040 series 12V 90* n* 175974-2)
Mounted on printed circuit board by brass rivets

View of terminals



NB : all electrical characteristics refer to the relative terminals of the device.

Air conditioning unit for cabs

The air conditioning system will operate only with engine running and the 3-speed electric fan unit switched on.

Technical specifications

type of refrigerant		R 134 a
inlet circuit pressure (*):	high bar	2,5
-	normal bar	0,8 ÷ 2,5
-	low bar	0,8
delivery circuit pressure	bar	see page 350
quantity of refrigerant in circuit	g	950
quantity of oil	cc	110
type of oil		Suniso SP20
minimum pressure switch setting	bar	2
maximum pressure switch setting	bar	27
electric fan pressure switch setting		close at 15 bar - open at 11 bar
tightening torque for pipeline fittings	kgm (Nm)	6 (58)

(*) *Pressure is influenced by ambient temperature; in normal operating conditions, with a temperature of 27 °C, pressure in the delivery circuit will be 12 bar.*

Pressure in the same system will register at around 16 bar with an ambient temperature of 38 °C.

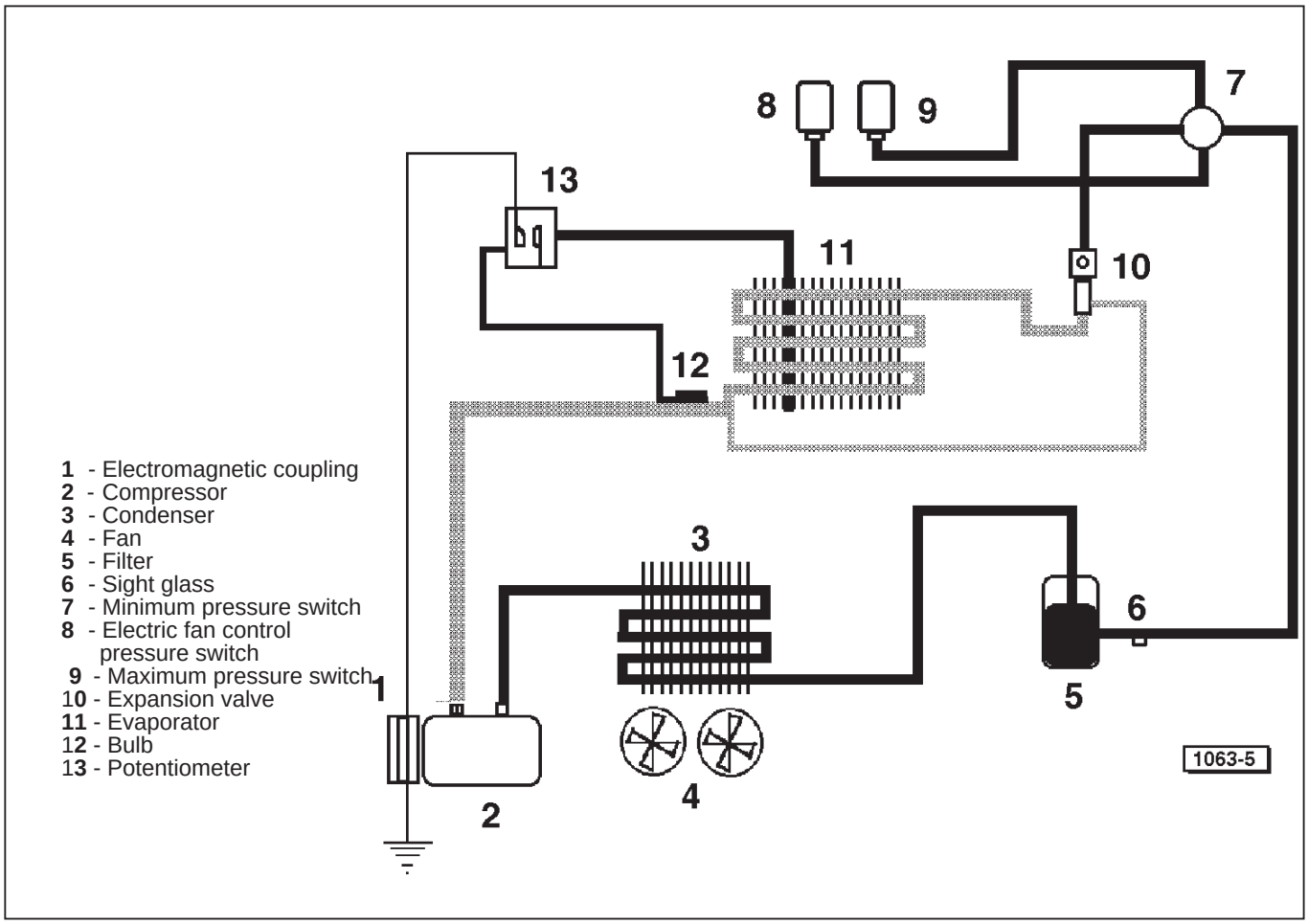


Fig. 4 - Diagram showing operation of air conditioning system.

Operation and maintenance of the air-conditioning system

Air-conditioning process
The cooler is provided with a three-speed fan which aspirates the air from the cab interior thereafter forcing it across the radiant block, thus making it cool down.

The heat exchange occurring between the gaseous coolant and the air lapping the walls of the cooler radiant block results in air cooling.

The conditioned air undergoes a humidification process, as the low temperature inside the cooler causes a great deal of condensate to deposit on the external walls of the cooler. This condensate is subsequently conveyed under liquid state outside the cab through a special drainage system.

The coolant at the cooler exit is still under gaseous state and has a temperature of 5 to 10°C, in spite of the thermal exchange that has taken place, and then it is aspirated by the compressor again.

How to operate the system

Before starting the air-conditioning system set the electric fan onto the desired speed. If the electric fan is switched off, the conditioning system will not start.

The air-conditioning system is turned on by progressively rotating the control potentiometer clockwise, which is located in the front top section of the cab compartment. Rotate it anticlockwise to turn the system off.

Warning: Before starting the engine, always make sure the air-conditioning system is switched off so that the battery will not be overloaded.

When the air-conditioning system is operating, the pivotable air diffusers should never be completely closed.
To obtain a quick cab environment cooling the following is recommended:

— fully open the air recirculation grating and the pivotable air diffusers;

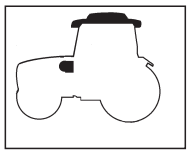
— turn first the fan control and then the potentiometer control all the way out;

— open the doors a few seconds so as to let the hot air out of the cab compartment if the tractor has stood under the sun radiation for a long period;

— adjust the system according to personal preference when the desired temperature has been attained.

Checking the air-conditioning system charge

System fully charged (refrigerant entirely in liquid state) System fully discharged (refrigerant entirely in gaseous state)	**Sight glass clear**
System not fully charged Refrigerant in part gaseous and part liquid state	**Sight glass cloudy (Bubbles or foam visible)**



Switching on the air conditioning

The system is activated by way of the 3 speed electric fan switch.

Once the potentiometer is moved beyond the initial contact, a voltage of 12 V is generated between terminal B (red wire) and the brown wire terminal of the potentiometer.

Electromagnetic coupling

Engagement and disengagement is piloted by the temperature control potentiometer, and by the pressure switch located at the demoisturizing filter which cuts in respectively at: 2 bar and 27 bar.

Pressure switches

Pressure switch (positioned on demoisturizing filter)
Comprises 3 switches:
— 1 controlling minimum pressure
— 1 controlling maximum pressure
— 1 controlling the electric fan of the condenser. The electric fan unit is interlocked to the pressure switch, which is set:
- to close the circuit at 15 bar
- to open the circuit at 11 bar

Expansion valve

Temperature sensor

Electric fan unit

By-pass the pressure switch (place jumpers across the two terminals of the pressure switch and the 2 leads of the fan unit).

If the fan operates: the pressure switch is defective

If the fan does not operate: **a**) check the fuse;
b) check the relay.

Tracing faults in the electric fan unit

Fault	Causes	Action
Fan does not start	Defective pressure switch	*Replace pressure switch*
	Defective relay	*Replace relay*
	Defective electrical or mechanical motorino difettosa	*Replace fan unit*

Relay (see electrical diagram)

Place a jumper across terminals **30** and **87a**; if the fan comes into operation, the relay needs to be replaced.

Water dripping from the points at which condensate drain lines are connected to the conditioning unit

The problem can be overcome by loosening the clip that fastens the drain line to the cab upright and pulling the tube downwards, so that the water can run away freely.

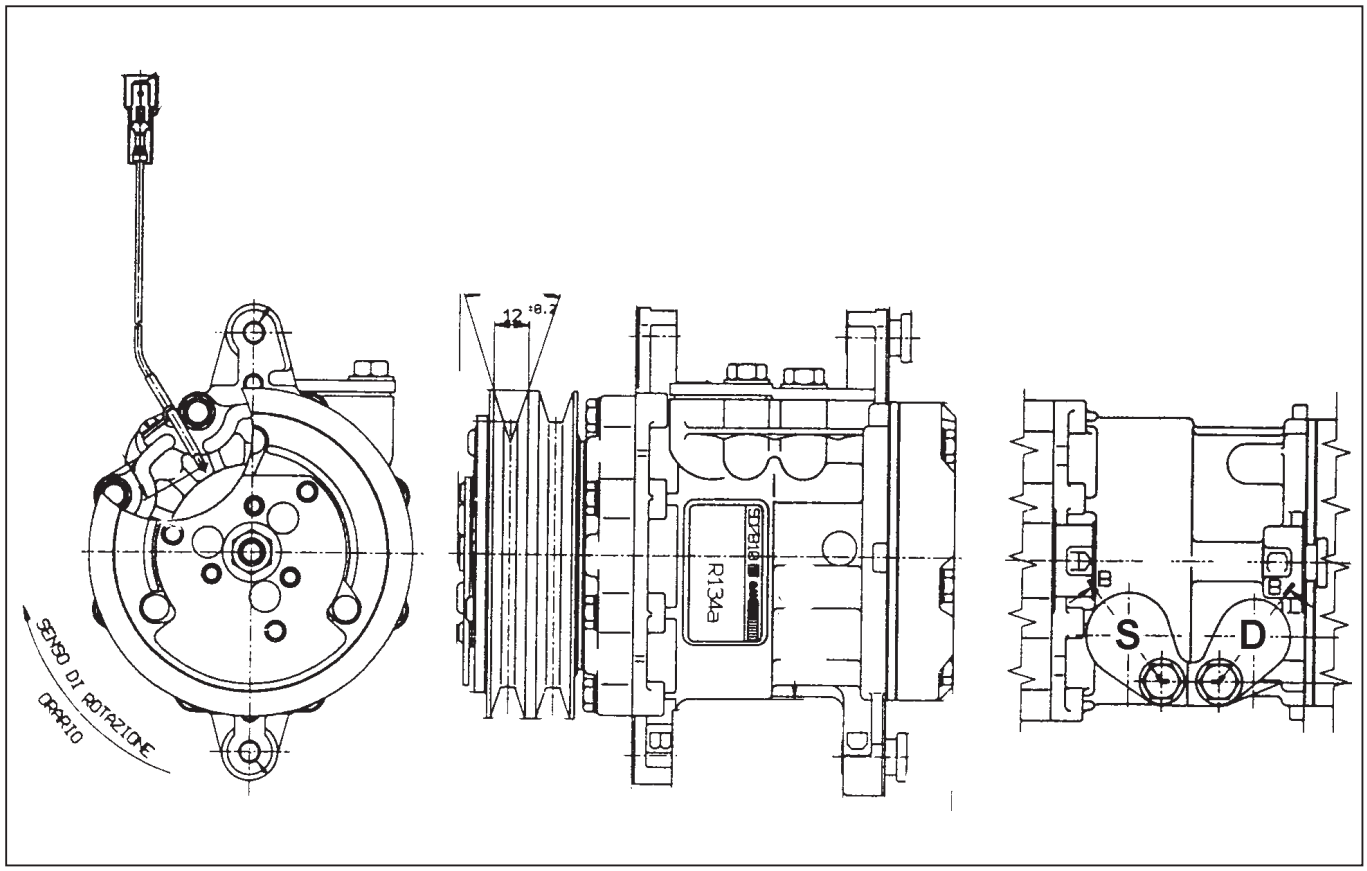


Fig. 5 - Compressor, air conditioning system.

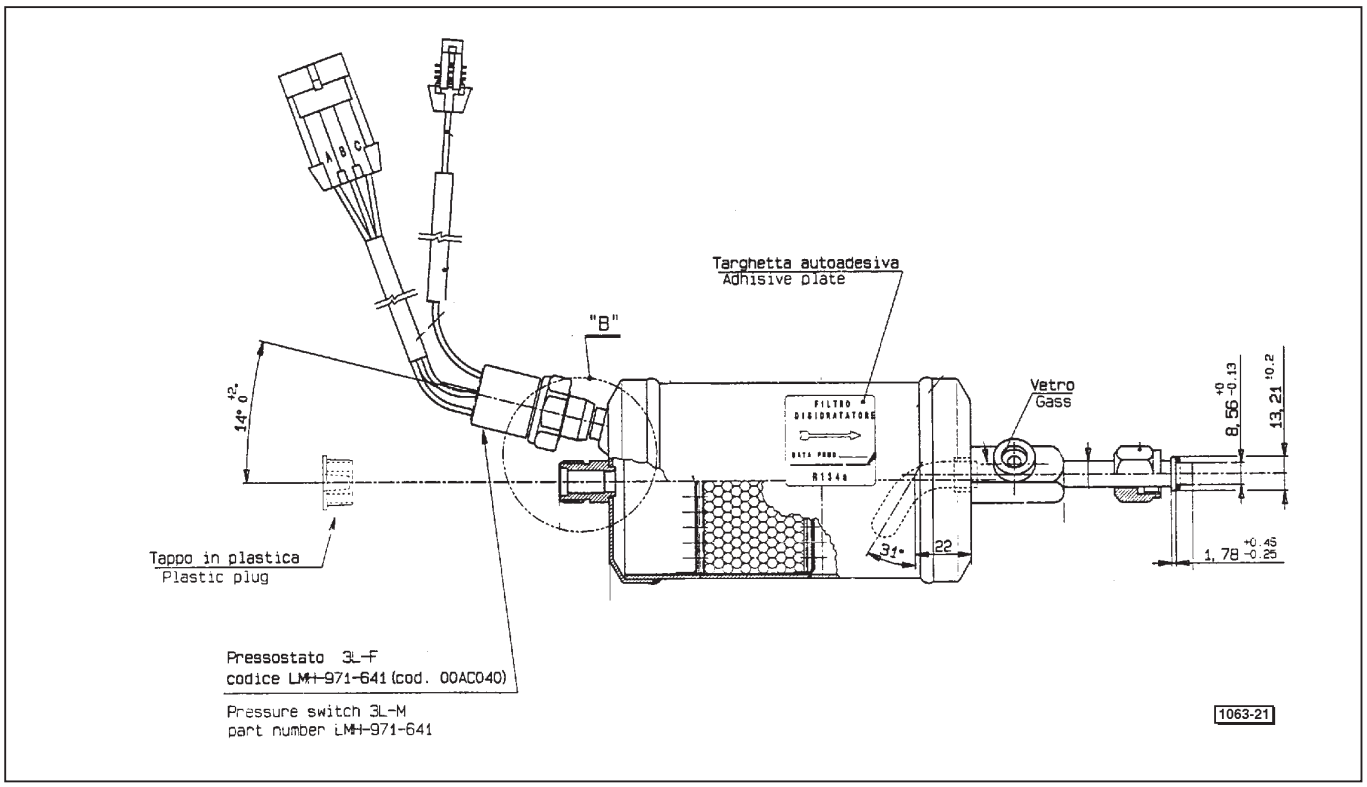
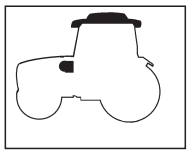


Fig. 6 - Demoisturizing filter, air conditioning system.



Checking system

Check belt tension: under finger pressure applied to the mid point between the two pulleys, belt should not deflect more than 8 to 10 mm maximally.

Always keep condenser fins duly clean.

Cleaning should be carried out by a water or air jet (be careful not to bend the fins, use the special combing tool to straighten them again).

Ascertain the compressor is firmly secured to tractor and the pulleys are properly aligned.

Warning: When removing the strainer or the air-conditioning unit it is required to plug the inlet and the outlet pipes at once, this is to prevent both dust and moisture from entering.

Even a few minutes' exposure to the environmental humidity makes it necessary to replace the filter.

System safety elements

Minimum pressure switch (7 Fig. 4)**.**
Switches the system off when a failure causes the pressure to drop below 2 bar within the high pressure circuit.

Maximum pressure switch (9 Fig. 4).
Switches the system off when pressure exceeds 27 bar inside the circuit as a result of an excessive setting or a failure (high pressure area).

Condenser fan control pressure switch (8 Fig. 4)
Switches the condenser fan on when coolant attains the same pressure of the switch setting pressure.
Note: The above three safety devices are grouped inside the thermostatic bulb located on the dehydrator filter. (Fig. 6 page 346).

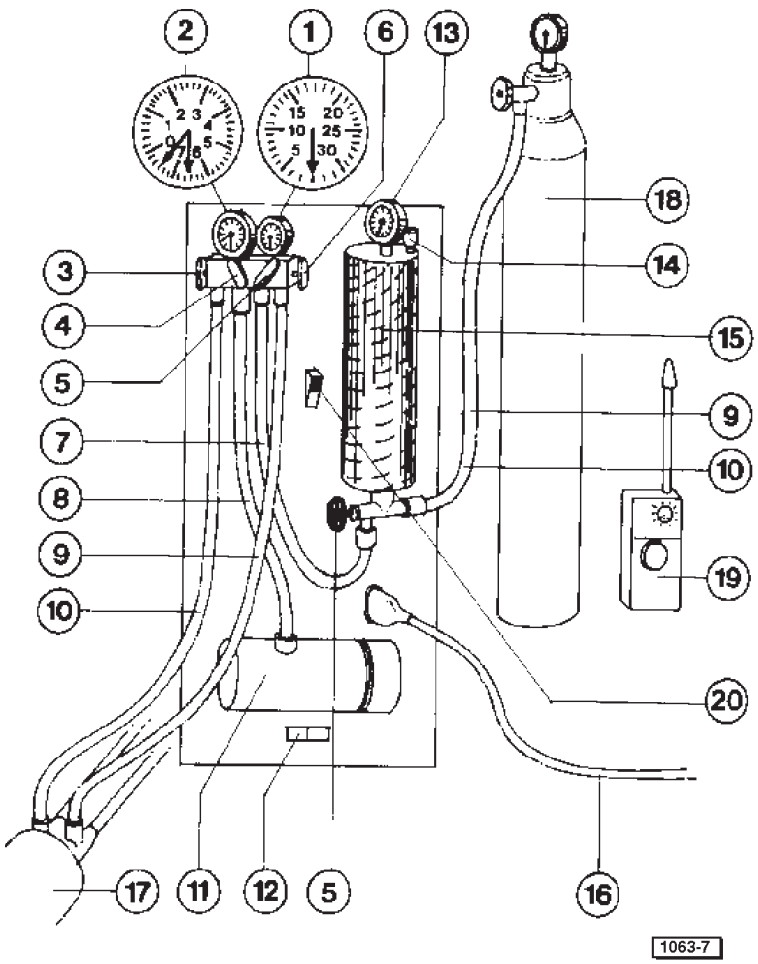
Temperature regulation

Temperature regulation is obtained through the potentiometer, which automatically regulates the temperature of the air flow from the radiator according to the environmental temperature, through a combined action with the fan.

A pressure switch located on the electric circuit opens and closes the electric circuit of the compressor electromagnetic coupling as a consequence of the potentiometer operation. When the circuit is open the pulley will be idling, on the other hand, when the circuit is closed, this will operate integrally with the compressor shaft.

Charging the system

with recharging station p/n 5.9030.508.6



1 - High pressure gauge
2 - Low pressure gauge
3 - Low pressure valve
4 - Vacuum pump valve
5 - Metering pipeline valve
6 - Recharge pipeline valve
7 - Metering pipeline
8 - Vacuum pump pipeline
9 - Low pressure circuit recharge pipeline (blue)
10 - High pressure circuit recharge pipeline (red)
11 - Vacuum pump
12 - Electric switch
13 - Metering gauge
14 - Air bleed valves
15 - Metering unit
16 - Electrical power supply
17 - Compressor
18 - Refrigerant bottle
19 - Electric leak detector
20 - Electric heater element for refrigerant

Fig. 7 - Charging station for air conditioning circuit.

Filling the metering unit

Connect the refrigerant bottle to the metering unit by means of the high or low pressure hose (9 - 10 Fig. 7).
Make certain that the valve (5 Fig. 7) is securely shut, then open the valve of the bottle.
As the refrigerant flows into the metering unit, open and shut the air bleed valve (14, Fig. 7) intermittently.
Read the value given by the hand of the metering unit pressure gauge (13, Fig. 7) and position the value against the indicator.
Once 950 grams of refrigerant have been transferred into the metering unit, close the valve of the bottle and detach the hose.

Refilling the system with oil

Connect the high pressure hose (9, Fig. 7) of the recharge station by way of the smaller diameter valve **A** (Fig. 8) positioned on the pipeline connecting with fitting **D** (Fig. 5) at the compressor (mounted in the engine compartment), and the low pressure hose (10, Fig. 7) of the station by way of the larger diameter valve **B** (Fig. 8) positioned on the pipeline connected with fitting **S** (Fig. 5) at the compressor; then activate the pump for 30 minutes approx to establish a vacuum in the circuit.
The valves (3 - 4 - 5 - 6, Fig. 7) must remain open during this operation.

Filling the system with oil
Detach the low pressure hose from the recharge station and immerse the end in a graduated vessel containing 110 g approx of SUNISO SP 20 or equivalent oil. Allow 200 grams of the oil to be drawn into the circuit.
Reconnect the hose to the station.

Filling the system with R134a refrigerant
When the low pressure gauge (2, Fig. 7) registers a minimum value (760 mm Hg) that remains stable for between 5 and 10 minutes, close the valve (5 Fig. 7).
Open the filler valve (5, Fig. 7) and wait until the specified quantity (1800 g) of refrigerant has entered the system.
In the event that the flow of refrigerant contained in the metering unit should be interrupted, warm the unit by means of the electric heater (20, Fig. 7), rechecking the pressure on the relative gauge.

Should this not be sufficient, start the engine and throttle up to 1200 rev/min approx. Switch on the air conditioning system, and the R 134a refrigerant will be drawn directly into the circuit by the compressor.

Close the filler valve (5, Fig. 7), and check with the leak detector that there is no escape of fluid from the fittings of the circuit.
Should any leaks be detected, these must be eliminated, and the refrigerant reintroduced.

Caution: In the event of oil leaks being detected, the circuit must be emptied completely and the entire recharge procedure recommenced.

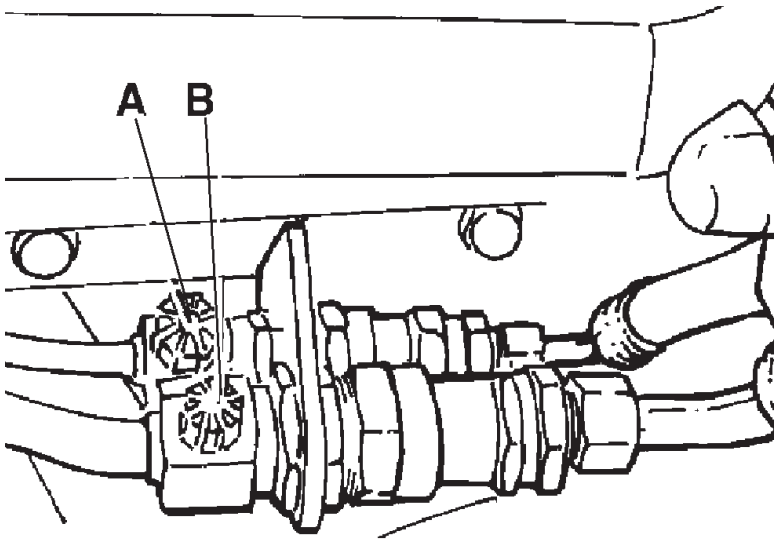


Fig. 8 - Valves for connection of the tractor to the air conditioning recharge and service station.
A - “HIGH PRESSURE” valve
B - “LOW PRESSURE” valve

Final checks
Once all the refrigerant has been put into the air conditioning circuit, check (with the engine running and the system in operation) that the high pressure gauge shows a value of between 15 and 20 bar, and the low pressure gauge a value between 1.5 and 2.5 bar.

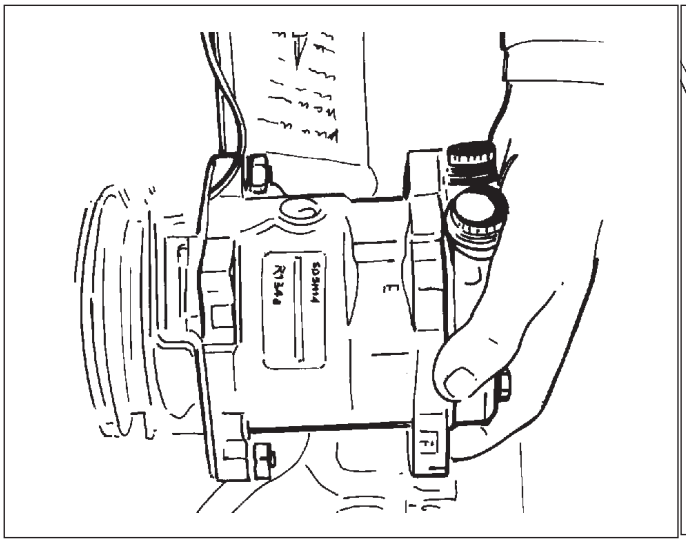
VERIFYING OPERATION OF THE SYSTEM AFTER RECHARGING

The system can be regarded as performing to its full capabilites when the delivery pressure gauge shows the following values:

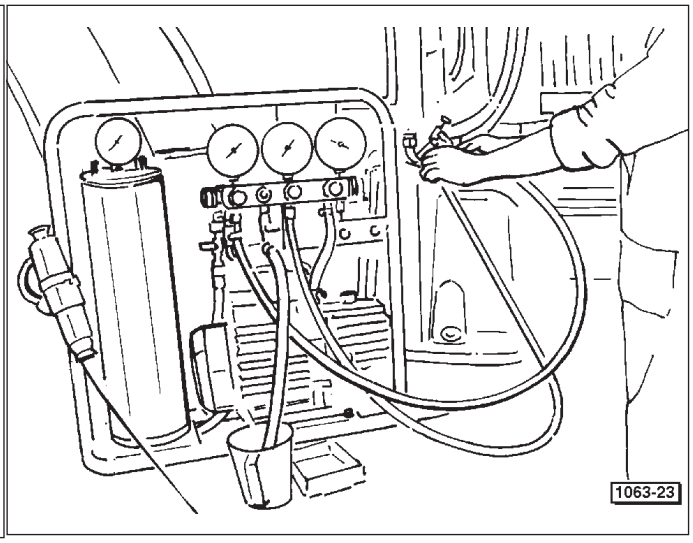
Ambient temperature °C	Delivery circuit pressure
27	12 bar
32	14 bar
35	15 bar
38	17 bar
40	18 bar
43	20 bar

The temperature values indicated are those likely to be encountered in the course of a typical twelve month period.

Having established that the system is operating correctly, detach the high and lower pressure hoses of the station from the tractor and check again with the leak detector that there is no escape of fluid from any part of the system.



CAUTION: use only R 134a refrigerant as indicated on the R data plate attached to the compressor of the air conditioning system.

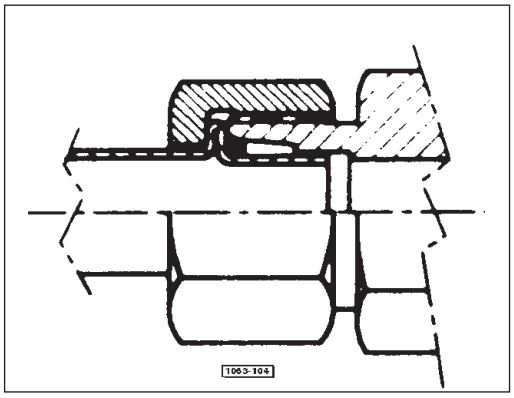


Connecting the tractor to the air conditioning system recharge and service station.

Directions for tightening air conditioning system pipeline fittings.

All fittings have an internal seal. Screw the two parts of the fitting together by hand until the nut is finger tight, then holding the one part steady with a fixed wrench, apply a torque wrench to the nut and tighten to the value indicated in the table.

Type	Thread	Tightening torque
6	5/8" - 18 UNF	13,6 ÷ 20,3 Nm (1,4 ÷ 2,1 kgm)
7	3/4" - 16 UNF	33,5 ÷ 40,0 Nm (3,3 ÷ 4,0 kgm)
8	7/8" - 16 UNF	35,5 ÷ 42,0 Nm (3,6 ÷ 4,3 kgm)
9	1 1/16" - 16 UNF	40,3 ÷ 47,5 Nm (4.1 ÷ 4,8 kgm)



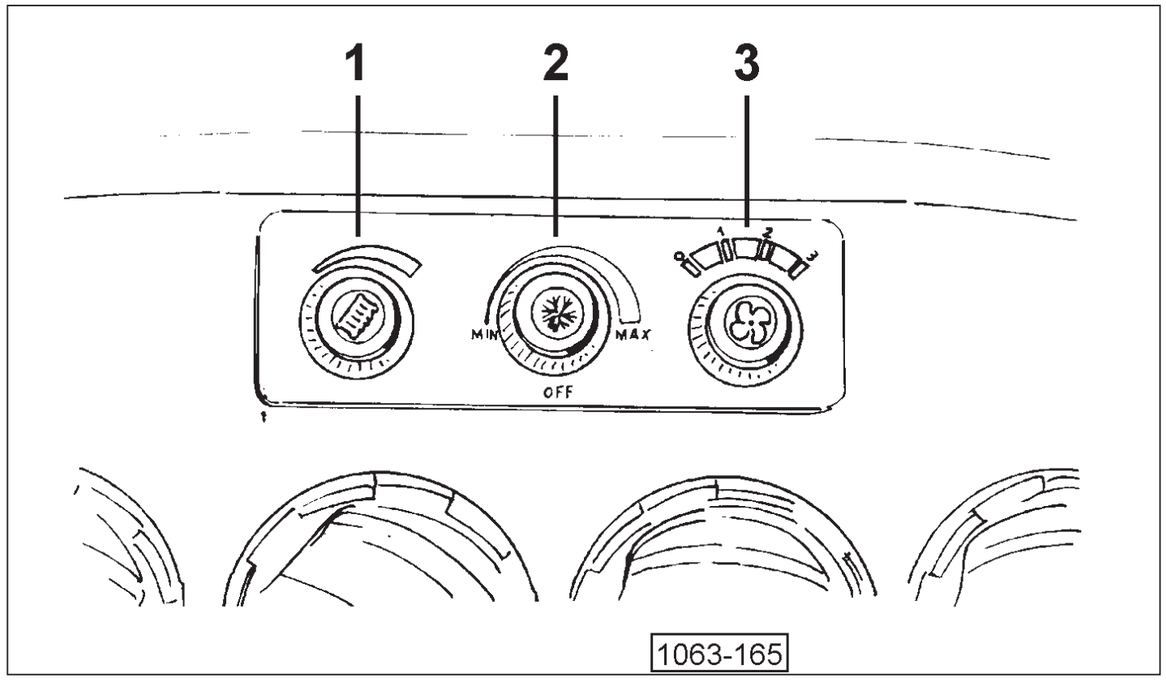


Fig. 9 - Control switches
1 - Heating control.
2 - Air conditioning control.
3 - Ventilation control.

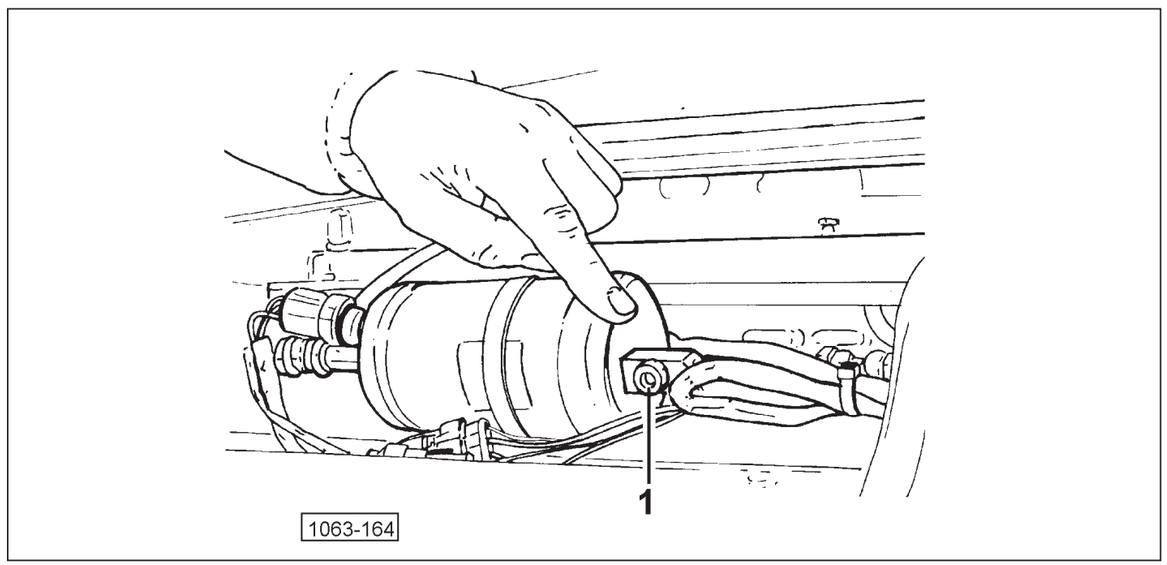


Fig. 10 - Receiver-drier located in cab roof.
1 - Level sight glass.

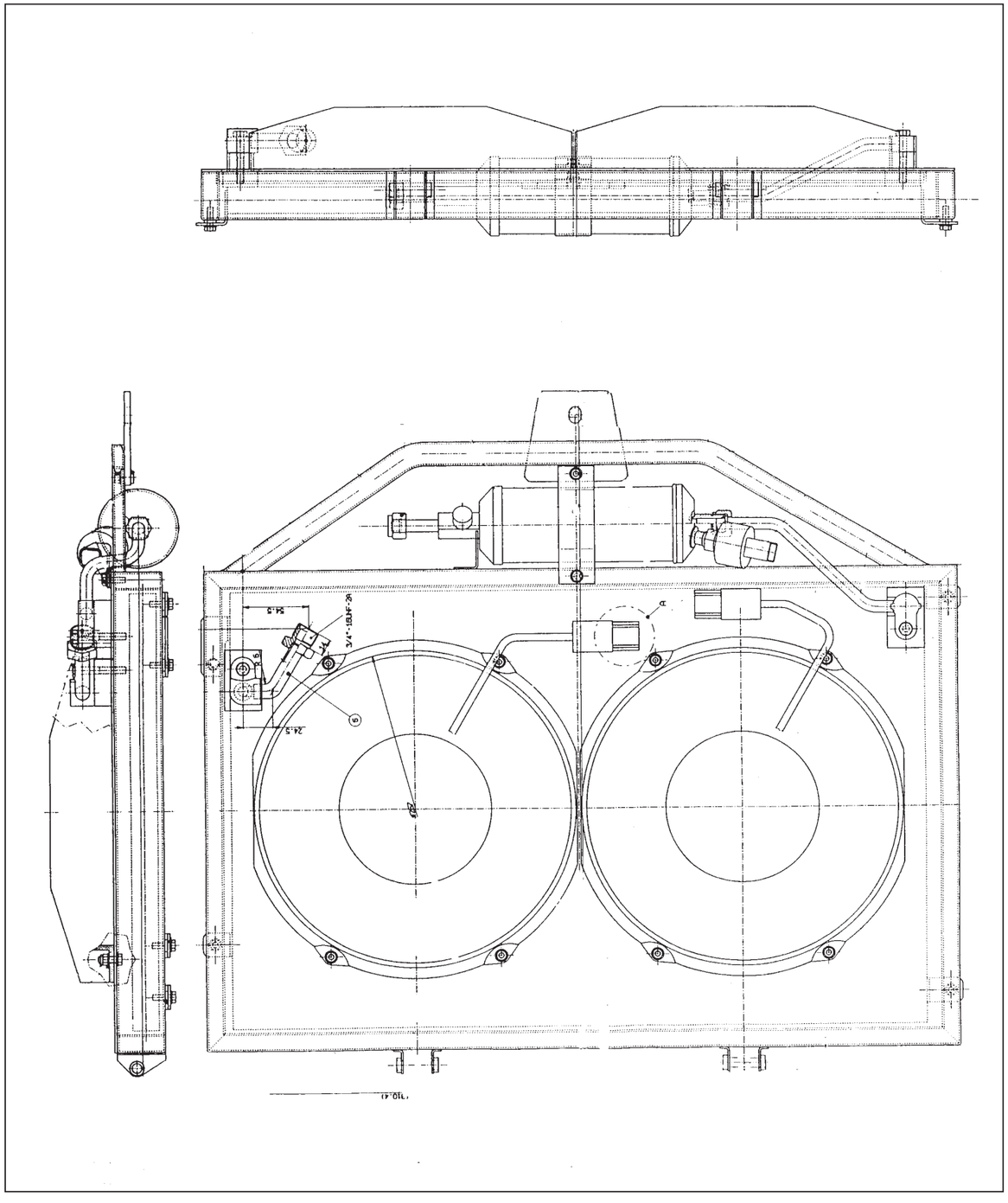
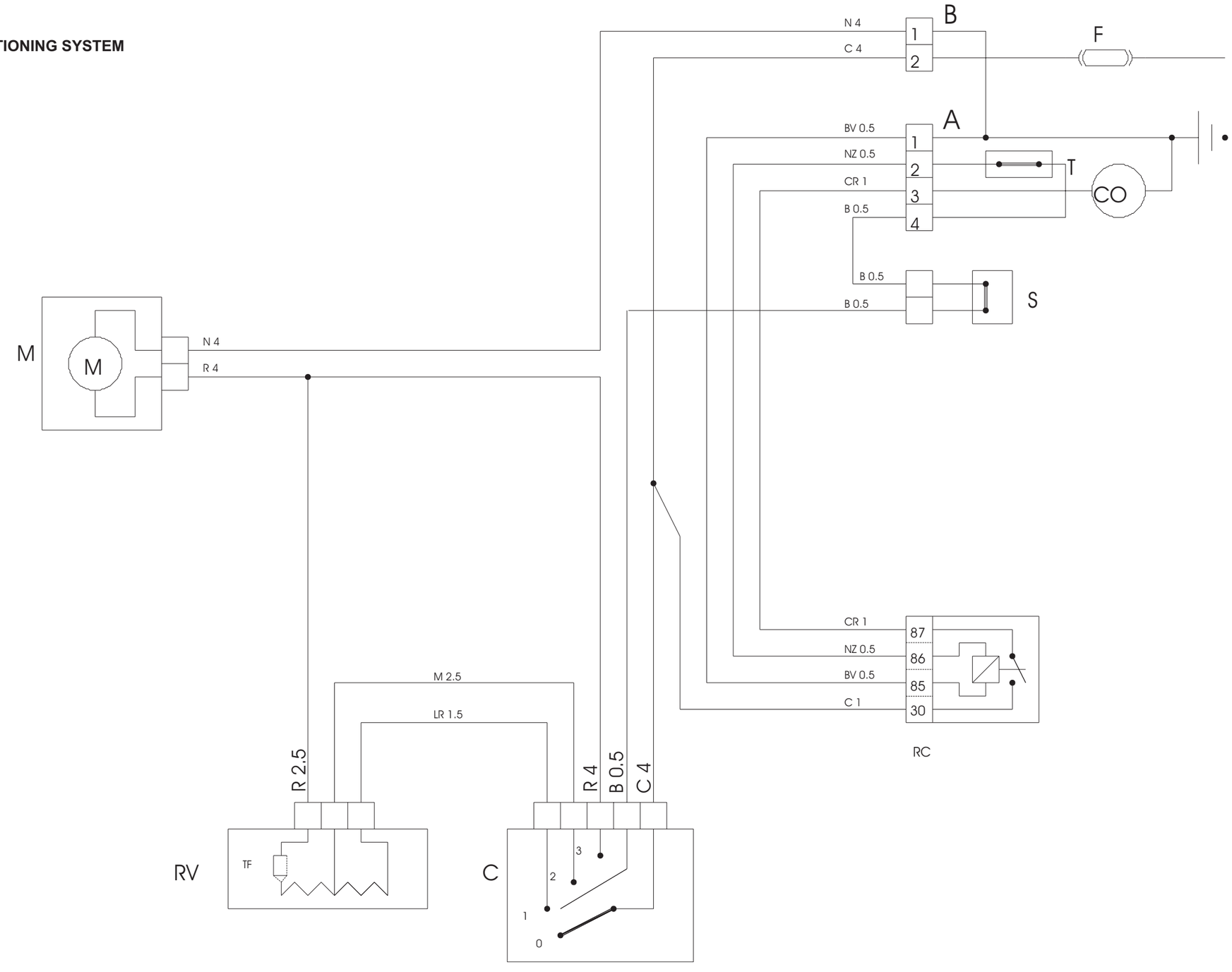


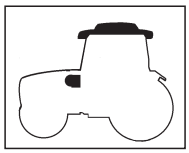
Fig. 11 - Air conditioning unit.

COMPONENTS LIST

ELECTRICAL DIAGRAM FOR HEATING AND AIR CONDITIONING SYSTEM

M - BLOWER FAN
RV - FAN SPEED RESISTOR
C - FAN SPEED CONTROL SWITCH
RC - AIR CONDITIONING RELAY
F - POWER FUSE
T - AIR CONDITIONING PRESSURE SWITCH
CO - COMPRESSOR
S - DEFROSTING THERMOSTAT
B - POWER SUPPLY CONNECTOR
A - CLIMATE CONTROL CONNECTOR





Diagnosing malfunctions

LOW PRESSURE CIRCUIT

excessive pressure	excessive pressure in the high pressure circuit	compressor not operatin smoothly	overcharged circuit	release system pressure through the compressor needle valves
	too a low pressure in the high pressure circuit	pressures inside both high and low pressure circuits tend to balance	expansion valve jammed up	replace the valve
normal pressure	excessive pressure in the high pressure circuit	bubbles are visible through the filter inspection glass	air in the circuit	dehydrate the circuit
	normal pressure in the high pressure circuit	poor system efficiency	Out-of-setting compressor needle valves	replace the valves
excessively low pressure		after some working time	humidity inside the circuit	replace the filter and bleed the air. Recharge the circuit
	normal pressure in the high pressure circuit	expansion valve jammed up	replace the expansion valve	
		blocked circuit	change the part and replace the filter	
	low pressure in the high pressure circuit	discharged system	recharge system	
system fails to operate	burnt fuse	replace the fuse		
	disconnected wires	connect the wires		
poor system efficiency	coolant leaks	inspect with a leak-finder	stop the leakage and recharge the system	

Hydraulic system

KEY TO FIGURE 1:

1 - Delivery line pressure test point
2 - Hydrostatic steering - maximum hydrostatic steering pressure (see instructions on p. 290).
3 - 4WD steering cylinder pressure test point
4 - 2WD steering cylinder pressure test point
5 - Gearbox oil cooler
6 - Electro-hydraulic control unit (see instructions on p. 196)
7 - Hydraulic power unit pressure test point (see instructions on p. 263)
8 - Lubrication oil pressure test point
9 - MULTISPEED
10 - Hydraulic unloader for equalising oil level in gearboxes
11 - Gearbox lubrication
12 - Rear P.T.O. (see instructions on p. 196)
13 - Diff. lock hydraulic actuators
14 - 4WD engagement hydraulic actuator
15 - Trailer braking hydraulic control valve
16 - Pressure test point for trailer braking hydraulic control valve
17 - 8-way auxiliary control valve (see instructions on pages 362-376)
18 - Pressure relief valve oil pressure test point
19 - Kick-out oil pressure test point (see instructions on page 365)
20 - Hydraulic ways oil pressure test point
21 - Connected implement
22 - Rear lift
23 - Front lift antishock valve and accumulator
24 - Implement coupled to rear lift
25 - Lifting cylinder
26 - Rear lift antishock valve
27 - Front lift (see instruction on p. 362)
28 - Lifting cylinder pressure test point
29 - Front lift antishock valve
30 - Shut-off valve for oil supply to front lift
31 - Supplementary hydraulic control valves
32 - Operating pressure test point for supplementary hydraulic control valve
33 - Pressure test point for kick-out of supplementary hydraulic control valve
34 - Implement connected to supplementary hydraulic control valve
35 - Pressure relief valve setting pressure test point
36 - Supplementary flow divider valve
37 - Hydraulic power unit for 3-point linkage with hydraulic adjustments
38 - Rh lift rod with hydraulic adjustment
39 - Rh stabiliser
40 - Lh stabiliser
41 - Free oil return line on right-hand side of tractor (for hydraulic motors)
42 - Hydraulic ways on right-hand side
43 - Free oil return line on rear of tractor (for hydraulic motors)
44 - Front P.T.O. control valve (see instructions on p. 281).

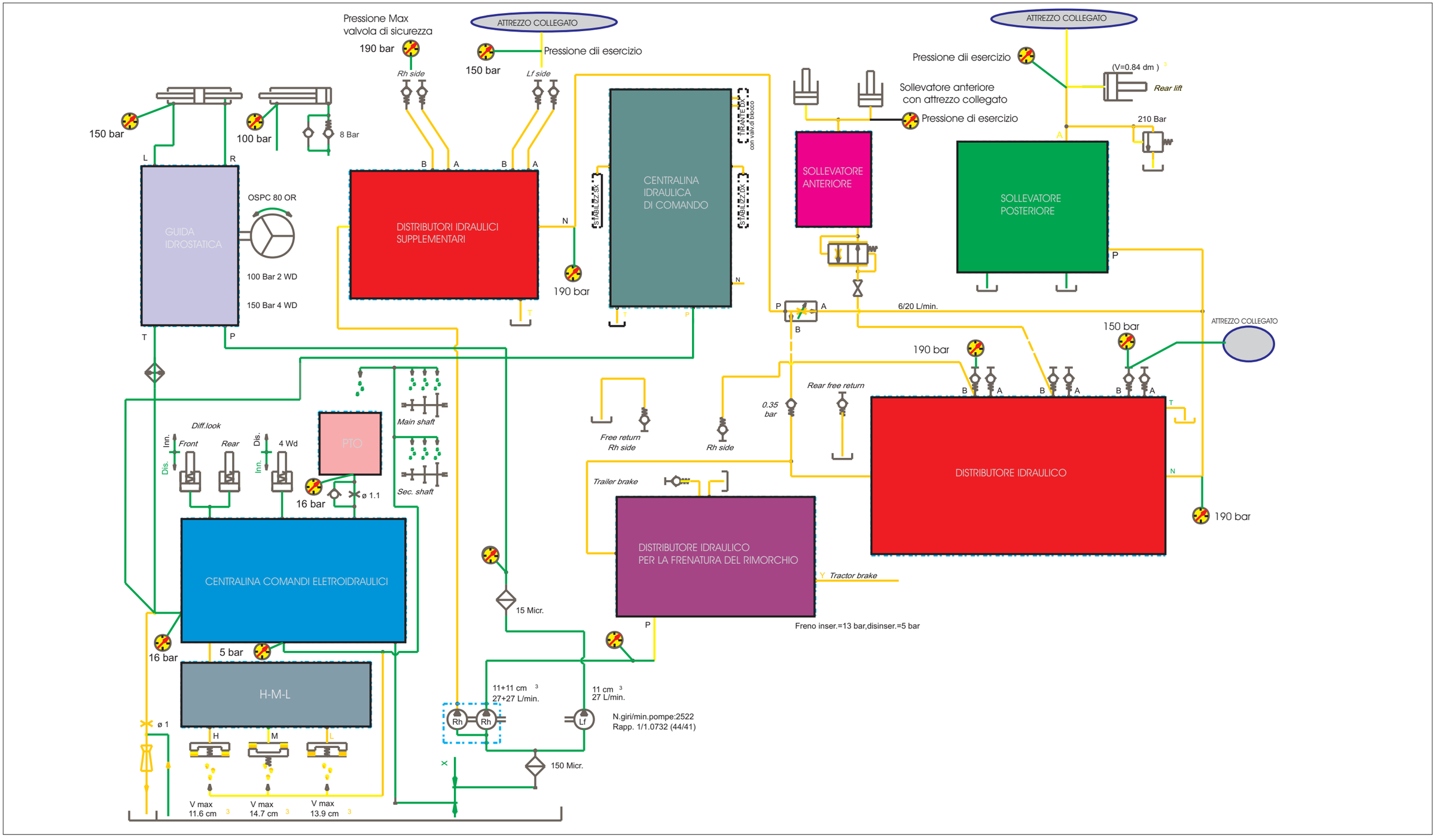


Fig. 1 - Points provided for measurement of oil pressures in the tractor's hydraulic system (see references on page 354).

Oil filters

The tractor is equipped with 15 micron filter installed on the delivery line from the hydraulic pump to the hydrostatic steering system. This filter is located on the right-hand side of the tractor and has an interchangeable cartridge.
This filter cartridge must be renewed:
after the first 50 operating hours;
after the first 150 operating hours;
and every 300 operating hours thereafter.

In any case, the filter cartridge must be renewed when the "filter clogging" warning light on the console stays on after the engine has reached normal operating temperature.
It is, however, perfectly normal for the light to stay on for a few minutes after starting the engine.

The tractor is equipped with a second filter with a filtration capacity of 160 microns. This second filter is located on the first section of the oil suction line on the right-hand side of the tractor.
This filter cartridge is to be renewed **after the first 150 operating hours** and thereafter **every 1200 operating hours**.
To renew the suction line filter cartridge, loosen the two straps securing the filter.
To renew the filter cartridge on the hydrostatic steering delivery line, simply unscrew the old cartridge and fit a new one.

The filter clogging pressure switch is set to 2.4 bar; if the pressure switch loses its setting it must be replaced with another switch with the same characteristics. Check that the calibration setting of 2.4 bar is stamped on the body of the new switch.

IMPORTANT - If the "filter clogging" warning light comes on during operation of the tractor, replace the filters.

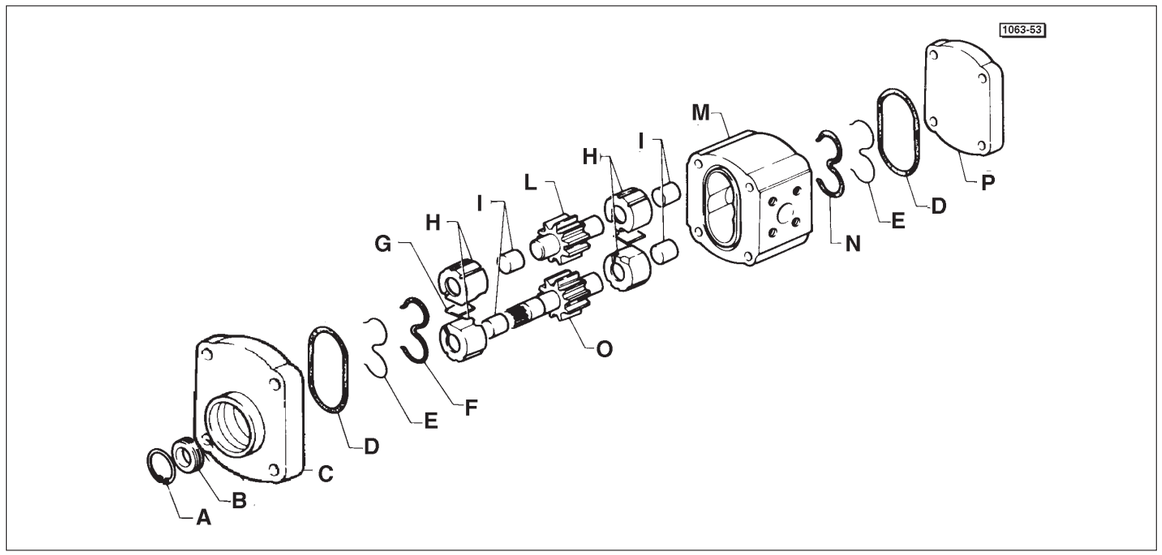


Fig. 2 - Component parts of the hydraulic pump assembly

A - Snap-ring
B - Seal ring
C - Cover
D - Gasket
E - Gasket
F - Gasket
G - Feather key
H - Bearing
I - Bushes
L - Driver gear
M - Pump case
N - Gasket
O - Driver gear
P - Cover

Hydraulic pumps

The hydraulic pumps require no servicing or adjustment even after a long period of operation, although the splines of the pump shaft must be inspected with care: there must be no signs of wear or damage.

Checking the relief valves of the hydraulic lift system

Using tool 5.9030.520.4, check that the circuit pressure relief valve incorporated into the auxiliary spool valves is set at 190 bar and the system pressure shock valve at 210 bar.
Remember that for the lift to function correctly, operating pressure in the circuit should not exceed 150 bar.
The operating pressure can be measured by coupling a gauge to the oil pressure line between the directional control valve and the lift; for machines with assistor cylinders, the gauge can be coupled to the pressure line connecting with the assistors.

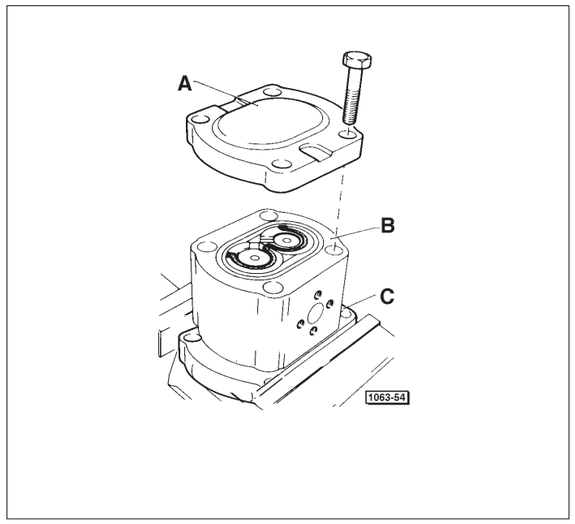


Fig. 3 - Pump assembly cover.
A - Cover.
B - Pump case.
C - Base.

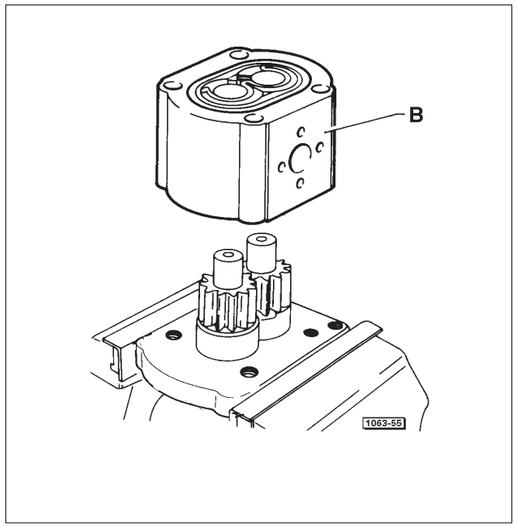


Fig. 4 - Pump case.
B - Pump case.

Stripping the hydraulic pump

Removal

The pump is secured to the tractor by 2 of the 4 bolts holding the pump assembly together. These two bolts are diagonally opposite. Having removed the pump from the vehicle, the components are freed by unscrewing the two remaining bolts which keep the cover **A** (see fig 3) clamped to the frame **C** and the pump body **B**; this done, remove the washers and then the cover.

Using a proper pointed tool remove gasket **E** and then take the bearings from their seats by pushing from inside outwards and being careful the connecting pin neither be lost nor damaged.

Mark with a reference the position of the bearings with respect to the pump case; then remove paying attention the connecting pin neither be lost nor damaged.

Block the pump case in a vice as shown in figure and then remove snap ring **F**.

Remove the seal ring.

WARNING - If pump bearings, pinions or casing are found to be damaged or worn, these parts cannot be repaired because of their construction tolerances.

When performing checks, in the warranty period, because of an oil leakage or an excessive and irregular delivery pressure, only the gaskets indicated in the spare parts lists can be replaced.

After checking the pump for wear due to abrasion from impurities and for any other visible damage, when reassembling use, in any case, a new set of gaskets.

It is also indispensable to mark each single part to reinstall correctly.

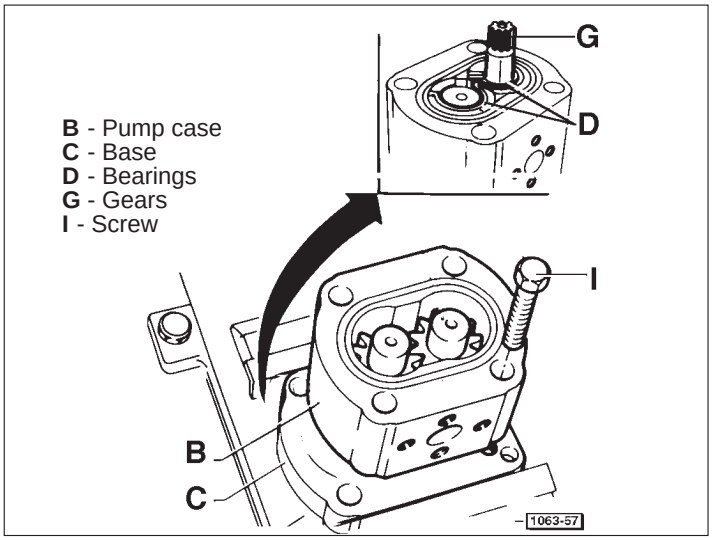


Fig. 5 - Pump disassembly.

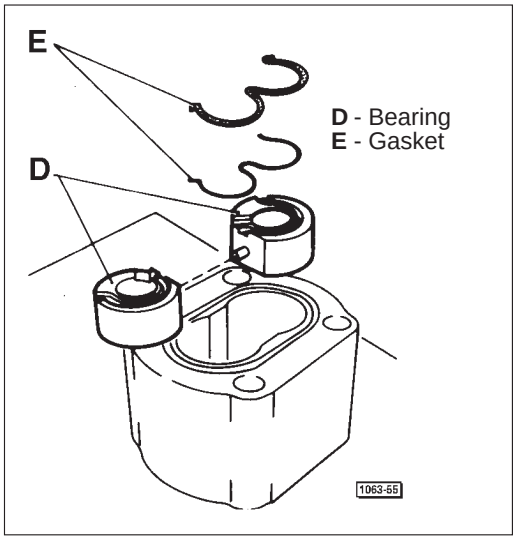


Fig. 6 - Internal pump case parts.

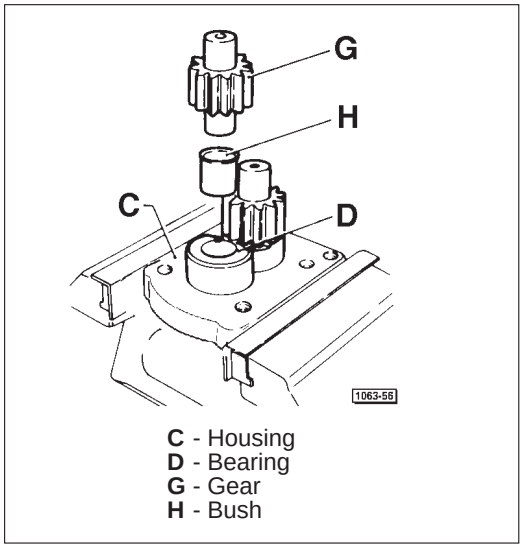


Fig. 7 - Pump gears.

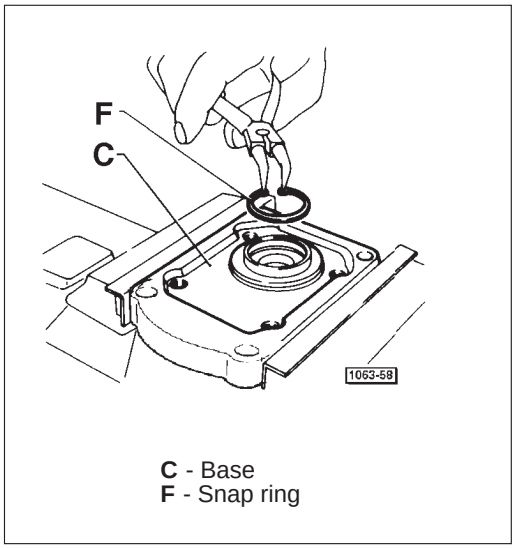
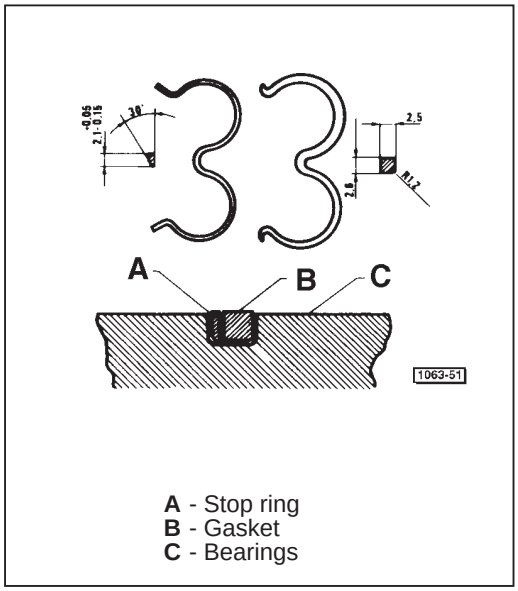


Fig. 8 - Snap ring removal.



A - Stop ring
B - Gasket
C - Bearings

Fig. 9 - Inside seals.

Inspections and checks
After removing all gaskets, both on the covers and in the pump casing, carefully clean all parts using a proper solvent.
Carefully dry all parts with compressed air; this is to avoid that any solvent residuals may damage the gaskets on reassembly.
Visually check the pump case internally and ascertain it is not damaged.
Check both pinions and bearings for wear, abrasion due to foreign matters or cavitation.

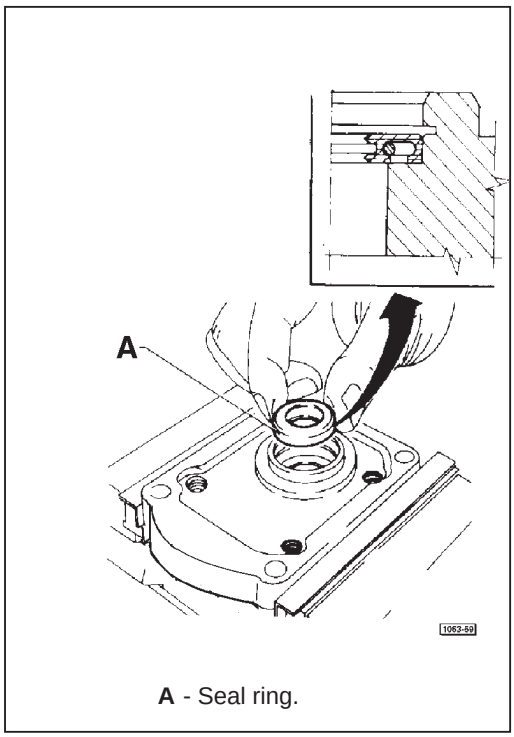
NOTE: Should any damage, wear or in any case a component replacement be required, with the exception of the gaskets, the entire pump must be replaced.
WARNING: Do not press onto the stop ring to prevent any damage.
Use a dial gauge to make sure the stop ring height be $2{,}1^{+0,05}_{-0,15}$ mm, otherwise replace taking care the new stop ring installed be measured accordingly.
If reading is above specification fit the new ring in reversed position and perform a grinding using a very fine emery cloth.

Assembly
Before assembly all parts shall be oiled.
Correctly position the gaskets in their seats paying particular attention stop ring **A** and gasket **B** be correctly fitted into bearings **C**.
Coat the cover inner side with a slight grease layer then apply to pump case and tightening the four securing screws to the recommended torque.

WARNING: The securing screws should be tightened gradually making also sure that rotors are not hampered in their movements.

Tightening torques
Pump cover securing screws:
- M8 2 kgm (20 Nm)
- M10 5 kgm (49 Nm)

Restore the recommended oil level and then bleed the air from the hydraulic circuit.



A - Seal ring.

Fig. 10 - Seal ring.

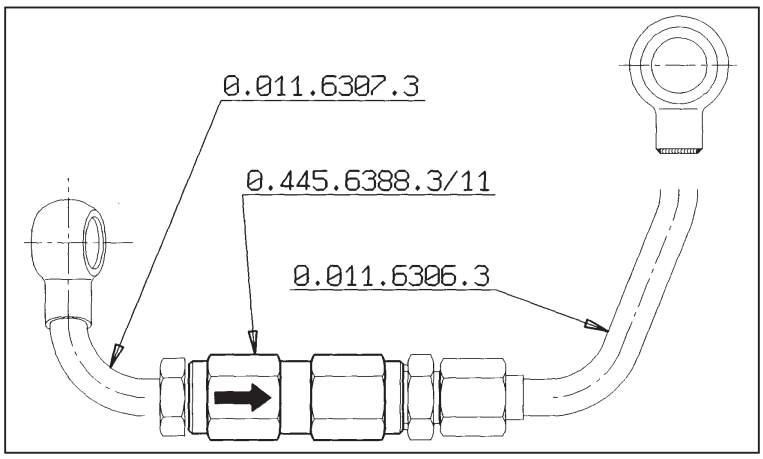
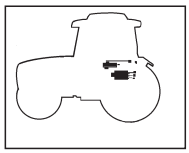


Fig. 11 - For tractors with flow control; fit the check valve observing the direction of the arrow in the figure.



Auxiliary hydraulic spool valves

Auxiliary spool valves are used to control hydraulically operated external implements.The controls consist in 2 levers for a 4-way valve, or 3 levers for a 6-way valve.The levers are located on the driver's right hand side.

Auxiliary hydraulic spool valves:

Version 1: with 4 ways (2 double-acting ways convertible to single-acting and equipped with detent).
Version 2: with 6 ways (2 double-acting ways convertible to single-acting and with kick-out; 2 double-acting ways convertible to single-acting and with detent; 2 double-acting ways convertible to single-acting with float position).

Version 3: with 10 ways (version 2 + 4 ways in mid-position of tractor, 2 on left and 2 on right).

- Furthermore, there is a special version for those users who require additional controls for a wider range of applications.

Version 4: 12 ways ***(version 2 + 6-way control valve)*** located in the middle of the tractor on the right-hand side with electro-hydraulically operated control valve.

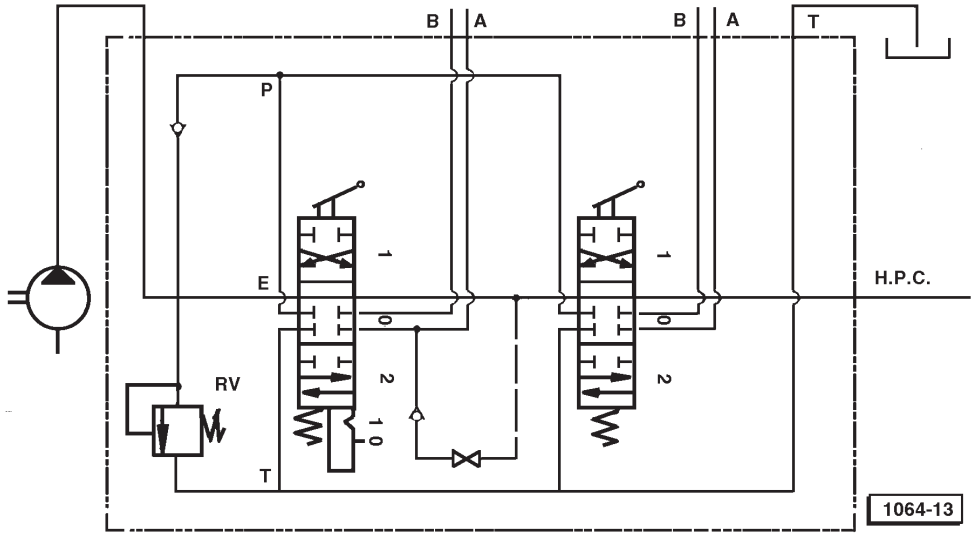
- 6 ways connected to the control valve located at the rear of the tractor and operated by way of the levers mounted on the mudguard to the right of the driver (two double-acting 1-1 ways convertible to single-acting and with kick-out, two double-acting 2-2 ways convertible to single-acting and with detent, two double-acting 3-3 ways with float position).

- 2 ways controlled by rocker switch with float position (mid-mounted ways number 4-4 on right-hand side of tractor) connected to the electro-hydraulically controlled control valve located in the middle of the tractor on right-hand side.

- 4 ways controlled by joystick (mid-mounted ways number 6-6 on left-hand side of tractor and mid-mounted ways number 5-5 on right-hand side of tractor) The joystick is equipped with a lock-release button; when this button is released when the joystick is in the position with the yellow marker, Float position is obtained. The ways are connected to the electro-hydraulic control valve located in the middle of the tractor on the right-hand side.

NB: with the auxiliary spool valve in use (unless a Flow Divider is incorporated), the operation of the hydraulic lift system remains inhibited (except in FLOAT position).

Important: the lever should be recentred to the neutral position as soon as the external actuator has reached its stroke limit, to ensure that the hydraulic circuit does not remain pressurized any longer than strictly necessary; prolonged pressure will result ultimately in the hydraulic system being subjected to hazardous levels of stress.
Auxiliary spool valves levers must always be in the neutral position when external services are not in use.



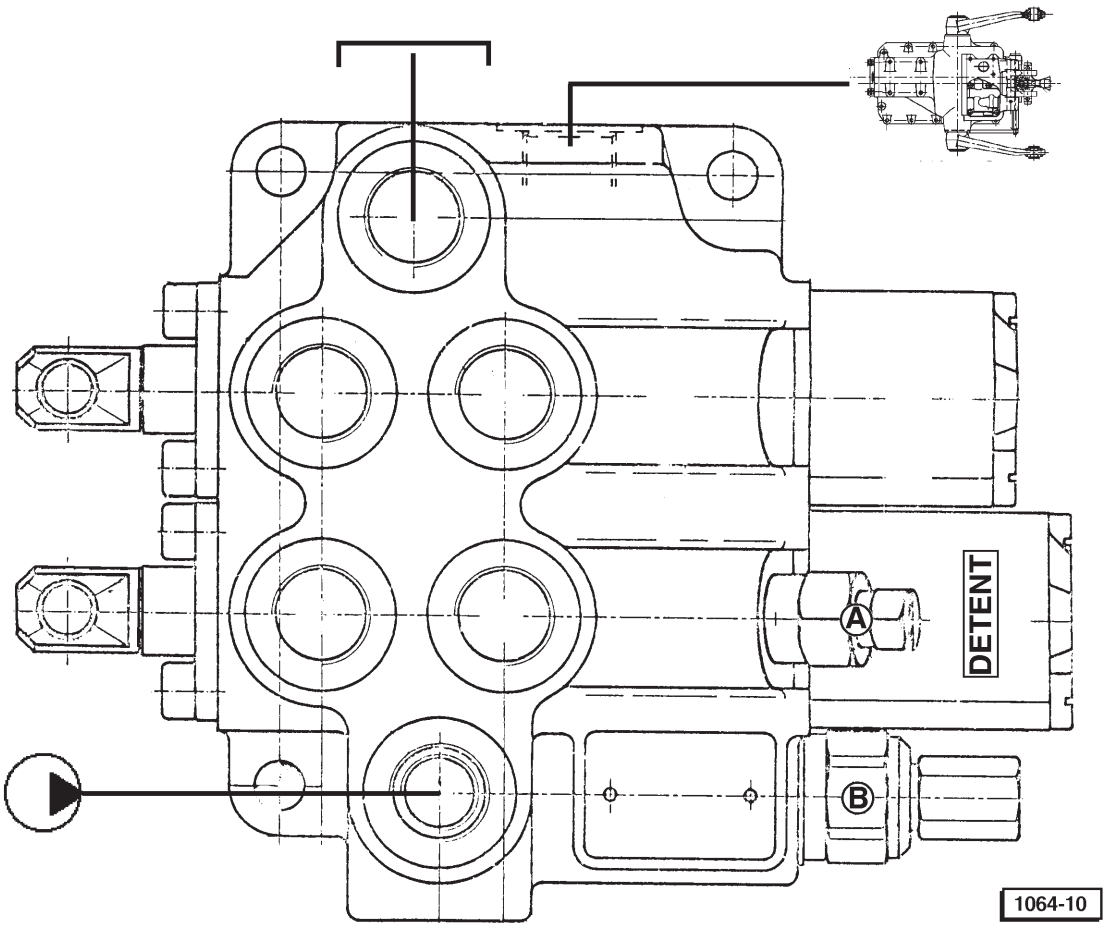


Fig. 1 - 4-way auxiliary control valves.
A - Quick coupling for free discharge
B - Pressure relief valve.

NOTE : The 6-way auxiliary spool valve incorporates a **FLOW DIVIDER** which ensures that oil can be delivered at a constant flow rate of between 3 and 42 l/min to the 2-way kickout section (first valve alongside the flow divider). The tap has set in the cab, to left hand of the place of guide.

NB: Even when the flow divider is shut off by turning the knob fully in the direction marked (**-**), a residual flow of oil will continue to reach the valve at a steady rate of 3 l/min.

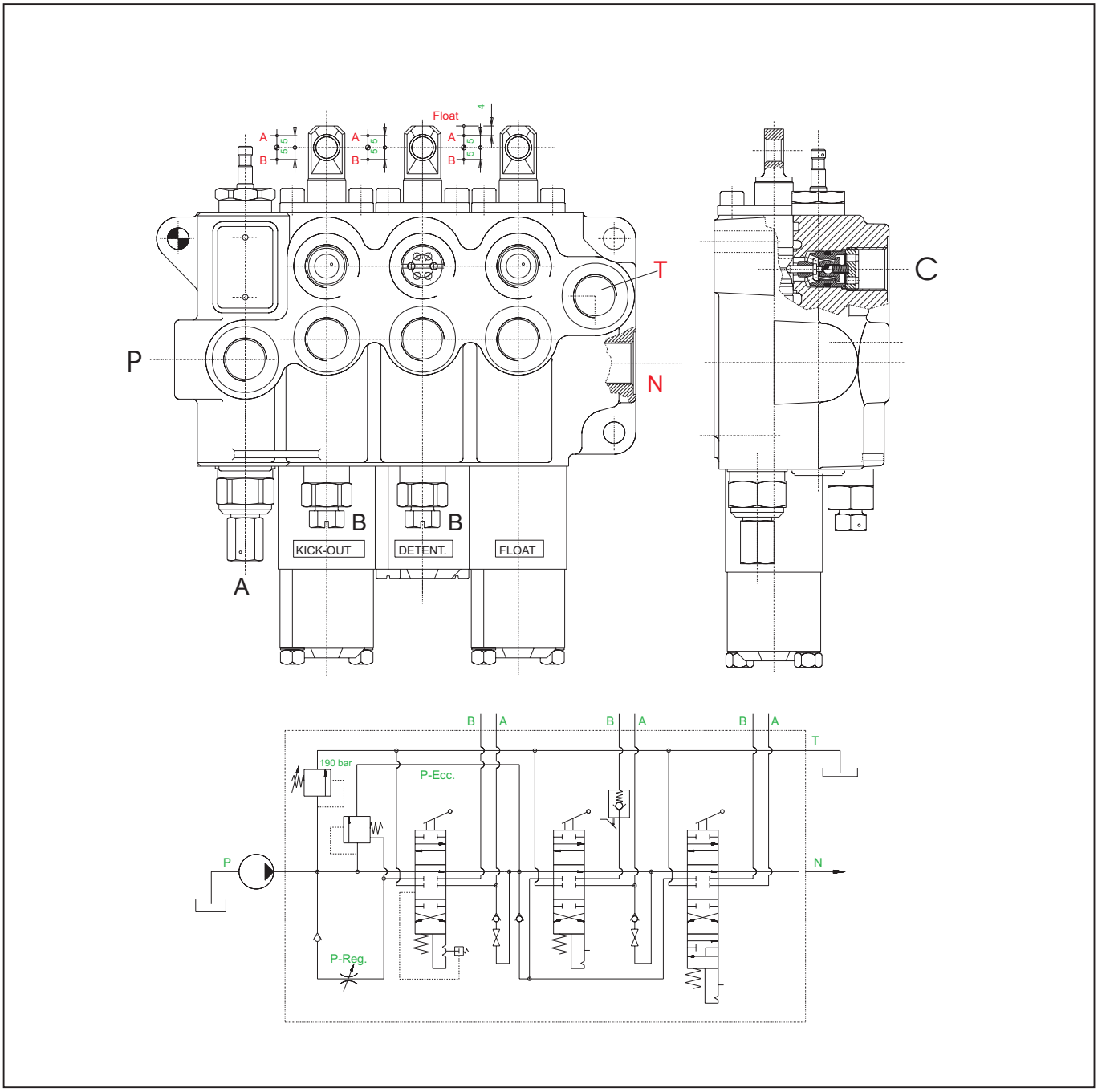


Fig. 2 - 6-way hydraulic control valve.
A - Pressure relief valve
B - Screws to convert from double-acting to single-acting control valve
C - Perfect seal valve
P - Hydraulic pump
N - To rear lift
T - Return

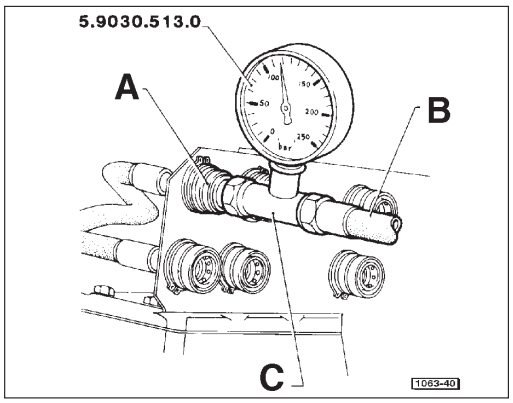


Fig. 4 - Checking the operating pressure of auxiliary spool valves.
A - Quick coupler
B - Implement oil line
C - Tee

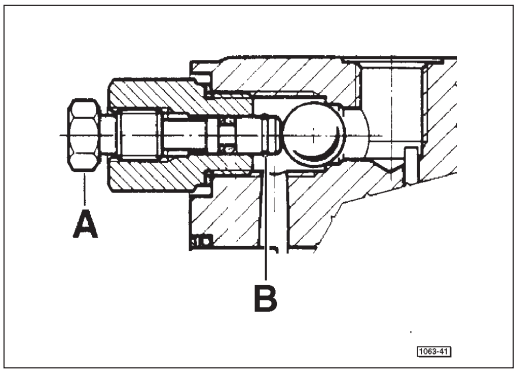


Fig. 5 - **A** conversion screw allowing conversion from double to single acting operation; **B** conversion screw lock washer.

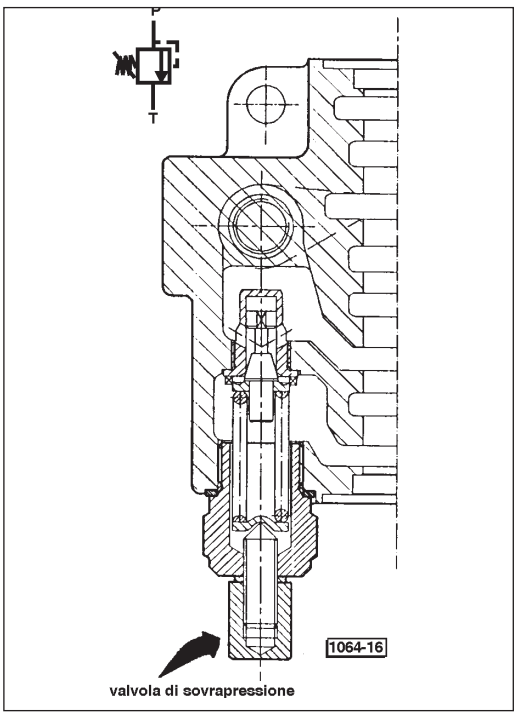


Fig.6 - Longitudinal section through 4/6-way spool valve.

Checking the pressure relief valve setting

The setting of the pressure relief valve must always be 190 bar, and can be checked by fitting gauge p/n 5.9030.513.0 to either of the valve work ports and operating the relative lever. If the gauge reading is not correct, adjust the pressure relief valve see pages 360 - 361.

NOTE: Remember that the purpose of the relief valve is to maintain pressure in the hydraulic lift circuit within a given safety limit, in this instance 190 bar.

Checking the operating pressure (see fig 4)

To verify the operating pressure of an implement connnected to the work ports of the auxiliary spool valves, proceed as follows:

— Connect a Tee between quick coupler **A** and the implement pressure line **B**.
— Connect pressure gauge p/n 5.9030.513.0 to the Tee.
— Start the engine, apply the parking brake in the interests of safety, and operate the spool valve connected to the implement.
— The gauge will show the operating pressure currently registering in the circuit.
The operating pressure depends on the weight of the hitched implement, and should not normally exceed 150 bar.

Conversion of auxiliary spool valves from double acting to single acting operation (see fig 5)

Where particular requirements dictate, a double acting spool valve can be converted to operate in single acting mode.

To effect the conversion from double to single acting, hold the fitting secure and loosen the screw **A** by between 4 and 5 turns.
NOTE: Once slack, the screw is best rotated by hand to avoid applying excessive force and unseating the lock washer **B**; if this were to happen, the washer would find its way into the hydraulic circuit and disallow normal operation of the lift system.

The conversion from single to double acting operation is effected by tightening the screw **A** fully.

Checking the surface of the valve spools

Check that the machined surface of the spool is free of scoring or any other kind of wear.

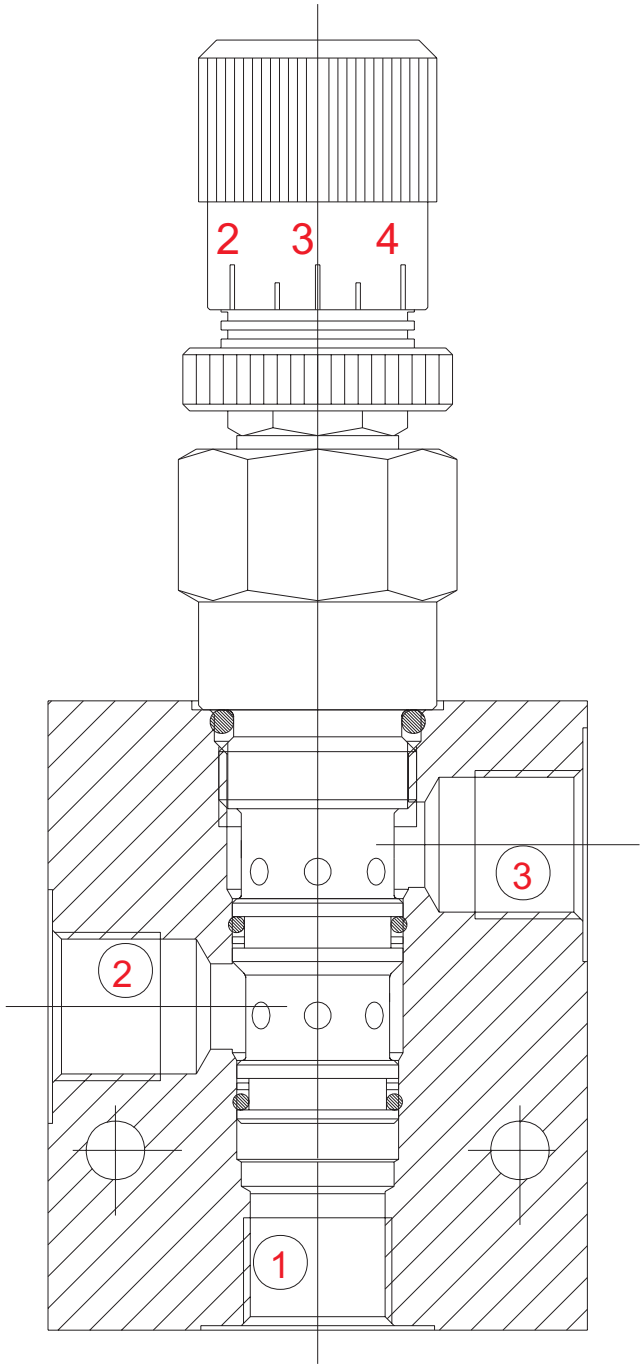


Fig. 7 - FLOW DIVIDER valve for adjustment of oil flow rate.
1 - Oil from hydraulic pump
2 - Oil to auxiliary control valves
3 - Oil to hydraulic lift

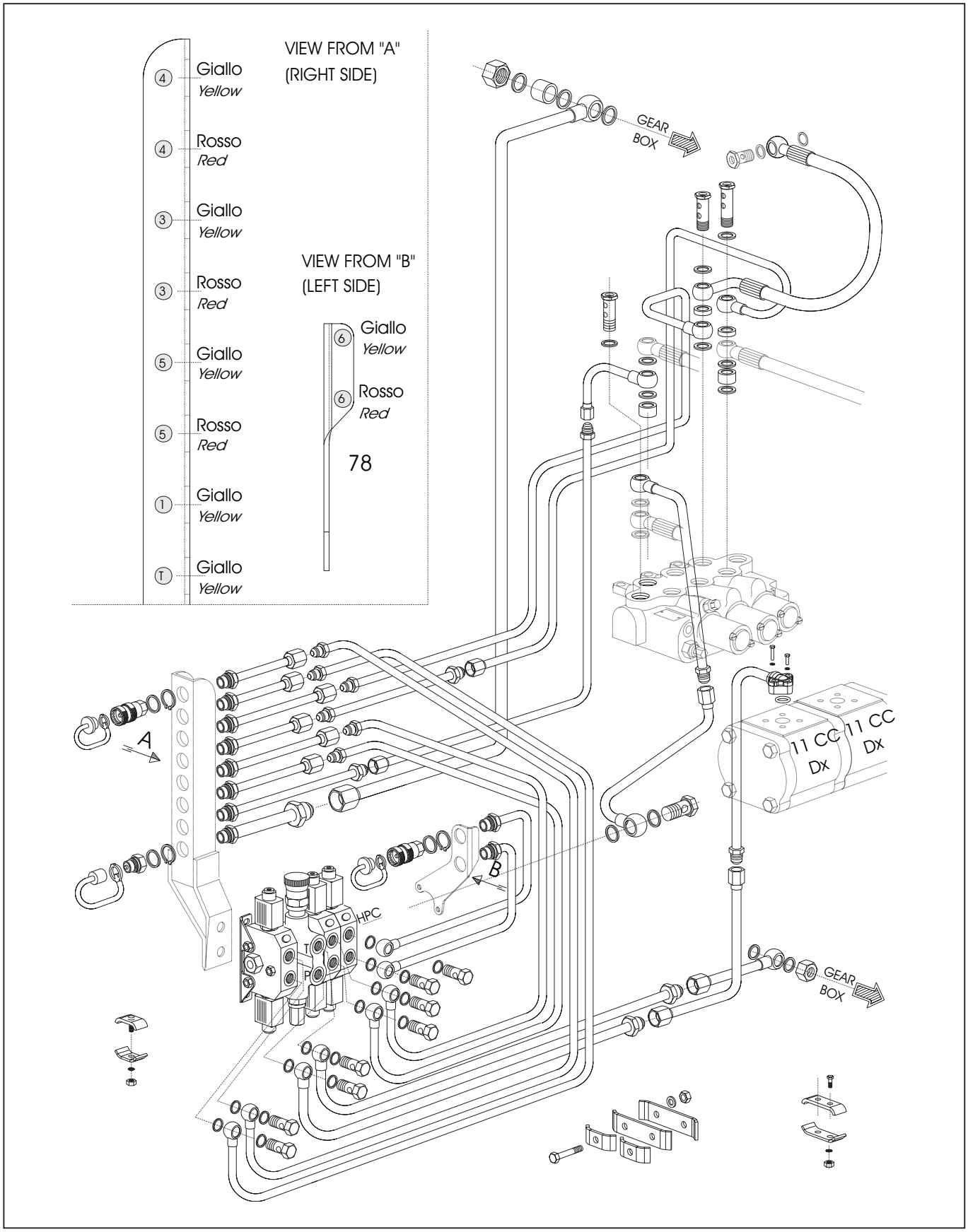


Fig. 8 - Hydraulic diagram for electro-hydraulically operated control valve.

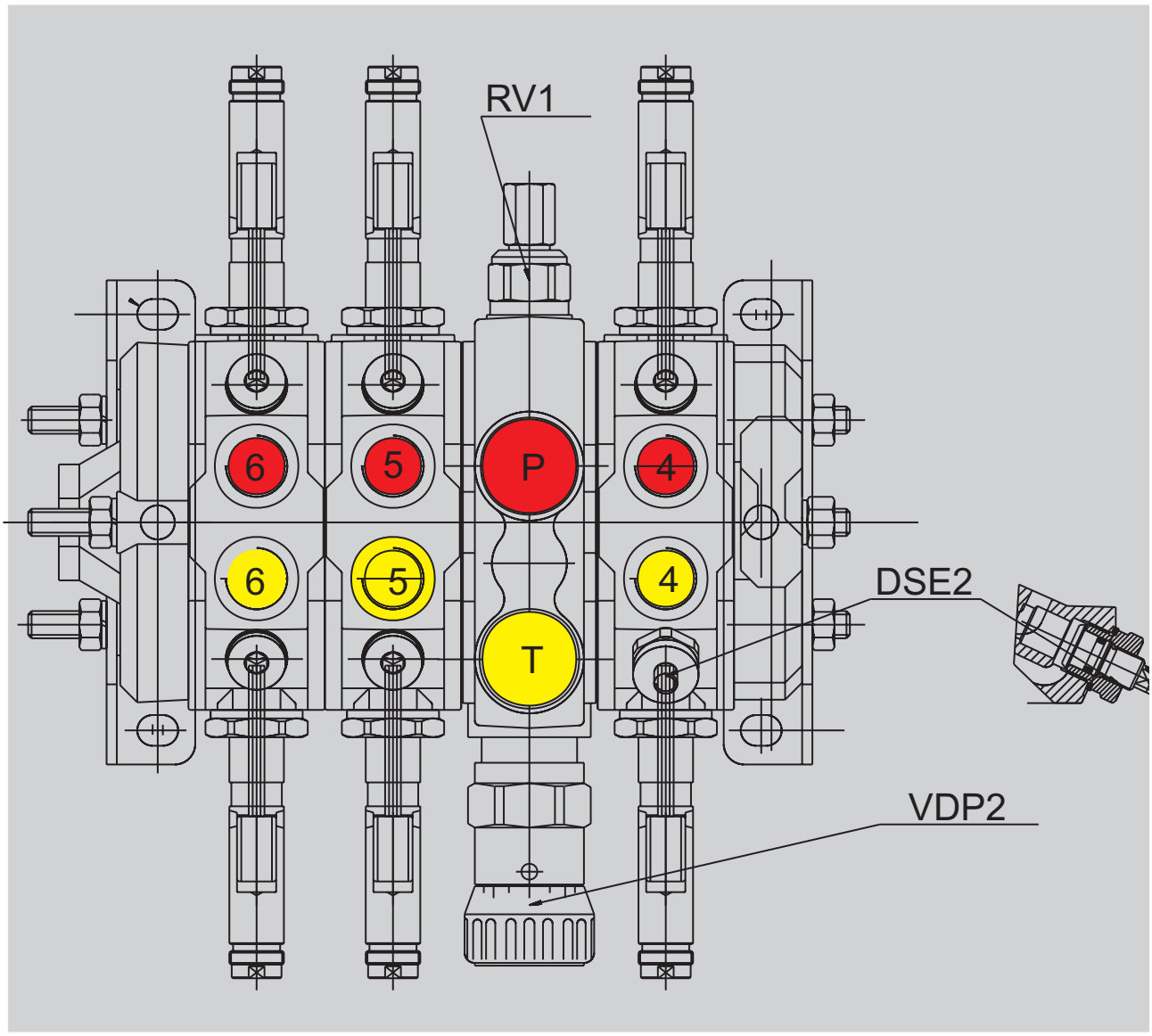


Fig. 9 - 6-way auxiliary control valve with electro-hydraulic controls.

DSE 2 - Screw to convert from double-acting to single-acting control valve
VDP 2 - FLOW DIVIDER valve for adjustment of oil flow rate.
RV 1 - Safety valve
4 - 4 - Hydraulic ways located in middle of tractor on right-hand side; controlled by rocker switch located on left-hand side of console.
5 - 5 - Hydraulic ways located in middle of tractor on right-hand side; controlled by joystick.
6 - 6 - Hydraulic ways located in middle of tractor on left-hand side; controlled by joystick.

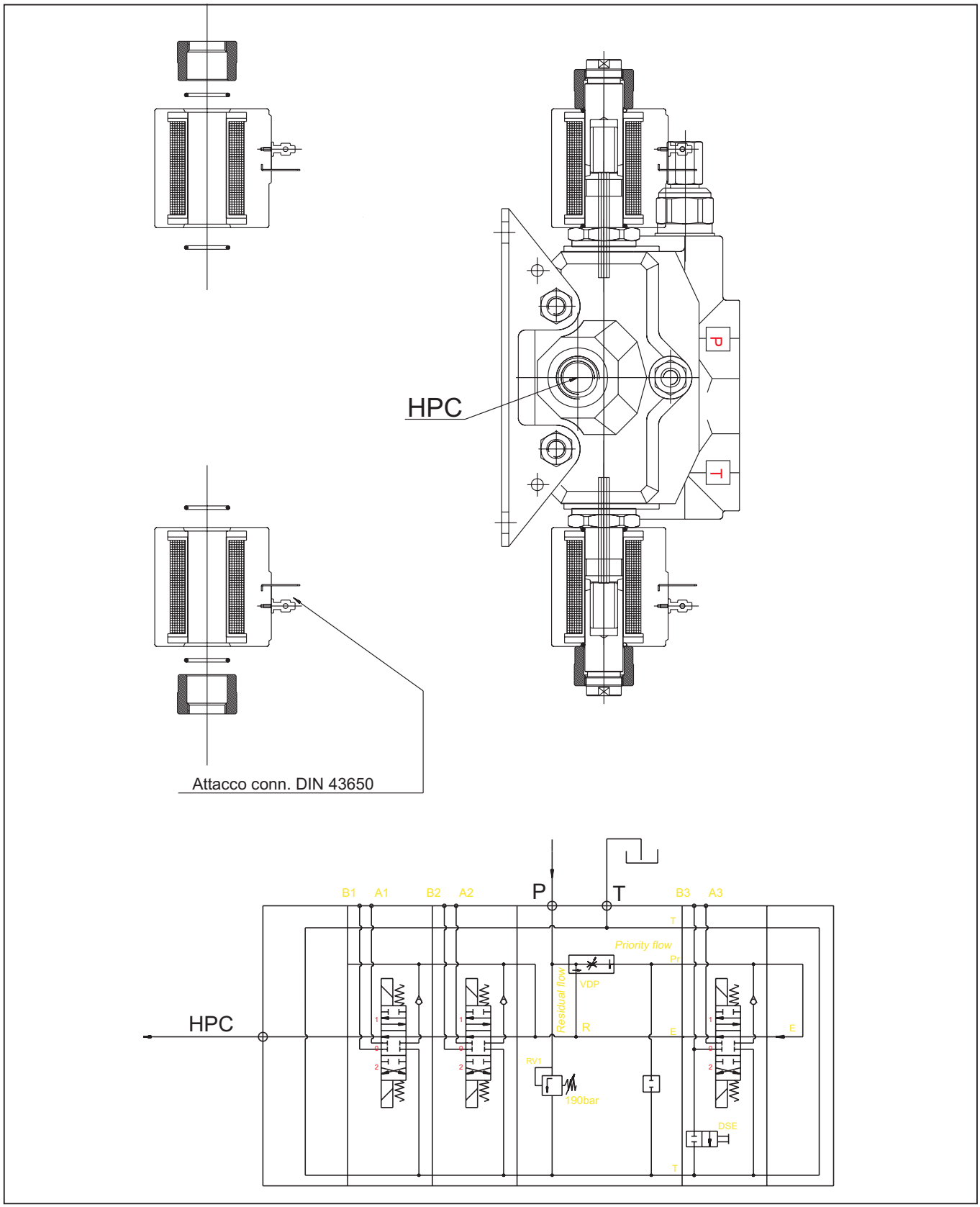
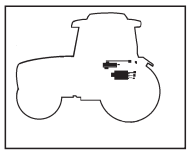


Fig. 10 - Solenoid valves for control of auxiliary service control valves (control valve located in centre of tractor on right-hand side)

HPC – Oil flow to electrohydraulic control unit for stabilisers and lift rod adjustment.



Trailer hydraulic braking system

The tractor can be equipped with a trailer hydraulic braking system on request.
For trailers with **"safety brake"** (ITALY version), the system is equipped with a specific distributor.

The oil used is drawn from the tractor's main hydraulic system by a special distributor.

This distributor is activated by a valve which is hydraulically connected to the tractor hydrostatic braking system control.

Trailer hydraulic braking distributor unit

Specifications (see Fig. 2, pag. 370)

P	supply
N	services connections
B	trailer brake connection
T	oil discharge
Y	trailer braking system connection (in series)
E	parking brake signaling pressure switch

Supply	l/min	20 ÷ 80
Constant pressure in **B** (only for distributor ITALY version)	bar	12,5 ÷ 2
Maximum pressure in **B** (only for distributor ITALY version)	bar	135 ÷ 5
Maximum operating pressure in **N**	bar	200
Working temperature	°C	-20 +100
Supply in **P**		with hydraulic mineral oil
Pilot **Y** (supplied from below)		with hydraulic mineral oil

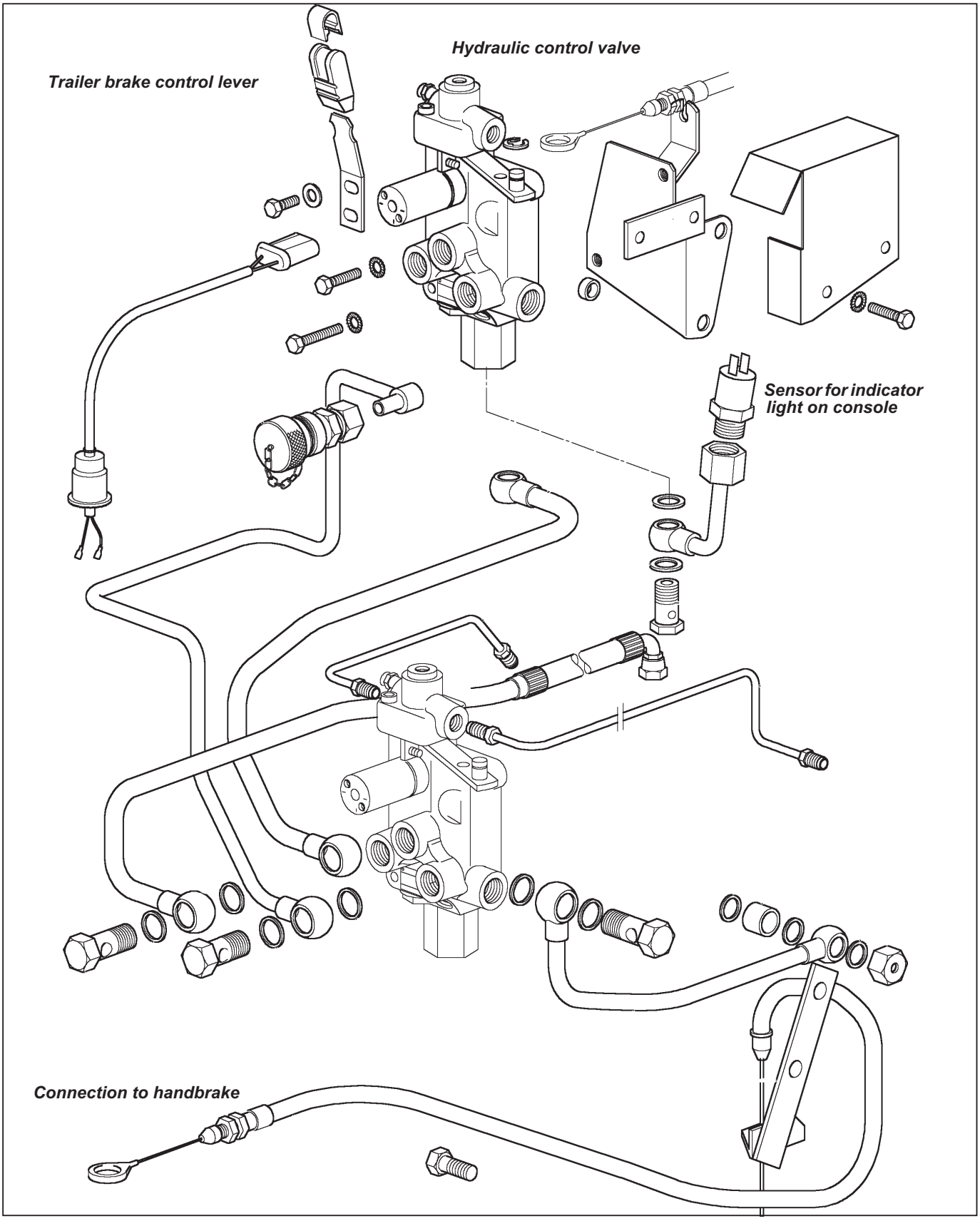
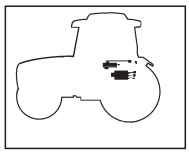


Fig. 1 - Hydraulic system for hydraulic trailer braking - ITALY version.



Use of the tractor with CUNA 341/01 hydraulic trailer braking

The trailer hitched to a tractor equipped with this type of trailer braking requires a **constant pressure of 12.5 ± 2.5 bar** in order to release the parking brake (spring operated).

Therefore, before hitching or unhitching the trailer, this pressure must be eliminated at the coupler by switching off the engine, or better still, by applying the tractor handbrake.

The **female quick-action** couplers are specially-designed to remain connected if the internal pressure exceeds 1.5 bar.

Unless the driver waits the few seconds required to eliminate the pressure, the coupling or uncoupling operation will be extremely difficult if not impossible.

The correct type of female coupler inserts are marked **FASTER VFBO/1815F** and can also be identified by the release ring, which is 44 mm in diameter and 38 mm long.

When uncoupling is performed to hastily, the pressure in the trailer's hydraulic system will lock the coupling, and in addition to compromising the efficiency of the **emergency brake**, it will prevent re-coupling.

In this event:

— Uncouple the push-pull coupler which passes through the reservoir mounted on the trailer, by pulling on the pipe connecting the trailer to the tractor.

— Discharge the pressure.

— Re-couple

Special recommendations

In order to change from parking brake applied from the trailer (tractor handbrake in lowered position) to efficient service braking, the tractor needs to send large quantities of oil to the cylinders.
Some time is therefore required before trailer braking reaches full efficiency and therefore, **should it be necessary to use the service brake urgently, after having released the parking brake, accelerate the engine to maximum speed before releasing the tractor handbrake.**

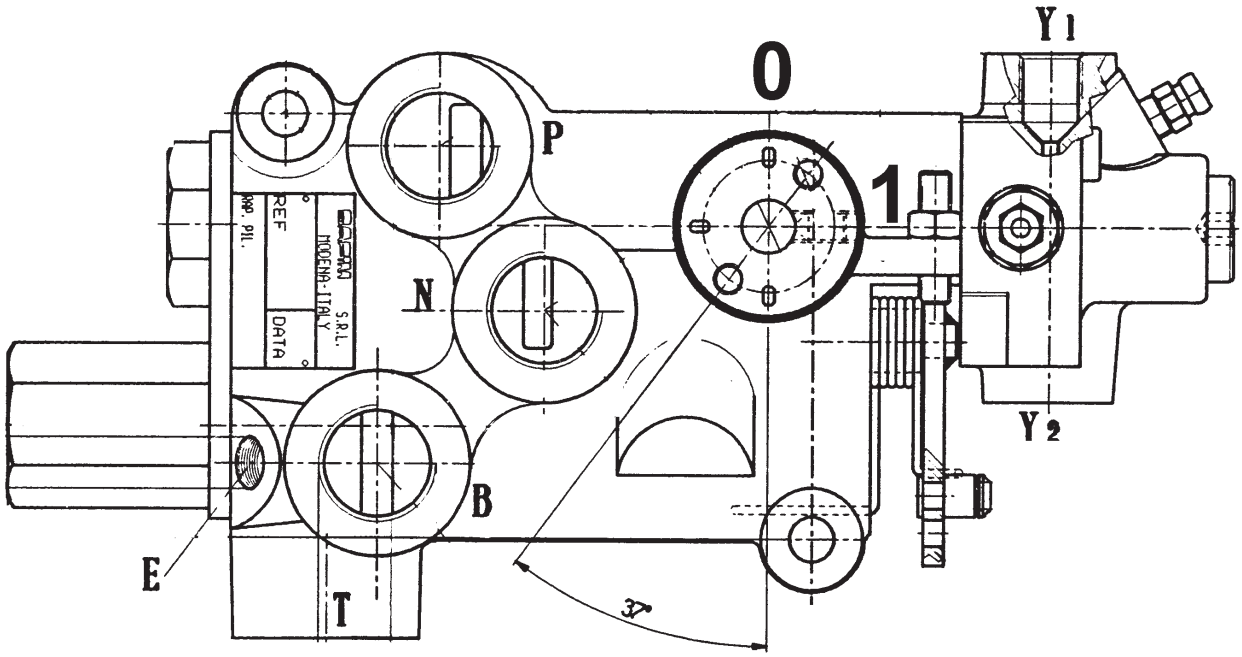
In general, remember that if the engine revs are high, braking will be more responsive.

Users who have trailers with "France" type non-type approved hydraulic braking may adapt the SAFIM brake valve to this type of operation by unscrewing the hex socket screw on the pilot side cover by three turns (6 mm Allen wrench).
In this way, the 12.5 bar pressure is eliminated, the red light remains on, only going out on braking.

It is dangerous to leave the valve by-pass lever in the braking position; trailers which do not conform to the Cuna 341/01 standard are not equipped any protection against this.

The lever on the valve serves to completely by-pass the valve when trailers are not to be hitched to the tractor for long periods (on trailers which conform to the Cuna 341/01 standard, even if the lever is left in the OFF position, the trailer parking brake is not released).

Starting



1 - Trailer parking brake control knob
(present only on mechanically operated valves)
A - Rotation for engagement
B - Rotation for disengagement

Fig. 2 - Control valve for hydraulic trailer braking.

No special precautions are required, simply start the engine and allow time(10-15 seconds) for the hydraulic pipes to fill with oil before operating the brake pedal.

Checking operation

The red indicator light illuminates when the pressure at the trailer brake coupling is less than 10 bar; the light therefore gives an indication of whether the system is functioning correctly.
During normal operation, providing the handbrake is released and the brake pedal is not depressed, the red light should be off.
When the handbrake is applied, the light will stay on even if the brake pedal is pressed.
The light also comes on if the engine stalls under load, as there will be no pressure at the trailer brake coupling.
In normal operation with the handbrake released, on pressing the brake pedal it is possible to hear the pump as it increases the pressure.
To test the pressures, apply a pressure gauge with a minimum full-scale value of 160 bar to the trailer brake coupling.

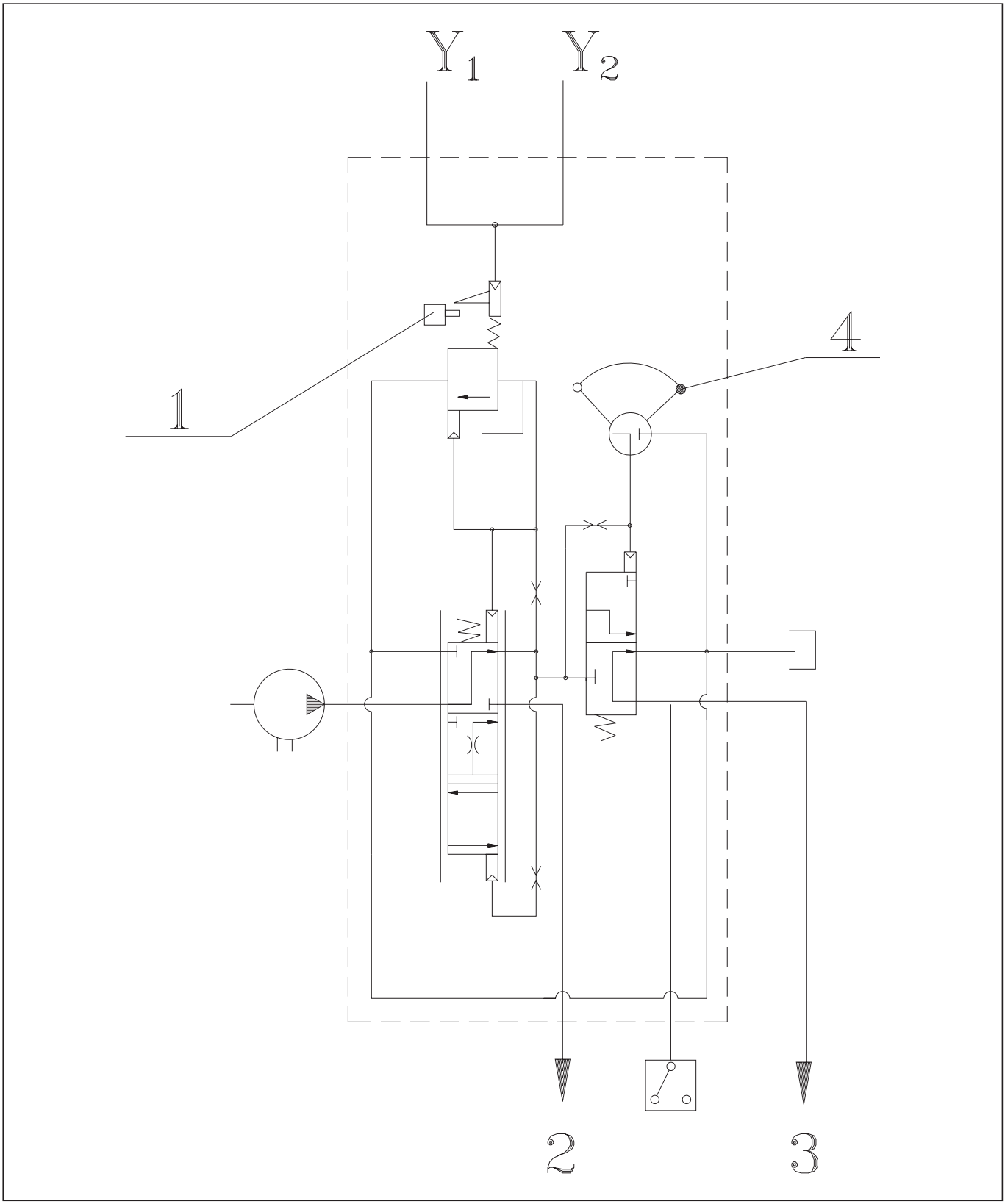


Fig. 3 - Operation of the hydraulic control valve
1 - brake ON/OFF control
2 - lift
3 - trailer brake
4 - trailer brake lever

The pressure readings to be taken are:

— Engine running with handbrake released: 12.5 ± 2.5 bar (10 to15 bar)
— With handbrake applied: 0 bar
— With handbrake released and brake pedal depressed, the pressure will rise from 12.5 to 130 ± 10 bar (120 to 140 bar).

N.B.: the valve is factory-set by Safim.

( ) values in brackets refer to pressure limits specified by the Cuna standard.

Malfunctions

Insufficient pressure

Check the position of the valve by-pass lever, which should be aligned with the axis of the valve.

For valves with mechanical parking brake control

Release the handbrake lever.
Check that the lever on the valve returns to the limit stop when the handbrake is lowered.

For electrically controlled valves

Check that the microswitch is functioning correctly (operated when the handbrake is lowered).

Check the power fuse and the power supply in general.

Check the connections to the coil on the two opposite poles.

If the coil is energised, by loosening the retaining nut it is possible to feel the pull of the magnet; the coil will be very hot.

The valve pressure rises very slowly

Clean the holes of the main piston; to access the piston, remove the plug on the opposite side to the pilot side (27 mm wrench).
Withdraw the piston and blow clean.
Close the hex socket adapter screw (France type).

The parking brake is slow to release

Clean the holes (by washing and blowing) the rapid discharge piston, which can be accessed by first removing the solenoid valve and the adapter.
Warning: the valve has been tested and calibrated correctly prior to delivery.
Before attempting to adjust the springs and the shims, to solve pressure problems, clean all the holes of the piston and solenoid valve with petrol and compressed air.

Low and high pressure settings

The pressure settings are regulated by way of shims, which can be accessed by removing the pilot valve and withdrawing the 25 mm diameter piston.
The internal spring regulates the pressure of 12.5 ± 2.5 bar, while the outer spring regulates the pressure of 130 ± 10 bar.

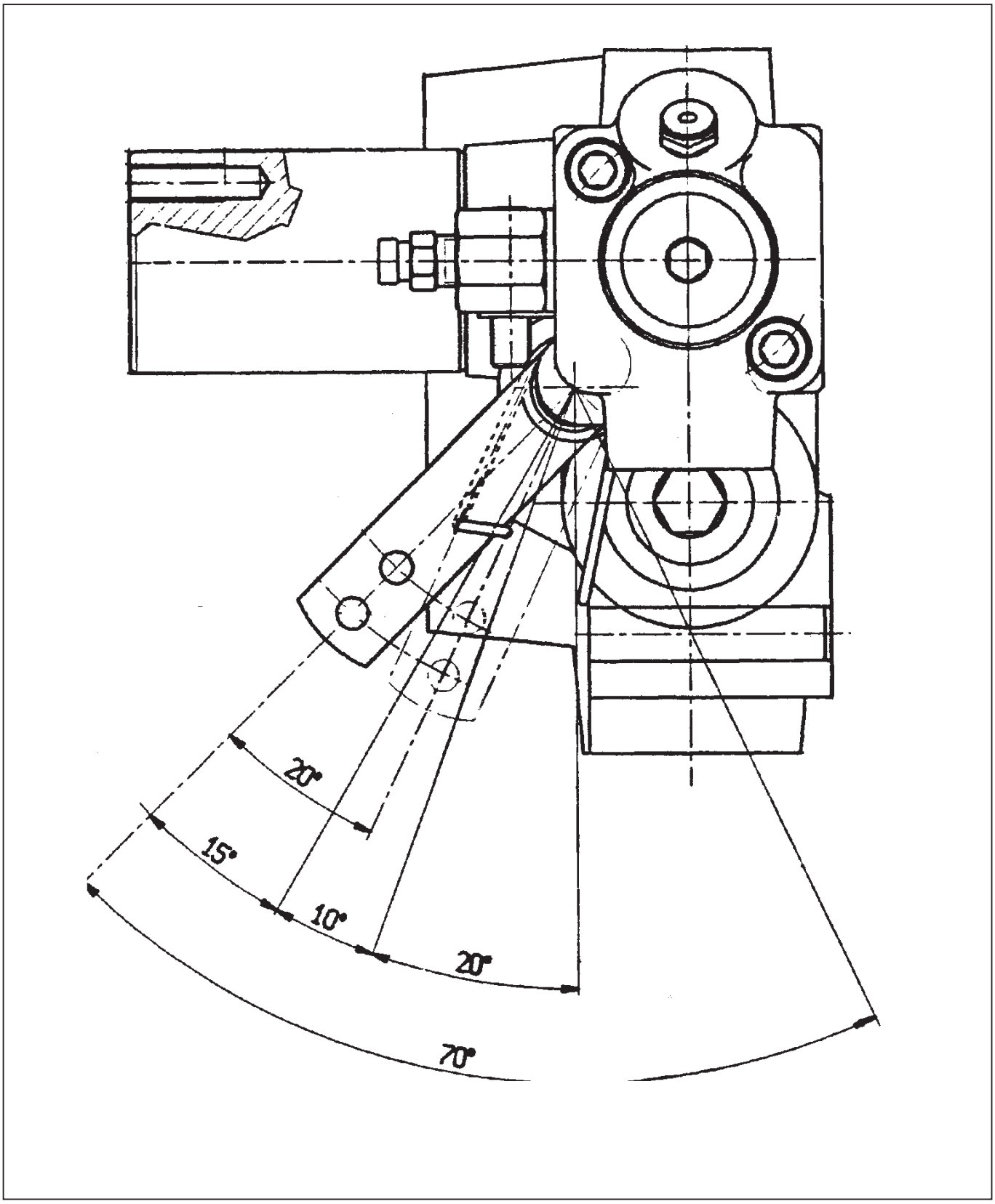


Fig. 4 - View of the hydraulic control valve.

1 - Lever position with handbrake fully applied
2 - Start of trailer braking
3 - Lever position with handbrake released

Installing the hydraulic braking valve for trailers equipped with “safety brake”

(ITALY version)

For correct valve and related controls installation, it is necessary to proceed as follows:

— Secure the distributor to the tractor and connect the various oil tubes;

— Bring the tractor parking brake lever to position **A** (4th notch of the toothed sector) and carry out the parking brake adjustment so that, in this position, the tractor is braked. Then bring the parking brake lever to position **B** (stop notch).

— Keeping the parking brake lever in position **B**, hook the hose fork to lever **Y**, pushing it against stop screw **X**. Then connect the hose to lock plate **C** by means of the special adjusting nuts. Under these conditions, acting on the parking brake hand lever of the tractor synchronizes the tractor and trailer stationary braking.

— Connect the wiring of the pressure switch, that is screwed on the distributor, to the specific warning light installed on the instrument panel of the dashboard.
Then, when engaging the parking brake, the tractor warning light and the trailer warning light will light up at the same time.

CAUTION:
When the trailer is hitched to the tractor the knob on the control valve must always be turned counter-clockwise to position **"1"**. When a trailer is not hitched or during trailer hitching/unhitching operations, the lever on the control valve must always be in the position **"0".**

On the hydraulic distributor is placed a sensor (**E** Fig, 5). This sensor active the warning light on the dashboard, when the trailer brakes is operated.

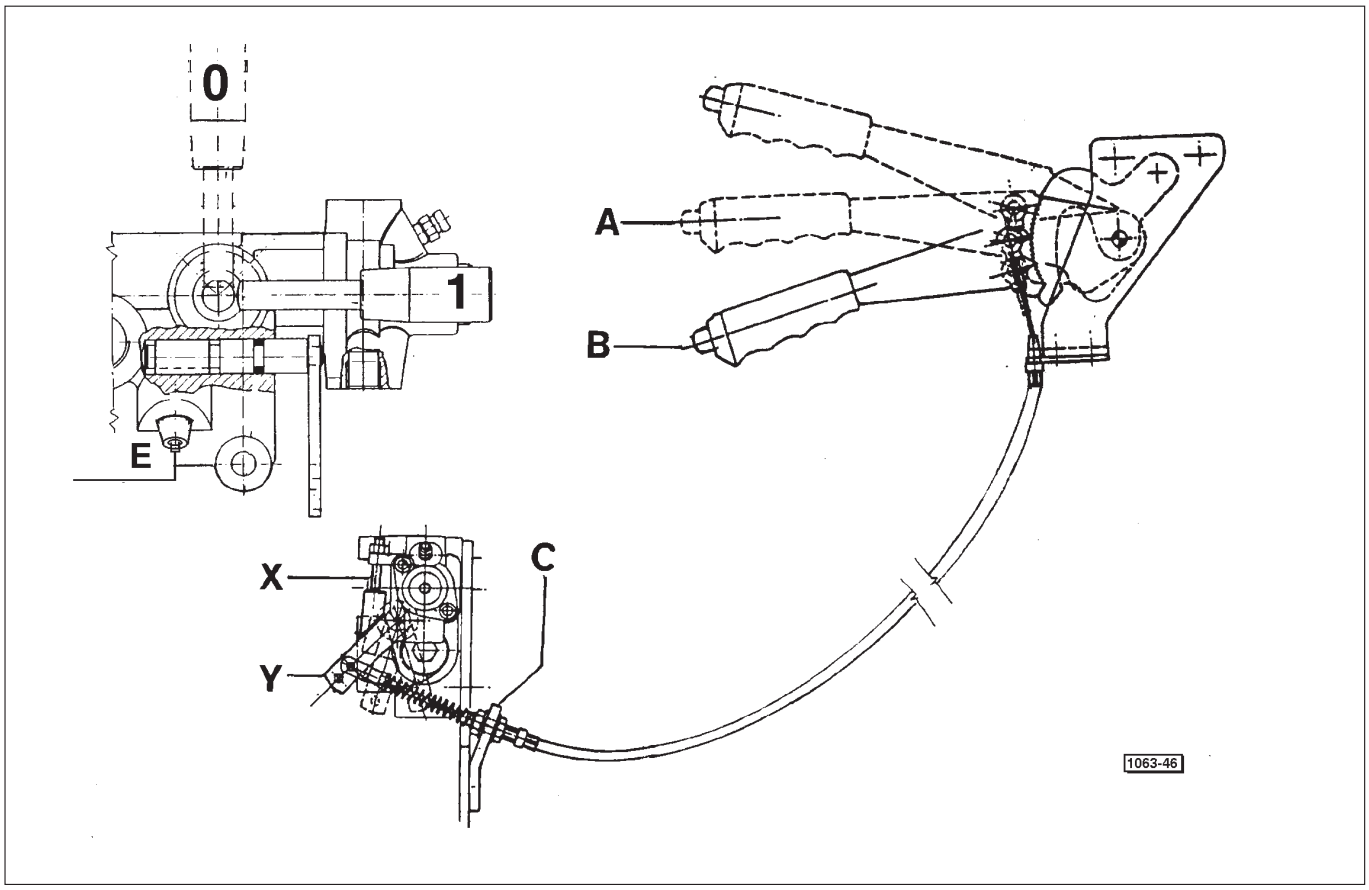


Fig. 5 - Diagram illustrating adjustment of brake control valve for trailer equipped with **safety brake** (ITALY version)

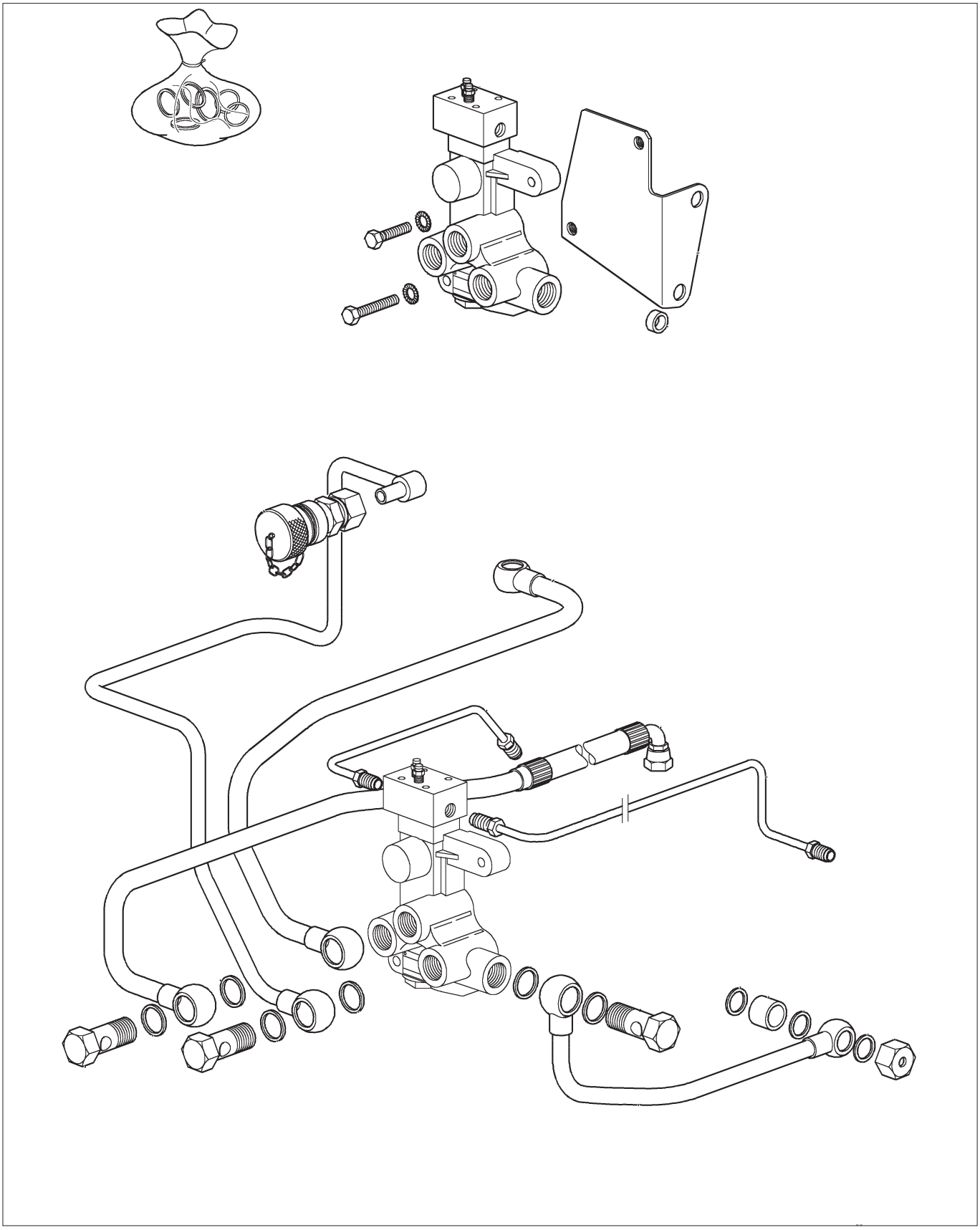


Fig. 6 - Components of the hydraulic control valve - FRANCE version.

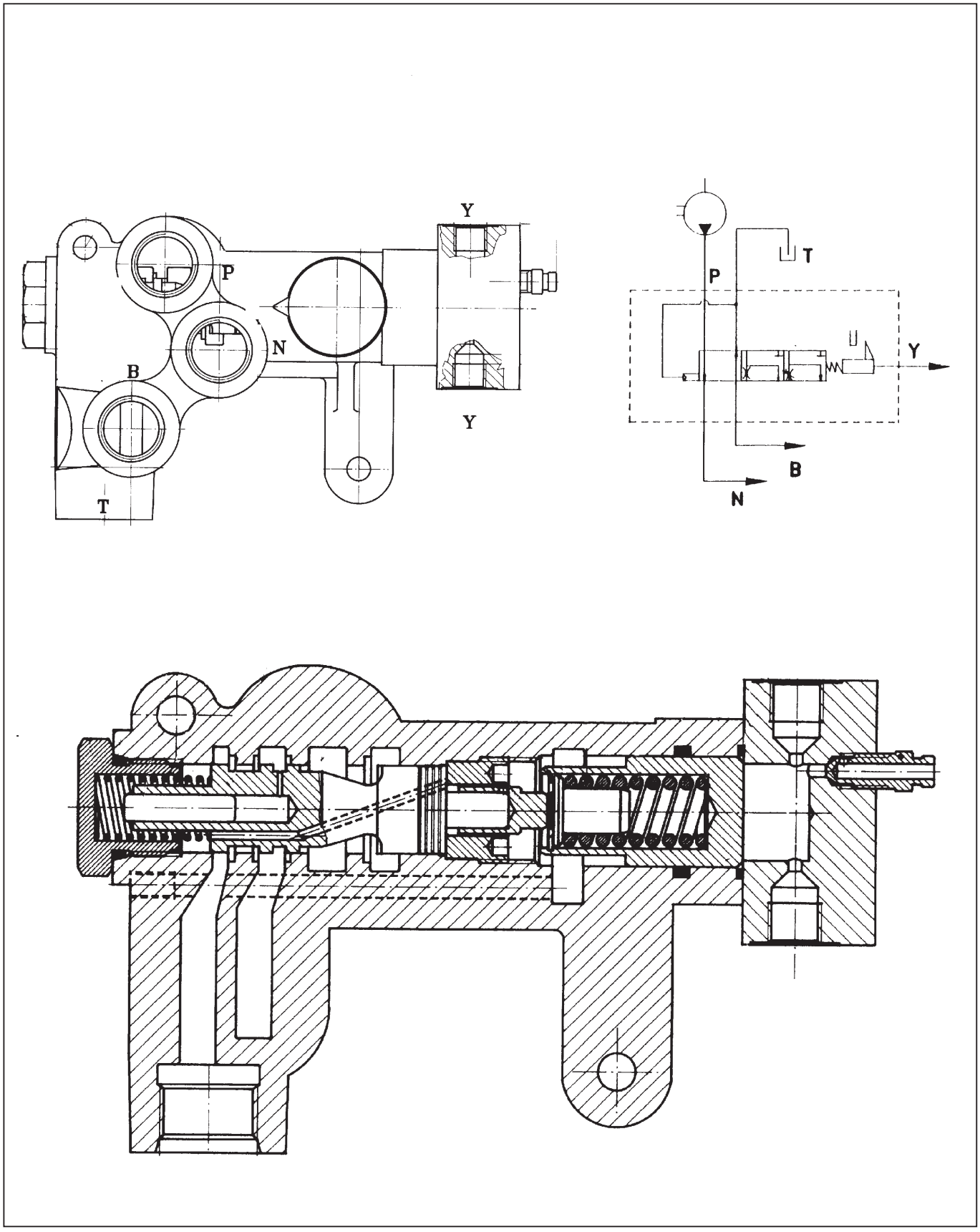
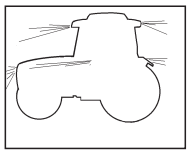


Fig. 7 - Components of the hydraulic control valve - FRANCE version



General specifications

The electrical system is fully equipped for use on the road, with sidelights, main/dipped beam headlamps, direction indicators doubling as hazard warning lights, and work lights, all fully in compliance with legal requirements as embodied in the highway code.

There are four essential components and assemblies: battery, alternator, starter motor, and auxiliaries.

The system operates at a nominal 12 Volts (the maximum voltage in operation is 14,3 ÷ 0,5 V).

The battery charge level is indicated by a relative meter on the instrument panel.

specifications

battery

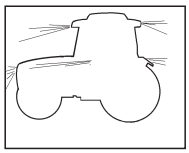
type		maintenance free
voltage	V	12
discharge capacity over 20 hours	Ah	100
rated current	A	470

starter motor

		60 -65 - 70 CV	75 - 80 - 90 CV
code	(BOSCH)	2.9619.300.0/10	2.9619.290.0/10
	(ISKRA)	2.9619.360.0	2.9619.350.0
voltage	V	12	12
nominal rated power	cv	3,4 (2,6 kW)	4 (3 kW)
electromagnetic control		electro-magnetic with translation and rotation of pinion	

alternator

RECHARGE SYSTEM		
code	65A MARELLI	2.9439.400.0
	65A BOSCH	2.9439.410.0
	65A ISKRA	2.9439.420.0/10
	ISKRA 85A	2.9439.460.0/10
type		three-phase
nominal rated voltage	V	12
pulley nut tightening torque	kgm	5,5 (62 Nm)



8 Systems

84 Electrical system

GENERAL SAFETY DIRECTIONS

PRECAUTIONS AGAINST DAMAGE TO ELECTRICAL AND ELECTRONIC EQUIPMENT

— Disconnect any electronic equipment from the system when the tractor is placed in any environment where temperatures are likely to exceed 80 °C.

— Remove battery connections when carrying out servicing or repair operations on the electrical system.

— Avoid overvoltages in the electrical system. Always remove battery and alternator connections before carrying out welding operations or when recharging the battery from the mains.

— Never disconnect the battery while the engine is running.

— In the event that the battery needs topping up, use only distilled or demineralized water; top up only when the battery is AT REST and at AMBIENT TEMPERATURE (allow time for cooling after operation and following recharge).

— Never carry out servicing operations on the electric heating system with the engine running.

— When washing the tractor, ensure that the alternator and starter motor are well protected. These components must not be sprayed or splashed directly with water.

PRECAUTIONS AGAINST ACCIDENT AND INJURY

WHEN COMMENCING ANY KIND OF WORK:

— Turn off the main ignition switch.

— Disconnect the battery; remember that electric shocks discharged through the human body will damage pacemakers.

— The battery generates powerful and inflammable gases when recharging, which if ignited can cause the battery itself to explode.

— Do not make temporary or makeshift connections to the battery, as these can cause dangerous sparks.

— The electrolyte in the battery is a solution of dilute sulphuric acid; if splashed on the skin, wash straight away with plenty of water and sodium bicoarbonate.

— Seek attention immediately from a doctor if electrolyte is splashed into the eyes.

— Never allow the battery near a naked flame.

— Keep a safe distance from a recharging battery when testing or measuring, and always wear suitable eye protection.

— Make certain that the electrical system and all its components are inspected and serviced periodically; any faults must be eliminated without delay, and parts subject to everyday wear replaced at the recommended intervals.

— Ensure that trailers are fitted with regulation lights.

IMPORTANT REMINDERS

The maker declines all liability in respect of any damages resulting from operations performed other than in the recommended manner and/or by unskilled hands.

— To ensure smooth operation of the machine, use only genuine replacement parts.

JUMP START UTILIZING ANOTHER BATTERY

NB: Emergency starting with the aid of a battery charger is not recommended.

PRECAUTIONS
The battery may be that of another tractor, but must in any event have the same rated voltage (and therefore an identical number of cells).

1 - Make certain the two vehicles are not in contact with one another.

2 - Ensure that all auxiliary services are switched off.

3 - Make certain the flat battery is securely connected to earth, that the filler caps are properly tightened and that the electrolyte is up to the correct level.

PROCEDURE:
A - Connect the two batteries as indicated in figure 1 (part **A**).

B - If the jump battery is that of another tractor, start the engine and run at approximately 1/4 of full throttle.

C - Turn the ignition key to start the engine of the stranded tractor, following the procedure given in the operator's handbook.
D - Remove the jump leads following the sequence as indicated in figure 1 (part **B**).

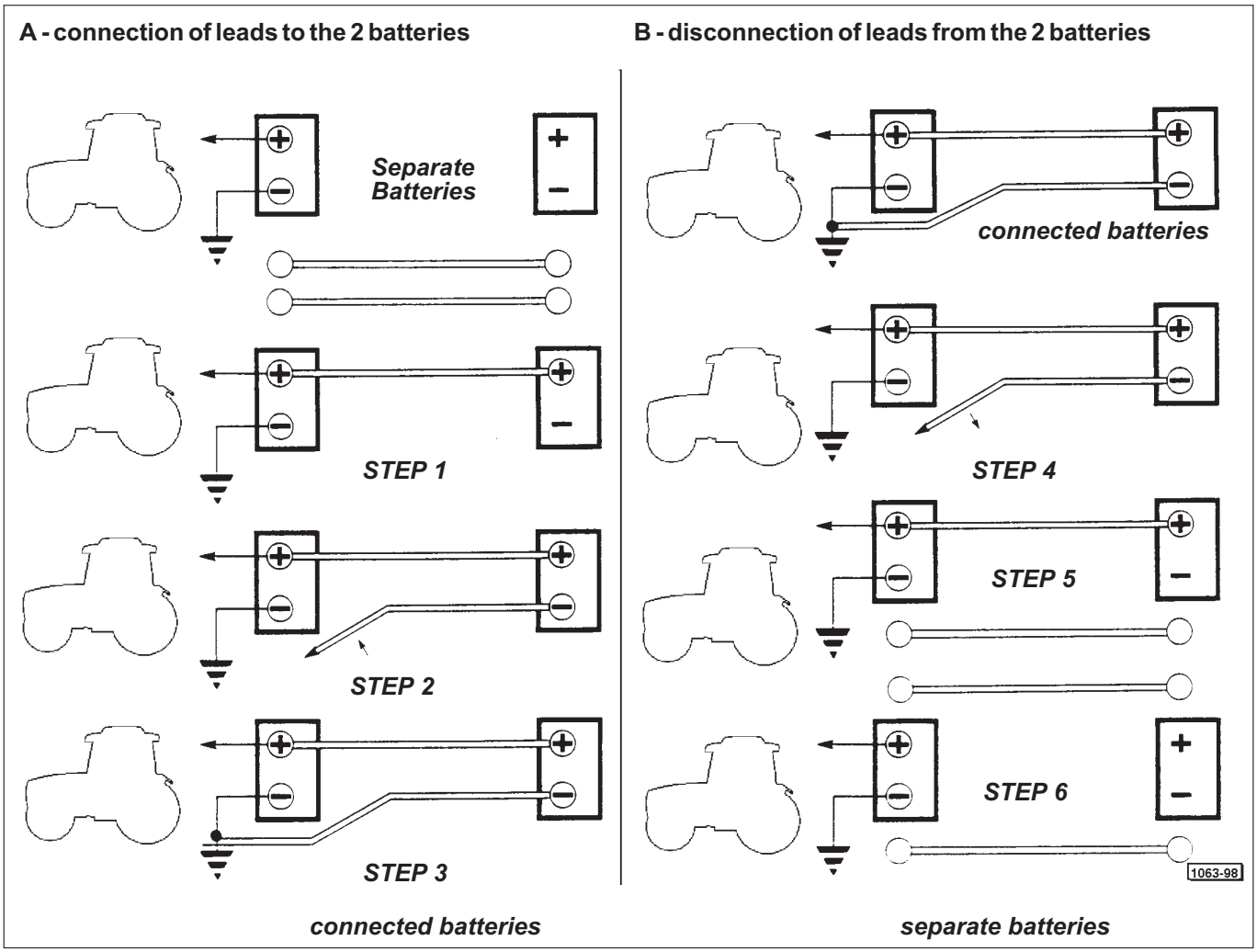


Fig. 1 - Diagram of battery connections and relative leads.

PRACTICAL HINTS

Before installing a battery, and periodically thereafter:
— Check over the electrical system, giving particular attention to connections; make certain there are no traces of oxidation and that there is no loss of current.
— Check the alternator and the voltage regulator setting. If the charging voltage is inordinately high, distilled water will be consumed at a high rate, whereas a low voltage will not charge the battery sufficiently.
— Check the charge level of a new battery with a density indicator or voltmeter; if the density is less than 1.25 kg/l or the no-load voltage after at least 3 - 4 hours at rest is lower than 12.5 V, proceed to recharge.
Install the battery in the machine with care, utilizing the anchorages provided.

When connecting the battery to the system:
— secure the clamp.
— before connecting the negative, make certain there is no sparking on contact with the corresponding terminal of the battery. Sparks are due to insulation defects that must be located and eliminated.
— Tighten the clamps on the terminals, smearing with petroleum jelly.

TABLE OF WIRING COLOUR CODES

BK = *BLACK*
BL = *BLUE*
BR = *BROWN*
DK BL = *DARK BLUE*
DK GN = *DARK GREEN*
GN = *GREEN*
GY = *GREY*
OR = *ORANGE*
PK = *PINK*
RD = *RED*
VL = *VIOLET*
WH = *WHITE*
YL = *YELLOW*

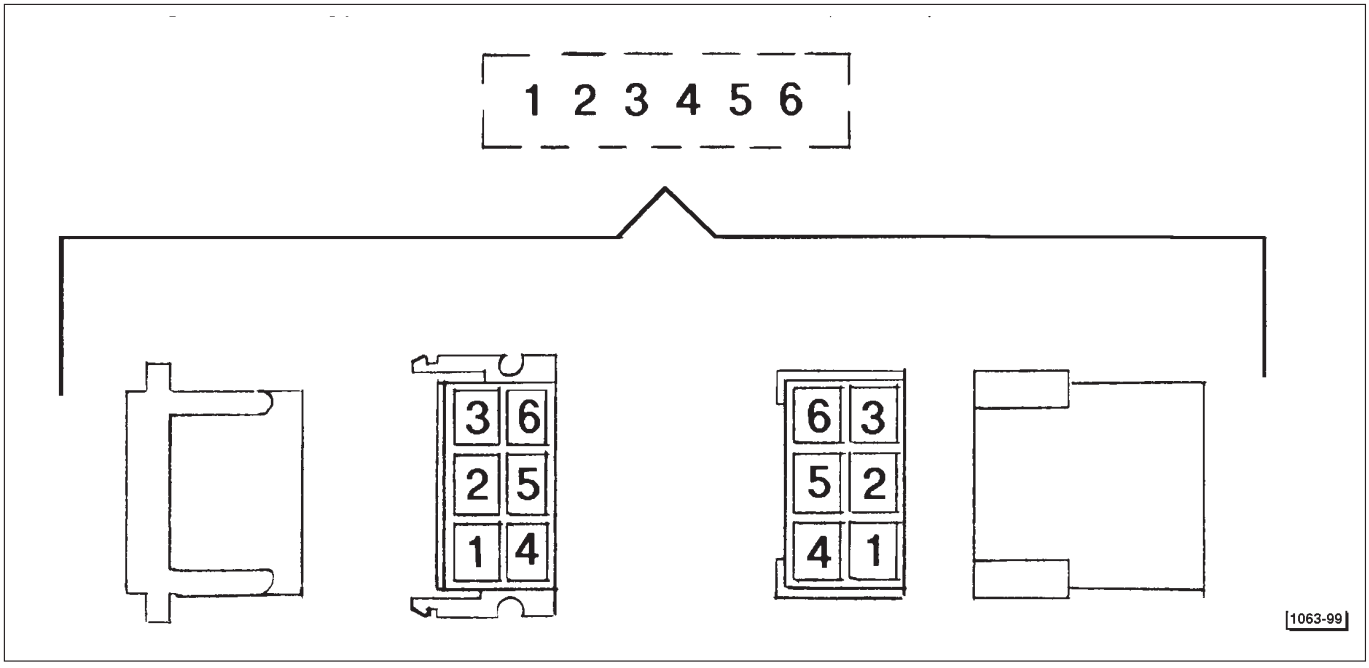
EXAMPLE:

DK BL/WH 1 680

DK BL/WH = *DARK BLUE WITH WHITE LINE*

1 = *CROSS SECTION OF WIRE (mm)*
680 = *LENGTH OF WIRE (mm)*

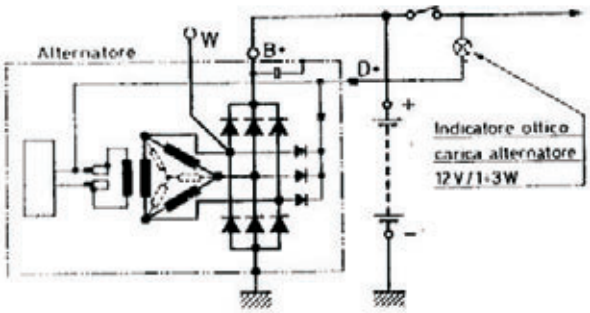
EXAMPLE OF CONNECTORS (CAN BE 4-PIN, 6-PIN OR 8-PIN)

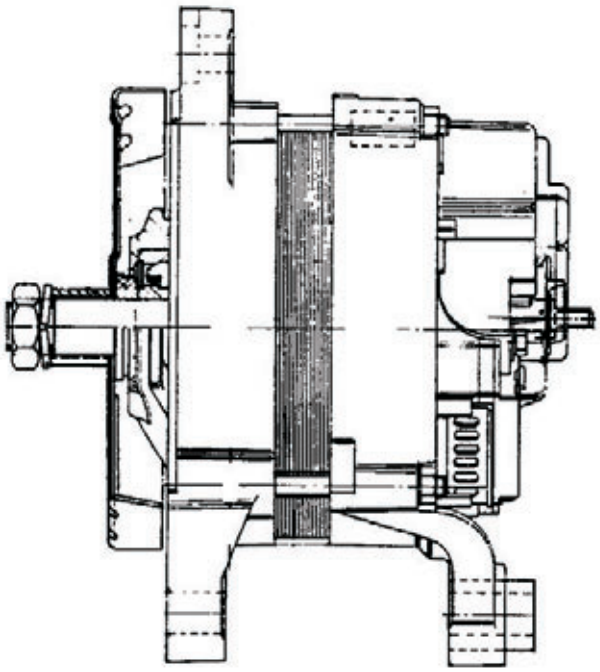


RECHARGE SYSTEM

65A ALTERNATOR

COD. 29439.400.0 Marelli
COD. 29439.410.0 Bosch
COD. 29439.420.0/10 Iskra





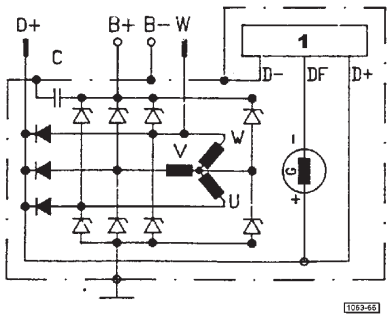
RECHARGE SYSTEM

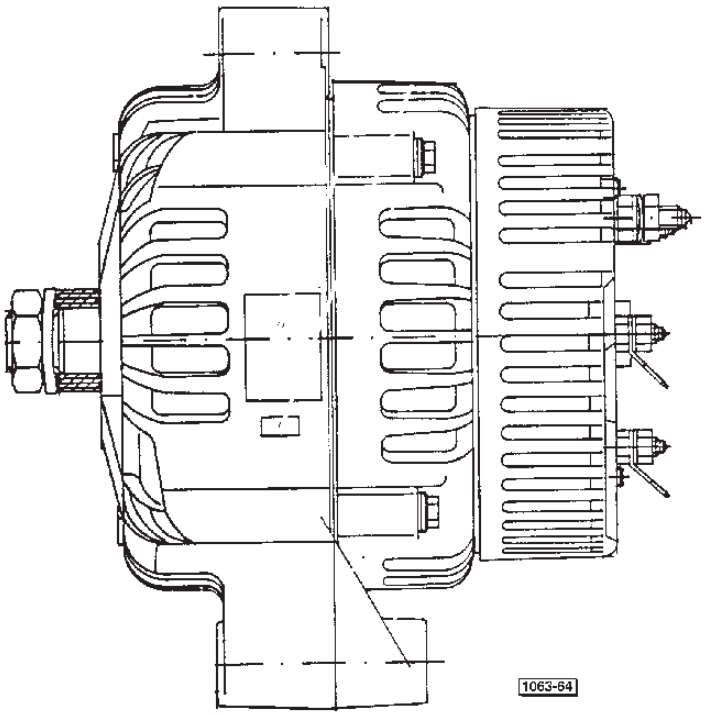
65A ALTERNATOR

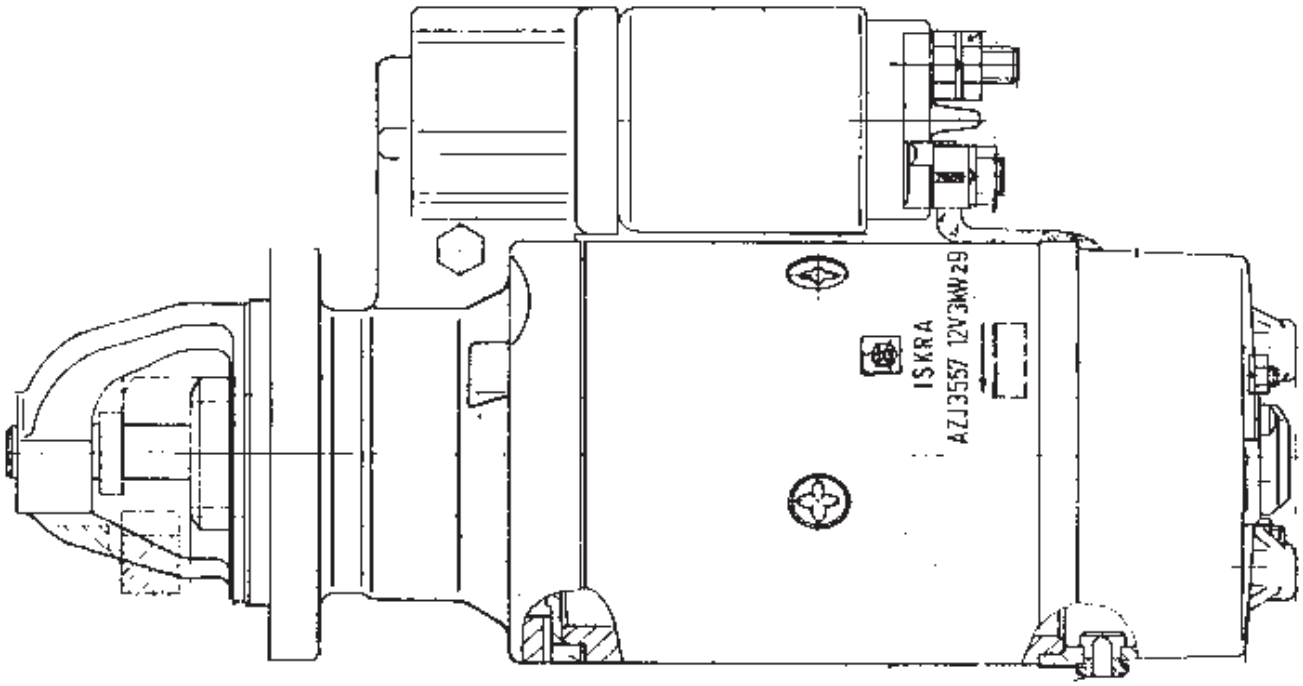
ISKRA AAK5114 14V 85A

COD. 29439.460.0/10

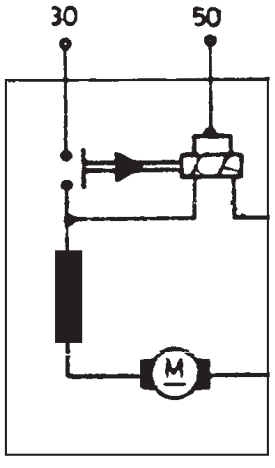
1 - Regolator







1063-72



1063-73

BOSCH

COD. 2.9619.300.0/10 per 60 - 70 CV
2.9619.3290.0/10 per 75 - 80 - 90 CV

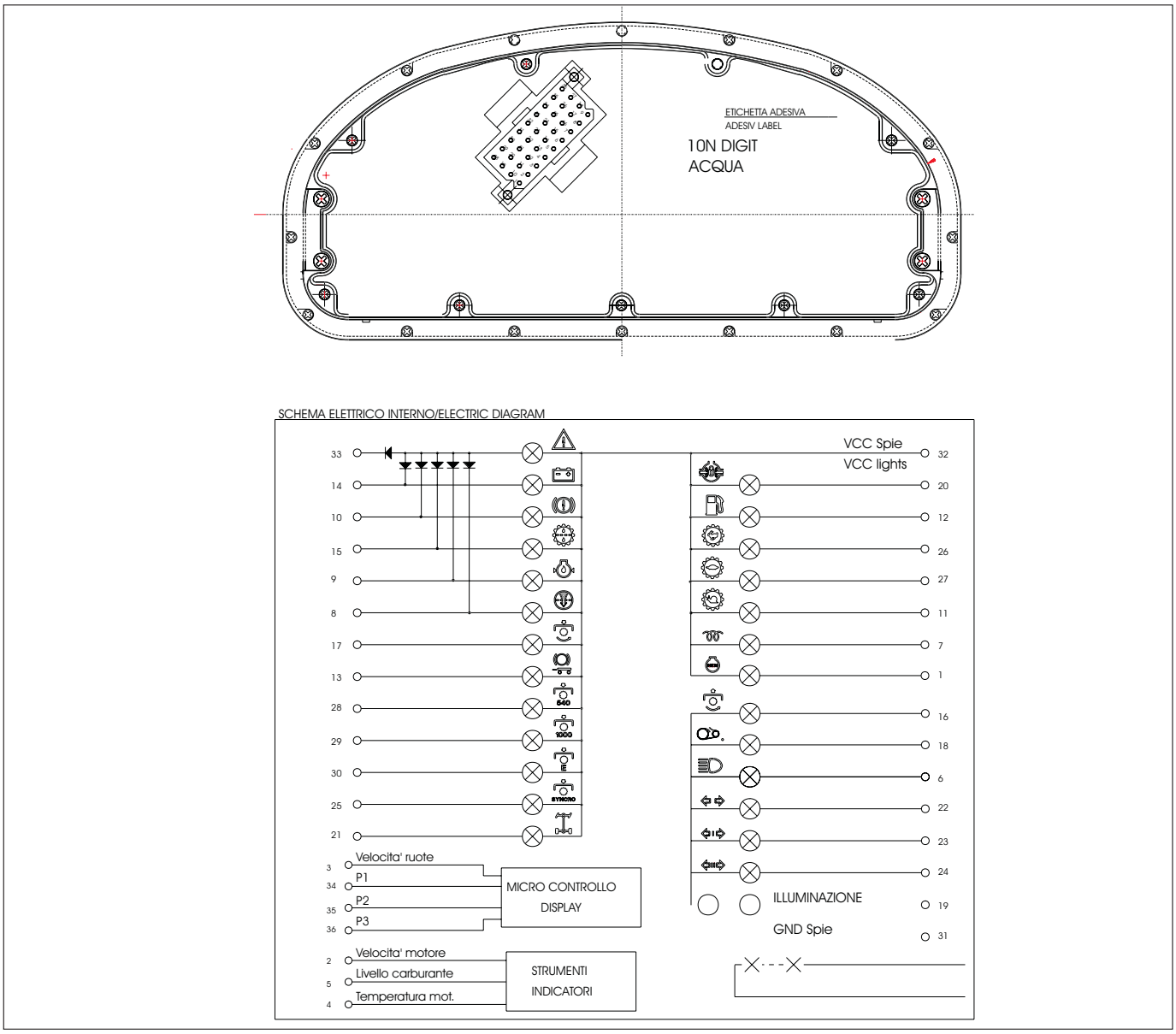


Fig. 2 - Check panel with digital display

Ignition switch

STOP

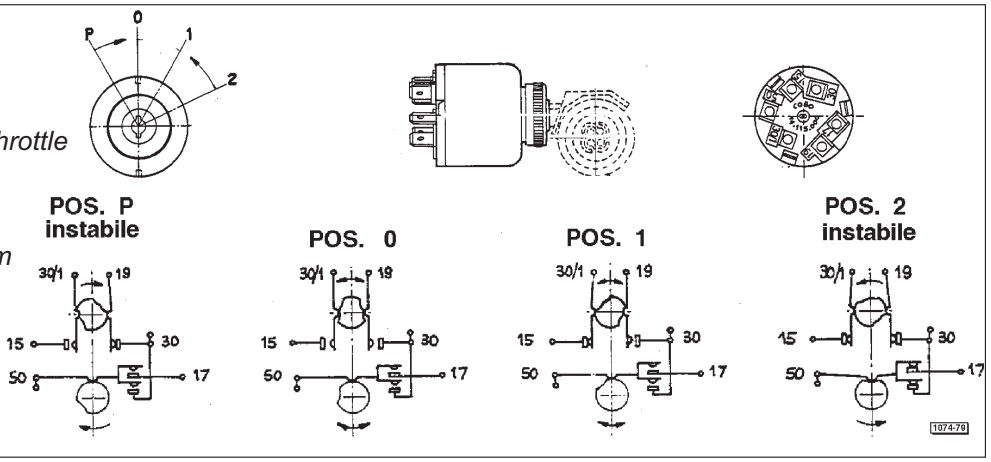
Electtrical system deenergized - Key removable
(On version with electronic RPM control, the engine shuts off automatically;
On versio with machanical throttle linkage, the engine is stopped by pulling the knob on the right of the instrument panel).

Active electric circyit: - Lamps indicating operation of the alternator and low engine oil pressure light up on the instrument panel. A lamp indicating pre-heating lights up when ambient temperature is below 10°C; wait until the lamp has gone out (approx. 55 seconds) before turning the key to the next position. If the ignition key is turned to the start position with the thermostart indicator still alight, the heater cuts out and power is supplied to the starter motor.

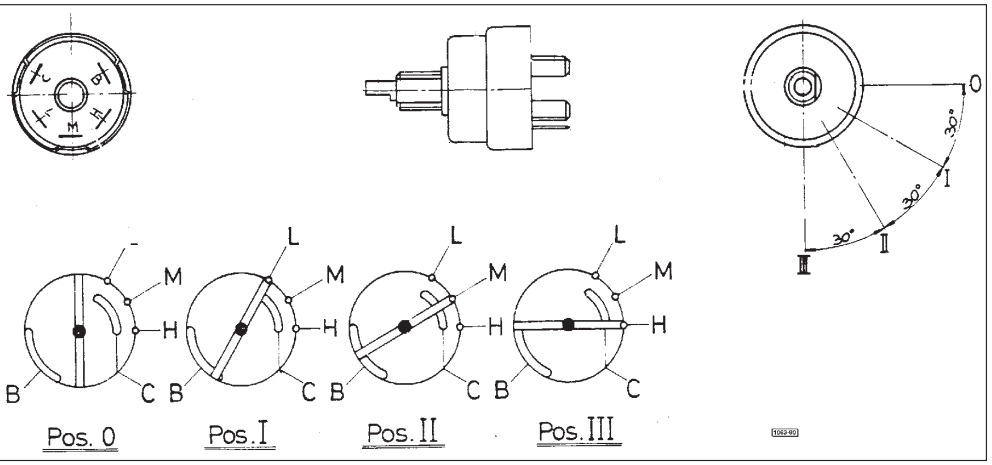
Engine starting

IGNITION KEY
machines with mechanical throttle linkage
p/n 2.7659.071.0/30

machines with electronic rpm
cod. 2.7659.127.0



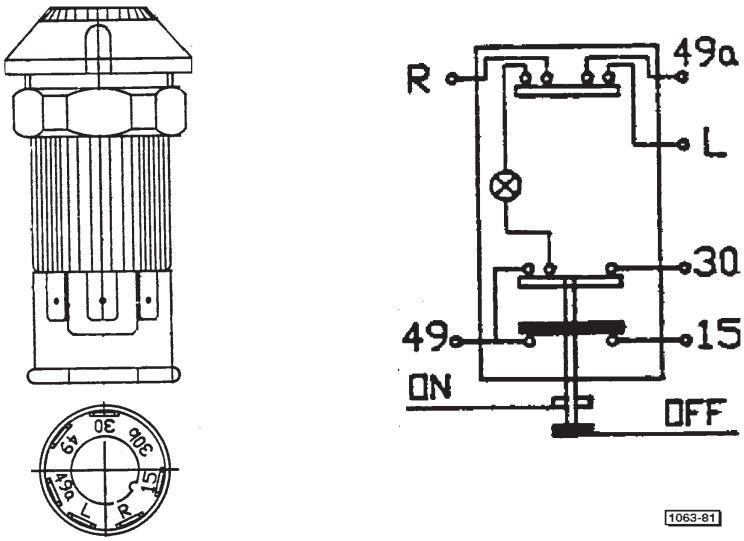
VENTILATION CONTROL
machines with cab



PUSH BUTTON CONTROL

CODE 2.7659.110.0

HAZARD WARNING LIGHTS

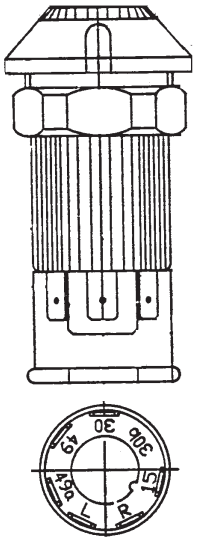


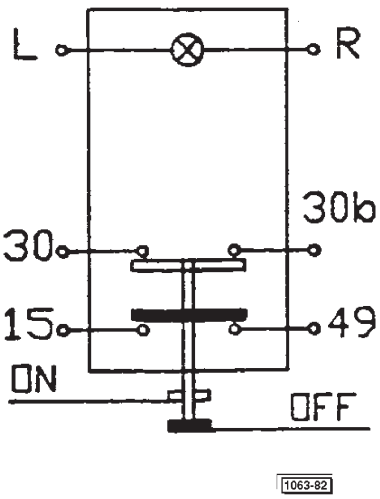
PUSH BUTTON CONTROL

P.T.O. CLUTCH
CODE 2.7659.108.0/10

DIFFERENTIAL LOCK
CODE 008.1335.0

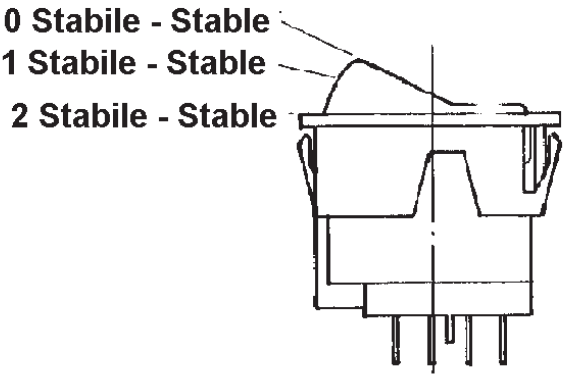
4WD COUPLER
CODE 007.5945.0/20

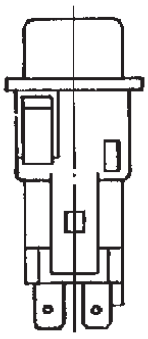


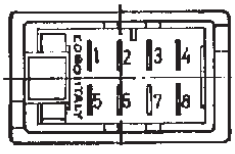


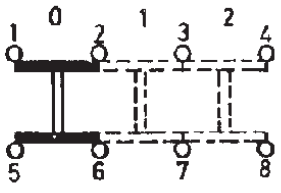
2-SPEED WINDSCREEN WIPER SWITCH

CODE 2.7659.092.0





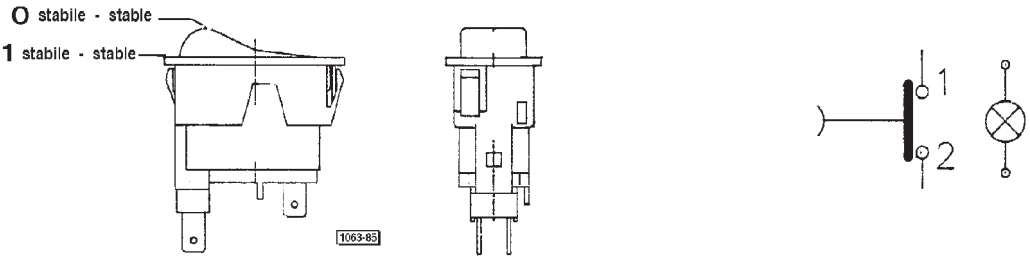




WORK LIGHTS

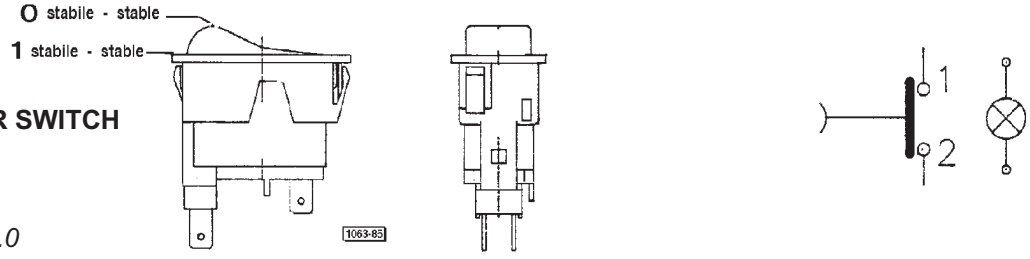
FRONT
CODE 2.7659.078.0

REAR
COD. 2.7659.079.0



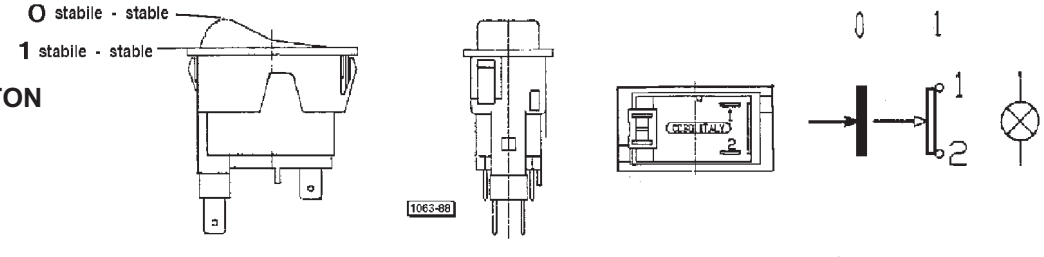
SCREEN WASHER SWITCH

FRONT
CODE 2.7659.091.0



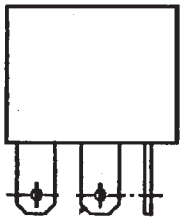
BEACON PUSH BUTTON

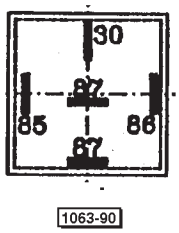
CODE 2.7659.126.0

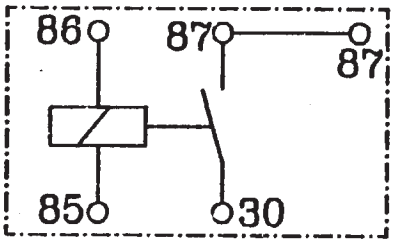


RELAY

CODE 010.2089.2



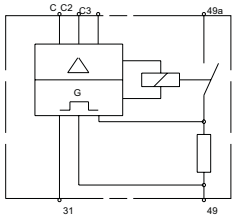




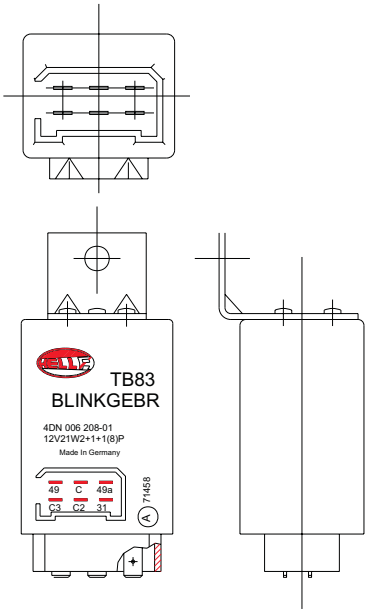
ELECTRONIC FLASHER UNIT

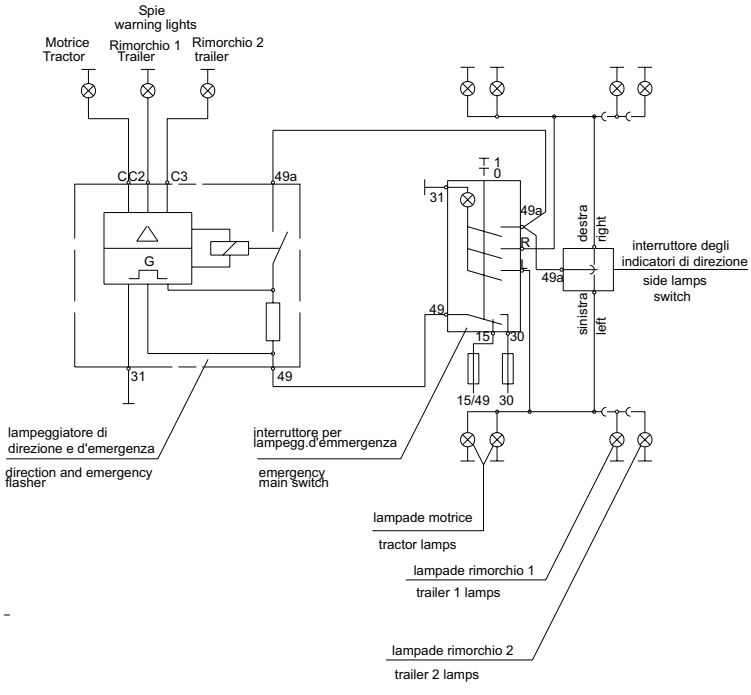
CODE 2.8639.007.0

Schema-funzionale
Functional diagram

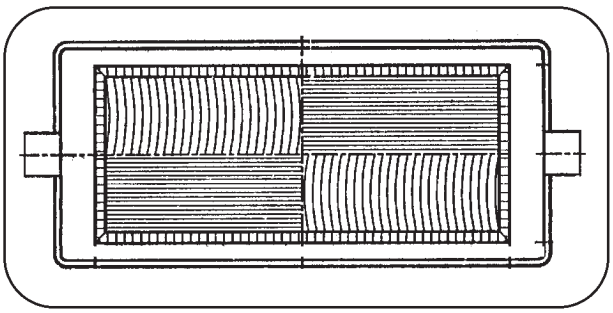


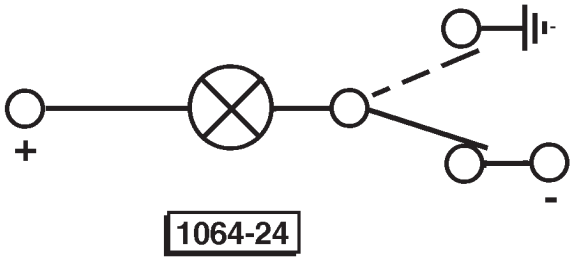
SCHEMA DI COLLEGAMENTO
WIRING DIAGRAM





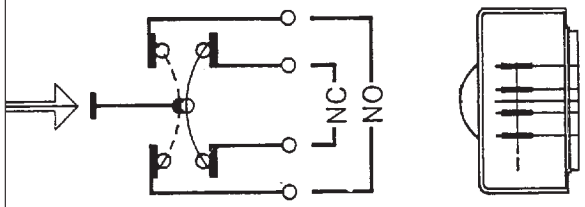
INTERIOR ROOF LIGHT *CODE 2.8339.032.0*

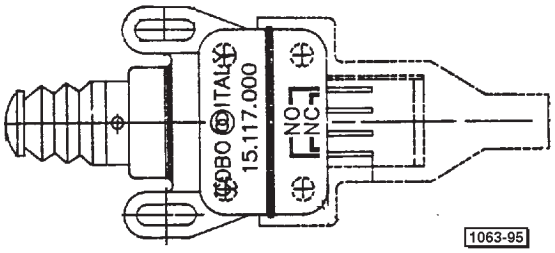




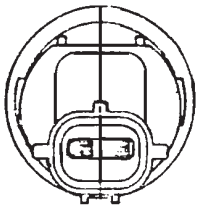
SWITCH CONTROLLING: differential lock - P.T.O. clutch - 4RM - 540 1000 rpm/min P.T.O. speed selector - Economy P.T.O. - Live P.t.o. - electric starter system.

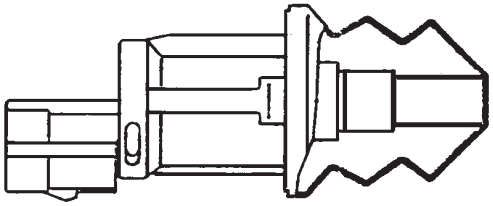
CODE 2.7659.097.0

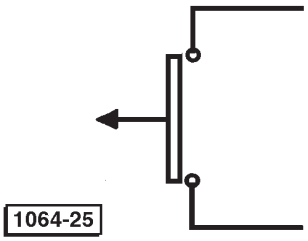




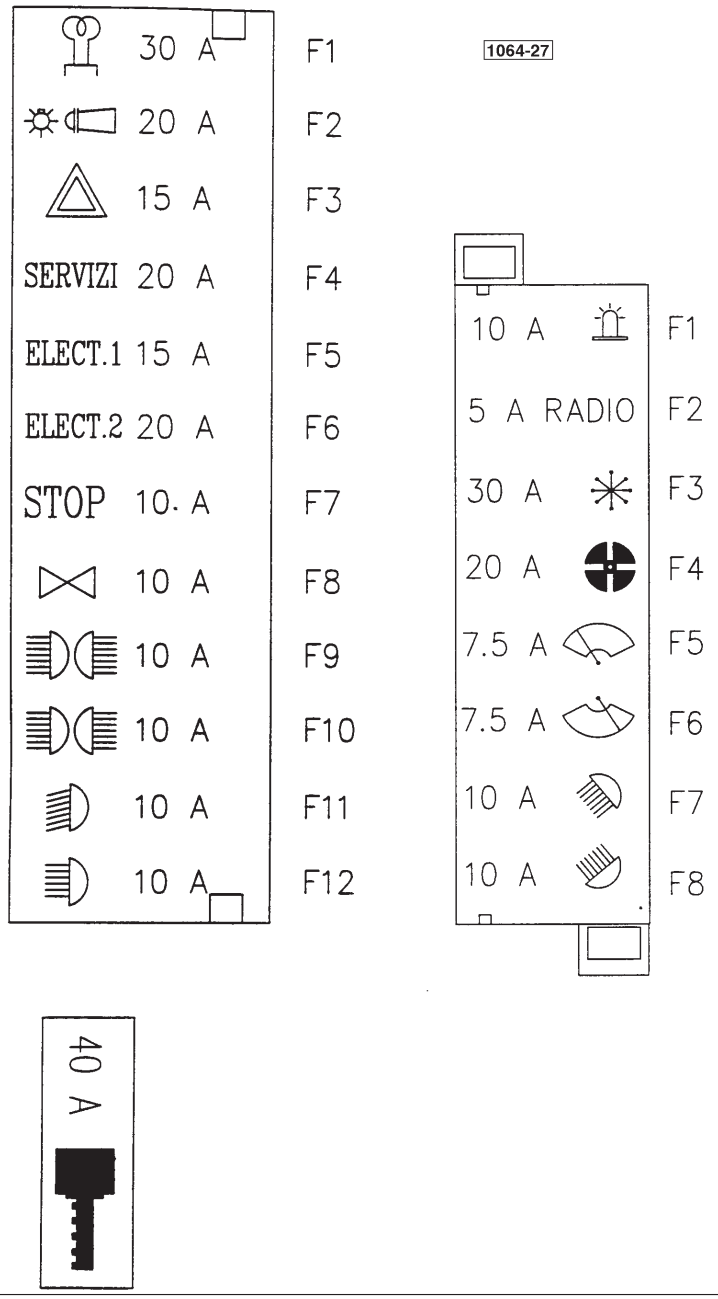
SWITCH FOR EMERGENCY BRAKE *CODE 2.7659.501.0*







FUSE BOX



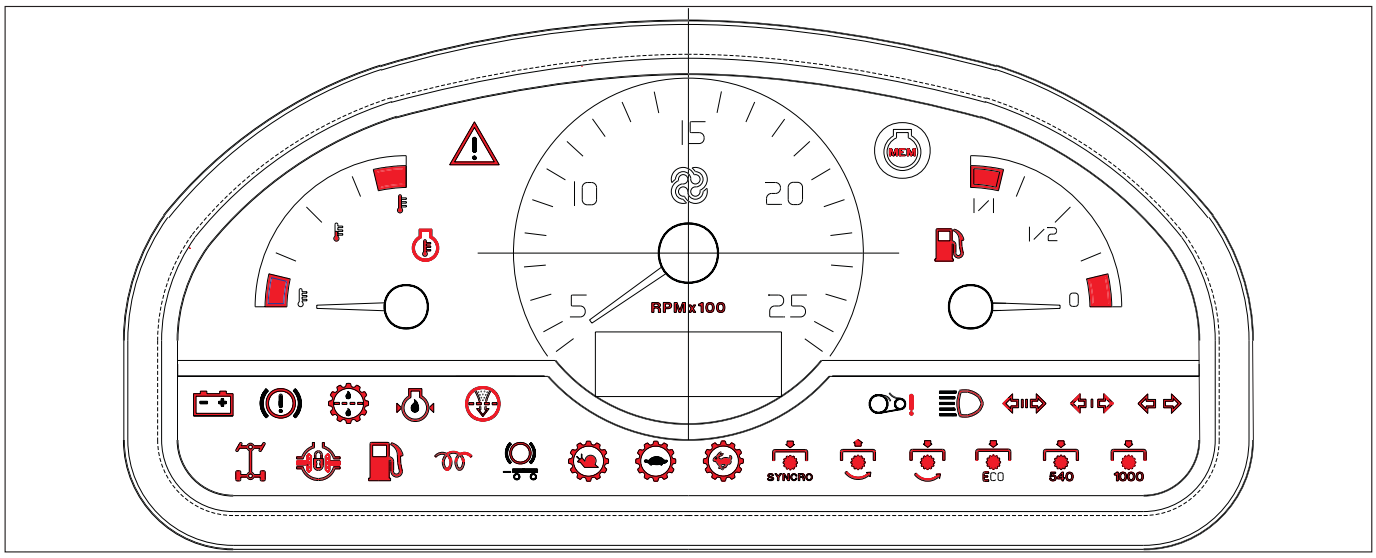
tractor fuse box

F1 - Thermostart
F2 - Side/tail lights - horn
F3 - Hazard warning lights
F4 - Work light - 12V socket
F5 - Engine RPM control unit
F6 - Thermostart/panel/HML
F7 - Engine STOP
F8 - Flashers
F9 - Side/tail light
F10 - Side/tail light
F11 - Dipped beam
F12 - Main beam

cab fuse box

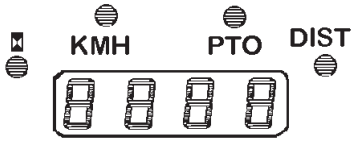
F1 - Flasher unit
F2 - Radio
F3 - Air conditioning
F4 - Blower/heater
F5 - Front wiper
F6 - Rear wiper
F7 - Front work lights
F8 - Rear work lights

Instrument panel with digital display



The basic panel is the same as described above, except that the dial is replaced by a 4-digit display showing:

— Lapsed time (hours)
— Ground speed (speedometer)
— P.T.O. speed
— Trip meter (distance in metres)



There are also 4 high brightness Leds which indicate the function currently being displayed.
Any specifications not individually listed below will be the same as for the basic panel.

Operation
The selection and content of the functions displayed are governed by pressing buttons, denoted **P1**, **P2** and **P3**.

Button **P1** - Red - Positioned to right of panel.
Button **P2** - Red - Positioned to right of panel.
Button **P3** - Red - Positioned near fuse box (cover must be removed to gain access).

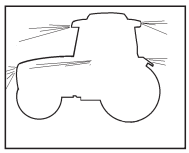
Auxiliary electrical systems on (turn the ignition key to position) the current total of working hours lapsed will be displayed.

Then, turn the ignition key to position , when the engine is started, the display sets itself to indicate ground speed (speedometer).

To display a function other than ground speed, simply press P1 repeatedly to select the function required.

1 - Lapsed work time indicated by 4 digits, to the nearest hour.

2 - Ground speed indicated by 3 digits, one of which after the decimal point (XX.X), in km/h. In the event that bigger or smaller rear tyres are fitted, the new size must be entered (see directions point 5 page 391).



IMPORTANT: If a function other than speedometer is currently selected, the panel will revert automatically to displaying ground speed if 20 km/h is exceeded.

3 - P.T.O. speed is indicated by 4 digits (XXX0), in revolutions per minute.

4 - The trip meter uses 4 digits (XXXX), indicating the distance covered by the tractor in metres. The trip meter can be reset; this is done by selecting the function then pressing and holding P2 for at least 3 seconds, during which time the figures will blink before returning to zero. If P2 is released before the 3 seconds have passed, there will be no reset.

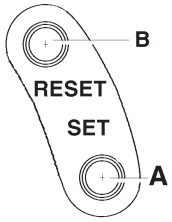
5 - **Entering the speed constant according to the tyre size**.
This operation becomes necessary only if the rear tyres are replaced with others of a different size.

With the engine at standstill and the ignition key in the off position

, press and hold button P3 while turning the key to the auxiliary position (without starting the engine), then release the button only when the tyre code appears in the display. This code will correspond to the type of rear tyre currently fitted.
Now press button P1 repeatedly to view the permissible tyre sizes, which will be displayed in sequence.
When the size displayed matches that of the new tyres, press P3 again to memorize.
If the key is returned to the off position without completing this last step, the tyre size code will not be updated.

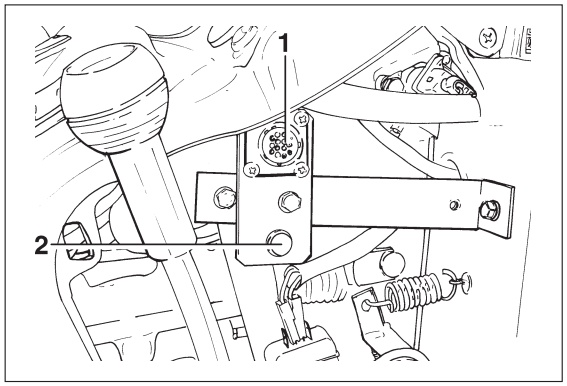
Listed below are the tyre options specified for the tractor. Each size has a relative display code which corresponds to the number of impulses per 100 metres picked up by the speedometer sensor.

Tyre	Impulses for 100 metres
380/70R20	5792
12.4R24	5768
380/70R20	5768
13.6R24	5560
380/70R24	5560
420/70/24	5280
12.4R28	5280
360/70R28	5280
13.6R28	5104
360/70R28	5104
420/70R28	4864



Button P 1, used to select the function displayed - Colour red.
The 3 available functions (speedometer, P.T.O. speed, trip meter) are displayed cyclically by pressing P1 repeatedly.
The function selected is indicated by a relative Led on the instrument panel.

Button P 2, used to reset the function selected - Colour red
Press this to select another function.



1 - Diagnostics socket (on version with electronic RPM control only)
3 - Tyre size selection button (on version with the Instrument panel with digital display)

ENGINE STOP OPERATION WITH A TYPE 2MH ENGINE CONTROL UNIT

The operation of the engine control unit in the presence of a STOP signal is very simple.

The control unit starts to monitor the STOP signal 2 seconds after the engine has been started, i.e. when it has attained idling speed.

The control unit checks for a STOP signal which remains active for at least 1 second.

On detection of a STOP signal, the control unit switches off the engine.

The engine control unit signals operation of the engine stop by 7 flashes of the HOLD led on the keypad.

The engine control unit will ignore the STOP signal if the engine is switched off. For example, if the engine stalls (under excessive load), a STOP signal may be generated but it will be ignored.

Using the ALL ROUND TESTER it is possible to check that the STOP signal is connected correctly.
One of the following messages will also appear on the monitor:

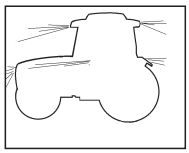
1) STOP NOT CONNECTED
2) STOP CONNECTED
3) STOP ACTIVE

On models equipped with STOP control units, the valid messages are nos. 2 and 3. Should message no. 1 “STOP NOT CONNECTED” appear, check the connection between pin 5 of the alarm control unit and pin 11 of the engine control unit.

If none of the above messages appears, this means that the engine control unit does not have the correct program.

DIAGNOSTICS

To enable the operator to identify the problem which caused the intervention of the engine stop, the control unit provides a diagnostic code by means of the oil pressure warning light, **but only when the handbrake is released**.
The diagnostic code is only given after the engine has been re-started and the control unit has given the first acoustic signal, typically 7 seconds after the alternator warning light goes out.
The diagnostic code consists of a series of flashes each with a duration of 0.3 seconds at 1 second intervals.
The oil pressure input does not provide a diagnostic code as the illumination of the warning light is forced by the pressure switch.



Diagnostic codes

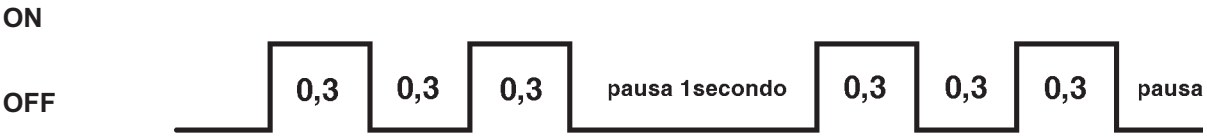
Below is a list of the diagnostic codes and their meanings:

flashes

WARNING LIGHT

2 Broken fanbelt sensor
3 Alternator signal D+
4 Auxiliary input
5 Engine STOP output

example of code 2



Description of faults

This chapter provides useful information for fault tracing.

Code 2 - Broken fanbelt sensor

This fault is signalled when the input signal from the fanbelt sensor fails to arrive at the control unit.
If this fault occurs :
- Check that the battery voltage is present between pins 1 and 2 of the sensor connector.
- Check that pin 3 of the sensor connector is connected to pin 6 of the alarm control unit.

Code 3 - Alternator signal D+

This fault is signalled when the alternator signal D+ is no longer present after the alarm control unit has entered into operation.
If this fault occurs:
- Check the alternator warning light: if on, the problem concerns the alternator itself; if off, the wire carrying the signal to the control unit may have come disconnected.

Code 4 - Auxiliary input

This fault is signalled when the voltage on pin 8 of the control unit falls below 12V.
This input is not currently utilised, consequently if this fault is signalled, check that pin 8 of the control unit is at +12V. If necessary, remove the contact from the connector and test again.

Code 5 - Engine STOP output

This fault is signalled when the control unit does not detect any connection to its STOP output.
If this fault occurs:
- Check that pin 5 of the alarm control unit is connected to pin 11of the engine control unit.

Broken belt alarm system and diagnostic codes. (If the light stays on continuously this indicates low engine oil pressure).

ELECTRICAL WIRING

PLATFORM WIRING DIAGRAMS

DIAGRAM 1 - functional diagram of tractor electrical system	**010.4368.4/30 dwg. 6**
DIAGRAM 2 - functional diagram of tractor electrical system	**010.4368.4/30 dwg. 5**
DIAGRAM 3 - central wiring for tractors with electronic governor - 1st version	**010.4368.4/30 dwg. 4**
DIAGRAM 4 - central wiring for tractors with electronic governor - 1st version	**010.4368.4/30 dwg. 3**
DIAGRAM 5 - central wiring for tractors with electronic governor - 1st version	**010.4368.4/30 dwg. 2**
DIAGRAM 6 - central wiring for tractors with electronic governor - 1st version	**010.4368.4/30 dwg. 1**
DIAGRAM 7 - central wiring for tractors with electronic governor - 2nd version	**011.4347.4/40 dwg. 6**
DIAGRAM 8 - central wiring for tractors with electronic governor - 2nd version	**011.4347.4/40 dwg. 5**
DIAGRAM 9 - central wiring for tractors with electronic governor - 2nd version	**011.4347.4/40 dwg. 4**
DIAGRAM 10 - central wiring for tractors with electronic governor - 2nd version	**011.4347.4/40 dwg. 3**
DIAGRAM 11 - central wiring for tractors with electronic governor - 2nd version	**011.4347.4/40 dwg. 2**
DIAGRAM 12 - central wiring for tractors with electronic governor - 2nd version	**011.4347.4/40 dwg. 1**
DIAGRAM 13 - central wiring for tractors with mechanical governor	**010.8683.4/80 dwg. 1bis**
DIAGRAM 14 - rear lights wiring (codes: 1st version with roll bar 010.9058.3, with cab 010.9057.3)	**010.6410.3/10**
DIAGRAM 15 - front lights wiring	**010.5582.3**
DIAGRAM 16 - HML control lever wiring	**010.5582.3**
DIAGRAM 17 - HML wiring (1st version)	**010.8388.3 (dwg. 1)**
DIAGRAM 18 - HML wiring (1st version)	**010.8388.3 (dwg. 2)**

TRACTOR WIRING

DIAGRAM 19 - complete front wiring	**010.6411.4/40 dwg. 1**
DIAGRAM 20 - complete front wiring	**010.6411.4/40 dwg. 2**
DIAGRAM 21 - battery wiring	**010.6708.3**
DIAGRAM 22 - power lift wiringe	**011.2025.4/20 dwg. 1**
DIAGRAM 23 - power lift wiring	**011.2025.4/20 dwg. 2**
DIAGRAM 24 - stabilisers and rh lift rod adjustment wiring	**012.0101.4 dwg. 1**
DIAGRAM 25 - stabilisers and rh lift rod adjustment wiring	**012.0101.4 dwg. 2**
DIAGRAM 26 - wiring for front control valves with electro-hydraulic control	**012.0102.4 dwg. 1**
DIAGRAM 27 - wiring for front control valves with electro-hydraulic control	**012.0102.4 dwg. 2**
DIAGRAM 28 - cab roof wiring	**010.7262.4/20 dwg. 1**
DIAGRAM 29 - cab roof wiring	**010.7262.4/20 dwg. 2**

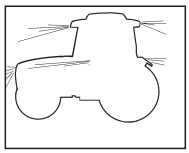
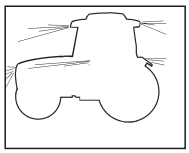


DIAGRAM 30 - cab roof wiring	**010.7262.4/20 dwg. 3**
DIAGRAM 31 - work lights and screen wiper wiring	**010.7263.4/20 dwg. 1**
DIAGRAM 32 - work lights and screen wiper wiring	**010.7263.4/20 dwg. 2**
DIAGRAM 33 - windscreen wiper wiring	**010.7264.4/10**
DIAGRAM 34 - electric heating wiring	**010.7265.4**
DIAGRAM 36 - cab air conditioning wiring	**011.0644.3/20 dwg. 1**
DIAGRAM 37 - cab air conditioning wiring	**011.0644.3/20 dwg. 2**

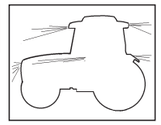
COMPONENTS

DIAGRAM 38 - steering column switch unit	**010.1173.4**
DIAGRAM 39 - clutch angle sensor	**2.7099.740.0**
DIAGRAM 40 - hand throttle with electronic governor (1st version)	**009.7620.4**
DIAGRAM 41 - engine control unit with belts alarm, HML and preheating - engine control unit with belts alarm, HML and preheating (010.7260.4 for 30 km/h 1st version) - engine control unit with belts alarm, HML and preheating (010.7261.4 for 40 km/h 1st version)	**011.4353.4/10**
DIAGRAM 42 - HML control unit	**010.8869.4/20**
DIAGRAM 43 - P.T.O. switch	**2.7659.194.0**
DIAGRAM 44 - rocker switch for stabiliser adjustment	**2.7659.218.0**
DIAGRAM 45 - switch for rh lift rod adjustment	**2.7659.219.0**
DIAGRAM 46 - switch for hydraulic ways 3 - 3	**2.7659.220.0**
DIAGRAM 47 - switch for hydraulic ways 4 - 4	**2.7659.221.0**
DIAGRAM 48 - lift control switch	**011.2991.0**



8 Systems

84 Electrical system

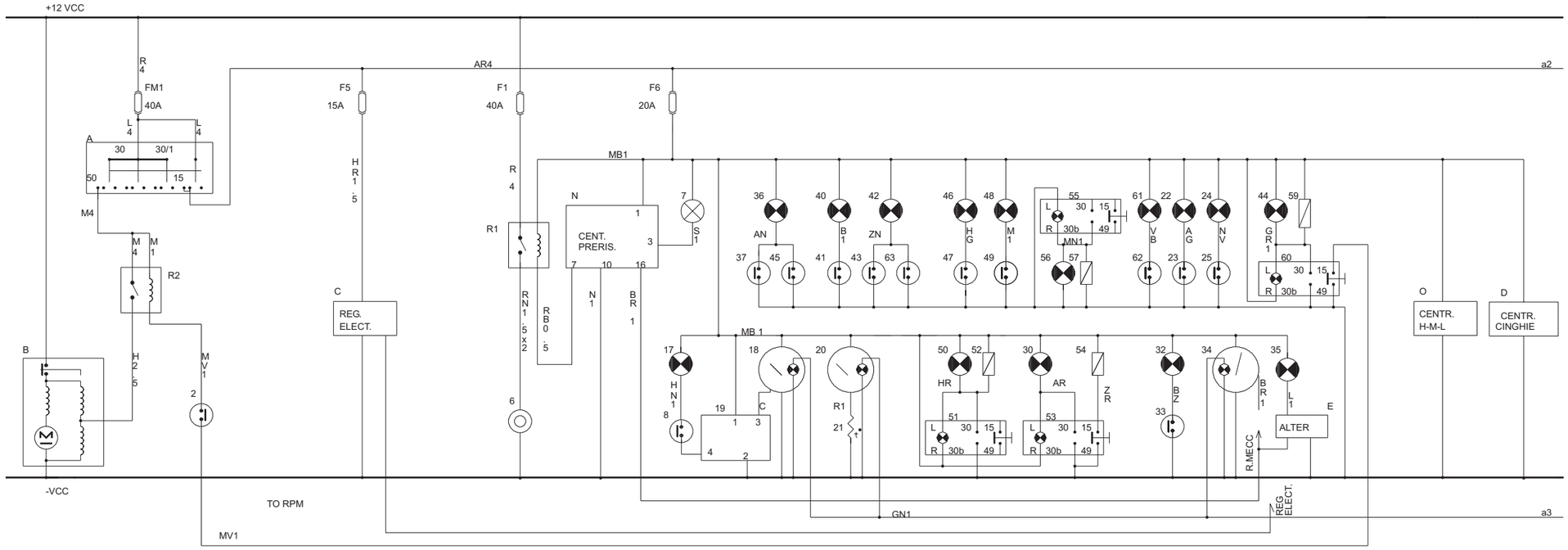


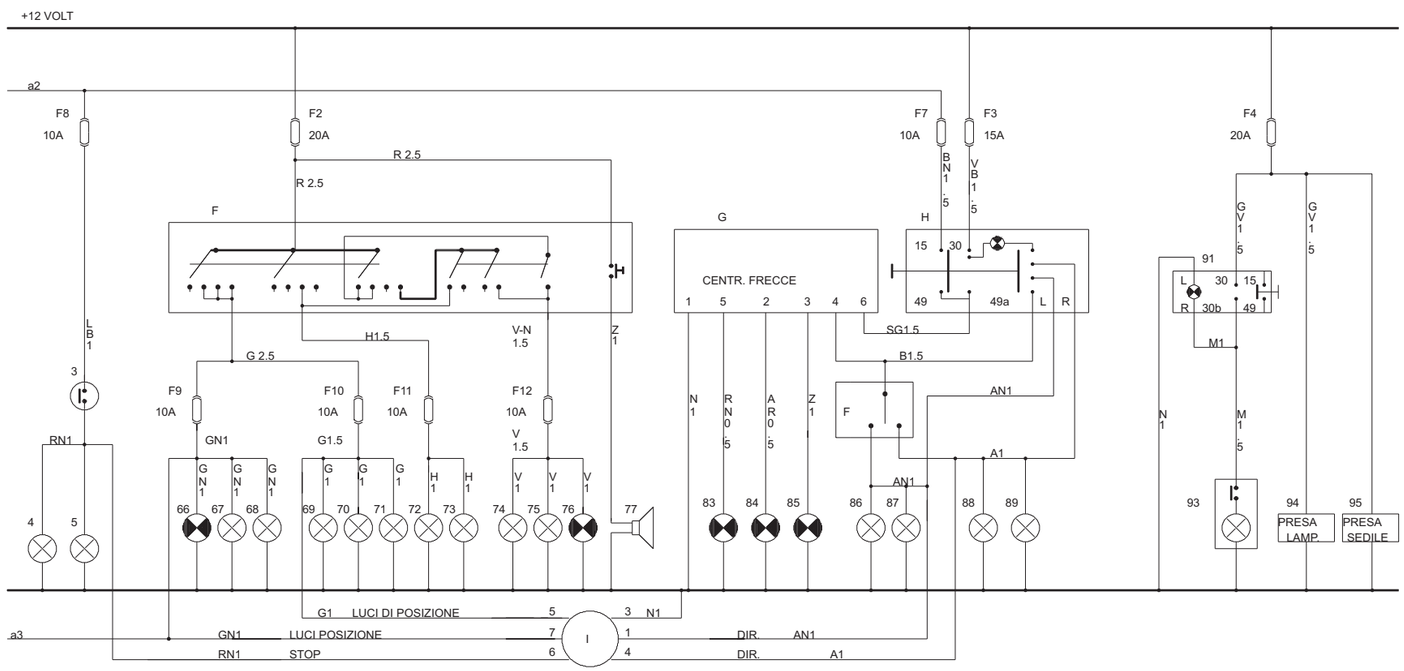
KEY TO FUNCTIONAL DIAGRAM OF TRACTOR ELECTRICAL SYSTEM

A STARTER SWITCH
B STARTER MOTOR
C ELECTRONIC GOVERNOR CONTROL UNIT
D BELTS ALARM CONTROL UNIT
E 65A ALTERNATOR WITH SOCKET W
F HORN AND DIRECTION INDICATORS SWITCH
G DIRECTION INDICATORS CONTROL UNIT
I TRAILER SOCKET
L LIFT SOCKET
S LINK ARMS VERSION
R1 PREHEATING RELAY
R2 RELAY FOR ENABLING STARTING
R3 RELAY CONTROLLED BY IGNITION SWITCH

2 2 CLUTCH STARTING ENABLING SENSOR
3 BRAKES SENSOR
4-5 BRAKE LIGHTS
7 PREHEATING INDICATOR LIGHT
8 LOW FUEL WARNING LIGHT
9 FUEL LEVEL INDICATOR
10 SENSOR FOR FUEL LEVEL INDICATOR
11 OIL TEMPERATURE INDICATOR
12 SENSOR FOR OIL/WATER TEMPERATURE INDICATOR
13 540 RPM REAR P.T.O. INDICATOR LIGHT
14 540 RPM REAR P.T.O. ENGAGEMENT SENSOR
15 1000 RPM REAR P.T.O. INDICATOR LIGHT
16 1000 RPM REAR P.T.O. ENGAGEMENT SENSOR
17 4WD INDICATOR LIGHT
32 TRAILER BRAKING INDICATOR LIGHT
33 TRAILER BRAKING SENSOR
34 MULTIMETER
35 ALTERNATOR INDICATOR LIGHT
36 HYDRAULIC OIL FILTER CLOGGING INDICATOR LIGHT
37 HYDRAULIC OIL FILTER CLOGGING SENSOR
38 AIR CLEANER CLOGGING INDICATOR LIGHT
39 AIR CLEANER CLOGGING SENSOR
40 BRAKE FLUID LEVEL INDICATOR LIGHT
41 BRAKE FLUID LEVEL SENSOR
42 REAR P.T.O. ROTATION INDICATOR LIGHT
43 SERVICES SENSOR (LOW OIL PRESSURE SOLENOID VALVES)
44 GROUNDSPEED P.T.O. INDICATOR LIGHT
45 GROUNDSPEED P.T.O. SENSOR
46 LOW ENGINE OIL PRESSURE INDICATOR LIGHT
47 LOW ENGINE OIL PRESSURE SENSOR
48 DIFFERENTIAL LOCK INDICATOR LIGHT
49 DIFFERENTIAL LOCK CONTROL SWITCH
50 DIFFERENTIAL LOCK SOLENOID
51 4WD ENGAGEMENT SWITCH
52 4WD ENGAGEMENT SOLENOID
53 FRONT P.T.O. CONTROL SWITCH
54 FRONT P.T.O. ENGAGEMENT INDICATOR LIGHT
55 FRONT P.T.O. SOLENOID
59 REAR P.T.O. SOLENOID
60 REAR P.T.O. CONTROL SWITCH/START ENABLE
61 ECONOMY P.T.O. INDICATOR LIGHT
62 ECONOMY P.T.O. SENSOR
63 HANDBRAKE SENSOR
64 SIDE LIGHTS ON INDICATOR LIGHT
65 FRONT LEFT SIDE LIGHT
66 REAR RIGHT SIDE LIGHT
67 FRONT RIGHT SIDE LIGHT
68 REAR LEFT SIDE LIGHT
69 NUMBER PLATE LIGHT
70 RIGHT LOW BEAM HEADLAMP
71 LEFT LOW BEAM HEADLAMP
72 LEFT HIGH BEAM HEADLAMP
73 RIGHT HIGH BEAM HEADLAMP
74 HIGH BEAM INDICATOR LIGHT
75 HORN
83 DIRECTION INDICATORS WARNING LIGHT
84 DIRECTION INDICATORS WARNING LIGHT - 1ST TRAILER
85 DIRECTION INDICATORS WARNING LIGHT - 2ND TRAILER
86 FRONT LEFT DIRECTION INDICATOR
87 REAR LEFT DIRECTION INDICATOR
88 FRONT RIGHT DIRECTION INDICATOR
89 REAR RIGHT DIRECTION INDICATOR
90 WORK LIGHT- "SWITZERLAND" SPEC.
93 REAR WORK LIGHTS
94 FLASHER SOCKET

PLATFORM WIRING DIAGRAM 1 - FUNCTIONAL DIAGRAM OF TRACTOR ELECTRICAL SYSTEM **010.4368.4/30 dwg. 6**





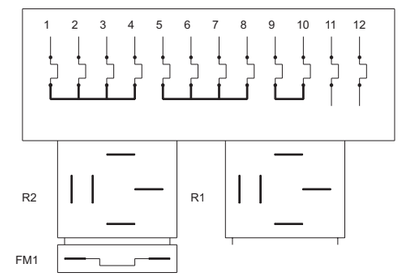


TABELLA COLORI / COLOURS TABLE			
M	Marrone/Brown	C	Arancio/Orange
V	Verde/Green	A	Azzurro/Blue
Z	Viola/Violet	B	Bianco/White
N	Nero/Black	L	Blu/Dark Blue
S	Rosa/Pink	G	Giallo/Yellow
R	Rosso/Red	H	Grigio/Gray

PLATFORM WIRING — DIAGRAM 2 - FUNCTIONAL DIAGRAM OF TRACTOR ELECTRICAL SYSTEM — **010.4368.4/30 dwg. 5**

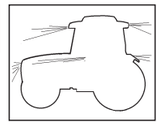
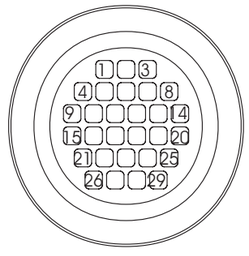


TABELLA COLORI / COLOURS TABLE			
Marrone/Brown	C	Arancio/Orange
Verde/Green	A	Azzurro/Blue
Viola/Violet	B	Bianco/White
Nero/Black	L	Blu/Dark Blue
Rosa/Pink	G	Giallo/Yellow
Rosso/Red	H	Grigio/Gray

AL CENTRALE
TO CENTRAL

A

1	PTO SYNCRO	SYNCRO PTO
2	PTO 540	PTO 540
3	PTO 1000	PTO 1000
4	PTO ECONOMICA	ECONOMIC PTO
5	F.OLIO/SERVICE	OIL FILTER/SERVICE
6	PRESS.RIMORCHIO	TRAILER PRESS.
7	FRENO MANO	HAND BRAKE
8	VELOCITA'RUOTE	SPEED
9	VEL.RUOTE GND	SPEED GND
10		
11	SPIA DT	LIGHT DT
12	EV PTO	EV PTO
13	EV DF	EV DF
14	EV DT	EV DT
15	COM DT/DF/PTO	COM DT/DF/PTO
16		
17	EV H	EV H
18	EV L	EV L
20	GALL RISERVA	EMPTY SENSOR
21	SPIA RISERVA	EMPTY LIGHT
22	GND	GND



G 1
AG 1
VN 1
VB 1
CN 1
AN 1
H 1
MV 1
GV 1
AR05
GR 1
HR 1
RV 1
MB 1
ZB 1
ZV 1
HN 1
HM 1
N 1

X2 GV 1

X 1

X 4

EV H
H EV
H
ZB 1 MB 1

EV L
L EV
L
ZV 1 MB 1

EV PTO
PTO EV
E5
GR 1 MB 1

EV DIFF
DIFF EV
E4
HR 1 MB 1

EV DT
EV DT
E3
RV 1 MB 1

SERVICE
SERVICE
V7
CN 1

FILTRO OLIO
OIL FILTER
V8
CN 1 N 1

V10
FRENO A MANO
HANDE BRAKE
H 1 N 1

P4
V.RUOTE
SPEED
GV 1 MV 1

P2
PTO SYNCRO
SYNCRO PTO
N 1 G 1

P3
DOPPIA TRAZIONE
4WDT
N 1 AR05

V13
II RISERVA
II EMPTY
N 1 HN 1 HM 1

V14
HM 1

V12
SYNCRO
SYNCRO
G1

V5
PTO 540
540 PTO
VN 1

V2
PTO ECONOMICA
ECONOMY PTO
VB 1

V6
PTO 1000
1000 PTO
AG 1

V11
PRESS.RIMORCHIO
TRAILER PRESSURE
AN 1 N 1

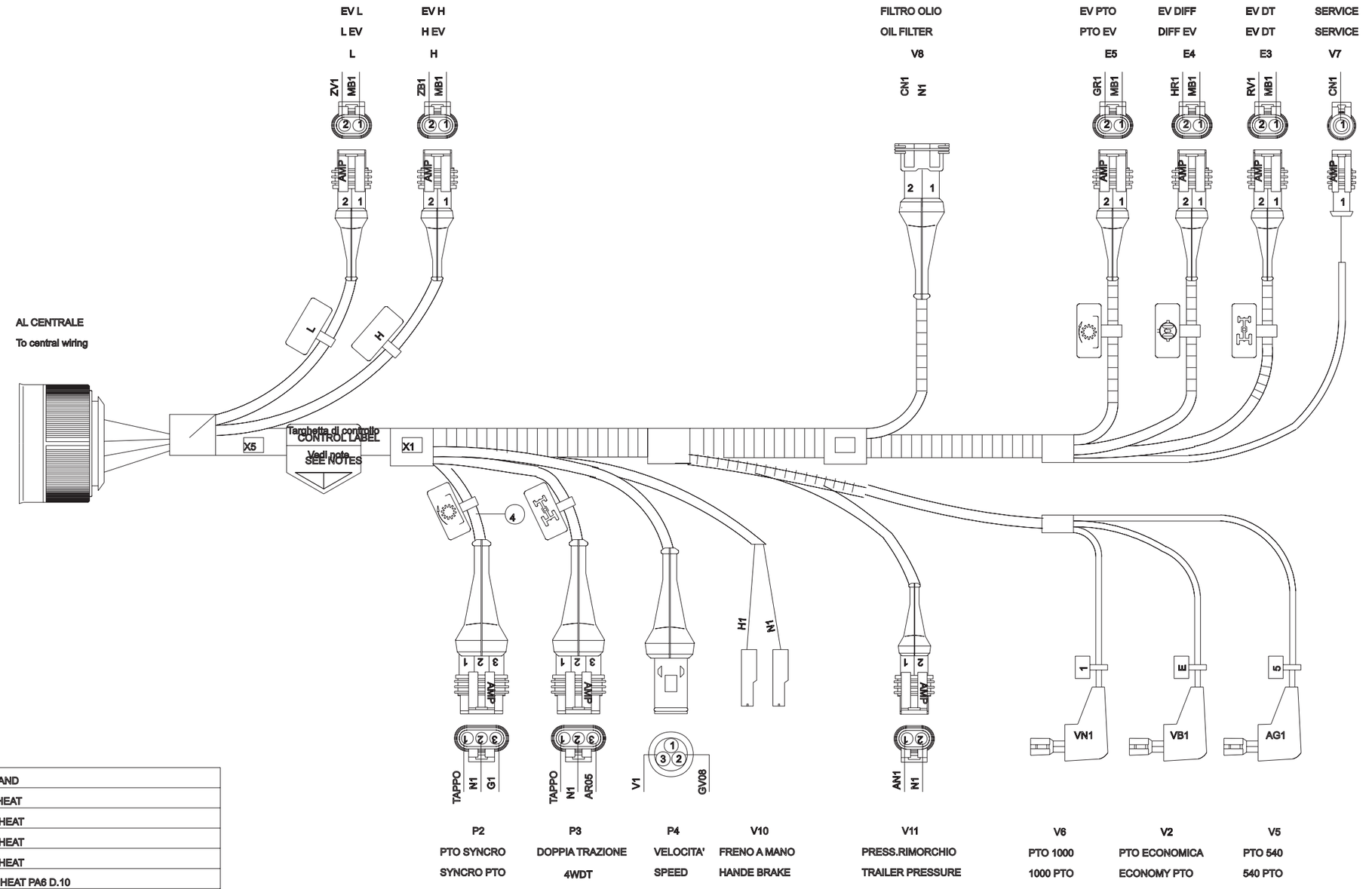
PLATFORM WIRING

DIAGRAM 3 - CENTRAL WIRING FOR TRACTORS WITH ELECTRONIC GOVERNOR - 1ST VERSION

010.4368.4/30 dwg. 4

TABELLA COLORI / COLOURS TABLE			
M	Marrone/Brown	C	Arancio/Orange
V	Verde/Green	A	Azzurro/Blue
Z	Viola/Violet	B	Bianco/White
N	Nero/Black	L	Blu/Dark Blue
S	Rosa/Pink	G	Giallo/Yellow
R	Rosso/Red	H	Grigio/Gray

1	PTO SYNCRO	SYNCRO PTO
2	PTO 540	PTO 540
3	PTO 1000	PTO 1000
4	PTO ECONOMICA	ECONOMIC PTO
5	F.OLIO/SERVICE	OIL FILTER/SERVICE
6	PRESS.RIMORCHIO	TRAILER PRESS.
7	FRENO MANO	HAND BRAKE
8	VELOCITA' RUOTE	SPEED
9	MASSA VELOCITA'	GND SPEED
10		
11	SPIA DT	LIGHT DT
12	EV PTO	EV PTO
13	EV DF	EV DF
14	EV DT	EV DT
15	COM DT/DF/PTO	COM DT/DF/PTO
16		
17		EV H
18		EV LO
19		
20		
22	GND	GND



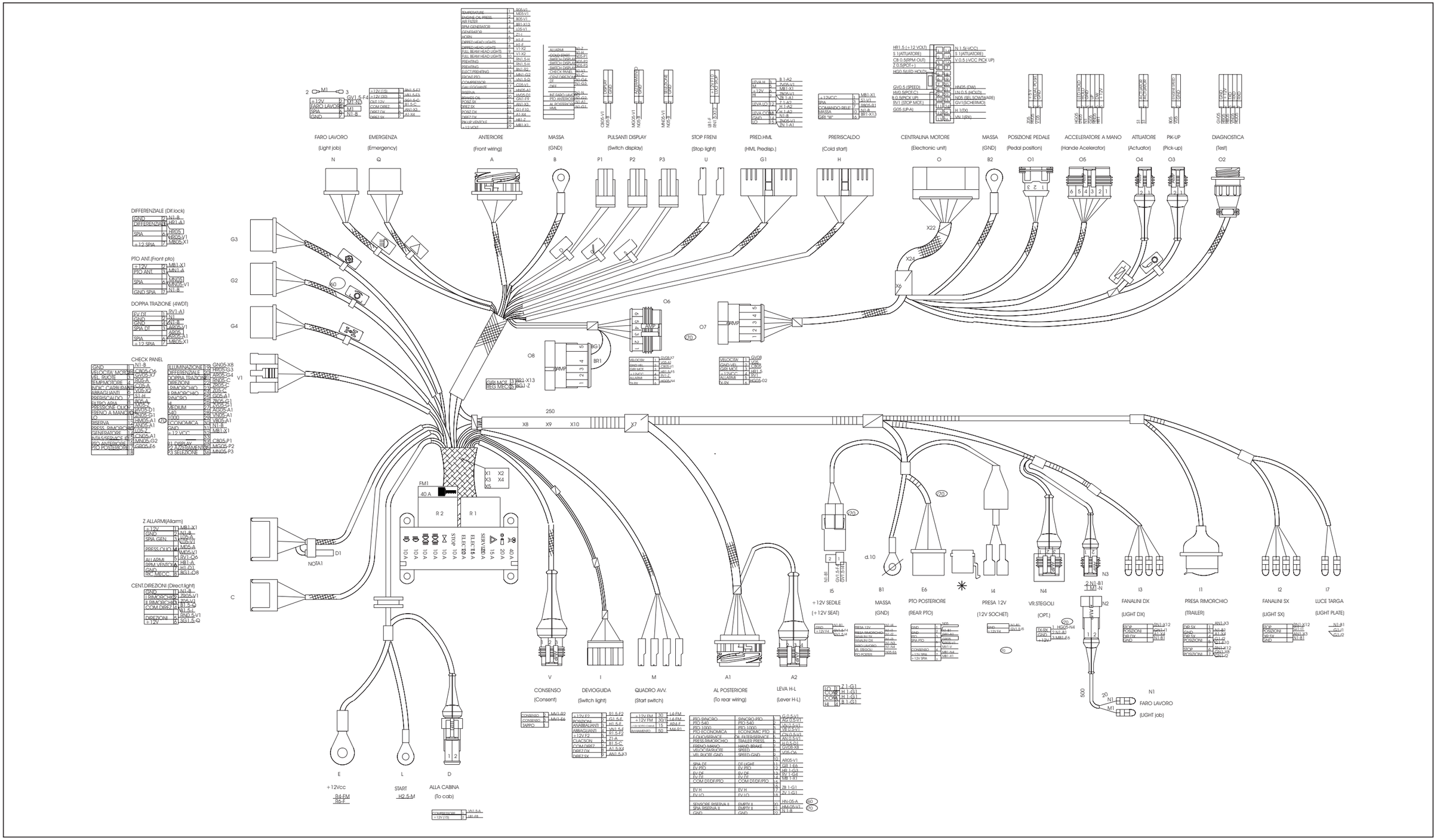
1	NASTRO IN PVC NERO	PVC BLCK BAND
2	GUAINA TERMORESTRINGENTE	SCRINKIN SHEAT
3	GUAINA IN PVC 105 C D.12	PVC 105 C SHEAT
4	GUAINA IN PVC 105 C D.5	PVC 105 C SHEAT
5	GUAINA IN PVC 105 C D.4	PVC 105 C SHEAT
6	TUBO CURRUGATO PA6 D.10	WRINKLED SHEAT PA6 D.10
7	TUBO CURRUGATO PA6 D.8.5	WRINKLED SHEAT PA6 D.8.5
8	TUBO CURRUGATO PA6 D.7.5	WRINKLED SHEAT PA6 D.7.5
9	FASCETTE SERIGRAFATE	FASCETTE SERIGRAFATE
10	FORNIRE 20 FASCETTE SCIOLTE	FORNIRE 20 FASCETTE SCIOLTE
11	TUBO CURRUGATO PA6 D.4.5	WRINKLED SHEAT PA6 D.4.5

PLATFORM WIRING — DIAGRAM 4 - CENTRAL WIRING FOR TRACTORS WITH ELECTRONIC GOVERNOR - 1ST VERSION — **010.4368.4/30 dwg. 3**

PLATFORM WIRING

DIAGRAM 5 - CENTRAL WIRING FOR TRACTORS WITH ELECTRONIC GOVERNOR - 1ST VERSION

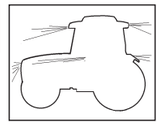
010.4368.4/30 dwg. 2



PLATFORM WIRING

DIAGRAM 6 - CENTRAL WIRING FOR TRACTORS WITH ELECTRONIC GOVERNOR - 1ST VERSION

010.4368.4/30 dwg. 1



KEY TO FUNCTIONAL DIAGRAM OF TRACTOR ELECTRICAL SYSTEM

A STARTER SWITCH
B STARTER MOTOR
C ELECTRONIC GOVERNOR CONTROL UNIT
D BELTS ALARM CONTROL UNIT
E 65A ALTERNATOR WITH SOCKET W
F HORN AND DIRECTION INDICATORS SWITCH
G DIRECTION INDICATORS CONTROL UNIT
H HAZARD WARNING LIGHTS SWITCH
I TRAILER SOCKET
L LIFT SOCKET
S LINK ARMS VERSION
R1 PREHEATING RELAY
R2 RELAY FOR ENABLING STARTING
R3 RELAY CONTROLLED BY IGNITION SWITCH

2 CLUTCH STARTING ENABLING SENSOR
3 BRAKES SENSOR
4-5 BRAKE LIGHTS
7 PREHEATING INDICATOR LIGHT
8 FLOAT SENSOR
17 LOW FUEL WARNING LIGHT
18 FUEL LEVEL INDICATOR
19 SENSOR FOR FUEL LEVEL INDICATOR
20 OIL TEMPERATURE INDICATOR
21 SENSOR FOR OIL/WATER TEMPERATURE INDICATOR
22 540 RPM REAR P.T.O. INDICATOR LIGHT
23 540 RPM REAR P.T.O. ENGAGEMENT SENSOR
24 1000 RPM REAR P.T.O. INDICATOR LIGHT
25 1000 RPM REAR P.T.O. ENGAGEMENT SENSOR
30 4WD INDICATOR LIGHT
32 TRAILER BRAKING INDICATOR LIGHT
33 TRAILER BRAKING SENSOR
34 MULTIMETER
35 ALTERNATOR INDICATOR LIGHT
36 HYDRAULIC OIL FILTER CLOGGING INDICATOR LIGHT
37 HYDRAULIC OIL FILTER CLOGGING SENSOR
40 AIR CLEANER CLOGGING INDICATOR LIGHT
41 AIR CLEANER CLOGGING SENSOR
42 BRAKE FLUID LEVEL INDICATOR LIGHT
43 BRAKE FLUID LEVEL SENSOR
44 REAR P.T.O. ROTATION INDICATOR LIGHT
44 SERVICES SENSOR (LOW OIL PRESSURE SOLENOID VALVES)
45 GROUNDSPEED P.T.O. INDICATOR LIGHT
46 GROUNDSPEED P.T.O. SENSOR
47 LOW ENGINE OIL PRESSURE INDICATOR LIGHT
48 LOW ENGINE OIL PRESSURE SENSOR
49 DIFFERENTIAL LOCK INDICATOR LIGHT
50 DIFFERENTIAL LOCK CONTROL SWITCH
51 DIFFERENTIAL LOCK SOLENOID
52 4WD ENGAGEMENT SWITCH
53 4WD ENGAGEMENT SOLENOID
54 FRONT P.T.O. CONTROL SWITCH
55 FRONT P.T.O. ENGAGEMENT INDICATOR LIGHT
56 FRONT P.T.O. SOLENOID
57 REAR P.T.O. SOLENOID
58 REAR P.T.O. CONTROL SWITCH/START ENABLE
59 ECONOMY P.T.O. INDICATOR LIGHT
60 ECONOMY P.T.O. SENSOR
61 HANDBRAKE SENSOR
64 SIDE LIGHTS INDICATOR LIGHT
65 FRONT LEFT SIDE LIGHT
66 REAR RIGHT SIDE LIGHT
67 FRONT RIGHT SIDE LIGHT
68 REAR LEFT SIDE LIGHT
69 NUMBER PLATE LIGHT
72 RIGHT LOW BEAM HEADLAMP
73 LEFT LOW BEAM HEADLAMP
74 LEFT HIGH BEAM HEADLAMP
75 RIGHT HIGH BEAM HEADLAMP
76 HIGH BEAM INDICATOR LIGHT
77 HORN
83 DIRECTION INDICATORS WARNING LIGHT
84 DIRECTION INDICATORS WARNING LIGHT - 1ST TRAILER
85 DIRECTION INDICATORS WARNING LIGHT - 2ND TRAILER
86 FRONT LEFT DIRECTION INDICATOR
87 REAR LEFT DIRECTION INDICATOR
88 FRONT RIGHT DIRECTION INDICATOR
89 REAR RIGHT DIRECTION INDICATOR
91 WORK LIGHT- "SWITZERLAND" SPEC.
93 REAR WORK LIGHTS
94 FLASHER SOCKET
95 SEAT SOCKET

PLATFORM WIRING — DIAGRAM 7 - CENTRAL WIRING FOR TRACTORS WITH ELECTRONIC GOVERNOR - 2ND VERSION — **011.4347.4/40 dwg. 6**

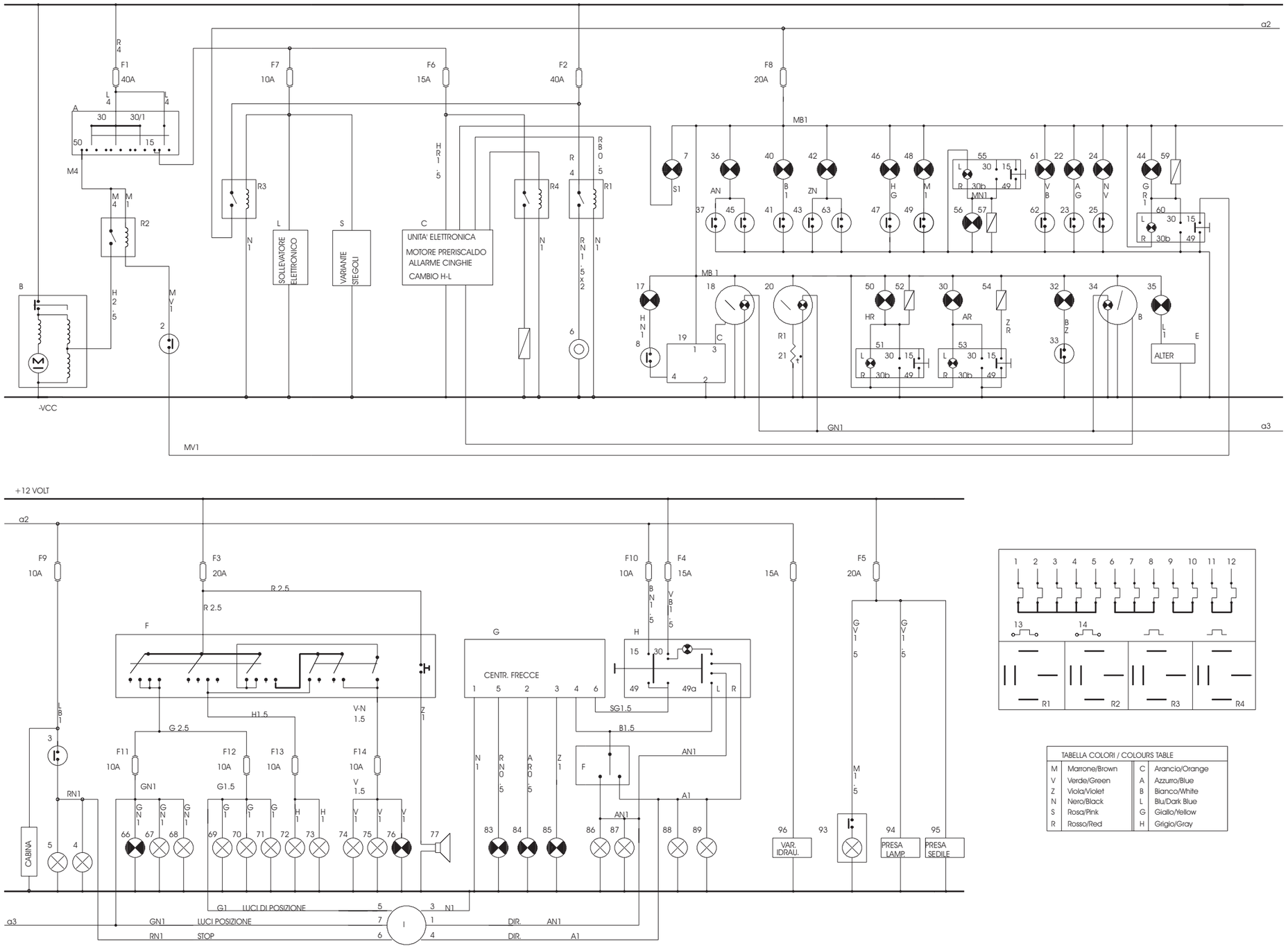


TABELLA COLORI / COLOURS TABLE			
M	Marrone/Brown	C	Arancio/Orange
V	Verde/Green	A	Azzurro/Blue
Z	Viola/Violet	B	Bianco/White
N	Nero/Black	L	Blu/Dark Blue
S	Rosa/Pink	G	Giallo/Yellow
R	Rosso/Red	H	Grigio/Gray

PLATFORM WIRING — DIAGRAM 8 - CENTRAL WIRING FOR TRACTORS WITH ELECTRONIC GOVERNOR - 2ND VERSION — **011.4347.4/40 dwg. 5**

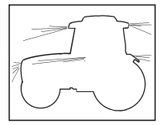


TABELLA COLORI / COLOURS TABLE			
Marrone/Brown	C	Arancio/Orange
Verde/Green	A	Azzurro/Blue
Viola/Violet	B	Bianco/White
Nero/Black	L	Blu/Dark Blue
Rosa/Pink	G	Giallo/Yellow
Rosso/Red	H	Grigio/Gray

AL CENTRALE
TO CENTRAL

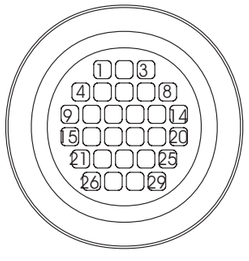
A

1	PTO SYNCRO	SYNCRO PTO
2	PTO 540	PTO 540
3	PTO 1000	PTO 1000
4	PTO ECONOMICA	ECONOMIC PTO
5	F.OLIO/SERVICE	OIL FILTER/SERVICE
6	PRESS.RIMORCHIO	TRAILER PRESS.
7	FRENO MANO	HAND BRAKE
8	VELOCITA'RUOTE	SPEED
9	VEL.RUOTE GND	SPEED GND
10		
11	SPIA DT	LIGHT DT
12	EV PTO	EV PTO
13	EV DF	EV DF
14	EV DT	EV DT
15	COM DT/DF/PTO	COM DT/DF/PTO
16		
17	EV H	EV H
18	EV L	EV L
20	GALL RISERVA	EMPTY SENSOR
21	SPIA RISERVA	EMPTY LIGHT
22	GND	GND

G 1
AG 1
VN 1
VB 1
CN 1
AN 1
H 1
MV 1
GV 1 — GV 1 — X2
AR05
GR 1
HR 1
RV 1
MB 1
ZB 1
ZV 1
HN 1
HM 1
N 1
X 1
X 4

EV H / H EV / H — ZB 1, MB 1
EV L / L EV / L — ZV 1, MB 1
EV PTO / PTO EV / E5 — GR 1, MB 1
EV DIFF / DIFF EV / E4 — HR 1, MB 1
EV DT / EV DT / E3 — RV 1, MB 1
SERVICE / SERVICE / V7 — CN 1
FILTRO OLIO / OIL FILTER / V8 — CN 1, N 1

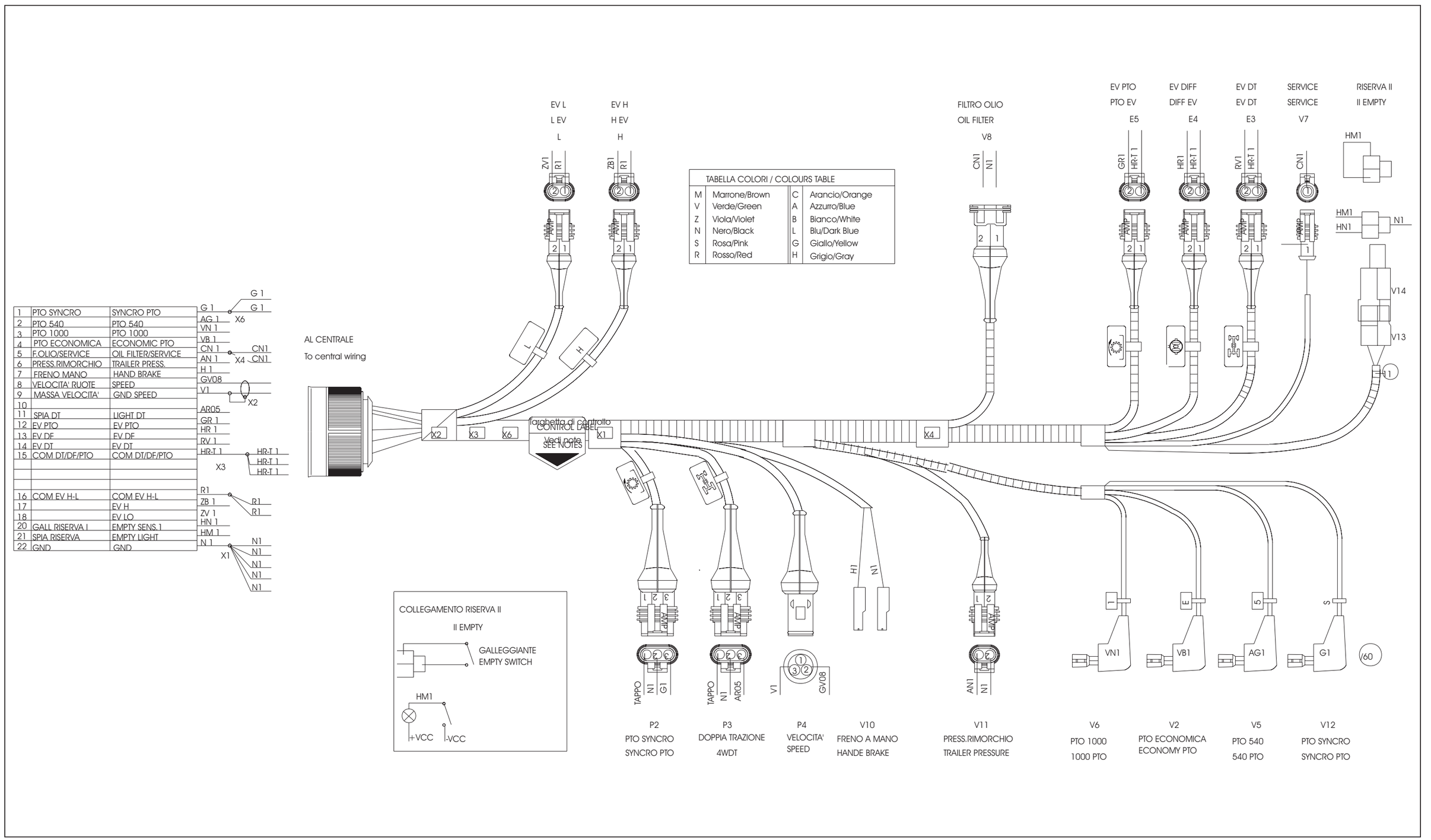
V10 — FRENO A MANO / HANDE BRAKE — H 1, N 1
P4 — V.RUOTE / SPEED — GV 1, MV 1
P2 — PTO SYNCRO / SYNCRO PTO — N 1, G 1
P3 — DOPPIA TRAZIONE / 4WDT — N 1, AR05
V13 — II RISERVA / II EMPTY — N 1, HN 1, HM 1
V14 — HM 1
V12 — SYNCRO / SYNCRO — G1
V5 — PTO 540 / 540 PTO — VN 1
V2 — PTO ECONOMICA / ECONOMY PTO — VB 1
V6 — PTO 1000 / 1000 PTO — AG 1
V11 — PRESS.RIMORCHIO / TRAILER PRESSURE — AN 1, N 1



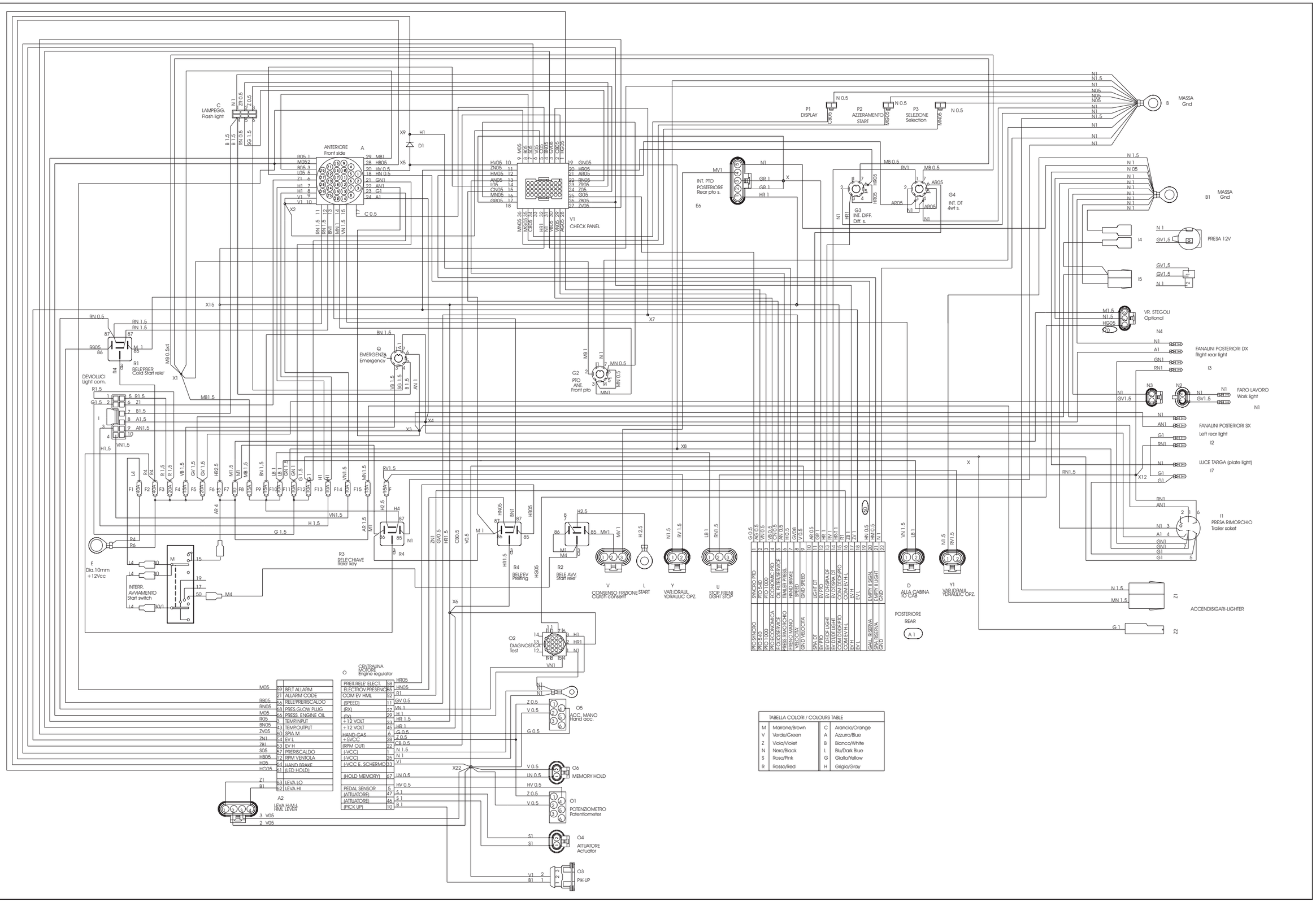
PLATFORM WIRING

DIAGRAM 9 - CENTRAL WIRING FOR TRACTORS WITH ELECTRONIC GOVERNOR - 2ND VERSION

011.4347.4/40 dwg. 4



PLATFORM WIRING — DIAGRAM 10 - CENTRAL WIRING FOR TRACTORS WITH ELECTRONIC GOVERNOR - 2ND VERSION — **011.4347.4/40 dwg. 3**



PLATFORM WIRING

DIAGRAM 11 - CENTRAL WIRING FOR TRACTORS WITH ELECTRONIC GOVERNOR - 2ND VERSION

011.4347.4/40 dwg. 2

TEMPERATURE	1	R05-Q
ENGINE OIL PRESS.	2	M05-V1
AIR FILTER	3	R05-V1
RPM GENERATOR	4	BR1-X13
GENERATOR	5	L05-V1
HORN	6	Z1-I
DIPPED HEAD LIGHTS	7	H1-F
DIPPED HEAD LIGHTS	8	H1-F
FULL BEAM HEAD LIGHTS	9	V1-X2
FULL BEAM HEAD LIGHTS	10	V1-X2
PREHEATING	11	RN1.5-H
PREHEATING	12	RN1.5-H
ELECT.PREHEATING	13	BN1-R3
FRONT PTO	14	MN1-G2
COMPRESSOR	15	VN1.5-D
GALLEGGIANTE	16	C05-V1
RISERVA	17	HN05-A1
BRAKES OIL	18	HV05-D1
POSIZ.SX	01	GN1-F9
DIREZ.SX	02	AN1-X3
POSIZ.DX	03	G1-F10
DIREZ.DX	04	A1-X4
PIK-UP VENTOLA	05	HB1-Z
+12 VOLT	09	MB1-X1

ANTERIORE

(Front wiring)

+12V (15)	1	BN1.5-F7
+12V (30)	3	VB1.5-F3
OUT 12V	4	SG1.5-C
COM.DIREZ.	5	B1.5-C
DIREZ.DX	6	AN1-X3
DIREZ.SX	7	A1-X4

EMERGENZA

(Emergency)

ALLARME		N1-7
CON.D.START		N1-H
SWITCH DISPLAY		M05-P1
SWITCH DISPLAY		M05-P2
SWITCH DISPLAY		M05-P3
CHECK PANEL		N1-V1
CENT.DIREZIONE		N1-C
DT		N1-G4
DIFF		N1-G3
INT.FARO LAVORO		N1-N
PTO ANTERIORE		N1-G2
AL POSTERIORE		N1-A1
HML		N1-G1
GND		N1

MASSA

(GND)

PULSANTI DISPLAY

(Switch display)

DISPLAY	1	C805-V1
GND	2	M05-B
ALTERNATORE	1	MG05-V1
GND	2	M05-B
SELETTORE	1	MV05-V1
GND	2	M05-B
LIGHT	1	LB1-F
GND	2	(IN1.5-D12)

(Stop light)

ACCENDISIGARI-LIGHTER

N.1 — MN 1.5 — G 1

CENTRALINA MOTORE

Engine regulator

PREIT.RELE' ELECT.	58	HR05
ELECTROV.PRESENCE	66	HN05
COM.EV HML	52	B1
(SPEED)	11	GV05
(RX)	23	VN1
(TX)	29	B1
+12 VOLT	13	HB1.5
+12 VOLT	65	HR1
HAND.GAS	6	G05
+5VCC	38	Z05
(RPM OUT)	22	C805
(-VCC)	1	N1.5
(+VCC)	25	N1
(-VCC E. SCHERMO)	33	V1
(HOLD MEMORY)	67	LN05
PEDAL SENSOR	5	HV05
(ATTUATORE)	47	S1
(ATTUATORE)	46	S1
(PICK UP)	10	B1
BELT ALLARM	59	M05
ALLARM CODE	21	
RELE'PRERISCALDO	56	RR05
PRES.GLOW PLUG	68	HN05
PRESS. ENGINE OIL	44	M05
TEMP.INPUT	3	I05
TEMP.OUTPUT	43	RN05
SPIA M	60	VD5
EV L	54	BN1
EV H	53	B1
PRERISCALDO	57	I05
RPM VENTOLA	12	RR05
HAND BRAKE	64	I05
(LED HOLD)	61	HG05
LEVA LO	63	Z1
LEVA HI	62	B1

CENTRALINA MOTORE

(Electronic unit)

FORNIRE SCIOLTO

SUPPLY FREE

GND		GV05
+12V		HR05
TXD		VD5
RXD		HG05

DIAGNOSTICA

(Test)

O2

+5V		Z05
GND		V05
PEDAL POSIT.		HG05

MASSA

(GND)

B2

POSIZIONE PEDALE

(Pedal position)

O1

+5VCC		Z05
GND		V05
PEDAL		HG05

ACCELERATORE A MANO

(Handle Acelerator)

O5 O6

MEMORY HOLD		LN05
GND		V05

ATTUATORE

(Actuator)

S1

PICK SENSOR		S1
PICK SENSOR		S1

PIK-UP

(Pick-up)

O3

EXCITER SIREN		B1
GND		O1

DIFFERENZIALE (Dif.lock)

GND	2	N1-8
DIFFERENZIALE	3	HR1-A1
	1	HR05
SPIA	4	MR05-V1
+12 SPIA	7	MB05-X1

DOPPIA TRAZIONE (4WDT)

EV DT	1	RV1-A1
GND	2	N1
GND	4	A1-8
SPIA DT	3	AR05-V1
	1	AD05
SPIA	4	AR05-A1
+12 SPIA	7	MR05-X1

PTO ANT.(Front pto)

+12V	5	MB1-X1
PTO ANT.	3	MN1-A
	6	MN05
SPIA	4	MN05-V1
GND SPIA	7	N1-8

G2

CHECK PANEL

HOLD	1	HG05-0
VELOCITA' MOTORE	2	C805-O6
VEL. RUOTE	3	GV05-X7
TEMP.MOTORE	4	BN05-O
FUNZ.CABINA.PANTE	5	C05-A
ABBAGLIANTI	6	V05-X2
PRERISCALDO	7	505-H
FILTRO ARIA	8	B05-A
PRESSIONE OLIO	9	M05-Z
FRENO A MANO/F.F.	10	HV05-D1
LO	11	ZN05-X
RISERVA	12	HM05-A1
PRESS. RIMORCHIO	13	AN05-A1
GENERATORE	14	I05-7
MASSERVICE EL.	15	CN05-A1
PTO ANTERIORE	16	MN05-G2
PTO POSTERIORE	17	GR06-X
ALLARME CINGHIE	18	

ILLUMINAZIONE	19	GN05-X8
DIFFERENZIALE	20	HR05-G3
DOPPIA TRAZIONE	21	AR05-G4
DIREZIONI	22	RN05-C
1.RIMORCHIO	23	Z605-C
2.RIMORCHIO	24	Z05-C
SYNCRO.	25	LG05-A1
H	26	ZB05-X
MEDIUM	27	ZV05-O
540	28	AG05-A1
1000	29	VN05-A1
ECONOMICA	30	VB05-A1
GND	31	N1-8
+12 VCC	32	HB1-X15
ALLARM CODE	33	C805-P1
P1 DISPLAY	34	MG05-P2
P2 AZIENDAMENTO	35	MN05-P3
P3 SELEZIONE	36	

GND	1	N1.4
+12VCC	2	RV1.5

VARIANTE IDRAULICA (YDRAULIC)

GND	1	N1-8
I.RIMORCHIO	2	Z9X5-V1
II.RIMORCHIO	3	Z05-V1
COM.DIREZ.	4	B1.5-G
	8	N1.5-L
DIREZIONI	5	SN1.5-V1
+12V	6	SG1.5-G

CENT.DIREZIONI (Direct.light)

15 A — 15 A — ELECTROVALVE — ELECT.1 — STOP — START — R4

X1 X2 X3 X4 X5 X6 X15

10A	10A	10A	10A	STOP	10A	ELECT.1	15A	15A	SERVIZI	20A	15A	40A	40A

15 A — 10 A — 10 A

RELE' EV PRERISCALDO

(Ev cold start)

COMANDO	85	HR05
GND	86	V05
EV.PRERISCALDO	87	BN1
+12VCC	30	HR1.5

CONSENSO

(Consent)

CONSENSO	2	MV1-R2
CONSENSO	3	MV1-E6
TAPPO	1	

DEVIOGUIDA

(Switch light)

+12V.F2	1	R1.5-F2
POSIZIONI	2	G1.5-F
ANABBAGLIANTI	3	H1.5-F
ABBAGLIANTI	4	VN1.5-F
+12V.F2	5	R1.5-F2
CLACSON	6	Z1-A
COM.DIREZ.	7	B1.5-C
DIREZ.DX	8	A1.5-X4
DIREZ.SX	9	AN1.5-X3

QUADRO AVV.

(Start switch)

+12V.FM	30	I4-FM
+12V.	30	I-4
+12V.	30/1	I-4
+12V.SOTTO CHIAVE	15	AR4-F
AVVIAMENTO	50	M4-B1

AL POSTERIORE

(To rear witing)

PTO SYNCRO.	SYNCRO.PTO.	G 0.5-V1
PTO 540	PTO. 540	AG 0.5-V1
PTO 1000	PTO. 1000	VN 0.5-V1
PTO ECONOMICA	ECONOMIC.PTO.	VB 0.5-V1
FOLIOVERSVICE	OIL FILTERSERVICE	CN 0.5-V1
PRESS.RIMORCHIO	TRAILER PRESS.	AN 0.5-V1
FRENO MANO	HAND BRAKE	H 0.5-D1
VELOCITASUOTE	SPEED	GW36-X8
VEL.RUOTE GND	SPEED GND	V05-O6
		A805-V1
SPIA DT	DT.LIGHT	GR 1-X
EV.PTO	EV.PTO	HR 1-G3
EV DF.	EV DF.	RV 1-G4
EV DF.	EV DF.	HB1.1-X15
COM.DT/DF/PTO	COM.DT/DF/PTO	B1-O
COM.EV.H-L	COM.EV.H-L	Z8 1-X
EV H	EV H	Z1 1-X
EV LO	EV LO	
RISERVA II	EMPTY II	HM-05-A
RISERVA II	EMPTY II	HN-05-V1
GND	GND	N 1-8

LEVA H-L

(Lever H-L)

LO	1	Z 1-G1
COM	2	H 1-G1
COM	3	H 1-G1
HI	4	B 1-G1

+12Vcc — R4-FM, R6-F

START — J42.5-M

ALLA CABINA

(To cab)

COMPRESSORE	1	VN1.5-A
+12V.(15)	2	HB1-F8

+12V SEDILE

(+12V SEAT)

GND		N1-B1
+12V.F4		GV1.5-F4
		GV1.5-I4

MASSA

(GND)

PRESA 12V		N1-I4
PRESA RIMORCHIO		N1-I1
FANALINI SX		N1-I2
FANALINI DX		N1-I3
FARO LAVORO		N1-N3
VR. STEGOLI		N1-N4
PTO POSTER.		N05-E6

PTO POSTERIORE

(REAR PTO)

+12 SPIA	1	HR 1 (X15)
PTO	2	GR 1 (X)
PTO	3	GR 1 (X)
CONSENSO	4	MV 1 (V)
COMUNE	5	N 1 (B1)

PRESA 12V

(12V SOCHET)

GND		N1-B1
+12V.F4		GV1.5

VR.YD.

(OPT.)

1 N1.5
2 RV1.5

VR.STEGOLI

(OPT.)

DX-RX	1	HG05-N4
GND	2	N1.5-B1
+12V	3	M1.5

2 N1
1 GV1.5

FANALINI DX

(LIGHT DX)

STOP		RN1-X12
POSIZIONI		GN1-U
DIR.DX		A1-X4
GND		N1-B1

PRESA RIMORCHIO

(TRAILER)

			AN1-X3
DIR.SX	1	N1-B1	
GND	3	A1-X4	
DIR.SX	4	G1-I2	
POSIZIONI	5	G1.5-I10	
STOP	6	RN1-X12	
POSIZIONI	7	GN1-X8	G1-I7

FANALINI SX

(LIGHT SX)

STOP	RN1-X12	
POSIZIONI	G1-I1	
DIR.SX	AN1-X3	
GND	N1-B1	

LUCE TARGA

(LIGHT PLATE)

N1-B1
G1-I1
G1-I2

GR 1 (A1) — GR O.5 (V1) — X — GR 1 (E6) — GR 1 (E6)

N1 — GV1.5 — FARO LAVORO (LIGHT job)

PTO ANT.(Front pto)

+12V	2	MB1-X1
PTO ANT.	3	MN1-A
SPIA	5	MN05
		VN05-V1
GND SPIA	7	N1-B

EMERGENZA

(Emergency)

+12V(15)	1	BN1-5-F7
+12V(30)	4	VB1-5-F3
OUT 12V	4	SG1-5-C
COM.DIREZ.	5	B1-5-C
DIREZ.DX	6	AN1-X3
DIREZ.SX	7	A1-X4

ANTERIORE

(Front wiring)

TEMPERATURE	1	R05-V1
ENGINE OIL PRESS.	2	M05-V1
AIR FILTER	3	B05-V1
RPM GENERATOR	4	BR1-X13
GENERATOR	5	U05-V1
VCSW	6	Z1-I
DIPPED HEAD LIGHTS	7	N1-F
DIPPED HEAD LIGHTS	8	N1-F
FULL BEAM HEAD LIGHTS	9	V1-X2
FULL BEAM HEAD LIGHTS	10	V1-X2
PREHITING	11	BN1-5-H
PREHITING	12	BN1-P2
ELECT.PREHITING	13	MN1-G2
FRONT PTO	14	VN1-5-D
COMPRESSOR	15	C05-V1
CALLEGGIANTE	17	HN05-A1
RISERVA	18	HV05-D1
BRAKES OIL	20	GN1-F9
POSIZ.SX	21	AN1-X3
DIREZ.SX	22	G1-F10
POSIZ.DX	23	A1-X4
DIREZ.DX	24	HB1-Z
PIK-UP VENTOLE	28	MB1-X1
+12 VOLT	29	

MASSA

(GND)

ALLARME	N1-Z
COLD START	N1-H
GND	N1
GND	N1
GND	N1
CHECK PANEL	N1-V1
CENT.DIREZIONI	N1-C
DF	N1-G4
DIFF	N1-G3
FT	N1-N
INT.FARO LAVORO	N1-G2
PTO ANTERIORE	N1-A1
AL POSTERIORE	N1-G1
HML	

STOP FRENI

(Stop light)

	+ 12V STOP	
	+ 12V STOP	
LED F.		
BN1.5-X12		

PRED.HML

(HML Predisp.)

LEVA H	7	B 1-A2
M	2	ZV05-V1
+12V	12	MB1-X1
HI	11	78 1-X
COM.EV.	3	R1-A1
LEVA LO.	6	2 1-A2
		H 1-X
LEVA COM.	5	N1-B
GND	1	N1-B
LO	9	7N 1-X
GND	4	N1-B

GND		
+ 12V		
		N1
		MN05

PRERISCALDO

(Cold start)

+12V/VCC	1	MB1-X1
	2	S1-V1
SPIA	5	
COMANDO RELE'	7	BR05-R1
MASSA	10	N1-B
GRI 'W'	14	BR1-X13

DIFFERENZIALE (Dif.lock)

GND	2	N1-B
DIFFERENZIALE	3	MB1-A1
		HR05
SPIA	6	VR05-V1
		MR05-X1
+12 SPIA	7	

DOPPIA TRAZIONE (4WDT)

EV DT	1	RV1-A1
GND	2	N1
GND	4	N1-B
SPIA DT	8	AR05-V1
		AR05
SPIA	5	AR05-A1
+12 SPIA	7	MR05-X1

CHECK PANEL

GND	1	N1-B	ILLUMINAZIONE	19	GN05-X8
VELOCITA' MOTORE	2	RR05-X13	DIFFERENZIALE	20	HR05-G3
	3		DOPPIA TRAZIONE	21	AR05-G4
TEMPM/OTORE	4	B05-A	DIREZIONI	22	BN05-C
INDIC. CARBURANTE	5	C05-A	RIMORCHIO	23	ZR05-C
ABBAGLIANTI	6	V1/LV2	RIMORCHIO	24	Z05-C
RISERVA	7	S1-H	RIMORCHIO	25	G05-A1
RISCALDO	8	B05-A	SINCRO	26	ZR05-G1
FILTRO ARIA	9	M05-Z	MEDIUM	27	ZV05-G1
PRESSIONE OLIO	10	HV05-D1	LOW	28	AG05-A1
FRENO A MANCO F.	11	ZN05-G1	ECONOMICA	29	VN05-A1
LO	12	HM05-A1	GND	30	VB05-A1
RISERVA	13	AN05-A1	+12 VCC	31	N1-B
PRESS. RIMORCHIO	14	I05-Z		32	MB1-X1
GENERATORE	15	CN05-A1		33	
N.DISSERVICE ID.	16	MN05-G2		34	
PTO ANTERIORE	17	GR05-X		35	
PTO POSTERIORE	18			36	

V1

ACCENDISIGARI-LIGHTER

CLOSE 1
GND 2 — N1-A
+12VCC 3 — RV1-5

VARIANTE IDRAULICA (YDRAULIC)

Z ALLARM(Allarm)

+12V	1	MB1-X1
GND	2	N1-B
SPIA GEN.	3	U05-A
		V05-V1
PRESS.OLIO M.	4	M05-A
		M05-V1
ALI ARIA	5	RG1
RPM VENTOLA	6	HB1-A
GND	7	N1-D1
RIC.MECC.	8	RG1

CENT.DIREZIONI (Direct.light)

GND	1	N1-B
RIMORCHIO	2	RR05-V1
RIMORCHIO	3	R05-V1
COM.DIREZ.	4	B1-5-D
		N1-5-I
DIREZIONI	5	BN1-5-V1
+12V	6	VG1-5-G

CONSENSO

(Consent)

CONSENSO	2	MV1-B2
CONSENSO	3	MV1-E6
TAPPO	1	

DEVIOGUIDA

(Switch light)

+12V.E2	1	B1-5-E2
POSIZIONI	2	G1-5-F
ANABBAGLIANTI	3	H1-5-E
ABBAGLIANTI	4	VN1-5-F
+12V.F2	5	B1-5-F2
CLACSON	6	Z1-A
COM.DIREZ.	7	B1-5-C
DIREZ.DX	8	A1-5-X4
DIREZ.SX	9	AN1-5-X3

QUADRO AVV.

(Start switch)

+12V.FM	30	I-4-FM
+12V	30	I-4
+12V	30/1	L-2
+12V SOTTO PEDALE	50	A04-F
AVVIAMENTO 50		M4-R1

AL POSTERIORE

(To rear wiring)

LEVA H-L

(Lever H-L)

PTO SYNCRO	SYNCRO PTO	1	G 0.5-V1
PTO 540	PTO 540	2	AG 0.5-V1
PTO 1000	PTO 1000	3	VN 0.5-V1
PTO ECONOMICA	ECONOMIC. PTO	4	VB 0.5-V1
CONSERVATA	OIL FILTER RESERVE	5	CN 0.5-V1
PRESS.RIMORCHIO	TRAILER PRESS.	6	AN 0.5-V1
FRENO MANO	HAND BRAKE	7	H 0.5-D1
		8	
		9	
SPIA DT	DT LIGHT	10	
EV PTO	EV PTO	11	AR05-V1
EV DF	EV DF	12	GR 1-X
EV DT	EV DT	13	HR 1-G3
COM.DIDIEVRSO	COM.DIDIEVRSO	14	RV 1-G4
COM.EV.H-L	COM.EV.H-L	15	MR 1-X1
EV.H	EV.H	16	91-G1
EV.LO	EV.LO	17	ZB 1-G1
		18	ZV 1-G1
SENSORE RISERVA	EMPTY SENSOR	19	HN 0.5
SPIA RISERVA	EMPTY LIGHT	20	HM 0.5
GND	GND	21	N1-B
		22	

LO		Z 1-G1
COM		H 1
COM		H 1
HI		B 1-G1

GR 1 (A1) — GR O.5 (V1) — X — GR 1 (E6) — GR 1 (E6)

+12Vcc — R4-FM, R6-F

START — H2 5-M

ALLA CABINA

(To cab)

COMPRESSORE	1	VN1.5-A
+12V (15)	2	LB1-F8

+12V SEDILE

(+12V SEAT)

GND		N1-B1
+12V.F4		GV1.5-F4
		GV1.5-I4

MASSA

(GND)

PRESA 12V	N1-I4
PRESA RIMORCHIO	N1-I1
FANALINI SX	N1-I2
FANALINI DX	N1-I3
FARO LAVORO	N1-N3
VD. STEGOLI	N1-N4
PTO POSTER.	N05-F6

PTO POSTERIORE

(REAR PTO)

+12V SPIA	1	MB 1 X1
PTO	2	GR 1 X
SPIA PTO	3	GR 1 X
CONSENSO	4	MV 1 V3
GND	5	N 1 B1

PRESA 12V

(12V SOCHET)

GND		N1-B1
+12V.F4		GV1.5 -I5

VR.YD.

(OPT.)

1 N1-5
2 RV1-5

VR.SOLL.

(OPT.)

D.UP	1	HG05
GND	2	N1-B1
+12V	3	N1

2-GV1.5
1 M1-N

FANALINI DX

(LIGHT DX)

STOP		BN1-X12
POSIZIONI		GN1-I1
DIR.DX		A1-X4
GND		N1-B1

PRESA RIMORCHIO

(TRAILER)

DIR.SX	1	AN1-X3
GND	3	N1-B1
DIR.SX	4	A1-X4
POSIZIONI	5	G1-I5
		G1-F10
STOP	6	BN1-X12
POSIZIONI	7	GN1-X8
		GN1-I2

FANALINI SX

(LIGHT SX)

STOP		BN1-X12
POSIZIONI		GN1-I1
DIR.SX		AN1-X3
GND		N1-B1

LUCE TARGA

(LIGHT PLATE)

N1-B1
G1-I1
G1-I2

N1 — FARO LAVORO (LIGHT JOB) — GV1.5

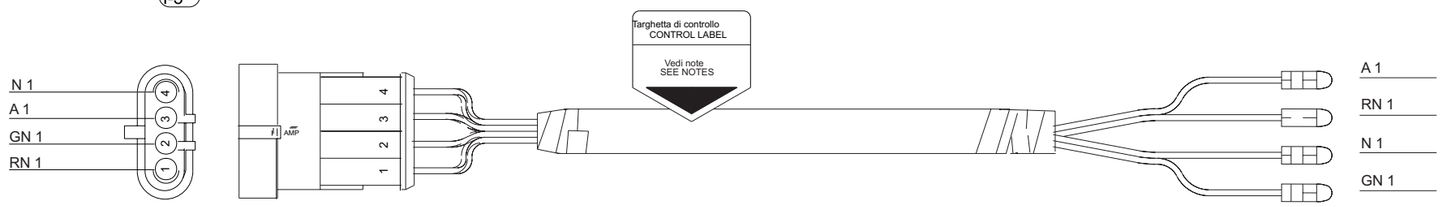
PLATFORM WIRING

DIAGRAM 13 - CENTRAL WIRING FOR TRACTORS WITH MECHANICAL GOVERNOR

010.8683.4/80 drw. 1bis

AL CABL.SOTTOTAPPETO (DX)
TO UNDER-CARPET WIRING (RIGHT)

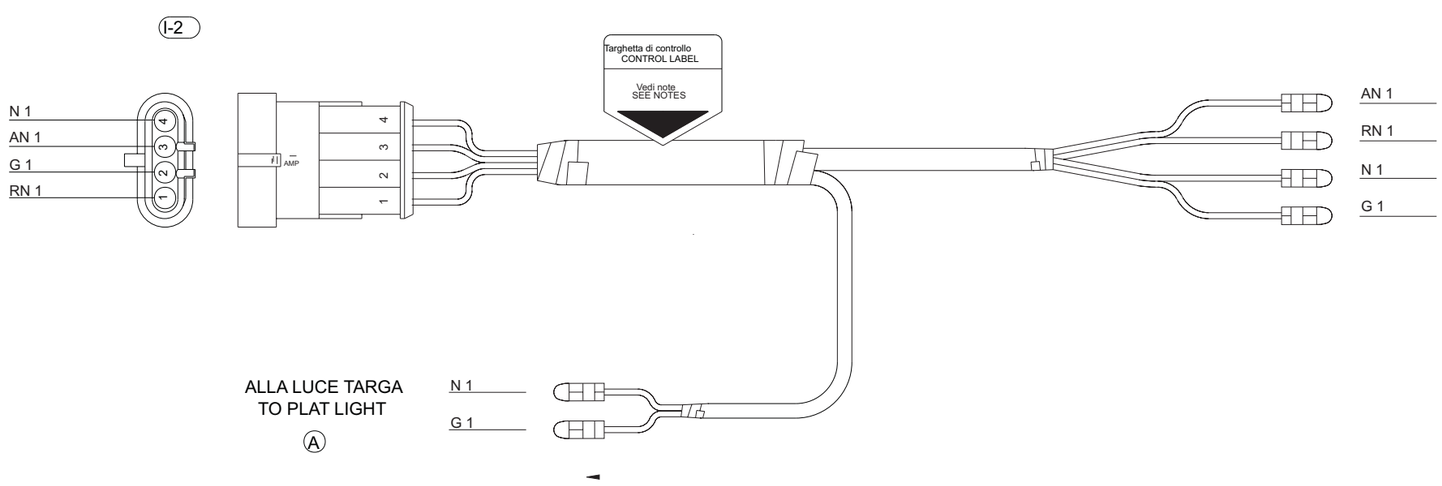
CONNETTORE (CONNECTOR)	POSIZIONE (POSITION)	QUOTA TOTALE (TOTAL QUOTA)	CONNETTORE (CONNECTOR)	POSIZIONE (POSITION)
I-3	1	RN 1	D	
I-3	2	GN 1	D	
I-3	3	A 1	D	
I-3	4	N 1	D	



AL FARO POSTERIORE DX
TO RIGHT BACK LIGHT

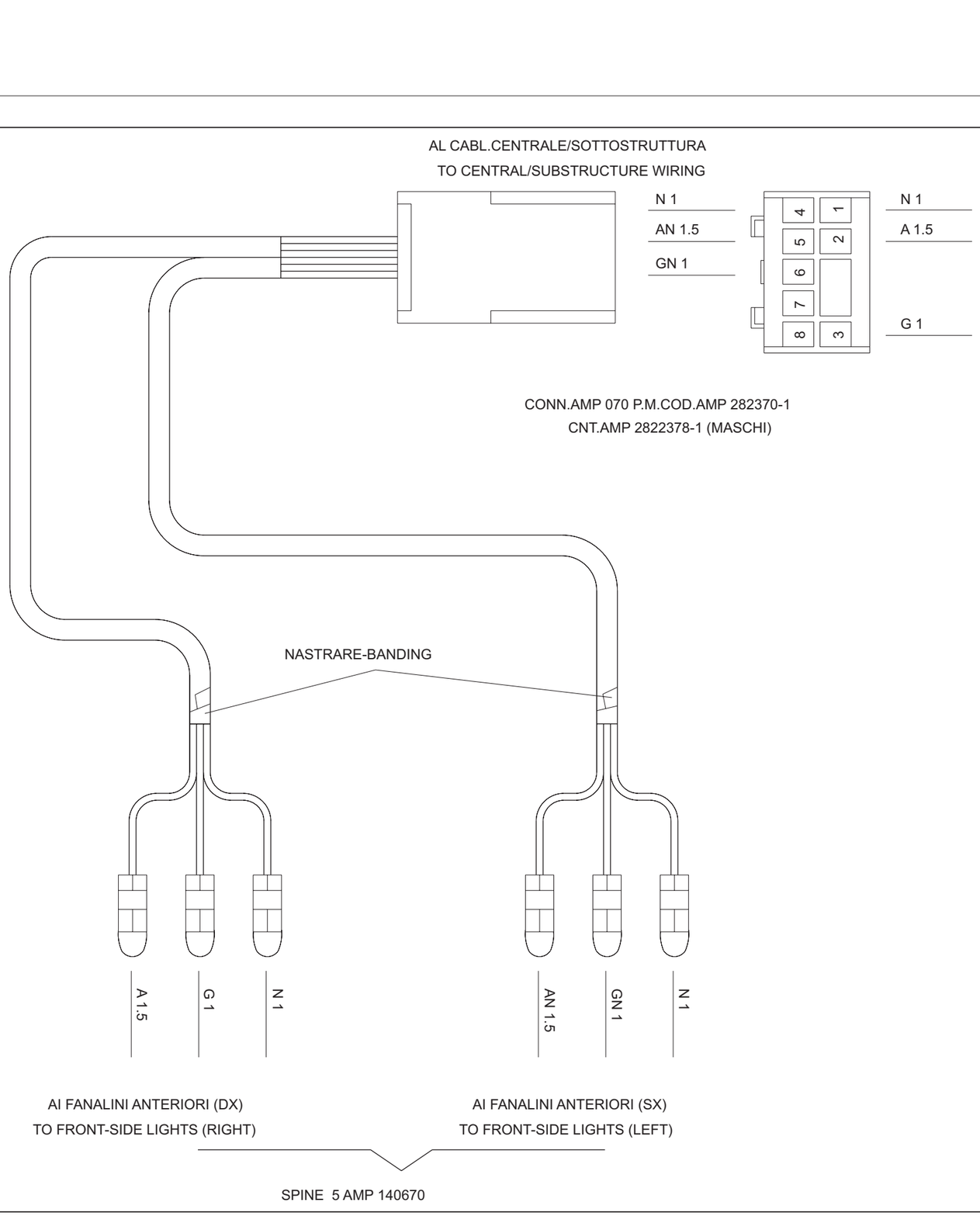
AL CABL.SOTTOTAPPETO (SX)
TO UNDER-CARPET WIRING (LEFT)

CONNETTORE (CONNECTOR)	POSIZIONE (POSITION)	QUOTA TOTALE (TOTAL QUOTA)	CONNETTORE (CONNECTOR)	POSIZIONE (POSITION)
I-2	1	RN 1	B	
I-2	2	G 1 G 1 S.U.2 G 1	B A	
I-2	3	AN 1	B	
I-2	4	N 1 N 1 S.U.3 N 1	A B	

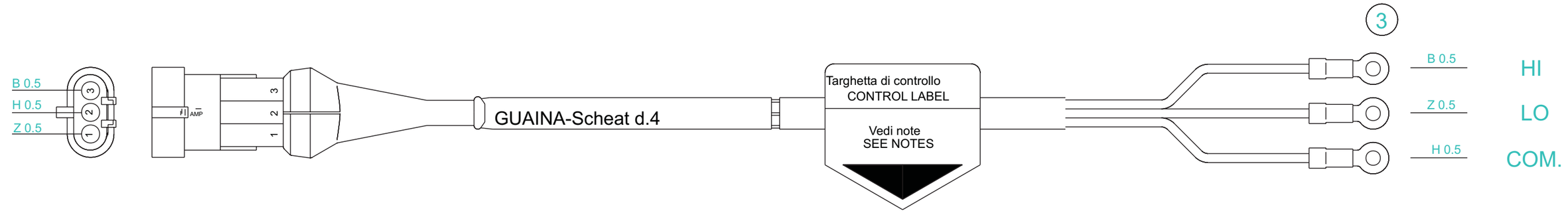


AL FARO POSTERIORE SX
TO LEFT BACK LIGHT

TABELLA COLORI / COLOURS TABLE			
M	Marrone/Brown	C	Arancio/Orange
V	Verde/Green	A	Azzurro/Blue
Z	Viola/Violet	B	Bianco/White
N	Nero/Black	L	Blu/Dark Blue
S	Rosa/Pink	G	Giallo/Yellow
R	Rosso/Red	H	Grigio/Gray



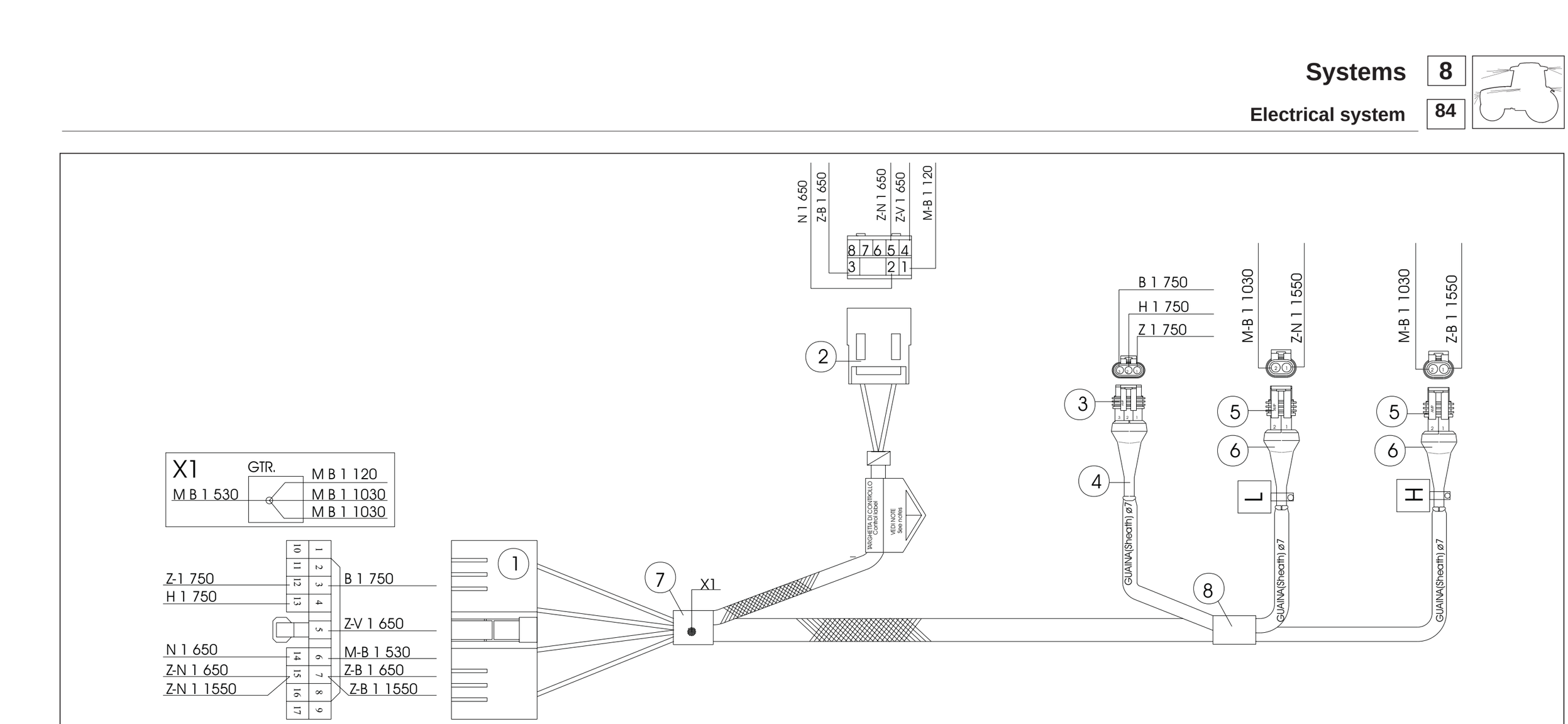
PLATFORM WIRING DIAGRAM 15 - FRONT LIGHTS WIRING **010.5582.3**



PLATFORM WIRING

DIAGRAM 16 - HML CONTROL LEVER WIRING
2nd version equipped with pushbuttons code 2.7659.183.0

010.8400.3

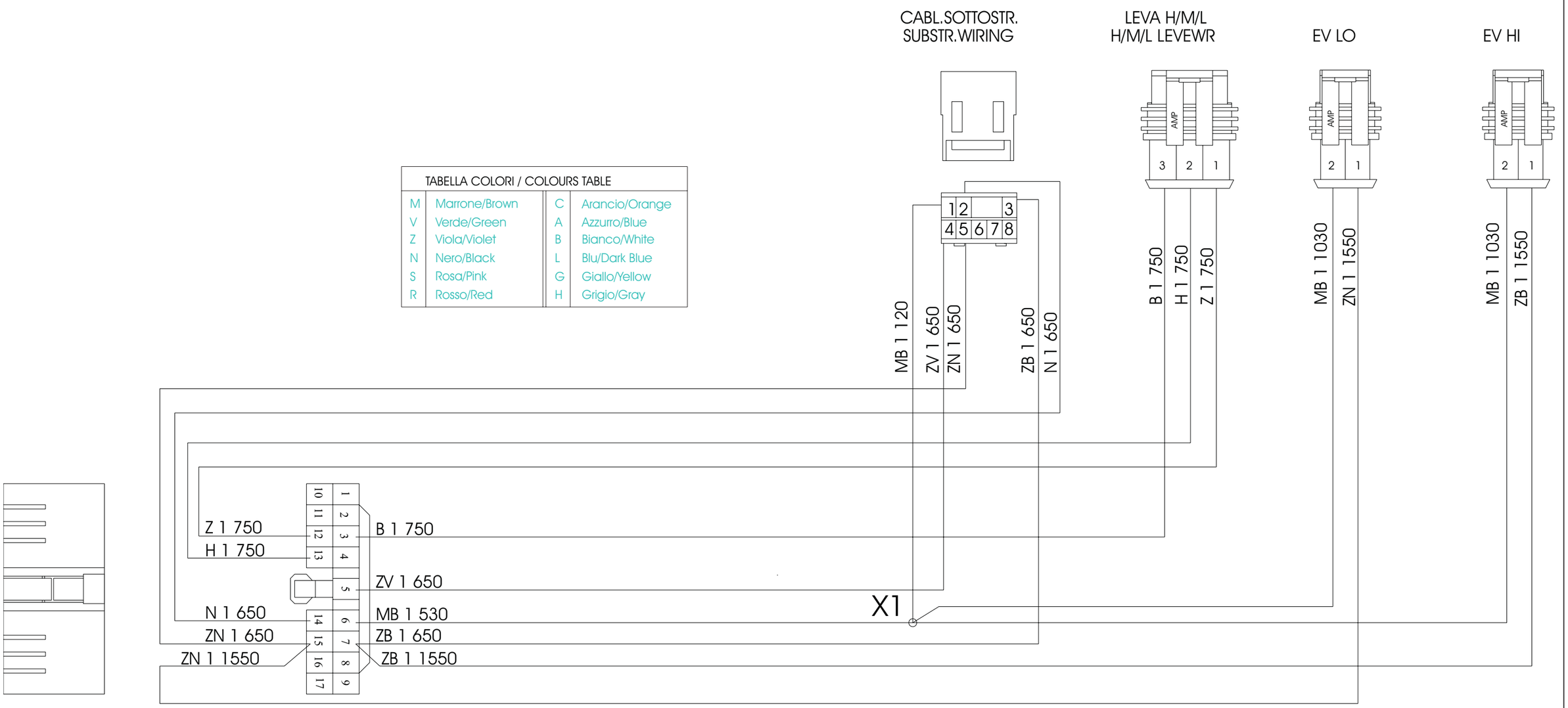


3 LEVA HI	12 LEVA LO
5 SPIA M	13 LEVA COM.
6 +12 Vcc	14 -Vcc
7 EV HI	15 EV LO

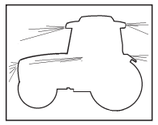
TABELLA COLORI / COLOURS TABLE			
M	Marrone/Brown	C	Arancio/Orange
V	Verde/Green	A	Azzurro/Blue
Z	Viola/Violet	B	Bianco/White
N	Nero/Black	L	Blu/Dark Blue
S	Rosa/Pink	G	Giallo/Yellow
R	Rosso/Red	H	Grigio/Gray

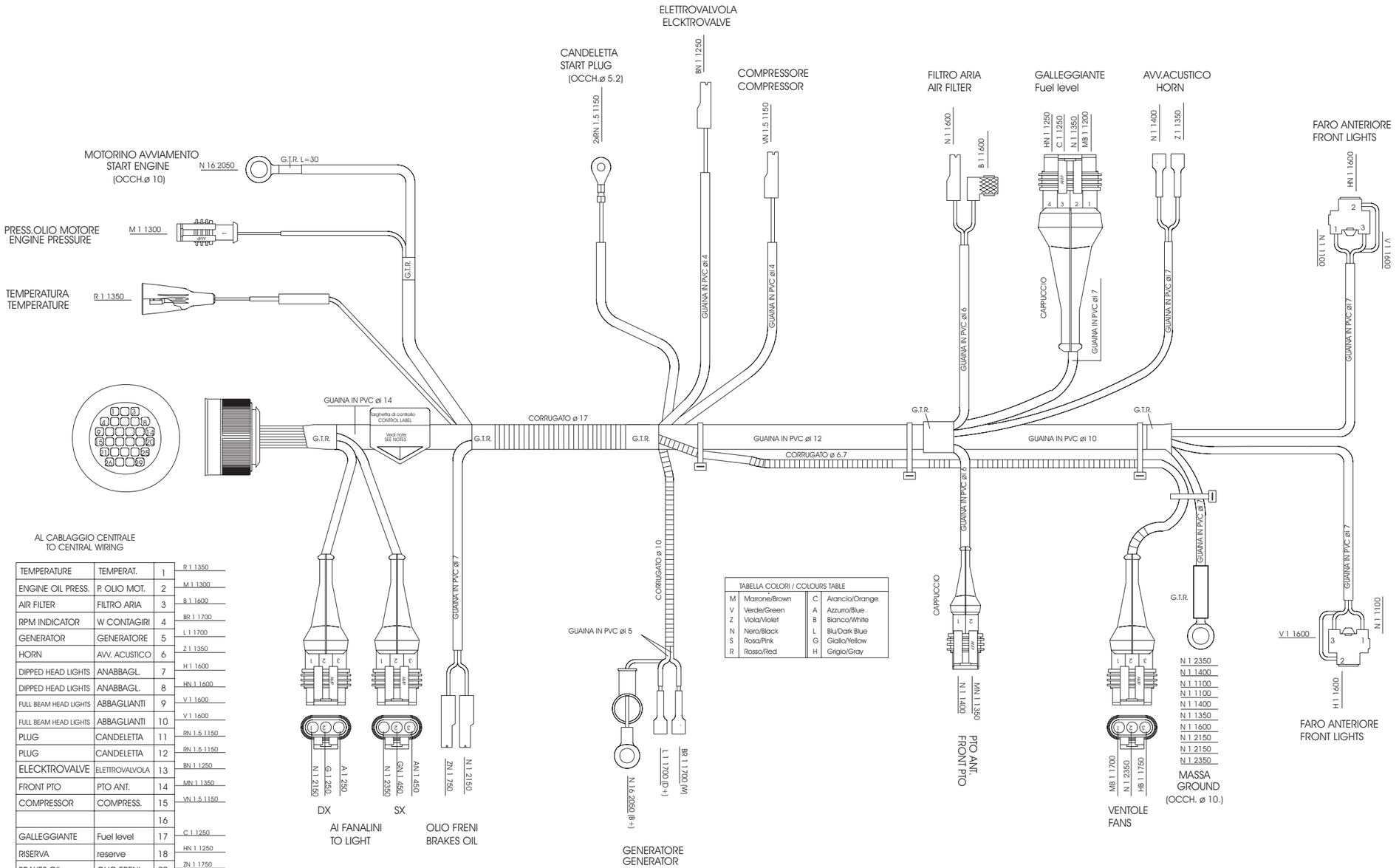
PLATFORM WIRING — DIAGRAM 17 - HML WIRING (1ST VERSION) — **010.8388.3 (dwg. 1)**

TABELLA COLORI / COLOURS TABLE			
M	Marrone/Brown	C	Arancio/Orange
V	Verde/Green	A	Azzurro/Blue
Z	Viola/Violet	B	Bianco/White
N	Nero/Black	L	Blu/Dark Blue
S	Rosa/Pink	G	Giallo/Yellow
R	Rosso/Red	H	Grigio/Gray



PLATFORM WIRING DIAGRAM 18 - HML WIRING (1ST VERSION) **010.8388.3 (dwg. 2)**





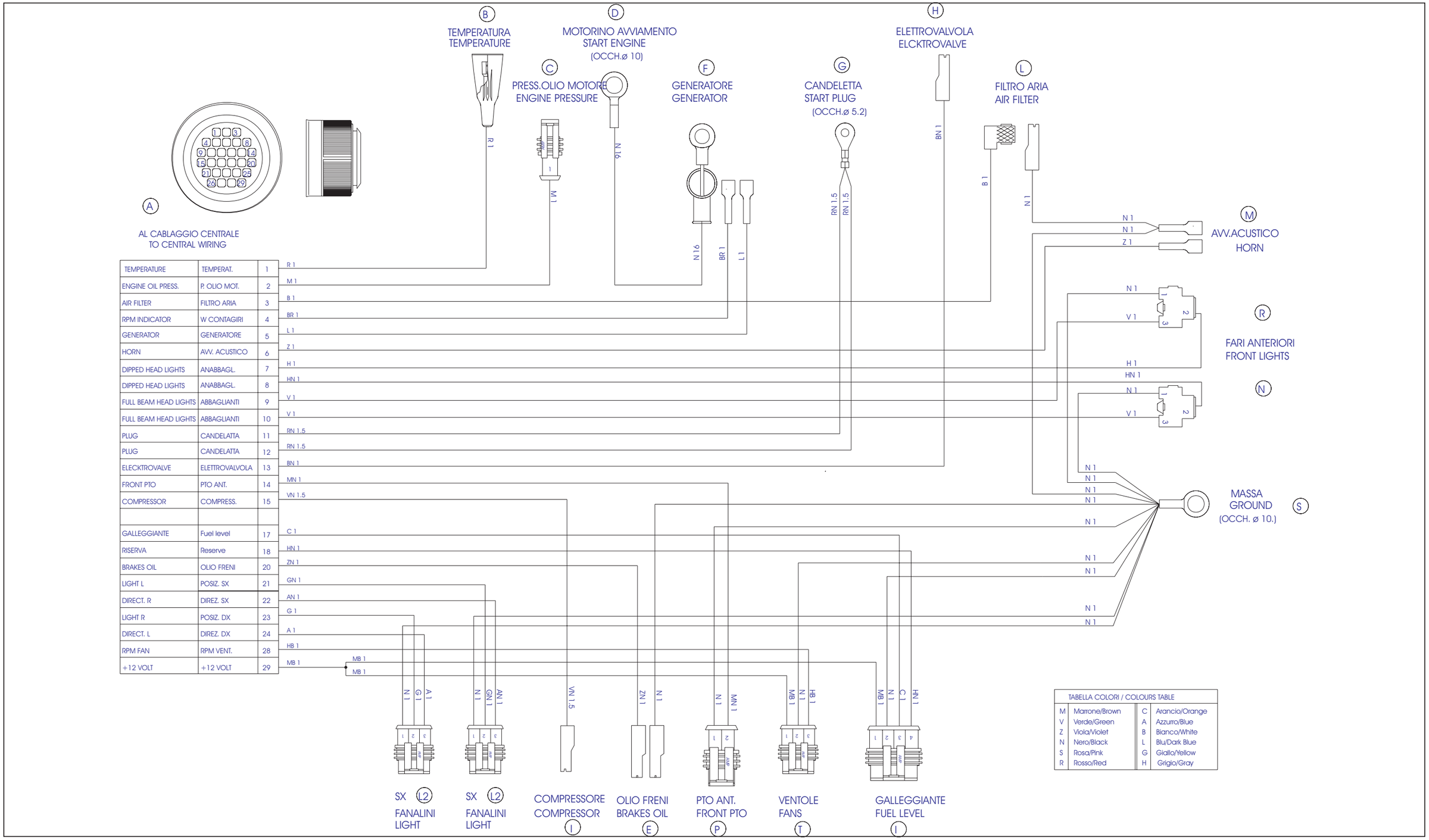
AL CABLAGGIO CENTRALE
TO CENTRAL WIRING

TEMPERATURE	TEMPERAT.	1	R 1 1350
ENGINE OIL PRESS.	P. OLIO MOT.	2	M 1 1300
AIR FILTER	FILTRO ARIA	3	B 1 1600
RPM INDICATOR	W CONTAGIRI	4	BR 1 1700
GENERATOR	GENERATORE	5	L 1 1700
HORN	AVV. ACUSTICO	6	Z 1 1350
DIPPED HEAD LIGHTS	ANABBAGL.	7	H 1 1600
DIPPED HEAD LIGHTS	ANABBAGL.	8	HN 1 1600
FULL BEAM HEAD LIGHTS	ABBAGLIANTI	9	V 1 1600
FULL BEAM HEAD LIGHTS	ABBAGLIANTI	10	V 1 1600
PLUG	CANDELETTA	11	RN 1.5 1150
PLUG	CANDELETTA	12	RN 1.5 1150
ELECKTROVALVE	ELETTROVALVOLA	13	BN 1 1250
FRONT PTO	PTO ANT.	14	MN 1 1350
COMPRESSOR	COMPRESS.	15	VN 1.5 1150
		16	
GALLEGGIANTE	Fuel level	17	C 1 1250
RISERVA	reserve	18	HN 1 1250
BRAKES OIL	OLIO FRENI	20	ZN 1 1750
LIGHT L	POSIZ.SX	21	GN 1 450
DIRECT. L	DIREZ. SX	22	AN 1 450
LIGHT R	POS. DX	23	G 1 250
DIRECT. R	DIREZ. DX	24	A 1 250
RPM FAN	RPM VENT.	28	HB 1 1750
+12 VOLT	+12 VOLT	29	MB 1 100 → MB 1 1700, MB 1 1200

TABELLA COLORI / COLOURS TABLE

M	Marrone/Brown	C	Arancio/Orange
V	Verde/Green	A	Azzurra/Blue
Z	Viola/Violet	B	Bianco/White
N	Nero/Black	L	Blu/Dark Blue
S	Rosa/Pink	G	Giallo/Yellow
R	Rosso/Red	H	Grigio/Gray

TRACTOR WIRING — DIAGRAM 19 - COMPLETE FRONT WIRING — **010.6411.4/40 dwg. 1**



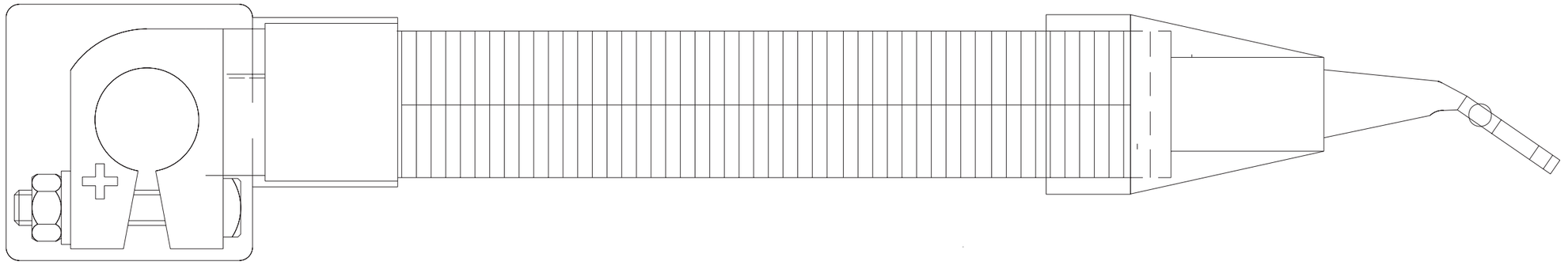
TEMPERATURE	TEMPERAT.	1
ENGINE OIL PRESS.	P. OLIO MOT.	2
AIR FILTER	FILTRO ARIA	3
RPM INDICATOR	W CONTAGIRI	4
GENERATOR	GENERATORE	5
HORN	AVV. ACUSTICO	6
DIPPED HEAD LIGHTS	ANABBAGL.	7
DIPPED HEAD LIGHTS	ANABBAGL.	8
FULL BEAM HEAD LIGHTS	ABBAGLIANTI	9
FULL BEAM HEAD LIGHTS	ABBAGLIANTI	10
PLUG	CANDELATTA	11
PLUG	CANDELATTA	12
ELECKTROVALVE	ELETTROVALVOLA	13
FRONT PTO	PTO ANT.	14
COMPRESSOR	COMPRESS.	15
GALLEGGIANTE	Fuel level	17
RISERVA	Reserve	18
BRAKES OIL	OLIO FRENI	20
LIGHT L	POSIZ. SX	21
DIRECT. R	DIREZ. SX	22
LIGHT R	POSIZ. DX	23
DIRECT. L	DIREZ. DX	24
RPM FAN	RPM VENT.	28
+12 VOLT	+12 VOLT	29

TABELLA COLORI / COLOURS TABLE			
M	Marrone/Brown	C	Arancio/Orange
V	Verde/Green	A	Azzurro/Blue
Z	Viola/Violet	B	Bianco/White
N	Nero/Black	L	Blu/Dark Blue
S	Rosa/Pink	G	Giallo/Yellow
R	Rosso/Red	H	Grigio/Gray

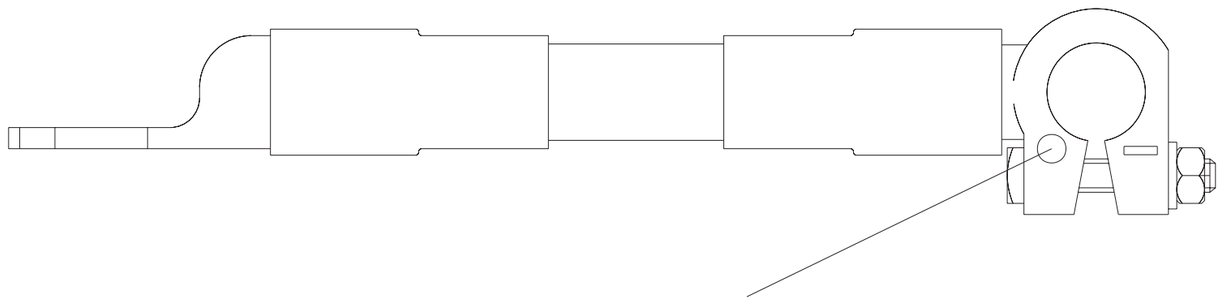
TRACTOR WIRING DIAGRAM 20 - COMPLETE FRONT WIRING **010.6411.4/40 dwg. 2**

MORSETTO POSITIVO BATTERIA (POSITIVE BATTERY)

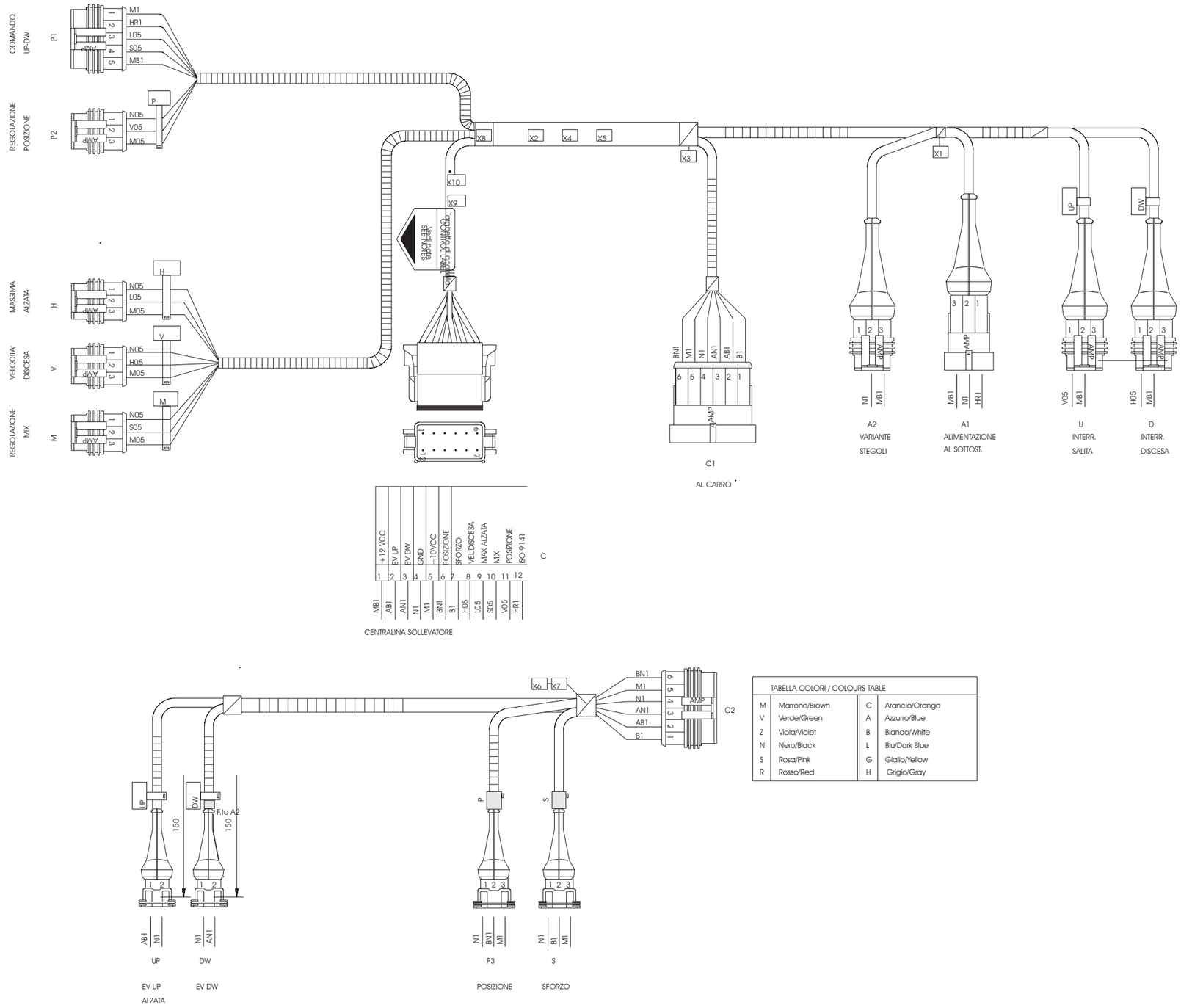
TERMINALE INTEGRALE ø 10.5



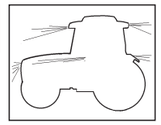
TERMINALE INTEGRALE ø 10.5 (INTEGRAL CONT.)



MORSETTO NEGATIVO BATTERIA (NEGATIVE BATTERY)



TRACTOR WIRING DIAGRAM 22 - POWER LIFT WIRING **011.2025.4/20 dwg. 1**



CENTRALINA SOLLEVATORE

C

6	5	4	3	2	7		9		8		10		11		12		1
POSIZIONE	+10VCC	GND	EV DW	EV UP	SFORZO		MAX ALZATA		VEL.DISCESA		MIX		POSIZIONE		ISO 9141		+12 VCC

BN1 M1 N1 AN1 AV1 B1 V05 H05 S05 L1 HR1 MB1

X1 X2 X3 X4 X5

C1 — AL CARRO

6	5	4	3	2	1
BN1	M1	N1	AN1	AV1	B1
POSIZIONE	+10VCC	GND	EV DW	EV UP	SFORZO

H — MASSIMA ALZATA

1	2	3
N05	V05	M05
GND	MAX ALZATA	+10VCC

V — VELOCITA' DISCESA

1	2	3
N05	H05	M05
GND	VEL.DISCESA	+10VCC

M — REGOLAZIONE MIX

1	2	3
N05	S05	M05
GND	MIX	+10VCC

P2 — REGOLAZIONE POSIZIONE

1	2	3
N05	L1	M05
GND	POSIZIONE	+10VCC

P1 — COMANDO UP-DW

1	2	3	4	5
M1	HR1	N1	M1	L05
+10VCC	ISO 9141	DW	UP	COMUNE

A1 — ALIMENTAZIONE AL SOTTOST.

3	2	1
MB1	N1	HR1
+12 VCC	GND	ISO 9141

A2 — VARIANTE STEGOLI

1	2	3
N1	MB1	
GND	+12 VCC	

C2

6	5	4	3	2	1
BN1	M1	N1	AN1	AV1	B1
POSIZIONE	+10VCC	GND	EV DW	EV UP	SFORZO

X6 X7

UP — EV UP ALZATA

1	2
AV1	N1

DW — EV DW

1	2
N1	AN1

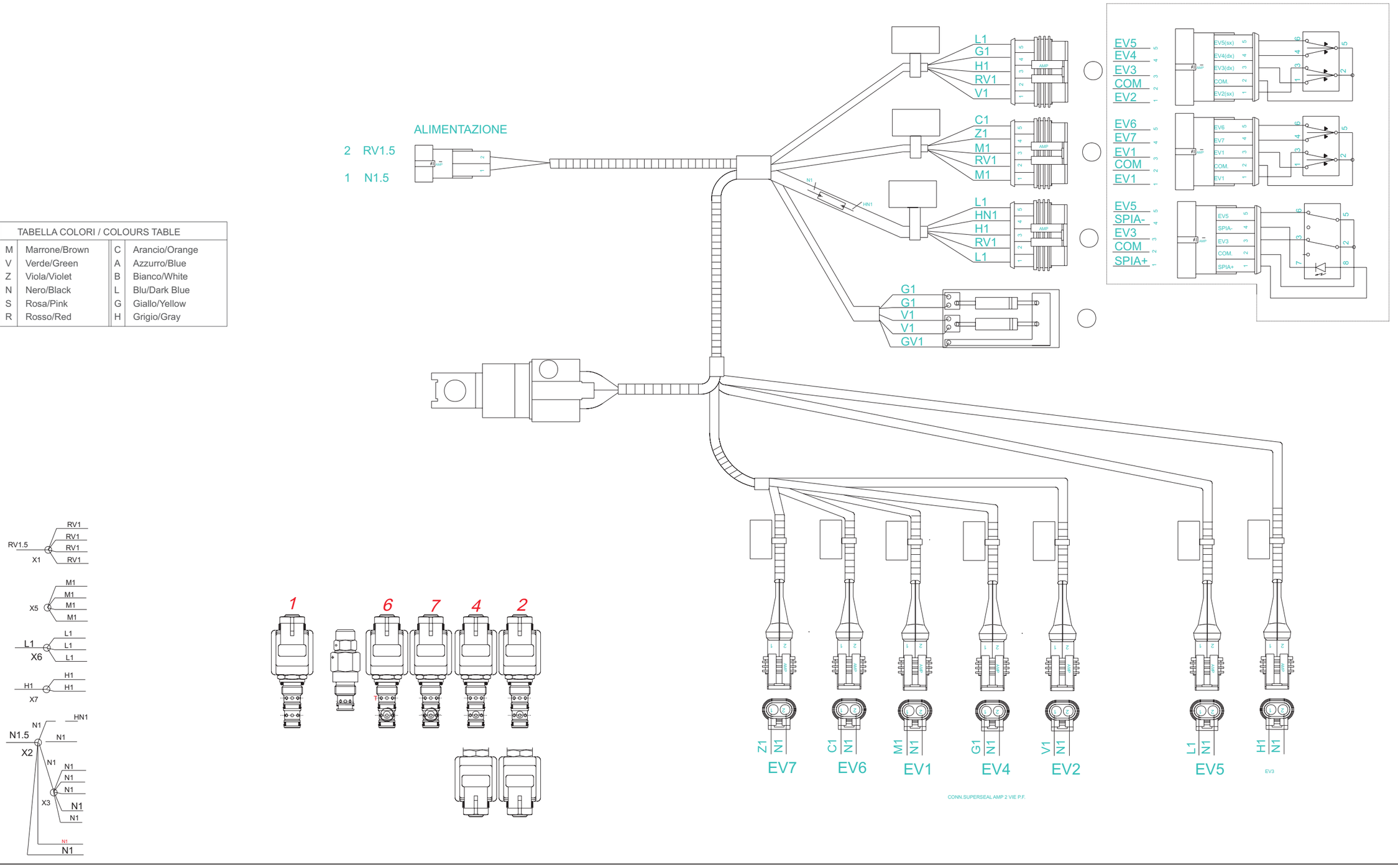
P3 — POSIZIONE

1	2	3
N1	BN1	M1

S — SFORZO

1	2	3
N1	B1	M1

TABELLA COLORI / COLOURS TABLE			
M	Marrone/Brown	C	Arancio/Orange
V	Verde/Green	A	Azzurro/Blue
Z	Viola/Violet	B	Bianco/White
N	Nero/Black	L	Blu/Dark Blue
S	Rosa/Pink	G	Giallo/Yellow
R	Rosso/Red	H	Grigio/Gray



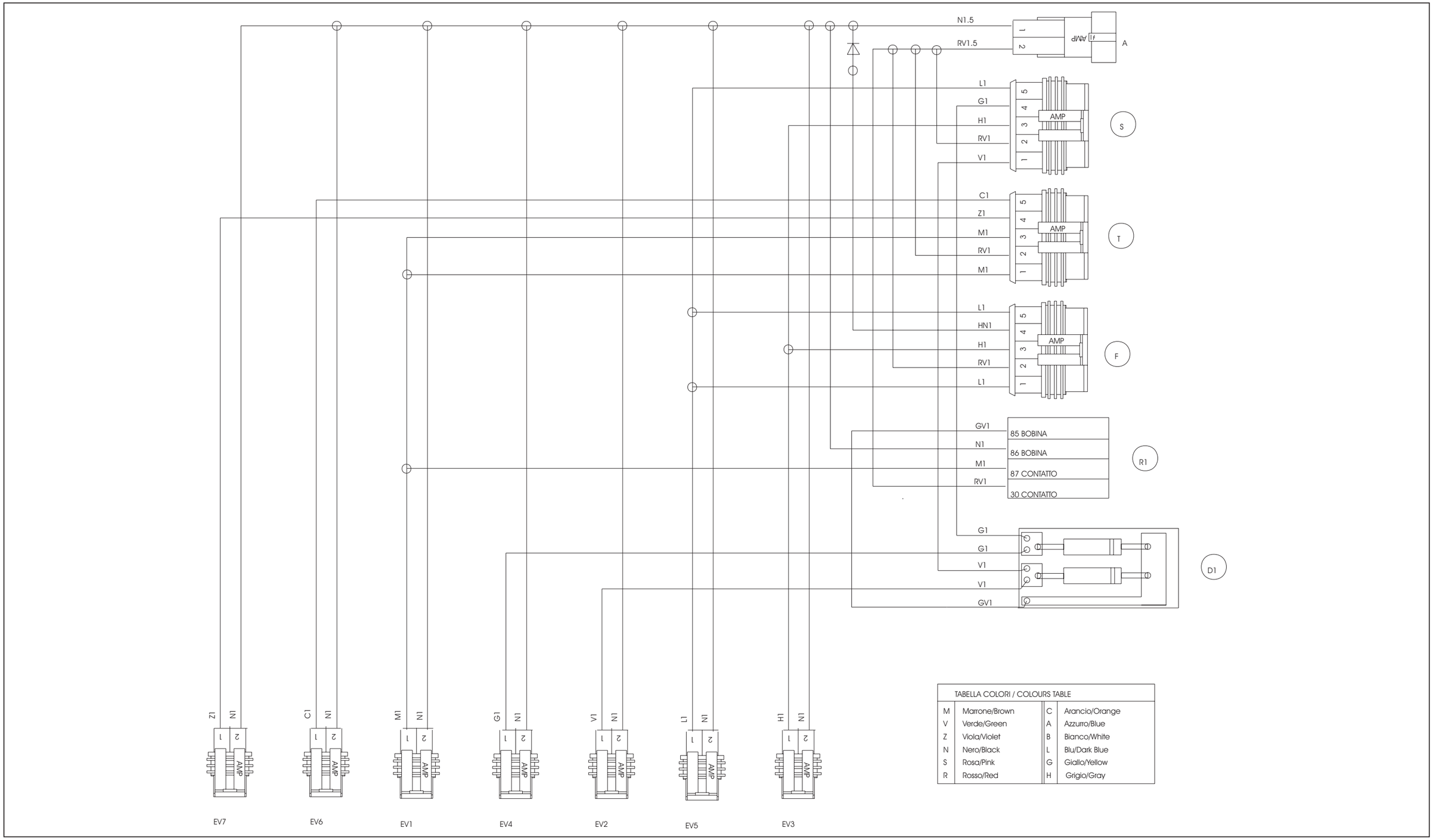


TABELLA COLORI / COLOURS TABLE			
M	Marrone/Brown	C	Arancio/Orange
V	Verde/Green	A	Azzurro/Blue
Z	Viola/Violet	B	Bianco/White
N	Nero/Black	L	Blu/Dark Blue
S	Rosa/Pink	G	Giallo/Yellow
R	Rosso/Red	H	Grigio/Gray

TRACTOR WIRING — DIAGRAM 25 - STABILISERS AND RH LIFT ROD ADJUSTMENT WIRING — **012.0101.4 dwg. 2**

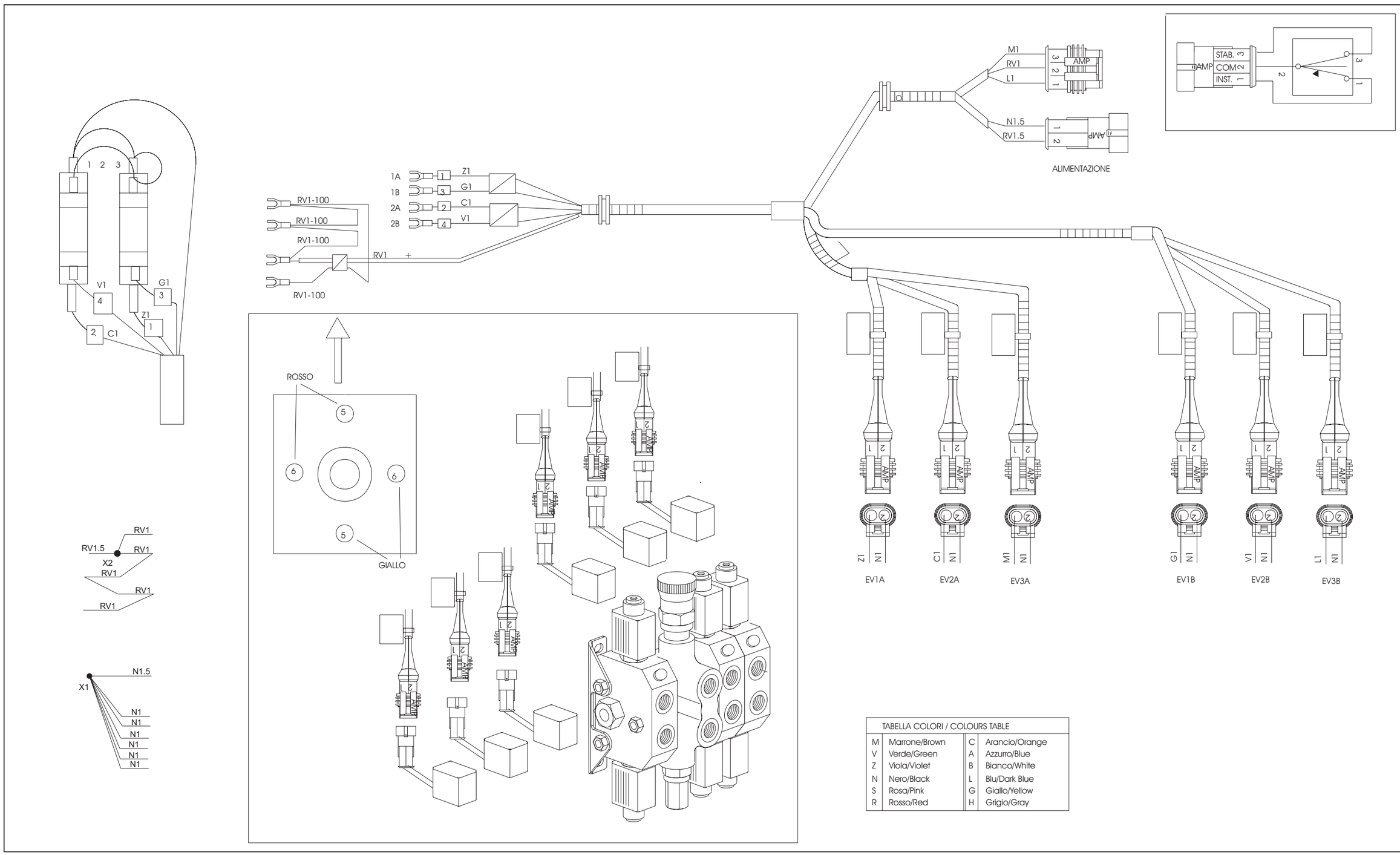


TABELLA COLORI / COLOURS TABLE			
M	Marrone/Brown	C	Arancio/Orange
V	Verde/Green	A	Azzurro/Blue
Z	Viola/Violet	B	Bianco/White
N	Nero/Black	L	Blu/Dark Blue
S	Rosa/Pink	G	Giallo/Yellow
R	Rosso/Red	H	Grigio/Gray

TRACTOR WIRING — DIAGRAM 26 - WIRING FOR FRONT CONTROL VALVES WITH ELECTRO-HYDRAULIC CONTROL — **012.0102.4 dwg. 1**

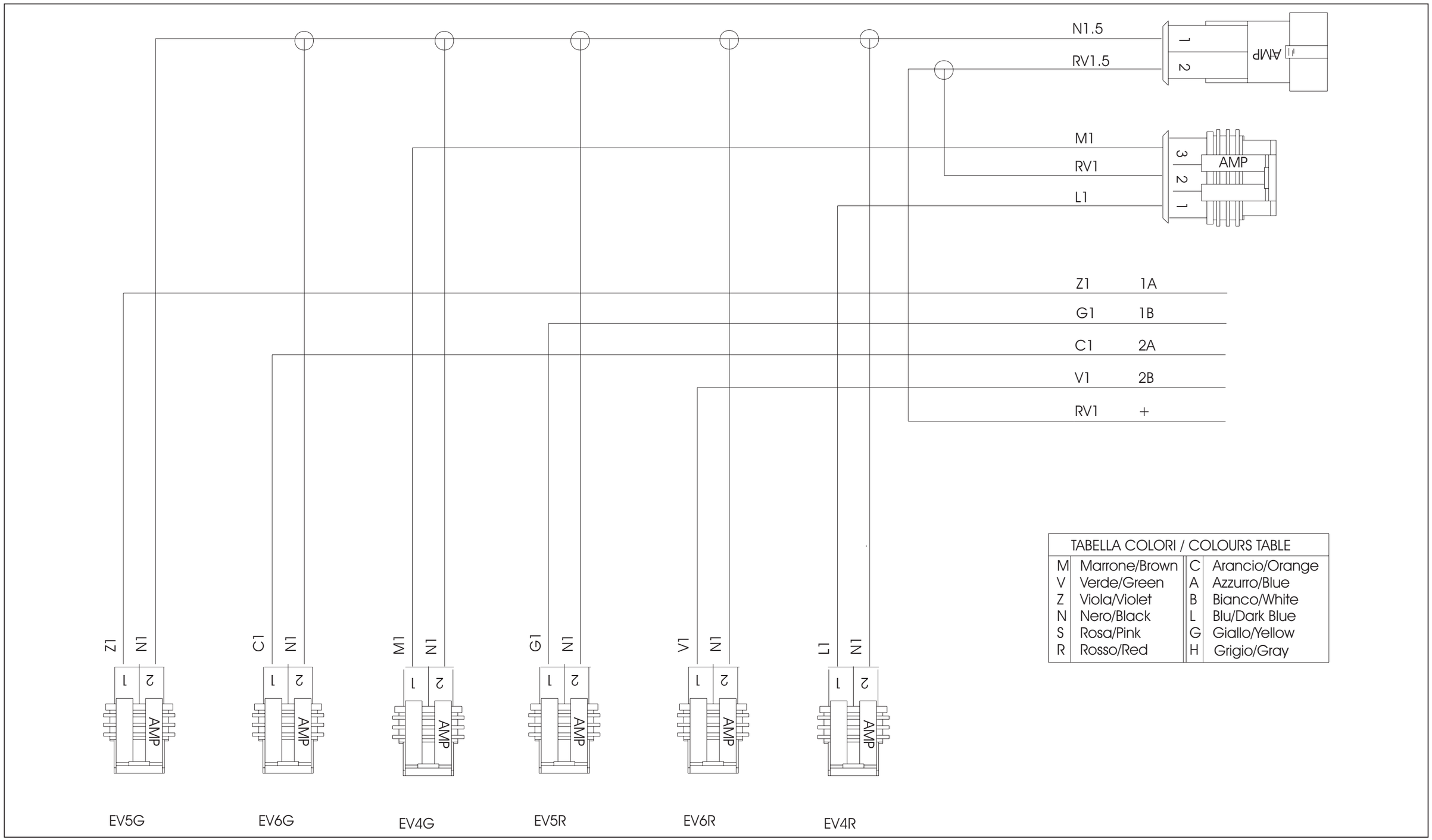
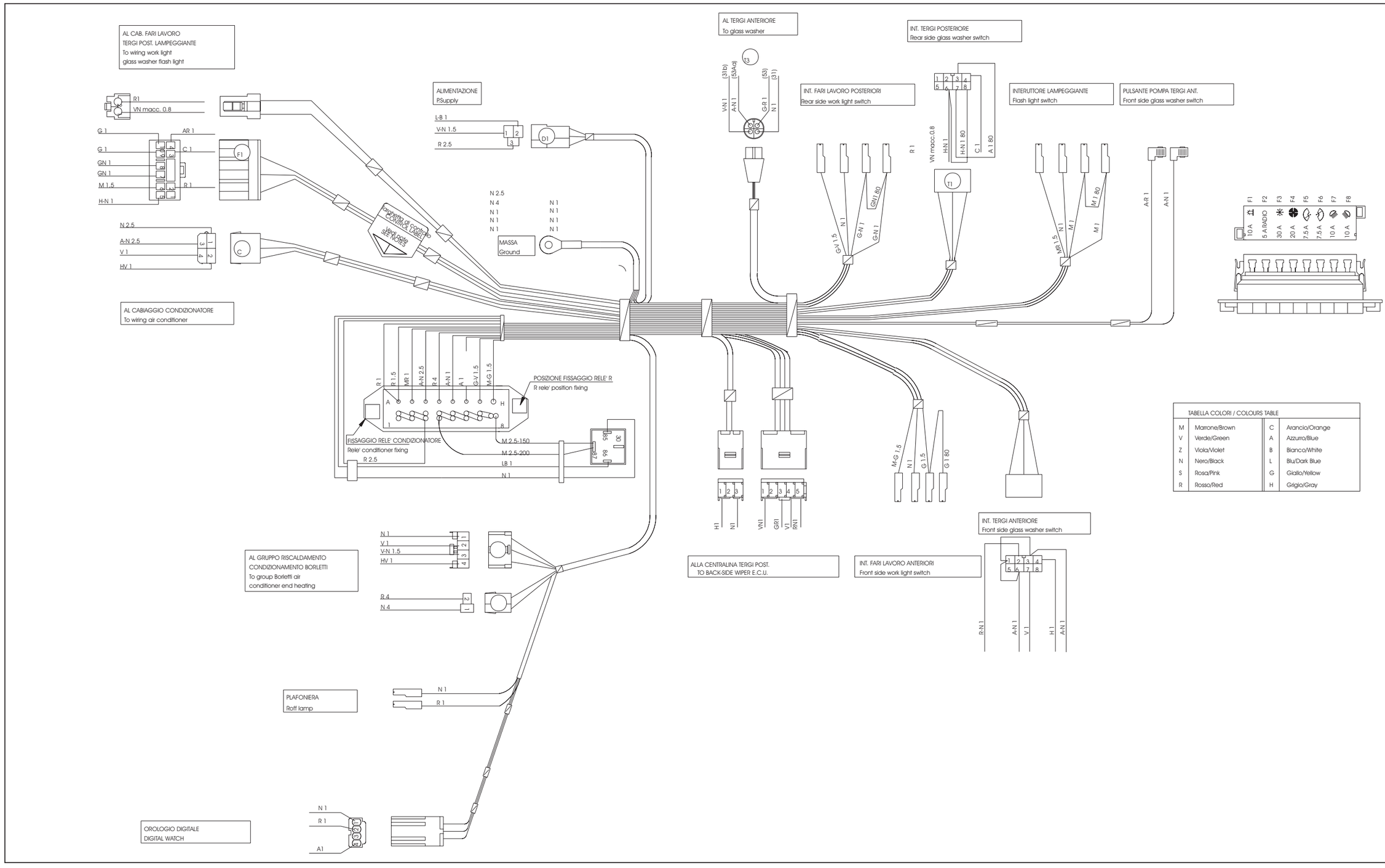


TABELLA COLORI / COLOURS TABLE			
M	Marrone/Brown	C	Arancio/Orange
V	Verde/Green	A	Azzurro/Blue
Z	Viola/Violet	B	Bianco/White
N	Nero/Black	L	Blu/Dark Blue
S	Rosa/Pink	G	Giallo/Yellow
R	Rosso/Red	H	Grigio/Gray

TRACTOR WIRING

DIAGRAM 27 - WIRING FOR FRONT CONTROL VALVES WITH ELECTRO-HYDRAULIC CONTROL

012.0102.4 dwg. 2



TRACTOR WIRING DIAGRAM 28 - CAB ROOF WIRING **010.7262.4/20 dwg. 1**

TABELLA COLORI / COLOURS TABLE			
M	Marrone/Brown	C	Arancio/Orange
V	Verde/Green	A	Azzurro/Blue
Z	Viola/Violet	B	Bianco/White
N	Nero/Black	L	Blu/Dark Blue
S	Rosa/Pink	G	Gialla/Yellow
R	Rossa/Red	H	Grigio/Gray

INT. FARI LAVORO ANTERIORI
Front side work light switch

INT. FARI LAVORO POSTERIORI
Rear side work light switch

INTERUTTORE LAMPEGGIANTE
Flash light switch

INT. TERGI ANTERIORE
Front side glass washer switch

PULSANTE POMPA TERGI ANT.
Front side glass washer switch

INT. TERGI POSTERIORE
Rear side glass washer switch

CENTRALINA TERGI
Elect. tergi

OROLOGIO DIGITALE
DIGITAL WATCH

AL CAB. FARI LAVORO
TERGI POST. LAMPEGGIANTE
To wiring work light
glass washer flash light

AL CABIAGGIO CONDIZIONATORE
To wiring air conditioner

AL GRUPPO RISCALDAMENTO
CONDIZIONAMENTO BORLETTI
To group Borletti air
conditioner end heating

AL TERGI ANTERIORE
To glass washer

AL CABLAGGIO ALIMENTAZIONE
To power supply

MASSA
Ground

PLAFONIERA
Roff lamp

RELE'

TRACTOR WIRING

DIAGRAM 29 - CAB ROOF WIRING

010.7262.4/20 dwg. 2

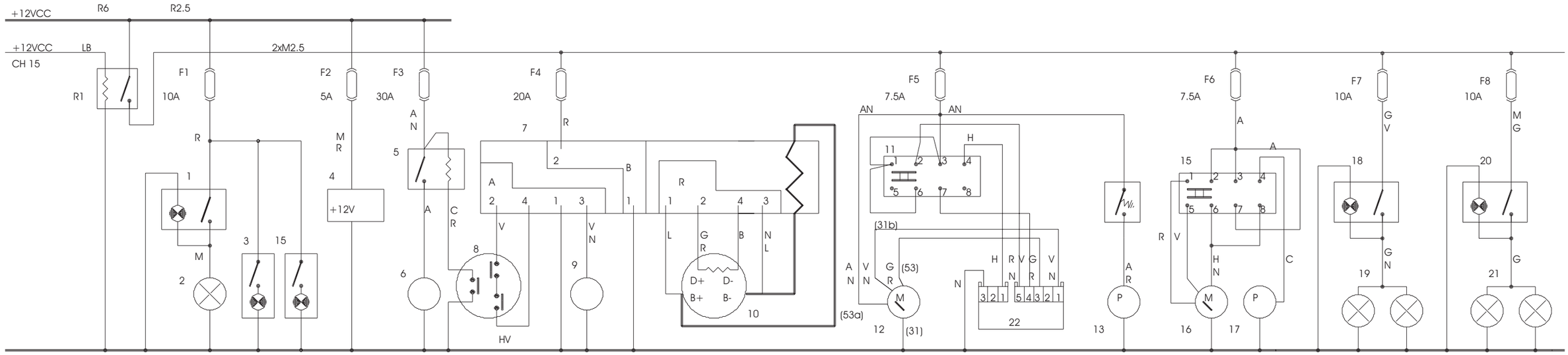


TABELLA COLORI / COLOURS TABLE			
M	Marrone/Brown	C	Arancio/Orange
V	Verde/Green	A	Azzurro/Blue
Z	Viola/Violet	B	Bianco/White
N	Nero/Black	L	Blu/Dark Blue
S	Rosa/Pink	G	Giallo/Yellow
R	Rosso/Red	H	Grigio/Gray

TRACTOR WIRING — DIAGRAM 30 - CAB ROOF WIRING — **010.7262.4/20 dwg. 3**

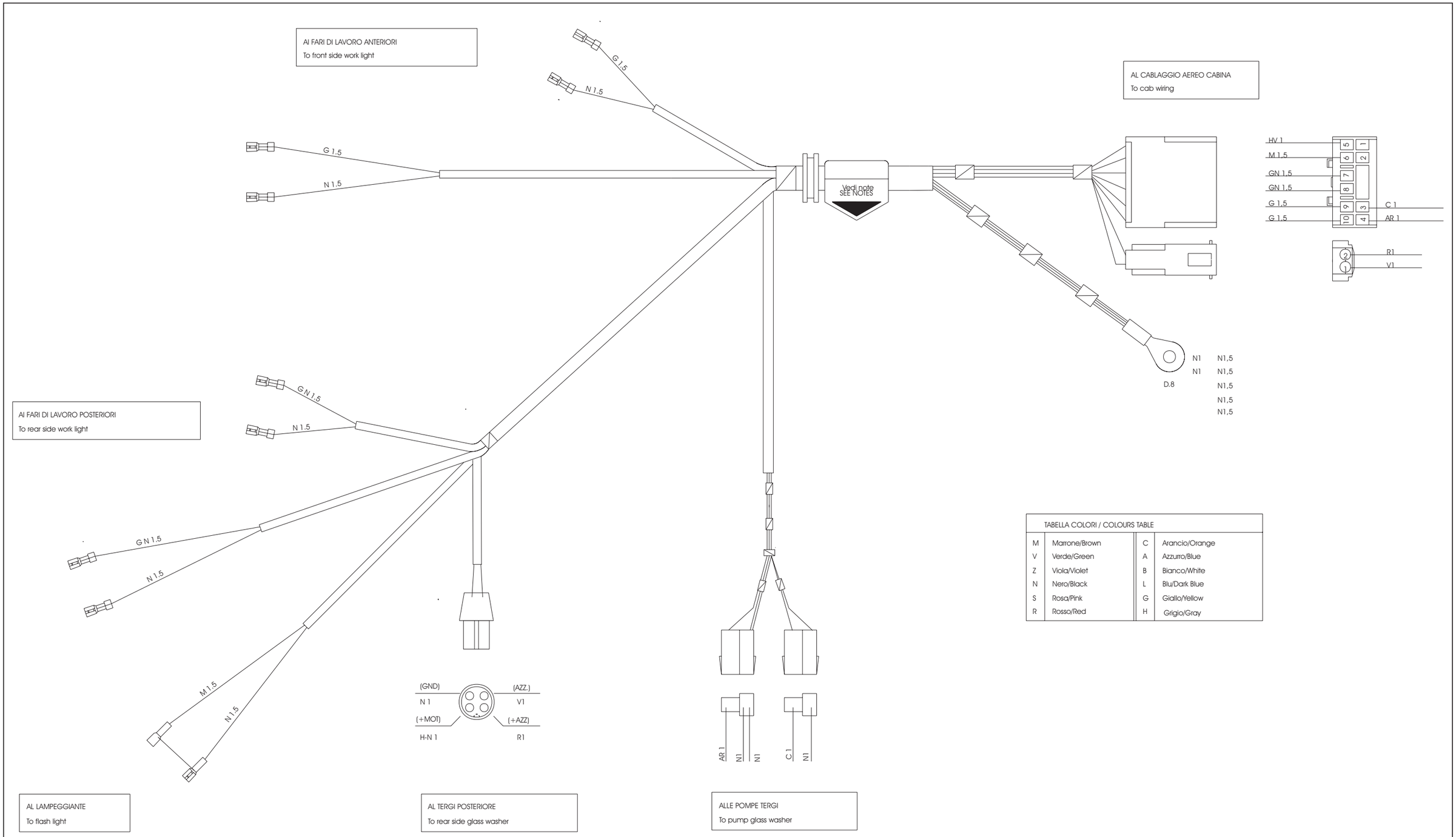


TABELLA COLORI / COLOURS TABLE

M	Marrone/Brown	C	Arancio/Orange
V	Verde/Green	A	Azzurro/Blue
Z	Viola/Violet	B	Bianco/White
N	Nero/Black	L	Blu/Dark Blue
S	Rosa/Pink	G	Giallo/Yellow
R	Rosso/Red	H	Grigio/Gray

TRACTOR WIRING — DIAGRAM 31 - WORK LIGHTS AND SCREEN WIPER WIRING — **010.7263.4/20 dwg. 1**

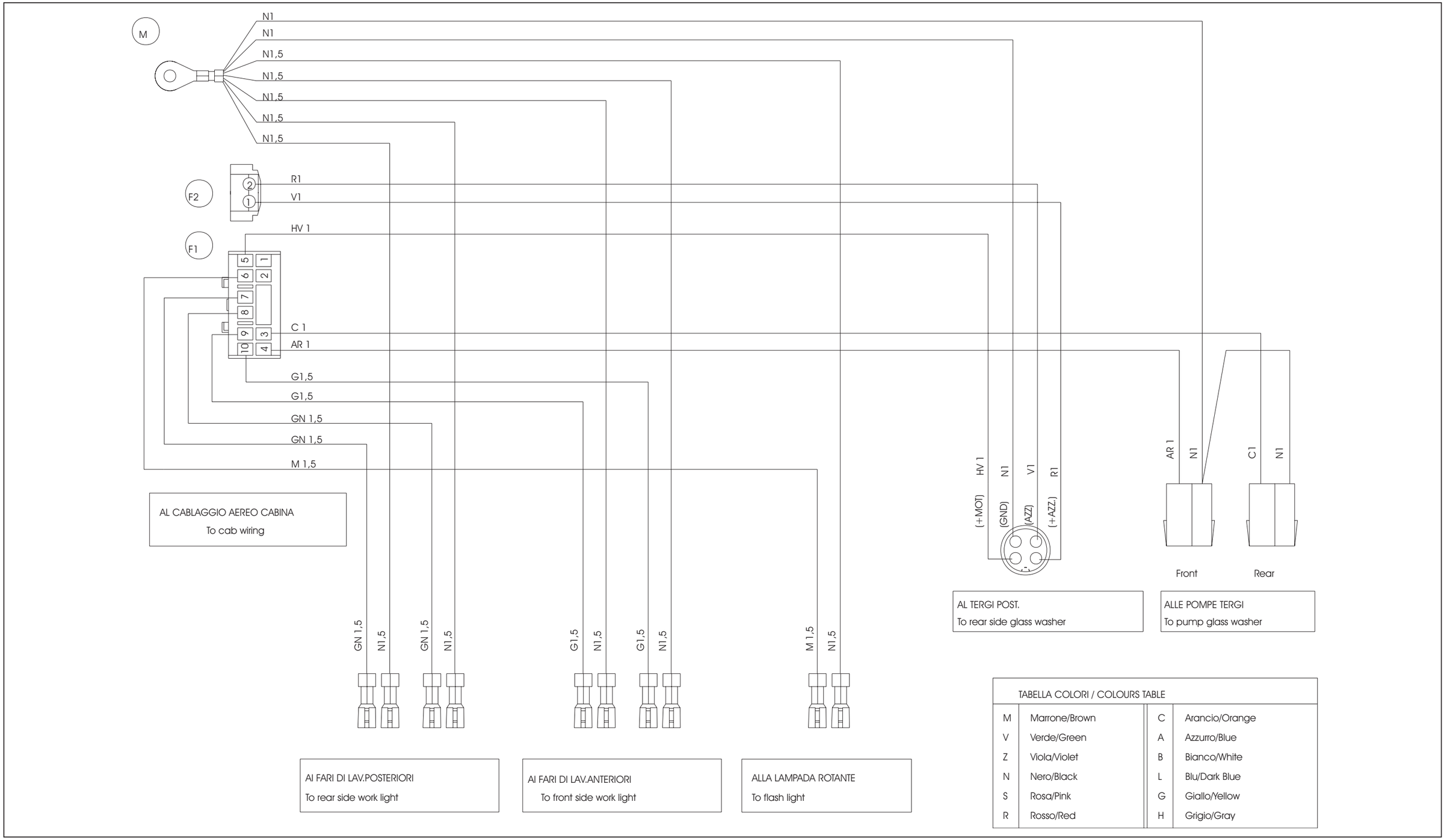
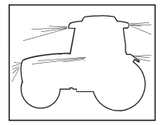


TABELLA COLORI / COLOURS TABLE

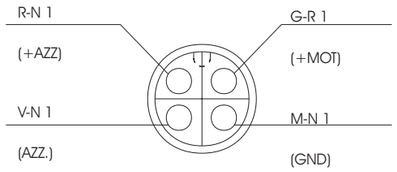
M	Marrone/Brown	C	Arancio/Orange
V	Verde/Green	A	Azzurro/Blue
Z	Viola/Violet	B	Bianco/White
N	Nero/Black	L	Blu/Dark Blue
S	Rosa/Pink	G	Giallo/Yellow
R	Rosso/Red	H	Grigio/Gray

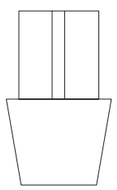
TRACTOR WIRING — DIAGRAM 32 - WORK LIGHTS AND SCREEN WIPER WIRING — **010.7263.4/20 dwg. 2**



FORNIRE SCIOLTI
Free (not connect)

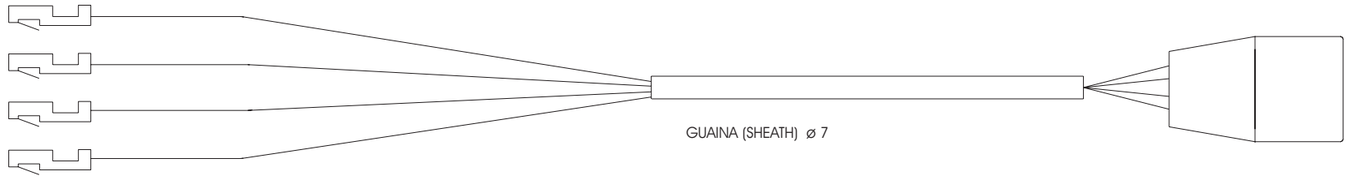
CONN. AMP M-L 4 VIE P.F.
COD. AMP 0-925076-0





AL TERGI ANTERIORE
To front glass wiper

CA. AMP163300-2



P.M. M-L 4VIE-WIRES
AMP 0-925075-0
TERM. AMP. 163301-2

AL CABLAGGIO
AEREO
To cab wiring
0.010.7262.4/20

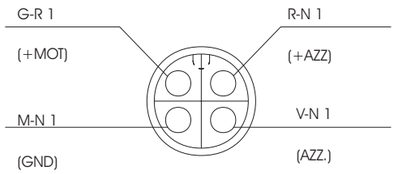
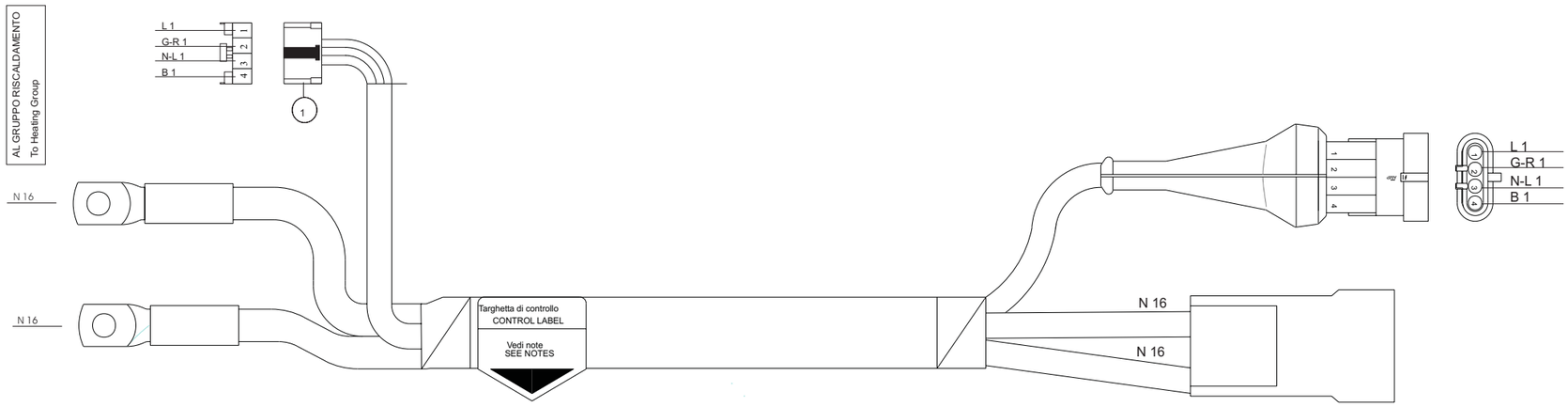


TABELLA COLORI / COLOURS TABLE			
M	Marrone/Brown	C	Arancio/Orange
V	Verde/Green	A	Azzurro/Blue
Z	Viola/Violet	B	Bianco/White
N	Nero/Black	L	Blu/Dark Blue
S	Rosa/Pink	G	Giallo/Yellow
R	Rosso/Red	H	Grigio/Gray

CORRISPONDENZA TRA CAVI CABLAGGIO E CAVI TERGI ANTERIORE

CABLAGGIO WIRES			MOTORINO TERGI Wiper
G-R	(+MOT)	G	
M-N	(GND)	M	
R-N	(+AZZ)	V-N	
V-N	(AZZ.)	N	

TRACTOR WIRING — DIAGRAM 33 - WINDSCREEN WIPER WIRING — **010.7264.4/10**



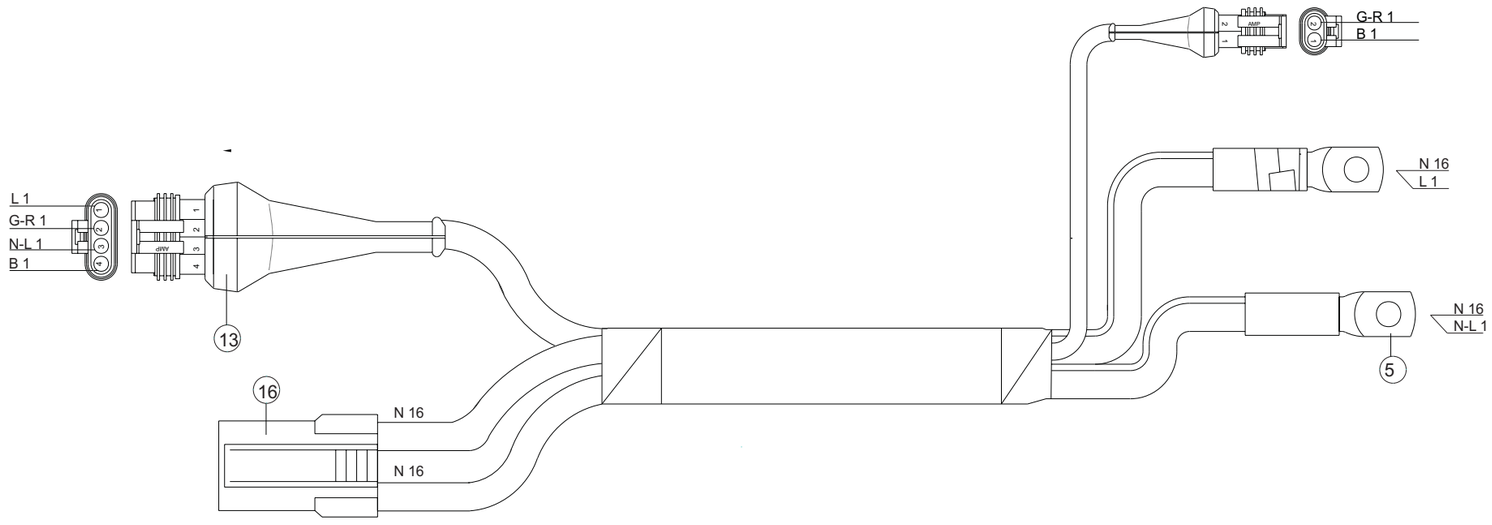
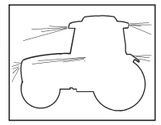


TABELLA COLORI			
M	Marrone/Brown	C	Arancio/Orange
V	Verde/Green	A	Azzurro/Blue
Z	Viola/Violet	B	Bianco/White
N	Nero/Black	L	Blu/Dark Blue
S	Rosa/Pink	G	Giallo/Yellow
R	Rosso/Red	H	Grigio/Gray

TRACTOR WIRING DIAGRAM 34 - ELECTRIC HEATING WIRING **010.7265.4**



AL CABLAGGIO CABINA
To cab wiring

L-B 1
V-N 1.5
R 2.5

2 1 3

R 6
R 2.5

AMP 1 2

L B 1
V-N 1.5

1 2

Targhetta di controllo
CONTROL LABEL

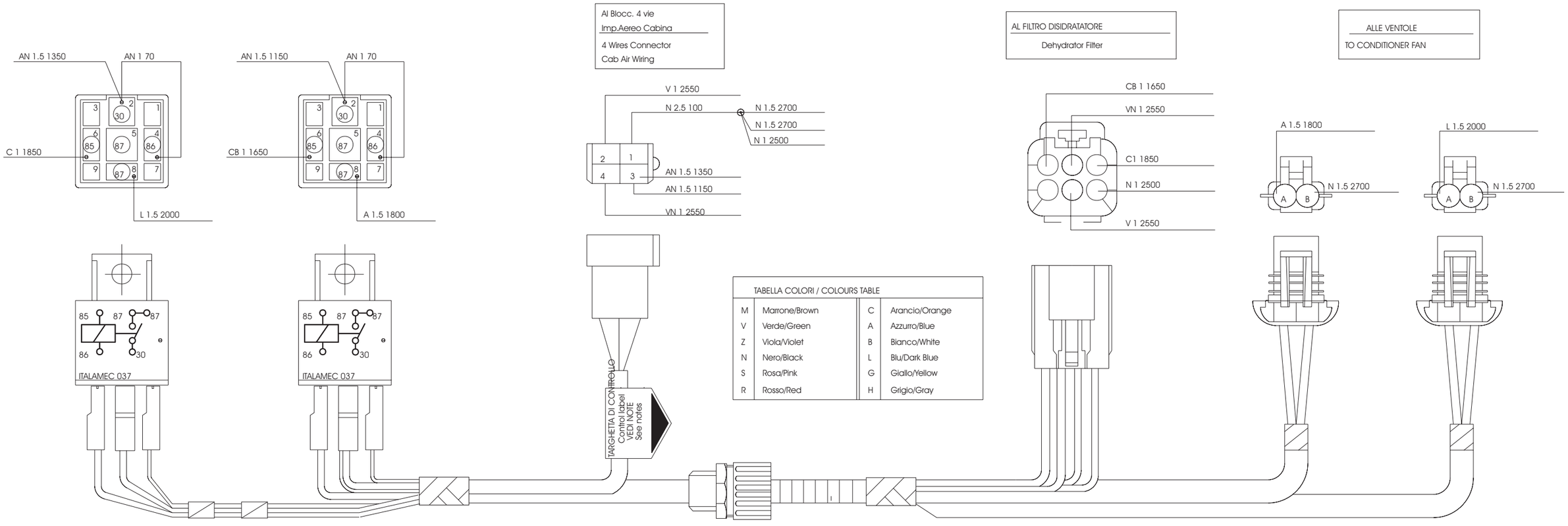
Vedi note
SEE NOTES

AL CABLAGGIO SOTTOSTRUTTURA
To sottostrutture wiring

AL POSITIVO
To positive

AL RELE'
To rele'

TABELLA COLORI / COLOURS TABLE			
M	Marrone/Brown	C	Arancio/Orange
V	Verde/Green	A	Azzurro/Blue
Z	Viola/Violet	B	Bianco/White
N	Nero/Black	L	Blu/Dark Blue
S	Rosa/Pink	G	Giallo/Yellow
R	Rosso/Red	H	Grigio/Gray



TRACTOR WIRING — DIAGRAM 36 - CAB AIR CONDITIONING WIRING — **011.0644.3/20 dwg. 1**

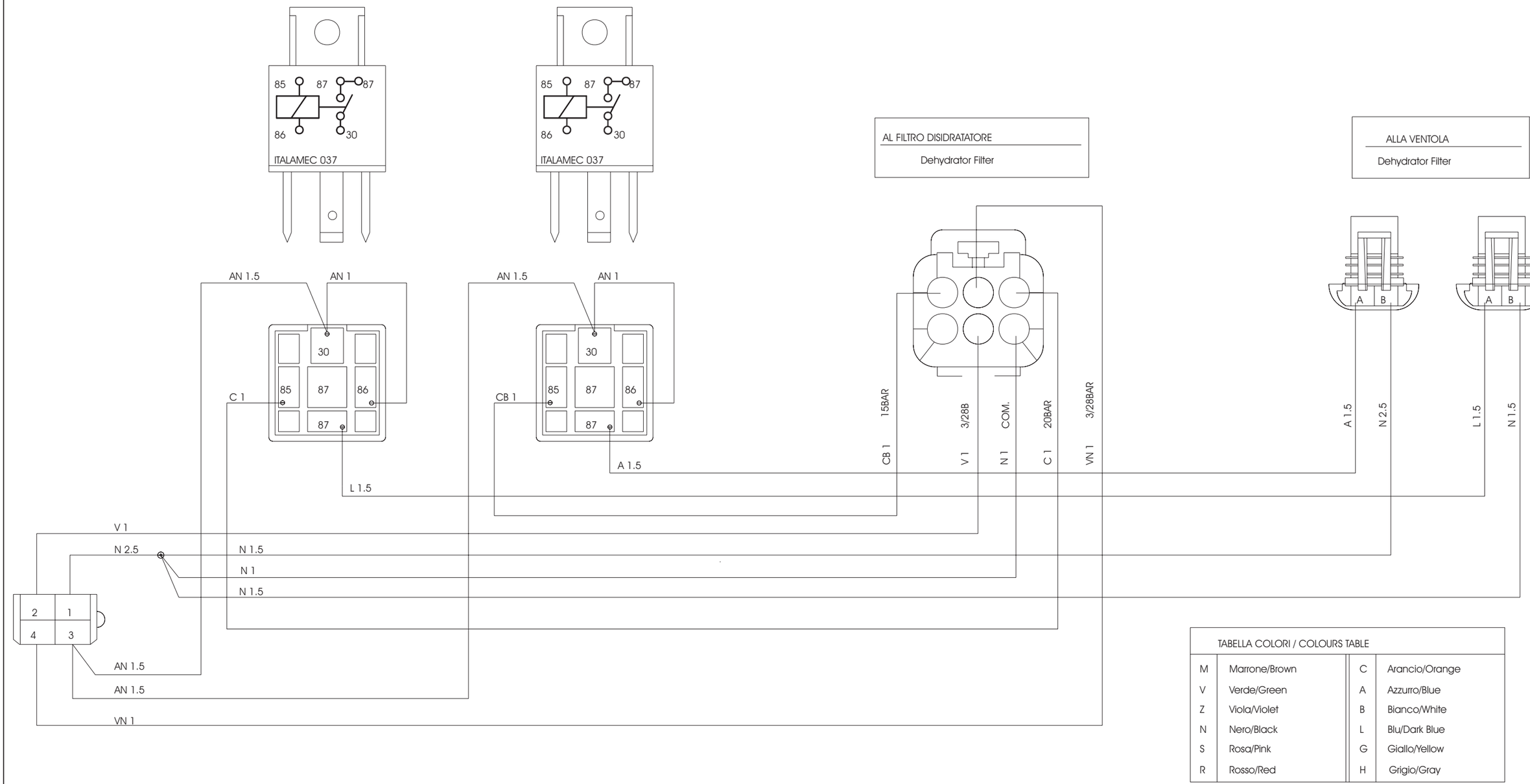
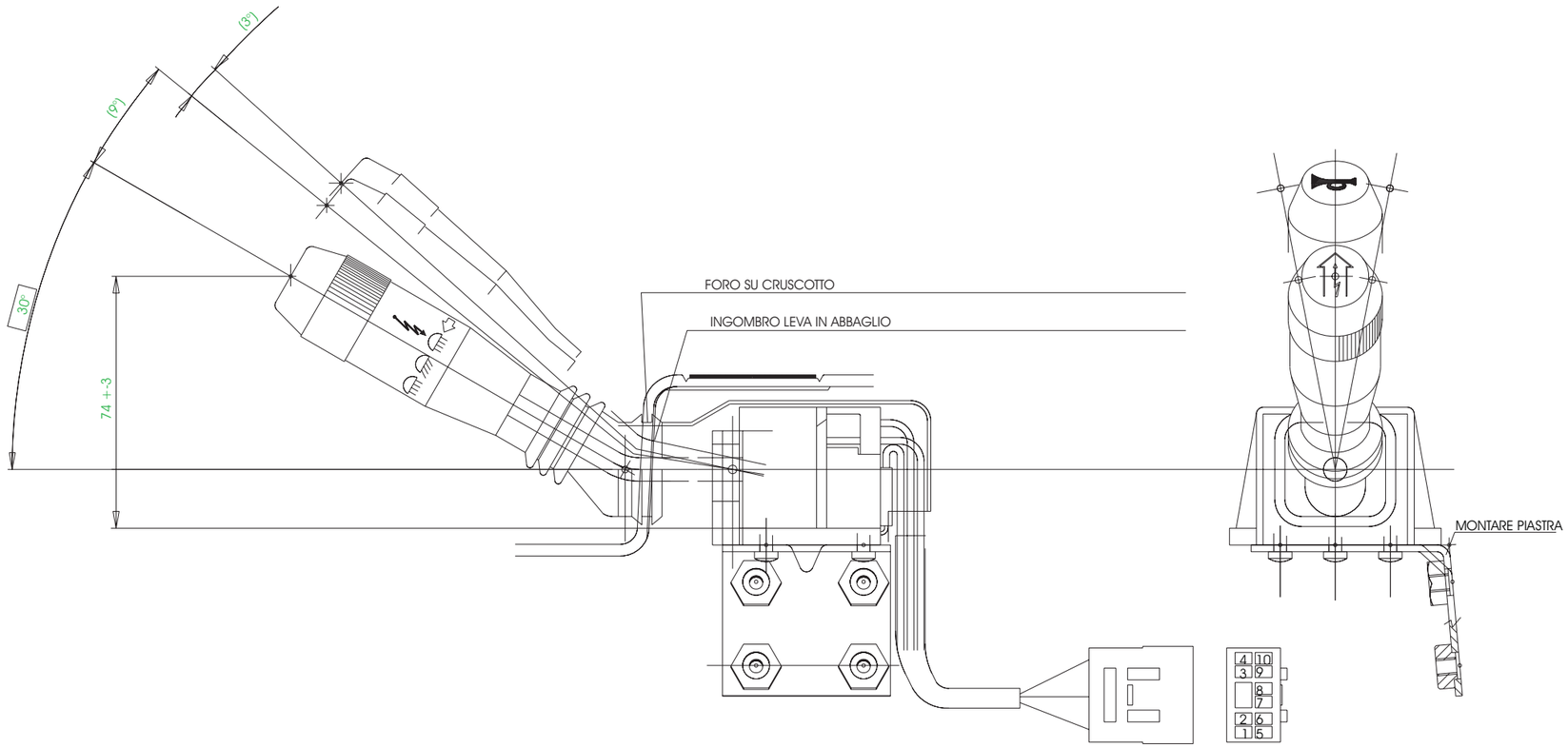
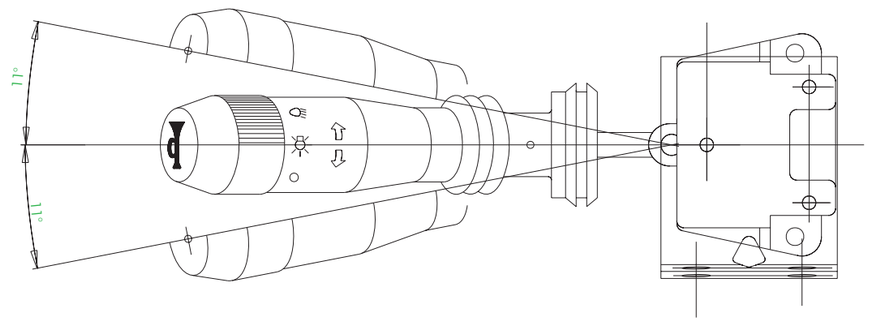


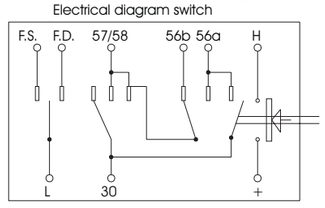
TABELLA COLORI / COLOURS TABLE			
M	Marrone/Brown	C	Arancio/Orange
V	Verde/Green	A	Azzurro/Blue
Z	Viola/Violet	B	Bianco/White
N	Nero/Black	L	Blu/Dark Blue
S	Rosa/Pink	G	Giallo/Yellow
R	Rosso/Red	H	Grigio/Gray

TRACTOR WIRING — DIAGRAM 37 - CAB AIR CONDITIONING WIRING — **011.0644.3/20 dwg. 2**

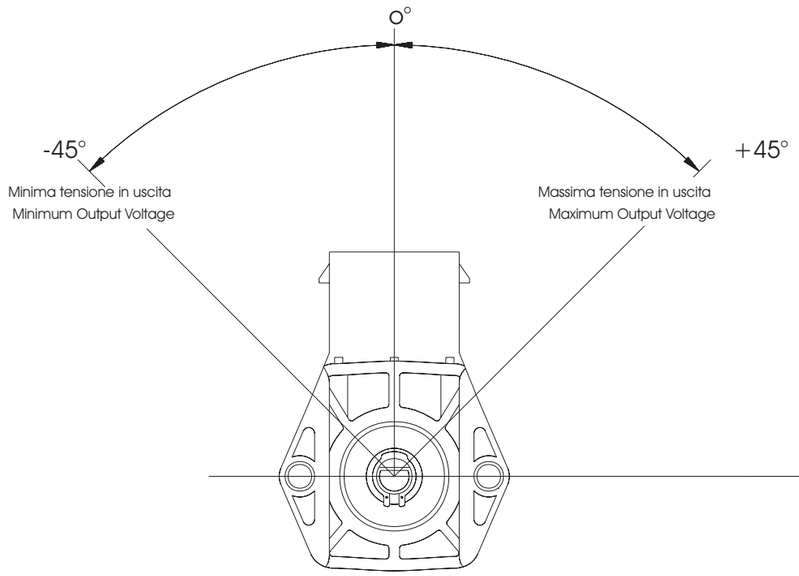


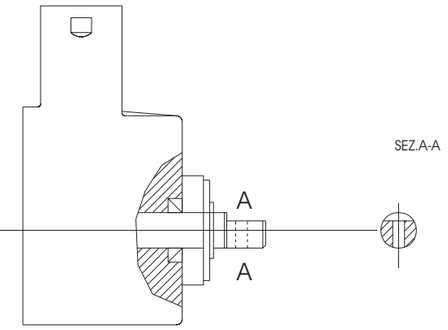


SCHEMA ELETTRICO DEVIOGUIDA
Electrical diagram switch



COMPONENTS DIAGRAM 38 - STEERING COLUMN SWITCH UNIT **010.1173.4**

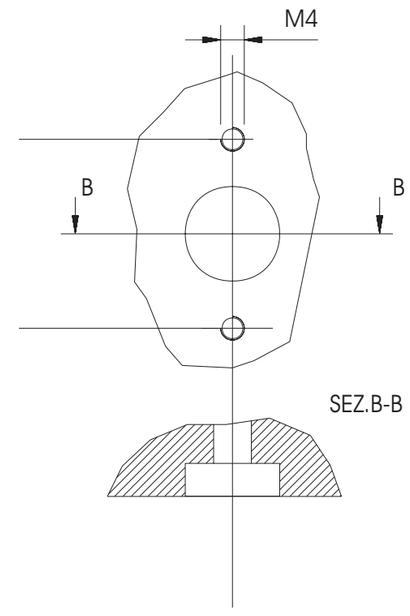


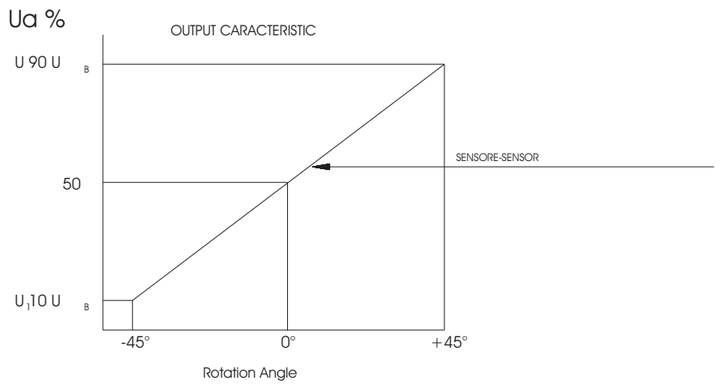


	SENSORE 1
+	POS.1
-	POS.2
OUT	POS.4

SEDE SI APPLICAZIONE

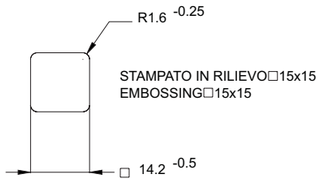
SITE OF APPLICATION





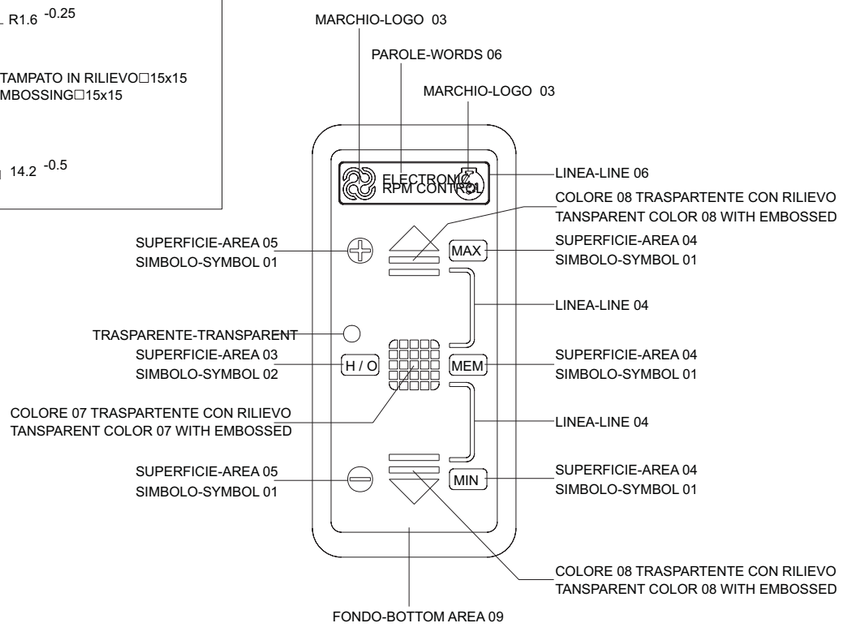
TERMOFORMATURA TASTI

KEYS EMBOSSING



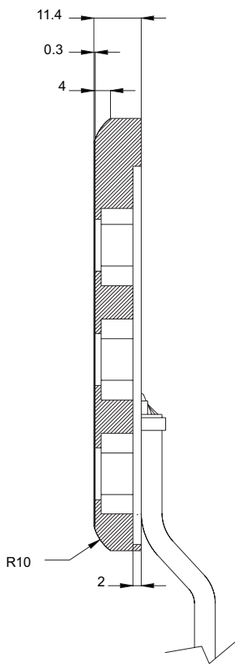
VISTA FRONTALE

FRONT VIEW



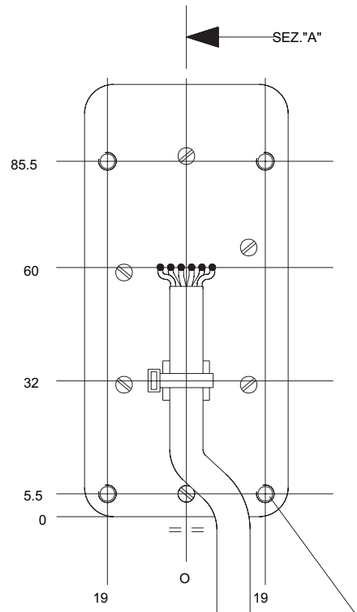
SEZIONE "A"

SECTION "A"



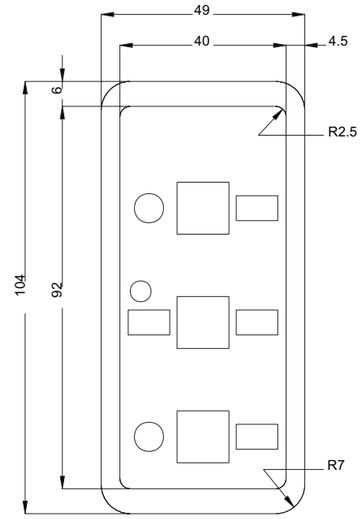
VISTA POSTERIORE

BACK-SIDE VIEW

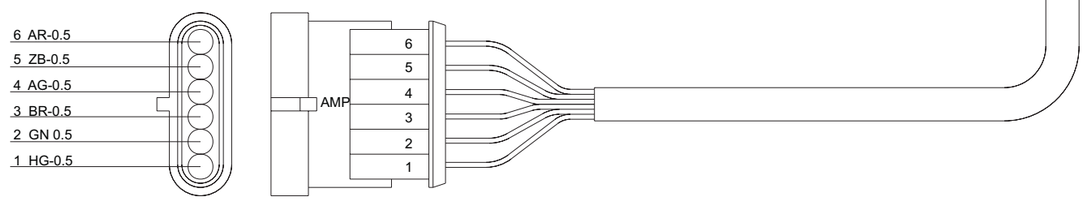


VISTA FRONTALE SENZA MASCHERINA SERIGRAFATA

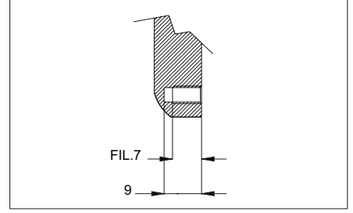
FRONT VIEW WITHOUT SERIGRAFHATED MASK



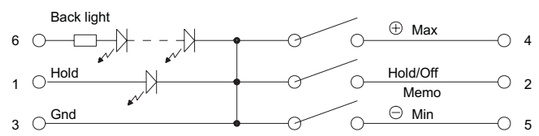
CODICE SIMBOLI SIMBOLS CODE	SPECIFICA COLORI - COLOURS SPECIFICATION
09	PANTONE COOL GRAY 11
08	RAL 6024 TRAFFIC GREEN TRANSPARENT
07	RAL 3020 TRAFFIC RED TRANSPARENT
06	RAL 9010 PURE WHITE
05	RAL 6024 TRAFFIC GREEN
04	RAL 1016 SULFUR YELLOW
03	RAL 3020 TRAFFIC RED
02	RAL 9010 PURE WHITE
01	RAL 9017 TRAFFIC BLACK



N.4 FORI M 4 PER FISSAGGIO TASTIERA
N.4 HOLES M 4 TO FIX KEYBOARD



SCHEMA ELETTRICO - ELECTRICAL DIAGRAM



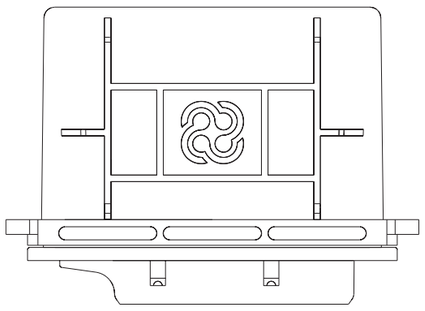
USARE CAVI "FASE III" - USER CABLES "FASE III"

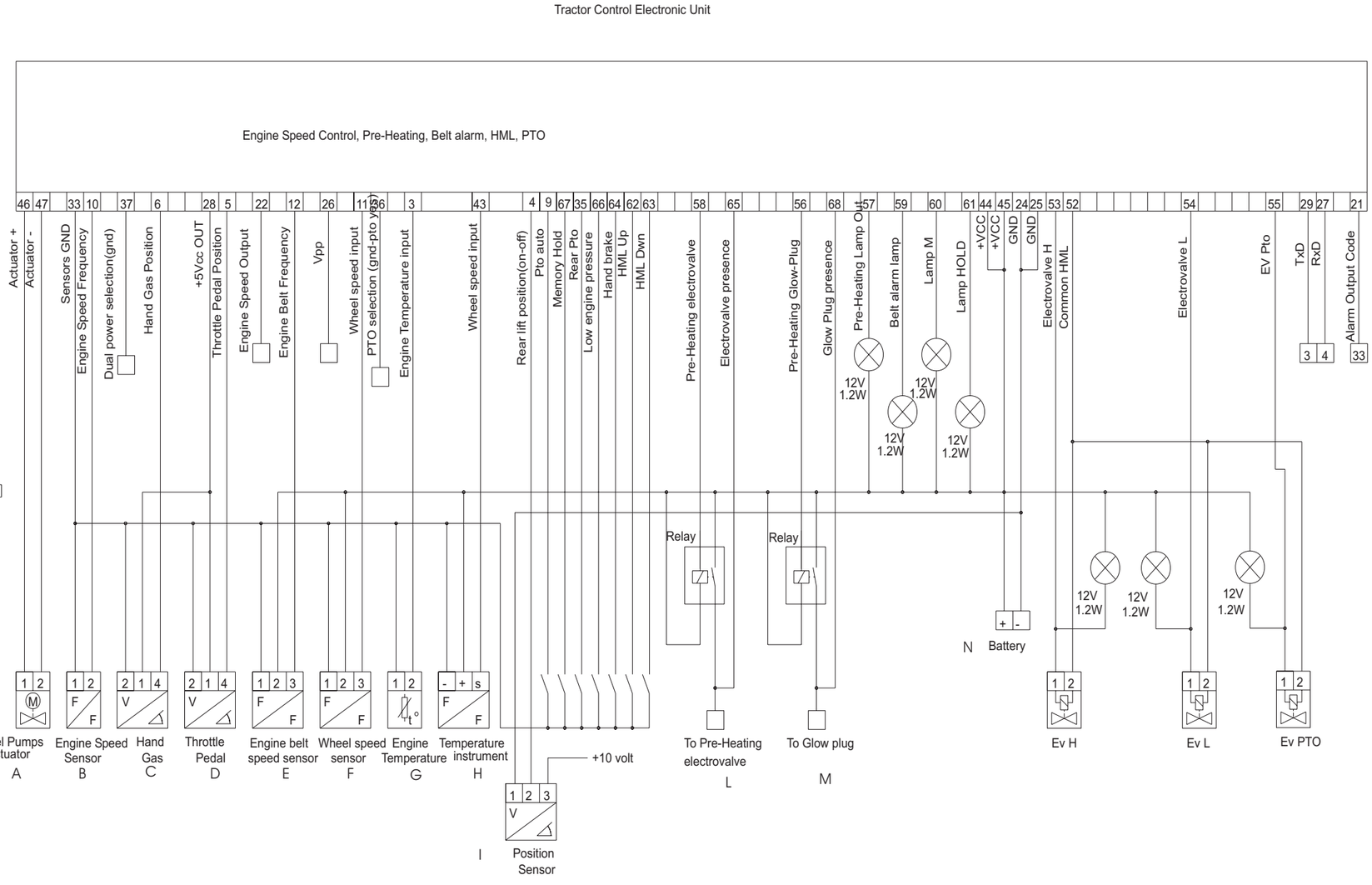
TABELLA COLORI / COLOURS TABLE

M	Marrone/Brown	C	Arancio/Orange
V	Verde/Green	A	Azzurro/Blue
Z	Viola/Violet	B	Bianco/White
N	Nero/Black	L	Blu/Dark Blue
S	Rosa/Pink	G	Giallo/Yellow
R	Rosso/Red	H	Grigio/Gray

46 Alternator+
47 Alternator-
33 Sensors GND
10 Engine speed frequency
37 Dual Power selection (GND)
6 Hand throttle position
28 +5 Vdc OUT
5 Accelerator pedal position
22 Engine speed output
12 Belts rotation frequency
26 Vpp
11 Wheel speed input
36 P.T.O. selection (P.T.O. GND)
3 Engine temperature input
43 Wheel speed input
4 Rear lift (ON-OFF)
9 Auto P.T.O.
67 Memorised engine speed
35 Rear P.T.O.
66 Low engine oil pressure
64 Parking brake
62 HML - UP
63 HML - DOWN
58 Preheating solenoid
65 Solenoid
56 Preheating Glow-plug
68 Glow-plug solenoid
57 Preheating light OUT
59 Broken belt light
60 M light
61 Hold light
44 +Vdc
45 +Vdc
24 GND
25 GND
53 H solenoid
52 HML control
54 L solenoid
55 P.T.O. solenoid
29 TxD
27 RxD
21 Alarm code output

A - Injection pump actuator
B - Engine speed sensor
C - Hand throttle
D - Accelerator pedal
E - Broken belt alarm sensor
F - Wheel speed sensor
G - Engine temperature
H - Temperature unit
I - Position sensor
L - Solenoid
M - Glow plug
N - battery

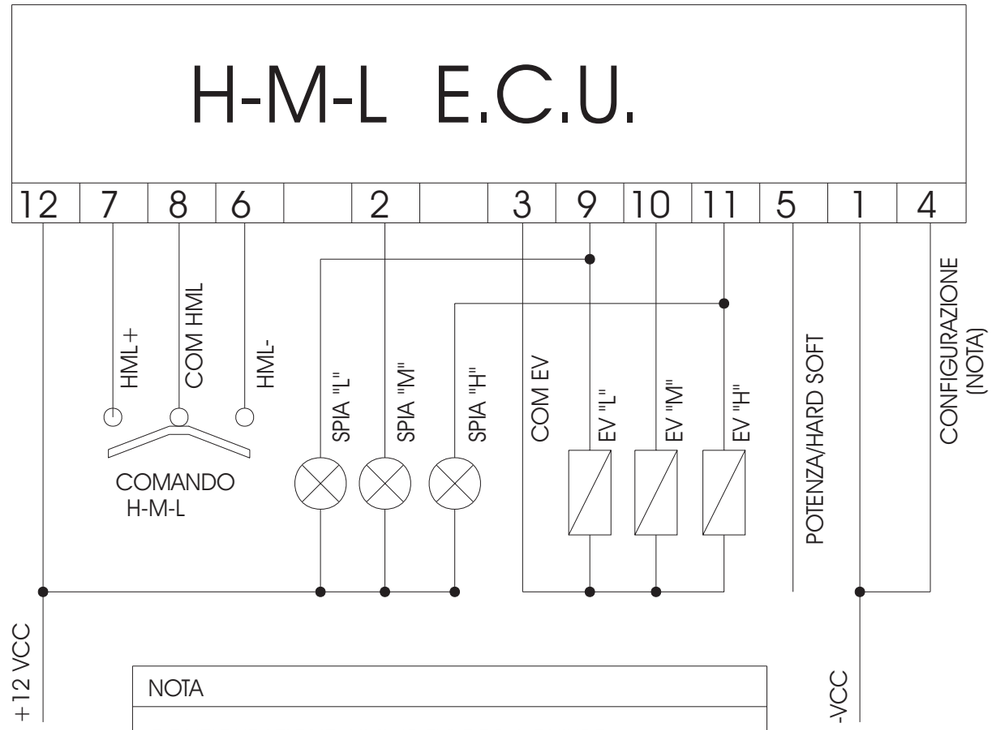




COMPONENTS

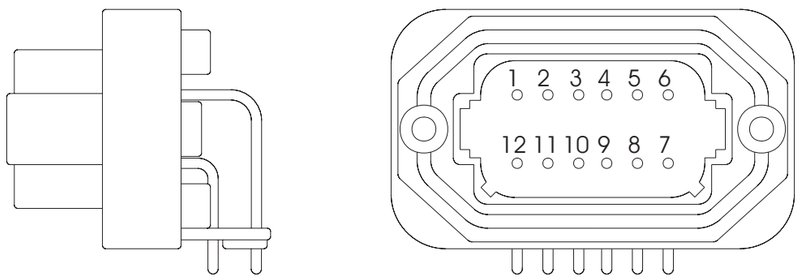
DIAGRAM 41 - ENGINE CONTROL UNIT WITH BELTS ALARM, HML AND PREHEATING **011.7027.4/10** (011.4353.4/10 for some models only)
ENGINE CONTROL UNIT WITH BELTS ALARM, HML AND PREHEATING (010.7260. 4 for 30 km/h 1st version)
ENGINE CONTROL UNIT WITH BELTS ALARM, HML AND PREHEATING (010.7261. 4 for 40 km/h 1st version)

PIN OUT

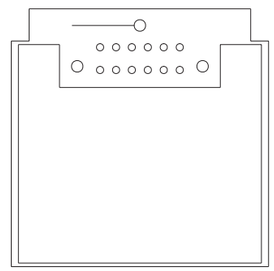


NOTA

CONFIGURAZIONE: HML (2 EV) POS.4 CHIUSO.
CONFIGURAZIONE: HML (3 EV) POS.4 APERTO.

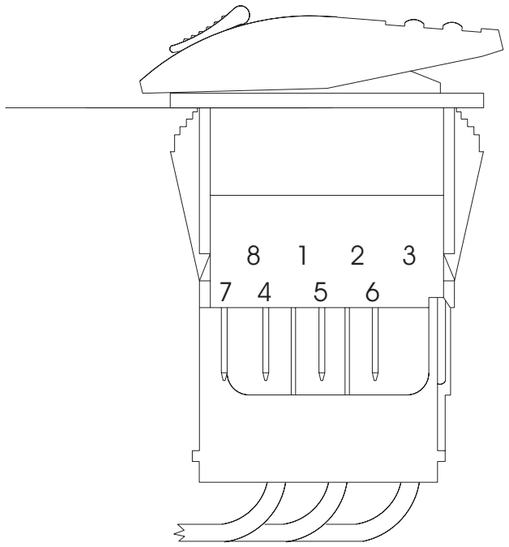


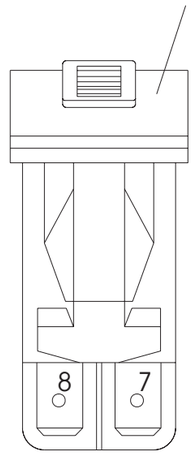
CONNETTORE



COMPONENTS DIAGRAM 42 - HML CONTROL UNIT **010.8869.4/20**

Sp PANNELLO (Panel's thinkness) : 1 : 3 mm





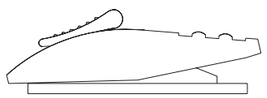
VETRINO BIANCO (White lens)
SERIGRAFIA NERA (Black serigraphy)

SCHEMA ELETTRICO
Electrical draw

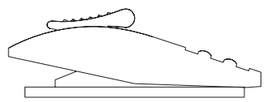
NB:SCHEMA RIF. A POS 1
NB:Electrical draw for pos.1

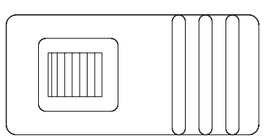
BLOCCO IN POSIZIONE 0

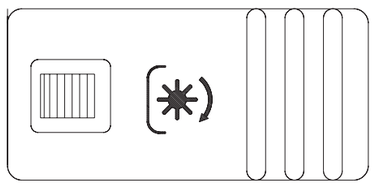
1 POSIZIONE STABILE
1 Stable position

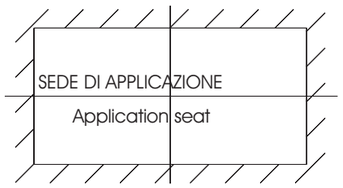


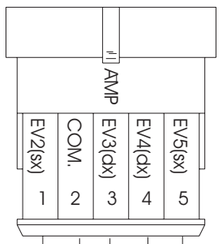
O POSIZIONE STABILE
O Stable position



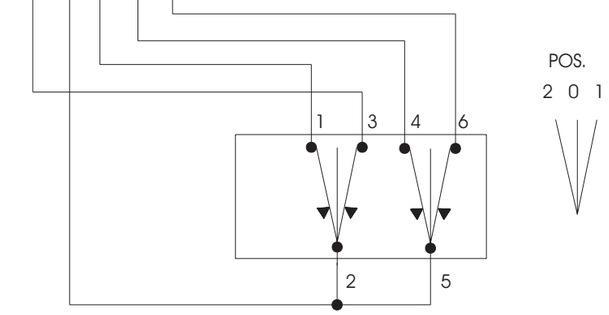




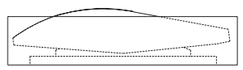




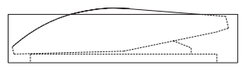
SCHEMA ELETTRICO
Electrical draw



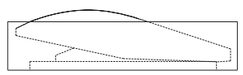
NUMERI CORRISPONDENTI AL CONNETTORE 5 VIE SUPERSEAL



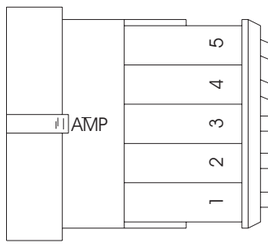
0 POSIZIONE STABILE
0 Stable position

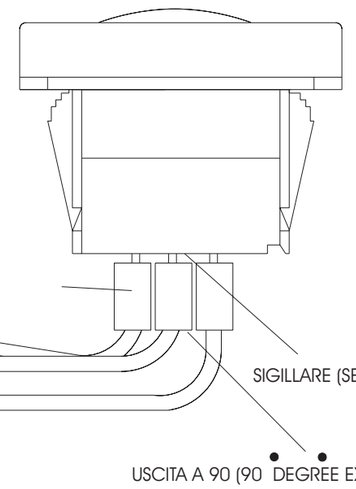


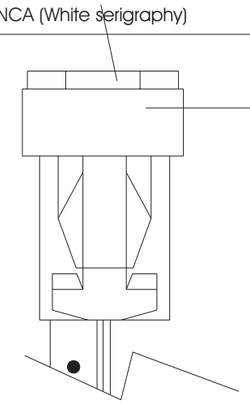
1 POSIZIONE INSTABILE
1 Momentary position

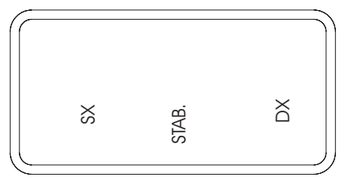


2 POSIZIONE INSTABILE
2 Momentary position



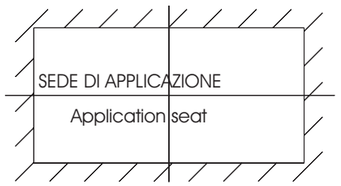


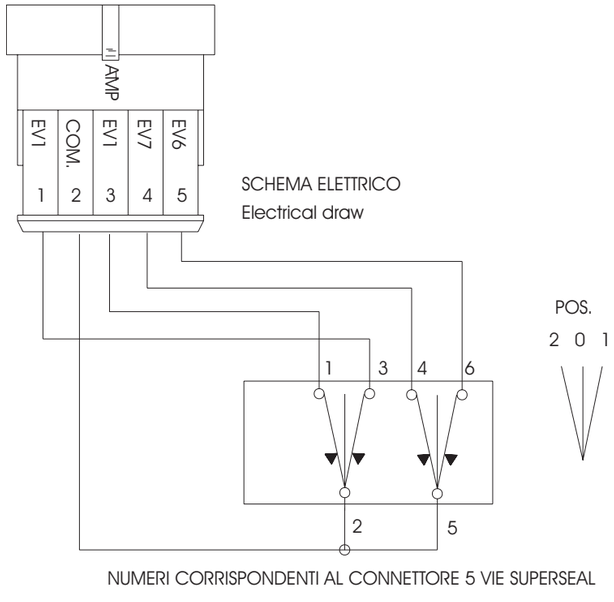




SIGILLARE (SEALED)

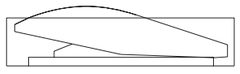
Sp PANNELLO (Panel's thinkness) : 1 : 3 mm

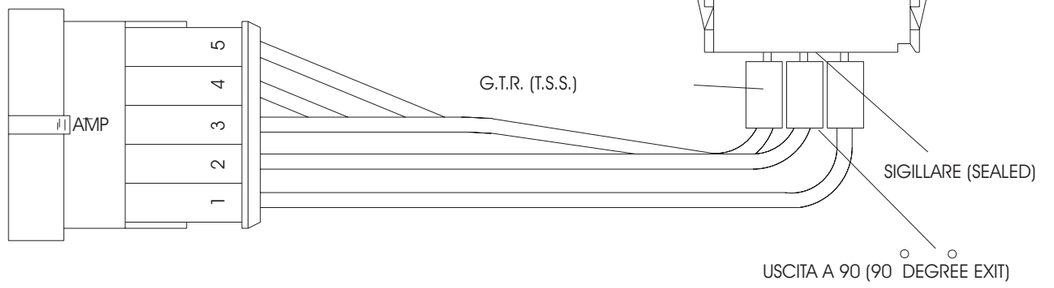


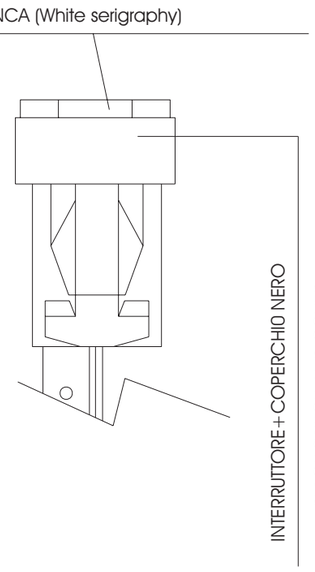


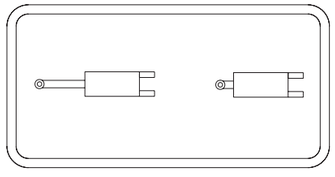
0 POSIZIONE STABILE
0 Stable position

1 POSIZIONE INSTABILE
1 Momentary position


2 POSIZIONE INSTABILE
2 Momentary position





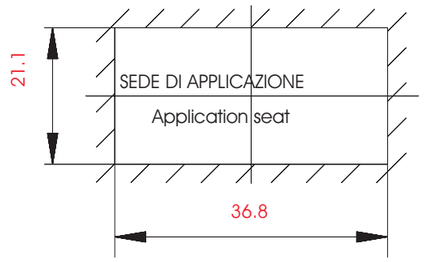


COMPONENTS

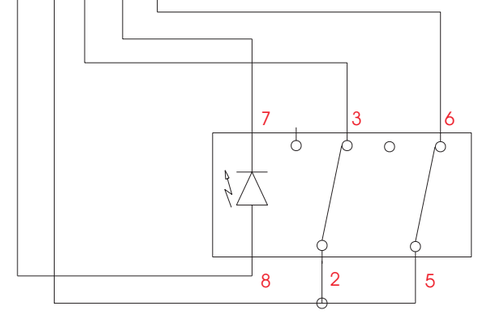
DIAGRAM 45 - SWITCH FOR RH LIFT ROD ADJUSTMENT

2.7659.219.0

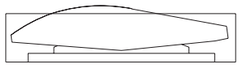
Sp PANNELLO (Panel's thinkness) : 1 : 3 mm



SCHEMA ELETTRICO
Electrical draw

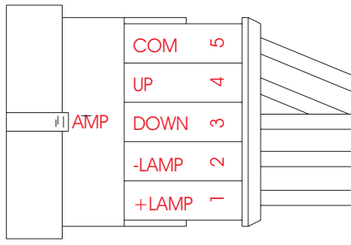


POS.
0 1



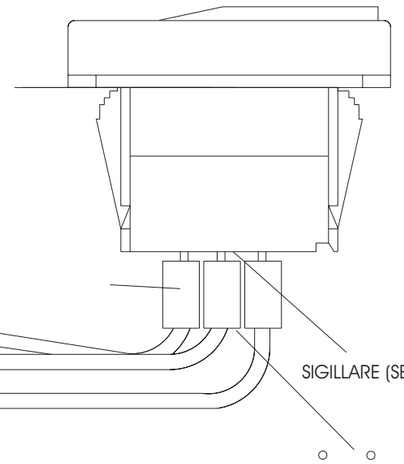
0 POSIZIONE STABILE
0 Stable position

1 POSIZIONE STABILE
1 Stable position



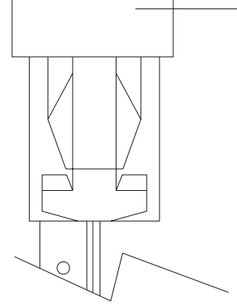
G.T.R. (T.S.S.)

SERIGRAFIA BIANCA (White serigraphy)



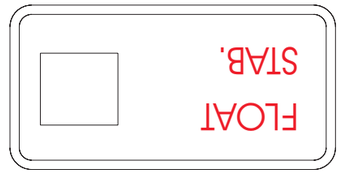
SIGILLARE (SEALED)

USCITA A 90 (90 DEGREE EXIT)



INTERRUTTORE+COPERCHIO NERO
BLACK SWITCH AND COVER

SIGILLARE (SEALED)

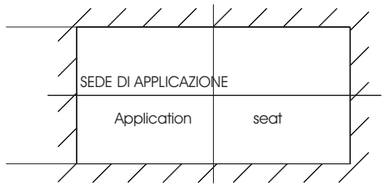


COMPONENTS

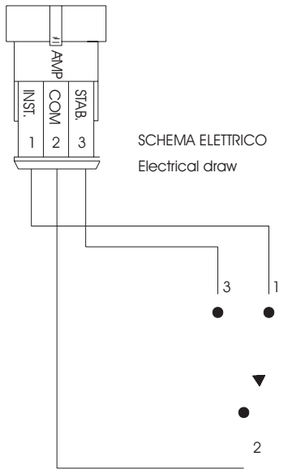
DIAGRAM 46 - SWITCH FOR HYDRAULIC WAYS 3 - 3

2.7659.220.0

Sp PANNELLO (Panel's thinkness) : 1 : 3 mm



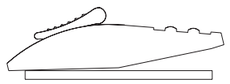
BLOCCO SOLO PER PASSARE DA POS.0 A POS 1



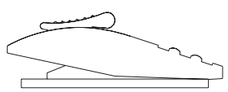
SCHEMA ELETTRICO
Electrical draw

POS.
1 0 2

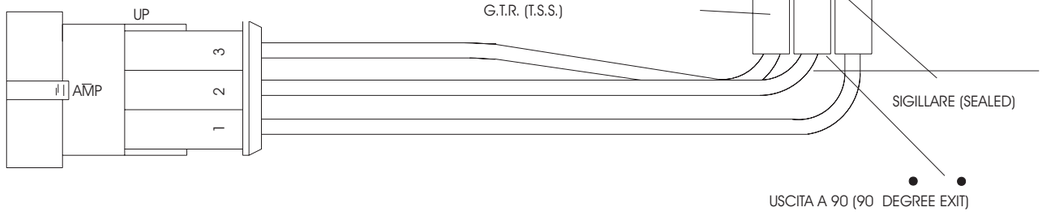
0 POSIZIONE STABILE
0 Stable position

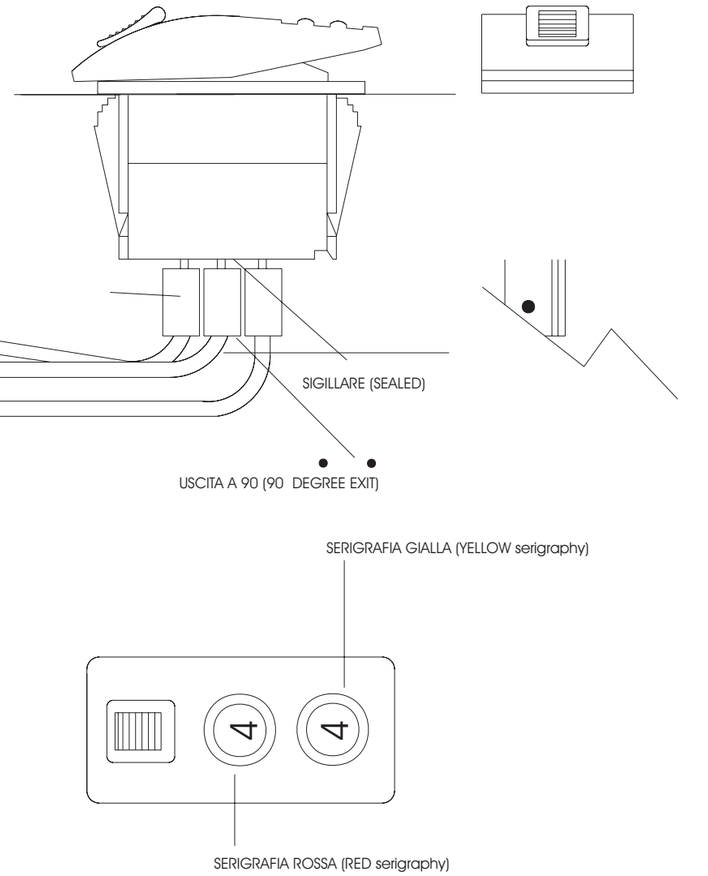


1 POSIZIONE STABILE
1 Momentary position

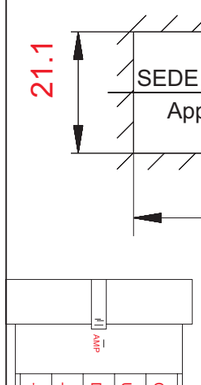


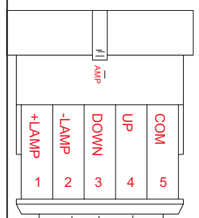
2 POSIZIONE INSTABILE
2 Momentary position



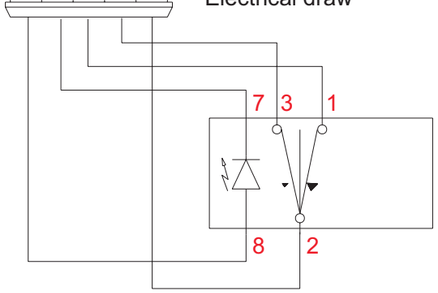


INTERRUTTORE + COPERCHIO NERO
BLACK SWITCH AND COVER

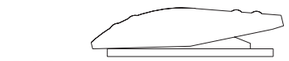




SCHEMA ELETTRICO
Electrical draw



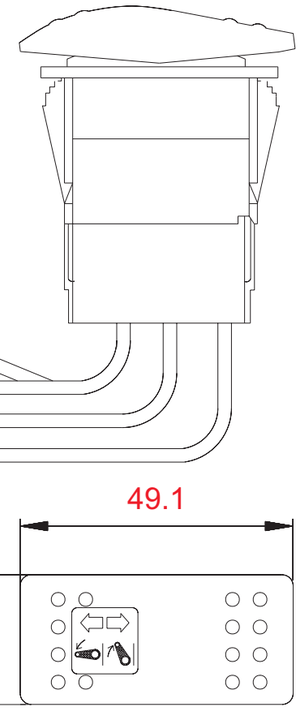
0 POSIZIONE STABILE
0 Stable position

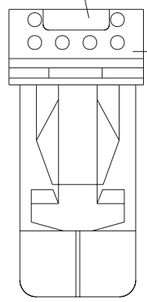


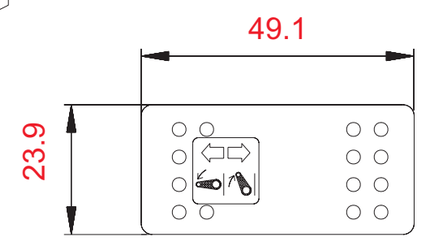
POS. 2 0 1

1 POSIZIONE INSTABILE
1 Momentary position

2 POSIZIONE INSTABILE
2 Momentary position







APPENDIX

ENGINE ELECTRONIC CONTROL UNIT

POWER LIFT TESTER (version 1.24A)

ENGINE ELECTRONIC CONTROL UNIT

Appendix

Engine electronic control unit

Engine electronic control unit

TESTER manual

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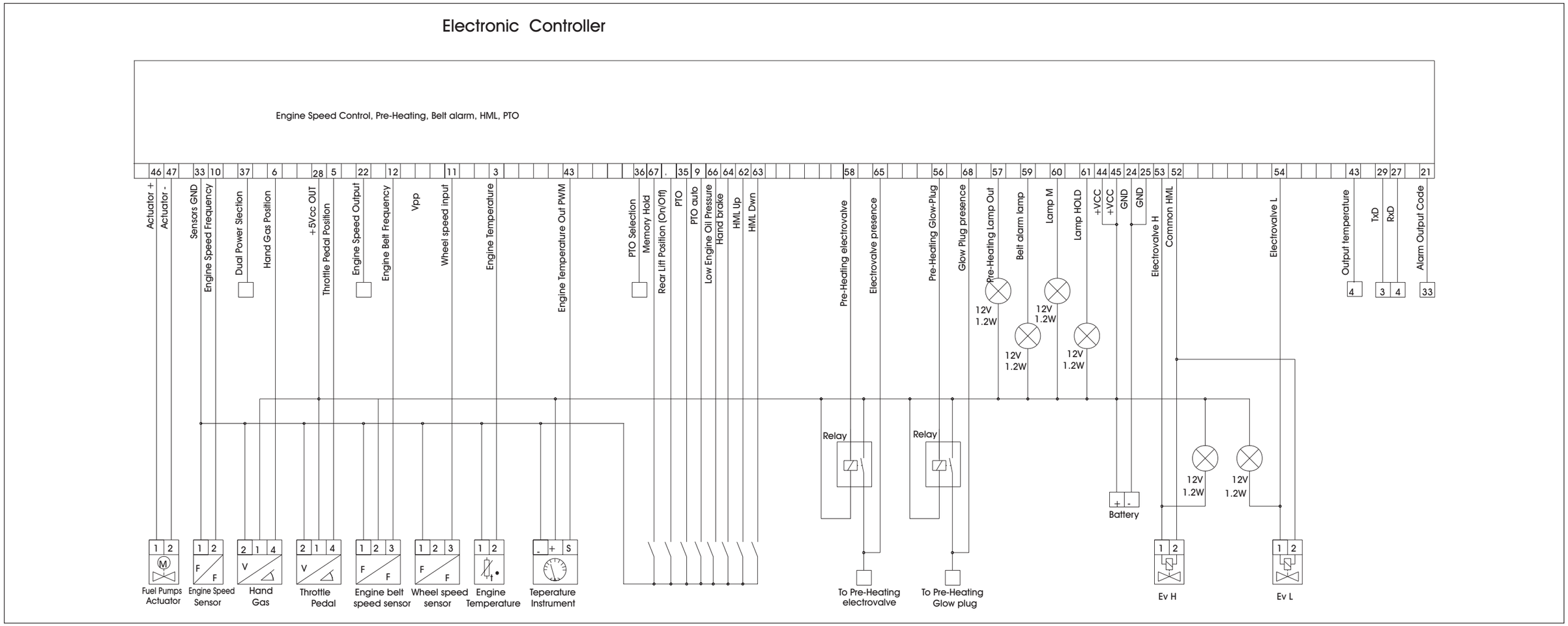
Scope

The scope of this document is:

— To present the diagnostic information that can be accessed through the All Round Tester (ART®)

— To list the alarm conditions that the electronic control unit is able to diagnose, along with the possible causes of the alarms and the relative actions of the electronic control unit.



ELECTRONIC CONTROL UNIT

46 - actuator+ 47 - actuator - 33 - sensor GND 10 - engine speed frequency 37 - dual power selection 6 - hand throttle position 28 - 5Vdc OUT
5 - accelerator pedal position 22 - engine speed output 12 - belts rotation frequency 11 - engine speed input 3 - engine temperature input 43 - engine temperature OUT PWM
36 - P.T.O. selection 67 - memorised engine speed

Appendix

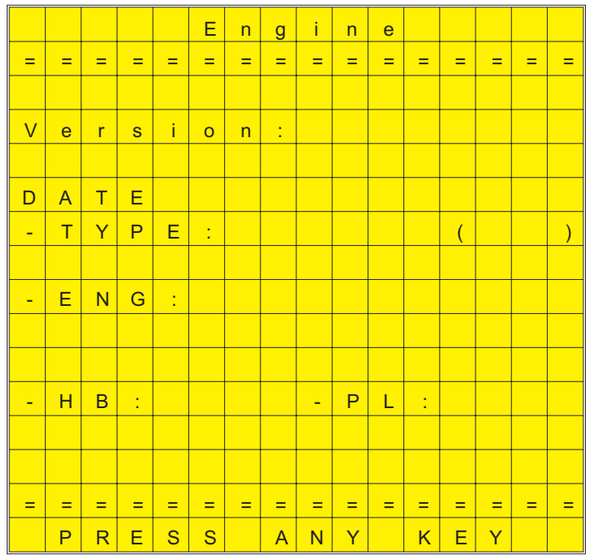
Engine electronic control unit

Presentation screen

The presentation screen displays preliminary information about the electronic control unit:

— Software version
— Date of software issue

The screen appears as follows:

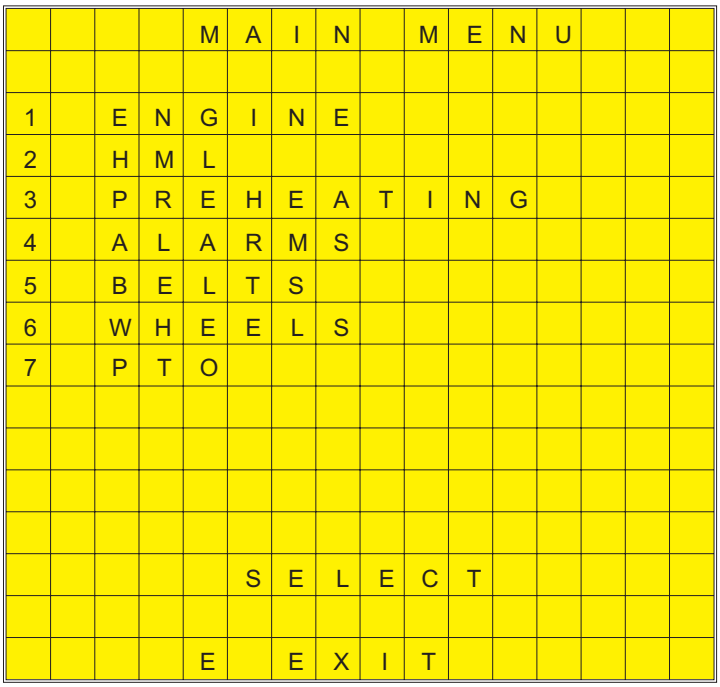


Where:

Version: software version
Date: date of software issue
Type: engine table
Eng: type of engine installed on tractor
HB: hardware batch
PL: production launch

Main Menu

From this menu you can select the component on which diagnosis is to be performed or display the general list of alarms detected up until that point.



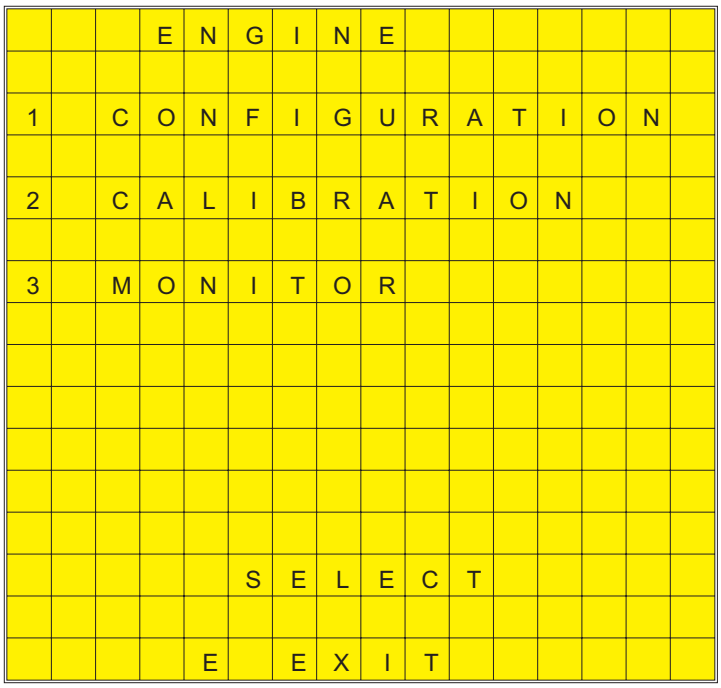
By selecting one of these options you access the menus described in the following pages.

Appendix

Engine electronic control unit

Engine menu

From this menu you can check the operation of the engine.



The various functions are described below.

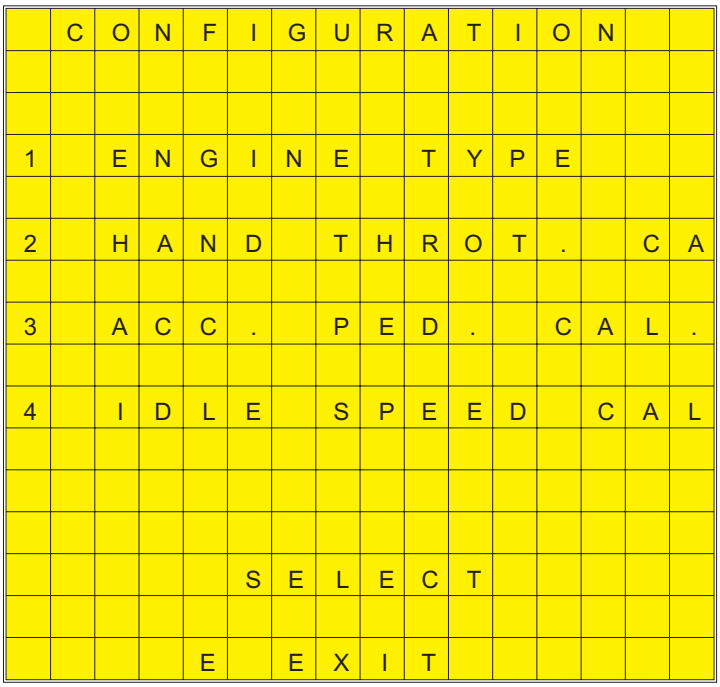
Engine menu

Configuration menu

From this menu you can configure and calibrate the following parameters:

1 engine type
2 hand throttle calibration
3 accelerator pedal calibration
4 idle speed calibration

The following menu is displayed on the ART.



The various functions are described below.

Appendix

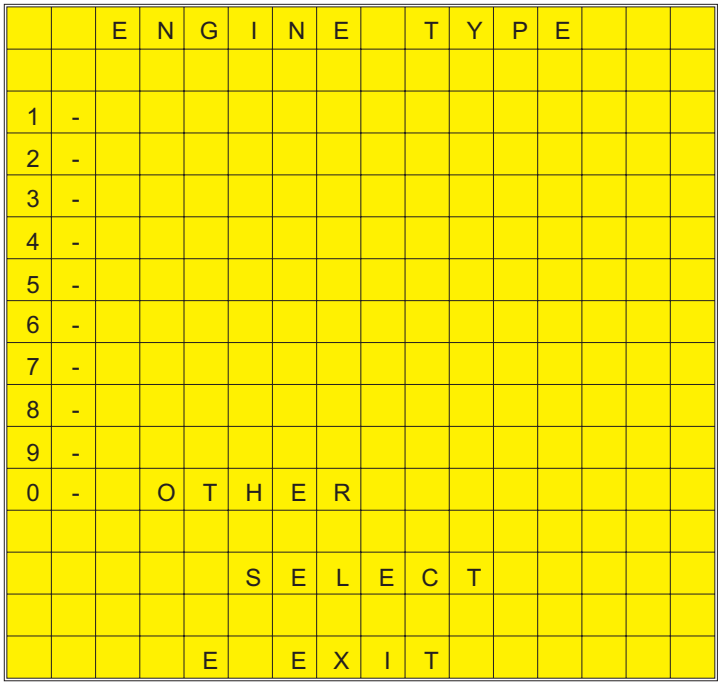
Engine electronic control unit

Engine menu

Configuration menu

Engine type

This screen p. lists all the possible engine types that can be controlled by the ECU. The operator should select the type of engine installed on the tractor.



Once you have selected the engine type, the program automatically displays the parameters menu which contains the parameters stored in memory for the selected engine type:

		P	R	O	G	R	A	M	M	I	N	G		
-	E	N	G											
	E	N	G	I	N	E		T	Y	P	E			
M	I	N	.		R	P	M					X	X	X	X
M	A	X	.		R	P	M					X	X	X	X
D	R	O	O	P									X	X	X
M	I	N	.	C	U	R	R	E	N	T		X	X	X	X
C		C	O	N	F	I	R	M						
				E		E	X	I	T					

Appendix

Engine electronic control unit

Engine menu

Configuration menu

Hand throttle calibration

In this menu you can calibrate the potentiometer of the hand throttle.

The following menu is displayed on the ART:

This screen displays the following values:

· Value potentiometer voltage in mV

To calibrate the potentiometer, proceed as follows:

1- Press **B** to select the parameter you wish to calibrate:

· minimum position
· maximum position

the value in mV corresponding to the selected position will be highlighted in reverse video.

2- Move the hand throttle to the correct position, as instructed on the screen.

Press "**C**" to save the value in memory.

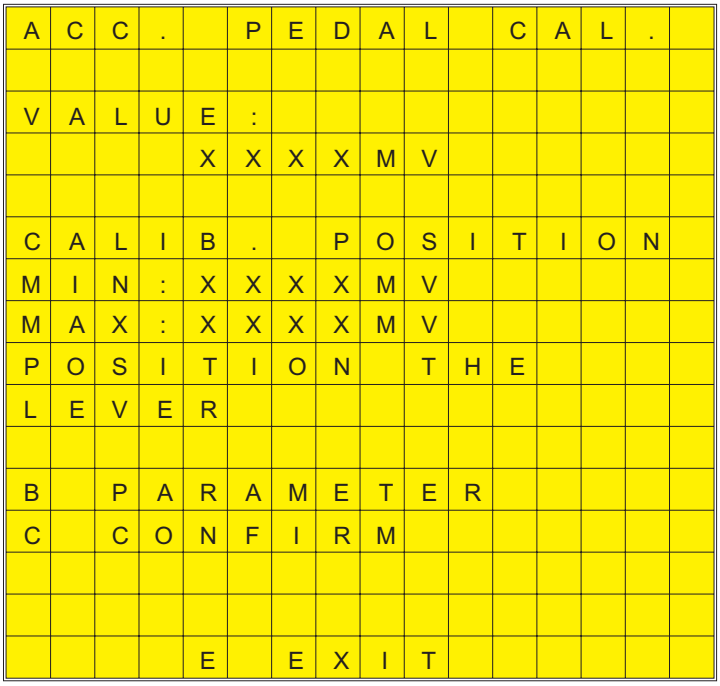
Engine menu

Configuration menu

Accelerator pedal calibration

In this menu you can calibrate the potentiometer of the of the accelerator pedal.

The following menu is displayed on the ART:



This screen displays the following values:

· Value Potentiometer voltage in mV

To calibrate the potentiometer, proceed as follows:

1 Press **B** to select the parameter you wish to calibrate:

· minimum position
· maximum position

the value in mV corresponding to the selected position will be highlighted in reverse video.

2 Move the accelerator pedal to the correct position, as instructed on the screen.

Press "**C**" to save the value in memory.

Appendix

Engine electronic control unit

Engine menu

Configuration menu

Idle speed calibration

In this menu you can calibrate the engine idle speed.

The following menu is displayed on the ART:

Where:

Engine rpm rpm detected by engine speed sensor
Rpm diff. Difference between required rpm and actual rpm

This operation, as indicated on the ART screen, must be carried out with the engine in no-load condition, i.e. without hydraulic or electrical loads.

Calibration procedure:

1 Wait until the rpm difference value stabilises.
2 Press "C" on the ART to save the value in EEprom.
3 There are three possible outcomes:

Parameter correctly programmed: the ART displays the following message:

	P	A	R	A	M	E	T	E	R				
	P	R	O	G	R	A	M	M	E	D			

If the rpm difference exceeds 40, the following message is displayed:

	R	P	M		D	I	F	F	.		4	0	

If the electronic control unit has problems writing to the EEprom, the calibration operation is not performed and the following message is displayed:

	E	e	P	R	O	M		F	A	U	L	T	

Appendix

Engine electronic control unit

Engine menu

Calibration menu

In this menu you can calibrate the operating parameters of the Engine Governor.

The following menu is displayed on the ART:

		C	A	L	I	B	R	A	T	I	O	N		
W	I	T	H		E	N	G	.	R	U	N	N	I	G
1		M	I	N	.		R	P	M					
2		M	I	N	.		C	U	R	R	.			
3		D	R	O	O	P								
A		D	E	F	A	U	L	T		V	A	L	U	E
F		I	N	C	R	E	A	S	E		V	A	L	U	E
D		D	E	C	R	E	A	S	E		V	A	L	U	E
B		S	E	L	E	C	T		P	A	R	A	M	.	
C		C	O	N	F	I	R	M							
				E		E	X	I	T						

Once an engine type has been saved in memory, it is possible to permanently modify certain parameters.

Min. rpm: minimum engine speed

Min. curr: current supplied to actuator to maintain engine rotation

Droop: parameter that regulates variations in engine speed according to the load

The selected parameter is highlighted in reverse video. Press "**B**" to scroll through the parameters. Press "**F**" or "**D**" to respectively increase or decrease the value of the selected parameter. Press "A" to restore the default factory-set value for the selected parameter.

The ART may display the following messages:

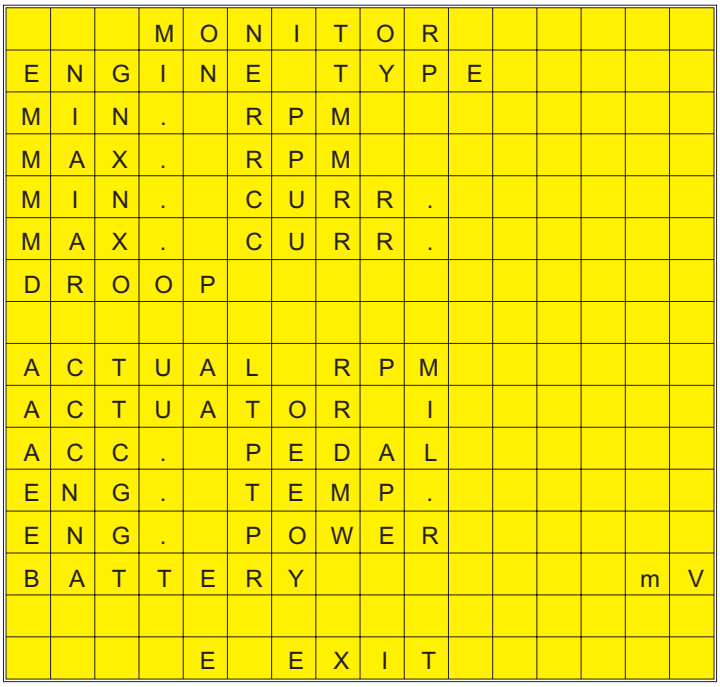
			P	A	R	A	M	E	T	E	R			
		P	R	O	G	R	A	M	M	E	D			

Engine menu

Monitor menu

From this menu you can monitor the operating parameters of the engine governor.

The following menu is displayed on the ART:



Where:

Engine type Engine type code

Min rpm Minimum engine speed given in engine characteristics table

Max rpm Maximum engine speed given in engine characteristics table

Min. Curr.. Current supplied to actuator to maintain engine at idle speed

MAX. CURR.. Maximum engine electronic control unit current (factory setting)

Droop Parameter that regulates variations in engine speed according to the load

Actual rpm Actual engine speed

Actuator I Current supplied to actuator

Acc. Pedal. Position of accelerator pedal

Eng. Temp. Engine temperature

Battery Battery voltage

Appendix

Engine electronic control unit

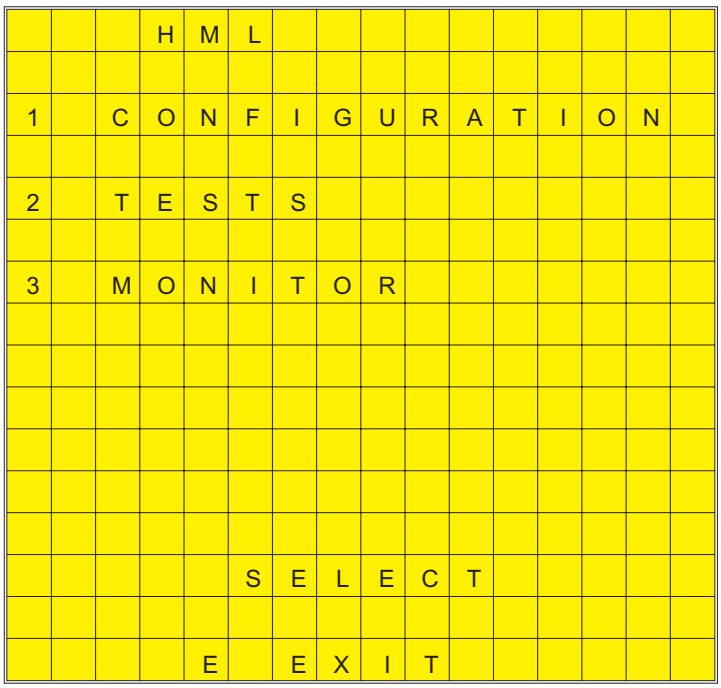
HML menu

From this menu you can

configure
test
monitor

the HML clutches control system.

The following menu is displayed on the ART:



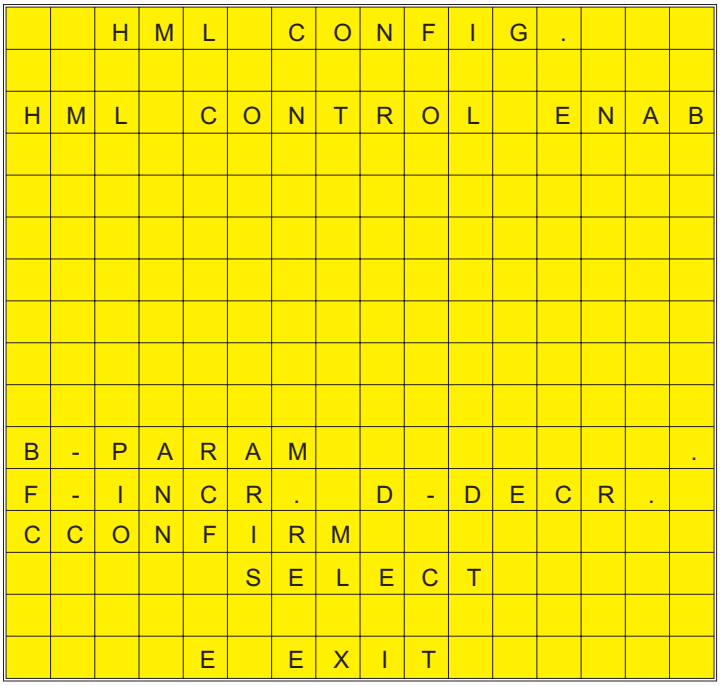
The various submenus are described below.

HML menu

Configuration menu

From this menu you can configure the HML control.

The following menu is displayed on the ART:



Press "**B**" to select the HML configuration parameter:

press "**F**" or "**D**" to enable or disable the HML function

To confirm and save the new configuration, press "**C**".

Appendix

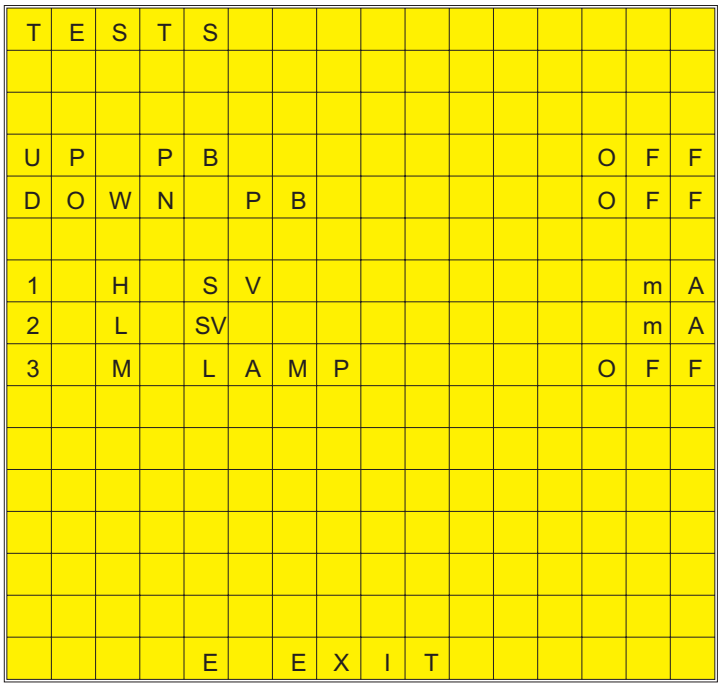
Engine electronic control unit

HML menu

Tests menu

From this menu you can test the operation of the HML system.

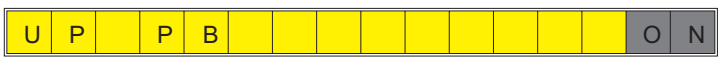
The following menu is displayed on the ART:



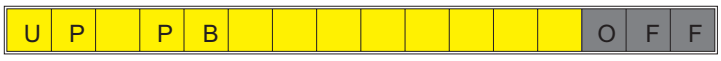
The following components of the HML system may be tested directly:

UP pushbutton Press this pushbutton to test the electrical continuity and mechanical integrity of the pushbutton.

With pushbutton on joystick pressed:

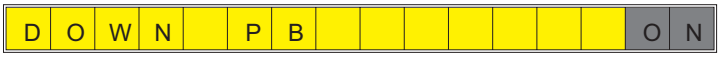


With pushbutton released



DOWN pushbutton Press this pushbutton to test the electrical continuity and mechanical integrity of the pushbutton.

With pushbutton on joystick pressed:

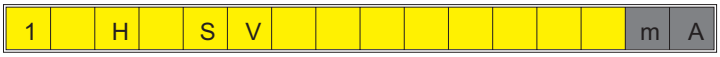


With pushbutton released

H SV H clutch solenoid valve
By pressing "1" on the ART you can energise or de-energise the solenoid valve of the H clutch.
The ART display is as follows:

Solenoid energised

Solenoid de-energised



M lamp M clutch indicator lamp
by pressing "**3**" on the ART you can test the operation of the M clutch indicator lamp

Appendix

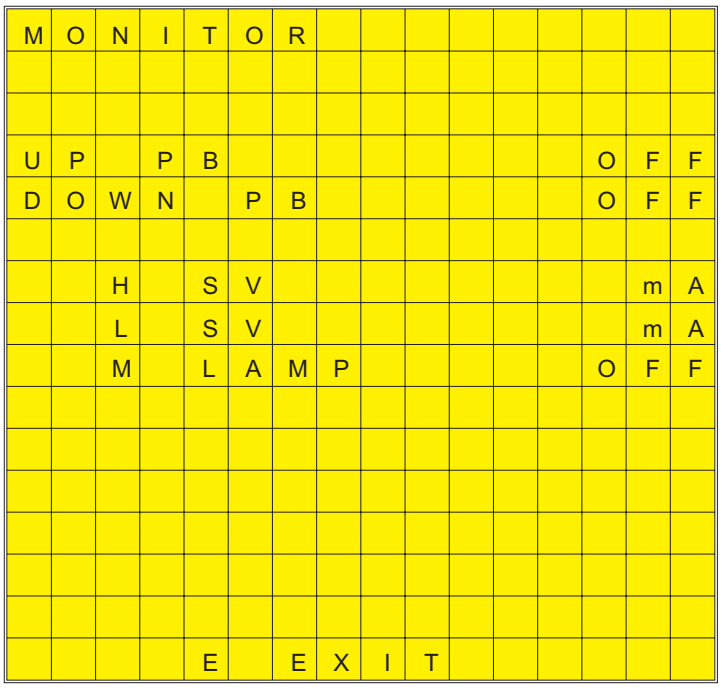
Engine electronic control unit

HML menu

Monitor menu

From this menu you can monitor the operating status of the HML system.

The following menu is displayed on the ART:



The screen is similar to that described in the previous paragraph, the sole difference being that this menu does not provide the possibility to control the solenoid valves.

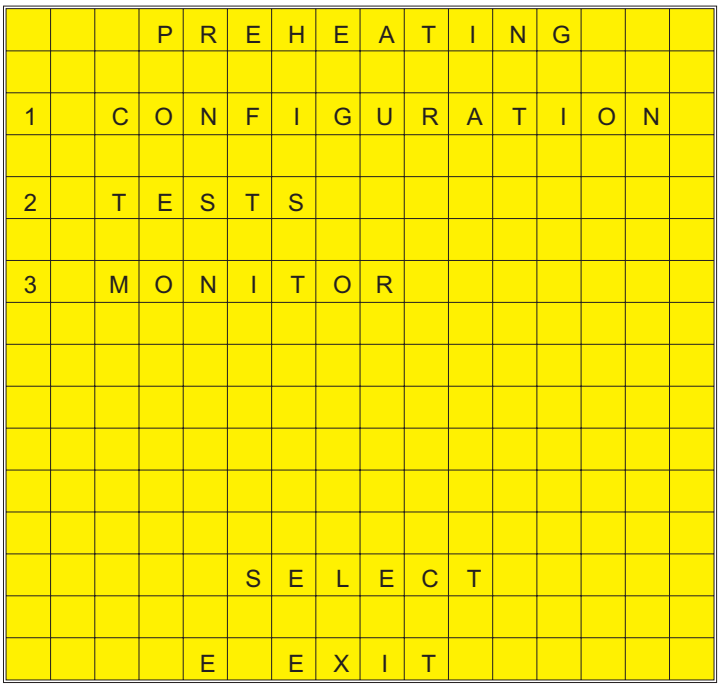
This screen displays the operating status of the HML system components.

Preheating menu

From this menu you can
- configure
- test
- monitor

the preheating system.

The following menu is displayed on the ART:



The various submenus are described below.

Appendix

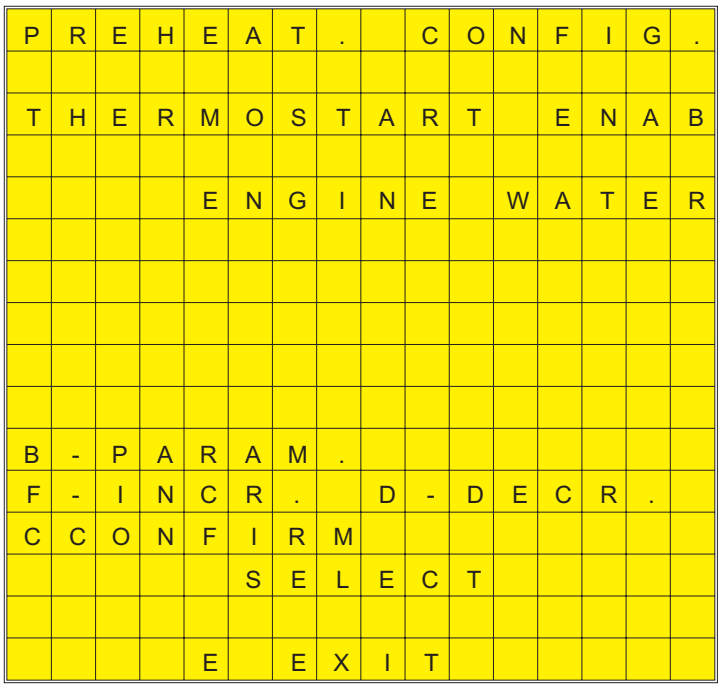
Engine electronic control unit

Preheating menu

Configuration menu

From this menu you can configure the preheating function.

The following menu is displayed on the ART:



Thermostarter configuration.

Press "**B**" to select the thermostarter configuration parameter:

press "F" or "D" to enable or disable the function

To confirm and save the new configuration, press "**C**".

Engine cooling configuration

Engine electronic control unit

The type of engine cooling (water or air) determines the type of sensor used to measure the engine temperature.

Press "**B**" to select the engine cooling configuration parameter. The ART screen appears as follows:

press "**F**" or "**D**" to change the cooling type

Appendix

Engine electronic control unit

Preheating menu

Tests menu

From this menu you can test the operation of the preheating system.

The following menu is displayed on the ART:

	P	R	E	H	E	A	T	I	N	G				
T	E	S	T	S										
T	E	M	P	.	S	E	N	S	.					C
1		G	L	O	W	P	L	U	G		R	.	O	F	F
2		P	R	E	H	E	A	T	.	S	V	.	O	F	F
3		P	R	E	H	E	A	T	.		L	.	O	F	F
		T	E	S	T	S									
A		W	I	T	H		E	N	G	I	N	E			
		O	F	F											
				E		E	X	I	T						

The various functions are described below:

Temp. Sens. engine temperature expressed in degrees centigrade

Glowplug rel. directly controls the glow plug relay.
By pressing "1" on the ART you can energise or de-energise the glow plug relay.
The corresponding line of the display changes as follows:

Relay not energised

Relay energised

Preheat. SV — Directly controls the preheating solenoid valve.
By pressing "1" on the ART you can energise or de-energise the preheating solenoid.
The corresponding line of the display changes as follows:

Preheating solenoid valve not energised:

Preheating solenoid valve energised:

Preheat. lamp — Directly controls the preheating lamp.
By pressing "1" on the ART you can energise or de-energise the preheating lamp.
The corresponding line of the display changes as follows:

Preheating lamp not energised:

Preheating lamp energised:

WARNING: all these tests are to be performed with the engine switched off, as indicated on the ART display.

Appendix

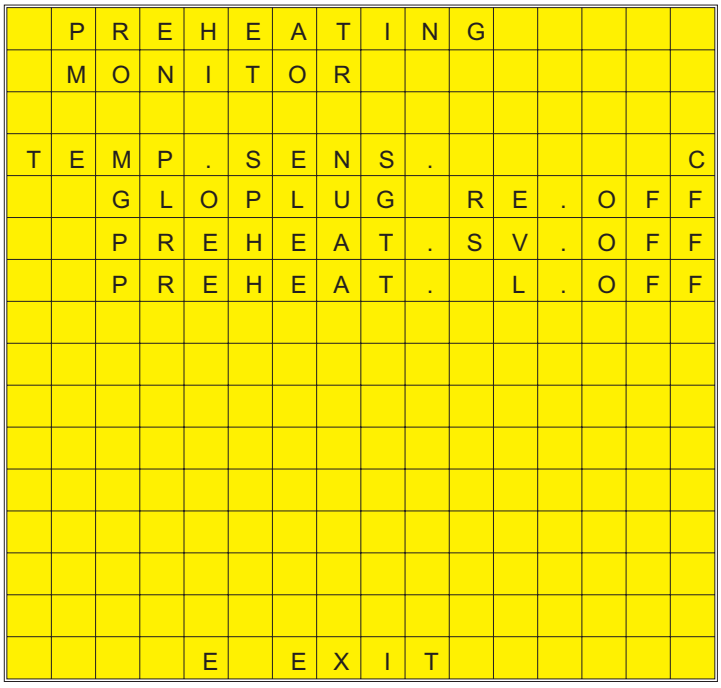
Engine electronic control unit

Preheating menu

Monitor menu

In this menu you can monitor the operation of the preheating system.

The following menu is displayed on the ART:



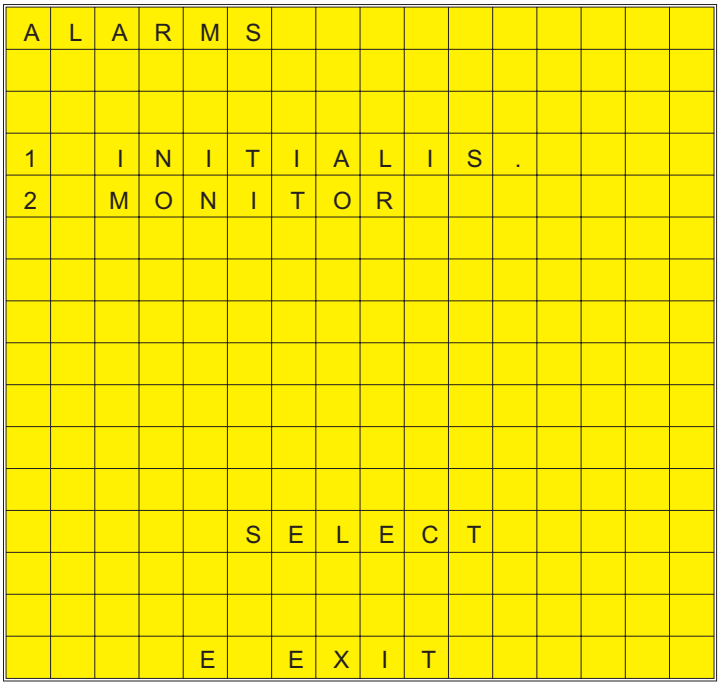
The screen is similar to that described in the previous paragraph, the sole difference being that this menu does not provide the possibility to control the various functions.

This screen displays the operating status of the preheating system components.

Alarms menu

From this menu you can:
initialise the alarms
display the alarms log.

The following menu is displayed on the ART:



The various functions are described below.

Appendix

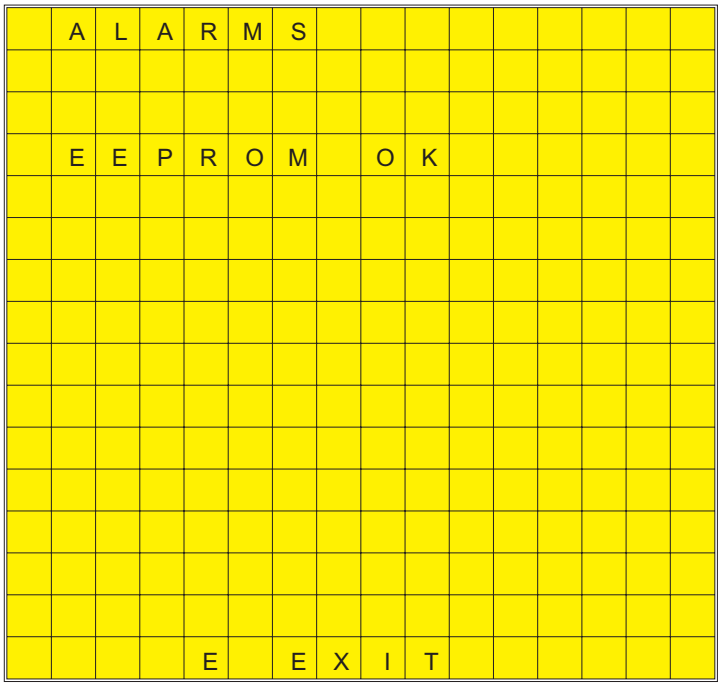
Engine electronic control unit

Alarms menu

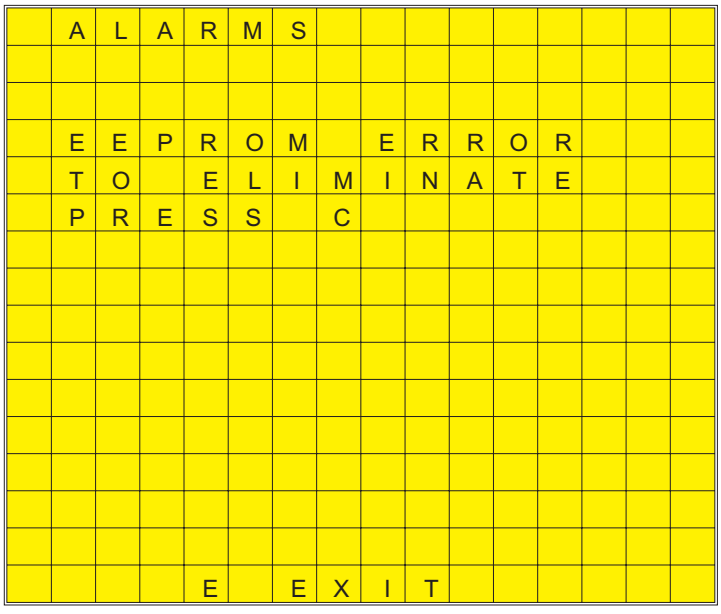
Initialisation menu

This operation is only required as part of the putting into the service procedure, generally after installing a later version of the software.

If the new and previous software versions are compatible, the ART will display the following screen:

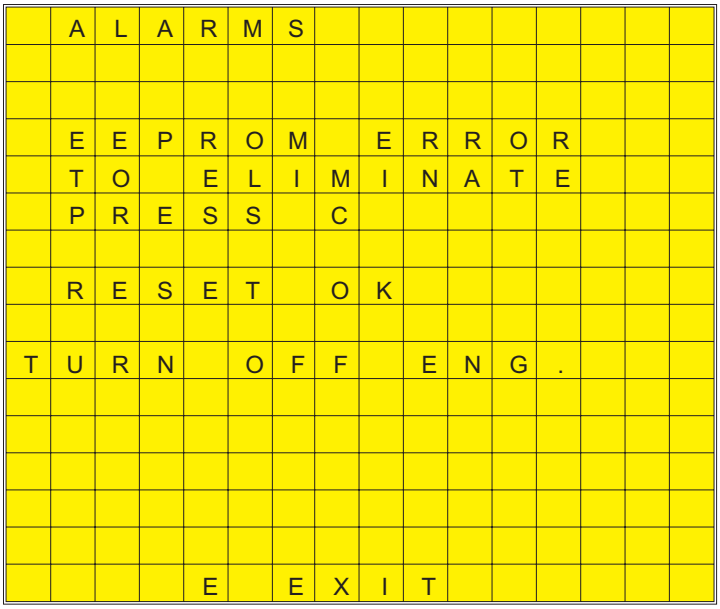


Otherwise:



On pressing **"C**" on the ART, the electronic control unit will reset this alarm.

The ART will display the following screen:



Appendix

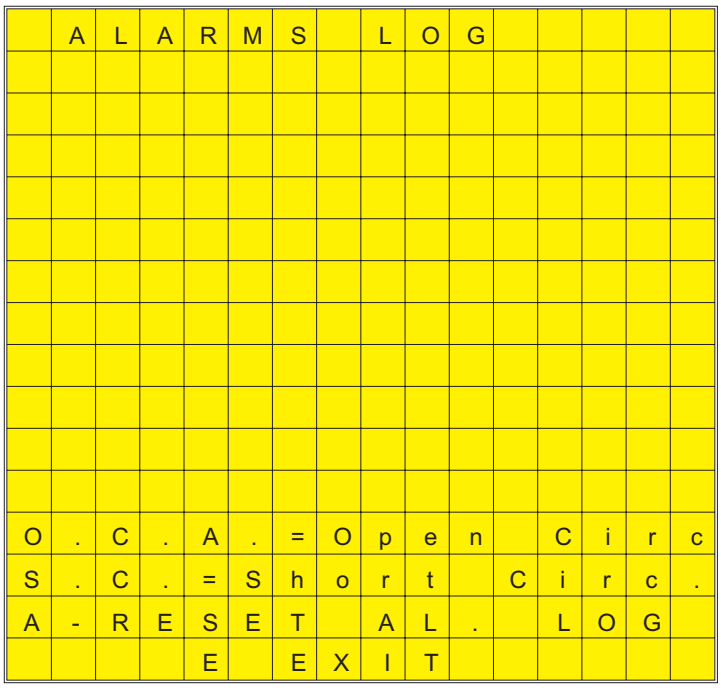
Engine electronic control unit

Alarms menu

Monitor menu

This screen displays the list of alarms detected by the system. The most recent alarm is first in the list. Pressing "A" cancels the list and resets the alarm log.

The screen displayed on the ART is as follows:



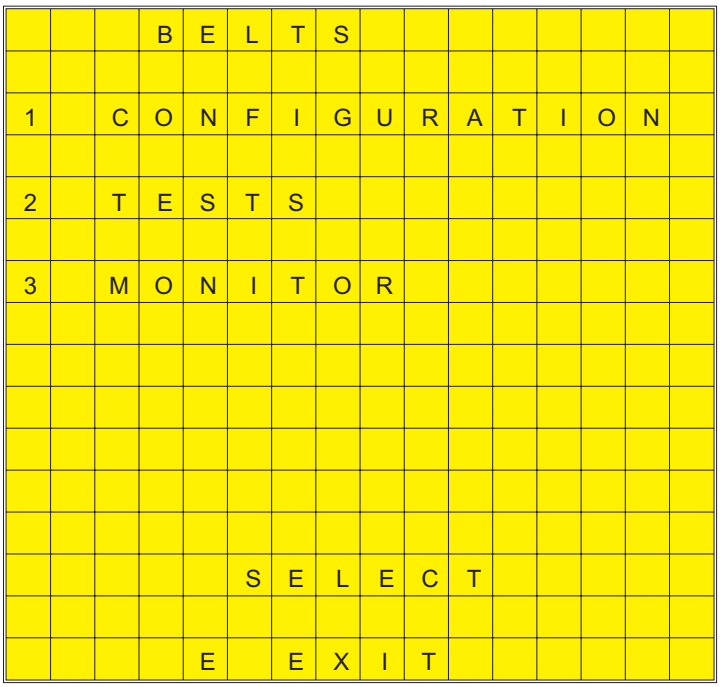
Heading 4 contains a list of all the alarms that can be detected by the electronic control unit.

Belts menu

From this menu you can
- configure
- test
- monitor

the drive belts control system.

The following menu is displayed on the ART:



The various submenus are described below.

Appendix

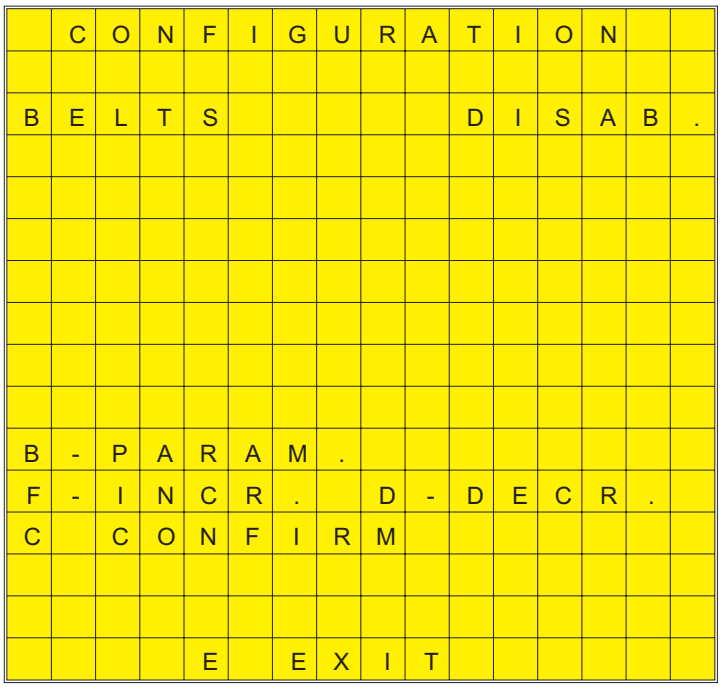
Engine electronic control unit

Belts menu

Configuration menu

From this menu you can configure the Belts Alarm function.

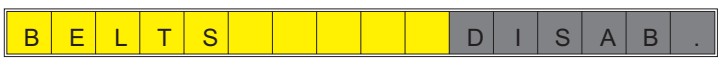
The following menu is displayed on the ART:



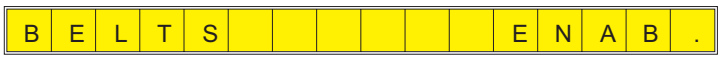
The various functions are described below:

Configuration::
By pressing "**B**" on the ART you can enable or disable the broken belts alarm system. The ART display will change as follows:

On pressing "**D**"



On pressing "**F**"



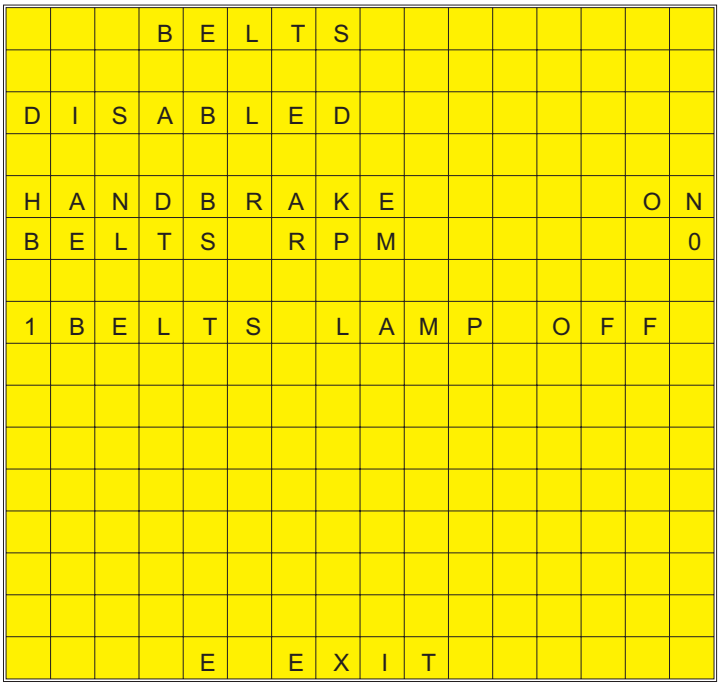
Saving:
Press "**C**" to confirm and save the new configuration.

Belts Menu

Tests Menu

From this menu you can test the operation of the Belts Alarm system.

The following menu is displayed on the ART:



The various functions are described below:

Handbrake	Handbrake with tractor at standstill
Belts rpm	Drive belt rotation speed
Belts lamp	Directly controls the belts alarm lamp. By pressing "1" on the ART you can energise or de-energise the broken belt warning lamp. The corresponding line of the display changes as follows:

Lamp not energised

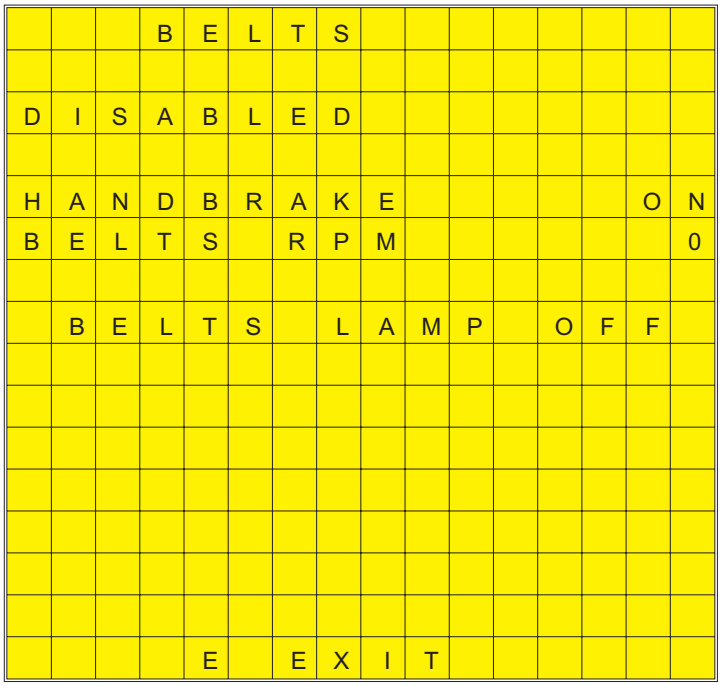
Lamp energised

Appendix

Engine electronic control unit

Belts menu

Monitor menu



The screen is similar to that described in the previous paragraph, the sole difference being that this menu does not provide the possibility to control the various functions.

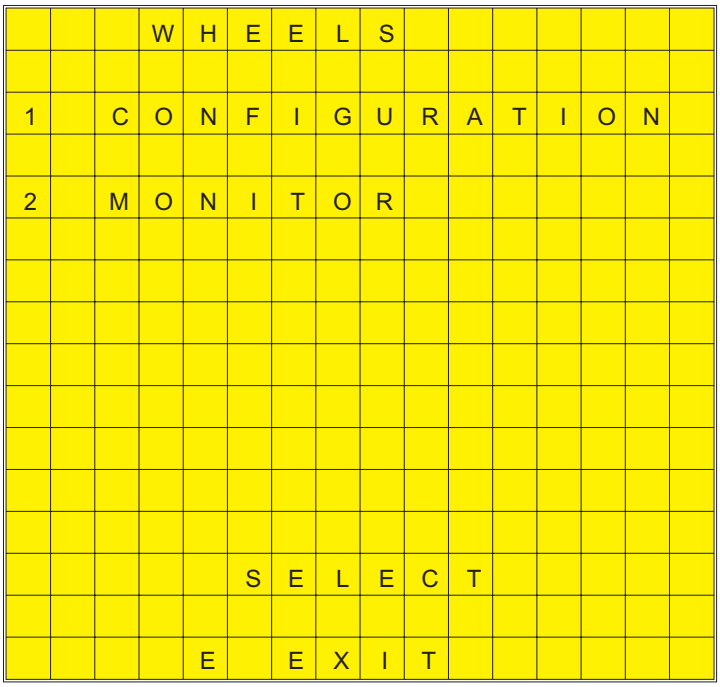
This screen displays the operating status of the belts alarm system components.

Wheels menu

From this menu you can:
· configure
· monitor

the Wheel Speed system.

The following menu is displayed on the ART:



The submenus are described below.

Appendix

Engine electronic control unit

Wheels menu

Configuration menu

From this menu you can configure the Maximum Speed Control function.
The following menu is displayed on the ART:

	C	O	N	F	I	G	U	R	A	T	I	O	N	
L	.	3	0		K	m	/	h		E	N	A	B	.
W	H	E	E	L		C	A	L	I	B	,	X	X	X	X
A		R	E	S	E	T		B	-	P	A	R	A	M	.
F	-	I	N	C	R	.		D	-	D	E	C	R	.	
C		C	O	N	F	I	R	M							
				E		E	X	I	T						

The various functions are described below:

by pressing "**B**" on the ART you can select the parameter to be modified. The ART display changes as follows:

on pressing "**B**" the selected parameter is highlighted in reverse video

L	.	3	0		K	m	/	h		E	N	A	B	.	

press "**F**" to enable the 30 km/h limit

L	.	3	0		K	m	/	h		E	N	A	B	.	

or press "**D**" to disable the 30 km/h limit

L	.	3	0		K	m	/	h		D	I	S	A	B	.

Only if the 30 km/h limit is enabled will it be necessary to calibrate the wheel diameter by following either the putting into service procedure or the following instructions.

Modification of the wheels parameter:

press "**A**" on the ART; the value displayed will be set to zero:

Enter the wheel parameter value K, which is listed in the paragraph "Engine type and Wheels Parameter" in this manual.

Press "**C**" to confirm and save the change.

Then press "E" to exit.

Appendix

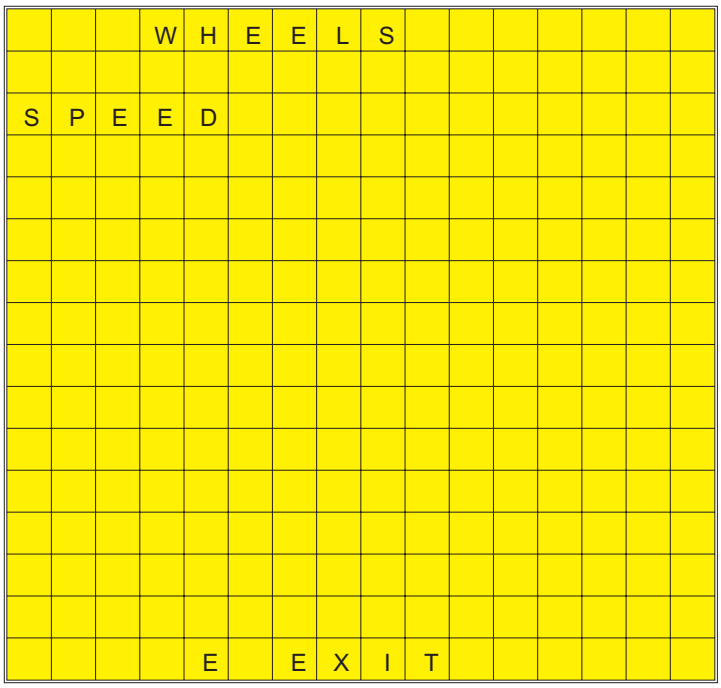
Engine electronic control unit

Wheels menu

Monitor menu

In this menu you can monitor the operation of the Maximum Speed Control function.

The menu displayed on the ART is as follows:



Where:

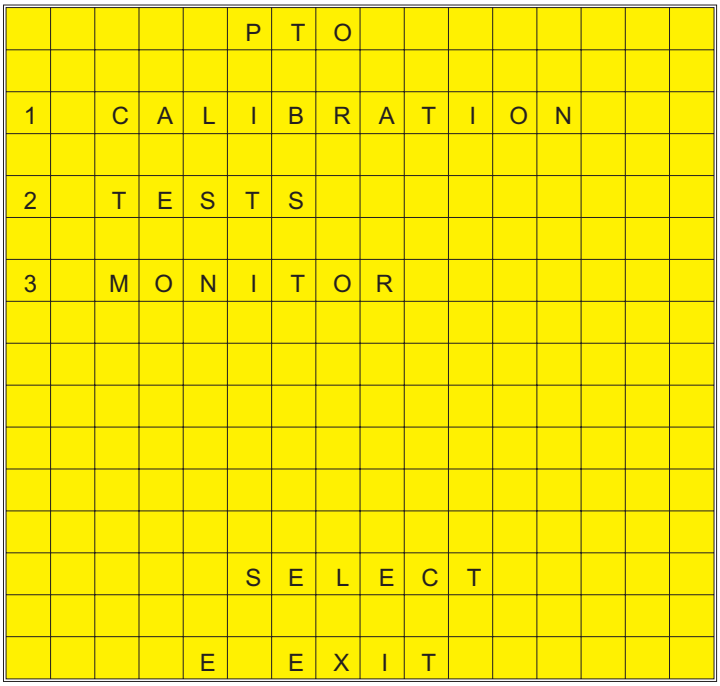
SPEED Tractor speed

P.T.O. menu

From this menu you can:
- calibrate
- configure
- monitor

the Automatic P.T.O. control system.

The following menu is displayed on the ART:



The various submenus are described below.

WARNING
This system may be used on tractors equipped with electronic lift control and electronic engine governor.

Appendix

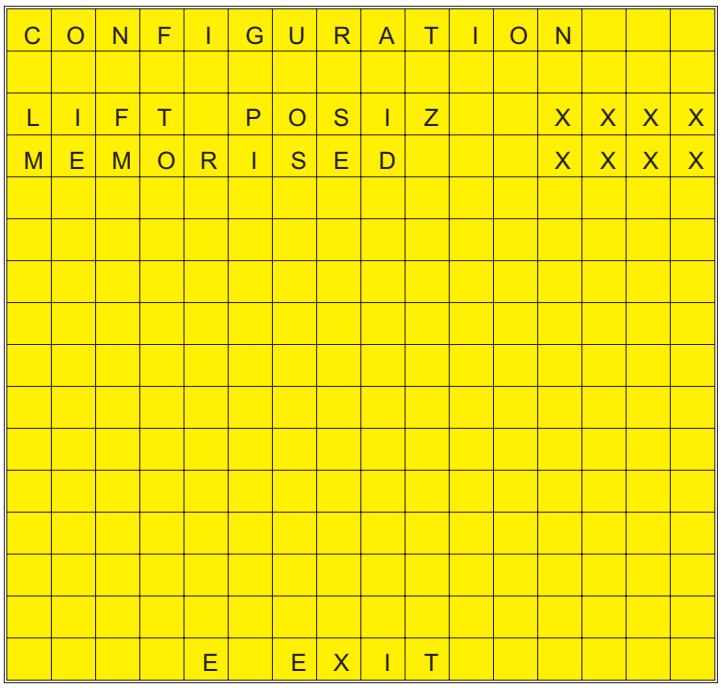
Engine electronic control unit

P.T.O. menu

Configuration menu

From this menu you can configure the Automatic P.T.O. function.

The menu is displayed on the ART as follows:

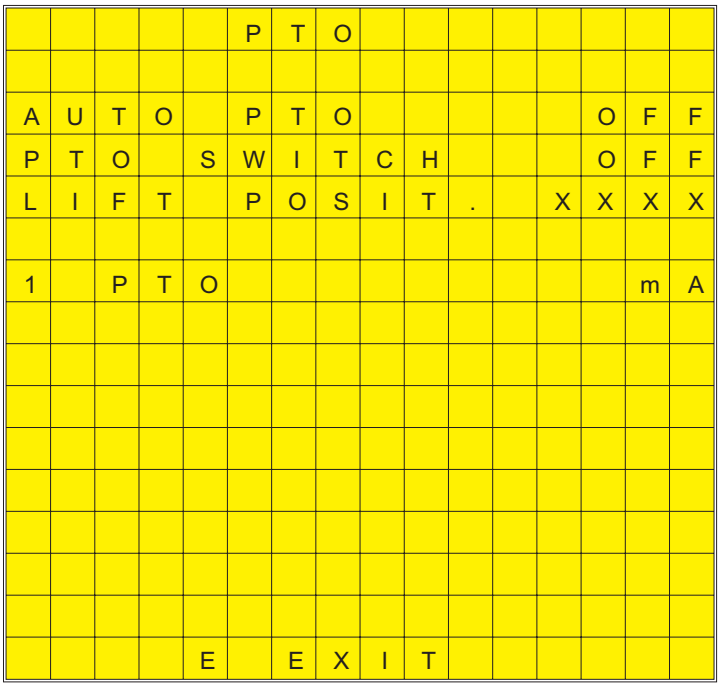


To configure this function, refer to the putting into service procedure.
See point 4 - AUTO P.T.O. calibration.

P.T.O. menu

Tests menu

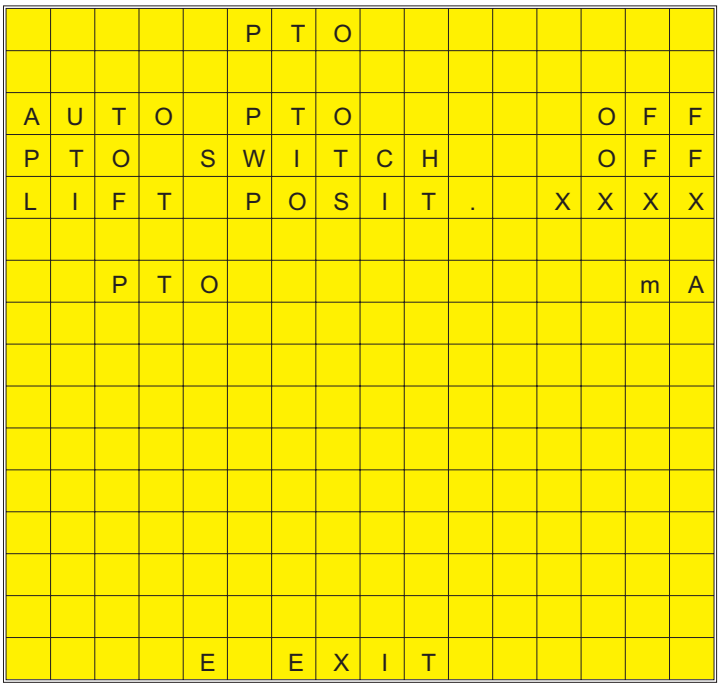
From this menu you can test the operation of the Automatic P.T.O. system.
The menu is displayed on the ART as follows:



P.T.O. menu

Monitor menu

In this menu you can monitor the operating status of the Automatic P.T.O. system.
The menu is displayed on the ART as follows:



Appendix

Engine electronic control unit

The screen is similar to that described in the previous paragraph, the sole difference being that this menu does not provide the possibility to control the various functions.
This screen displays the operating status of the belts alarm system components.

Engine Type/Wheels Constant tables

Engine type configuration

POWER	ENGINE	
55 A/W-60W	1000.3A7	11
55 A/W-60W	1000.3W2	11
55 A/W-60W	1000.3A2	11
55 A/W-60W	1000.3W1*E	11
63-A/63-W	1000.3AT2	26
63-A/63-W	1000.3WT2	26
67 CV A (3C)	1000.3AT1	29
70 CV W (4C)	1000.3WT1*E	29
70 CV W (3C)	1000.4W4	21
75 CV A (4C-DO)	1000.4A3*E	10
80 CV W (DO)	1000.4W3	24
85 CV A (GO)	1000.4AT1*E	16
85 CV A (DO)	1000.4AT3	46
85 CV W (DO)	1000.4WT4	44
87 CV W	1000.4WT1*E	31

Wheels Constant

TYRE model	**K**
320/70 R24	1512
12.4 R24	1442
360/70 R24	1442
13.6 R24	1390
380/70 R24	1390
12.4 R28	1320
420/70 R24	1320
360/70 R28	1320
13.6 R28	1276
380/70 R28	1276
420/70 R28	1216
380/70 R20	1448
11.2 R24	1448

Fault signalling by flashing of the HOLD indicator light

code	*ALARM TYPE*	*EFFECT*	*ECU action*
1	OVER SPEED	• - engine switched off • - fault signalled	• - ENGINE SPEED OVER 3200 RPM
2	ACTUATOR	• - engine switched off • - fault signalled	• - SHORT CIRCUIT • - OPEN CIRCUIT
3	ACCELERATOR PEDAL	• - normal operation • - fault signalled	• - SHORT CIRCUIT • - OPEN CIRCUIT
4	ENGINE PICKUP	• -engine switched off • - fault signalled	• - OPEN CIRCUIT • - INTERMITTENT PULSES
5	EEPROM	• - operation with default parameters • - fault signalled	• - CHKSUM ERROR • - WRITING ERROR
6	WHEELS PICKUP	• - max engine speed 1500 rpm • - fault signalled	• - OPEN CIRCUIT • - INTERMITTENT PULSES
7	BELTS	• - handbrake applied: engine switched off and fault signalled • - no handbrake • - fault signalled	• - BELT BROKEN • - NO PULSES FROM BELT SENSOR • - LOW OIL PRESSURE AND ENGINE RUNNING
8	HAND THROTTLE	• - NORMAL OPERATION • - FAULT SIGNALLED	• - SHORT CIRCUIT • - OPEN CIRCUIT
9	HML SOLENOID VALVES	• - NORMAL ENGINE OPERATION • - M GEAR • - FAULT SIGNALLED	• - SHORT CIRCUIT ON SOLENOID • - SOLENOID NOT CONNECTED • - COMMAND DOES NOT CORRESPOND TO OUTPUT STATUS • - NO VOLTAGE ON HML COMMON

Appendix

Engine electronic control unit

Tester alarm messages and possible causes of malfunction

Tester message	Possible cause of fault	ECU action
Overspeed	RPM OVER 3200.	ACTUATOR CONTROL REMOVED. RESTART ENGINE.
Actuator O.C.	ACTUATOR NOT CONNECTED. WIRING ERROR. CHECK PINS 46 47	ACTUATOR NOT CONTROLLED. RESTART ENGINE.
Actuator S.C.	ACTUATOR FAULTY. WIRING ERROR CHECK PINS 46 47	ACTUATOR CONTROL REMOVED. RESTART ENGINE
RPM sensor O.C.	PICKUP NOT CONNECTED. CHECK PIN 10	ACTUATOR CONTROL REMOVED. RESTART ENGINE.
RPM sensor interm.	PICKUP FAULTY. POOR CONTACT. CHECK PIN 10.	ACTUATOR CONTROL REMOVED. RESTART ENGINE.
Hand throt. S.C.	Hand throttle potentiometer short circuit. WIRING ERROR. CHECK PIN 6	Loss of hand throttle function.
Acc. manuale C.A.	Hand throttle potentiometer not connected. Wiring error. CHECK PIN 6.	Loss of hand throttle function.
Acc. pedal S.C.	Accelerator pedal potentiometer short circuit. Wiring error. CHECK PIN 5.	Loss of accelerator pedal function.

Tester message	Possible cause of fault	ECU action
Acc. pedal O.C.	Accelerator pedal potentiometer not connected. Wiring error. CHECK PIN 5.	Loss of accelerator pedal function.
Hold lamp O.C.	Lamp disconnected. Lamp faulty. Check pin 61.	No limitations.
Speed sens. O.C.	Wheel speed sensor disconnected. Sensor faulty. Check pin 11.	Speed limited.
Speed sens. interm.	Sensor faulty. Poor contact. Check pin 11.	Speed limited.
Glowplug rel. O.C.	Electrical system fault. Relay faulty. ECU faulty. Check pin 56.	Preheating system operation limited.
Glowplug O.C.	Electrical system fault. Glow plug faulty. ECU faulty. Check pin 68.	Preheating system operation limited.
Glowplug rel. fault	Electrical system fault. Relay faulty. ECU faulty. Check pin 68.	Preheating system operation limited.

Tester message	Possible cause of fault	ECU action
Preheat.rel. O.C.	Electrical system fault. Relay faulty. ECU faulty. Check pin 58.	Preheating system operation limited.
Preheat. S.V. O.C.	Electrical system fault. Solenoid faulty. ECU faulty. Check pin 65.	Preheating system operation limited.
Preheat.rel. fault	Electrical system fault. Relay faulty. ECU faulty. Check pin 65.	Preheating system operation limited.
Preheat. lamp O.C.	Electrical system fault. Lamp faulty. ECU faulty. Check pin 57.	No limitations.
Temp.sens. O.C.	Electrical system fault. Temperature sensor faulty. ECU faulty. Check pin 3.	Preheating permanently activated.
Lift.pos.sens. O.C.	Electrical system fault. Position sensor faulty. ECU faulty. Check pin 4.	Operation of automatic P.T.O. system limited.

Tester message	Possible cause of fault	ECU action
Temp.sens. S.C.	Electrical system fault. Temperature sensor faulty. ECU faulty. Check pin 3.	Preheating permanently activated. Incorrect indication on console.
Belt speed sens. O.C.		Not utilised.
Belt lamp O.C.	Electrical system fault. Belts warning lamp faulty. ECU faulty. Check pin 59.	No limitations.
H SV S.C.	Electrical system fault. H clutch solenoid faulty. ECU faulty. Check pin 53.	H gear cannot be utilised.
H SV O.C.	Electrical system fault. H clutch solenoid faulty. ECU faulty. Check pin 53.	HML and P.T.O. systems not functioning.
L SV S.C.	Electrical system fault. L clutch solenoid faulty. ECU faulty. Check pin 54.	L gear cannot be utilised.
L SV O.C.	Electrical system fault. L clutch solenoid faulty. ECU faulty. Check pin 54.	HML and P.T.O. systems not functioning.

Appendix

Engine electronic control unit

Tester message	Possible cause of fault	ECU action
M lamp O.C.	Electrical system fault. M lamp faulty. ECU faulty. Check pin 50.	No limitations.
P.T.O. SV S.C.	Electrical system fault. P.T.O. solenoid faulty. ECU faulty. Check pin 55.	P.T.O. system not functioning.
P.T.O. SV O.C.	Electrical system fault. P.T.O. solenoid faulty. ECU faulty. Check pin 55.	P.T.O. and HML systems not functioning.
HL P.T.O. SV lock	Electrical system fault. Solenoid faulty. ECU faulty. Check pin 52.	P.T.O. and HML systems not functioning.
Flash fault	ECU faulty. Software error.	All systems inhibited.
EEPROM fault	ECU faulty. Updated software version.	Default values used. Parameters for default engine type used.
General alarm	ECU faulty.	LAMPS AND RELAYS DISABLED.
EEP B1B2 bench (SAME only)	ECU faulty. Updated software version.	Use of CORRECT BENCH values

Tester message	Possible cause of fault	ECU action
Low oil press.	Low oil pressure. Electrical system fault. Oil pressure sensor faulty. ECU faulty. Check pin 66.	Belts warning light on. If handbrake on, engine switched off.
Instr. temp. S.C.	Electrical system fault. Sensor faulty. ECU faulty. Check pin 43.	No limitations.
EEPROMparamLimit	ECU faulty. Incorrect calibration value. Updated software version.	Default values utilised.

POWER LIFT ELECTRONIC CONTROL UNIT

Appendix

Power lift tester

Power Lift Electronic Control Unit

TESTER manual

CONTENTS

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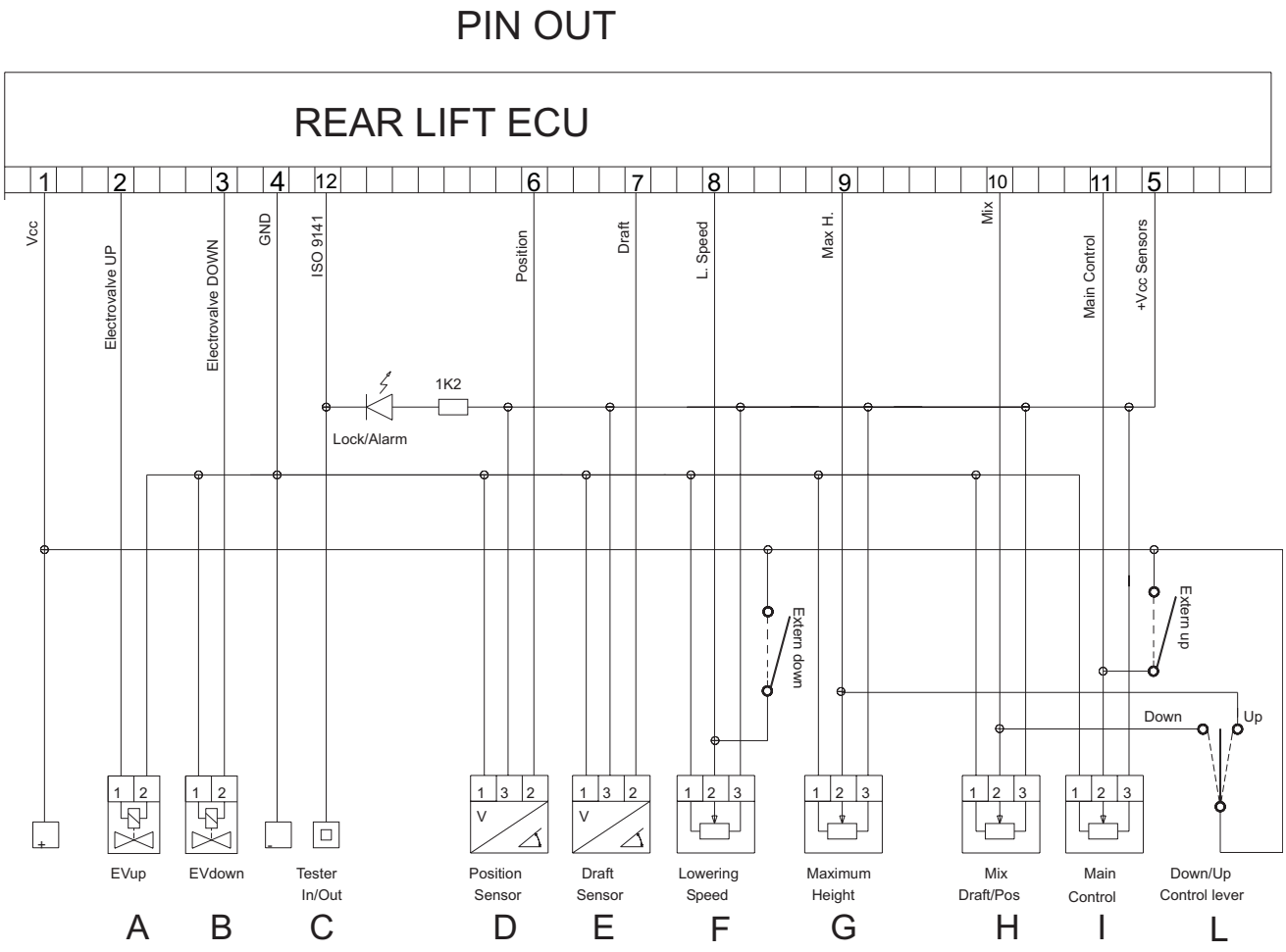
1 Calibrations menu ..... *LIII*
2 Monitor menu ..... *LVI*
1 Lift menu ..... *LVII*
2 Draft menu ..... *LIX*
3 Position menu ..... *LX*
3 Position menu ..... *LXI*
3 Tests menu ..... *LXII*
1 Sensors menu ..... *LXIII*
2 Power supply menu ..... *LXIV*
3 Inputs menu
4 Alarms menu ..... *LXVI*

·**Scope**

The scope of this document is:

Pto present the diagnostic information that can be accessed through the All Round Tester (ART®);

Pto list the alarm conditions that the electronic control unit is able to diagnose, along with the possible causes of the alarms and the relative actions of the electronic control unit.



Functional diagram of the lift ECU.

1 - Vdc
2 - UP solenoid valve
3 - DOWN solenoid valve
4 - GND
5 - Sensor Vdc
6 - Position
7 - Draft
8 - Speed limit
9 - Maximum height
10 - Mix
11 - Main control
12 - ISO 9141

A - UP solenoid valve
B - DOWN solenoid valve
C - Tester IN/OUT
D - Position sensor
E - Draft sensor
F - Lowering speed
G - Maximum height
H - DRAFT/POSITION MIX control
I - Main control
L - UP/DOWN control

Appendix

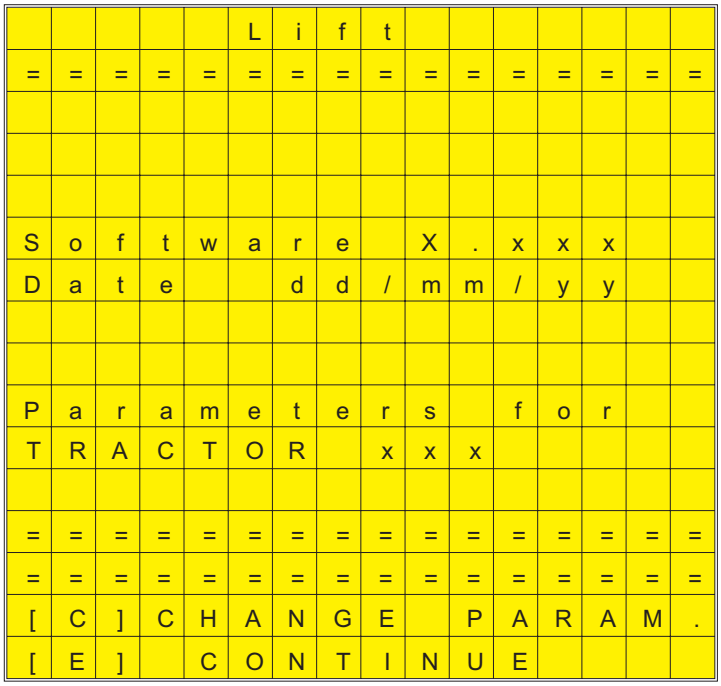
Power lift tester

Presentation screen

The presentation screen displays preliminary information about the electronic control unit:

- Software version
- Date of software issue

The screen appears as follows:



Where:
Version software version
Date: date of software issue

The various functions are described below:

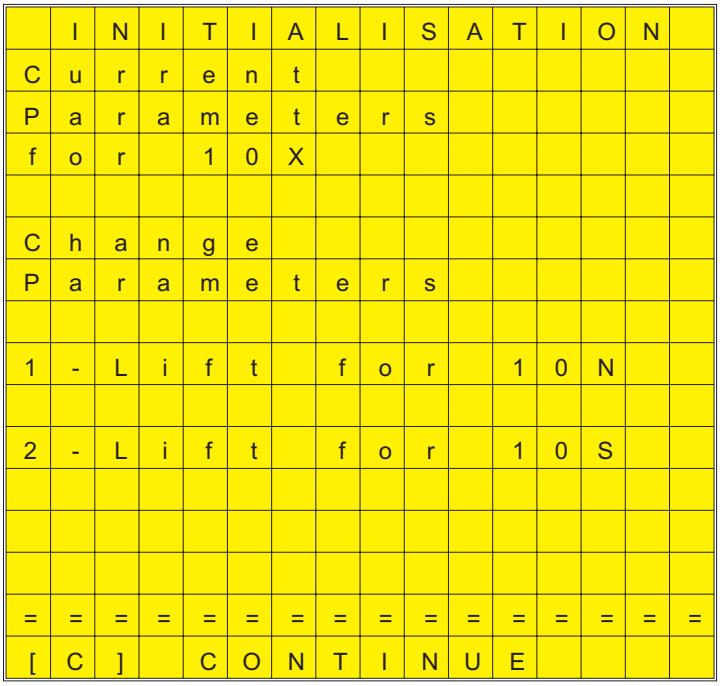
Pressing "C"
displays the initialisation menu for the tractor type:
·lift for tractor 10N
·lift for tractor 10S
As described on page 6

Pressing "E"
displays the menu described on page

Initialisation menu

From the Initialisation menu you can configure the lift in accordance wit the type of tractor.

The menu is displayed on the ART as follows:



Where:

Current parameters defines the configuration currently saved in memory

To change and save the configuration, press "1" or "2".

Press "1" to configure the lift for the 10N tractor.
Press "2" to configure the lift for the 10S tractor.

Press "C" to pass to the main menu.

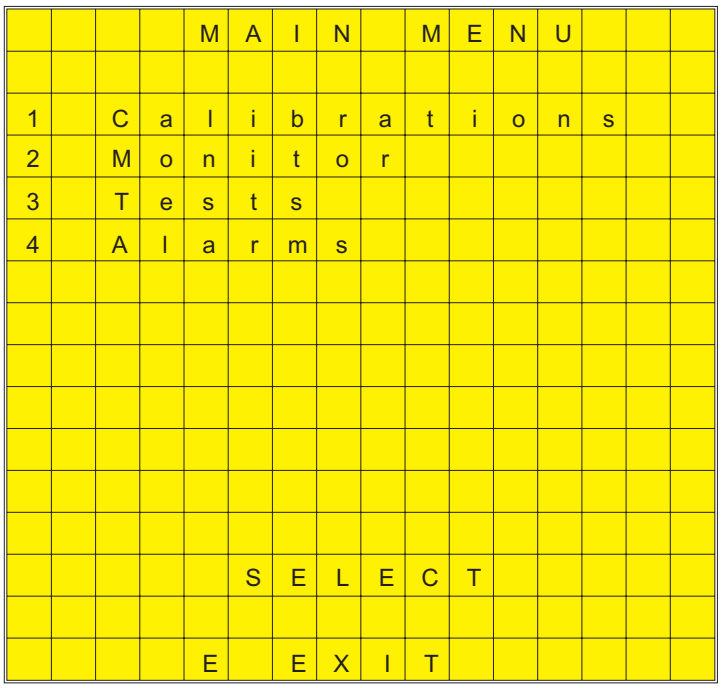
Appendix

Power lift tester

1 - Main menu

From this menu you can select the component on which diagnosis is to be performed or display the general list of alarms detected up until that point.

The menu displayed on the ART is as follows:

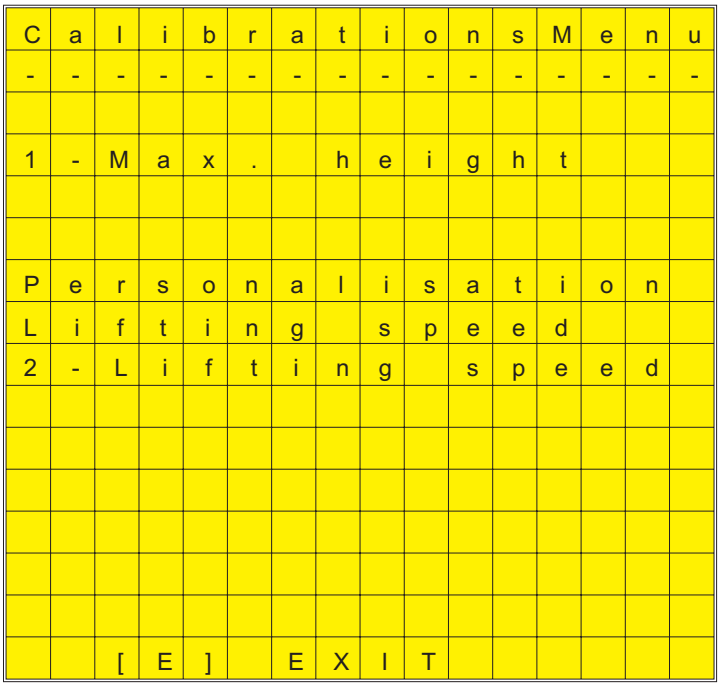


The various menus and submenus are described below.

1 - Calibrations menu

From this menu you can calibrate the lift system.

The menu displayed on the ART is as follows:



These operations are to be performed with the lift lock released.

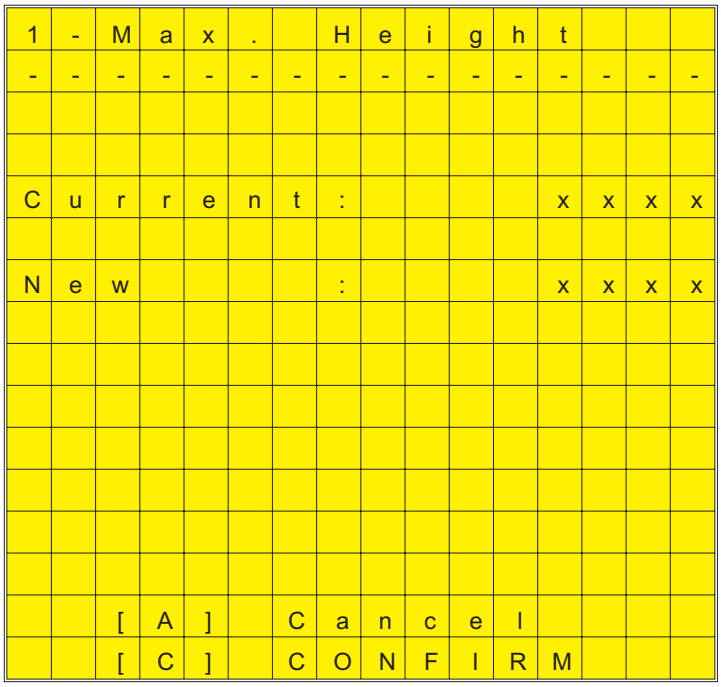
The various calibration functions are described below.

Appendix

Power lift tester

Maximum height calibration:

On pressing "1" on the ART, the following screen is displayed:



Where:

Current The position value between 0 and 1000 saved in memory.
New The position value between 0 and 1000 read by the position sensor.

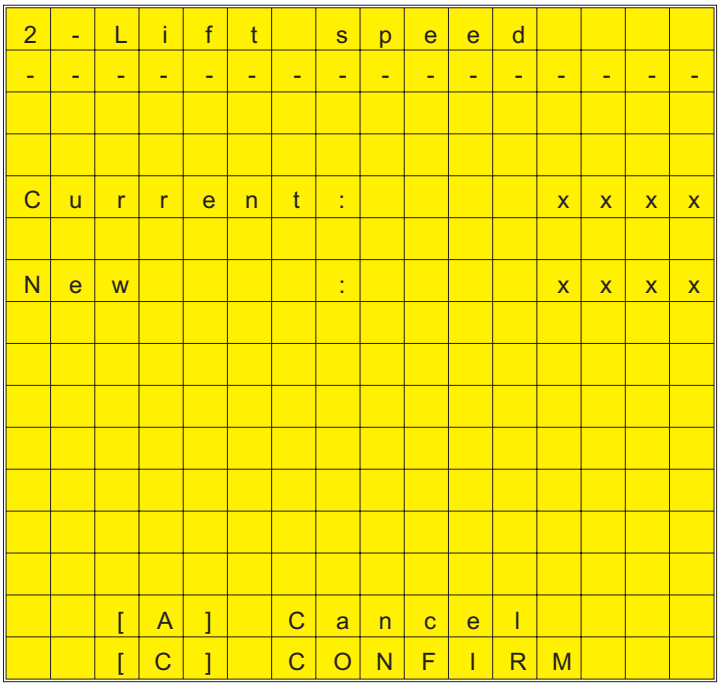
Press:

- "A" to cancel the operation;
- "C" to save the new value in the Eeprom.

Implement lifting speed

This parameter regulates the speed at which the lift is raised when using the manual controls only.

On pressing "2" on the ART, the following screen is displayed:



Where:

Current The speed value between 300 and 400 saved in memory.
New The actual speed value between 300 and 400.

Press:

- "A" to cancel the operation;
- "C" to save the new value in the Eeprom.

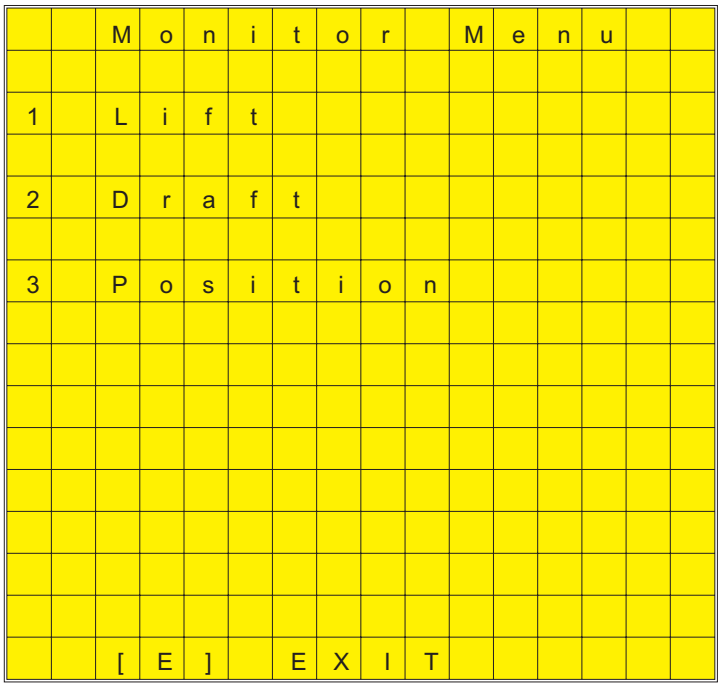
Appendix

Power lift tester

2 - Monitor menu

From this menu you can display the operating parameters of the lift system.

The menu displayed on the ART is as follows:



The various functions are described below.

2 - Monitor Menu

1 - Lift menu

From this menu you can display the following values:
Set Point
Position
Draft
Maximum height

The menu displayed on the ART is as follows:

	L	i	f	t										
-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
P	o	t	e	n	t	i	o	m	e	t	e	r	s	
S	e	t	P	o	i	n	t	.	.	.	.	x	x	x	x
M	a	x	.		h	e	i	g	h	t	.	.	.	x	X
L	o	w	.		s	p	e	e	d	.	.	.	.	x	X
M	i	x	.	.	.	.	.	.	.	.	.	x	x	x	X
			S	E	N	S	O	R	S						
P	o	s	i	t	i	o	n	.	.	.	.	.	x	x	X
D	r	a	f	t	.	.	.	.	.	.	.	x	x	x	X
			P	W	M		O	U	T	P	U	T			
P	w	m											x	x	X
L	I	F	T		S	T	A	T	U	S	.	.			
S	t	a	t	u	s	.	.	.	.	.	.	.	.	x	X
				E		E	X	I	T						

POTENTIOMETERS

Set Point Reference value for automatic control of the lift.
On turning the potentiometer, the value displayed varies between 0 and 1000 (this operation must be performed with the lift lock released).

Max. height Represents the maximum height to which the lift can be raised.
On turning the potentiometer, the value displayed varies between 0 and 100.

Lowering speed Represents the lowering speed as a percentage of the maximum speed.
On turning the potentiometer, the value displayed varies between 0 and 100.

Mix This parameter allows you to select the type of control, passing continuously from position to draft control.
On turning the potentiometer, the value displayed varies between 0 and 1000.

Appendix

Power lift tester

- **SENSORS**

Position Signal read by the position sensor; values range from 0 to 100.
Draft Signal read by the draft sensor; values range from 0 to 1000.

- **PWM OUTPUT**

PWM Value that controls the current supplied to the proportional solenoid valves.
Value ranges 290 to 600.

- **LIFT STATUS**

Status Displays the operating status of the lift system.
Press the control button to view the following status information
LOCK lift locked
STOP lift operative awaiting command
CNTRL lift operating in automatic control mode
FLOAT lift in float mode

2 - Monitor menu

2 - Draft menu

From this menu you can display the following values:

- Pwm Base Up
- PWM Base Down
- PWM Max Up
- PWM Max Down
- Gain Up
- GainDown

The menu displayed on the ART is as follows:

			D	r	a	f	t								
-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
1	-	P	W	M	B	a	s	e	U	p			x	x	x
2	-	P	W	M	B	a	s	e	D	o	w	n	x	x	x
3	-	P	W	M	M	a	x		U	p			x	x	x
4	-	P	W	M	M	a	x		D	o	w	n	x	x	x
5	-	G	a	i	n		U	p						x	x
6	-	G	a	i	n		D	o	w	n				x	X
		[	E	]		E	X	I	T						

These are the characteristic values of the electronic lift control system.
All these values range between 290 and 600.
The default values are as follows:

Values	10N	10S
PWM BaseUp	390	360
PWM BaseDown	400	420
PWM MaxUp	600	600
PWM MaxDown	600	600
Gain Up	25	18
Gain Down	45	35

Appendix

Power lift tester

2 - Monitor Menu

3 - Position Menu

From this menu you can display the following values:

- Pwm Base Up
- PWM Base Down
- PWM Max Up
- PWM Max Down
- Gain Up
- GainDown

The menu displayed on the ART is as follows:

			P	o	s	i	t	i	o	n					
-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
1	-	P	W	M	B	a	s	e	U	p					
2	-	P	W	M	B	a	s	e	D	o	w	n			
3	-	P	W	M	M	a	x		U	p					
4	-	P	W	M	M	a	x		D	o	w	N			
5	-	G	a	i	n		U	p							
6	-	G	a	i	n		D	o	w	n					
		[	E	]		E	X	I	T						

These are the characteristic values of the electronic lift control system.
All these PWM values range between 290 and 600 and the gain values range from 1 to 99.

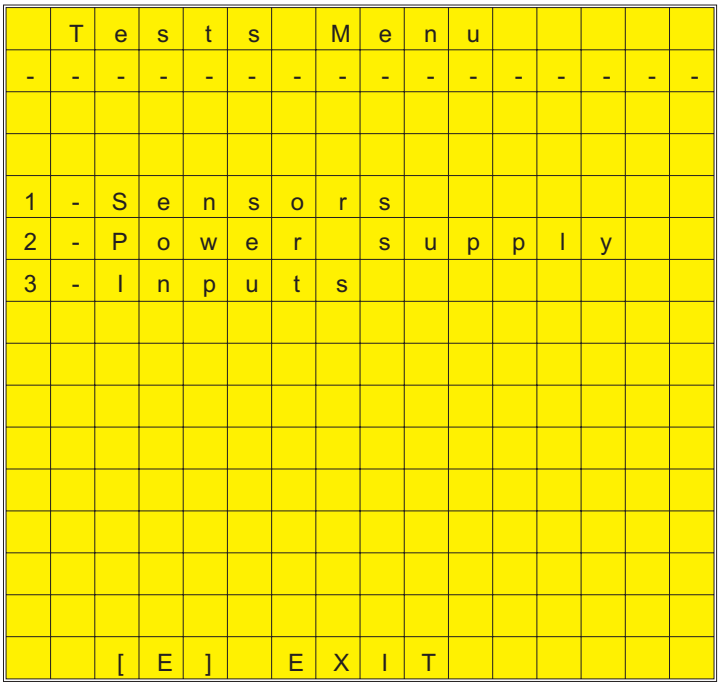
The default values are as follows:

Values	10N	10S
PWM BaseUp	315	315
PWM BaseDown	310	320
PWM MaxUp	600	600
PWM MaxDown	600	600
Gain Up	10	10
Gain Down	50	50

3 - Tests Menu

From this menu you can test the operation of the sensors and actuators connected to the ECU.

The menu displayed on the ART is as follows:



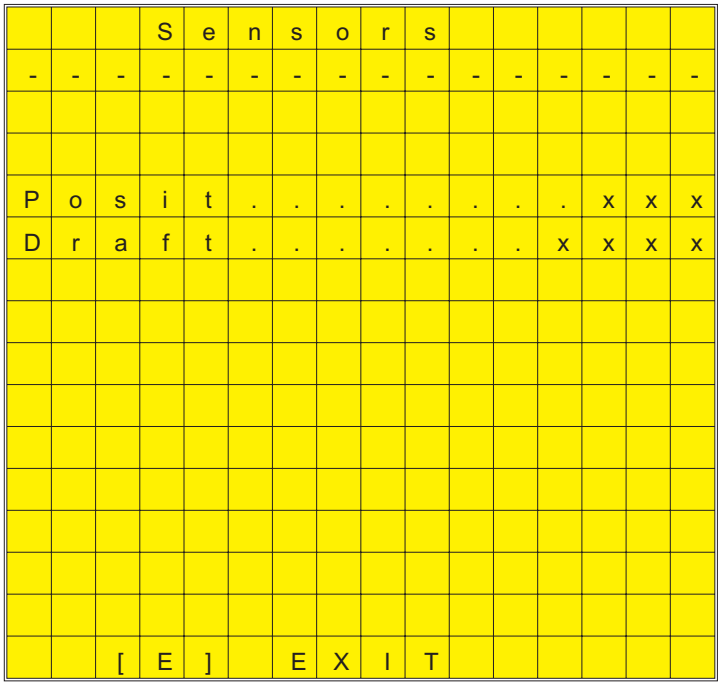
The various functions are described.

3 - Tests Menu

1 - Sensors menu

In this menu you can check the operation of the sensors.

The menu displayed on the ART is as follows:



Where:

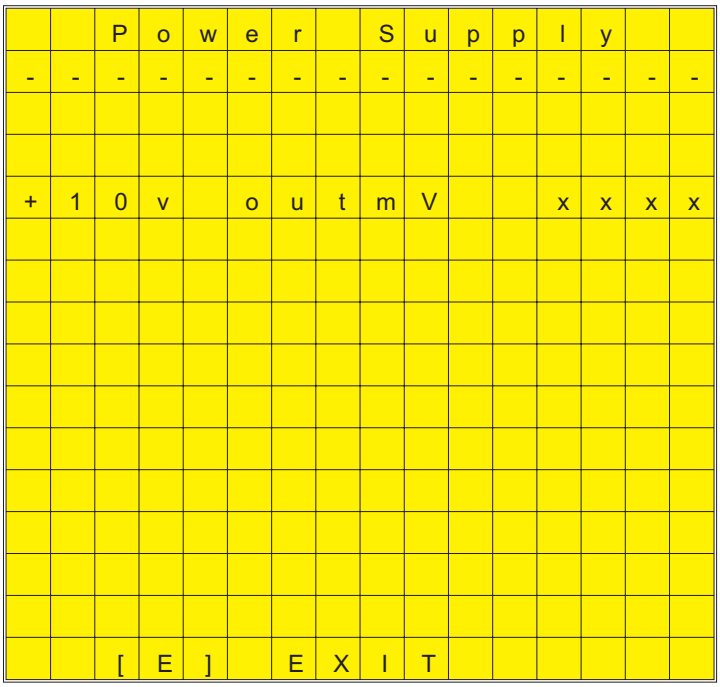
Position Position sensor value ranging from 0 to 100
Draft Draft sensor value ranging from 0 to 1000

3 - Tests Menu

2 - Power supply

In this menu you check the power supply to the ECU.

The menu displayed on the ART is as follows:



Where:

+10V represents the output voltage value in the range 0 to 1000 mV.

Appendix

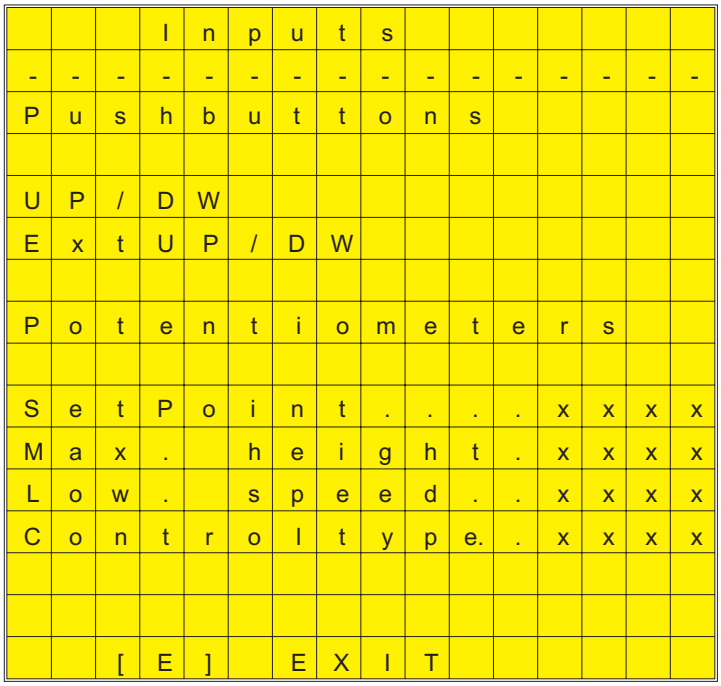
Power lift tester

3 - Tests menu

3 - Inputs

In this menu you can check the operation of the ECU inputs.

The menu displayed on the ART is as follows:

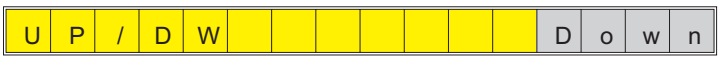


Where:

Up/Down rocker switch on console

On pressing the UP side of the rocker switch with the lift lock released, the display will appear as follows:

On pressing the DOWN side of the rocker switch with the lift lock released, the display will appear as follows:



ExtUP/DW ground-operated pushbuttons

On pressing the button with the UP arrow when the lift lock is released, the display will appear as follows

On pressing the button with the DOWN arrow when the lift lock is released, the display will appear as follows

Set Point on turning the potentiometer, the value varies in the range 0 to 1000

Max. height on turning the potentiometer, the value varies in the range 0 to 100

Lowering speed on turning the potentiometer, the value varies in the range 0 to 100

CONTROL TYPE on pressing the control pushbutton, the following conditions are displayed

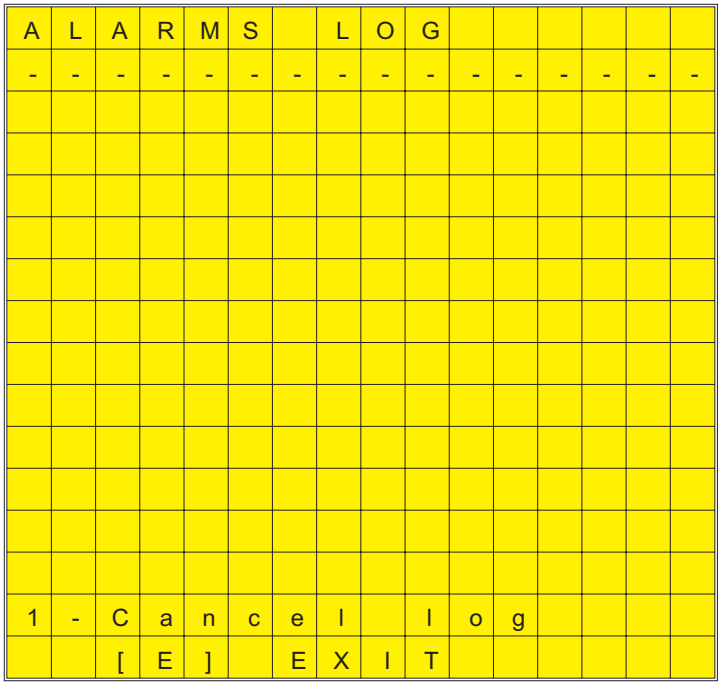
Appendix

Power lift tester

5 - Alarms menu

Displays alarm log.

The screen displayed on the ART is as follows:



The various functions are described below.

Tester message	Possible cause of fault	ECU action	N° of flashes
UPSV DISCONN.	*UP proportional solenoid valve disconnected.* **- Wiring error.** Check electrical continuity **-Solenoid faulty.** Check solenoid **-ECU faulty.**	Loss of lift system function.	11
UPSV S.C.	Short circuit of UP proportional solenoid valve. **- Wiring error.** Check electrical continuity. **- Solenoid faulty.** Check solenoid. **- ECU faulty.**	Loss of lift system function.	12
DWSV DISCONN.	DOWN proportional solenoid valve disconnected. **- Wiring error.** Check electrical continuity. **- Solenoid faulty.** Check solenoid. **- ECU faulty.**	Loss of lift system function.	13
DWSV S.C.	*Short circuit of DOWN proportional solenoid valve.* **- Wiring error.** Check electrical continuity. **- Solenoid faulty.** Check solenoid. **- ECU faulty.**	Loss of lift system function.	14
EPROM CHECK	*EPROM error.* **- ECU faulty.**	Total lockup of lift system	15
POS.SENS.S.C.	*Short circuit of position sensor.* **- Wiring error.** Check electrical continuity. **- Sensor faulty.** Check sensor. **- ECU faulty.**	Loss of lift system function.	21

Appendix

Power lift tester

Tester message	Possible cause of fault	ECU action	N° of flashes
POS.SENS. DIS.	*Position sensor disconnected.* **- Wiring error.** Check electrical continuity. **- Sensor faulty.** Check sensor. **- ECU faulty.**	Loss of lift system function.	22
GEN.FAIL.CPU	General failure of CPU. **- ECU faulty.**	Loss of lift system function.	23
DRAFT SENS. DIS.	*Draft sensor not connected.* **- Wiring error.** Check electrical continuity. **- Sensor faulty.** Check sensor. **- ECU faulty.**	Loss of draft control function.	41
DRAFT SENS. S.C.	Draft sensor short circuit. **- Wiring error.** Check electrical continuity. **- Sensor faulty.** Check sensor. **- ECU faulty.**	Loss of draft control function.	42
EEPROM CHECK	*EEPROM error.* **- ECU faulty**	Default values (for 10N) uti-lised	45
NO V. SENSOR	*No voltage at sensor.* **- Wiring error.** Check electrical continuity. **- ECU faulty.** Check using Tests menu on ART.	Loss of lift system function.	50

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NOTES

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